

THE ISEGORIA MATHEMATICS SERIES · VOLUME II

Structure Before Algebra

A complete textbook of solution technique

Integrals · Derivatives · Differentials · Limits · Logic

A hard problem is hard because of what you fail to notice in the first thirty seconds, not because of the algebra that follows.

Structure Before Algebra

A complete textbook of solution technique in calculus and logic.

Volume II of the Isegoria mathematics series. Volume I is *The Complete Daily Integral Field Guide*.

This book was assembled from a working corpus of solution archetypes: the ninety-six core generator families and the several hundred technique variants catalogued from daily problem sets, deduplicated into the technique inventory that organises these chapters. Every closed form, limit, derivative, solution and count asserted in the text was verified numerically or by exhaustive enumeration before it was typeset, and a number of errors in the source catalogue were corrected in the process.

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How to use this book

This is a book about recognition, not about theory. It assumes you can differentiate, integrate, and manipulate an inequality, and it spends none of its pages proving that you can. What it does instead is catalogue, as completely as we could manage, the moves that actually solve problems, and attach to each one the surface feature that should make you reach for it.

Every technique in the book appears as a numbered block with the same five parts. The **CUE** is the visual or structural trigger: what you see on the page that should make this technique come to mind. The **METHOD** is the mechanism, stated in as few sentences as it takes. The worked example is one full instance, chosen to be hard enough to be worth the space. The trap box names the specific way that technique is most often botched, which is usually more valuable than the method itself. And the see-also line names siblings, because techniques come in families and knowing the family is most of knowing when to switch.

Chapter numbers and technique numbers are stable: technique 22.7 is the seventh technique of Chapter 22, and it is referred to that way in the index. Every chapter closes with a recognition matrix, a compressed table of surface features against first moves, and Chapter 53 collects the most useful rows of all of them into five cross-domain tables. If you own the book for one thing, own it for those tables.

There are three reasonable ways to read it.

Front to back, which works: the parts are ordered so that each one uses the previous. Limits come before derivatives because differentiability is a limit; derivatives come before integrals because half of integration is recognising a derivative; differential equations come after both; and the logic part stands alone but shares the book's organising idea, which is that difficulty is camouflage.

By domain, treating each part as a self-contained manual. Nothing in the integral chapters depends on the logic chapters, and the cross-references are by name rather than by number so that they survive being read out of order.

By cue, which is how the book is designed to be used once you have read it once. You have a problem; you look at the recognition matrices; you find the row that matches the surface; you go to the chapter that owns the move. The index at the back is by technique name for the same purpose.

A word on what the book leaves out. There are no exercises. This is deliberate: problems are abundant and technique catalogues are not, and a reader who wants practice has a daily supply of it. There are also no proofs of the standard theorems, only of the facts a solver actually uses, and where a result is quoted rather than derived the text says so.

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PART I

The method

Reading a problem

What this chapter is for. This chapter teaches the thirty seconds before the first line of algebra. Every problem in this book was built by someone: a constant was chosen, an exponent was inserted, an interval was selected, and the choices were made so that one particular structure would appear and one particular cancellation would happen. The chapter gives you five questions to interrogate an expression with, a protocol that orders those questions, and a vocabulary for the kinds of camouflage a problem-setter uses to hide the structure. At the end you should be able to look at an unfamiliar integral, limit, derivative, differential equation, or puzzle and say, before computing anything, what family it belongs to, what is decoration, and what single move would collapse it.

The first thirty seconds

A hard problem is almost never hard because of the algebra. The algebra is downstream. A problem is hard because of what you failed to notice while you were still reading it, and everything after that failure is expensive: three pages of correct manipulation that arrive nowhere, a substitution that is legal and useless, an integration by parts that reproduces the original integrand with the sign unchanged.

The corrective is to treat the statement itself as evidence. Consider what it takes to write down

$$\int_0^1 \frac{x^4(1-x)^4}{1+x^2} dx.$$

Nobody arrives at this by accident. The exponent 4 is not 3 or 5; the numerator is not $x^4(1+x)^4$; the interval is not $[0, 2]$. Somebody divided $x^4(1-x)^4$ by x^2+1 , saw that the remainder was the constant 4, and built the problem backwards from that fact. Once you know a division was intended, the problem is a two-minute exercise; the answer is $\frac{22}{7} - \pi$, roughly 0.0012645, which is the classical proof that $22/7$ overestimates π . Once you do not know it, no amount of trigonometric substitution will save you.

This is the central claim of the book, and it is worth stating flatly.

THE DESIGN PRINCIPLE

A constructed problem carries the fingerprints of its construction. An arbitrary-looking exponent, an interval endpoint that matches a constant inside the integrand, a coefficient that makes a discriminant vanish, a starting configuration whose entries sum to something divisible by the number of entries: these are not coincidences, they are the author's working shown through the paper. Reading them is not cleverness. It is literacy.

There is a sharp asymmetry of cost here. Reading a problem well costs thirty seconds and, at worst, wastes them. Reading it badly costs the whole attempt, because a wrong classification is self-reinforcing: having decided that an integral needs parts, you will keep doing parts, and each round will look like progress. The expected value of the first thirty seconds is therefore enormous, and the discipline this chapter asks for is simply to spend them deliberately instead of accidentally.

It also changes what practice means. Practising calculation means doing many problems of a type you already recognise. Practising recognition means looking at many problems you do not yet know how to solve, saying out loud what kind of object each one is, and only then checking. The second is uncomfortable and much faster.

Two objections arrive immediately. The first is that this only works on artificial problems. It is true that a randomly generated integral is usually unsolvable in closed form and that a randomly generated puzzle is usually false; but the problems you will actually be asked, in examinations, in competitions, in this book, and in the applied settings where somebody has already done the modelling, are constructed objects. The second objection is that recognition cannot be taught, only accumulated. This is half right. What can be taught is the list of things to look at, and the order to look at them in. Recognition is then a matter of having looked at enough expressions with the right questions in hand.

Five questions to ask before you compute

Before touching an expression, ask it five things. They are ordered so that the cheap questions come first, and so that a positive answer to an early one usually makes the later ones unnecessary.

1. **What is the domain of the object?** Is this an elementary problem (the answer is built from powers, exponentials, logarithms, and trigonometric functions), a special-function problem (the answer needs Γ , B , ψ , Li_2 , erf , Si , or a ζ value), or a discrete problem (the answer is a count, a residue, a parity, or a strategy)? Naming the domain tells you which toolbox to open and, just as importantly, which one to keep shut.
2. **What symmetry does it have?** Parity in x ; reflection $x \mapsto a + b - x$ across an interval; the involution $x \mapsto 1/x$ on $(0, \infty)$; periodicity; invariance under swapping two parameters; invariance of a discrete process under a relabelling. Symmetry is the cheapest structure to test and the most likely to be the intended route.
3. **What is engineered about it?** Hunt for the suspiciously clean feature: a bound that is exactly attainable, an exponent that could have been anything, a constant that makes a numerator divisible by a denominator, a coefficient that makes a quadratic a perfect square, sixteen numbers whose exponents sum to a multiple of sixteen.
4. **What does it reduce to as raw algebra?** Before any calculus, expand, cancel, divide, apply the Pythagorean and logarithm laws, and rationalise. A surprising proportion of frightening expressions are a short expression wearing a costume.
5. **What would make it trivial?** Name, explicitly, the one substitution or identity that would collapse the problem to something you can finish in a line. Then ask whether the problem was built to admit exactly that move. This question is run backwards from the answer, and it is the most powerful of the five.

The rest of this section makes each question into a technique. These are meta-techniques: they do not compute anything themselves, they decide what will be computed. Every one of them recurs in every part of the book.

CORE

1.1 Simplify as raw algebra first

CUE An expression whose difficulty is visual: stacked fractions, a numerator and a denominator that share obvious factors, trigonometric squares in both, nested exponentials and logarithms, a polynomial quotient with numerator degree at least that of the denominator.

METHOD Do the algebra you would do if calculus did not exist. Expand and cancel; divide polynomials until the numerator has lower degree; apply $\sin^2 + \cos^2 = 1$ and $\sec^2 - \tan^2 = 1$ to numerator and denominator *separately*; collapse $e^{\ln u}$, $\ln(ab)$, and $a^{\log_a b}$; rationalise a difference of radicals by its conjugate. Only then classify the problem. The design principle behind this technique is that a setter hides a simple object by multiplying it by something that cancels.

WORKED EXAMPLE

$\int_0^1 \frac{x^4(1-x)^4}{1+x^2} dx$. Expanding, $x^4(1-x)^4 = x^8 - 4x^7 + 6x^6 - 4x^5 + x^4$, and dividing by $x^2 + 1$ gives

$$\frac{x^4(1-x)^4}{1+x^2} = x^6 - 4x^5 + 5x^4 - 4x^2 + 4 - \frac{4}{1+x^2}.$$

Every term is now a power. Integrating on $[0, 1]$: $\frac{1}{7} - \frac{4}{6} + 1 - \frac{4}{3} + 4 - \pi = \frac{22}{7} - \pi \approx 0.0012645$.

WHERE THIS GOES WRONG

Cancelling a factor without checking that it is nonzero on the interval, and cancelling $\cos^2 x$ from numerator and denominator on an interval containing $\pi/2$. The cancellation is an identity of functions only where both sides are defined.

SEE ALSO Polynomial long division; trigonometric identity pre-processing; conjugate multiplication.

Simplification tells you what the object *is*. The next question is what *kind* of object it is, because the same surface can belong to three different worlds and the worlds have disjoint toolkits. The single most common waste of effort in calculus is hunting for an elementary antiderivative that provably does not exist.

1.2 Name the domain before choosing a tool

CUE Any expression you are about to attack. Specifically: a logarithm divided by a linear or quadratic factor, an integrand with e^{-x^2} , $\sin x/x$, or x^s with s non-integer, a sum of reciprocal squares, a factorial inside a root.

METHOD Decide, in one sentence, whether the answer lives among elementary functions, among the named special functions, or among integers. Log-over-linear and log-over-quadratic on $[0, 1]$ or $(0, \infty)$ almost always land on $\zeta(2)$, $\zeta(3)$, Catalan's constant, or a Li_2 value; $x^{s-1}e^{-x}$ lands on Γ ; $x^{m-1}(1-x)^{n-1}$ lands on B ; a count or a probability lands on an integer or a rational. Then commit to that toolkit.

WORKED EXAMPLE

$\int_0^1 \frac{\ln x}{1+x} dx$. There is no elementary antiderivative, and hunting for one is the failure mode. The domain is the ζ family, so expand the geometric series and integrate termwise: $\int_0^1 x^n \ln x dx = -\frac{1}{(n+1)^2}$, hence

$$\int_0^1 \frac{\ln x}{1+x} dx = \sum_{n \geq 0} (-1)^n \cdot \frac{-1}{(n+1)^2} = - \sum_{m \geq 1} \frac{(-1)^{m-1}}{m^2} = -\frac{\pi^2}{12} \approx -0.822467.$$

WHERE THIS GOES WRONG

Declaring the special-function domain too early. Many integrals with $\ln$ and $\arctan$ in them are elementary after one reflection; check the symmetry question before you reach for a polylogarithm.

SEE ALSO Termwise integration of a Maclaurin series; the log-power moment; Gamma-function moment.

Having named the domain, test the cheapest structure. Symmetry costs one substitution to check and, when it is present, usually finishes the problem outright. The specific form to look for is an *involution*: a map of the domain to itself which, applied twice, is the identity. Reflection $x \mapsto a+b-x$ on $[a, b]$, negation $x \mapsto -x$ on $[-a, a]$, and inversion $x \mapsto 1/x$ on $(0, \infty)$ are the three that account for most of the definite integrals in this book.

CORE

1.3 Look for the involution

CUE A definite integral whose interval is fixed by an obvious map: $[-a, a]$ under $x \mapsto -x$, $[a, b]$ under $x \mapsto a+b-x$, $(0, \infty)$ under $x \mapsto 1/x$, $[0, \pi/2]$ under $x \mapsto \pi/2 - x$. In discrete problems: a process invariant under reversing the order of moves, or a bijection pairing each configuration with a partner.

METHOD Apply the involution, call the result I as well, and *add*. If $g(x) + g(\sigma(x))$ is constant, or if the two copies differ by a sign, the integral evaluates without an antiderivative. The three outcomes are: the sum is constant (answer is length times constant over two), the sum is zero (answer is zero), or the two are equal (you have gained an identity to feed to a later step).

WORKED EXAMPLE

$J = \int_0^\infty \frac{\ln x}{1+x^2} dx$. Under $x \mapsto 1/x$ the interval is fixed, $dx \mapsto dx/x^2$, and $\frac{\ln x}{1+x^2} dx \mapsto -\frac{\ln x}{1+x^2} dx$. So $J = -J$ and $J = 0$. No antiderivative is needed, and none is pleasant.

WHERE THIS GOES WRONG

Forgetting that the involution must fix the *interval*, not merely the integrand. On $(0, 1)$ the map $x \mapsto 1/x$ sends the interval to $(1, \infty)$; that is a useful move, but it is a change of region, not a symmetry, and the two pieces must then be recombined.

SEE ALSO King's rule; parity splitting on a symmetric interval; complementary sine and cosine.

The involution question is really a special case of a broader habit: asking which features of the problem are load-bearing and which are decoration. A setter who wants to hide a reflection will bury it under an exponent chosen to look forbidding. The counter-move is to replace the forbidding thing by a symbol and watch it fail to appear in the answer.

1.4 Delete the ugly constant

CUE A constant that has no business being there: an exponent of $\sqrt{2}$ or π or 2026, a coefficient like $1/7$ attached to one term, an evaluation point such as $x = 67$, a modulus of 2027, a nested function like $\cos(x^{99})$ that is doing nothing near the point of interest.

METHOD Replace the offending constant by a letter, or delete the offending factor entirely, and solve the resulting skeleton problem. If the skeleton has a clean answer and the letter never appears in it, the constant was decoration and the skeleton answer is the answer. If the letter does appear, you have learned that the constant is structural, and you should then ask what identity it completes.

WORKED EXAMPLE

$\lim_{x \rightarrow 0} \frac{e^{x^3} - 1 - x^3}{x^6 \cos(x^{99})}$. Delete $\cos(x^{99})$: it tends to 1 and is continuous, so it cannot affect the value. The skeleton is $\lim_{u \rightarrow 0} \frac{e^u - 1 - u}{u^2}$ with $u = x^3$, which is $\frac{1}{2}$ by the second-order expansion of e^u . The limit is $\frac{1}{2}$; numerically the quotient at $x = 0.03$ is 0.5000045.

WHERE THIS GOES WRONG

Deleting a constant that is structural. The exponent 2026 paired with a modulus of 2027 is not noise: it completes a cyclotomic identity, since $\prod_{k=1}^{2026} (x^2 - 2x \cos \frac{k\pi}{2027} + 1) = \frac{x^{4054} - 1}{x^2 - 1}$. Test the constant, do not assume it.

SEE ALSO Parameter differentiation; the decoy evaluation point; exponent-independent reflection integrals.

The last of the five questions inverts the direction of work. Instead of pushing the given expression forwards, you postulate the shape of the answer and differentiate, expand, or count backwards to see what would produce it. This is how antiderivatives that look impossible are found in one line, and it is how a setter's intentions become visible.

1.5 Work backwards from a clean answer

CUE An integrand of the form $e^x \cdot (\text{rational})$, $e^x(f + f')$, or $\frac{u'v - uv'}{v^2}$; a sum whose terms are differences of a shifted quantity; an answer that the problem has told you is an integer, or a nice multiple of π , or independent of a parameter.

METHOD Guess the family the answer belongs to, write the general member with undetermined pieces, and differentiate (or difference, or count) it forwards. Matching one or two coefficients then pins it down. The reverse product rule $\int e^x(f(x) + f'(x)) dx = e^x f(x) + C$ and the reverse quotient rule are the two workhorses; for series, the corresponding move is to guess $a_k = b_k - b_{k+1}$ and solve for b .

WORKED EXAMPLE

$\int_0^1 \frac{x e^x}{(1+x)^2} dx$. Guess that the antiderivative is e^x over something linear, and try $\frac{e^x}{1+x}$. Differentiating,

$$\frac{d}{dx} \frac{e^x}{1+x} = \frac{e^x(1+x) - e^x}{(1+x)^2} = \frac{x e^x}{(1+x)^2},$$

which is the integrand exactly. So the value is $\frac{e}{2} - 1 \approx 0.359141$.

WHERE THIS GOES WRONG

Guessing a family that is too narrow and concluding that no antiderivative exists. If $e^x \cdot \frac{P}{Q}$ fails with P constant, allow P to be a general polynomial of the degree of Q before abandoning the guess.

SEE ALSO Reverse product and quotient rule; telescoping sums; engineering the antiderivative v .

The order of attack

The five questions become a protocol when you fix their order. The order is chosen so that each step is cheaper than the one after it, and so that a success at any step sends you back to the beginning with a smaller problem.

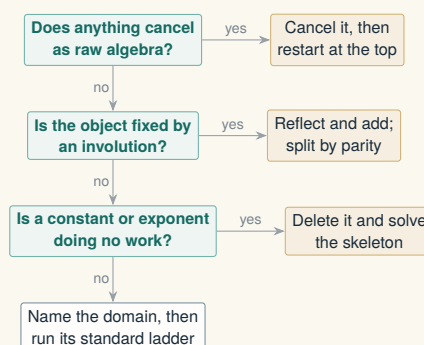
1. **Read the question, not the label.** Decide what is actually being asked for: a value, a closed form, an existence claim, a bound, a strategy, a count. A problem filed under "derivatives" whose content is a Vieta relation is a Vieta problem.
2. **Simplify as raw algebra.** If anything cancels, cancel it, and restart at step 1 with the reduced object.
3. **Name the domain.** Elementary, special-function, or discrete. Commit.

4. **Test the symmetries.** Parity, reflection, inversion, periodicity, parameter exchange, and, in discrete problems, invariance of a candidate quantity under the move.
5. **Identify the decoration.** Replace every arbitrary-looking constant by a letter. Solve the skeleton.
6. **Ask what would make it trivial.** Name the one substitution or identity, and check whether the problem was built to admit it.
7. **Only now execute.** Choose the standard technique for the family you have identified, and run it.
8. **Verify.** By a method independent of the one that produced the answer. This is Chapter 2, and it is not optional.

The first six steps should take under a minute on a problem of ordinary difficulty. If they take longer, that is usually a sign that step 2 was done carelessly.

Two refinements. The protocol is a loop, not a list: every success returns you to step 1 with a new object, and the new object deserves the same interrogation as the old one. A substitution that turns an integral over $(0, \infty)$ into an integral over $[0, \pi/2]$ has handed you a fresh interval with fresh symmetries, and failing to re-ask the symmetry question at that moment is the commonest way to solve half a problem and stall.

The second refinement concerns what to do when nothing triggers. If steps 2 through 6 all come back empty, the problem is probably in the special-function or parameter-method families, where the structure is not visible in the statement but is created by you: insert a parameter where none was given and differentiate in it, or introduce an integral representation of one factor and exchange the order. That is a deliberate escalation, and it belongs after the cheap questions rather than instead of them. It is also the point at which a genuinely hard problem announces itself: hard problems are those where you must manufacture the structure, not merely notice it.



Difficulty is layers, not depth

It is tempting to rank problems by the sophistication of the technique they need. That ranking is close to useless. A problem requiring Feynman's trick and a Beta-function reflection can be routine, because both steps are visible from the statement. A problem requiring nothing beyond a polynomial division can be brutal, because the division is hidden under four layers of disguise. Difficulty, in a constructed problem, measures the number of layers of camouflage between the surface and the structure.

WHAT DIFFICULTY MEASURES

Depth is how advanced the decisive technique is. Camouflage is how many transformations separate the statement from the form in which that technique is visible. A problem's difficulty tracks camouflage far more closely than it tracks depth. A three-layer problem solved by long division is harder than a one-layer problem solved by contour integration.

Camouflage comes in a small number of species, and they are worth naming, because a named thing is a thing you can look for.

Exponents that do not matter. An exponent is inserted precisely because it looks like it must be handled. In $\int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$ the square roots are inert: under $x \mapsto \pi/2 - x$ the integrand becomes its own complement, the two copies sum to 1, and the value is $\pi/4$ for the square root, the cube root, or any exponent at all. The rule of thumb: an exponent that appears identically in every occurrence of a function is usually decoration.

Parameters that cancel. A parameter is inserted so that the reader will try to track it. In $\int_{-1}^1 \frac{x^2}{1+e^x} dx$ the exponential looks like the difficulty. Writing $g(x) = \frac{1}{1+e^x}$, the identity $g(x) + g(-x) = 1$ means that reflecting and adding leaves $\int_{-1}^1 x^2 dx$

over two copies, that is $\int_0^1 x^2 dx = \frac{1}{3}$. The exponential never appears in the answer. Any g with $g(x) + g(-x) = 1$ would do.

Integrands that are secretly a derivative. The reverse product and quotient rules generate integrands of arbitrary ugliness from clean antiderivatives. If a rational or exponential integrand has a squared denominator and a numerator of one degree less than you expect, differentiate the obvious candidate before doing anything else. The demonstration in the “work backwards” block above is exactly this species.

Sums that are secretly telescoping. $\sum_{k \geq 1} \arctan \frac{1}{k^2 + k + 1}$ looks like it needs a transform. It does not: $\arctan(k+1) - \arctan k = \arctan \frac{1}{1+k(k+1)}$, so the sum telescopes to $\frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$. The signature is a term whose denominator is a shifted product, or whose form matches a subtraction identity for arctan, ln, or a partial fraction.

Expressions that are secretly constant. The extreme case of camouflage. A construction with nested trigonometry, a geometric series, and a conjugate logarithm can collapse to x : for instance $x \cdot \frac{1+\cos 2x}{2} \cdot \sum_{n \geq 0} \left(\frac{\tan^2 x}{1+\tan^2 x} \right)^n = x \cos^2 x \cdot \sec^2 x = x$, so its derivative is 1 everywhere and the elaborate evaluation point in the problem statement is decoration. Before differentiating a monster, evaluate it at two convenient points and see whether it moves.

Endpoints chosen to make a symmetry exist. The most delicate species, because the camouflage is in what is *not* written. An integral over $[0, \pi/4]$ rather than $[0, \pi/2]$, or over $[1, e]$ rather than $[1, 2]$, is announcing which reflection or substitution was intended: $\theta \mapsto \pi/4 - \theta$ in the first case, $x \mapsto e/x$ or $u = \ln x$ in the second. When an endpoint is an odd-looking constant, ask what map fixes the interval it defines, and what substitution sends that interval to something standard.

Two closing remarks on camouflage. First, the layers compose: a real problem will hide a telescoping sum inside a substitution inside an algebraic simplification, and peeling one layer is supposed to reveal the next, not the answer. If a first move produces something that is merely a different mess, that is normal; ask the five questions again of the new object. Second, camouflage is detectable in the aggregate even when it is invisible in a single term. Symptoms are: an answer promised to be an integer, a problem that specifies far more constants than the answer could plausibly depend on, and an instruction to evaluate at a point that no honest calculation would prefer.

The protocol in action

Three problems, from three different parts of the book. Each is solved in a few lines, and in each case the lines are short only because the reading was done first. Every value below has been checked numerically.

An integral

$$I = \int_0^1 \frac{\ln(1+x)}{1+x^2} dx.$$

Domain. The logarithm over a quadratic on $[0, 1]$ suggests the Li_2 world, which would be a long road. Hold that thought.

Symmetry. The interval $[0, 1]$ is not fixed by any obvious involution, and the integrand has no parity. But $1 + x^2$ in a denominator is the signature of $x = \tan \theta$, and that substitution maps $[0, 1]$ to $[0, \pi/4]$, an interval whose reflection $\theta \mapsto \pi/4 - \theta$ is an involution.

Engineering. The 1 in $\ln(1+x)$ and the upper limit 1 are the same 1. That is the fingerprint: change either and the collapse dies.

Substituting $x = \tan \theta$, with $dx = \sec^2 \theta d\theta$ and $1 + x^2 = \sec^2 \theta$,

$$I = \int_0^{\pi/4} \ln(1 + \tan \theta) d\theta.$$

Now reflect. Since $\tan(\frac{\pi}{4} - \theta) = \frac{1 - \tan \theta}{1 + \tan \theta}$,

$$1 + \tan\left(\frac{\pi}{4} - \theta\right) = \frac{2}{1 + \tan \theta},$$

so the reflected integrand plus the original is $\ln 2$, a constant. Adding the two copies, $2I = \frac{\pi}{4} \ln 2$ and

$$I = \frac{\pi \ln 2}{8} \approx 0.2721983.$$

The special-function domain was a decoy: the answer is elementary because the reflection was available.

A limit

$$L = \lim_{n \rightarrow \infty} \frac{1}{n} \left(\frac{(2n)!}{n!} \right)^{1/n}.$$

Domain. Factorials with an n th root read as “Stirling”. Stirling would work, and it would be four lines of asymptotic book-keeping.

Reduction as raw algebra. $\frac{(2n)!}{n!} = \prod_{k=1}^n (n+k)$, so the whole expression is $\left(\prod_{k=1}^n \frac{n+k}{n} \right)^{1/n}$: the n th root of a product of n factors, each of which depends on k only through k/n . That is a geometric mean, and the $1/n$ prefactor was placed exactly to make it one.

What would make it trivial. A logarithm, which converts the geometric mean into an arithmetic mean, which is a Riemann sum. So

$$\ln L = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \ln\left(1 + \frac{k}{n}\right) = \int_0^1 \ln(1+x) dx = 2 \ln 2 - 1,$$

and therefore $L = e^{2 \ln 2 - 1} = \frac{4}{e} \approx 1.4715178$. At $n = 5000$ the expression equals 1.47162, converging from above as expected for a concave integrand.

A puzzle

The sixteen numbers $2^0, 2^1, \dots, 2^{15}$ are written on a board. Repeatedly, two of them are erased and replaced by their arithmetic mean and their harmonic mean. The sixteen values converge to a common limit L . Find L .

Read the question. It asks for a terminal state under a repeated move, and the answer is claimed to be independent of which pairs are chosen and in what order. Order-independence is the loudest possible signal that an invariant governs the process.

Hunt the invariant. Test the candidates against the move $\{a, b\} \mapsto \{\frac{a+b}{2}, \frac{2ab}{a+b}\}$. The count is trivially preserved. The sum is not: it changes unless $a = b$. The product is preserved exactly, because

$$\frac{a+b}{2} \cdot \frac{2ab}{a+b} = ab.$$

So $\prod_i a_i$ is invariant across the whole board.

Engineering. The initial product is $2^{0+1+\dots+15} = 2^{120}$, and 120 is divisible by 16. The setter chose sixteen consecutive powers precisely so that $120/16$ would be a half-integer with a clean radical.

Existence of the limit. The invariant alone constrains a limit that might not exist, so this step is obligatory: since $AM \geq GM \geq HM$, each move weakly contracts the spread between the largest and smallest entries and never moves an entry outside the current range, so the values converge to a common L . Then $L^{16} = 2^{120}$, giving

$$L = 2^{15/2} = 128\sqrt{2} \approx 181.0193.$$

Simulating four hundred thousand random moves from the given start indeed drives every entry to 181.01934.

Recognition matrix

If you see	First move	If that fails
A quotient with $\deg(\text{num}) \geq \deg(\text{den})$	Divide; read the remainder	Partial fractions
Integral over $[-a, a]$	Split into even and odd parts	Reflect and add
$f(x) + f(a + b - x)$ in a denominator	Reflection $x \mapsto a + b - x$	Parameter differentiation
Integral over $(0, \infty)$ with x and $1/x$ balanced	Inversion $x \mapsto 1/x$	Glasser or Cauchy-Schlömilch
An exponent that is $\sqrt{2}$, π , or a big integer	Replace it by p ; expect it to cancel	Test $p = 1$ numerically
ln over a linear or quadratic factor on $[0, 1]$	Expand geometrically; expect ζ or Li_2	Feynman in a parameter
Factorials or $n!$ under an n th root	Take logs; look for a Riemann sum	Stirling
$\frac{1}{n} \sum_{k=1}^n f(k/n)$ in any disguise	Recognise the Riemann sum	Squeeze between integrals
A product of many factors, derivative wanted	Logarithmic differentiation	Check whether the product telescopes
An evaluation point of 0, 1, or $\pi/2$	Find the factor that vanishes there	Expand to first surviving order
Numerator one degree below a squared denominator	Reverse quotient rule	Parts with $dv = dx$
$e^x(f(x) + f'(x))$	Reverse product rule: answer is $e^x f(x)$	Tabular integration
Term of the form $b_k - b_{k+1}$	Telescope	Abel summation
$\frac{dy}{dx} = F(y/x)$	Substitute $v = y/x$	Test exactness
$\frac{dy}{dx} + P(x)y = Q(x)$	Integrating factor $e^{\int P}$	Variation of parameters
A move replacing two items by two items	Test sum, product, parity, sum of squares	Monovariant plus termination
"Prove this is impossible"	Colour, or take residues mod k	Extremal principle
A statement about its own truth	Encode as a biconditional	Truth table on 2^n rows
An answer claimed independent of a parameter	Set the parameter to its easiest value	Differentiate in the parameter
A limit in two variables	Test the paths $y = mx$ and $y = x^2$	Polar-coordinate squeeze

Knowing that you are right

What this chapter is for. A solution you cannot check is a guess with working attached. This chapter treats verification as a technique with its own cues, its own methods, and its own traps, exactly like substitution or parts. It presents eight checks ordered by cost, from a five-second scaling argument to a full recomputation by an independent route, and it names the specific errors each one is built to catch. At the end you should have a reflex: on finishing any problem, you reach for the cheapest check that could possibly falsify the answer you just wrote, and you run it before you believe yourself.

Verification is a technique, not a virtue

Checking your work is usually taught as a moral instruction, which is why it is usually ignored. It is better understood as a decision problem. You have an answer produced by a chain of ten steps, each of which you believe with probability 0.98; the answer is therefore about 80 percent likely to be right, which is not good enough. A check that would fail on most wrong answers and passes on this one moves that number a long way for very little effort. The question is never whether to check, but which check has the best ratio of discriminating power to cost.

That framing has two consequences. The first is that a check must be capable of failing. Recomputing the same algebra a second time is not verification, because the error that survived the first pass survives the second; the mind that made the slip is the mind rechecking it. A real check uses different information: a bound, a special case, a numerical value, a symmetry, a second technique. The second consequence is that checks should be run in order of cost, because a cheap check that kills the answer saves you from paying for the expensive one.

THE DISCRIMINATING-CHECK RULE

A check is worth running in proportion to the fraction of plausible wrong answers it would reject. Confirming that a positive integral came out positive is nearly free but rejects only the sign errors; confirming that it agrees with numerical quadrature to six digits rejects essentially everything. Spend the first ten seconds on the checks that cost nothing, and the next two minutes on the one check that would be decisive.

The verification ladder

The eight checks below are arranged by cost. The first four cost seconds and should be automatic; the last four cost minutes and are what you run on an answer that matters. On any given problem two or three of them will apply, and applying two independent ones is worth much more than applying one twice.

2.1 Dimensional and scaling sanity

CORE

CUE An answer containing a parameter that carries a scale: a length a , a rate k , an upper limit T . Also any integral over a scale-invariant domain such as $(0, \infty)$.

METHOD Substitute $x = \lambda u$ and see how both sides must transform. If the integrand is homogeneous of degree d in x and a jointly and the domain is scale-invariant, then the answer is a^{d+1} times a pure number, and the power of a is forced before any computation. Any claimed closed form with the wrong power is wrong.

WORKED EXAMPLE

Claim: $\int_0^\infty \frac{dx}{(x^2 + a^2)^2} = \frac{\pi}{4a^3}$ for $a > 0$. Substituting $x = au$ gives $\frac{1}{a^3} \int_0^\infty \frac{du}{(u^2 + 1)^2}$, so the answer is forced to be $c a^{-3}$; the constant $c = \pi/4$ then follows from the single case $a = 1$. A claimed answer of $\pi/(4a^2)$ or $\pi/(2a^3)$ is refuted by the first half of that argument alone.

Using the scaling argument on a domain that is not scale-invariant. On $[0, 1]$ the substitution $x = au$ moves the upper limit, so it yields a relation between two different integrals, not a constraint on one.

SEE ALSO Degenerate and limiting cases; symmetry of the answer.

Scaling constrains the shape of the answer. The next check constrains its value, by pushing a parameter to a place where you already know what the answer must be.

CORE

2.2 Degenerate and limiting cases

CUE Any closed form containing a parameter, an index, a dimension, or a count. Also any general formula you have just derived and are about to trust.

METHOD Set the parameter to 0, to 1, or let it tend to ∞ , and check that the formula degenerates to something you know independently. Prefer the extreme case to the convenient one: at $a = 1$ many wrong answers coincide with the right one, whereas at $a \rightarrow 0$ and $a \rightarrow \infty$ they separate.

WORKED EXAMPLE

Claim: $\int_0^1 x^m (1-x)^n dx = \frac{m! n!}{(m+n+1)!}$ for non-negative integers m, n . At $n = 0$ the integral is $\frac{1}{m+1}$ and the formula gives $\frac{m! 0!}{(m+1)!} = \frac{1}{m+1}$. At $m = n = 0$ both sides are 1. At $m = 2, n = 5$ the formula gives $\frac{2 \cdot 120}{40320} = \frac{1}{168} = 0.0059524$, and Simpson's rule on the integral returns 0.0059524.

WHERE THIS GOES WRONG

Choosing a degenerate case that the wrong answer also satisfies. If a suspected error is a missing factor of $\frac{1}{2}$, then $m = n$ will not expose it; pick the case where the suspected error is maximal.

SEE ALSO Small-case brute force; dimensional and scaling sanity.

If a parameter is not available, evaluate. Numerical evaluation is the most powerful cheap check in the book, and the only one that essentially never passes a wrong answer.

CORE

2.3 Numerical evaluation to several digits

CUE Any definite integral, any limit, any series with a claimed closed form, any probability.

METHOD Get four to six significant figures. Simpson's rule for smooth integrands; the midpoint rule for periodic ones and for anything with an endpoint singularity, since it never evaluates the endpoint; a substitution that removes the singularity when the midpoint rule converges too slowly. For a limit, evaluate at $n = 10, 10^2, 10^3, 10^4$ and read the trend, not the last value.

WORKED EXAMPLE

Claim: $\int_0^{\pi/2} \frac{dx}{1 + \sin^2 x} = \frac{\pi}{2\sqrt{2}} = 1.1107207$. The midpoint rule with only eight subintervals returns 1.1107207, agreeing to seven decimals, because the integrand extends to a smooth periodic function and the midpoint rule then converges geometrically. Eight function evaluations settle the question.

WHERE THIS GOES WRONG

Accepting agreement to two digits. A wrong answer differing from the right one by a factor of $\frac{\pi^2}{10}$ or by $\frac{101}{100}$ passes two digits comfortably. Insist on four, and on more than that when the suspected error is a small rational factor.

SEE ALSO Sign and order of magnitude; small-case brute force.

For an antiderivative or a differential equation, there is a check that is not merely strong but complete: run the problem backwards. Differentiation is algorithmic in a way that integration is not, which is exactly what makes it a reliable inverse.

2.4 Differentiate the answer back

CUE An indefinite integral, a solution to an ODE, an inverse function, an implicit relation, or any claim of the form “ F is an antiderivative of f ”.

METHOD Differentiate the claimed answer and simplify to the integrand. Then check the constant separately against an initial condition or a single evaluation point, and check the interval on which the identity holds, since an antiderivative valid on $(0, \infty)$ may be invalid across a singularity.

WORKED EXAMPLE

Claim: $\int \sec x \, dx = \ln |\sec x + \tan x| + C$. Differentiating,

$$\frac{d}{dx} \ln |\sec x + \tan x| = \frac{\sec x \tan x + \sec^2 x}{\sec x + \tan x} = \frac{\sec x (\tan x + \sec x)}{\sec x + \tan x} = \sec x.$$

As a numerical cross-check, the central difference of $\ln |\sec x + \tan x|$ at $x = 0.7$ is 1.3074593, and $\sec 0.7 = 1.3074593$.

WHERE THIS GOES WRONG

Verifying the derivative and forgetting that the constant of integration is constant only on each connected piece of the domain. Across a pole of $\tan x$ the two branches may carry different constants, and an ODE solution assembled from $\ln |y|$ inherits the same defect.

SEE ALSO The independent second method; degenerate and limiting cases.

The next check applies when there is nothing to differentiate and no parameter to push: bound the object crudely, and see whether the claimed number is even in the right region.

2.5 Sign and order of magnitude

CUE Any numerical answer produced by a long chain of substitutions, especially one whose sign changed at some point in the chain.

METHOD Bound the object with the crudest available inequality. A positive integrand forces a positive integral. A monotone integrand on $[a, b]$ forces a value between $(b - a)f(a)$ and $(b - a)f(b)$. A convex one is bracketed below by the midpoint rectangle and above by the trapezoid. For a sum, compare with the integral of the same expression.

WORKED EXAMPLE

Any claimed value for $\int_0^1 \frac{dx}{\sqrt{1-x^4}}$ must lie in $[1, 2]$. Below: the integrand exceeds 1 on $(0, 1)$, so the value exceeds 1.

Above: $1 - x^4 = (1 - x)(1 + x + x^2 + x^3) \geq 1 - x$, so the integrand is at most $(1 - x)^{-1/2}$, whose integral over $[0, 1]$ is 2. The true value is 1.3110288, and the two inequalities took ten seconds.

WHERE THIS GOES WRONG

Applying a positivity argument to an integrand that changes sign on the interval. Then the crude bound controls $\int |f|$ only, and the integral itself may be anywhere between $-\int |f|$ and $\int |f|$.

SEE ALSO Numerical evaluation to several digits; squeeze arguments.

Symmetry is the check that costs nothing when it applies, because the symmetry was usually already noticed while solving. Its power is that it constrains the whole answer at once rather than one number.

2.6 Symmetry of the answer

CUE A problem invariant under exchanging two parameters, two variables, or two roles; a definite integral fixed by an involution; a count invariant under relabelling.

METHOD If the statement is unchanged by a transformation, the answer must be unchanged too. A closed form that is not symmetric under a swap the problem is symmetric under is wrong, without any further computation. The same argument run forwards is often the fastest derivation.

Claim: $\int_0^\infty \frac{dx}{(1+x^2)(1+x^a)} = \frac{\pi}{4}$ for every real a . The substitution $x \mapsto 1/x$ fixes $(0, \infty)$, fixes $\frac{dx}{1+x^2}$, and sends $a \mapsto -a$; so the value is even in a . Adding the two copies gives $\int_0^\infty \frac{dx}{1+x^2} = \frac{\pi}{2}$, hence the value is $\frac{\pi}{4}$ independently of a . Numerically the integral is 0.7853982 at $a = 3$, at $a = 0.7$, and at $a = \pi$. Any claimed answer depending on a dies at the first line.

Asserting a symmetry the problem does not have. Frullani's integral is antisymmetric, not symmetric, under exchanging its two parameters, so its answer $\ln(b/a)$ must change sign under the swap; a proposed answer symmetric in a and b is wrong for the opposite reason.

SEE ALSO Look for the involution; dimensional and scaling sanity.

For discrete answers there is a check with no analogue in the continuous world, and it is close to decisive: compute the thing directly for small inputs.

2.7 Small-case brute force

CUE A count, a probability, the closed form of a recurrence, a combinatorial identity, or any claim beginning “in general”.

METHOD Instantiate at $n = 0, 1, 2, 3, 4$ and compare with direct enumeration. Also check the type constraints: a count must be a non-negative integer, a probability must lie in $[0, 1]$, a family of probabilities must sum to 1, and the growth rate must match intuition. Choose at least one case that was not used in deriving the formula, and at least one degenerate case.

Claim: the number of derangements of n objects is $D_n = n! \sum_{k=0}^n \frac{(-1)^k}{k!}$. The formula gives 0, 1, 2, 9, 44, 265 for $n = 1, \dots, 6$; exhaustive enumeration of all 720 permutations of six objects gives 265 fixed-point-free ones, and the smaller cases match by hand. A formula that survives $n \leq 4$ against brute force is usually right; one that fails at $n = 1$ is always wrong.

Testing only $n = 1$ and $n = 2$, where almost every plausible wrong formula agrees with the right one, and testing only the case the formula was fitted to.

SEE ALSO Degenerate and limiting cases; hand enumeration with an exhaustiveness statement.

The last rung is the most expensive and the only one that is close to conclusive. Two derivations that share no decisive step are two independent measurements of the same quantity.

2.8 The independent second method

CUE Any answer that matters; any derivation containing a step you were unsure of, especially an interchange of limits, a contour deformation, or a rearrangement.

METHOD Recompute by a route that shares no step with the first. Reflection versus parameter differentiation; series expansion versus a substitution; a bijection versus a recursion. The point is independence: if both routes rely on the same interchange of summation and integration, they fail together and prove nothing.

$\int_0^1 \frac{x-1}{\ln x} dx$. Method A, parameter differentiation: with $I(s) = \int_0^1 \frac{x^s-1}{\ln x} dx$ we get $I'(s) = \int_0^1 x^s dx = \frac{1}{s+1}$ and $I(0) = 0$, so $I(s) = \ln(s+1)$ and $I(1) = \ln 2$. Method B, integral representation: $\frac{x-1}{\ln x} = \int_0^1 x^t dt$, so exchanging the order gives $\int_0^1 \frac{dt}{t+1} = \ln 2$. Numerically the integral is 0.6931472.

A “second method” that reuses the first method's decisive step. Differentiating under the integral sign twice, in two different-looking parametrisations, is one method run twice.

SEE ALSO Numerical evaluation to several digits; symmetry of the answer.

What the checks actually catch

Verification is worth more when you know which errors you are hunting. The six failure modes below account for most wrong answers in calculus, and each has a check on the ladder that is built for it.

Bounds not transformed under a substitution. In $\int_0^1 x\sqrt{1-x^2} dx$ with $u = 1 - x^2$ and $du = -2x dx$, the honest form is $-\frac{1}{2} \int_1^0 \sqrt{u} du = \frac{1}{2} \int_0^1 \sqrt{u} du = \frac{1}{3}$. Keeping the minus sign and the limits 0 to 1 together gives $-\frac{1}{3}$. The sign check kills this in one second, since the integrand is non-negative on $[0, 1]$; the same slip on an integrand of mixed sign needs numerical evaluation instead.

A lost absolute value in $\ln|u|$. The antiderivative of $1/x$ is $\ln|x|$, so $\int_{-2}^{-1} \frac{dx}{x} = \ln 1 - \ln 2 = -\ln 2 \approx -0.693147$; written as $\ln x$ it is not defined anywhere on the interval. The same applies to $\int \tan x dx = -\ln|\cos x| + C$ and to every separable ODE that produces $\ln|y|$. Differentiating the answer back catches it, provided you also ask on which interval the identity holds.

A lost solution from dividing by something that can vanish. Separating $\frac{dy}{dx} = y^2 - y$ requires dividing by $y^2 - y$, which silently discards the constant solutions $y \equiv 0$ and $y \equiv 1$. The general family that emerges is $y = \frac{1}{1 - Ce^x}$, which recovers $y \equiv 1$ at $C = 0$ but never produces $y \equiv 0$: that solution is genuinely lost. The same slip in trigonometry turns $\sin 2x = \sin x$ into $\cos x = \frac{1}{2}$, discarding every $x = k\pi$. The check is to substitute each discarded root back into the original equation before dividing.

A wrongly chosen branch of an inverse function. Applying $t = \tan \frac{x}{2}$ to $\int_0^{2\pi} \frac{dx}{2 + \cos x}$ and reading the limits as $t = 0$ to $t = 0$ gives the answer 0; the substitution is illegitimate because t has a pole at $x = \pi$ and is not a continuous bijection on the interval. The true value is $\frac{2\pi}{\sqrt{3}} \approx 3.627599$. The order-of-magnitude check is decisive: the integrand is at least $\frac{1}{3}$ everywhere, so the integral is at least $\frac{2\pi}{3} \approx 2.094$. The general rule is that $\arctan(\tan \theta) = \theta$ only on $(-\frac{\pi}{2}, \frac{\pi}{2})$, and every Weierstrass substitution must be split at the poles of the tangent.

A rearranged conditionally convergent series. The alternating harmonic series sums to $\ln 2 \approx 0.693147$, but taking two positive terms for each negative one rearranges it to $\frac{3}{2} \ln 2 \approx 1.039721$, and by Riemann's theorem some rearrangement converges to any prescribed value. The check is to test absolute convergence before rearranging, reordering, or interchanging a sum with an integral, and then to compare partial sums of the series as actually written against the claimed value.

A limit computed along one path only. For $f(x, y) = \frac{xy}{x^2 + y^2}$ the value along either axis is 0 and the value along $y = x$ is $\frac{1}{2}$, so the joint limit at the origin does not exist even though both iterated limits are 0. Two paths is the minimum; the polar form $f = \frac{1}{2} \sin 2\theta$, independent of r , settles it at once. The analogous one-sided error in a single variable is computing a limit as $x \rightarrow a^+$ and reporting it as the limit.

A wrong answer, caught three ways

Consider the claim

$$\int_0^{\pi/2} \ln(\sin x) dx \stackrel{?}{=} -\pi \ln 2.$$

This is not a careless answer. It comes from a real and largely correct derivation: the Fourier expansion $\ln(2 \sin \frac{\theta}{2}) = -\sum_{k \geq 1} \frac{\cos k\theta}{k}$ integrates termwise over $[0, \pi]$ to give $\int_0^\pi \ln(2 \sin \frac{\theta}{2}) d\theta = -\sum_{k \geq 1} \frac{\sin k\pi}{k^2} = 0$, hence $\int_0^\pi \ln \sin \frac{\theta}{2} d\theta = -\pi \ln 2$. The solver then substitutes $x = \theta/2$ and forgets that $d\theta = 2 dx$. Everything except one factor of two is right, and the answer has the correct sign, the correct constant $\ln 2$, and the correct multiple of π . Three checks kill it.

Check one: a crude bound, ten seconds. On $(0, \frac{\pi}{2})$ concavity of the sine puts the graph above its chord, so $\sin x \geq \frac{2x}{\pi}$ and therefore $\ln \sin x \geq \ln \frac{2x}{\pi}$. Integrating the lower bound with $u = 2x/\pi$,

$$\int_0^{\pi/2} \ln \sin x dx \geq \frac{\pi}{2} \int_0^1 \ln u du = -\frac{\pi}{2} \approx -1.570796.$$

The claimed value $-\pi \ln 2 \approx -2.177586$ lies below that bound, so it is impossible. No computation, no numerics, one inequality.

Check two: numerical evaluation. The integrand has an integrable logarithmic singularity at 0, so use the midpoint rule, which never evaluates the endpoint. With a thousand subintervals it returns -1.08825 ; a tanh-sinh quadrature returns -1.0887930 . The claim is off by a factor of exactly 2, which is the signature of a lost Jacobian and tells you where to look.

Check three: an independent second method. Reflection gives $\int_0^{\pi/2} \ln \cos x \, dx = I$, where I is the integral in question. Then

$$2I = \int_0^{\pi/2} \ln(\sin x \cos x) \, dx = \int_0^{\pi/2} \ln \frac{\sin 2x}{2} \, dx = -\frac{\pi}{2} \ln 2 + \frac{1}{2} \int_0^{\pi} \ln \sin u \, du,$$

where the substitution $u = 2x$ has been done carefully. Since $\sin u$ is symmetric about $u = \pi/2$, the last integral is $2I$, so $2I = -\frac{\pi}{2} \ln 2 + I$ and

$$I = -\frac{\pi}{2} \ln 2 \approx -1.0887930.$$

This route uses no Fourier series and no termwise integration, so it fails independently of the first derivation. It agrees with the numerics to seven digits.

A NORMALISATION WORTH MEMORISING

The correct value says that $\int_0^{\pi/2} \ln(2 \sin x) \, dx = 0$, and likewise $\int_0^{\pi} \ln(2 \sin x) \, dx = 0$. Any answer in the log-sine family can be tested against this in one line, and the factor of 2 inside the logarithm is precisely what a lost Jacobian destroys.

Recognition matrix

Answer shape	Check that would catch an error	What it catches
A definite integral of a positive integrand	Sign, plus a crude upper bound	Sign slips, lost factors
A closed form containing a parameter a	Set $a \rightarrow 0$, $a = 1$, $a \rightarrow \infty$	Wrong power, missing constant
An answer over a scale-invariant domain	Rescale $x = au$ and count powers	Wrong homogeneity degree
An indefinite integral	Differentiate it and simplify	Chain-rule factor, lost absolute value
An ODE solution	Substitute into the equation and the initial condition	Lost constant solution, wrong C
A closed form containing $\ln u $	Test a point where $u < 0$	The dropped absolute value
An inverse-trig or Weierstrass result	Evaluate at an interior point and at both endpoints	Wrong branch, pole inside the interval
An answer expected symmetric in m and n	Swap m and n	Asymmetric algebra slip
An answer claimed independent of a	Evaluate numerically at two values of a	A parameter that did not cancel
A count, a probability, or a recurrence	Brute force at $n \leq 5$	Off-by-one, wrong family
A value involving π , $\ln 2$, $\zeta(2)$	Numerical agreement to six digits	Rational factors, lost Jacobians
A sum of a conditionally convergent series	Partial sums in the original order	Illegitimate rearrangement
A limit in more than one variable	A second path, then polar coordinates	Path dependence
A limit obtained by expansion	Recompute with one more Taylor order	Premature truncation
Anything you are about to submit	Recompute by a technique sharing no step	Everything the others missed

PART II

Limits, asymptotics, sequences and series

The asymptotic hierarchy and the indeterminate forms

What this chapter is for. Almost every limit that a course or a competition will put in front of you is decided by a single question: which term wins? A quotient at infinity is settled by the fastest-growing term on the top and the fastest-growing term on the bottom; a quotient at a point is settled by the lowest-order term of each. The apparatus of “indeterminate forms” is not a separate subject. It is a lookup table whose only purpose is to tell you which algebraic rewrite converts an unrecognisable expression into a dominance question you can already answer. This chapter installs the growth hierarchy as a memorised total order, teaches the four ways of exploiting it (divide by the dominant term, replace factors by their asymptotic equivalents, carry explicit remainders, and rescale logarithmically), and then classifies the seven indeterminate forms by the single conversion move each one demands. At the end you should be able to look at a limit, name its form in two seconds, name the rewrite in two more, and know before you start what the answer will depend on.

The hierarchy, and why it beats reflexive L'Hopital

Everything in this chapter descends from one displayed statement. Write $f \ll g$ to mean $\lim_{x \rightarrow \infty} f(x)/g(x) = 0$. Then, for every fixed $M > 0$, every $\varepsilon > 0$, and all $0 < a < b$ and $c > 0$,

$$1 \ll \ln \ln x \ll \ln x \ll (\ln x)^M \ll x^\varepsilon \ll x^a \ll x^b \ll e^{cx} \ll e^{x^2} \ll n! \ll n^n.$$

Read it as a chain of strict classes, not a list of examples. The three facts worth internalising are these. First, the relation is transitive, so a single glance places any two of these objects. Second, no fixed power rescues a slower class: $(\ln x)^{100} \ll x^{1/100}$, even though the crossover does not occur until x is around $e^{1.2 \times 10^5}$. Third, the last two entries are sequence-only, since $n!$ and n^n are not defined for continuous x without invoking Γ .

The chain has a mirror image at the origin, and you will use it as often. Writing $f \ll g$ now for $\lim_{x \rightarrow 0^+} f/g = 0$,

$$e^{-1/x} \ll x^b \ll x^a \ll \frac{1}{\ln(1/x)} \ll 1 \quad (0 < a < b).$$

The two ends are the interesting ones. On the left, $e^{-1/x}$ vanishes faster than every power, which is why the standard example of a smooth non-analytic function is built from it. On the right, $1/\ln(1/x)$ vanishes more slowly than every power, which is why logarithmic factors are so hard to dislodge. In between, the rule is the reverse of the rule at infinity: at a point, the *lowest* power dominates, because it is the one that vanishes most slowly. Every technique in this chapter is one of these two readings applied to a specific shape.

The pedagogical claim of this chapter is that the hierarchy should be your first instrument and L'Hopital's rule your third. There are three reasons, and they are not matters of taste.

Cost. L'Hopital reduces the exponent of a power by one per application. On $(\ln x)^{100}/x^{1/100}$ it needs one hundred differentiations, each producing a longer expression than the last, to reach a conclusion the hierarchy delivers in one line. On a ratio of cubics it needs three rounds where factoring x^3 needs none.

Failure. L'Hopital can cycle. On $\sqrt{1+x^2}/x$ as $x \rightarrow \infty$, differentiating gives $x/\sqrt{1+x^2}$, and differentiating that gives $\sqrt{1+x^2}/x$ again; the rule loops forever on a limit whose value, 1, follows immediately from dividing by x . It can also report “no limit” when a limit exists, which is the subject of the next chapter's final technique.

Information. Dominance reasoning tells you *why* the answer is what it is, and therefore tells you what happens when a coefficient or an exponent in the problem is perturbed. A stack of derivatives tells you a number.

WHY CLASS BEATS EXPONENT

Substitute $x = e^t$ and the whole left half of the chain collapses into a single comparison. The claim $(\ln x)^M \ll x^\varepsilon$ becomes $t^M \ll e^{\varepsilon t}$, which is one fact about one pair of functions. The right half comes from the crude bound $n! \geq (n/2)^{n/2}$, obtained by discarding the smaller half of the factors: taking logarithms, $(n/2) \ln(n/2) > cn$ as soon as $n > 2e^{2c}$, so $e^{cn} \ll n!$ for every c . The hierarchy is two facts, not eleven.

It is worth watching the cost difference once, on a limit small enough to write out in full. Take $\lim_{x \rightarrow \infty} \frac{x^3 + 5x}{2x^3 - x^2 + 7}$. By hierarchy: divide by x^3 and read $\frac{1 + 5x^{-2}}{2 - x^{-1} + 7x^{-3}} \rightarrow \frac{1}{2}$, one line, and the line also tells you that the answer depends only on the leading coefficients. By L'Hopital: $\frac{3x^2 + 5}{6x^2 - 2x}$, still $\frac{\infty}{\infty}$, then $\frac{6x}{12x - 2}$, still $\frac{\infty}{\infty}$, then $\frac{6}{12} = \frac{1}{2}$. Three rounds, three form checks, and at no point does the calculation reveal that the constant 7 and the linear term $5x$ were irrelevant from the start. Now replace the exponent 3 by 30 and the second method costs thirty rounds while the first costs the same single line. None of this makes L'Hopital useless. It makes it a specialised tool with a narrow best case, which is exactly how the next chapter treats it. Here we build the instrument that replaces it in the majority of cases.

EASY

3.1 The growth hierarchy

CUE A quotient at infinity whose numerator and denominator are drawn from *different* growth classes: a logarithm against a power, a power against an exponential, an exponential against a factorial.

METHOD Place both against the chain above and read off the answer: if the denominator's class strictly dominates, the limit is 0; if the numerator's does, the limit is $\pm\infty$ according to sign; only same-class competitors require further work. The useful strengthening is that class beats exponent, so $(\ln x)^M \ll x^\varepsilon$ and $x^M \ll e^{\varepsilon x}$ hold for every fixed M however large and every fixed $\varepsilon > 0$ however small.

WORKED EXAMPLE

Two limits that look like they need work and do not.

$$\lim_{x \rightarrow \infty} \frac{(\ln x)^{100}}{x^{1/100}} = 0, \quad \lim_{n \rightarrow \infty} \frac{n^{10} 2^n}{n!} = 0.$$

The first is a logarithm against a power, so the power wins regardless of the exponents; substituting $x = e^t$ turns it into $t^{100}/e^{t/100}$ if you want to see the mechanism. The second is a product of a power and an exponential against a factorial, and the factorial is above both in the chain. Numerically the second is already 1.4×10^{-20} at $n = 40$, whereas the first is still enormous at $x = e^{200}$ and does not fall below 1 until $\ln x$ exceeds about 1.17×10^5 .

A third, to show how to place an object that is not literally in the chain:

$$\lim_{x \rightarrow \infty} \frac{x^x}{e^{x^2}} = 0.$$

Write $x^x = e^{x \ln x}$ and compare exponents. Since $x \ln x \ll x^2$, the quotient is $e^{x \ln x - x^2} \rightarrow 0$. The general principle: to place two exponentials, compare their exponents; to place an exponential against anything else, take logarithms of both.

WHERE THIS GOES WRONG

Believing that a huge exponent on the slower function wins. It only delays the crossover; it never changes the limit. The companion error is reaching for L'Hopital on the first of these, where one hundred differentiations are needed to say what one line of hierarchy says.

SEE ALSO Divide by the dominant term; logarithmic rescaling; L'Hopital's failure modes.

A remark on how to use the chain when an object is not literally in it. Every function you meet can be placed by one of three moves: rewrite it as an exponential and compare exponents, as was done above with x^x ; take logarithms of both competitors and compare the results, which is safe because $\ln$ is increasing; or bound it between two entries of the chain, which is enough whenever both bounds give the same verdict. Nothing else is needed, and the chain is short enough to be memorised in an afternoon.

The hierarchy answers questions in which the two sides live in different classes, and it answers them without any arithmetic at all. When the two sides live in the *same* class the answer is finite and nonzero, and it is a ratio of coefficients. Extracting that ratio is pure algebra, and the algebra always takes the same shape: identify the one term that dominates each sum, divide it out, and let the hierarchy annihilate the remainder.

3.2 Divide by the dominant term

CUE A ratio of sums as $x \rightarrow \infty$ or $n \rightarrow \infty$ in which one term of the numerator and one term of the denominator visibly outgrow everything else in their own sum.

METHOD Factor the single dominant term out of the top and out of the bottom, then kill the rest with the hierarchy. For polynomials this is

$$\frac{a_m x^m + \dots}{b_n x^n + \dots} = x^{m-n} \cdot \frac{a_m + O(1/x)}{b_n + O(1/x)} \rightarrow \begin{cases} 0, & m < n, \\ a_m/b_n, & m = n, \\ \pm\infty, & m > n, \end{cases}$$

and the same move works verbatim when the “terms” are exponentials, factorials, or logarithms rather than powers.

WORKED EXAMPLE

$$\lim_{x \rightarrow \infty} \frac{\sqrt{x^4 + x^2 + 1} + \ln x}{3x^2 - \arctan x}.$$

The dominant term upstairs is $\sqrt{x^4} = x^2$, since $\ln x \ll x^2$; downstairs it is $3x^2$, since $\arctan x$ is bounded. Divide top and bottom by x^2 :

$$\frac{\sqrt{1 + x^{-2} + x^{-4}} + x^{-2} \ln x}{3 - x^{-2} \arctan x} \rightarrow \frac{1 + 0}{3 - 0} = \frac{1}{3}.$$

Confirmed numerically: the quotient is 0.3333333334 at $x = 10^6$.

The same move at a sequence, where the dominant terms are factorials:

$$\lim_{n \rightarrow \infty} \frac{n! + 3^n}{(n+1)! - 3^{n+1}} = \lim_{n \rightarrow \infty} \frac{1 + 3^n/n!}{(n+1) - 3^{n+1}/n!} = 0,$$

after dividing top and bottom by $n!$ and using $3^n \ll n!$ from the hierarchy. The value at $n = 50$ is 0.0196, which is $1/51$ to four figures, exactly as the surviving $(n+1)$ downstairs predicts.

WHERE THIS GOES WRONG

Applying L'Hopital instead. On a ratio of cubics it works but costs three rounds and produces no insight; on $\sqrt{1+x^2}/x$ it cycles forever, because each differentiation swaps the two functions back. The second trap is dividing by x^m from the numerator's degree rather than by a single consistent power, which scrambles the bookkeeping when $m \neq n$. The third is forgetting that “dominant” is decided separately in the numerator and in the denominator: in $\frac{e^x + x^{100}}{x^{100} + \ln x}$ the dominant terms are e^x and x^{100} , from different classes, and the limit is $+\infty$.

SEE ALSO The growth hierarchy; radicals at infinity; expansion at infinity via $t = 1/x$.

The same move works when the competing terms are exponentials rather than powers, and it is worth seeing once in that setting because the notation obscures it. In $(2^n + 3^n)^{1/n}$ the dominant term is 3^n , so the expression equals $3(1 + (2/3)^n)^{1/n}$ and the limit is 3; the bracket tends to 1 because $(2/3)^n \rightarrow 0$ and the exponent tends to 0. Nothing about the exponent $1/n$ changed the analysis. It was still a question of which term dominates.

Dominance at infinity has an exact mirror at a point: as $x \rightarrow 0$ the lowest power wins rather than the highest, and the statement “ $\sin x$ behaves like x ” is the same kind of statement as “ $3x^3 - x$ behaves like $3x^3$ ”. The formal device that lets you use it is asymptotic equivalence, and the whole of its difficulty lies in knowing which operations preserve it.

3.3 Asymptotic equivalence and its legal algebra

CUE A product or quotient in which every factor has a known leading behaviour: $\sin u \sim u$, $\tan u \sim u$, $e^u - 1 \sim u$, $\ln(1+u) \sim u$, $\arcsin u \sim u$, $\arctan u \sim u$, $\sinh u \sim u$, $1 - \cos u \sim u^2/2$, all as $u \rightarrow 0$.

METHOD Write $f \sim g$ for $\lim f/g = 1$. Equivalence is closed under multiplication, division, and fixed real powers, so in a pure product or quotient you may replace each factor by its equivalent independently:

$$\lim \frac{\prod_i f_i}{\prod_j g_j} = \lim \frac{\prod_i \tilde{f}_i}{\prod_j \tilde{g}_j} \quad \text{whenever } f_i \sim \tilde{f}_i, g_j \sim \tilde{g}_j.$$

It is *not* closed under addition, subtraction, or substitution into an outer function.

$$\lim_{x \rightarrow 0} \frac{(e^{3x} - 1) \arcsin 2x}{\tan 5x \ln(1 + 4x)} = \lim_{x \rightarrow 0} \frac{(3x)(2x)}{(5x)(4x)} = \frac{6}{20} = \frac{3}{10}.$$

Every factor is a composition of a standard function with a linear argument, so each contributes only its multiplier. A second instance shows the same mechanism cancelling a circular function against its hyperbolic sibling:

$$\lim_{x \rightarrow 0^+} \frac{\sin(\sqrt{x}) \arcsin(x^2)}{\sinh(\sqrt{x}) \operatorname{arsinh}(x^2)} = 1,$$

because $\sin u$ and $\sinh u$ share the leading term u , as do $\arcsin$ and arsinh , and the four leading terms cancel pairwise. Fixed powers are legal too, and that is often the whole trick:

$$\lim_{x \rightarrow 0} \frac{(1 - \cos x)^{3/2}}{x^3} = \lim_{x \rightarrow 0} \frac{(x^2/2)^{3/2}}{x^3} = 2^{-3/2} = \frac{1}{2\sqrt{2}} \approx 0.353553,$$

verified as 0.3535533 at $x = 10^{-3}$. Note that $1 - \cos x \sim x^2/2$ is not a leading term of the form cu ; equivalence does not require the equivalent to be linear, only that the ratio tends to 1.

WHERE THIS GOES WRONG

Substituting equivalents inside a difference. From $\tan x \sim x$ and $\sin x \sim x$ one “deduces” $\tan x - \sin x \sim 0$; the truth is $\tan x - \sin x \sim \frac{1}{2}x^3$. Every difference of same-order quantities requires full expansion, not equivalence.

SEE ALSO Big-O and little-o bookkeeping; order bookkeeping across a subtraction; the standard expansion table.

EQUIVALENCE IS A STATEMENT ABOUT RATIOS, NOT ABOUT DIFFERENCES

$f \sim g$ asserts that $f/g \rightarrow 1$, which controls the *relative* error and says nothing whatever about the absolute one. Two functions can be equivalent while their difference grows without bound: $x^2 + x \sim x^2$, and yet the difference is x . That single observation explains both halves of the closure rule. Ratios of equivalent things are equivalent because relative errors multiply; differences of equivalent things are uncontrolled because relative errors say nothing about what survives a cancellation.

Composition deserves a sentence of its own, because the failure there is less familiar than the failure under subtraction and just as fatal. From $x^2 + x \sim x^2$ it does not follow that $e^{x^2+x} \sim e^{x^2}$: the ratio is e^x , which tends to infinity. The reason is that $\sim$ controls a ratio, and the exponential converts a ratio into a difference, which $\sim$ never controlled. Two compositions are safe. Fixed real powers are safe, since $f \sim g$ gives $f^r \sim g^r$ directly. Logarithms of quantities tending to infinity are safe, since $\ln f = \ln g + \ln(f/g)$ and the correction tends to 0 while $\ln g$ does not. Everything else must be checked, and the practical rule is that if the outer function is not a power or a logarithm, expand instead of substituting.

That trap is the reason the next technique exists. Equivalence throws away all information beyond the leading term, which is fine in a product and fatal in a difference. The repair is to keep a named remainder rather than an ellipsis, so that the moment a leading term cancels you still know what is left.

3.4 Big-O and little-o bookkeeping

MEDIUM

CUE You are about to truncate a series, and you need to certify that the discarded tail cannot affect the answer.

METHOD Carry an explicit remainder and use its algebra: $x^m O(x^n) = O(x^{m+n})$, $O(x^m) \pm O(x^n) = O(x^{\min(m,n)})$, $O(x^m) O(x^n) = O(x^{m+n})$, and $O(x^n)/x^k \rightarrow 0$ whenever $n > k$. The operational rule that follows: expand every ingredient to the same *absolute order*, never to the same number of terms.

WORKED EXAMPLE

$$\lim_{x \rightarrow 0} \frac{1 - \cos x - \frac{1}{2}x^2}{x^4}.$$

Write $\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} + O(x^6)$. Then $1 - \cos x - \frac{1}{2}x^2 = -\frac{x^4}{24} + O(x^6)$ exactly, and dividing by x^4 gives $-\frac{1}{24} + O(x^2)$. The limit is $-\frac{1}{24}$. The remainder is what licenses the conclusion: without it, “the next terms are small” is an assertion, and with it, $O(x^6)/x^4 = O(x^2) \rightarrow 0$ is a proof. Numerically the quotient is -0.0416665 at $x = 10^{-2}$ against $-1/24 = -0.0416667$.

A second instance, chosen to expose the term-count error:

$$\lim_{x \rightarrow 0} \frac{e^x - 1 - \ln(1 + x)}{x^2} = 1.$$

Three terms of e^x reach order 2; three terms of $\ln(1+x)$ reach order 3. Expanded to a common order 2, the numerator is $(x + \frac{x^2}{2}) - (x - \frac{x^2}{2}) + O(x^3) = x^2 + O(x^3)$, so the limit is 1; a solver who matched term counts rather than orders would have carried an unnecessary cubic through the whole calculation.

A third instance, showing the remainder algebra doing real work inside a product:

$$\lim_{x \rightarrow 0} \frac{\sin x \ln(1+x) - x^2}{x^3} = -\frac{1}{2}.$$

Target order 3, so expand each factor through order 2 and keep the O : $\sin x = x + O(x^3)$ and $\ln(1+x) = x - \frac{x^2}{2} + O(x^3)$. The product is $x^2 - \frac{x^3}{2} + O(x^4)$, since $x \cdot O(x^3) = O(x^4)$ and $O(x^3) \cdot O(x) = O(x^4)$ absorb everything discarded. Subtracting x^2 and dividing by x^3 gives $-\frac{1}{2} + O(x)$. Verified as -0.49983 at $x = 10^{-3}$.

WHERE THIS GOES WRONG

Expanding “three terms of each” rather than “to order x^4 of each”. Different functions reach a given order after different numbers of terms, so a term-count rule silently drops a surviving contribution from the function whose series is sparse. Here $\cos$ needs three terms to reach order 4 while $\ln(1+x)$ would need four.

SEE ALSO Asymptotic equivalence and its legal algebra; expand to the first surviving order; order bookkeeping across a subtraction.

Those four techniques are the complete toolkit for a limit whose ingredients are already in a comparable shape. The next section deals with the three shapes that are not, and its message is that each one needs a specific piece of preprocessing before dominance can be read off at all. Skipping the preprocessing does not produce a harder calculation; it produces a wrong sign, an overflow, or a confident answer that is off by a term nobody looked for.

THE TWO-SECOND PROTOCOL

Before writing anything, answer three questions. Where is the limit point, 0 or a finite a or $\pm\infty$? What does direct substitution produce, a number or one of the seven symbols? And which single term dominates on the top and which on the bottom, remembering that at a point the lowest order dominates and at infinity the highest does? A solver who answers these three has already chosen the method. A solver who starts differentiating has not.

Two things the hierarchy does not settle

The chain sorts classes. Inside a class it says nothing, and there are two situations where that silence matters.

The first is a comparison that is finer than the chain. All of $\ln x$, $\ln(3x)$, $\ln(x^2)$ and $\ln(x^{100})$ sit in the same class, but they are not interchangeable: $\ln(3x)/\ln x \rightarrow 1$, so those two are equivalent, whereas $\ln(x^2)/\ln x = 2$ exactly, so those two are not. The rule is that a constant *factor* inside a logarithm is asymptotically invisible while a constant *exponent* is not, because $\ln(cx) = \ln x + \ln c$ but $\ln(x^k) = k \ln x$. The same distinction runs through the exponential class: e^{x+1} and e^x differ by the fixed factor e , while e^{2x} and e^x are in different classes altogether. Whenever a problem is engineered so that a ratio is finite and nonzero, this is usually where the engineering happened.

The second is oscillation. Nothing in the chain applies to $\sin x$, which neither grows nor decays, and the correct instrument there is the squeeze theorem. Three examples fix the pattern: $\frac{\sin x}{x} \rightarrow 0$ as $x \rightarrow \infty$ because a bounded numerator over an unbounded denominator is squeezed between $\pm 1/x$; $\frac{x + \sin x}{x} \rightarrow 1$ because the perturbation is bounded and the main term is not; and $\frac{x \sin x}{x^2} = \frac{\sin x}{x} \cdot 1$ has no limit at all, because the oscillation is not damped relative to the denominator. Dominance reasoning tells you which of these three situations you are in, and then hands the problem to the squeeze.

Radicals, logarithms, and the named asymptotics

The three techniques in this section are dominance reasoning applied to the three shapes that most often disguise it: a radical, which hides an absolute value; a factorial or a tower, which is too large to compare directly and must be examined on a logarithmic scale; and a named special function, whose growth is standard but which no table of elementary limits covers. In all three, the analysis is the analysis of the previous section. What changes is the preprocessing needed before the hierarchy can be read off, and each of the three has a characteristic failure attached to it: a lost minus sign, an overflowed calculation, and a numerical probe that lies.

3.5 Radicals at infinity, and the sign of the dominant power

CUE A square root or k -th root of a polynomial inside a limit as $x \rightarrow \pm\infty$.

METHOD Pull the dominant power out through the radical, and write it as an absolute value:

$$\sqrt{a_n x^n + \dots} = |x|^{n/2} \sqrt{a_n + O(1/x)}, \quad \sqrt{x^2} = |x| = \begin{cases} x, & x \rightarrow +\infty, \\ -x, & x \rightarrow -\infty. \end{cases}$$

The absolute value is the entire content of the technique. When $x \rightarrow -\infty$, substitute $x = -t$ with $t \rightarrow +\infty$ and the sign takes care of itself.

WORKED EXAMPLE

$$\lim_{x \rightarrow -\infty} \frac{\sqrt{4x^2 + 1}}{x + 3}.$$

Here $\sqrt{4x^2 + 1} = |x| \sqrt{4 + x^{-2}} = -x \sqrt{4 + x^{-2}}$, so the quotient equals $-\sqrt{4 + x^{-2}} / (1 + 3/x) \rightarrow -2$. The same expression tends to $+2$ as $x \rightarrow +\infty$. A second instance, where the sign decides more than a sign: with $x = -t$,

$$\lim_{x \rightarrow -\infty} (x + \sqrt{x^2 + 3x}) = \lim_{t \rightarrow \infty} (t\sqrt{1 - 3/t} - t) = \lim_{t \rightarrow \infty} t\left(-\frac{3}{2t} + O(t^{-2})\right) = -\frac{3}{2},$$

verified as -1.50000001 at $x = -10^8$.

Finally, a quotient that mixes an odd root with an even one, where only one of the two carries a sign:

$$\lim_{x \rightarrow -\infty} \frac{\sqrt[3]{x^3 + 2x}}{\sqrt{x^2 + 1}} = \lim_{x \rightarrow -\infty} \frac{x\sqrt[3]{1 + 2x^{-2}}}{|x|\sqrt{1 + x^{-2}}} = \lim_{x \rightarrow -\infty} \frac{x}{-x} = -1,$$

verified as -1.00000000 at $x = -10^6$. The cube root contributed x , the square root contributed $-x$, and the entire answer is the mismatch.

WHERE THIS GOES WRONG

Cancelling $\sqrt{x^2}$ against x and reporting $+2$ on both sides. This is the single most common sign error in the subject, and it is invisible in the algebra because $\sqrt{x^2}$ and x are typeset differently but read identically. The defence is mechanical: whenever the limit point is $-\infty$, substitute $x = -t$ on the first line and never write $|x|$ at all.

SEE ALSO Divide by the dominant term; conjugate multiplication; expansion at infinity via $t = 1/x$.

The absolute value appears only for even roots. For odd k the function $\sqrt[k]{\cdot}$ is defined and increasing on the whole line, so $\sqrt[k]{x^3} = x$ with no case split, and the technique reduces to factoring. The practical consequence is that a problem mixing $\sqrt{\cdot}$ with $\sqrt[k]{\cdot}$ at $-\infty$ has exactly one place where a sign can be lost, and you should mark it before doing anything else.

Radicals hide a sign. Factorials and towers hide a scale: they are so large that direct comparison overflows both arithmetic and intuition, and the problem itself is usually posed on a logarithmic scale to compensate.

HARD

3.6 Logarithmic rescaling of a monstrous expression

CUE Factorials, towers, or products so large that the question is posed logarithmically: $\log_n(\cdot)$, or $\ln(\cdot)/n^2$, or the n -th root of a product.

METHOD Take logarithms first, insert Stirling's expansion

$$\ln(n!) = n \ln n - n + \frac{1}{2} \ln(2\pi n) + O(1/n),$$

keep only what survives the stated normalisation, and read off the answer. Discard subdominant terms only after the normalisation is applied, since a term that is negligible before dividing by n^2 may not be negligible after.

WORKED EXAMPLE

$$\lim_{n \rightarrow \infty} \log_n \left(\ln \frac{(n^2)!}{(n!)^n} \right).$$

Stirling gives $\ln(n^2)! = 2n^2 \ln n - n^2 + O(\ln n)$ and $n \ln(n!) = n^2 \ln n - n^2 + O(n \ln n)$, so the inner logarithm is $n^2 \ln n (1 + o(1))$. Then

$$\log_n(n^2 \ln n) = 2 + \frac{\ln \ln n}{\ln n} \rightarrow 2.$$

At $n = 4000$ the true value is 2.25505 and the predicted correction $\ln \ln n / \ln n$ is 2.25507: the agreement of the *residual*, not the closeness of the raw number to 2, is what confirms the analysis.

The same preprocessing settles the standard root-of-a-factorial limit:

$$\lim_{n \rightarrow \infty} \frac{\sqrt[n]{n!}}{n} = \exp \left(\lim_{n \rightarrow \infty} \left[\frac{\ln(n!)}{n} - \ln n \right] \right) = \exp \left(\lim_{n \rightarrow \infty} \left[\ln n - 1 + \frac{\ln(2\pi n)}{2n} - \ln n \right] \right) = e^{-1}.$$

Verified as 0.36808 at $n = 10^4$ against $e^{-1} = 0.36788$; the residual decays like $\ln n / (2n)$, which is 0.00046 at that n , and that too matches.

WHERE THIS GOES WRONG

Probing a doubly-logarithmic limit numerically and concluding that the answer is 2.25. Convergence at rate $1/\ln n$ is invisible to floating point: even $n = 10^{40}$ leaves an error above 0.05. Any limit whose correction term contains $\ln \ln n$ must be settled symbolically. The second trap is arithmetic rather than analytic: computing $(n^2)!$ directly overflows every floating-point format for n above about 7, so the logarithm must be taken symbolically before any number is evaluated, using $\ln(m!) = \sum_{k \leq m} \ln k$ or Stirling, never $\ln(\text{a computed factorial})$.

SEE ALSO The growth hierarchy; Stirling's approximation; numerically deceptive limits.

Stirling is the first entry in a small catalogue of asymptotics you are expected to quote rather than derive. The rest of that catalogue is the subject of the next block, and it shares Stirling's defining hazard: the corrections decay so slowly that numerical evidence is worthless.

HARD

3.7 Asymptotics of the named special functions

CUE A harmonic number H_n , a Lambert W , a digamma ψ , a $\Gamma(1+x)$ near zero, or a polylogarithm appears, and the problem plainly does not want an exact evaluation.

METHOD Substitute the headline asymptotic and stop. The four that carry most problems are

$$\begin{aligned} H_n &= \ln n + \gamma + \frac{1}{2n} + O(n^{-2}), & W(x) &= \ln x - \ln \ln x + o(1), \\ \psi(x) &= \ln x - \frac{1}{2x} + O(x^{-2}), & \Gamma(1+x) &= 1 - \gamma x + O(x^2) \quad (x \rightarrow 0). \end{aligned}$$

WORKED EXAMPLE

Two limits with the same moral. First,

$$\lim_{x \rightarrow \infty} \frac{\ln x}{W(x)} = 1,$$

since $W(x) = \ln x - \ln \ln x + o(1) = \ln x (1 - \ln \ln x / \ln x + o(1))$ and $\ln \ln x \ll \ln x$. Second, a case where two terms of the expansion are needed:

$$\lim_{n \rightarrow \infty} n(H_{2n} - H_n - \ln 2) = \lim_{n \rightarrow \infty} n \left(\ln 2 + \frac{1}{4n} - \frac{1}{2n} + O(n^{-2}) \right) - n \ln 2 = -\frac{1}{4},$$

verified as -0.2499969 at $n = 2 \times 10^4$. Note that the leading terms $\ln n + \gamma$ cancelled and the answer came entirely from the $1/(2n)$ correction.

The small-argument entry works the same way and is worth having by reflex:

$$\lim_{x \rightarrow 0} \frac{\Gamma(1+x) - 1}{x} = -\gamma \approx -0.5772156,$$

since $\Gamma(1+x) = 1 - \gamma x + O(x^2)$. Verified as -0.577206 at $x = 10^{-5}$. This is nothing more than the statement $\Gamma'(1) = -\gamma$, which is where γ comes from in the first place.

WHERE THIS GOES WRONG

Trusting a numerical probe on the first of these. The ratio $\ln x / W(x)$ is still 1.0193 at $x = e^{300}$ and 1.0069 at $x = e^{1000}$; a machine will never show you the limit. Every asymptotic in this list carries an $O(\ln \ln x / \ln x)$ or $O(1/\ln n)$ error, so 10^{100} is not large.

SEE ALSO Logarithmic rescaling; Stirling's approximation; Stolz-Cesaro for harmonic sums.

The two hard techniques just given share a discipline that is worth stating separately, because it is the only defence against a whole class of wrong answers. When an asymptotic expansion is substituted, decide before substituting how many correction terms the normalisation can see. If the problem divides by n , then an $O(1/n)$ correction contributes and an $O(n^{-2})$ correction does not. If the problem divides by $\ln n$, then a $\ln \ln n$ term contributes and a constant does not. Substituting one term more than necessary is harmless; substituting one term fewer produces an answer that is wrong by a finite amount and looks entirely plausible.

WHEN A NUMERICAL PROBE IS EVIDENCE AND WHEN IT IS NOISE

A probe is evidence when the correction term decays like a power: if the true behaviour is $L + c/n$, then computing at n and at $10n$ and watching the discrepancy shrink by a factor of ten confirms both L and c . A probe is noise when the correction decays like $1/\ln n$ or $\ln \ln n/\ln n$, because then a factor of ten in n buys almost nothing: the correction $\ln \ln n/\ln n$ falls only from 0.2551 at $n = 4 \times 10^3$ to 0.2315 at $n = 2 \times 10^4$. The discipline is to predict the correction first and then check that the residual matches it, rather than to read the limit off the raw number.

The seven indeterminate forms and their conversions

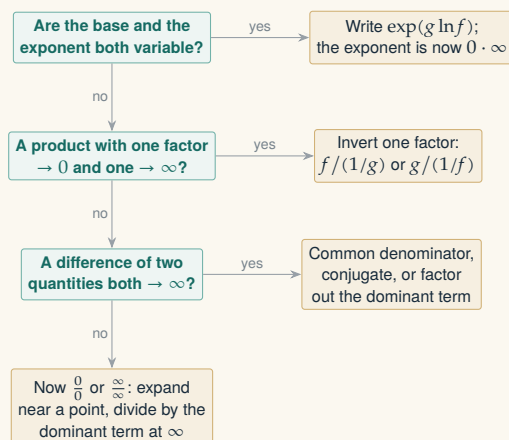
Direct substitution either produces a number, in which case you are finished, or produces one of exactly seven symbols. Those seven are

$$\frac{0}{0}, \quad \frac{\infty}{\infty}, \quad 0 \cdot \infty, \quad \infty - \infty, \quad 1^\infty, \quad 0^0, \quad \infty^0.$$

Nothing else is indeterminate. In particular $0^\infty = 0$, $\infty^\infty = \infty$, $0^{-\infty} = \infty$, $\infty + \infty = \infty$ and $\infty \cdot \infty = \infty$ are determinate, and treating them as though they were not is a reliable way to introduce an error into a problem that had none.

There is a reason the list has exactly seven entries, and it is worth stating because it makes the list memorable rather than arbitrary. A form is indeterminate precisely when it combines two competing monotonicities: one ingredient pushing the value up and the other pushing it down, at rates that the symbol does not record. In $0 \cdot \infty$ one factor is shrinking while the other grows. In 1^∞ the base's excess over 1 is shrinking while the exponent that amplifies it grows. By contrast 0^∞ has both ingredients pushing the same way, base below 1 and exponent large, so the answer is forced. Run that test over the possible combinations of 0, 1, ∞ in a quotient, a product, a difference and a power, and exactly seven survive.

The taxonomy then earns its place because each of the seven has exactly one standard conversion, and every conversion terminates in $\frac{0}{0}$ or $\frac{\infty}{\infty}$, which is to say in a dominance question. The three power forms all go through the logarithm; the two remaining non-quotient forms go through algebra. The decision tree below is the whole system.



Read the tree top-down and note that the arrows out of the three leaves on the right all point, conceptually, back into the bottom box. That is the organising claim of this chapter in one picture: the forms are not seven problems, they are one problem with six pieces of packaging.

One walk-through, to show that the tree is meant to be executed rather than admired. Take $\lim_{x \rightarrow 0^+} \left(\frac{1}{x}\right)^{\tan x}$. First test: base and exponent both variable, so take the right branch and write $\exp(\tan x \cdot \ln \frac{1}{x}) = \exp(-\tan x \ln x)$. The exponent is now a product with $\tan x \rightarrow 0$ and $\ln x \rightarrow -\infty$, which is the second test's yes branch: invert the algebraic factor to get $-\frac{\ln x}{\cot x}$.

That is $\frac{\infty}{\infty}$, and since $\tan x \sim x$ the exponent behaves like $-x \ln x \rightarrow 0$. The answer is $e^0 = 1$. Three tests, three conversions, no differentiation.

One caution before the blocks. A form is a statement about a specific approach, and a two-sided limit can present different forms from the two sides. As $x \rightarrow 0^+$ the expression $x^{1/x}$ has form 0^∞ and value 0; as $x \rightarrow 0^-$ it is not even defined for most x , since a negative base with a non-integer exponent is not real. Similarly $e^{1/x}$ has limit $+\infty$ from the right and 0 from the left, so anything built from it must be treated one side at a time. Whenever the expression contains $1/x$, a logarithm, or a non-integer power, decide which side you are on before naming the form.

EASY

3.8 Classifying the form before touching it

CUE Direct substitution produces a symbol rather than a number.

METHOD Name the form out loud before writing anything: $\frac{0}{0}$, $\frac{\infty}{\infty}$, $0 \cdot \infty$, $\infty - \infty$, 1^∞ , 0^0 , or ∞^0 . If the substitution produced anything else, the expression is determinate and must simply be evaluated. If it produced one of the seven, apply the conversion the tree assigns and do not improvise.

WORKED EXAMPLE

Classify, then act.

$$\lim_{x \rightarrow 0^+} x^{1/x} = 0.$$

The base tends to 0 and the exponent to $+\infty$, so the form is 0^∞ , which is *not* on the list: a quantity below $\frac{1}{2}$ raised to an unboundedly large power tends to 0, and no machinery is required. Contrast

$$\lim_{x \rightarrow 0^+} (\cot x)^{1/\ln x} = e^{-1},$$

where base $\rightarrow \infty$ and exponent $\rightarrow 0$, form ∞^0 , genuinely indeterminate. Taking logarithms, $\ln \cot x \sim -\ln x$ because $\cot x \sim 1/x$, so the exponent limit is $\lim(-\ln x)/\ln x = -1$. Verified: the value is 0.3678794 at $x = 10^{-6}$ against $e^{-1} = 0.3678794$.

One more pair, to show that the classification is about the pair of behaviours and not about the notation. In $\lim_{x \rightarrow 0^+} x^x$ the base and the exponent both tend to 0: form 0^0 , indeterminate, and the answer is 1. In $\lim_{x \rightarrow 0^+} x^{1/\ln x}$ the base tends to 0 and the exponent to 0^- : also form 0^0 , and the answer is e , because $x^{1/\ln x} = \exp(\ln x / \ln x) = e$ for every $x > 0$. Two limits of the same form with different values is what “indeterminate” means, stated concretely.

WHERE THIS GOES WRONG

Reflexively logarithmising or differentiating a determinate form. The reverse error is worse: assuming that because the base tends to 1 the answer is 1. It is the co-occurrence of a base tending to 1 with an exponent tending to ∞ that makes 1^∞ indeterminate, and the three limits $(1+x)^{1/x} \rightarrow e$, $(1+x)^{1/x^2} \rightarrow \infty$, $(1+x^2)^{1/x} \rightarrow 1$, all as $x \rightarrow 0^+$, show that every answer in $[1, \infty]$ is available. The same demonstration works for 0^0 : $x^x \rightarrow 1$ while $x^{1/\ln x} = e$ identically, so two limits of the same form give different values.

SEE ALSO The $0 \cdot \infty$ conversion; the $\infty - \infty$ conversion; the exponential conversion for power forms.

THE DETERMINATE COUSINS, AND WHY THEY ARE THEOREMS

$0^\infty = 0$, $0^{-\infty} = +\infty$, $\infty^\infty = \infty$, $\infty^{-\infty} = 0$, $\infty + \infty = \infty$, $\infty \cdot \infty = \infty$, $c/\infty = 0$, and $c/0^+ = +\infty$ for $c > 0$. None of these is a convention. Each is a one-line consequence of monotonicity, and each can be proved by the same logarithmic conversion used for the indeterminate cases, with the difference that the exponent limit comes out as $-\infty$, $+\infty$ or a definite number rather than as a further indeterminate form.

Of the four non-quotient forms, the product is the easiest, because there is only one thing to do and the only decision is which of two ways to do it.

EASY

3.9 $0 \cdot \infty$: move one factor into the denominator

CUE A product $f \cdot g$ with $f \rightarrow 0$ and $g \rightarrow \infty$.

METHOD Rewrite as $\frac{f}{1/g}$, which is $\frac{0}{0}$, or as $\frac{g}{1/f}$, which is $\frac{\infty}{\infty}$. Both are legal; choose the one whose denominator is a plain power, because that is the one Taylor expansion or a single L'Hopital round will finish. As a rule of thumb, put the transcendental factor on top and the algebraic factor underneath.

$$\lim_{x \rightarrow 0^+} x \ln x = \lim_{x \rightarrow 0^+} \frac{\ln x}{1/x} = \lim_{x \rightarrow 0^+} \frac{1/x}{-1/x^2} = \lim_{x \rightarrow 0^+} (-x) = 0.$$

The choice mattered: the reciprocal $\frac{x}{1/\ln x}$ has derivative $-1/(x \ln^2 x)$ downstairs and gets worse at every round. A second instance where the product hides at infinity:

$$\lim_{x \rightarrow \infty} x \left(\frac{\pi}{2} - \arctan x \right) = \lim_{x \rightarrow \infty} \frac{\frac{\pi}{2} - \arctan x}{1/x} = \lim_{x \rightarrow \infty} \frac{-1/(1+x^2)}{-1/x^2} = \lim_{x \rightarrow \infty} \frac{x^2}{1+x^2} = 1,$$

confirmed as 0.9999999836 at $x = 10^7$.

WHERE THIS GOES WRONG

Choosing the wrong reciprocal and then persevering. If after one round the expression is longer than it was, undo the choice rather than differentiating again; the two options are genuinely unequal in cost, often by a factor of five. A subtler failure is converting a $0 \cdot \infty$ product in which *neither* factor is transcendental, for instance $x \cdot x^{-1}$ dressed up: there the product simplifies algebraically and no conversion is needed at all. Always cancel before converting.

SEE ALSO Classifying the form before touching it; the exponential conversion for power forms; choosing the profitable conversion.

The difference form is harder, because it admits three distinct moves and the wrong one produces a correct but unusable expression. It is also the form where the naive guess is most often wrong, and the reason is structural: in a product or a quotient the leading terms multiply, so the leading behaviour of the answer is visible immediately, whereas in a difference the leading terms cancel by construction and the answer is hiding one or more orders down. A solver who has internalised that fact will reach for expansion the moment a difference appears, and will not be tempted to guess.

MEDIUM

3.10 $\infty - \infty$: common denominator, conjugate, or dominant factor

CUE A difference of two quantities each tending to ∞ , most often two reciprocals of vanishing quantities, or two radicals that both grow.

METHOD Three moves, in order of preference. (i) Put the difference over a common denominator, producing $\frac{0}{0}$. (ii) Multiply and divide by the conjugate if the difference is of radicals, using $\sqrt{A} - \sqrt{B} = (A - B)/(\sqrt{A} + \sqrt{B})$. (iii) Factor out the dominant term: $f - g = f(1 - g/f)$, which turns the difference into a product and refers you back to the previous technique.

WORKED EXAMPLE

$$\lim_{x \rightarrow 0} \left(\frac{1}{x^2} - \frac{1}{\sin^2 x} \right) = \lim_{x \rightarrow 0} \frac{\sin^2 x - x^2}{x^2 \sin^2 x}.$$

The denominator is $x^4 + O(x^6)$. The numerator needs the cube term of sin: from $\sin x = x - \frac{x^3}{6} + O(x^5)$ we get $\sin^2 x = x^2 - \frac{x^4}{3} + O(x^6)$, so the numerator is $-\frac{x^4}{3} + O(x^6)$ and the limit is $-\frac{1}{3}$. Verified as -0.3333334 at $x = 10^{-3}$. A radical instance handled by move (iii):

$$\lim_{x \rightarrow \infty} (\sqrt[3]{x^3 + x^2} - \sqrt{x^2 - x}) = \lim_{x \rightarrow \infty} \left[\left(x + \frac{1}{3} \right) - \left(x - \frac{1}{2} \right) \right] + o(1) = \frac{5}{6},$$

verified as 0.8333333 at $x = 10^7$.

Move (i) at a finite point other than the origin, which is the commonest examination form of all:

$$\lim_{x \rightarrow 1} \left(\frac{1}{\ln x} - \frac{1}{x-1} \right) = \lim_{h \rightarrow 0} \frac{h - \ln(1+h)}{h \ln(1+h)} = \lim_{h \rightarrow 0} \frac{h^2/2 + O(h^3)}{h^2 + O(h^3)} = \frac{1}{2},$$

after setting $h = x - 1$. Verified as 0.4999985 at $x = 1 + 10^{-5}$.

WHERE THIS GOES WRONG

"Both blow up at the same rate, so the difference is 0." Equal rates guarantee only that the difference is finite. Its value is the next-order coefficient, which is precisely the quantity you have not yet computed. The second trap is applying the conjugate to a difference of cube roots as though it were a difference of square roots; there the correct factor is $A^{2/3} + A^{1/3}B^{1/3} + B^{2/3}$.

SEE ALSO Choosing the profitable conversion; conjugate multiplication; when the third order is genuinely needed.

The three power forms are the only ones with a mandatory first move, and it is always the same move.

3.11 $1^\infty, 0^0, \infty^0$: the $\exp(g \ln f)$ conversion

CUE A variable base raised to a variable exponent, where substitution gives 1^∞ , 0^0 , or ∞^0 .

METHOD Always write

$$f(x)^{g(x)} = \exp(g(x) \ln f(x)),$$

evaluate the exponent limit (which is now $0 \cdot \infty$, and refers you back two techniques), and exponentiate at the very end. In the 1^∞ case there is a shortcut worth memorising: if $f \rightarrow 1$ and $g \rightarrow \infty$ then $\ln f \sim f - 1$, so $f^g \rightarrow \exp(\lim g(f - 1))$, which removes the logarithm entirely.

WORKED EXAMPLE

$$\lim_{x \rightarrow 0^+} (\sin x)^x = \exp\left(\lim_{x \rightarrow 0^+} x \ln \sin x\right) = \exp\left(\lim_{x \rightarrow 0^+} (x \ln x + x \ln \frac{\sin x}{x})\right) = e^0 = 1.$$

The split isolates the only dangerous piece, $x \ln x \rightarrow 0$, while $\ln(\sin x/x) \rightarrow 0$ is harmless. A 1^∞ instance using the shortcut:

$$\lim_{x \rightarrow 0} \left(\frac{\arcsin x}{x}\right)^{1/x^2} = \exp\left(\lim_{x \rightarrow 0} \frac{1}{x^2} \left(\frac{\arcsin x}{x} - 1\right)\right) = \exp\left(\lim_{x \rightarrow 0} \frac{x^2/6 + O(x^4)}{x^2}\right) = e^{1/6},$$

verified as 1.18137 at $x = 10^{-2}$ against $e^{1/6} = 1.18136$.

And a sequence version, where the shortcut removes all the work:

$$\lim_{n \rightarrow \infty} \left(\cos \frac{a}{\sqrt{n}}\right)^n = \exp\left(\lim_{n \rightarrow \infty} n \left(\cos \frac{a}{\sqrt{n}} - 1\right)\right) = \exp\left(\lim_{n \rightarrow \infty} n \cdot \left(-\frac{a^2}{2n}\right)\right) = e^{-a^2/2}.$$

For $a = 1.3$ this predicts 0.4295574, and the value at $n = 10^6$ is 0.4295573.

WHERE THIS GOES WRONG

Reporting the exponent as the answer. Having worked out that $g \ln f \rightarrow \frac{1}{6}$, it is startlingly easy to write $\frac{1}{6}$ and stop. The other classic is taking limits of base and exponent separately: that is precisely the operation whose invalidity makes 1^∞ indeterminate.

SEE ALSO The $0 \cdot \infty$ conversion; the standard e limit; classifying the form before touching it.

One technique remains, and it is the metaskill the rest of the chapter has been building towards. Naming the form tells you which conversions are legal; it does not tell you which is cheap. That is a separate judgement, and it is where a competent solver is distinguished from a fast one. The judgement has a single criterion, and it is worth stating before the block that formalises it: a conversion is good when it leaves a plain power of x , or of $x - a$, in the denominator, because a plain power is the one denominator that a series can be divided by term for term. Every other denominator, however innocuous it looks, forces you either to expand it as well or to differentiate it repeatedly.

MEDIUM

3.12 Choosing the profitable conversion

CUE You have converted a form once and the result is worse than what you started with: a denominator that is a product of three transcendental factors, or a numerator whose derivative will need the product rule twice.

METHOD Rank the available rewrites by what the denominator becomes, and prefer the one whose denominator is a plain power of x or of $x - a$, because that is the version a single Taylor expansion resolves. If no rewrite achieves that, substitute $t = 1/x$ or $h = x - a$ first so that a power is available to divide by. Two minutes of re-planning routinely saves six lines of quotient rule.

WORKED EXAMPLE

$$\lim_{x \rightarrow 0} \left(\frac{1}{\operatorname{arsinh} x} - \frac{1}{\ln(1+x)} \right).$$

The obvious move, a common denominator, gives $\frac{\ln(1+x) - \operatorname{arsinh} x}{\operatorname{arsinh} x \ln(1+x)}$: a $\frac{0}{0}$ form whose denominator is a product of two transcendental functions, so L'Hopital on it is unpleasant. The profitable move is to expand each reciprocal separately about its pole. Since $\operatorname{arsinh} x = x(1 - \frac{x^2}{6} + O(x^4))$ and $\ln(1+x) = x(1 - \frac{x}{2} + \frac{x^2}{3} + O(x^3))$,

$$\frac{1}{\operatorname{arsinh} x} = \frac{1}{x} + \frac{x}{6} + O(x^3), \quad \frac{1}{\ln(1+x)} = \frac{1}{x} + \frac{1}{2} - \frac{x}{12} + O(x^2).$$

The $1/x$ poles cancel on sight and the answer is $-\frac{1}{2}$, verified as -0.499975 at $x = 10^{-4}$.

The same judgement applied to a power form: for $\lim_{x \rightarrow 0} (\frac{\sin x}{x})^{1/x^2}$, the conversion $\exp(\ln(\sin x/x)/x^2)$ leaves a logarithm of a quotient, whereas the 1^∞ shortcut $\exp(\lim x^{-2}(\sin x/x - 1))$ leaves a plain power downstairs and finishes at once with $e^{-1/6}$. Both are legal; only one has a denominator you can divide into a series.

WHERE THIS GOES WRONG

Grinding on the first rewrite that came to mind because you have already invested three lines in it. Sunk cost is not a mathematical argument. The diagnostic is simple: if the expression after a conversion has more distinct transcendental factors in its denominator than before, the conversion was the wrong one. A second diagnostic, for power forms: if after taking logarithms you are looking at a logarithm of a quotient rather than at a plain power, the 1^∞ shortcut was available and you did not use it.

SEE ALSO The $\infty - \infty$ conversion; series division and reciprocal series; expansion at a point other than zero.

A classification drill

The whole chapter compresses into a habit: name the form, choose the conversion, compare orders. The point of the drill below is speed, so the target is two lines per item and no differentiation anywhere. Read each statement, decide the form before reading the solution, and check whether your first instinct was the conversion that was actually used.

- (i) $\lim_{x \rightarrow \infty} (\frac{x+1}{x-1})^x$. Form 1^∞ . The base exceeds 1 by $\frac{2}{x-1}$, so the shortcut gives $\exp(\lim x \cdot \frac{2}{x-1}) = e^2$. Verified as 7.389056 at $x = 10^6$.
- (ii) $\lim_{x \rightarrow \infty} (x - \ln(e^x + 1))$. Form $\infty - \infty$; factor the dominant term out of the logarithm. Since $\ln(e^x + 1) = x + \ln(1 + e^{-x})$, the difference is $-\ln(1 + e^{-x}) \rightarrow 0$.
- (iii) $\lim_{x \rightarrow \infty} x^{1/x}$. Form ∞^0 , indeterminate, so convert: $\exp(\ln x/x)$, and $\ln x \ll x$ by the hierarchy, so the answer is $e^0 = 1$.
- (iv) $\lim_{x \rightarrow \infty} x^2(e^{1/x} - 1 - \frac{1}{x})$. Form $\infty \cdot 0$; set $t = 1/x$ to convert it into $\lim_{t \rightarrow 0^+} (e^t - 1 - t)/t^2 = \frac{1}{2}$. Verified as 0.49996 at $x = 10^6$.
- (v) $\lim_{x \rightarrow 0^+} x^{1/x}$. Base $\rightarrow 0$, exponent $\rightarrow +\infty$: form 0^∞ , determinate, answer 0. No conversion, no logarithm, no rule.
- (vi) $\lim_{x \rightarrow 0^+} (\cot x)^{1/\ln x}$. Base $\rightarrow \infty$, exponent $\rightarrow 0$: form ∞^0 , indeterminate. Convert, use $\cot x \sim 1/x$ so that $\ln \cot x \sim -\ln x$, and get e^{-1} . Items (v) and (vi) are typographically almost the same object and are settled by opposite reasoning: in (v) the base and the exponent push the value the same way, so nothing is in doubt; in (vi) they compete, so everything is.

In every case the conversion was chosen before any calculation began, and in every case it produced a denominator that was a plain power. That is the diagnostic to carry into the next chapter, where the tool that resolves those powers is built properly.

THE ONE SENTENCE TO CARRY FORWARD

Every indeterminate form is converted into $\frac{0}{0}$ or $\frac{\infty}{\infty}$, and every $\frac{0}{0}$ or $\frac{\infty}{\infty}$ is answered by comparing orders: lowest order wins at a point, highest order wins at infinity.

What the chapter has built is a single reflex with two halves. The first half is dominance: at a point the lowest order wins, at infinity the highest order wins, and the growth hierarchy tells you which is which without computation. The second half is the conversion table: each of the seven indeterminate forms has one standard rewrite whose purpose is to expose a dominance question, and the quality of a solution is largely decided by whether the rewrite left a plain power in the denominator. The next chapter supplies the instrument that answers the dominance question at a point in full generality, namely Taylor expansion, and demotes L'Hopital's rule to the special case it actually is.

The matrix below is a compression of the chapter, not a substitute for it. Read the middle column as the first move only. Every row terminates, after that move, in a comparison of orders, and the third column is what to do when the first move produces something worse than it started with.

Recognition matrix

If you see	Reach for	If that fails
Log against power, power against exponential, exponential against factorial	The growth hierarchy; the answer is 0 or ∞	Substitute $x = e^t$ and re-read the chain
Ratio of sums, $x \rightarrow \infty$	Divide by the dominant term of each sum	Expansion at infinity via $t = 1/x$
Pure product or quotient of standard functions, $x \rightarrow 0$	Replace each factor by its equivalent	Full Taylor expansion
A difference whose leading terms cancel	Explicit $O(\cdot)$ bookkeeping to one order past the cancellation	Push to the next order and recheck
$\sqrt{\cdot}$ with $x \rightarrow -\infty$	$\sqrt{x^2} = x = -x$; substitute $x = -t$	Conjugate multiplication
$n!$, n^n , or $\log_n(\cdot)$	Take logarithms, insert Stirling	Normalise first, discard second
H_n , $W(x)$, $\psi(x)$, $\Gamma(1+x)$	Quote the headline asymptotic	Keep the next correction term
Substitution gives $0 \cdot \infty$	Invert the algebraic factor	Invert the other factor instead
Substitution gives $\infty - \infty$	Common denominator, then expand	Conjugate; or factor out the dominant term
Substitution gives $1^\infty, 0^0, \infty^0$	$\exp(g \ln f)$, then treat the exponent as $0 \cdot \infty$	For 1^∞ , use $\exp(\lim g(f-1))$
Substitution gives $0^\infty, \infty^\infty, \infty + \infty$	Nothing: the form is determinate	Recheck the one-sided behaviour of the base
A conversion that made the expression worse	Re-rank the rewrites by what the denominator becomes	Substitute $t = 1/x$ or $h = x - a$ first
A bounded oscillation added to an unbounded term	Squeeze; the oscillation cannot change the class	Bound above and below, then compare
In of a product, quotient or power	Log laws first, so that only $\ln x$ remains	Compare $k \ln x$ against $\ln x$ honestly

Expansion as the primary limit tool

What this chapter is for. A Taylor expansion converts a transcendental function into a polynomial plus a controlled remainder, and once every ingredient of a limit is a polynomial the limit is arithmetic. That is why expansion is not one technique among several: it is the general-purpose engine, and almost every other method at a point is either a special case of it or a slower route to the same coefficient. This chapter installs the standard Maclaurin table, teaches order bookkeeping so that you decide in advance how many terms each factor needs rather than discovering it after three failed attempts, and shows how to move the expansion point to a finite a or to infinity. It closes with L'Hopital's rule treated honestly: what its hypotheses actually require, where repeated application is genuinely the fastest route, where it cycles, and the standard limits at which the rule reports failure while the limit exists perfectly well. At the end you should reach for expansion first, use L'Hopital deliberately rather than reflexively, and never be surprised by how far you had to expand.

The engine, and the table it runs on

The mechanism is one line. If f is n times differentiable at 0, then

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(0)}{k!} x^k + o(x^n) \quad (x \rightarrow 0),$$

and if f is $(n+1)$ times differentiable near 0 the remainder can be written in Lagrange form as $\frac{f^{(n+1)}(\xi)}{(n+1)!} x^{n+1}$ for some ξ between 0 and x . For limit work the little- o form is the one you need, because the only question ever asked is whether the discarded tail dies faster than the denominator.

The reason this dominates the subject is structural. A limit at a point is a question about the first few coefficients of the functions involved, and Taylor's theorem is the statement that those coefficients exist and are computable. Every competing technique extracts the same coefficients less directly. L'Hopital's rule computes $f^{(k)}(0)$ and $g^{(k)}(0)$ one at a time by differentiating the whole expression, which reintroduces the product and quotient rules at every step. Asymptotic equivalence extracts only the first coefficient and then has nothing further to say. Expansion extracts all of them at once, from a table you already know, and the bookkeeping is addition of polynomials.

L'HOPITAL IS A COROLLARY, NOT A RIVAL

Suppose f and g vanish at a and are twice differentiable there. Expanding both,

$$\frac{f(x)}{g(x)} = \frac{f'(a)(x-a) + \frac{1}{2}f''(a)(x-a)^2 + o((x-a)^2)}{g'(a)(x-a) + \frac{1}{2}g''(a)(x-a)^2 + o((x-a)^2)} = \frac{f'(a) + \frac{1}{2}f''(a)(x-a) + o(x-a)}{g'(a) + \frac{1}{2}g''(a)(x-a) + o(x-a)},$$

and if $g'(a) \neq 0$ the limit is $f'(a)/g'(a)$. That is exactly L'Hopital's conclusion, obtained by cancelling one power of $(x-a)$ from a pair of expansions. One round of the rule is therefore the same operation as truncating both series at order 1; k rounds is truncating at order k . The difference is that expansion performs all k truncations in a single pass, from a table, whereas the rule performs them one at a time, by differentiating expressions that grow at every step. Everything in the last section of this chapter follows from that observation.

You cannot use the engine without the fuel. The following table is the minimum, given to the order that actually comes up. The final column is the leading deviation from the linear or constant part, which is the number that decides most problems and the number most often misremembered.

Function	Maclaurin series, to the order that matters	Signature term
e^x	$1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + O(x^5)$	$+\frac{x^2}{2}$
$\ln(1+x)$	$x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + O(x^5)$	$-\frac{x^2}{2}$
$(1+x)^a$	$1 + ax + \frac{a(a-1)}{2}x^2 + \frac{a(a-1)(a-2)}{6}x^3 + O(x^4)$	$+\frac{a(a-1)}{2}x^2$
$\sin x$	$x - \frac{x^3}{6} + \frac{x^5}{120} + O(x^7)$	$-\frac{x^3}{6}$
$\cos x$	$1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720} + O(x^8)$	$-\frac{x^2}{2}$
$\tan x$	$x + \frac{x^3}{3} + \frac{2x^5}{15} + \frac{17x^7}{315} + O(x^9)$	$+\frac{x^3}{3}$
$\sec x$	$1 + \frac{x^2}{2} + \frac{5x^4}{24} + \frac{61x^6}{720} + O(x^8)$	$+\frac{x^2}{2}$
$\arcsin x$	$x + \frac{x^3}{6} + \frac{3x^5}{40} + \frac{5x^7}{112} + O(x^9)$	$+\frac{x^3}{6}$
$\arctan x$	$x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + O(x^9)$	$-\frac{x^3}{3}$
$\sinh x$	$x + \frac{x^3}{6} + \frac{x^5}{120} + O(x^7)$	$+\frac{x^3}{6}$
$\cosh x$	$1 + \frac{x^2}{2} + \frac{x^4}{24} + \frac{x^6}{720} + O(x^8)$	$+\frac{x^2}{2}$
$\tanh x$	$x - \frac{x^3}{3} + \frac{2x^5}{15} - \frac{17x^7}{315} + O(x^9)$	$-\frac{x^3}{3}$
$\operatorname{arsinh} x$	$x - \frac{x^3}{6} + \frac{3x^5}{40} + O(x^7)$	$-\frac{x^3}{6}$

Three structural facts make this table smaller than it looks. The hyperbolic entries are the circular ones with the alternating signs removed, so $\sinh$, $\cosh$ and $\tanh$ cost nothing extra once $\sin$, $\cos$ and $\tan$ are known. The signature term of a function and of its inverse always have opposite signs and, for the sine family, equal magnitude: $\sin$ has $-\frac{x^3}{6}$ and $\arcsin$ has $+\frac{x^3}{6}$, $\tan$ has $+\frac{x^3}{3}$ and $\arctan$ has $-\frac{x^3}{3}$, $\sinh$ has $+\frac{x^3}{6}$ and arsinh has $-\frac{x^3}{6}$. And $(1+x)^a$ subsumes every radical, so no separate root entries are needed.

EASY

4.1 The standard expansion table

CUE Any limit at all as $x \rightarrow 0$, and any limit that a substitution can move to 0.

METHOD Know the table above cold, to the orders shown. In use, replace each transcendental factor by its polynomial and carry an explicit remainder. The single most valuable line of the table is the group of four functions that all begin $x \pm \frac{x^3}{k}$, since differences within that group are the commonest engineered limits in the subject.

WORKED EXAMPLE

$$\lim_{x \rightarrow 0} \frac{\arcsin x - \arctan x}{x^3}.$$

The linear terms cancel by construction, and the signature terms are $+\frac{x^3}{6}$ and $-\frac{x^3}{3}$. So the numerator is $(\frac{1}{6} + \frac{1}{3})x^3 + O(x^5) = \frac{x^3}{2} + O(x^5)$ and the limit is $\frac{1}{2}$, verified as 0.4999875 at $x = 10^{-2}$. The whole calculation is two table lookups and one addition; L'Hopital would need three rounds, and the third derivative of $\arcsin$ is not a pleasant object. The same two lookups settle the most-asked limit in the family:

$$\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3} = \left(\frac{1}{3} - \left(-\frac{1}{6}\right)\right) = \frac{1}{2},$$

verified as 0.5000001 at $x = 10^{-3}$. Note what this says about the trap in the previous chapter: $\tan x \sim x$ and $\sin x \sim x$ are both true, and their difference is nonetheless of order 3 with coefficient $\frac{1}{2}$. Equivalence could not have found this; the table found it in one line.

WHERE THIS GOES WRONG

Confusing the signs within the $x \pm \frac{x^3}{k}$ family. The reliable rule is the inverse-function rule above: $\tan$ and $\arcsin$ bend up, $\arctan$ and $\tanh$ bend down, $\sin$ is negative and $\sinh$ is positive. The second trap is quoting $\arcsin x = x + \frac{x^3}{3}$, borrowing $\tan$'s denominator; it is 6, not 3.

SEE ALSO Expand to the first surviving order; the binomial series for radicals; composite and nested expansion.

The table covers functions of x . Two extensions cover essentially everything else: an arbitrary real power, which absorbs all radicals, and a function of a function, which absorbs all the composites. Take the radicals first, because they are the ones most often mishandled by stopping one term too early.

4.2 The binomial series for radicals

CUE A square root, cube root, or any fractional power of something near 1; equivalently, any radical whose argument can be written as $c(1+u)$ with $u \rightarrow 0$.

METHOD Use

$$(1+u)^a = 1 + au + \frac{a(a-1)}{2}u^2 + \frac{a(a-1)(a-2)}{6}u^3 + O(u^4),$$

where u is the *whole* small perturbation and may itself be a series. Force the argument into the form $1+u$ first, pulling out constants and dominant powers, since the series is valid only there.

WORKED EXAMPLE

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+2x} - \sqrt[3]{1+3x}}{x^2}.$$

With $a = \frac{1}{2}$, $u = 2x$: $\sqrt{1+2x} = 1 + x - \frac{x^2}{2} + O(x^3)$, since $\frac{a(a-1)}{2}u^2 = \frac{(1/2)(-1/2)}{2} \cdot 4x^2 = -\frac{x^2}{2}$. With $a = \frac{1}{3}$, $u = 3x$: $\sqrt[3]{1+3x} = 1 + x - x^2 + O(x^3)$, since $\frac{(1/3)(-2/3)}{2} \cdot 9x^2 = -x^2$. The linear terms cancel and the limit is $-\frac{1}{2} - (-1) = \frac{1}{2}$. Verified as 0.49884 at $x = 10^{-3}$.

A version at infinity, where the perturbation has to be manufactured:

$$\lim_{x \rightarrow \infty} x^3 (\sqrt[3]{x^3 + 3x} - \sqrt{x^2 + 2}) = \lim_{x \rightarrow \infty} x^3 \left[\left(x + \frac{1}{x} - \frac{1}{x^3}\right) - \left(x + \frac{1}{x} - \frac{1}{2x^3}\right) \right] = -\frac{1}{2},$$

both radicals having been written as $x(1+u)^a$ with $u = 3x^{-2}$ and $u = 2x^{-2}$. Verified to six figures at $x = 10^3$ with 50-digit arithmetic; ordinary floating point cannot see it, because the two radicals agree to twelve significant figures and the difference is pure round-off.

WHERE THIS GOES WRONG

Using $\sqrt{1+u} \approx 1 + \frac{u}{2}$ when the problem needs the $-\frac{u^2}{8}$ term. A radical limit whose denominator is x^2 always needs second order, and a limit like the one at infinity above needs third. The other error is applying the series to $\sqrt{x^2+2}$ directly, as though x^2 were small; the perturbation must be the small quantity, which is why the x is pulled out first.

SEE ALSO The standard expansion table; composite and nested expansion; expansion at infinity via $t = 1/x$.

Radicals were the first composition. The general case is a series inside a series, and it looks more frightening than it is, because the inner function vanishes at the origin and that fact alone bounds the work.

MEDIUM

4.3 Composite and nested expansion

CUE A series inside a series: $\ln(\cos x)$, $e^{\sin x}$, $\sin(\tan x)$, $\sqrt{1+\sin^2 x}$, $\tan(\ln(1+x))$.

METHOD Expand the inner function first, call the result $u = u(x)$ with $u = O(x)$, then substitute into the outer expansion in powers of u and discard every term whose total degree in x exceeds the target. The work is bounded because the u^k term of the outer series contributes only from degree k upward: if the target is degree N and u has order 1, only the first N terms of the outer series can matter, and usually far fewer.

WORKED EXAMPLE

$$\lim_{x \rightarrow 0} \frac{\sqrt[3]{1+\sin^2 x} - \sqrt{\cos x}}{x^2}.$$

Target degree 2. Inner functions: $\sin^2 x = x^2 + O(x^4)$ and $\cos x - 1 = -\frac{x^2}{2} + O(x^4)$, both of order 2, so only the linear term of each outer binomial can contribute. Hence $\sqrt[3]{1+\sin^2 x} = 1 + \frac{1}{3}x^2 + O(x^4)$ and $\sqrt{\cos x} = 1 - \frac{1}{4}x^2 + O(x^4)$, and the numerator is $(\frac{1}{3} + \frac{1}{4})x^2 + O(x^4)$. The limit is $\frac{7}{12}$, verified as 0.5833331 at $x = 10^{-3}$.

A composite where the outer function must be exploited rather than expanded:

$$\lim_{x \rightarrow 0} \frac{e^{\sin x} - e^{\tan x}}{x^3} = \lim_{x \rightarrow 0} e^{\sin x} \cdot \frac{1 - e^{\tan x - \sin x}}{x^3} = \lim_{x \rightarrow 0} \frac{-(\tan x - \sin x)}{x^3} = -\frac{1}{2},$$

since $\tan x - \sin x = \frac{x^3}{2} + O(x^5)$ and $e^u - 1 \sim u$. Verified as -0.50504 at $x = 10^{-2}$, converging to -0.5 . Factoring out the common exponential turned a two-series problem into a one-series problem.

A nesting where the outer series must be carried to u^2 :

$$\lim_{x \rightarrow 0} \frac{\ln(\cos x) + \frac{x^2}{2}}{x^4} = -\frac{1}{12}.$$

Here $u = \cos x - 1 = -\frac{x^2}{2} + \frac{x^4}{24} + O(x^6)$ has order 2 and the target degree is 4, so the outer series $\ln(1 + u) = u - \frac{u^2}{2} + O(u^3)$ must be taken to u^2 and no further, since $u^3 = O(x^6)$. Then $\ln(\cos x) = -\frac{x^2}{2} + \frac{x^4}{24} - \frac{1}{2} \cdot \frac{x^4}{4} + O(x^6) = -\frac{x^2}{2} - \frac{x^4}{12} + O(x^6)$. Verified as -0.0833356 at $x = 10^{-2}$ against $-1/12 = -0.0833333$.

WHERE THIS GOES WRONG

Substituting the entire inner series into the entire outer series and multiplying out. That generates dozens of irrelevant high-degree terms and buries the two that matter. Decide the target degree first, then note the order of u , then expand the outer series only as far as $u^{\lfloor N/\text{ord } u \rfloor}$.

SEE ALSO The binomial series for radicals; order bookkeeping across a subtraction; when the third order is genuinely needed.

The same limit four ways

It is worth doing one limit by every available method, because the comparison is the argument of this chapter in miniature.

Take $L = \lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3}$.

By the table. The signature terms are $+\frac{x^3}{3}$ and $-\frac{x^3}{6}$, so the numerator is $\frac{x^3}{2} + O(x^5)$ and $L = \frac{1}{2}$. Two lookups.

By factoring, then the table. $\tan x - \sin x = \sin x(\sec x - 1) = \frac{\sin x(1 - \cos x)}{\cos x}$, and now the three factors have known leading behaviour x , $\frac{x^2}{2}$ and 1, so the numerator is $\frac{x^3}{2} + O(x^5)$ again. This route is the one to prefer when the table has been forgotten, since $1 - \cos x \sim \frac{x^2}{2}$ is the only fact used.

By L'Hopital. Three rounds. First $\frac{\sec^2 x - \cos x}{3x^2}$, still $\frac{0}{0}$; then $\frac{2 \sec^2 x \tan x + \sin x}{6x}$, still $\frac{0}{0}$; then $\frac{2 \sec^4 x + 4 \sec^2 x \tan^2 x + \cos x}{6} \rightarrow \frac{2 + 0 + 1}{6} = \frac{1}{2}$. Correct, and three chances to lose a factor of two.

By asymptotic equivalence. $\tan x \sim x$ and $\sin x \sim x$, so the numerator is “ $\sim x - x = 0$ ” and $L = 0$. This is wrong. Equivalence is not closed under subtraction, and the failure is silent: nothing in the calculation signals that an illegal step was taken.

Four methods, three correct, one fast, and one that fails without warning. That ranking is stable across the whole subject, and it is why the table comes first.

Order bookkeeping: deciding in advance how far to expand

Beginners expand, discover the answer is $0/0$ again, expand further, and repeat. That loop is avoidable, and avoiding it is most of what separates a fluent solver from a slow one. The rule is arithmetic on orders, where the *order* of a function is the degree of its lowest nonvanishing Maclaurin term: $\text{ord}(\sin x) = 1$, $\text{ord}(1 - \cos x) = 2$, $\text{ord}(\tan x - x) = 3$. Three facts govern that arithmetic. Orders add under multiplication, $\text{ord}(fg) = \text{ord } f + \text{ord } g$, with no exceptions. Orders multiply under composition, $\text{ord}(f \circ g) = \text{ord } f \cdot \text{ord } g$ when $g(0) = 0$, which is why $\text{ord}(\sin(x^2)) = 2$ and $\text{ord}(1 - \cos(x^2)) = 4$. And under addition $\text{ord}(f \pm g) \geq \min(\text{ord } f, \text{ord } g)$, with equality unless the leading terms cancel, which is the one place where the arithmetic cannot be done by inspection and where every difficult problem in this section lives.

HOW MANY TERMS, DECIDED IN ADVANCE

Let N/D be the quotient and let $d = \text{ord } D$. Expand N through degree d inclusive, carrying $O(x^{d+1})$. If $\text{ord } N < d$ the limit is infinite; if $\text{ord } N = d$ the limit is the ratio of the two leading coefficients; if N vanishes through degree d the limit is 0. For a numerator that is a product $N = \prod_i f_i$ with $\text{ord } f_i = m_i$ and total order $m = \sum_i m_i$, factor f_i must be expanded only through degree

$$d - (m - m_i),$$

because the other factors contribute at least $m - m_i$ degrees for free. That formula is the whole of order bookkeeping.

4.4 Expand to the first surviving order, and no further

CUE A quotient $N(x)/x^k$ or $N(x)/D(x)$ as $x \rightarrow 0$, where N is a product or a sum of standard functions.

METHOD Compute $d = \text{ord } D$, apply the budget formula above to each factor of N , expand exactly that far, and stop. The identity underneath is that a function of order m satisfies $f(x) = \frac{f^{(m)}(0)}{m!}x^m + O(x^{m+1})$, so the answer is a single coefficient and everything beyond it is waste.

WORKED EXAMPLE

$$\lim_{x \rightarrow 0} \frac{(e^x - 1)^2 (\ln(1 + x) - \sin x)^2}{x^6}.$$

Here $d = 6$. The factors are $f_1 = (e^x - 1)^2$ with $m_1 = 2$ and $f_2 = (\ln(1 + x) - \sin x)^2$ with $m_2 = 4$, so $m = 6$ and the budgets are $6 - (6 - 2) = 2$ for f_1 and $6 - (6 - 4) = 4$ for f_2 . Within budget: $e^x - 1 = x + O(x^2)$ gives $f_1 = x^2 + O(x^3)$, and $\ln(1 + x) - \sin x = (x - \frac{x^2}{2} + \frac{x^3}{3}) - (x - \frac{x^3}{6}) + O(x^4) = -\frac{x^2}{2} + O(x^3)$ gives $f_2 = \frac{x^4}{4} + O(x^5)$. The product is $\frac{x^6}{4} + O(x^7)$ and the limit is $\frac{1}{4}$, verified as 0.2475 at $x = 10^{-2}$.

A quotient where the denominator is not a bare power, so d must be computed:

$$\lim_{x \rightarrow 0} \frac{(x - \sin x)(e^x - 1)}{x^2(1 - \cos x)}.$$

The denominator has order $2 + 2 = 4$, so $d = 4$. The numerator's factors have orders 3 and 1, total 4, so each needs only its leading term: $x - \sin x = \frac{x^3}{6} + O(x^5)$ and $e^x - 1 = x + O(x^2)$, giving $\frac{x^4}{6} + O(x^5)$. The denominator's leading coefficient is $\frac{1}{2}$, so the limit is $\frac{1/6}{1/2} = \frac{1}{3}$, verified as 0.333500 at $x = 10^{-3}$.

WHERE THIS GOES WRONG

Expanding one factor to order 5 and another to order 2 by instinct. The budget formula exists precisely so that this is a computation rather than a guess. The reverse error is equally costly: expanding every factor to order 6 here would mean carrying the fifth Maclaurin coefficient of $\ln(1 + x)$, which cannot affect the answer. A third failure is applying the budget formula with a stale m_i : if a factor turns out to have higher order than you assumed, because its leading terms cancelled, every other factor's budget shrinks and must be recomputed.

SEE ALSO The standard expansion table; order bookkeeping across a subtraction; series division and reciprocal series.

The budget formula assumes you know the order of each factor. When the numerator is a difference of two functions with the same leading behaviour, you do not: the order of the difference is exactly what the cancellation determines, and getting it wrong is the standard way to produce a confident wrong answer.

MEDIUM

4.5 Order bookkeeping across a subtraction

CUE Two expressions with identical leading terms are being subtracted, and the answer lives entirely in what survives.

METHOD Decide in advance how many orders will cancel, then expand both sides one order past that. If the denominator is x^k , expand everything through x^k inclusive with an explicit $O(x^{k+1})$, and then confirm that the surviving coefficient is genuinely nonzero before reporting it. When two functions agree to order r , their difference has order at least $r + 1$, and the denominator of the problem tells you what order the setter intends.

WORKED EXAMPLE

$$\lim_{x \rightarrow 0} \frac{(W_0(x) - x + x^2)(\sin x - x)}{x^6},$$

where W_0 is the principal Lambert function, $W_0(x) = x - x^2 + \frac{3}{2}x^3 - \frac{8}{3}x^4 + O(x^5)$. The first factor is engineered to cancel two orders, leaving $\frac{3}{2}x^3 + O(x^4)$, so $m_1 = 3$. The second is $\sin x - x = -\frac{x^3}{6} + O(x^5)$, so $m_2 = 3$ and $m = 6 = d$: each factor needs only its leading term. The product is $\frac{3}{2} \cdot (-\frac{1}{6})x^6 + O(x^7)$ and the limit is $-\frac{1}{4}$. Verified as -0.24564 at $x = 10^{-2}$ and -0.24867 at $x = 3 \times 10^{-3}$, converging to -0.25 from above as the $O(x^7)$ correction dies.

A shorter instance of the same discipline, with no exotic functions:

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - e^{x/2}}{x^2} = -\frac{1}{4}.$$

Both functions equal $1 + \frac{x}{2} + \dots$, so one order cancels and the denominator says to expand through order 2. From the table, $\sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + O(x^3)$ and $e^{x/2} = 1 + \frac{x}{2} + \frac{x^2}{8} + O(x^3)$: the quadratic coefficients differ only in sign, which is the whole design. The difference is $-\frac{x^2}{4} + O(x^3)$ and the limit is $-\frac{1}{4}$, verified as -0.2499584 at $x = 10^{-3}$.

WHERE THIS GOES WRONG

The cancellation runs deeper than expected and the term you kept was itself cancelled, so the reported answer is a spurious 0 or ∞ . Always check that the surviving coefficient is nonzero as a number, not as a symbol: if it is a difference of two fractions, evaluate it. In the example above, a solver who wrote $W_0(x) = x - x^2 + O(x^3)$ would have concluded that the first factor is $O(x^3)$ with an unknown coefficient, which is true but useless.

SEE ALSO Expand to the first surviving order; engineered-cancellation limits; when the third order is genuinely needed.

Some problems make the cancellation the entire point, and once you see the design the answer can be read off without any expansion at all.

4.6 Engineered-cancellation limits

MEDIUM

CUE The numerator visibly subtracts off the first few Maclaurin terms of a standard function, or two standard functions are averaged so that their low-order terms annihilate.

METHOD Recognise the design. If the numerator is $f(x)$ minus the first m terms of f , and the denominator is x^{m+1} or x^{m+2} , the answer is simply the next Taylor coefficient of f , that is $f^{(m+1)}(0)/(m+1)!$ or $f^{(m+2)}(0)/(m+2)!$ as the case may be, with 0 if the intervening coefficient vanishes.

WORKED EXAMPLE

The bare form first:

$$\lim_{x \rightarrow 0} \frac{1 - \cos x - \frac{1}{2}x^2}{x^4} = -\frac{1}{24},$$

because the numerator is exactly $-\frac{x^4}{24} + O(x^6)$: the constant and quadratic terms of $\cos$ have been removed by design and the quartic coefficient is what remains. Now the same design wearing a 1^∞ costume:

$$\lim_{x \rightarrow 0} \left(\frac{\cosh x + \cos x}{2} \right)^{1/x^4} = e^{1/24}.$$

The averaging kills the x^2 terms, since $\cosh$ contributes $+\frac{x^2}{2}$ and $\cos$ contributes $-\frac{x^2}{2}$, while the quartic terms reinforce: both contribute $+\frac{x^4}{24}$, so the base is $1 + \frac{x^4}{24} + O(x^8)$. The 1^∞ shortcut gives $\exp(\lim x^{-4} \cdot \frac{x^4}{24}) = e^{1/24}$. Verified as 1.0425469 at $x = 0.05$ against $e^{1/24} = 1.0425469$.

WHERE THIS GOES WRONG

Attacking the first form with four rounds of L'Hopital. It works, but the fourth derivative of the numerator has to be produced without error and the form has to be rechecked at each of the four stages. Reading one coefficient off the table is not merely faster, it is a different order of reliability.

SEE ALSO Expand to the first surviving order; the standard expansion table; the exponential conversion for power forms.

READ THE DENOMINATOR AS A HINT

In a constructed problem the exponent of the denominator is not arbitrary: it is chosen so that the limit is finite and nonzero, which means it equals the order of the numerator. So the denominator tells you in advance how deep the cancellation goes, and therefore how far to expand. If your expansion produces a numerator of a different order than the denominator, the likeliest explanation is an arithmetic slip, not an infinite or zero limit. Check before believing.

The engineered cases are generous: the denominator announces how deep the cancellation goes. The remaining case is the ungenerous one, where the cancellation is deeper than any reasonable person would guess, and the only defence is to expand honestly and keep going.

4.7 When the third order is genuinely needed

CUE The denominator is x^3 or higher, or the numerator is a difference of two functions that agree to second order and possibly much further.

METHOD Push both expansions to exactly the order the denominator demands, using the budget rule, and do not stop early because several orders have cancelled. Some classics need far more than three: $\sin(\tan x)$ and $\tan(\sin x)$ agree through order 6, so anything built from their difference lives at order 7.

WORKED EXAMPLE

$$\lim_{x \rightarrow 0} \frac{\sin(\tan x) - \tan(\sin x)}{x^7}.$$

Both composites expand to $x + \frac{x^3}{6} - \frac{x^5}{40} + O(x^7)$, and the two agree at orders 1, 3 and 5; the even orders vanish because both functions are odd. The first surviving difference is at order 7, where the coefficients differ by $-\frac{1}{30}$, so the limit is $-\frac{1}{30}$. The numerical evidence has to be read carefully: the quotient is -0.033472 at $x = 0.06$ and -0.033368 at $x = 0.03$, approaching -0.033333 quadratically in x , exactly as an $O(x^2)$ relative correction predicts.

WHERE THIS GOES WRONG

Concluding after two orders of cancellation that “the limit is 0” and stopping. Cancellation through order k says that the answer is the coefficient at order $k + 1$; it says nothing about that coefficient being zero. The second trap here is numerical: at $x = 0.3$ the quotient reads -0.0370 , which rounds to neither $-1/30$ nor anything else recognisable, and a solver who trusts it will chase a nonexistent closed form.

The deep agreement in that example is not an accident, and knowing why saves you from expecting it where it does not occur. Both $\sin \circ \tan$ and $\tan \circ \sin$ are odd, so all even coefficients vanish for free; and the order-3 and order-5 coefficients of a composite $u \circ v$ built from two members of the $x \pm \frac{x^3}{k}$ family depend only on the sum of the two signature coefficients, which is symmetric under swapping the functions. Order 7 is the first place where the composition order can be felt. The same structure gives

$$\lim_{x \rightarrow 0} \frac{\arcsin(\arctan x) - \arctan(\arcsin x)}{x^7} = -\frac{1}{30},$$

the same value, verified as -0.033318 at $x = 0.03$. Whenever you meet a pair of composites of this shape, budget for order 7 before you start.

SEE ALSO Order bookkeeping across a subtraction; composite and nested expansion; repeated L'Hopital with simplification.

An order-counting drill

For each quotient, state $d = \text{ord } D$ and the budget for each numerator factor before computing anything. The point is that all of this is decidable by inspection.

- (i) $\frac{\sin x - x}{x^3}$. Here $d = 3$; one factor, budget 3; answer $-\frac{1}{6}$.
- (ii) $\frac{(1 - \cos x) \ln(1 + x)}{x^3}$. Here $d = 3$, orders 2 and 1 summing to 3, so each factor needs only its leading term; answer $\frac{1}{2}$.
- (iii) $\frac{\tan x - \arctan x}{x^3}$. Here $d = 3$ and the linear terms cancel, so both must go to order 3; the signature terms are $+\frac{1}{3}$ and $-\frac{1}{3}$, giving $\frac{2}{3}$.
- (iv) $\frac{e^{x^2} - \cos x}{x^2}$. Here $d = 2$; the inner argument x^2 has order 2, so $e^{x^2} = 1 + x^2 + O(x^4)$ needs only one term of the outer series; against $\cos x = 1 - \frac{x^2}{2} + O(x^4)$ the answer is $\frac{3}{2}$.
- (v) $\frac{x^2(e^x - 1)}{\sin x(1 - \cos x)}$. Here $d = 1 + 2 = 3$, the numerator has order $2 + 1 = 3$, and both leading coefficients are available at once; the answer is $\frac{1}{1 \cdot \frac{1}{2}} = 2$.
- (vi) $\frac{1 - \cos(x^2)}{x^2 \sin^2 x}$. Composition multiplies orders, so the numerator has order 4 with leading term $\frac{x^4}{2}$; the denominator has order $2 + 2 = 4$ with leading coefficient 1; the answer is $\frac{1}{2}$. Nobody needs to expand anything beyond one term.

- (vii) $\frac{\arctan x - \operatorname{arsinh} x}{x^3}$. Both are $x \mp \frac{x^3}{k}$ types with signature terms $-\frac{1}{3}$ and $-\frac{1}{6}$, so $d = 3$ and the answer is $-\frac{1}{3} + \frac{1}{6} = -\frac{1}{6}$.
The linear terms cancel; the cubic terms do not.

Not one of these required a decision to be revised halfway through, and that is the whole benefit. A solver who counts orders first never expands twice.

Moving the expansion point, and dividing series

Everything so far assumed the limit point was 0. That assumption costs nothing, because a substitution always restores it, but the substitution has to be made explicitly rather than assumed. The three techniques here move the expansion point to a finite a , to infinity, and then handle the one algebraic operation that expansions do not support directly, namely division by a series with no constant term.

THE THREE SUBSTITUTIONS

$h = x - a$ moves a finite limit point to the origin. $t = 1/x$ moves $\pm\infty$ to the origin. $u = 1/(x - a)$ moves a point where the expression blows up to infinity, which is occasionally what a problem wants. In every case the substitution must be applied to the *whole* expression, exponents included, and the direction of approach must be tracked: $x \rightarrow a^+$ becomes $h \rightarrow 0^+$, and $x \rightarrow -\infty$ becomes $t \rightarrow 0^-$, which is where signs are lost.

4.8 Expansion at a point other than zero

MEDIUM

CUE $x \rightarrow a$ with $a \neq 0$: most often $x \rightarrow 1$, $x \rightarrow \pi/2$, or $x \rightarrow e$.

METHOD Substitute $h = x - a$ and expand in h :

$$f(a + h) = f(a) + f'(a)h + \frac{f''(a)}{2}h^2 + O(h^3).$$

For the standard functions it is usually quicker to use an addition formula than to differentiate: $\sin(a + h) = \sin a \cos h + \cos a \sin h$, then expand $\cos h$ and $\sin h$ from the table. Once the substitution is made, every technique for $x \rightarrow 0$ applies verbatim.

WORKED EXAMPLE

$$\lim_{x \rightarrow 1} \left\{ \frac{-ax + \sin(x - 1) + a}{x + \sin(x - 1) - 1} \right\}^{(1-x)/(1-\sqrt{x})}, \quad a < 1.$$

Put $h = x - 1$. The numerator is $a(1 - x) + \sin(x - 1) = -ah + (h - \frac{h^3}{6} + O(h^5)) = (1 - a)h + O(h^3)$, and the denominator is $h + \sin h = 2h + O(h^3)$, so the base tends to $\frac{1-a}{2}$, a positive number. The exponent factors rather than expands: $\frac{1-x}{1-\sqrt{x}} = \frac{(1-\sqrt{x})(1+\sqrt{x})}{1-\sqrt{x}} = 1 + \sqrt{x} \rightarrow 2$. The limit is therefore $\left(\frac{1-a}{2}\right)^2$. Verified at $x = 1 + 10^{-6}$: the value is 0.2500000 for $a = 0$ and 4.000001 for $a = -3$.

A second, shorter instance:

$$\lim_{x \rightarrow \pi/2} \frac{1 - \sin x}{(x - \pi/2)^2} = \lim_{h \rightarrow 0} \frac{1 - \cos h}{h^2} = \frac{1}{2},$$

using $\sin(\pi/2 + h) = \cos h$. Verified as 0.5000000 at $h = 10^{-4}$.

And one where the addition formula does the work:

$$\lim_{x \rightarrow \pi/4} \frac{\tan x - 1}{x - \pi/4} = \lim_{h \rightarrow 0} \frac{\frac{1 + \tan h}{1 - \tan h} - 1}{h} = \lim_{h \rightarrow 0} \frac{2 \tan h}{h(1 - \tan h)} = 2,$$

using $\tan(\pi/4 + h) = \frac{1 + \tan h}{1 - \tan h}$ and $\tan h \sim h$. This is of course $\sec^2(\pi/4) = 2$, the derivative of $\tan$ at $\pi/4$, which is the point: a difference quotient at a is an expansion at a truncated at first order.

WHERE THIS GOES WRONG

Expanding $\sin(x - 1)$ "around $x = 0$ " because that is the table you memorised. The expansion point must be the limit point. A related error is expanding the base around a but leaving the exponent in terms of x and then taking limits of the two separately, which is only legal when the base has a limit different from 1, as it does here.

SEE ALSO Composite and nested expansion; expansion at infinity via $t = 1/x$; the exponential conversion for power forms.

The limit point ∞ is handled the same way, by a substitution that manufactures a small quantity where none was visible.

4.9 Expansion at infinity via $t = 1/x$

EASY

CUE $x \rightarrow \infty$ with no single dominant term, or a difference of two quantities that both grow and nearly cancel.

METHOD Set $t = 1/x$ and expand in $t \rightarrow 0^+$. The workhorse case is

$$\sqrt{x^2 + ax + b} = x\sqrt{1 + at + bt^2} = x\left(1 + \frac{a}{2}t + \frac{4b - a^2}{8}t^2 + O(t^3)\right) = x + \frac{a}{2} + \frac{4b - a^2}{8x} + O(x^{-2}),$$

valid for $x \rightarrow +\infty$; for $x \rightarrow -\infty$ pull out $|x| = -x$ instead.

WORKED EXAMPLE

$$\lim_{x \rightarrow \infty} (\sqrt{x^2 + x} - x) = \lim_{x \rightarrow \infty} \left(x + \frac{1}{2} + O(x^{-1}) - x\right) = \frac{1}{2},$$

directly from the display with $a = 1, b = 0$. Verified as 0.4999999 at $x = 10^6$. A case that needs the next order:

$$\lim_{x \rightarrow \infty} x^{3/2}(\sqrt{x+1} + \sqrt{x-1} - 2\sqrt{x}).$$

Write $\sqrt{x \pm 1} = \sqrt{x}(1 \pm t)^{1/2}$ with $t = 1/x$. The linear terms $\pm \frac{1}{2}t$ cancel in the sum, and the quadratic terms reinforce: $(1+t)^{1/2} + (1-t)^{1/2} = 2 - \frac{t^2}{4} + O(t^4)$. So the bracket is $-\frac{\sqrt{x}}{4x^2} + O(x^{-7/2})$ and the limit is $-\frac{1}{4}$, verified as -0.2500000 at $x = 10^4$.

WHERE THIS GOES WRONG

Expanding $\sqrt{x^2 + x}$ as though x were small. The binomial series needs a small perturbation, and manufacturing one is the entire purpose of the substitution. The second trap is dropping the $O(t^2)$ term in the first example because “it goes to zero”: it does, but only after multiplication by the x that was pulled out, so its order must be tracked through that multiplication.

SEE ALSO The binomial series for radicals; divide by the dominant term; radicals at infinity.

For a difference of exactly two square roots there is a shortcut worth keeping: multiplying by the conjugate turns $\sqrt{A} - \sqrt{B}$ into $(A - B)/(\sqrt{A} + \sqrt{B})$, and since $A - B$ is usually a polynomial the whole problem collapses without any series at all. On $\sqrt{x^2 + x} - x$ this gives $x/(\sqrt{x^2 + x} + x) \rightarrow \frac{1}{2}$ immediately. The substitution route is the general one, however, because it survives cube roots, three-term differences, and the case where the polynomial difference is itself of high degree, none of which the conjugate handles.

One operation remains unsupported. A Maclaurin series exists only for a function finite at the origin, so a quotient whose denominator vanishes has no series until the vanishing factor is removed. Removing it is mechanical.

4.10 Series division and reciprocal series

MEDIUM

CUE A quotient of two vanishing series, or the reciprocal of a series with zero constant term, or a difference of two such reciprocals.

METHOD Factor the leading power out of the denominator and expand the reciprocal of what remains geometrically, using

$$\frac{1}{1+w} = 1 - w + w^2 - \dots \text{ with } w \text{ the sum of all the non-constant terms:}$$

$$\frac{1}{x(1 + c_1x + c_2x^2 + \dots)} = \frac{1}{x} \left(1 - c_1x + (c_1^2 - c_2)x^2 + \dots\right).$$

The extracted $1/x$ is a genuine pole, and in a difference of two such reciprocals the poles are what cancel.

WORKED EXAMPLE

Taking $\ln(1+x) = x(1 - \frac{x}{2} + \frac{x^2}{3} - \dots)$, so $c_1 = -\frac{1}{2}$ and $c_2 = \frac{1}{3}$:

$$\frac{1}{\ln(1+x)} = \frac{1}{x} \left(1 + \frac{x}{2} + \left(\frac{1}{4} - \frac{1}{3}\right)x^2 + \dots\right) = \frac{1}{x} + \frac{1}{2} - \frac{x}{12} + O(x^2),$$

verified numerically: $(1/\ln(1+x) - 1/x - \frac{1}{2})/x$ is -0.083318 at $x = 10^{-4}$ against $-1/12 = -0.083333$. The immediate payoff:

$$\lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{e^x - 1} \right) = \lim_{x \rightarrow 0} \left(\frac{1}{x} - \left(\frac{1}{x} - \frac{1}{2} + \frac{x}{12} + O(x^2) \right) \right) = \frac{1}{2},$$

using $e^x - 1 = x(1 + \frac{x}{2} + \frac{x^2}{6} + \dots)$ and the same reciprocal formula. Verified as 0.4999917 at $x = 10^{-4}$.

The same extraction applied to the cotangent, which is where it is most often needed:

$$\cot x = \frac{\cos x}{\sin x} = \frac{1 - \frac{x^2}{2} + O(x^4)}{x(1 - \frac{x^2}{6} + O(x^4))} = \frac{1}{x} \left(1 - \frac{x^2}{2} \right) \left(1 + \frac{x^2}{6} \right) + O(x^3) = \frac{1}{x} - \frac{x}{3} + O(x^3),$$

so $\lim_{x \rightarrow 0} \frac{\cot x - 1/x}{x} = -\frac{1}{3}$, verified as -0.3333334 at $x = 10^{-3}$. Every $\infty - \infty$ limit built from $\cot$, $\csc$, or $1/\ln(1+x)$ is this calculation wearing different clothes.

WHERE THIS GOES WRONG

Trying to “expand $1/\ln(1+x)$ ” as a Maclaurin series. It has a pole at 0 and no such series exists; the pole must be extracted first. The second trap is forgetting the c_1^2 term in the reciprocal formula: $1/(1+w)$ is $1 - w + w^2$, and w^2 contributes at order 2 whenever $c_1 \neq 0$. In the $\ln(1+x)$ computation above, omitting it turns $\frac{1}{4} - \frac{1}{3} = -\frac{1}{12}$ into $-\frac{1}{3}$, and the resulting answer for any $\infty - \infty$ limit built from it is wrong by a factor of four.

SEE ALSO Choosing the profitable conversion; the $\infty - \infty$ conversion; expand to the first surviving order.

L'Hopital's rule, in its proper place

The rule is a genuine theorem and it is occasionally the shortest path. It is also the most over-applied result in elementary analysis, and the reason is that its statement is easy to remember while its hypotheses are easy to forget. This section states it with the hypotheses attached, shows the one shape where it beats expansion, shows how to keep it from exploding, and then catalogues the three ways it fails.

Keep the keynote from the start of this chapter in view while reading it. One round of the rule is one order of truncation of two Taylor series, so the rule can never produce information that expansion could not. What it can do is produce that information without requiring the series to be known, which matters exactly when the functions are not in the table: at infinity, where Maclaurin series do not apply; at a finite point where the functions are arbitrary rather than standard; and in the rare case where a single differentiation happens to simplify an expression rather than complicate it. Those three cases are the legitimate domain of the rule, and they are narrower than most courses suggest.

4.11 The rule, with its hypotheses actually checked

EASY

CUE Substitution gives $\frac{0}{0}$ or $\frac{\infty}{\infty}$, and both functions are differentiable near, though not necessarily at, the point.

METHOD The theorem: if $f, g \rightarrow 0$ (or both $\rightarrow \pm\infty$) as $x \rightarrow a$, if $g' \neq 0$ on a punctured neighbourhood of a , and if $\lim f'/g'$ exists finitely or as $\pm\infty$, then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}.$$

Verify the form at *every* application, not only the first. Differentiate numerator and denominator separately; the quotient rule has no role here.

WORKED EXAMPLE

$$\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2}.$$

Form $\frac{0}{0}$; $g' = 2x$ is nonzero on a punctured neighbourhood. One round gives $\frac{e^x - 1}{2x}$, which is again $\frac{0}{0}$, and the hypothesis holds again. A second round gives $\frac{e^x}{2} \rightarrow \frac{1}{2}$. Since that limit exists, both applications are justified in retrospect, which is the correct logical order: the rule concludes backwards from the existence of the derivative limit. Verified as 0.5000167 at $x = 10^{-4}$. Compare the expansion route: $e^x - 1 - x = \frac{x^2}{2} + O(x^3)$, one line.

A case where the rule genuinely wins, because there is no Maclaurin series to use:

$$\lim_{x \rightarrow \infty} \frac{\ln x}{\sqrt[3]{x}} = \lim_{x \rightarrow \infty} \frac{1/x}{\frac{1}{3}x^{-2/3}} = \lim_{x \rightarrow \infty} \frac{3}{x^{1/3}} = 0.$$

The form is $\frac{\infty}{\infty}$, $g' = \frac{1}{3}x^{-2/3}$ is nonzero throughout, and one round converts the problem into a determinate one. The growth hierarchy gives the same answer even faster, but this is the shape in which a single round of the rule costs nothing and settles everything.

WHERE THIS GOES WRONG

Differentiating a quotient that is no longer indeterminate. After one round the form is often $\frac{c}{0}$ or $\frac{c}{a}$, and applying the rule there produces nonsense: on $\frac{x}{x+1}$ at $x \rightarrow 0$, the rule would give 1 instead of 0. The other classic is differentiating the quotient rather than the two functions separately.

SEE ALSO The point-value chain; repeated L'Hopital with simplification; the rule's failure modes.

There is one shape where a single round of the rule is genuinely the best move: a limit at a finite point where the functions are not standard ones with memorised series, so that expansion would mean computing derivatives anyway.

4.12 The point-value chain

MEDIUM

CUE A limit at a specific finite point a where the numerator's vanishing is not obvious, and where the functions are not in the Maclaurin table.

METHOD Apply the rule once and evaluate $f'(a)$ and $g'(a)$ honestly as numbers. The interest of these problems is usually that $f'(a)$ turns out to vanish as well, which settles the limit at once, or that $g'(a)$ is a clean nonzero number, in which case the answer is $f'(a)/g'(a)$ and there is nothing further to do.

WORKED EXAMPLE

$$\lim_{x \rightarrow e} \frac{e^x - x^e}{e - x}.$$

Both numerator and denominator vanish at $x = e$, since $e^e = e^e$. Differentiating, $f'(x) = e^x - e x^{e-1}$ and $g'(x) = -1$. At $x = e$,

$$f'(e) = e^e - e \cdot e^{e-1} = e^e - e^e = 0,$$

so the limit is $0/(-1) = 0$. Verified: the quotient is -2.8×10^{-6} at $x = e + 10^{-6}$, which is consistent with a limit of 0 approached linearly. The whole problem was the observation that e^x and x^e are tangent at $x = e$.

When one round is not enough, expanding about a is usually better than a second round. For

$$\lim_{x \rightarrow 1} \frac{x^x - x}{x - 1 - \ln x},$$

put $h = x - 1$. Then $x \ln x = (1+h)(h - \frac{h^2}{2} + O(h^3)) = h + \frac{h^2}{2} + O(h^3)$, so $x^x = e^{h+h^2/2+\dots} = 1 + h + h^2 + O(h^3)$ and the numerator is $h^2 + O(h^3)$. The denominator is $h - \ln(1+h) = \frac{h^2}{2} + O(h^3)$. The limit is 2, verified as 2.0000224 at $x = 1 + 10^{-5}$. Two rounds of L'Hopital would require differentiating x^x twice.

WHERE THIS GOES WRONG

Mis-differentiating x^e as $x^e \ln x$, which is the rule for e^x , or e^x as $x e^{x-1}$, which is the rule for x^e . The problem is engineered around exactly this confusion, and the two wrong derivatives give two different wrong answers, neither of which is 0.

SEE ALSO The rule, with its hypotheses actually checked; expansion at a point other than zero; repeated L'Hopital with simplification.

When one round is not enough, the rule's cost grows faster than most solvers expect, because each differentiation of a product multiplies the number of terms. Controlling that growth is a technique in itself.

4.13 Repeated application, with simplification between rounds

CUE One round of the rule has left you still at $\frac{0}{0}$, and the expression is a product or a quotient of several factors.

METHOD Before differentiating again, simplify and cancel. Pull out any factor whose limit exists and is finite and nonzero, and evaluate it separately, using $\lim(uv) = \lim u \cdot \lim v$ when both limits exist. Apply trigonometric identities to collapse the derivative you just produced. Each such step keeps the expression from doubling in length.

WORKED EXAMPLE

$$\lim_{x \rightarrow 0} \frac{\tan x - x}{x^3}.$$

Raw, this takes three rounds and the third derivative of $\tan$. With one identity it takes one round: differentiating gives $\frac{\sec^2 x - 1}{3x^2}$, and $\sec^2 x - 1 = \tan^2 x$, so the expression is $\frac{1}{3} \left(\frac{\tan x}{x} \right)^2 \rightarrow \frac{1}{3}$. Verified as 0.3333335 at $x = 10^{-3}$. The expansion route is shorter still: $\tan x - x = \frac{x^3}{3} + O(x^5)$ from the table, so the answer is the signature coefficient of $\tan$ and nothing needs to be differentiated at all.

A second instance, where the useful simplification is factoring rather than an identity:

$$\lim_{x \rightarrow 0} \frac{e^x \sin x - x - x^2}{x^3} = \frac{1}{3}.$$

Three rounds of the rule on $e^x \sin x$ generate eight terms by the product rule. Expanding instead, $e^x \sin x = (1 + x + \frac{x^2}{2} + \frac{x^3}{6})(x - \frac{x^3}{6}) + O(x^4) = x + x^2 + (\frac{1}{2} - \frac{1}{6})x^3 + O(x^4)$, so the numerator is $\frac{x^3}{3} + O(x^4)$. Verified as 0.3333333 at $x = 10^{-3}$.

WHERE THIS GOES WRONG

Never simplifying. By the third derivative of a product of four factors you have thirty terms and the limit is lost in the bookkeeping. This is precisely the regime where expansion is strictly better, because a Taylor coefficient is a lookup while a third derivative is a computation.

SEE ALSO Expand to the first surviving order; the rule, with its hypotheses actually checked; the rule's failure modes.

Finally, the failures. They are not exotic pathologies invented to embarrass students. Two of the three appear in ordinary problems, and the third is the reason the theorem's statement is phrased as a one-way implication.

HARD

4.14 The rule's failure modes

CUE $\lim f'/g'$ does not exist, or reproduces the quotient you started with, or the hypothesis on g' is violated.

METHOD Recognise three distinct failures and the correct response to each. (a) *Cycling*. $\lim_{x \rightarrow \infty} \frac{\sqrt{1+x^2}}{x}$ differentiates to $\frac{x}{\sqrt{1+x^2}}$, which differentiates back to $\frac{\sqrt{1+x^2}}{x}$: an infinite loop. Response: divide by the dominant term, giving $\sqrt{x^{-2} + 1} \rightarrow 1$. (b) *The derivative limit does not exist although the original does*. Response: the rule is silent, not negative; use squeeze or dominance instead. (c) *The hypothesis on g' is violated*, because g' vanishes on a sequence approaching the point. Response: the rule does not apply at all, and no conclusion of any kind may be drawn from f'/g' .

WORKED EXAMPLE

The standard instance of (b):

$$\lim_{x \rightarrow \infty} \frac{x + \sin x}{x + \cos x} = 1,$$

by dividing by x : the quotient is $\frac{1 + x^{-1} \sin x}{1 + x^{-1} \cos x}$ and both perturbations are squeezed to 0. Verified as 1.0000000130 at $x = 10^8$. The derivative quotient, however, is $\frac{1 + \cos x}{1 - \sin x}$, which oscillates forever and, worse, has zero denominator at every $x = \frac{\pi}{2} + 2k\pi$, so hypothesis (c) fails too.

An instance at a finite point, combining (b) and (c):

$$\lim_{x \rightarrow 0} \frac{x^2 \sin(1/x)}{\sin x} = 0,$$

since $|x^2 \sin(1/x)| \leq x^2$ and $\sin x \sim x$, so the quotient is squeezed between $\pm|x|$. The derivative quotient is $\frac{2x \sin(1/x) - \cos(1/x)}{\cos x}$, whose numerator oscillates between -1 and 1 without settling: it reads -0.872 at $x = 10^{-2}$ and -0.561 at $x = 10^{-3}$. The rule reports nothing; the limit is 0 regardless.

Failure (c) in isolation, and it is the sharpest of the three. Take $f(x) = x + \sin x \cos x$ and $g(x) = e^{\sin x}(x + \sin x \cos x)$ as $x \rightarrow \infty$. Both tend to ∞ , so the form is right. But $f/g = e^{-\sin x}$, which oscillates forever between e^{-1} and e and has no limit. Meanwhile $f' = 1 + \cos 2x = 2 \cos^2 x$ and $g' = e^{\sin x} \cos x(x + \sin x \cos x + 2 \cos x)$, so

$$\frac{f'}{g'} = \frac{2 \cos x}{e^{\sin x}(x + \sin x \cos x + 2 \cos x)} \rightarrow 0.$$

The derivative quotient converges beautifully to 0 and the original quotient has no limit at all. The rule is not violated, because g' vanishes at every $x = \frac{\pi}{2} + k\pi$, so the hypothesis was never satisfied. Numerically: f/g reads 1.659 at $x = 100$ and 0.437 at $x = 1000$, while f'/g' reads 0.0283 and 0.00049 at the same points. The moral is that “ $g' \neq 0$ on a punctured neighbourhood” is a condition to be checked, not a formality, and that a well-behaved derivative quotient is no evidence whatever that the checking can be skipped.

WHERE THIS GOES WRONG

Concluding from “ $\lim f'/g'$ does not exist” that “ $\lim f/g$ does not exist”. The theorem is a one-way implication and its converse is false, as the two examples above show. The practical rule: if the derivative quotient is not visibly better than what you started with, abandon the rule immediately rather than differentiating again in hope.

SEE ALSO Divide by the dominant term; the squeeze theorem; repeated L'Hopital with simplification.

THE STANDING ORDER

Expand first. Use L'Hopital when the functions are not in the table and a single differentiation produces a number, as in the point-value chain. Never apply it more than twice without asking what the corresponding Taylor coefficient would have been.

The chapter has one thesis and one procedure. The thesis is that a limit at a point is a question about the first few Taylor coefficients, so the method that produces those coefficients directly is the right method, and every other method is either a way of computing the same coefficients more slowly or a way of computing fewer of them. The procedure is: substitute so that the limit point is 0 ; compute the order of the denominator; budget each factor of the numerator against that order; expand exactly that far with an explicit remainder; read off the coefficient. L'Hopital's rule enters only when the functions are not in the table, and it leaves as soon as the derivative quotient stops improving.

Recognition matrix

If you see	Reach for	If that fails
$\frac{0}{0}$ at $x = 0$ with standard functions	The Maclaurin table, expanded to ord of the denominator	Push one order further and recheck
A radical of something near 1	$(1 + u)^a$ with u the whole perturbation	Second order; radicals over x^2 need it
A function of a function	Expand the inner to u , substitute into the outer	Factor out a common exponential first
A product in the numerator	Budget: expand f_i through degree $d - (m - m_i)$	Recompute m_i after any cancellation
Two functions with the same leading term, subtracted	Expand both one order past the expected cancellation	Confirm the surviving coefficient is nonzero
$f(x)$ minus its own first m Taylor terms	Read off the next coefficient of f	Check whether the intervening one vanishes
Cancellation past order 3 , or odd composites	Expand to order 7 ; do not conclude 0 early	Read numerics as a rate, not a value
$x \rightarrow a$ with $a \neq 0$	$h = x - a$, then an addition formula	Differentiate at a directly
$x \rightarrow \infty$ with near-cancellation	$t = 1/x$, then the binomial series	Conjugate multiplication
A reciprocal of a vanishing series	Extract the pole, then expand $1/(1 + w)$	Common denominator instead
$\frac{0}{0}$ with non-tabulated functions	One L'Hopital round, then evaluate $f'(a)$, $g'(a)$	Expand f and g about a by hand
The derivative quotient repeats itself	Stop; divide by the dominant term	Substitute $t = 1/x$
The derivative quotient oscillates	Stop; the rule is silent, use squeeze	Bound above and below and compare
g' vanishes arbitrarily near the point	The rule does not apply; draw no conclusion	Direct estimate or squeeze

Algebraic surgery, the exponential machine, and the squeeze

What this chapter is for. Three whole families of limit yield without a single derivative, a single series, and a single appeal to L'Hôpital. The first family is *surgery on the expression*: conjugates, factorisation of a removable singularity, common denominators, and the deliberate insertion of a bridging term, all of which turn an indeterminate shape into a determinate one by pure algebra. The second is the *exponential machine*: every power-form indeterminacy (1^∞ , 0^0 , ∞^0) becomes a product the moment you write $f^g = \exp(g \ln f)$, and the whole *e*-family falls out as one corollary. The third is the *squeeze*, which is the only tool in the book that works when the limit exists but nothing inside the expression converges at all. By the end you should be able to look at an indeterminate form and say, in a few seconds, which of the three families it belongs to, and you should be able to execute all three without hesitation and without reaching for calculus you do not need.

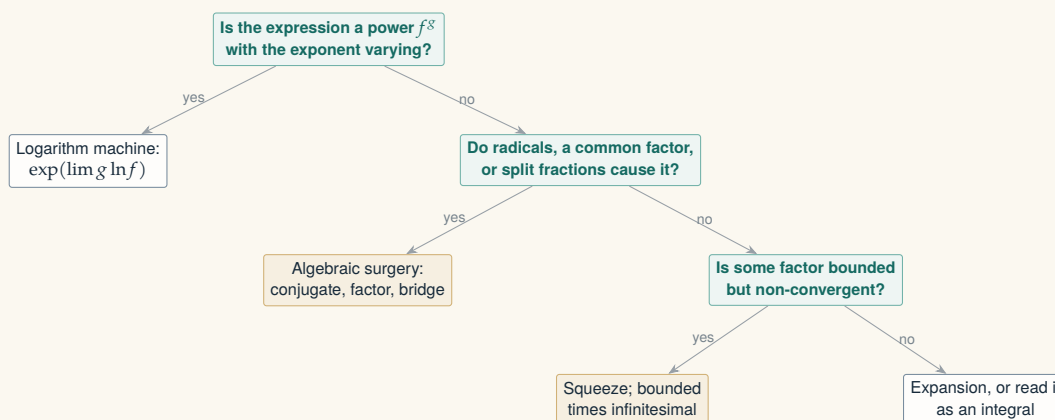
Choosing the family

An indeterminate form is not one problem but a symptom shared by several. Before choosing a technique, ask what is actually producing the indeterminacy. If it is a *cancellation* that the algebra is hiding, whether a common factor, a difference of radicals, or two fractions that must be combined, then the cure is surgery and nothing else is needed. If it is an *exponent* fighting a base, the cure is the logarithm, which demotes the fight to a product. If the expression contains a piece that simply does not converge, such as $\sin(1/x)$ near 0, or a fractional part, or a maximum over a growing index set, then no amount of rewriting will produce a limit for that piece, and the only honest move is to bound the whole expression from both sides.

The three questions are worth asking in that order, and the order is not arbitrary. Surgery is cheap and often removes the problem entirely, leaving a value you can read off by substitution. The logarithm is a mechanical transform that never fails but always costs you a second limit, so it is worth paying for only when the first question has been answered “no”. The squeeze demands that you invent bounds, which is the only genuinely creative step of the three, and inventing bounds is work you should not do until you know the other two will not serve.

It also helps to name the form precisely rather than calling everything “indeterminate”. There are seven: $\frac{0}{0}$, $\frac{\infty}{\infty}$, $0 \cdot \infty$, $\infty - \infty$, 1^∞ , 0^0 , and ∞^0 . The last three are all powers and all go to the same place, the logarithm. The middle two are one algebraic step from a quotient, since $0 \cdot \infty$ becomes $\frac{0}{0}$ on inverting either factor and $\infty - \infty$ becomes a quotient on taking a common denominator or a conjugate. So the seven forms are really three: a quotient, a power, and a nonconvergent piece. That is the whole taxonomy this chapter needs.

There is a fourth answer that this chapter deliberately postpones. If the expression is a sum of n terms with a $1/n$ in front, or a difference quotient in disguise, the right reading is neither algebraic nor exponential nor a squeeze: it is an integral or a derivative, and that is the subject of the next chapter. The decision tree below sends you there when none of the three families fits.



Surgery on the expression

Every technique in this section rests on one observation: an indeterminate form is a statement about the *written expression*, not about the function. The function $\frac{x^2 - 4}{x - 2}$ is the function $x + 2$ with one point deleted, and the $\frac{0}{0}$ at $x = 2$ is an artefact of how somebody chose to write it down. Surgery is the art of rewriting until the artefact is gone. The five moves below are the complete toolkit for the rational and radical cases, and they are ordered by how visible the hidden cancellation is: a common polynomial factor is the most visible, a bridging term the least.

It is worth being explicit about what “determinate” means here, because that is the target of every rewrite. A quotient is determinate at a when the denominator does not vanish there and both parts are continuous, so that substitution is legal. Every move in this section aims at that state, and you should stop rewriting the instant you reach it. The most common failure of technique in this whole chapter is continuing to manipulate after the problem has already been solved.

THE STANDARD LIMITS THESE MOVES REDUCE TO

Surgery is only useful because a small list of limits is known outright. Memorise these six and you will never need a series expansion for a first-order question:

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{\sin x}{x} &= 1, & \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} &= \frac{1}{2}, & \lim_{x \rightarrow 0} \frac{e^x - 1}{x} &= 1, \\ \lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} &= 1, & \lim_{x \rightarrow 0} \frac{a^x - 1}{x} &= \ln a, & \lim_{x \rightarrow 0} \frac{(1+x)^\alpha - 1}{x} &= \alpha.\end{aligned}$$

The last of these is what a conjugate computation reproduces by hand for $\alpha = \frac{1}{2}$ and $\alpha = \frac{1}{3}$; the conjugate is the elementary proof, not a shortcut past it.

WHEN SURGERY IS THE WRONG CHOICE

Surgery fails, and should be abandoned quickly, when the cancelling factor is transcendental rather than algebraic. In $\frac{e^x - 1 - x}{x^2}$ there is no factor to divide out and no conjugate to apply: the numerator is not a difference of powers of anything. That limit is $\frac{1}{2}$, and it belongs to the expansion techniques, not here. The diagnostic is simple. If every vanishing piece is a polynomial or a root of one, surgery will finish the job; if any vanishing piece is e^x , $\ln$, or a trigonometric function set against its own linearisation, it will not.

Start with radicals, because a difference of radicals is the most common way to disguise a cancellation. The mechanism is always the same: a difference of n th roots is hard, but a difference of n th powers is easy, so multiply by whatever converts one into the other.

5.1 Conjugate multiplication for a difference of square roots

EASY

CUE Any $\sqrt{A} - \sqrt{B}$ in a numerator or a denominator, where $A - B$ is visibly simpler than either radical. The classic signatures are $\sqrt{x+h} - \sqrt{x}$, $\sqrt{x^2 + ax} - x$, and a denominator that vanishes because two radicals become equal in the limit.

METHOD Multiply the whole fraction by $\frac{\sqrt{A} + \sqrt{B}}{\sqrt{A} + \sqrt{B}}$. The identity is

$$\sqrt{A} - \sqrt{B} = \frac{A - B}{\sqrt{A} + \sqrt{B}},$$

which trades a difference that is $0 - 0$ for a quotient whose numerator carries the cancelling factor explicitly and whose denominator tends to a harmless nonzero constant. If the radicals sit in the denominator, the same multiplication rationalises it and exposes the factor that must be cancelled from the numerator.

WORKED EXAMPLE

$$\lim_{x \rightarrow 0} \frac{\sin^2 x}{\sqrt{2} - \sqrt{1 + \cos x}}.$$

Both parts vanish. Multiply top and bottom by $\sqrt{2} + \sqrt{1 + \cos x}$; the denominator becomes $2 - (1 + \cos x) = 1 - \cos x$, so the quotient is

$$\frac{\sin^2 x (\sqrt{2} + \sqrt{1 + \cos x})}{1 - \cos x} = (1 + \cos x)(\sqrt{2} + \sqrt{1 + \cos x}),$$

using $\sin^2 x = (1 - \cos x)(1 + \cos x)$ to cancel the vanishing factor outright. Now substitute: the value is $2 \cdot 2\sqrt{2} = 4\sqrt{2} \approx 5.65685$. No series, no derivatives, and note that the notorious $\frac{1-\cos x}{x^2}$ never had to be invoked.

The same identity handles $\infty - \infty$ at infinity, where it is the standard first move and where the arithmetic is if anything easier. Three specimens worth carrying:

$$\sqrt{x^2 + x + 1} - \sqrt{x^2 - x + 1} = \frac{2x}{\sqrt{x^2 + x + 1} + \sqrt{x^2 - x + 1}} \rightarrow 1,$$

because the denominator is $2x(1 + o(1))$; likewise $x(\sqrt{x^2 + 1} - x) = \frac{x}{\sqrt{x^2 + 1} + x} \rightarrow \frac{1}{2}$, and $\sqrt{x^2 + ax} - x \rightarrow \frac{a}{2}$. In each case the conjugate converts a cancellation of leading terms, which floating-point arithmetic also finds catastrophic, into an honest quotient.

WHERE THIS GOES WRONG

Multiplying only the denominator by its conjugate. You must multiply the *whole fraction* by $\frac{\text{conjugate}}{\text{conjugate}}$, which is multiplication by 1; touching one storey only changes the function. The second common error occurs at $x \rightarrow -\infty$, where $\sqrt{x^2} = |x| = -x$, so factoring x out of a radical carries a sign that is easy to lose and that flips the answer.

SEE ALSO The cube-root conjugate; add and subtract a bridging term; change of variable to move the limit point.

Square roots are the easy case only because $a^2 - b^2$ factors in one step. For cube roots the same idea works, but the multiplier is different, and guessing it is where most attempts die.

MEDIUM

5.2 The cube-root conjugate

CUE A difference $\sqrt[3]{A} - \sqrt[3]{B}$, or a cube root minus a constant, inside a $\frac{0}{0}$ quotient. Also any mixed $\sqrt[3]{A} - \sqrt{B}$, which is handled by first splitting it with a bridging term and then applying this move to the cube-root half.

METHOD Set $a = \sqrt[3]{A}$, $b = \sqrt[3]{B}$ and use

$$a - b = \frac{a^3 - b^3}{a^2 + ab + b^2} = \frac{A - B}{a^2 + ab + b^2}.$$

The denominator tends to $3L^2$ when both roots tend to a common nonzero L , so it is inert and all the content sits in $A - B$. The general n th-root version is

$$a - b = \frac{a^n - b^n}{a^{n-1} + a^{n-2}b + \dots + ab^{n-2} + b^{n-1}},$$

with n terms in the denominator, each tending to L^{n-1} .

WORKED EXAMPLE

$$\lim_{x \rightarrow 0} \frac{\sqrt[3]{8 + 3x - x^2} - 2}{x + x^2}.$$

With $a = \sqrt[3]{8 + 3x - x^2}$ and $b = 2$ we have $a^3 - b^3 = 3x - x^2 = x(3 - x)$, while $a^2 + 2a + 4 \rightarrow 4 + 4 + 4 = 12$. Hence the quotient equals

$$\frac{x(3 - x)}{(x + x^2)(a^2 + 2a + 4)} = \frac{3 - x}{(1 + x)(a^2 + 2a + 4)} \rightarrow \frac{3}{12} = \frac{1}{4}.$$

A mixed instance, resolved by the same identity applied twice:

$$\lim_{x \rightarrow 0} \frac{\sqrt[3]{1 + x} - \sqrt{1 + x}}{x} = \frac{1}{3} - \frac{1}{2} = -\frac{1}{6} \approx -0.166667,$$

where each root is compared with 1 separately, the cube root by the identity above and the square root by the previous technique.

WHERE THIS GOES WRONG

Guessing the wrong multiplier. The cube conjugate of $a - b$ is $a^2 + ab + b^2$, with all plus signs; $a^2 - ab + b^2$ is the multiplier for $a + b$. Writing the wrong one produces an expression that is simply not equal to the original, and the error is silent because the answer still looks plausible. Check the multiplier by expanding it against $a - b$ before you use it.

SEE ALSO Conjugate multiplication for a difference of square roots; add and subtract a bridging term; change of variable to move the limit point.

When no radicals are present, the hidden cancellation is a common polynomial factor, and finding it is a division problem rather than an identity problem. This is the most elementary technique in the chapter and also the one most often skipped in favour of machinery.

5.3 Factor out the removable singularity

EASY

CUE A rational expression, or one that factors, evaluating to $\frac{0}{0}$ at $x = a$. Equally: a difference of two fractions, each of which blows up at a , whose combination over a common denominator is finite.

METHOD If numerator and denominator both vanish at $x = a$, then $(x - a)$ divides both, by the factor theorem. Divide it out, by inspection or by synthetic division, and repeat until at least one of the two no longer vanishes at a . Then substitute. When the expression is a difference of fractions, combine over a common denominator *first*; the cancellation only becomes visible once there is a single fraction to look at.

WORKED EXAMPLE

$$\lim_{x \rightarrow 2} \frac{x^4 - 6x^3 + 13x^2 - 12x + 4}{x^3 - 5x^2 + 8x - 4}.$$

Both vanish at $x = 2$, and both still vanish after one division, so $x = 2$ is at least a double root of each. In fact the numerator is $(x - 1)^2(x - 2)^2$ and the denominator is $(x - 1)(x - 2)^2$, so on a punctured neighbourhood the quotient is exactly $x - 1$, and the limit is 1.

A second shape, the split fraction, where the common denominator is the whole trick:

$$\lim_{x \rightarrow 1} \left(\frac{1}{1 - x} - \frac{3}{1 - x^3} \right) = \lim_{x \rightarrow 1} \frac{(1 + x + x^2) - 3}{1 - x^3} = \lim_{x \rightarrow 1} \frac{(x - 1)(x + 2)}{-(x - 1)(1 + x + x^2)} = -1.$$

Neither fraction has a limit at $x = 1$; their difference does, and only the combined form shows it.

A third, worth knowing as a formula: $\lim_{x \rightarrow 1} \frac{x^m - 1}{x^n - 1} = \frac{m}{n}$, since both factor as $(x - 1)$ times a sum of m or n terms each tending to 1.

WHERE THIS GOES WRONG

Reaching for L'Hôpital. It gives the same answer here, but it destroys the structural information: the point of the calculation is that the singularity was *removable* and the function extends continuously across $x = a$, a fact that continuity arguments and radius-of-convergence arguments later depend on. The second error is stopping after one division when the root is repeated, as above, and concluding that the limit is still indeterminate.

SEE ALSO The cube-root conjugate; change of variable to move the limit point; the limit is a derivative.

The three moves so far all require a shared structure to exploit: a conjugate needs matching radicals, a factorisation needs a common root. What if there is none, because the two competing pieces have nothing algebraically in common? Then you manufacture the structure, by splitting the difference at its common limit value. This is telescoping applied to a difference of two terms rather than to a long sum, and it is the reason the technique looks like an accounting trick rather than an identity.

5.4 Add and subtract a bridging term

MEDIUM

CUE A difference of two dissimilar expressions that tend to the same value: $\sqrt[3]{f(x)} - \sqrt{g(x)}$ with $f, g \rightarrow 1$, or $e^{u(x)} - \cos v(x)$, or any $A - B$ where A and B are each tractable against a constant but not against each other.

METHOD Insert the common limit L and telescope the difference:

$$A - B = (A - L) - (B - L).$$

Each bracket now has a single radical or a single transcendental function set against a constant, so each yields to its own conjugate or its own standard limit. Take the two limits separately and subtract. The same device extends to three or more pieces, bridging at L each time.

$$\lim_{x \rightarrow 0} \frac{\sqrt[3]{1 + \sin^2 x} - \sqrt{\cos x}}{x^2}.$$

Both radicals tend to 1, so bridge at $L = 1$:

$$\frac{\sqrt[3]{1 + \sin^2 x} - 1}{x^2} + \frac{1 - \sqrt{\cos x}}{x^2}.$$

For the first, the cube conjugate gives $\frac{\sin^2 x}{x^2(a^2 + a + 1)} \rightarrow \frac{1}{3}$ with $a = \sqrt[3]{1 + \sin^2 x} \rightarrow 1$. For the second, the square conjugate gives $\frac{1 - \cos x}{x^2(1 + \sqrt{\cos x})} \rightarrow \frac{1/2}{2} = \frac{1}{4}$. The limit is $\frac{1}{3} + \frac{1}{4} = \frac{7}{12} \approx 0.583333$.

The same split on a cleaner pair confirms the bookkeeping: $\frac{\sqrt[3]{1 + 3x} - \sqrt{1 + 2x}}{x^2} \rightarrow -\frac{1}{2}$, since $\sqrt[3]{1 + 3x} = 1 + x - x^2 + \dots$ and $\sqrt{1 + 2x} = 1 + x - \frac{x^2}{2} + \dots$, so the linear parts cancel and the quadratic parts do not.

WHERE THIS GOES WRONG

Splitting a limit into two pieces that individually diverge. The identity $\lim(P + Q) = \lim P + \lim Q$ requires both right-hand limits to exist finitely. Bridging at the common value L is exactly the device that guarantees this, which is why L must be the shared limit of A and B and not merely some convenient constant. If A and B tend to different values, the original difference is not indeterminate and there is nothing to bridge.

SEE ALSO Conjugate multiplication for a difference of square roots; the cube-root conjugate; log-transform a product or a nested power.

All four moves are easier to see when the limit point is 0. That is not a coincidence: every standard limit in the keynote above is written at 0, and every series expansion in the next chapter is written at 0. The last move of this section is the bookkeeping that puts you there, and it is also the move that turns competing radical orders into competing polynomial degrees.

EASY

5.5 Change of variable to move the limit point

CUE The limit point is awkward ($x \rightarrow 1$, $x \rightarrow \pi/2$, $x \rightarrow \infty$), or the expression is a function of a compound argument such as $\sqrt[12]{x}$ or $\ln x$, or several roots of different orders appear together.

METHOD Substitute so that the limit point becomes 0 or ∞ and the roots become polynomial: $h = x - a$, $t = 1/x$, $u = \pi/2 - x$, or $x = t^m$ with $m = \text{lcm of the root orders}$. State the new direction of approach explicitly, because a substitution can convert a two-sided limit into a one-sided one and can reverse orientation.

WORKED EXAMPLE

$$\lim_{x \rightarrow 1} \frac{\sqrt[3]{x} - 1}{\sqrt[4]{x} - 1}.$$

The orders are 3 and 4, so put $x = t^{12}$; as $x \rightarrow 1$ through positive values, $t \rightarrow 1$. The quotient becomes a ratio of polynomials with a visible common factor:

$$\frac{t^4 - 1}{t^3 - 1} = \frac{(t - 1)(t^3 + t^2 + t + 1)}{(t - 1)(t^2 + t + 1)} \rightarrow \frac{4}{3}.$$

A second instance, moving a trigonometric limit point to the origin: $\lim_{x \rightarrow \pi/2} \frac{\cos x}{x - \pi/2}$ becomes, under $u = x - \pi/2$,

$$\lim_{u \rightarrow 0} \frac{\cos(\frac{\pi}{2} + u)}{u} = \lim_{u \rightarrow 0} \frac{-\sin u}{u} = -1.$$

A third, moving infinity to the origin: $\lim_{x \rightarrow \infty} x \ln\left(1 + \frac{1}{x}\right)$ becomes $\lim_{t \rightarrow 0^+} \frac{\ln(1 + t)}{t} = 1$ under $t = 1/x$, where the standard limit applies directly.

WHERE THIS GOES WRONG

Substituting without transforming the direction of approach. Under $t = 1/x$ the one-sided limit $x \rightarrow 0^-$ becomes $t \rightarrow -\infty$, not $t \rightarrow +\infty$, and every sign in the answer depends on it. Under $x = t^2$ the variable t ranges over one side only, so a genuinely

two-sided limit in x must either be checked for t of both signs or the substitution restricted to $x > 0$ with the other side treated separately.

SEE ALSO Factor out the removable singularity; exponent simplification before base analysis; Riemann sum recognition.

The five moves are meant to be combined, and a problem of any weight will need two or three of them in sequence. A short illustration of the protocol in full:

$$\lim_{x \rightarrow 0} \frac{\sqrt[3]{1+x} - \sqrt[3]{1-x}}{\sqrt{1+x} - \sqrt{1-x}}.$$

The form is $\frac{0}{0}$. There is no common factor to divide out, so factorisation is not the move. Both storeys are differences of radicals, so apply the appropriate conjugate to each: the numerator becomes $\frac{2x}{a^2 + ab + b^2}$ with $a = \sqrt[3]{1+x}$ and $b = \sqrt[3]{1-x}$, both tending to 1, and the denominator becomes $\frac{2x}{\sqrt{1+x} + \sqrt{1-x}}$. The factors of $2x$ cancel, which is the removable singularity making itself visible, and what remains is

$$\frac{\sqrt{1+x} + \sqrt{1-x}}{a^2 + ab + b^2} \rightarrow \frac{2}{3} \approx 0.666667.$$

Three moves, no calculus, and the answer is exact. Note the order: conjugate first to expose the factor, cancel second, substitute third, and stop.

The exponential machine

A power $f(x)^{g(x)}$ with both f and g varying is not directly attackable: there is no arithmetic rule for the limit of a power when the base tends to 1 and the exponent to ∞ , because $1^n = 1$ pulls one way and c^∞ pulls the other, and both pulls are real. The resolution is not a special theorem but a change of representation. For $f > 0$,

$$f^g = \exp(g \ln f),$$

and $\exp$ is continuous on all of $\mathbb{R}$. Continuity is the entire content of the method: if $g \ln f \rightarrow L$ with L finite, then $f^g \rightarrow e^L$; if $g \ln f \rightarrow -\infty$ then $f^g \rightarrow 0$; if $g \ln f \rightarrow +\infty$ then $f^g \rightarrow \infty$. Nothing is indeterminate about $g \ln f$ except the product $0 \cdot \infty$, and a product is a shape the previous section already knows how to break: divide one factor by the reciprocal of the other, or expand the logarithm.

THE TRANSFORM, ONCE AND FOR ALL

Every power-form indeterminacy is the same problem. 1^∞ gives $\infty \cdot 0$ in the exponent, ∞^0 gives $0 \cdot \infty$, and 0^0 gives $0 \cdot (-\infty)$. In all three cases write

$$\lim f^g = \exp(\lim g \ln f),$$

and solve the product. The only hypotheses are that $f > 0$ on a punctured neighbourhood of the point and that the inner limit exists in the extended reals. Nothing else about f or g is needed, and in particular neither has to be differentiable.

The two forms other than 1^∞ deserve a moment, because they are easier than they look and are usually dispatched in one line. For 0^0 , take $\lim_{x \rightarrow 0^+} x^x$: the exponent is $x \ln x \rightarrow 0$, since $\ln x$ diverges only logarithmically while x vanishes linearly, so the limit is $e^0 = 1$. For ∞^0 , take $\lim_{x \rightarrow \infty} x^{1/x}$: the exponent is $\frac{\ln x}{x} \rightarrow 0$, so again the limit is 1. In both cases the decision was made by comparing growth rates inside the product, which is the only skill those two forms require. The reason 1^∞ gets techniques of its own is that there the product is a genuine contest between two quantities of the same order, and the answer depends on the constant that survives.

The rest of this section is a graded sequence: the general form first, the shortcut that covers most cases second, the memorised e -limits third as a corollary of the shortcut rather than as separate facts, and then two structural techniques that use the logarithm for what it does to products rather than to exponents.

5.6 The standard 1^∞ form

CUE The base tends to 1 and the exponent tends to ∞ or to $-\infty$. Typical shapes: $(\frac{\sin x}{x})^{1/x^2}$, $(\cos x)^{1/x^2}$, $(\frac{a^x+b^x}{2})^{1/x}$, and every problem in which a sequence $(1 + \varepsilon_n)^{N_n}$ appears with $\varepsilon_n \rightarrow 0$ and $N_n \rightarrow \infty$.

METHOD Write $\lim f^g = \exp(\lim g \ln f)$ and expand the logarithm about 1: with $u = f - 1 \rightarrow 0$,

$$\ln f = \ln(1 + u) = u - \frac{u^2}{2} + \frac{u^3}{3} - \dots$$

Keep as many terms of u , and as many terms of this expansion, as the exponent's blow-up rate demands: if g grows like x^{-m} , you need $\ln f$ correct to order x^m inclusive, and no further.

WORKED EXAMPLE

$$\lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{1/x^2}.$$

Here $u = \frac{\sin x}{x} - 1 = -\frac{x^2}{6} + \frac{x^4}{120} - \dots$, so $u^2 = O(x^4)$ and the quadratic term of the logarithm cannot reach order x^2 :

$$\ln \frac{\sin x}{x} = u - \frac{u^2}{2} + \dots = -\frac{x^2}{6} + O(x^4).$$

Multiplying by $g = 1/x^2$ gives $-\frac{1}{6} + O(x^2) \rightarrow -\frac{1}{6}$, so the limit is $e^{-1/6} \approx 0.8464817$, and direct evaluation at $x = 10^{-3}$ returns 0.8464817.

Two siblings, computed the same way and worth holding in memory as a set:

$$\lim_{x \rightarrow 0} (\cos x)^{1/x^2} = e^{-1/2} \approx 0.606531, \quad \lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right)^{1/x^2} = e^{1/3} \approx 1.395612,$$

since $\ln \cos x = -\frac{x^2}{2} + O(x^4)$ and $\ln \frac{\tan x}{x} = \frac{x^2}{3} + O(x^4)$. In every case the answer is e^c where c is simply the coefficient of x^2 in $\ln f$. There is no pattern connecting the three constants; each has to be computed from its own expansion, which is precisely why the method is a method and not a table.

The exponent decides how much work the expansion has to do, and this is the only genuine judgement call in the technique. With $g = 1/x^2$ only the x^2 coefficient of $\ln f$ survives, and everything else is either infinite (there is nothing of lower order, so the limit is finite) or vanishing. With $g = 1/x^4$ the x^4 coefficient becomes the answer, and any surviving x^2 coefficient becomes an infinity that decides the limit to be 0 or ∞ before the x^4 term is even reached. The tidy case, and the one problem-setters like, is when the x^2 term is absent by design.

A FOURTH-ORDER EXAMPLE WORTH MEMORISING

$$\lim_{x \rightarrow 0} \left(\frac{\cosh x + \cos x}{2} \right)^{1/x^4} = e^{1/24} \approx 1.0425469.$$

The x^2 terms of $\cosh x = 1 + \frac{x^2}{2} + \frac{x^4}{24} + \dots$ and $\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \dots$ cancel exactly, leaving the base equal to $1 + \frac{x^4}{24} + O(x^8)$. Since $u = O(x^4)$, the term $u^2/2$ is $O(x^8)$ and may be dropped, so $g \ln f \rightarrow \frac{1}{24}$. Compare $(\cos x)^{1/x^4}$, where the x^2 term survives, $g \ln f \rightarrow -\infty$, and the limit is 0.

WHERE THIS GOES WRONG

Stopping at $\ln f \approx f - 1$ when the exponent is $1/x^2$ or worse and the quadratic term still contributes. Match the expansion order to the exponent: with $g = x^{-m}$, an error of size x^m in $\ln f$ is an error of size 1 in the answer's logarithm, which is a whole factor of e in the answer. The symptom is an answer that is off by a clean exponential factor rather than by a little.

SEE ALSO The exponential shortcut without a logarithm; the e limit family; log-transform a product or a nested power.

In practice the logarithm is often unnecessary, because to first order $\ln(1 + u)$ and u agree. It is worth knowing exactly when the shortcut is legitimate, and the honest statement is both simpler and stronger than the one usually quoted.

5.7 The $\exp(\lim g(f-1))$ shortcut

CUE A clean 1^∞ in which $g(f-1)$ visibly tends to a finite limit. This is the common case, and recognising it saves you from writing a logarithm you would only expand again.

METHOD If $f \rightarrow 1$, $g \rightarrow \infty$ and $g(f-1) \rightarrow L$ with L finite, then

$$\lim f^g = \exp(\lim g(f-1)) = e^L.$$

The proof is one line. Write $\ln f = (f-1) - \frac{1}{2}(f-1)^2(1+o(1))$; then

$$g(f-1)^2 = \frac{(g(f-1))^2}{g} \rightarrow \frac{L^2}{\infty} = 0,$$

so $g \ln f$ and $g(f-1)$ have the same limit.

WORKED EXAMPLE

$$\lim_{x \rightarrow 0} (1 + \sin 2x)^{\cot 3x}.$$

Here $f-1 = \sin 2x \rightarrow 0$ and $g = \cot 3x \rightarrow \infty$, so the form is 1^∞ and the shortcut applies once the product is shown to converge:

$$g(f-1) = \frac{\cos 3x \sin 2x}{\sin 3x} = \cos 3x \cdot \frac{\sin 2x}{2x} \cdot \frac{3x}{\sin 3x} \cdot \frac{2}{3} \rightarrow 1 \cdot 1 \cdot 1 \cdot \frac{2}{3} = \frac{2}{3}.$$

The limit is $e^{2/3} \approx 1.9477340$; numerical evaluation at $x = 10^{-6}$ gives 1.9477327.

THE CONDITION IS AUTOMATIC, AND THAT IS THE POINT

Textbooks state the hypothesis as $g(f-1)^2 \rightarrow 0$, which looks like a condition to be checked. It is not. Whenever $g \rightarrow \infty$ and $g(f-1)$ converges to a finite L , the quantity $g(f-1)^2 = (g(f-1))^2/g$ tends to 0 all by itself. The shortcut is therefore valid exactly when it produces a finite answer, and the only real danger is the opposite case, in which $g(f-1)$ diverges and the discarded term is no longer negligible.

WHERE THIS GOES WRONG

Using the shortcut when $g(f-1) \rightarrow \pm\infty$ and you need more than the sign. Take $f = 1 + \frac{1}{n}$ and $g = n^2$. The shortcut reports $\exp(n)$, but the truth is

$$\left(1 + \frac{1}{n}\right)^{n^2} = \exp\left(n^2\left(\frac{1}{n} - \frac{1}{2n^2} + \frac{1}{3n^3} - \dots\right)\right) = \exp\left(n - \frac{1}{2} + O\left(\frac{1}{n}\right)\right),$$

so $\left(1 + \frac{1}{n}\right)^{n^2} e^{-n} \rightarrow e^{-1/2} \approx 0.60653$ and not 1; at $n = 10^6$ the ratio is 0.6065309. The dropped $-\frac{1}{2}$ is a genuine factor. When the exponent diverges, expand the logarithm.

SEE ALSO The standard 1^∞ form; the e limit family; exponent simplification before base analysis.

Everything usually presented as “the e limits” is a special case of the shortcut, and it is worth watching the list collapse, because the collapse tells you which of the many memorised forms is the one you actually need. There is only one: the product $g(f-1)$.

EASY

5.8 The $e = \lim(1 + 1/n)^n$ family

CUE Shapes such as $(1 + \frac{a}{n})^{bn}$, $(\frac{n+a}{n+b})^n$, $(1+ax)^{b/x}$, or any base written as $1 + (\text{small})$ with an exponent that is the reciprocal of that small quantity, up to a constant factor.

METHOD Apply the shortcut. The master forms are

$$\lim_{n \rightarrow \infty} \left(1 + \frac{a}{n}\right)^{bn} = e^{ab}, \quad \lim_{x \rightarrow 0} (1+ax)^{b/x} = e^{ab},$$

both because $g(f-1) = bn \cdot \frac{a}{n} = ab$ identically, with no expansion at all. For $(\frac{n+a}{n+b})^n$, rewrite the base as $1 + \frac{a-b}{n+b}$ and read off e^{a-b} ; for example $(\frac{n+3}{n+1})^n \rightarrow e^2 \approx 7.389056$. When the deviation is of order n^{-2} the exponent must be of order n^2 to match: $(\frac{n^2+1}{n^2-2})^{n^2} = (1 + \frac{3}{n^2-2})^{n^2} \rightarrow e^3 \approx 20.0855$, which at $n = 10^3$ evaluates to 20.0864.

$$\lim_{x \rightarrow 0} \left(\frac{a^x + b^x}{2} \right)^{1/x}, \quad a, b > 0.$$

The base tends to $\frac{1+1}{2} = 1$, so the form is 1^∞ , and

$$g(f-1) = \frac{1}{x} \cdot \frac{(a^x - 1) + (b^x - 1)}{2} = \frac{1}{2} \left(\frac{a^x - 1}{x} + \frac{b^x - 1}{x} \right) \rightarrow \frac{\ln a + \ln b}{2} = \ln \sqrt{ab},$$

using the standard limit $\frac{a^x - 1}{x} \rightarrow \ln a$. The limit is therefore $\sqrt{ab}$: the arithmetic mean of the exponentials produces the geometric mean of the bases. At $a = 2, b = 5$ this is $\sqrt{10} \approx 3.162278$, which is what numerical evaluation at $x = 10^{-6}$ returns.

WHERE THIS GOES WRONG

Mismatching the two rates. In $(1 + \frac{a}{n^2})^n$ the product $g(f-1) = a/n \rightarrow 0$, so the limit is 1, not e^a . The exponent must be the reciprocal of the base's deviation from 1 to within a constant factor; if the exponent is smaller in order the answer is 1, and if it is larger the answer is 0 or ∞ . Always compute the product rather than pattern-matching the shape.

SEE ALSO The standard 1^∞ form; the exponential shortcut without a logarithm; the n th root of a sum tends to the maximum.

Before invoking any of this machinery, look at the exponent on its own. A surprising number of frightening powers are not indeterminate at all once the exponent has been simplified, and running the logarithm on a determinate form wastes half a page and invites the sign errors that come with it.

MEDIUM

5.9 Exponent simplification before base analysis

CUE A power whose *exponent* is itself an indeterminate quotient, typically at $x \rightarrow 1$ with radicals present: exponents like $\frac{1-x}{1-\sqrt{x}}, \frac{\ln x}{x-1}$, or $\frac{x^2-1}{x-1}$.

METHOD Resolve the exponent first, using the surgery of the previous section. If it tends to a finite constant c and the base tends to a positive $B \neq 1$, the whole limit is simply B^c , by continuity of $(b, t) \mapsto b^t$ on $(0, \infty) \times \mathbb{R}$. Only if the base also tends to 1 do you need the logarithm, and by then the exponent is a known constant and the remaining work is halved.

WORKED EXAMPLE

$$\lim_{x \rightarrow 1} \left(\frac{x}{1+x^2} \right)^{\frac{1-x}{1-\sqrt{x}}}.$$

The exponent factors: $1-x = (1-\sqrt{x})(1+\sqrt{x})$, so $\frac{1-x}{1-\sqrt{x}} = 1+\sqrt{x} \rightarrow 2$. The base tends to $\frac{1}{2}$, which is neither 0 nor 1, so the form is determinate and the answer is $(\frac{1}{2})^2 = \frac{1}{4}$, confirmed numerically at $x = 1 - 10^{-7}$.

Change the base to $\frac{2x}{1+x^2}$, which tends to 1, and the same exponent analysis leaves a genuine 1^∞ ; here the base is $1 - \frac{(1-x)^2}{1+x^2}$, so $g(f-1) \rightarrow 2 \cdot 0 = 0$ and the limit is 1. The point is that in both cases the exponent was settled before any decision about machinery was made.

WHERE THIS GOES WRONG

Logarithmising a form that is not indeterminate. If the base tends to $\frac{1}{2}$ and the exponent to 2, then $\ln f \rightarrow -\ln 2$ and $g \ln f \rightarrow -2 \ln 2$: correct, but three unnecessary lines, and along the way it is easy to write $\lim g \ln f = \lim g \cdot \lim \ln f$ in a situation where that product rule genuinely does not apply, because the exponent's own limit had not yet been established.

SEE ALSO Change of variable to move the limit point; the standard 1^∞ form; factor out the removable singularity.

The logarithm does more than tame exponents. Its structural role is that it converts multiplication into addition, and sums are the objects for which we have the largest arsenal: telescoping, comparison, integral bounds, and (in the next chapter) recognition as integrals. Any time a limit contains a product of many factors, the first move is to take its logarithm and see what kind of sum appears.

5.10 Log-transform a product or a nested power

CUE An infinite product, an n th root of a product of n factors, a tower of powers, or a sum of logarithms that would be a product if you read it backwards.

METHOD Use $\ln \prod_k c_k = \sum_k \ln c_k$. Roots of products become averages of logarithms, infinite products become series, and the whole apparatus for sums becomes available. Exponentiate only at the very end. Before taking logarithms, confirm that the factors are eventually positive, and if finitely many are not, split them off as a separate constant factor.

WORKED EXAMPLE

A finite warm-up in which the product telescopes and no analysis is needed:

$$\prod_{k=2}^n \left(1 - \frac{1}{k^2}\right) = \prod_{k=2}^n \frac{(k-1)(k+1)}{k^2} = \frac{1}{2} \cdot \frac{n+1}{n} \rightarrow \frac{1}{2},$$

where the numerators and denominators cancel in two interleaved chains.

A second finite product, telescoping through a trigonometric identity rather than an arithmetic one. Repeated use of $\sin 2\theta = 2 \sin \theta \cos \theta$ gives

$$\prod_{k=1}^n \cos \frac{x}{2^k} = \frac{\sin x}{2^n \sin(x/2^n)} \rightarrow \frac{\sin x}{x} \quad (n \rightarrow \infty),$$

because $2^n \sin(x/2^n) \rightarrow x$. At $x = 1$ the partial products agree with $\sin 1 = 0.8414710$ to fifteen digits. This is Viète's product, and it is the cleanest example in the book of a product whose logarithm you should *not* take: the telescoping is visible in the product itself.

Now the serious case. Euler's product $\frac{\sin x}{x} = \prod_{n \geq 1} \left(1 - \frac{x^2}{n^2 \pi^2}\right)$ converges for every x , and for $0 < |x| < \pi$ every factor is positive, so taking logarithms is legitimate:

$$\sum_{n=1}^{\infty} \ln \left(1 - \frac{x^2}{n^2 \pi^2}\right) = \ln \frac{\sin x}{x} = -\frac{x^2}{6} + O(x^4).$$

Dividing by x^2 ,

$$\lim_{x \rightarrow 0} \frac{1}{x^2} \sum_{n=1}^{\infty} \ln \left(1 - \frac{x^2}{n^2 \pi^2}\right) = -\frac{1}{6},$$

and evaluating the series at $x = 10^{-3}$ gives -0.1666667 . Reading the same identity termwise, $\ln(1-t) = -t + O(t^2)$ turns the sum into $-\frac{x^2}{\pi^2} \sum n^{-2}$, so the answer is $-\frac{1}{6}$ precisely because $\sum n^{-2} = \pi^2/6$. The two routes agree, which is Euler's original argument run backwards.

WHERE THIS GOES WRONG

Exchanging $\ln$ with an infinite sum, or a limit with an infinite sum, without a convergence argument. The termwise reading above is legitimate because $\sum x^2/(n^2 \pi^2)$ converges uniformly on compact subsets of $(-\pi, \pi)$. The second failure is silent: $\ln$ of a negative factor is undefined, so a product whose early factors are negative must have them split off before the logarithm is applied, and a product with a zero factor has no logarithm at all.

SEE ALSO Riemann sum via the logarithm of a product; the standard 1^∞ form; nested limits taken strictly in order.

The last technique of this section is a discipline rather than a manipulation, and it becomes unavoidable the moment a problem contains two limit operators. It is placed here because the commonest source of a hidden double limit is precisely an inner 1^∞ whose value is a function of an outer variable, so the two halves of the problem use the two halves of this section.

5.11 Nested limits, taken strictly in order

CUE Two limit symbols, an inner limit that defines a function of the outer variable, or a sequence of functions whose limit is then evaluated at a moving point.

METHOD Complete the inner limit *fully*, obtaining a closed form in the outer variable, and only then take the outer limit. The two orders generally disagree, and neither may be interchanged without a uniformity hypothesis such as uniform convergence or monotonicity.

Let $g(x) = \lim_{r \rightarrow 0} ((x+1)^{r+1} - x^{r+1})^{1/r}$ for $x > 0$, and evaluate $\lim_{x \rightarrow \infty} \frac{g(x)}{x}$.

Inner limit. Put $h(r) = (x+1)^{r+1} - x^{r+1}$, so $h(0) = (x+1) - x = 1$: the inner form is 1^∞ in r . Because $\ln h(0) = 0$, the quotient $\frac{\ln h(r)}{r}$ is itself a difference quotient, and

$$\lim_{r \rightarrow 0} \frac{\ln h(r)}{r} = (\ln h)'(0) = \frac{h'(0)}{h(0)} = h'(0) = (x+1) \ln(x+1) - x \ln x,$$

since $h'(r) = (x+1)^{r+1} \ln(x+1) - x^{r+1} \ln x$. Exponentiating,

$$g(x) = \exp((x+1) \ln(x+1) - x \ln x) = \frac{(x+1)^{x+1}}{x^x}.$$

Outer limit. With the closed form in hand the second limit is routine:

$$\frac{g(x)}{x} = \frac{(x+1)^{x+1}}{x^{x+1}} = \left(1 + \frac{1}{x}\right)^{x+1} \rightarrow e,$$

by the e limit family. Numerically $g(10^5)/10^5 = 2.71859$ against $e = 2.71828$.

WHERE THIS GOES WRONG

Collapsing both limits at once, or swapping them. The standard warning is

$$\lim_{m \rightarrow \infty} \lim_{n \rightarrow \infty} \frac{m}{m+n} = 0, \quad \lim_{n \rightarrow \infty} \lim_{m \rightarrow \infty} \frac{m}{m+n} = 1.$$

In the worked example, letting $r \rightarrow 0$ and $x \rightarrow \infty$ together loses the fact that the inner answer depends on x through $(x+1) \ln(x+1) - x \ln x$, and that dependence is exactly what produces e rather than 1. A sharper pair, where both iterated limits exist and disagree by more than a constant: with $a_{m,n} = (1 + \frac{1}{m})^n$, the inner limit over n is ∞ for every fixed m , while the inner limit over m is 1 for every fixed n . Neither order is wrong; they are answers to different questions.

SEE ALSO Log-transform a product or a nested power; the standard 1^∞ form; iterated versus joint limits.

One closing observation ties the section together. Every technique in it was an application of a single sentence: a power is an exponential of a product, and $\exp$ is continuous. The generalisation of the geometric-mean example is immediate and worth doing yourself once,

$$\lim_{x \rightarrow 0} \left(\frac{a_1^x + a_2^x + \cdots + a_m^x}{m} \right)^{1/x} = (a_1 a_2 \cdots a_m)^{1/m},$$

for positive a_i , because $g(f-1) = \frac{1}{m} \sum_i \frac{a_i^x - 1}{x} \rightarrow \frac{1}{m} \sum_i \ln a_i$. At $m = 3$ with $(2, 3, 7)$ the value is $\sqrt[3]{42} \approx 3.476027$, matched by direct evaluation at $x = 10^{-6}$. Nothing new was needed for the general case, which is the sign that the machine, and not the individual answers, is the thing to learn.

The squeeze

The first two sections both assume that the expression can be transformed into one whose parts converge. Some expressions cannot. If $f(x) = x^2 \sin(1/x)$, then $\sin(1/x)$ has no limit at 0 and no rewriting will give it one; the limit of f nevertheless exists, and the only way to see it is to trap f between two functions that do converge. This is the squeeze, and it is the tool of last resort in exactly the sense that it is the tool that never needs the expression to cooperate.

The squeeze also has a second life as a technique for sums whose length grows with the limit variable, where it competes directly with the Riemann-sum recognition of the next chapter. The rule of thumb is worth stating in advance, because it decides which chapter a problem belongs to: if the crude bounds, meaning the smallest term times the count and the largest term times the count, already agree in the limit, use the squeeze and stop. If they disagree, the sum has genuine internal structure across its range and you need the integral.

5.12 The squeeze theorem template

CUE An expression that resists exact evaluation but is visibly trapped between two computable bounds. Oscillating factors, floor and fractional-part functions, and anything defined by an inequality rather than a formula are the standard signatures.

METHOD Produce $\ell(x) \leq f(x) \leq u(x)$ on a punctured neighbourhood of a , show $\lim_{x \rightarrow a} \ell = \lim_{x \rightarrow a} u = L$, and conclude $\lim_{x \rightarrow a} f = L$. The bounds may be crude; crudeness costs nothing provided both ends still converge to the same value. The theorem holds verbatim for sequences, for one-sided limits, and for $a = \pm\infty$.

WORKED EXAMPLE

$$\lim_{x \rightarrow 0^+} x \left\lfloor \frac{1}{x} \right\rfloor.$$

The defining inequality of the floor is $\frac{1}{x} - 1 < \left\lfloor \frac{1}{x} \right\rfloor \leq \frac{1}{x}$ for every $x > 0$. Multiplying through by the positive number x preserves both directions:

$$1 - x < x \left\lfloor \frac{1}{x} \right\rfloor \leq 1.$$

Both ends tend to 1, so the limit is 1, even though $\lfloor 1/x \rfloor$ jumps infinitely often in every neighbourhood of 0 and the function is discontinuous at infinitely many points of any interval $(0, \delta)$.

The baby case, and the one to reach for when an oscillation is bounded: $-x^2 \leq x^2 \sin \frac{1}{x} \leq x^2$ gives $\lim_{x \rightarrow 0} x^2 \sin \frac{1}{x} = 0$, with no claim whatsoever about $\sin(1/x)$ itself.

A sequence version, where the bound is built by comparing consecutive terms. For fixed $a > 0$ choose an integer $m > 2a$. For $n > m$ every further factor multiplies by $a/k < \frac{1}{2}$, so

$$0 < \frac{a^n}{n!} \leq \frac{a^m}{m!} \left(\frac{1}{2}\right)^{n-m} \rightarrow 0,$$

and therefore $a^n/n! \rightarrow 0$ for every a . At $a = 20$, $n = 300$ the value is about 6.7×10^{-225} , though the sequence increases until $n = 20$ before it turns. The squeeze is indifferent to that early growth, which is exactly why it is the right tool.

WHERE THIS GOES WRONG

Bounds that converge to different values prove nothing at all, and bounds valid only on one side of a prove only a one-sided limit. The multiplication step above is the other classic failure: multiplying an inequality by x reverses it when $x < 0$, so the corresponding limit as $x \rightarrow 0^-$ needs its own argument with the inequalities written the other way round.

SEE ALSO Bounded times infinitesimal; squeeze on a sum whose length is the limit variable; integral bounds on a sum.

THE SQUEEZE PROVES THE FACTS EVERYTHING ELSE ASSUMES

$\lim_{n \rightarrow \infty} n^{1/n} = 1$. Write $n^{1/n} = 1 + h_n$ with $h_n > 0$ for $n \geq 2$. The binomial theorem gives $n = (1 + h_n)^n \geq \binom{n}{2} h_n^2 = \frac{n(n-1)}{2} h_n^2$, so $0 < h_n \leq \sqrt{\frac{2}{n-1}} \rightarrow 0$. There is no way to obtain this by algebra on $n^{1/n}$ itself, and every use of the root test later in the book depends on it. The same argument gives $a^{1/n} \rightarrow 1$ for every $a > 0$.

The single most frequent application of the squeeze is so common that it deserves its own name and its own reflex, because the alternative reasoning that students reach for is not merely inefficient but invalid.

THE SQUEEZE RUN BACKWARDS PROVES NONEXISTENCE

The squeeze establishes a limit by trapping; the same idea run backwards refutes one. If two sequences $x_n \rightarrow a$ and $y_n \rightarrow a$ (with $x_n, y_n \neq a$) give $f(x_n) \rightarrow L_1$ and $f(y_n) \rightarrow L_2$ with $L_1 \neq L_2$, then $\lim_{x \rightarrow a} f(x)$ does not exist. For $f(x) = \sin(1/x)$ at $a = 0$, take $x_n = \frac{1}{2\pi n}$ and $y_n = \frac{1}{\pi/2 + 2\pi n}$: the values are 0 and 1 for every n . This is why $x^2 \sin(1/x) \rightarrow 0$ has to be proved by bounding and cannot be proved by a product rule: the second factor genuinely has no limit, and the sequence argument shows it in two lines.

5.13 Bounded times infinitesimal

CUE A product $b(x) \varepsilon(x)$ where b is bounded, typically an oscillating sin, cos, fractional part, or sgn, and $\varepsilon \rightarrow 0$. Also a quotient in which such a bounded term is added to something that dominates it.

METHOD If $|b| \leq M$ on a punctured neighbourhood and $\varepsilon \rightarrow 0$, then $b\varepsilon \rightarrow 0$. The proof is $0 \leq |b\varepsilon| \leq M|\varepsilon| \rightarrow 0$ followed by the squeeze applied to $|b\varepsilon|$. Crucially, no assumption is made that $\lim b$ exists, and none is available in the interesting cases.

$$\lim_{x \rightarrow \infty} \frac{x + \sin x}{x + \cos x} = \lim_{x \rightarrow \infty} \frac{1 + \frac{\sin x}{x}}{1 + \frac{\cos x}{x}} = \frac{1 + 0}{1 + 0} = 1.$$

Dividing through by x is what converts the two non-convergent terms into bounded-times-infinitesimal products; each tends to 0 by the bound $|\sin x| \leq 1$, and only after that step is the quotient rule for limits legal, because only then do numerator and denominator separately have limits.

The same bound settles fractional parts, where the oscillation is not even continuous. Writing $\{t\} = t - [t]$ for the fractional part, $0 \leq \{1/x\} < 1$ for all $x \neq 0$, so

$$\lim_{x \rightarrow 0} x \left\{ \frac{1}{x} \right\} = 0 \quad \text{and} \quad \lim_{x \rightarrow 0} \left(x \left\{ \frac{1}{x} \right\} + x \left[\frac{1}{x} \right] \right) = \lim_{x \rightarrow 0} x \cdot \frac{1}{x} = 1,$$

the second because the two pieces reassemble into something exact. The first limit is bounded times infinitesimal; the second is a reminder that the decomposition, not the individual pieces, is what carries the information.

WHERE THIS GOES WRONG

Writing $\lim_{x \rightarrow \infty} \frac{\sin x}{x} = \lim_{x \rightarrow \infty} \sin x \cdot \lim_{x \rightarrow \infty} \frac{1}{x}$. The product rule for limits requires both factors to have limits; here the first has none, and the correct justification is the boundedness estimate. The conclusion happens to be right, which is what makes the habit dangerous: the same reasoning applied to $\frac{x \sin x}{x}$ would produce 0 instead of the correct answer, which is that the limit does not exist.

SEE ALSO The squeeze theorem template; squeeze on a sum whose length is the limit variable; the n th root of a sum tends to the maximum.

Both of the last two blocks share a feature worth naming: the bound was found by discarding information deliberately. Replacing $\sin x$ by 1, or $[1/x]$ by $1/x$, throws away everything that made the expression hard, and the limit survives the loss. That is the whole art. A bound that keeps too much detail is as useless as no bound, because it will be as hard to evaluate as the original; a bound that throws away too much will not pinch. Aim first for the crudest bound imaginable, check whether both ends agree, and only sharpen if they do not.

Sums whose number of terms grows are the second natural home of the squeeze. Every individual term tends to 0, so the termwise reading gives 0; the sum does not. Bounding all the terms by the two extreme ones resolves the tension in a single step, whenever it works at all, and the diagnostic for whether it works is worth internalising because it also tells you when to turn the page to the Riemann sums.

MEDIUM

5.14 Squeeze on a sum whose length is the limit variable

CUE $S_n = \sum_{k=1}^n a_{n,k}$ where the number of terms grows with n and each individual term tends to 0. The tell is that the terms vary only slightly across k , so that the first and last are nearly equal.

METHOD Bound every summand by the extremes:

$$n \cdot \min_{1 \leq k \leq n} a_{n,k} \leq S_n \leq n \cdot \max_{1 \leq k \leq n} a_{n,k}.$$

If both bounds share a limit you are finished, and no integral is needed. If they do not, the sum's internal variation is of the same order as the sum itself, and the Riemann-sum machinery of the next chapter is required instead.

WORKED EXAMPLE

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{\sqrt{n^2 + k}}.$$

The terms decrease in k , so the largest is at $k = 1$ and the smallest at $k = n$:

$$\frac{n}{\sqrt{n^2 + n}} \leq S_n \leq \frac{n}{\sqrt{n^2 + 1}}.$$

Both ends tend to 1, so the limit is 1; at $n = 10^6$ the sum is 0.9999997. The reason the squeeze pinches is that $k \leq n$ is negligible beside n^2 , so the ratio of the extreme terms tends to 1. The same test passes for $\sum_{k=1}^n \frac{k}{n^2 + k}$, whose extreme

terms are $\frac{1}{n^2+1}$ and $\frac{n}{n^2+n}$: here the crude bounds do *not* pinch, but the sum is $\frac{1}{n^2} \sum k$ to within $O(1/n)$, so the limit is $\frac{1}{2}$, and at $n = 10^6$ the value is 0.5000002. Read that as a warning: failing the ratio test for the crude bounds means only that these particular bounds fail, not that the sum is hard.

Contrast $\sum_{k=1}^n \frac{1}{\sqrt{n^2+k^2}}$, where the same bounds give only $\frac{1}{\sqrt{2}} \leq S_n \leq 1$ and refuse to close. Here k enters at the same scale as n , the extreme terms differ by a factor of $\sqrt{2}$ in the limit, and the answer is the integral

$$\int_0^1 \frac{du}{\sqrt{1+u^2}} = \ln(1+\sqrt{2}) \approx 0.881374.$$

WHERE THIS GOES WRONG

Taking the limit termwise and concluding $0 + 0 + \dots = 0$. The sum rule for limits extends to a *fixed* number of summands only, and here the number of summands is the limit variable itself. The diagnostic for which tool applies is the ratio of the largest term to the smallest: if it tends to 1, the squeeze pinches; if it tends to a constant greater than 1, it cannot, and no refinement of the same two bounds will help.

SEE ALSO Riemann sum recognition; the squeeze theorem template; integral bounds on a sum.

When the summand is monotone in k but its variation across the range is too large for the crude bounds, there is still a squeeze available, with the two bounds supplied by integrals rather than by extreme terms. This is a sharper instrument, it is the source of most discrete asymptotics, and it is the bridge to the next chapter: the same picture of rectangles against a curve reappears there with the roles of sum and integral exchanged.

5.15 Integral bounds on a sum

MEDIUM

CUE A sum of the values of a monotone function, where you need the asymptotic size rather than the exact value: $\sum k^p$, $\sum 1/k$, $\sum \ln k$, or a tail $\sum_{k>n} f(k)$ whose size controls a remainder.

METHOD For f decreasing on $[1, \infty)$,

$$\int_1^{n+1} f(x) dx \leq \sum_{k=1}^n f(k) \leq f(1) + \int_1^n f(x) dx,$$

and for f increasing the inequalities reverse in the obvious way, becoming $\int_0^n f \leq \sum_{k=1}^n f(k) \leq \int_1^{n+1} f$. This is simultaneously the integral test for convergence and the source of $H_n = \ln n + \gamma + O(1/n)$.

WORKED EXAMPLE

$$\lim_{n \rightarrow \infty} \frac{1}{n^{p+1}} \sum_{k=1}^n k^p, \quad p > 0.$$

Since x^p increases, the rectangle of height k^p and width 1 sits above $\int_{k-1}^k x^p dx$ and below $\int_k^{k+1} x^p dx$, so summing,

$$\frac{n^{p+1}}{p+1} = \int_0^n x^p dx \leq \sum_{k=1}^n k^p \leq \int_1^{n+1} x^p dx = \frac{(n+1)^{p+1} - 1}{p+1}.$$

Dividing by n^{p+1} , both ends tend to $\frac{1}{p+1}$. At $p = 3$ the limit is $\frac{1}{4}$, and the sum at $n = 2 \times 10^5$ gives 0.2500025.

The same inequality with $f(x) = 1/x$, now decreasing, gives $\ln(n+1) \leq H_n \leq 1 + \ln n$, so $H_n - \ln n$ is trapped in $[0, 1]$; the sequence is decreasing and its limit is $\gamma \approx 0.5772157$. At $n = 10^6$ the difference is 0.5772162. With $f(x) = \ln x$ the same picture gives $\ln(n!) = n \ln n - n + O(\ln n)$, which is Stirling's formula without its constant; at $n = 10^6$ the discrepancy $\ln(n!) - (n \ln n - n)$ is 7.8267, against $\frac{1}{2} \ln(2\pi n) = 7.8267$.

The same inequality run on a tail rather than a head is how remainders are estimated. For f decreasing and positive,

$$\int_{n+1}^{\infty} f \leq \sum_{k>n} f(k) \leq \int_n^{\infty} f,$$

so for $f(x) = x^{-2}$ the tail $\sum_{k>n} k^{-2}$ is trapped between $\frac{1}{n+1}$ and $\frac{1}{n}$, giving $n \sum_{k>n} k^{-2} \rightarrow 1$. The two bounds differ by a term of order n^{-2} , which is what makes the estimate useful rather than merely true.

Using the wrong endpoint convention and losing a whole term. Draw the rectangles: for a decreasing f , the rectangle of width 1 and height $f(k)$ sits *above* $\int_k^{k+1} f$ and *below* $\int_{k-1}^k f$, and the direction of both inequalities flips when f increases. A misplaced endpoint changes H_n by 1, which is the difference between a convergent and a divergent conclusion in some comparison arguments.

SEE ALSO Squeeze on a sum whose length is the limit variable; Riemann sum recognition; the integral test.

The final technique of the chapter is the purest illustration of the squeeze's power: an expression involving several competing exponentials, with no algebraic simplification available and no logarithm that helps, whose limit falls out of two lines of bounding.

EASY

5.16 The n th root of a sum tends to the maximum

CUE $(a_1^n + \dots + a_m^n)^{1/n}$ with m fixed and all $a_i \geq 0$; equivalently the L^p norm of a fixed finite list as $p \rightarrow \infty$.

METHOD Let $M = \max_i a_i > 0$. Then $M^n \leq \sum_i a_i^n \leq m M^n$, since every term is at most M^n and at least one equals it. Taking n th roots,

$$M \leq \left(\sum_{i=1}^m a_i^n \right)^{1/n} \leq M m^{1/n} \rightarrow M,$$

since $m^{1/n} = e^{(\ln m)/n} \rightarrow 1$ for fixed m . The squeeze does all the work; the largest base swallows the rest.

WORKED EXAMPLE

$\lim_{n \rightarrow \infty} (2^n + 3^n)^{1/n} = 3$, because $3 \leq (2^n + 3^n)^{1/n} \leq 3 \cdot 2^{1/n} \rightarrow 3$; at $n = 200$ the value already agrees with 3 to fifteen digits. Equally $(1^n + 2^n + \dots + 7^n)^{1/n} \rightarrow 7$, with the crude upper bound $7 \cdot 7^{1/n}$. Constants in front change nothing: $\left(\frac{a^n + b^n}{2}\right)^{1/n} \rightarrow \max(a, b)$, since the factor $2^{-1/n} \rightarrow 1$ is absorbed by the same argument. Compare this with the geometric mean produced by $\left(\frac{a^x + b^x}{2}\right)^{1/x}$ as $x \rightarrow 0$: identical expression, opposite limit point, and the answers are the two classical means.

The continuous analogue is the same argument: for a step function taking finitely many values, $(\int_0^1 |f|^p)^{1/p} \rightarrow \|f\|_\infty$ as $p \rightarrow \infty$, with m playing the role of the number of levels and the measure of each level playing the role of the constant that $m^{1/n}$ absorbs.

WHERE THIS GOES WRONG

Letting m grow with n . The factor $m^{1/n}$ then need not tend to 1 and the conclusion fails outright: take $m = 2^n$ copies of $a_i = 1$, so the sum is 2^n and its n th root is 2, while the maximum is 1. The hypothesis that m is fixed is doing real work, and any problem in which the number of competing terms grows must be handled by the sum techniques instead.

SEE ALSO The squeeze theorem template; the e limit family; the root test for series.

Stand back from the three sections and the division of labour is clean. Surgery works when the expression is lying about itself, and it succeeds by producing an equal expression that tells the truth. The exponential machine works when the expression is honest but the operation is wrong, and it succeeds by replacing that operation with one that limits respect. The squeeze works when the expression contains something that will never converge, and it succeeds by refusing to look inside. Nothing in this chapter needed a derivative, a Taylor series, or L'Hôpital's rule, and a solver who reaches for those first will spend three times as long on most of these problems and will still get the structural questions wrong.

Recognition matrix

If you see	Reach for	If that fails
$\sqrt{A} - \sqrt{B}$ in a $\frac{0}{0}$ or $\infty - \infty$	Conjugate: $\frac{A - B}{\sqrt{A} + \sqrt{B}}$	Bridging term at the common limit
$\sqrt[3]{A} - \sqrt[3]{B}$	Cube conjugate $a^2 + ab + b^2$	Binomial expansion of $(1 + u)^{1/3}$
$\sqrt[3]{f} - \sqrt[3]{g}$, both $\rightarrow 1$	Bridge at 1, then two conjugates	Expand both to order x^2
Polynomial $\frac{0}{0}$ at $x = a$	Divide out $(x - a)$, possibly twice	Synthetic division, then substitute
Two fractions each blowing up	Common denominator first	Change of variable $h = x - a$
Roots of orders m and n at $x = 1$	Substitute $x = t^{\text{lcm}(m, n)}$	Difference-quotient reading
Limit point $\pi/2, 1$, or ∞	Move it to 0: $u = x - a$ or $t = 1/x$	Track the direction of approach
Base $\rightarrow 1$, exponent $\rightarrow \infty$	$\exp(\lim g(f - 1))$	$\exp(\lim g \ln f)$, $\ln(1 + u)$ expanded

If you see	Reach for	If that fails
Base $\rightarrow 1$, exponent like x^{-4}	Expand $\ln f$ to order x^4	Check whether the x^2 term cancels
Exponent is itself a $\frac{0}{0}$ quotient	Simplify the exponent first	Then test whether the base is 1
0^0 or ∞^0	Same transform: $\exp(g \ln f)$	Compare growth rates in the product
A product, or an n th root of one	Take logarithms: $\ln \prod = \sum \ln$	Read the sum as an integral
Two limit operators	Inner limit to closed form, then outer	Check that the orders disagree
A bounded but non-convergent factor	Bounded times infinitesimal	Squeeze with explicit ℓ and u
$\sum_{k=1}^n a_{n,k}$, terms nearly equal	$n \min \leq S_n \leq n \max$	Riemann sum, or integral bounds
$\sum f(k)$ with f monotone	Integral bounds on the sum	Euler and Maclaurin correction terms
$(a_1^n + \dots + a_m^n)^{1/n}$, m fixed	Factor out the maximum	Check that m really is fixed

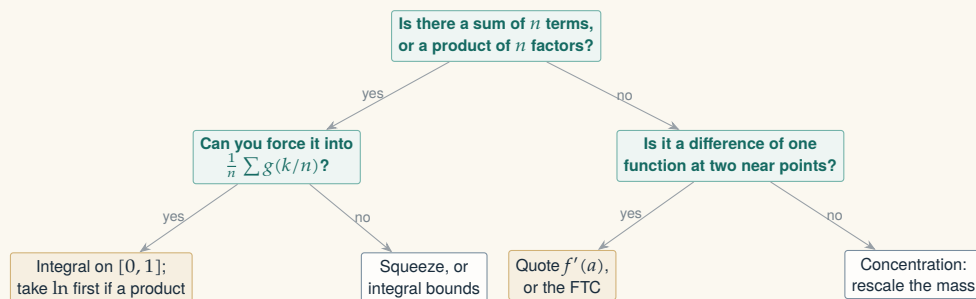
Limits that are secretly integrals or derivatives

What this chapter is for. Two recognitions in this chapter are worth more than any other pair in the limits part of the book. The first: a sum of n terms carrying a factor of $\frac{1}{n}$, in which the index k appears only through the ratio k/n , is a Riemann sum, and its limit is an integral. The second: a quotient whose numerator is the difference of one function evaluated at two nearby points is a derivative, and you should quote it rather than compute it. Both are pattern-matching skills, so this chapter teaches the matching explicitly: how to read the interval and the integrand off a sum, what to do when the interval is not $[0, 1]$, when the partition is geometric rather than uniform, and when the object is a product that must be logged before it becomes a sum. The chapter closes with the concentration limits, where an n th power inside an integral collapses the integral onto a single point.

Reading a sum as an integral

The definition of the Riemann integral is usually presented as a definition and then abandoned in favour of the fundamental theorem. That is a mistake of emphasis. The definition is a *recognition tool*: it says that a particular kind of limit of a particular kind of sum equals a particular integral, and it can be read from left to right (to define the integral) or from right to left (to evaluate a limit). This chapter reads it from left to right, and the reading is worth more in competition and in analysis than most of the integration technique that follows it.

The pattern to match is small and rigid. You need three things at once: exactly n terms (or cn terms, or n terms shifted), a factor of $\frac{1}{n}$ multiplying the whole sum, and every occurrence of k appearing inside a single function of the ratio k/n . If all three are present, the limit is an integral. If any one is missing, either it can be manufactured by algebra, in which case do so, or it cannot, in which case the sum belongs to the squeeze and integral-bound techniques of the previous chapter.



6.1 Riemann sum recognition

MEDIUM

CUE A factor $\frac{1}{n}$, or $\frac{b-a}{n}$, multiplying a sum of n terms in which k appears only through the ratio k/n . The factor is often hidden: it may sit inside a radical, or appear as an n in a numerator that has to be divided into the summand before it becomes visible.

METHOD For f Riemann integrable on $[0, 1]$,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n f\left(\frac{k}{n}\right) = \int_0^1 f(x) \, dx,$$

and more generally $\frac{b-a}{n} \sum_{k=1}^n f\left(a + \frac{k(b-a)}{n}\right) \rightarrow \int_a^b f$. The left endpoint version, with k running from 0 to $n-1$, has the same limit; so does any choice of sample point inside each subinterval, which is what makes the recognition robust.

One. Count the terms and locate the mesh factor $\frac{1}{n}$. If you cannot see one, factor the largest power of n out of the summand until one appears. **Two.** Substitute $u = k/n$ everywhere and read off $f(u)$. **Three.** Verify that no free n survives inside f . A single leftover n means the summand is not a fixed function sampled at n points, and the conversion is invalid.

WORKED EXAMPLE

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{n}{n^2 + k^2}.$$

There is no visible $\frac{1}{n}$. Divide numerator and denominator by n^2 : the summand becomes $\frac{1/n}{1 + (k/n)^2}$, so the sum is $\frac{1}{n} \sum_{k=1}^n \frac{1}{1 + (k/n)^2}$ and $f(u) = \frac{1}{1 + u^2}$. Hence

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{n}{n^2 + k^2} = \int_0^1 \frac{dx}{1 + x^2} = \frac{\pi}{4} \approx 0.785398,$$

and the sum at $n = 10^6$ is 0.7853979. Note that the crude bounds of the previous chapter give only $\frac{1}{2} \leq S_n \leq 1$, so the squeeze cannot resolve this one and the integral is genuinely needed.

The interval $[0, 1]$ is a convention, not a constraint, and the two ways of leaving it are worth separating. Extending the range of k moves the right endpoint; shifting the range of k moves both endpoints. Both are read off the same way: the smallest and largest values of k/n are the limits of integration.

WORKED EXAMPLE

A longer range. $\sum_{k=1}^{3n} \frac{n}{n^2 + k^2} = \frac{1}{n} \sum_{k=1}^{3n} \frac{1}{1 + (k/n)^2}$ now samples $u = k/n$ across $[0, 3]$, so the limit is $\int_0^3 \frac{dx}{1 + x^2} = \arctan 3 \approx 1.249046$; the sum at $n = 10^6$ is 1.2490453.

A shifted range. $\sum_{k=n+1}^{2n} \frac{1}{k}$. Put $k = n + j$ with j from 1 to n : the summand is $\frac{1}{n+j} = \frac{1}{n} \cdot \frac{1}{1+j/n}$, so the limit is $\int_0^1 \frac{dx}{1+x} = \ln 2 \approx 0.693147$. Equivalently, read $u = k/n$ directly on $[1, 2]$ and get $\int_1^2 \frac{dx}{x}$, which is the same number, as it must be.

Partitions do not have to be uniform, and the case that actually occurs is the geometric one, where the sample points are $a^{k/n}$ rather than k/n . It looks unfamiliar and is not: the substitution $x = a^t$ converts it into a uniform partition, and you may work on whichever side is more convenient.

GEOMETRIC PARTITIONS ARE UNIFORM IN THE EXPONENT

The points $x_k = a^{k/n}$, $k = 0, \dots, n$, partition $[1, a]$ with ratio $a^{1/n}$ between consecutive points, so the mesh $x_k - x_{k-1} = a^{(k-1)/n}(a^{1/n} - 1)$ tends to 0 and

$$\sum_{k=1}^n f(x_k)(x_k - x_{k-1}) \rightarrow \int_1^a f(x) dx.$$

Under $x = a^t$ this is the uniform statement $\frac{1}{n} \sum_{k=1}^n g(k/n) \rightarrow \int_0^1 g$ with $g(t) = f(a^t)a^t \ln a$. Concretely,

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{2^{k/n} - 2^{(k-1)/n}}{2^{k/n}} = \int_1^2 \frac{dx}{x} = \ln 2,$$

verified as 0.693146 at $n = 2 \times 10^5$; while the same content in uniform form is

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \frac{1}{1 + 2^{k/n}} = \int_0^1 \frac{dt}{1 + 2^t} = \frac{1}{\ln 2} \ln \frac{4}{3} \approx 0.415037,$$

where the substitution $x = 2^t$ turns the integral into $\frac{1}{\ln 2} \int_1^2 \frac{dx}{x(1+x)}$.

WHERE THIS GOES WRONG

The $\frac{1}{n}$ is hidden, and you conclude too early that there is none. It may be inside a radical ($\sqrt{n^2 + k^2}$ contributes an n when factored), or it may be an n in a numerator that has to be divided into the summand. Conversely, if after honest effort you cannot

manufacture an explicit $\frac{1}{n} \sum g(k/n)$, then it is not a Riemann sum, and the squeeze or the integral bounds of the previous chapter are the correct tools.

SEE ALSO Riemann sum via the logarithm of a product; Riemann sums that need algebraic massage first; squeeze on a sum whose length is the limit variable.

A product of n factors is a sum of n logarithms, so every product whose factors depend on k/n is a Riemann sum waiting for one line of preparation. This is the single most reliable way to evaluate limits of products, and the n th root that usually accompanies them is exactly the $\frac{1}{n}$ the recognition needs.

MEDIUM

6.2 Riemann sum via the logarithm of a product

CUE An n th root of a product of n factors, each depending on k only through k/n ; or a factorial expression such as $\frac{1}{n} (n!)^{1/n}$ or $\frac{1}{n} [(n+1) \cdots (2n)]^{1/n}$, which is the same shape once the factorials are written as products.

METHOD Take logarithms, so that the n th root becomes the factor $\frac{1}{n}$ and the product becomes a sum:

$$\ln \left[\prod_{k=1}^n c_k \right]^{1/n} = \frac{1}{n} \sum_{k=1}^n \ln c_k.$$

Arrange each c_k to depend on k/n alone, evaluate the resulting integral, and exponentiate at the end. The outer normalising factor (usually a $\frac{1}{n}$ in front of the root) must be distributed across all n factors as $n^{-1} = (n^{-n})^{1/n}$ before the substitution is made.

WORKED EXAMPLE

$$\lim_{n \rightarrow \infty} \frac{1}{n} [(n+1)(n+2) \cdots (2n)]^{1/n}.$$

Distribute the outer $\frac{1}{n}$ into the root, one factor of n per term:

$$\frac{1}{n} \prod_{k=1}^n (n+k)^{1/n} = \prod_{k=1}^n \left(\frac{n+k}{n} \right)^{1/n} = \prod_{k=1}^n \left(1 + \frac{k}{n} \right)^{1/n} =: P_n.$$

Now take logarithms and recognise the Riemann sum:

$$\ln P_n = \frac{1}{n} \sum_{k=1}^n \ln \left(1 + \frac{k}{n} \right) \rightarrow \int_0^1 \ln(1+x) dx = [(1+x) \ln(1+x) - x]_0^1 = 2 \ln 2 - 1.$$

Exponentiating, the limit is $e^{2 \ln 2 - 1} = \frac{4}{e} \approx 1.4715178$, and the product at $n = 2 \times 10^5$ gives 1.4715203.

Two variations are worth having in hand. First, the factorial itself: $\frac{(n!)^{1/n}}{n} = \left[\prod_{k=1}^n \frac{k}{n} \right]^{1/n}$, whose logarithm is $\frac{1}{n} \sum \ln(k/n) \rightarrow \int_0^1 \ln x dx = -1$, so the limit is e^{-1} . The integral is improper at 0 but convergent, and the sum's smallest term $\frac{1}{n} \ln \frac{1}{n} \rightarrow 0$, so the recognition survives. Second, a product with no factorial interpretation at all:

WORKED EXAMPLE

$$\lim_{n \rightarrow \infty} \left[\prod_{k=1}^n \left(1 + \frac{k^2}{n^2} \right) \right]^{1/n}.$$

Logarithm first: $\frac{1}{n} \sum_{k=1}^n \ln(1 + (k/n)^2) \rightarrow \int_0^1 \ln(1+x^2) dx$. Integrate by parts with $u = \ln(1+x^2)$:

$$\int_0^1 \ln(1+x^2) dx = [x \ln(1+x^2) - 2x + 2 \arctan x]_0^1 = \ln 2 - 2 + \frac{\pi}{2}.$$

So the limit is $\exp(\ln 2 - 2 + \frac{\pi}{2}) = 2e^{\pi/2-2} \approx 1.3020546$, against 1.3020569 for the product at $n = 2 \times 10^5$.

WHERE THIS GOES WRONG

Forgetting to distribute the outer $\frac{1}{n}$ across all n factors, or losing it altogether. In the worked example, $\frac{1}{n}$ becomes one factor of $\frac{1}{n}$ inside each of the n roots, and omitting it leaves a product that diverges. The check is dimensional: after the distribution, every factor must tend to a finite nonzero limit, so that the geometric mean of the factors is finite.

SEE ALSO Riemann sum recognition; log-transform a product or a nested power; Riemann sums that need algebraic massage first.

The recognition is exact only after the algebra is exact, and problems designed to be hard are designed so that the Riemann structure appears only after two or three terms have been combined. The technique below is the recognition applied with the algebra of the previous chapter placed in front of it.

6.3 Riemann sums that need algebraic massage first

HARD

CUE A sum that is *almost* $\frac{1}{n} \sum f(k/n)$ but has two or more terms inside, or a mismatched index range, or a residual n that looks fatal. Also any expression in which a Riemann sum is buried under an outer operation such as a reciprocal, a power, or a root.

METHOD Combine the terms over a common denominator first, then substitute $u = k/n$ symbolically and confirm that every n has vanished except the overall $\frac{1}{n}$. Only then name the integral. Shifted or lengthened ranges give integrals over shifted or lengthened intervals, and outer operations are applied to the integral's value at the very end.

WORKED EXAMPLE

$$\lim_{n \rightarrow \infty} \left[\sum_{k=1}^n \left(\frac{n}{\pi \sqrt{n^4 + k^4}} - \frac{2nk^2}{\pi(n^2 + k^2)\sqrt{n^4 + k^4}} \right) \right]^{-2}.$$

Work on the bracket, writing $u = k/n$. For the first term, $\sqrt{n^4 + k^4} = n^2 \sqrt{1 + u^4}$, so it equals $\frac{1}{n} \cdot \frac{1}{\pi \sqrt{1 + u^4}}$. For the second, $n^2 + k^2 = n^2(1 + u^2)$ and $k^2 = n^2 u^2$, so it equals $\frac{1}{n} \cdot \frac{2u^2}{\pi(1 + u^2)\sqrt{1 + u^4}}$. Every n has gone except the mesh factor. Combining over the common denominator,

$$\frac{1}{\pi} \cdot \frac{(1 + u^2) - 2u^2}{(1 + u^2)\sqrt{1 + u^4}} = \frac{1}{\pi} \cdot \frac{1 - u^2}{(1 + u^2)\sqrt{1 + u^4}},$$

so the bracket tends to $\frac{1}{\pi} \int_0^1 \frac{1 - u^2}{(1 + u^2)\sqrt{1 + u^4}} du$.

The integral yields to Glasser's substitution. Divide numerator and denominator by u^2 and set $w = u + \frac{1}{u}$, so that $dw = (1 - \frac{1}{u^2}) du$ and $u^2 + \frac{1}{u^2} = w^2 - 2$:

$$\int_0^1 \frac{(\frac{1}{u^2} - 1) du}{(u + \frac{1}{u})\sqrt{u^2 + \frac{1}{u^2}}} = \int_2^\infty \frac{dw}{w\sqrt{w^2 - 2}} = \frac{1}{\sqrt{2}} \left[\operatorname{arcsec} \frac{w}{\sqrt{2}} \right]_2^\infty = \frac{1}{\sqrt{2}} \left(\frac{\pi}{2} - \frac{\pi}{4} \right) = \frac{\pi}{4\sqrt{2}},$$

the orientation reversing because w decreases from ∞ to 2 as u increases from 0 to 1. Hence the bracket tends to $\frac{1}{\pi} \cdot \frac{\pi}{4\sqrt{2}} = \frac{1}{4\sqrt{2}}$, and the answer is $(4\sqrt{2})^2 = 32$. Numerically the bracket at $n = 4 \times 10^5$ is 0.1767763 against $\frac{1}{4\sqrt{2}} = 0.1767767$, and its inverse square is 32.0001.

WHERE THIS GOES WRONG

Declaring "Riemann sum" before checking that the residual n -dependence really cancels. A single stray n inside f invalidates the whole conversion, because then the summand is not one fixed function sampled at n points but a different function for each n . The habit that prevents this is to perform the substitution $u = k/n$ literally, in writing, and to look at what is left.

SEE ALSO Riemann sum recognition; Riemann sum via the logarithm of a product; squeeze on a sum whose length is the limit variable.

Reading a quotient as a derivative

The second recognition of the chapter is smaller than the first and even cheaper to execute. A derivative is defined by a limit; therefore any limit of that shape has already been evaluated, and quoting the answer is not a shortcut but the definition read in the direction it was written.

6.4 The limit is a derivative in disguise

CUE $\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$ or $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$, possibly with the constant $f(a)$ written as a number so that the structure is hidden. The signature is a numerator that is one function evaluated at two points approaching each other, over the distance between them.

METHOD Do not differentiate anything by rule inside the limit: recognise the expression as the *definition* of $f'(a)$ and quote the derivative. The scaled and symmetric variants follow from the same definition:

$$\lim_{h \rightarrow 0} \frac{f(a + \alpha h) - f(a + \beta h)}{h} = (\alpha - \beta)f'(a), \quad \lim_{h \rightarrow 0} \frac{f(a+h) - 2f(a) + f(a-h)}{h^2} = f''(a),$$

the second requiring f'' to exist and be continuous at a .

WORKED EXAMPLE

$$\lim_{x \rightarrow 0} \frac{\sqrt[5]{32+x} - 2}{x}.$$

Write $g(t) = t^{1/5}$ and note $g(32) = 2$, so the quotient is $\frac{g(32+x) - g(32)}{x}$, which is the difference quotient for g at 32. Since $g'(t) = \frac{1}{5}t^{-4/5}$ and $32^{-4/5} = \frac{1}{16}$, the limit is $\frac{1}{80} = 0.0125$.

A second disguise, at $h = 1/n$: $\lim_{n \rightarrow \infty} n(a^{1/n} - 1) = \lim_{h \rightarrow 0} \frac{a^h - a^0}{h} = \ln a$, since $\frac{d}{dt}a^t = a^t \ln a$. At $a = 5$ and $n = 10^7$ the value is 1.6094380 against $\ln 5 = 1.6094379$.

A third, a second difference: $\lim_{h \rightarrow 0} \frac{\sin(a+2h) - 2\sin(a+h) + \sin a}{h^2} = -\sin a$, matching $f''(a)$ for $f = \sin$; at $a = 0.7$, $h = 10^{-4}$ the quotient is -0.644294 against $-\sin 0.7 = -0.644218$.

WHERE THIS GOES WRONG

Using L'Hôpital, which requires differentiating f and is therefore circular when f' is precisely what is being defined. This is the standard fallacy in "proving" $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ by L'Hôpital: the derivative of $\sin$ at 0 is that limit. The second error is applying the symmetric second-difference formula to a function whose second derivative does not exist; the symmetric quotient can converge when $f''(a)$ does not, and then it is not $f''(a)$.

SEE ALSO Mean-value forms and FTC-driven limits; factor out the removable singularity; the standard limits at zero.

The integral analogue of the difference quotient is the fundamental theorem, and it produces the same kind of instant evaluation for expressions in which an integral over a shrinking interval is divided by the length of that interval.

MEDIUM

6.5 Mean-value forms and FTC-driven limits

CUE An integral with a variable endpoint divided by a power of that endpoint, or divided by the length of the interval of integration. Also any $\frac{0}{0}$ quotient in which one storey is an integral.

METHOD For f continuous at a ,

$$\lim_{h \rightarrow 0} \frac{1}{h} \int_a^{a+h} f(t) dt = f(a),$$

which is the fundamental theorem, or equivalently the integral mean value theorem: the average of f over a shrinking interval is a value of f inside it. For a $\frac{0}{0}$ quotient of integrals, L'Hôpital combined with the Leibniz rule differentiates the bounds:

$$\frac{d}{dx} \int_{u(x)}^{v(x)} f(t) dt = f(v(x))v'(x) - f(u(x))u'(x).$$

WORKED EXAMPLE

$$\lim_{x \rightarrow 0} \frac{1}{x^3} \int_0^x \frac{t^2}{1+t^4} dt.$$

Both storeys vanish, and the numerator is differentiable by the FTC. One application of L'Hôpital gives

$$\lim_{x \rightarrow 0} \frac{x^2/(1+x^4)}{3x^2} = \frac{1}{3}.$$

confirmed at $x = 10^{-3}$ by numerical quadrature. The structural reading is the same: the integrand is $t^2(1 + O(t^4))$, so the integral is $\frac{x^3}{3} + O(x^7)$.

A case where the chain-rule factors matter:

$$\lim_{x \rightarrow 0^+} \frac{1}{x^3} \int_0^{x^2} \sin \sqrt{t} \, dt.$$

Here $v(x) = x^2$, so the derivative of the numerator is $\sin(\sqrt{x^2}) \cdot 2x = 2x \sin x$ for $x > 0$, and the limit is $\lim_{x \rightarrow 0^+} \frac{2x \sin x}{3x^2} = \frac{2}{3}$, matching 0.666666 by quadrature at $x = 10^{-3}$.

WHERE THIS GOES WRONG

Differentiating the integral without the chain-rule factors v' and u' , or dropping the minus sign on the lower limit. In the second worked example, omitting the factor $2x$ turns the answer from $\frac{2}{3}$ into an infinite limit. The other frequent error is applying L'Hôpital to a quotient that is not indeterminate, which happens whenever the integrand fails to vanish at the endpoint.

SEE ALSO The limit is a derivative in disguise; endpoint concentration; Riemann sum recognition.

When the mass concentrates

The last three techniques share one mechanism and it is worth stating before any of them. An integral $\int_0^1 \varphi_n(x)g(x) \, dx$ in which φ_n carries an n th power does not converge by anything happening across the whole interval. The power kills the integrand everywhere except on a shrinking window, and the entire value of the limit comes from that window. The technique in each case is the same: find the window, rescale so that the window becomes fixed, and identify the limiting profile. What differs between the three is where the window is (an endpoint, an interior maximum, a level set) and how wide it is ($1/n$, $1/\sqrt{n}$, or vanishing).

The warning that goes with all three is also common. Passing the limit inside the integral before rescaling gives 0, because $x^n \rightarrow 0$ pointwise on $[0, 1)$; the outside factor of n or $\sqrt{n}$ is exactly what compensates, and its presence in the problem is the signal that pointwise convergence is the wrong lens.

HARD

6.6 Endpoint concentration: $n \int_0^1 x^n f(x) \, dx \rightarrow f(1)$

CUE A high power x^n (or $\sin^n$, $\cos^n$) weighting a continuous function on an interval where the power attains its maximum at an endpoint, with a factor of n or $\sqrt{n}$ outside the integral.

METHOD The weight x^n concentrates its mass in a window of width $O(1/n)$ at $x = 1$, and $n x^n$ acts as an approximate identity there:

$$\lim_{n \rightarrow \infty} n \int_0^1 x^n f(x) \, dx = f(1) \quad \text{for } f \text{ continuous on } [0, 1].$$

The substitution $x = 1 - s/n$ makes this explicit: the integral becomes $\int_0^n (1 - \frac{s}{n})^n f(1 - \frac{s}{n}) \, ds$, whose integrand is dominated by $e^{-s} \sup |f|$ and converges pointwise to $e^{-s} f(1)$, and $\int_0^\infty e^{-s} \, ds = 1$.

WORKED EXAMPLE

$$\lim_{n \rightarrow \infty} \left(n + \frac{1}{2}\right) \int_0^1 x^n e^x \, dx = e.$$

The extra $\frac{1}{2}$ changes nothing in the limit, since $\frac{n+1/2}{n} \rightarrow 1$, and it improves the rate: integrating by parts, $\int_0^1 x^n e^x \, dx = \frac{e}{n+1} - \frac{1}{n+1} \int_0^1 x^{n+1} e^x \, dx$, so $n \int \rightarrow e$ with an error of order $1/n$. The convergence is genuinely slow: quadrature gives 2.6982 at $n = 200$ and 2.71826 at $n = 2 \times 10^5$, against $e = 2.718282$.

A cleaner instance of the same statement: $\lim_{n \rightarrow \infty} n \int_0^1 \frac{x^n}{1+x} \, dx = \frac{1}{1+1} = \frac{1}{2}$, with 0.499996 at $n = 5 \times 10^4$.

WHERE THIS GOES WRONG

Passing the limit inside the integral, which gives 0 because $x^n \rightarrow 0$ pointwise on $[0, 1)$. The n prefactor is exactly what rescues a finite answer, and the correct interchange happens only *after* the substitution $x = 1 - s/n$ has made the window fixed. A second

error is assuming continuity of f only on $[0, 1)$; the value $f(1)$ is the answer, so f must be continuous at the endpoint where the mass sits.

SEE ALSO Wallis and Laplace concentration; convergence of an n th power to an indicator; mean-value forms and FTC-driven limits.

When the maximum of the weight is interior rather than at an endpoint, the window is wider (of order $n^{-1/2}$ rather than n^{-1}) and the limiting profile is Gaussian rather than exponential. This is Laplace's method, and its most familiar instance is the Wallis integral.

HARD

6.7 Wallis and Laplace concentration

CUE $\sin^n x$ or $\cos^n x$ integrated over an interval containing a peak of the weight, with a $\sqrt{n}$ normalisation outside; more generally $\int e^{-n\varphi(x)} g(x) dx$ with φ attaining an interior minimum.

METHOD The mass concentrates in a window of width $O(n^{-1/2})$ around the peak, and a quadratic expansion there gives the profile. For $\sin^n$ on $[0, \pi/2]$, put $x = \frac{\pi}{2} - t/\sqrt{n}$, so $\sin^n x = \cos^n(t/\sqrt{n}) = \exp(n \ln \cos \frac{t}{\sqrt{n}}) \rightarrow e^{-t^2/2}$, whence

$$\int_0^{\pi/2} \sin^n x dx \sim \frac{1}{\sqrt{n}} \int_0^\infty e^{-t^2/2} dt = \sqrt{\frac{\pi}{2n}}.$$

The general statement is $\int_a^b e^{-n\varphi(x)} g(x) dx \sim g(x_0) e^{-n\varphi(x_0)} \sqrt{\frac{2\pi}{n\varphi''(x_0)}}$ for an interior minimum x_0 with $\varphi''(x_0) > 0$.

WORKED EXAMPLE

$$\lim_{n \rightarrow \infty} \sqrt{n} \int_0^\pi \sin^n x dx.$$

On $[0, \pi]$ the weight $\sin^n$ peaks at the interior point $\pi/2$, and by symmetry $\int_0^\pi \sin^n = 2 \int_0^{\pi/2} \sin^n$. Using the asymptotic above,

$$\sqrt{n} \int_0^\pi \sin^n x dx \sim 2\sqrt{n} \cdot \sqrt{\frac{\pi}{2n}} = 2\sqrt{\frac{\pi}{2}} = \sqrt{2\pi} \approx 2.506628.$$

Quadrature at $n = 400$ gives 2.505062, the discrepancy being the expected $O(1/n)$ correction. The half-interval sibling is $\lim_{n \rightarrow \infty} \sqrt{n} \int_0^{\pi/2} \cos^n x dx = \sqrt{\pi/2} \approx 1.253314$, with 1.252531 at $n = 400$.

WHERE THIS GOES WRONG

Forgetting the factor 2 from $\int_0^\pi = 2 \int_0^{\pi/2}$, which silently halves the answer, and its mirror image, forgetting that a peak sitting at an *endpoint* contributes only half the Gaussian and therefore half the constant. Decide where the peak is before you write the constant, and count how much of the window lies inside the interval of integration.

SEE ALSO Endpoint concentration; convergence of an n th power to an indicator; Stirling's approximation.

The third form of concentration produces not a point mass but a set. When the n th power sits in a denominator, the integrand does not vanish everywhere: it tends to 1 where the base is below 1 and to 0 where the base is above, so the limit measures a region rather than sampling a point.

HARD

6.8 Convergence of an n th power to an indicator

CUE $\int \frac{dx}{1 + g(x)^n}$ or a similar expression in which an n th power appears in a denominator and g crosses the value 1 somewhere in the domain.

METHOD Pointwise, $\frac{1}{1 + g(x)^n} \rightarrow \mathbf{1}\{g(x) < 1\}$, with the value $\frac{1}{2}$ on the crossing set, which has measure zero when g is strictly monotone through the crossing. Since the integrand is bounded by 1 on a set of finite measure, dominated convergence applies and the limit is the *measure* of $\{x : g(x) < 1\}$.

WORKED EXAMPLE

$$\lim_{n \rightarrow \infty} \int_0^1 \frac{dx}{1 + (x \sec x)^n}.$$

On $[0, 1]$ we have $\cos x > 0$, since $1 < \pi/2$, and $x/\cos x$ is strictly increasing from 0 to $1/\cos 1 \approx 1.851$. So $x \sec x < 1$ exactly when $x < \cos x$, that is on $[0, x^*)$ where x^* is the unique root of $\cos x = x$, the Dottie number. The integrand is bounded by 1, which supplies the dominating function, so

$$\lim_{n \rightarrow \infty} \int_0^1 \frac{dx}{1 + (x \sec x)^n} = |\{x \in [0, 1] : x < \cos x\}| = x^* \approx 0.7390851.$$

Fixed-point iteration on $x \mapsto \cos x$ gives $x^* = 0.7390851332$, and quadrature at $n = 200$ gives 0.7390851.

WHERE THIS GOES WRONG

Assuming the pointwise limit may be integrated without a dominating function. Here $0 \leq \text{integrand} \leq 1$ on a set of finite measure supplies one, but that must be *said*, and on an unbounded domain it would not be enough. The second error is ignoring the crossing set: if $g \equiv 1$ on an interval, the limit picks up $\frac{1}{2}$ of that interval's length, and the answer is not the measure of $\{g < 1\}$ alone.

SEE ALSO Endpoint concentration; Wallis and Laplace concentration; dominated convergence.

Seen together, the eight techniques of this chapter are one idea applied at three scales. A Riemann sum is an integral read at the scale of the whole interval, where every part of the domain contributes and the mesh is the only thing that shrinks. A difference quotient is an integral's inverse read at the scale of a single point, where the interval itself shrinks. A concentration limit sits between them: the domain stays fixed but the weight shrinks the effective interval to a window, and the answer depends on how fast. When you meet a limit involving many terms or an integral with a parameter, the first question is which of the three scales the problem lives at, because that question decides the technique before any calculation begins.

Recognition matrix

If you see	Reach for	If that fails
$\frac{1}{n} \sum_{k=1}^n f(k/n)$	$\int_0^1 f$	Check for a stray n inside f
$\sum_{k=1}^n \frac{n}{n^2 + k^2}$, no visible $\frac{1}{n}$	Divide by n^2 to expose the mesh	Squeeze if the mesh will not appear
k ranging to cn , or from n to $2n$	Integral over $[0, c]$ or $[1, 2]$	Reindex $k = n + j$ and re-read
Sample points $a^{k/n}$	Geometric partition of $[1, a]$	Substitute $x = a^t$ for uniformity
$[\prod_{k=1}^n c_k]^{1/n}$	ln first, then Riemann sum	Distribute the outer $\frac{1}{n}$ into the root
$\frac{1}{n} (n!)^{1/n}$ or $[(n+1) \cdots (2n)]^{1/n}$	ln gives $\frac{1}{n} \sum \ln(k/n)$	Watch the improper endpoint at 0
Two summands, neither a clean $f(k/n)$	Common denominator, then substitute	Split into two separate integrals
$\frac{f(a+h) - f(a)}{h}$, however disguised	Quote $f'(a)$	Rewrite as $g(t)$ with $t = a + h$
$\frac{f(a + \alpha h) - f(a + \beta h)}{h}$	$(\alpha - \beta)f'(a)$	Bridge through $f(a)$
$n(a^{1/n} - 1)$	Derivative of a^t at $t = 0$: $\ln a$	Logarithm machine
$\frac{1}{h} \int_a^{a+h} f$	$f(a)$, by the FTC	Mean value theorem for integrals
Integral with variable bounds over x^p	L'Hôpital plus the Leibniz rule	Expand the integrand and integrate
$n \int_0^1 x^n f(x) dx$	$f(1)$; substitute $x = 1 - s/n$	Integrate by parts for the rate
$\sqrt{n} \int \sin^n$ or $\int e^{-nq} g$	Laplace: expand at the peak	Count which half of the peak is inside
$\int \frac{dx}{1 + g(x)^n}$	Measure of $\{g < 1\}$	Name the dominating function

Sequences, recursion, and discrete asymptotics

What this chapter is for. A recursively defined sequence $x_{n+1} = f(x_n)$ poses two entirely separate questions, and almost every wrong answer in this area comes from answering only the second. The first question is whether the sequence converges at all; the second is what it converges to. Existence is settled by an inequality argument (monotone and bounded, or a contraction estimate); the value is then settled by solving $L = f(L)$ and discarding the roots the sequence can never reach. The second half of the chapter turns to the asymptotic side: how to extract the growth rate of a sequence you cannot solve, using Stolz-Cesaro, averaging, n th-root and ratio limits, and Stirling's approximation. By the end you should be able to look at a recursion and name, within a few seconds, which existence argument it wants, and to read the size of a_n off its shape when no closed form is available.

Two questions, not one

Here is the error this chapter exists to kill. A student is given $x_1 = 1$ and $x_{n+1} = \sqrt{2 + x_n}$ and asked for $\lim x_n$. They write: let L be the limit, then $L = \sqrt{2 + L}$, so $L^2 - L - 2 = 0$, so $L = 2$. They box the answer and move on. The algebra is correct and the answer is correct, and the argument is worth almost nothing, because it never established that a limit exists. The identical manipulation applied to $x_1 = 1$, $x_{n+1} = 3x_n - 2$ yields $L = 3L - 2$, hence $L = 1$; but that sequence is constantly 1 only if it starts at 1, and from $x_1 = 2$ it runs away to $+\infty$ while the fixed-point equation sits there unchanged, still offering $L = 1$. The equation $L = f(L)$ is a *consequence* of convergence, not a cause of it.

So the discipline is fixed and it does not vary from problem to problem. Step one: prove that $\lim x_n$ exists, by an argument that never mentions its value. Step two: use continuity of f to pass to the limit in $x_{n+1} = f(x_n)$, obtaining $L = f(L)$, and then select among its roots using what step one told you about the range of the sequence. Step one is where the mathematics is. Step two is bookkeeping.

There is a practical corollary about how to write the answer. The sentence “let $L = \lim x_n$ ” is a claim, not a definition, and it should not appear on the page until it has been earned. Write the two inductions first, then the sentence, then the algebra. Examiners award the marks in that order because the mathematics is in that order, and a script that opens with “let L be the limit” has spent its first line asserting the thing it was asked to prove.

EXISTENCE BEFORE VALUE
 $L = f(L)$ is what a limit must satisfy if it exists. It is silent on whether it exists. A solution that solves the fixed-point equation and stops is not a partial solution; it is a solution to a different problem.

Why is step one enough? Because of completeness. A bounded increasing sequence has a supremum S , and for any $\varepsilon > 0$ some term exceeds $S - \varepsilon$; monotonicity then keeps every later term in $(S - \varepsilon, S]$, so $x_n \rightarrow S$. That is the whole proof, and it is worth carrying in your head, because it explains why the theorem gives you no information about the value: the supremum is produced by an axiom, not by a formula. The fixed-point equation is the only route to the number, and it is available only once the supremum is known to be a limit.

The bound you need is never a mystery either. Solve $L = f(L)$ first, informally, as reconnaissance; the roots tell you where the sequence is trying to go, and the relevant root is the bound to attempt in the induction. Doing the algebra first is fine. Presenting it first, as the argument, is not.

There are only two existence arguments in wide use, and they split by the shape of f . If f is increasing on the interval where the sequence lives, the sequence is automatically monotone, and you need only a bound: that is the monotone convergence theorem. If f is not increasing, or the bound is awkward, you instead show that f shrinks distances near the fixed point, which is the contraction argument. Everything else in the first half of this chapter is a refinement of one of those two.

7.1 Monotone convergence: proving that the limit exists

CUE A recursion $x_{n+1} = f(x_n)$ with no closed form, where f is increasing on the range of the sequence, and the problem asks for $\lim x_n$.

METHOD Prove two statements by induction, separately: (i) the sequence is bounded (guess the bound from the fixed point), and (ii) it is monotone. A bounded monotone real sequence converges; this is the completeness of $\mathbb{R}$ in its most usable form. Only after both inductions are complete may you write $L = f(L)$.

For monotonicity, the cleanest route when f is increasing is to check the first step only. If $x_2 > x_1$, then applying the increasing f to both sides gives $x_3 > x_2$, and induction carries it forever. The sign of $x_2 - x_1$ decides the direction of the whole sequence.

WORKED EXAMPLE

Let $x_1 = 1$ and $x_{n+1} = \sqrt{2 + x_n}$.

Bounded above by 2. True for x_1 . If $x_n < 2$ then $x_{n+1} = \sqrt{2 + x_n} < \sqrt{4} = 2$.

Increasing. Since all terms are positive, compare squares:

$$x_{n+1}^2 - x_n^2 = 2 + x_n - x_n^2 = (2 - x_n)(1 + x_n) > 0$$

because $0 < x_n < 2$. So $x_{n+1} > x_n$.

Bounded and increasing, so $L = \lim x_n$ exists and lies in $[1, 2]$. Now $f(t) = \sqrt{2 + t}$ is continuous, so $L = \sqrt{2 + L}$, giving $L^2 - L - 2 = (L - 2)(L + 1) = 0$. The root $L = -1$ is impossible since $L \geq x_1 = 1$. Hence $L = 2$.

The same recursion from above. Start instead at $x_1 = 3$. Now $x_{n+1}^2 - x_n^2 = (2 - x_n)(1 + x_n) < 0$, so the sequence decreases, and $x_n > 2$ is preserved: $x_{n+1} = \sqrt{2 + x_n} > \sqrt{4} = 2$. Decreasing and bounded below, so it converges, and the limit still satisfies $L = \sqrt{2 + L}$, giving $L = 2$ again. The terms are 2.2361, 2.0582, 2.0145, 2.0036, 2.0009. The lesson: f increasing means the sign of $x_2 - x_1$ sets the direction, and the fixed point is approached from whichever side you started on.

WHERE THIS GOES WRONG

Solving $L = f(L)$ with no existence argument. The fixed-point equation for $x_{n+1} = 3x_n - 2$ has the root $L = 1$ for every starting value, including the ones that diverge. A second, quieter failure: proving boundedness and forgetting monotonicity, which leaves open the oscillating case.

SEE ALSO Contraction and the derivative criterion; subsequence arguments; solving the fixed-point equation and selecting the admissible root.

Notice how little of that argument was computation. Two inductions, one continuity remark, one root rejected on sign: the entire content is inequalities. This is typical, and it is why the technique is tiered easy despite being the load-bearing result of the chapter. What makes it fail in practice is not difficulty but omission.

Once existence is in hand, the remaining work is to choose correctly among the roots of $L = f(L)$. That sounds trivial, and for a quadratic with one positive and one negative root it is. It stops being trivial when both roots are admissible-looking, which is exactly the situation the next technique is built for. Note also that a great many problems are *posed* in the self-similar form: nested radicals, periodic continued fractions and towers are all presented as static objects rather than as sequences, and the first move is always to name the sequence of truncations whose limit the object is defined to be. Until you have done that there is nothing to prove convergence of.

MEDIUM

7.2 Solve $L = f(L)$, then select the admissible root

CUE A recursion whose tail reproduces the whole expression: a periodic continued fraction, an infinite nested radical, an iterated exponential tower, a self-similar product. The defining feature is that the object contains a copy of itself.

METHOD Name the value L , write the self-similarity as $L = f(L)$, and solve. Then discard every root that the sequence's own range forbids, using the bound you proved in step one. Sign, monotonicity and the interval $[\inf x_n, \sup x_n]$ are the three selectors.

WORKED EXAMPLE

Evaluate the tower $t^{t^{\dots}}$ at $t = \sqrt{2}$, meaning $\lim x_n$ for $x_1 = \sqrt{2}$, $x_{n+1} = t^{x_n}$.

Existence. $f(u) = (\sqrt{2})^u$ is increasing. If $x_n \leq 2$ then $x_{n+1} \leq (\sqrt{2})^2 = 2$, so the sequence is bounded above by 2; and $x_2 = (\sqrt{2})^{\sqrt{2}} \approx 1.6325 > \sqrt{2} = x_1$, so it increases. The limit L exists and $L \in [\sqrt{2}, 2]$.

Value. $L = (\sqrt{2})^L$, that is $L^{1/L} = \sqrt{2}$. This equation has *two* positive roots: $L = 2$, since $2^{1/2} = \sqrt{2}$, and $L = 4$, since $4^{1/4} = \sqrt{2}$. The bound from step one kills $L = 4$ outright. Hence $L = 2$; numerically the iteration from $x_1 = \sqrt{2}$ reaches 2.0000000000000004 in double precision.

A periodic continued fraction. Evaluate $x = 1 + \frac{1}{2 + \frac{1}{2 + \dots}}$, that is the limit of the convergents of $[1; 2, 2, 2, \dots]$. Writing

y for the repeating tail $2 + 1/(2 + \dots)$ gives $y = 2 + 1/y$, so $y^2 - 2y - 1 = 0$ and $y = 1 + \sqrt{2}$ (the negative root is impossible since every convergent exceeds 2). Then $x = 1 + 1/y = 1 + (\sqrt{2} - 1) = \sqrt{2}$. The convergents are alternately above and below, so each consecutive pair brackets the answer, and the iteration confirms 1.41421356237.

WHERE THIS GOES WRONG

Reporting both roots, or the wrong one. The tower at $t = \sqrt{2}$ is the standard demonstration: 4 satisfies the equation and is not the limit, because the sequence never gets past 2. The fixed-point equation is necessary and not sufficient, and it is the sequence's range, established independently, that does the selecting.

SEE ALSO Monotone convergence; contraction and the derivative criterion; subsequence arguments.

Recursions that are secretly explicit

Before reaching for any existence theorem, spend ten seconds asking whether the recursion can be solved outright, because a closed form makes every question in this chapter trivial. Three substitutions account for most of the solvable cases.

Reciprocals linearise Moebius recursions. If $x_{n+1} = \frac{x_n}{1 + x_n}$ then $\frac{1}{x_{n+1}} = \frac{1}{x_n} + 1$, so $1/x_n$ is an arithmetic progression: from $x_1 = 1$ we get $1/x_n = n$ and $x_n = 1/n$ exactly. No induction, no fixed point, no rate estimate; the answer is 0 and the rate is $1/n$. More generally any $x_{n+1} = (ax_n + b)/(cx_n + d)$ becomes linear in $1/(x_n - \lambda)$ where λ is a fixed point.

Logarithms linearise power recursions. If $x_{n+1} = x_n^2$ then $\ln x_{n+1} = 2 \ln x_n$, so $x_n = x_1^{2^{n-1}}$: the sequence tends to 0 for $|x_1| < 1$, to 1 at $x_1 = 1$, and to ∞ beyond. The fixed-point equation $L = L^2$ offers 0 and 1 and cannot distinguish these three regimes on its own.

Trigonometric substitution linearises Chebyshev recursions. If $x_{n+1} = 2x_n^2 - 1$ and $|x_1| \leq 1$, write $x_1 = \cos \theta$; the double-angle formula gives $x_n = \cos(2^{n-1}\theta)$ exactly, as the values 0.7648, 0.1700, -0.9422, 0.7756 from $\theta = 0.7$ confirm. The sequence therefore has no limit for generic θ , and the reason is visible: the angle doubles forever. The fixed-point equation $L = 2L^2 - 1$ has the perfectly good roots 1 and $-\frac{1}{2}$, neither of which the sequence approaches.

That last example is the sharpest warning in the chapter. A recursion can be completely understood, have two real fixed points, and converge to neither.

Rates, and the local picture at a fixed point

The self-similarity move has one further use worth stating, because it is where the technique earns its hard tier: it applies to objects that are not obviously sequences at all. A doubly infinite product, an integral defined by a functional equation, a probability defined by conditioning on the first step, all reduce to $L = f(L)$ plus a range argument. The range argument is the part that gets skipped, and it is the part that is doing the work.

Knowing that $x_n \rightarrow L$ is often not the end of the question. Numerical work needs to know how many steps buy a decimal digit; qualitative work needs to know whether the approach is one-sided or oscillating, since a one-sided approach means every term is a bound on the answer. Both are read off the derivative of f at the fixed point, and both can be seen before any calculation by drawing the cobweb.

7.3 Reading a recursion off the cobweb

CUE You want the qualitative behaviour of $x_{n+1} = f(x_n)$ (monotone, alternating, escaping) before committing to an inductive proof, or the induction is not obviously going to close.

METHOD Draw $y = f(x)$ and $y = x$ on the same axes. Starting at x_1 on the horizontal axis, go vertically to the curve (this is x_2), horizontally to the diagonal (this transfers x_2 to the horizontal axis), and repeat. The path staircases or spirals; the fixed points are the crossings, and the slope of f at a crossing dictates the local picture. For $0 < f'(L) < 1$ the path is a monotone staircase inward, so convergence is one-sided; for $-1 < f'(L) < 0$ it spirals inward and the terms alternate about L ; for $f'(L) > 1$ the staircase runs outward and the sequence escapes; for $f'(L) < -1$ the spiral widens and the sequence oscillates without bound.

The structural consequence, which you should extract and use even without drawing anything: if f is increasing the sequence is monotone, and if f is decreasing the even and odd subsequences are each monotone, in opposite directions.

WORKED EXAMPLE

$x_1 = 1$, $x_{n+1} = \cos x_n$. Here $f = \cos$ is decreasing on $(0, 1)$, so the cobweb spirals and the terms alternate around the crossing L of $y = \cos x$ with $y = x$. Numerically $L = 0.7390851332$, and $\cos'(L) = -\sin L = -0.6736$. The signed errors $x_n - L$ do alternate, with successive ratios $-0.762, -0.596, -0.716, -0.641, -0.693, -0.660$, settling slowly on -0.6736 . Every consecutive pair of terms brackets the answer: $x_{2k} < L < x_{2k+1}$.

WHERE THIS GOES WRONG

Reading alternation as divergence. A spiral that tightens is convergence with a bonus: consecutive terms bracket the limit, which a monotone sequence never gives you. What signals divergence is a spiral that widens, that is $|f'(L)| > 1$.

SEE ALSO Contraction and the derivative criterion; subsequence arguments; monotone convergence.

Two structural facts fall straight out of the picture and deserve to be stated without it. First, if f is increasing the sequence is monotone, so the monotone convergence theorem is available and you need only a bound. Second, if f is decreasing then $f \circ f$ is increasing, so the even-indexed and odd-indexed subsequences are each monotone, in opposite directions, and the question becomes whether their two limits agree. Those two sentences dispose of the qualitative behaviour of every one-step recursion with monotone f , which is most of them.

The cobweb is a picture of an inequality, and the inequality is the mean value theorem. Making it quantitative gives the single most useful rate statement in the subject.

MEDIUM

7.4 Contraction and the $|f'| < 1$ criterion

CUE A smooth recursion $x_{n+1} = f(x_n)$ with a known or computable fixed point L , and a question about the rate, or a monotonicity induction that refuses to close.

METHOD Subtract the fixed-point equation and apply the mean value theorem: $x_{n+1} - L = f(x_n) - f(L) = f'(\xi_n)(x_n - L)$ for some ξ_n between x_n and L . If $|f'| \leq k < 1$ on an interval I containing L with $f(I) \subseteq I$, then

$$|x_n - L| \leq k^{n-1} |x_1 - L|,$$

so convergence is geometric with ratio $|f'(L)|$, and it happens for *every* starting point in I . If $|f'(L)| > 1$ the fixed point repels and no starting point other than L itself converges to it. If $f'(L) = 0$ the linear term vanishes and convergence is faster than any geometric rate.

WORKED EXAMPLE

Two rates from the same idea.

Geometric. For $x_{n+1} = \sqrt{2 + x_n}$ with $L = 2$, $f'(t) = \frac{1}{2\sqrt{2+t}}$ so $f'(2) = \frac{1}{4}$. The errors from $x_1 = 1$ are 0.2679, 0.06815, 0.01711, 0.004282, 0.001071, with successive ratios 0.2543, 0.2511, 0.2503, 0.2501. One digit per 1.66 steps, forever.

Quadratic. Newton's iteration for $\sqrt{k}$ is $x_{n+1} = \frac{1}{2}(x_n + \frac{k}{x_n})$, and $f'(t) = \frac{1}{2}(1 - \frac{k}{t^2})$ vanishes at $t = \sqrt{k}$. With $k = 2$ and $x_1 = 2$ the errors are 8.58×10^{-2} , 2.45×10^{-3} , 2.12×10^{-6} , 1.59×10^{-12} : each error is roughly the square of the previous one, which is what $f'(L) = 0$ buys.

Repelling. $x_{n+1} = x_n^2$ has fixed points 0 and 1, with $f'(0) = 0$ and $f'(1) = 2$. The first is superattracting and the second repels: from $x_1 = 0.99$ the sequence falls to 0, from $x_1 = 1.01$ it runs to ∞ , and only the exact value 1 stays. No inequality argument will ever produce 1 as a limit, and the fixed-point equation is powerless to say so.

Convert either statement into digits with logarithms. Geometric convergence with ratio k buys $-\log_{10} k$ decimal digits per step, so the ratio $\frac{1}{4}$ above gives 0.602 digits per step, or one digit per 1.66 steps, indefinitely. Quadratic convergence doubles the digit count each step, so the number of steps to reach any accuracy grows like $\log \log$ of it, which is why Newton's method is what calculators use.

WHERE THIS GOES WRONG

The borderline $|f'(L)| = 1$, where the criterion is silent and the sequence may converge, diverge, or crawl; the next technique handles it. A second failure is applying the estimate on an interval that is not forward invariant: $|f'(L)| < 1$ is a statement at a point, and without $f(I) \subseteq I$ nothing stops x_2 from landing outside I where the bound does not hold.

THE DERIVATIVE IS THE RATE

At an attracting fixed point the error is multiplied by $|f'(L)|$ each step, so $|f'(L)|$ is not a qualitative label but the actual convergence ratio. Its sign tells you the pattern (positive means one-sided, negative means alternating with brackets), its magnitude tells you the speed, and its vanishing tells you the speed is not geometric at all but quadratic.

SEE ALSO Sublinear decay when the derivative equals one; reading a recursion off the cobweb; monotone convergence.

The borderline case is not rare. It is exactly what happens whenever $f(x) = x - cx^p + \dots$, which is the generic form of a recursion whose fixed point is at 0 and whose first-order behaviour is the identity. Every such sequence converges, and none of them converges geometrically. The rate lives one order beyond the derivative, and there is a standard trick for extracting it.

HARD

7.5 Sublinear decay: the rate when $f'(L) = 1$

CUE $x_{n+1} = x_n - cx_n^p + o(x_n^p)$ with $c > 0$ and $p > 1$, after shifting the fixed point to 0. The linearisation is the identity and gives no rate.

METHOD Do not study x_n ; study the reciprocal power $y_n = x_n^{1-p}$, which grows linearly. Expanding,

$$x_{n+1}^{1-p} = x_n^{1-p} (1 - cx_n^{p-1} + \dots)^{1-p} = x_n^{1-p} + c(p-1)x_n^{1-p} + o(1),$$

so the increments of y_n tend to $c(p-1)$. By Stolz-Cesaro, $y_n/n \rightarrow c(p-1)$, hence

$$x_n \sim (c(p-1)n)^{-1/(p-1)}.$$

The decay is a power of n , never a power of a constant.

WORKED EXAMPLE

$x_1 = 1, x_{n+1} = \sin x_n$. Since $\sin t = t - \frac{t^3}{6} + O(t^5)$ we have $c = \frac{1}{6}$ and $p = 3$, so

$$x_n \sim \left(\frac{1}{6} \cdot 2 \cdot n\right)^{-1/2} = \sqrt{\frac{3}{n}}.$$

The check: at $n = 10^6$, $x_n \sqrt{n/3} = 0.9999946$.

The same machinery on $x_{n+1} = x_n - x_n^2$ gives $c = 1, p = 2$, hence $x_n \sim 1/n$, which you can also see directly from $1/x_{n+1} - 1/x_n \rightarrow 1$.

And on $x_{n+1} = \ln(1 + x_n)$, from $\ln(1 + t) = t - \frac{t^2}{2} + O(t^3)$ we read $c = \frac{1}{2}, p = 2$, so $x_n \sim \frac{2}{n}$. The check at $n = 10^6$: $nx_n/2 = 1.0000023$. Three different recursions, three different constants, one method.

WHERE THIS GOES WRONG

Expanding only to first order, seeing $x_{n+1} \approx x_n$, and concluding either that the sequence fails to converge or that it converges geometrically. Both are wrong: it converges, and it converges like $n^{-1/(p-1)}$, which is far slower than any geometric rate. Halving the error in $x_{n+1} = \sin x_n$ costs a factor of four in n .

SEE ALSO Contraction and the derivative criterion; Stolz-Cesaro; Cesaro means and the averaging principle.

When the sequence does not converge, and when it is linear

Everything so far has assumed the answer is yes. Some problems are posed precisely because the answer is no, and the standard demand then is a proof of non-existence, which is a different kind of object: you must produce evidence, not an

estimate. An estimate can only ever say that the terms are somewhere; to deny a limit you must produce two places they keep returning to.

The right general framework is $\limsup$ and $\liminf$, which exist for every bounded sequence and are the largest and smallest subsequential limits. They convert the qualitative question into an equation: $\lim x_n$ exists exactly when $\liminf x_n = \limsup x_n$. In practice you rarely compute them; you exhibit two subsequences with different limits, which is the same thing done concretely.

MEDIUM

7.6 Subsequence arguments and $\limsup$, $\liminf$

CUE You suspect a sequence does *not* converge; or the recursion alternates, so the monotone argument is unavailable directly.

METHOD To disprove convergence, exhibit two subsequences with different limits; that is the entire proof, since every subsequence of a convergent sequence inherits the limit. To handle an alternating recursion, note that if f is decreasing then $f \circ f$ is increasing, so the even terms $x_2, x_4, \dots$ and the odd terms $x_1, x_3, \dots$ are each monotone under $g = f \circ f$; find both limits and the original sequence converges if and only if they agree. Always available as a fallback: $\liminf x_n \leq \limsup x_n$, with equality exactly when the limit exists.

WORKED EXAMPLE

Disproof. $a_n = \cos(n\pi/3)$ has $a_{6k} = 1$ and $a_{6k+3} = -1$. Two subsequences, two limits, no limit.

Alternating recursion, convergent. $x_1 = 0, x_{n+1} = \frac{6}{1+x_n}$. Here f is decreasing, so the terms alternate about the positive fixed point: $L(1+L) = 6$ gives $L = 2$ or $L = -3$, and positivity selects $L = 2$. Since $f'(2) = -\frac{6}{(1+2)^2} = -\frac{2}{3}$ and $|\frac{2}{3}| < 1$, the even and odd subsequences both converge to 2, so $x_n \rightarrow 2$.

Alternating recursion, divergent. $x_{n+1} = 3 - 2x_n$ has the same fixed point $L = 1$, but $f' = -2$, so from any $x_1 \neq 1$ the even and odd subsequences run to $+\infty$ and $-\infty$. The fixed-point equation is identical in both examples; only the derivative separates them.

The $\limsup$ and $\liminf$ are worth keeping in reserve for sequences too irregular to describe by subsequences you can write down. For $a_n = \sin n$ the values are dense in $[-1, 1]$, because the multiples of 1 modulo 2π are equidistributed, so $\liminf a_n = -1$ and $\limsup a_n = 1$; there is no single subsequence one would naturally exhibit, and the two-number summary is the entire available description.

WHERE THIS GOES WRONG

Showing that one subsequence converges and calling the sequence convergent. The even terms of $(-1)^n$ converge to 1 and prove nothing. Convergence of a single subsequence is evidence of nothing; convergence of the even and the odd subsequences to the same value is a proof.

SEE ALSO Monotone convergence; reading a recursion off the cobweb; contraction and the derivative criterion.

One family escapes all of this machinery, because it can be solved outright. When f is linear and the recursion reaches back more than one step, the closed form is available and the limit question becomes a question about the size of two numbers.

MEDIUM

7.7 Linear recurrences by characteristic root

CUE $a_{n+1} = pa_n + qa_{n-1}$ with constant p, q , or a sequence built explicitly from Fibonacci or Lucas numbers.

METHOD Solve the characteristic equation $t^2 = pt + q$. For distinct roots λ_1, λ_2 the general solution is $a_n = A\lambda_1^n + B\lambda_2^n$, with A, B fixed by two initial values; for a repeated root λ it is $a_n = (A + Bn)\lambda^n$. If $|\lambda_1| > |\lambda_2|$ the dominant term wins, so $a_{n+1}/a_n \rightarrow \lambda_1$ and $a_n^{1/n} \rightarrow |\lambda_1|$.

WORKED EXAMPLE

The Fibonacci recursion $F_{n+1} = F_n + F_{n-1}$ has $t^2 = t + 1$, roots $\varphi = \frac{1+\sqrt{5}}{2}$ and $\bar{\varphi} = \frac{1-\sqrt{5}}{2}$, with $|\bar{\varphi}| = 0.618 < \varphi$. Hence $F_n/F_{n-1} \rightarrow \varphi$. Then

$$\lim_{n \rightarrow \infty} \left(\frac{F_n}{F_{n-1}} - \frac{F_{n-1}}{F_n} \right) = \varphi - \frac{1}{\varphi} = \varphi - (\varphi - 1) = 1,$$

using the defining relation $\varphi^2 = \varphi + 1$ in the form $1/\varphi = \varphi - 1$. At $n = 60$ the double-precision value of the bracket is 1.0 to machine accuracy.

Repeated root: $a_{n+1} = 4a_n - 4a_{n-1}$ has $t^2 - 4t + 4 = 0$, so $\lambda = 2$ twice and $a_n = (A + Bn)2^n$. With $a_0 = 0, a_1 = 2$ this is $a_n = n2^n$, and indeed $a_2 = 4 \cdot 2 - 4 \cdot 0 = 8 = 2 \cdot 2^2$.

The dominant-root statement is worth unpacking, since it is the only place in the chapter where a limit is obtained by a cancellation rather than by an estimate. Writing $a_n = A\lambda_1^n + B\lambda_2^n$ with $|\lambda_1| > |\lambda_2|$ and $A \neq 0$,

$$\frac{a_{n+1}}{a_n} = \lambda_1 \cdot \frac{1 + (B/A)(\lambda_2/\lambda_1)^{n+1}}{1 + (B/A)(\lambda_2/\lambda_1)^n} \rightarrow \lambda_1,$$

because $(\lambda_2/\lambda_1)^n \rightarrow 0$. The convergence is geometric with ratio $|\lambda_2/\lambda_1|$, which for Fibonacci is $0.618/1.618 = 0.382$, about 0.42 decimal digits per step. The proviso $A \neq 0$ matters: the Lucas-type sequence built to have $A = 0$ is a pure multiple of λ_2^n and its ratio tends to λ_2 instead.

One extension is worth knowing because it appears constantly in disguise. If the recursion carries a constant, $a_{n+1} = pa_n + q$ with $p \neq 1$, subtract the fixed point $c = q/(1-p)$: the shifted sequence $a_n - c$ satisfies the homogeneous recursion $b_{n+1} = pb_n$, so $a_n = c + (a_1 - c)p^{n-1}$. This converges exactly when $|p| < 1$, and to c , which is the contraction criterion for a linear f with $f' = p$. From $a_1 = 1, a_{n+1} = 2a_n + 1$ we get $c = -1$ and $a_n = 2^n - 1$, a divergent case where the fixed-point equation nonetheless has the perfectly respectable root -1 .

WHERE THIS GOES WRONG

Writing $A\lambda^n + B\lambda^n$ for a repeated root. That is a one-parameter family wearing two letters, and it cannot match two independent initial conditions. The correct second solution carries the factor n .

SEE ALSO Solving the fixed-point equation and selecting the admissible root; root and ratio limits for sequences; contraction and the derivative criterion.

One recursion, all four questions

It is worth seeing the whole apparatus applied once, in order, to a single problem, because in an examination all four questions are usually asked at once and the marks are allocated to the steps rather than to the answer.

Problem. $x_1 = 0, x_{n+1} = \sqrt{6 + x_n}$. Show the sequence converges, find the limit, give the rate, and say whether the terms over- or under-estimate the limit.

Existence. $f(t) = \sqrt{6 + t}$ is increasing on $[0, \infty)$. Bound: if $0 \leq x_n < 3$ then $x_{n+1} = \sqrt{6 + x_n} < \sqrt{9} = 3$, and $x_1 = 0 < 3$, so $x_n < 3$ for all n . Monotone: $x_2 = \sqrt{6} = 2.449 > 0 = x_1$, and f increasing propagates this, so the sequence increases. Bounded and increasing, hence convergent.

Value. Passing to the limit in the recursion, $L = \sqrt{6 + L}$, so $L^2 - L - 6 = (L - 3)(L + 2) = 0$. Since $L \geq x_1 = 0$ we take $L = 3$.

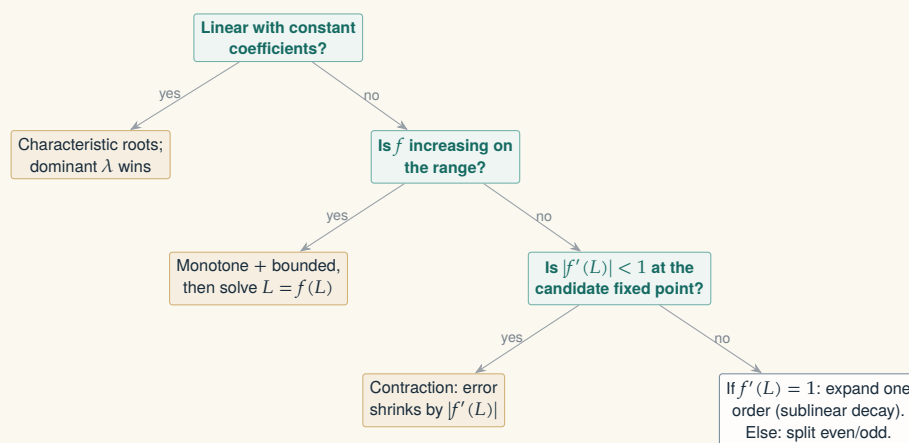
Rate. $f'(t) = \frac{1}{2\sqrt{6 + t}}$, so $f'(3) = \frac{1}{6}$. The errors are 0.5505, 0.09320, 0.015574, 0.0025967, 0.00043282, with successive ratios 0.1835, 0.1693, 0.1671, 0.16674, 0.166679, settling on $\frac{1}{6}$ as promised. That is $\log_{10} 6 = 0.778$ decimal digits per step.

Pattern. $f'(3) > 0$, so the approach is one-sided, from below. Every term is therefore a rigorous lower bound for 3, and no term is ever an upper bound. Had $f'(3)$ been negative, consecutive terms would have bracketed the limit instead, which is strictly more useful information.

Four questions, four different tools, and the only one that touches the fixed-point equation is the second. That proportion is the point of the chapter.

Choosing the convergence argument

The five recursion techniques above are not interchangeable, and the choice is forced by the shape of f , not by taste. The tree below is the order in which to test.



The tree is ordered by cost, not by power. Testing whether the recursion is linear costs one glance; testing whether f is increasing costs one derivative; computing $f'(L)$ costs finding L first, which is why it sits third even though it is the sharpest of the three. If two branches apply, take the earlier one: a linear recursion should never be attacked with a contraction estimate, and an increasing f with an obvious bound should never be attacked with the mean value theorem.

A caution about the third question. Writing down $|f'(L)|$ presumes you already know L , which presumes the fixed-point equation has been solved and the right root chosen. That is legitimate here in a way it was not in the opening section, because the derivative test is a genuine existence proof: if $|f'| \leq k < 1$ on a forward-invariant interval containing L , convergence follows for every start in that interval, and L is then the limit rather than a candidate for it. The order of operations is still solve, then verify, then conclude, but here the verification does the work that the induction did before.

Two remarks on using it honestly. First, the leftmost yes branch at the second question still owes you a bound; “ f is increasing” delivers monotonicity for free but nothing else. Second, the tree tells you which existence argument to attempt, and the value of the limit is in every case obtained afterwards from $L = f(L)$. There is no branch on which the fixed-point equation is the argument.

Discrete asymptotics

The rest of the chapter changes the question. Instead of a recursion whose limit is a number, we are given an explicit but unwieldy a_n (a partial sum, a factorial, an n th root) and asked for its size. The tools here are transfer principles: each one converts a limit you cannot compute into one you can, at the cost of a hypothesis that must be checked.

Before the tools, the hierarchy. Every asymptotic argument in this section is ultimately a statement about where a quantity sits in the chain

$$1 \ll \ln \ln n \ll \ln n \ll n^\alpha \ll c^n \ll n! \ll n^n \quad (\alpha > 0, c > 1),$$

where $u_n \ll v_n$ means $u_n/v_n \rightarrow 0$. Each step is strict and each gap is enormous: $\ln n$ passes 100 only past $n = 10^{43}$, while $n!$ passes 10^{100} before $n = 70$. When a limit has the form u_n/v_n and the two sides sit at different links of the chain, the answer is 0 or ∞ and no technique is needed. The techniques below are for the cases where they sit at the same link and the constant, or the polynomial correction, is what is being asked for.

One transfer that belongs here but is developed elsewhere in this book deserves a signpost, because it is the first thing to try on a limit of the shape $\frac{1}{n} \sum_{k \leq n} g(k/n)$: that is a Riemann sum, and its limit is $\int_0^1 g$. The signature is a sum whose summand depends on k only through the ratio k/n . When the summand instead depends on k and n separately, the Riemann reading fails and the techniques below are what remain.

The first is the discrete L'Hopital, and it earns its keep on any ratio whose numerator is a sum. Differences are to sequences what derivatives are to functions, and a sum has a trivially simple difference.

7.8 Stolz-Cesaro

CUE A ratio a_n/b_n where b_n is strictly monotone and unbounded: the discrete ∞/∞ . Almost always a_n is a partial sum, or something whose consecutive difference is much simpler than itself.

METHOD If b_n is strictly increasing and unbounded and

$$\lim_{n \rightarrow \infty} \frac{a_{n+1} - a_n}{b_{n+1} - b_n} = L,$$

then $\lim a_n/b_n = L$ too. This is the exact sequence analogue of L'Hopital's rule and, like it, is a one-way implication: the difference ratio may fail to have a limit while the original ratio converges perfectly well.

WORKED EXAMPLE

$$\lim_{n \rightarrow \infty} \frac{1^p + 2^p + \cdots + n^p}{n^{p+1}} \quad (p > 0).$$

Take a_n the sum and $b_n = n^{p+1}$, which is strictly increasing and unbounded. Then

$$\frac{a_{n+1} - a_n}{b_{n+1} - b_n} = \frac{(n+1)^p}{(n+1)^{p+1} - n^{p+1}}.$$

The denominator is $(p+1)n^p + O(n^{p-1})$ by the binomial theorem, so the ratio tends to $\frac{1}{p+1}$, and that is the answer.

At $p = 3$ and $n = 10^4$ the partial value is 0.2500500 against $\frac{1}{4}$.

A second, more instructive instance: $a_n = H_n = \sum_{k \leq n} 1/k$ and $b_n = \ln n$. The difference ratio is $\frac{1/(n+1)}{\ln(1+1/n)} \rightarrow 1$, so $H_n/\ln n \rightarrow 1$. Note how slowly: at $n = 10^7$ the ratio is still 1.0358, because $H_n = \ln n + \gamma + o(1)$ and the correction only dies like $\gamma/\ln n$. Stolz-Cesaro gives the limit and says nothing about the rate.

There is a second, quieter use of the same theorem. Because it turns a sum into its summand, it is the standard way to convert a statement about increments into a statement about size: if $a_{n+1} - a_n \rightarrow c$ then $a_n/n \rightarrow c$, so $a_n \sim cn$. That is the step used to extract the rate in the sublinear decay technique, applied there to $y_n = x_n^{1-p}$, and it is the reason that technique works at all.

One more instance, to show the range. Let $a_n = \ln(n!)$ and $b_n = n \ln n$. The difference ratio is

$$\frac{\ln(n+1)}{(n+1)\ln(n+1) - n \ln n},$$

whose denominator is $\ln n + 1 + o(1)$, so the ratio tends to 1 and $\ln(n!) \sim n \ln n$. That is the crude half of Stirling's approximation, obtained with no analysis at all beyond a difference and a limit. What Stolz-Cesaro cannot give you is the next term, the $-n$; for that you need the real thing.

WHERE THIS GOES WRONG

Applying it when b_n is not monotone; the hypothesis is not decoration. And, exactly as with L'Hopital, concluding that the original ratio has no limit because the difference ratio has none. Take $a_n = (-1)^n$, $b_n = n$: the difference ratio oscillates between ± 2 and has no limit, while $a_n/b_n \rightarrow 0$.

SEE ALSO Cesaro means and the averaging principle; sublinear decay when the derivative equals one; root and ratio limits for sequences.

It is worth being explicit about the direction of the implication, because it is the same asymmetry as L'Hopital and it is misused the same way. Stolz-Cesaro says: if the difference ratio converges, so does the original. It never says the reverse. In practice this means Stolz-Cesaro can only ever produce an answer; it can never justify the sentence "the limit does not exist". For that you need the subsequence machinery.

The special case $b_n = n$ deserves its own name, because it says something structural: averaging is a smoothing operation. It preserves limits and it can manufacture them out of oscillation, which is why the implication runs one way only.

7.9 Cesaro means and the averaging principle

CUE An arithmetic mean $\frac{1}{n} \sum_{k \leq n} a_k$, or a geometric mean $(a_1 a_2 \cdots a_n)^{1/n}$, or any expression that is secretly one of these.

METHOD If $a_n \rightarrow L$ then the arithmetic means converge to L ; if in addition $a_n > 0$ then the geometric means converge to L as well. (The geometric statement is the arithmetic one applied to $\ln a_n$.) The averaging is a contraction on oscillation: it can only destroy irregularity, never create it.

WORKED EXAMPLE

$$\lim_{n \rightarrow \infty} \frac{n^2}{H_n \sum_{k=1}^n \frac{k}{H_k}}.$$

Let $S_n = \sum_{k \leq n} k/H_k$. Apply Stolz-Cesaro with $b_n = n^2/H_n$, which is strictly increasing and unbounded. Since $H_{n+1} - H_n = 1/(n+1)$ is negligible against H_n , the increment of b_n is $(2n+1)/H_{n+1} + o(n/H_n)$, while the increment of S_n is $(n+1)/H_{n+1}$. The ratio tends to $\frac{1}{2}$, so $S_n \sim \frac{n^2}{2H_n}$ and the required limit is

$$\frac{n^2}{H_n \cdot n^2/(2H_n)} = 2.$$

The numerical trend is 1.891, 1.913, 1.928, 1.938 at $n = 10^4, 10^5, 10^6, 10^7$: monotone increasing toward 2 with an $O(1/\ln n)$ correction, which is precisely what an H_n in the denominator predicts.

The geometric version is the one that does real work, because so many discrete limits are products in disguise. A second instance: to evaluate

$$\lim_{n \rightarrow \infty} \left(\frac{(2n)!}{n! n^n} \right)^{1/n},$$

set $a_n = (2n)!/(n! n^n)$ and compute the ratio, $\frac{a_{n+1}}{a_n} = \frac{2(2n+1)}{n+1} \left(\frac{n}{n+1} \right)^n \rightarrow \frac{4}{e}$. The geometric-mean principle then delivers the root limit $4/e = 1.47152$, confirmed numerically as 1.47152 at $n = 2 \times 10^5$. Doing this directly, by expanding both factorials, is a much longer computation for the same answer.

WHERE THIS GOES WRONG

Reading the converse. If the Cesaro means converge, the sequence need not: $a_n = (-1)^n$ has means tending to 0 and no limit at all. Averaging is where information goes to die, so you may push limits through it but never pull them back.

SEE ALSO Stolz-Cesaro; root and ratio limits for sequences; Stirling's approximation.

The geometric-mean half of the last technique has a consequence that is used far more often than the technique itself: it converts a ratio limit into a root limit. Since ratios of consecutive terms are usually easy and n th roots are usually not, the transfer runs in the useful direction.

7.10 Root and ratio limits for sequences

CUE An n th root of a rapidly growing quantity (a factorial, a product, a power), or a ratio of consecutive terms that simplifies.

METHOD If $a_n > 0$ and $\lim a_{n+1}/a_n = L$, then $\lim \sqrt[n]{a_n} = L$. The proof is the geometric-mean principle applied to the telescoped product $a_n = a_1 \prod_{k < n} (a_{k+1}/a_k)$. The converse is false, so the root limit is strictly the stronger statement: it exists in cases where the ratio limit does not.

WORKED EXAMPLE

$$\lim_{n \rightarrow \infty} \frac{n}{\sqrt[n]{n!}}.$$

Set $a_n = n^n/n!$, so that $n/\sqrt[n]{n!} = \sqrt[n]{a_n}$. The ratio is

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)^{n+1}}{(n+1)!} \cdot \frac{n!}{n^n} = \frac{(n+1)^n}{n^n} = \left(1 + \frac{1}{n}\right)^n \rightarrow e.$$

Hence the root limit is e as well. At $n = 2 \times 10^5$ the value is 2.71819 against $e = 2.718282$.

The transfer also explains a fact used constantly without comment: $\sqrt[n]{n} \rightarrow 1$, $\sqrt[n]{n!} \rightarrow \infty$ like n/e , and $\sqrt[n]{c} \rightarrow 1$ for any fixed $c > 0$. Each is a ratio computation of one line. And it is the sequence statement underneath the root and ratio tests for series, where exactly the same asymmetry appears: the root test decides every series the ratio test decides, and some that it does not.

A second instance, worth doing because the direct approach is genuinely unpleasant: $\lim \sqrt[n]{\binom{2n}{n}}$. The ratio is

$$\frac{\binom{2n+2}{n+1}}{\binom{2n}{n}} = \frac{(2n+1)(2n+2)}{(n+1)^2} = \frac{2(2n+1)}{n+1} \rightarrow 4,$$

so the root limit is 4. Numerically $\sqrt[n]{\binom{2n}{n}} = 3.998$ at $n = 10^4$, approaching 4 from below at the leisurely rate $4(\pi n)^{-1/(2n)}$, which is what the next technique will make precise.

WHERE THIS GOES WRONG

Concluding that the root limit fails because the ratio limit does. Take $a_n = 2^{n+(-1)^n}$: the ratio $a_{n+1}/a_n = 2^{1-2(-1)^n}$ alternates between 8 and $\frac{1}{2}$ and has no limit, yet $\sqrt[n]{a_n} = 2^{1+(-1)^n/n} \rightarrow 2$ without difficulty. The ratio test is a sufficient condition, not a diagnostic.

SEE ALSO Cesaro means and the averaging principle; Stirling's approximation; linear recurrences by characteristic root.

The previous technique gets the exponential scale of $n!$ right and throws away everything else. When the normalisation is finer, you need the full asymptotic, including the constant, and that is Stirling.

7.11 Stirling's approximation

CUE A factorial under a root, inside a logarithm, or divided by a power of n . Also central binomial coefficients, which are factorials in disguise.

METHOD Use

$$n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n, \quad \text{equivalently} \quad \ln(n!) = n \ln n - n + \frac{1}{2} \ln(2\pi n) + o(1).$$

Take logarithms *first*, substitute, keep only the terms that survive the normalisation you are given, then exponentiate. Deciding which terms survive is the entire skill.

WORKED EXAMPLE

Three normalisations of the same asymptotic.

The constant dies. $\frac{\sqrt[n]{n!}}{n}$: taking logs, $\frac{1}{n} \ln(n!) - \ln n = -1 + \frac{1}{2n} \ln(2\pi n) + o(1/n) \rightarrow -1$, so the limit is $e^{-1} = 0.36788$.

Verified: 0.36789 at $n = 2 \times 10^5$. The factor $\sqrt{2\pi n}$ contributes $(2\pi n)^{1/(2n)} \rightarrow 1$ and vanishes.

The constant is the whole answer. $\frac{n! e^n}{n^n \sqrt{n}} \rightarrow \sqrt{2\pi} = 2.50663$. Here the exponential parts cancel exactly by construction and only the constant is left. At $n = 1000$ the value is 2.50684.

A mixed case. $\binom{2n}{n} = \frac{(2n)!}{(n!)^2} \sim \frac{\sqrt{4\pi n} (2n/e)^{2n}}{2\pi n (n/e)^{2n}} = \frac{4^n}{\sqrt{\pi n}}$: the constant survives but in the combination $\sqrt{\pi}$, not $\sqrt{2\pi}$. At $n = 1000$ the ratio of the two sides is 0.999875.

A fourth normalisation, which appears whenever double factorials do: $(2n)!! = 2^n n!$ and $(2n-1)!! = \frac{(2n)!}{2^n n!}$, so

$$\frac{(2n-1)!!}{(2n)!!} = \frac{(2n)!}{4^n (n!)^2} = \frac{1}{4^n} \binom{2n}{n} \sim \frac{1}{\sqrt{\pi n}}.$$

At $n = 10^4$ the ratio of the two sides is 0.9999875. This single asymptotic decides an entire family of series, and it is the reason those series sit exactly at the ratio test's blind spot.

The log form is also the practical form. The number of decimal digits of $n!$ is $\lfloor \log_{10}(n!) \rfloor + 1$, and Stirling gives $\log_{10}(100!) \approx (100 \ln 100 - 100 + \frac{1}{2} \ln(200\pi)) / \ln 10 = 157.9696$ against the exact 157.9700, so 100! has 158 digits. Four correct significant figures from a formula with one term of error.

If more accuracy is wanted, the next term of the asymptotic series is available and is worth memorising because it is so cheap:

$$n! = \sqrt{2\pi n} \left(\frac{n}{e}\right)^n \left(1 + \frac{1}{12n} + O(n^{-2})\right).$$

At $n = 10$ the uncorrected formula gives 3598696 against the exact 3628800, an error of 0.83 per cent; the single correction factor $1 + \frac{1}{120}$ brings it to 3628685, an error of 0.003 per cent. Even at $n = 5$ the corrected value is 119.99 against 120. Stirling is not merely asymptotic; it is accurate immediately.

WHERE THIS GOES WRONG

Keeping $\sqrt{2\pi n}$ when the normalisation kills it, or dropping it when it survives. The rule is mechanical once you take logarithms: the factor contributes $\frac{1}{2} \ln(2\pi n)$, and you ask whether that term is o of the normalisation. In $\sqrt[n]{n!}/n$ it is divided by n and dies; in $n!e^n/(n^n \sqrt{n})$ it is all that remains.

SEE ALSO Root and ratio limits for sequences; Cesaro means and the averaging principle; Stolz-Cesaro.

SUMS OR PRODUCTS

Stolz-Cesaro and the Cesaro means handle sums, because differencing a sum is trivial. Root and ratio limits handle products, because a logarithm turns a product into a sum. Stirling handles the one product that occurs more often than all the others together. When a discrete limit resists, the productive question is not "which clever manipulation" but "is the object underneath a sum or a product", because that single distinction selects the transfer principle.

One last connection back to the first half of the chapter. The asymptotic techniques and the recursion techniques are not two subjects. The sublinear decay technique is Stolz-Cesaro applied to a reciprocal power; the contraction estimate is the statement that a certain sequence of errors is geometric, which is a statement about where it sits in the growth hierarchy; the linear recurrence technique is the observation that a sum of two exponentials is asymptotically the larger one. In every case the recursion supplies increments and the asymptotic machinery converts increments into size. If you find yourself with a recursion you cannot solve and a rate you need, difference something and apply Stolz-Cesaro; it is the move that works more often than any other in this chapter.

Read the matrix by shape, not by topic. The left column lists what is visible on the page before any work has been done: the form of f , the form of a_n , the presence of a factorial or a partial sum. The middle column is the technique that shape selects, and the right column is what to do when the hypothesis of that technique fails to check out, which is usually the same technique's nearest neighbour rather than a fresh idea.

Recognition matrix

If you see	Reach for	If that fails
$x_{n+1} = f(x_n)$, f increasing	Monotone + bounded induction, then $L = f(L)$	Contraction estimate on an invariant interval
$x_{n+1} = f(x_n)$, f decreasing	Even and odd subsequences under $f \circ f$	Cobweb, then $ f'(L) $
Nested radical, continued fraction, tower	$L = f(L)$, then discard roots outside the range	Monotone convergence for the bound
Two roots both satisfying $L = f(L)$	The proved bound on x_n selects	Sign and monotonicity of the sequence

If you see	Reach for	If that fails
$x_{n+1} = x_n - cx_n^p, p > 1$	Reciprocal power: $x_n^{1-p} \sim c(p-1)n$	Stolz-Cesaro on x_n^{1-p}
$a_{n+1} = pa_n + qa_{n-1}$	Characteristic roots; dominant λ	Repeated root gives $(A + Bn)\lambda^n$
Partial sum over a power of n	Stolz-Cesaro	Recognise it as a Riemann sum
$\frac{1}{n} \sum a_k$ or $(\prod a_k)^{1/n}$	Cesaro means	Stolz-Cesaro directly
$\sqrt[n]{a_n}$ with a_n a product	Ratio-to-root transfer	Logarithms plus Cesaro means
$n!$ under a root, log, or power	Stirling, in log form	Ratio-to-root transfer if the constant dies
Suspected divergence	Two subsequences, two limits	$\liminf < \limsup$

Series: deciding whether it converges

What this chapter is for. Deciding whether $\sum a_n$ converges is not a creative act. There are seven or eight tests, each with a hypothesis that fits one recognisable shape of general term, and the whole skill is matching the term to the test before writing anything down. A student who reaches for the ratio test on $\sum (n^2 + 3)/(n^4 - n + 1)$ has not made a small error: the ratio there tends to 1 for structural reasons, the test is inconclusive by construction, and no amount of algebra afterwards rescues the attempt. This chapter gives an explicit selection protocol, works each test in the situation it was designed for, and then treats the two things students most often get wrong once they can run the tests: that $a_n \rightarrow 0$ proves nothing, and that conditional convergence is a genuinely weaker property than absolute convergence, with consequences.

Test selection is the whole skill

The tests are not equally powerful, and they are not in competition. Each one is a machine that consumes a particular structural feature: the ratio test consumes factorials and n th powers because those are the things whose consecutive ratio simplifies; limit comparison consumes rational and algebraic terms because those have a clean dominant power; the integral test consumes terms that are values of a function you can antidifferentiate. Applied to the shape it was built for, each test is two lines. Applied to any other shape, it either fails to conclude or turns into an unbounded computation.

It is worth being precise about what “fails” means, because the three failure modes are different and call for different repairs. A test can be *inapplicable*, when its hypothesis is false: the integral test on a term that is not decreasing, the alternating test on magnitudes that do not decrease monotonically. A test can be *inconclusive*, when its hypothesis holds but its verdict is the neutral value: ratio or root giving $L = 1$, limit comparison giving $c = 0$ or $c = \infty$. And a test can be *unusable*, when it applies and would conclude but the computation does not close: the integral test on f with no elementary antiderivative. Inapplicable means you have made an error; inconclusive means you must switch tests; unusable means you must switch tests for a different reason. Only the first is your fault.

So the failure mode in this chapter is not arithmetic. It is selection. Almost every lost mark comes from picking a test whose hypothesis does not fit, grinding through it correctly, and arriving at $L = 1$ or an incomputable integral. The remedy is to spend the first ten seconds looking at the *shape* of a_n and nothing else.

There is a second reason to take selection seriously, beyond economy. Several of the tests are not merely inefficient outside their domain; they are silent there in a way that is easy to mistake for evidence. A ratio limit of 1 feels like a near miss, as though a little more care would push it below 1. It is not a near miss. It is the test reporting that it cannot see the distinction you are asking about, and the correct response is to put it down. Students who do not internalise this spend their time refining a computation that was structurally incapable of answering the question.

LOOK AT THE SHAPE FIRST

Factorials and n th powers go to the ratio or root test. Rational and algebraic terms go to limit comparison with a p -series. A decreasing $f(n)$ with a computable $\int f$ goes to the integral test. Alternating with decreasing magnitude goes to Leibniz. An oscillating factor times a monotone one goes to Dirichlet. The shape selects the test; you do not.

8.1 The selection protocol

EASY

CUE Any $\sum a_n$ whose convergence is in question. This is the technique you run before every other technique in the chapter.

METHOD Work the checklist in order and stop at the first line that applies. (1) Does $a_n \rightarrow 0$? If not, the series diverges and you are finished. (2) Is it geometric or a p -series? Quote the answer. (3) Does a_n contain a factorial or an n th power? Ratio or root test. (4) Is a_n a rational or algebraic function of n ? Limit comparison against the p -series of matching degree. (5) Is $a_n = f(n)$ with f positive, decreasing and integrable in closed form? Integral test. (6) Does a_n alternate? Leibniz for convergence, then test $\sum |a_n|$ separately. (7) Is a_n a product of an oscillating factor and a monotone one? Dirichlet or Abel.

Six series, six lines, no computation off the checklist.

$\sum \frac{n^2}{2n^2 + 1}$: step (1). The term tends to $\frac{1}{2} \neq 0$, so it diverges. Stop.

$\sum \frac{n^2 + 3}{n^4 - n + 1}$: step (4). Dominant powers give $n^2/n^4 = n^{-2}$; limit comparison with $\sum n^{-2}$ gives a finite nonzero ratio, so it converges.

$\sum \frac{(n!)^2}{(2n)!}$: step (3). The ratio is $\frac{(n+1)^2}{(2n+1)(2n+2)} \rightarrow \frac{1}{4} < 1$, so it converges. (Its value is $\frac{4}{3} + \frac{2\pi\sqrt{3}}{27} = 1.7363999$, but the test was never asked for that.)

$\sum_{n \geq 2} \frac{\sin n}{\ln n}$: steps (1) to (6) all fail; step (7) applies, and Dirichlet gives convergence.

$\sum_{n \geq 2} \frac{1}{n(\ln n)^{3/2}}$: step (5). The term is decreasing and $\int \frac{dx}{x(\ln x)^{3/2}} = -\frac{2}{\sqrt{\ln x}}$ is finite at ∞ , so it converges. (Equivalently, benchmark three with $p = \frac{3}{2} > 1$.)

$\sum \frac{(-1)^n}{\sqrt{n} + 1}$: step (6). The magnitudes decrease to 0, so Leibniz gives convergence; but $\sum \frac{1}{\sqrt{n} + 1}$ diverges by comparison with $\sum n^{-1/2}$, so the convergence is conditional and must be reported as such.

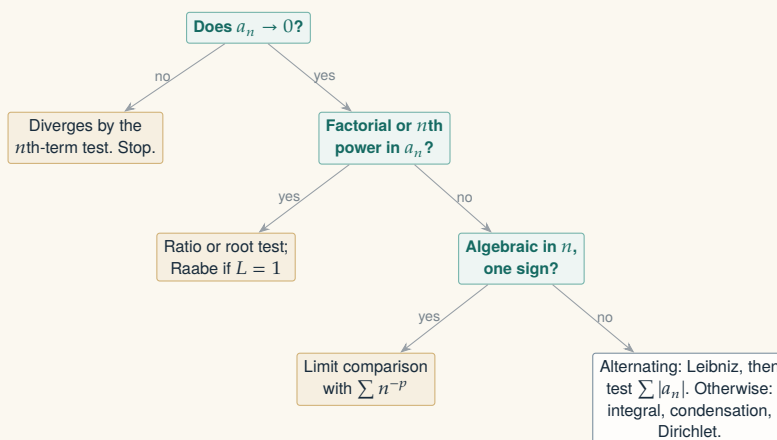
Two operational notes on running the checklist. First, the steps are ordered by cost, and the first one is free: glancing at $\lim a_n$ takes no work and ends a nontrivial fraction of all problems on the spot, usually the ones designed to look hardest. Second, the checklist is a routing table, not a hierarchy of power; a series routed to limit comparison is not thereby an easy series, and one routed to Dirichlet is not thereby a hard one. What the routing buys is that the test you run is the test that can answer the question, so the work you do is work that terminates.

WHERE THIS GOES WRONG

Reaching for the ratio test on an algebraic term. If a_n is a rational function of n , then $a_{n+1}/a_n \rightarrow 1$ identically, whatever the degrees: the test cannot distinguish $\sum 1/n$ from $\sum 1/n^2$ because both give $L = 1$. The ratio test is blind to polynomial scale by design, and running it on a rational term is a guaranteed waste of a page.

SEE ALSO The n th-term test and the benchmark series; limit comparison; ratio and root tests.

The protocol renders as a decision tree, and it is worth reading the tree rather than the list, because the tree makes the exclusions visible: each “no” branch is a shape you have ruled out.



Read the tree left to right and notice what it refuses to do: there is no branch on which you evaluate anything. Convergence tests are deliberately insensitive to the value of the series, which is what makes them cheap, and any attempt to sum the thing first is a much harder problem introduced for no reason. The one exception is a geometric series or a telescoping one, where the closed form of the partial sum is easier than any test; those are handled by inspection at step (2) and never reach the rest of the tree.

The other thing the tree encodes is precedence. Several branches can apply at once. For $\sum n!/n^n$ both the ratio test and the root test work; for $\sum 1/(n^2 + 1)$ both the integral test and limit comparison work. When that happens, take the branch that appears first, because the ordering is by cost: an n th-term check is free, a comparison is one limit, an integral is an antiderivative, and Dirichlet is a partial-sum bound you may have to construct.

The benchmarks and the two comparisons

Every convergence test in the positive-term world is ultimately a comparison with something known. The benchmarks are therefore not optional background; they are the entire supply of known answers, and a series is decided by being matched against one of them.

8.2 The n th-term test and the benchmark series

EASY

CUE The first look at any series, and the last resort when nothing else fits.

METHOD If $a_n \not\rightarrow 0$ the series diverges, because convergence of the partial sums forces $a_n = S_n - S_{n-1} \rightarrow 0$. The converse is false and its failure is the single most common error in the subject. Memorise the benchmarks: $\sum_{n \geq 0} r^n$ converges exactly when $|r| < 1$, to $\frac{1}{1-r}$; $\sum n^{-p}$ converges exactly when $p > 1$; and $\sum_{n \geq 2} \frac{1}{n(\ln n)^p}$ converges exactly when $p > 1$.

WORKED EXAMPLE

$\sum \frac{1}{n}$ diverges even though $\frac{1}{n} \rightarrow 0$: the partial sums are $H_n \sim \ln n$, unbounded. $\sum \frac{1}{n^2}$ converges, to $\frac{\pi^2}{6} = 1.6449341$. The two general terms differ only in an exponent, they both tend to 0, and they have opposite fates. That is the content of the benchmark list: the boundary between convergence and divergence sits at $p = 1$ and the n th-term test cannot see it.

The divergence of $\sum 1/n$ deserves to be seen once rather than quoted, because the argument is the model for the condensation test later in the chapter. Group the terms in blocks of doubling length:

$$\underbrace{\frac{1}{2}}_{\geq 1/2} + \underbrace{\frac{1}{3} + \frac{1}{4}}_{\geq 1/2} + \underbrace{\frac{1}{5} + \cdots + \frac{1}{8}}_{\geq 1/2} + \cdots$$

Each block of length 2^k has smallest term $2^{-(k+1)}$, so contributes at least $\frac{1}{2}$. Hence $H_{2^k} \geq 1 + \frac{k}{2}$, which is unbounded. (At $k = 14$, $H_{2^{14}} = 10.281$ against the guaranteed 8.) The growth is logarithmic and therefore invisible numerically: you would need about 10^{43} terms to reach a partial sum of 100, which is why no amount of computation can settle a borderline series and why the tests exist.

The third benchmark refines the boundary. $\sum \frac{1}{n \ln n}$ diverges while $\sum \frac{1}{n(\ln n)^2}$ converges, to 2.10974. So between the divergent $\sum n^{-1}$ and every convergent $\sum n^{-p}$ there is a further scale of series, and the boundary is not attained from either side.

One structural fact underlies every positive-term test and is worth stating separately: for $a_n \geq 0$ the partial sums are increasing, so $\sum a_n$ converges if and only if the partial sums are *bounded*. Every test in the next three techniques is a device for producing that bound by comparison with something whose partial sums are known. This is also why sign matters so much: once the terms change sign the partial sums stop being monotone, boundedness stops being sufficient, and an entirely different set of tools is needed.

WHERE THIS GOES WRONG

" $a_n \rightarrow 0$, therefore the series converges." This one error accounts for more wrong answers than every other in the chapter combined. The n th-term test is a disproof instrument only: it can end a problem with the word diverges, and it can never end one with the word converges.

SEE ALSO The selection protocol; direct comparison; the integral test with an explicit tail bound.

With benchmarks in hand, the two comparison tests do the work. Direct comparison is sharper and more demanding, since it wants a genuine inequality; limit comparison is blunter and far more usable, since it wants only matching asymptotics. In practice you reach for limit comparison first and fall back to direct comparison at the boundary cases where limit comparison goes silent.

8.3 Direct comparison

EASY

CUE Non-negative terms that are visibly bounded above, or below, by a benchmark, usually because a bounded factor such as $|\sin n|$ or $\arctan n$ can simply be thrown away.

METHOD If $0 \leq a_n \leq b_n$ for all large n and $\sum b_n$ converges, then $\sum a_n$ converges. If $a_n \geq b_n \geq 0$ for all large n and $\sum b_n$ diverges, then $\sum a_n$ diverges. Only the tail matters: finitely many terms can be anything at all.

$\sum \frac{|\sin n|}{n^2}$. Since $|\sin n| \leq 1$, the terms are at most n^{-2} , and $\sum n^{-2}$ converges; hence so does this, and its partial sum to 10^6 is 1.279889. Note what the argument avoids: any attempt to understand the distribution of $\sin n$, which is a genuinely hard question and completely irrelevant here.

In the other direction, $\sum_{n \geq 2} \frac{1}{\ln n}$ diverges, because $\ln n < n$ gives $\frac{1}{\ln n} > \frac{1}{n}$ and $\sum \frac{1}{n}$ diverges.

Two more where the discarded piece is a nuisance rather than a difficulty. $\sum \frac{1}{2^n + n} < \sum \frac{1}{2^n} = 1$, so it converges (to 0.697277); the $+n$ can only help and is simply dropped. And $\sum_{n \geq 1} \frac{1}{n!}$ converges because $n! \geq 2^{n-1}$ for $n \geq 1$, so the terms are at most 2^{1-n} and the sum is at most 2; the true value is $e - 1 = 1.718282$.

The reason direct comparison is worth keeping, given that limit comparison is easier to apply, is that it survives where limit comparison dies. Limit comparison requires the ratio a_n/b_n to have a limit, and for terms carrying a bounded oscillating factor it has none: $|\sin n|/n^2$ divided by n^{-2} is $|\sin n|$, which oscillates forever. Direct comparison needs only an inequality, and inequalities are indifferent to oscillation. As a rule: if the term contains $\sin$, $\cos$, $\{n\alpha\}$ or any other bounded factor with no limit, go straight to direct comparison and throw the factor away.

WHERE THIS GOES WRONG

Comparing in the useless direction. Bounding a suspected-convergent series *below* by a convergent series proves nothing, and bounding a suspected-divergent series *above* by a divergent one proves nothing. Half of all direct-comparison attempts produce a true inequality pointing the wrong way.

SEE ALSO Limit comparison; the n th-term test and the benchmark series; the integral test with an explicit tail bound.

8.4 Limit comparison

EASY

CUE Terms that are asymptotically a benchmark but algebraically messy: roots, sums of unlike powers, differences that nearly cancel.

METHOD If $a_n, b_n > 0$ and $\lim a_n/b_n = c$ with $0 < c < \infty$, then $\sum a_n$ and $\sum b_n$ converge or diverge together. Choose b_n by keeping only the dominant power in the numerator and in the denominator and discarding everything else.

WORKED EXAMPLE

$\sum \frac{2n + \sqrt{n}}{n^3 - 5}$. The dominant powers are $2n$ over n^3 , so take $b_n = 2/n^2$. Then

$$\frac{a_n}{b_n} = \frac{(2n + \sqrt{n})n^2}{2(n^3 - 5)} \rightarrow 1,$$

(numerically 1.0158 at $n = 1000$), so the series converges with $\sum n^{-2}$.

Two more where the asymptotics have to be extracted rather than read off. $\sum \sin \frac{1}{n}$: since $\sin t/t \rightarrow 1$ as $t \rightarrow 0$, the ratio of $\sin(1/n)$ to $1/n$ tends to 1 (it is 0.99833 already at $n = 10$), so the series diverges with the harmonic series, despite every term being smaller than $1/n$. And $\sum (\sqrt[n]{n} - 1)$: writing $\sqrt[n]{n} - 1 = e^{(\ln n)/n} - 1 \sim \frac{\ln n}{n}$, limit comparison with $\sum \frac{\ln n}{n}$ (divergent, being termwise larger than the harmonic series past $n = 3$) gives divergence.

A case where the shape hides the answer: $\sum \frac{1}{n^{1+1/n}}$. The ratio test is useless here. Limit comparison with $\sum \frac{1}{n}$ gives $a_n/b_n = n^{-1/n} = e^{-(\ln n)/n} \rightarrow 1$, so the series *diverges*, despite every term being strictly smaller than $1/n$. Numerically the ratio is 0.99999 at $n = 10^6$.

WHERE THIS GOES WRONG

The boundary cases. If $c = 0$ then convergence of $\sum b_n$ still gives convergence of $\sum a_n$, but divergence of $\sum b_n$ gives nothing; if $c = \infty$ the implications reverse. Both boundary cases are one-directional and must be run as direct comparisons, so quoting “limit comparison” with $c = 0$ and drawing a two-way conclusion is a genuine error.

There is a useful reformulation for terms that are messy differences rather than messy quotients. If $a_n = g(1/n)$ for a function g with $g(0) = 0$, expand g in a Taylor series about 0 and read off the first nonvanishing term: $a_n \sim g^{(m)}(0)n^{-m}/m!$, so the series converges exactly when $m \geq 2$. This settles $\sum (1 - \cos \frac{1}{n})$ (first term $\frac{1}{2n^2}$, convergent), $\sum (\frac{1}{n} - \sin \frac{1}{n})$ (first term

$\frac{1}{6n^3}$, convergent) and $\sum(e^{1/n} - 1)$ (first term $\frac{1}{n}$, divergent) without any algebraic rearrangement at all. Taylor expansion is the standard way to produce the b_n that limit comparison needs.

SEE ALSO Direct comparison; the selection protocol; the n th-term test and the benchmark series.

A word on how to pick b_n in practice, since this is where limit comparison is actually lost. Do not simplify the term; extract its dominant behaviour factor by factor. In a quotient, take the highest power in the numerator over the highest power in the denominator and discard all constants you like (constants cannot change c from finite-and-nonzero). Bounded factors with a limit, such as $\arctan n \rightarrow \pi/2$, may be replaced by that limit. Bounded factors with no limit may not, and send you to direct comparison instead. Logarithms may never be discarded and may never be compared to a power: $\ln n$ grows faster than every constant and slower than every positive power, so a term containing one is not settled by any p -series comparison, and you should be reaching for condensation or the integral test.

Tests driven by the shape of the term

The comparisons handle anything algebraic. Factorials and n th powers are not algebraic, and they need tests that look at the multiplicative structure instead of the polynomial degree. The distinction is worth naming precisely, because it is what the selection protocol is keying on. An algebraic term has $a_{n+1}/a_n \rightarrow 1$: the terms change by a vanishing proportion at each step, and their size is a power of n . A term built multiplicatively has $a_{n+1}/a_n \rightarrow L \neq 1$: the terms change by a fixed proportion at each step, and their size is exponential. The comparisons resolve the first kind because the p -series are indexed by exactly the parameter that varies; the ratio and root tests resolve the second because the ratio itself is the parameter. Neither can do the other's job, and the boundary between the two families, where the ratio tends to 1 but the decay is faster than any power, is exactly where the remaining tests in this section live.

8.5 Ratio and root tests

EASY

CUE Factorials, n th powers, products of consecutive integers, or any term built by multiplying something on at each step.

METHOD Compute $L = \lim |a_{n+1}/a_n|$ (ratio) or $L = \lim \sqrt[n]{|a_n|}$ (root). Both give absolute convergence when $L < 1$, divergence when $L > 1$, and **nothing at all** when $L = 1$. Divergence for $L > 1$ is not an accident of the proof: it follows because $|a_n|$ is eventually increasing, so $a_n \not\rightarrow 0$. The root test succeeds strictly more often than the ratio test, since a ratio limit always forces the same root limit but not conversely.

WORKED EXAMPLE

Ratio. $\sum \frac{n!}{n^n}$ has

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{n!} = \left(1 + \frac{1}{n}\right)^{-n} \rightarrow \frac{1}{e} < 1,$$

so the series converges (its value is 1.879854).

Root. $\sum \left(\frac{n+1}{3n+2}\right)^n$ has $\sqrt[n]{a_n} = \frac{n+1}{3n+2} \rightarrow \frac{1}{3} < 1$, so it converges. The ratio here is a mess; the root test reads the answer off the base.

A constant that matters. $\sum \frac{c^n n!}{n^n}$ has ratio $c(1 + \frac{1}{n})^{-n} \rightarrow \frac{c}{e}$, so it converges for $c < e$ and diverges for $c > e$. At $c = 2$ the sum is 12.94895; at $c = 3$ the terms grow without bound. The borderline $c = e$ is exactly the case the test cannot decide, and it in fact diverges, as Raabe or Stirling will show.

WHERE THIS GOES WRONG

Treating $L = 1$ as divergence. Both $\sum \frac{1}{n}$ (divergent) and $\sum \frac{1}{n^2}$ (convergent) give $L = 1$ under both tests, so $L = 1$ carries exactly zero information and you must start again with a different test. The correct response to $L = 1$ is to look at the shape again, not to push harder.

These two tests are also the engine of power series. For $\sum c_n x^n$ the root test gives $\limsup \sqrt[n]{|c_n|} |x| < 1$, that is $|x| < R$ with $1/R = \limsup \sqrt[n]{|c_n|}$, and the interval of convergence follows immediately. The two endpoints $x = \pm R$ are precisely where $L = 1$, which is why every power series problem ends with two separate boundary investigations run by other tests. The blind spot of the ratio and root tests is not a defect to be patched; it is where all the interesting behaviour lives.

SEE ALSO Raabe's test at the ratio-test boundary; the selection protocol; Cauchy condensation.

The instruction “start again” is unsatisfying when the term really is a product of factorials, so that nothing but a ratio test is natural, and the ratio nonetheless tends to 1. That happens whenever the term decays like a power of n while being built multiplicatively. There is a sharper test for exactly this case: it measures how fast the ratio approaches 1.

HARD

8.6 Raabe's test at the ratio-test boundary

CUE A term built from factorials or double factorials whose ratio tends to 1, so that the ratio and root tests are both inconclusive, and the decay is evidently a power of n .

METHOD Compute

$$R = \lim_{n \rightarrow \infty} n \left(\frac{a_n}{a_{n+1}} - 1 \right).$$

If $R > 1$ the series converges; if $R < 1$ it diverges; $R = 1$ is again inconclusive. The interpretation is direct: R is the effective power p in $a_n \approx Cn^{-p}$, so Raabe's test simply reads off which p -series the term matches.

WORKED EXAMPLE

Decide $\sum_{n \geq 1} \left(\frac{(2n-1)!!}{(2n)!!} \right)^p$ for $p > 0$, where $(2n)!! = 2 \cdot 4 \cdots (2n)$ and $(2n-1)!! = 1 \cdot 3 \cdots (2n-1)$.

The ratio is $\left(\frac{2n+1}{2n+2} \right)^p \rightarrow 1$ for every p , so the ratio test is dead on arrival. Raabe:

$$n \left(\left(\frac{2n+2}{2n+1} \right)^p - 1 \right) = n \left(\left(1 + \frac{1}{2n+1} \right)^p - 1 \right) \rightarrow \frac{p}{2}.$$

So the series converges exactly when $p > 2$. Numerically, with the sum truncated at $n = 2000$: $p = 1$ gives 49.47 and is climbing like $\sqrt{n}$, $p = 2$ gives 2.486 and is climbing like $\ln n$, and $p = 3$ gives 0.385 and has stopped moving. The Raabe values at $p = 3$ are 1.4977, 1.49998, 1.4999998 at $n = 10, 10^2, 10^3$, converging to $\frac{3}{2} > 1$ as predicted.

Raabe's test is the first rung of a ladder. If $R = 1$ as well, Gauss's test looks at the next term, expanding $a_n/a_{n+1} = 1 + \frac{h}{n} + O(n^{-1-\delta})$ and concluding convergence exactly when $h > 1$; beyond that lies Bertrand's test with its $\ln n$ correction. Each rung resolves the previous rung's boundary and creates a new one, and there is no last rung: no single test decides every series. That is a theorem, not an accident, and it is the reason this chapter is a protocol rather than a formula.

Why does the test look the way it does? If $a_n = Cn^{-p}$ exactly, then

$$\frac{a_n}{a_{n+1}} = \left(1 + \frac{1}{n} \right)^p = 1 + \frac{p}{n} + O(n^{-2}),$$

so $n(a_n/a_{n+1} - 1) \rightarrow p$. Raabe's test is therefore nothing more than the p -series comparison, carried out on the ratio rather than on the term itself, which is what makes it usable when the term is a product of factorials with no closed form. Read R as the exponent and the convergence criterion $R > 1$ is the familiar $p > 1$.

WORKED EXAMPLE

A second instance, and the one that explains the ratio test's blind spot for central binomial coefficients. Decide

$$\sum_{n \geq 1} \frac{1}{4^n} \binom{2n}{n}.$$

The ratio is $\frac{a_{n+1}}{a_n} = \frac{(2n+1)(2n+2)}{4(n+1)^2} = \frac{2n+1}{2n+2} \rightarrow 1$: inconclusive, and the shape offers nothing else. Raabe:

$$n \left(\frac{2n+2}{2n+1} - 1 \right) = \frac{n}{2n+1} \rightarrow \frac{1}{2} < 1,$$

so the series diverges. The reading is exact: $R = \frac{1}{2}$ says the term behaves like $n^{-1/2}$, and indeed $4^{-n} \binom{2n}{n} \sim (\pi n)^{-1/2}$, and the two sides agree to within 1.3×10^{-5} at $n = 10^4$.

WHERE THIS GOES WRONG

Getting the fraction upside down. The quantity is $n(a_n/a_{n+1} - 1)$, with the *earlier* term on top, so that a convergent series gives a positive R . Writing $n(a_{n+1}/a_n - 1)$ flips every sign and reverses the conclusion.

SEE ALSO Ratio and root tests; limit comparison; the integral test with an explicit tail bound.

Both of the last two tests answer yes or no. Often the real question is quantitative: given that $\sum a_n$ converges, how far is S_{100} from S ? The integral test is the one positive-term test that answers this, because its proof is a pair of inequalities rather than a limit.

8.7 The integral test with an explicit tail bound

MEDIUM

CUE $a_n = f(n)$ where f is positive, continuous and decreasing on $[N, \infty)$, and $\int f$ is computable in closed form. Logarithms in the denominator are the standard signal.

METHOD The series and the integral converge together. Better, the rectangles that prove it give a two-sided bound on the tail:

$$\int_{N+1}^{\infty} f(x) dx \leq \sum_{n>N} f(n) \leq \int_N^{\infty} f(x) dx,$$

which converts a partial sum into an interval containing the true value.

WORKED EXAMPLE

Convergence. $\sum_{n \geq 2} \frac{1}{n \ln n}$ diverges, since

$$\int_2^{\infty} \frac{dx}{x \ln x} = [\ln \ln x]_2^{\infty} = \infty.$$

Replacing $\ln n$ by $(\ln n)^2$ makes the antiderivative $-1/\ln x$, which is finite, so $\sum 1/(n(\ln n)^2)$ converges.

A term with an antiderivative and nothing else. $\sum \frac{1}{n^2 + 1}$ is decreasing and $\int_1^{\infty} \frac{dx}{x^2 + 1} = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$ is finite, so it converges (to 1.0766739). Limit comparison with $\sum n^{-2}$ also settles it in one line; the point of doing it this way is that the integral test additionally hands you the tail bracket, which limit comparison never does.

Error bound. For $\sum n^{-2}$ with $N = 10$: the partial sum is $S_{10} = 1.5497677$ and $\int_N^{\infty} x^{-2} dx = 1/N$, so

$$1.5497677 + \frac{1}{11} \leq S \leq 1.5497677 + \frac{1}{10}, \quad \text{i.e.} \quad 1.6406768 \leq S \leq 1.6497677.$$

The true value $\pi^2/6 = 1.6449341$ sits inside, and ten terms have located it to within 0.01.

Two more uses of the bracket. First, as a stopping rule: to compute $\sum n^{-2}$ to within 10^{-6} you need $\int_N^{\infty} x^{-2} dx = 1/N \leq 10^{-6}$, so $N = 10^6$ terms, and you know this before summing anything. Second, as a convergence test on a term the comparisons cannot reach: $\sum \frac{\ln n}{n^2}$ has an increasing numerator, so no benchmark comparison is immediate, but $\int_1^{\infty} \frac{\ln x}{x^2} dx = \left[-\frac{1 + \ln x}{x} \right]_1^{\infty} = 1$ is finite (the function is decreasing past $x = e$), so the series converges; its value is 0.9375483.

WHERE THIS GOES WRONG

Believing the integral's value is the series' value. Here the integral from 1 is 1 and the series is 1.6449; they are not close and were never meant to be. Only the convergence verdict and the tail bracket transfer, and the bracket is what you should actually quote.

The same rectangles give more than a verdict and more than a bracket. For f positive and decreasing, the difference $D_N = \sum_{n \leq N} f(n) - \int_1^N f$ is itself decreasing and bounded below by 0, hence convergent, whether or not the series converges. For $f(x) = 1/x$ this is the statement that $H_N - \ln N$ tends to a limit, the Euler-Mascheroni constant $\gamma = 0.5772157$ (the value at $N = 10$ is already 0.6264). So the integral test does not only compare a series to an integral; it measures the gap, and the gap is often the interesting object.

SEE ALSO Cauchy condensation; the n th-term test and the benchmark series; the alternating series test and its error bound.

The integral test needs an antiderivative. When the term is decreasing but the integral is unavailable, or when you want the logarithmic benchmarks proved rather than memorised, there is a purely discrete substitute that samples the sequence at powers of 2.

8.8 Cauchy condensation

CUE A positive decreasing a_n with iterated logarithms, or any term where doubling n has a clean effect. The signal is that a_{2n}/a_n simplifies.

METHOD For $a_n \geq 0$ decreasing,

$$\sum_{n \geq 1} a_n \text{ converges} \iff \sum_{k \geq 0} 2^k a_{2^k} \text{ converges.}$$

Group the terms in blocks of length 2^k and bound each block above and below by its endpoint. The effect is to turn a slowly varying series into a geometric-looking one.

WORKED EXAMPLE

The p -series in one line. With $a_n = n^{-p}$, the condensed series is $\sum_k 2^k 2^{-kp} = \sum_k (2^{1-p})^k$, geometric with ratio 2^{1-p} , convergent exactly when $2^{1-p} < 1$, that is $p > 1$.

A second level of logarithm. With $a_n = \frac{1}{n \ln n (\ln \ln n)^p}$ the condensed term is

$$2^k \cdot \frac{1}{2^k (k \ln 2) (\ln(k \ln 2))^p} = \frac{1}{\ln 2} \cdot \frac{1}{k (\ln k + \ln \ln 2)^p},$$

which is the previous benchmark again and converges exactly when $p > 1$. Condensation handles the whole iterated-logarithm tower by repeating one move, where the integral test needs a new substitution at every level.

The logarithmic benchmark. With $a_n = \frac{1}{n(\ln n)^p}$ the condensed series is

$$\sum_k 2^k \cdot \frac{1}{2^k (k \ln 2)^p} = \frac{1}{(\ln 2)^p} \sum_k \frac{1}{k^p},$$

convergent exactly when $p > 1$. So the third benchmark is the second benchmark in disguise, and the same trick applied again gives $\sum \frac{1}{n \ln n (\ln \ln n)^p}$, convergent exactly when $p > 1$.

The economy of the method is worth appreciating. Condensation replaces 2^k terms by one, so it collapses a series whose terms vary slowly (which is exactly when convergence is delicate) into one whose terms vary geometrically (which is exactly when convergence is obvious). It is the discrete shadow of the substitution $x = 2^u$ in the integral test, and it needs no antiderivative, which is its whole advantage: it settles the entire iterated-logarithm family at a stroke, where the integral test needs a fresh substitution at every level.

WHERE THIS GOES WRONG

Applying it without monotonicity. The block-bounding argument replaces every term in a block by an endpoint, which is legitimate only if the terms are ordered. For a non-monotone a_n the condensed series can converge while the original diverges.

SEE ALSO The integral test with an explicit tail bound; the n th-term test and the benchmark series; direct comparison.

Signs, and what convergence is worth

Everything above assumed a single sign, so that partial sums are monotone and the only question is boundedness. Mixed signs change the subject. Cancellation can rescue a series whose magnitudes are far too large to sum, and the rescued object turns out to be a weaker thing than it looks.

The strategic point is that with mixed signs there are now two questions where before there was one, and they must be answered in the right order. Question one: does $\sum |a_n|$ converge? That is a positive-term problem and everything above applies to it unchanged. If the answer is yes you are finished, because absolute convergence implies convergence and carries every good property with it. Only if the answer is no does the second question arise: does $\sum a_n$ nonetheless converge by cancellation? That is what the alternating test, Dirichlet and Abel are for, and a positive answer must always be reported with the word conditionally attached, because the qualifier is what tells the next reader which manipulations are forbidden.

8.9 The alternating series test and its error bound

CUE $\sum (-1)^n b_n$ with $b_n > 0$, where b_n is decreasing to 0.

METHOD If b_n is *eventually decreasing* and $b_n \rightarrow 0$, the series converges. Moreover the partial sums straddle the answer, $S_{2k} < S < S_{2k+1}$ or the reverse, and therefore

$$|S - S_N| \leq b_{N+1} :$$

the first omitted term bounds the error. No other elementary test hands you an error bound this cheaply.

WORKED EXAMPLE

$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = \ln 2$. The magnitudes decrease to 0, so it converges, and after N terms the error is at most $1/(N+1)$: to get three decimals you need a thousand terms, which is a bound worth knowing before you start computing.

$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n}} = 0.6048986$. The magnitudes decrease to 0, so it converges, but very slowly: after 10^6 terms the partial sum is 0.6043986, an error of 5.0×10^{-4} , comfortably inside the guaranteed $b_{N+1} = 10^{-3}$ and nowhere near enough for four decimals. The error bound is honest about how bad the situation is, which is its main value.

$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^3} = \frac{3\zeta(3)}{4} = 0.9015427$. To force the error below 10^{-3} take $b_{N+1} = (N+1)^{-3} \leq 10^{-3}$, that is $N \geq 9$. And indeed $S_9 = 0.9021165$, an error of 5.74×10^{-4} , comfortably inside the promised 10^{-3} .

Two refinements of the hypothesis. “Eventually decreasing” is genuinely enough: the first thousand terms may do anything, since altering finitely many terms changes the sum but not its convergence. And when the magnitudes are given by a formula, the cleanest monotonicity proof is to differentiate the corresponding function of a real variable: $b_n = n/(n^2 + 1)$ is decreasing for $n \geq 1$ because $t/(t^2 + 1)$ has derivative $(1 - t^2)/(t^2 + 1)^2 \leq 0$ there, which is faster than any inequality manipulation.

It is worth knowing why the bound is so clean, because the reason also tells you when it is sharp. With b_n decreasing, the partial sums $S_1, S_3, S_5, \dots$ decrease and $S_2, S_4, S_6, \dots$ increase, and every odd partial sum exceeds every even one; the limit is trapped in every one of the nested intervals $[S_{2k}, S_{2k+1}]$, whose length is exactly b_{2k+1} . So the bound $|S - S_N| \leq b_{N+1}$ is not an estimate that could be improved by working harder: it is the length of an interval you have actually produced, and the true error is typically about half of it.

WHERE THIS GOES WRONG

Skipping the monotonicity check. That $b_n \rightarrow 0$ is not enough. Take $b_{2k} = \frac{1}{k}$ and $b_{2k+1} = \frac{1}{k^2}$, both tending to 0; then pairing the terms gives $\sum_k (\frac{1}{k} - \frac{1}{k^2})$, which diverges. Alternation plus decay to zero does not suffice; the decay must be monotone.

SEE ALSO Absolute versus conditional convergence; Dirichlet and Abel tests; the selection protocol.

A series that converges by the alternating test may or may not converge when you delete the signs, and the difference is not bookkeeping. It determines which manipulations are legal.

8.10 Absolute versus conditional convergence

CUE Any series with mixed signs, once you have established that it converges.

METHOD Test $\sum |a_n|$ as a separate problem, with the positive-term machinery. If it converges the original is *absolutely* convergent, and you may rearrange the terms freely, multiply two such series by the Cauchy product, and split the sum into positive and negative parts. If $\sum a_n$ converges while $\sum |a_n| = \infty$, the convergence is *conditional*, and none of those operations is licensed.

The structural statement behind all of this is a decomposition. Write $a_n^+ = \max(a_n, 0)$ and $a_n^- = \max(-a_n, 0)$, so that $a_n = a_n^+ - a_n^-$ and $|a_n| = a_n^+ + a_n^-$. Then $\sum a_n$ converges absolutely exactly when both $\sum a_n^+$ and $\sum a_n^-$ converge, and converges conditionally exactly when both diverge to $+\infty$ while their difference settles down. There is no third possibility: if one half converges and the other does not, the partial sums are unbounded and the series diverges. Conditional convergence is therefore always a cancellation between two infinities, which is exactly why it is fragile under reordering.

$\sum \frac{(-1)^{n+1}}{n}$ converges to $\ln 2$ but $\sum \frac{1}{n}$ diverges: conditional. $\sum \frac{(-1)^{n+1}}{n^2}$ converges to $\frac{\pi^2}{12} = 0.8224670$ and $\sum \frac{1}{n^2}$ converges: absolute.

Here is what conditional costs. Rearrange the alternating harmonic series to take two positive terms for every negative one:

$$1 + \frac{1}{3} - \frac{1}{2} + \frac{1}{5} + \frac{1}{7} - \frac{1}{4} + \dots$$

Summing 3×10^5 blocks numerically gives 1.0397199, and the exact value is $\frac{3}{2} \ln 2 = 1.0397208$. The same terms, reordered, sum to something else. Riemann's theorem says more: a conditionally convergent series can be rearranged to converge to *any* prescribed real number, or to diverge. Absolute convergence is exactly the hypothesis that rules this out.

The Cauchy product failure is worth seeing concretely, since it is the one that appears in analysis rather than in trick questions. Let $a_n = b_n = \frac{(-1)^n}{\sqrt{n+1}}$, both series convergent by Leibniz and neither absolutely. Their Cauchy product has terms

$$c_n = \sum_{k=0}^n a_k b_{n-k} = (-1)^n \sum_{k=0}^n \frac{1}{\sqrt{(k+1)(n-k+1)}},$$

and the sum is a Riemann sum for $\int_0^1 \frac{dt}{\sqrt{t(1-t)}} = \pi$. So $|c_n| \rightarrow \pi$: the product's terms do not tend to 0 and the product diverges by the n th-term test, even though both factors converge. Absolute convergence of one factor is exactly what Mertens' theorem needs to rule this out.

Riemann's rearrangement theorem is worth understanding constructively rather than as a slogan, because the construction shows exactly which hypothesis fails. If $\sum a_n$ converges conditionally, then the positive terms alone sum to $+\infty$ and the negative terms alone sum to $-\infty$; if either were finite, both would be, and the series would converge absolutely. Given a target t , take positive terms until the partial sum first exceeds t , then negative terms until it first falls below t , then positives again, and so on. Each switch overshoots by at most the last term used, and the terms tend to 0, so the overshoots tend to 0 and the rearranged series converges to t . Every step of that construction uses the divergence of the two halves. Absolute convergence is precisely the hypothesis that makes both halves finite and the construction impossible.

WHERE THIS GOES WRONG

Manipulating a conditionally convergent series as if it were absolutely convergent: reordering it, splitting it into $\sum(\text{positives}) - \sum(\text{negatives})$ (both halves are $+\infty$), or forming a Cauchy product. Each of these silently assumes the property you failed to check.

SEE ALSO The alternating series test and its error bound; Dirichlet and Abel tests; the selection protocol.

One family of mixed-sign series defeats the whole battery so far: the signs do not alternate regularly, so Leibniz does not apply, and the magnitudes are not summable, so absolute convergence is unavailable. The resolution is to stop treating the oscillation as a sign pattern and start treating it as a factor with bounded partial sums.

8.11 Dirichlet and Abel tests

CUE $\sum a_n b_n$ where b_n is monotone and a_n oscillates with bounded partial sums, typically $a_n = \sin n\theta$ or $\cos n\theta$.

METHOD Dirichlet: if the partial sums $A_N = \sum_{n \leq N} a_n$ are bounded and b_n decreases monotonically to 0, then $\sum a_n b_n$ converges. **Abel:** if $\sum a_n$ converges and b_n is monotone and bounded (not necessarily to 0), then $\sum a_n b_n$ converges. Both are proved by summation by parts, which is the discrete integration by parts: it moves the difference off b_n and onto the bounded A_N .

The mechanism deserves to be written once, because both tests are corollaries of it and because it is the discrete analogue of integration by parts. With $A_n = \sum_{k \leq n} a_k$ and $A_0 = 0$, substituting $a_n = A_n - A_{n-1}$ and regrouping gives Abel's summation formula

$$\sum_{n=1}^N a_n b_n = A_N b_N + \sum_{n=1}^{N-1} A_n (b_n - b_{n+1}).$$

The left side is a sum of products with no structure; the right side is a bounded quantity times b_N , plus a sum whose terms are $|A_n|$ times the increments of b_n . If $|A_n| \leq M$ and b_n decreases, that second sum is dominated by $M \sum (b_n - b_{n+1})$, which telescopes to $M(b_1 - b_N)$ and is therefore absolutely convergent. Everything in Dirichlet and Abel is contained in that one line.

WORKED EXAMPLE

$\sum_{n=1}^{\infty} \frac{\sin n}{n}$. Take $a_n = \sin n$, $b_n = 1/n$. The partial sums are computable in closed form,

$$\sum_{n=1}^N \sin n = \frac{\sin(N/2) \sin((N+1)/2)}{\sin(1/2)}, \quad \text{so} \quad \left| \sum_{n=1}^N \sin n \right| \leq \frac{1}{\sin(1/2)} = 2.0858$$

for every N (the observed maximum over $N \leq 10^5$ is 1.9582). Since $1/n$ decreases to 0, Dirichlet applies and the series converges. Its value is $\frac{\pi - 1}{2} = 1.0707963$; the partial sum at 10^7 terms is 1.0707964. Note that $\sum |\sin n|/n$ diverges, so this convergence is conditional and purely a cancellation effect.

The same bound settles a whole family at once. Since $b_n = 1/\ln n$ also decreases to 0, Dirichlet gives convergence of $\sum_{n \geq 2} \frac{\sin n}{\ln n}$, which no other test in this chapter can touch: the magnitudes are nowhere near summable, the signs do not alternate in any regular pattern, and there is no monotone f to integrate. Dirichlet is the test for series that are held together by cancellation alone.

Abel, by contrast, upgrades a known convergence: since $\sum (-1)^{n+1}/n$ converges and $b_n = n^{1/n}$ is monotone and bounded for $n \geq 3$, the series $\sum (-1)^{n+1} n^{1/n}/n$ converges as well, to 0.573252.

Both tests are stated for real sequences but are used constantly in complex form. Taking $a_n = z^n$ with $|z| = 1$ and $z \neq 1$ gives $|A_N| = \left| \frac{z(z^N - 1)}{z - 1} \right| \leq \frac{2}{|z - 1|}$, bounded, so $\sum b_n z^n$ converges for every b_n decreasing to 0 and every such z . Taking real and imaginary parts recovers the convergence of $\sum b_n \cos n\theta$ and $\sum b_n \sin n\theta$ for all θ that are not multiples of 2π , in one line, and shows precisely where the single excluded case comes from: $z = 1$ is the one point where the geometric partial sums fail to be bounded.

WHERE THIS GOES WRONG

Using Dirichlet when the partial sums are unbounded. If θ is a multiple of 2π then $\sum \cos n\theta = N$ grows linearly, the hypothesis fails, and the conclusion fails with it: $\sum \cos(2\pi n)/n = \sum 1/n$ diverges. Always produce the bound on A_N explicitly; it is the only hypothesis that is ever in doubt.

SEE ALSO The alternating series test and its error bound; absolute versus conditional convergence; the selection protocol.

A final word on what a convergence verdict is worth. Nothing in this chapter produces the value of a series; the tests are built to be insensitive to it, which is exactly why they are cheap. The next chapter takes up evaluation, and it is a different subject with a different set of moves, none of which can be started until the question settled here has been settled. Attempting to sum a divergent series is the one error in this material that cannot be partially credited.

Four series, start to finish

The protocol is only worth anything if it runs quickly under pressure, so here it is on four terms chosen to look similar and behave differently. In each case the first move is to name the shape.

$\sum_{n \geq 1} \frac{n^3 + 2n}{n^5 - n^2 + 7}$. Algebraic, one sign eventually. Degree gap $5 - 3 = 2$, so limit comparison with $\sum n^{-2}$: the ratio tends to 1. Converges.

$\sum_{n \geq 1} \frac{3^n (n!)^2}{(2n)!}$. Factorials: ratio test. The ratio is $\frac{3(n+1)^2}{(2n+1)(2n+2)} \rightarrow \frac{3}{4} < 1$. Converges.

$\sum_{n \geq 2} \frac{1}{\sqrt{n} \ln n}$. Algebraic times a logarithm, positive and decreasing. Limit comparison against $\sum n^{-p}$ is inconclusive because

$\ln n$ sits between every pair of powers, so use condensation: the condensed series is $\sum_k 2^k \cdot \frac{1}{2^{k/2} k \ln 2} = \frac{1}{\ln 2} \sum_k \frac{2^{k/2}}{k}$, whose terms grow without bound. Diverges. (Direct comparison also works, since $\ln n < \sqrt{n}$ gives terms exceeding $1/n$.)

$\sum_{n \geq 1} \frac{2 + \cos n}{n}$. Mixed-looking but positive, since $2 + \cos n \geq 1$. So it is a positive-term series with $a_n \geq 1/n$, and direct comparison gives divergence. The oscillating factor is a decoy: Dirichlet is not applicable because the partial sums of $2 + \cos n$ grow linearly, and no cancellation is available because there are no sign changes.

$\sum_{n \geq 1} \frac{(-1)^n n}{n^2 + 1}$. Mixed signs, so first check absolute convergence: $\frac{n}{n^2 + 1} \sim \frac{1}{n}$, so $\sum |a_n|$ diverges. Then Leibniz: the magnitudes decrease (the derivative of $t/(t^2 + 1)$ is negative for $t > 1$) and tend to 0. Converges conditionally, and the answer must say “conditionally”, because that is the part with content.

Five series, five different tests, and in each case the test was fixed by the shape before any computation began. That is what the chapter is asking you to be able to do. Notice too how short each argument is once the routing is right: two lines, sometimes one. Length is a symptom. If a convergence argument is running past half a page, the usual cause is not that the series is hard but that the test is wrong, and the right response is to go back to the general term and look at its shape again.

THE VERDICT IS NOT THE VALUE

No test in this chapter computes a sum, and none is meant to. Two of them do produce numbers: the integral test brackets the tail between two integrals, and the alternating test bounds the truncation error by the first omitted term. If a problem demands a numerical answer to a stated accuracy, those two are the only tools here that finish it.

Recognition matrix

If you see	Reach for	If that fails
$a_n \not\rightarrow 0$	n th-term test: diverges, stop	Nothing else is needed
$n!, (2n)!, c^n, n^n$	Ratio test	Root test, then Raabe
$(g(n))^n$	Root test on the base	Logarithms, then compare
Rational or algebraic in n	Limit comparison with $\sum n^{-p}$, p the degree gap	Direct comparison at $c = 0$ or $c = \infty$
Ratio or root gives $L = 1$	Raabe's test, or limit comparison	Condensation
$\ln n$ or $\ln \ln n$ in the denominator	Cauchy condensation	Integral test
$f(n)$ decreasing with $\int f$ known	Integral test, and quote the tail bracket	Condensation
Bounded factor times a benchmark	Direct comparison	Limit comparison
$(-1)^n b_n$, b_n decreasing to 0	Leibniz; error $\leq b_{N+1}$	Test $\sum a_n $ separately
$\sin n\theta$ or $\cos n\theta$ times monotone b_n	Dirichlet, with an explicit bound on A_N	Abel if $\sum a_n$ already converges
Converges, signs mixed	Test $\sum a_n $ before rearranging	Assume nothing: treat as conditional

Series: finding what it converges to

What this chapter is for. A convergence test is an existence proof and nothing more: it certifies that $\sum a_n$ has a sum without producing one digit of it. Evaluation is a separate craft, and a surprisingly small one. There are essentially five moves. Collapse the series, so that almost every term cancels against a neighbour. Manufacture it from the geometric series by differentiating or integrating. Recognise it as a Taylor series read at a particular point. Read it off a Fourier expansion at a well chosen point, which is where $\zeta(2)$, $\zeta(4)$ and the Leibniz series come from. Or convert it into an object you can already handle, by an integral representation, a generating function, or summation by parts. This chapter treats each move as a recognisable technique with a cue, a mechanism and a characteristic failure, and it closes with the one structural hazard that runs through all of them: a conditionally convergent series has no fixed value under regrouping, and treating it as though it does will produce a confident wrong answer.

Collapse it

The cheapest exact evaluation in mathematics is the one where you never sum anything. If the general term is the difference of consecutive values of a single sequence, the partial sum has only boundary terms in it, and the infinite sum is a limit of two or three numbers rather than of infinitely many. Everything in this section is a variation on that observation, and the variations differ only in how well the difference is hidden.

Telescoping deserves to come first for a second reason. It is the one method whose answer needs no analytic licence at all: the partial sum is an identity between finitely many rational numbers, so the only limit taken is the harmless one at the end. Every other technique in this chapter needs a theorem to justify some interchange. Telescoping needs none.

There is also a standing procedural point. Almost every series in this chapter was built backwards by whoever set the problem, and the fastest route to the answer is often to reconstruct the construction. If a denominator factors into two pieces that are one expression at k and at $k+1$, somebody arranged that; if a numerator is a peculiar quadratic, somebody chose it to be the numerator of a difference. Reading a problem as a piece of engineering rather than as a natural object is the habit this chapter is trying to install.

EASY

9.1 Telescoping

CUE The general term can be written as $c_k - c_{k+1}$ for some explicit sequence c , or as $c_{k-1} - c_{k+1}$ for a two step telescope. The tell is a denominator that factors into two pieces which are the same expression evaluated at k and at $k+1$.

METHOD Identify c , then write

$$\sum_{k=1}^n (c_k - c_{k+1}) = c_1 - c_{n+1}, \quad \sum_{k=1}^n (c_{k-1} - c_{k+1}) = c_0 + c_1 - c_n - c_{n+1},$$

and take $n \rightarrow \infty$. The whole discipline is bookkeeping: a one step telescope leaves one term at each end, a two step telescope leaves two.

WORKED EXAMPLE

$$S = \sum_{r=1}^{\infty} \frac{4r^2 - 2}{4r^4 + 1}.$$

The Sophie Germain identity factors the denominator, $4r^4 + 1 = (2r^2 - 2r + 1)(2r^2 + 2r + 1)$, and the two factors are one sequence at consecutive arguments: if $g(r) = 2r^2 - 2r + 1$ then $g(r+1) = 2r^2 + 2r + 1$. Set

$$d_r = \frac{2r-1}{2r^2-2r+1}, \quad \text{so} \quad d_{r+1} = \frac{2r+1}{2r^2+2r+1}.$$

Then $d_r - d_{r+1}$ has numerator $(2r-1)(2r^2+2r+1) - (2r+1)(2r^2-2r+1) = (4r^3+2r^2-1) - (4r^3-2r^2+1) = 4r^2-2$, which is exactly the numerator we were handed. Hence the partial sum is $d_1 - d_{n+1}$, and $d_1 = 1$, $d_{n+1} \rightarrow 0$, so $S = 1$.

You are rarely handed c_k . Factor the denominator as $g(k)g(k+1)$, then look for $c_k = P(k)/g(k)$ with P of degree one less than g , and fix the coefficients of P by matching numerators in $c_k - c_{k+1}$. For $g(k) = 2k^2 - 2k + 1$ the ansatz $P(k) = \alpha k + \beta$ produces the numerator $2\alpha k^2 + 2\beta k + \dots$, and comparing with $4k^2 - 2$ forces $\alpha = 2, \beta = -1$. Two unknowns, two equations, and no cleverness required once the factorisation is in hand.

Two failures dominate. The first is dropping a surviving boundary term: in the two step telescope the answer is $c_0 + c_1$, and writing only c_1 costs you a whole term. The second is telescoping a series that is only conditionally convergent, where regrouping is not a legal operation and the collapse is an illusion.

SEE ALSO Partial fractions engineered to telescope; arctangent difference telescoping; infinite products.

The hard part is never the collapse; it is seeing that a collapse is available. In the example above the numerator $4r^2 - 2$ was engineered backwards from the difference, and the reader's job was to run the engineering in reverse. The next two techniques are the two standard factories for that engineering: rational terms, where partial fractions expose the difference, and arctangent terms, where the subtraction formula does.

9.2 Partial fractions engineered to telescope

CUE A rational general term whose denominator factors into consecutive integers, or into an arithmetic progression such as $2n - 1$ and $2n + 1$. The presence of two or three factors spaced by a constant is the signature.

METHOD Decompose, but decompose into *blocks that already differ by a shift*, not into individual simple fractions. The two identities worth memorising are

$$\frac{1}{n(n+1)(n+2)} = \frac{1}{2} \left[\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)} \right], \quad \frac{1}{4n^2 - 1} = \frac{1}{2} \left[\frac{1}{2n-1} - \frac{1}{2n+1} \right].$$

In the first, the bracketed pieces are one function of n at consecutive arguments; in the second they are one function at arguments two apart, which is why the factor $\frac{1}{2}$ appears in front. The general statement behind the first is worth carrying: for a run of $m + 1$ consecutive factors,

$$\frac{1}{n(n+1) \cdots (n+m)} = \frac{1}{m} \left[\frac{1}{n(n+1) \cdots (n+m-1)} - \frac{1}{(n+1)(n+2) \cdots (n+m)} \right],$$

so the sum from $n = 1$ collapses to $\frac{1}{m \cdot m!}$. The case $m = 2$ gives $\frac{1}{4}$; $m = 1$ gives 1; $m = 3$ gives $\frac{1}{18}$.

Both standard sums fall out at once. From the first identity,

$$\sum_{n=1}^N \frac{1}{n(n+1)(n+2)} = \frac{1}{2} \left[\frac{1}{1 \cdot 2} - \frac{1}{(N+1)(N+2)} \right] \rightarrow \frac{1}{4}.$$

From the second,

$$\sum_{n=1}^N \frac{1}{4n^2 - 1} = \frac{1}{2} \left[1 - \frac{1}{2N+1} \right] \rightarrow \frac{1}{2}.$$

A third, with a gap of two rather than one: $\frac{1}{n(n+2)} = \frac{1}{2} \left[\frac{1}{n} - \frac{1}{n+2} \right]$ telescopes with *two* surviving head terms, giving $\frac{1}{2} \left(1 + \frac{1}{2} \right) = \frac{3}{4}$. Notice how the gap governs the answer. With a gap of 1 one head term survives, with a gap of 2 two survive, and with a gap of q the value is $\frac{1}{q} \left(1 + \frac{1}{2} + \cdots + \frac{1}{q} \right) = \frac{H_q}{q}$. Checking that formula against the two computations above is a thirty second audit of the whole technique.

Decomposing $\frac{1}{n(n+1)(n+2)}$ as $\frac{A}{n} + \frac{B}{n+1} + \frac{C}{n+2}$ and then summing the three pieces separately. Each piece diverges, so the split is meaningless as a statement about infinite series; it is legal only inside a partial sum, where the divergences are finite and cancel. Group first, split second.

SEE ALSO Telescoping; digamma and harmonic number closed forms; arctangent difference telescoping.

Rational terms are the obvious factory. The less obvious one is inverse trigonometric, where the subtraction formula for arctan turns a difference of two simple angles into a single ugly one. Reading that formula backwards is the entire technique, and it is worth isolating because the ugly form is what a problem will actually hand you.

9.3 Arctangent difference telescoping

MEDIUM

CUE An arctan of a rational function of n . The diagnostic denominator is $n^2 + n + 1$, and its relatives $n^2, 2n^2, n^2 + n$, all of which are $1 + ab$ for a simple pair a, b .

METHOD Use $\arctan a - \arctan b = \arctan \frac{a-b}{1+ab}$ from right to left. With $a = \frac{1}{n}$ and $b = \frac{1}{n+1}$ the right side is $\arctan \frac{1}{n^2 + n + 1}$; with $a = n + 1$ and $b = n - 1$ it is $\arctan \frac{2}{n^2}$. Once the term is a difference of two arctangents, telescope and use $\arctan t \rightarrow \pi/2$.

WORKED EXAMPLE

$$\sum_{n=1}^{\infty} \arctan \frac{2}{n^2}.$$

Here $\frac{(n+1) - (n-1)}{1 + (n+1)(n-1)} = \frac{2}{n^2}$, so the term is $\arctan(n+1) - \arctan(n-1)$, a two step telescope. The partial sum is

$$\arctan(N+1) + \arctan N - \arctan 1 - \arctan 0 \rightarrow \frac{\pi}{2} + \frac{\pi}{2} - \frac{\pi}{4} = \frac{3\pi}{4}.$$

The one step cousin is $\sum_{n=1}^{\infty} \arctan \frac{1}{n^2 + n + 1} = \arctan 1 - \lim \arctan \frac{1}{N+1} = \frac{\pi}{4}$.

WHERE THIS GOES WRONG

Ignoring the branch. The subtraction formula as written requires $ab > -1$; outside that range the true value differs from the principal arctan by $\pm\pi$. In the example this is safe because $(n+1)(n-1) = n^2 - 1 \geq 0$ for all $n \geq 1$, but a problem with a sign flip in it will punish an unchecked application.

SEE ALSO Telescoping; partial fractions engineered to telescope.

Differentiate or integrate the parent

When nothing collapses, the next question is whether the series is a known one in disguise. The single most productive parent is the geometric series, because term by term differentiation and integration are legal inside the radius of convergence and each operation moves a factor of n up or down. An entire family of closed forms descends from one line.

9.4 Differentiating or integrating a known power series

MEDIUM

CUE Coefficients that carry a factor of n, n^2 or $\frac{1}{n}$ relative to a geometric progression. A sum such as $\sum n/2^n$ or $\sum x^n/n$ is the geometric series with one derivative or one integral applied.

METHOD Start from $\sum_{n \geq 0} x^n = \frac{1}{1-x}$ on $|x| < 1$. Differentiate and multiply by x to raise the power of n ; integrate from 0 to x to lower it:

$$\sum_{n \geq 1} nx^n = \frac{x}{(1-x)^2}, \quad \sum_{n \geq 1} n^2 x^n = \frac{x(1+x)}{(1-x)^3}, \quad \sum_{n \geq 1} \frac{x^n}{n} = -\ln(1-x).$$

The operator $x \frac{d}{dx}$ is the machine: applying it to $\frac{1}{1-x}$ once, twice, three times generates the whole ladder.

WORKED EXAMPLE

Evaluate $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$. Put $x = \frac{1}{2}$ into the second identity:

$$\frac{x(1+x)}{(1-x)^3} = \frac{\frac{1}{2} \cdot \frac{3}{2}}{(\frac{1}{2})^3} = \frac{3/4}{1/8} = 6.$$

The same machine gives $\sum n/2^n = 2$ and, after one more application of $x \frac{d}{dx}$, $\sum n^3/2^n = 26$ from $\frac{x(1+4x+x^2)}{(1-x)^4}$.

A limit that looks like something else entirely: $\lim_{n \rightarrow \infty} n \sum_{k=0}^{\infty} \frac{k}{n^k}$. The inner sum is the first identity at $x = 1/n$, namely $\frac{1/n}{(1-1/n)^2}$, so the whole expression is $\frac{1}{(1-1/n)^2} \rightarrow 1$.

THE OPERATOR $x \frac{d}{dx}$

Write $\theta = x \frac{d}{dx}$. Then θ multiplies the n th coefficient by n , so $\sum_{n \geq 0} P(n)x^n = P(\theta)[\frac{1}{1-x}]$ for any polynomial P . Two consequences are worth stating. First, the answer is always a rational function with denominator $(1-x)^{\deg P+1}$, so you can predict its shape before computing. Second, the inverse operation, $\int_0^x \frac{dt}{t}$, divides coefficients by n and is what produces $-\ln(1-x)$, $\text{Li}_2(x)$, and the whole polylogarithm ladder from the same parent.

WHERE THIS GOES WRONG

Differentiating termwise outside the radius of convergence, or evaluating the result at the endpoint $|x| = 1$ without comment. Term by term differentiation is legal on the open disc only; $\sum x^n/n$ at $x = -1$ does equal $-\ln 2$, but that equality is Abel's theorem, not termwise integration.

SEE ALSO Known Maclaurin series evaluated at a point; generating functions and binomial closed forms; Abel summation.

The geometric parent handles coefficients that are polynomial in n . Factorials need a different parent, and there the recognition problem is sharper: the reader must match a numerator that has been deliberately roughened against a clean standard series. The remedy is almost always algebraic surgery on the numerator before any matching is attempted.

MEDIUM

9.5 Known Maclaurin series evaluated at a point

CUE Factorials of the form $n!$, $(2n)!$ or $(2n+1)!$ in the denominator, usually with an alternating sign and a low degree polynomial in n upstairs.

METHOD Match against the three standard expansions

$$e^x = \sum_{n \geq 0} \frac{x^n}{n!}, \quad \cos x = \sum_{n \geq 0} \frac{(-1)^n x^{2n}}{(2n)!}, \quad \sin x = \sum_{n \geq 0} \frac{(-1)^n x^{2n+1}}{(2n+1)!},$$

evaluated at $x = 1$ or another convenient point. If the numerator is a polynomial in n , split it into pieces that each cancel part of the factorial: write the polynomial in terms of $(2n+1)$, or of n so that $n/n! = 1/(n-1)!$.

WORKED EXAMPLE

$$T = \sum_{n=0}^{\infty} (-1)^n \frac{n+1}{(2n+1)!}.$$

Write $n+1 = \frac{1}{2}(2n+1) + \frac{1}{2}$. The first half gives $\frac{(2n+1)}{2(2n+1)!} = \frac{1}{2(2n)!}$, and the second gives $\frac{1}{2(2n+1)!}$. So

$$T = \frac{1}{2} \sum_{n \geq 0} \frac{(-1)^n}{(2n)!} + \frac{1}{2} \sum_{n \geq 0} \frac{(-1)^n}{(2n+1)!} = \frac{\cos 1 + \sin 1}{2} \approx 0.690887.$$

A second instance, with $n!$ rather than $(2n+1)!$ and no alternation:

$$\sum_{n=0}^{\infty} \frac{n^2+1}{n!}.$$

Here the useful rewriting is $n^2 = n(n-1) + n$, because $\frac{n(n-1)}{n!} = \frac{1}{(n-2)!}$ and $\frac{n}{n!} = \frac{1}{(n-1)!}$. Every piece is then e , and the sum is $e + e + e = 3e \approx 8.154845$. The general rule: against $n!$, expand the numerator in *falling factorials* $n(n-1) \cdots (n-j+1)$, not in powers of n .

WHERE THIS GOES WRONG

Trying to force the entire sum onto a single standard series. The algebraic split is not a step in the technique, it is the technique; a numerator of $n+1$ against $(2n+1)!$ has no chance of matching anything until it is rewritten in terms of $2n+1$.

SEE ALSO Differentiating or integrating a known power series; known zeta and eta values; special function series identities.

Fourier series, and where the zeta values come from

Power series are read at points inside a disc. Fourier series are read at points of an interval, and because the coefficients of a Fourier expansion involve $1/n$ or $1/n^2$, evaluating one at a well chosen point is the standard manufacturing route for the closed forms of $\sum n^{-2}$, $\sum n^{-4}$ and the alternating odd reciprocals. A reader who knows only that these values exist is helpless; a reader who knows where they come from can generate the next one.

EASY

9.6 Known ζ and η values

CUE A clean power of n downstairs, possibly with alternating signs, possibly restricted to odd indices. There is nothing to manipulate; the task is to recognise and to perform the odd/even split correctly.

METHOD Quote the table

$$\zeta(2) = \frac{\pi^2}{6}, \quad \zeta(4) = \frac{\pi^4}{90}, \quad \eta(1) = \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} = \ln 2, \quad \sum_{n \geq 0} \frac{(-1)^n}{2n+1} = \frac{\pi}{4},$$

and generate the restricted variants by splitting. Since the even indexed terms of $\sum n^{-s}$ are 2^{-s} times the whole sum,

$$\sum_{n \text{ odd}} \frac{1}{n^s} = (1 - 2^{-s})\zeta(s), \quad \eta(s) = \sum_{n \geq 1} \frac{(-1)^{n+1}}{n^s} = (1 - 2^{1-s})\zeta(s).$$

In particular $\sum_{k \geq 0} (2k+1)^{-2} = \frac{3}{4}\zeta(2) = \frac{\pi^2}{8}$ and $\sum (-1)^{n+1} n^{-2} = \frac{1}{2}\zeta(2) = \frac{\pi^2}{12}$.

WORKED EXAMPLE

Where $\zeta(2)$ comes from. On $[-\pi, \pi]$ the function x^2 has the cosine expansion

$$x^2 = \frac{\pi^2}{3} + 4 \sum_{n \geq 1} \frac{(-1)^n}{n^2} \cos nx.$$

At $x = \pi$ every $\cos n\pi = (-1)^n$, so the signs cancel and $\pi^2 = \frac{\pi^2}{3} + 4\zeta(2)$, giving $\zeta(2) = \frac{\pi^2}{6}$.

Now apply Parseval to the same expansion. With $a_0 = \frac{2\pi^2}{3}$ and $a_n = \frac{4(-1)^n}{n^2}$,

$$\frac{1}{\pi} \int_{-\pi}^{\pi} x^4 dx = \frac{a_0^2}{2} + \sum_{n \geq 1} a_n^2 \Rightarrow \frac{2\pi^4}{5} = \frac{2\pi^4}{9} + 16\zeta(4),$$

so $16\zeta(4) = 2\pi^4(\frac{1}{5} - \frac{1}{9}) = \frac{8\pi^4}{45}$ and $\zeta(4) = \frac{\pi^4}{90}$.

The Leibniz series is the same trick on a square wave: the odd function equal to 1 on $(0, \pi)$ has expansion $\frac{4}{\pi} \sum_{k \geq 0} \frac{\sin(2k+1)x}{2k+1}$, and reading it at $x = \pi/2$ gives $1 = \frac{4}{\pi}(1 - \frac{1}{3} + \frac{1}{5} - \dots)$, that is $\frac{\pi}{4}$.

WHICH POINT TO READ AT

The choice of evaluation point is the only creative step. Read a cosine expansion at an endpoint of the interval, where $\cos n\pi = (-1)^n$ collapses an alternating series into a positive one. Read a sine expansion at the midpoint, where $\sin \frac{n\pi}{2}$ kills the even terms and leaves the odd ones with alternating signs. If neither point gives the series you want, apply Parseval instead: it squares the coefficients and therefore doubles the exponent, taking you from $\zeta(2)$ to $\zeta(4)$, and from $\zeta(4)$ to $\zeta(8)$.

WHERE THIS GOES WRONG

Inventing a closed form for $\zeta(3)$, or for any odd zeta value. None is known. If a computation lands on $\zeta(3)$ and the problem promises an elementary answer, the error is upstream. The second trap is misapplying the even/odd split: the factor is $1 - 2^{-s}$ for the odd part and $1 - 2^{1-s}$ for the alternating sum, and confusing them costs a factor of two.

SEE ALSO Known Maclaurin series evaluated at a point; integral representation and interchange; special function series identities.

Fourier expansions also work in the other direction. If a series has a $\cos n\theta$ or $\sin n\theta$ in it, or acquires one after an integral is performed inside the summand, the closed form may be a Fourier series read backwards. That is the second half of the next technique, whose first half is the more common manoeuvre: replacing a reciprocal by an integral so that the sum becomes geometric.

9.7 Integral representation and sum integral interchange

CUE A denominator that is linear or quadratic in the index, most often n , $2n + 1$ or $3n + 1$, and no visible telescope. Such a denominator is an integral: $\frac{1}{n} = \int_0^1 t^{n-1} dt$ and $\frac{1}{2r+1} = \int_0^1 t^{2r} dt$.

METHOD Substitute the integral for the reciprocal, exchange sum and integral, sum the resulting geometric series in closed form, and integrate what is left. The exchange needs a licence: monotone convergence for non-negative terms, dominated convergence otherwise, or uniform convergence on compact subsets followed by a limit at the endpoint.

WORKED EXAMPLE

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{3n+1} = \sum_{n=0}^{\infty} (-1)^n \int_0^1 t^{3n} dt = \int_0^1 \sum_{n=0}^{\infty} (-t^3)^n dt = \int_0^1 \frac{dt}{1+t^3}.$$

Partial fractions on $1+t^3 = (1+t)(t^2-t+1)$ then give

$$\int_0^1 \frac{dt}{1+t^3} = \frac{\ln 2}{3} + \frac{\pi}{3\sqrt{3}} \approx 0.835649,$$

which agrees with the alternating sum to six places. The interchange is licensed on $[0, 1-\epsilon]$ by uniform convergence and extended to $t = 1$ by Abel's theorem.

A harder instance, where an integral is already present. Consider

$$\lim_{n \rightarrow \infty} \sum_{r=0}^n \frac{1}{2r+1} \int_0^1 \sin((2r+1)x) dx.$$

The inner integral is $\frac{1 - \cos(2r+1)}{2r+1}$, so the object is $\sum_{r=0}^n \frac{1 - \cos(2r+1)}{(2r+1)^2}$. Split it: the constant part is $\frac{\pi^2}{8}$, and the cosine part is a Fourier series read at $t = 1$, namely $\sum_{r=0}^n \frac{\cos(2r+1)t}{(2r+1)^2} = \frac{\pi(\pi-2t)}{8}$ on $[0, \pi]$. Hence the value is

$$\frac{\pi^2}{8} - \frac{\pi(\pi-2)}{8} = \frac{\pi}{4}.$$

WHERE THIS GOES WRONG

Swapping sum and integral without justification. For a family with mixed signs and no dominating function the interchange is not a technicality that can be waved through; it is sometimes simply false, and the two orders give different numbers. Note also that the second example is engineered so that a reader who evaluates only the constant part reports $\pi^2/8$ and stops one step early.

SEE ALSO Known zeta and eta values; double sums and order interchange; generating functions and binomial closed forms.

Generating functions, products, and summation by parts

The remaining techniques all convert the series into a different kind of object. A generating function turns it into a function of a dummy variable to be evaluated at a point; a product turns into a sum of logarithms; summation by parts turns it into a different series with better behaviour. The common cost is that all three routinely need an evaluation on the boundary of a disc, which is exactly where Abel's theorem is required and exactly where careless work fails.

9.8 Generating functions and binomial closed forms

CUE Coefficients that are combinatorial rather than analytic: central binomials $\binom{2n}{n}$, Catalan numbers, harmonic numbers H_n , or anything defined by a recurrence.

METHOD Attach the coefficients to x^n , identify the resulting function, evaluate at the required x . The three workhorses are

$$\sum_{n \geq 0} \binom{2n}{n} x^n = \frac{1}{\sqrt{1-4x}} \quad (|x| < \frac{1}{4}), \quad \sum_{n \geq 1} H_n x^n = \frac{-\ln(1-x)}{1-x}, \quad \sum_{n \geq 0} C_n x^n = \frac{1 - \sqrt{1-4x}}{2x}.$$

The harmonic one is worth deriving rather than memorising: H_n is the n th partial sum of $\sum 1/k$, so its generating function is the Cauchy product of $-\ln(1-x)$ with $\frac{1}{1-x}$.

$$\sum_{n=1}^{\infty} \frac{H_n}{2^n} = \left. \frac{-\ln(1-x)}{1-x} \right|_{x=1/2} = \frac{\ln 2}{1/2} = 2 \ln 2 \approx 1.386294.$$

A deeper one, where the generating function is read at the edge of its disc. The identity

$$\sum_{n \geq 1} \frac{(2x)^{2n}}{n^2 \binom{2n}{n}} = 2 \arcsin^2 x$$

at $x = \frac{1}{2}$ gives $\sum_{n \geq 1} \frac{1}{n^2 \binom{2n}{n}} = 2 \left(\frac{\pi}{6} \right)^2 = \frac{\pi^2}{18} \approx 0.548311.$

And a Catalan sum, which is nothing more than the generating function at a point inside its disc:

$$\sum_{n \geq 0} \frac{C_n}{8^n} = \left. \frac{1 - \sqrt{1-4x}}{2x} \right|_{x=1/8} = \frac{1 - \sqrt{1/2}}{1/4} = 4 - 2\sqrt{2} \approx 1.171573.$$

WHERE THIS GOES WRONG

Evaluating a generating function at a boundary point of its disc of convergence without invoking Abel's theorem. The central binomial series has radius $\frac{1}{4}$, and $x = \frac{1}{4}$ is on the circle, where $1/\sqrt{1-4x}$ blows up while the series diverges: the identity survives the limit only where both sides do.

SEE ALSO Differentiating or integrating a known power series; infinite products; Abel summation and summation by parts.

A product over k is a sum of logarithms, and the previous techniques therefore apply to it verbatim. In practice the more efficient move is the reverse: recognise that the product itself telescopes, usually because the factors split into two families each of which telescopes separately, and evaluate without ever taking a logarithm.

9.9 Infinite products: Euler, Viete, and telescoping

HARD

CUE A product over k , or a sum of logarithms, which is the same object. Factorable numerators and denominators are the cue for a telescope; a sin or cos in the problem is the cue for a named product.

METHOD Either factor and telescope, or match one of

$$\frac{\sin x}{x} = \prod_{n \geq 1} \left(1 - \frac{x^2}{n^2 \pi^2} \right), \quad \prod_{k \geq 1} \cos \frac{x}{2^k} = \frac{\sin x}{x},$$

the second of which is Viete's formula, obtained by applying $\sin 2\theta = 2 \sin \theta \cos \theta$ repeatedly to get $\prod_{k=1}^n \cos \frac{x}{2^k} = \frac{\sin x}{2^n \sin(x/2^n)}$ and letting $n \rightarrow \infty$.

WORKED EXAMPLE

$$P = \prod_{k=2}^{\infty} \frac{k^3 - 1}{k^3 + 1}.$$

Factor both: $k^3 - 1 = (k-1)(k^2 + k + 1)$ and $k^3 + 1 = (k+1)(k^2 - k + 1)$. Two families telescope independently. The linear family gives

$$\prod_{k=2}^n \frac{k-1}{k+1} = \frac{1 \cdot 2 \cdots (n-1)}{3 \cdot 4 \cdots (n+1)} = \frac{2}{n(n+1)}.$$

For the quadratic family put $g(k) = k^2 - k + 1$; then $k^2 + k + 1 = g(k+1)$, so the product is $\frac{g(n+1)}{g(2)} = \frac{n^2 + n + 1}{3}$.

Multiplying,

$$\prod_{k=2}^n \frac{k^3 - 1}{k^3 + 1} = \frac{2(n^2 + n + 1)}{3n(n+1)} \rightarrow \frac{2}{3}.$$

Viete's formula at $x = \pi/2$ gives the companion $\prod_{i \geq 1} \cos \frac{\pi}{2^{i+1}} = \frac{\sin(\pi/2)}{\pi/2} = \frac{2}{\pi}$, equivalently $\sum_{i \geq 1} \ln \cos \frac{\pi}{2^{i+1}} = \ln \frac{2}{\pi} \approx -0.451583.$

Telescoping one of the two factor families and quietly discarding the boundary constants of the other. In the example the limit $\frac{2}{3}$ is manufactured entirely out of those constants, $\frac{2}{n(n+1)}$ against $\frac{n^2+n+1}{3}$; the tails contribute nothing. Losing a boundary constant here does not perturb the answer, it replaces it.

SEE ALSO Telescoping; known zeta and eta values; generating functions and binomial closed forms.

Summation by parts is the last of the conversion techniques and the most general. It is the discrete integration by parts, and like its continuous original it is used in two different modes: to evaluate, by trading a hard sum for an easy one, and to prove convergence, since Dirichlet's and Abel's tests are corollaries of the identity rather than independent results.

HARD

9.10 Abel summation and summation by parts

CUE A sum $\sum a_n b_n$ in which one factor has simple *partial sums* and the other is monotone or smooth. The classic pattern is a_n oscillating with bounded partial sums against b_n decreasing to zero.

METHOD With $A_N = \sum_{n \leq N} a_n$,

$$\sum_{n=1}^N a_n b_n = A_N b_N - \sum_{n=1}^{N-1} A_n (b_{n+1} - b_n).$$

Choose a to be the factor whose partial sums you can write down. If A_n is bounded and b_n decreases to 0, the boundary term vanishes and the new series converges absolutely, which is Dirichlet's test.

WORKED EXAMPLE

Accelerate $\sum_{n \geq 1} \frac{(-1)^{n+1}}{n}$. Take $a_n = (-1)^{n+1}$, so $A_n = 1$ for odd n and $A_n = 0$ for even n , and $b_n = \frac{1}{n}$, so $b_{n+1} - b_n = -\frac{1}{n(n+1)}$. The identity gives

$$\sum_{n=1}^N \frac{(-1)^{n+1}}{n} = \frac{A_N}{N} + \sum_{n=1}^{N-1} \frac{A_n}{n(n+1)} = \frac{A_N}{N} + \sum_{\substack{n < N \\ n \text{ odd}}} \frac{1}{n(n+1)}.$$

Letting $N \rightarrow \infty$ kills the boundary term and leaves

$$\ln 2 = \sum_{k \geq 0} \frac{1}{(2k+1)(2k+2)} = \frac{1}{1 \cdot 2} + \frac{1}{3 \cdot 4} + \frac{1}{5 \cdot 6} + \cdots,$$

whose terms decay like $\frac{1}{4k^2}$ instead of $\frac{1}{n}$. Ten terms of the new series beat ten thousand of the old.

TWO TESTS, ONE IDENTITY

Both standard tests for non-absolute convergence are readings of the same line. Dirichlet: if A_N is bounded and b_n decreases to 0, then $A_N b_N \rightarrow 0$ and $\sum |A_n| |b_{n+1} - b_n| \leq M \sum (b_n - b_{n+1}) < \infty$, so $\sum a_n b_n$ converges. Abel: if $\sum a_n$ converges and b_n is monotone and bounded, the same two estimates apply with A_N convergent rather than merely bounded. Memorising the identity makes both tests unmemorable in the good sense: you can rederive them in a line.

WHERE THIS GOES WRONG

Dropping the boundary term $A_N b_N$ before checking that it vanishes, and mis-indexing the second sum, which runs to $N-1$ and not to N because the shift $b_{n+1} - b_n$ consumes one index. Both errors are silent: they produce a plausible number.

SEE ALSO Digamma and harmonic number closed forms; telescoping; rearrangement and its dangers.

9.11 Digamma and harmonic number closed forms

CUE Reciprocals of an arithmetic progression, or a sign pattern that repeats with period 3, 4 or 6. Individually the pieces diverge; only the grouped combination converges.

METHOD Write partial sums through the digamma function $\psi(x) = \frac{d}{dx} \ln \Gamma(x)$, using

$$\sum_{k=0}^K \frac{1}{qk+a} = \frac{1}{q} \left[\psi\left(K+1+\frac{a}{q}\right) - \psi\left(\frac{a}{q}\right) \right].$$

Because $\psi(K+c) = \ln K + O(1/K)$ for any fixed c , the K dependent digammas cancel exactly when the signs in the group sum to zero, leaving a combination of special values. The ones to know are $\psi(1) = -\gamma$, $\psi(\frac{1}{2}) = -\gamma - 2\ln 2$, and the reflection formula $\psi(1-x) - \psi(x) = \pi \cot \pi x$, whence $\psi(\frac{3}{4}) - \psi(\frac{1}{4}) = \pi$.

WORKED EXAMPLE

$$S = \sum_{k=1}^{\infty} \left(\frac{1}{4k+1} + \frac{1}{4k+2} - \frac{1}{4k+3} - \frac{1}{4k+4} \right).$$

Let T be the same sum started at $k=0$, so $S = T - (1 + \frac{1}{2} - \frac{1}{3} - \frac{1}{4}) = T - \frac{11}{12}$. With $q=4$ the four digamma arguments are $K+1+\frac{1}{4}$, $K+1+\frac{1}{2}$, $K+1+\frac{3}{4}$, $K+2$, carrying signs $+, +, -, -$; each is $\ln K + o(1)$, so they cancel and

$$T = \frac{1}{4} \left[-\psi\left(\frac{1}{4}\right) - \psi\left(\frac{1}{2}\right) + \psi\left(\frac{3}{4}\right) + \psi(1) \right] = \frac{1}{4} \left[\pi + (-\gamma + \gamma + 2\ln 2) \right] = \frac{\pi}{4} + \frac{\ln 2}{2}.$$

Hence $S = \frac{\pi}{4} + \frac{\ln 2}{2} - \frac{11}{12} \approx 0.215305$, confirmed numerically as 0.2153045 after 4×10^5 groups.

SPLITTING IS NOT REARRANGING

T can also be found without digamma: its odd indexed terms are $1 - \frac{1}{3} + \frac{1}{5} - \dots = \frac{\pi}{4}$ and its even indexed terms are $\frac{1}{2}(1 - \frac{1}{2} + \frac{1}{3} - \dots) = \frac{\ln 2}{2}$. This is legal, even though T is only conditionally convergent, because we are not reordering anything: the partial sum T_{4m} is written as $P_{2m} + Q_{2m}$ where both P and Q converge, and a sum of two convergent sequences converges to the sum of the limits. The illegal operation is reordering, not splitting a partial sum.

WHERE THIS GOES WRONG

Breaking the group of four into four separate infinite sums. Each of $\sum \frac{1}{4k+1}$, $\sum \frac{1}{4k+2}$ and the rest diverges logarithmically; only their signed combination is finite. The grouping is load bearing and must survive to the end of the calculation.

SEE ALSO Abel summation and summation by parts; partial fractions engineered to telescope; special function series identities.

9.12 Special function series identities

CUE A named constant appearing as the *coefficient* of the summand: $\zeta(n)$, a dilogarithm value, $\ln \Gamma$, or a Lambert W . The sum is then a Taylor series of a special function, usually read at the edge of its disc.

METHOD Recognise the expansion. Two that recur:

$$\ln \Gamma(1+x) = -\gamma x + \sum_{n \geq 2} \frac{(-1)^n \zeta(n)}{n} x^n \quad (|x| < 1), \quad \text{Li}_2(-x) + \text{Li}_2(-1/x) = -\frac{\pi^2}{6} - \frac{1}{2} \ln^2 x \quad (x > 0).$$

The second is the dilogarithm inversion formula, and its usual role is to cancel a divergence that has been engineered into a limit.

WORKED EXAMPLE

Put $x=1$ in the $\ln \Gamma$ expansion. The left side is $\ln \Gamma(2) = \ln 1 = 0$, so

$$0 = -\gamma + \sum_{n \geq 2} \frac{(-1)^n \zeta(n)}{n} \quad \Rightarrow \quad \sum_{n \geq 2} \frac{(-1)^n \zeta(n)}{n} = \gamma \approx 0.5772157.$$

The series converges slowly, since $\zeta(n) \rightarrow 1$ and the terms behave like $(-1)^n/n$; splitting off $\sum_{n \geq 2} (-1)^n/n = 1 - \ln 2$ and summing $\sum_{n \geq 2} (-1)^n (\zeta(n) - 1)/n$, whose terms fall like 2^{-n} , confirms the value in a dozen terms.

The dilogarithm identity, used as a cancellation: as $x \rightarrow \infty$, $-\text{Li}_2(-x) = \frac{\pi^2}{6} + \frac{1}{2} \ln^2 x + \text{Li}_2(-1/x)$, and $\frac{1}{2} \ln^2(x+1) - \frac{1}{2} \ln^2 x \rightarrow 0$, so

$$\lim_{x \rightarrow \infty} \left(-\text{Li}_2(-x) - \frac{1}{2} \ln^2(x+1) \right) = \frac{\pi^2}{6}.$$

Both divergent $\ln^2$ pieces were placed there to annihilate each other.

The dilogarithm's other functional equation, $\text{Li}_2(x) + \text{Li}_2(1-x) = \frac{\pi^2}{6} - \ln x \ln(1-x)$, evaluates the one dilogarithm value a reader is likely to meet: at $x = \frac{1}{2}$ the two sides are equal, so

$$\text{Li}_2\left(\frac{1}{2}\right) = \sum_{n \geq 1} \frac{1}{n^2 2^n} = \frac{\pi^2}{12} - \frac{\ln^2 2}{2} \approx 0.582241.$$

WHERE THIS GOES WRONG

Substituting the boundary point without checking that the series converges there. The $\ln \Gamma(1+x)$ expansion has radius exactly 1, and $x = 1$ sits on the circle; the substitution is legitimate only because the series converges at that point and $\ln \Gamma$ is continuous up to it.

SEE ALSO Known zeta and eta values; infinite products; digamma and harmonic number closed forms.

Two indices, and the order you take them in

Everything so far has been a single sum. Once there are two indices, or one index and one integration variable, a new question appears that has no analogue in the one dimensional case: whether the answer depends on the order in which the limits are taken. It usually does not, and there is a clean theorem saying when. When the theorem's hypotheses fail, the answer usually does depend on the order, and the failure is not detectable by inspecting either order alone.

9.13 Double sums and order interchange

HARD

CUE A visible $\sum_m \sum_n$, or a single sum whose summand is itself a sum or an integral, or a limit in which two parameters go to infinity in a stated order.

METHOD Swap freely when the terms are non-negative (Tonelli) or when $\sum_{m,n} |a_{mn}| < \infty$ (Fubini). Choose the order in which the inner sum has a closed form. If the index ranges are dependent, redraw the region and rewrite the limits before swapping.

WORKED EXAMPLE

$$L = \lim_{m \rightarrow \infty} \lim_{n \rightarrow \infty} \sum_{k=1}^m \sum_{r=kn+1}^{(k+1)n} \frac{n}{r^2}.$$

Fix k and take $n \rightarrow \infty$ first. Writing $r = kn + j$ with $1 \leq j \leq n$,

$$\sum_{r=kn+1}^{(k+1)n} \frac{n}{r^2} = \frac{1}{n} \sum_{j=1}^n \frac{1}{(k + \frac{j}{n})^2} \rightarrow \int_k^{k+1} \frac{dx}{x^2} = \frac{1}{k} - \frac{1}{k+1}.$$

The inner limit is a Riemann sum in disguise. The outer sum then telescopes: $\sum_{k=1}^m \left(\frac{1}{k} - \frac{1}{k+1} \right) = 1 - \frac{1}{m+1} \rightarrow 1$, so $L = 1$.

A genuine interchange, where the swap is the whole idea:

$$\sum_{n=2}^{\infty} (\zeta(n) - 1) = \sum_{n \geq 2} \sum_{k \geq 2} \frac{1}{k^n} = \sum_{k \geq 2} \sum_{n \geq 2} \frac{1}{k^n} = \sum_{k \geq 2} \frac{1/k^2}{1 - 1/k} = \sum_{k \geq 2} \frac{1}{k(k-1)} = 1.$$

Every term is positive, so Tonelli licenses the swap without further comment, the inner sum becomes geometric, and the outer sum telescopes. Three of this chapter's techniques in one line.

Swapping a doubly indexed family with mixed signs and no absolute summability. The standard counterexample is $a_{mn} = 1$ when $m = n$, $a_{mn} = -1$ when $n = m + 1$, and 0 otherwise: every row sums to 0, so one order gives 0, while every column sums to 0 except the first, which sums to 1, so the other order gives 1. Nothing in either single computation reveals the problem.

SEE ALSO Integral representation and interchange; telescoping; rearrangement and its dangers.

The counterexample above is the two index version of a one index hazard, and it is worth meeting the one index version head on, because it is the single most common way a correct evaluation is turned into an incorrect one. A conditionally convergent series does not have a sum in the way an absolutely convergent one does. It has a sum *in the given order*, and reordering it destroys that number completely.

MEDIUM

9.14 Rearrangement and its dangers

CUE A conditionally convergent series being reordered, regrouped across sign changes, or split into two infinite subsums. Also any manipulation that treats $\sum a_n$ as though addition were commutative in the infinite case.

METHOD Riemann's theorem: if $\sum a_n$ converges but $\sum |a_n|$ diverges, then for *any* prescribed $L \in \mathbb{R} \cup \{\pm\infty\}$ there is a rearrangement converging to L . The construction is greedy and completely explicit: add positive terms in order until the running total first exceeds L , then add negative terms until it first drops below L , and repeat forever. It works because the positive terms alone sum to $+\infty$ and the negative terms alone to $-\infty$, so each phase must terminate, while $a_n \rightarrow 0$ forces the overshoot at each turn to 0.

WORKED EXAMPLE

Rearrange $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$, whose sum is $\ln 2 \approx 0.693147$, to converge to 2. Run the greedy algorithm on the odd and even reciprocals:

- $1 + \frac{1}{3} + \frac{1}{5} + \dots + \frac{1}{15} = 2.021800$, the first time the total exceeds 2 (eight terms);
- subtract $\frac{1}{2}$: 1.521800;
- add $\frac{1}{17} + \frac{1}{19} + \dots + \frac{1}{41} =$ thirteen more odd terms, reaching 2.004063;
- subtract $\frac{1}{4}$: 1.754063, and so on.

The overshoots are 0.021800, then 0.004063, then smaller still, and the running total converges to 2. Every term of the original series is used exactly once.

The structured version is cleaner. Taking r positive terms for each negative term gives

$$\sum_{j=1}^{rm} \frac{1}{2j-1} - \sum_{j=1}^m \frac{1}{2j} \approx \left(\frac{1}{2} \ln(rm) + \ln 2 + \frac{\gamma}{2} \right) - \frac{1}{2} (\ln m + \gamma) = \ln 2 + \frac{1}{2} \ln r.$$

So $r = 2$ gives $\frac{3}{2} \ln 2 \approx 1.039721$, confirmed numerically as 1.0397195, and not $\ln 2$.

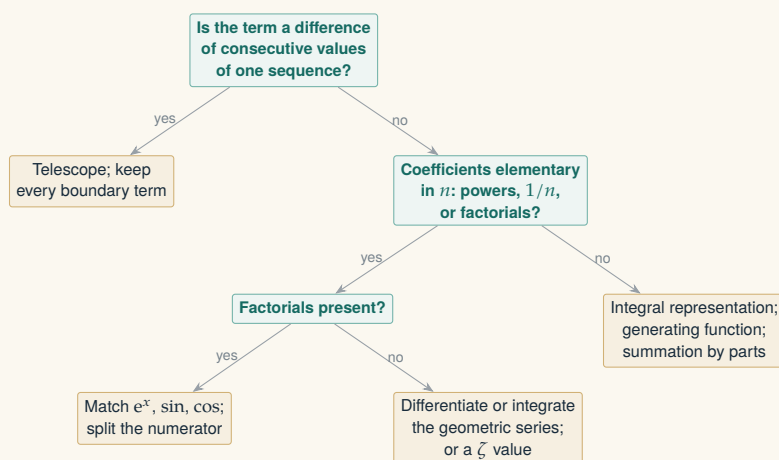
WHERE THIS GOES WRONG

Writing $\sum \frac{(-1)^{n+1}}{n} = \sum_{\text{odd}} \frac{1}{n} - \sum_{\text{even}} \frac{1}{n}$. Neither sum on the right exists, so the identity is not false, it is meaningless. Contrast the digamma computation earlier in this chapter, where a partial sum was split into two convergent partial sums: that is legal, and this is not.

SEE ALSO Double sums and order interchange; Abel summation and summation by parts; telescoping.

Choosing the move

The techniques in this chapter are close to disjoint in their cues, so the choice is nearly mechanical. Look at the general term and ask, in order: does it collapse, is it geometric with polynomial decoration, is it a standard expansion read at a point. Only if all three fail do you spend effort on a conversion. The tree below is that order, and it is worth internalising because the expensive techniques at the bottom are also the ones with hypotheses to check.



One caution about the tree. It routes on the shape of the term, not on difficulty, and the left branch is not always the easy one: a telescope can be very deeply buried, as the Sophie Germain example shows. What the tree guarantees is that you will not reach for a generating function before you have checked whether the sum simply collapses.

A second caution concerns the bottom right box, which is not one technique but three, ordered by cost. An integral representation is cheapest and should be tried first: it needs only that some factor of the term be writable as $\int_0^1 t^\alpha dt$. A generating function is next, and requires that you recognise the coefficients as a named combinatorial family. Summation by parts is last, because it does not by itself produce a closed form; it produces a different series, and you still have to evaluate that.

CONFIRM BEFORE YOU COMMIT

Every closed form in this chapter can be checked in under a minute by summing a few thousand terms numerically. Do it. The characteristic error in series evaluation is not a wrong method but a lost constant, a dropped boundary term, or a factor of two from a mishandled odd/even split, and all three show up instantly as a mismatch in the third decimal place. If the partial sums approach your answer but do not reach it, compare the residual against the expected tail: a discrepancy of size $1/N$ is a truncation artefact, a discrepancy that stops shrinking is a mistake.

Recognition matrix

If you see	Reach for	If that fails
Denominator factoring as $g(k)g(k+1)$	Telescope with $c_k = (\text{linear})/g(k)$	Partial fractions, then group
$\frac{1}{n(n+1)(n+2)}, \frac{1}{4n^2-1}$	Partial fractions into blocks that shift	Digamma partial sums
arctan of a rational function	$\arctan a - \arctan b$ read backwards	Check the branch, $ab > -1$
n, n^2 or $1/n$ against x^n	Apply $x \frac{d}{dx}$ or $\int_0^x$ to $\frac{1}{1-x}$	Generating function of the coefficients
$n!, (2n)!, (2n+1)!$ downstairs	Split the numerator, match $e^x, \sin, \cos$	Integral representation
$1/n^2, 1/n^4$, odd or alternating	$\zeta(2), \zeta(4)$ and the $1-2^{-s}$ split	Fourier series at a chosen point
$1/(qn+a)$ with no telescope	$\int_0^1 t^{qn+a-1} dt$, then swap	Digamma special values
$\binom{2n}{n}, C_n, H_n$	Named generating function	Abel's theorem at the edge
A product over k	Factor into two telescoping families	Take logs; Viète or Euler
$\sum a_n b_n, A_N$ simple	Summation by parts	Dirichlet's test for convergence only
$\zeta(n)$ or Li_2 as a coefficient	Taylor series of $\ln \Gamma$ or Li_2	Inversion formula to cancel divergences
$\sum_m \sum_n$ with mixed signs	Test absolute summability first	Compute both orders and compare
Any reordering of an alternating sum	Stop: Riemann's theorem applies	Restore the original order

Several variables, rigour, and limits built to punish reflexes

What this chapter is for. Three things go wrong once a reader is fluent with one variable limits, and this chapter treats all three. First, in two variables a limit must agree along every approach at once, so the craft splits cleanly in two: disproof is a hunt for two paths with different values, and proof is almost always a polar bound that must be uniform in the angle. Second, the ε - δ definition stops being ceremony and becomes the only instrument that settles a genuinely ambiguous case; writing one is mechanical once you know the order of the moves. Third, a whole class of limits is engineered so that the first reflex gives a confident wrong answer: L'Hôpital loops back on itself, a floor function is treated as continuous, a parameter hides a case split, or an asymptotic replacement is smuggled across a subtraction. By the end you should be able to decide within thirty seconds whether a two variable limit is a path problem or a polar problem, produce a correct ε - δ proof for any elementary function, and recognise every one of the standard traps by its shape rather than by having been burnt by it.

Two variables: the limit must survive every approach

In one variable a point has two sides, and checking both is a finite job. In the plane a point has infinitely many, and they are not merely directions: a limit at the origin ranges over every curve, every spiral, every sequence of points tending to 0 . The definition is unchanged in form, $0 < \|(x, y)\| < \delta$ implies $|f(x, y) - L| < \varepsilon$, but its content is enormously stronger, because the punctured disc contains approaches that no one-variable intuition supplies.

That asymmetry organises the whole subject. The two questions, does the limit exist and what is it, are answered by two different and non-interchangeable techniques. To show that a limit fails you need exactly two approaches with different values, and finding them is a search. To show that a limit exists you need a single bound valid on the whole punctured disc, and no number of successful paths will substitute for it. Every mistake in this section is a case of using one tool for the other job.

The practical consequence is that you should decide which job you are doing before touching the algebra. A quick way to decide: substitute $y = mx$ and look at what happens to m . If m survives into the answer, you are done in one line and the limit does not exist. If m cancels, you have learned nothing yet, and the next move is either a cleverer path or a polar bound, according to the diagnostic developed below.

10.1 The path test for non-existence

MEDIUM

CUE A limit at the origin of a ratio built from monomials, where numerator and denominator have comparable size along the axes. Any $f(x, y) = P(x, y)/Q(x, y)$ with $Q(0, 0) = 0$ deserves the test before anything else is attempted.

METHOD Approach along a family of curves and watch the parameter. Straight lines $y = mx$ come first, together with the vertical approach $x = 0$, which the family $y = mx$ does not contain. If the value depends on m , the limit does not exist and you are finished. If every line gives the same value, escalate the degree: try $y = kx^2$, then $y = kx^3$. The rule for choosing the exponent is written into the denominator, as the worked examples show.

WORKED EXAMPLE

The standard counterexample first.

$$f(x, y) = \frac{x^2 y}{x^4 + y^2}.$$

Along $y = mx$ with $m \neq 0$, $f = \frac{mx^3}{x^4 + m^2 x^2} = \frac{mx}{x^2 + m^2} \rightarrow 0$; along $y = 0$ and along $x = 0$ the function is identically 0. Every straight line gives 0. But along $y = kx^2$,

$$f(x, kx^2) = \frac{kx^4}{x^4 + k^2 x^4} = \frac{k}{1 + k^2},$$

a constant depending on k : the value is $\frac{1}{2}$ on $y = x^2$, $\frac{2}{5}$ on $y = 2x^2$, 0 in the limit $k \rightarrow \infty$. Two parabolas with different values, so the limit does not exist.

The denominator dictated the exponent. In $x^4 + y^2$ the two terms balance when $y^2 \sim x^4$, that is on $y = kx^2$, and that is exactly the family on which the ratio stops decaying. Push the same construction up one degree:

$$g(x, y) = \frac{x^3 y}{x^6 + y^2}.$$

Here x^6 balances y^2 on $y = kx^3$, where $g = k/(1 + k^2)$. On $y = kx$ the value is $kx^2/(x^4 + k^2) \rightarrow 0$, and on $y = kx^2$ it is $kx/(x^2 + k^2) \rightarrow 0$: every polynomial path of degree below three is defeated, and only the cubic separates. Numerically, on $y = 5x$ the values at $x = 10^{-1}, 10^{-2}, 10^{-4}$ are 0.0020, 2.0×10^{-5} , 2.0×10^{-9} , while on $y = x^3$ they are 0.5 throughout.

WHERE THIS GOES WRONG

Concluding existence from all straight lines. The function $x^2 y / (x^4 + y^2)$ exists precisely to punish that inference, and a reader who has not held it in their hands will make the mistake once. A second, quieter failure: testing $y = mx$ and forgetting $x = 0$, which is not a member of that family and is sometimes the only path that misbehaves.

SEE ALSO Polar squeeze for existence; continuity and iterated limits; the squeeze theorem template.

FOUR SPECIMENS WORTH MEMORISING

The examinable stock of two variable counterexamples is small, and knowing it by sight replaces most of the searching. $\frac{xy}{x^2 + y^2}$ is constant on every line through the origin, so the lines alone kill it. $\frac{x^2 y}{x^4 + y^2}$ survives all lines and dies on $y = kx^2$. $\frac{x^3 y}{x^6 + y^2}$ survives every polynomial path of degree below three. And $\frac{x^3}{x^2 + y^2}$ is the one that genuinely converges, so it is the specimen against which you calibrate a polar proof. A problem set at this level is almost always one of the four with the variables renamed or a harmless factor attached.

Paths can only ever destroy a limit, never establish one. However many curves agree, the punctured disc contains uncountably many more, and the two examples above show that the disagreeing one can be hidden at any polynomial degree you like. So when the paths all agree, stop searching and switch instruments: convert to polar coordinates, where the whole disc is described by one small parameter and one bounded one.

10.2 Polar squeeze for existence

MEDIUM

CUE Every path you have tried gives the same value L , and you now want a proof rather than more evidence. Also any f whose numerator visibly carries more total degree than its denominator.

METHOD Set $x = r \cos \theta$, $y = r \sin \theta$, so that $(x, y) \rightarrow \mathbf{0}$ is exactly $r \rightarrow 0^+$, and try to produce a bound

$$|f(r \cos \theta, r \sin \theta) - L| \leq g(r) \quad \text{for every } \theta, \quad g(r) \rightarrow 0.$$

The quantifier “for every θ ” is the entire content of the technique. A bound whose constant depends on θ proves nothing at all, because the approach is free to send θ and r to their limits together. The workhorse inequalities are $|\cos \theta| \leq 1$, $|\sin \theta| \leq 1$ and $|\cos \theta \sin \theta| \leq \frac{1}{2}$, or in Cartesian form $|x| \leq r$, $|y| \leq r$ and $|xy| \leq \frac{1}{2}(x^2 + y^2)$.

WORKED EXAMPLE

Three that succeed and one that fails, so the diagnostic is visible.

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^3}{x^2 + y^2} = \frac{r^3 \cos^3 \theta}{r^2} = r \cos^3 \theta, \quad |r \cos^3 \theta| \leq r \rightarrow 0,$$

so the limit is 0, and the bound r contains no θ .

$$\frac{x^2 y^2}{x^2 + y^2} = r^2 \cos^2 \theta \sin^2 \theta \leq \frac{r^2}{4}, \quad (x^2 + y^2) \ln(x^2 + y^2) = r^2 \ln r^2 = 2r^2 \ln r \rightarrow 0.$$

In the last one θ does not appear at all, which is the ideal case and a signal that the function was really a function of r in disguise.

Now the failure. For $f = x^2y/(x^4 + y^2)$ the polar form is

$$f = \frac{r \cos^2 \theta \sin \theta}{r^2 \cos^4 \theta + \sin^2 \theta}.$$

Fix θ with $\sin \theta \neq 0$ and let $r \rightarrow 0$: the expression is $O(r)$ and tends to 0. That computation is worthless. It is the line test wearing polar clothing, and the bound it produces, $r \cos^2 \theta / |\sin \theta|$, blows up as $\theta \rightarrow 0$. Indeed choosing $\theta = \theta(r)$ with $\sin \theta = r \cos^2 \theta$, which is the parabola $y = x^2$ again, makes the denominator $2 \sin^2 \theta$ and the whole expression exactly $\frac{1}{2}$ for every r .

UNIFORM IN θ , OR IT IS NOT A PROOF

Read the bound aloud with the quantifiers in place: for every $\varepsilon > 0$ there is $\delta > 0$ such that for every $r < \delta$ and every θ , the deviation is below ε . If your δ came out as a formula containing θ , you have proved something about each ray separately and nothing about the disc. The practical test takes five seconds: strike out every trigonometric factor by replacing it with its worst case, 1 for a sine or cosine upstairs and, downstairs, whatever value makes the denominator smallest. If what survives still tends to 0, the proof is sound.

WHERE THIS GOES WRONG

Taking $r \rightarrow 0$ with θ held fixed and calling that a proof. It is the line test in disguise, and it certifies the false limit 0 for $x^2y/(x^4 + y^2)$. A related error is cancelling r from numerator and denominator and then dropping a factor whose reciprocal is unbounded in θ .

SEE ALSO The path test for non-existence; the squeeze theorem template; bounded times infinitesimal.

Existence of the joint limit is one question; continuity is that question plus a comparison with the function's actual value; and the iterated limits are a third object entirely, which beginners routinely mistake for the first. The relationship between them is not symmetric, and knowing which implication runs which way saves a great deal of wasted work.

MEDIUM

10.3 Continuity, and iterated versus joint limits

CUE A piecewise definition with a special value assigned at the origin, or a problem that writes $\lim_{x \rightarrow 0} \lim_{y \rightarrow 0} f(x, y)$ with the inner limit taken first.

METHOD f is continuous at $\mathbf{a}$ exactly when the joint limit exists and equals $f(\mathbf{a})$, so continuity is never established by computing iterated limits. The logic runs one way only: if the joint limit is L , then each iterated limit equals L *provided it exists at all*. The converse fails, and it fails in both available ways. Two iterated limits can exist and agree while the joint limit does not; and the joint limit can exist while an iterated limit does not.

WORKED EXAMPLE

Both failure directions, on two lines each.

$$f(x, y) = \frac{xy}{x^2 + y^2} : \quad \lim_{x \rightarrow 0} (\lim_{y \rightarrow 0} f) = \lim_{x \rightarrow 0} 0 = 0, \quad \lim_{y \rightarrow 0} (\lim_{x \rightarrow 0} f) = 0.$$

The iterated limits both exist and agree. Yet along $y = x$ the function is constantly $\frac{1}{2}$, and along $y = -x$ it is $-\frac{1}{2}$, so there is no joint limit and no assignment of $f(0, 0)$ makes f continuous. In polar form $f = \cos \theta \sin \theta$, a function of θ alone, which is the cleanest possible certificate of failure: it does not decay in r at all.

Now the other direction.

$$h(x, y) = y \sin \frac{1}{x} \quad (x \neq 0), \quad h(0, y) = 0.$$

Since $|h| \leq |y| \leq r$, the joint limit at the origin is 0. But for fixed $y \neq 0$ the inner limit $\lim_{x \rightarrow 0} y \sin(1/x)$ does not exist, so the iterated limit $\lim_{y \rightarrow 0} \lim_{x \rightarrow 0} h$ is not even defined. The joint limit is the stronger statement; the iterated ones are shadows of it that may be missing.

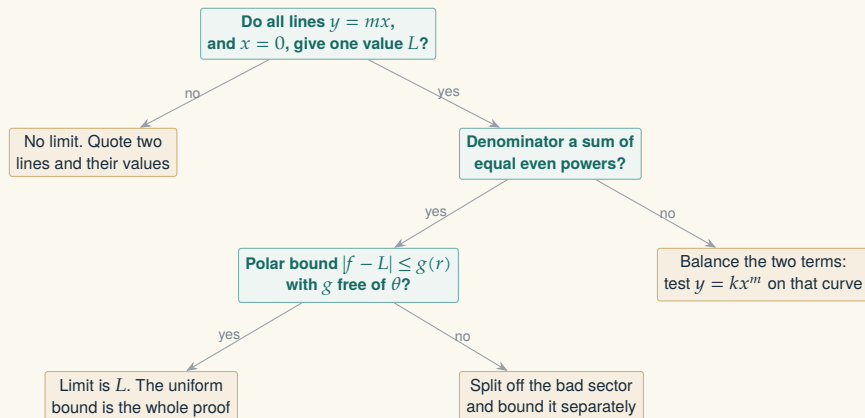
A third specimen worth carrying, where the iterated limits exist and *disagree*: $(x^2 - y^2)/(x^2 + y^2)$ gives 1 taking $y \rightarrow 0$ first and -1 taking $x \rightarrow 0$ first. Disagreement is conclusive: no joint limit can exist.

Finally, a continuity question posed the way an examiner poses it. Define $F(x, y) = \frac{x^3 - y^3}{x^2 + y^2}$ away from the origin and $F(0, 0) = c$. For which c is F continuous at the origin? In polars the quotient is $r(\cos^3 \theta - \sin^3 \theta)$, bounded in absolute value by $2r$ for every θ , so the joint limit is 0 and the answer is $c = 0$, uniquely. The whole exercise is one uniform bound; the iterated limits, both 0, would have suggested the same value without proving anything.

Computing an iterated limit and reporting it as the limit. This is the two variable form of the ordering hazard in nested limits, and it is more dangerous here because the notation $\lim_{(x,y) \rightarrow (0,0)}$ looks so much like the notation $\lim_{x \rightarrow 0} \lim_{y \rightarrow 0}$ that the substitution passes unnoticed.

SEE ALSO The path test for non-existence; nested limits, taken strictly in order; polar squeeze for existence.

The two techniques are complementary, and the choice between them is nearly mechanical once you have looked at the denominator. If the denominator is a sum of two even powers with the *same* exponent, such as $x^2 + y^2$ or $x^4 + y^4$, polar coordinates will decide the matter, because every direction is then on an equal footing and the trigonometric factors are bounded. If the exponents differ, as in $x^4 + y^2$ or $x^6 + y^2$, the denominator has a preferred curve along which its two terms balance, and that curve is where the counterexample lives. The tree records this.



The exponent to test is not guesswork. Assign x the weight 1 and y the weight m , so that the curve $y = kx^m$ becomes the statement that both variables have the same weighted size. A monomial $x^i y^j$ then has weighted degree $i + jm$. Choose m so that the two terms of the denominator have *equal* weighted degree; on that curve neither term dominates and the ratio stops decaying. For $x^4 + y^2$ the equation is $4 = 2m$, giving $m = 2$; for $x^6 + y^2$ it is $6 = 2m$, giving $m = 3$; for $x^2 + y^2$ it is $m = 1$, which is why the lines already decide that case and why polar coordinates work there. Then compare the numerator's weighted degree with the denominator's: equal degrees mean a k -dependent constant and no limit, a larger numerator degree means the limit survives that family, and a smaller one means the ratio blows up.

The right hand branch is the productive one for problems that were set rather than found. If a problem hands you $x^4 + y^2$ rather than $x^2 + y^2$, somebody chose the mismatch, and the only reason to choose it is to make the parabola work. Reading the exponents as an instruction is faster than any amount of experimenting.

Epsilon and delta, used as a tool

The definition is usually taught as a rite of passage and then abandoned. That is the wrong reading of it. It is the instrument of last resort, the thing that settles a case where every computational rule has run out: a function defined by cases, a limit that some authority asserts and you doubt, a claim about a function too irregular for expansion. It earns its place by being the only definition of what a limit is; everything else in this book is a theorem downstream of it.

Writing one is also completely mechanical, which is the part most treatments obscure by presenting the finished proof rather than the search. The search runs backwards and the proof runs forwards, and separating those two movements is the whole method. On scratch paper you start from the inequality you want, $|f(x) - L| < \varepsilon$, and manipulate it into a statement about $|x - a|$. On the page you reverse the arrows.

10.4 The ε - δ template

CUE The words “prove that” attached to a limit, or any situation where you must certify rather than compute.

METHOD The definition: for every $\varepsilon > 0$ there exists $\delta > 0$ such that $0 < |x - a| < \delta$ implies $|f(x) - L| < \varepsilon$. To find δ , factor $f(x) - L$ so that $(x - a)$ appears explicitly, giving $|f(x) - L| = |x - a| \cdot |C(x)|$. Then impose a *preliminary* restriction, conventionally $|x - a| < 1$, which confines x to a bounded interval and lets you replace the messy factor by a constant bound $|C(x)| \leq M$. Finally take

$$\delta = \min\left\{1, \frac{\varepsilon}{M}\right\},$$

and write the argument out in the forward direction.

WORKED EXAMPLE

A polynomial, then a quotient, then a radical.

For $f(x) = x^2$ at $a = 3$: $|x^2 - 9| = |x - 3||x + 3|$. If $|x - 3| < 1$ then $2 < x < 4$, so $|x + 3| < 7$ and $|x^2 - 9| < 7|x - 3|$.

Take $\delta = \min\{1, \varepsilon/7\}$.

For $f(x) = 1/x$ at $a = 2$, the difficulty is a factor that is unbounded near 0:

$$\left|\frac{1}{x} - \frac{1}{2}\right| = \frac{|2 - x|}{2|x|}.$$

The preliminary restriction must keep x away from 0: with $|x - 2| < 1$ we get $x > 1$, so the bound is $|x - 2|/2$, and $\delta = \min\{1, 2\varepsilon\}$ works.

For $f(x) = \sqrt{x}$ at $a > 0$ no preliminary restriction is needed, because rationalising produces a factor that is bounded automatically:

$$|\sqrt{x} - \sqrt{a}| = \frac{|x - a|}{\sqrt{x} + \sqrt{a}} \leq \frac{|x - a|}{\sqrt{a}}, \quad \delta = \varepsilon\sqrt{a}.$$

The template transfers to two variables unchanged, and doing it once shows what the polar squeeze of the previous section was really producing. To prove $\lim_{(x,y) \rightarrow (0,0)} x^3/(x^2 + y^2) = 0$, note that $|x| \leq \sqrt{x^2 + y^2}$ gives

$$\left|\frac{x^3}{x^2 + y^2}\right| \leq \frac{(x^2 + y^2)^{3/2}}{x^2 + y^2} = \sqrt{x^2 + y^2},$$

so $\delta = \varepsilon$ works, with no case analysis and no mention of θ . A uniform polar bound $g(r)$ is nothing other than a recipe for δ : solve $g(\delta) = \varepsilon$.

BACKWARDS TO FIND IT, FORWARDS TO WRITE IT

The scratch work and the proof are different documents. Scratch: assume $|f(x) - L| < \varepsilon$ and solve for $|x - a|$, taking whatever liberties you like. Proof: declare δ , assume $0 < |x - a| < \delta$, and derive $|f(x) - L| < \varepsilon$ by a chain of inequalities that never runs the wrong way. A proof that presents the scratch work as the argument has assumed its conclusion, and graders are right to reject it even when the δ is correct.

WHERE THIS GOES WRONG

Letting δ depend on x . The quantifiers fix ε first, then δ , then x ; a δ containing x is not a number and the proof says nothing. The usual route into this error is bounding $|C(x)|$ by an expression in x instead of by a constant, which is why the preliminary restriction is not optional decoration.

SEE ALSO The variants at infinity and one-sided; piecewise matching at a junction; the squeeze theorem template.

NEGATING THE DEFINITION, AND WHAT A PATH TEST REALLY IS

To prove that $\lim_{x \rightarrow a} f \neq L$ you negate the quantifiers in order: there exists $\varepsilon_0 > 0$ such that for every $\delta > 0$ there is some x with $0 < |x - a| < \delta$ and $|f(x) - L| \geq \varepsilon_0$. In practice you exhibit a *sequence* $x_n \rightarrow a$ with $x_n \neq a$ along which $f(x_n)$ stays away from L , since supplying one point in each shrinking neighbourhood is exactly supplying such a sequence. That is what the path test of the previous section is doing, in the plane rather than on the line: the parabola $y = kx^2$ is a family of points, one in every disc about the origin, on which f is pinned at $k/(1 + k^2)$. Seeing the path test as an instance of the negated definition is what makes it a proof rather than a rule of thumb.

The template above covers a finite limit at a finite point. Every other kind of limit in this book is the same idea with the quantifiers rewired, and the rewiring is worth doing explicitly once, because the commonest error at this level is not a bad estimate but a mismatched template: a δ deployed where an N was needed, or a punctured neighbourhood invented at infinity where none exists.

10.5 The variants: ε - N , at infinity, and one-sided

CUE A sequence, a limit as $x \rightarrow \infty$, an infinite limit, or a limit as $x \rightarrow a^\pm$, in each case with a proof demanded.

METHOD Keep the shape “for every tolerance there is a threshold” and change only what the threshold constrains.

- Sequences: for every $\varepsilon > 0$ there is N with $|a_n - L| < \varepsilon$ for all $n > N$.
- At infinity: for every $\varepsilon > 0$ there is M with $|f(x) - L| < \varepsilon$ for all $x > M$. There is no punctured neighbourhood here, because ∞ is not a point.
- Infinite limits: for every K there is $\delta > 0$ with $f(x) > K$ whenever $0 < |x - a| < \delta$. The tolerance is now a height, not a width.
- One-sided: replace $0 < |x - a| < \delta$ by $a < x < a + \delta$, or by $a - \delta < x < a$.

WORKED EXAMPLE

A sequence, then a limit at infinity, then an infinite limit.

$$\lim_{n \rightarrow \infty} \frac{2n+1}{n+3} = 2 : \quad \left| \frac{2n+1}{n+3} - 2 \right| = \left| \frac{2n+1-2n-6}{n+3} \right| = \frac{5}{n+3} < \varepsilon \iff n > \frac{5}{\varepsilon} - 3,$$

so $N = 5/\varepsilon$ suffices, and no absolute value survives because the quantity is positive.

$$\lim_{x \rightarrow \infty} \frac{3x^2+1}{x^2-5} = 3 : \quad \left| \frac{3x^2+1}{x^2-5} - 3 \right| = \frac{16}{x^2-5} \quad (x > \sqrt{5}),$$

which is below ε once $x^2 > 5 + 16/\varepsilon$, so $M = \sqrt{5 + 16/\varepsilon}$ works. Note that the preliminary restriction $x > \sqrt{5}$ plays exactly the role that $|x - a| < 1$ played before: it makes the denominator positive so the absolute value can be removed.

$$\lim_{x \rightarrow 1} \frac{1}{(x-1)^2} = +\infty : \quad \frac{1}{(x-1)^2} > K \iff |x-1| < \frac{1}{\sqrt{K}},$$

so $\delta = 1/\sqrt{K}$, and here the equivalence is exact in both directions, which is unusually convenient. Take $K > 0$ without loss, since a threshold that works for a large K works for every smaller one.

One-sided, where the restriction to a side is what makes the algebra legal at all: $\lim_{x \rightarrow 0^+} \sqrt{x} = 0$ requires $\sqrt{x} < \varepsilon$ for $0 < x < \delta$, and squaring gives $\delta = \varepsilon^2$. The two-sided statement is not false here, it is undefined, because $\sqrt{x}$ has no left neighbourhood of 0 in its domain. Naming the domain before writing the quantifiers avoids a whole family of vacuous arguments.

WHERE THIS GOES WRONG

Mixing templates. Writing δ for a sequence limit is the visible version; the invisible version is asserting a punctured neighbourhood at infinity, or forgetting that an infinite limit has no ε in it at all. A third: at a one-sided limit, using $|x - a| < \delta$ and then quietly assuming $x > a$ in the algebra.

SEE ALSO The ε - δ template; piecewise matching at a junction; expansion at infinity via $t = 1/x$.

Piecewise functions are where the one-sided machinery pays for itself. They are also where the standard shortcut, matching derivative formulas, is wrong often enough to be worth a technique of its own, because the shortcut is a theorem with hypotheses and the hypotheses are exactly what a well set problem removes.

10.6 Piecewise matching at a junction

CUE A function given by different formulas on either side of $x = a$, with unknown parameters, and a demand for continuity or differentiability there.

METHOD Continuity is one equation: $\lim_{x \rightarrow a^-} f = \lim_{x \rightarrow a^+} f = f(a)$. Differentiability adds a second condition on the one-sided *difference quotients*,

$$f'(a^-) = \lim_{h \rightarrow 0^-} \frac{f(a+h) - f(a)}{h}, \quad f'(a^+) = \lim_{h \rightarrow 0^+} \frac{f(a+h) - f(a)}{h},$$

which must exist and be equal. When both pieces are differentiable on a neighbourhood of a and f is already continuous at a , the mean value theorem lets you replace this by the cheaper test $\lim_{x \rightarrow a^-} f'(x) = \lim_{x \rightarrow a^+} f'(x)$. Solve continuity first, always.

Take $f(x) = x^2$ for $x \leq 1$ and $f(x) = ax + b$ for $x > 1$. Continuity at 1 forces $1 = a + b$. Differentiability, with continuity already secured, forces $2 = a$, hence $b = -1$. The unique answer is $a = 2$, $b = -1$, which is to say the linear piece is the tangent line to x^2 at $x = 1$. That is the geometric content of every problem of this shape.

Now the reason the order matters. Let $g(x) = x^2$ for $x < 1$ and $g(x) = x^2 + 5$ for $x \geq 1$. The derivative formulas are $2x$ on both sides and agree perfectly at $x = 1$. But g jumps by 5 there, so it is not even continuous, let alone differentiable; the difference quotient $(g(1+h) - g(1))/h$ behaves like $-5/h$ as $h \rightarrow 0^-$ and diverges. Matching f' formulas without continuity certifies nothing.

The difference quotient must sometimes be computed by hand, because f' has no one-sided limit. For $f(x) = x^2 \sin(1/x)$ with $f(0) = 0$, the quotient is $h \sin(1/h) \rightarrow 0$, so $f'(0) = 0$ exists; yet $f'(x) = 2x \sin(1/x) - \cos(1/x)$ has no limit at 0. The shortcut would report “not differentiable” and be wrong.

WHERE THIS GOES WRONG

Matching the derivative formulas before securing continuity, as in the g above. The shortcut $f'(a^-) = f'(a^+)$ is a corollary of the mean value theorem applied to a *continuous* function; strip the continuity hypothesis and the corollary is false. The second trap is the reverse: declaring non-differentiability because $\lim f'$ fails, when the difference quotient itself converges.

SEE ALSO The variants at infinity and one-sided; the limit is a derivative in disguise; floor and fractional-part limits.

Three points are worth extracting from this section before leaving it. The preliminary restriction is not a trick but the mechanism by which a variable quantity is turned into a constant, and it recurs under other names throughout analysis. The quantifier order is the content of every definition here, so reciting it in the correct order is half of any proof. And a definition is what you use when the rules run out: if you find yourself unable to decide whether a limit exists, the reason is almost always that you have been trying to compute it instead of estimating it.

Limits built to punish reflexes

Everything so far has been about what a limit is. This section is about how limits are weaponised. A problem setter who wants to distinguish understanding from fluency does it by building an expression whose most automatic response is wrong, and there are only a handful of ways to do that. The rule you reach for loops instead of resolving; the operation you want to exchange with the limit is not continuous in the relevant sense; the numerics look convincing and are not; a discontinuous function is treated as continuous; a parameter conceals a case split; or an asymptotic replacement is carried across a subtraction where it is illegal. Each of the six has a signature you can learn.

What follows is therefore not a list of curiosities. A trap limit is a diagnostic instrument: it is constructed so that the fast answer and the correct answer differ, which means that getting it right is evidence of something the fast answer cannot supply. Treat each one as a statement about a hypothesis you were in the habit of skipping, and the section becomes an audit of your own reflexes rather than a museum of oddities.

10.7 Limits where L'Hôpital loops forever

MEDIUM

CUE Applying the rule reproduces the original quotient, or its reciprocal, up to constants. The families that do this are built from e^x and e^{-x} , or from x and $\sqrt{x^2 + c}$, or from $\sinh$ and $\cosh$.

METHOD Recognise the loop after *one* repetition and abandon the rule. Switch to dividing by the dominant term, which is the correct tool for a quotient of comparable exponentials, or to the substitution that linearises the pair. Continuing the differentiation is not merely slow; it cannot terminate, because the loop is an algebraic identity, not an accident of the first two rounds.

WORKED EXAMPLE

$$\lim_{x \rightarrow \infty} \frac{e^x + e^{-x}}{e^x - e^{-x}}.$$

Both parts tend to ∞ , so the rule applies and gives $\frac{e^x - e^{-x}}{e^x + e^{-x}}$, the reciprocal; a second application returns the original. Dividing by e^x instead is immediate:

$$\frac{1 + e^{-2x}}{1 - e^{-2x}} \rightarrow 1.$$

The same structure with radicals: $\lim_{x \rightarrow \infty} \frac{x}{\sqrt{x^2 + 1}}$ sends L'Hôpital to $\sqrt{x^2 + 1}/x$ and back again, while dividing numerator and denominator by x gives $1/\sqrt{1 + x^{-2}} \rightarrow 1$.

THE LOOP IS NOT ALWAYS USELESS

In both examples the rule reports $L = 1/L$. If you already know that the limit exists and is positive, that equation alone finishes the problem: $L = 1$. The caveat is real and must be stated. L'Hôpital's conclusion runs one way, from the existence of the new limit to the value of the old one, so the reciprocal identity licenses $L = 1/L$ only once existence is established independently, for instance by monotonicity or by the one-line division above. Used as a check on an answer you already have, it is free; used as the proof, it is circular.

WHERE THIS GOES WRONG

Continuing past the second round in the hope that it resolves. It cannot. The clean diagnostic is to compare the third expression with the first: if they match, stop and divide by the dominant term.

SEE ALSO Divide by the dominant term; the rule's failure modes; illegal asymptotic substitution inside a difference.

L'Hôpital loops because a rule was applied outside the situation it was built for. The next trap is the same disease at a higher level: an operation is exchanged with a limit without checking that the exchange is legal, and the operation in question is usually one nobody thinks of as an operation.

HARD

10.8 When the leading term lies: non-uniform convergence

CUE A limit of a *functional* of a sequence of functions: an arc length, a total variation, a maximum, an integral of a derivative. The tell is that f_n appears inside something that is not just $\int f_n$.

METHOD Do not push the limit inside. Identify the limiting object geometrically, then ask whether the functional is continuous under the mode of convergence you actually have. Pointwise convergence controls almost nothing; uniform convergence controls integrals and continuity but still not arc length, since arc length involves f'_n ; and f'_n is not controlled by any convergence of f_n alone.

WORKED EXAMPLE

The arc length of $y = x^n$ on $[-1, 1]$ as $n \rightarrow \infty$. The graphs converge pointwise to the function that is 0 on $(-1, 1)$ and 1 at the endpoints, so a naive interchange inside

$$\Lambda_n = \int_{-1}^1 \sqrt{1 + n^2 x^{2n-2}} \, dx$$

replaces the integrand by 1 and predicts $\Lambda_n \rightarrow 2$. The truth is 4. The limiting curve is not the pointwise limit function but a polyline: the graph flattens onto the segment from $(-1, 0)$ to $(1, 0)$, of length 2, and then climbs vertically at each end from $y = 0$ to $y = 1$, contributing 1 on each side. Numerically $\Lambda_{50} = 3.8356$, $\Lambda_{500} = 3.9740$, $\Lambda_{5000} = 3.9965$, $\Lambda_{50000} = 3.99956$: slow, monotone, and unmistakably heading for 4 rather than 2.

Uniform convergence does not rescue the situation either, which is the point most often missed. Let s_n be the sawtooth on $[0, 1]$ with n^2 teeth of amplitude $1/n$. Then $\|s_n\|_\infty = 1/n \rightarrow 0$, so $s_n \rightarrow 0$ uniformly, yet the graph of s_n has length $\sqrt{1 + 4n^2} \approx 2n \rightarrow \infty$. Arc length depends on s'_n , and uniform control of a function says nothing whatever about its derivative.

A smaller instance of the same disease, where the functional is an ordinary integral. Let $f_n(x) = nxe^{-nx^2}$ on $[0, 1]$. For each fixed $x > 0$, $f_n(x) \rightarrow 0$, and $f_n(0) = 0$, so $f_n \rightarrow 0$ pointwise. Yet

$$\int_0^1 nxe^{-nx^2} \, dx = \frac{1}{2}(1 - e^{-n}) \rightarrow \frac{1}{2}.$$

The mass does not vanish; it escapes into a spike of height $\sim \sqrt{n}$ and width $\sim 1/\sqrt{n}$ near $x = 0$. Pointwise convergence never sees the spike.

WHERE THIS GOES WRONG

Assuming that pointwise convergence of f_n implies convergence of $\int F(f_n, f'_n)$. Arc length is not continuous under pointwise convergence, and neither is $\int f_n$ without a domination hypothesis. When a problem asks for the limit of a geometric quantity, draw the limiting picture before writing any integral.

SEE ALSO Endpoint concentration; convergence of an n th power to an indicator; numerically deceptive limits.

The arc length example was at least honest about its slowness: every partial value was visibly far from 2. The next family is worse, because the numbers look like they have converged and they have not.

10.9 Numerically deceptive limits

HARD

CUE A limit whose approach carries a correction of order $n^{-1/2}$, $1/\ln n$, or $\ln \ln n / \ln n$. Any limit involving a central limit theorem, a logarithm of a logarithm, or a sum of $\sqrt{n}$ comparable terms is a candidate.

METHOD Prove it, do not probe it. Establish the exact leading behaviour analytically and read the constant off that; then use numerics in the only role they are fit for, checking the *residual* against the predicted correction. If the residual halves when n quadruples, you have confirmed an $n^{-1/2}$ correction and, with it, your constant.

WORKED EXAMPLE

$$\lim_{n \rightarrow \infty} e^{-n} \sum_{k=0}^n \frac{n^k}{k!} = \frac{1}{2}.$$

The sum is exactly $\mathbb{P}(X \leq n)$ for X Poisson with mean n . Such an X is a sum of n independent Poisson(1) variables, so it is asymptotically normal with mean n and standard deviation $\sqrt{n}$, and the central limit theorem puts half the mass on each side of the mean. Hence the limit is $\frac{1}{2}$, with an error of order $n^{-1/2}$ from the lattice correction.

Now the deception. The values are 0.52656, 0.50841, 0.50266, 0.50119 at $n = 10^2, 10^3, 10^4, 5 \times 10^4$. A reader who computes only $n = 1000$ sees 0.5084 and is invited to guess a wrong closed form near that value. A reader who computes two of them sees the residuals 0.02656 and 0.00841, whose ratio is $3.16 \approx \sqrt{10}$, and immediately recognises $\frac{1}{2} + c/\sqrt{n}$.

A second and even slower specimen: $\ln x/x^{0.01} \rightarrow 0$ as $x \rightarrow \infty$. At $x = 10^6$ it is 12.03; at $x = 10^{30}$ it is 34.62, still increasing. It peaks at $x = e^{100}$ with value $100/e = 36.79$, and does not fall back below 1 until roughly $x = e^{650} \approx 10^{282}$. No amount of computation would suggest the correct answer.

WHERE THIS GOES WRONG

Naive dominance reasoning gives 0 or 1 for the Poisson sum, depending on which term you decide dominates, and naive numerics at $n = 1000$ suggests 0.508. Both are wrong for the same reason: the sum is a probability that concentrates at a boundary, and boundaries are exactly where dominance arguments have no content.

SEE ALSO Logarithmic rescaling of a monstrous expression; asymptotics of the named special functions; when the leading term lies.

Deception by slow convergence is at least a quantitative failure. The floor function fails qualitatively: it is not continuous anywhere it matters, so substituting the limit into it is not a slightly inaccurate move but a meaningless one.

10.10 Floor and fractional-part limits

MEDIUM

CUE A $[\cdot]$, $\lceil \cdot \rceil$ or $\{\cdot\}$ anywhere inside the limit, including inside a summation index.

METHOD Use the universal bracket

$$t - 1 < [t] \leq t \quad \text{for all real } t,$$

and squeeze. Multiplying it by a negative quantity reverses both inequalities, which is where one-sided answers diverge if they are going to. For averages of fractional parts, Weyl equidistribution supplies the answer directly: for irrational α the sequence $\{k\alpha\}$ is equidistributed in $[0, 1)$, so $\frac{1}{n} \sum_{k=1}^n \{k\alpha\} \rightarrow \frac{1}{2}$.

WORKED EXAMPLE

$$\lim_{x \rightarrow 0^+} x \left\lfloor \frac{1}{x} \right\rfloor = 1.$$

From the bracket with $t = 1/x$ and $x > 0$: $x(\frac{1}{x} - 1) < x \lfloor \frac{1}{x} \rfloor \leq 1$, that is $1 - x < x \lfloor 1/x \rfloor \leq 1$, and the squeeze closes. For $x < 0$ multiplication reverses the inequalities and gives $1 \leq x \lceil 1/x \rceil < 1 - x$, so the left-hand limit is also 1 and the two-sided limit exists. That coincidence is worth noticing precisely because it does not always happen.

A summation version, where the bracket is applied n times:

$$\lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{k=1}^n [kx] = \frac{x}{2}.$$

Summing $kx - 1 < [kx] \leq kx$ over k gives $x \frac{n(n+1)}{2} - n < \sum [kx] \leq x \frac{n(n+1)}{2}$, and dividing by n^2 squeezes both ends to $x/2$. At $x = \sqrt{2}$ and $n = 10^5$ the sum divided by n^2 is 0.7071089 against $x/2 = 0.7071068$; at $x = 3.7$ it is 1.850014 against 1.85.

The Weyl average, checked at $\alpha = \sqrt{2}$ with $n = 2 \times 10^6$, gives 0.4999999.

WHERE THIS GOES WRONG

Treating $[\cdot]$ as continuous and substituting the limit. That reflex turns $x[1/x]$ into $x \cdot \infty$ and produces nothing. The second trap is assuming the one-sided answers agree: they do for $x[1/x]$ at 0, but $\lim_{x \rightarrow 0} [x]$ genuinely differs on the two sides, being -1 from the left and 0 from the right, so the two-sided limit does not exist.

SEE ALSO The squeeze theorem template; parameter-dependent limits that branch into cases; the variants at infinity and one-sided.

A floor function announces its discontinuity in the notation. A parameter does not announce anything: the expression looks like a single problem, and it is several, joined at values of the parameter where the dominant term changes hands.

HARD

10.11 Parameter-dependent limits that branch into cases

CUE A symbol a , p or t in the expression with the answer requested as a function of it; or, in the harder direction, a limit value supplied and the parameter to be found.

METHOD Locate the critical values where the dominant term changes, and case-split there. In the inverse direction, matching asymptotic orders on numerator and denominator converts the limit into a plain algebraic equation in the parameter, and that equation may have several admissible roots. Report all of them, then check each against any stated restriction on the parameter.

WORKED EXAMPLE

The direct kind first. For $x \geq 0$,

$$F(x) = \lim_{n \rightarrow \infty} \frac{x^{2n} - 1}{x^{2n} + 1} = \begin{cases} -1, & 0 \leq x < 1, \\ 0, & x = 1, \\ +1, & x > 1, \end{cases}$$

three cases governed by whether x^{2n} tends to 0, stays at 1, or tends to ∞ . Note in passing that a limit of continuous functions has produced a discontinuous one.

The inverse kind, which is where the real difficulty lies. For $|a| > 1$ solve

$$\lim_{n \rightarrow \infty} \frac{\sum_{r=1}^n r^{1/3}}{n^{7/3} \sum_{r=1}^n (an + r)^{-2}} = 54.$$

The numerator is a sum of a power, so $\sum_{r \leq n} r^{1/3} \sim \frac{3}{4}n^{4/3}$. The inner sum in the denominator is a Riemann sum after one factor of n^{-2} is extracted:

$$\sum_{r=1}^n \frac{1}{(an + r)^2} = \frac{1}{n} \cdot \frac{1}{n} \sum_{r=1}^n \frac{1}{(a + \frac{r}{n})^2} \rightarrow \frac{1}{n} \int_0^1 \frac{du}{(a + u)^2} = \frac{1}{n} \cdot \frac{1}{a(a+1)}.$$

So the denominator behaves like $n^{4/3} / (a(a+1))$, the powers $n^{4/3}$ cancel, and the limit is $\frac{3}{4}a(a+1)$. Setting that to 54 gives $a^2 + a - 72 = 0$, hence

$$a = \frac{-1 \pm 17}{2} = 8 \text{ or } -9,$$

and both satisfy $|a| > 1$, so both are admissible. Direct summation at $n = 2 \times 10^6$ returns 54.00002 for $a = 8$ and 54.00001 for $a = -9$.

WHERE THIS GOES WRONG

Finding one root and stopping. The quadratic was engineered to have two admissible solutions and the negative one is the one that gets dropped. A second trap: reading the stated range $|a| > 1$ as decoration. It is what keeps $a + u$ away from 0 on $[0, 1]$, so that the Riemann sum has a non-singular integrand; without it the interchange is invalid and the answer changes.

SEE ALSO Riemann sum recognition; integral bounds on a sum; floor and fractional-part limits.

The last trap is the most common of all, and the most invisible, because it is a correct technique used one step outside its domain of validity. Asymptotic replacement is legal for factors and illegal for summands, and the boundary between those two situations is easy to cross without noticing.

10.12 Illegal asymptotic substitution inside a difference

MEDIUM

CUE You are about to replace a factor by its leading equivalent, and that factor sits inside a sum or difference of comparable terms rather than in a product or quotient.

METHOD Replacement of f by g with $f \sim g$ is legal for factors of a product or a quotient, and illegal across a $+$ or $-$ sign whenever the leading terms cancel. Inside a difference, expand *both* sides to a common order, far enough that the first surviving term appears, and only then divide.

WORKED EXAMPLE

$$\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3}.$$

The illegal move is to substitute $\tan x \sim x$ and $\sin x \sim x$, obtaining $(x - x)/x^3 = 0$. The leading terms cancelled exactly, which is precisely the situation the substitution rule excludes. Expanding both to order x^3 ,

$$\tan x = x + \frac{x^3}{3} + O(x^5), \quad \sin x = x - \frac{x^3}{6} + O(x^5), \quad \tan x - \sin x = \frac{x^3}{2} + O(x^5),$$

so the limit is $\frac{1}{2}$. Numerically the quotient is 0.50001250 at $x = 10^{-2}$ and 0.50000013 at $x = 10^{-3}$.

A second, where the equivalent being smuggled in is a constant:

$$\lim_{x \rightarrow 0} \frac{(1+x)^{1/x} - e}{x}.$$

Substituting $(1+x)^{1/x} \rightarrow e$ gives $0/x$ and then 0, which is wrong. Write $(1+x)^{1/x} = \exp(\frac{1}{x} \ln(1+x)) = \exp(1 - \frac{x}{2} + \frac{x^2}{3} - \dots) = e(1 - \frac{x}{2} + O(x^2))$, so the quotient is $-\frac{e}{2} + O(x)$ and the limit is $-e/2 \approx -1.359141$. Numerically, at $x = 10^{-5}$ the quotient is -1.359127 .

WHERE THIS GOES WRONG

The error is invisible because the same substitution is correct in the neighbouring problem $\frac{\tan x \cdot \sin x}{x^2} \rightarrow 1$, where the two factors multiply rather than subtract. Readers generalise from the product case, and nothing in the notation warns them. The rule to carry: before substituting, ask whether the terms you are replacing will cancel to leading order. If they will, you must expand.

SEE ALSO Asymptotic equivalence and its legal algebra; order bookkeeping across a subtraction; add and subtract a bridging term.

THE COMMON SHAPE OF ALL SIX TRAPS

Every trap in this section is a legal operation applied where its hypothesis fails. L'Hôpital needs the new limit to exist. Interchanging a limit with a functional needs the functional to be continuous in the right topology. Numerical evidence needs a known error order to be interpretable. Substituting a value needs continuity at that value. A single answer needs the parameter to have a single admissible value. Asymptotic replacement needs a product. The remedy is uniform: name the hypothesis before you use the tool, and check it. That habit costs a few seconds per problem and removes an entire class of confident errors.

One last remark on how these three families relate. They are not three topics but three places where the same gap opens: between what a formula lets you compute and what a definition actually asserts. In two variables the gap is between finitely many paths and the whole punctured disc. In the ε - δ section it is between a value you can evaluate and a claim you can certify. In the trap section it is between a rule and its hypotheses. A reader who closes that gap by habit will find that the hard problems in later chapters are hard for reasons of technique rather than of rigour, which is the position you want to be in.

The matrix below is arranged in the order of this chapter, and it is meant to be read down the first column only. Almost every problem in the three families is identified by a surface feature: a denominator with mismatched exponents, the words “prove that”, a floor bracket, a free parameter, a subtraction of two things that are individually equivalent to the same thing. Identification is the expensive step, and it is the one the matrix removes.

Recognition matrix

If you see	Reach for	If that fails
$\lim_{(x,y) \rightarrow (0,0)} P/Q$, monomials	Lines $y = mx$, and $x = 0$ separately	Escalate to $y = kx^2$, then $y = kx^3$
Denominator $x^2 + y^2$ or $x^4 + y^4$	Polar, bound uniform in θ	Split off the sector that resists
Denominator $x^{2m} + y^2$, $m \geq 2$	Path $y = kx^m$; the terms balance there	No uniform polar bound exists
A polar bound whose constant holds θ	Reject it: that is the line test	Strike out every trigonometric factor
$\lim_{x \rightarrow 0} \lim_{y \rightarrow 0} f$ written out	An iterated limit, nothing more	Joint limit needs its own argument
"Prove that $\lim_{x \rightarrow a} f = L$ "	Factor out $(x - a)$; restrict $ x - a < 1$	$\delta = \min\{1, \varepsilon/M\}$
A sequence, or $x \rightarrow \infty$	ε - N or ε - M ; no punctured ball	Solve the inequality for n or x
Piecewise formulas with parameters	Continuity equation first, then f'	Compute the difference quotient by hand
L'Hôpital returns the same quotient	Divide by the dominant term	$L = 1/L$, once existence is known
Arc length, variation, $\int F(f_n, f'_n)$	Draw the limiting object first	Check continuity of the functional
Numerics drifting at the third digit	Prove the constant, then fit the residual	Ratio test on residuals: $\sqrt{10}$ per decade
$\lfloor \cdot \rfloor$, $\lceil \cdot \rceil$ or $\{ \cdot \}$	$t - 1 < \lfloor t \rfloor \leq t$, then squeeze	Weyl equidistribution for averages
A parameter with the value prescribed	Match asymptotic orders; solve; keep all roots	Test each root against the stated range
A parameter with the answer requested	Case-split at the critical value	Sketch the limit function
Replacement across $+$ or $-$	Expand both sides to a common order	Conjugate, or a bridging term

PART III

Derivatives

Simplify before you differentiate

What this chapter is for. Differentiation is the one genuinely mechanical operation in calculus: every elementary function has a derivative, the rules compose without exception, and no ingenuity is required to apply them. It follows that a derivative problem cannot be hard *as a derivative problem*. If a problem is hard, the difficulty was placed in the algebra, and the algebra is what a good solver attacks first. This chapter teaches the opening move: read the expression as an algebraist for ten seconds before reaching for any rule. Log laws split a monster into a sum of independent terms; a trig identity turns a product into a single sinusoid and makes the seventh derivative a one-liner; a conjugate pair reveals that the function was a constant all along; a radical is a power in disguise; a series, a product, or an infinite nesting has a closed form that must be found before anything is differentiated. At the end you should be unable to start a differentiation without first asking what the expression really is.

Ten seconds of algebra

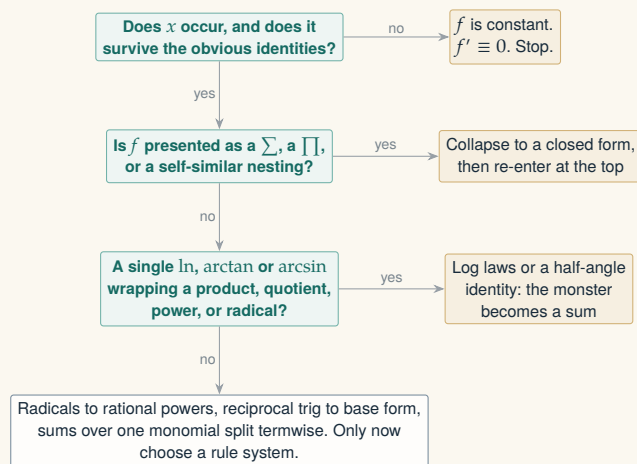
Here is the asymmetry that organises the whole chapter. Integration is genuinely hard: elementary functions have non-elementary antiderivatives, and there is no algorithm you can run without thought. Differentiation has none of that structure. The derivative of an elementary function is elementary, the rules are finitely many, and applying them is bookkeeping. Nobody sets a problem whose difficulty is “can you apply the chain rule three times”. What they set is a problem whose difficulty is “can you see that this expression is $\frac{1}{2} \sin 4x$ ”, dressed up so that the mechanical route exists but costs two pages.

So a hard derivative problem is a disguised algebra problem, and the disguise is always one of a short list of moves. The list is short because the setter has to be able to write the disguised form down, which means the disguise is an identity applied backwards: a sum was recombined into a single logarithm, a sinusoid was expanded into a product, a constant was written as a difference of two nonconstant pieces, a rational power was written as a nest of radicals. Undoing the disguise is the same identity applied forwards, and there are perhaps eight of them.

THE TEN-SECOND SCAN

Before writing a single prime, answer four questions about the expression on the page. Where does x actually occur? Is there a wrapper (a logarithm, an inverse trig function) that an identity will remove? Are there radicals, reciprocal trig functions, or a quotient with one term downstairs, all of which are powers in disguise? And is the function given to you as a *process* (a sum, a product, an infinite nesting) rather than as a formula? Four questions, ten seconds, and they decide whether the problem takes two lines or two pages. The cost of skipping them is not a wrong answer; it is the right answer obtained slowly, with five opportunities to drop a sign.

The scan has an order, because the moves are not independent. Constancy is checked first, since a constant function makes every other question moot. Wrapper removal comes next, because it typically converts one product-and-quotient monster into a sum of terms that then need their own scan. Normalisation to powers comes third. And the process forms are checked last only because they are the rarest, not because they are the least profitable: they are the ones where the payoff is largest.



Read the tree as a filter, not as a classifier. Most expressions fall through every test and land in the bottom box, which is the honest outcome and the subject of the next chapter. The tree earns its keep on the minority that do not, and on those the saving is not marginal: it is the difference between a line and a page.

Removing the wrapper

The commonest disguise is a single outer function wrapped around an algebraic mess. Logarithms are the classic case because the log laws are exactly a dictionary between multiplicative structure inside and additive structure outside, and additive structure is what differentiation likes. Inverse trig functions are the same phenomenon one step further along: the identity that removes the wrapper is a half-angle or double-angle formula rather than a log law, but the effect is identical.

The identities that do this work are few enough to tabulate, and tabulating them is worth half a page because recognition, not derivation, is what fails under pressure. The right-hand column is not decoration: every one of these collapses holds only on an interval, and the interval is where the traps live.

Written as	Collapses to	Valid where
$\ln A + \ln B$	$\ln(AB)$	$A > 0, B > 0$
$\ln \frac{A}{B}, \ln A^p$	$\ln A - \ln B, p \ln A$	$A > 0, B > 0$
$\ln(\sqrt{1+x^2} + x) + \ln(\sqrt{1+x^2} - x)$	0	all x
$\ln(\sec x + \tan x) + \ln(\sec x - \tan x)$	0	$\cos x > 0$
$\arctan x + \arctan(1/x)$	$\pi/2$	$x > 0$
$\arcsin x + \arccos x$	$\pi/2$	$-1 \leq x \leq 1$
$\arctan \frac{x}{1 + \sqrt{1+x^2}}$	$\frac{1}{2} \arctan x$	all x
$\arcsin \frac{2x}{1+x^2}, \arctan \frac{2x}{1-x^2}$	$2 \arctan x$	$ x \leq 1, x < 1$
$\sqrt{\frac{1-\cos x}{1+\cos x}}$	$\tan \frac{x}{2}$	$0 < x < \pi$
$\sin^2 x, \cos^2 x$	$\frac{1}{2}(1 \mp \cos 2x)$	all x
$\sin A \cos B$	$\frac{1}{2}[\sin(A+B) + \sin(A-B)]$	all A, B
$1/\csc^2 x, \sec^2 x - \tan^2 x$	$\sin^2 x, 1$	$\sin x \neq 0, \cos x \neq 0$
$\sqrt[m]{x^k}, \ln(\cosh x + \sinh x)$	$x^{k/m}, x$	$x > 0$, all x

Read down the second column and the pattern is uniform: a two-term expression becomes a one-term expression, a transcendental becomes an algebraic, or an expression becomes a number. That is the whole of the chapter's first half, and the rest is knowing which row you are looking at.

11.1 Log-law expansion before differentiating

CUE A single $\ln$ or $\log$ wraps a product, a quotient, a power, or a radical: $\ln(\cos^3 \sqrt{x})$, $\ln(x^x \tan x e^{-x^2/\ln x})$, and $\ln[(x^2 + 1)^3 \sqrt{x-2} (x+4)^{-5} e^{-x}]$.

METHOD Apply $\ln(AB) = \ln A + \ln B$, $\ln(A/B) = \ln A - \ln B$ and $\ln(A^p) = p \ln A$ *before* differentiating, converting one chain-and-quotient monster into a sum of independent terms. Then differentiate termwise with $\frac{d}{dx} \ln u = u'/u$. The point is that the quotient rule never appears: division inside the logarithm has already become subtraction outside it.

WORKED EXAMPLE

Take $f(x) = \ln(\cos^3 \sqrt{x})$ and ask for $f'(\pi^2/16)$. Expanding first, $f(x) = 3 \ln \cos \sqrt{x}$, so

$$f'(x) = 3 \cdot \frac{-\sin \sqrt{x}}{\cos \sqrt{x}} \cdot \frac{1}{2\sqrt{x}} = -\frac{3 \tan \sqrt{x}}{2\sqrt{x}}.$$

At $x = \pi^2/16$ we have $\sqrt{x} = \pi/4$ and $\tan(\pi/4) = 1$, so $f'(\pi^2/16) = -\frac{3}{2 \cdot \pi/4} = -\frac{6}{\pi}$, numerically -1.90986 . Note that the evaluation point was chosen so that the tangent is 1; that is the setter telling you the expansion was intended. A quotient case, where the saving is larger:

$$f(x) = \ln \frac{(x^2 + 1)^3 \sqrt{x-2}}{(x+4)^5 e^x} = 3 \ln(x^2 + 1) + \frac{1}{2} \ln(x-2) - 5 \ln(x+4) - x,$$

whence $f'(x) = \frac{6x}{x^2 + 1} + \frac{1}{2(x-2)} - \frac{5}{x+4} - 1$ and

$$f'(3) = \frac{18}{10} + \frac{1}{2} - \frac{5}{7} - 1 = \frac{126 + 35 - 50 - 70}{70} = \frac{41}{70},$$

numerically 0.585714. Differentiating the unexpanded form means a chain rule outside a quotient rule outside two product rules, and the e^x never cancels visibly.

One more, to show that the technique does not need the answer to be pretty. Take

$$f(x) = \ln(x^x \tan x e^{-x^2/\ln x}) = x \ln x + \ln \tan x - \frac{x^2}{\ln x}.$$

Three independent terms, each with its own one-line derivative. The first gives $\ln x + 1$ by the product rule. The second gives $\frac{\sec^2 x}{\tan x} = \frac{2}{\sin 2x}$. The third gives $\frac{2x \ln x - x}{(\ln x)^2}$ by the quotient rule, which here is applied to a genuinely irreducible two-block quotient and is therefore the right tool. So

$$f'(x) = \ln x + 1 + \frac{2}{\sin 2x} - \frac{2x \ln x - x}{(\ln x)^2},$$

and at $x = e$, where $\ln x = 1$, this is $2 + \frac{2}{\sin 2e} - e = -3.388343$. Verified against a symmetric difference quotient at step 10^{-6} : -3.388343 . Without the expansion, the variable exponent x^x alone would have forced a separate logarithmic differentiation inside a chain rule inside a quotient rule.

WHERE THIS GOES WRONG

Expanding *after* differentiating. A solver who writes $f' = (\text{argument})'/(\text{argument})$ first has committed to a quotient rule on the argument, and the resulting expression is correct but will not simplify without the very log laws that should have been used at the start. The second trap is silent: the split $\ln(A/B) = \ln A - \ln B$ requires $A > 0$ and $B > 0$ separately, whereas $\ln(A/B)$ needs only $A/B > 0$. On the domain where both are negative you must carry absolute values, and $\ln((x-1)(x-2))$ at $x = 0$ is defined while $\ln(x-1)$ is not.

SEE ALSO Conjugate log collapse; logarithmic differentiation of many-factor products; algebraic normalisation.

Log-law expansion assumes the sum you produce is worth having. Sometimes the sum collapses further, and in the extreme case it collapses to nothing at all. The next technique is that extreme case, and it deserves its own block because the failure to notice it is the most expensive failure in the chapter.

11.2 Conjugate log collapse: the function is constant

CUE Two logarithms whose arguments are algebraic conjugates, added: $\ln(\sqrt{1+x^2}+x) + \ln(\sqrt{1+x^2}-x)$, or $\ln(\sec x + \tan x) + \ln(\sec x - \tan x)$. More generally, any expression built so that a Pythagorean or difference-of-squares identity annihilates it.

METHOD Combine into $\ln(AB)$ and evaluate the product before doing anything else. Conjugate pairs multiply to a difference of squares, and the whole design of these problems is that the difference of squares is 1:

$$(\sqrt{1+x^2}+x)(\sqrt{1+x^2}-x) = (1+x^2) - x^2 = 1, \quad \sec^2 x - \tan^2 x = 1.$$

So $f \equiv \ln 1 = 0$ and $f' \equiv 0$ everywhere the expression is defined.

WORKED EXAMPLE

$f(x) = \ln(\sqrt{1+x^2}+x) + \ln(\sqrt{1+x^2}-x)$; find $f'(2)$.

Combine: $f(x) = \ln[(\sqrt{1+x^2}+x)(\sqrt{1+x^2}-x)] = \ln 1 = 0$ for every real x . Hence $f'(2) = 0$, and indeed $f'(x) = 0$ identically.

It is worth seeing what the honest grind produces, because the cancellation is instructive. The first logarithm is $\operatorname{arsinh} x$, whose derivative is $1/\sqrt{1+x^2}$. The second is $-\operatorname{arsinh} x$, since $\sqrt{1+x^2}-x = 1/(\sqrt{1+x^2}+x)$. So the two derivatives are $+1/\sqrt{1+x^2}$ and $-1/\sqrt{1+x^2}$ and they cancel. Three lines of correct algebra to reach a number that one line of identity gives for free.

The circular twin behaves the same way: $\ln(\sec x + \tan x) + \ln(\sec x - \tan x) = \ln(\sec^2 x - \tan^2 x) = 0$, so its derivative vanishes on every interval where both factors are positive, even though each logarithm separately has the perfectly lively derivative $\pm \sec x$.

WHERE THIS GOES WRONG

Differentiating both logarithms and grinding two conjugate radical derivatives. The arithmetic is correct and the answer comes out, but a solver who does this under time pressure has spent three minutes and two sign decisions on a problem whose answer was visible. The deeper trap is the psychological one: the expression *looks* expensive, so the reflex is to start computing rather than to start reading, and the design of the problem depends on that reflex.

SEE ALSO Log-law expansion; constant scan; inverse-trig identity collapse.

Why is the identically constant function such an effective trap? Because every visual cue points the other way. The expression contains x in four places; it contains a radical; it contains two transcendental functions. Nothing on the page announces constancy, and the solver's pattern-matching runs on surface complexity. The defence is not vigilance but procedure: make "is this constant?" the first question rather than a question you might get to, which is exactly why it sits at the top of the decision tree. And test it numerically when you can. Two evaluations at unrelated points, say $x = 0.3$ and $x = 1.7$, settle the matter in seconds; if the values agree, stop differentiating and start proving.

SPOT-CHECK BEFORE YOU COMMIT

Two evaluations cost nothing and settle more than they should. If $f(0.3)$ and $f(1.7)$ agree to full precision, the function is constant and you are done. If a claimed closed form for f' disagrees with a symmetric difference quotient $(f(a+h) - f(a-h))/2h$ at $h = 10^{-5}$ in the fourth digit, you have dropped a term. Every worked example in this chapter was checked that way, most by complex-step differentiation, which computes $f'(a) = \operatorname{Im} f(a+ih)/h$ to machine precision with no cancellation at all. The habit is not a substitute for algebra; it is what tells you which algebra to trust.

The same trap has a family. $\arctan x + \arctan(1/x)$ is $\pi/2$ for $x > 0$ and $-\pi/2$ for $x < 0$, so its derivative is zero on each half-line while the naive computation gives $\frac{1}{1+x^2} + \frac{-1/x^2}{1+1/x^2}$, which is indeed zero after simplification. Likewise $\arcsin x + \arccos x = \pi/2$, $\sin^2 x + \cos^2 x = 1$, and $e^{\ln x} = x$ for $x > 0$. Every one of them can be buried inside a larger expression, and every one of them collapses a subexpression to a constant that then factors out of the whole problem.

Inverse trig wrappers behave like logarithms with a different dictionary. The identity that removes them is a substitution, performed mentally rather than on paper.

11.3 Inverse-trig and half-angle identity collapse

CUE An arctan or arcsin whose argument is a rational expression in x and radicals: $\arctan \frac{x}{1 + \sqrt{1 + x^2}}$, $\arctan \sqrt{\frac{1 - \cos x}{1 + \cos x}}$, $\arcsin \frac{2x}{1 + x^2}$.

METHOD Substitute $x = \tan \theta$ (for the $1 + x^2$ shapes) or $x = \sin \theta$ (for $1 - x^2$) in your head. The argument becomes a half-angle or double-angle expression, the inverse function cancels the direct one, and the whole wrapper collapses to a linear function of $\arctan x$ or of x :

$$\arctan \frac{x}{1 + \sqrt{1 + x^2}} = \frac{1}{2} \arctan x, \quad \sqrt{\frac{1 - \cos x}{1 + \cos x}} = \tan \frac{x}{2} \text{ on } (0, \pi), \quad \arcsin \frac{2x}{1 + x^2} = 2 \arctan x \text{ on } [-1, 1].$$

Differentiating a linear function of $\arctan x$ is then immediate.

WORKED EXAMPLE

$f(x) = \arctan \frac{x}{1 + \sqrt{1 + x^2}}$, find $f'(\sqrt{2})$. Put $x = \tan \theta$ with $\theta \in (-\pi/2, \pi/2)$, so $\sqrt{1 + x^2} = \sec \theta$ and

$$\frac{x}{1 + \sqrt{1 + x^2}} = \frac{\tan \theta}{1 + \sec \theta} = \frac{\sin \theta}{\cos \theta + 1} = \tan \frac{\theta}{2},$$

the last step being the standard half-angle formula. Hence $f(x) = \theta/2 = \frac{1}{2} \arctan x$ and $f'(x) = \frac{1}{2(1 + x^2)}$, giving $f'(\sqrt{2}) = \frac{1}{2 \cdot 3} = \frac{1}{6}$.

Now the cos version. On $(0, \pi)$, $1 - \cos x = 2 \sin^2(x/2)$ and $1 + \cos x = 2 \cos^2(x/2)$, so the radical is $|\tan(x/2)| = \tan(x/2)$ there, and $g(x) = \arctan \tan(x/2) = x/2$. Therefore $g'(1) = \frac{1}{2}$, and in fact $g' \equiv \frac{1}{2}$ on the whole interval. Verified numerically: the symmetric difference quotient at $x = 1$ with step 10^{-5} returns 0.5000000.

And the double-angle version: $\arcsin \frac{2x}{1 + x^2} = 2 \arctan x$ for $|x| \leq 1$, so the derivative is $\frac{2}{1 + x^2}$, which at $x = 1/\sqrt{3}$ equals $\frac{2}{4/3} = \frac{3}{2}$.

WHERE THIS GOES WRONG

Applying the identity on the wrong branch. Every one of these collapses is valid only on an interval, because the outer inverse function has a restricted range. Outside $(0, \pi)$ the radical $\sqrt{(1 - \cos x)/(1 + \cos x)}$ equals $|\tan(x/2)|$, so on $(\pi, 2\pi)$ the function is $\pi - x/2$ and the slope is $-\frac{1}{2}$, not $+\frac{1}{2}$. Similarly $\arcsin(2x/(1 + x^2))$ equals $2 \arctan x$ only for $|x| \leq 1$; for $x > 1$ it is $\pi - 2 \arctan x$ and the derivative changes sign. Check which interval the evaluation point lies in before you quote the collapsed form.

SEE ALSO Conjugate log collapse; trig identity reduction before high-order differentiation; log-law expansion.

Identities and normal forms

The wrapper techniques remove an outer function. The next two remove interior structure: a product of trig functions becomes a single sinusoid, and a tangle of radicals and reciprocals becomes a sum of powers. Both are about putting the expression into a normal form on which one rule, and only one rule, acts.

The trig case is where the saving is most spectacular, because it compounds with the derivative order. Differentiating a product of trig functions once is mildly annoying; differentiating it seven times is a catastrophe, unless you first notice that it was a single sine wave the whole time.

11.4 Trig identity reduction before high-order differentiation

CUE A product or power of trigonometric functions, with a derivative of order two or more requested: $f^{(7)}$ of $\sin^2 x$, f'' of $\sin 2x \cos 2x$, $f^{(5)}$ of $\sin x \cos^3 x$.

METHOD Collapse to a sum of pure sinusoids using power reduction, double angle, or product to sum:

$$\sin^2 x = \frac{1}{2} - \frac{1}{2} \cos 2x, \quad \sin 2x \cos 2x = \frac{1}{2} \sin 4x, \quad \sin A \cos B = \frac{1}{2} [\sin(A+B) + \sin(A-B)].$$

Then every order at once from the master formula

$$\frac{d^n}{dx^n} \sin(ax+b) = a^n \sin\left(ax+b+\frac{n\pi}{2}\right), \quad \frac{d^n}{dx^n} \cos(ax+b) = a^n \cos\left(ax+b+\frac{n\pi}{2}\right).$$

Each differentiation multiplies by the frequency and advances the phase by a quarter period; that is the entire content of the trigonometric derivative table.

THE QUARTER TURN

The four-step cycle $\sin \rightarrow \cos \rightarrow -\sin \rightarrow -\cos \rightarrow \sin$ is usually memorised as a list. It is better understood as a single statement: differentiation advances the phase by $\pi/2$ and multiplies by the frequency. That is why $\sin$ and $\cos$ need no separate table, why the cycle has period 4, and why $f^{(n)}$ for large n costs nothing more than f' once the function is a sum of pure sinusoids. It also tells you which n the setter will choose: $n \equiv 1$ or $3 \pmod{4}$ to swap sine for cosine, so that a point like $\pi/4$ evaluates to something clean.

WORKED EXAMPLE

$f(x) = \sin^2 x$; find $f^{(7)}(\pi/4)$. Power reduction gives $f(x) = \frac{1}{2} - \frac{1}{2} \cos 2x$. The constant contributes nothing for $n \geq 1$, and

$$f^{(7)}(x) = -\frac{1}{2} \cdot 2^7 \cos\left(2x + \frac{7\pi}{2}\right) = -2^6 \cos\left(2x - \frac{\pi}{2}\right) = -64 \sin 2x,$$

using $\cos(\theta + 7\pi/2) = \cos(\theta - \pi/2) = \sin \theta$. At $x = \pi/4$, $\sin(\pi/2) = 1$, so $f^{(7)}(\pi/4) = -64$. Verified by contour quadrature of the seventh Cauchy coefficient: -64.0000000 .

A harder one, to show the product-to-sum step doing real work. Take $f(x) = \sin x \cos^3 x$ and ask for $f^{(5)}$. Write $\cos^2 x = \frac{1}{2}(1 + \cos 2x)$ and $\sin x \cos x = \frac{1}{2} \sin 2x$, so

$$f(x) = (\sin x \cos x) \cos^2 x = \frac{1}{2} \sin 2x \cdot \frac{1}{2} (1 + \cos 2x) = \frac{1}{4} \sin 2x + \frac{1}{4} \sin 2x \cos 2x = \frac{1}{4} \sin 2x + \frac{1}{8} \sin 4x.$$

Now the master formula, with $\sin(\theta + 5\pi/2) = \cos \theta$:

$$f^{(5)}(x) = \frac{1}{4} \cdot 2^5 \cos 2x + \frac{1}{8} \cdot 4^5 \cos 4x = 8 \cos 2x + 128 \cos 4x.$$

At $x = \pi/4$ this is $8 \cos(\pi/2) + 128 \cos \pi = 0 - 128 = -128$. Verified: -128.0000000 . The alternative is five rounds of the product rule on a four-factor product, which generates dozens of terms that must then be collapsed by the very identities you declined to use at the start.

WHERE THIS GOES WRONG

Grinding the product rule n times. The term count roughly doubles per round, so the seventh derivative of a two-factor product is a hundred-term expression before cancellation, and the cancellation is the identity you skipped. The second trap is subtler: after power reduction the constant term is invisible in every derivative, so a solver who reduces correctly but then also drops it from f itself will get $f(\pi/4)$ wrong while getting every $f^{(n)}(\pi/4)$ right. Reduce for the derivatives; keep the constant for the function.

SEE ALSO Algebraic normalisation; the cyclic n -th derivative pattern; log-law expansion.

Trig identities are one normal form. The other, less glamorous and far more frequently useful, is the reduction of every algebraic decoration to a rational power. This is the technique that removes the largest number of unnecessary quotient rules from a working solver's life.

11.5 Algebraic normalisation: roots to powers, quotients to sums

CUE Radicals, nested radicals, reciprocal trig functions, or a fraction whose numerator is a *sum* sitting over a single monomial denominator: $\frac{x^3 + \sqrt{x} - 1}{x^2}$, $\sqrt[3]{x^5}$, $\sqrt{x\sqrt{x\sqrt{x}}}$, $\frac{1}{\csc^2 x}$.

METHOD Rewrite every root as a rational power, $\sqrt[m]{x^k} = x^{k/m}$, and split every sum-over-monomial quotient termwise, $\frac{a_1 + \dots + a_n}{g} = \frac{a_1}{g} + \dots + \frac{a_n}{g}$ when g is a single term. Send reciprocal trig functions back to base form: $1/\csc^2 x = \sin^2 x$, $1/\sec x = \cos x$, and use $\sec^2 x - \tan^2 x = 1$ wherever it appears. After normalisation the power rule $\frac{d}{dx} x^p = px^{p-1}$ usually does all the work, and the quotient and chain rules never enter.

WORKED EXAMPLE

Nested radicals first, because they look worst and cost least:

$$f(x) = \sqrt{x\sqrt{x\sqrt{x}}} = (x \cdot (x \cdot x^{1/2})^{1/2})^{1/2} = (x \cdot x^{3/4})^{1/2} = x^{7/8},$$

so $f'(x) = \frac{7}{8}x^{-1/8}$ and $f'(2) = \frac{7}{8} \cdot 2^{-1/8} = 0.802379$. Three chain rules avoided by one exponent computation.

Now an expression engineered to look like a quotient-rule problem and to be nothing of the kind:

$$f(x) = \frac{(\sin x + e^{\cos x})(\cos^2 x + \sin^2 x)}{\csc^2 x}.$$

The middle factor is 1 and the denominator is $\sin^{-2} x$, so $f(x) = (\sin x + e^{\cos x}) \sin^2 x$: a two-factor product with no quotient anywhere. Then

$$f'(x) = (\cos x - \sin x e^{\cos x}) \sin^2 x + (\sin x + e^{\cos x}) \cdot 2 \sin x \cos x.$$

At $x = \pi/2$: $\cos x = 0$, $\sin x = 1$, $e^{\cos x} = 1$. The first bracket is $0 - 1 = -1$ and multiplies 1; the second term carries a factor $\cos(\pi/2) = 0$. So $f'(\pi/2) = -1$. Verified by complex-step differentiation: -1.0000000 .

Finally the everyday case, a sum over a monomial:

$$f(x) = \frac{3x^4 - 2x^2 + 5}{x^3} = 3x - \frac{2}{x} + 5x^{-3}, \quad f'(x) = 3 + \frac{2}{x^2} - 15x^{-4},$$

and $f'(1) = 3 + 2 - 15 = -10$. The quotient rule reaches the same place through a numerator of degree six over x^6 , with a cancellation of x^3 that must be performed by hand.

WHERE THIS GOES WRONG

Splitting a quotient whose *denominator* is a sum. The move $\frac{a+b}{g} = \frac{a}{g} + \frac{b}{g}$ is legal; the move $\frac{1}{\frac{a}{b} + \frac{c}{d}} = \frac{1}{\frac{a}{b}} + \frac{1}{\frac{c}{d}}$ is not, and it is the single commonest algebraic error in this family. The test is mechanical: you may split over a sum upstairs, never over a sum downstairs. A second, quieter error is losing the domain: $\sqrt{x^2} = |x|$, not x , so normalising $\sqrt{x^2}$ to x^1 silently changes the derivative from $\operatorname{sgn} x$ to 1.

SEE ALSO Log-law expansion; when the quotient rule is the wrong tool; trig identity reduction.

Reading what is actually variable

Normalisation and identity reduction share a governing idea worth naming. Both replace an expression by an equivalent one on which *fewer rules* act. A nest of radicals needs three chain rules; the same function written as $x^{7/8}$ needs one power rule. A four-factor trig product needs a product rule per differentiation; the same function written as $\frac{1}{4} \sin 2x + \frac{1}{8} \sin 4x$ needs none, only a phase advance. The measure of a good normal form is exactly this: how many distinct rules survive contact with it. Aim for one.

Every technique so far assumed the whole expression depends on x and asked how to reorganise it. The next assumes almost none of it does. This is a different kind of reading; not algebraic manipulation but a census of where the variable actually appears, and it is the fastest technique in the book when it applies.

11.6 Constant scan: find the one factor that depends on x

CUE A wall of intimidating symbols with at most one place where x actually occurs: γ , e , π , Γ of a number, $\cos^{1000}(\pi/e)$, $\sum_{n=1}^{1000} n$, Ω , all arranged so the expression looks like it needs everything you know.

METHOD Before writing any rule, scan every sub-expression for the literal variable. Named constants, factorials and Gamma values of numbers, finite numeric sums, and transcendental functions evaluated at constants are all numbers: they differentiate to zero and factor out of the problem. Differentiate only the surviving term. If nothing survives, $f' \equiv 0$.

WORKED EXAMPLE

$$f(x) = \frac{9 (\sin \gamma)^{99999 \cos(1/x)}}{e^2 + 2\pi - 2026e + \cos^{1000}(\pi/e)} + \sum_{n=1}^{1000} n.$$

The census: γ is Euler's constant, a number, so $\sin \gamma$ is a number. The entire denominator is a number; call it K . The finite sum $\sum_{n=1}^{1000} n = 500500$ is a number. The only occurrence of x in the whole display is inside $\cos(1/x)$, sitting in the exponent. So with $a = \sin \gamma$ and $E(x) = 99999 \cos(1/x)$ we have $f(x) = \frac{9}{K} a^{E(x)} + 500500$ and

$$f'(x) = \frac{9}{K} a^{E(x)} \ln a \cdot E'(x) = \frac{9}{K} (\sin \gamma)^{99999 \cos(1/x)} \ln(\sin \gamma) \cdot 99999 \cdot \frac{\sin(1/x)}{x^2},$$

since $\frac{d}{dx} \cos(1/x) = -\sin(1/x) \cdot (-x^{-2}) = \sin(1/x)/x^2$. One application of the exponential rule and one chain rule, on a display that appears to need six.

A constant that is doing structural work, for contrast. In $f(x) = \cos^{1000}(x/1000)$ the exponent and the divisor are both large numbers and neither is decoration: $f'(x) = -1000 \cos^{999}(x/1000) \sin(x/1000) \cdot \frac{1}{1000} = -\cos^{999}(x/1000) \sin(x/1000)$, and the two thousands cancel exactly. A solver who deletes them as noise loses nothing here and everything in the next problem, where they do not cancel.

A shorter instance of the same reading: $\frac{d}{dx} e^{\sin^2 x + \cos^2 x} = \frac{d}{dx} e = 0$, and $\frac{d}{dx} (x^0 + \Gamma(7) + \arctan(\tan 1)) = 0$ for $x \neq 0$. Both are trivial once you notice that no exponent, no argument, and no index depends on x .

WHERE THIS GOES WRONG

Treating γ or Ω as a function of x because the symbol looks exotic. The reverse error is worse and rarer: assuming that a large decorative constant is inert when it is doing structural work. In the display above the exponent 99999 multiplies the answer by 99999; deleting it as decoration costs five orders of magnitude. Read the constants, then keep them.

SEE ALSO Conjugate log collapse; the general power rule; algebraic normalisation.

The constant scan and the conjugate collapse are two faces of one idea: the derivative is a local statement about variation, and anything that does not vary is invisible to it. Where they differ is in how the constancy is hidden. In the constant scan it is hidden by volume: there is so much symbol on the page that the eye never audits it. In the conjugate collapse it is hidden by structure: every symbol genuinely contains x , and the constancy emerges only after multiplication. The first is defeated by a census, the second by an identity, and a numerical spot check defeats both.

When the function is given as a process

The last two techniques handle a different presentation entirely. So far the function has been a formula, however disguised. Now it is a recipe: a series to be summed, a product to be multiplied out, a nesting to be continued forever. In every such case the differentiation is trivial once the recipe is resolved, and essentially impossible before. Attacking these termwise is not merely slower; it is usually a different and harder problem, and it quietly assumes analytic facts that nobody has checked.

11.7 Sum the hidden series, product, or continued fraction first

CUE The function is presented as a $\sum$, a $\prod$, or a continued fraction, and a single derivative *value* is requested. The request for one value is the tell: nobody wants a series for the answer.

METHOD Recognise and collapse the closed form before differentiating. The recurring engines are few. Geometric series, $\sum_{k \geq 0} r^k = \frac{1}{1-r}$ for $|r| < 1$. Doubling products, $\prod_{k=0}^n (1+x^{2^k}) = \frac{1-x^{2^{n+1}}}{1-x}$, which is just the binary expansion of every exponent from 0 to $2^{n+1}-1$. The cotangent telescope, $\cot \theta - 2 \cot 2\theta = \tan \theta$, which sums halving tangent series. And recognising a listed Maclaurin series as an elementary function. Two further engines appear less often but are unmistakable when they do: the cyclotomic factorisation

$$x^{2^n} - 1 = (x^2 - 1) \prod_{k=1}^{n-1} \left(x^2 - 2x \cos \frac{k\pi}{n} + 1 \right),$$

which turns a product of $n-1$ quadratics into a binomial (checked at $n=3$, $x=1.7$: both sides equal 23.137569), and

Gauss's continued fraction $\tan x = \frac{x}{1 - \frac{x^2}{3 - \frac{x^2}{5 - \dots}}}$, which is the only sane reading of a continued fraction whose partial denominators run 1, 3, 5, ...

WORKED EXAMPLE

A geometric tail inside a radical:

$$f(x) = \sqrt{x + 1 + \frac{1}{x} + \frac{1}{x^2} + \dots} \quad (x > 1).$$

The bracket is geometric with first term x and ratio $1/x$, so it sums to $\frac{x}{1-1/x} = \frac{x^2}{x-1}$ and $f(x) = \frac{x}{\sqrt{x-1}} = x(x-1)^{-1/2}$. Then

$$f'(x) = (x-1)^{-1/2} - \frac{x}{2}(x-1)^{-3/2} = (x-1)^{-3/2} \left[(x-1) - \frac{x}{2} \right] = \frac{x-2}{2(x-1)^{3/2}},$$

so $f'(2) = 0$: the function has a minimum exactly at the integer point you were asked about. Verified: 0.0000000.

A telescoping product, done two ways as a check. Let $f(x) = (1+x)(1+x^2)(1+x^4)(1+x^8)$ and find $f'(1)$. Closed form

first: multiplying by $(1-x)$ telescopes, giving $f(x) = \frac{1-x^{16}}{1-x} = 1+x+x^2+\dots+x^{15}$, so $f'(1) = 0+1+2+\dots+15 = 120$.

As a check, the logarithmic derivative of the product form at $x=1$ is $\frac{1}{2} + \frac{2}{2} + \frac{4}{2} + \frac{8}{2} = \frac{15}{2}$ and $f(1) = 2^4 = 16$, giving $16 \cdot \frac{15}{2} = 120$ again. Verified: 120.0000000.

Finally the tangent telescope, which is the hardest of the three and the most satisfying. Using $\tan \theta = \cot \theta - 2 \cot 2\theta$ with $\theta = x/2^n$, the sum $\sum_{n \geq 1} 2^{-n} \tan(x/2^n)$ telescopes to $\frac{1}{x} - \cot x$, since $2^{-n} \cot(x/2^n) \rightarrow 1/x$. Hence

$$f(x) = \sum_{n \geq 1} \frac{1}{2^n} \tan \frac{x}{2^n} = \frac{1}{x} - \cot x, \quad f'(x) = -\frac{1}{x^2} + \csc^2 x,$$

and $f'(\pi/3) = -\frac{9}{\pi^2} + \frac{1}{3/4} = \frac{4}{3} - \frac{9}{\pi^2}$, numerically 0.421443. The identity itself was checked to nine digits at $x=1.1$ against $1/x - \cot x$.

WHERE THIS GOES WRONG

Differentiating termwise and then trying to sum the derivative series. This is usually a much harder sum than the original, and it silently assumes that the differentiated series converges uniformly on a neighbourhood of the point, which is exactly the hypothesis that licenses termwise differentiation. For the tangent series above, termwise differentiation produces $\sum 4^{-n} \sec^2(x/2^n)$, which converges but whose closed form nobody recognises. Sum first, then differentiate.

SEE ALSO Self-similar fixed-point resolution; algebraic normalisation; logarithmic differentiation of many-factor products.

A function given as a series, a product, or an infinite nesting is not a harder differentiation problem. It is a problem that has not been stated yet. The statement is the closed form, and finding it is algebra, summation, or fixed-point reasoning, none of which is calculus. Once it is found the differentiation is routine, which is the tell: if the derivative you are computing is itself hard, you have almost certainly not finished the previous step. This is also why these problems ask for a single *value*. A value can be checked; a series cannot.

Summing a series requires a formula for the partial sums. Infinite nestings do not have partial sums in any useful sense, but they have something better: they contain themselves, and that self-reference converts an infinite object into a finite equation in one line.

11.8 Self-similar fixed-point resolution

HARD

CUE An infinite nesting that repeats itself: a power tower $x^{x^{x^{\cdots}}}$, a nested radical $\sqrt{x + \sqrt{x + \cdots}}$, an infinite logarithmic tower $\ln(x \ln(x \ln \cdots))$, or the limit of an iterated map.

METHOD Name the whole object t and exploit self-similarity: the thing inside the outermost operation is the whole object again. That gives a finite equation $t = \Phi(x, t)$. Solve it if you can; otherwise differentiate it implicitly, which is almost always enough. For the tower, $t = x^t$; for the log tower, $t = \ln(xt)$, that is $e^t = xt$; for an iterated map with limit L , solve $L = \varphi(L)$ and note that L is then a genuine constant and the calculus is over.

WORKED EXAMPLE

The log tower, where the inverse function is the clean object. Let $f(x) = \ln(x \ln(x \ln(\cdots)))$. Self-similarity gives $f = \ln(xf)$, hence $e^f = xf$. Now solve for x rather than for f : if $y = f(x)$ then $x = \frac{e^y}{y}$, so the inverse function is explicit,

$$f^{-1}(y) = \frac{e^y}{y}, \quad (f^{-1})'(y) = \frac{e^y y - e^y}{y^2} = \frac{e^y(y-1)}{y^2},$$

and at $y = \frac{3}{2}$ this is $\frac{e^{3/2} \cdot \frac{1}{2}}{\frac{9}{4}} = \frac{2e^{3/2}}{9}$, numerically 0.995931. Verified by complex-step differentiation of e^y/y .

The power tower, done implicitly. Let $t = x^{x^{x^{\cdots}}}$, so $t = x^t$ and $\ln t = t \ln x$. Differentiating both sides in x ,

$$\frac{t'}{t} = t' \ln x + \frac{t}{x} \implies t' \left(\frac{1}{t} - \ln x \right) = \frac{t}{x} \implies t' = \frac{t^2}{x(1 - t \ln x)}.$$

At $x = \sqrt{2}$ the tower converges to $t = 2$, since $(\sqrt{2})^2 = 2$, and $t \ln x = 2 \cdot \frac{1}{2} \ln 2 = \ln 2$. So $t'(\sqrt{2}) = \frac{4}{\sqrt{2}(1 - \ln 2)} = 9.21754$. Verified by iterating the tower four thousand times and taking a symmetric difference quotient with step 10^{-6} : 9.21754.

The nested radical, which is the gentlest member of the family and the best place to see the mechanism. Let $t = \sqrt{x + \sqrt{x + \sqrt{x + \cdots}}}$. Everything under the outermost radical is x plus the whole object again, so $t^2 = x + t$, a quadratic. Taking the root that is positive for $x > 0$,

$$t = \frac{1 + \sqrt{1 + 4x}}{2}, \quad \frac{dt}{dx} = \frac{1}{\sqrt{1 + 4x}},$$

or, without solving at all, differentiate $t^2 = x + t$ implicitly to get $2tt' = 1 + t'$, hence $t' = \frac{1}{2t-1}$, which is the same thing. At $x = 2$ the fixed point is $t = 2$ and $t' = \frac{1}{3}$. Verified by iterating the radical two thousand times and taking a symmetric difference quotient: 0.333333. The implicit route is the one to prefer, because it works when the fixed-point equation cannot be solved in closed form.

WHERE THIS GOES WRONG

Forgetting to ask which fixed point the construction converges to, and whether it converges at all. The tower $x^{x^{x^{\cdots}}}$ converges only for $e^{-e} \leq x \leq e^{1/e}$, and the equation $t = x^t$ has two roots on part of that range; at $x = \sqrt{2}$ the roots are $t = 2$ and $t = 4$, and the tower converges to 2. Writing t' in terms of the wrong root gives a confident wrong number. The second trap is the reverse of the usual one: after x is replaced by a specific number the nesting may collapse to a constant, so the calculus becomes trivial rather than harder, and a solver braced for difficulty can walk straight past that.

SEE ALSO Sum the hidden series, product, or continued fraction first; power towers and iterated exponentials; conjugate log collapse.

The scan applied end to end

Every technique above has been shown in isolation. Real problems stack them, and the stacking is the point: a setter who wants a hard problem applies three identities backwards, not one. Here is a single expression that needs four of the chapter's moves in sequence, and which takes two lines once they are made and rather more than two pages if they are not. For $x > 0$, let

$$f(x) = \ln \left(\frac{(\sin 2x \cos 2x)^3 \sqrt[4]{x^6}}{e^{\arctan x + \arctan(1/x)}} \right).$$

Move one, log laws. The outer logarithm wraps a product, a quotient, and two powers, so expand:

$$f(x) = 3 \ln(\sin 2x \cos 2x) + \frac{6}{4} \ln x - (\arctan x + \arctan(1/x)),$$

the exponential in the denominator having been annihilated by $\ln e^u = u$.

Move two, the constant scan. The last bracket is $\pi/2$ for every $x > 0$. It contributes nothing to f' and can be struck out now.

Move three, a trig identity. $\sin 2x \cos 2x = \frac{1}{2} \sin 4x$, so the first term is $3 \ln \frac{1}{2} + 3 \ln \sin 4x$, and the constant $3 \ln \frac{1}{2}$ also dies.

Move four, radicals to powers. $\sqrt[4]{x^6} = x^{3/2}$, so the middle term is $\frac{3}{2} \ln x$, which was already the case but is now visibly so.

What remains is $f(x) = 3 \ln \sin 4x + \frac{3}{2} \ln x + \text{const}$, and one differentiation finishes it:

$$f'(x) = 3 \cdot \frac{4 \cos 4x}{\sin 4x} + \frac{3}{2x} = 12 \cot 4x + \frac{3}{2x}.$$

At $x = \pi/16$ we have $4x = \pi/4$ and $\cot(\pi/4) = 1$, so

$$f'\left(\frac{\pi}{16}\right) = 12 + \frac{3}{2 \cdot \pi/16} = 12 + \frac{24}{\pi} = 19.639437.$$

Verified by complex-step differentiation: 19.639437. Count what was avoided: a chain rule around a quotient rule around two product rules around a chain rule, and an arctan pair whose derivatives cancel only after being written out. The evaluation point $\pi/16$ was, as always, the setter's signature: it is the point at which the surviving cotangent is 1.

One closing remark on procedure, since the chapter has been a list of moves and the moves are worth nothing without the habit that triggers them. The habit is cheap to install: after reading the problem and before writing anything, say out loud what the expression *is*. "It is a logarithm of a quotient." "It is a product of two sinusoids." "It is a constant." The sentence forces the census that the decision tree formalises, and in the overwhelming majority of cases it also names the technique. When the sentence comes out as "it is a genuinely irreducible composition of three functions", you have learned something too, and the next chapter is about what to do then.

The chapter's claim, stated once more in its sharpest form: there is no such thing as a hard derivative, only a hard expression. Every technique here is a way of making the expression easier before the derivative is taken, and every one of them is an identity you already know, applied in the direction the setter did not. What the chapter adds is the order in which to try them and the discipline to try them at all. The next chapter takes up the residue, the expressions that genuinely do not collapse, and asks a different question: given that you must differentiate, which rule system should you use?

Recognition matrix

If you see	Reach for	If that fails
In of a product, quotient, power or radical	Log laws first, then differentiate termwise	Check positivity of each split factor
Two logs with conjugate arguments, added arctan or arcsin of a rational-in-radicals argument	Combine: the product is usually 1, so $f' \equiv 0$ $x = \tan \theta$ or $x = \sin \theta$; a half-angle collapse	Evaluate f at two points and compare Identify the interval; the branch may flip the sign
$f^{(n)}$ of a product or power of trig functions	Power reduction to $\sum c_k \sin(ax + b_k)$	Master formula: $a^n \sin(ax + b + n\pi/2)$
Radicals or nested radicals of powers of x	Collapse the exponents; then the power rule alone	Watch $\sqrt{x^2} = x $
A sum over a single monomial denominator	Split termwise; never split over a sum downstairs	Write uv^{-1} and use product + chain
$1/\csc^2 x$, $\sec^2 x - \tan^2 x$, $\cos^2 x + \sin^2 x$	Reciprocal trig to base form; Pythagorean identities	Substitute a numeric x to confirm the value
A wall of γ , π , $\Gamma(7)$, $\sum_{n=1}^{1000} n$	Census for the literal x ; everything else is a number	If no x survives, $f' \equiv 0$
$\arctan x + \arctan(1/x)$, $\arcsin x + \arccos x$	A constant on each interval; derivative 0	Confirm the constant's value per branch
f given as $\sum$ or $\prod$, one value wanted	Sum or telescope to closed form, then differentiate	Log-differentiate the finite product instead

If you see	Reach for	If that fails
$\prod_{k=0}^n (1 + x^{2^k})$ $\sum 2^{-n} \tan(x/2^n)$ and relatives $x^{x^{x^{\dots}}}$ or $\sqrt{x + \sqrt{x + \dots}}$ An infinite nesting at a specific x Nothing collapses after the four questions	Multiply by $(1 - x)$: it telescopes Cotangent telescope $\tan \theta = \cot \theta - 2 \cot 2\theta$ Set $t = \Phi(x, t)$, differentiate implicitly Check convergence and which fixed point Go to the rule systems: chain, product, quotient	Logarithmic derivative of the product Check the identity numerically first Solve for x in terms of t and invert The nesting may be a constant there Logarithmic differentiation as a solvent

The rule systems, and logarithmic differentiation

What this chapter is for. When the algebra of the previous chapter runs out, you differentiate, and differentiating means choosing a rule. The choice is not free. The product, quotient and chain rules are adapted to three different shapes, and applying one to a shape it was not built for produces algebra that never ends: a correct answer buried under a cancellation you now have to perform by hand. This chapter gives explicit selection guidance, including the cases where the quotient rule is the wrong tool and should be replaced by a termwise split or by a negative power, and the generalised product rule for n factors, which is what people reach for two-at-a-time and lose terms doing. It then develops logarithmic differentiation as the general solvent: take $\ln$ of both sides, differentiate implicitly, multiply back by y . That single manoeuvre handles many-factor products, the variable-base-variable-exponent form $f(x)^{g(x)}$ where neither the power rule nor the exponential rule applies, iterated exponential towers, and multiplicative functional equations. At the end you should be able to look at a shape and name its rule system before writing anything.

The rules are typed by shape

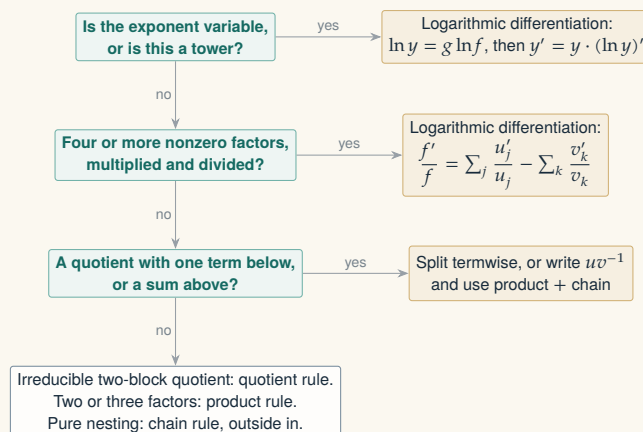
Every elementary expression is a tree. At its root sits one operation: an addition, a multiplication, a division, or a composition. The differentiation rules are typed by that root operation, and the discipline is to identify the root before doing anything else. $\sin(x) \cos(x)$ has multiplication at the root; $\sin(x \cos x)$ has composition at the root and multiplication one level down. The two expressions share every symbol and need entirely different opening moves.

This sounds obvious, and it is the source of most wasted work in the subject. What actually happens is that a solver sees a fraction bar, reaches for the quotient rule reflexively, and only later discovers that the denominator was a single monomial and the whole thing was a sum of three powers. Or a solver sees four factors, applies the two-factor product rule three times in a nested way, and drops a term. Or a solver sees x^x , recognises neither a power nor an exponential, and produces $x \cdot x^{x-1} = x^x$, which is wrong by a factor that happens to be $1 + \ln x$.

THE FOUR SHAPES AND THEIR SYSTEMS

Nesting, $f(g(x))$: chain rule, peeled outside in, each derivative evaluated at the remaining inner stack. *Juxtaposition*, $u_1 u_2 \cdots u_n$: the generalised product rule, one term per factor. *Division*, u/v : the quotient rule only if the division is irreducible; otherwise split termwise or write uv^{-1} . *Variable exponent*, $f(x)^{g(x)}$: neither the power rule nor the exponential rule, but logarithmic differentiation, which is the only system that covers all four shapes at once. When two or more shapes are present, work from the root down, and re-read the tree after every step.

Logarithmic differentiation deserves its position at the end of that list. It is not a fourth rule alongside the others; it is a change of representation that turns multiplication into addition, division into subtraction, and exponentiation into multiplication, so that the product, quotient and power rules all become the sum rule. That is why it is the general solvent, and why the second half of this chapter is devoted to it.



Two of the four leaves say “logarithmic differentiation”, which is a fair reflection of how often it is the right answer once an expression is genuinely multiplicative. The bottom box is the ordinary case and it is not a failure: most derivatives really are one product rule or one chain rule, executed cleanly. What the tree prevents is the two expensive misreadings, namely using the quotient rule on a reducible quotient and using a power or exponential rule on a variable-exponent form.

The three rules, used deliberately

Start with composition, because it is the one rule that never has an alternative. Products can be log-differentiated and quotients can be rewritten, but a genuine nesting must be peeled, and the only question is whether you peel it in a disciplined order.

12.1 Outside-in chain-rule discipline

EASY

CUE Three or more genuine layers with no available simplification: $\sin(\cos(1 + x^2))$, $e^{\tan(\ln x)}$, $\operatorname{arsinh} \sqrt{1 + e^{3x}}$.

METHOD Peel from the outermost function inward, never from the inside out:

$$\frac{d}{dx} f_1(f_2(\cdots f_k(x))) = \prod_{j=1}^k f'_j(f_{j+1}(\cdots f_k(x))),$$

where each factor is evaluated at the *remaining inner stack*, not at x . Write the layers as a numbered list before differentiating if there are more than three, and substitute the numerical evaluation point only at the very end.

WORKED EXAMPLE

Three layers, chosen so the point is kind. $f(x) = e^{\tan(\ln x)}$ has layers exp, tan, ln, so

$$f'(x) = e^{\tan(\ln x)} \cdot \sec^2(\ln x) \cdot \frac{1}{x}.$$

At $x = 1$: $\ln 1 = 0$, $\tan 0 = 0$, $e^0 = 1$, $\sec^2 0 = 1$, so $f'(1) = 1 \cdot 1 \cdot 1 = 1$. Verified: 1.0000000.

Four layers, with a radical in the middle. $f(x) = \operatorname{arsinh} \sqrt{1 + e^{3x}}$. With $u = \sqrt{1 + e^{3x}}$,

$$f'(x) = \frac{u'}{\sqrt{1 + u^2}}, \quad u' = \frac{3e^{3x}}{2\sqrt{1 + e^{3x}}}, \quad 1 + u^2 = 2 + e^{3x}.$$

At $x = 0$: $u = \sqrt{2}$, $u' = \frac{3}{2\sqrt{2}}$, $\sqrt{1 + u^2} = \sqrt{3}$, so $f'(0) = \frac{3}{2\sqrt{6}} = \frac{\sqrt{6}}{4} = 0.612372$. Verified to seven digits.

And one where the point is deliberately unkind, to show that this is allowed. $f(x) = \sin(\cos(1 + x^2))$ gives $f'(x) = -2x \cos(\cos(1 + x^2)) \sin(1 + x^2)$, and at $x = \pi - 1$ this is

$$-2(\pi - 1) \cos(\cos(1 + (\pi - 1)^2)) \sin(1 + (\pi - 1)^2) = 1.979199.$$

There is no simplification. The evaluation point contains π purely to make you hunt for one.

And, for balance, a nest that should never have been peeled at all. Consider $f(x) = \arctan(\sinh(\ln x))$ for $x > 0$. Peeling gives

$$f'(x) = \frac{1}{1 + \sinh^2(\ln x)} \cdot \cosh(\ln x) \cdot \frac{1}{x} = \frac{\cosh(\ln x)}{x \cosh^2(\ln x)} = \frac{1}{x \cosh(\ln x)},$$

using $1 + \sinh^2 = \cosh^2$, and since $\cosh(\ln x) = \frac{1}{2}(x + 1/x)$ this is $\frac{2}{x^2 + 1}$. So $f'(1) = 1$ and $f'(2) = \frac{2}{5}$, both verified to seven digits. The chain rule got there, but the previous chapter's scan would have noticed $\sinh(\ln x) = \frac{1}{2}(x - 1/x)$ at the start and reduced the whole thing to a single arctan of a rational function. Run the scan first; the chain rule is what you use when it comes back empty.

WHERE THIS GOES WRONG

Evaluating an outer derivative at x instead of at the inner block: writing $\cos(x)$ where the formula demands $\cos(\cos(1 + x^2))$. This is the defining error of the chain rule and it survives into every later technique, because Faa di Bruno, implicit differentiation and the inverse-function rule are all chain-rule bookkeeping. The second trap is assuming that an ugly evaluation point *must* trigger a hidden collapse; sometimes the answer is simply transcendental, and the hunt costs more than the arithmetic.

SEE ALSO Composition bookkeeping and functional inversion; the generalised product rule for n factors; logarithmic differentiation of many-factor products.

Composition has one rule and no alternative. Multiplication has two: the product rule, which is exact and always available, and logarithmic differentiation, which is faster once the factor count rises but fails where a factor vanishes. It is worth stating the product rule in its n -factor form first, because the two-factor form applied repeatedly is where terms get lost.

EASY

12.2 Generalised product rule for n factors

CUE Three or more factors multiplied, each nontrivial: $x^2 e^x \sin x$, $\ln(x^2 + 3x) e^{\tan x} (x^3 + 1)^{-1}$, $x(x-1)(x-2)(x-3)$.

METHOD One term per factor, each term differentiating exactly one slot and leaving the rest alone:

$$(u_1 u_2 \cdots u_n)' = \sum_{j=1}^n u_j' \prod_{k \neq j} u_k.$$

The result has exactly n terms, which is the checksum: count them before simplifying. When every factor is nonzero at the point, dividing through by f gives $\frac{f'}{f} = \sum_j \frac{u_j'}{u_j}$, which is logarithmic differentiation and is usually the faster route from four factors up.

WORKED EXAMPLE

$f(x) = x^2 e^x \sin x$, three factors, so three terms:

$$f'(x) = 2x e^x \sin x + x^2 e^x \cos x + x^2 e^x \sin x.$$

At $x = \pi$ the first two terms carry $\sin \pi = 0$ and only the third survives: $f'(\pi) = \pi^2 e^\pi \cos \pi = -\pi^2 e^\pi = -228.3895$. Verified to seven digits. The choice of π is the setter arranging for two of the three terms to die, which is a common design and a reason to keep the terms unexpanded until the point is substituted.

The four-factor case, where the product rule beats logarithmic differentiation outright. Let $f(x) = x(x-1)(x-2)(x-3)$ and find $f'(0)$. The generalised rule gives four terms, and three of them retain the factor x , which vanishes at 0. Only the term that differentiates the first slot survives:

$$f'(0) = 1 \cdot (0-1)(0-2)(0-3) = -6.$$

Verified: -6.0000000 . Logarithmic differentiation cannot be used here, because $f(0) = 0$ and f'/f is undefined at the very point requested. Note how little work the product rule did: at a zero of one factor, $n-1$ of the n terms disappear. A three-factor case with a chain rule inside one slot, to show the two rules composing. With $u_1 = \ln(x^2 + 3x)$, $u_2 = e^{\tan x}$, $u_3 = (x^3 + 1)^{-1}$,

$$\frac{f'}{f} = \frac{2x+3}{(x^2+3x)\ln(x^2+3x)} + \sec^2 x - \frac{3x^2}{x^3+1},$$

and at $x = 1$ the three contributions are 0.901686, 3.425519 and -1.5 , with $f(1) = 3.290024$, giving $f'(1) = 9.301566$. Verified by complex-step differentiation.

WHERE THIS GOES WRONG

Nesting the two-factor rule and losing a term. Applying $(uv)' = u'v + uv'$ to $u = u_1$, $v = u_2 u_3$ is correct but requires a second application inside, and the standard failure is to differentiate $u_2 u_3$ as $u_2' u_3'$. Count terms: an n -factor product must produce exactly n terms before simplification. The other error is bracketing so that a chain-rule factor generated in one slot silently multiplies a neighbouring term.

SEE ALSO Logarithmic differentiation of many-factor products; when the quotient rule is the wrong tool; outside-in chain-rule discipline.

That leaves division, which is the shape most often mishandled, not because the quotient rule is hard but because it is applied when it should not be. The quotient rule is the correct tool for exactly one situation: an irreducible ratio of two genuinely composite blocks. Everything else is cheaper another way, and the difference is not stylistic.

12.3 When the quotient rule is the wrong tool

CUE A quotient whose denominator is a monomial, a constant, a single exponential, or a power of one factor; or a quotient whose numerator is a sum; or a quotient of two messy nonzero products.

METHOD Three cheaper routes, in order of preference. First, split: $\frac{a_1 + \dots + a_n}{g} = \sum_j \frac{a_j}{g}$ whenever g is a single term, then differentiate a sum. Second, rewrite $\frac{u}{v} = uv^{-1}$ and use the product rule with a chain rule on v^{-1} , which avoids squaring the denominator. Third, when u and v are both nonzero products, log-differentiate. Reserve

$$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$$

for a genuinely irreducible two-block quotient such as $\frac{x^2 + 1}{x^2 - 1}$ or $\frac{\ln x}{\sin x}$.

WORKED EXAMPLE

The split, on the standard example:

$$f(x) = \frac{x^3 + \sqrt{x} - 1}{x^2} = x + x^{-3/2} - x^{-2}, \quad f'(x) = 1 - \frac{3}{2}x^{-5/2} + 2x^{-3},$$

so $f'(4) = 1 - \frac{3}{2} \cdot \frac{1}{32} + \frac{2}{64} = 1 - \frac{3}{64} + \frac{2}{64} = \frac{63}{64}$. Verified: 0.984375. The quotient rule reaches the same place through a numerator of degree 4 over x^4 and a cancellation of x^2 that must be done by hand, with the half-integer power differentiated inside a bracket.

The negative-power rewrite, where the split is unavailable because the denominator is transcendental. Take $f(x) = \frac{e^x + x^2}{e^x}$. Here the denominator *is* a single term, so the split does apply and is best: $f(x) = 1 + x^2e^{-x}$, and

$$f'(x) = 2xe^{-x} - x^2e^{-x} = x(2 - x)e^{-x},$$

whence $f'(2) = 0$. Verified: 0.0000000. The quotient rule gives $\frac{(e^x + 2x)e^x - (e^x + x^2)e^x}{e^{2x}}$, which is the same number after cancelling e^x from every term, but the cancellation is invisible until it is performed.

And a case where the quotient rule is genuinely right, so that the technique is a selection rule and not a prohibition.

For $f(x) = \frac{x^2 + 1}{x^2 - 1}$ neither block factors usefully and neither is a single term, so

$$f'(x) = \frac{2x(x^2 - 1) - (x^2 + 1)2x}{(x^2 - 1)^2} = \frac{-4x}{(x^2 - 1)^2},$$

and $f'(2) = -\frac{8}{9}$. Verified: -0.888889.

WHERE THIS GOES WRONG

Sign order in the numerator. It is $u'v - uv'$, never $uv' - u'v$, and the mnemonic that survives pressure is that the numerator's derivative comes first, exactly as in the product rule. The second standard error is forgetting to square the denominator. The third is the algebraic one from the previous chapter, splitting over a sum *downstairs*: $\frac{1}{a+b} \neq \frac{1}{a} + \frac{1}{b}$.

SEE ALSO Algebraic normalisation: roots to powers, quotients to sums; the generalised product rule for n factors; variable-base logarithms.

One derivative, three rule systems

It is worth doing a single derivative by every route on offer, because the comparison is this chapter's argument in miniature. Take

$$f(x) = \frac{x^2\sqrt{x+1}}{(x-2)^3}, \quad \text{find } f'(3).$$

By the quotient rule. The numerator is itself a product, so a product rule is needed inside: $u = x^2\sqrt{x+1}$ gives $u' = 2x\sqrt{x+1} + \frac{x^2}{2\sqrt{x+1}}$, and $v = (x-2)^3$ gives $v' = 3(x-2)^2$. Then $f' = \frac{u'v - uv'}{v^2}$ with a sixth-degree denominator that must be cancelled

down to a cube. At $x = 3$: $u = 9 \cdot 2 = 18$, $u' = 6 \cdot 2 + \frac{9}{4} = \frac{57}{4}$, $v = 1$, $v' = 3$, so $f'(3) = \frac{57}{4} - 54 = -\frac{159}{4}$. Correct, and four places to lose a factor.

By the product rule with a negative power. Write $f = x^2(x+1)^{1/2}(x-2)^{-3}$, a three-factor product, and use the generalised rule:

$$f' = 2x(x+1)^{1/2}(x-2)^{-3} + \frac{1}{2}x^2(x+1)^{-1/2}(x-2)^{-3} - 3x^2(x+1)^{1/2}(x-2)^{-4}.$$

At $x = 3$ the three terms are 12, $\frac{9}{4}$ and -54 , summing to $-\frac{159}{4}$. No denominator was ever squared.

By logarithmic differentiation. Three nonzero factors and a specific point, so

$$\frac{f'}{f} = \frac{2}{x} + \frac{1}{2(x+1)} - \frac{3}{x-2},$$

which at $x = 3$ is $\frac{2}{3} + \frac{1}{8} - 3 = -\frac{53}{24}$. With $f(3) = \frac{9 \cdot 2}{1} = 18$,

$$f'(3) = 18 \cdot \left(-\frac{53}{24}\right) = -\frac{159}{4} = -39.75.$$

Verified by complex-step differentiation: -39.750000 . Three routes, one answer, and a clear ranking: the third is two lines and involves no differentiation of anything but $\ln$ of linear and radical factors; the second is three lines and needs no cancellation; the first is correct and costs a page.

The ranking flips if the question changes. Asked for f' as a *function* in simplified form, logarithmic differentiation ends with a multiplication by f that reassembles the whole expression, and the product-rule route with a common factor pulled out is cleaner. Asked for $f'(2)$, where f is undefined, none of the three applies and the question is about a pole. Read what is asked before choosing.

One more piece of rule discipline before the solvent. The chain rule is usually read forwards, from the composition to its derivative. Read backwards it answers a different and commonly set question: you are told about a composition and asked about one of its ingredients.

MEDIUM

12.4 Composition bookkeeping and functional inversion

CUE You are given q and $p \circ q$ but asked for p' at an *output* value; or you are asked for $(f \circ g^{-1})'(c)$ with g^{-1} not available in closed form.

METHOD The chain rule solved for the unknown factor. From $(p \circ q)'(a) = p'(q(a))q'(a)$: first solve $q(a) = \text{target}$ for a , then

$$p'(\text{target}) = \frac{(p \circ q)'(a)}{q'(a)}.$$

For the inverse form, find a with $g(a) = c$, which is a root-finding step and not a differentiation step, and then $(f \circ g^{-1})'(c) = \frac{f'(a)}{g'(a)}$. The whole difficulty is bookkeeping: knowing which argument each derivative is evaluated at.

WORKED EXAMPLE

Given $q(x) = 2x + 1$ and $(p \circ q)(x) = 4x^2 - 2x + 3$, find $p'(12)$. The target is an output of q , so solve $q(a) = 12$: $2a + 1 = 12$ gives $a = \frac{11}{2}$. Then $(p \circ q)'(x) = 8x - 2$, so $(p \circ q)'(\frac{11}{2}) = 44 - 2 = 42$, and $q' \equiv 2$. Hence

$$p'(12) = \frac{42}{2} = 21.$$

As a check, p is available here: $p(u) = (p \circ q)(\frac{u-1}{2}) = (u-1)^2 - (u-1) + 3 = u^2 - 3u + 5$, so $p'(u) = 2u - 3$ and $p'(12) = 21$.

Now the inverse form, where g^{-1} is not worth computing. Let $f(x) = \ln(x^2 + 2x + 4)$ and $g(x) = \frac{4}{1 + e^{-2x}}$, and find $(f \circ g^{-1})'(2)$. First locate a with $g(a) = 2$: the logistic is 2 at its midpoint, where $e^{-2a} = 1$, so $a = 0$. Then

$$f'(x) = \frac{2x+2}{x^2+2x+4} \implies f'(0) = \frac{2}{4} = \frac{1}{2}, \quad g'(x) = \frac{8e^{-2x}}{(1+e^{-2x})^2} \implies g'(0) = \frac{8}{4} = 2,$$

so $(f \circ g^{-1})'(2) = \frac{1/2}{2} = \frac{1}{4}$. Verified by complex-step differentiation of both functions at 0: $f'(0) = 0.5000000$ and $g'(0) = 2.0000000$.

Evaluating p' at a rather than at $q(a)$. In the first example that would give $p'(\frac{11}{2}) = 8$ instead of $p'(12) = 21$, and nothing in the arithmetic complains. The companion error is dividing by q' evaluated at the target instead of at a ; here q' is constant so the error hides, which is exactly why the example is a poor test of understanding and a good place to build the habit. Always write the argument of every derivative explicitly.

SEE ALSO Outside-in chain-rule discipline; the general power rule for $f(x)^{g(x)}$; variable-base logarithms.

Logarithmic differentiation as the general solvent

The three rules and their selection guidance are now complete, and between them they differentiate any expression built from sums, products, quotients and compositions. What they do not touch is the fourth shape, the variable exponent, and they handle the third shape, division by a many-factor product, only expensively. Both gaps are closed by one manoeuvre, which is the subject of the rest of the chapter.

Here is the manoeuvre in three lines. Given $y = f(x)$ with f nonzero near the point, take logarithms of both sides, differentiate implicitly using $\frac{d}{dx} \ln y = \frac{y'}{y}$, and multiply back:

$$\ln y = \ln f(x) \implies \frac{y'}{y} = (\ln f)'(x) \implies y' = f(x) \cdot (\ln f)'(x).$$

Nothing here is deep. What makes it a solvent is what the logarithm does to structure on the way in: products become sums, quotients become differences, powers become products, and towers lose a storey. Every one of those is a reduction in the complexity the differentiation rules have to face. The quantity f'/f has a name, the logarithmic derivative, and it is worth thinking of as an operator $\mathcal{L}[f] = f'/f$ with the property $\mathcal{L}[fg] = \mathcal{L}[f] + \mathcal{L}[g]$: it converts the multiplicative structure of functions into the additive structure of their derivatives.

$\mathcal{L}$ TURNS MULTIPLICATION INTO ADDITION

The logarithmic derivative $\mathcal{L}[f] = f'/f$ satisfies

$$\mathcal{L}[fg] = \mathcal{L}[f] + \mathcal{L}[g], \quad \mathcal{L}[f/g] = \mathcal{L}[f] - \mathcal{L}[g], \quad \mathcal{L}[f^a] = a \mathcal{L}[f], \quad \mathcal{L}[c] = 0, \quad \mathcal{L}[cf] = \mathcal{L}[f].$$

Those five lines are the whole method. The first says a product rule becomes a sum; the second says a quotient rule becomes a difference; the third says a power rule becomes a multiplication; the fourth and fifth say constants are invisible, which is why $\mathcal{L}$ never carries the clutter that f' does. Note also $\mathcal{L}[f] = (\ln|f|)'$, so the operator is well defined wherever $f \neq 0$, sign notwithstanding.

Two conditions govern its use. First, f must be nonzero at and near the point, since f'/f is otherwise undefined; at a zero, fall back on the product rule, which handles zeros without complaint. Second, when factors are negative you are really working with $\ln|f|$, whose derivative is still f'/f , so the final formula is unaffected, but any intermediate reasoning about $\ln$ of a negative factor is not licensed.

12.5 Logarithmic differentiation of many-factor products and quotients

CORE

CUE Four or more nonzero factors multiplied and divided, especially many linear factors, with a *specific* evaluation point requested.

METHOD Apply $\mathcal{L}$ to the whole expression:

$$\frac{f'}{f} = \sum_j \frac{u'_j}{u_j} - \sum_k \frac{v'_k}{v_k}.$$

Evaluate that sum of reciprocals at the point, then multiply by f at the point. For a product of linear factors it collapses to $\frac{f'}{f} = \sum_j \frac{1}{x - r_j}$, a sum of simple reciprocals with no differentiation left in it at all.

WORKED EXAMPLE

$$f(x) = \frac{(x-1)(x-2)(x-3)}{(x-4)(x-5)}, \quad \text{find } f'(0).$$

All five factors are nonzero at 0, so

$$\frac{f'(0)}{f(0)} = \frac{1}{-1} + \frac{1}{-2} + \frac{1}{-3} - \frac{1}{-4} - \frac{1}{-5} = -\frac{11}{6} + \frac{9}{20} = -\frac{83}{60},$$

and $f(0) = \frac{(-1)(-2)(-3)}{(-4)(-5)} = -\frac{3}{10}$. Hence

$$f'(0) = \left(-\frac{3}{10}\right) \left(-\frac{83}{60}\right) = \frac{83}{200} = 0.415.$$

Verified by complex-step differentiation: 0.4150000. The quotient rule on this expression requires a product rule on a cubic numerator, a product rule on a quadratic denominator, and a fourth-degree denominator to square.

Scale it up and the advantage becomes absurd. For $f(x) = \prod_{k=1}^{10} (x - k)$ at $x = 0$,

$$\frac{f'(0)}{f(0)} = -\sum_{k=1}^{10} \frac{1}{k} = -H_{10} = -\frac{7381}{2520}, \quad f(0) = (-1)^{10} 10! = 3\,628\,800,$$

so $f'(0) = -3\,628\,800 \cdot \frac{7381}{2520} = -1440 \cdot 7381 = -10\,628\,640$. Verified to nine significant figures. The product rule would produce ten terms of nine factors each.

WHERE THIS GOES WRONG

Using it where a factor vanishes at the evaluation point. If $f(x) = x(x-1)(x-2)(x-3)$ and you want $f'(0)$, then $f(0) = 0$ and f'/f is undefined, so the method reports nothing while the answer is a perfectly ordinary -6 . The rule is: a zero of f at the point sends you back to the generalised product rule, where $n-1$ of the n terms vanish and the computation is a one-liner. The second, smaller trap is worrying about $\ln$ of a negative factor. Work with $\ln|u_j|$ throughout; the formula $f'/f = \sum u'_j/u_j$ is unchanged.

SEE ALSO The generalised product rule for n factors; the logarithmic derivative as an operator on functional equations; the general power rule for $f(x)^{g(x)}$.

Products were the easy case, since the product rule was always available as an alternative. The variable-exponent form has no alternative at all, and it is the place where students most reliably produce a confidently wrong answer, because two familiar rules both almost apply.

MEDIUM

12.6 The general power rule: $f(x)^{g(x)}$

CUE A variable base raised to a variable exponent: $(\sin x)^{\tan x}$, $x^{\sin x}$, $(1+x)^{\ln x}$, $(3 \sin x)^{1/(x^2+1)}$, $\Gamma(x)^{W(x)}$.

METHOD Neither elementary rule applies, and the reason is worth stating precisely. The power rule $\frac{d}{dx} u^n = nu^{n-1}u'$ is proved for constant n and its proof uses the binomial theorem with a fixed exponent. The exponential rule $\frac{d}{dx} a^v = a^v \ln a \cdot v'$ is proved for constant base a and its proof uses $a^v = e^{v \ln a}$ with $\ln a$ a number. When both vary, write $y = e^{g \ln f}$ or equivalently take logarithms, $\ln y = g \ln f$, and differentiate:

$$\frac{y'}{y} = g' \ln f + \frac{g f'}{f}, \quad \text{so} \quad y' = f^g \left(g' \ln f + \frac{g f'}{f} \right).$$

The two terms are exactly the two wrong answers: the first is what you get pretending the base is constant, the second is what you get pretending the exponent is constant. Both are always present.

THE TWO WRONG ANSWERS TO x^x

Ask three solvers for $\frac{d}{dx} x^x$ and you will get $x \cdot x^{x-1} = x^x$ from the one who saw a power, $x^x \ln x$ from the one who saw an exponential, and $x^x (\ln x + 1)$ from the one who saw neither. The third is right, and it is exactly the sum of the first two. That is not a coincidence: $g' \ln f$ is the exponential contribution and $g f'/f$ is the power contribution, and the general formula says to add them. Whenever you catch yourself about to apply one familiar rule to a variable-exponent form, write both terms instead.

WORKED EXAMPLE

$y = (\sin x)^{\tan x}$ at $x = \pi/4$. Here $f = \sin x$, $g = \tan x$, so

$$\frac{y'}{y} = \sec^2 x \ln \sin x + \tan x \cdot \frac{\cos x}{\sin x} = \sec^2 x \ln \sin x + 1,$$

the second term collapsing because $\tan x \cot x = 1$. At $x = \pi/4$: $\sec^2 = 2$, $\ln \sin(\pi/4) = \ln \frac{1}{\sqrt{2}} = -\frac{1}{2} \ln 2$, and $y = (\frac{1}{\sqrt{2}})^1 = \frac{1}{\sqrt{2}}$. So

$$y' \left(\frac{\pi}{4} \right) = \frac{1}{\sqrt{2}} (1 - \ln 2) = 0.216978.$$

Verified to seven digits.

A cleaner design, where the point kills one term outright. For $y = x^{\sin x}$ at $x = \pi$: $\frac{y'}{y} = \cos x \ln x + \frac{\sin x}{x}$, and at π the second term vanishes while $y(\pi) = \pi^0 = 1$. Hence $y'(\pi) = -\ln \pi = -1.144730$. Verified. Setters engineer this constantly: choose the point so that $\ln f = 0$ (that is, $f = 1$) or $g = 0$, and one of the two terms disappears. The same design with a special function. For $f(x) = \Gamma(x)^{W(x)}$ at $x = 1$: $f(1) = \Gamma(1)^{W(1)} = 1$, and

$$\frac{f'(1)}{f(1)} = W'(1) \ln \Gamma(1) + W(1) \frac{\Gamma'(1)}{\Gamma(1)} = W'(1) \cdot 0 + \Omega \cdot \psi(1) = -\gamma \Omega,$$

using $\Gamma(1) = 1$, $\psi(1) = -\gamma$, and $W(1) = \Omega = 0.5671433$, the omega constant. So $f'(1) = -\gamma \Omega = -0.3273640$. The unknown quantity $W'(1)$ never had to be computed, because $\ln \Gamma(1) = 0$ deleted it.

A case where the other term dies instead. For $y = (3 \sin x)^{1/(x^2+1)}$ at $x = \pi/2$, the exponent is $g = (x^2 + 1)^{-1}$ and the base is $f = 3 \sin x$, so

$$\frac{y'}{y} = \frac{-2x}{(x^2 + 1)^2} \ln(3 \sin x) + \frac{1}{x^2 + 1} \cdot \cot x.$$

At $x = \pi/2$ the cotangent vanishes, $\sin x = 1$, and $y = 3^{1/(\pi^2/4+1)} = 1.372783$, giving

$$y' \left(\frac{\pi}{2} \right) = 1.372783 \cdot \frac{-\pi \ln 3}{(\pi^2/4 + 1)^2} = -0.394083.$$

Verified to seven digits. Here it was the base's contribution that died and the exponent's that survived, the mirror image of the Γ^W example, and in both cases the point was chosen for exactly that purpose.

WHERE THIS GOES WRONG

Writing $gf^{g-1}f'$, which is valid only for constant g , or $f^g \ln f \cdot g'$, which is valid only for constant f . Test both against $y = x^x$, whose true derivative is $x^x(\ln x + 1)$: the first gives $x \cdot x^{x-1} = x^x$, the second gives $x^x \ln x$, and their sum is the correct answer, which is not a coincidence but the two terms of the formula. A subtler trap: $\ln f$ requires $f > 0$, so $(-2)^x$ and $(\cos x)^x$ near $\cos x < 0$ need care, and the expression may not be real at all.

SEE ALSO Power towers and iterated exponentials; exponent arithmetic before log-differentiating; logarithmic differentiation of many-factor products.

One variable exponent needs one pass of logarithmic differentiation. Two stacked variable exponents need two, and the only new difficulty is bookkeeping: knowing what counts as the base, what counts as the exponent, and which way the tower associates.

12.7 Power towers and iterated exponentials

HARD

CUE Two or more stacked variable exponents: x^{x^x} , $x^{x^{\sin x}}$, $(x^x)^{\ln x}$, or a finite tower of any height.

METHOD Log-differentiate *outermost first*, treating the entire inner tower as a single block g . For $f = x^g$ we get $\ln f = g \ln x$ and

$$\frac{f'}{f} = g' \ln x + \frac{g}{x},$$

which is the general power rule with $f = x$. Now g is itself a tower and needs its own pass. Recurse until the exponent is elementary. Before any of this, settle the association: towers are read top-down, $x^{x^x} = x^{(x^x)}$, and in particular $(x^x)^{\ln x} = x^{x \ln x}$ is a different function from $x^{(x \ln x)}$.

The bracketed case first, because it collapses. $f(x) = (x^x)^{\ln x} = x^{x \ln x} = e^{x(\ln x)^2}$, so the exponent is elementary and one pass suffices:

$$\frac{f'}{f} = (\ln x)^2 + x \cdot 2 \ln x \cdot \frac{1}{x} = (\ln x)^2 + 2 \ln x.$$

At $x = e$: $(\ln e)^2 + 2 \ln e = 1 + 2 = 3$ and $f(e) = e^e$, so $f'(e) = 3e^e = 45.46279$. Verified to seven digits.

Now the genuine two-storey tower, $f(x) = x^{x^x}$. Outermost pass with $g = x^x$:

$$\frac{f'}{f} = g' \ln x + \frac{g}{x}.$$

Inner pass: $g = x^x$ has $\frac{g'}{g} = \ln x + 1$, so $g' = x^x (\ln x + 1)$. Substituting,

$$f'(x) = x^{x^x} (x^x (\ln x + 1) \ln x + x^{x-1}).$$

At $x = 1$: $\ln 1 = 0$ kills the first term, $x^{x-1} = 1$, and $f(1) = 1$, so $f'(1) = 1$. At $x = 2$: $x^x = 4$, $\ln 2 = 0.693147$, $f(2) = 2^4 = 16$, and

$$f'(2) = 16(4(1 + \ln 2) \ln 2 + 2) = 64 \ln 2 (1 + \ln 2) + 32 = 107.11041.$$

Both verified to seven digits.

WHERE THIS GOES WRONG

Reading the tower with the wrong association. $x^{x^2} = x^{(x^2)} = e^{x^2 \ln x}$, whereas $(x^x)^2 = x^{2x} = e^{2x \ln x}$: different functions, different derivatives, identical-looking notation if the brackets are dropped. The second trap is stopping after one pass while g' is still an unevaluated symbol; the answer is not finished until every storey has been differentiated. The third is losing the term g/x , which is the contribution of the base and is easy to drop when the exponent looks like all the action.

SEE ALSO The general power rule for $f(x)^{g(x)}$; exponent arithmetic before log-differentiating; self-similar fixed-point resolution.

Towers are one way to make an exponent complicated. The other is to make it long: a product of many powers of x , where the exponents must be summed before anything else happens. This is a chapter-eleven move applied inside a chapter-twelve technique, and it is worth its own block because the collapse is invisible if you start differentiating.

12.8 Exponent arithmetic before log-differentiating

MEDIUM

CUE A finite product of powers of the same base, or stacked radicals of such powers: $(\prod_{i=1}^n x^i)^2$, or a chain like $\frac{x^{10x^2}}{\sqrt{x^{x^3}}}$.

$$\frac{\sqrt[3]{x^{x^3}}}{x^{5x^{3/2}}}.$$

METHOD All the work is in the exponent. Collapse $\prod_i x^{e_i} = x^{\sum_i e_i}$ using whatever summation the exponents invite: the arithmetic series $\sum_{i=1}^n i = \frac{n(n+1)}{2}$, a geometric sum, or a telescoping cancellation. Only then apply the general power rule. A useful special value: at $x = 1$ every such f has $f(1) = 1$ and $\ln 1 = 0$, so $f'(1) = E(1)$, the exponent evaluated at 1.

WORKED EXAMPLE

Take $f(x) = (\prod_{i=1}^n x^i)^2$ with the index bound set equal to the variable, as these puzzles are usually posed. The inner product is $x^{1+2+\dots+n} = x^{n(n+1)/2}$, the outer square doubles the exponent, and with $n = x$,

$$f(x) = x^{x(x+1)} = x^{x^2+x}.$$

Now one pass of the general power rule with $E(x) = x^2 + x$:

$$\frac{f'}{f} = E'(x) \ln x + \frac{E(x)}{x} = (2x + 1) \ln x + x + 1.$$

At $x = 2$: $E(2) = 6$, so $f(2) = 2^6 = 64$, and $f'/f = 5 \ln 2 + 3$. Hence

$$f'(2) = 64(3 + 5 \ln 2) = 413.8071.$$

Verified to seven digits. At $x = 1$ the shortcut applies: $f(1) = 1$ and $f'(1) = E(1) = 2$, confirmed numerically as 2.0000000.

The stacked-radical chain works the same way. Writing $\sqrt{x^3} = x^{3/2}$ and $\sqrt[3]{x^3} = x^{3/3}$, the four displayed factors contribute exponents $10x^2$, $-\frac{1}{2}x^3$, $+\frac{1}{3}x^3$ and $-\frac{5}{2}x^3$, so $E(x) = 10x^2 - \frac{8}{3}x^3$ and one pass finishes it. The point is that no differentiation happened until the exponent arithmetic was complete.

WHERE THIS GOES WRONG

Differentiating factor by factor as a giant product and never noticing that the exponents telescope. With n factors that is n applications of the general power rule and n terms to add, in place of one arithmetic series. The other error is forgetting that an outer square or root multiplies *every* exponent: $(x^{n(n+1)/2})^2 = x^{n(n+1)}$, not $x^{n(n+1)/2+2}$ and not $x^{2n(n+1)/2 \cdot 2}$.

SEE ALSO Power towers and iterated exponentials; the general power rule for $f(x)^{g(x)}$; sum the hidden series, product, or continued fraction first.

Logarithms have appeared so far as a tool applied to the outside of an expression. They can also appear inside it, and when the *base* of a logarithm depends on x the change-of-base formula is compulsory rather than optional.

12.9 Variable-base logarithms $\log_{u(x)} v(x)$

MEDIUM

CUE A logarithm whose base is not a constant: $\log_x 2$, $\log_x(x^2 + 1)$, $\log_{x^2+3x}(x^3 + \sin x)$.

METHOD Change base first, to natural logarithms, and only then differentiate:

$$\log_u v = \frac{\ln v}{\ln u} \quad \Rightarrow \quad \frac{d}{dx} \log_u v = \frac{\frac{v'}{v} \ln u - \frac{u'}{u} \ln v}{(\ln u)^2}.$$

This is one of the few places where the quotient rule is genuinely the right tool: after the change of base the expression is an irreducible ratio of two transcendental blocks, and neither splitting nor a negative power helps.

WORKED EXAMPLE

A short one to fix the shape. $f(x) = \log_x 2 = \frac{\ln 2}{\ln x}$ has a constant numerator, so

$$f'(x) = \ln 2 \cdot \left(-\frac{1}{(\ln x)^2} \right) \cdot \frac{1}{x} = -\frac{\ln 2}{x(\ln x)^2},$$

and $f'(e) = -\frac{\ln 2}{e} = -0.2549946$. Verified.

The full two-term case. Let $f(x) = \log_{x^2+3x}(x^3 + \sin x)$ and evaluate $f'(3)$. Here $u = x^2 + 3x$ with $u(3) = 18$ and $u'(3) = 2x + 3 = 9$; $v = x^3 + \sin x$ with $v(3) = 27 + \sin 3 = 27.141120$ and $v'(3) = 3x^2 + \cos x = 27 + \cos 3 = 26.010007$. Then

$$\frac{v'}{v} = 0.958331, \quad \ln u = \ln 18 = 2.890372, \quad \frac{u'}{u} = \frac{1}{2}, \quad \ln v = 3.301207,$$

so the numerator is $0.958331 \times 2.890372 - 0.5 \times 3.301207 = 2.769737 - 1.650604 = 1.119133$, and dividing by $(\ln 18)^2 = 8.354250$ gives

$$f'(3) = 0.1339905.$$

Verified by complex-step differentiation of $\ln(x^3 + \sin x) / \ln(x^2 + 3x)$: 0.1339905.

WHERE THIS GOES WRONG

Treating the base as constant and writing only $\frac{v'}{v \ln u}$. In the example above that term alone gives 0.331565, against a true value of 0.133990: the neglected second term is more than half the answer. The error is easy to make because $\log_a$ with a numeric a is so much more familiar, and because the notation puts the base in small type.

SEE ALSO When the quotient rule is the wrong tool; logarithmic differentiation of many-factor products; the general power rule for $f(x)^{g(x)}$.

The last technique in the chapter uses the logarithmic derivative not as a computational device but as an operator with an algebraic property, applied to a function that is never written down. This is the most abstract use and the most powerful: it solves problems where no formula for f exists.

12.10 The logarithmic derivative as an operator on functional equations

CUE f is given only through a multiplicative functional equation, such as $f(x+1) = xf(x)$ or $f(2x) = f(x)^2g(x)$, and the question involves $g = f'/f$ rather than f itself.

METHOD Apply $\mathcal{L}[f] = f'/f$ to both sides. Because $\mathcal{L}[fg] = \mathcal{L}[f] + \mathcal{L}[g]$ and $\mathcal{L}[c] = 0$ for constants, a multiplicative recurrence in f becomes an *additive* recurrence in g , which telescopes. From $f(x+1) = xf(x)$, taking logarithms gives $\ln f(x+1) = \ln x + \ln f(x)$ and differentiating gives

$$g(x+1) = g(x) + \frac{1}{x},$$

which is the digamma recurrence $\psi(x+1) = \psi(x) + 1/x$. Never identify f ; just telescope.

WORKED EXAMPLE

Suppose f satisfies $f(x+1) = xf(x)$ on $x > 0$ and $g = f'/f$. Compute $12(g(5) - g(1))$.

Applying $\mathcal{L}$ to both sides gives $g(x+1) - g(x) = 1/x$, so telescoping from 1 to 5,

$$g(5) - g(1) = \sum_{k=1}^4 \frac{1}{k} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} = \frac{25}{12},$$

and therefore $12(g(5) - g(1)) = 25$. Verified numerically against the digamma function, which is one solution of the recurrence: $\psi(5) - \psi(1) = 2.083313$ from a truncated series, against $25/12 = 2.083333$.

Note what was not needed. No formula for f , no value of f anywhere, no convergence argument, and no special-function theory. The multiplicative recurrence was converted into an additive one and the additive one was summed. The same move handles $f(2x) = f(x)^2$, which gives $2g(2x) = 2g(x)$, that is $g(2x) = g(x)$, so the logarithmic derivative is invariant under doubling.

The general telescope is worth recording, since it is the shape every such problem takes. For any integer $n \geq 1$,

$$g(n) = g(1) + \sum_{k=1}^{n-1} \frac{1}{k} = g(1) + H_{n-1},$$

so the entire integer behaviour of g is fixed by one constant. If f is normalised to be Γ , that constant is $g(1) = \psi(1) = -\gamma$ and the formula reads $\psi(n) = -\gamma + H_{n-1}$, the standard evaluation of the digamma function at integers. But note the direction of the logic: the recurrence gave every *difference* for free, and the normalisation was needed only to fix the additive constant. Questions that ask for a difference, as this one did, never need it.

WHERE THIS GOES WRONG

Trying to identify f explicitly as Γ . The functional equation $f(x+1) = xf(x)$ does not determine f : multiply any solution by a period-one function such as $e^{\sin 2\pi x}$ and you get another. Pinning down Γ requires an extra hypothesis, logarithmic convexity in the Bohr and Mollerup theorem, which the problem has not given you and does not need. Work with the difference equation and stop.

SEE ALSO Logarithmic differentiation of many-factor products; the general power rule for $f(x)^{g(x)}$; composition bookkeeping and functional inversion.

A closing word on when *not* to use the solvent. Logarithmic differentiation costs a logarithm, a derivative, and a multiplication back by f , and that overhead is only repaid when the expression is genuinely multiplicative. On a sum it is useless, since $\ln(u+v)$ has no expansion. On a two-factor product it is slower than the product rule. At a zero of f it is undefined. And when the question asks for f' as a function rather than at a point, the final multiplication by f reassembles all the mess you just dissolved, which is sometimes worse than never dissolving it. The technique earns its name on many-factor products evaluated at a point, on variable exponents where nothing else works, and on functional equations where f is unavailable. Those three cases are most of what this chapter has been about.

Two habits carry all of it. The first is naming the root operation before differentiating, so that the rule is chosen rather than defaulted to. The second is asking, whenever the expression is multiplicative, whether taking a logarithm would make the problem additive; if it would, and if f does not vanish at the point, it almost always should be taken. Between them they eliminate the two failure modes this chapter opened with: algebra that never ends, and a confident answer that is missing one of its two terms.

Recognition matrix

If you see	Reach for	If that fails
Three or more genuine layers, no simplification	Chain rule outside in; each factor at the inner stack	List the layers before differentiating
An ugly evaluation point in a nested composition	Peel first, substitute last	Accept a transcendental answer
Three or more nontrivial factors	Generalised product rule: exactly n terms	Check the term count as a checksum
A product evaluated where one factor vanishes	Product rule: $n - 1$ terms die	Never f'/f ; it is undefined there
A quotient with a monomial denominator	Split termwise into powers, then the power rule	Rewrite as uv^{-1}
A quotient with a sum on top	Split; you may split above, never below	Product + chain on uv^{-1}
An irreducible two-block quotient	Quotient rule $\frac{u'v - uv'}{v^2}$; square the denominator	Log-differentiate if both blocks are products
Given $p \circ q$ and q , asked for p' at an output $(f \circ g^{-1})'(c)$	Solve $q(a) = \text{target}$, then divide by $q'(a)$ Find a with $g(a) = c$; answer is $f'(a)/g'(a)$	Recover p explicitly and check Inverse-function rule on g alone
Four or more nonzero factors, a point requested	$\frac{f'}{f} = \sum \frac{u'_j}{u_j} - \sum \frac{v'_k}{v_k}$, then multiply by f	Product rule if any factor vanishes
Many linear factors	$\frac{f'}{f} = \sum_j \frac{1}{x - r_j}$; no differentiation left	Harmonic sums appear at integer points
$f(x)^{g(x)}$ with both variable	$y' = f^g(g' \ln f + gf'/f)$; both terms always	Pick the point that kills one term
A stacked tower x^{x^x}	Log-differentiate outermost first, then recurse	Fix the association before starting
A long product of powers of x	Sum the exponents first, then one power-rule pass	At $x = 1, f'(1) = E(1)$
$\log_{u(x)} v(x)$	Change base to $\ln v / \ln u$, then the quotient rule	Both terms matter; the base is not inert
$f(x+1) = xf(x)$, or any multiplicative recurrence	Apply $\mathcal{L}[f] = f'/f$; it becomes additive	Telescope; do not identify f

Implicit, inverse, parametric and polar structure

What this chapter is for. Four presentations of a curve look like four different subjects and are one. An implicit equation $F(x, y) = 0$, an inverse relation $f(y) = x$, a parametrisation $x = x(t)$, $y = y(t)$, and a polar equation $r = r(\theta)$ all share the property that y is never handed to you as a formula in x . In every case the response is identical: differentiate the relation you were actually given, let the chain rule record the fact that y depends on x , and solve the resulting linear equation for $\frac{dy}{dx}$. This chapter installs that one move, then works through what changes in each costume: how to get a second derivative implicitly without freezing y' into a constant, why the inverse-function theorem turns problems that look unsolvable into one line of arithmetic, how the inverse trigonometric and inverse hyperbolic derivatives are derived rather than memorised, and the two-step formula for $\frac{d^2y}{dx^2}$ on a parametrised curve that almost everyone gets wrong the first time. At the end you should be able to look at any relation between two variables and produce a derivative without ever solving for y .

Four costumes, one idea

The usual complaint is that there is nothing to differentiate, because y has not been given as a function of x . The complaint is false. You have been given a *relation*, and a relation can be differentiated exactly as a formula can, provided you keep track of what depends on what.

Assume there is a differentiable function $y = y(x)$ satisfying the relation near the point of interest. That assumption is what the implicit function theorem supplies, and its hypothesis will reappear below as the condition $F_y \neq 0$. Under that assumption every appearance of y in the relation is a composite function of x , so differentiating it emits a factor y' by the chain rule. The relation is an identity in x , so its derivative is an identity in x too. That identity contains y' linearly, because y' enters only through single applications of the chain rule, and a linear equation in one unknown is solved by rearrangement.

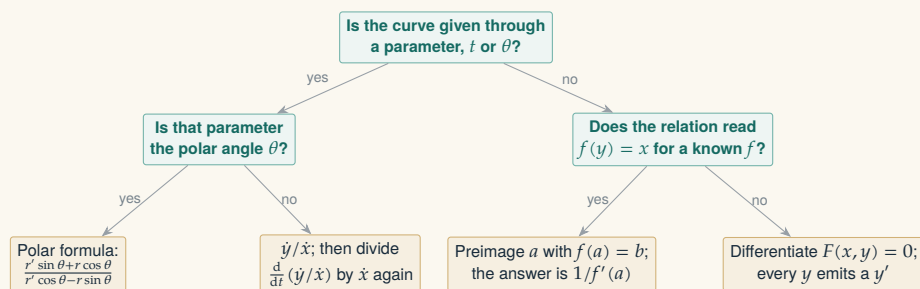
Here are the four presentations, and here is the same move in each.

- **Implicit.** $F(x, y) = 0$. Differentiate: $F_x + F_y y' = 0$, so $y' = -F_x/F_y$.
- **Inverse.** $y = f^{-1}(x)$ means $f(y) = x$, which is the implicit relation $F(x, y) = f(y) - x = 0$. Differentiate: $f'(y)y' = 1$, so $y' = 1/f'(y)$.
- **Parametric.** $x = x(t)$, $y = y(t)$. The relation is that both coordinates are tied to a common t . Differentiate $y(t) = y(x(t))$: $\dot{y} = \frac{dy}{dx}\dot{x}$, so $\frac{dy}{dx} = \dot{y}/\dot{x}$.
- **Polar.** $r = r(\theta)$ is the parametric case with $x = r(\theta) \cos \theta$ and $y = r(\theta) \sin \theta$, so nothing new is needed at all.

THE SINGLE MOVE

Differentiate the relation you were given, with respect to x , treating y as an unknown function of x . Collect the y' terms on one side. Divide. Everything else in this chapter is bookkeeping about which relation you were given and what you do after the first derivative is in hand. In particular the inverse-function theorem is not a separate theorem: it is implicit differentiation applied to the relation $f(y) = x$, where the collection step is trivial because only one term contains y' .

The four costumes do differ in one respect, and it is worth naming before the techniques begin. Implicit and inverse relations are *unparametrised*: they describe a set of points, and a derivative is attached to a point of that set. Parametric and polar descriptions are *parametrised*: they describe a motion, and a derivative can be attached to a time as well as to a point. That distinction is why a parametrised curve can cross itself and still have two perfectly well defined tangents at the crossing, one for each visit, while $F(x, y) = 0$ at a crossing has $F_x = F_y = 0$ and offers you nothing.



There is a fifth costume, mentioned here and then left to the chapter on applications, because recognising it early saves confusion later. If both x and y depend on a time t and a relation $F(x, y) = 0$ holds throughout the motion, differentiating the relation with respect to t rather than x gives $F_x \dot{x} + F_y \dot{y} = 0$. That is a related-rates problem, and it is implicit differentiation with the independent variable renamed. On $x^2 + y^2 = 25$ with $\dot{x} = 3$ at the point $(3, 4)$, the relation gives $6 \cdot 3 + 8\dot{y} = 0$, so $\dot{y} = -\frac{9}{4}$; and $\frac{dy}{dx} = \dot{y}/\dot{x} = -\frac{3}{4}$, the slope computed three other ways below. Nothing new happened. The only change was which letter the derivatives were taken with respect to.

The same tangent, three ways

It is worth doing one tangent by every available description, because the agreement is the argument of this chapter in miniature. Take the circle of radius 5 at the point $(3, 4)$.

As an implicit relation. $x^2 + y^2 = 25$ differentiates to $2x + 2yy' = 0$, so $y' = -x/y = -3/4$.

As a parametrisation. $x = 5 \cos t$, $y = 5 \sin t$, with $(3, 4)$ at the value of t where $\cos t = 3/5$ and $\sin t = 4/5$. Then $\dot{x} = -5 \sin t$, $\dot{y} = 5 \cos t$, and $\frac{dy}{dx} = -\cot t = -\frac{3/5}{4/5} = -3/4$.

As a polar curve. $r(\theta) = 5$, a constant, so $r' = 0$ and the polar formula degenerates to

$$\frac{dy}{dx} = \frac{0 + r \cos \theta}{0 - r \sin \theta} = -\cot \theta,$$

which is the parametric answer, as it must be, since $r' = 0$ is exactly the statement that the parametrisation is the circular one. Numerically $-\cot(\arctan(4/3)) = -0.750000$.

As an explicit function. $y = \sqrt{25 - x^2}$ on the upper semicircle, so $y' = -x/\sqrt{25 - x^2} = -3/4$. This is the only one of the four that required a branch choice, and it is the only one that fails at $(5, 0)$.

Four descriptions, one tangent, and the differences are entirely in the bookkeeping. The lesson generalises: when a curve admits two descriptions, use the one whose derivative is cheapest, and use the other as a check. The astroid $x^{2/3} + y^{2/3} = 1$ is a good second exercise in that habit. Implicitly, $\frac{2}{3}x^{-1/3} + \frac{2}{3}y^{-1/3}y' = 0$ gives $y' = -(y/x)^{1/3}$; parametrically, $x = \cos^3 t$, $y = \sin^3 t$ gives $\dot{y}/\dot{x} = 3 \sin^2 t \cos t / (-3 \cos^2 t \sin t) = -\tan t$. The two agree because $(y/x)^{1/3} = \tan t$, and at $t = 0.7$ both return -0.842288 . The parametric form is the one to use, because it never divides by zero at the cusps.

Implicit relations

Start with the general case, because the other three are special cases of it. The mechanical instruction is: differentiate every term of $F(x, y) = 0$ with respect to x ; a term in x alone differentiates normally; a term in y alone differentiates and picks up a factor y' ; a mixed term needs the product rule and then picks up a factor y' on the part that came from y . Then collect and divide.

The formula $y' = -F_x/F_y$ is the same statement in partial-derivative notation, and it is worth having both. The term-by-term version is faster on a small relation and needs no multivariable vocabulary. The partial version is faster on a large one, is the only sensible route to the second derivative in closed form, and makes the failure condition visible: the recipe breaks exactly where $F_y = 0$.

13.1 Implicit differentiation at a point

CUE x and y are algebraically entangled in a single equation that cannot be solved for y , or can be solved only at the cost of a radical with a branch choice, and $\frac{dy}{dx}$ is wanted at a specific point on the curve.

METHOD Differentiate every term with respect to x , treating $y = y(x)$ so that each appearance of y emits a y' ; collect the y' terms and solve the resulting linear equation. Equivalently

$$\frac{dy}{dx} = -\frac{F_x(x, y)}{F_y(x, y)} \quad \text{for } F(x, y) = 0.$$

Substitute the numerical coordinates of the point *before* solving whenever a numerical answer is all that is wanted: it turns a symbolic rearrangement into arithmetic.

WORKED EXAMPLE

The folium of Descartes, $x^3 + y^3 = 6xy$, at the point $(3, 3)$.

Differentiating term by term, $3x^2 + 3y^2y' = 6y + 6xy'$. Collecting, $y'(3y^2 - 6x) = 6y - 3x^2$, so

$$y' = \frac{6y - 3x^2}{3y^2 - 6x} = \frac{2y - x^2}{y^2 - 2x}.$$

At $(3, 3)$ this is $\frac{6 - 9}{9 - 6} = -1$. The tangent there is $y - 3 = -(x - 3)$, that is $x + y = 6$, which is the line of symmetry of the folium reflected: the curve meets its own axis of symmetry $y = x$ at $(3, 3)$ at right angles, and $y' = -1$ says exactly that.

Note how much work the point saved. Substituting $x = y = 3$ into the differentiated identity gives $27 + 27y' = 18 + 18y'$ immediately, hence $9y' = -9$ and $y' = -1$, without ever writing the general quotient.

A second instance where solving for y is genuinely impossible. On $e^y + xy = e$ at the point $(0, 1)$: differentiating gives $e^yy' + y + xy' = 0$, and at $(0, 1)$ this reads $e y' + 1 + 0 = 0$, so $y' = -1/e$. The relation cannot be inverted in closed form; the derivative took one line.

A third instance, where the whole curve is interrogated rather than one point. On the ellipse $x^2 + xy + y^2 = 3$, differentiation gives $2x + y + xy' + 2yy' = 0$, so

$$y' = -\frac{2x + y}{x + 2y}.$$

Horizontal tangents are the zeros of the numerator, $y = -2x$; substituting into the relation gives $x^2 - 2x^2 + 4x^2 = 3$, so $x = \pm 1$ and the points are $(1, -2)$ and $(-1, 2)$. Vertical tangents are the zeros of the denominator, $x = -2y$, giving $(2, -1)$ and $(-2, 1)$. A numerical slope on the implicit branch at $(1, -2)$ returns 1.1×10^{-10} . Note that the two lists are obtained from the same quotient by reading it twice, once from the top and once from the bottom, which is the standard economy of the implicit form.

WHERE THIS GOES WRONG

Differentiating a y -term as though y were a constant, so that $\frac{d}{dx}y^3$ comes out as $3y^2$ instead of $3y^2y'$. The reliable check is that every term of the differentiated identity must contain either a y' or no y at all, apart from the ones the product rule split. The second trap is dividing by F_y without checking $F_y \neq 0$: on the folium $F_y = 3y^2 - 6x$ vanishes at $(0, 0)$ and at $(2\sqrt[3]{4}, 2\sqrt[3]{2})$, and at the latter the tangent is vertical rather than nonexistent, so the correct report is $\frac{dx}{dy} = 0$, not “undefined”.

SEE ALSO Second derivatives implicitly; the implicit variable exponent; the inverse-function derivative.

WHAT THE HYPOTHESIS $F_y \neq 0$ IS DOING

The implicit function theorem says that if F is continuously differentiable near (x_0, y_0) , if $F(x_0, y_0) = 0$, and if $F_y(x_0, y_0) \neq 0$, then near that point the solution set of $F = 0$ is the graph of a differentiable function $y = y(x)$, and its derivative is $-F_x/F_y$. Every implicit calculation silently assumes this. When $F_y = 0$ but $F_x \neq 0$, swap the roles: the solution set is locally the graph of $x = x(y)$ with $\frac{dx}{dy} = -F_y/F_x = 0$, and the tangent is vertical. When both vanish the point is singular, the curve may cross itself or form a cusp, and no first-order statement is available: the folium of Descartes at the origin is the standard example, where two branches cross and the two tangents are the coordinate axes.

Once y' is in hand as an expression in x and y , a concavity or curvature question asks for y'' , and this is where implicit differentiation acquires its one genuine subtlety. The expression for y' still contains y , and y still depends on x , so differentiating it a second time produces fresh y' factors that must then be eliminated using the first-derivative result. The order of those two steps is the whole difficulty.

13.2 Second and higher derivatives implicitly

CUE An implicit curve with $\frac{d^2y}{dx^2}$ requested, usually as part of a concavity, inflection, or curvature question, and occasionally a third derivative.

METHOD Differentiate the *expression* already obtained for y' , again with respect to x , remembering that every y inside that expression still carries a y' ; then substitute the known y' and simplify using the original relation. In closed form,

$$\frac{d^2y}{dx^2} = -\frac{F_{xx}F_y^2 - 2F_{xy}F_xF_y + F_{yy}F_x^2}{F_y^3}.$$

The three-term numerator is the quadratic form F_{yy}, F_{xy}, F_{xx} evaluated against the vector $(F_x, -F_y)$, which is how to remember its signs.

WORKED EXAMPLE

The circle $x^2 + y^2 = r^2$ first, because the answer is checkable by eye. Differentiating, $2x + 2yy' = 0$ gives $y' = -x/y$. Differentiating that quotient,

$$y'' = -\frac{d}{dx} \frac{x}{y} = -\frac{y - xy'}{y^2} = -\frac{y + x^2/y}{y^2} = -\frac{y^2 + x^2}{y^3} = -\frac{r^2}{y^3}.$$

The substitution $y' = -x/y$ happened at the second step, after the differentiation, and the last simplification used the original relation. On the upper semicircle $y > 0$ and $y'' < 0$: concave down, as it must be. At $r = 5$ and $x = 3$ the formula gives $-25/64 = -0.390625$, and a central second difference of $\sqrt{25 - x^2}$ at $x = 3$ returns -0.39062 .

Now the folium at $(3, 3)$, where nothing is checkable by eye. Differentiate the identity $3x^2 + 3y^2y' = 6y + 6xy'$ once more:

$$6x + 6y(y')^2 + 3y^2y'' = 6y' + 6y' + 6xy''.$$

Only now substitute $x = y = 3$ and $y' = -1$: $18 + 18 + 27y'' = -12 + 18y''$, so $9y'' = -48$ and

$$y'' = -\frac{16}{3}.$$

A central second difference on the implicit branch through $(3, 3)$ returns -5.33333 , and $-16/3 = -5.\bar{3}$. The closed form agrees: with $F = x^3 + y^3 - 6xy$ we have $F_x = F_y = 9$, $F_{xx} = F_{yy} = 18$, $F_{xy} = -6$ at $(3, 3)$, so the numerator is $18 \cdot 81 + 2 \cdot 6 \cdot 81 + 18 \cdot 81 = 3888$ and $-3888/729 = -16/3$.

Third derivatives need no new idea, only stamina and the same discipline: differentiate the identity again, and substitute the now-known y' and y'' at the very end. On the circle, differentiating $y'' = -r^2y^{-3}$ gives $y''' = 3r^2y^{-4}y' = 3r^2y^{-4}(-x/y) = -3r^2x/y^5$, which vanishes only on the vertical axis, as the symmetry of the circle demands.

WHERE THIS GOES WRONG

Substituting the numerical value of y' into the first-derivative expression *before* the second differentiation. That freezes y' into a constant, so its own derivative is lost, and the reported y'' is missing the $6y(y')^2$ term entirely. The rule is: differentiate first, substitute second. The related slip is differentiating $y' = -x/y$ with the quotient rule and writing $\frac{d}{dx}y = 1$; the denominator's derivative is y' , not 1.

SEE ALSO Implicit differentiation at a point; the parametric second derivative; curvature.

A word on what y'' is usually wanted for. Inflection points of an implicit curve are the points where y'' changes sign, and the closed form makes them computable without ever separating the variables: set the numerator $F_{xx}F_y^2 - 2F_{xy}F_xF_y + F_{yy}F_x^2$ to zero and intersect that new curve with $F = 0$. Two equations, two unknowns, no branch choice anywhere. Curvature is the same numerator wearing a different denominator, $\kappa = |y''| / (1 + (y')^2)^{3/2}$, so an implicit curvature question is answered by the same computation with one extra substitution at the end.

One family of implicit relations deserves separate treatment, not because the method changes but because the first step is invisible until you have seen it once. When the unknown appears in its own exponent, the relation is not polynomial in y and term-by-term differentiation produces something unusable. Taking logarithms first converts the exponent into a product, and the relation becomes ordinary.

13.3 Implicit variable exponent: $y = x^y$ and relatives

CUE The unknown appears in its own exponent: $y = x^y$, $x^y = y^x$, $x^y = e^{x-y}$, $y^y = x$.

METHOD Take logarithms of the whole relation, which turns every exponent into a multiplier, then differentiate implicitly as usual. For $y = x^y$: $\ln y = y \ln x$, so $\frac{y'}{y} = y' \ln x + \frac{y}{x}$, and collecting,

$$y' = \frac{y^2}{x(1 - y \ln x)}.$$

Always check whether the logged relation happens to be solvable for y ; sometimes it is, and then no implicit work is needed at all.

WORKED EXAMPLE

$x^y = e^{x-y}$, and $\frac{dy}{dx}$ at $x = e$.

Logging both sides gives $y \ln x = x - y$, which is linear in y , so this relation *is* solvable:

$$y = \frac{x}{1 + \ln x}.$$

Then the quotient rule gives

$$\frac{dy}{dx} = \frac{(1 + \ln x) - x \cdot \frac{1}{x}}{(1 + \ln x)^2} = \frac{\ln x}{(1 + \ln x)^2},$$

and at $x = e$ this is $\frac{1}{4}$. A central difference of $x/(1 + \ln x)$ at $x = e$ with step 10^{-5} returns 0.2500000.

The implicit route reaches the same place and is the one to use when the logged relation is not solvable. From $y \ln x = x - y$, differentiate: $y' \ln x + \frac{y}{x} = 1 - y'$, so $y'(1 + \ln x) = 1 - \frac{y}{x}$. At $x = e$ the relation gives $y = e/2$, hence $y' \cdot 2 = 1 - \frac{1}{2}$ and $y' = \frac{1}{4}$ again.

For a relation that genuinely cannot be solved, take $y = x^y$ at $x = 1.2$. The relation has root $y = 1.2577$, and the formula gives $y' = y^2/(x(1 - y \ln x)) = 1.71048$; a central difference of the numerically continued branch returns 1.71048.

The most-set member of the family is $x^y = y^x$, whose logged form is $y \ln x = x \ln y$. Differentiating,

$$y' \ln x + \frac{y}{x} = \ln y + \frac{x}{y} y', \quad \text{so} \quad y' = \frac{y(x \ln y - y)}{x(y \ln x - x)}.$$

The curve has two components: the line $y = x$, on which the formula is 0/0 and useless, and a nontrivial branch through $(2, 4)$, since $2^4 = 4^2 = 16$. There the formula gives

$$y' = \frac{4(2 \ln 4 - 4)}{2(4 \ln 2 - 2)} = -3.177399,$$

and a central difference along the numerically continued branch returns -3.177399 . The 0/0 on the diagonal is the algebra telling you that the two components cross, and that a point on the diagonal belongs to both.

WHERE THIS GOES WRONG

Forgetting the $y' \ln x$ term, that is, treating the y in the exponent as a constant while remembering the y elsewhere. The other failure is dividing by $1 - y \ln x$ without asking whether it vanishes: $1 - y \ln x = 0$ together with $y = x^y$ occurs at $x = e^{1/e}$, which is precisely the fold where the two branches of $y = x^y$ meet and the tangent turns vertical. That is a feature of the curve, not an accident of the algebra.

SEE ALSO Logarithmic differentiation of many-factor products; implicit differentiation at a point; the general power rule.

Inverse functions

An inverse relation is an implicit relation with one term. Writing $y = f^{-1}(x)$ is the same as writing $f(y) = x$, and differentiating that gives $f'(y)y' = 1$ at once. The consequence is a formula so short that its value is easy to miss.

If f is differentiable at a , $f(a) = b$, and $f'(a) \neq 0$, then f^{-1} is differentiable at b and

$$(f^{-1})'(b) = \frac{1}{f'(a)}.$$

The derivative of the inverse at b is evaluated using the derivative of f at the *preimage* a , never at b . Geometrically, the graph of f^{-1} is the graph of f reflected in $y = x$, reflection swaps rise and run, and a slope becomes its reciprocal. Note also what the hypothesis $f'(a) \neq 0$ costs: where $f'(a) = 0$ the reflected tangent is vertical and f^{-1} is not differentiable at b at all.

The reason this is more than a curiosity is that problems are set on functions whose inverse has no closed form whatsoever. If $f(x) = x^5 + x^3 + 2x$, then f^{-1} cannot be written down: a general quintic has no radical solution. But $f(1) = 4$, so $(f^{-1})'(4) = 1/f'(1) = 1/10$, and the problem that looked impossible is one line. The work in such problems is never the differentiation. It is finding the preimage.

CORE

13.4 The inverse-function derivative and preimage hunting

CUE $(f^{-1})'(b)$ is requested, or the derivative of an inverse branch, and the evaluation point b has been written in a form designed to hide the preimage.

METHOD Use $(f^{-1})'(b) = 1/f'(a)$ with $f(a) = b$. Spend all the effort on finding a exactly: try small integers first, then rational points, then the algebraic identity that the value of b was built from. Conjugates and surd norms crack most of the engineered cases. When several branches exist the theorem applies to each separately, so name the branch before answering.

WORKED EXAMPLE

$f(x) = \sqrt{x} + \sqrt{x+1}$ on $x \geq 0$. Find $(f^{-1})'(s)$ where $s = (1 + \sqrt{2})^2 = 3 + 2\sqrt{2}$.

The function is strictly increasing, so the inverse exists and there is exactly one preimage. The decisive structural fact is the conjugate identity

$$\sqrt{x+1} - \sqrt{x} = \frac{(x+1) - x}{\sqrt{x+1} + \sqrt{x}} = \frac{1}{f(x)}.$$

So if $f(a) = s$, then subtracting the identity from the definition gives $2\sqrt{a} = s - \frac{1}{s}$. With $s = 3 + 2\sqrt{2}$ we have $1/s = 3 - 2\sqrt{2}$, so $2\sqrt{a} = 4\sqrt{2}$, $\sqrt{a} = 2\sqrt{2}$, and $a = 8$. Check: $\sqrt{8} + \sqrt{9} = 2\sqrt{2} + 3 = 5.82843 = s$.

Now the easy half. $f'(x) = \frac{1}{2\sqrt{x}} + \frac{1}{2\sqrt{x+1}}$, so $f'(8) = \frac{1}{4\sqrt{2}} + \frac{1}{6}$, and

$$(f^{-1})'(s) = \frac{1}{f'(8)} = \frac{24\sqrt{2}}{6 + 4\sqrt{2}} = \frac{12\sqrt{2}}{3 + 2\sqrt{2}} = 12\sqrt{2}(3 - 2\sqrt{2}) = 36\sqrt{2} - 48.$$

Numerically 2.911688, and $1/f'(8)$ evaluated directly is 2.911688. The rationalisation used $(3 + 2\sqrt{2})(3 - 2\sqrt{2}) = 9 - 8 = 1$, which is the same surd norm that made s a natural choice of evaluation point in the first place. That is the tell: when b is presented as $(1 + \sqrt{2})^2$ rather than as 5.828, the setter is telling you which identity to use.

A second instance, chosen because the answer can be checked against a known inverse. Let $f(x) = \ln(x + \sqrt{x^2 + 1})$ and find $(f^{-1})'(\ln 2)$. The preimage solves $\ln(a + \sqrt{a^2 + 1}) = \ln 2$, that is $\sqrt{a^2 + 1} = 2 - a$, which squares to $a^2 + 1 = 4 - 4a + a^2$ and gives $a = \frac{3}{4}$. Then

$$f'(x) = \frac{1 + \frac{x}{\sqrt{x^2 + 1}}}{x + \sqrt{x^2 + 1}} = \frac{1}{\sqrt{x^2 + 1}}, \quad f'(\tfrac{3}{4}) = \frac{1}{5/4} = \frac{4}{5},$$

so the answer is $\frac{5}{4}$. The check: $f = \operatorname{arsinh}$, so $f^{-1} = \sinh$, and $\sinh'(\ln 2) = \cosh(\ln 2) = \frac{1}{2}(2 + \frac{1}{2}) = \frac{5}{4}$. Squaring introduced the constraint $a \leq 2$, which the root satisfies, so no spurious solution was created.

WHERE THIS GOES WRONG

Writing $(f^{-1})'(b) = 1/f'(b)$. The derivative is taken at the preimage, and the two agree only in the accidental case $f(b) = b$. The second failure is answering without naming a branch when f is not injective: $f(x) = x^3 - 3x$ has three preimages of $b = 1$, and the three inverse branches have three different derivatives there.

SEE ALSO Inverse trigonometric and inverse hyperbolic derivatives; Lagrange inversion; composition bookkeeping.

The multi-branch case is worth one further remark, because it has a surprising and useful closed form. When the preimages of b under a polynomial p are the roots $r_1, \dots, r_n$ of $p(x) - b = 0$, the sum of the derivatives of all the inverse branches is $\sum_i 1/p'(r_i)$, and that sum is computable without finding a single root.

ALL THE BRANCHES AT ONCE

For a polynomial p of degree $n \geq 2$ with distinct roots $r_1, \dots, r_n$, partial fractions of $1/p$ give $\frac{1}{p(x)} = \sum_i \frac{1}{p'(r_i)(x - r_i)}$; multiplying by x and letting $x \rightarrow \infty$ forces

$$\sum_{i=1}^n \frac{1}{p'(r_i)} = 0.$$

Applied to $p(x) = x^3 - 3x - 1$, whose roots are the three preimages of $b = 1$ under $f(x) = x^3 - 3x$, the sum of $(f^{-1})'(1)$ over all three branches is 0 exactly, and a numerical root-find confirms $\sum_i 1/(3r_i^2 - 3) = -4 \times 10^{-16}$. The companion identity

$\sum_i \frac{1}{r_i - c} = -\frac{p'(c)}{p(c)}$ handles the weighted versions of the same question.

The second derivative of an inverse is obtained by the same one move, and its shape is worth noticing. Differentiating $y' = 1/f'(y)$ with respect to x gives $y'' = -f''(y)y'/f'(y)^2$, and substituting $y' = 1/f'(y)$ leaves

$$(f^{-1})''(b) = -\frac{f''(a)}{f'(a)^3}, \quad f(a) = b.$$

A cube in the denominator, exactly as in the parametric second derivative, and for exactly the same reason: converting a y -derivative into an x -derivative costs one factor of f' , and doing it twice through a quotient costs three. For $f(x) = x^3 + x$ and $b = 10$ the preimage is $a = 2$, so the formula gives $-12/13^3 = -0.0054620$, and a numerical second difference of the inverted function returns -0.0054620 .

The inverse trigonometric and inverse hyperbolic derivatives are the inverse-function theorem applied seven times, and they are worth deriving once rather than memorising, for two reasons. The derivation takes ten seconds, which is faster than recovering a half-remembered sign under pressure; and it explains the shape of the answers, which is what makes them stick.

EASY

13.5 Inverse trigonometric and inverse hyperbolic derivatives

CUE Any of arcsin, arccos, arctan, arcsec, arsinh, arcosh, artanh appearing as an outer or an inner layer of a composite.

METHOD Derive, do not recall. For $y = \arcsin x$ write $\sin y = x$, differentiate to get $y' \cos y = 1$, and eliminate $\cos y = \sqrt{1 - \sin^2 y} = \sqrt{1 - x^2}$ using the Pythagorean identity, taking the positive root because arcsin has range $[-\pi/2, \pi/2]$ where cosine is nonnegative. The same three lines give every entry. With the chain rule,

$$\frac{d}{dx} \arcsin u = \frac{u'}{\sqrt{1 - u^2}}, \quad \frac{d}{dx} \arctan u = \frac{u'}{1 + u^2}, \quad \frac{d}{dx} \operatorname{arsinh} u = \frac{u'}{\sqrt{u^2 + 1}}, \quad \frac{d}{dx} \operatorname{artanh} u = \frac{u'}{1 - u^2}.$$

g	Relation	Eliminate	$g'(x)$
$\arcsin x$	$\sin y = x$	$\cos y = \sqrt{1 - x^2}$	$\frac{1}{\sqrt{1 - x^2}}$
$\arccos x$	$\cos y = x$	$\sin y = \sqrt{1 - x^2}$	$-\frac{1}{\sqrt{1 - x^2}}$
$\arctan x$	$\tan y = x$	$\sec^2 y = 1 + x^2$	$\frac{1}{1 + x^2}$
$\operatorname{arccot} x$	$\cot y = x$	$\csc^2 y = 1 + x^2$	$-\frac{1}{1 + x^2}$
$\operatorname{arcsec} x$	$\sec y = x$	$\tan y = \pm \sqrt{x^2 - 1}$	$\frac{1}{ x \sqrt{x^2 - 1}}$
$\operatorname{arsinh} x$	$\sinh y = x$	$\cosh y = \sqrt{x^2 + 1}$	$\frac{1}{\sqrt{x^2 + 1}}$
$\operatorname{arcosh} x$	$\cosh y = x$	$\sinh y = \sqrt{x^2 - 1}$	$\frac{1}{\sqrt{x^2 - 1}}$
$\operatorname{artanh} x$	$\tanh y = x$	$\operatorname{sech}^2 y = 1 - x^2$	$\frac{1}{1 - x^2}$

Three structural facts compress the table. The co-functions carry a minus sign and nothing else, because $\arccos x + \arcsin x = \pi/2$ is constant, so their derivatives must cancel. The hyperbolic column is the circular column with the sign

inside the radical or the denominator flipped, which is the usual circular-to-hyperbolic correspondence. And arcsec is the only entry needing an absolute value, because sec is even on its two branches while its derivative is odd.

The hyperbolic entries admit a second derivation which is worth knowing, because it is a check that uses none of the same steps. Solving $\sinh y = x$ by the quadratic in e^y gives the closed form $\operatorname{arsinh} x = \ln(x + \sqrt{x^2 + 1})$, and differentiating that logarithm directly,

$$\frac{d}{dx} \ln(x + \sqrt{x^2 + 1}) = \frac{1 + x/\sqrt{x^2 + 1}}{x + \sqrt{x^2 + 1}} = \frac{1}{\sqrt{x^2 + 1}},$$

because the numerator is $(x + \sqrt{x^2 + 1})/\sqrt{x^2 + 1}$. The same manoeuvre gives $\operatorname{artanh} x = \frac{1}{2} \ln \frac{1+x}{1-x}$ and hence $1/(1-x^2)$ by partial fractions. Nothing analogous exists for the circular functions, which is the real reason the hyperbolic entries feel easier.

WORKED EXAMPLE

$f(x) = e^{\arcsin x} \cosh(2x + 2)$, and $f'(1/2)$.

This is a product, each factor a composite. The product rule gives

$$f'(x) = e^{\arcsin x} \left[\frac{\cosh(2x + 2)}{\sqrt{1 - x^2}} + 2 \sinh(2x + 2) \right].$$

At $x = \frac{1}{2}$ the two special values do the rest: $\arcsin \frac{1}{2} = \frac{\pi}{6}$ and $\sqrt{1 - \frac{1}{4}} = \frac{\sqrt{3}}{2}$, while $2x + 2 = 3$. Hence

$$f'(\tfrac{1}{2}) = e^{\pi/6} \left[\frac{2}{\sqrt{3}} \cosh 3 + 2 \sinh 3 \right] \approx 53.4465.$$

A central difference of f at $x = 0.5$ returns 53.4465. Notice that the evaluation point was chosen so that arcsin lands on a special angle and the radical lands on a surd; that is the standard design of these problems, and spotting it tells you the point is not arbitrary.

A second instance, where the table is the slow route. Differentiate $g(x) = \arctan\left(\frac{2x}{1-x^2}\right)$. Mechanically, with $u = 2x/(1-x^2)$,

$$g' = \frac{u'}{1+u^2}, \quad u' = \frac{2(1-x^2) + 2x \cdot 2x}{(1-x^2)^2} = \frac{2(1+x^2)}{(1-x^2)^2}, \quad 1+u^2 = \frac{(1+x^2)^2}{(1-x^2)^2},$$

so $g' = \frac{2}{1+x^2}$, and at $x = 0.3$ that is 1.834862, matching a central difference to seven figures. But the double-angle identity $\tan 2\alpha = 2 \tan \alpha / (1 - \tan^2 \alpha)$ says $g(x) = 2 \arctan x$ outright for $|x| < 1$, and the derivative is one line. Both routes agree, which is the point: recognising the identity is faster, and the mechanical route is the insurance.

WHERE THIS GOES WRONG

Sign confusion between arcsin and arccos, and between arctan (denominator $1+u^2$) and artanh (denominator $1-u^2$). Derive rather than recall and both vanish. The domain trap is arcsec: the derivative is $1/(|u|\sqrt{u^2-1})u'$, and dropping the absolute value gives the wrong sign on the whole branch $u \leq -1$.

SEE ALSO The inverse-function derivative; inverse-trig identity collapse; conjugate log collapse.

Parametric and polar curves

A parametrised curve replaces one relation between x and y by two relations to a common parameter. Nothing about the tangent line changes: the tangent to a curve at a point is a geometric object and cannot depend on how the curve was described. What changes is the route to it. The chain rule applied to $y(t) = y(x(t))$ gives $\dot{y} = \frac{dy}{dx} \cdot \dot{x}$, and dividing is the first derivative. The second derivative is where nearly everyone loses a step.

13.6 Parametric first and second derivatives

CUE The curve is given as $x = x(t)$, $y = y(t)$, and a slope, a tangent line, or a concavity is wanted at a value of the parameter.

METHOD First derivative: $\frac{dy}{dx} = \dot{y}/\dot{x}$, valid wherever $\dot{x} \neq 0$. For the second derivative, differentiate that quotient *with respect to t* and then divide by $\dot{x}$ once more, because converting a t -derivative into an x -derivative always costs one factor of $\dot{x}$:

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{\dot{y}}{\dot{x}}\right)}{\dot{x}} = \frac{\ddot{y}\dot{x} - \dot{y}\ddot{x}}{\dot{x}^3}.$$

In practice the left-hand form is faster, because $\dot{y}/\dot{x}$ usually simplifies before you differentiate it.

WORKED EXAMPLE

$x = t^2 - 1$, $y = t^3 - 3t$, at $t = 2$.

First, $\dot{x} = 2t$ and $\dot{y} = 3t^2 - 3$, so

$$\frac{dy}{dx} = \frac{3t^2 - 3}{2t} = \frac{3t}{2} - \frac{3}{2t},$$

which at $t = 2$ is $\frac{9}{4}$. Now differentiate that simplified form with respect to t : $\frac{3}{2} + \frac{3}{2t^2} = \frac{3t^2 + 3}{2t^2}$. Divide by $\dot{x} = 2t$:

$$\frac{d^2y}{dx^2} = \frac{3t^2 + 3}{4t^3},$$

and at $t = 2$ this is $\frac{15}{32} = 0.46875$. A numerical differentiation of the slope function divided by $\dot{x}$ returns 0.468750.

Note that the curve crosses itself at $t = \pm\sqrt{3}$, both visits landing on $(2, 0)$, with slopes $3/t = \pm\sqrt{3}$; a parametrised curve carries one tangent per visit, which is exactly the information a single equation $F(x, y) = 0$ cannot record.

The cycloid, where simplifying first pays even better: $x = t - \sin t$, $y = 1 - \cos t$. Then

$$\frac{dy}{dx} = \frac{\sin t}{1 - \cos t} = \frac{2 \sin \frac{t}{2} \cos \frac{t}{2}}{2 \sin^2 \frac{t}{2}} = \cot \frac{t}{2}.$$

Differentiating that with respect to t gives $-\frac{1}{2} \csc^2 \frac{t}{2}$, and dividing by $\dot{x} = 1 - \cos t = 2 \sin^2 \frac{t}{2}$ gives

$$\frac{d^2y}{dx^2} = -\frac{1}{4 \sin^4(t/2)},$$

negative everywhere, so every arch of the cycloid is concave down. At $t = 1.1$ the formula gives -3.34944 and the numerical value is -3.34944 . Attempting this without the half-angle simplification means differentiating $\sin t/(1 - \cos t)$ by the quotient rule and then simplifying a four-term numerator, which is three times the work for the same answer.

WHERE THIS GOES WRONG

Computing $\frac{d^2y}{dx^2}$ as $\ddot{y}/\ddot{x}$. This is the single most common error in the chapter and it is not a slip of the pen: it is a false analogy with the correct first-derivative formula $\dot{y}/\dot{x}$. On the example above it would give $6t/2 = 3t$, which is 6 at $t = 2$ rather than $15/32$.

The extra division by $\dot{x}$ is mandatory, and the reason is that $\frac{d}{dx} = \frac{1}{\dot{x}} \frac{d}{dt}$ as operators, so applying it twice produces two factors of $1/\dot{x}$ and a derivative of the first one.

SEE ALSO Second derivatives implicitly; polar tangents; curvature.

The condition $\dot{x} \neq 0$ is not a technicality to be waved past. Where $\dot{x} = 0$ and $\dot{y} \neq 0$ the tangent is vertical and the curve is perfectly smooth; where both vanish the point is a candidate cusp and the slope must be recovered as a limit of $\dot{y}/\dot{x}$. On the cycloid, $\dot{x} = 1 - \cos t$ and $\dot{y} = \sin t$ both vanish at $t = 2k\pi$, and the limit of $\cot(t/2)$ there is infinite from one side and infinite with the opposite sign from the other: those are the cusps where the wheel touches the ground. On $x = t^2$, $y = t^3$ the same test at $t = 0$ gives $\dot{y}/\dot{x} = 3t/2 \rightarrow 0$, so that famous cusp has a horizontal tangent rather than no tangent at all. The test costs one line and settles which of the two pictures you are looking at.

WHY THE PARAMETRIC SECOND DERIVATIVE HAS A CUBE

Converting a t -derivative into an x -derivative is an operator identity:

$$\frac{d}{dx} = \frac{1}{\dot{x}} \frac{d}{dt}.$$

Apply it once to y and you get $\dot{y}/\dot{x}$. Apply it a second time to that quotient and you pay two prices: a fresh factor $1/\dot{x}$ out front, and the quotient rule acting on $\dot{y}/\dot{x}$, which contributes $\dot{x}^2$ in its own denominator. That is where the $\dot{x}^3$ comes from. The same reasoning gives the third derivative without any new idea, and explains why parametric derivatives of order k carry $\dot{x}^{2k-1}$ in the denominator: the powers accumulate, they do not repeat.

Polar coordinates need no new theory at all, only the discipline to remember that $r = r(\theta)$ is not a graph of y against x . The polar equation is a parametrisation in disguise, with θ as the parameter, and the moment you write $x = r(\theta) \cos \theta$ and $y = r(\theta) \sin \theta$ the previous technique applies verbatim.

13.7 Polar tangents

MEDIUM

CUE A curve given as $r = r(\theta)$, and a slope, a horizontal tangent, a vertical tangent, or a tangent at the pole is wanted.

METHOD Treat θ as the parameter. Both $\dot{x}$ and $\dot{y}$ need the product rule, since r and the trigonometric factor both depend on θ :

$$\frac{dy}{dx} = \frac{\dot{y}}{\dot{x}} = \frac{r'(\theta) \sin \theta + r(\theta) \cos \theta}{r'(\theta) \cos \theta - r(\theta) \sin \theta}.$$

Horizontal tangents are the zeros of the numerator, vertical tangents the zeros of the denominator, and a point where both vanish needs a limit. At the pole, where $r(\theta_0) = 0$ and $r'(\theta_0) \neq 0$, the formula collapses to $\frac{dy}{dx} = \tan \theta_0$: the tangent line at the pole is the ray $\theta = \theta_0$ itself.

WORKED EXAMPLE

The cardioid $r = 1 + \cos \theta$. Find its horizontal and vertical tangents.

Here $r' = -\sin \theta$, so the numerator is

$$-\sin^2 \theta + (1 + \cos \theta) \cos \theta = \cos \theta + \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta + \cos \theta - 1 = (2 \cos \theta - 1)(\cos \theta + 1),$$

using $\sin^2 = 1 - \cos^2$. It vanishes when $\cos \theta = \frac{1}{2}$, that is $\theta = \pm\pi/3$, and when $\cos \theta = -1$, that is $\theta = \pi$, where $r = 0$ as well and the point is the pole. So the honest horizontal tangents are at $\theta = \pm\pi/3$, where $r = \frac{3}{2}$ and the point is $(\frac{3}{4}, \pm\frac{3\sqrt{3}}{4})$. A numerical slope at $\theta = \pi/3$ returns 0 to fourteen digits, and the denominator there is $-\sqrt{3} \neq 0$, so the tangent really is horizontal and not indeterminate.

The denominator is

$$-\sin \theta \cos \theta - (1 + \cos \theta) \sin \theta = -\sin \theta (2 \cos \theta + 1),$$

vanishing at $\theta = 0$, at $\theta = \pi$, and where $\cos \theta = -\frac{1}{2}$, that is $\theta = \pm 2\pi/3$. At $\theta = 0$ the point is $(2, 0)$, the nose of the cardioid, and the tangent is vertical. At $\theta = \pm 2\pi/3$ the radius is $r = \frac{1}{2}$ and the tangents are vertical there too. At $\theta = \pi$ both numerator and denominator vanish; that is the cusp at the pole, and the tangent-at-the-pole rule gives the direction $\theta = \pi$, the negative x -axis, which is the cusp axis.

The pole rule is worth one clean instance of its own. The four-petalled rose $r = \sin 2\theta$ has $r = 0$ at $\theta = 0, \pi/2, \pi, 3\pi/2$, and $r' = 2 \cos 2\theta$ is nonzero at each. So the curve passes through the origin four times, and on each visit the tangent is the ray $\theta = \theta_0$: the two coordinate axes, each traversed twice. Every petal is therefore tangent to an axis at the origin, which is exactly what the picture shows, and the general formula never had to be evaluated.

WHERE THIS GOES WRONG

Writing $\frac{dy}{dx} = r'(\theta)$ or $\frac{dy}{dx} = \frac{dr}{d\theta}$. Neither is a slope in the Cartesian sense; r' measures how fast the radius grows, which is a different question. The second trap is reporting a horizontal tangent at a θ where the numerator vanishes without checking the denominator: on the cardioid $\theta = \pi$ kills both, and calling that a horizontal tangent misses the cusp.

SEE ALSO Parametric first and second derivatives; implicit differentiation at a point; related rates.

THE ANGLE BETWEEN THE RADIUS AND THE TANGENT

Polar problems often ask not for the Cartesian slope but for ψ , the angle from the radius vector to the tangent line. The parametric formula rearranges into a much simpler statement:

$$\tan \psi = \frac{r(\theta)}{r'(\theta)},$$

with ψ measured at the point θ . The Cartesian slope is then $\tan(\theta + \psi)$, which is often the fastest route to it. The formula also identifies the one curve on which ψ is constant: r/r' constant means $r'/r = 1/a$, so $r = e^{\theta/a}$, the logarithmic spiral. For $r = e^{0.4\theta}$ at $\theta = 1$ the numerical angle between radius and tangent is 1.190290, and $\arctan(1/0.4) = 1.190290$. That constancy is why the logarithmic spiral is called equiangular, and why it is the shape a moth flying at a fixed angle to a light traces out.

One last member of the family sits between inverse functions and series. When the inverse is defined by a simple equation but a *high* derivative of it is wanted, the inverse-function theorem gives only the first order and repeated implicit differentiation becomes unmanageable. The right tool then is coefficient matching, which converts the whole question into linear algebra on a power series.

HARD

13.8 Lagrange inversion and series reversion

CUE The inverse is defined implicitly by a compact equation, such as $x = y + ay^3$ or $we^w = x$, and a high derivative of the inverse at 0 is wanted.

METHOD Posit a power series for y in x that respects the parity or symmetry of the relation, substitute it into the relation, and match coefficients order by order; each order determines one new coefficient from those already found. Then read the derivative off with $y^{(n)}(0) = n! [x^n]y$. For a formula rather than a recursion, Lagrange inversion gives

$$[x^n]f^{-1}(x) = \frac{1}{n} [y^{n-1}] \left(\frac{y}{f(y)} \right)^n.$$

WORKED EXAMPLE

$x = y + ay^3$ defines $y = y(x)$ near the origin with $y(0) = 0$. Find $y^{(5)}(0)$.

The relation is odd in the pair (x, y) : replacing y by $-y$ replaces x by $-x$. So y is an odd function of x and only odd powers survive. Write $y = x + c_3x^3 + c_5x^5 + O(x^7)$, the leading coefficient being 1 because $x = y + O(y^3)$. Substituting,

$$y + ay^3 = (x + c_3x^3 + c_5x^5) + a(x^3 + 3c_3x^5) + O(x^7) = x + (c_3 + a)x^3 + (c_5 + 3ac_3)x^5 + O(x^7),$$

where $y^3 = x^3 + 3c_3x^5 + O(x^7)$ came from the binomial expansion. Matching against x forces $c_3 = -a$ and then $c_5 = -3ac_3 = 3a^2$. Hence

$$y = x - ax^3 + 3a^2x^5 - 12a^3x^7 + \dots, \quad y^{(5)}(0) = 5! \cdot 3a^2 = 360a^2.$$

At $a = 0.3$ that is 32.4; evaluating the fifth derivative of the numerically inverted function by the Cauchy integral formula on a circle of radius 0.25 returns 32.400000. The same computation at order 7 gives $7! \cdot (-12a^3) = -1632.96$, confirmed to six figures.

The named case is $we^w = x$, whose solution is the Lambert function W . Here Lagrange inversion is worth using as a formula rather than as a recursion. With $f(w) = we^w$ we have $w/f(w) = e^{-w}$, so

$$[x^n]W(x) = \frac{1}{n} [w^{n-1}]e^{-w} = \frac{1}{n} \cdot \frac{(-n)^{n-1}}{(n-1)!} = \frac{(-n)^{n-1}}{n!},$$

and therefore $W^{(n)}(0) = n! [x^n]W = (-n)^{n-1}$. That is a startlingly clean answer for a function with no closed form: $W'(0) = 1$, $W''(0) = -2$, $W'''(0) = 9$, $W^{(4)}(0) = -64$, $W^{(5)}(0) = 625$. The Cauchy integral formula applied to a numerically inverted W on a circle of radius 0.15 returns 1.000, -2.000, 9.000, -64.000, 625.000. The series itself, $W(x) = x - x^2 + \frac{3}{2}x^3 - \frac{8}{3}x^4 + \dots$, is the one quoted in the chapter on expansion; this is where it comes from.

WHERE THIS GOES WRONG

Reporting the coefficient $3a^2$ instead of $n!$ times it. The other failure is not exploiting parity: without the observation that y is odd, the matching has to carry c_2 and c_4 through every step, and the bookkeeping doubles for coefficients that are all zero.

SEE ALSO Maclaurin coefficient extraction; the inverse-function derivative; Faa di Bruno for composites.

Choosing the description

The four costumes are not equally convenient, and a curve presented in one form can often be moved to another. Three rules of thumb decide the move.

Convert to parametric when the implicit form has singular points you care about. A self-intersection, a cusp, or a vertical tangent is invisible to $-F_x/F_y$, which simply reports a division by zero, but a parametrisation walks through the point with a well defined velocity and tells you which branch you are on. The folium of Descartes is the model case: implicitly it has an unusable origin, and under the rational parametrisation $x = 6t/(1+t^3)$, $y = 6t^2/(1+t^3)$ the origin is visited at $t = 0$ and at $t = \infty$, giving the two tangents separately.

Convert to implicit when the parameter is a nuisance and only a relation is wanted. If $x = \cos t$ and $y = \cos 2t$, do not differentiate twice; note that $y = 2x^2 - 1$ and the curve is a parabolic arc, whose slope is $4x$ everywhere.

Do not convert at all when the question is about a specific point and one description already reaches it. Most of the work lost in this chapter is lost by solving for y when nobody asked for y . The derivative at a point needs one linear equation, and that equation is available in whatever form the curve arrived in.

Recognition matrix

If you see	Reach for	If that fails
One equation in x and y , not solvable for y	Differentiate term by term; every y emits a y'	$y' = -F_x/F_y$
An implicit curve with $\frac{d^2y}{dx^2}$ wanted	Differentiate the expression for y' , then substitute	The closed form in F_{xx}, F_{xy}, F_{yy}
$F_y = 0$ at the point	Vertical tangent: report $\frac{dx}{dy} = 0$	Parametrise the curve near the point
The unknown in its own exponent	Log the relation, then differentiate implicitly	Check whether the logged form solves for y
$(f^{-1})'(b)$ requested	Find a with $f(a) = b$; answer $1/f'(a)$	Conjugate or surd identity to crack a
b written as $(1 + \sqrt{2})^2$ or similar	The surd norm is the hint; use it	Small integer preimages first
Several inverse branches	One answer per branch	$\sum_i 1/p'(r_i) = 0$ for the total
$\arcsin, \arctan, \operatorname{arcsec}$ as a layer	Derive from $\sin y = x$ and a Pythagorean identity	The table, with $ u $ for arcsec
$x = x(t), y = y(t)$, slope wanted	$\dot{y}/\dot{x}$	Eliminate t only if it is easy
$x = x(t), y = y(t)$, concavity wanted	$\frac{d}{dt}(\dot{y}/\dot{x})$, then divide by $\dot{x}$	$(\ddot{y}\dot{x} - \dot{y}\ddot{x})/\dot{x}^3$
$r = r(\theta)$	$\frac{r' \sin \theta + r \cos \theta}{r' \cos \theta - r \sin \theta}$	Convert to x, y and use the parametric rule
$r(\theta_0) = 0$, tangent wanted	The tangent at the pole is $\theta = \theta_0$	Take a limit of the quotient
High derivative of an implicit inverse	Match coefficients in a series ansatz	Lagrange inversion
$f^{(n)}(0)$ from a reverted series	$n! [x^n]y$, not $[x^n]y$	Use parity to halve the work
Angle between radius and tangent	$\tan \psi = r/r'$	Cartesian slope is $\tan(\theta + \psi)$
$(f^{-1})''(b)$	$-f''(a)/f'(a)^3$ at the preimage a	Differentiate $y' = 1/f'(y)$ by hand
$\dot{x} = \dot{y} = 0$ at the parameter value	Candidate cusp: take the limit of $\dot{y}/\dot{x}$	Expand $x(t)$ and $y(t)$ about t_0
Both x and y moving in time	Differentiate the relation in t : $F_x \dot{x} + F_y \dot{y} = 0$	Solve for the wanted rate
$w e^w = x$	Lagrange inversion: $W^{(n)}(0) = (-n)^{n-1}$	Match coefficients directly
Inflections of an implicit curve	Set the y'' numerator to zero, intersect with $F = 0$	Parametrise and use $\ddot{y}\dot{x} - \dot{y}\ddot{x}$
A self-intersection of the curve	Parametrise: one tangent per visit	Factor F near the singular point

Higher-order and pattern derivatives

What this chapter is for. A request for $f^{(50)}$, or for $f^{(20)}(0)$, is never an instruction to differentiate fifty times. It is an instruction to find the pattern, and the order was chosen precisely to make the mechanical route impossible and the structural route short. There are exactly four places the pattern can live, and identifying which one is the whole of the work. It can live in a cyclic four-step orbit, as it does for $\sin$ and $\cos$, where differentiation advances a phase by $\pi/2$ and the answer depends only on $n \bmod 4$. It can live in a factorial-times-power orbit, as it does for $1/(ax+b)$ and $\ln(ax+b)$ and therefore for every rational function once it has been split into partial fractions. It can live in a Leibniz expansion, where the binomial sum for $(uv)^{(n)}$ has $n+1$ terms of which all but two or three vanish. Or it can live in a Maclaurin coefficient, where $f^{(n)}(0) = n!a_n$ reads the answer off a series you already know. That last one is the highest-value move in the chapter, and the one a setter builds against when a problem is meant to be hard. Along the way this chapter installs the complex-exponential form that collapses $e^{ax} \sin bx$ to a single power, the parity argument that answers half of these problems in one line, Faà di Bruno's formula for composites, and the summation and fractional-order variants. At the end you should treat a large n as a hint rather than a threat.

Four homes for the pattern

Nobody differentiates twenty times. The reason is not stamina but growth: each application of the product rule to a two-factor expression doubles the term count, so a fourth derivative of $e^x \sin x \cos x$ already has more terms than a page holds, and the twentieth has more than there are atoms in the room. Every problem of this kind therefore has a structural solution, and the shape of the problem tells you which one.

To see why, count. Differentiating a sum of m terms, each a product of two factors, produces up to $2m$ terms, and nothing collapses them unless the factors are related. Starting from the two-term product $e^x \sin x$ the counts run 2, 4, 8, 16, ..., so the twentieth derivative has up to 2^{20} terms before simplification, which is over a million. Every technique below works by refusing to enter that doubling at all.

Home one: a cyclic orbit. Differentiating $\sin(ax+b)$ multiplies by a and advances the phase by $\pi/2$, so after four steps the function returns to itself scaled by a^4 . The orbit is finite, the answer depends on $n \bmod 4$, and the entire computation is one reduction modulo four. Any expression that can be linearised into a sum of $\sin(ax+b)$ and $\cos(ax+b)$ lives here, which is why $\sin^2 x$ and $\cos^3 x$ must be linearised before anything else happens to them.

Home two: a factorial-times-power orbit. Differentiating $(ax+b)^{-1}$ multiplies by $-a$ and raises the power of the denominator by one, so the n -th derivative is $(-1)^n n! a^n (ax+b)^{-n-1}$. The orbit is infinite but the pattern is explicit, and $\ln(ax+b)$ joins it after one differentiation. Every rational function lives here once partial fractions have split it, and no rational function lives anywhere else.

Home three: a Leibniz expansion that truncates. When $f = uv$ and u is a polynomial of degree d , the binomial sum $\sum_k \binom{n}{k} u^{(k)} v^{(n-k)}$ has $n+1$ terms of which only the first $d+1$ are nonzero, and evaluating at a well chosen point usually kills all but one of those. A twentieth derivative then costs three terms, or one.

Home four: a Maclaurin coefficient. At the origin, derivatives and coefficients are the same information in different units, related by $f^{(n)}(0) = n!a_n$. If you can name the series, you can name the derivative, and naming a series is often possible where finding a derivative pattern is not. This is the most general of the four and the one to reach for when the other three fail to apply, which is exactly the situation a hard problem is built to produce.

THE BRIDGE BETWEEN DERIVATIVES AND SERIES

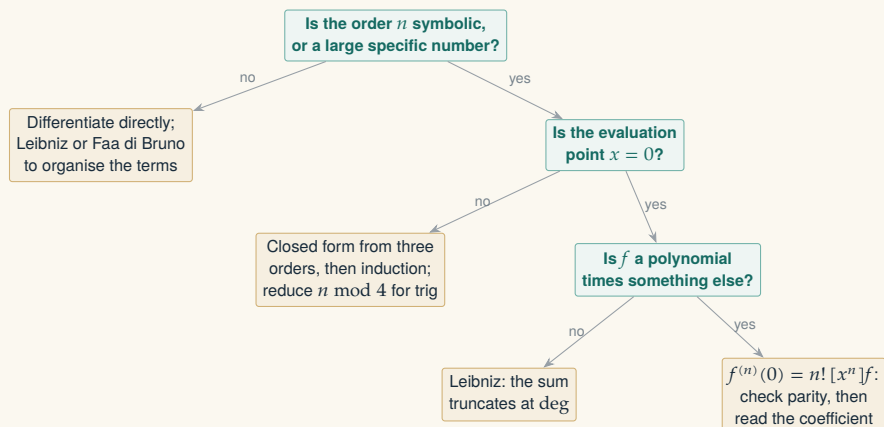
If $f(x) = \sum_{k \geq 0} a_k x^k$ near 0, then

$$f^{(n)}(0) = n! a_n = n! [x^n] f(x).$$

Differentiating n times kills every term below x^n , leaves $n! a_n$ from the x^n term, and sends every term above it to zero at $x = 0$. That is the entire content, and it runs in both directions: a hard derivative becomes an easy coefficient, and a hard coefficient becomes an easy derivative. The factor $n!$ is where almost every error in this chapter lives. Two symptoms of the intended route:

an order in the teens or higher, and an answer normalised by $n!$ or $\Gamma(n)$, which is the setter cancelling the factorial back out so that the printed answer is small.

Two free wins sit above all four homes and should be checked first, because each can end the problem before it starts. The first is parity: at the origin, an odd function has all even derivatives zero and an even function has all odd derivatives zero. The second is the evaluation point: a large part of the expression may vanish at the point in question, and the terms that survive are often a single one.



Closed forms and cyclic patterns

The elementary families all have closed-form n -th derivatives, and there are only four of them worth memorising. Everything else is reduced to these four by an identity, a partial fraction, or a complex exponential.

14.1 n -th derivative closed forms and the cyclic trig pattern

CORE

CUE $f^{(n)}$ is wanted symbolically in n , or $f^{(n)}(a)$ for a large specific n , and f is or can be reduced to one of the elementary families.

METHOD Establish the pattern from three orders, then prove it by induction. The standard table:

$$\begin{aligned} \frac{d^n}{dx^n} \sin(ax + b) &= a^n \sin\left(ax + b + \frac{n\pi}{2}\right), & \frac{d^n}{dx^n} e^{ax} &= a^n e^{ax}, \\ \frac{d^n}{dx^n} \frac{1}{ax + b} &= \frac{(-1)^n n! a^n}{(ax + b)^{n+1}}, & \frac{d^n}{dx^n} \ln(ax + b) &= \frac{(-1)^{n-1} (n-1)! a^n}{(ax + b)^n}. \end{aligned}$$

The trigonometric entry is periodic with period 4 in n , so reduce $n \bmod 4$ before evaluating. Reduce any rational function to a sum of the third form by partial fractions *before* differentiating anything.

WORKED EXAMPLE

$f(x) = \sin^2 x$, and $f^{(7)}(\pi/4)$.

Do not differentiate $\sin^2 x$. Reduce it first: $\sin^2 x = \frac{1}{2} - \frac{1}{2} \cos 2x$. The constant dies at the first derivative, and the cosine is in the table with $a = 2$ and a phase, so for $n \geq 1$

$$f^{(n)}(x) = -\frac{1}{2} \cdot 2^n \cos\left(2x + \frac{n\pi}{2}\right) = -2^{n-1} \cos\left(2x + \frac{n\pi}{2}\right).$$

At $n = 7$ and $x = \pi/4$ the argument is $\frac{\pi}{2} + \frac{7\pi}{2} = 4\pi$, so $f^{(7)}(\pi/4) = -2^6 \cos 4\pi = -64$. The Cauchy integral formula applied to $\sin^2 z$ on a circle of radius 0.6 about $\pi/4$ returns -64.0000000 .

Now a rational function, where the partial fraction step is not optional. Take $f(x) = \frac{1}{x^2 - 5x + 6}$ and find $f^{(n)}(0)$.

Factor and split:

$$f(x) = \frac{1}{(x-2)(x-3)} = \frac{1}{x-3} - \frac{1}{x-2},$$

and each piece is the third table entry with $a = 1$. So

$$f^{(n)}(x) = (-1)^n n! \left[\frac{1}{(x-3)^{n+1}} - \frac{1}{(x-2)^{n+1}} \right],$$

and at $x = 0$ the two signs $(-1)^n$ and $(-1)^{n+1}$ multiply to -1 , leaving

$$f^{(n)}(0) = n! \left[\frac{1}{2^{n+1}} - \frac{1}{3^{n+1}} \right].$$

At $n = 0$ this is $\frac{1}{2} - \frac{1}{3} = \frac{1}{6} = f(0)$, a useful sanity check. At $n = 12$ it is 58171.43299, and contour differentiation returns 58171.43299. Attempting the same by twelve applications of the quotient rule to $1/(x^2 - 5x + 6)$ is not a calculation anybody finishes.

A third specimen, where the pattern starts late. For $f(x) = x \ln x$ the first derivative is $\ln x + 1$ and the second is $1/x$, at which point the table takes over: for $n \geq 2$,

$$f^{(n)}(x) = \frac{d^{n-2}}{dx^{n-2}} \frac{1}{x} = \frac{(-1)^{n-2}(n-2)!}{x^{n-1}} = \frac{(-1)^n(n-2)!}{x^{n-1}}.$$

At $x = 1$ that gives $1, -1, 2, -6, 24, \dots$ for $n = 2, 3, 4, 5, 6$, and contour differentiation of $z \ln z$ about $z = 1$ returns $1.000, -1.000, -6.000$ at $n = 2, 3, 5$. The lesson is that a closed form need not be valid from $n = 0$; state the range it holds on, and handle the first one or two orders separately.

WHERE THIS GOES WRONG

Getting the logarithm family off by one: the n -th derivative of $\ln(ax + b)$ carries $(n-1)!$, not $n!$, because the first differentiation produces $a/(ax+b)$ with no factorial at all and the table for $1/(ax+b)$ then starts from $n-1$. The second standard slip is losing the alternation of signs, which is best avoided by checking $n = 1$ against a direct computation every time. The third is differentiating a rational function before splitting it: the quotient rule squares the denominator at every step, and after four rounds the expression is unrecognisable while the partial-fraction form is still two terms.

SEE ALSO Complex exponentials for high derivatives; Maclaurin coefficient extraction; trig identity reduction before high-order differentiation.

The induction that justifies the trigonometric entry is worth writing once, because the same three lines justify all four entries and because the phase-shift form is what makes the induction trivial. Suppose $\frac{d^n}{dx^n} \sin(ax+b) = a^n \sin(ax+b+\frac{n\pi}{2})$. Differentiating once more gives $a^{n+1} \cos(ax+b+\frac{n\pi}{2})$, and $\cos \theta = \sin(\theta+\frac{\pi}{2})$ turns that into $a^{n+1} \sin(ax+b+\frac{(n+1)\pi}{2})$, which is the statement at $n+1$. The base case $n = 0$ is trivial. The whole force of the phase-shift notation is that differentiation becomes addition of $\pi/2$, so the induction step is an identity rather than a case analysis on $n \bmod 4$.

REDUCE $n \bmod 4$, BUT REDUCE THE FUNCTION FIRST

Since $\sin(\theta+\frac{n\pi}{2})$ depends only on $n \bmod 4$, the derivatives of a sinusoid cycle with period 4 up to the factor a^n . But the reduction is only useful after the function has been brought into the form $\sin(ax+b)$ or $\cos(ax+b)$. Powers and products of sinusoids must be linearised first, by $\sin^2 x = \frac{1}{2}(1-\cos 2x)$, $\cos^2 x = \frac{1}{2}(1+\cos 2x)$, $\sin x \cos x = \frac{1}{2} \sin 2x$, and their relatives. A solver who differentiates $\cos^3 x$ four times without linearising will get the right answer and will take twenty times as long; a solver who tries it eight times will not get the right answer.

INDUCTION HYGIENE

A conjectured $f^{(n)}$ is worth nothing until it has been checked at both ends. Check the base case at the smallest n for which the formula is claimed, which is often 1 or 2 rather than 0; the $x \ln x$ example above holds only for $n \geq 2$, and a solver who asserts it at $n = 1$ produces $(-1)!$ and a nonsense answer. Check the step by differentiating the claimed $f^{(n)}$ symbolically and matching it against the claim at $n+1$, not by testing another numerical case. And check the sign at one odd and one even order, since a lost factor of $(-1)^n$ is invisible at even orders.

One extension of the second home is needed often enough to state separately. Partial fractions do not always produce distinct linear factors; a repeated root leaves a power $(x-c)^{-m}$ with $m \geq 2$, and the table entry generalises without difficulty, since each differentiation multiplies by the current exponent and lowers it by one:

$$\frac{d^n}{dx^n} (x-c)^{-m} = (-1)^n \frac{(m+n-1)!}{(m-1)!} (x-c)^{-m-n}.$$

At $m = 1$ this is the table entry, since $(m + n - 1)!/(m - 1)! = n!$. As a specimen, take $f(x) = \frac{2x + 1}{(x - 1)^2(x + 2)}$ and find $f^{(n)}(0)$. The partial fractions are $\frac{1/3}{x - 1} + \frac{1}{(x - 1)^2} - \frac{1/3}{x + 2}$, checked at $x = 0$ where the three pieces give $-\frac{1}{3} + 1 - \frac{1}{6} = \frac{1}{2}$ against $f(0) = \frac{1}{2}$. Applying the two formulas at $x = 0$, with $(0 - 1)^{-n-1} = (-1)^{n+1}$ and $(0 - 1)^{-n-2} = (-1)^n$,

$$f^{(n)}(0) = (n + 1)! - \frac{n!}{3} - \frac{(-1)^n n!}{3 \cdot 2^{n+1}}.$$

At $n = 0$ that is $1 - \frac{1}{3} - \frac{1}{6} = \frac{1}{2}$, and at $n = 1, 2, 3$ it gives 1.75, 5.25, 22.125, all confirmed by contour differentiation. The $(n + 1)!$ from the double pole dominates everything else, which is the general lesson: the order of the worst pole sets the growth rate of the derivatives, and the poles nearest the origin set the size of the terms.

Products of an exponential and a sinusoid do not linearise, and their derivatives do not cycle: each differentiation mixes the two terms. The standard escape is to stop treating them as two real functions and start treating them as one complex one, at which point the pattern reappears in its simplest possible form.

14.2 Complex exponentials for $e^{ax} \sin(bx)$ -type high derivatives

MEDIUM

CUE A product of a real exponential and a sinusoid, differentiated many times, typically evaluated at 0.

METHOD Write the function as the real or imaginary part of a single complex exponential, $e^{ax} \sin bx = \text{Im } e^{(a+ib)x}$ and $e^{ax} \cos bx = \text{Re } e^{(a+ib)x}$. Differentiation of e^{cx} is multiplication by c , so

$$\left. \frac{d^n}{dx^n} (e^{ax} \sin bx) \right|_{x=0} = \text{Im}[(a + ib)^n],$$

and the power is evaluated in polar form, $(a + ib)^n = r^n e^{in\theta}$ with $r = \sqrt{a^2 + b^2}$ and $\theta = \arg(a + ib)$. Taking real and imaginary parts commutes with differentiation because a and b are real constants. Keeping x instead of setting it to zero gives the general n -th derivative in amplitude-phase form, which is the statement worth memorising:

$$\frac{d^n}{dx^n} (e^{ax} \sin bx) = r^n e^{ax} \sin(bx + n\theta), \quad \frac{d^n}{dx^n} (e^{ax} \cos bx) = r^n e^{ax} \cos(bx + n\theta).$$

Each differentiation scales the amplitude by r and advances the phase by θ : the derivatives spiral rather than cycle. When $a = 0$ the modulus is $r = b$ and the phase step is $\theta = \pi/2$, the spiral closes after four steps, and the formula degenerates to the cyclic entry of the previous technique. That is the precise sense in which the first home is a special case of this one.

WORKED EXAMPLE

$f(x) = e^x \sin x$, and $f^{(7)}(0)$.

Here $c = 1 + i$, so $r = \sqrt{2}$ and $\theta = \pi/4$. Then $c^7 = 2^{7/2} e^{7i\pi/4}$, and $e^{7i\pi/4} = \frac{\sqrt{2}}{2}(1 - i)$, so $c^7 = 8 - 8i$ and

$$f^{(7)}(0) = \text{Im}(8 - 8i) = -8.$$

Contour differentiation returns -8.0000000 . The binomial route, expanding $(1 + i)^7$ into eight terms, gets there too, but the polar form generalises: $f^{(n)}(0) = 2^{n/2} \sin(n\pi/4)$ for every n , which is the symbolic answer the binomial route never produces.

A second instance with a real part and an unequal pair. For $f(x) = e^{2x} \cos 3x$ the constant is $c = 2 + 3i$ and

$$f^{(5)}(0) = \text{Re}[(2 + 3i)^5] = 122,$$

which contour differentiation confirms as 122.0000000 . Here the binomial expansion is actually competitive, since $(2 + 3i)^2 = -5 + 12i$ and squaring again gives $-119 - 120i$, so $(2 + 3i)^5 = (-119 - 120i)(2 + 3i) = 122 - 597i$. Repeated squaring is the practical method for a specific moderate n ; polar form is the method for symbolic n .

A third instance, evaluated away from the origin and with $a < 0$, because that is where the branch of $\arg$ starts to matter. For $f(x) = e^{-x} \sin x$ the constant is $c = -1 + i$, so $r = \sqrt{2}$ and $\theta = \arg(-1 + i) = 3\pi/4$, not $\arctan(1/(-1)) = -\pi/4$. The amplitude-phase form gives

$$f^{(n)}(x) = 2^{n/2} e^{-x} \sin\left(x + \frac{3n\pi}{4}\right),$$

so $f^{(4)}(0.7) = 4e^{-0.7} \sin(0.7 + 3\pi) = -4e^{-0.7} \sin(0.7) = -1.2796361$, and contour differentiation about $x = 0.7$ returns -1.2796361 . Had the argument been taken as $-\pi/4$, the phase at $n = 4$ would have been $-\pi$ rather than 3π ,

which is the same angle modulo 2π only by luck; at $n = 5$ the two choices differ by 5π and the sign of the answer flips.

WHERE THIS GOES WRONG

Taking Re when the function was a sine, or Im when it was a cosine. Write the identification down before computing anything. The second trap is a wrong branch of $\arg$: for $a < 0$ the argument is not $\arctan(b/a)$ but $\arctan(b/a) \pm \pi$, and a sign error there flips the answer for every odd n .

SEE ALSO n -th derivative closed forms; parity annihilation at the origin; summing all derivatives at a point.

Before any of these machines is started, one free check should be run, because it settles a large fraction of problems at the origin without a single differentiation.

EASY

14.3 Parity annihilation at the origin

CUE A high derivative at $x = 0$ of a function visibly assembled from odd and even pieces.

METHOD If f is odd, every even-order derivative vanishes at 0; if f is even, every odd-order derivative vanishes at 0. Determine the parity by the multiplication rules (odd times even is odd, odd times odd is even, and composition with an even inner function is even) before computing anything. The reason is the bridge: an odd function has only odd-degree Maclaurin coefficients, and $f^{(n)}(0) = n!a_n$.

WORKED EXAMPLE

$f(x) = \sinh x \cos x$, and $f^{(10)}(0)$.

$\sinh$ is odd, $\cos$ is even, so f is odd, so all even derivatives vanish at 0 and $f^{(10)}(0) = 0$. Contour differentiation returns -4×10^{-9} , which is zero to the accuracy of the quadrature. The mechanical alternative is a ten-fold product rule with 2^{10} terms before cancellation.

The same one-line argument handles orders that no other method could reach: $f(x) = \frac{x}{1+x^2}$ is odd, so $f^{(2026)}(0) = 0$.

And it composes with the other routes rather than competing with them: for $g(x) = x^2 e^{-x^2} \sin x$, the factors are even, even, odd, so g is odd and every even derivative at 0 vanishes; the odd ones still need the series, but half the work has already been deleted.

WHERE THIS GOES WRONG

Applying parity at a point other than the centre of symmetry. Oddness of f says nothing whatever about $f^{(10)}(1)$. If the function is symmetric about $x = c$ rather than 0, substitute $t = x - c$ first and then apply the rule at $t = 0$. The other trap is a misremembered parity: $\sinh$ is odd and $\cosh$ is even, matching $\sin$ and $\cos$, and $\tanh$ is odd.

SEE ALSO Maclaurin coefficient extraction; complex exponentials for high derivatives; symmetry kills the term.

PARITY IS THE CASE $k = 2$ OF A SHARPER STATEMENT

If $f(x) = g(x^k)$ for some integer $k \geq 2$, then the series of f contains only powers that are multiples of k , so

$$f^{(n)}(0) = 0 \text{ unless } k \mid n, \quad \text{and} \quad f^{(n)}(0) = n! [x^{n/k}]g \text{ when } k \mid n.$$

Evenness is the case $k = 2$ with g arbitrary. The gain grows with k : for $f(x) = \sin(x^3)$, two out of every three derivatives vanish at the origin, and the surviving ones are read straight off the sine series. At $n = 15$, $[x^{15}]f = [x^5]\sin = 1/120$, so $f^{(15)}(0) = 15!/120 = 10897286400$, which contour differentiation confirms. Likewise e^{x^2} has $f^{(9)}(0) = 0$ and $f^{(20)}(0) = 20!/10! = 670442572800$, since $[x^{20}]e^{x^2} = [x^{10}]e^x = 1/10!$. Whenever the variable appears only as x^k , substitute $t = x^k$ before doing anything else.

THE EVALUATION POINT IS PART OF THE PROBLEM

Parity is the sharpest instance of a more general economy: the point at which the derivative is evaluated usually annihilates most of what you would otherwise compute. In the Leibniz expansion of $x^2 e^{3x}$ below, the general n -th derivative has three terms and the value at 0 has one. In a Faà di Bruno expansion, only the terms whose inner factors survive at the point contribute. The discipline is to carry the evaluation point through the computation rather than saving it for the end: substitute the point as early as the algebra permits, and let it delete terms before you write them down rather than after.

Reading the coefficient

The four homes are not equals. The fourth, the Maclaurin coefficient, is the general one, and the first three are the special cases where the coefficient happens to be available in closed form: a cyclic orbit is a series with period-four coefficients, a factorial-times-power orbit is a geometric series with the factorial divided back out, and a truncating Leibniz sum is a Cauchy product with finitely many terms. That is why a setter who wants a genuinely hard high-derivative problem builds it out of a function whose series you know and whose derivative pattern you do not. Coefficient extraction is accordingly the highest-value technique in the chapter: it is the one that still works when the other three have nothing to say.

Coefficient extraction is only as good as the series you can name, so it is worth having the working set in one place. The following are the coefficients, not the series, because the coefficient is what the technique consumes.

$f(x)$	$[x^n]f$	$f^{(n)}(0)$
$\frac{1}{1-x}$	a^n	$n! a^n$
$\frac{1}{(1-x)^k}$	$\binom{n+k-1}{k-1}$	$n! \binom{n+k-1}{k-1}$
e^{ax}	$\frac{a^n}{n!}$	a^n
$\ln(1+x)$	$\frac{(-1)^{n-1}}{(n-1)!}$ ($n \geq 1$)	$(-1)^{n-1} (n-1)!$
$\sin x$	$\frac{(-1)^m}{(2m+1)!}$ at $n = 2m+1$	$\sin(n\pi/2)$
$\arctan x$	$\frac{(-1)^m}{(2m+1)!}$ at $n = 2m+1$	$(-1)^m (2m)!$
$(1+x)^\alpha$	$\binom{\alpha}{n}$	$\alpha(\alpha-1)\cdots(\alpha-n+1)$

Two operations extend the table indefinitely. Multiplying by $1/(1-x)$ replaces each coefficient by the partial sum of all coefficients up to it. Multiplying by x^m shifts the coefficients right by m . Between them, and the Cauchy product, almost every function a setter can build out of the table is reachable.

CORE

14.4 Maclaurin coefficient extraction: $f^{(n)}(0) = n! [x^n]f(x)$

CUE An absurdly high derivative order at 0, such as $n = 15, 20, 100, 2026$; or an answer normalised by $n!$ or $\Gamma(n)$; or a function built visibly out of series you already know.

METHOD Expand f as a power series only as far as x^n , read off the coefficient, and multiply by $n!$. For rational functions get the coefficient by partial fractions plus the geometric family

$$\frac{1}{1-x} = \sum_{n \geq 0} x^n, \quad \frac{1}{(1-x)^2} = \sum_{n \geq 0} (n+1)x^n, \quad \frac{1}{(1-x)^k} = \sum_{n \geq 0} \binom{n+k-1}{k-1} x^n.$$

For products, use the Cauchy product: $[x^n](fg) = \sum_{j=0}^n [x^j]f \cdot [x^{n-j}]g$. The rule runs backwards too: a printed series with recognisable coefficients is often an elementary function in disguise, and identifying it answers questions about f at points other than 0.

WORKED EXAMPLE

Let $f(x) = \frac{u - ux - x}{(1-x)^2}$ with u a constant, and evaluate $\frac{f^{(15)}(0)}{\Gamma(15)}$.

The normalisation by $\Gamma(15) = 14!$ is the tell. Split the fraction: the numerator is $u(1-x) - x$, so

$$f(x) = \frac{u}{1-x} - \frac{x}{(1-x)^2}.$$

The first piece contributes u to every coefficient. The second is $x \sum_{n \geq 0} (n+1)x^n = \sum_{n \geq 1} nx^n$, contributing n with a minus sign. So $[x^{15}]f = u - 15$, and

$$\frac{f^{(15)}(0)}{\Gamma(15)} = \frac{15!(u-15)}{14!} = 15u - 225.$$

At $u = \pi$ that is $15\pi - 225 = -177.87611$. Contour differentiation of f at $u = \pi$ on a circle of radius 0.5 gives $f^{(15)}(0)/14! = -177.87611$. The factorials cancelling down to a single factor of 15 is the whole design of the problem; the answer is small precisely because the setter divided the 15! back out.

Now the reverse direction. A function is specified by

$$f(x) = x + x^2 - \frac{x^3}{6} - \frac{x^4}{6} + \frac{x^5}{120} + \frac{x^6}{120} - \dots,$$

and $f'(6)$ is wanted. The coefficients come in equal pairs $(1, 1), (-\frac{1}{6}, -\frac{1}{6}), (\frac{1}{120}, \frac{1}{120})$, which are the coefficients of $\sin x$ repeated. Pairing a series with its own shift is multiplication by $1 + x$, so $f(x) = (1 + x) \sin x$. Then $f'(x) = \sin x + (1 + x) \cos x$ and

$$f'(6) = \sin 6 + 7 \cos 6.$$

No coefficient was ever extracted; the series was read as a product. This is the direction most often missed.

A rational specimen, to show the geometric family in action. For $f(x) = \frac{1}{1 - 3x + 2x^2}$, factor the denominator as $(1 - x)(1 - 2x)$ and split:

$$f(x) = \frac{2}{1 - 2x} - \frac{1}{1 - x} = \sum_{n \geq 0} (2^{n+1} - 1)x^n,$$

so $f^{(n)}(0) = n!(2^{n+1} - 1)$. At $n = 10$ that is $3628800 \cdot 2047 = 7428153600$, and contour differentiation returns 7.4281536×10^9 . Note that the split was written in the form $1/(1 - ax)$ rather than $1/(x - r)$, because that is the form whose geometric expansion is immediate; converting between the two is where signs are lost.

Finally an algebraic specimen, to show that the table's binomial row is not decoration. For $f(x) = (1 - 4x)^{-1/2}$ the general binomial series gives

$$[x^n]f = \binom{-1/2}{n}(-4)^n = \binom{2n}{n},$$

the standard evaluation of the central binomial coefficient, so

$$f^{(n)}(0) = n! \binom{2n}{n} = \frac{(2n)!}{n!}.$$

At $n = 8$ that is $40320 \cdot 12870 = 518918400$, and contour differentiation on a circle of radius 0.15 returns 518918400.0. The radius has to be small here: the nearest singularity is the branch point at $x = 1/4$, and a contour outside it computes nothing. That constraint is worth carrying into every numerical check in this chapter.

WHERE THIS GOES WRONG

Reporting the coefficient instead of $n!$ times it. This is the single most punished slip in the whole chapter, and it is invisible in the working: every line is correct until the last one. The countermeasure is to write $f^{(n)}(0) = n! a_n$ on the page before computing a_n . The second trap is expanding further than needed: the target is one coefficient, and every term beyond x^n is waste that also creates opportunities for error.

SEE ALSO Summing all derivatives at a point; Lagrange inversion and series reversion; the special-function ODE recursion.

HOW LARGE SHOULD THE ANSWER BE

The coefficients of a function analytic at 0 satisfy $|a_n| \approx C/R^n$, where R is the distance from the origin to the nearest singularity of f in the complex plane. Through the bridge that becomes a magnitude estimate for the derivative itself,

$$|f^{(n)}(0)| \approx \frac{C n!}{R^n},$$

and it is the cheapest possible check on a long computation. For $f = 1/(x^2 - 5x + 6)$ the poles are at 2 and 3, so $R = 2$ and the estimate at $n = 12$ is about $12!/2^{13} = 58471.9$, against the exact 58171.43: agreement to half a percent, which is more than enough to catch a dropped factorial or a sign. For $\ln(1 + x)/(1 - x)$ both singularities sit at distance 1, so $f^{(20)}(0)$ should be a modest multiple of $20!$, and 1.63×10^{18} against $20! = 2.43 \times 10^{18}$ is exactly that. An answer that misses this estimate by a factor of n or of $n!$ has a located error, not a mysterious one.

A twentieth derivative, in full

It is worth doing one of these completely, on a function whose twentieth derivative is genuinely hopeless by any other method, because the contrast is the argument of this chapter. Take

$$f(x) = \frac{\ln(1 + x)}{1 - x}, \quad \text{and find } f^{(20)}(0).$$

The direct route is not merely long, it is impossible in practice. The first derivative is $\frac{1}{(1+x)(1-x)} + \frac{\ln(1+x)}{(1-x)^2}$, already two terms; the second has four, the third has eight, and the logarithm never disappears, so no cancellation ever reduces the count. Nor does the closed-form route work: neither factor is in the elementary table in a way that survives the product, and Leibniz does not truncate because neither factor is a polynomial. Parity is no help either, since f is neither odd nor even.

What is available is that both factors have famous series. From the standard table,

$$\ln(1+x) = \sum_{k \geq 1} \frac{(-1)^{k-1}}{k} x^k, \quad \frac{1}{1-x} = \sum_{m \geq 0} x^m,$$

both valid for $|x| < 1$, which is a neighbourhood of 0 and therefore enough. The Cauchy product of the two says that the coefficient of x^n in the product is the sum of (coefficient of x^k in the first) $\times$ (coefficient of x^{n-k} in the second) over all k . Every coefficient of the second series is 1, so the product collapses to a partial sum:

$$[x^n]f = \sum_{k=1}^n \frac{(-1)^{k-1}}{k} \cdot 1 = 1 - \frac{1}{2} + \frac{1}{3} - \cdots + \frac{(-1)^{n-1}}{n}.$$

Multiplying a series by $1/(1-x)$ is exactly the operation “replace each coefficient by the sum of all coefficients up to it”, and recognising that saves the Cauchy product entirely.

That partial sum has a closed form. Splitting the alternating sum at $n = 2m$ into its positive and negative halves,

$$\sum_{k=1}^{2m} \frac{(-1)^{k-1}}{k} = \sum_{k=1}^{2m} \frac{1}{k} - 2 \sum_{j=1}^m \frac{1}{2j} = H_{2m} - H_m,$$

where H_m is the m -th harmonic number. At $n = 20$, so $m = 10$,

$$[x^{20}]f = H_{20} - H_{10} = \frac{155685007}{232792560} = 0.66877140.$$

The bridge finishes it:

$$f^{(20)}(0) = 20! (H_{20} - H_{10}) = \frac{2432902008176640000 \cdot 155685007}{232792560} = 1627055289796608000,$$

the division being exact because $232792560 = \text{lcm}(1, 2, \dots, 20)$ divides $20!$, with quotient 10450944000. Numerically that is 1.6270553×10^{18} , and the Cauchy integral formula

$$f^{(20)}(0) = \frac{20!}{2\pi i} \oint_{|z|=0.55} \frac{f(z)}{z^{21}} dz$$

evaluated by the trapezoid rule on 65536 points returns 1.6270553×10^{18} , agreeing to eleven digits.

WHAT THE EXAMPLE IS SHOWING

Three things, each general. First, the two factors were chosen so that neither is separately hard and the product is, which is how these problems are always built: look for the factorisation before looking for a derivative. Second, multiplication by $1/(1-x)$ is partial summation of coefficients, and multiplication by $1/(1-x)^2$ is partial summation twice; those two facts alone handle most rational multipliers. Third, the answer came out as $20!$ times a rational number, which is what a twentieth derivative at 0 of any function with rational series coefficients must look like. If your answer to such a problem is not of that shape, it is wrong.

The same three moves handle the Fibonacci generating function, which is the other standard specimen. Since $\frac{x}{1-x-x^2} = \sum_{n \geq 1} F_n x^n$ with F_n the Fibonacci numbers, the twentieth derivative at 0 is $20! F_{20} = 20! \cdot 6765$, about 1.646×10^{22} . The recursion $F_n = F_{n-1} + F_{n-2}$ is just the statement that multiplying the series by $1-x-x^2$ leaves x , read one coefficient at a time. There is no closed-form n -th derivative here at all, and there does not need to be.

One more, because it produces an answer worth recognising. For $g(x) = \frac{e^x}{1-x}$, partial summation of the exponential coefficients gives

$$[x^n]g = \sum_{k=0}^n \frac{1}{k!}, \quad g^{(n)}(0) = n! \sum_{k=0}^n \frac{1}{k!}.$$

That is an integer, and it is the nearest integer to $n!e$ for every $n \geq 1$, since the tail $n! \sum_{k > n} 1/k!$ is less than $1/2$. At $n = 10$ the value is 9864101, while $10!e = 9864101.0991$; contour differentiation returns 9864101.0. A problem whose answer is “the nearest integer to $10!e$ ” is one nobody solves by differentiating.

The same derivative three ways

Three of the four homes are open to one and the same function, and it is worth walking into all three once. Take $f(x) = x^2 e^{3x}$ and $f^{(20)}(0)$.

By the closed form. Differentiate three times and look: $f' = (2x + 3x^2)e^{3x}$, $f'' = (2 + 12x + 9x^2)e^{3x}$, $f''' = (18 + 54x + 27x^2)e^{3x}$. The pattern is $(\alpha_n + \beta_n x + 3^n x^2)e^{3x}$ with $\beta_n = 2n \cdot 3^{n-1}$ and $\alpha_n = n(n-1)3^{n-2}$, provable by a one-line induction. At $x = 0$ only α_n survives.

By Leibniz. Three nonzero terms, as in the block below, and only the $k = 2$ term survives at $x = 0$: $2\binom{n}{2}3^{n-2}$.

By the coefficient. $e^{3x} = \sum_{k \geq 0} 3^k x^k / k!$, so $f = x^2 e^{3x} = \sum_{k \geq 0} 3^k x^{k+2} / k!$ and $[x^{20}]f = 3^{18} / 18!$. The bridge gives $f^{(20)}(0) = 20! \cdot 3^{18} / 18! = 20 \cdot 19 \cdot 3^{18}$.

All three give 147219785820, and contour differentiation agrees. The comparison is instructive because the three differ in what they generalise to. The closed form is the only one that answers a question about $f^{(20)}(1)$ rather than $f^{(20)}(0)$. Leibniz is the only one that works when the second factor is unknown and only combinations of its derivatives are given. Coefficient extraction is the only one that survives when the second factor is replaced by something with no closed-form high derivative, such as $\ln(1+x)$. Choose by asking which of those three futures the problem is heading towards.

Products, composites, and the sums that truncate

Coefficient extraction assumes you can get the series. When you cannot, but the function is a product with one simple factor, the binomial theorem for derivatives does the job directly, and it does it in finitely many terms.

LEIBNIZ'S RULE AND THE CAUCHY PRODUCT ARE ONE STATEMENT

Take the Cauchy product $[x^n](uv) = \sum_{j=0}^n [x^j]u \cdot [x^{n-j}]v$ and translate every coefficient into a derivative by the bridge, $[x^m]w = w^{(m)}(0)/m!$. Multiplying through by $n!$ turns the coefficient $n!/(j!(n-j)!)$ into $\binom{n}{j}$ and leaves

$$(uv)^{(n)}(0) = \sum_{j=0}^n \binom{n}{j} u^{(j)}(0) v^{(n-j)}(0),$$

which is Leibniz's rule. So the Leibniz expansion and the Maclaurin coefficient are not competitors: they are the same identity written in two systems of units, derivatives on one side and coefficients on the other. Use whichever side the data arrive in. If you are handed derivative values, use Leibniz; if you are handed series, use the product.

MEDIUM

14.5 Leibniz's rule for $(fg)^{(n)}$

CUE A product differentiated to high order, especially a polynomial times an exponential or a sinusoid; or a product where the data given are certain *combinations* of one factor's derivatives rather than the factor itself.

METHOD Use

$$(uv)^{(n)} = \sum_{k=0}^n \binom{n}{k} u^{(k)} v^{(n-k)}.$$

When u is a polynomial of degree d , every term with $k > d$ vanishes and the sum has $d+1$ terms regardless of n . When $v = e^x$, all $v^{(j)}$ are equal and the sum becomes $e^x \sum_k \binom{n}{k} u^{(k)}$. Choose which factor plays u by asking which one runs out of derivatives first.

WORKED EXAMPLE

$f(x) = x^2 e^{3x}$, and $f^{(20)}(0)$.

Take $u = x^2$ and $v = e^{3x}$. Then $u = x^2$, $u' = 2x$, $u'' = 2$, and $u^{(k)} = 0$ for $k \geq 3$, so the sum has three terms:

$$f^{(n)}(x) = \binom{n}{0} x^2 \cdot 3^n e^{3x} + \binom{n}{1} 2x \cdot 3^{n-1} e^{3x} + \binom{n}{2} 2 \cdot 3^{n-2} e^{3x}.$$

At $x = 0$ the first two terms die and only the third survives:

$$f^{(n)}(0) = 2 \binom{n}{2} 3^{n-2} = n(n-1)3^{n-2}, \quad f^{(20)}(0) = 20 \cdot 19 \cdot 3^{18} = 147219785820.$$

Contour differentiation returns 1.4721979×10^{11} . Note that the evaluation point did half the work: the general $f^{(n)}(x)$ has three terms, and $f^{(n)}(0)$ has one.

Now the version where Leibniz is the only route, because the factor g is never given. Let $f(x) = e^x g(x)$, where all that is known about g is

$$g(0) + g'''(0) = 5, \quad g'(0) + g''(0) = 2.$$

Find $f'''(0)$. With $v = e^x$ every $v^{(j)}(0) = 1$, so

$$f'''(0) = \sum_{k=0}^3 \binom{3}{k} g^{(3-k)}(0) = g'''(0) + 3g''(0) + 3g'(0) + g(0).$$

Regrouping into the combinations supplied, this is $(g(0) + g'''(0)) + 3(g'(0) + g''(0)) = 5 + 3 \cdot 2 = 11$. The binomial coefficients 1, 3, 3, 1 are symmetric, which is exactly why the two given combinations were enough; the problem was designed around that symmetry.

A third instance, at an order chosen to be absurd. Let $f(x) = x^3 \sin x$ and find $f^{(50)}(0)$. Take $u = x^3$, so $u(0) = u'(0) = u''(0) = 0$ and $u'''(0) = 6$: of the four surviving Leibniz terms, three are killed by the evaluation point and one remains,

$$f^{(50)}(0) = \binom{50}{3} \cdot 6 \cdot \sin^{(47)}(0) = 19600 \cdot 6 \cdot \sin\left(\frac{47\pi}{2}\right) = -117600,$$

using $47 \equiv 3 \pmod{4}$ so that $\sin(47\pi/2) = \sin(3\pi/2) = -1$. Two homes were used at once: Leibniz truncated the sum and the cyclic orbit evaluated the surviving factor. The fourth home confirms it independently, since $x^3 \sin x = \sum_{m \geq 0} (-1)^m x^{2m+4} / (2m+1)!$ gives $[x^{50}]f = -1/47!$ and hence $f^{(50)}(0) = -50!/47! = -50 \cdot 49 \cdot 48 = -117600$. The same computation at $n = 12$ predicts $\binom{12}{3} \cdot 6 \cdot \sin(9\pi/2) = 1320$, and contour differentiation returns 1320.0000.

WHERE THIS GOES WRONG

Reversing which factor carries k and which carries $n - k$. With symmetric binomial coefficients this is harmless, and solvers therefore never learn to be careful, which is fatal the first time the two factors differ in an unsymmetric way. The second trap is forgetting the truncation and writing out all $n + 1$ terms of a sum in which only three are nonzero.

SEE ALSO The generalised product rule for n factors; Faa di Bruno for composites; Maclaurin coefficient extraction.

Leibniz handles products. It says nothing about composites, and for good reason: there is no binomial-style formula for $(g \circ f)^{(n)}$, because the chain rule mixes orders rather than distributing them. The correct generalisation exists, is more complicated, and is mostly useful as a structural statement rather than as a computational recipe.

14.6 Faa di Bruno's formula for composites

HARD

CUE A high derivative of a composite $g(f(x))$ where Leibniz does not apply: $e^{\sin x}$, $\ln(1 + \sin x)$, $\cos(x^2)$, to fourth or fifth order.

METHOD The formula is

$$\frac{d^n}{dx^n} g(f(x)) = \sum_{k=1}^n g^{(k)}(f(x)) B_{n,k}(f'(x), f''(x), \dots, f^{(n-k+1)}(x)),$$

where $B_{n,k}$ are the partial Bell polynomials. For $n \leq 4$ it is faster to differentiate directly and collect, or to compose the two Maclaurin series; the formula earns its keep for structural arguments, for deciding which terms can survive at a given point, and for organising a computation that would otherwise be an unlabelled pile of terms. Written out, the first four orders are

$$\begin{aligned} (g \circ f)' &= g'f', & (g \circ f)'' &= g''(f')^2 + g'f'', \\ (g \circ f)''' &= g'''(f')^3 + 3g''f'f'' + g'f''', \\ (g \circ f)'''' &= g''''(f')^4 + 6g'''(f')^2f'' + g''(3(f'')^2 + 4f'f''') + g'f'''' \end{aligned}$$

with every $g^{(k)}$ evaluated at $f(x)$. The coefficients 1; 1, 1; 1, 3, 1; 1, 6, 3+4, 1 are counts of set partitions, not binomial coefficients, and that is the structural difference from Leibniz: a product distributes derivatives between two slots, while a composite distributes them among an unknown number of copies of f .

WORKED EXAMPLE

$f(x) = e^{\sin x}$, and the first four derivatives at 0.

Series composition is the practical route. With $u = \sin x = x - \frac{x^3}{6} + O(x^5)$,

$$e^u = 1 + u + \frac{u^2}{2} + \frac{u^3}{6} + \frac{u^4}{24} + O(u^5),$$

and to degree 4 we need $u = x - \frac{x^3}{6}$, $u^2 = x^2 - \frac{x^4}{3}$, $u^3 = x^3$, $u^4 = x^4$. Collecting,

$$e^{\sin x} = 1 + x + \frac{x^2}{2} + \left(-\frac{1}{6} + \frac{1}{6}\right)x^3 + \left(-\frac{1}{6} + \frac{1}{24}\right)x^4 + O(x^5) = 1 + x + \frac{x^2}{2} + 0 \cdot x^3 - \frac{x^4}{8} + O(x^5).$$

The bridge gives $f'(0) = 1$, $f''(0) = 1$, $f'''(0) = 0$, and $f^{(4)}(0) = 24 \cdot (-\frac{1}{8}) = -3$. Contour differentiation returns -3.0000000 . The vanishing third derivative is the interesting entry, and the formula above says exactly why. At $x = 0$ the inner derivatives are $f' = 1$, $f'' = 0$, $f''' = -1$, $f^{(4)} = 0$, and every $g^{(k)}$ equals $e^{\sin 0} = 1$. The third-order line therefore reads $1 \cdot 1 + 3 \cdot 1 \cdot 1 \cdot 0 + 1 \cdot (-1) = 0$: the $k = 3$ term and the $k = 1$ term cancel exactly, and the middle term was already dead because $\sin''(0) = 0$. The fourth-order line reads $1 + 6 \cdot 1 \cdot 0 + (3 \cdot 0 + 4 \cdot 1 \cdot (-1)) + 0 = -3$, matching the series computation. Away from the origin nothing cancels: at $x = 0.4$ the same four-term line gives -5.6262947 , which contour differentiation confirms.

Two more of the same kind. For $\ln(1 + \sin x)$, the same substitution into $\ln(1 + u) = u - \frac{u^2}{2} + \frac{u^3}{3} - \frac{u^4}{4}$ gives $x - \frac{x^2}{2} + \frac{x^3}{6} - \frac{x^4}{12}$, so $f^{(4)}(0) = 24 \cdot (-\frac{1}{12}) = -2$, confirmed numerically as -2.0000000 . For $\cos(x^2) = 1 - \frac{x^4}{2} + O(x^8)$, only every fourth derivative is nonzero at 0, and $f^{(4)}(0) = 24 \cdot (-\frac{1}{2}) = -12$, confirmed as -12.0000000 .

WHERE THIS GOES WRONG

Writing $(g \circ f)^{(n)} = g^{(n)}(f) \cdot (f')^n$. That is only the $k = n$ term of the formula, and every lower k contributes. The error is seductive because it is correct at $n = 1$ and gives a plausible-looking answer at every higher n . Check against $n = 2$, where the truth is $g''(f)(f')^2 + g'(f)f''$ and the missing second term is usually the larger one.

SEE ALSO Outside-in chain-rule discipline; Maclaurin coefficient extraction; Leibniz's rule for a product.

Sums of derivatives, defining equations, and non-integer orders

Three further shapes complete the chapter. The first asks not for one derivative but for all of them added together, which sounds worse and is usually easier.

MEDIUM

14.7 Summing all derivatives at a point

CUE A sum $\sum_{n \geq 0} f^{(n)}(a)$, or a sum of derivatives of all orders evaluated at one point.

METHOD Two outcomes, and deciding which one applies is the whole technique. If f is a polynomial of degree d , the sum truncates at $n = d$ and is a finite computation. If f is self-similar under differentiation, meaning $f^{(n)} = c^n f$ or a scaled phase shift of f , the sum is geometric. For $f = e^{ax}$ with $|a| < 1$, $\sum_{n \geq 0} f^{(n)}(0) = \sum_{n \geq 0} a^n = \frac{1}{1-a}$. For a sinusoid, write the phase-shift form and sum a complex geometric series using $\cos(\phi + \frac{n\pi}{2}) = \operatorname{Re}(e^{i\phi} i^n)$.

WORKED EXAMPLE

$$f(x) = \sqrt{1 + \cos x}, \text{ and } \sum_{n \geq 0} f^{(n)}(\pi/2).$$

The radical is the obstruction, and the half-angle identity removes it: $1 + \cos x = 2 \cos^2(x/2)$, so $f(x) = \sqrt{2} |\cos(x/2)|$, and near $x = \pi/2$ the cosine is positive, so $f(x) = \sqrt{2} \cos(x/2)$ with no absolute value. Numerically $\sqrt{1 + \cos 1.3} = 1.1258325$ and $\sqrt{2} \cos(0.65) = 1.1258325$.

Now the table entry applies with $a = \frac{1}{2}$:

$$f^{(n)}(x) = \sqrt{2} 2^{-n} \cos\left(\frac{x}{2} + \frac{n\pi}{2}\right), \quad f^{(n)}(\pi/2) = \sqrt{2} 2^{-n} \cos\left(\frac{\pi}{4} + \frac{n\pi}{2}\right).$$

Contour differentiation at $n = 0, \dots, 6$ returns $1, -0.5, -0.25, 0.125, 0.0625, -0.03125, -0.015625$, matching the formula exactly. Summing, with $\cos(\frac{\pi}{4} + \frac{n\pi}{2}) = \operatorname{Re}(e^{i\pi/4} i^n)$,

$$\sum_{n \geq 0} f^{(n)}(\pi/2) = \sqrt{2} \operatorname{Re} \sum_{n \geq 0} e^{i\pi/4} \left(\frac{i}{2}\right)^n = \sqrt{2} \operatorname{Re} \frac{e^{i\pi/4}}{1 - i/2} = \operatorname{Re} \frac{1 + i}{1 - i/2} = \operatorname{Re}(0.4 + 1.2i) = \frac{2}{5}.$$

The ratio $i/2$ has modulus $\frac{1}{2} < 1$, so the series converges. Summing the closed-form terms numerically to $n = 300$ gives 0.40000000.

The exponential case in one line: for $f(x) = e^{x/3}$, $f^{(n)}(0) = 3^{-n}$ and $\sum_{n \geq 0} 3^{-n} = \frac{3}{2}$.

A damped oscillation, to show the complex geometric sum in its natural setting. For $f(x) = e^{-x/2} \cos(x/2)$ write $f = \operatorname{Re} e^{cx}$ with $c = \frac{-1+i}{2}$, so $f^{(n)}(0) = \operatorname{Re}(c^n)$ and $|c| = \frac{1}{\sqrt{2}} < 1$, which settles convergence before any summing. Then

$$\sum_{n \geq 0} f^{(n)}(0) = \operatorname{Re} \frac{1}{1-c} = \operatorname{Re} \frac{1}{\frac{3}{2} - \frac{i}{2}} = \operatorname{Re} \frac{2}{3-i} = \operatorname{Re} \frac{3+i}{5} = \frac{3}{5}.$$

Summing the terms numerically to $n = 400$ gives 0.600000000, and the individual check $f'''(0) = \operatorname{Re}(c^3) = \frac{1}{4}$ is confirmed by contour differentiation. Note that the damping is what makes the sum converge: for the undamped $\cos x$ the derivatives at 0 run $1, 0, -1, 0, 1, \dots$ and the series has no sum at all.

The polynomial case is the one people overlook, because it looks too easy to be the intended reading. For $f(x) = x^4 - 2x^3 + 7$, every derivative past the fourth is identically zero, so the sum has five terms. At $a = 1$: $f(1) = 6$, $f'(1) = 4 - 6 = -2$, $f''(1) = 12 - 12 = 0$, $f'''(1) = 24 - 12 = 12$, $f^{(4)}(1) = 24$, and the total is 40. No convergence question arises, because the series was finite from the start.

WHERE THIS GOES WRONG

Assuming that $\sum_{n \geq 0} f^{(n)}(a)$ equals $f(a+1)$ by Taylor's theorem. It does not, in general: Taylor's series at a with $x-a = 1$ has terms $f^{(n)}(a)/n!$, not $f^{(n)}(a)$, so the two sums agree only in the special circumstances where the missing factorials cancel against the growth of the derivatives. The other check to run before summing is convergence: for e^{ax} the sum is $1/(1-a)$ only when $|a| < 1$, and at $a = 1$ the sum of $f^{(n)}(0) = 1$ diverges outright.

SEE ALSO n -th derivative closed forms; Maclaurin coefficient extraction; sum the hidden series first.

The next shape is a function that is not given by a formula at all but by a defining integral or integral representation. Differentiating such a thing repeatedly by brute force is out of the question. Converting it to a differential equation is not.

14.8 Special functions via the defining ODE recursion

HARD

CUE A function defined by an integral or an integral representation, with a specific $f^{(n)}(0)$ wanted: Dawson's function $F(x) = e^{-x^2} \int_0^x e^{t^2} dt$, or Bessel's $J_0(x) = \frac{1}{\pi} \int_0^\pi \cos(x \sin \theta) d\theta$.

METHOD Differentiate the definition once, using the product rule and the fundamental theorem of calculus, to obtain a differential equation with no integral in it. Then feed a power series into the equation and match coefficients, which produces a recursion; parity halves the work. When the representation is an integral over a parameter, differentiating under the integral sign and evaluating a Wallis integral is often faster still.

WORKED EXAMPLE

Dawson's function $F(x) = e^{-x^2} \int_0^x e^{t^2} dt$, and $F^{(5)}(0)$.

Differentiate the product. The exponential gives $-2xe^{-x^2} \int_0^x e^{t^2} dt = -2xF(x)$, and the fundamental theorem gives $e^{-x^2} \cdot e^{x^2} = 1$. So

$$F' = 1 - 2xF, \quad F(0) = 0,$$

and the integral has disappeared. Now substitute $F = \sum_{k \geq 0} a_k x^k$. Matching coefficients of x^m gives $(m+1)a_{m+1} = [m=0] - 2a_{m-1}$, that is $a_1 = 1$ and $a_{m+1} = -\frac{2a_{m-1}}{m+1}$ for $m \geq 1$. Since F is odd (the integrand is even, so the integral is odd, and e^{-x^2} is even), all even coefficients vanish and only the odd chain matters:

$$a_1 = 1, \quad a_3 = -\frac{2}{3}, \quad a_5 = -\frac{2a_3}{5} = \frac{4}{15}, \quad a_7 = -\frac{2a_5}{7} = -\frac{8}{105}.$$

Hence $F(x) = x - \frac{2}{3}x^3 + \frac{4}{15}x^5 - \frac{8}{105}x^7 + \dots$, and the bridge gives

$$F^{(5)}(0) = 5! \cdot \frac{4}{15} = 32.$$

Numerical quadrature of the defining integral at $x = 0.3$ gives $F(0.3) = 0.2826317$ against the truncated series value 0.2826313, the discrepancy being the x^9 term.

Bessel's J_0 takes the other route. Differentiating under the integral sign four times brings down $\sin^4 \theta$ with sign $(+1)$:

$$J_0^{(4)}(0) = \frac{1}{\pi} \int_0^\pi \sin^4 \theta \, d\theta = \frac{1}{\pi} \cdot \frac{3\pi}{8} = \frac{3}{8},$$

using the Wallis value $\int_0^\pi \sin^4 = \frac{3\pi}{8}$. The check is independent: the series $J_0(x) = \sum_{k \geq 0} (-1)^k (x/2)^{2k} / (k!)^2$ has $[x^4] = \frac{1}{64}$, so the bridge gives $4!/64 = \frac{3}{8}$ as well, and contour differentiation of the integral representation on $|z| = 1$ returns 0.3750000.

WHERE THIS GOES WRONG

Trying to differentiate the integral representation five times by brute force, which reintroduces the integral at every step. Differentiate once, convert to an equation, and never look at the integral again. The second waste is computing the even coefficients of an odd function: the parity observation should come before the recursion, not after.

SEE ALSO Differentiating under the integral sign; Maclaurin coefficient extraction; the logarithmic derivative as an operator.

The last shape is a curiosity that is nonetheless completely standard once the right basis is chosen. If differentiation is multiplication by c on the exponential e^{cx} , there is no obstacle to multiplying by $c^{1/2}$ instead, and the whole theory of fractional derivatives follows from taking that remark seriously.

HARD

14.9 Fractional derivatives via the exponential basis

CUE A differentiation order that is not an integer: $\frac{d^{1/2}}{dx^{1/2}}$, or an order named as a parameter α .

METHOD Work on the basis where differentiation is diagonal. On exponentials, $\frac{d^\alpha}{dx^\alpha} e^{cx} = c^\alpha e^{cx}$, with the Liouville convention whose lower limit is $-\infty$; decompose the function into exponentials using $\sin x = \frac{e^{ix} - e^{-ix}}{2i}$ and $a^x = e^{x \ln a}$. On monomials the corresponding rule is

$$\frac{d^\alpha}{dx^\alpha} x^p = \frac{\Gamma(p+1)}{\Gamma(p-\alpha+1)} x^{p-\alpha},$$

which reproduces the integer rule because $\Gamma(m+1) = m!$.

WORKED EXAMPLE

$f(x) = \pi^x \sin x$, and $\frac{d^{1/2}}{dx^{1/2}} f$ evaluated at $x = 0$.

Write both factors as exponentials. With $L = \ln \pi$,

$$f(x) = e^{Lx} \cdot \frac{e^{ix} - e^{-ix}}{2i} = \frac{e^{(L+i)x} - e^{(L-i)x}}{2i},$$

a difference of two pure exponentials, each with positive real part in the exponent, so the Liouville integral converges. Half-differentiating multiplies each by the square root of its own constant, and at $x = 0$ both exponentials equal 1:

$$\left. \frac{d^{1/2}}{dx^{1/2}} f \right|_0 = \frac{(L+i)^{1/2} - (L-i)^{1/2}}{2i} = \operatorname{Im}[(L+i)^{1/2}],$$

the last step because the two constants are complex conjugates and $(\bar{z})^{1/2} = \overline{z^{1/2}}$ on the principal branch. With $L = 1.1447299$, the modulus of $L+i$ is 1.520002 and its argument is 0.718019, so the principal square root has modulus 1.232884 and argument 0.359009, giving

$$\left. \frac{d^{1/2}}{dx^{1/2}} f \right|_0 = 1.232884 \sin(0.359009) = 0.4331699.$$

Evaluating the Liouville integral $\frac{1}{\Gamma(1/2)} \frac{d}{dx} \int_0^\infty s^{-1/2} f(x-s) \, ds$ numerically at $x = 0$ returns 0.4331699.

The monomial rule deserves one check of its own, because it is the fastest way to see that the definition is the right one. Half-differentiating x gives $\frac{\Gamma(2)}{\Gamma(3/2)} x^{1/2} = \frac{2}{\sqrt{\pi}} x^{1/2}$, and half-differentiating that again gives $\frac{2}{\sqrt{\pi}} \cdot \frac{\Gamma(3/2)}{\Gamma(1)} x^0 = \frac{2}{\sqrt{\pi}} \cdot \frac{\sqrt{\pi}}{2} = 1$, which is $\frac{d}{dx} x$. Two half-derivatives compose to one whole derivative, exactly as the notation promises.

Similarly $\frac{d^{1/2}}{dx^{1/2}} x^{3/2} = \frac{\Gamma(5/2)}{\Gamma(2)} x = \frac{3\sqrt{\pi}}{4} x = 1.3293404 x$.

Choosing the wrong branch of the complex square root, which flips the sign of the answer. Stay on the principal branch, where the argument lies in $(-\pi, \pi]$, and check that conjugate inputs produce conjugate outputs. The other trap is mixing conventions: the Liouville form used here has lower limit $-\infty$ and makes exponentials eigenfunctions, while the Riemann-Liouville form with lower limit 0 and the Caputo form do not, and the three disagree by terms that depend on the initial values. State which one you are using.

SEE ALSO Complex exponentials for high derivatives; n -th derivative closed forms; Maclaurin coefficient extraction.

A fiftieth derivative, in full

One closing problem, chosen because it needs three of the four homes in sequence and is hopeless without them. Find $f^{(50)}(0)$ for

$$f(x) = \frac{\sin(x^2)}{1-x^2}.$$

First move: the variable appears only as x^2 . Substitute $u = x^2$, so $f(x) = g(u)$ with $g(u) = \frac{\sin u}{1-u}$. Every odd derivative at the origin vanishes, and $f^{(50)}(0) = 50! [u^{25}]g$. A fiftieth derivative has become a twenty-fifth coefficient, and half the possible answers have been deleted without a computation. The check runs the other way too: $f^{(49)}(0) = 0$, and contour differentiation returns -5×10^{52} against a scale of 10^{64} , which is zero to the precision of the quadrature.

Second move: the multiplier $1/(1-u)$ is partial summation. The coefficients of g are the partial sums of the coefficients of $\sin u$, so

$$[u^{25}]g = \sum_{\substack{j \leq 25 \\ j \text{ odd}}} [u^j] \sin u = \sum_{m=0}^{12} \frac{(-1)^m}{(2m+1)!} = 1 - \frac{1}{3!} + \frac{1}{5!} - \cdots + \frac{1}{25!}.$$

Third move: recognise the partial sum. That is the Maclaurin series for $\sin 1$, truncated after thirteen terms. The first omitted term is $1/27! \approx 9.2 \times 10^{-29}$, so the partial sum agrees with $\sin 1 = 0.8414709848078965$ in every digit a calculator will print. Exactly,

$$[u^{25}]g = \frac{13052233190723859821607001}{25!},$$

and therefore

$$f^{(50)}(0) = \frac{50!}{25!} \cdot \frac{13052233190723859821607001}{25!} = 2.5592577 \times 10^{64},$$

an integer because $25!$ divides $50!$. Contour differentiation on a circle of radius 0.75 returns 2.5592577×10^{64} , and the magnitude estimate agrees as well: the nearest singularities of f are the poles at $x = \pm 1$, so $R = 1$ and the answer should be a modest fraction of $50! = 3.04 \times 10^{64}$, which it is.

What was never done is any differentiation at all. The three moves were a substitution, a partial summation, and a recognition, and each of them is one line. That is what the whole chapter has been arguing: a request for a fiftieth derivative is a request to locate the pattern, and the pattern is always in one of four places.

Recognition matrix

If you see	Reach for	If that fails
$f^{(n)}$ wanted symbolically in n	Three orders, conjecture, induction	Reduce f by an identity first
$f^{(20)}(0), f^{(100)}(0), f^{(2026)}(0)$	$f^{(n)}(0) = n! [x^n]f$	Parity first: the answer may be 0
An answer divided by $n!$ or $\Gamma(n)$	Coefficient extraction; the factorials cancel	Read the normalisation as the hint
$\sin^2 x, \cos^3 x, \sin x \cos x$	Linearise, then $a^n \sin(ax + b + n\pi/2)$	Complex exponentials
A rational function, high order	Partial fractions, then $(-1)^n n! a^n (ax + b)^{-n-1}$	Geometric series coefficients
$\ln(ax + b)$ to order n	$(-1)^{n-1} (n-1)! a^n (ax + b)^{-n}$	Differentiate once, then the previous row
$e^{ax} \sin bx$ or $e^{ax} \cos bx$	Im or Re of $(a + ib)^n$	Repeated squaring for a specific n
Odd times even, evaluated at 0	Parity: half the derivatives vanish	Shift to the centre of symmetry first
Polynomial times anything	Leibniz; the sum stops at deg	Series product instead
Only combinations of $g^{(k)}(0)$ given	Leibniz, then regroup into those combinations	Check the binomial symmetry
$g(f(x))$ to order 4 or 5	Compose the two Maclaurin series	Faa di Bruno with Bell polynomials
$\sum_{n \geq 0} f^{(n)}(a)$	Polynomial: truncates. Self-similar: geometric	Phase-shift form plus Re

If you see	Reach for	If that fails
f defined by an integral	Differentiate once into an ODE, then recurse	Differentiate under the sign
An order that is not an integer	$\frac{d^\alpha}{dx^\alpha} e^{cx} = c^\alpha e^{cx}$	$\Gamma(p+1)/\Gamma(p-\alpha+1)$ on monomials
$\ln(1+x)$ or $\arctan x$ as a factor	Cauchy product with the other factor's series	Times $1/(1-x)$ is partial summation
A generating function with a recursion	Read $[x^n]$ off the recursion, then times $n!$	Partial fractions if the denominator factors
An answer that is not $n!$ times a rational	Suspect an arithmetic slip	Recheck the factorial at the last step
$f(x) = g(x^k)$, derivative at 0	Zero unless $k \mid n$; else $n! [x^{n/k}]g$	Substitute $t = x^k$ first
A repeated factor in the denominator	$(-1)^n \frac{(m+n-1)!}{(m-1)!} (x-c)^{-m-n}$	The worst pole sets the growth
$e^{ax} \sin bx$ at general x	$r^n e^{ax} \sin(bx + n\theta), r = \sqrt{a^2 + b^2}$	Set $x = 0$ and take $\text{Im}(a + ib)^n$
An answer of the wrong order of magnitude	Compare with $n!/R^n$, R to the nearest pole	Recount the poles of f
$(1-4x)^{-1/2}$ or a surd factor	Binomial series: $[x^n] = \binom{2n}{n}$	$\binom{\alpha}{n}$ in general

Differentiating integrals, and derivatives from the definition

What this chapter is for. Every rule you have met so far assumes that the function in front of you is an explicit formula, assembled from elementary pieces by arithmetic and composition. Two large families of problems supply no such formula, and in both of them the rules are not merely inconvenient but inapplicable. In the first family the function is defined by an integral: the variable sits in a limit, or inside the integrand, or in both places at once, and the antiderivative is typically unavailable and never needed. In the second family the function is assembled from pieces, or has a corner, or is assigned a value by fiat at the one point where its formula degenerates; there the difference quotient is the only instrument that answers the question. This chapter builds the full Leibniz integral rule from one multivariable chain rule so that its three terms stop being three things to memorise, then returns to the definition of the derivative and works through matching, one-sided limits, kinks, cusps, and the standard counterexamples that separate “ $f'(a)$ exists” from “ f is continuous at a ”. At the end you should be able to differentiate an integral-defined function on sight, and to recognise instantly when a problem is asking you to go back to first principles.

Functions defined by an integral

Let f be continuous and set $F(x) = \int_a^x f(t) dt$. The Fundamental Theorem says $F'(x) = f(x)$. Read that statement carefully, because its content is the engine of this entire section: integration of a continuous function produces a differentiable function, and differentiating undoes it *pointwise*, with no reference whatever to a closed form for F . The function $\int_0^x e^{-t^2} dt$ has no elementary antiderivative, and its derivative is e^{-x^2} , known exactly and forever. That asymmetry between what is computable about F and what is computable about F' is precisely why integral-defined functions get set as derivative problems.

Everything in this section is that one theorem plus a chain rule. The order of presentation is deliberate: a variable upper limit, then two variable limits, then a variable inside the integrand as well. Each stage adds exactly one term, and the third stage is the general rule of which the first two are special cases. Learn the general rule and read the specialisations off it. Memorising three separate formulas is how the third term goes missing.

CORE

15.1 The Fundamental Theorem with a variable upper limit

CUE $F(x) = \int_a^{u(x)} f(t) dt$ with a constant lower limit and a non-trivial upper limit; typically an integrand with no elementary antiderivative, which is the setter’s way of telling you not to integrate.

METHOD Compose the Fundamental Theorem with the chain rule. Write G for any antiderivative of f , so that $F(x) = G(u(x)) - G(a)$; then $F'(x) = G'(u(x))u'(x)$, that is,

$$\frac{d}{dx} \int_a^{u(x)} f(t) dt = f(u(x)) u'(x).$$

The symbol G is scaffolding: it appears in the justification and never in the calculation. If instead the *lower* limit carries the variable, reverse the orientation first, $\int_{u(x)}^a f = -\int_a^{u(x)} f$, so the derivative is $-f(u(x))u'(x)$.

WORKED EXAMPLE

Take $F(x) = \int_0^{x^3} e^{-t^2} dt$, with $u(x) = x^3$ and $f(t) = e^{-t^2}$:

$$F'(x) = e^{-(x^3)^2} \cdot 3x^2 = 3x^2 e^{-x^6}.$$

At $x = 1.2$ the formula gives 0.21811592, against a central difference of the numerically integrated F of 0.21811592. No attempt to evaluate F was made, and none would have succeeded in closed form.

Now the variable in the lower limit. Let $H(x) = \int_{\sin x}^2 \ln(1+t^2) dt$. Flip the orientation first:

$$H(x) = -\int_2^{\sin x} \ln(1+t^2) dt, \quad H'(x) = -\ln(1+\sin^2 x) \cos x.$$

At $x = \pi/4$ this is $-\ln \frac{3}{2} \cdot \frac{\sqrt{2}}{2} = -0.2867071$, confirmed by central difference to seven digits.

Finally, a question that only looks like an integration problem. Where does $F(x) = \int_0^x (t^2 - 3t + 2)e^t dt$ have local extrema? Do not integrate. The Fundamental Theorem gives $F'(x) = (x^2 - 3x + 2)e^x = (x-1)(x-2)e^x$, and since $e^x > 0$ the sign of F' is the sign of $(x-1)(x-2)$: positive, negative, positive. So F has a local maximum at $x = 1$ and a local minimum at $x = 2$, obtained without ever writing down F .

WHERE THIS GOES WRONG

Omitting $u'(x)$, which is the commonest error in the whole chapter, and mis-signing when the variable is downstairs: $\frac{d}{dx} \int_{u(x)}^a f = -f(u(x))u'(x)$, with the minus coming from the orientation flip and not from anything about f . A third, subtler failure is trying to evaluate F first. If the integrand has no elementary antiderivative, that attempt cannot terminate, and it was never required.

SEE ALSO Both limits moving; the full Leibniz integral rule; outside-in chain-rule discipline.

Nothing changes structurally when the lower limit starts moving too. Split the integral at any convenient constant c inside the interval, apply the previous technique to each half, and read off the answer. The reason the result is worth its own block is not the derivation but the design of the problems: setters choose the two limits so that the two contributions combine through an identity, and a solver who evaluates each piece numerically instead of looking for that identity misses the point of the question.

EASY

15.2 Both limits moving

CUE Both limits carry x , very often reciprocally or as a power: $\int_{1/x}^x, \int_x^{x^2}, \int_{-x}^x$.

METHOD Split at a constant c between them, $\int_{a(x)}^{b(x)} = \int_c^{b(x)} - \int_c^{a(x)}$, and differentiate each piece:

$$\frac{d}{dx} \int_{a(x)}^{b(x)} f(t) dt = f(b(x))b'(x) - f(a(x))a'(x).$$

The upper limit enters with a plus and the lower limit with a minus, always. When the limits are reciprocal, expect a complementary identity such as $\arctan x + \arctan(1/x) = \pi/2$ for $x > 0$ to collapse the two terms into one; look for it before reaching for a calculator.

WORKED EXAMPLE

Let $f(x) = \int_{1/x}^x \frac{\arctan t}{t} dt$ for $x > 0$, and find $f'(3)$. Here $b(x) = x, b' = 1, a(x) = 1/x, a' = -1/x^2$, so

$$f'(x) = \frac{\arctan x}{x} - \frac{\arctan(1/x)}{1/x} \cdot \left(-\frac{1}{x^2}\right) = \frac{\arctan x}{x} + \frac{\arctan(1/x)}{x} = \frac{1}{x} \cdot \frac{\pi}{2}.$$

So $f'(x) = \frac{\pi}{2x}$ for every $x > 0$, and $f'(3) = \frac{\pi}{6} = 0.5235988$, matching a central difference of the numerically integrated f to seven digits. Note the three separate cancellations that made this clean: the $1/t$ in the integrand became x when evaluated at $t = 1/x$, that x cancelled against the $1/x^2$ from a' , and the two arctangents summed to $\pi/2$. All three were engineered.

A second, where no identity is available and none is needed. With $K(x) = \int_x^{x^2} \ln t dt$ for $x > 1$,

$$K'(x) = \ln(x^2) \cdot 2x - \ln x \cdot 1 = 4x \ln x - \ln x = (4x - 1) \ln x,$$

so $K'(e) = 4e - 1 = 9.873127$, verified to seven digits.

A third, symmetric case, which is the one worth recognising on sight. For $G(x) = \int_{-x}^x f(t) dt$ the rule gives

$$G'(x) = f(x) \cdot 1 - f(-x) \cdot (-1) = f(x) + f(-x),$$

so an even integrand doubles and an odd integrand cancels to zero, recovering the standard symmetry facts as a one-line corollary. With $f(t) = t^2 e^t$, which is neither even nor odd, $G'(x) = x^2(e^x + e^{-x}) = 2x^2 \cosh x$, and $G'(1) = 2 \cosh 1 = 3.0861613$, matching a central difference to seven digits.

The lower-limit sign. In the first example the rule's minus and the negative $a'(x) = -1/x^2$ multiply to a plus, and the visible plus sign in the middle expression is exactly what makes a nervous solver “correct” it back to a minus. Write b' and a' down explicitly, with their signs, before substituting. The other error is splitting at a constant outside the interval of integration, which is legitimate for the algebra but can put you outside the domain of f .

SEE ALSO The Fundamental Theorem with a variable upper limit; the full Leibniz integral rule; inverse-trigonometric complementary identities.

So far the variable has appeared only in the limits. The general case lets it appear in the integrand as well, and the resulting formula has three terms rather than two. It is not worth memorising as a formula. It is worth deriving once, because the derivation makes the three terms inevitable rather than arbitrary, and because seeing where each term comes from is what stops you dropping one.

Introduce a function of three independent slots,

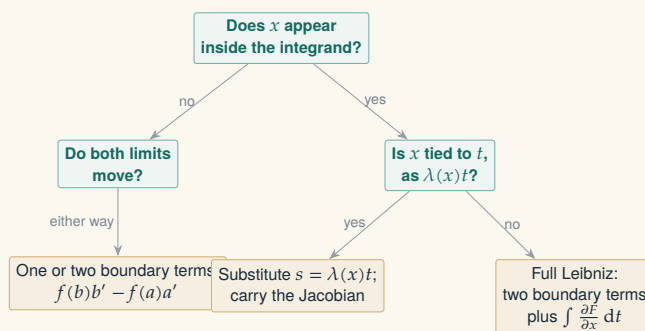
$$\Phi(x, y, z) = \int_y^z F(x, t) dt,$$

and observe that the object we want is $\Phi(x, a(x), b(x))$, a composition in which x enters through three different doors. The multivariable chain rule therefore produces three terms, one per door. Through the third slot, $\partial\Phi/\partial z = F(x, z)$ by the Fundamental Theorem, multiplied by $b'(x)$. Through the second slot, $\partial\Phi/\partial y = -F(x, y)$, because y is the lower limit, multiplied by $a'(x)$. Through the first slot the limits are frozen and only the integrand varies, giving $\int_y^z \partial F/\partial x dt$. Adding them yields the rule.

THREE DOORS, THREE TERMS

The variable x can enter an integral in exactly three ways: through the upper limit, through the lower limit, and through the integrand. Each entry contributes one term, and no term may be skipped merely because another one is present. If x appears in all three places, your answer has three pieces; if in two, two; if in one, one. Counting the doors before differentiating is the checksum for this whole section.

The tree below is that count, drawn. It has no branch labelled “look up a formula”, because there is only one formula and the branches merely record how many of its terms survive.



15.3 The full Leibniz integral rule

MEDIUM

CUE Moving limits *and* the variable present as a parameter inside the integrand: $\int_x^{x^2} e^{xt}/t dt$, $\int_0^x \sin(x-t)g(t) dt$.

METHOD All three contributions, every time:

$$\frac{d}{dx} \int_{a(x)}^{b(x)} F(x, t) dt = F(x, b(x))b'(x) - F(x, a(x))a'(x) + \int_{a(x)}^{b(x)} \frac{\partial F}{\partial x}(x, t) dt.$$

The two boundary terms are evaluations, requiring no integration; the third is a fresh integral, usually easier than the original because differentiating in x tends to simplify the t -dependence. Hypotheses: F and $\partial F/\partial x$ continuous on a rectangle containing the swept region, and a, b differentiable.

WORKED EXAMPLE

Let $f(x) = \int_x^{x^2} \frac{e^{xt}}{t} dt$ and find $f'(1)$. Identify the pieces: $F(x, t) = e^{xt}/t$, $b(x) = x^2$, $a(x) = x$, $\partial F/\partial x = e^{xt}$, the t in the denominator having been consumed.

Upper boundary: $F(x, x^2)b'(x) = \frac{e^{x^3}}{x^2} \cdot 2x = \frac{2e^{x^3}}{x}.$

Lower boundary: $-F(x, x)a'(x) = -\frac{e^{x^2}}{x}.$

Interior: $\int_x^{x^2} e^{xt} dt = \left[\frac{e^{xt}}{x} \right]_{t=x}^{t=x^2} = \frac{e^{x^3} - e^{x^2}}{x}.$

At $x = 1$ the three pieces are $2e$, $-e$ and 0 , so $f'(1) = e = 2.7182818$, confirmed by central difference to seven digits. The interior term vanishing at $x = 1$ is a coincidence of the evaluation point, not a feature of the problem: at $x = 1.2$ the three pieces are 9.3823064 , -3.5172465 and 1.1739067 , summing to 7.0389667 against a central difference of 7.0389667 , and dropping the third would be a 16.7 per cent error.

A case with fixed limits, where only the third door is open. Let $S(x) = \int_0^1 \frac{\sin(xt)}{t} dt$. Both boundary terms are zero because $a' = b' = 0$, and

$$S'(x) = \int_0^1 \cos(xt) dt = \frac{\sin x}{x}.$$

At $x = 1.7$ this is 0.5833322 , against a central difference of 0.5833322 . The original integrand has no elementary antiderivative in t ; its x -derivative has an immediate one. That reversal is typical and is the reason the third term is usually the easy one.

Finally, the case that explains why this rule matters outside examination questions. Let

$$y(x) = \int_0^x \sin(x-t)g(t) dt,$$

a convolution with x in the upper limit *and* in the integrand. The boundary term is $\sin(0)g(x) = 0$ and the interior term is $\int_0^x \cos(x-t)g(t) dt$, so $y'(x) = \int_0^x \cos(x-t)g(t) dt$. Differentiating once more, the boundary term is now $\cos(0)g(x) = g(x)$ and the interior term is $-y(x)$, giving

$$y'' + y = g, \quad y(0) = y'(0) = 0.$$

The integral is the solution of a forced oscillator. Check it on $g(t) = t$: the integral evaluates to $y = x - \sin x$, so $y' = 1 - \cos x$ and $y'' = \sin x$, and indeed $y'' + y = x$. Numerically at $x = 1.3$ the integral is 0.3364418 and $x - \sin x = 0.3364418$.

WHERE THIS GOES WRONG

Stopping after the two boundary terms because “the parameter is already inside the integral” and therefore feels accounted for. It is not: the boundary terms record how the *region* moves, the interior term records how the *integrand* changes, and these are independent effects. The second error is writing $\frac{dF}{dx}$ where the formula demands $\frac{\partial F}{\partial x}$, and then differentiating t as though it depended on x . Inside the third term, t is a bound dummy and is held fixed.

SEE ALSO Both limits moving; decoupling substitution before Leibniz; differentiating under the integral sign.

The three-term rule is general but it is not always the cheapest route. When the parameter and the dummy variable are algebraically tied together, so that the same expression in x appears both in a limit and multiplying t , a substitution can move all of the x -dependence into the limits and a prefactor. What was a three-term Leibniz problem becomes a one-term Fundamental Theorem problem wrapped in a product or quotient rule, which is strictly less error-prone.

HARD

15.4 Decoupling substitution before Leibniz

CUE The same function of x appears in a limit *and* inside the integrand, tied to t : $\int_0^{x^2} \ln(1 + tx^2) dt, \int_0^x g(t/x) dt$.

METHOD Substitute $s = \lambda(x)t$, where λ is the offending factor. Then $dt = ds/\lambda(x)$, the prefactor $1/\lambda(x)$ comes outside, the integrand becomes a function of s alone, and every trace of x now lives in the upper limit and in that prefactor. Differentiate what remains with the product or quotient rule plus one Fundamental Theorem application. Use this whenever the substitution is available; fall back on the three-term rule when it is not.

Let $g(x) = \int_0^{x^2} \ln(1 + tx^2) dt$ for $x > 0$, and find $g'(1)$. The tie is the factor x^2 multiplying t , so put $s = tx^2$, giving $dt = ds/x^2$ and $t = x^2 \Rightarrow s = x^4$:

$$g(x) = \frac{1}{x^2} \int_0^{x^4} \ln(1 + s) ds = \frac{(1 + x^4) \ln(1 + x^4) - x^4}{x^2},$$

using $\int \ln(1 + s) ds = (1 + s) \ln(1 + s) - s$. A numerical check before differentiating: at $x = 1.3$ the original integral is 1.3895324 and the closed form is 1.3895324.

Now differentiate the closed form. Write $N(x) = (1 + x^4) \ln(1 + x^4) - x^4$, so $N'(x) = 4x^3 \ln(1 + x^4) + 4x^3 - 4x^3 = 4x^3 \ln(1 + x^4)$, a pleasant collapse. Then

$$g'(x) = \frac{N'(x)x^2 - 2xN(x)}{x^4} = \frac{4x^3 \ln(1 + x^4)}{x^2} - \frac{2N(x)}{x^3}.$$

At $x = 1$: $N(1) = 2 \ln 2 - 1$ and $N'(1) = 4 \ln 2$, so

$$g'(1) = 4 \ln 2 - 2(2 \ln 2 - 1) = 4 \ln 2 - 4 \ln 2 + 2 = 2.$$

Verified: 2.0000000. The logarithms cancel exactly, which is the signature of a designed problem; had you run the three-term rule you would have produced $\ln(1 + x^4) \cdot 2x$, 0, and $\int_0^{x^2} \frac{2xt}{1+tx^2} dt$, and the same cancellation would have happened one stage later and less visibly.

The extreme case, where decoupling removes the calculus entirely. Let $\Psi(x) = \int_0^x \sin \frac{t}{x} dt$ for $x > 0$. Put $u = t/x$, so $dt = x du$ and the limits become 0 and 1:

$$\Psi(x) = x \int_0^1 \sin u du = x(1 - \cos 1).$$

The function is linear, and $\Psi'(x) = 1 - \cos 1 = 0.4596977$ for every x , confirmed numerically at $x = 2.7$ where the integral itself is $1.2411838 = 2.7(1 - \cos 1)$. The three-term rule would have produced $\sin 1$ from the boundary and $-\frac{1}{x} \int_0^x \frac{t}{x} \cos \frac{t}{x} dt$ from the interior, which is correct and takes a further integration by parts to reach the same constant. Any integrand of the form $g(t/x)$ decouples this way.

WHERE THIS GOES WRONG

Forgetting the Jacobian. The factor $1/\lambda(x)$ is not decoration: it depends on x and is exactly what forces the quotient rule at the end. Dropping it produces an answer that is wrong by a smooth factor, which no sanity check on magnitude will catch. The second trap is substituting when $\lambda(x)$ can vanish on the domain, which destroys the change of variable at that point.

SEE ALSO The full Leibniz integral rule; algebraic normalisation before differentiating; differentiating under the integral sign.

There is a second reason to differentiate an integral, and it inverts the logic of everything above. Instead of being handed a function defined by an integral and asked for its derivative, you are handed a definite integral depending on a parameter, and differentiating in that parameter is the fastest way to *evaluate* it. The limits are fixed, so only the third door is open, and the rule reduces to a single interior term. This is the technique usually named after Feynman, though it is Leibniz's rule with two terms switched off.

15.5 Differentiating under the integral sign

HARD

CUE A definite integral over fixed limits, often infinite, whose integrand carries a parameter a ; and a question about $\frac{d}{da}$ of it, or an evaluation that resists every substitution.

METHOD With fixed limits the boundary terms vanish and

$$\frac{d}{da} \int_{a_0}^{b_0} F(a, t) dt = \int_{a_0}^{b_0} \frac{\partial F}{\partial a}(a, t) dt.$$

Two uses. Forwards: differentiate a known parametric identity to get a new one for free. Backwards: choose the parameter so that $\partial_a F$ is elementary, evaluate the easy integral to get $I'(a)$, then integrate in a and fix the constant from a value of a where I is obvious. On an infinite interval the interchange needs justification: uniform convergence of $\int \partial_a F$ on a neighbourhood of a , or a dominating integrable function.

Forwards, on a standard identity. For $a \geq 0$,

$$\int_0^\infty \frac{\ln(1+a^2x^2)}{1+x^2} dx = \pi \ln(1+a),$$

which at $a = 2$ evaluates to $\pi \ln 3 = 3.4513923$ against a numerical value of 3.4513. Differentiating the closed form gives $\frac{\pi}{1+a}$, so the a -derivative at $a = 2$ is $\frac{\pi}{3} = 1.0471976$. Nobody should differentiate the integrand here: the closed form is already available and the derivative of the answer is one line, whereas $\int_0^\infty \frac{2ax^2}{(1+a^2x^2)(1+x^2)} dx$ needs partial fractions to reach the same place.

Backwards, where the closed form is what we are after. Let $J(a) = \int_0^\infty e^{-ax} \frac{\sin x}{x} dx$ for $a > 0$. Differentiating under the sign kills the awkward $1/x$:

$$J'(a) = - \int_0^\infty e^{-ax} \sin x dx = -\frac{1}{1+a^2},$$

so $J(a) = -\arctan a + C$. As $a \rightarrow \infty$ the integrand is crushed and $J \rightarrow 0$, forcing $C = \pi/2$, hence $J(a) = \arctan(1/a)$. At $a = 1.5$ the formula gives 0.5880026 and the numerical integral gives 0.5880026; the derivative there is $-1/(1+2.25) = -0.3076923$. Letting $a \rightarrow 0^+$ recovers $\int_0^\infty \frac{\sin x}{x} dx = \pi/2$ as a bonus.

The moment generator, which is the technique's most reused instance. From

$$I(a) = \int_0^\infty e^{-ax^2} dx = \frac{1}{2} \sqrt{\frac{\pi}{a}} \quad (a > 0),$$

differentiating in a brings down a factor $-x^2$ on the left and differentiates $\frac{1}{2} \sqrt{\pi} a^{-1/2}$ on the right:

$$\int_0^\infty x^2 e^{-ax^2} dx = \frac{1}{4} \sqrt{\frac{\pi}{a^3}}.$$

At $a = 1$ that is $\frac{\sqrt{\pi}}{4} = 0.4431135$, against a numerical value of 0.4431135. Differentiating again gives $\int_0^\infty x^4 e^{-ax^2} dx = \frac{3}{8} \sqrt{\pi/a^5}$, which at $a = 1$ is 0.6646702, again confirmed numerically. Every even moment of the Gaussian falls out of one identity and repeated differentiation, and the parameter a was introduced solely to have something to differentiate with respect to.

WHERE THIS GOES WRONG

Swapping the derivative and the integral on an improper integral without a word of justification. The interchange can genuinely fail, and the standard counterexamples have an integrand whose mass escapes to infinity as the parameter moves. The other, more practical, trap is answering the wrong question: if the problem asks only for $I'(a)$, evaluating $I(a)$ first is wasted labour, and the derivative is often far simpler than the function.

SEE ALSO The full Leibniz integral rule; special functions defined by a differential equation; partial derivatives and mixed partials.

The last member of this family turns the machinery around once more. The unknown is now the function inside the integral, and the integral relation itself is the equation to be solved. Differentiation is the tool that converts an integral equation into something local, and the classical case where the naive differentiation fails is worth seeing, because it shows what the hypotheses of the Fundamental Theorem are actually protecting.

HARD

15.6 A function defined implicitly by an integral equation

CUE The unknown y appears only inside an integral with a variable limit: $\int_0^x K(x-t)y(t) dt = h(x)$, a Volterra convolution equation. The Abel kernel $K(u) = u^{-1/2}$ is the classic case.

METHOD If K is continuous, differentiate the whole equation in x by the Leibniz rule; the boundary term is $K(0)y(x)$ and the interior term is $\int_0^x K'(x-t)y(t) dt$, converting the integral equation into a simpler one or into an ODE. If K is singular at $u = 0$, as the Abel kernel is, that step is illegal and you use the Laplace transform instead: convolution becomes multiplication, so $\widehat{K}(s)\widehat{y}(s) = \widehat{h}(s)$, and with $\widehat{u^{-1/2}}(s) = \sqrt{\pi}/s$ the equation inverts in one line.

Solve $\int_0^x \frac{y(t)}{\sqrt{x-t}} dt = x^2$ for $x > 0$, and evaluate $\pi y'(4)$.

The kernel is singular at $t = x$, so transform. With $\hat{h}(s) = 2/s^3$ and $\hat{K}(s) = \sqrt{\pi/s}$,

$$\hat{y}(s) = \frac{2}{s^3} \sqrt{\frac{s}{\pi}} = \frac{2}{\sqrt{\pi}} s^{-5/2}.$$

Inverting with $s^{-\nu} \mapsto x^{\nu-1}/\Gamma(\nu)$ and $\Gamma(5/2) = \frac{3}{4}\sqrt{\pi}$,

$$y(x) = \frac{2}{\sqrt{\pi}} \cdot \frac{x^{3/2}}{\frac{3}{4}\sqrt{\pi}} = \frac{8x^{3/2}}{3\pi}.$$

Check it directly rather than trusting the transform: substituting $t = x - u^2$ turns the left side into $2 \int_0^{\sqrt{x}} y(x - u^2) du$, which at $x = 2$ evaluates numerically to 4.0000000, exactly x^2 . Then $y'(x) = \frac{8}{3\pi} \cdot \frac{3}{2}x^{1/2} = \frac{4\sqrt{x}}{\pi}$, so $y'(4) = \frac{8}{\pi}$ and $\pi y'(4) = 8$.

Contrast the regular case, where differentiation is legal and finishes the job on the spot. Solve $\int_0^x e^{x-t} y(t) dt = x$. Here $K(u) = e^u$ is continuous, so apply the Leibniz rule to both sides: the boundary term is $K(0)y(x) = y(x)$, the interior term is $\int_0^x e^{x-t} y(t) dt$, which the original equation says equals x , and the right side differentiates to 1. Hence

$$y(x) + x = 1, \quad y(x) = 1 - x,$$

with no transform and no series. Substituting back, $\int_0^x e^{x-t}(1-t) dt = e^x \cdot xe^{-x} = x$, which at $x = 1.7$ evaluates numerically to 1.7000000. The lesson is the dividing line: a continuous kernel yields to differentiation, a kernel singular at $u = 0$ does not.

WHERE THIS GOES WRONG

Differentiating under the sign when the kernel blows up at the endpoint. For the Abel equation the interior term $\int_0^x \partial_x(x-t)^{-1/2} y(t) dt$ diverges and the boundary term $K(0)y(x)$ is infinite, so the manipulation produces nonsense that looks formally correct. Either smooth first by integrating once, or transform. The second trap is forgetting that a Volterra equation carries its own initial condition, so no constant of integration is free to be chosen later.

SEE ALSO The full Leibniz integral rule; differentiating under the integral sign; the Fundamental Theorem with a variable upper limit.

Inverses of integral-defined functions

A function defined by an integral of a positive integrand is strictly increasing, hence invertible, and questions about the inverse are then answerable even though neither the function nor its inverse can be written down. The tool is the inverse-function rule, $(G^{-1})'(y) = \frac{1}{G'(x)}$ evaluated at the x with $G(x) = y$, and the Fundamental Theorem supplies G' for free.

Let $G(x) = \int_0^x e^{t^2} dt$. The integrand is positive, so G is strictly increasing on $\mathbb{R}$ and G^{-1} exists. Since $G(0) = 0$,

$$(G^{-1})'(0) = \frac{1}{G'(0)} = \frac{1}{e^0} = 1.$$

At a less convenient point the only extra work is locating x . Numerically $G(1) = 1.4626517$, so

$$(G^{-1})'(1.4626517) = \frac{1}{e^1} = 0.3678794.$$

The second derivative follows from the standard formula $(G^{-1})'' = -G''/(G')^3$, and here $G''(x) = 2xe^{x^2}$, which vanishes at $x = 0$; so G^{-1} has an inflection at $y = 0$, a fact about a function nobody can write down.

The same reading explains a familiar object. With $L(x) = \int_1^x dt/t$ the Fundamental Theorem gives $L'(x) = 1/x$, so $(L^{-1})'(y) = 1/L'(x) = x = L^{-1}(y)$: the inverse is its own derivative, which is the definition of the exponential. The logarithm defined as an integral and the exponential defined as its inverse are constructed exactly this way.

Second derivatives, and shape without a formula

Nothing stops you differentiating twice. If $F(x) = \int_a^x f(t) dt$ then $F' = f$ and $F'' = f'$, so the entire apparatus of the next chapter, monotonicity from the sign of the first derivative and concavity from the sign of the second, applies to a function

you cannot write down. This is worth isolating because it is how integral-defined functions appear in shape questions, and because the answers are usually cleaner than the function.

Take $F(x) = \int_0^x (t^2 - 3t + 2)e^t dt$ again. We already have $F'(x) = (x-1)(x-2)e^x$, so F rises on $(-\infty, 1)$, falls on $(1, 2)$, and rises thereafter. For concavity, differentiate again by the product rule:

$$F''(x) = (2x-3)e^x + (x^2-3x+2)e^x = (x^2-x-1)e^x,$$

which vanishes at $x = \frac{1 \pm \sqrt{5}}{2}$, that is at 1.6180340 and -0.6180340 . Both are simple roots of the quadratic, so the sign genuinely changes and both are inflection points. Note that neither coincides with a critical point, and that the golden ratio has appeared out of an integrand chosen for its ordinariness.

When the upper limit is a function of x , the second derivative needs a product rule as well as a chain rule. From $F(x) = \int_a^{u(x)} f$, differentiating $F' = f(u)u'$ gives

$$F''(x) = f'(u(x))u'(x)^2 + f(u(x))u''(x),$$

which is Faa di Bruno's formula at order two and is the shape you should expect: a term with the inner derivative squared, and a term with the inner second derivative. For $F(x) = \int_0^{x^3} e^{-t^2} dt$ this reads $F''(x) = -2x^3e^{-x^6} \cdot 9x^4 + e^{-x^6} \cdot 6x$, that is $6xe^{-x^6}(1-3x^6)$, so the graph has inflections at $x = 0$ and at $x = \pm 3^{-1/6}$.

Back to the definition

The rest of this chapter is about the other kind of failure. In the integral family the rules did not apply because the function had no formula; here the function has a formula, or several, but the formula is not differentiable-by-rule at the point in question. The remedy is always the same and it is always available:

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h},$$

computed as an honest limit. This is not a fallback for the unlucky. It is the definition, and there are questions that only it can answer.

Four situations force you back to it. The function is defined by cases and a is a junction point. The function is defined by a rule that degenerates at a , with $f(a)$ supplied separately. The function involves absolute values, floors, or fractional powers whose derivative formula fails at a . Or the function is pathological on purpose, as in the counterexamples that distinguish differentiability from continuous differentiability. The first technique below covers the raw computation; the rest specialise it.

15.7 The derivative from first principles

MEDIUM

CUE The problem says “from the definition”; or the function is defined by cases, by conditions on rationality, or by a limit; or the rules produce an expression that is undefined at the very point requested.

METHOD Form the difference quotient and evaluate the limit honestly. The workhorse supporting limits are $\lim_{h \rightarrow 0} \frac{\sin h}{h} = 1$, $\lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1$, $\lim_{h \rightarrow 0} \frac{\ln(1+h)}{h} = 1$, and $\lim_{h \rightarrow 0} \frac{(1+h)^r - 1}{h} = r$. When the numerator cannot be factored, squeeze: if $|f(a+h) - f(a)| \leq C|h|^{1+\epsilon}$ for some $\epsilon > 0$, then $f'(a) = 0$ immediately, with no algebra at all.

WORKED EXAMPLE

The squeeze in its purest form. Let D be the Dirichlet function, equal to 1 at rational points and 0 elsewhere, and set $f(x) = x^2D(x)$. This function is discontinuous at every point except 0, so no rule of any kind applies anywhere. At the origin, however, $f(0) = 0$ and

$$\left| \frac{f(h) - f(0)}{h} \right| = \left| \frac{h^2 D(h)}{h} \right| \leq |h| \rightarrow 0,$$

so $f'(0) = 0$ exists. A function can be differentiable at exactly one point.

A case where the two sides disagree, so the answer is that the derivative does not exist. Let $f(x) = \frac{x}{1 + e^{1/x}}$ for $x \neq 0$ with $f(0) = 0$. The quotient is

$$\frac{f(h) - f(0)}{h} = \frac{1}{1 + e^{1/h}}.$$

As $h \rightarrow 0^+$, $1/h \rightarrow +\infty$ and $e^{1/h} \rightarrow \infty$, so the quotient tends to 0. As $h \rightarrow 0^-$, $1/h \rightarrow -\infty$ and $e^{1/h} \rightarrow 0$, so the quotient tends to 1. Both one-sided derivatives exist and are finite, $f'_+(0) = 0$ and $f'_-(0) = 1$, and they differ: f has a corner at the origin. Observe that f is continuous there, since $|f(x)| \leq |x|$; continuity was never the issue.

And a case where the algebra has to be done rather than squeezed. Let $f(x) = |x|^3$ and compute $f'(0)$ and $f''(0)$ from the definition. First,

$$\frac{f(h) - f(0)}{h} = \frac{|h|^3}{h} = h|h| \rightarrow 0,$$

so $f'(0) = 0$. Away from 0 the rules give $f'(x) = 3x|x|$, which is continuous at 0 with value 0, so f is C^1 . Repeating,

$$\frac{f'(h) - f'(0)}{h} = \frac{3h|h|}{h} = 3|h| \rightarrow 0,$$

so $f''(0) = 0$ and $f''(x) = 6|x|$, again continuous. But $f'''(x) = 6 \operatorname{sgn} x$ has a genuine jump at 0, so f is C^2 and no better. The bars did not obstruct anything until the third derivative.

WHERE THIS GOES WRONG

Assuming that a function built from differentiable pieces must be differentiable at the junction, and evaluating a piece's derivative formula at the junction instead of taking a one-sided limit of the difference quotient. The second trap is cancelling h out of the numerator before checking that a factor of h is genuinely there; if the numerator is $O(h^{1/2})$, the quotient blows up and the derivative does not exist.

SEE ALSO Recognising a limit as a derivative in disguise; one-sided derivatives, corners and cusps; piecewise matching for continuity and differentiability.

The definition can also be run backwards, and doing so is one of the highest-value recognitions in elementary calculus. A limit that looks intractable is very often a difference quotient wearing a disguise, and once it is recognised the answer is a single derivative evaluation. The move costs nothing and it works in cases where L'Hopital's rule is not licensed, since the definition needs g differentiable only *at* the point, while L'Hopital needs g' to exist in a whole punctured neighbourhood.

15.8 Recognising a limit as a derivative in disguise

EASY

CUE A limit of shape $\lim_{x \rightarrow a} \frac{g(x) - g(a)}{x - a}$, or $\lim_{h \rightarrow 0} \frac{g(a + \alpha h) - g(a + \beta h)}{h}$, or $\lim_{n \rightarrow \infty} n(g(a + c/n) - g(a))$; typically an indeterminate $0/0$ in which one specific number keeps reappearing as an argument.

METHOD Match the shape to the definition and identify g and a . The symmetric two-point version follows by adding and subtracting $g(a)$:

$$\lim_{h \rightarrow 0} \frac{g(a + \alpha h) - g(a + \beta h)}{h} = (\alpha - \beta) g'(a).$$

The $n \rightarrow \infty$ version is the same statement with $h = 1/n$. Read it in whichever direction is cheaper: forwards it evaluates a hard limit by one differentiation, backwards it computes a derivative that the rules cannot reach.

WORKED EXAMPLE

Evaluate $\lim_{h \rightarrow 0} \frac{\sin(\frac{\pi}{6} + 3h) - \sin(\frac{\pi}{6} - h)}{h}$. The repeated argument is $\pi/6$, so $g = \sin$, $a = \pi/6$, $\alpha = 3$, $\beta = -1$, and the limit is

$$(3 - (-1)) \cos \frac{\pi}{6} = 4 \cdot \frac{\sqrt{3}}{2} = 2\sqrt{3} = 3.4641016,$$

against a numerical quotient at $h = 10^{-6}$ of 3.4640996. Expanding both sines with the addition formula reaches the same answer in six more lines.

A sequence version. Find $\lim_{n \rightarrow \infty} n(2^{1/n} - 1)$. Write $2^{1/n} = g(1/n)$ with $g(u) = 2^u$ and note $g(0) = 1$, so the expression is $\frac{g(1/n) - g(0)}{1/n} \rightarrow g'(0) = \ln 2 = 0.6931472$. Numerically at $n = 10^7$ the value is 0.6931472.

A version with a tangent, where the coefficient does the work. Consider $\lim_{h \rightarrow 0} \frac{\tan(\frac{\pi}{4} + h) - \tan(\frac{\pi}{4} - h)}{h}$. The repeated argument is $\pi/4$, so $g = \tan$, $\alpha = 1$, $\beta = -1$, and the limit is $2 \sec^2(\pi/4) = 2 \cdot 2 = 4$, against a numerical quotient of 4.0000000. Expanding both tangents with the addition formula produces a ratio of quadratics in $\tan h$ and takes half a page.

And backwards, to compute something the rules handle badly. What is $\frac{d}{dx}x^x$ at $x = 3$? The rules do reach this by logarithmic differentiation, but the definition reaches it too:

$$\lim_{h \rightarrow 0} \frac{(3+h)^{3+h} - 3^3}{h} = 3^3(1 + \ln 3) = 56.662532,$$

matching a symmetric difference of 56.662532. The point is not that the definition is faster here, but that it is the same object: a derivative and this limit are two names for one number, and a problem may hand you either.

WHERE THIS GOES WRONG

Reaching for L'Hopital's rule reflexively. It is heavier than the definition and it has a hypothesis the definition does not: g' must exist near a , not just at a . On a function like $x^2 D(x)$ at the origin, L'Hopital is simply unavailable while the definition answers instantly. The second trap is misreading α and β : the coefficient is $\alpha - \beta$, so a minus sign inside the second argument becomes a plus in the answer.

SEE ALSO The derivative from first principles; the derivative at a point of continuous extension; linearisation and differentials.

A harder variant arises when the formula for f is not merely awkward at a but indeterminate there, with $f(a)$ assigned by continuous extension. Now the difference quotient is a $0/0$ limit whose numerator is itself the difference of an indeterminate form from its limiting value, and the only reliable instrument is a series expansion carried far enough.

HARD

15.9 The derivative at a point of continuous extension

CUE f has a removable indeterminacy at exactly the point where f' is requested, usually 1^∞ or ∞^0 , with $f(a)$ supplied by fiat to fill the hole.

METHOD Do not differentiate the formula. Expand. Take logarithms and produce the Taylor expansion of $\ln f$ about a to first order in $(x - a)$: if

$$\ln f(x) = c_0 + c_1(x - a) + O((x - a)^2), \quad \text{then} \quad f(a) = e^{c_0}, \quad f'(a) = f(a) c_1.$$

The bookkeeping point is that an exponent of the form $1/(x - a)$ divides the expansion by $(x - a)$, so to obtain c_1 you must expand the bracket to *second* order. Truncating at first order loses exactly the term you need.

WORKED EXAMPLE

Let $f(x) = \left(\frac{\ln x}{x-1}\right)^{1/(x-1)}$ for $x > 0, x \neq 1$, with $f(1) = e^{-1/2}$. Find $f'(1)$.

Set $u = x - 1$. Then $\ln x = \ln(1 + u) = u - \frac{u^2}{2} + \frac{u^3}{3} - \dots$, so

$$\frac{\ln x}{x-1} = 1 - \frac{u}{2} + \frac{u^2}{3} - \dots = 1 + w, \quad w = -\frac{u}{2} + \frac{u^2}{3} + O(u^3).$$

Now $\ln(1 + w) = w - \frac{w^2}{2} + O(w^3)$, and $w^2 = \frac{u^2}{4} + O(u^3)$, so

$$\ln \frac{\ln x}{x-1} = -\frac{u}{2} + \frac{u^2}{3} - \frac{u^2}{8} + O(u^3) = -\frac{u}{2} + \frac{5u^2}{24} + O(u^3).$$

Dividing by u gives $\ln f = -\frac{1}{2} + \frac{5u}{24} + O(u^2)$, so $c_0 = -\frac{1}{2}$, confirming $f(1) = e^{-1/2}$, and $c_1 = \frac{5}{24}$. Hence

$$f'(1) = e^{-1/2} \cdot \frac{5}{24} = 0.1263606,$$

against a central difference of 0.1263606. Note where the $5/24$ came from: $\frac{1}{3}$ from the cubic term of $\ln(1 + u)$ and $-\frac{1}{8}$ from the quadratic correction in $\ln(1 + w)$. Stop either expansion one order early and you get $\frac{1}{3}$ or $-\frac{1}{8}$ alone, both wrong.

A shorter instance of the same machine. Let $f(x) = (1 + x)^{1/x}$ for $x > -1, x \neq 0$, with $f(0) = e$. Then

$$\ln f = \frac{\ln(1 + x)}{x} = \frac{1}{x} \left(x - \frac{x^2}{2} + \frac{x^3}{3} - \dots \right) = 1 - \frac{x}{2} + \frac{x^2}{3} - \dots,$$

so $c_0 = 1$ and $c_1 = -\frac{1}{2}$, giving $f(0) = e$ as stated and

$$f'(0) = e \cdot \left(-\frac{1}{2}\right) = -\frac{e}{2} = -1.3591409,$$

against a central difference of -1.3591409 . Here the second-order term of $\ln(1+x)$ is what produced c_1 , again one order deeper than a careless expansion would go.

WHERE THIS GOES WRONG

Expanding to first order only. The division by $(x-a)$ demotes every order by one, so first-order accuracy in f' demands second-order accuracy in the bracket. The same arithmetic explains why applying L'Hopital once to $\frac{f(x)-f(a)}{x-a}$ typically stalls: the rule must be applied twice, and after the first application the expression is worse, not better.

SEE ALSO The derivative from first principles; Maclaurin coefficients by series manipulation; recognising a limit as a derivative in disguise.

Matching, corners, and the failure of the rules

The three techniques of the previous section all computed a derivative that existed. The remaining question is when one exists at all, and the answer is always decided by comparing two one-sided limits. That comparison is the organising idea for everything below: matching problems choose parameters to force the two sides to agree, corner and cusp problems ask you to observe that they do not, and the final counterexample family shows that agreement of the one-sided derivatives is a strictly weaker condition than continuity of f' .

The most common examination form of the definition is not exotic at all. A function is given by two formulas meeting at a point, with unknown constants, and the constants must be chosen so that the join is smooth. This is two equations in two unknowns, and the only subtlety is which two equations.

15.10 Piecewise matching for continuity and differentiability

CORE

CUE f given by two formulas meeting at $x = c$, with parameters a, b to be determined so that f is differentiable (or twice differentiable) at c .

METHOD Impose the conditions in order, because the second presupposes the first. *Continuity:* $\lim_{x \rightarrow c^-} f(x) = \lim_{x \rightarrow c^+} f(x) = f(c)$. *Differentiability:* $f'_-(c) = f'_+(c)$. Solve the resulting 2×2 system. The shortcut $\lim_{x \rightarrow c^-} f'(x) = \lim_{x \rightarrow c^+} f'(x)$ is legitimate only when f is already continuous at c and both one-sided limits of f' exist; that is the usual situation with polynomial and exponential pieces, and it is what makes the method quick. When either hypothesis fails, go back to one-sided difference quotients.

WORKED EXAMPLE

The archetype first. Let $f(x) = x^2$ for $x \leq 1$ and $f(x) = ax + b$ for $x > 1$. Continuity at 1: $1 = a + b$. Differentiability: the left derivative is $2x$ at $x = 1$, namely 2, and the right derivative is a , so $a = 2$ and therefore $b = -1$. The smooth continuation of x^2 by a line is its own tangent, $y = 2x - 1$, which is the geometric content of the answer.

A version where the parameters sit on the other side and the second derivative is also matched. Let

$$f(x) = \begin{cases} ax^2 + bx + 1, & x \leq 0, \\ e^{2x} \cos 3x, & x > 0. \end{cases}$$

Continuity: the left limit is 1 and the right limit is $e^0 \cos 0 = 1$, so continuity holds for every a and b ; the setter has spent that equation. Differentiability: the right piece has $y' = e^{2x}(2 \cos 3x - 3 \sin 3x)$, so $f'_+(0) = 2$, while $f'_-(0) = b$. Hence $b = 2$. That exhausts the first-derivative condition and leaves a free, so the question must be asking for C^2 . Differentiating again,

$$y'' = e^{2x}(4 \cos 3x - 6 \sin 3x) + e^{2x}(-6 \sin 3x - 9 \cos 3x) = e^{2x}(-5 \cos 3x - 12 \sin 3x),$$

so $f''_+(0) = -5$, while $f''_-(0) = 2a$. Therefore $a = -\frac{5}{2}$, $b = 2$. Numerically, the right piece has $y'(0) = 2.0000000$ and $y''(0) = -5.0000001$ by central differences. The general pattern: matching k derivatives consumes $k + 1$ equations, counting continuity as the zeroth.

And a case where the derivative shortcut is illegal and must be replaced. Let $f(x) = x^2 \sin(1/x)$ for $x < 0$ and $f(x) = ax + b$ for $x \geq 0$. Continuity at 0: the left limit is 0 by the squeeze $|f| \leq x^2$, and the right value is b , so $b = 0$. Differentiability: the right derivative is plainly a . For the left, the shortcut fails outright, because

$$\lim_{x \rightarrow 0^-} f'(x) = \lim_{x \rightarrow 0^-} \left(2x \sin \frac{1}{x} - \cos \frac{1}{x} \right)$$

does not exist. Go to the definition instead:

$$\left| \frac{f(h) - f(0)}{h} \right| = |h \sin(1/h)| \leq |h| \rightarrow 0,$$

so $f'_-(0) = 0$ and therefore $a = 0$, $b = 0$. The matched function is $f(x) = x^2 \sin(1/x)$ patched by the zero function, differentiable at the junction with f' discontinuous there.

COUNTING THE CONDITIONS

Matching to order k at a junction imposes $k + 1$ equations, one for each of $f, f', \dots, f^{(k)}$, and therefore needs $k + 1$ free parameters to be solvable in general. If a problem hands you two parameters and the continuity equation turns out to be automatic, the intended condition is C^2 , not C^1 : count the equations you actually get, not the ones you expected.

WHERE THIS GOES WRONG

Imposing differentiability without continuity. A function discontinuous at c is never differentiable there, so a system built from the derivative condition alone can have a solution that solves nothing. The mirror error is matching derivatives by substituting c into the two derivative formulas when one of the pieces has no one-sided derivative at c : for $f(x) = x^2 \sin(1/x)$ patched to $f(0) = 0$, the formula for f' has no limit at 0 although $f'(0)$ exists.

SEE ALSO One-sided derivatives, corners, cusps and vertical tangents; differentiable but not continuously differentiable; the derivative from first principles.

Piecewise definitions are often disguised. An absolute value is a two-case definition, a maximum of two functions is a two-case definition, and a fractional power is a formula whose derivative simply ceases to exist at one point. It is worth having a vocabulary for the three ways a derivative can fail while the function stays continuous, because problems ask you to name them.

FOUR WAYS A CONTINUOUS FUNCTION FAILS TO BE DIFFERENTIABLE

Corner: both one-sided derivatives finite and unequal, as $|x|$ at 0; the graph has two distinct tangent directions. **Cusp:** both infinite with opposite signs, as $x^{2/3}$; the two branches leave vertically on the same side. **Vertical tangent:** both infinite with the same sign, as $x^{1/3}$; the graph is smooth as a curve but not as a function of x . **Oscillation:** neither one-sided limit exists, as $x \sin(1/x)$; there is no tangent direction at all. Continuity is assumed throughout, and a discontinuity is a fifth and cruder failure that needs no analysis.

MEDIUM

15.11 One-sided derivatives: corners, cusps and vertical tangents

CUE Absolute values, max or min of two functions, floors, or fractional powers: $|x^2 - 4|$, $\min(x, x^2)$, $x^{2/3}$, $\sqrt[3]{x}$.

METHOD Compute the two one-sided limits separately,

$$f'_-(a) = \lim_{h \rightarrow 0^-} \frac{f(a+h) - f(a)}{h}, \quad f'_+(a) = \lim_{h \rightarrow 0^+} \frac{f(a+h) - f(a)}{h},$$

and conclude that $f'(a)$ exists exactly when both exist finitely and agree. Then name the failure. *Corner:* both finite, unequal, as $|x|$ at 0. *Cusp:* both infinite with opposite signs, as $x^{2/3}$ at 0. *Vertical tangent:* both infinite with the same sign, as $x^{1/3}$ at 0. For $|g(x)|$ the failures sit exactly at the *sign-changing* zeros of g , and at those points $f'_\pm = \pm g'$, so the jump is $2|g'|$.

WORKED EXAMPLE

Let $f(x) = |x^2 - 4|$. The inner function changes sign at $x = \pm 2$ and nowhere else, so those are the only candidates. Away from them $f'(x) = 2x \operatorname{sgn}(x^2 - 4)$. At $x = 2$, with $g(x) = x^2 - 4$ and $g'(2) = 4$:

$$f'_-(2) = -g'(2) = -4, \quad f'_+(2) = g'(2) = 4,$$

a corner with a jump of 8. Similarly $f'_-(-2) = 4$ and $f'_+(-2) = -4$.

Now the contrast that decides whether a sign change matters. Take $F(x) = |x^2 - 4| (x - 2)$ near $x = 2$: the factor $(x - 2)$ vanishes there, and

$$\frac{F(2+h) - F(2)}{h} = \frac{|(2+h)^2 - 4| \cdot h}{h} = |4h + h^2| \rightarrow 0,$$

so $F'(2) = 0$ exists despite the absolute-value bars. The bars are not the obstruction; an unmatched pair of one-sided slopes is.

And the three named failures, side by side at the origin. For $x^{2/3}$, $\frac{h^{2/3}}{h} = h^{-1/3}$, which tends to $+\infty$ from the right and $-\infty$ from the left: a cusp. For $x^{1/3}$, $\frac{h^{1/3}}{h} = h^{-2/3} \rightarrow +\infty$ from both sides: a vertical tangent, and the curve is smooth as a set even though f' does not exist. For $\min(x, x^2)$, the pieces cross at $x = 0$ and $x = 1$; at 0 the slopes are 1 and $2x = 0$, a corner, and at 1 they are 1 and 2, another corner.

One with a floor in it, to show that the method is unchanged. Let $f(x) = \lfloor x \rfloor + (x - \lfloor x \rfloor)^2$. On each interval $[n, n+1]$ this is $n + (x - n)^2$, which rises from n to $n+1$, so f is continuous at every integer despite the floor: the left limit at n is $(n-1) + 1 = n$ and the value is n . Differentiability is another matter. On $(n-1, n)$ the derivative is $2(x - n + 1)$, tending to 2 as $x \rightarrow n^-$; on $(n, n+1)$ it is $2(x - n)$, tending to 0 as $x \rightarrow n^+$. So $f'_-(n) = 2$ and $f'_+(n) = 0$: a corner at every integer, with f continuous everywhere and differentiable off $\mathbb{Z}$.

WHERE THIS GOES WRONG

Declaring $|x|^3$ non-differentiable at 0 because of the bars. In fact $|x|^3$ has derivative $3x|x|$ and second derivative $6|x|$, so it is C^2 at the origin and fails only at the third derivative. The kink comes from a sign change of the inside, never from the notation, and at a zero of even order there is no sign change. The reverse error is checking only the zeros of g and forgetting that $|g|$ can also fail where g itself is not differentiable.

SEE ALSO Piecewise matching for continuity and differentiability; differentiable but not continuously differentiable; sign charts for monotonicity and concavity.

One final distinction, and it is the one that most often goes unexamined. “ $f'(a)$ exists” and “ f' is continuous at a ” are different statements, and the second is strictly stronger. The standard family separating them is $x^k \sin(1/x)$, and it repays being worked through carefully once, because the same computation settles a whole row of examination questions about whether a limit of f' says anything about $f'(a)$.

HARD

15.12 Differentiable but not C^1

CUE A function of the form $x^k \sin(1/x)$ with $f(0) = 0$, and a question about $f'(0)$, about $\lim_{x \rightarrow 0} f'(x)$, or about whether f is C^1 .

METHOD Two separate computations, kept apart. First, $f'(0)$ from the definition, by squeezing the difference quotient. Second, $f'(x)$ for $x \neq 0$ by the rules, followed by an examination of $\lim_{x \rightarrow 0} f'(x)$. The exponent decides everything: $k = 1$ gives no $f'(0)$, $k = 2$ gives $f'(0) = 0$ with f' discontinuous at 0, and each further power buys one more order of smoothness. Note also Darboux’s theorem: a derivative has the intermediate value property, so f' can never have a jump discontinuity, only an oscillatory one, which is exactly what this family provides.

WORKED EXAMPLE

Let $f(x) = x^2 \sin(1/x)$ for $x \neq 0$ and $f(0) = 0$. From the definition,

$$\left| \frac{f(h) - f(0)}{h} \right| = |h \sin(1/h)| \leq |h| \rightarrow 0, \quad \text{so } f'(0) = 0.$$

Away from the origin the rules give

$$f'(x) = 2x \sin \frac{1}{x} - \cos \frac{1}{x},$$

whose first term tends to 0 but whose second oscillates between -1 and 1 forever. So $\lim_{x \rightarrow 0} f'(x)$ does not exist, while $f'(0) = 0$ does. The function is differentiable on all of $\mathbb{R}$ and its derivative is discontinuous at one point.

The neighbouring cases. For $k = 1$, the quotient is $\sin(1/h)$, which oscillates without settling, so $f'(0)$ does not exist at all; here f is continuous but not differentiable. For $k = 3$, $f'(x) = 3x^2 \sin(1/x) - x \cos(1/x) \rightarrow 0 = f'(0)$, so f is C^1 ;

but the second difference quotient $\frac{f'(h)-f'(0)}{h} = 3h \sin(1/h) - \cos(1/h)$ has no limit, so $f''(0)$ does not exist. Each unit of k buys one derivative, and the pattern continues.

WHERE THIS GOES WRONG

Concluding that $f'(0)$ fails to exist because $\lim_{x \rightarrow 0} f'(x)$ fails to exist. Those are different statements and this family is the standard counterexample. The valid implication runs only one way: if f is continuous at a and $\lim_{x \rightarrow a} f'(x) = L$ exists, then $f'(a) = L$. The converse is false, and any argument that assumes it has assumed C^1 without saying so.

SEE ALSO The derivative from first principles; one-sided derivatives, corners, cusps and vertical tangents; Rolle and mean-value arguments as tools.

A protocol for differentiability questions

Almost every question in this half of the chapter yields to the same four-step reading, and it is worth carrying explicitly, because the steps have to happen in this order.

First, is f continuous at a ? If not, stop: no derivative exists and no matching condition can rescue it. This step is free and it eliminates a whole class of wrong answers.

Second, does a rule reach a ? A rule reaches a when f agrees with a single elementary formula on a whole neighbourhood of a . A junction point, an absolute value at a sign change, an assigned value filling a hole, and a fractional power at its branch point all fail this test, and each is an instruction to use the definition.

Third, compute both one-sided difference quotients. Not the one-sided limits of f' : those agree with the one-sided derivatives only when they exist, and the whole point of the hard cases is that they may not. Where a factor can be bounded, squeeze rather than expand.

Fourth, say precisely what was found. “ $f'(a)$ exists and equals L ” is a different statement from “ f is continuous at a ”, which is different again from “ $\lim_{x \rightarrow a} f'(x) = L$ ”. A question that mentions C^1 , or asks whether f' is continuous, is asking the second; a question that asks for $f'(a)$ is asking the first. Answering the wrong one is the commonest way to lose a mark while doing correct mathematics.

Two habits carry this chapter. The first is counting the doors: whenever x appears in an integral, ask through how many of the three entrances it enters, and produce exactly that many terms. The second is knowing when to stop looking for a rule. A junction point, an assigned value at a removable singularity, an absolute value at a sign change, and a deliberately oscillatory patch are all announcements that the rule systems have run out, and each one is answered by the same difference quotient, computed one side at a time.

Recognition matrix

If you see	Reach for	If that fails
$\int_a^{u(x)} f(t) dt$	$f(u(x))u'(x)$; never find the antiderivative	Split at a constant if both limits move
$\int_{u(x)}^a f(t) dt$	Flip orientation first: $-f(u(x))u'(x)$	Write b' and a' with signs before substituting
$\int_{a(x)}^{b(x)} f(t) dt$	$f(b)b' - f(a)a'$	Look for a complementary identity in the two terms
Reciprocal limits $\int_{1/x}^x$	Expect $\arctan x + \arctan(1/x) = \pi/2$ or a sibling	Differentiate and simplify before evaluating
x in the limits and in the integrand	Full Leibniz: two boundary terms plus $\int \frac{\partial F}{\partial x}$	Count the doors; three doors, three terms
The same $\lambda(x)$ in a limit and multiplying t	Substitute $s = \lambda(x)t$; carry the Jacobian	Fall back on the three-term rule
Fixed limits, a parameter in the integrand	Differentiate under the sign; boundary terms vanish	Recognise a known parametric identity first
An integral you cannot evaluate, with a parameter	Get $I'(a)$, integrate back, fix C at an easy a	Check uniform convergence before swapping
$\int_0^x K(x-t)y(t) dt = h(x)$	Differentiate if K is regular; Laplace if K is singular	$u^{-1/2} = \sqrt{\pi/s}$ for the Abel kernel
“From the definition”	$\lim_{h \rightarrow 0} \frac{f(a+h)-f(a)}{h}$, honestly	Squeeze: $ \Delta f \leq C h ^{1+\epsilon}$ gives $f'(a) = 0$
$\lim \frac{g(a+\alpha h)-g(a+\beta h)}{h}$	$(\alpha - \beta)g'(a)$	Identify g by the argument that repeats
$n(g(a+c/n) - g(a))$ as $n \rightarrow \infty$	Same limit with $h = 1/n$: answer $cg'(a)$	Not L'Hopital; it needs g' near a
$f(a)$ assigned to fill a 1^∞ hole	Expand $\ln f$ to second order; $f'(a) = f(a)c_1$	First order is one order too few

If you see	Reach for	If that fails
Two formulas meeting at c , unknown a, b	Continuity first, then $f'_-(c) = f'_+(c)$	Matching k derivatives needs $k + 1$ equations
$ g(x) $ $x^{2/3}, x^{1/3}$ at 0	Non-differentiable only at sign-changing zeros of g Cusp: $\pm\infty$ opposite; vertical tangent: same sign	Jump in slope is $2 g' $ there Both are one-sided limits, computed separately
$x^k \sin(1/x)$ with $f(0) = 0$ $\lim_{x \rightarrow a} f'(x)$ does not exist	Definition for $f'(0)$; rules for $f'(x)$, $x \neq 0$ Says nothing about $f'(a)$	$k = 1$ none, $k = 2$ exists but f' jumps Darboux: a derivative has no jump discontinuity
$G(x) = \int_a^x f$ with $f > 0$, inverse wanted Concavity of $\int_a^{u(x)} f$	$(G^{-1})'(y) = 1/f(x)$ at the x with $G(x) = y$ $F'' = f'(u)u'^2 + f(u)u''$	Locating x is root-finding, not calculus Two terms; the second dies only if u is affine
A floor or fractional part in a formula	Rewrite as cases on $[n, n + 1)$, then one-sided limits	Continuity at the joins is not automatic

Applied structure: rates, optimisation, approximation

What this chapter is for. Applied derivative problems are won or lost before any differentiation happens. The calculus in a related-rates problem is one chain rule; the calculus in a constrained optimisation is one derivative and one sign check; the calculus in a linearisation is one evaluation. What separates a solved problem from an unsolved one is the setup: which quantities get names, which relation among them is true at every instant rather than only at the instant of interest, which variable the constraint eliminates, and what the domain actually is. This chapter therefore teaches setup as an explicit discipline, family by family: related rates, constrained optimisation, tangents and normals, linearisation and differentials, Newton's method, curvature, sign charts, and mean-value arguments used as instruments rather than as theory. Each block states its protocol first and its differentiation second, because that is the order in which the work is done and the order in which it goes wrong. At the end you should be able to turn a paragraph of English into a correctly parameterised equation on the first attempt, and to know before differentiating what your answer will be a function of.

Related rates: the relation comes first

A related-rates problem hands you a physical situation, one known rate, and one wanted rate. The mechanism is trivial, being nothing but the chain rule applied to a relation between quantities. Everything difficult lies in producing that relation, and there is a discipline that makes it reliable.

The reason the mechanism is trivial deserves saying plainly, because it is what licenses the whole procedure. If Q and u are both functions of time and $Q = g(u)$ holds *identically* in t , then differentiating the identity gives $\dot{Q} = g'(u)\dot{u}$ at every t , including the instant of interest. The word “identically” carries the entire load. An equation that happens to hold at one moment, such as $r = 18$, is not an identity and must not be differentiated; an equation that holds at all moments, such as $V = \frac{4}{3}\pi r^3$, may be. Every error in this family is a violation of that distinction, either by differentiating something only instantaneously true, or by treating as constant a quantity that is in fact varying.

THE FIVE-STEP PROTOCOL, IN ORDER

- 1. Name.** Give every varying quantity a symbol and say out loud that it is a function of t . Draw the figure and label it with symbols, never with numbers.
- 2. Relate.** Write one equation among those symbols that is true at *every* time. This step decides the problem.
- 3. Reduce.** Use the geometry to eliminate every quantity whose rate is neither known nor wanted.
- 4. Differentiate.** Apply $\frac{d}{dt}$ to both sides, chain rule on every symbol.
- 5. Substitute last.** Only now put in the instantaneous numbers, and only now reconcile the units.

Step five carries the warning that gives the family its character. A number substituted before differentiation becomes a constant, and constants have zero derivative, so an early substitution silently deletes a term. If the radius of a sphere is 18 at the instant asked about, writing $V = \frac{4}{3}\pi(18)^3$ makes V constant and $\dot{V} = 0$, which is the exact opposite of the truth.

16.1 Related rates: the standard pipeline

CORE

CUE Language of the form “at what rate”, “how fast is ... changing”, with two or more quantities varying in time, one rate given and another wanted.

METHOD Run the five steps above. The differentiation itself is always the same shape: from $Q = g(u)$ follows $\dot{Q} = g'(u)\dot{u}$. Power relations are the commonest, and for $Q = cu^k$ the relative form

$$\frac{\dot{Q}}{Q} = k \frac{\dot{u}}{u}$$

is often the fastest route, since it needs neither c nor the absolute size of u . Check units at the end and convert only then.

WORKED EXAMPLE

A spherical balloon deflates at $72 \text{ cm}^3/\text{s}$. How fast is the radius shrinking when $r = 18 \text{ cm}$?

Name $r(t)$ and $V(t)$. Relate: $V = \frac{4}{3}\pi r^3$, true at all times. Differentiate: $\dot{V} = 4\pi r^2 \dot{r}$. Substitute last, with $\dot{V} = -72$ and $r = 18$:

$$\dot{r} = \frac{-72}{4\pi(18)^2} = \frac{-72}{1296\pi} = -\frac{1}{18\pi} \text{ cm/s} = -0.0176839 \text{ cm/s},$$

which in centimetres per minute is $-\frac{60}{18\pi} = -\frac{10}{3\pi} = -1.0610330$. The relative form gets there in one line: $\dot{V}/V = 3\dot{r}/r$ with $V = \frac{4}{3}\pi(18)^3$.

A second, where the reduction step does real work. Water pours into an inverted cone of top radius 5 and height 10 at $8 \text{ m}^3/\text{min}$. How fast is the depth rising at $h = 6$?

Name $h(t)$, $r(t)$, $V(t)$. Relate: $V = \frac{1}{3}\pi r^2 h$, but this has two unknown rates. Reduce using similar triangles, $r/h = 5/10$, so $r = h/2$ at all times and

$$V = \frac{\pi}{3} \cdot \frac{h^2}{4} \cdot h = \frac{\pi h^3}{12}, \quad \dot{V} = \frac{\pi h^2}{4} \dot{h}.$$

At $h = 6$: $8 = \frac{36\pi}{4} \dot{h} = 9\pi \dot{h}$, so $\dot{h} = \frac{8}{9\pi} = 0.2829421 \text{ m/min}$. Had the reduction been skipped, the answer would have needed $\dot{r}$, which was never given.

The ladder, which exists to punish early substitution. A ladder of length 10 leans against a wall; its foot is pulled away at 1 m/s . How fast does the top descend when the foot is 6 from the wall?

Name $x(t)$ and $y(t)$. Relate: $x^2 + y^2 = 100$, true at every instant because the ladder is rigid. Differentiate: $2x\dot{x} + 2y\dot{y} = 0$, so $\dot{y} = -\frac{x\dot{x}}{y}$. Substitute last: at $x = 6$ we get $y = 8$ from the relation, and

$$\dot{y} = -\frac{6 \cdot 1}{8} = -0.75 \text{ m/s}.$$

Now watch the error. Writing the relation as $6^2 + y^2 = 100$ at the outset, because $x = 6$ “now”, gives $2y\dot{y} = 0$ and hence $\dot{y} = 0$: the top of the ladder never moves. The relation was true at that instant and false as an identity, and differentiating it destroyed the problem. As $x \rightarrow 10$ the formula gives $\dot{y} \rightarrow -\infty$, which is the physical statement that the top accelerates without bound as the ladder flattens.

WHERE THIS GOES WRONG

Substituting the instantaneous numbers before differentiating, which turns a variable into a constant and kills its rate. Next in frequency are the unit traps: per second against per minute, and diameter against radius. Third is differentiating a relation that holds only at the instant in question, such as the Pythagorean relation for a ladder written with the current lengths already inserted.

SEE ALSO Related rates by geometric reduction; implicit differentiation; the multivariable chain rule.

The pipeline is not where these problems are lost. They are lost at step two, and the reason is that the obvious relation usually has too many variables in it. What distinguishes a fluent solver is a stock of geometric reductions that collapse a three-variable picture into a two-variable one before any calculus starts.

RELATIVE RATES

For any power relation $Q = cu^k$, dividing the differentiated equation by the original gives $\dot{Q}/Q = k\dot{u}/u$, a statement about *percentages* that needs neither c nor the absolute size of anything. A sphere whose radius grows by one per cent gains three per cent of volume and two per cent of surface area, at any radius. Reach for this form whenever the data or the question is phrased as a rate of change relative to current size, and for products and quotients generally, where it becomes a sum of relative rates.

MEDIUM

16.2 Related rates by geometric reduction

CUE A configuration whose naive relation carries more variables than you have rates for: a conical tank, a ladder, a shadow, a hinged pair of covers, a triangle with a changing vertex angle, a chain of inscribed and circumscribed figures.

METHOD Find the *single* clean formula before differentiating anything. The reductions that recur are worth knowing by name: similar triangles, giving a constant ratio such as r/h ; the chord $d = 2R \sin(\theta/2)$ for a fixed-length hinge; the law of sines $R = \frac{a}{2 \sin A}$ for a circumradius; and ratio chaining through an intermediate quantity, where each link contributes its own chain-rule factor. Choose the parameter that makes the relation shortest, which is frequently an angle rather than a length.

Two book covers each of length 25 cm are hinged at the spine, and the opening angle θ increases at 0.1 rad/s. How fast do the free edges separate when $\theta = \pi/2$?

The law of cosines would give $d^2 = 25^2 + 25^2 - 2 \cdot 25^2 \cos \theta$, which is correct and clumsy. The chord formula reduces it at sight: the free edges lie on a circle of radius 25 centred at the hinge, so

$$d = 2 \cdot 25 \sin \frac{\theta}{2} = 50 \sin \frac{\theta}{2}, \quad \dot{d} = 25 \cos \frac{\theta}{2} \dot{\theta}.$$

At $\theta = \pi/2$: $\dot{d} = 25 \cdot \frac{\sqrt{2}}{2} \cdot 0.1 = 1.25\sqrt{2} = 1.7677670$ cm/s.

A chaining example. A lamp 6 m high, a walker 1.8 m tall moving away at 1.5 m/s: how fast does the tip of the shadow move, and how fast does the shadow lengthen?

Let x be the walker's distance from the lamp base and s the shadow length, so the tip is at $x + s$. Similar triangles give $\frac{s}{x+s} = \frac{1.8}{6}$, hence $6s = 1.8x + 1.8s$ and $s = \frac{3}{7}x$ at all times. Then $\dot{s} = \frac{3}{7}(1.5) = \frac{9}{14} = 0.6428571$ and the tip moves

at $\dot{x} + \dot{s} = \frac{10}{7}(1.5) = \frac{15}{7} = 2.1428571$ m/s. The two answers are different questions and are routinely confused; the tip moves faster than the walker, the shadow grows more slowly.

An angular reduction, where choosing the angle as the parameter is what makes the problem short. A camera stands 50 m from a straight track and turns to keep a car in frame; the car travels at 30 m/s. How fast does the camera rotate when the car is 50 m from the point of closest approach?

Let x be the car's displacement from that point and θ the camera's angle from the perpendicular. Then $\tan \theta = \frac{x}{50}$ at all times, and differentiating,

$$\sec^2 \theta \dot{\theta} = \frac{\dot{x}}{50}.$$

At $x = 50$ the angle is $\pi/4$ and $\sec^2 \theta = 2$, so $\dot{\theta} = \frac{30}{50 \cdot 2} = 0.3$ rad/s. Solving instead for $\theta = \arctan(x/50)$ and differentiating gives $\dot{\theta} = \frac{50\dot{x}}{2500 + x^2}$, the same number by a longer route; the implicit form avoids the arctangent entirely.

WHERE THIS GOES WRONG

Grinding the law of cosines when the chord formula is available, and more generally differentiating a relation before checking whether a shorter one exists. The second trap belongs to chaining: when a rate passes through an intermediate variable, every link contributes a factor, and dropping one link produces an answer with the right units and the wrong size.

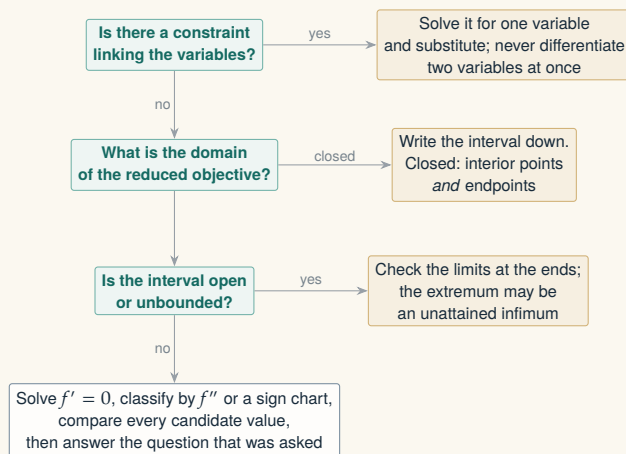
SEE ALSO Related rates: the standard pipeline; parametric derivatives; constrained optimisation by one-variable reduction.

ANSWER THE QUESTION THAT WAS ASKED

A configuration usually contains several rates, and they are not interchangeable. The shadow's *length* and the shadow's *tip* move at different speeds; the water's depth and the water's surface radius do too; the ladder's top and the area of the triangle it forms do too. Before substituting, reread the sentence and identify which single quantity the question names. Producing a correct rate for the wrong quantity is the most expensive way to be right.

Constrained optimisation

Optimisation problems have the same character: the calculus is one derivative, and the difficulty is in reducing a description to a single-variable function together with the interval on which it lives. Two things get forgotten with remarkable regularity, and both are part of the setup rather than the calculus. The first is the constraint, which is what makes the problem one-dimensional. The second is the domain, which is what makes the answer global rather than merely local.



The tree's lowest box is the only place any differentiation occurs, and that is the point of drawing it. Two of the three questions above it are about bookkeeping, and they are where marks are lost: a constraint left unused leaves a function of two variables that a single derivative cannot handle, and a domain left unstated turns a correct critical point into an unjustified claim of a global optimum.

THE OPTIMISATION PROTOCOL

1. Name the variables and write the objective, in as many variables as it naturally needs. 2. Write the constraint as an equation. 3. Solve the constraint for one variable and substitute, obtaining a function of one variable. 4. State the domain explicitly, deduced from the physical or algebraic restrictions, not guessed. 5. Solve $f' = 0$ inside the domain. 6. Evaluate the objective at every interior critical point *and* at every endpoint, and compare the numbers. 7. Answer the question actually asked, which may be the optimising variable, the optimal value, or a ratio.

16.3 Constrained optimisation by one-variable reduction

CORE

CUE Maximise or minimise a quantity subject to a stated constraint: a wire cut into two shapes, a box of fixed volume, the closest point of a curve to a given point, a rectangle inscribed in a region.

METHOD Eliminate one variable with the constraint, differentiate the resulting one-variable objective, solve $f' = 0$, and classify with f'' or a sign chart. Then compare against the endpoints: an interior critical point is not automatically a global optimum, and on an open region the extremum may be an infimum that is never attained. Two standing simplifications: minimise squared distance rather than distance, since $t \mapsto \sqrt{t}$ is increasing and the square root only complicates the derivative; and substitute $u = x^2$ when the objective is even, which halves the degree. Lagrange multipliers, $\nabla f = \lambda \nabla g$, are the multivariable alternative when the constraint will not solve cleanly.

WORKED EXAMPLE

A wire of length 16 is cut into two pieces, one bent into a circle and the other into a square. Minimise the total area. Let c be the circle's perimeter, so the square's is $16 - c$ and $0 \leq c \leq 16$, a closed interval: both endpoints are legal and must be checked. The circle has radius $c/2\pi$ and area $c^2/4\pi$, so

$$A(c) = \frac{c^2}{4\pi} + \left(\frac{16-c}{4}\right)^2, \quad A'(c) = \frac{c}{2\pi} - \frac{16-c}{8} = 0.$$

Clearing denominators, $8c = 2\pi(16 - c)$, so $c = \frac{16\pi}{4 + \pi} = 7.0384135$. Since $A'' = \frac{1}{2\pi} + \frac{1}{8} > 0$ everywhere, this is the global minimum on the interval, with

$$A_{\min} = \frac{64}{4 + \pi} = 8.9615865,$$

and the square receives $16 - c = \frac{64}{4 + \pi}$, numerically equal to $A_{\min}$ by coincidence of this data, and the larger of the two pieces. The endpoints tell the other half of the story: $A(0) = 16$, all wire to the square, and $A(16) = \frac{256}{4\pi} = 20.3718327$, all wire to the circle. So the *maximum* sits at an endpoint, where $A' \neq 0$, and no derivative would ever have found it.

Now the squared-distance simplification. Find the point of $y = x^2$ closest to $(0, 3)$. Minimise

$$D(x) = x^2 + (x^2 - 3)^2,$$

not its square root. Substituting $u = x^2 \geq 0$ gives $D = u + (u - 3)^2 = u^2 - 5u + 9$, a parabola with minimum at $u = \frac{5}{2}$, which lies in the domain $u \geq 0$. Then $D = \frac{11}{4}$ and the distance is $\frac{\sqrt{11}}{2} = 1.6583124$, attained at $x = \pm\sqrt{5/2}$: two closest points, by symmetry. Verified by a grid search: 2.7500000.

A third, where the constraint is a volume and the objective is material. An open-topped box has a square base of side x and height h , with volume 32000 cm^3 ; minimise the surface area. The constraint $x^2h = 32000$ gives $h = 32000/x^2$, and the objective, base plus four sides, becomes

$$S(x) = x^2 + 4xh = x^2 + \frac{128000}{x}, \quad x > 0.$$

Then $S'(x) = 2x - \frac{128000}{x^2} = 0$ gives $x^3 = 64000$, so $x = 40$ and $h = 20$: the optimal box is twice as wide as it is tall, a ratio independent of the volume. The minimum is $S = 1600 + 3200 = 4800 \text{ cm}^2$, confirmed by a grid search. The domain here is the open ray $(0, \infty)$, on which $S \rightarrow \infty$ at both ends, so the single interior critical point must be the global minimum and no endpoint comparison is possible or needed.

A fourth, on a closed interval where an endpoint is genuinely a candidate. Inscribe a rectangle with its base on the x -axis under the parabola $y = 12 - x^2$. With half-width x , the height is $12 - x^2$ and

$$A(x) = 2x(12 - x^2) = 24x - 2x^3, \quad 0 \leq x \leq 2\sqrt{3}.$$

Then $A'(x) = 24 - 6x^2 = 0$ gives $x = 2$, and $A(2) = 32$. The endpoints give $A(0) = 0$ and $A(2\sqrt{3}) = 0$, degenerate rectangles of zero width and zero height respectively, so the interior point wins and the maximum area is 32. The endpoint check took one line and is the difference between a proof and an assertion.

And the open-domain warning. The condition $\operatorname{Re}(z^2) > 0$ says $x^2 > y^2$, that is $|x| > |y|$, an *open* double wedge. The least value of $|z + i|$ over that set is the distance from $(0, -1)$ to the line $y = x$, namely $\frac{\sqrt{2}}{2} = 0.7071068$, but that value is an infimum and is not attained, because the boundary line is excluded. A problem asking for the minimum has no answer; a problem asking for the greatest lower bound has this one.

WHERE THIS GOES WRONG

Reporting an interior critical point as the global optimum without checking the endpoints, which in the wire problem would have you announce a maximum area that is actually the minimum. Second, ignoring a stated restriction such as $-8 < x < 8$, which can move the answer to a boundary or destroy it. Third, minimising distance rather than squared distance, which is not wrong but manufactures an unnecessary chain rule and a denominator that can hide a sign.

SEE ALSO Optimising a power constraint via logarithms; sign charts for monotonicity and concavity; common tangents and prescribed-angle tangents.

Classifying a critical point, and when the second derivative is silent

Once $f'(c) = 0$ has been solved, something must decide whether c is a maximum, a minimum, or neither. Three instruments are available and they are not equally good in all situations.

The *second-derivative test* says that $f''(c) > 0$ gives a local minimum and $f''(c) < 0$ a local maximum. It is fast when f'' is cheap, and it is completely silent when $f''(c) = 0$: at the origin x^4 has a minimum, $-x^4$ has a maximum, and x^3 has neither, and all three have vanishing first and second derivatives there. A vanishing f'' is not evidence of an inflection; it is the absence of evidence.

The *sign chart* of f' never fails and needs no second derivative, which makes it the right choice when f' arrives already factored or when f'' is unpleasant. It also handles endpoints and points where f' does not exist, both invisible to the second-derivative test.

The *global comparison* is the only one that answers the question actually asked in an optimisation problem. Local classification tells you the shape near c ; the problem wants the best value on the whole domain. Evaluate the objective at every interior critical point and at every endpoint, then compare the numbers. In the wire problem of the previous block, $A'' > 0$

everywhere settled the interior point instantly, but the maximum lived at an endpoint where the second-derivative test had nothing to say.

A convexity shortcut is worth keeping in reserve. If $f'' > 0$ on the whole domain, any critical point is the unique global minimum and no comparison is needed; the same with the signs reversed for a concave objective. This is what makes the wire, box and closest-point problems collapse to a single line of justification.

Reduction by substitution assumes the constraint can be solved. When the constraint is exponential, with the unknowns in both base and exponent, the direct solution is unusable and the right first move is not calculus at all: take logarithms, which linearises the constraint and makes the elimination trivial.

16.4 Optimising a power constraint via logarithms

HARD

CUE The constraint is itself a power relation, $x^y = C$ or $x^2y^3 = C$, with a product or a sum to be optimised.

METHOD Take logarithms of the constraint, turning it into a linear relation among $\ln x$ and $\ln y$, solve for one of them, and reduce to a single variable. One structural payoff recurs so often it is worth memorising:

$$\frac{d}{dx} \frac{x}{\ln x} = \frac{\ln x - 1}{(\ln x)^2} = 0 \text{ at } x = e,$$

independently of every constant in the problem. Keep track of the domain that the logarithm imposes, which is usually $x > 1$ so that $\ln x > 0$.

WORKED EXAMPLE

Let $x^y = 2026$ with $x, y > 0$. Minimise xy .

Taking logarithms, $y \ln x = \ln 2026$, so $y = \frac{\ln 2026}{\ln x}$, which requires $x > 1$ for $y > 0$. The objective becomes a one-variable function on $(1, \infty)$:

$$P(x) = xy = \ln 2026 \cdot \frac{x}{\ln x}.$$

The constant $\ln 2026 > 0$ is inert, so the minimiser is the minimiser of $x/\ln x$, which by the displayed identity is $x = e$. Checking the ends, $P \rightarrow \infty$ as $x \rightarrow 1^+$ and as $x \rightarrow \infty$, so $x = e$ is the global minimum, and

$$\min(xy) = e \ln 2026 = 20.6965050,$$

confirmed by a grid search over $(1, 21)$ giving 20.6965050. Note the structure of the answer: it is $e \ln C$ for every $C > 1$, and the optimising x never depends on C at all. The corresponding y is $\ln 2026 = 7.6138$.

A second instance, where the logarithm is a convenience rather than a necessity but still shortens the work. Maximise x^2y^3 subject to $x + y = 5$ with $x, y > 0$. Taking logarithms turns the product into a sum, so maximise

$$\Lambda(x) = 2 \ln x + 3 \ln(5 - x), \quad 0 < x < 5,$$

whose derivative $\frac{2}{x} - \frac{3}{5-x}$ vanishes when $2(5-x) = 3x$, that is $x = 2$, and then $y = 3$. The maximum value is $2^23^3 = 108$, confirmed by a grid search over $(0, 5)$. Note the shape of the answer: the constraint total 5 is split in the ratio 2 : 3 of the exponents, which is the general rule for maximising $x^p y^q$ on $x + y = c$, and the logarithm is what makes it visible.

WHERE THIS GOES WRONG

Differentiating x^y with y treated as a constant, which is the general power rule applied where both base and exponent vary. Second, losing the branch condition: on $0 < x < 1$ the logarithm is negative, $y < 0$, and the constraint has no solution in the intended range, so the domain is $(1, \infty)$ and not $(0, \infty)$.

SEE ALSO Constrained optimisation by one-variable reduction; the general power rule for $f(x)^{g(x)}$; implicit differentiation with a variable exponent.

Lines attached to a curve

Tangents and normals are the simplest applied objects and they are also the connective tissue of this chapter: the linearisation is the tangent read as an approximation, Newton's method is the tangent read as a root-finder, and the mean value theorem is a statement that some tangent is parallel to a chord. Getting the two formulas exactly right, including which point each ingredient is evaluated at, pays repeatedly.

16.5 Tangent lines and normal lines

CUE “The tangent at $x = a$ ”, “the normal at P ”, an intercept of one of them, or unknown parameters constrained by values and slopes simultaneously.

METHOD The tangent is $y = f(a) + f'(a)(x - a)$. The normal has slope $-1/f'(a)$, so $y = f(a) - \frac{x - a}{f'(a)}$. Two intercepts are worth knowing outright: the tangent meets the x -axis at

$$a - \frac{f(a)}{f'(a)},$$

which is exactly Newton’s step, and the normal meets it at $a + f(a)f'(a)$. When parameters are unknown, the value conditions and the slope conditions together form a small system; solve it before differentiating anything else.

WORKED EXAMPLE

Let $f(x) = e^{ax^2+bx+1}$ and suppose $f(1) = f(0) = f'(0)$. Find the normal at $x = 1$ and its x -intercept.

The exponential never vanishes, so $f'(x) = (2ax + b)f(x)$. At $x = 0$: $f(0) = e$ and $f'(0) = be$, and the condition $f'(0) = f(0)$ gives $b = 1$. The condition $f(1) = f(0)$ reads $e^{a+b+1} = e$, so $a + b + 1 = 1$ and $a = -1$. Then

$$f'(1) = (2a + b)f(1) = (-2 + 1)e = -e.$$

The normal at $(1, e)$ therefore has slope $-1/(-e) = 1/e$ and equation $y - e = \frac{x - 1}{e}$. Setting $y = 0$ gives $x - 1 = -e^2$, so the x -intercept is $1 - e^2 = -6.3890561$. The shortcut formula agrees: $a + f(a)f'(a) = 1 + e(-e) = 1 - e^2$.

A tangent from an external point, which is the same formula with the unknown moved. Find every line through $(0, -1)$ tangent to $y = x^2$. The point of tangency is unknown, so call it a ; the tangent there is $y = a^2 + 2a(x - a)$, and requiring it to pass through $(0, -1)$ gives

$$-1 = a^2 - 2a^2 = -a^2,$$

so $a = \pm 1$ and the two tangents are $y = 2x - 1$ and $y = -2x - 1$. Note what was *not* done: no attempt was made to write the line as $y = mx - 1$ and impose a repeated root, which works for a quadratic and for nothing else. Parametrising by the point of tangency is the move that generalises.

A lemma worth carrying, since tangent problems on cubics recur. The tangent to $y = x^3$ at $x = a$ meets the curve again at $x = -2a$. To see it, subtract:

$$x^3 - (a^3 + 3a^2(x - a)) = x^3 - 3a^2x + 2a^3 = (x - a)^2(x + 2a),$$

where the double factor $(x - a)^2$ is the algebraic signature of tangency and the remaining linear factor names the second intersection. At $a = 2$ the tangent is $y = 12x - 16$ and it recrosses at $x = -4$, as the identity predicts. The same subtract-and-factor move identifies tangency for any polynomial: a tangent produces a repeated root, and the leftover factor tells you where else the line goes.

WHERE THIS GOES WRONG

Writing $-1/f'(a)$ when $f'(a) = 0$. There the tangent is horizontal and the normal is the vertical line $x = a$, which has no slope; the formula must be abandoned, not patched. The mirror case is a vertical tangent, where the normal is horizontal. Second, confusing the point on the curve with the point of evaluation: every ingredient of both formulas is evaluated at a , and $f(a)$ is a number, not a function.

SEE ALSO Common tangents and prescribed-angle tangents; linearisation and differentials; Newton’s method as one step of an iteration.

One line and one curve is a single condition. One line and two curves is a system, and it is the standard way a tangent problem is made hard. The essential recognition is that the two points of tangency are independent unknowns, so there are two of them and you need two equations.

16.6 Common tangents and prescribed-angle tangents

CUE One line tangent to *two* curves; or a line required to meet a given tangent, or another curve, at a stated angle.

METHOD For a common tangent, let the line touch f at a and g at b , with a and b unrelated, and impose two conditions:

$$f'(a) = g'(b) \quad (\text{equal slopes}), \quad f'(a) = \frac{g(b) - f(a)}{b - a} \quad (\text{collinearity}).$$

Solve for (a, b) ; expect more than one solution. For a prescribed angle, use

$$\tan \theta = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right|,$$

solved for the unknown slope, and remember that the angle between two *curves* means the angle between their tangents at the intersection point.

WORKED EXAMPLE

Find every common tangent to $y = x^2$ and $y = -(x - 2)^2$.

Touch the first at a and the second at b . Equal slopes: $2a = -2(b - 2)$, so $a = 2 - b$, that is $b = 2 - a$. Collinearity, with $f(a) = a^2$ and $g(b) = -(b - 2)^2 = -a^2$ after substituting $b - 2 = -a$:

$$2a = \frac{-a^2 - a^2}{b - a} = \frac{-2a^2}{2 - 2a} = \frac{-a^2}{1 - a}.$$

Cross-multiplying, $2a(1 - a) = -a^2$, so $2a - 2a^2 = -a^2$ and $a^2 = 2a$, giving $a = 0$ or $a = 2$. For $a = 0$: $b = 2$, the line is $y = 0$, tangent to the parabola at its vertex and to the downward parabola at its vertex. For $a = 2$: $b = 0$, slope 4, through $(2, 4)$, so $y = 4x - 4$; checking on the second curve, $g(0) = -4$ and $g'(0) = 4$, and the line gives $y(0) = -4$. Both check exactly. Note that the two tangency points are different in both solutions, which is the whole reason two unknowns were needed.

A prescribed angle. Let $f(x) = \sin\left(\frac{\pi x}{2}\right) \cos\left(\frac{\pi x}{4}\right)$. Then

$$f'(x) = \frac{\pi}{2} \cos \frac{\pi x}{2} \cos \frac{\pi x}{4} - \frac{\pi}{4} \sin \frac{\pi x}{2} \sin \frac{\pi x}{4},$$

and at $x = 4$ the second term dies because $\sin 2\pi = 0$, while the first is $\frac{\pi}{2} \cdot 1 \cdot (-1)$, so $f'(4) = -\frac{\pi}{2} = -1.5707963$, verified by central difference. Now find the line through that point of tangency meeting the tangent at 60° with the larger slope. With $m_1 = -\pi/2$ and $\tan 60^\circ = \sqrt{3}$, dropping the absolute value in the direction that gives $m_2 > m_1$,

$$m_2 = \frac{\sqrt{3} + m_1}{1 - \sqrt{3} m_1} = \frac{\sqrt{3} - \pi/2}{1 + \sqrt{3} \pi/2} = 0.0433398,$$

and substituting back, $\frac{m_2 - m_1}{1 + m_1 m_2} = 1.7320508 = \sqrt{3}$ exactly as required.

The angle between two curves is the same formula with both slopes known. The curves $y = x^2$ and $y = x^3$ meet where $x^2 = x^3$, at $x = 0$ and $x = 1$. At the origin the slopes are 0 and 0, so the curves are tangent and the angle is zero. At $(1, 1)$ the slopes are 2 and 3, so

$$\tan \theta = \left| \frac{3 - 2}{1 + 6} \right| = \frac{1}{7}, \quad \theta = \arctan \frac{1}{7} = 0.1418971 \text{ rad} = 8.13^\circ.$$

Nothing about the curves is used beyond their derivatives at the shared point, which is why the notion is well defined at all.

WHERE THIS GOES WRONG

Assuming a common tangent must touch both curves at the same x . It need not, and imposing $a = b$ turns a solvable system into an inconsistent one. Second, taking only one root of the angle equation: the absolute value produces two candidate slopes, symmetric about the given line, and the problem usually specifies which by a side condition.

SEE ALSO Tangent lines and normal lines; differentiating an area function and the tangent-chord lemmas; constrained optimisation by one-variable reduction.

Approximation: the tangent as a numerical instrument

The tangent line is the best linear model of a function near a point, and two entire techniques are nothing but that sentence used in opposite directions. Linearisation asks what f is worth near a , given $f(a)$ and $f'(a)$. Newton's method asks where f vanishes near a , given the same two numbers. Both are exact statements about the tangent and approximate statements about f , and the error in both is governed by f'' .

EASY

16.7 Linearisation and differentials

CUE “Approximate”, “estimate the change in”, a small perturbation Δx , a propagated measurement error, or a percentage change.

METHOD Use $L(x) = f(a) + f'(a)(x - a)$ with a the nearest point where f is known exactly. In differential notation, $dy = f'(x) dx$ so $\Delta y \approx f'(x)\Delta x$. For relative error, divide:

$$\frac{dy}{y} = \frac{f'(x)}{f(x)} dx,$$

which is the logarithmic derivative, so for $y = cx^k$ a relative error of ϵ in x produces $k\epsilon$ in y . The size of the error is controlled by the Lagrange remainder $\frac{1}{2}f''(\xi)(x - a)^2$, and its *sign* is the sign of f'' : a convex f lies above its tangent, so L underestimates.

WORKED EXAMPLE

Estimate $\sqrt{101}$. Take $f(x) = \sqrt{x}$ and $a = 100$, where $f(a) = 10$ and $f'(a) = \frac{1}{20}$:

$$\sqrt{101} \approx 10 + \frac{1}{20} = 10.05.$$

The true value is 10.0498756, so the error is 1.244×10^{-4} . The remainder bound predicts this well: $f''(x) = -\frac{1}{4}x^{-3/2}$, so $|\frac{1}{2}f''(\xi)| \leq \frac{1}{8} \cdot 100^{-3/2} = 1.25 \times 10^{-4}$. And the sign is right: $\sqrt{\cdot}$ is concave, $f'' < 0$, so the tangent lies above the curve and 10.05 is an overestimate.

Error propagation, where the relative form does all the work. A sphere's radius is measured as 12 cm with uncertainty 0.05 cm. From $V = \frac{4}{3}\pi r^3$,

$$\frac{dV}{V} = 3 \frac{dr}{r} = 3 \cdot \frac{0.05}{12} = 1.25\%,$$

so the volume is uncertain by 1.25 per cent, and in absolute terms $dV = 4\pi(12)^2(0.05) = 90.4778656 \text{ cm}^3$. No value of V was needed for the percentage, which is the advantage of the logarithmic form.

The comparison question, in full. For $f(x) = \sqrt{x}$ at $a = 100$ with $\Delta x = 1$: the differential is $dy = f'(100) \cdot 1 = 0.0500000$, while the true change is $\Delta y = \sqrt{101} - 10 = 0.0498756$. So $dy > \Delta y$ here, and the reason is structural rather than numerical: $f'' < 0$ makes f concave, the tangent lies above the curve, and the linear estimate of the increase therefore overshoots. Reverse the concavity and the inequality reverses with it. Any question of the form “is dy larger or smaller than Δy ” is a question about the sign of f'' and can be answered without computing either quantity.

And the refinement, since the remainder term is not merely a bound but the leading correction. Adding the quadratic term of the Taylor expansion,

$$\sqrt{101} \approx 10 + \frac{1}{20} + \frac{f''(100)}{2}(1)^2 = 10.05 - \frac{1}{8000} = 10.0498750,$$

against the true 10.0498756: the error has fallen from 1.24×10^{-4} to 6×10^{-7} , roughly the cube of the displacement over the appropriate scale. That is the general pattern, and it is the reason a linearisation is trusted near a and abandoned away from it.

WHERE THIS GOES WRONG

Using the estimate far from a , where the quadratic remainder is no longer small; the linearisation is a local statement and its error grows like the square of the displacement. Second, confusing dy with Δy when the question asks which is larger. They differ by the remainder, so the comparison is decided entirely by the sign of f'' , and answering without checking it is a coin flip.

SEE ALSO Newton's method as one step of an iteration; tangent lines and normal lines; the total differential.

Reading the same tangent line the other way turns approximation into root-finding. It is worth presenting Newton's method as a fixed-point iteration rather than as a formula, because that framing explains both why it converges so fast and exactly when it stops doing so.

16.8 Newton's method as one step of an iteration

CUE “One iteration of Newton’s method”, “improve the approximation to the root”, or a recursion presented without its origin.

METHOD The iteration is

$$x_{n+1} = N(x_n), \quad N(x) = x - \frac{f(x)}{f'(x)},$$

which is the x -intercept of the tangent at x_n . Read it as a fixed-point iteration for N : a root r of f is a fixed point of N , and

$$N'(x) = \frac{f(x)f''(x)}{f'(x)^2},$$

which vanishes at a simple root because $f(r) = 0$ while $f'(r) \neq 0$. A fixed-point iteration with $N'(r) = 0$ converges quadratically, which is the whole explanation. At a root of multiplicity m the limit of N' is $1 - 1/m \neq 0$, so convergence degrades to linear with exactly that ratio. It fails outright when $f'(x_n)$ is near zero.

WORKED EXAMPLE

Take $f(x) = x^2 - 2$ and $x_0 = 1$. Then $f'(x) = 2x$ and

$$x_1 = 1 - \frac{1-2}{2} = \frac{3}{2}, \quad x_2 = \frac{3}{2} - \frac{\frac{9}{4} - 2}{3} = \frac{3}{2} - \frac{1}{12} = \frac{17}{12} = 1.4166667,$$

against $\sqrt{2} = 1.4142136$: the error falls from 8.6×10^{-2} to 2.5×10^{-3} , roughly squaring, as promised.

Now a multiple root, to see the degradation concretely. Take $f(x) = (x-1)^3$ and $x_0 = 2$. Here $f/f' = (x-1)^3/3(x-1)^2 = (x-1)/3$, so the iteration is exactly

$$x_{n+1} = x_n - \frac{x_n - 1}{3} = \frac{2x_n + 1}{3},$$

whose error satisfies $e_{n+1} = \frac{2}{3}e_n$ precisely. That is $1 - 1/m$ with $m = 3$. Starting from $e_0 = 1$, four steps give $x_4 = 1.1975309$ with $e_4 = (2/3)^4 = 0.1975309$: linear, not quadratic, and no amount of iterating changes the ratio.

A failure, so that the hypotheses are not decorative. Take $f(x) = x^3 - 2x + 2$ with $x_0 = 0$. Then $f(0) = 2$ and $f'(0) = -2$, so $x_1 = 0 - \frac{2}{-2} = 1$; and $f(1) = 1$ with $f'(1) = 1$, so $x_2 = 1 - \frac{1}{1} = 0$. The iteration cycles between 0 and 1 forever and never approaches the real root near -1.7693 . Nothing is wrong with the arithmetic: N simply has a two-cycle, and a fixed-point iteration is under no obligation to converge from an arbitrary start. Quadratic convergence is a statement about starting *near* a simple root, not about starting anywhere.

The fixed-point framing pays a second dividend: it tells you what to expect from iterations that are *not* Newton's. Any rearrangement of $f(x) = 0$ into $x = g(x)$ gives an iteration, and it converges near a fixed point r exactly when $|g'(r)| < 1$, at a linear rate equal to $|g'(r)|$. Solving $x = \cos x$ by the naive iteration $x_{n+1} = \cos x_n$ from $x_0 = 1$ converges to $r = 0.7390851$, because $g' = -\sin$ and $|g'(r)| = 0.6736120 < 1$; each step buys about 0.17 of a decimal digit. Newton's method on $f(x) = x - \cos x$ uses $N(x) = x - \frac{x - \cos x}{1 + \sin x}$ and reaches 0.7390851 from the same start in four steps, with the iterates 0.7503639, 0.7391129, 0.7390851. The difference is entirely accounted for by $N'(r) = 0$ against $g'(r) = -0.67$.

WHERE THIS GOES WRONG

The sign of the correction. It is $x - f/f'$, and writing $x + f/f'$ produces an iteration that runs away from the root at exactly the same speed. Second, claiming quadratic convergence at a repeated root: the observed ratio is $1 - 1/m$, and reporting the number of correct digits as doubling per step is then simply false. Third, iterating without checking $f'(x_n)$: a near-horizontal tangent throws the next iterate arbitrarily far away.

SEE ALSO Linearisation and differentials; tangent lines and normal lines; fixed-point iteration and contraction.

Linearisation uses f' and treats f'' as an error term. Curvature does the opposite: it is the invariant built from f'' that says how far the curve departs from its tangent, and it is the natural second-order companion to the two techniques above.

THE ERROR HIERARCHY

Three statements about the same tangent, in increasing strength. *Zeroth order*: $f(x) \approx f(a)$, error $O(x-a)$, controlled by f' . *First order*: $f(x) \approx L(x)$, error $\frac{1}{2}f''(\xi)(x-a)^2$, controlled by f'' . *Second order*: add $\frac{1}{2}f''(a)(x-a)^2$, error controlled by f''' . Each extra derivative buys one more power of the displacement, which is why linearisation is excellent nearby and worthless far away, and why curvature, the next block's subject, is the first quantity that says anything the tangent line does not.

16.9 Curvature and the osculating circle

CUE “Radius of curvature”, “osculating circle”, “where does the curve bend most sharply”, or a request for the circle that best fits a curve at a point.

METHOD For a graph $y = f(x)$,

$$\kappa = \frac{|y''|}{(1 + (y')^2)^{3/2}}, \quad R = \frac{1}{\kappa}.$$

For a parametrised curve,

$$\kappa = \frac{|\dot{x}\ddot{y} - \dot{y}\ddot{x}|}{(\dot{x}^2 + \dot{y}^2)^{3/2}},$$

whose numerator and denominator are exactly the ingredients of the parametric second-derivative formula. The osculating circle has radius R , lies on the concave side, and has centre at distance R along the normal. At a point where $y' = 0$ the graph formula collapses to $\kappa = |y''|$, which is worth choosing the coordinates to arrange.

WORKED EXAMPLE

For $y = x^2$ at the origin, $y' = 0$ and $y'' = 2$, so $\kappa = 2$ and $R = \frac{1}{2}$: the osculating circle is centred at $(0, \frac{1}{2})$ with radius $\frac{1}{2}$. Every parabola $y = ax^2$ has $R = 1/2|a|$ at its vertex. The centre is found by walking a distance R from the point along the normal, towards the concave side: here the normal at the origin is the y -axis and the curve opens upward, so the centre is at $(0, \frac{1}{2})$ and the osculating circle is $x^2 + (y - \frac{1}{2})^2 = \frac{1}{4}$. Solving the circle for its lower branch, $y = \frac{1}{2} - \frac{1}{2}\sqrt{1 - 4x^2} = x^2 + x^4 + \dots$, which agrees with the parabola to second order and first differs at fourth order; that agreement to second order is what “osculating” means.

Parametrically, take the ellipse $x = 3 \cos t$, $y = 2 \sin t$ at $t = 0$, the end of the major axis. Then $\dot{x} = -3 \sin t = 0$, $\dot{y} = 2 \cos t = 2$, $\ddot{x} = -3 \cos t = -3$, $\ddot{y} = -2 \sin t = 0$, so

$$\kappa = \frac{|0 \cdot 0 - 2 \cdot (-3)|}{(0 + 4)^{3/2}} = \frac{6}{8} = \frac{3}{4}, \quad R = \frac{4}{3},$$

in agreement with the classical value $b^2/a = 4/3$ at the end of the major axis. The Cartesian formula would first require solving for y and would meet a vertical tangent at exactly this point.

Where does $y = e^x$ bend most sharply? With $u = e^x$, $\kappa = u(1 + u^2)^{-3/2}$, and

$$\frac{d\kappa}{du} = (1 + u^2)^{-5/2}(1 + u^2 - 3u^2) = 0 \quad \text{at } u = \frac{1}{\sqrt{2}},$$

that is $x = -\frac{1}{2} \ln 2 = -0.3465736$, giving $\kappa_{\max} = \frac{2}{3\sqrt{3}} = 0.3849002$; a grid search over $[-3, 3]$ returns 0.3849002.

WHERE THIS GOES WRONG

Omitting the exponent $3/2$ or the absolute value in the numerator, both of which produce a quantity that is not curvature and is not even dimensionally right. Second, applying the Cartesian formula to a parametric curve without first eliminating the parameter, which is invalid and, at a vertical tangent, impossible. Third, forgetting that curvature is not invariant under rescaling the axes, so a picture cannot be used to check it.

SEE ALSO Parametric derivatives; higher implicit derivatives; sign charts for monotonicity and concavity.

Extracting shape, and the mean value theorems as tools

Everything so far produced a number: a rate, an optimal value, an approximation, a curvature. What remains asks instead for a description: how many extrema, on which intervals f increases, how many zeros f' can have. That change of goal changes the technique. Where a numerical answer wants f' evaluated, a qualitative answer wants f' factored, because sign information lives in the factors and is destroyed by expansion. And where no formula for f is supplied at all, the mean value theorems replace computation with a bound or a count.

The two instruments below are therefore the sign chart and the pair of mean value theorems, and both are used here as working tools rather than as items of theory.

16.10 Sign charts: monotonicity, concavity, and counting critical points

CUE f' arrives in *factored* form, or the question asks for a count of extrema, a sum of stationary values, or the integers satisfying an inequality built from f, f' and f'' .

METHOD Keep f' factored; never expand. A factor changes sign only at a root of *odd* multiplicity, and at such a root $+$ to $-$ marks a local maximum and $-$ to $+$ a local minimum. Positive factors may be discarded outright: since $e^g > 0$ always, $\text{sgn}(e^g)' = \text{sgn } g'$, so e^g has exactly the extrema of g . When the derivative carries an oscillatory phase, stop drawing and start counting lattice points: extrema of $\cos \varphi(x)$ occur where $\varphi(x) = k\pi$, so tally the integers k lying in the image interval $\varphi([a, b])$.

WORKED EXAMPLE

Let $f(x) = \cos(3^{x+1})$ on $[0, 3]$. Then

$$f'(x) = -3^{x+1} \ln 3 \sin(3^{x+1}),$$

and the prefactor $-3^{x+1} \ln 3$ never vanishes, so the critical points are exactly the solutions of $3^{x+1} = k\pi$. The phase $\varphi(x) = 3^{x+1}$ is increasing and maps $[0, 2)$ onto $[3, 27)$ and $[2, 3]$ onto $[27, 81]$. Counting integers k with $k\pi$ in each: from $3/\pi = 0.955$ to $27/\pi = 8.594$ gives $k = 1, \dots, 8$, so 8 extrema; from $27/\pi$ to $81/\pi = 25.783$ gives $k = 9, \dots, 25$, so 17. The ratio is $8/17$, and no equation was ever solved.

A nested case, where the sign chart is about which factor can vanish. Let $f(x) = \tan(\cos(\sin x))$ on $[-\pi/2, \pi/2]$. The chain rule gives

$$f'(x) = \sec^2(\cos \sin x) \cdot (-\sin(\sin x)) \cdot \cos x.$$

The first factor is never zero. The second vanishes when $\sin(\sin x) = 0$, that is $\sin x = 0$, that is $x = 0$. The third vanishes at $x = \pm\pi/2$. So the stationary points are $x = 0, \pm\pi/2$, and the corresponding values are $f(0) = \tan(\cos 0) = \tan 1$ and $f(\pm\pi/2) = \tan(\cos 1)$ twice. Hence

$$\sum f = \tan 1 + 2 \tan(\cos 1) = 1.5574077 + 2(0.5998406) = 2.7570890.$$

The pure multiplicity count deserves an instance of its own, since it is the form in which the technique is usually tested. Suppose

$$f'(x) = x^2(x-1)^3(x-2)(x+3)^4.$$

How many local extrema does f have? Four roots, but only those of odd multiplicity switch the sign: $x = 1$ with multiplicity 3 and $x = 2$ with multiplicity 1. The roots at 0 and -3 have even multiplicity and are stationary points that are not extrema. So the answer is two, and their types follow from one sign test each: at $x = 3$ the product is positive, at $x = 1.5$ it is negative, and at $x = 0.5$ it is positive, so f has a local maximum at $x = 1$ and a local minimum at $x = 2$. Expanding f' into a degree-ten polynomial would have destroyed every bit of this in one step.

Two further reductions belong here because they save whole pages. First, an inert positive wrapper: to locate the extrema of e^{x^3-3x} , do not differentiate the exponential at all. Since $e^g > 0$, the derivative $e^g g'$ has the sign of $g' = 3x^2 - 3$, so the extrema sit at $x = \pm 1$, a maximum at -1 and a minimum at 1 , exactly as for g itself. The same argument disposes of any strictly increasing wrapper: $\ln$, $\arctan$, odd powers, and $\sinh$ all preserve the location and the type of every extremum.

Second, poles. To solve $\frac{(x-1)(x-4)^2}{x-3} > 0$, mark the boundary points 1, 3, 4 and remember that 3 is a pole rather than a zero but still separates sign regions, while 4 is a double zero and therefore does *not*. Testing one point in each region gives positive on $(-\infty, 1)$, negative on $(1, 3)$, positive on $(3, 4)$ and positive on $(4, \infty)$, so the solution set is $(-\infty, 1) \cup (3, 4) \cup (4, \infty)$, with 4 removed because the expression vanishes there. A chart that lists only the zeros of the numerator gets this wrong twice over: it invents a sign change at 4 and misses the one at 3.

WHERE THIS GOES WRONG

Counting a double root as a sign change. The factor $(x-2)^2$ produces a stationary point that is not an extremum, and expanding f' destroys exactly the information needed to see that. Second, forgetting that a rational expression changes sign at its *poles* as well as its zeros, so an inequality such as $\frac{(x-1)}{(x-3)} > 0$ has its boundary at 3 even though nothing vanishes there.

SEE ALSO Constrained optimisation by one-variable reduction; one-sided derivatives, corners and cusps; Rolle and mean-value arguments as tools.

Sign charts need a formula for f' . The mean value theorems need none, which is exactly why they are the tool of choice when the problem withholds one and asks a question about counting or bounding anyway.

16.11 Rolle and mean-value arguments as tools

CUE A conclusion about how many zeros some derivative-built expression has, or a bound on $f(b)$ from bounds on f' , with no explicit formula for f anywhere in the problem.

METHOD Two statements, used in opposite directions. *Mean value theorem*: $\frac{f(b) - f(a)}{b - a} = f'(c)$ for some $c \in (a, b)$, so bounds on f' convert directly into bounds on an endpoint value. *Rolle*: between consecutive zeros of g lies a zero of g' , so n zeros of g force at least $n - 1$ zeros of g' . The strong form handles first-order operators: for $(1 + \lambda D)g$, multiply by the integrating factor $e^{x/\lambda}$ and note that

$$(e^{x/\lambda}g)' = \frac{e^{x/\lambda}}{\lambda}(g + \lambda g'),$$

so since $e^{x/\lambda}g$ has exactly the zeros of g , Rolle gives $(1 + \lambda D)g$ at least $n - 1$ zeros. Iterate for powers of the operator.

WORKED EXAMPLE

A bound from the mean value theorem. Suppose f is differentiable on $[0, 3]$ with $f(0) = 10$ and $-6.5 \leq f' \leq 8$ throughout. Applying the theorem on $[0, 3]$,

$$-6.5 \leq \frac{f(3) - 10}{3} \leq 8 \implies -19.5 \leq f(3) - 10 \leq 24 \implies -9.5 \leq f(3) \leq 34.$$

The integers available to $f(3)$ run from -9 to 34 , and there are 44 of them. Both bounds are attained by linear functions, so none can be improved.

A count from Rolle, in operator form. Let

$$F = f + 6f' + 12f'' + 8f'''.$$

Recognise the coefficients: $1, 6, 12, 8$ are $\binom{3}{k}2^k$ for $k = 0, 1, 2, 3$, so $F = (1 + 2D)^3f$ with $D = \frac{d}{dx}$. Suppose f has 5 zeros. Each application of $(1 + 2D)$ costs at most one zero by the integrating-factor argument with $\lambda = 2$, so $(1 + 2D)f$ has at least 4, then at least 3, then at least 2. Therefore F has at least 2 zeros. The binomial recognition is the whole trick; expanded, the expression looks like an arbitrary linear combination and nothing follows from it.

The mean value theorem also proves inequalities, which is its most useful everyday role. For any a, b , applying it to $\sin$ gives $\sin b - \sin a = (b - a) \cos c$ for some c , and since $|\cos c| \leq 1$,

$$|\sin b - \sin a| \leq |b - a|,$$

so $\sin$ is a contraction in the weak sense, with no trigonometric identity used anywhere. Similarly, for $0 < a < b$

apply it to $\ln$: $\ln b - \ln a = \frac{b - a}{c}$ with $a < c < b$, and since $\frac{1}{b} < \frac{1}{c} < \frac{1}{a}$,

$$\frac{b - a}{b} < \ln \frac{b}{a} < \frac{b - a}{a}.$$

Setting $b = 1 + x$ and $a = 1$ recovers $\frac{x}{1 + x} < \ln(1 + x) < x$ for $x > 0$, two of the standard logarithm bounds, from one application of the theorem.

And Rolle used as an upper bound, which is the direction people forget. How many real roots can $p(x) = x^3 - 3x + c$ have? Its derivative $3x^2 - 3$ has exactly two zeros, so by Rolle p can have at most three, and the count is settled by comparing c with the two stationary values $p(\pm 1) = c \mp 2$.

One structural consequence is worth recording, because it constrains what a derivative can look like and therefore what answers are possible. Darboux's theorem says that f' has the intermediate value property on any interval where it exists, even when f' is discontinuous. So a derivative can never jump: a function defined piecewise with slopes 1 on the left and 2 on the right is not the derivative of anything, and a problem asserting that it is has an error in it. The proof is a Rolle argument applied to $f(x) - cx$, which is exactly the kind of use this block is about.

WHERE THIS GOES WRONG

Asserting an exact zero count when Rolle supplies only a lower bound in one direction and only an upper bound in the other. "At least $n - 1$ " and "at most $k + 1$ " are different claims and neither is "exactly". Second, applying the mean value theorem without checking its hypotheses: continuity on the *closed* interval and differentiability on the *open* one. On $[-1, 1]$ the function $|x|$ satisfies neither conclusion nor hypothesis, and the failure is instructive.

SEE ALSO Sign charts for monotonicity and concavity; the Leibniz rule for higher derivatives and binomial recognition; differentiable but not C^1 .

The last technique in the chapter is where the two halves of Part III meet: a quantity defined as an area, so that differentiating it needs the Leibniz rule, and an optimisation whose answer is a statement about tangents. Both of the lemmas below are worth carrying, because they turn a page of integration into a line.

16.12 Differentiating an area function; tangent-chord lemmas

HARD

CUE A quantity defined as an area that depends on a moving point, with its rate wanted; sometimes with the moving point itself parametrised by a further variable.

METHOD Write the area as a definite integral with the moving point in the limits *and* inside the integrand, then apply the full Leibniz rule. The boundary term usually vanishes, because a tangent meets its curve at the very point in question. Two lemmas result. *First*: for the region between a curve and its tangent at x_p , cut off by the y -axis,

$$\frac{dA}{dx_p} = \frac{1}{2}x_p^2 f''(x_p).$$

Second, Archimedes' lemma: among points of a smooth arc, the one maximising the area of the triangle with two fixed endpoints is where the tangent is parallel to the chord, which is the mean value theorem read geometrically.

WORKED EXAMPLE

Derive the first lemma, since its sign is the part that gets lost. With $T(x) = f(p) + f'(p)(x - p)$ the tangent at $x_p = p$, put $A(p) = \int_0^p (f(x) - T(x)) dx$. Leibniz gives a boundary term $f(p) - T(p) = 0$ and an interior term $-\int_0^p \frac{\partial T}{\partial p} dx$, where

$$\frac{\partial T}{\partial p} = f'(p) + f''(p)(x - p) - f'(p) = f''(p)(x - p).$$

Hence

$$\frac{dA}{dp} = -f''(p) \int_0^p (x - p) dx = -f''(p) \left(\frac{p^2}{2} - p^2 \right) = \frac{p^2}{2} f''(p).$$

Sanity check on $f(x) = x^2$: the tangent at p is $2px - p^2$, the gap is $(x - p)^2$, so $A(p) = p^3/3$ and $A'(p) = p^2$, matching $\frac{1}{2}p^2 \cdot 2$. Numerically at $p = 1.3$ the derivative is 1.6900000 both ways. The sign is positive for a convex f , as it must be, since the region grows as p moves right.

Now the parametrised version. Suppose $f''(x) = \arctan(3^x)$ and the moving point is $x_p = \sin t$; find $\frac{dA}{dt}$ at $t = \pi/6$. Chain the lemma with $\frac{dx_p}{dt} = \cos t$:

$$\frac{dA}{dt} = \frac{x_p^2}{2} f''(x_p) \cos t.$$

At $t = \pi/6$: $x_p = \frac{1}{2}$, $f''(\frac{1}{2}) = \arctan \sqrt{3} = \frac{\pi}{3}$, and $\cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}$, so

$$\frac{dA}{dt} = \frac{1}{2} \cdot \frac{1}{4} \cdot \frac{\pi}{3} \cdot \frac{\sqrt{3}}{2} = \frac{\pi\sqrt{3}}{48} = 0.1133625.$$

Nothing about f beyond its second derivative was ever needed, which is the point of having the lemma.

Archimedes' lemma, proved and used. Fix $A = (a, f(a))$ and $B = (b, f(b))$ on a smooth arc and let $P = (p, f(p))$ move between them. The triangle ABP has base AB fixed, so its area is maximal exactly when the distance from P to the line AB is maximal. That distance is proportional to

$$D(p) = f(p) - \left(f(a) + \frac{f(b) - f(a)}{b - a}(p - a) \right),$$

and $D'(p) = 0$ precisely when $f'(p) = \frac{f(b) - f(a)}{b - a}$: the tangent at P is parallel to the chord AB . That is the mean value theorem, and it guarantees such a p exists. For the parabola $f(x) = x^2$ on $[a, b]$ the condition reads $2p = a + b$, so the optimal P is always at the midpoint abscissa, whatever a and b are; the resulting triangle has area $\frac{1}{8}(b - a)^3$, which is Archimedes' quadrature constant of $4/3$ away from the parabolic segment.

WHERE THIS GOES WRONG

Dropping the chain factor $\frac{dx_p}{dt}$ when the moving point is itself parametrised, which is the single commonest failure here and costs a factor of $\cos t$. Second, substituting the numerical value of the moving point into the area formula before differentiating, which is the related-rates error in a new costume. Third, getting the sign backwards by integrating tangent minus curve when the region is curve minus tangent; check it against $f(x) = x^2$, where the answer must be positive.

SEE ALSO The full Leibniz integral rule; constrained optimisation by one-variable reduction; common tangents and prescribed-angle tangents.

The same chain rule, four times over

It is worth seeing that the families in this chapter are not four subjects but one, differently packaged. In every case the object is a relation between quantities and the instrument is the chain rule; what changes is which variable is doing the driving.

In a related-rates problem the driving variable is time, which appears nowhere in the geometry. The relation $Q = g(u)$ is timeless, and $\dot{Q} = g'(u)\dot{u}$ merely records that both sides are being watched as t advances. In a constrained optimisation the driving variable is the one surviving degree of freedom, and the constraint is what reduced the count to one; the derivative that is set to zero is a derivative along the constraint. In a linearisation the driving variable is the displacement from a base point, and the chain rule appears as the statement that $\Delta y \approx f'\Delta x$ composes correctly through several stages. In a curvature or area calculation the driving variable may itself be parametrised, and then a further factor $\frac{dx_p}{dt}$ appears, which is the error people make most often here.

The practical consequence is a single diagnostic question, applicable to all four: *what is varying, and with respect to what?* Answer it in writing before differentiating, and the correct chain rule follows mechanically. Leave it unanswered, and you will substitute a number where a function belonged, or differentiate along the wrong variable, or omit a factor that was never named.

One habit unifies the chapter, and it is worth stating as a sentence rather than a slogan: before differentiating, write down what is varying, what is fixed, and on what set. In a related-rates problem that means naming every time-dependent quantity and refusing to insert an instantaneous number. In an optimisation problem it means using the constraint to reach one variable and then stating the interval. In a linearisation it means choosing the base point and knowing the sign of f'' . In a counting problem it means keeping f' factored and asking which factors can actually vanish. The differentiation that follows is, in every one of these families, the shortest part of the work.

Recognition matrix

If you see	Reach for	If that fails
"At what rate", two quantities in time	Name, relate, reduce, differentiate, substitute last	Never insert the instant before differentiating
A relation with more variables than rates	Reduce by geometry first: similar triangles, chord, law of sines	Chain through an intermediate, one factor per link
Two hinged segments of fixed length	Chord $d = 2R \sin(\theta/2)$; parametrise by the angle	Law of cosines, if the chord will not serve
A power relation $Q = cu^k$	Relative rates: $\dot{Q}/Q = k\dot{u}/u$	No constant c is needed
"Maximise subject to"	Eliminate with the constraint, then one derivative	State the domain before solving $f' = 0$
A closed interval of feasible values	Compare interior critical points <i>and</i> both endpoints	The maximum often sits where $f' \neq 0$
An open or unbounded feasible set	Take limits at the ends; expect an unattained infimum	Report an infimum, not a minimum
"Closest point on a curve"	Minimise squared distance; substitute $u = x^2$ if even	Expect a symmetric pair of minimisers
A constraint $x^y = C$	Take logarithms; $x/\ln x$ is minimised at $x = e$	Keep the branch $x > 1$
"The normal at P "	Slope $-1/f'(a)$; x -intercept $a + f(a)f'(a)$	If $f'(a) = 0$ the normal is $x = a$
One line tangent to two curves	Two unknowns a, b : equal slopes and collinearity	The touch points are generally different
A stated angle between lines	$\tan \theta = (m_2 - m_1)/(1 + m_1 m_2) $	Two roots; a side condition picks one
"Approximate" or "estimate the change"	$L(x) = f(a) + f'(a)(x - a)$; error $\frac{1}{2}f''(\xi)(x - a)^2$	Sign of f'' decides over or under
A percentage or propagated error	$dy/y = (f'/f) dx$; for cx^k multiply by k	No absolute values are needed
"One Newton iteration"	$x_{n+1} = x_n - f(x_n)/f'(x_n)$	At a root of multiplicity m , ratio $1 - 1/m$
"Radius of curvature"	$\kappa = y'' /(1 + y'^2)^{3/2}$; $R = 1/\kappa$	Parametric: $ \dot{x}\ddot{y} - \dot{y}\ddot{x} /(\dot{x}^2 + \dot{y}^2)^{3/2}$
f' given factored	Sign chart; odd multiplicity only; discard positive factors	Never expand f'
$\cos \varphi(x)$ or $\sin \varphi(x)$ on an interval	Count integers k with $k\pi$ in the image of φ	Solving the equation is unnecessary
Bounds on f' , a value of f wanted	Mean value theorem on the whole interval	Both bounds are attained by lines
n zeros of g , zeros of $g + \lambda g'$ wanted	Rolle on $e^{x/\lambda}g$; at least $n - 1$	Recognise $\binom{n}{k}\lambda^k$ as $(1 + \lambda D)^n$
Area between a curve and its tangent	$\frac{dA}{dx_p} = \frac{1}{2}x_p^2 f''(x_p)$	Chain in $\frac{dx_p}{dt}$ if the point is parametrised
Largest triangle on an arc	Archimedes: tangent parallel to the chord	The mean value theorem, read geometrically

If you see	Reach for	If that fails
An angle tracking a moving object $f'(c) = 0$ and $f''(c) = 0$	Relate by $\tan \theta$, differentiate implicitly The second-derivative test is silent; use a sign chart	$\sec^2 \theta$ is cheaper than an arctangent $x^4, -x^4, x^3$ all look alike here
A strictly increasing wrapper around g	Extrema of $e^g, \ln g, \arctan g$ are those of g	Differentiate the wrapper only if you must
A rational inequality	Mark zeros <i>and</i> poles; even multiplicity does not switch	Test one point per region

Partial derivatives and the multivariable chain

What this chapter is for. Nothing you learned about differentiation in one variable is repealed when a second variable appears. The rules are the same rules, the table of derivatives is the same table, and the algebra is the same algebra. What changes is the bookkeeping: which symbols are frozen while you differentiate, how many terms a chain rule must have, and which of several possible derivatives the problem is actually asking for. Almost every error in this chapter is a bookkeeping error, not a calculus error, and the techniques below are all devices for making the bookkeeping visible so that it can be checked. You will learn to hold variables fixed deliberately rather than by habit, to draw the dependency tree of a composite and read the chain rule off it, to use the gradient as the single object that answers every question about direction, to differentiate an implicit relation in one line by $-F_x/F_y$, to propagate small errors through a formula, and to test a pair of functions for exactness, which is the last technique in this part and the first technique in the part on differential equations.

Holding the others fixed

A partial derivative is an ordinary derivative taken along a coordinate line. To compute $\frac{\partial f}{\partial x}$ at a point you restrict f to the horizontal line through that point, which turns it into a function of the single variable x , and differentiate that function in the ordinary way. Every rule from the single-variable theory therefore applies verbatim: product, quotient, chain, the derivatives of $\sin$ and $\ln$ and $\arctan$. The only new instruction is that y is now a constant, and the only new failure is forgetting that it is one, or forgetting that it stops being one.

That instruction is easy to state and surprisingly hard to obey, because the mind does not treat all constants alike. Nobody differentiating $7x^3$ writes anything but $21x^2$, but a large number of readers differentiating $y e^{xy}$ with respect to x write $y e^{xy}$ and stop, because the y in the exponent is doing two jobs at once and only one of them registers. The discipline that fixes this is mechanical: before differentiating, rewrite the expression with the frozen variable replaced by a letter that looks like a constant. Differentiating $y e^{xy}$ becomes differentiating $c e^{cx}$, which is $c^2 e^{cx}$, so $\frac{\partial}{\partial x}(y e^{xy}) = y^2 e^{xy}$. The substitution costs five seconds and removes the entire error class.

CORE

17.1 Partial differentiation with the others frozen

CUE Any $f(x, y)$ or $f(x, y, z)$ with f_x, f_{xy} , or $\frac{\partial^{n+m} f}{\partial x^n \partial y^m}$ requested.

METHOD Differentiate with respect to the named variable, treating every other variable as a numerical constant. For a mixed partial, do the derivatives one at a time in whichever order is cheaper: Clairaut's theorem says $f_{xy} = f_{yx}$ wherever both are continuous, so the order is yours to choose. Two structural facts do most of the work. Any term depending on only one of the variables is annihilated by any mixed partial. And any term whose degree in x is less than n is annihilated by ∂_x^n , no matter what it does in the other variables.

WORKED EXAMPLE

$f(x, y) = 10x^4y^3 - e^{3xy} + 1$, evaluated at $(2, 0)$.

Take f_x with y frozen: $f_x = 40x^3y^3 - 3y e^{3xy}$. Every term carries a factor of y , so $f_x(2, 0) = 0$. Take f_y with x frozen: $f_y = 30x^4y^2 - 3x e^{3xy}$, and at $(2, 0)$ the first term dies and the second is -6 . For the mixed partial, differentiate f_x with respect to y , remembering that $-3y e^{3xy}$ needs the product rule:

$$f_{xy} = 120x^3y^2 - 3e^{3xy} - 9xy e^{3xy},$$

so $f_{xy}(2, 0) = 0 - 3 - 0 = -3$. Differentiating f_y with respect to x instead gives $120x^3y^2 - 3e^{3xy} - 9xy e^{3xy}$ as well, which is the theorem doing its job. Central differences confirm $f_y(2, 0) = -6.0000$ and $f_{xy}(2, 0) = -3.0000$.

Now the same function with an order nobody intends you to reach by hand: $\frac{\partial^9 f}{\partial x^5 \partial y^4}$ at $(2, 0)$. Two of the three terms die before any work. The polynomial $10x^4y^3$ has degree 4 in x and is annihilated by ∂_x^5 , and the constant 1 is annihilated by everything. Only $-e^{3xy}$ is left, and with x the active variable and y frozen it is a pure exponential, so

$$\partial_x^5(-e^{3xy}) = -(3y)^5 e^{3xy}.$$

Every remaining derivative is in y , and Leibniz on the product $(3y)^5 \cdot e^{3xy}$ with only four y -derivatives available can lower the exponent of y^5 to at most y^1 :

$$\partial_y^4[(3y)^5 e^{3xy}] = \sum_{j=0}^4 \binom{4}{j} \frac{5!}{(5-j)!} 3^5 y^{5-j} (3x)^{4-j} e^{3xy},$$

and every term carries a positive power of y . At $y = 0$ they all vanish, so the ninth mixed partial is exactly 0 at $(2, 0)$, and at every other point of the x -axis as well. Finite differences on the lower analogue $\partial_x^3 \partial_y^2 e^{3xy}$ at $(2, 0)$ return 3.44 at step 0.05 and 0.52 at step 0.02, shrinking like h^2 towards the exact zero. The whole computation was two annihilation rules and one count of how many y -derivatives were on offer.

WHERE THIS GOES WRONG

Letting a frozen variable participate in the chain rule anyway. Differentiating x^{zy} with respect to y is not a power rule: with x and z frozen the function is c^{zy} in the exponent, so the derivative is $x^{zy} \ln x \cdot z$. The factor $\ln x$ appears in the y -partial and in the z -partial and does not appear in the x -partial, where the power rule genuinely applies and gives $zy x^{zy-1}$. A single expression can require three different rules for its three partials.

SEE ALSO The dependency tree; mixed-partial coefficient extraction; the exactness test.

Clairaut's theorem is worth more than a footnote, because it is the only licence you have to reorder work, and because the hypothesis it carries is the kind that is usually true and occasionally decisive. The statement is that if f_{xy} and f_{yx} both exist and are continuous on a neighbourhood of a point, they are equal there. In practice this means: choose the order that kills the most terms first, or the order that avoids a product rule.

WHERE THE MIXED PARTIALS DISAGREE

Continuity is a real hypothesis. Let

$$f(x, y) = \frac{xy(x^2 - y^2)}{x^2 + y^2}, \quad f(0, 0) = 0.$$

Along the y -axis, $f(h, y) - f(0, y) = f(h, y)$, so $f_x(0, y) = \lim_{h \rightarrow 0} \frac{y(h^2 - y^2)}{h^2 + y^2} = -y$; by the antisymmetry of f under swapping x and y , $f_y(x, 0) = x$. Differentiating those one-variable functions gives

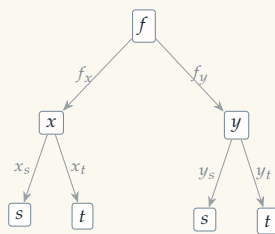
$$f_{xy}(0, 0) = \frac{d}{dy}(-y) = -1, \quad f_{yx}(0, 0) = \frac{d}{dx}(x) = +1.$$

Numerically, nested difference quotients with an inner step of 10^{-6} and an outer step of 10^{-4} return -0.999800 and $+0.999800$. Both second partials exist at the origin and they differ by 2. What fails is continuity: f_{xy} has no limit at the origin, since its value depends on the direction of approach. Every function you will meet in a problem is smooth enough for Clairaut, but the reason to know this example is that it tells you what the theorem is actually asserting, which is a statement about the second derivatives near the point and not merely at it.

The dependency tree

The chain rule in one variable has one term because a composite $f(g(x))$ has one path from x to f . That is the whole content of the rule, and it generalises without modification: the derivative of a composite is a sum over paths, one term per route from the differentiating variable up to the output, and each term is the product of the derivatives encountered along that route. Draw the routes and the formula writes itself. Refuse to draw them and you will omit one, which is the single commonest error in this chapter and the reason the tree is taught here as a technique rather than as an illustration.

The picture for $f(x(s, t), y(s, t))$ is this.



There are exactly two paths from s to f : through x , contributing $f_x x_s$, and through y , contributing $f_y y_s$. Hence $f_s = f_x x_s + f_y y_s$, and by the same count $f_t = f_x x_t + f_y y_t$. Reading the tree is the technique. Counting the leaves labelled s tells you how many terms $\frac{\partial f}{\partial s}$ has, before you compute anything, which is the check that catches the omission.

CORE

17.2 The chain rule as a sum over paths

CUE $z = f(x, y)$ with $x = x(t)$ and $y = y(t)$; or $f(u(s, t), v(s, t))$; or any composite whose inner map has more than one component.

METHOD Draw the dependency tree from the output down to the independent variables. For each independent variable, sum over all paths reaching it, multiplying the derivative on each edge. With one independent variable the derivatives at the bottom are total and the ones at the top are partial:

$$\frac{dz}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}, \quad \frac{\partial z}{\partial s} = f_u u_s + f_v v_s.$$

The one-variable chain rule and the related-rates rule are the cases where the tree has a single branch.

WORKED EXAMPLE

Take $z = x^2 y$ along the unit circle, $x = \cos t$, $y = \sin t$. The tree has two paths, so

$$\frac{dz}{dt} = 2xy \cdot (-\sin t) + x^2 \cdot \cos t = -2 \cos t \sin^2 t + \cos^3 t = \cos t (\cos^2 t - 2 \sin^2 t).$$

At $t = 1.1$ this reads -0.6272108 ; differentiating $\cos^2 t \sin t$ directly and evaluating gives -0.6272108 .

Now two independent variables. Let $z = e^x \sin y$ with $x = st$ and $y = s^2 - t^2$. The tree is the one drawn above, so each of z_s and z_t has two terms:

$$z_s = e^x \sin y \cdot t + e^x \cos y \cdot 2s, \quad z_t = e^x \sin y \cdot s + e^x \cos y \cdot (-2t).$$

At $(s, t) = (1, 2)$ we have $x = 2$, $y = -3$, and $z_s = e^2(2 \sin(-3) + 2 \cos(-3)) = -16.71571$, $z_t = e^2(\sin(-3) - 4 \cos(-3)) = 28.21770$. Central differences on $e^{st} \sin(s^2 - t^2)$ return -16.71571 and 28.21770 .

Finally the case that generates the notational trap, where a variable appears both as an independent variable and inside the intermediates. Let $w = f(x, y, t) = xy + t^2 x$ with $x = \cos t$ and $y = e^t$. The tree now has three edges from w , and t appears as a leaf under x , under y , and directly under w : three paths, three terms.

$$\frac{dw}{dt} = f_x \frac{dx}{dt} + f_y \frac{dy}{dt} + f_t = (y + t^2)(-\sin t) + x e^t + 2tx.$$

At $t = 0.7$ this is 0.9980203 , and differencing $w(t) = e^t \cos t + t^2 \cos t$ directly gives 0.9980203 . The single term $f_t = 2tx$ evaluates to 1.0707791 on its own, a different number entirely. Both are legitimate derivatives of the same function with respect to the same letter; only one of them is the rate of change of w along the path.

WHERE THIS GOES WRONG

Writing one term where the tree has two. This is the path-of-least-resistance error and it is silent: the answer looks like a chain rule, it is dimensionally sensible, and it is wrong. The second trap is notational. If $w = f(x, y, t)$ with $x = x(t)$ and $y = y(t)$, then $\frac{\partial f}{\partial t}$ means the derivative through the explicit t slot only, while $\frac{dw}{dt} = f_x x' + f_y y' + f_t$ collects every path. The two are different numbers written with confusingly similar symbols, and a problem that uses both is testing exactly this.

SEE ALSO Partial differentiation with the others frozen; the gradient and directional derivatives; the total differential.

Second derivatives through the tree

The tree does not stop at first order. To compute z_{ss} you differentiate $z_s = f_x x_s + f_y y_s$ again, and the essential observation is that f_x and f_y are themselves composite functions of s and t , so each of them must be run back through the same tree.

The result is a product rule on every term, and the count doubles: a first derivative with two paths produces a second derivative with four second-order terms plus two first-order ones. Nearly every mistake at second order is a dropped x_{ss} , and it is dropped because the solver differentiated the coefficients and forgot the factors.

The canonical instance is the Laplacian in polar coordinates, worth doing once in full because every other second-order change of variable follows the same pattern. With $x = r \cos \theta$ and $y = r \sin \theta$,

$$z_r = z_x \cos \theta + z_y \sin \theta,$$

and differentiating again in r requires sending z_x and z_y back down the tree, while $\cos \theta$ and $\sin \theta$ are constants in r :

$$z_{rr} = z_{xx} \cos^2 \theta + 2z_{xy} \sin \theta \cos \theta + z_{yy} \sin^2 \theta.$$

In θ the coefficients are not constant, so the product rule contributes the extra terms:

$$z_\theta = -r z_x \sin \theta + r z_y \cos \theta, \quad z_{\theta\theta} = r^2(z_{xx} \sin^2 \theta - 2z_{xy} \sin \theta \cos \theta + z_{yy} \cos^2 \theta) - r z_r.$$

Adding z_{rr} to $r^{-2}z_{\theta\theta}$ makes the cross terms cancel and the squares combine, and the leftover $-r^{-1}z_r$ is what forces the middle term of the standard formula:

$$z_{xx} + z_{yy} = z_{rr} + \frac{1}{r}z_r + \frac{1}{r^2}z_{\theta\theta}.$$

For $z = x^2y$ at $r = 1.7$, $\theta = 0.9$, the polar side evaluates numerically to 2.663312 and the Cartesian side is $2y = 2r \sin \theta = 2.663311$.

THE MISSING TERM AT SECOND ORDER

If the inner functions are not linear, z_{ss} contains $f_x x_{ss} + f_y y_{ss}$ in addition to the four second-order terms. In the polar case $x_{rr} = y_{rr} = 0$, which is why the r -derivative is clean, and $x_{\theta\theta} = -x$, $y_{\theta\theta} = -y$, which is exactly where the $-r z_r$ comes from. Whenever a second-order change of variables produces a formula with a first-derivative term in it, that term is the second derivative of the substitution, and its absence from your working is the diagnostic that you differentiated the coefficients only.

The tree also explains why implicit differentiation works, why the total differential has the shape it has, and why the gradient is a useful object at all. Each of the next three techniques is the chain rule with a particular tree drawn in advance, which is why they can be reduced to formulas worth memorising. Take direction first.

Direction, and the one object that encodes it

Once you can differentiate along the two coordinate directions, the natural question is what happens along an arbitrary one. Fix a point P and a unit vector $\hat{\mathbf{v}}$, and walk from P along $\hat{\mathbf{v}}$ at unit speed: the position is $P + h\hat{\mathbf{v}}$, and the rate of change of f is a one-variable derivative in h . The tree for that composite has one path per coordinate, so the answer is $f_x v_1 + f_y v_2 + f_z v_3$, which is a dot product. Collecting the partials into a single vector is therefore not a convenience of notation: it is the statement that all directional information at a point is carried by three numbers.

MEDIUM

17.3 The gradient and directional derivatives

CUE “Rate of change in the direction of $\mathbf{v}$ ”; “direction of steepest ascent”; “normal to the level surface”; “tangent plane at P ”.

METHOD Assemble $\nabla f = (f_x, f_y, f_z)$ at the point. Then

$$D_{\mathbf{v}}f = \nabla f \cdot \hat{\mathbf{v}}, \quad \hat{\mathbf{v}} = \frac{\mathbf{v}}{\|\mathbf{v}\|}.$$

Three consequences answer almost every question asked about direction. The maximum of $D_{\mathbf{v}}f$ over unit $\hat{\mathbf{v}}$ is $\|\nabla f\|$, attained when $\hat{\mathbf{v}}$ points along ∇f ; the minimum is $-\|\nabla f\|$, along $-\nabla f$; and $D_{\mathbf{v}}f = 0$ exactly when $\hat{\mathbf{v}} \perp \nabla f$, which is why ∇f is normal to the level set through the point and why the tangent plane to $f = c$ at P is $\nabla f(P) \cdot (\mathbf{r} - P) = 0$.

WORKED EXAMPLE

A clean case first. For $f = x^2 + 2y^2 + 3z^2$ at $(1, 1, 1)$, $\nabla f = (2x, 4y, 6z) = (2, 4, 6)$. The steepest ascent is along $(1, 2, 3)$ with rate $\|\nabla f\| = \sqrt{56} = 2\sqrt{14} \approx 7.48331$, and along $\mathbf{v} = (1, 2, 2)$, whose norm is 3, the rate is $\frac{2+8+12}{3} = \frac{22}{3}$. The tangent plane to the ellipsoid $f = 6$ at that point is $2(x - 1) + 4(y - 1) + 6(z - 1) = 0$.

A case that exercises the frozen-variable trap. Let

$$f(x, y, z) = x^3z + 1 + \sin(xy) - x^{zy}$$

at $(3, 0, 1)$, along $\mathbf{v} = (2, 4, 3)$. The last term is the dangerous one, and it must be differentiated three different ways. In x the power rule applies: $\partial_x(x^{zy}) = zy x^{zy-1}$, which vanishes at $y = 0$. In y and in z the exponent is the variable, so each partial carries a factor $\ln x$: $\partial_y(x^{zy}) = x^{zy} \ln x \cdot z$ and $\partial_z(x^{zy}) = x^{zy} \ln x \cdot y$. Assembling,

$$f_x = 3x^2z + y \cos(xy) - zy x^{zy-1} = 27, \quad f_y = x \cos(xy) - z x^{zy} \ln x = 3 - \ln 3, \quad f_z = x^3 - y x^{zy} \ln x = 27.$$

With $\|\mathbf{v}\| = \sqrt{29}$,

$$D_{\mathbf{v}}f = \frac{2(27) + 4(3 - \ln 3) + 3(27)}{\sqrt{29}} = \frac{147 - 4 \ln 3}{\sqrt{29}} \approx 26.48119,$$

and a central difference along the unit vector returns 26.48119.

The level-surface questions, all answered by the same vector. Take $F(x, y, z) = xyz$ and the surface $F = 6$ through $(1, 2, 3)$. Then $\nabla F = (yz, xz, xy) = (6, 3, 2)$ at that point, and $\|\nabla F\| = \sqrt{36 + 9 + 4} = 7$. From that one vector: the tangent plane is $6(x - 1) + 3(y - 2) + 2(z - 3) = 0$, that is $6x + 3y + 2z = 18$; the normal line is $(1, 2, 3) + t(6, 3, 2)$; the steepest increase of F is at rate 7 along $(6, 3, 2)$; the directions in which F does not change to first order are exactly those perpendicular to $(6, 3, 2)$, a two-parameter family spanning the tangent plane; and the rate along any curve lying inside the surface is 0, since such a curve has $\mathbf{r}' \perp \nabla F$. Five distinct questions, one computation.

WHERE THIS GOES WRONG

Not normalising $\mathbf{v}$. This is the commonest slip in the chapter and it produces an answer that is off by the factor $\|\mathbf{v}\|$, which is invisible unless you check. If the problem says “in the direction of the vector $\mathbf{v}$ ” you must divide; if it says “in the direction of the unit vector” or gives an angle, you must not. The second slip is reporting ∇f itself as the answer to “what is the maximum rate of increase”. The direction is ∇f ; the rate is $\|\nabla f\|$.

SEE ALSO The chain rule as a sum over paths; implicit slope by $-F_x/F_y$; the exactness test.

The gradient being normal to level sets is not a separate fact to memorise. It is the previous technique applied to a curve that stays on a level set: if $\mathbf{r}(t)$ satisfies $f(\mathbf{r}(t)) = c$, differentiating gives $\nabla f \cdot \mathbf{r}' = 0$, so every tangent vector to the level set is orthogonal to the gradient. That one line is also the whole of implicit differentiation, and it deserves its own technique because the resulting formula is worth carrying in your head.

Curves given implicitly

17.4 Implicit slope by $-F_x/F_y$

CORE

CUE A relation $F(x, y) = 0$ with the slope wanted, especially when F is symmetric, or large, or has y buried in exponents and logarithms so that collecting y' by hand is error-prone.

METHOD Regard y as a function of x along the curve and differentiate $F(x, y(x)) = 0$. The tree has two paths, so $F_x + F_y y' = 0$, whence

$$\frac{dy}{dx} = -\frac{F_x}{F_y} \quad \text{wherever } F_y \neq 0.$$

That is the implicit function theorem in one line, and it is worth memorising because it converts an error-prone algebraic manipulation into two independent partial derivatives and a quotient. The three-variable version is identical: for

$$F(x, y, z) = 0, \quad \frac{\partial z}{\partial x} = -\frac{F_x}{F_z} \quad \text{and} \quad \frac{\partial z}{\partial y} = -\frac{F_y}{F_z}.$$

WORKED EXAMPLE

The folium of Descartes, $x^3 + y^3 - 6xy = 0$, at $(3, 3)$. Take $F = x^3 + y^3 - 6xy$, which indeed vanishes there since $27 + 27 - 54 = 0$. Then $F_x = 3x^2 - 6y = 9$ and $F_y = 3y^2 - 6x = 9$, so $y' = -1$. Tracing the branch numerically and differencing gives -1.0000000 .

Now a relation where hand-collecting is genuinely unpleasant: $x^y = y^x$ at $(2, 4)$, a point on the curve since $2^4 = 4^2 = 16$. Set $F = x^y - y^x$. Each term needs the frozen-variable discipline, because x is a base in one term and an exponent in the other:

$$F_x = y x^{y-1} - y^x \ln y, \quad F_y = x^y \ln x - x y^{x-1}.$$

At $(2, 4)$: $F_x = 4 \cdot 8 - 16 \ln 4 = 32 - 32 \ln 2$ and $F_y = 16 \ln 2 - 2 \cdot 4 = 16 \ln 2 - 8$. Hence

$$\frac{dy}{dx} = -\frac{32(1 - \ln 2)}{8(2 \ln 2 - 1)} = -\frac{4(1 - \ln 2)}{2 \ln 2 - 1} \approx -3.177399.$$

Solving $x^y = y^x$ numerically for y near 4 at $x = 2 \pm 10^{-5}$ and differencing gives -3.177399 . Doing this by logarithmic differentiation instead means writing $y \ln x = x \ln y$, differentiating to $y' \ln x + \frac{y}{x} = \ln y + \frac{x}{y} y'$, and then collecting; the answer is the same, and there are three more places to lose a term.

The three-variable version, for a surface. On $x^2 + y^2 + z^2 = 14$ at $(1, 2, 3)$, take $F = x^2 + y^2 + z^2 - 14$, so $F_x = 2x$, $F_y = 2y$, $F_z = 2z$ and

$$\frac{\partial z}{\partial x} = -\frac{2x}{2z} = -\frac{x}{z} = -\frac{1}{3}, \quad \frac{\partial z}{\partial y} = -\frac{y}{z} = -\frac{2}{3},$$

which agrees with differentiating the explicit branch $z = \sqrt{14 - x^2 - y^2}$ but never required solving for it. That matters as soon as the surface cannot be solved for z in closed form, which is the normal case.

WHERE THIS GOES WRONG

Forgetting the minus sign. The formula is a rearrangement of $F_x + F_y y' = 0$, so the sign is forced; if you cannot remember it, rederive it in that one line rather than guessing. The second failure is applying the formula at a singular point, where $F_x = F_y = 0$. The folium has such a point at the origin, where the curve crosses itself and two tangent lines exist; $-F_x/F_y$ there is $0/0$ and the implicit function theorem does not apply. A quotient of the form $0/0$ in this formula is a signal that the curve is not locally a graph, not an invitation to use L'Hopital.

SEE ALSO The gradient and directional derivatives; the chain rule as a sum over paths; the exactness test.

THE CYCLIC RELATION, AND WHY IT IS NOT +1

For a relation $F(x, y, z) = 0$ the three implicit partials multiply to -1 , not to 1 :

$$\left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y = \left(-\frac{F_y}{F_x}\right) \left(-\frac{F_z}{F_y}\right) \left(-\frac{F_x}{F_z}\right) = -1.$$

Each factor contributes one minus sign and the six partials cancel in pairs. On $x^2 + y^2 + z^2 = 14$ at $(1, 2, 3)$ the three factors are $-2, -\frac{3}{2}, -\frac{1}{3}$, and their product is -1 ; on the ideal gas law $PV = nRT$ the numerical product of $\frac{\partial P}{\partial V}$, $\frac{\partial V}{\partial T}$ and $\frac{\partial T}{\partial P}$ at $V = 0.02$, $T = 300$ is -1.0000008 , the discrepancy being finite-difference error. The subscripts are not decoration: each partial is taken with a different variable held fixed, and the identity is false if you hold the same one throughout.

The connection to the previous section is worth stating explicitly, because it makes the formula unforgettable. The curve $F = 0$ is a level set of F , so $\nabla F = (F_x, F_y)$ is normal to it. A normal vector (A, B) corresponds to a tangent line of slope $-A/B$. Substituting $(A, B) = (F_x, F_y)$ gives $-F_x/F_y$ immediately, and the condition $F_y \neq 0$ becomes the geometric statement that the tangent is not vertical.

The same slope three ways

It is worth taking one relation and differentiating it by every available route, because the comparison is the argument for memorising $-F_x/F_y$ in miniature. Take $x^y = y^x$ at the point $(2, 4)$.

By the formula. Set $F = x^y - y^x$, compute the two partials independently, divide, negate. Each partial is a single application of the frozen-variable rule to a two-term expression, and no equation is ever rearranged. The answer is $-\frac{4(1 - \ln 2)}{2 \ln 2 - 1}$.

By logarithmic differentiation. Write $y \ln x = x \ln y$ first, which is a genuine simplification, then differentiate with $y = y(x)$:

$$y' \ln x + \frac{y}{x} = \ln y + \frac{x}{y} y' \implies y' = \frac{\ln y - \frac{y}{x}}{\ln x - \frac{x}{y}} = \frac{\ln 4 - 2}{\ln 2 - \frac{1}{2}} = -3.177399.$$

Same number, and the intermediate expressions are smaller. This route wins when taking logarithms actually simplifies the relation, which here it does.

By direct implicit differentiation. Differentiate $x^y = y^x$ as it stands. Both sides need the general power rule, $\frac{d}{dx}(u^v) = u^v(\frac{vu'}{u} + v' \ln u)$, and since $x^y = y^x = 16$ the common factor cancels, leaving $\frac{y}{x} + y' \ln x = \ln y + \frac{x}{y} y'$, which is the same equation as the previous route reached more cheaply.

Three routes, one answer, and the ranking depends on the relation. Use $-F_x/F_y$ when F is large or symmetric and no simplification presents itself, because it never asks you to collect y' from both sides. Take logarithms first when the relation

is a product, a quotient, or a tower. Never differentiate a relation in place while also trying to simplify it; do one, then the other.

Small changes

So far every technique has produced an exact derivative. The next one uses derivatives to estimate, which is the form the question takes whenever real measurements are involved. The idea is one dimension of linearisation per variable, added.

MEDIUM

17.5 The total differential and error propagation

CUE A quantity computed from several measured inputs, each carrying a small uncertainty, with the uncertainty in the output wanted; or any question phrased as “by approximately how much does f change if x increases by ...”.

METHOD The linear approximation in several variables is

$$df = f_x dx + f_y dy, \quad \Delta f \approx f_x \Delta x + f_y \Delta y.$$

For a worst-case error bound take absolute values, $|\Delta f| \leq |f_x| |\Delta x| + |f_y| |\Delta y|$, which assumes the errors conspire. For independent random errors take them in quadrature, $\sqrt{f_x^2 \Delta x^2 + f_y^2 \Delta y^2}$, which assumes they do not. When f is a product or quotient of powers, take logarithms first: $d(\ln f) = df/f$ turns the whole computation into a weighted sum of relative errors, with the weights being the exponents.

WORKED EXAMPLE

A cylinder, $V = \pi r^2 h$. Logarithmic differentiation gives $\ln V = \ln \pi + 2 \ln r + \ln h$, so

$$\frac{dV}{V} = 2 \frac{dr}{r} + \frac{dh}{h},$$

and a 1% error in each input gives a worst case of 3%. The exact inflation, $(1.01)^2(1.01) - 1 = 0.030301$, confirms that the linear estimate is the leading term. In quadrature the answer would be $\sqrt{2^2 + 1^2} = 2.236\%$ instead, and the difference between 3% and 2.24% is entirely a modelling decision about whether the errors are correlated.

A case where the answer is not obvious in advance. Two resistors in parallel, $R = \frac{R_1 R_2}{R_1 + R_2}$, with $R_1 = 100 \pm 1$ and $R_2 = 200 \pm 2$, so 1% each. Here

$$\frac{\partial R}{\partial R_1} = \frac{R_2^2}{(R_1 + R_2)^2} = \frac{4}{9}, \quad \frac{\partial R}{\partial R_2} = \frac{R_1^2}{(R_1 + R_2)^2} = \frac{1}{9},$$

and $R = 66.\overline{66}$. The worst case is $\frac{4}{9}(1) + \frac{1}{9}(2) = \frac{2}{3}$, which is 1.000% of R . The relative error does not grow: a parallel combination of two components each known to 1% is itself known to 1%, because the two partials are weights that sum to 1 when scaled by the relative errors. Scaling both resistors by 1.01 exactly gives $R \mapsto 1.01R$, which is the structural reason. In quadrature the figure drops to 0.745%.

A case where an exponent is fractional and therefore reduces the error. The period of a pendulum is $T = 2\pi\sqrt{L/g}$, so $\ln T = \ln 2\pi + \frac{1}{2} \ln L - \frac{1}{2} \ln g$ and

$$\frac{dT}{T} = \frac{1}{2} \frac{dL}{L} - \frac{1}{2} \frac{dg}{g}.$$

A length known to 0.5% and a local g known to 0.2% give a worst case of $0.25\% + 0.10\% = 0.35\%$ in the period. Notice that the minus sign is irrelevant to the worst case, since the errors are bounded in absolute value and either sign is available, but it is decisive if the two errors have a known common cause. Notice also that measuring g from T and L inverts the weights: $dg/g = 2 dT/T - dL/L$, and now the timing error counts double.

WHERE THIS GOES WRONG

Mixing the two error models. Adding absolute values when the problem says the measurements are independent overstates the error by up to a factor of $\sqrt{\pi}$; using quadrature when the problem asks for a guaranteed bound understates it. Decide which is wanted from the wording, and say which you used. The second trap is dropping the exponent weight: in $V = \pi r^2 h$ the radius error counts twice, and in $T = 2\pi\sqrt{L/g}$ the length error counts half.

SEE ALSO The chain rule as a sum over paths; the gradient and directional derivatives; logarithmic differentiation.

The three formulas

$$df = f_x dx + f_y dy, \quad z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b), \quad \Delta f \approx \nabla f \cdot \Delta \mathbf{r}$$

are the same statement written for three audiences. The first is the calculus of small changes, the second is the geometry of the best affine approximation, and the third is the vector form used for directional questions. If you can move between them freely you will never have to remember which one a problem wants, because you can produce any of them from the list of partials at the point. The single fact underneath all three is differentiability: $f(\mathbf{a} + \mathbf{h}) = f(\mathbf{a}) + \nabla f(\mathbf{a}) \cdot \mathbf{h} + o(\|\mathbf{h}\|)$. Existence of the partials alone is not enough for that, which is the one place where the multivariable theory is genuinely harder than the one-variable theory rather than merely bulkier.

Exactness, and the bridge out of this part

Clairaut's theorem says that if a pair of functions are the two partials of a single f , they must satisfy one identity. Read forwards, that is a labour-saving device. Read backwards, it is a test: given a pair (M, N) , is there any f at all with $f_x = M$ and $f_y = N$? The necessary condition is immediate, and on a well-behaved domain it is also sufficient. This is the last technique of Part III and the first technique of the part on differential equations, because an equation written as $M dx + N dy = 0$ is solved in one step the moment the test passes.

17.6 The exactness test $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

HARD

CUE A pair (M, N) presented as a candidate gradient, a vector field, or a differential form $M dx + N dy$; or a first-order equation already arranged as $M dx + N dy = 0$; or a potential f requested from its partials.

METHOD On a simply connected domain, the form is exact, equivalently the field is a gradient, if and only if

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$$

When it passes, reconstruct f by integrating M in x and repairing the result:

$$f = \int M dx + g(y), \quad \text{then set } \frac{\partial f}{\partial y} = N \text{ to determine } g'(y).$$

The constant of integration is a function of the other variable, and finding it is the whole second half of the work. Any leftover parameters are fixed by point values or by extra conditions.

WORKED EXAMPLE

Suppose $\nabla f = (3x^2 + by^2, 2bxy + a)$ with unknown constants a, b , subject to $f(0, 0) = 0, f(1, 1) = 2$, and $f_{xxx} + \frac{1}{x}f_{yy} = 10$. Find $f(1, 2)$.

First the test: $M_y = 2by$ and $N_x = 2by$, so the pair is exact for every a and b , and a potential exists. Integrate M in x :

$$f = x^3 + bxy^2 + g(y).$$

Differentiate in y and match N : $f_y = 2bxy + g'(y) = 2bxy + a$, so $g'(y) = a$ and $g(y) = ay + C$. The condition $f(0, 0) = 0$ gives $C = 0$, so $f = x^3 + bxy^2 + ay$.

Now the two remaining conditions. $f_{xxx} = 6$ and $f_{yy} = 2bx$, so $f_{xxx} + \frac{1}{x}f_{yy} = 6 + 2b = 10$, giving $b = 2$. Then $f(1, 1) = 1 + b + a = 2$ forces $a = -1$. Hence $f = x^3 + 2xy^2 - y$ and

$$f(1, 2) = 1 + 8 - 2 = 7.$$

Differencing f numerically confirms $\nabla f(1.3, 0.7) = (6.05, 2.64)$, matching $(3x^2 + 2y^2, 4xy - 1)$ at that point.

The same machinery solves a differential equation. The equation $(2xy + y^2) dx + (x^2 + 2xy) dy = 0$ has $M_y = 2x + 2y = N_x$, so it is exact; integrating M gives $f = x^2y + xy^2 + g(y)$, and $f_y = x^2 + 2xy + g'(y) = N$ forces $g' = 0$. The general solution is the level set $x^2y + xy^2 = C$, obtained without ever separating variables or guessing an integrating factor.

Writing “+C” instead of “+g(y)” after the x -integration. Holding y fixed means the constant of integration may depend on y , and discarding that dependence throws away exactly the information the second step is designed to recover. The second trap is the domain hypothesis. On the punctured plane the field

$$M = \frac{-y}{x^2 + y^2}, \quad N = \frac{x}{x^2 + y^2}$$

satisfies $M_y = N_x = \frac{y^2 - x^2}{(x^2 + y^2)^2}$, verified as -0.2525040 for both at $(1.3, 0.7)$. Yet it has no potential on the punctured plane, because the only candidate is the polar angle, which cannot be made single-valued around the origin. The test is necessary always and sufficient only where the domain has no holes.

SEE ALSO Partial differentiation with the others frozen; the gradient and directional derivatives; implicit slope by $-F_x/F_y$.

$M_y = N_x$ is Clairaut, an exactness criterion, and a conservative-field criterion, and these are the same statement seen from three directions. Forwards it lets you choose the cheaper order of mixed differentiation. Backwards it decides whether a pair of functions is a gradient. Physically it decides whether the work done by a field is path independent. When you meet integrating factors in Part V, what you are doing is multiplying a form that fails this test by something that makes it pass.

When the test fails: integrating factors

A form that fails $M_y = N_x$ is not a dead end, because exactness is a property of the way an equation is written, not of the equation itself. Multiplying $M dx + N dy = 0$ by any nonzero $\mu(x, y)$ leaves its solutions untouched but changes both partials, and μ can often be chosen to make the test pass. The searchable case is when the correcting factor depends on one variable only:

$$\frac{M_y - N_x}{N} = h(x) \implies \mu = e^{\int h(x) dx}, \quad \frac{N_x - M_y}{M} = k(y) \implies \mu = e^{\int k(y) dy}.$$

Compute the first quotient; if the y cancels, you have $\mu(x)$. If not, compute the second; if the x cancels, you have $\mu(y)$. If neither cancels, the equation needs a different method, and Part V supplies them.

Take $(3xy + y^2) dx + (x^2 + xy) dy = 0$. Here $M_y = 3x + 2y$ and $N_x = 2x + y$, which differ, so the form is not exact. The first quotient is

$$\frac{M_y - N_x}{N} = \frac{x + y}{x^2 + xy} = \frac{x + y}{x(x + y)} = \frac{1}{x},$$

free of y , so $\mu = e^{\int dx/x} = x$. Numerically at $(1.3, 0.7)$ the quotient reads 0.769231 against $1/x = 0.769231$. Multiplying through gives $(3x^2y + xy^2) dx + (x^3 + x^2y) dy = 0$, whose partials are both $3x^2 + 2xy$, confirmed as 6.890000 each at that point. Integrating the new M in x gives $f = x^3y + \frac{1}{2}x^2y^2 + g(y)$, and matching $f_y = x^3 + x^2y$ forces $g' = 0$. The solution curves are

$$x^3y + \frac{1}{2}x^2y^2 = C.$$

Differencing f at $(1.3, 0.7)$ returns $f_x = 4.186000$ and $f_y = 3.380000$, matching the corrected M and N exactly. The whole of the theory of first-order exact equations is this test plus this repair, which is why the technique is placed at the end of a chapter on partial derivatives rather than at the start of a chapter on differential equations.

Very high mixed partials

The last technique in this chapter is the two-variable version of a trick from the chapter on higher-order derivatives: when the order requested is absurd, nobody expects you to differentiate. The requested number is a Taylor coefficient in disguise, and the way to get a Taylor coefficient is to expand, not to differentiate.

17.7 Mixed-partial coefficient extraction

CUE An enormous mixed partial at a specific point: $\frac{\partial^{134}f}{\partial x^{67}\partial y^{67}}, \frac{\partial^{100}f}{\partial x^{50}\partial y^{50}}$. The order is far too high for repeated differentiation to be intended.

METHOD At the origin, the identity is

$$\frac{\partial^{n+m}f}{\partial x^n \partial y^m} \Big|_{(0,0)} = n! m! [x^n y^m] f(x, y),$$

so expand f as a double series and read off the one coefficient you need. Two structural helps. A diagonal series such as $\sum c_k (xy)^k$ has only one term contributing to a symmetric mixed partial. And when f contains $\ln(x^2 + y^2)$, complexify: $\ln(x^2 + y^2) = \ln(x + iy) + \ln(x - iy)$, and each logarithm has an elementary high derivative in both variables, so the point need not be the origin.

WORKED EXAMPLE

The diagonal case. For $f = \frac{1}{1-xy} = \sum_{k \geq 0} (xy)^k$, the only term containing $x^{67}y^{67}$ is $k = 67$, whose coefficient is 1. So

$$\frac{\partial^{134}f}{\partial x^{67}\partial y^{67}} \Big|_{(0,0)} = 67! \cdot 67! = (67!)^2.$$

The analogue at $n = 1$ can be checked directly: a second difference gives $f_{xy}(0,0) = 1.0000 = (1!)^2$.

The complexified case, evaluated away from the origin:

$$\frac{\partial^{100}}{\partial x^{50}\partial y^{50}} \left[\arctan \frac{x}{y} + \arctan \frac{y}{x} + \ln(x^2 + y^2) \right]_{(1,1)}.$$

The two arctangents sum to $\pi/2$ for $x, y > 0$, a constant, so they contribute nothing. For the logarithm write $u = x + iy$. Then $\partial_x^n \ln u = (-1)^{n-1} (n-1)! u^{-n}$, and differentiating u^{-n} in y brings down a factor i each time, giving

$$\frac{\partial^{n+m}}{\partial x^n \partial y^m} \ln(x + iy) = (-1)^{n+m-1} i^m (n+m-1)! u^{-(n+m)},$$

and the conjugate factor gives the same with i replaced by $-i$ and u by $\bar{u}$. At $n = m = 50$ the sign factors are $(-1)^{99} i^{50} = (-1)(-1) = 1$ and likewise for the conjugate, so the value is

$$99! [(1+i)^{-100} + (1-i)^{-100}].$$

Since $(1+i)^2 = 2i$, we get $(1+i)^{100} = (2i)^{50} = 2^{50} i^{50} = -2^{50}$, and the same for the conjugate. Hence the answer is

$$99! \left(-\frac{1}{2^{50}} - \frac{1}{2^{50}} \right) = -\frac{99!}{2^{49}} \approx -1.6578 \times 10^{141}.$$

The general formula was checked against direct difference quotients at $n = m = 1, 2, 3$, where it returns $-1, -3$ and -30 against numerical values $-1.0000, -3.0000$ and -30.0000 .

Two more double series worth knowing, because they cover the non-diagonal cases. First,

$$\frac{1}{1-x-y} = \sum_{k \geq 0} (x+y)^k = \sum_{n,m \geq 0} \binom{n+m}{n} x^n y^m,$$

so $\frac{\partial^{n+m}f}{\partial x^n \partial y^m} \Big|_{(0,0)} = n! m! \binom{n+m}{n} = (n+m)!$, a striking collapse: the answer depends on n and m only through their sum. Difference quotients at $(n, m) = (1, 1), (2, 1), (2, 2), (3, 2)$ return 2.001, 6.006, 24.05 and 120.4 against 2, 6, 24 and 120. Second, $\sin(xy)$ is diagonal but with a parity gap: since $\sin u = u - \frac{u^3}{6} + \frac{u^5}{120} - \dots$, the coefficient of $x^n y^n$ is the coefficient of u^n in $\sin u$, which is 0 for even n and $(-1)^{(n-1)/2}/n!$ for odd n . Hence

$$\frac{\partial^{2n}}{\partial x^n \partial y^n} \sin(xy) \Big|_{(0,0)} = \begin{cases} 0, & n \text{ even,} \\ (-1)^{(n-1)/2} n!, & n \text{ odd,} \end{cases}$$

one factorial rather than two, because the series coefficient already carries a $1/n!$. Numerically at $n = 1, 2, 3$ the difference quotients give 1.000, 0.000 and -6.000 , against 1, 0 and -6 .

Forgetting that there are two factorials. The coefficient formula carries $n!m!$, not $(n+m)!$ and not $n!$ alone, and the error is invisible in the symmetric case only until you check magnitudes. The second waste is expanding the double series far past the single monomial you need: decide (n, m) first, then find that one coefficient.

SEE ALSO Maclaurin coefficient extraction; partial differentiation with the others frozen; the annihilated mixed partial.

Everything in this chapter has been the one-variable theory with more bookkeeping. The partial derivative is an ordinary derivative with the other letters frozen. The chain rule is the one-variable chain rule summed over the paths of a tree. The gradient is the list of partials, and every directional question is a dot product with it. The implicit formula is the chain rule solved for y' . The total differential is the tangent line, one dimension at a time. Exactness is Clairaut read backwards. And the high mixed partial is a Taylor coefficient, exactly as in one variable. If you find yourself doing something in this chapter that has no one-variable ancestor, you have probably made a bookkeeping mistake.

Recognition matrix

If you see	Reach for	If that fails
f_x of an expression with y in it	Freeze y ; rewrite it as a literal constant first	Rename y to c and redo
A variable appearing as base and exponent	Three different rules for the three partials	Take logarithms and differentiate
f_{xy} requested	Clairaut: pick the order that kills more terms	Check continuity if the function is piecewise
$\frac{\partial^{n+m} f}{\partial x^n \partial y^m}$ with n, m large	Coefficient extraction, $n!m! [x^n y^m]f$	Complexify $\ln(x^2 + y^2)$
A composite with a multi-component inner map	Draw the dependency tree; sum over paths	Count leaves to check the term count
Both $\frac{\partial f}{\partial t}$ and $\frac{df}{dt}$ in one problem	Explicit slot versus all paths; they differ	Write the tree and label the t leaves
"In the direction of $\mathbf{v}$ "	$\nabla f \cdot \mathbf{v} / \ \mathbf{v}\ $	Check whether $\mathbf{v}$ was already a unit vector
"Steepest ascent", "maximum rate"	Direction ∇f , rate $\ \nabla f\ $	Steepest descent is $-\nabla f$
"Normal to the level surface", tangent plane	$\nabla f(P) \cdot (\mathbf{r} - P) = 0$	Rewrite the surface as $F = 0$ first
$F(x, y) = 0$, slope wanted	$\frac{dy}{dx} = -F_x / F_y$	If $F_x = F_y = 0$, the point is singular
$F(x, y, z) = 0$, $\frac{\partial z}{\partial x}$ wanted	$-F_x / F_z$	Solve for z explicitly if possible
Measured inputs with tolerances	$df = f_x dx + f_y dy$; absolute values for worst case	Quadrature if the errors are independent
A product or quotient of powers	Logarithms: relative errors weighted by exponents	Differentiate directly and divide by f
A pair (M, N) , potential wanted	Test $M_y = N_x$, then integrate and repair	Check the domain is simply connected
$M dx + N dy = 0$	Exactness test; solution is the level set $f = C$	Look for an integrating factor
$M_y = N_x$ but no potential exists	The domain has a hole; suspect $\arctan(y/x)$	Restrict to a simply connected piece

Derivative problems that collapse

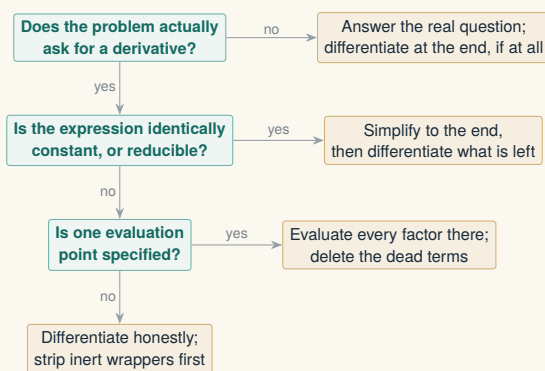
What this chapter is for. There is a family of problems, common in competitions and common in examinations built by people who write competition problems, whose correct solution is to do almost nothing. They are engineered so that the intimidating surface hides a structure that makes the calculation trivial: the point at which the derivative is requested kills most of the terms, or the expression is identically constant, or the requested order of differentiation exceeds the degree available, or the outer layer of a composite cannot affect the answer. A solver who differentiates on reflex will get the right answer after two pages, or, more often, get a wrong answer after two pages. This chapter teaches the diagnostic that comes before differentiating, which is to evaluate the pieces at the requested point and ask which of them survive. It then teaches the opposite discipline, because the second-commonest failure here is inventing a collapse that is not there and reporting a clean wrong answer where the honest ugly one was correct.

The diagnostic

Every technique in this chapter is a special case of one habit: read the whole question, including the evaluation point, before writing a single derivative. Differentiation is a local operation, and if the problem specifies where it wants the answer, that specification is information. Most of the time it is decisive information, because a problem that names a point has usually named it for a reason.

The diagnostic is three questions, asked in order, and it takes about twenty seconds.

First, is a derivative even being asked for? A problem can be filed under differentiation and still be a geometry problem, a number theory problem, or an algebra problem with one trivial differentiation at the end. *Second, does the expression simplify identically?* Baroque constructions built from trigonometric identities, conjugate logarithms, and geometric series are usually baroque on purpose, and the purpose is to hide a constant or a linear function. *Third, if a specific point is named, what dies there?* Evaluate every factor and every inner function at that point before applying any rule, and mark the ones that vanish. In a product of k factors the derivative has k terms, and a single vanishing factor can eliminate $k - 1$ of them.



The tree is not a decoration. Three of its four leaves are cheap, and the expensive leaf at the bottom is reached only after the cheap ones have been ruled out. That ordering is the entire content of the chapter.

Points that kill terms

Start with the simplest and most frequent case, where the collapse is caused by the evaluation point alone and requires no cleverness at all to detect, only the willingness to look.

18.1 Evaluation point where most terms vanish

CUE A product or quotient of three or more factors, with the derivative requested at one suspiciously convenient point: $x = 0$, $x = 1$, $x = \pi/2$, $x = \ln 2$.

METHOD Write the product rule symbolically before computing anything, as $f' = u'vw + uv'w + uvw'$, and then evaluate each factor at the point and delete every term containing a factor that is zero there. Note carefully which factor is differentiated in each term: if $u(a) = 0$, the surviving term is the one containing u' , because that is the term in which u itself does not appear. The standard annihilators are worth memorising. At $x = 0$: $\sin x$, $\tan x$, $\arcsin x$, $\arctan x$, $\ln(1 + x)$, $e^x - 1$, and every positive power x^k . At $x = \pi/2$: $\cos x$ and $\cot x$. At $x = 1$: $\ln x$ and $x - 1$.

WORKED EXAMPLE

The classical shape:

$$f(x) = \frac{e^{x^2} \sin x}{1 + x^3}, \quad f'(0) = ?$$

At $x = 0$ the numerator is 0 and the denominator is 1, so in the quotient rule $\frac{N'D - ND'}{D^2}$ the second term dies outright. Also $D = 1$ and $D' = 3x^2 = 0$ there, so $f'(0) = N'(0)$. And $N' = e^{x^2}(2x \sin x + \cos x)$, whose value at 0 is 1. Hence $f'(0) = 1$, confirmed by a central difference as 1.0000000. No quotient rule was ever executed. A polynomial version where the collapse is total:

$$f(x) = \prod_{k=1}^{10} (x - k), \quad f'(1) = ?$$

The product rule gives ten terms, each a product of nine factors. Nine of those ten terms still contain the factor $(x - 1)$, which vanishes at $x = 1$. Only the term in which $(x - 1)$ was differentiated survives, and it equals $\prod_{k=2}^{10} (1 - k) = (-1)(-2) \cdots (-9) = (-1)^9 9! = -362880$, confirmed numerically as -362880.00 . Expanding the degree-ten polynomial first would be a page of arithmetic for the same number.

When two factors vanish, the first derivative is zero and the question moves up an order. For

$$f(x) = \sin x \cdot \ln(1 + x) \cdot (e^x + 2),$$

two of the three factors vanish at 0, so $f'(0) = 0$ without computation, and the counting rule gives $f''(0) = 2! \cdot \sin'(0) \cdot (\ln(1 + x))'(0) \cdot (e^0 + 2) = 2 \cdot 1 \cdot 1 \cdot 3 = 6$. A second difference returns 6.0000000. Appending a fourth vanishing factor $\tan x$ pushes it to third order, and $f'''(0) = 3! \cdot 1 \cdot 1 \cdot 1 \cdot 3 = 18$, confirmed numerically as 18.000005. Nothing here required the product rule to be written out at all.

And one where the annihilator is $\ln x$ at $x = 1$:

$$f(x) = \ln x \cdot (x^3 + \arctan x) \cdot e^{x^2-1}, \quad f'(1) = \frac{1}{x}(x^3 + \arctan x)e^{x^2-1} \Big|_1 = 1 + \frac{\pi}{4},$$

since the other two terms both retain the factor $\ln x$. Numerically 1.7853982 against $1 + \pi/4 = 1.7853982$. The two remaining factors never had to be differentiated at all.

WHERE THIS GOES WRONG

Substituting the point into f and then differentiating. That gives 0 every time, and it gives 0 for the wrong reason. The point is substituted into f' , after the symbolic differentiation, never before. The subtler error is deleting the wrong term: with $u(a) = 0$ it is the terms containing u undifferentiated that die, so the surviving term is the one carrying u' . Students who delete “the term with the zero in it” half the time delete precisely the term that survives.

SEE ALSO The expression is secretly constant; the decoy evaluation point; the annihilated mixed partial.

COUNTING THE SURVIVORS BEFORE YOU COMPUTE THEM

For $f = u_1 u_2 \cdots u_k$ the product rule has k terms, and the j th term omits u_j and includes u'_j . So the number of surviving terms at a point a is decided by how many of the u_i vanish at a . If exactly one vanishes, say $u_m(a) = 0$, then exactly one term survives, the m th, and

$$f'(a) = u'_m(a) \prod_{i \neq m} u_i(a).$$

If two or more vanish, every term still contains at least one of them and $f'(a) = 0$ outright, with no computation at all. That dichotomy, one zero factor or two, decides the entire problem before a single derivative is taken, and it extends: if r factors vanish at a then the first $r - 1$ derivatives of f vanish there and $f^{(r)}(a) = r! \prod$ of the surviving pieces times the product of the $u'_i(a)$ for the vanishing ones.

The point $x = 0$ carries more structure than any other, and two further species live only there. The first is the Maclaurin reading: a derivative requested *only* at the origin is a series coefficient, since $f^{(n)}(0) = n! [x^n]f(x)$. For $f(x) = \arctan(x^3)$ the expansion is $x^3 - \frac{1}{3}x^9 + \frac{1}{5}x^{15} - \dots$, so $f^{(9)}(0) = 9! \cdot (-\frac{1}{3}) = -120960$ and $f^{(10)}(0) = 0$, and nobody differentiated anything nine times. The second is parity, which is a symmetry killing a term rather than a point killing one. If f is even, every odd derivative vanishes at 0; if f is odd, every even one does. So for the even function $f(x) = e^{-x^2} \cos(x^2)$, $f^{(101)}(0) = 0$ at no cost, and a problem that asks for exactly that is asking whether you noticed. Both species share this chapter's diagnostic: the point was named because the point does the work.

The vanishing-point collapse is local: it happens at the named point and nowhere else. The next kind is global. The expression is not merely convenient at one point, it is the same at every point, and every derivative beyond the first is zero.

Expressions that are constant

18.2 The expression is secretly constant

MEDIUM

CUE A baroque construction with a derivative requested at an arbitrary-looking point such as $x = \pi/5$ or $x = 67$: nested trigonometry, an infinite geometric series whose ratio is a trigonometric square, a product of conjugate logarithms, or a functional inequality asserted for all real arguments.

METHOD Test the hypothesis “this is identically constant” before doing anything else. Two cheap tests: substitute two convenient values and compare, or simplify one layer at a time from the inside out. The engines that produce these are a short list. $\sin^2 + \cos^2 = 1$ and $\sec^2 - \tan^2 = 1$; the reflection identity $\arctan x + \arctan(1/x) = \pi/2$ for $x > 0$; a product $\ln(a+b)\ln(a-b)$ arranged to conjugate; a geometric series whose sum is exactly the reciprocal of the prefactor; and a functional inequality valid for all arguments, where free choice of the arguments forces rigidity.

WORKED EXAMPLE

$$f(x) = x \cdot \frac{1 + \cos 2x}{2} \cdot \sum_{n \geq 0} \left(\frac{\tan^2 x}{1 + \tan^2 x} \right)^n, \quad f'(\pi/5) = ?$$

Peel from the inside. The ratio is $\frac{\tan^2 x}{\sec^2 x} = \sin^2 x$, so the series is geometric with ratio $\sin^2 x < 1$ and sums to $\frac{1}{1 - \sin^2 x} = \sec^2 x$. The middle factor is $\cos^2 x$ by the double-angle identity. So $f(x) = x \cos^2 x \sec^2 x = x$ identically, and $f' \equiv 1$. In particular $f'(\pi/5) = 1$. Evaluating the original expression numerically at $x = \pi/5$ returns 0.628318530718 = $\pi/5$ exactly, and the difference quotient returns 1.00000000.

The reflection identity deserves its own line, because it is the engine most often disguised. For $x > 0$, $\arctan x + \arctan \frac{1}{x} = \frac{\pi}{2}$, and for $x < 0$ the same sum is $-\frac{\pi}{2}$. Either way the function is locally constant, so its derivative is 0 everywhere except at $x = 0$, where it is undefined. Differentiating directly confirms it and shows why the cancellation is exact:

$$\frac{d}{dx} \left[\arctan x + \arctan \frac{1}{x} \right] = \frac{1}{1+x^2} + \frac{1}{1+\frac{1}{x^2}} \cdot \left(-\frac{1}{x^2} \right) = \frac{1}{1+x^2} - \frac{1}{x^2+1} = 0.$$

The useful lesson is in the constant: a problem that asks for the *value* rather than the derivative needs the sign of x , whereas a problem that asks for the derivative does not. Reading which is wanted is the whole difficulty.

A second engine, shorter and even more common, is the conjugate logarithm. The pair

$$f(x) = \ln(x + \sqrt{x^2 + 1}) + \ln(\sqrt{x^2 + 1} - x)$$

looks like two unrelated inverse-hyperbolic pieces, but the two arguments are conjugates, and their product is $(\sqrt{x^2 + 1})^2 - x^2 = 1$. So $f(x) = \ln 1 = 0$ identically, and $f' \equiv 0$ everywhere. Evaluated numerically at $x = 0.3$, 2 and 7, the expression returns 5.6×10^{-17} , 4.4×10^{-16} and 3.6×10^{-15} : zero, to floating-point precision. Differentiating the two logarithms separately is entirely feasible, produces $\frac{1}{\sqrt{x^2 + 1}}$ and $-\frac{1}{\sqrt{x^2 + 1}}$, and takes ten times as long to reach the same conclusion. The tell is always the same: two factors or two summands whose arguments differ by a sign in one place.

Now the functional-inequality version, which looks like a different subject. Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies

$$f(x+y) + f(y+z) + f(z+x) \geq 3f(x+2y+3z) \quad \text{for all real } x, y, z.$$

Find $f'(67)$. The substitution $u = x+y$, $v = y+z$, $w = z+x$ is invertible, since the coefficient matrix has determinant 2, and $x+2y+3z = 2v+w$. So the hypothesis says $f(u) + f(v) + f(w) \geq 3f(2v+w)$ for all real u, v, w . Choose $v = t$ and $w = -t$, so that $2v+w = t$:

$$f(u) + f(t) + f(-t) \geq 3f(t) \implies f(u) \geq 2f(t) - f(-t) \quad \text{for every } u.$$

The right side does not involve u , so f is bounded below; write $m = \inf f$, which is finite. Taking the infimum over u gives $m \geq 2f(t) - f(-t)$, and replacing t by $-t$ gives $m \geq 2f(-t) - f(t)$. Adding the two, $2m \geq f(t) + f(-t)$. But $f(t) \geq m$ and $f(-t) \geq m$, so $f(t) + f(-t) \geq 2m$. Both inequalities force equality, and since each summand is at least m , each equals m . Hence $f \equiv m$ and $f'(67) = 0$.

WHERE THIS GOES WRONG

Differentiating the intimidating form correctly, obtaining a mess that “should” simplify, and then failing to make it simplify. The work is not wrong; it is wasted, and it is wasted in a way that is hard to abandon because each individual step was correct. The rule is to spend twenty seconds testing for constancy before spending twenty minutes not testing for it. The reverse trap is declaring constancy from two agreeing sample points: two points prove nothing, and the identity must actually be exhibited.

SEE ALSO Evaluation point where most terms vanish; the decoy evaluation point; the outer wrapper is inert.

A constant function is the extreme case of a function with low degree. The next technique is the same collapse counted more finely: the expression is not constant, but it is of low degree in each variable separately, and the derivative requested is of high order in each.

Orders that exceed the degree

HARD

18.3 The annihilated mixed partial

CUE A very high mixed partial of an expression assembled from $\ln x$, $\ln y$, and terms linear in one of the variables, often presented as a determinant or a sum of a dozen pieces, at an elaborate evaluation point.

METHOD Expand the expression into a sum of terms first, then test each term against the two annihilation rules. A term depending on only one variable is killed by any mixed partial. A term of degree at most $k-1$ in x is killed by ∂_x^k , whatever else it contains, and a factor like $\ln y$ carries degree zero in x . If every term dies, the answer is 0 and the ornate evaluation point was decoration.

WORKED EXAMPLE

$$f(x, y) = \det \begin{pmatrix} \ln x & \ln y & 1 \\ \ln y & \ln x & 1 \\ x & y & 1 \end{pmatrix}, \quad \frac{\partial^{110} f}{\partial x^{90} \partial y^{20}} \Big|_{(67, 69)} = ?$$

Expand the determinant once, along the bottom row or by the rule of Sarrus:

$$f = (\ln x)^2 - (\ln y)^2 - x \ln x + x \ln y - y \ln x + y \ln y.$$

Now classify the six terms. Four of them, namely $(\ln x)^2$, $-x \ln x$, $-(\ln y)^2$ and $y \ln y$, depend on one variable only, so any mixed partial annihilates them. The two cross terms survive that test but fail the degree test: $x \ln y$ is degree 1 in x , and ∂_x^{90} needs degree at least 90; $-y \ln x$ is degree 1 in y , and ∂_y^{20} needs degree at least 20. So

$$\frac{\partial^{110} f}{\partial x^{90} \partial y^{20}} \equiv 0,$$

identically, and the point $(67, 69)$ was never relevant. Numerically, the fourth mixed difference $\partial_x^2 \partial_y^2 f$ at $(67, 69)$, taken at step 0.5, returns -2×10^{-13} , which is round-off against zero.

The threshold matters and is worth checking. The same function has $\partial_x^2 \partial_y f = 1/x^2$, which at $x = 67$ is 0.00022277, against a numerical value of 0.00022279. That derivative is not zero, because ∂_y^1 does not reach the degree threshold on the term $-y \ln x$. The collapse needs both orders to be at least 2, and asserting it without checking both is guessing.

Reaching for Jacobi's formula, $\frac{d}{dt} \det A = \text{tr}(\text{adj}(A)A')$, and applying it ninety times, instead of expanding one 3×3 determinant once. Any technique whose cost grows with the order of the derivative is the wrong technique when the order is in the dozens. The other error is asserting annihilation from the “looks linear” impression without expanding: an unexpanded determinant hides which terms are products of the two variables.

SEE ALSO Mixed-partial coefficient extraction; the expression is secretly constant; parity annihilation.

The three collapses so far are triggered by different features of the problem, but they share a shape: some quantity in the expression is smaller than the operation applied to it. In the first, the number of nonvanishing factors at the point is smaller than the number of terms in the product rule. In the second, the number of essentially distinct values of the function is one. In the third, the degree in each variable is smaller than the order of differentiation in that variable. In each case the correct response is to establish the inequality and stop, and in each case the establishing takes a few lines while ignoring it takes a few pages. What follows is the discipline that keeps this reflex from becoming a superstition.

The opposite error

Everything so far trains one reflex: look for the collapse. That reflex, left unbalanced, produces its own characteristic failure. A solver who has learned that named points are significant will start manufacturing significance, will decide that $x = \pi - 1$ must simplify because π appeared, and will report a clean wrong answer with confidence. The correct posture is two-sided, and the discipline is symmetric: a collapse must be exhibited, not assumed, and an honest transcendental answer is a perfectly good answer.

MEDIUM

18.4 The decoy point, and the large constant

CUE An evaluation point that looks as though it should trigger a named-angle simplification but does not, such as $x = \pi - 1$, $x = 2$, $x = 4$; or a large specific constant such as 2026 that might be structural or might be noise.

METHOD Two-sided discipline. Going one way: if the point does not simplify anything, accept that, and execute the chain rule cleanly, reporting the transcendental value. A wrong clean answer is worse than a right ugly one. Going the other way: before dismissing a large constant as arbitrary, check whether it makes an identity close. Constants adjacent to a prime, or one less than the index of a root of unity, or one more than a factorial, are the ones to suspect. The test is cheap: try the same construction with the constant replaced by 3 or 5 and see whether a pattern appears.

WORKED EXAMPLE

The decoy. Let $f(x) = \sin(\cos(1 + x^2))$ and find $f'(\pi - 1)$. The appearance of π invites the hope that $1 + x^2$ lands on a named angle. It does not: $1 + (\pi - 1)^2 = 5.5864191$, which is not a rational multiple of π . So differentiate honestly, outside in:

$$f'(x) = -2x \cos(\cos(1 + x^2)) \sin(1 + x^2),$$

and $f'(\pi - 1) = 1.9791992$, matching the central difference 1.9791992. That is the answer. There is nothing further to extract.

The structural constant. Let

$$f(x) = \prod_{k=1}^{2026} \left(x^2 - 2x \cos \frac{k\pi}{2027} + 1 \right), \quad f'(1) = ?$$

Here 2026 is not noise: it is one less than 2027, and each factor is $|x - e^{ik\pi/2027}|^2$ for a 4054th root of unity. The identity is

$$\prod_{k=1}^{n-1} \left(x^2 - 2x \cos \frac{k\pi}{n} + 1 \right) = \frac{x^{2n} - 1}{x^2 - 1} = \sum_{j=0}^{n-1} x^{2j},$$

checked numerically at $n = 3, 5, 8, 13$ and several values of x , all agreeing to ten digits. With $n = 2027$ the product is $\sum_{j=0}^{2026} x^{2j}$, a polynomial, so

$$f'(1) = \sum_{j=0}^{2026} 2j = 2 \cdot \frac{2026 \cdot 2027}{2} = 2026 \cdot 2027 = 4106702.$$

The general formula $f'(1) = n(n-1)$ was confirmed numerically at $n = 3, 5, 13$, returning 6.000, 20.000 and 156.000. Two thousand factors, and the derivative is a single multiplication.

The same constant, twice, once structural and once not. Consider $f(x) = x^{2026}e^{-x}$ and the request for $f'(2026)$. Logarithmic differentiation gives $\frac{f'}{f} = \frac{2026}{x} - 1$, which vanishes exactly at $x = 2026$. So $f'(2026) = 0$, and the constant in the exponent and the constant in the evaluation point were the same number for a reason: the point is the maximum of the integrand of $\Gamma(2027)$. Now consider $g(x) = \sin(2026x)$ and the request for $g'(1)$. Here $2026 \cos(2026) = -1918.4724$, and $\cos(2026) = -0.9469262$ is not any recognisable number. The constant is decoration. The two problems look alike on the page and only the cheap test tells them apart.

WHERE THIS GOES WRONG

Manufacturing a simplification that is not there. If you cannot write down the identity you are appealing to, you do not have one. The mirror-image trap is dismissing a large constant without the cheap test: substituting $n = 3$ or $n = 5$ into the same construction takes thirty seconds and either exposes the pattern or rules it out.

SEE ALSO Evaluation point where most terms vanish; the expression is secretly constant; the label is a decoy.

BOTH ERRORS HAVE THE SAME CURE

Assuming a collapse and missing a collapse are the same mistake made in opposite directions, and both are cured by the same twenty seconds of inspection. Evaluate the pieces. Write down the identity you think applies, in full. If it is there, the problem is finished; if it is not, you have lost twenty seconds and gained certainty that the ugly answer is the right one.

What counts as evidence

The two-sided discipline is only usable if you can say what evidence licenses a collapse, so here is the standard. An algebraic identity counts if you can write it down in full and check it at one numerical value. A vanishing factor counts if you have evaluated that factor at the point and got zero. A degree argument counts if you have expanded the expression into a sum of terms and inspected each one. A symmetry counts if you have named the map under which the expression is invariant.

What does not count: that the point contains π ; that the constant is large and specific; that the expression is too ugly to be the intended answer; that two sample values happened to agree. The first three are the reasons students most often give themselves, and all three are compatible with there being no collapse whatever. The fourth is worse than useless, because a construction can be constant on a set of measure zero, and because two points also lie on a line.

It is worth keeping the five signatures in one list, because they are what you are scanning for and a list of five is short enough to run through in a few seconds. One: a named evaluation point at which a factor, an inner function, or a denominator's derivative vanishes. Two: an expression assembled from identity pairs, conjugates, or a summable series, so that the whole thing reduces to a constant or to x . Three: a differentiation order that exceeds the degree available in the relevant variable. Four: a monotone outer wrapper on a question about location or sign. Five: a statement whose real content is not calculus at all. Every problem in this chapter is one of those five, and most engineered problems are one of the first two.

There is one asymmetry worth exploiting. Checking for a collapse is cheap and bounded: evaluate the factors, sum the series, try $n = 3$. Failing to check is unbounded, because the honest calculation of an engineered expression can run for pages. So the check is always worth making, even when you are fairly sure it will fail, and the only discipline required is to stop checking once it does.

Layers that cannot matter

The next collapse is not about the point or the constant but about the shape of a composite. Questions about *where* a derivative vanishes, or about its *sign*, are often insensitive to the outermost layer of the function, and recognising that saves the entire calculation.

18.5 The outer wrapper is inert

CUE A question about stationary points, monotonicity, or the sign of a derivative for $g(h(x))$, where g is strictly monotone: exp, ln, arctan, an odd power, or the Lambert W .

METHOD If g' is nonzero throughout the relevant range, then $(g \circ h)' = g'(h)h'$ vanishes exactly where h' vanishes and carries the sign of h' wherever $g' > 0$. Differentiate only the inside. The wrapper is inert for questions about location and sign, and it is emphatically *not* inert for the value of the derivative or for the domain of the composite, so check what the question is asking before discarding it.

WORKED EXAMPLE

Let $f(x) = W(\sin \Gamma(x))$ on $[2, \frac{7}{2}]$, where W is the principal Lambert function. Locate the stationary points.

The Lambert function satisfies $W'(u) = \frac{W(u)}{u(1+W(u))}$ on the principal branch, an expression with its own branch analysis, and it is never zero. So the stationary points of f are exactly those of $\sin \Gamma(x)$, whose derivative is $\cos(\Gamma(x))\Gamma'(x)$. Now $\Gamma' = \Gamma\psi$, and the only positive zero of the digamma function is at $x = 1.4616321$, which is outside $[2, \frac{7}{2}]$; on that interval $\psi > 0$, since $\psi(2) = 0.4227843$, so $\Gamma' > 0$ throughout and contributes no stationary point. What remains is $\cos \Gamma(x) = 0$, that is $\Gamma(x) = \pi/2$, whose solution on the interval is

$$x^* = 2.7209149, \quad \Gamma(x^*) = 1.5707963 = \frac{\pi}{2}.$$

That is the unique stationary point, and it is a maximum, with $f(x^*) = W(1) = 0.5671433$, the omega constant. Numerically $f'(2.6) = +0.0549981$ and $f'(2.85) = -0.0978350$, straddling it, and $f'(x^*) = 0$ to machine precision. At every sample point the sign of f' agrees with the sign of $(\sin \Gamma)'$, as the argument predicts.

The same argument in a much more ordinary setting: to maximise $x^{1/x}$ for $x > 0$, do not differentiate $x^{1/x}$. Since ln is strictly increasing, the maximiser of $x^{1/x}$ is the maximiser of $\frac{\ln x}{x}$, whose derivative is $\frac{1 - \ln x}{x^2}$, zero at $x = e$.

The maximum value is $e^{1/e} = 1.4446679$. Differentiating $x^{1/x}$ directly is possible, needs logarithmic differentiation anyway, and produces the factor $x^{1/x}$ that was never going to vanish. Every optimisation of a product, a quotient, or a power is an instance of this: take logarithms, because the logarithm is the archetypal inert wrapper.

Now the part the inertness argument does *not* cover. Extend the question to $[2, 4]$. The principal W requires its argument to be at least $-1/e = -0.3678794$, and $\sin \Gamma(x)$ drops below that: it crosses $-1/e$ downward at $x = 3.5512873$ and returns at $x = 3.9874729$, reaching -1 at $x = 3.8032322$, where $\Gamma(x) = 3\pi/2$. So f is undefined on an interval of length 0.436, and the stationary point of $\sin \Gamma$ at 3.8032322 is not a stationary point of f because f does not exist there. The wrapper cannot move a stationary point, but it can delete one by removing it from the domain.

WHERE THIS GOES WRONG

Computing W' , or Γ' in closed form, only to find that neither vanishes. The cost is high and the information gained is zero. The opposite trap is the one exposed above: treating the wrapper as inert for a question it does not govern. “Where is $f' = 0$ ” and “what is the sign of f' ” are insensitive to a monotone outer layer; “what is $f'(a)$ ”, “where is f defined”, and “how many stationary points does f have” are not.

SEE ALSO Sign charts and extremum counting; chain-rule discipline; the expression is secretly constant.

When the label lies

The last collapse is the most complete. Some problems filed under differentiation contain no function to differentiate, or contain one whose differentiation is a single line arriving after all the real work is done. The signature is that the statement is long and the word “derivative” appears once, near the end.

18.6 The label is a decoy: no derivative is required

CUE A problem headed “derivative” whose statement is mostly geometry, number theory, or algebra, and in which the object to be differentiated is not even named until the final clause; or a ratio of two evaluations of the same unknown polynomial.

METHOD Read the question, not the heading. Identify what is actually unknown and what the constraints actually determine. The recurring cores hiding under a derivative label are worth listing: Vieta relations on a quadratic whose roots are trigonometric functions of the angles of a triangle; complex-region geometry reduced to a point-to-line distance; a divisor count $\tau(N)$ feeding into Γ of a small integer; the reflection symmetry of $\ln x$ and e^x about the line $y = x$; and a fixed point of an iterated map that turns a tower of exponents into a constant. In every case the differentiation, when it comes, is one line.

WORKED EXAMPLE

Vieta under a derivative heading. In right triangle PQR with $\angle R = \pi/2$, the numbers $\tan \frac{P}{2}$ and $\tan \frac{Q}{2}$ are the roots of $Q(x) = a(x^2 - sx + p)$. Compute $\frac{Q'(1/2)}{Q(0)}$.

Nothing about this requires differentiating anything interesting. Since $P + Q = \pi/2$, we have $\frac{P+Q}{2} = \frac{\pi}{4}$, and the tangent addition formula gives

$$\tan \frac{P+Q}{2} = \frac{\tan \frac{P}{2} + \tan \frac{Q}{2}}{1 - \tan \frac{P}{2} \tan \frac{Q}{2}} = \frac{s}{1-p} = 1,$$

so $s = 1 - p$, that is $s + p = 1$. Now the one line of calculus: $Q'(x) = a(2x - s)$, so $Q'(1/2) = a(1 - s) = ap$. And $Q(0) = ap$. Hence the ratio is 1, independently of a and of the triangle. Random right triangles and random a reproduce $s + p = 1.0000000$ and the ratio 1.0000000 .

Geometry under a derivative heading. Find the minimum distance between the curves $y = \ln x$ and $y = e^x$. The tempting move is to set up the squared distance between (x, e^x) and $(t, \ln t)$ and solve a two-variable stationarity system. Do not. The curves are reflections of one another in the line $y = x$, so the segment realising the minimum is perpendicular to that line, and the problem reduces to minimising the distance from $y = e^x$ to the line, namely $\frac{e^x - x}{\sqrt{2}}$. One

derivative: $e^x - 1 = 0$ at $x = 0$, giving distance $\frac{1}{\sqrt{2}}$ from the curve to the line. Doubling, the distance between the

curves is $\sqrt{2}$, attained between $(0, 1)$ and $(1, 0)$. A brute-force descent over both curves converges to 1.4142136 with closest points $(0, 1)$ and $(1, 0)$, against $\sqrt{2} = 1.4142136$.

A fixed point under a derivative heading. Let $y = x^{x^{x^{\dots}}}$ and suppose $y = 2$. Find $\frac{dy}{dx}$ there. The infinite tower is the fixed point of $u \mapsto x^u$, so it satisfies $y = x^y$, and this is the whole content of the tower notation: the exponent becomes the value. With $y = 2$ the equation reads $2 = x^2$, so $x = \sqrt{2}$. Only now does any calculus happen. Taking logarithms of $y = x^y$ gives $\ln y = y \ln x$, and differentiating,

$$\frac{y'}{y} = y' \ln x + \frac{y}{x} \implies y' = \frac{y^2}{x(1 - y \ln x)}.$$

At $x = \sqrt{2}$, $y = 2$: $y \ln x = 2 \cdot \frac{1}{2} \ln 2 = \ln 2$, so $y' = \frac{4}{\sqrt{2}(1 - \ln 2)} = 9.2175367$. Solving $y = x^y$ numerically on either

side of $\sqrt{2}$ and differencing returns 9.2175367. Iterating the tower at $x = \sqrt{2}$ converges to 2.0000000, confirming that the branch chosen is the right one.

WHERE THIS GOES WRONG

Spending the whole time budget hunting for a derivative technique when the intended skill is elsewhere. The tell is that the statement contains a great deal of structure that no differentiation would use. A second and more specific error: when the extremum sits on an open boundary, an infimum exists but a minimum does not, and reporting a minimum is then wrong even though the number is right. Check that the optimising point is in the domain before naming it.

SEE ALSO Constrained optimisation; the expression is secretly constant; the self-similar fixed point.

Why these problems are worth studying

It is tempting to file this chapter under recreation: engineered problems, competition tricks, curiosities that will never appear in an engineering calculation. That would be a mistake, and the reason is methodological. An ordinary exercise requires recognition and computation together, and shares its marks between them, so a solver who finishes one never learns which of the two was carrying the answer. The problems here are built so that the computation is unaffordable and only recognition can pay. Each of them is therefore a controlled experiment with a single factor varied, and the thing it measures is whether you looked before you started.

The structures being detected are not artificial, which is why the training transfers. A factor that vanishes at the point of interest is what a boundary condition looks like. An expression that is secretly constant is what a conservation law looks like. A differentiation order that exceeds the available degree is what a truncated model looks like. A monotone outer wrapper is what a change of units, a link function, or a log-likelihood looks like. No setter invented any of these; a setter merely arranges for one of them to be decisive and for every other route to be a wall. Meeting them here, where a shortcut is guaranteed to exist, is cheap practice for meeting them where nothing guarantees it and where nobody tells you afterwards that one was available.

The thesis of this chapter is narrow and worth stating plainly. Differentiation is cheap in principle and expensive in practice, and the expense is almost entirely in the terms you did not need. Every technique here removes terms before they are written: the vanishing point removes all but one of them, constancy removes all of them, insufficient degree removes all of them again, the monotone wrapper removes an entire layer, and the mislabelled problem removes the differentiation altogether. The habit that produces all five is the same, and it is the habit this whole book is about. Look at the object before you apply a rule to it, and let the structure decide the method rather than the other way round.

Recognition matrix

If you see	Reach for	If that fails
Three or more factors, derivative at one point	Write the product rule, then delete terms with a vanishing factor	Take logarithms and differentiate
$x = 0$ in the statement	$\sin, \tan, \arcsin, \arctan, \ln(1+x), x^k$ all vanish	Expand to first order in x
$x = 1$ with a $\ln x$ factor present	Only the term carrying $1/x$ survives	Substitute $x = 1 + h$ and expand
$\prod_k (x - a_k)$, derivative at some a_j	$\prod_{k \neq j} (a_j - a_k)$; one term survives	Logarithmic differentiation
$f^{(n)}(0)$ for large n , only at the origin	Maclaurin coefficient: $n! [x^n]f$	Expand the outer function in its own variable
f even or odd, a derivative at 0	Parity kills the opposite half outright	Split f into even and odd parts
Nested trig over a geometric series	Sum the series; suspect an identity	Evaluate at two points and compare
Conjugate logs, or $\arctan x + \arctan \frac{1}{x}$	Constant; the derivative is 0	Simplify one layer at a time
A functional inequality for all arguments	Free substitution; expect rigidity, then $f' = 0$	Set the arguments equal, then opposite
$\frac{\partial^{n+m} f}{\partial x^n \partial y^m}$ with n, m huge	Expand first; kill one-variable terms and low degrees	Coefficient extraction, $n! m! [x^n y^m]f$
A determinant with a high mixed partial	Expand the determinant once, then classify terms	Check both orders exceed both degrees
A point like $\pi - 1$ or 2 that almost simplifies	Nothing; differentiate honestly	Report the transcendental value
A large constant such as 2026	Test whether $n - 1$ completes a root-of-unity identity	Rerun the construction with $n = 3$
$W(\cdot), \ln(\cdot), \arctan(\cdot)$ wrapped round the real content	Stationary points and signs come from the inside only	Check the wrapper's domain separately
"Derivative" but no function in the statement	Solve the real problem; differentiate at the end	Look for Vieta, symmetry, or a fixed point
Two curves that are mirror images in $y = x$	Distance to the line, then double	Check the contact point lies in the domain

PART IV

Integrals

Structural rewriting before integrating

What this chapter is for. An integrand that resists every technique in the book usually resists because it has not been simplified. This chapter covers the pass you make *before* choosing a method: cancel what cancels, divide when the numerator is too big, decompose what is rational, complete the square under every quadratic, multiply by a conjugate when two surds refuse to combine, and linearise every product of trigonometric functions. None of these operations is an integration technique. All of them decide which integration technique you will need, and several of them finish the problem outright. The organising fact is that the standard integral table is short, so the whole art is manoeuvring an expression into one of its entries. At the end you should be able to look at a rational function and know in advance exactly which of four shapes its antiderivative has, and you should never again attempt integration by parts on something that was secretly a constant.

The first move is never a technique

Beginners reach for a method. Experienced solvers reach for a pencil and rewrite. The reason is not aesthetic. Every technique in the remaining chapters of this part has a precondition: substitution needs a visible inner function, parts needs a factor that improves when differentiated, partial fractions needs a proper rational function. Rewriting is what establishes those preconditions, and an integrand that satisfies none of them is usually one algebraic step away from satisfying one.

There is a second reason, less obvious and more important. Problems are constructed, and the construction almost always runs backwards from a simple answer. A setter picks $\int_1^2 \ln x \, dx$, then multiplies numerator and denominator by $\cos^4 x$ written in three different disguises, then expands. The forward direction looks hopeless; the backward direction is one cancellation. If an integrand looks engineered, it was, and the design is nearly always a factor hidden on both sides of a fraction bar.

THE REWRITING PASS

Before choosing a method, run this list. *One:* are numerator and denominator sharing a factor once you expand and apply $\sin^2 + \cos^2 = 1$ to each separately? *Two:* do the exponent and logarithm laws collapse anything? *Three:* is the fraction improper, so that division comes first? *Four:* does every quadratic appear as a completed square? *Five:* is every product of trigonometric functions at distinct frequencies written as a sum? Only when all five answers are no have you earned the right to pick a technique.

There is a practical test for whether you are looking at an engineered problem, and it costs nothing. Evaluate the integrand at one or two convenient points and compare with something simple. If $f(1)$ and $f(2)$ both equal $\ln 1$ and $\ln 2$, you are looking at $\ln x$ in a costume. If the integrand is constant at three points, it is constant. This is not a proof and it is not meant to be; it is a way of deciding, in ten seconds, whether to spend the next ten minutes simplifying or the next ten minutes integrating.

PROBLEMS ARE BUILT BACKWARDS

A setter starts from an answer, not from a question. The usual construction is to take an easy integrand $h(x)$, choose a nonzero function φ , and present $\frac{h(x)\varphi(x)}{\varphi(x)}$ with numerator and denominator expanded and rearranged so that the shared factor is invisible. Every symptom of this construction is a symptom of a cancellation: an unusually high degree in both numerator and denominator, trigonometric squares at more than one frequency, a numerator whose terms all carry the same transcendental factor. When you see them, look for φ rather than for a technique.

19.1 Algebraic camouflage collapse

CUE An integrand engineered to look terrible: stacked trigonometric squares over a product of trigonometric factors, a tower of logarithms, a ratio of two near-identical polynomials. Nothing suggests a substitution because there is nothing to substitute.

METHOD Treat the integrand as raw algebra. Expand, apply the Pythagorean identities to the numerator and to the denominator *separately*, and cancel. The design principle behind such problems is that the numerator is secretly (denominator) $\times$ (something elementary), so the entire difficulty evaporates in one line and what remains is a table integral.

WORKED EXAMPLE

$$\int_1^2 \frac{\ln x - \ln x \sin^2 x - \frac{1}{4} \ln x \sin^2(2x)}{\cos^2 x (1 - \sin^2 x)} dx.$$

The denominator is $\cos^2 x \cdot \cos^2 x = \cos^4 x$. For the numerator, factor out $\ln x$ and use $\sin^2(2x) = 4 \sin^2 x \cos^2 x$:

$$\ln x (1 - \sin^2 x - \sin^2 x \cos^2 x) = \ln x (\cos^2 x - \sin^2 x \cos^2 x) = \ln x \cos^2 x (1 - \sin^2 x) = \ln x \cos^4 x.$$

Since $\cos x \neq 0$ on $[1, 2]$, the quotient is $\ln x$ and the integral is $[x \ln x - x]_1^2 = 2 \ln 2 - 1 \approx 0.386294$, confirmed by quadrature to twelve digits.

A second instance where the camouflage is a genuine identity rather than a manufactured factor:

$$\int_0^{\pi/2} (\sin^6 x + \cos^6 x + 3 \sin^2 x \cos^2 x) dx.$$

Cubing $\sin^2 x + \cos^2 x = 1$ gives $\sin^6 x + \cos^6 x = 1 - 3 \sin^2 x \cos^2 x$, so the integrand is identically 1 and the integral is $\pi/2$. Anyone who reaches for power reduction here will produce four correct pages and the same answer.

WHERE THIS GOES WRONG

Cancelling a factor without checking that it is nonzero on the interval of integration. Dividing by $\cos^4 x$ is legitimate on $[1, 2]$ because $\pi/2 \approx 1.5708$ lies outside it, but only just; on $[1, 2.5]$ the same cancellation would be removing a genuine singularity from both sides and the resulting integral would be improper. State the check, do not skip it.

SEE ALSO Exponent and logarithm law collapse; factorisation identities; trigonometric identity pre-processing.

The camouflage above was trigonometric. The same trick is played with exponents and logarithms, and there the collapse is even more complete, because the laws involved are unconditional identities rather than a single Pythagorean relation.

EASY

19.2 Exponent and logarithm law collapse

CUE The same base carrying two different exponents, several logarithm bases in one expression, $e^{-\ln A}$, or a constant exponent written in a deliberately irrational way.

METHOD Apply the laws before doing anything else. $x^a x^b = x^{a+b}$, and if $a + b = \sin^2 x + \cos^2 x$ the exponent is 1. Every logarithm reduces to $\ln$ via $\log_b x = \ln x / \ln b$, so mixed bases produce constants that cancel. And $e^{-\ln A} = 1/A$, $e^{k \ln A} = A^k$. The point is that these steps cost nothing and can reduce a transcendental-looking integrand to a polynomial.

WORKED EXAMPLE

$$\int_0^{\sqrt{2\pi}} x^{\sin^2 x} x^{\cos^2 x} dx = \int_0^{\sqrt{2\pi}} x^{\sin^2 x + \cos^2 x} dx = \int_0^{\sqrt{2\pi}} x dx = \frac{2\pi}{2} = \pi.$$

A second, with mixed bases:

$$\int_1^{e^2} \frac{\log_3 x}{x \log_3 e} dx = \int_1^{e^2} \frac{\ln x / \ln 3}{x (1/\ln 3)} dx = \int_1^{e^2} \frac{\ln x}{x} dx = \left[\frac{1}{2} (\ln x)^2 \right]_1^{e^2} = 2.$$

The factor $\ln 3$ was never going to matter, and noticing that in advance saves you from carrying it through a substitution. A third, one line long:

$$\int_0^1 e^{-\ln(1+x)} dx = \int_0^1 \frac{dx}{1+x} = \ln 2.$$

The collapse changes the domain. The expression $x^{f(x)}$ is defined only for $x > 0$, so $x^{\sin^2 x} x^{\cos^2 x}$ agrees with x on $(0, b]$ and is undefined at $x = 0$; the integral is improper at the left endpoint and converges because the integrand extends continuously by 0. Write the identity on the open interval, then take the limit.

SEE ALSO Algebraic camouflage collapse; exponential substitution; factorisation identities.

Both of the collapses so far relied on identities you already know. The third kind relies on recognising that the denominator in front of you is one factor of a familiar factorisation, and that the partner factor is sitting in the numerator waiting to cancel.

19.3 Factorisation identities

EASY

CUE A quotient whose denominator is a recognisable piece of a standard factorisation: $\frac{1-x^n}{1-x}$, $1-x^{1/3}+x^{2/3}$, x^2-x+1 , x^4+x^2+1 , $x^3 \pm 1$.

METHOD Identify the partner factor and cancel. The workhorses are the geometric identity $\frac{1-x^n}{1-x} = 1+x+\dots+x^{n-1}$; the sum and difference of cubes $a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$, which handles $x^2-x+1 = \frac{x^3+1}{x+1}$ and its cube-root analogues; and the Sophie Germain style split

$$x^4 + x^2 + 1 = (x^2 + x + 1)(x^2 - x + 1),$$

obtained by adding and subtracting x^2 . Each of these turns a fourth-degree or fractional-power denominator into something already in the table.

WORKED EXAMPLE

Take $\int_0^1 \frac{1+x}{1-x^{1/3}+x^{2/3}} dx$. Write $a = 1$, $b = x^{1/3}$, so $1+x = a^3+b^3 = (1+x^{1/3})(1-x^{1/3}+x^{2/3})$. The denominator cancels entirely and

$$\int_0^1 (1+x^{1/3}) dx = 1 + \frac{3}{4} = \frac{7}{4},$$

confirmed numerically as 1.750000000000.

Now the quartic split. In $\int_0^1 \frac{x^2+x+1}{x^4+x^2+1} dx$ the denominator factors as $(x^2+x+1)(x^2-x+1)$, so the integrand is $\frac{1}{x^2-x+1}$. Completing the square gives $(x-\frac{1}{2})^2 + \frac{3}{4}$ and

$$\int_0^1 \frac{dx}{x^2-x+1} = \left[\frac{2}{\sqrt{3}} \arctan \frac{2x-1}{\sqrt{3}} \right]_0^1 = \frac{2}{\sqrt{3}} \left(\frac{\pi}{6} + \frac{\pi}{6} \right) = \frac{2\pi\sqrt{3}}{9} \approx 1.209200.$$

Finally the geometric one: $\int_0^1 \frac{1-x^5}{1-x} dx = \int_0^1 (1+x+x^2+x^3+x^4) dx = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} = \frac{137}{60}$.

And the sum of cubes in its plainest form, where the whole integral disappears:

$$\int_0^1 \frac{x^3+1}{x^2-x+1} dx = \int_0^1 (x+1) dx = \frac{3}{2}.$$

Compare the work required if the factorisation is missed: the denominator is irreducible with $\Delta = -3$, the fraction is improper, and an honest solver would divide, decompose and produce a logarithm and an arctangent that must then cancel exactly.

WHERE THIS GOES WRONG

Treating $x = 1$ as a singularity of $\frac{1-x^5}{1-x}$ and splitting the integral there. The singularity is removable: the function equals a polynomial everywhere except at one point, and the definite integral is entirely ordinary. Splitting at a removable singularity is harmless but signals that you have not seen the factorisation, and on a quotient like $\frac{1-x^n}{1-x}$ it usually precedes an attempt at partial fractions on a denominator with n complex roots.

SEE ALSO Polynomial long division; partial fractions with distinct linear factors; completing the square.

So far the hidden structure was finite: a factor, an identity, a law. It can also be infinite, and then the collapse is a fixed-point calculation rather than an algebraic one. This is rarer, but the recognition is unmistakable once you have seen it once.

19.4 Self-similar collapse: nested radicals and continued fractions

CUE The integrand contains an object that repeats inside itself: $\sqrt{g(x) + \sqrt{g(x) + \dots}}$ continuing forever, an infinite continued fraction, or a power tower.

METHOD Name the repeating object y , write down the equation it satisfies by substituting the whole into a copy of itself, and solve. From $y = \sqrt{g + y}$ we get $y^2 - y - g = 0$, hence

$$y = \frac{1}{2}(1 + \sqrt{1 + 4g}),$$

taking the positive root because $y \geq 0$. A *finite* nesting is not a fixed point and must instead be collapsed bottom upwards as ordinary algebra; confusing the two changes the answer completely.

WORKED EXAMPLE

The infinite case first. For $x \in [0, 1]$,

$$\int_0^1 \sqrt{x + \sqrt{x + \sqrt{x + \dots}}} dx = \int_0^1 \frac{1 + \sqrt{1 + 4x}}{2} dx = \frac{1}{2} + \frac{1}{2} \left[\frac{(1 + 4x)^{3/2}}{6} \right]_0^1 = \frac{1}{2} + \frac{5\sqrt{5} - 1}{12},$$

which is 1.348361657292; direct quadrature on the truncated nesting agrees to twelve digits by the tenth level.

The finite case, where the fixed-point reflex would be wrong. The three-level fraction

$$\left(1 + \frac{1}{1 + \frac{1}{1 + x}} \right)^{-1}$$

collapses from the bottom, one level at a time: $1 + \frac{1}{1+x} = \frac{2+x}{1+x}$, its reciprocal is $\frac{1+x}{2+x}$, adding 1 gives $\frac{3+2x}{2+x}$, and the reciprocal of that is $\frac{2+x}{2x+3}$. Then, writing $\frac{2+x}{2x+3} = \frac{1}{2} \left(1 + \frac{1}{2x+3} \right)$,

$$\int_0^1 \frac{2+x}{2x+3} dx = \frac{1}{2} + \frac{1}{4} \ln \frac{5}{3} \approx 0.627706.$$

The corresponding infinite continued fraction is a different function entirely. If $z = x + \frac{1}{x + \frac{1}{x + \dots}}$ then $z = x + \frac{1}{z}$,

so $z^2 - xz - 1 = 0$ and $z = \frac{1}{2}(x + \sqrt{x^2 + 4})$, the positive root. Hence

$$\int_0^1 z dx = \frac{1}{4} + \frac{1}{2} \int_0^1 \sqrt{x^2 + 4} dx = \frac{1}{4} + \frac{\sqrt{5}}{4} + \ln \frac{1 + \sqrt{5}}{2} \approx 1.290229,$$

using $\int \sqrt{x^2 + a^2} dx = \frac{x}{2} \sqrt{x^2 + a^2} + \frac{a^2}{2} \operatorname{arsinh} \frac{x}{a}$. Nothing about the written expression distinguishes the finite three-level fraction from the infinite one except the word “infinite”, and the two answers are not close.

WHERE THIS GOES WRONG

Taking the wrong root of the fixed-point quadratic. The equation $y^2 = g + y$ has two solutions and only the one that is real, nonnegative and continuous on the whole interval is the limit of the nesting. The second error is treating a three-level fraction as though it were infinite: an infinite continued fraction $z = x + 1/z$ gives $z = \frac{1}{2}(x + \sqrt{x^2 + 4})$, which is nothing like the rational function a finite one produces.

SEE ALSO Rationalising a radical by substitution; algebraic camouflage collapse; conjugate multiplication.

The last of the collapse family concerns inverse trigonometric functions, where the simplification is a triangle rather than an identity, and where the branch restrictions are the whole difficulty.

19.5 Inverse-function and co-function collapse

CUE Nested inverse trigonometric functions: $\sin(\operatorname{arcsec} x)$, $\tan(\operatorname{arccot} u)$, $\arctan \frac{1-x}{1+x}$, $\arctan \frac{2x}{1-x^2}$, or $\arctan t$ next to $\arctan(1/t)$.

METHOD Peel with a right triangle and the standard identities: $\operatorname{arccot} z = \frac{\pi}{2} - \arctan z$; $\arctan a - \arctan b = \arctan \frac{a-b}{1+ab}$ when the result stays in $(-\frac{\pi}{2}, \frac{\pi}{2})$; $\arctan t + \arctan \frac{1}{t} = \frac{\pi}{2}$ for $t > 0$; $\sin(\operatorname{arcsec} x) = \frac{\sqrt{x^2-1}}{x}$ for $x \geq 1$; and $\arctan \frac{2x}{1-x^2} = 2 \arctan x$ for $|x| < 1$.

WORKED EXAMPLE

$$\int_1^2 \sin(\operatorname{arcsec} x) \, dx = \int_1^2 \frac{\sqrt{x^2-1}}{x} \, dx.$$

Differentiating $\sqrt{x^2-1} - \operatorname{arcsec} x$ gives $\frac{x}{\sqrt{x^2-1}} - \frac{1}{x\sqrt{x^2-1}} = \frac{x^2-1}{x\sqrt{x^2-1}} = \frac{\sqrt{x^2-1}}{x}$, so the value is $\sqrt{3} - \frac{\pi}{3} \approx 0.684853$.

The subtraction identity in action:

$$\int_0^1 \arctan \frac{1-x}{1+x} \, dx = \int_0^1 \left(\frac{\pi}{4} - \arctan x \right) \, dx,$$

valid because $\frac{1-x}{1+x} = \frac{\tan(\pi/4) - \tan(\arctan x)}{1 + \tan(\pi/4)\tan(\arctan x)}$ and both angles lie in $[0, \pi/4]$. Since $\int_0^1 \arctan x \, dx = \frac{\pi}{4} - \frac{1}{2} \ln 2$ by parts, the answer is $\frac{1}{2} \ln 2 \approx 0.346574$.

The doubling identity, used inside its branch:

$$\int_0^{1/2} \arctan \frac{2x}{1-x^2} \, dx = 2 \int_0^{1/2} \arctan x \, dx = 2 \left[x \arctan x - \frac{1}{2} \ln(1+x^2) \right]_0^{1/2} = \arctan \frac{1}{2} - \ln \frac{5}{4} \approx 0.240504.$$

The upper limit $\frac{1}{2}$ was chosen to stay inside $|x| < 1$; at $x = 1$ the argument $\frac{2x}{1-x^2}$ blows up and the identity changes form.

WHERE THIS GOES WRONG

Crossing a branch cut in the middle of the interval. The identity $\arctan \frac{2x}{1-x^2} = 2 \arctan x$ holds only on $|x| < 1$; for $x > 1$ the correct statement is $2 \arctan x - \pi$. An integral over $[0, 2]$ that uses the unrestricted form is wrong by π times the length of the offending piece, and nothing in the algebra warns you.

SEE ALSO Trigonometric identity pre-processing; conjugate multiplication; completing the square.

Rational integrands: divide, align, decompose

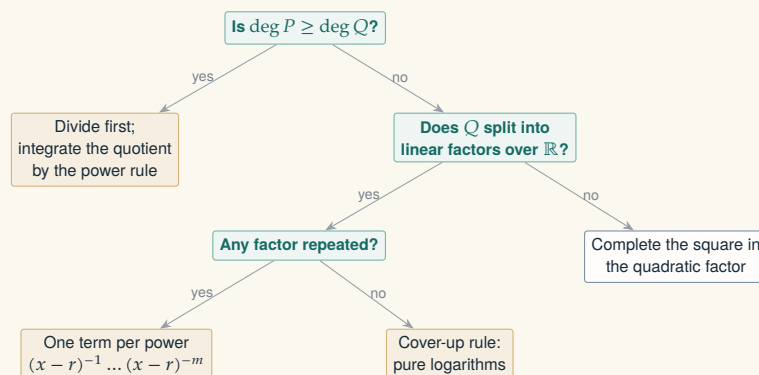
Every rational function has an elementary antiderivative, and there are exactly four shapes it can be built from: a polynomial, a logarithm $\ln|x-r|$, a negative power $(x-r)^{-k}$, and an arctangent. Knowing this in advance changes how you work, because it means the entire problem is bookkeeping: get the fraction proper, factor the denominator, allocate one term per factor, and read off which of the four shapes each term produces. Nothing is ever discovered along the way.

The order of operations is forced. Division comes first, because a partial-fraction ansatz on an improper fraction has no solution and the linear system silently returns nonsense. Factoring the denominator comes second, because the shape of the decomposition depends only on the factorisation. Only then do you solve for coefficients, and the cover-up rule makes most of that solving unnecessary. The tree below is the whole procedure.

The claim that there are only four shapes is worth stating precisely, because it is what makes the whole procedure mechanical. Factor the denominator over the reals into linear and irreducible quadratic pieces, which the fundamental theorem of algebra guarantees is possible. Then each piece contributes exactly the terms in the table below, and each of those terms has an antiderivative you already know.

Factor of Q	Terms allocated	Antiderivative produced
$x - r$, appearing once	$\frac{A}{x - r}$	$A \ln x - r $
$(x - r)^m$	$\frac{A_1}{x - r} + \dots + \frac{A_m}{(x - r)^m}$	one ln, plus $\frac{-A_k}{(k - 1)(x - r)^{k-1}}$
$x^2 + bx + c, \Delta < 0$	$\frac{Bx + C}{x^2 + bx + c}$	$\frac{B}{2} \ln(x^2 + bx + c)$ plus an arctan
$(x^2 + bx + c)^m, \Delta < 0$	$\sum_{k=1}^m \frac{B_k x + C_k}{(x^2 + bx + c)^k}$	negative powers, one ln, one arctan

Count the unknowns and you find exactly $\deg Q$ of them, which is exactly the number of equations that matching coefficients supplies, so the system is square and, because the decomposition is unique, nonsingular. That is the theoretical guarantee. In practice you almost never solve the square system: cover-up delivers the coefficients attached to the highest power of each factor for free, and the remaining ones fall out of one or two coefficient comparisons at convenient values of x .



EASY

19.6 Polynomial long division

CUE A rational integrand whose numerator has degree greater than or equal to that of its denominator.

METHOD Divide, obtaining $\frac{P(x)}{Q(x)} = S(x) + \frac{R(x)}{Q(x)}$ with $\deg R < \deg Q$. The polynomial part S integrates by the power rule and the proper remainder is what the rest of the section handles. For a linear divisor, synthetic division is faster than the long form; for $x^2 + 1$ as divisor, repeated use of $x^2 = (x^2 + 1) - 1$ is faster still.

WORKED EXAMPLE

$$\int_0^1 \frac{x^3}{x + 1} dx.$$

Dividing, $x^3 = (x + 1)(x^2 - x + 1) - 1$, so the integrand is $x^2 - x + 1 - \frac{1}{x + 1}$ and

$$\int_0^1 \left(x^2 - x + 1 - \frac{1}{x + 1} \right) dx = \frac{1}{3} - \frac{1}{2} + 1 - \ln 2 = \frac{5}{6} - \ln 2 \approx 0.140186.$$

With an even denominator the division is worth doing by the $x^2 = (x^2 + 1) - 1$ route:

$$\frac{x^4}{x^2 + 1} = \frac{(x^2 + 1)^2 - 2(x^2 + 1) + 1}{x^2 + 1} = x^2 - 1 + \frac{1}{x^2 + 1},$$

$$\text{so } \int_0^1 \frac{x^4}{x^2 + 1} dx = \frac{1}{3} - 1 + \frac{\pi}{4} = \frac{\pi}{4} - \frac{2}{3} \approx 0.118731.$$

WHERE THIS GOES WRONG

Setting up a partial-fraction ansatz on an improper fraction. Matching $\frac{x^3}{x + 1}$ against $\frac{A}{x + 1}$ gives $x^3 = A$, which has no solution, and a solver who instead “fits” the constant at $x = -1$ obtains $A = -1$ and an answer short by $\int_0^1 (x^2 - x + 1) dx$. Check degrees before writing a single letter.

SEE ALSO Numerator alignment; partial fractions with distinct linear factors; factorisation identities.

Division handles the case where the numerator is too big outright. There is a softer version of the same idea, used when the degrees are already fine but the numerator nearly matches the denominator, and it is the cheapest way to reduce a power in the denominator without setting up any system at all.

19.7 Numerator alignment: add and subtract the denominator

MEDIUM

CUE A numerator that almost is the denominator, or almost is a factor of it: x^2 over $(1 + x^2)^2$, $\sin^2 x$ over $1 + \sin^2 x$, x^2 over $x^2 + a^2$.

METHOD Force the denominator into the numerator and split. Writing $x^2 = (x^2 + a^2) - a^2$ gives the identity that does most of the work in this family,

$$\frac{x^2}{(x^2 + a^2)^2} = \frac{1}{x^2 + a^2} - \frac{a^2}{(x^2 + a^2)^2},$$

in which each piece has one power fewer or is already a table entry. Iterating drops the exponent one step at a time until you land on arctan or ln.

WORKED EXAMPLE

$$\int_0^1 \frac{x^2}{(1 + x^2)^2} dx = \int_0^1 \frac{dx}{1 + x^2} - \int_0^1 \frac{dx}{(1 + x^2)^2}.$$

The first is $\pi/4$. The second is the standard $\left[\frac{x}{2(1 + x^2)} + \frac{1}{2} \arctan x \right]_0^1 = \frac{1}{4} + \frac{\pi}{8}$. Subtracting, the value is $\frac{\pi}{8} - \frac{1}{4} \approx 0.142699$.

The same move works when the “denominator” is trigonometric:

$$\int_0^{\pi/2} \frac{\sin^2 x}{1 + \sin^2 x} dx = \int_0^{\pi/2} \left(1 - \frac{1}{1 + \sin^2 x} \right) dx = \frac{\pi}{2} - \frac{\pi}{2\sqrt{2}} \approx 0.460076,$$

using $\int_0^{\pi/2} \frac{dx}{1 + \sin^2 x} = \frac{\pi}{2\sqrt{2}}$, which the substitution $t = \tan x$ delivers in one line.

SPLITTING A NUMERATOR IS THE GENERAL MOVE

Alignment, the derivative-plus-constant split of $Bx + C$, and the decomposition of a partial fraction are three faces of one principle: a numerator should be written as a sum of pieces, each of which matches a table entry against the denominator in front of it. There are only three pieces worth aiming at. A multiple of $Q'(x)$ gives $\ln|Q|$. A multiple of $Q(x)$ itself cancels and drops the degree. Anything left over is a constant, and a constant over a completed square is an arctangent, an arcsine, or a negative power. Write every numerator as (multiple of Q') plus (multiple of Q) plus (remainder) and there is nothing left to decide.

WHERE THIS GOES WRONG

Splitting into two divergent pieces. On an infinite interval $\int_1^\infty \frac{x^2}{(1 + x^2)^2} dx$ converges, but the alignment produces $\int_1^\infty \frac{dx}{1 + x^2}$ and $\int_1^\infty \frac{dx}{(1 + x^2)^2}$, and here both happen to converge. Change the numerator to x^3 and the two fragments diverge individually while their difference does not. Verify convergence of each piece before separating.

SEE ALSO Polynomial long division; partial fractions with repeated and irreducible factors; completing the square.

Alignment is opportunistic. Partial fractions is the systematic version: it decomposes any proper rational function, once and for all, into pieces whose antiderivatives are known. The simplest case, distinct linear factors, also admits a shortcut that removes the linear system entirely.

19.8 Partial fractions with distinct linear factors, and the cover-up rule

CUE A denominator that factors into distinct linear pieces. Often the factorisation appears only after a substitution: a cubic in e^x , a quartic in $u = \sin x$, a quadratic in x^2 .

METHOD Write

$$\frac{P(x)}{\prod_k (x - r_k)} = \sum_k \frac{A_k}{x - r_k}, \quad A_k = \frac{P(r_k)}{\prod_{j \neq k} (r_k - r_j)}.$$

The formula for A_k is the cover-up rule: cover the factor $(x - r_k)$ with a finger and evaluate everything that remains at $x = r_k$. Each term integrates to $A_k \ln |x - r_k|$, and the sum should be combined into a single logarithm of a product before any limit is taken.

WORKED EXAMPLE

Cover-up on a small example first. For $\frac{2x+3}{(x-1)(x+2)}$: cover $(x-1)$ and evaluate $\frac{2x+3}{x+2}$ at $x = 1$, giving $\frac{5}{3}$; cover $(x+2)$ and evaluate $\frac{2x+3}{x-1}$ at $x = -2$, giving $\frac{-1}{-3} = \frac{1}{3}$. Hence

$$\int_2^3 \frac{2x+3}{(x-1)(x+2)} dx = \frac{5}{3} \ln 2 + \frac{1}{3} \ln \frac{5}{4} \approx 1.229626.$$

Now the case that shows why combining matters. With $u = e^x$,

$$\int_1^\infty \frac{6 du}{u(u+1)(u+2)(u+3)}.$$

The four cover-up values are $\frac{6}{1 \cdot 2 \cdot 3} = 1$ at $u = 0$, $\frac{6}{(-1)(1)(2)} = -3$ at $u = -1$, $\frac{6}{(-2)(-1)(1)} = 3$ at $u = -2$, and $\frac{6}{(-3)(-2)(-1)} = -1$ at $u = -3$: the signed binomial row $1, -3, 3, -1$, which is what a numerator of $3!$ over four consecutive shifts always produces. The antiderivative is

$$\ln u - 3 \ln(u+1) + 3 \ln(u+2) - \ln(u+3) = \ln \frac{u(u+2)^3}{(u+1)^3(u+3)},$$

whose argument tends to 1 as $u \rightarrow \infty$, so the upper limit contributes 0, and the value is $-\ln \frac{27}{32} = \ln \frac{32}{27} \approx 0.169899$.

THE DENOMINATOR MAY FACTOR ONLY AFTER A SUBSTITUTION

Partial fractions is an operation on rational functions, but the integrand does not have to be rational to start with. A great many trigonometric, exponential and radical integrals become rational under one substitution, and the decomposition then applies verbatim. The signals are a lone $\cos x$ beside a polynomial in $\sin x$, a lone e^x beside a polynomial in e^x , and a lone x beside a polynomial in x^2 . For instance $u = \sin x$ turns

$$\int_0^{\pi/2} \frac{\cos x dx}{\sin^2 x - 5 \sin x + 6} \quad \text{into} \quad \int_0^1 \frac{du}{(u-2)(u-3)} = \left[\ln \frac{|u-3|}{|u-2|} \right]_0^1 = 2 \ln 2 - \ln 3 = \ln \frac{4}{3},$$

and the fact that neither root lies in $[0, 1]$ is what makes the integral proper. Always ask what the denominator is a polynomial in, not merely what it is a polynomial in x in.

WHERE THIS GOES WRONG

Evaluating the four logarithms separately at the infinite endpoint. Each one diverges; only the combination converges, because the coefficients sum to $1 - 3 + 3 - 1 = 0$. That vanishing sum is exactly the condition for convergence at infinity, and it is worth checking before you integrate rather than after.

SEE ALSO Polynomial long division; partial fractions with repeated and irreducible factors; exponential substitution.

The cover-up rule computes a residue at a simple pole, so it stops working the moment a factor repeats or fails to be linear. Both cases have fixed remedies, and the remedy for the irreducible quadratic is the single most useful splitting move in the chapter.

19.9 Partial fractions with repeated and irreducible factors

CUE A denominator carrying $(x - r)^m$ with $m \geq 2$, or a quadratic $x^2 + bx + c$ with negative discriminant.

METHOD Allocate one term per power for a repeated root and a linear numerator for an irreducible quadratic:

$$\frac{A_1}{x - r} + \frac{A_2}{(x - r)^2} + \cdots + \frac{A_m}{(x - r)^m}, \quad \frac{Bx + C}{x^2 + bx + c}.$$

Cover-up still gives A_m instantly (cover $(x - r)^m$, evaluate at r) and, for the quadratic, the crucial step is to split $Bx + C$ into a multiple of the derivative of the denominator plus a constant. The first half integrates to a logarithm, the second to an arctangent after completing the square.

SPLIT THE NUMERATOR IN TWO

Given $\frac{Bx + C}{x^2 + bx + c}$, write

$$Bx + C = \frac{B}{2}(2x + b) + \left(C - \frac{Bb}{2}\right),$$

so the integral is $\frac{B}{2} \ln|x^2 + bx + c|$ plus $(C - \frac{Bb}{2})$ times the arctangent piece. This is the general form of the rule that a numerator should be broken into parts that each match a table entry, and it removes every guess from the irreducible-quadratic case. Compute $\Delta = b^2 - 4ac$ first: $\Delta > 0$ means the quadratic was not irreducible and you should have used two logarithms, $\Delta < 0$ means an arctangent, $\Delta = 0$ means a perfect square and a negative power.

COVER-UP WITH A DERIVATIVE

Cover-up is usually taught as a trick for simple poles, but it extends to a repeated factor and removes the linear system there too.

Set $h(x) = \frac{(x - r)^m P(x)}{Q(x)}$, which is what remains after the factor $(x - r)^m$ is covered. Then

$$A_{m-j} = \frac{1}{j!} h^{(j)}(r), \quad j = 0, 1, \dots, m - 1.$$

The case $j = 0$ is ordinary cover-up. The case $j = 1$ is the one worth learning, because a single differentiation replaces a whole coefficient comparison: for $\frac{x + 4}{(x + 1)(x + 2)^2}$ we have $h(x) = \frac{x + 4}{x + 1}$, so $h(-2) = -2$ gives $A_2 = -2$ and $h'(x) = \frac{-3}{(x + 1)^2}$ gives $A_1 = h'(-2) = -3$. Both coefficients of the repeated factor, with no system solved.

WORKED EXAMPLE

A repeated linear factor. For $\frac{x + 4}{(x + 1)(x + 2)^2}$, cover-up gives the coefficient of $\frac{1}{x + 1}$ as $\frac{-1 + 4}{(-1 + 2)^2} = 3$ and the coefficient of $\frac{1}{(x + 2)^2}$ as $\frac{-2 + 4}{-2 + 1} = -2$. The remaining coefficient follows from matching the x^2 term: $A_1 + B_1 = 0$ forces $B_1 = -3$. Then

$$\int_0^1 \frac{x + 4}{(x + 1)(x + 2)^2} dx = \left[3 \ln \frac{x + 1}{x + 2} + \frac{2}{x + 2} \right]_0^1 = 3 \ln \frac{4}{3} - \frac{1}{3} \approx 0.529713.$$

An irreducible quadratic. For $\frac{1}{(x + 1)(x^2 + 1)}$ the ansatz is $\frac{A}{x + 1} + \frac{Bx + C}{x^2 + 1}$; cover-up gives $A = \frac{1}{2}$, and matching the x^2 and constant terms gives $B = -\frac{1}{2}$, $C = \frac{1}{2}$. So

$$\int_0^1 \frac{dx}{(x + 1)(x^2 + 1)} = \frac{1}{2} \ln 2 - \frac{1}{4} \ln 2 + \frac{1}{2} \cdot \frac{\pi}{4} = \frac{1}{4} \ln 2 + \frac{\pi}{8} \approx 0.565986.$$

WHERE THIS GOES WRONG

Writing $\frac{A}{x^2 + 1}$ instead of $\frac{Bx + C}{x^2 + 1}$. The system then has one equation too few and is inconsistent, and the usual response is to force a solution by ignoring one equation, producing a confidently wrong answer. The mirror error is allocating $\frac{Bx + C}{(x - r)^2}$ to a repeated linear factor, which is not wrong but is redundant, since $\frac{Bx + C}{(x - r)^2} = \frac{B}{x - r} + \frac{Br + C}{(x - r)^2}$.

SEE ALSO Completing the square; numerator alignment; partial fractions with distinct linear factors.

When the irreducible factor itself repeats

A denominator such as $(x^2 + a^2)^m$ needs one linear numerator per power, so the ansatz carries $\frac{B_1x + C_1}{x^2 + a^2} + \dots + \frac{B_mx + C_m}{(x^2 + a^2)^m}$.

The B_kx halves are harmless: each is a constant times $\frac{d}{dx}(x^2 + a^2)$ over a power, so each integrates by the power rule to $-\frac{B_k}{2(k-1)(x^2 + a^2)^{k-1}}$, and only $k = 1$ produces a logarithm. The C_k halves need the reduction formula

$$I_m = \int \frac{dx}{(x^2 + a^2)^m} = \frac{x}{2a^2(m-1)(x^2 + a^2)^{m-1}} + \frac{2m-3}{2a^2(m-1)} I_{m-1}, \quad I_1 = \frac{1}{a} \arctan \frac{x}{a},$$

which is exactly numerator alignment applied repeatedly and then tidied with one integration by parts. For $m = 2$ and $a = 1$ it gives the entry used twice already, $I_2 = \frac{1}{2}(\frac{x}{1+x^2} + \arctan x)$.

Putting the pieces together on a full example: for $\frac{x^2 + x + 2}{(x^2 + 1)^2}$, matching coefficients in $(B_1x + C_1)(x^2 + 1) + B_2x + C_2 = x^2 + x + 2$ gives $B_1 = 0, C_1 = 1, B_2 = 1, C_2 = 1$. Hence

$$\int_0^1 \frac{x^2 + x + 2}{(x^2 + 1)^2} dx = \frac{\pi}{4} + \underbrace{\left[-\frac{1}{2(x^2 + 1)}\right]_0^1}_{\tilde{C}_1} + \underbrace{\left(\frac{1}{4} + \frac{\pi}{8}\right)}_{C_2} = \frac{3\pi}{8} + \frac{1}{2} \approx 1.678097,$$

which quadrature confirms. Note that the whole calculation was reading off four numbers and consulting two table entries; at no point was anything discovered.

Splitting a numerator to create a derivative

Alignment was described above as adding and subtracting the denominator. The wider principle is that you may add and subtract *anything* that makes one half of the numerator the derivative of something in sight, and the two most productive targets are a repeated linear factor and the biquadratic $x^4 + 1$.

Against a repeated linear factor the target is a power of the factor itself. To integrate $\frac{2x+1}{(x+1)^3}$, write the numerator in terms of $x+1$: since $2x+1 = 2(x+1) - 1$,

$$\frac{2x+1}{(x+1)^3} = \frac{2}{(x+1)^2} - \frac{1}{(x+1)^3}, \quad \int_0^1 \frac{2x+1}{(x+1)^3} dx = \left[-\frac{2}{x+1} + \frac{1}{2(x+1)^2}\right]_0^1 = \frac{5}{8}.$$

That is the entire partial-fraction decomposition of a repeated linear factor, obtained by rewriting a linear numerator rather than by solving anything. The same one-line move gives $\int_0^1 \frac{x dx}{(x+1)^3} = \int_0^1 \left(\frac{1}{(x+1)^2} - \frac{1}{(x+1)^3}\right) dx = \frac{1}{8}$. Whenever the denominator is $(x-r)^m$ and nothing else, expand the numerator in powers of $x-r$ and the problem is finished.

The biquadratic is the more interesting case, because there the split is engineered so that each half acquires a derivative it did not have. Neither $\int \frac{dx}{x^4+1}$ nor a partial-fraction attack on $x^4 + 1 = (x^2 + \sqrt{2}x + 1)(x^2 - \sqrt{2}x + 1)$ is pleasant. Instead split the constant numerator into two:

$$\frac{1}{x^4+1} = \frac{1}{2} \cdot \frac{x^2+1}{x^4+1} - \frac{1}{2} \cdot \frac{x^2-1}{x^4+1},$$

and divide each half above and below by x^2 . The first becomes $\frac{1+x^{-2}}{x^2+x^{-2}}$, whose numerator is the derivative of $x - x^{-1}$ and whose denominator is $(x - x^{-1})^2 + 2$; the second becomes $\frac{1-x^{-2}}{x^2+x^{-2}}$, whose numerator is the derivative of $x + x^{-1}$ and whose denominator is $(x + x^{-1})^2 - 2$. Each half is now a table entry:

$$\int \frac{x^2+1}{x^4+1} dx = \frac{1}{\sqrt{2}} \arctan \frac{x^2-1}{\sqrt{2}x} + C, \quad \int \frac{x^2-1}{x^4+1} dx = \frac{1}{2\sqrt{2}} \ln \frac{x^2 - \sqrt{2}x + 1}{x^2 + \sqrt{2}x + 1} + C.$$

On $[0, 1]$ the first gives $\frac{\pi}{2\sqrt{2}}$, the argument of the arctangent running from $-\infty$ to 0; the second gives $\frac{1}{2\sqrt{2}} \ln(3 - 2\sqrt{2}) = -\frac{\ln(1 + \sqrt{2})}{\sqrt{2}}$, using $3 - 2\sqrt{2} = (\sqrt{2} - 1)^2$. Halving and subtracting,

$$\int_0^1 \frac{dx}{x^4 + 1} = \frac{\pi + 2\ln(1 + \sqrt{2})}{4\sqrt{2}} \approx 0.866973,$$

which quadrature confirms to twelve digits. The lesson generalises past this one integrand: when a denominator is a polynomial in x^2 with no odd terms, dividing by x^2 and looking at $x \pm x^{-1}$ is the standard move, and the two signs are two different substitutions that must both be used.

One rational integrand from start to finish

The individual moves are easy. Running them in the right order on a single problem is the skill, so here is one integrand taken all the way through. Consider

$$\int_1^2 \frac{x^5 + 2}{x^3 + x} dx.$$

Step one, degrees. The numerator has degree 5 and the denominator degree 3, so the fraction is improper and division comes first. Since $(x^3 + x)(x^2 - 1) = x^5 - x$, we get $x^5 + 2 = (x^3 + x)(x^2 - 1) + (x + 2)$ and

$$\frac{x^5 + 2}{x^3 + x} = x^2 - 1 + \frac{x + 2}{x^3 + x}.$$

Step two, factor. $x^3 + x = x(x^2 + 1)$, one simple linear factor and one irreducible quadratic with $\Delta = -4$. The table above allocates $\frac{A}{x} + \frac{Bx + C}{x^2 + 1}$ and no more.

Step three, coefficients. Cover-up at $x = 0$ gives $A = \frac{0 + 2}{0 + 1} = 2$. Matching the x^2 coefficient in $2(x^2 + 1) + (Bx + C)x = x + 2$ gives $2 + B = 0$, so $B = -2$; matching the x coefficient gives $C = 1$.

Step four, split the quadratic numerator. Write $\frac{-2x + 1}{x^2 + 1} = -\frac{2x}{x^2 + 1} + \frac{1}{x^2 + 1}$, the first piece being g'/g and the second a table arctangent. Assembling,

$$\int_1^2 \left(x^2 - 1 + \frac{2}{x} - \frac{2x}{x^2 + 1} + \frac{1}{x^2 + 1} \right) dx = \left[\frac{x^3}{3} - x + 2\ln x - \ln(x^2 + 1) + \arctan x \right]_1^2,$$

which evaluates to $\frac{4}{3} + 3\ln 2 - \ln 5 + \arctan 2 - \frac{\pi}{4} \approx 2.125088$, matching quadrature to twelve digits. Four steps, no cleverness, and every step decided by the shape of the expression rather than by inspiration.

A shape-recognition drill

For each integrand, name the shape of the antiderivative before computing anything. The point is that this is decidable by inspection, and that deciding it first tells you whether you are about to make an error.

- (i) $\frac{x^4}{x^2 + 1}$. Improper, so a quadratic polynomial plus one arctangent. No logarithm can appear, because the denominator has no real root.
- (ii) $\frac{2x + 1}{x^2 + 4}$. Proper, irreducible quadratic; split as $\frac{2x}{x^2 + 4} + \frac{1}{x^2 + 4}$, giving one logarithm and one arctangent, namely $\ln \frac{5}{4} + \frac{1}{2} \arctan \frac{1}{2}$ on $[0, 1]$.
- (iii) $\frac{x^3}{(x^2 + 1)^2}$. Proper, repeated irreducible; but the numerator is odd, so $u = x^2$ reduces it to $\frac{1}{2} \int \frac{u du}{(u + 1)^2}$ and only logarithms and negative powers appear. On $[0, 1]$ the value is $\frac{1}{2}(\ln 2 - \frac{1}{2})$.
- (iv) $\frac{1}{x^3 + x}$. Simple linear times irreducible quadratic; two logarithms and no arctangent, since cover-up gives $A = 1$ and matching gives $B = -1, C = 0$.
- (v) $\frac{x + 4}{(x + 1)(x + 2)^2}$. Repeated linear; two logarithms and one reciprocal term. No arctangent is possible.

- (vi) $\frac{1}{x^4 + x^2 + 1}$. The denominator is not irreducible: it splits as $(x^2 + x + 1)(x^2 - x + 1)$, so the answer is two arctangents and two logarithms, and anyone who declares it irreducible will spend the afternoon on complex roots.
- (vii) $\frac{1 - x^5}{1 - x}$. Not a rational function in any interesting sense; the factor cancels and the answer is a polynomial.

Item (vi) is the one that catches people, and it is the reason the very first question about a quartic denominator should be whether it factors, not whether it looks factorable.

Quadratics under a radical, and conjugates

The arctangent case above needed the denominator written as a completed square. That is not a special trick for partial fractions: it is the universal normal form for a quadratic, and it is what converts the three standard radicals into the three standard substitutions of the next chapter. Anywhere a quadratic appears with a nonzero linear term, complete the square first and read the result.

MEDIUM

19.10 Completing the square

CUE A general quadratic $ax^2 + bx + c$ under a radical or in a denominator, with no obvious factorisation and a linear term present.

METHOD Write

$$ax^2 + bx + c = a\left(x + \frac{b}{2a}\right)^2 + \left(c - \frac{b^2}{4a}\right)$$

and shift with $u = x + \frac{b}{2a}$, transforming the bounds at the same moment. Three outcomes exhaust the cases: $\frac{1}{u^2 + k^2}$ gives $\arctan$, $\frac{1}{\sqrt{k^2 - u^2}}$ gives $\arcsin$, and $\frac{1}{\sqrt{u^2 \pm k^2}}$ gives an inverse hyperbolic function, equivalently a logarithm.

WORKED EXAMPLE

$$\int_1^2 \frac{dx}{\sqrt{2x - x^2}} = \int_1^2 \frac{dx}{\sqrt{1 - (x-1)^2}} = [\arcsin(x-1)]_1^2 = \frac{\pi}{2}.$$

The completed square also records the domain: $2x - x^2 = 1 - (x-1)^2$ is negative outside $[0, 2]$, so the integrand is real exactly where the square root of a nonnegative number is being taken, and $x = 2$ is an endpoint singularity that the arcsine handles without difficulty.

Two more, one of each remaining type:

$$\int_0^1 \frac{dx}{x^2 + 2x + 5} = \int_0^1 \frac{dx}{(x+1)^2 + 4} = \left[\frac{1}{2} \arctan \frac{x+1}{2}\right]_0^1 = \frac{1}{2} \left(\frac{\pi}{4} - \arctan \frac{1}{2}\right) \approx 0.160875,$$

$$\int_0^1 \frac{dx}{\sqrt{x^2 + 4x + 5}} = \int_0^1 \frac{dx}{\sqrt{(x+2)^2 + 1}} = [\operatorname{arsinh}(x+2)]_0^1 = \ln(3 + \sqrt{10}) - \ln(2 + \sqrt{5}) \approx 0.374811.$$

A leading coefficient other than 1, handled by pulling it out before completing:

$$\int_0^1 \frac{dx}{4x^2 + 4x + 2} = \int_0^1 \frac{dx}{(2x+1)^2 + 1} = \left[\frac{1}{2} \arctan(2x+1)\right]_0^1 = \frac{1}{2} \left(\arctan 3 - \frac{\pi}{4}\right) \approx 0.231824,$$

where the outer $\frac{1}{2}$ is the Jacobian of $u = 2x + 1$ and is precisely the factor that goes missing when the square is completed carelessly. Finally the case that trips people, a negative leading coefficient:

$$\int_0^1 \frac{dx}{\sqrt{3 + 2x - x^2}} = \int_0^1 \frac{dx}{\sqrt{4 - (x-1)^2}} = \left[\arcsin \frac{x-1}{2}\right]_0^1 = 0 - \arcsin\left(-\frac{1}{2}\right) = \frac{\pi}{6} \approx 0.523599.$$

Here $3 + 2x - x^2 = -(x^2 - 2x - 3) = -((x-1)^2 - 4)$, and the sign must be extracted before the square is completed, not after. A negative leading coefficient always produces the arcsine form, never the arctangent or the logarithm, because $k^2 - u^2$ is the only one of the three shapes it can reach.

Two omissions account for nearly every error here. The first is forgetting the overall $1/\sqrt{a}$ or $1/a$ when $a \neq 1$: $\sqrt{2}x^2 + 4x + 5 = \sqrt{2}\sqrt{(x+1)^2 + \frac{3}{2}}$, and the $\sqrt{2}$ must come out front. The second is leaving the bounds in terms of x after shifting to u . Shift the bounds in the same stroke as the variable, every time, and the error cannot occur.

SEE ALSO Sine substitution; hyperbolic substitution; partial fractions with repeated and irreducible factors.

A completed square handles one quadratic. When two irrational expressions sit next to each other and neither can be completed into anything useful, the move is to manufacture a difference of squares, which is what multiplication by a conjugate does.

19.11 Conjugate multiplication

MEDIUM

CUE A sum or difference of two surds in a numerator or a denominator, or the trigonometric analogues $\sec x \pm \tan x$ and $1 \pm \sin x$, with nothing cancelling.

METHOD Multiply above and below by the conjugate and use $(A+B)(A-B) = A^2 - B^2$. For surds this rationalises:

$\frac{1}{\sqrt{x+1} + \sqrt{x}} = \sqrt{x+1} - \sqrt{x}$, since the product of the two is 1. For trigonometric pairs it exploits the Pythagorean identities: $\sec^2 - \tan^2 = 1$ gives $\frac{1}{\sec x + \tan x} = \sec x - \tan x$, and $\frac{1}{1 - \sin u} = \frac{1 + \sin u}{\cos^2 u} = \sec^2 u + \sec u \tan u$.

WORKED EXAMPLE

$$\int_0^1 \frac{dx}{\sqrt{x+1} + \sqrt{x}} = \int_0^1 (\sqrt{x+1} - \sqrt{x}) dx = \left[\frac{2}{3}(x+1)^{3/2} - \frac{2}{3}x^{3/2} \right]_0^1 = \frac{4\sqrt{2}-4}{3} \approx 0.552285.$$

When the conjugate product is a constant other than 1, that constant is the only bookkeeping: $(\sqrt{x+2} - \sqrt{x})(\sqrt{x+2} + \sqrt{x}) = 2$, so

$$\int_0^1 \frac{dx}{\sqrt{x+2} - \sqrt{x}} = \frac{1}{2} \int_0^1 (\sqrt{x+2} + \sqrt{x}) dx = \frac{1}{3} (3\sqrt{3} - 2\sqrt{2} + 1) \approx 1.122575.$$

The trigonometric version, and the reason the combined form matters:

$$\int_0^{\pi/2} \frac{dx}{\sec x + \tan x} = \int_0^{\pi/2} (\sec x - \tan x) dx = [\ln(1 + \sin x)]_0^{\pi/2} = \ln 2 \approx 0.693147.$$

The antiderivative is found by noting that $\sec x - \tan x = \frac{1 - \sin x}{\cos x} = \frac{\cos x}{1 + \sin x}$, which is g'/g with $g = 1 + \sin x$. A third, in the same family:

$$\int_0^{\pi/2} \frac{dx}{1 + \sin x} = \int_0^{\pi/2} \frac{1 - \sin x}{\cos^2 x} dx = [\tan x - \sec x]_0^{\pi/2} = \left[\frac{\sin x - 1}{\cos x} \right]_0^{\pi/2} = 0 - (-1) = 1.$$

The companion, which the same conjugate handles and which the identity table also handles: $\frac{1}{1 + \cos x} = \frac{1 - \cos x}{\sin^2 x}$, so

$$\int_0^{\pi/2} \frac{dx}{1 + \cos x} = \left[\frac{1 - \cos x}{\sin x} \right]_0^{\pi/2} = 1,$$

the value at the lower endpoint being the limit 0. Alternatively $1 + \cos x = 2 \cos^2 \frac{x}{2}$ gives $\frac{1}{2} \int \sec^2 \frac{x}{2} dx = [\tan \frac{x}{2}]_0^{\pi/2} = 1$, and the agreement of the two routes is a useful check on both.

Finally the two antiderivatives that this technique exists to produce. Multiplying $\sec x$ by $\frac{\sec x + \tan x}{\sec x + \tan x}$ makes the numerator $\sec^2 x + \sec x \tan x$, which is exactly the derivative of the denominator, so

$$\int \sec x dx = \ln |\sec x + \tan x| + C, \quad \int \csc x dx = -\ln |\csc x + \cot x| + C,$$

the second by the same multiplication with $\csc x + \cot x$. Both are worth deriving once rather than memorising, because the derivation is the technique: on $[0, \pi/4]$ the first gives $\ln(\sqrt{2} + 1) \approx 0.881374$, and on $[\pi/4, \pi/2]$ the second gives the same number, as the substitution $x \mapsto \pi/2 - x$ requires.

Splitting $\int_0^{\pi/2} (\sec x - \tan x) dx$ into $\int \sec x dx - \int \tan x dx$. Both diverge at $\pi/2$ and only the difference converges. The same discipline saves the last example: $\tan x$ and $\sec x$ each blow up at $\pi/2$, and it is the combination $\frac{\sin x - 1}{\cos x}$, whose limit is 0, that must be evaluated. Never separate a conjugate pair after having formed it.

SEE ALSO Rationalising a radical by substitution; self-similar collapse; trigonometric identity pre-processing.

Trigonometric identities as pre-processing

Trigonometric integrands come in two states: linearised, meaning written as a sum of $\sin(kx)$ and $\cos(kx)$ terms, and unlinearised, meaning written as products and powers. Every table entry is in the first state, so the pre-processing pass is the work of getting there. Power reduction handles even powers, product-to-sum handles products at different frequencies, and the half-angle identities handle $1 \pm \cos x$ and the radicals built on it. These are not substitutions and they are not integration; they are the trigonometric analogue of expanding a bracket.

The identities worth having by heart are few, and they are listed here in the direction you actually use them, which is always from the product side to the sum side.

Identity	What it is for
$\sin^2 x = \frac{1}{2}(1 - \cos 2x), \quad \cos^2 x = \frac{1}{2}(1 + \cos 2x)$	every even power, by repetition
$\sin^3 x = \frac{1}{4}(3 \sin x - \sin 3x), \quad \cos^3 x = \frac{1}{4}(3 \cos x + \cos 3x)$	odd powers, without a substitution
$1 + \cos x = 2 \cos^2 \frac{x}{2}, \quad 1 - \cos x = 2 \sin^2 \frac{x}{2}$	making $\sqrt{1 \pm \cos x}$ elementary
$\sin x \cos x = \frac{1}{2} \sin 2x$	halving the frequency count
$\sin^4 x + \cos^4 x = 1 - \frac{1}{2} \sin^2 2x$	the engineered even-power sums
$\sin^6 x + \cos^6 x = 1 - \frac{3}{4} \sin^2 2x$	the same, one level up
$\tan \frac{x}{2} = \frac{1 - \cos x}{\sin x} = \frac{\sin x}{1 + \cos x}$	the bridge to the Weierstrass substitution

Two of these deserve comment. The odd-power identities remove the need for the substitution $u = \cos x$ on $\int \sin^3 x dx$, which matters when the odd power is buried inside a larger expression at a different frequency, as $\cos^3(2x)$ is. And the last line is not a pre-processing identity at all: it is the statement that the half-angle tangent is a rational function of $\sin x$ and $\cos x$, which is the whole reason the substitution of the next chapter works.

19.12 Trigonometric identity pre-processing

EASY

CUE $1 \pm \cos x$, $1 \pm \cos 2x$, $\sqrt{1 + \cos x}$, an even power of $\sin$ or $\cos$, or the combination $\sin^4 x + \cos^4 x$.

METHOD Apply the identity before choosing any method. The power reduction pair $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$ and $\cos^2 x = \frac{1}{2}(1 + \cos 2x)$ turns every even power into a linear combination of cosines by repetition. The half-angle forms $1 + \cos x = 2 \cos^2 \frac{x}{2}$ and $1 - \cos x = 2 \sin^2 \frac{x}{2}$ are what make $\sqrt{1 \pm \cos x}$ rational. And $\sin^4 x + \cos^4 x = 1 - \frac{1}{2} \sin^2 2x = \frac{1}{2}(1 + \cos^2 2x)$.

WORKED EXAMPLE

Power reduction, twice:

$$\int_0^{\pi/2} \sin^4 x dx = \int_0^{\pi/2} \left(\frac{1 - \cos 2x}{2} \right)^2 dx = \frac{1}{4} \int_0^{\pi/2} (1 - 2 \cos 2x + \frac{1 + \cos 4x}{2}) dx = \frac{1}{4} \left(\frac{\pi}{2} - 0 + \frac{\pi}{4} \right) = \frac{3\pi}{16},$$

matching 0.589049 numerically. The half-angle form, where the identity is what makes the radical elementary:

$$\int_0^{\pi} \sqrt{1 + \cos x} dx = \sqrt{2} \int_0^{\pi} \left| \cos \frac{x}{2} \right| dx = \sqrt{2} \int_0^{\pi} \cos \frac{x}{2} dx = 2\sqrt{2} \approx 2.828427,$$

the modulus being removable because $\cos \frac{x}{2} \geq 0$ on $[0, \pi]$. A harder one in which two identities cooperate:

$$\int_{-\pi/2}^{\pi/2} \sqrt{\cos x} \sqrt{1 + \cos x} dx = \sqrt{2} \int_{-\pi/2}^{\pi/2} \cos \frac{x}{2} \sqrt{\cos x} dx.$$

Now put $s = \sin \frac{x}{2}$, so $ds = \frac{1}{2} \cos \frac{x}{2} dx$ and $\cos x = 1 - 2s^2$, with s running from $-\frac{1}{\sqrt{2}}$ to $\frac{1}{\sqrt{2}}$. The integral becomes $2\sqrt{2} \int_{-1/\sqrt{2}}^{1/\sqrt{2}} \sqrt{1 - 2s^2} ds$, and $s = \frac{1}{\sqrt{2}} \sin \theta$ turns that into $2 \int_{-\pi/2}^{\pi/2} \cos^2 \theta d\theta = \pi$.

Writing $\sqrt{2\cos^2 \frac{x}{2}} = \sqrt{2}\cos \frac{x}{2}$ without the modulus. On $[0, \pi]$ this is harmless because the cosine is nonnegative; on $[0, 3\pi]$ it is fatal, and the resulting antiderivative decreases where the integrand is positive. The general rule is that $\sqrt{f^2} = |f|$ always, and you drop the bars only after checking the sign on the actual interval.

SEE ALSO Product to sum, and sum to product; the Weierstrass substitution; algebraic camouflage collapse.

Power reduction linearises powers at a single frequency. Products at different frequencies need the other half of the identity table, and its reverse direction, sum to product, is what cancels shared factors out of a ratio of trigonometric sums.

MEDIUM

19.13 Product to sum, and sum to product

CUE A product of trigonometric functions at different frequencies, such as $\sin 3x \cos 5x$ or $\cos^3(2x)$, or a quotient in which sums of trigonometric functions at nearby frequencies appear above and below.

METHOD In the linearising direction,

$$\begin{aligned}\cos A \cos B &= \frac{1}{2}[\cos(A - B) + \cos(A + B)], & \sin A \sin B &= \frac{1}{2}[\cos(A - B) - \cos(A + B)], \\ \sin A \cos B &= \frac{1}{2}[\sin(A + B) + \sin(A - B)].\end{aligned}$$

In the factoring direction, $\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$ and $\cos A - \cos B = -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2}$, so a common factor $\sin \frac{A+B}{2}$ cancels between numerator and denominator.

WORKED EXAMPLE

Linearising a product:

$$\int_0^{\pi/4} \sin 3x \cos 5x \, dx = \frac{1}{2} \int_0^{\pi/4} (\sin 8x - \sin 2x) \, dx = \frac{1}{2} \left[-\frac{\cos 8x}{8} + \frac{\cos 2x}{2} \right]_0^{\pi/4} = \frac{1}{2} \left(-\frac{1}{8} - \frac{3}{8} \right) = -\frac{1}{4}.$$

Linearising an odd power: $\cos^3 \theta = \frac{3}{4} \cos \theta + \frac{1}{4} \cos 3\theta$, so with $\theta = 2x$,

$$\int_0^{\pi/4} \cos^3(2x) \, dx = \frac{3}{4} \cdot \frac{1}{2} [\sin 2x]_0^{\pi/4} + \frac{1}{4} \cdot \frac{1}{6} [\sin 6x]_0^{\pi/4} = \frac{3}{8} - \frac{1}{24} = \frac{1}{3}.$$

Now the factoring direction, which is where signs are lost. With $A = 6x$ and $B = 7x$, the half-sum is $\frac{13x}{2}$ and the half-difference is $-\frac{x}{2}$. Therefore $\sin 6x + \sin 7x = 2 \sin \frac{13x}{2} \cos \frac{x}{2}$ and $\cos 6x - \cos 7x = -2 \sin \frac{13x}{2} \sin(-\frac{x}{2}) = +2 \sin \frac{13x}{2} \sin \frac{x}{2}$, so the quotient is $+\cot \frac{x}{2}$ and

$$\int_{\pi}^{3\pi/2} \frac{\sin 6x + \sin 7x}{\cos 6x - \cos 7x} \, dx = \left[2 \ln \left| \sin \frac{x}{2} \right| \right]_{\pi}^{3\pi/2} = 2 \ln \frac{1}{\sqrt{2}} = -\ln 2 \approx -0.693147,$$

which direct quadrature on the original quotient confirms to twelve digits.

One further use of the linearising direction is worth naming, because it is the mechanism behind Fourier series and it costs one line here. On a full or half period, integrals of $\cos mx \cos nx$ and $\sin mx \sin nx$ vanish whenever $m \neq n$, because product-to-sum turns them into cosines at the frequencies $m \pm n$, both of which integrate to zero over the period. So $\int_0^{\pi/2} \cos 2x \cos 4x \, dx = \frac{1}{2} \int_0^{\pi/2} (\cos 2x + \cos 6x) \, dx = 0$ without any computation beyond the identity. When the interval is not a period the cancellation is only partial and the arithmetic must be done: $\int_0^{\pi/6} \cos x \cos 3x \, dx = \frac{1}{2} \int_0^{\pi/6} (\cos 2x + \cos 4x) \, dx = \frac{\sqrt{3}}{8} + \frac{\sqrt{3}}{16} \approx 0.324760$.

WHERE THIS GOES WRONG

The double sign in $\cos A - \cos B$. The formula carries an explicit minus, and the half-difference $\frac{A-B}{2}$ is itself negative when $A < B$; the two cancel. Getting one of them and not the other turns $+\cot \frac{x}{2}$ into $-\cot \frac{x}{2}$ and flips the answer. The second hazard is cancelling $\sin \frac{13x}{2}$ without noticing that it vanishes inside $[\pi, 3\pi/2]$: the cancellation is valid off a finite set, which is enough for the integral, but you must say so.

SEE ALSO Trigonometric identity pre-processing; the Weierstrass substitution; algebraic camouflage collapse.

Repetition is the whole of power reduction

Power reduction is stated for squares, and readers therefore assume it stops there. It does not: applying it to the result of applying it is how every even power is linearised, and the only cost is bookkeeping. Take $\cos^6 x$. One application gives

$\left(\frac{1 + \cos 2x}{2}\right)^3 = \frac{1 + 3 \cos 2x + 3 \cos^2 2x + \cos^3 2x}{8}$. Now reduce the two surviving powers, $\cos^2 2x = \frac{1}{2}(1 + \cos 4x)$ and $\cos^3 2x = \frac{3}{4} \cos 2x + \frac{1}{4} \cos 6x$, and collect:

$$\cos^6 x = \frac{5}{16} + \frac{15}{32} \cos 2x + \frac{3}{16} \cos 4x + \frac{1}{32} \cos 6x.$$

Every term after the first integrates to a sine that vanishes at both ends of $[0, \pi/2]$ except $\cos 2x$, whose integral over that interval is 0 as well, so

$$\int_0^{\pi/2} \cos^6 x \, dx = \frac{5}{16} \cdot \frac{\pi}{2} = \frac{5\pi}{32} \approx 0.490874,$$

in agreement with the Wallis product $\frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2}$. The constant term of the linearised form is $\binom{6}{3}/2^6 = 20/64 = 5/16$, and that is the general rule: the mean value of $\cos^{2n} x$ over a period is $\binom{2n}{n}/4^n$, which is Wallis in disguise and a thirty second check on any power reduction you perform.

Five integrands that need no technique at all

Collected here, without commentary between them, are five problems whose entire content is the rewriting pass. Each looks as though it belongs to a later chapter and none of them does.

One. $\int_0^{\pi/4} \frac{\cos 2x}{\cos x + \sin x} \, dx$. Since $\cos 2x = \cos^2 x - \sin^2 x = (\cos x - \sin x)(\cos x + \sin x)$, the denominator cancels and

$$\int_0^{\pi/4} (\cos x - \sin x) \, dx = [\sin x + \cos x]_0^{\pi/4} = \sqrt{2} - 1 \approx 0.414214.$$

Two. $\int_1^2 \frac{x^4 + 3x^2 + 2}{x^3 + x} \, dx$. The numerator is $(x^2 + 1)(x^2 + 2)$ and the denominator is $x(x^2 + 1)$, so the quotient is $\frac{x^2 + 2}{x} = x + \frac{2}{x}$ and the value is $\left[\frac{x^2}{2} + 2 \ln x\right]_1^2 = \frac{3}{2} + 2 \ln 2 \approx 2.886294$. A solver who divides without factoring gets a quotient of x and a remainder of $2x^2 + 2$, then decomposes $\frac{2x^2 + 2}{x^3 + x}$ and arrives at the same place three steps later.

Three. $\int_0^{\pi/2} (\sin^4 x - \cos^4 x) \, dx$. The difference of two squares gives $(\sin^2 x - \cos^2 x)(\sin^2 x + \cos^2 x) = -\cos 2x$, so the integral is $-\left[\frac{1}{2} \sin 2x\right]_0^{\pi/2} = 0$. Power reduction on each term separately gives $\frac{3\pi}{16} - \frac{3\pi}{16}$, which is the same answer after four times the work.

Four. $\int_0^{\pi/4} \frac{1 - \tan^2 x}{1 + \tan^2 x} \, dx$. The denominator is $\sec^2 x$, so the quotient is $\cos^2 x - \sin^2 x = \cos 2x$ and the value is $\left[\frac{1}{2} \sin 2x\right]_0^{\pi/4} = \frac{1}{2}$. The Weierstrass substitution would also work and would take fifteen lines.

Five. $\int_0^{\pi/2} \frac{\sin x - \cos x}{\sqrt{1 + \sin 2x}} \, dx$. Since $1 + \sin 2x = (\sin x + \cos x)^2$ and $\sin x + \cos x > 0$ on $[0, \pi/2]$, the radical is $\sin x + \cos x$ and the integrand is $-\frac{g'}{g}$ for $g = \sin x + \cos x$. Hence the value is $-\left[\ln(\sin x + \cos x)\right]_0^{\pi/2} = -(\ln 1 - \ln 1) = 0$. Note that the answer is zero for a structural reason, not by cancellation of two computed halves: g takes the same value at both endpoints.

One more remark about order, because it is the only thing in this chapter that is genuinely easy to get wrong. Completing the square and factoring are in competition: $x^2 - x + 1$ can be completed into $(x - \frac{1}{2})^2 + \frac{3}{4}$, which is right when it sits in a denominator by itself, and can be recognised as $\frac{x^3 + 1}{x + 1}$, which is right when a numerator carrying $x^3 + 1$ is nearby. Look for the cancellation first, always, because completing the square destroys the visible factorisation while factoring destroys nothing. The same principle explains why linearising trigonometric products comes last: $\sin 6x + \sin 7x$ over $\cos 6x - \cos 7x$ must be factored before either half is expanded, and a solver who linearises first is left with four terms at four frequencies and no cancellation in sight.

Two habits carry everything in this chapter. The first is that you never integrate an expression you have not first tried to shorten, because a shorter expression is not merely more pleasant, it is often in the table. The second is that rewriting has a fixed order: collapse, divide, factor, decompose, complete, linearise. Run it in that order and the rational case is finished before it starts, the radical case is handed to the substitution families of the next chapter in its normal form $\sqrt{u^2 \pm k^2}$ or

$\sqrt{k^2 - u^2}$, and the trigonometric case arrives as a sum of $\sin(kx)$ and $\cos(kx)$ with nothing left to do. Run it out of order and you will complete the square inside something that was about to cancel.

Recognition matrix

If you see	Reach for	If that fails
An integrand engineered to look terrible $x^a x^b$ with $a + b$ simple, or mixed log bases $\frac{1 - x^n}{1 - x}, 1 - x^{1/3} + x^{2/3}, x^4 + x^2 + 1$	Expand; apply $\sin^2 + \cos^2 = 1$ to each side separately Exponent and logarithm laws, before anything The partner factor of a standard identity	Look for a common factor top and bottom Convert every log to $\ln$ and cancel Sum or difference of cubes; add and subtract x^2
An infinitely repeating radical or fraction $\sin(\operatorname{arcsec} x), \arctan \frac{1 - x}{1 + x}$ $\deg P \geq \deg Q$ in a quotient Numerator almost equal to the denominator Distinct linear factors below	Name it y , solve the fixed-point equation Right triangle, or the arctangent subtraction law Divide first, then treat the remainder Add and subtract the denominator, then split Cover-up rule, then one combined logarithm	Check the nesting is infinite, not finite Verify the branch on the whole interval Repeated use of $x^2 = (x^2 + 1) - 1$ Check each fragment converges Match coefficients if a numerator is awkward
$(x - r)^m$ below $x^4 + 1$, or any even polynomial below $x^2 + bx + c$ with $\Delta < 0$ below $(x^2 + a^2)^m$ below Any quadratic with a linear term Two surds added, or $\sec x \pm \tan x$ Even powers of $\sin$ or $\cos$ $\sqrt{1 \pm \cos x}$ A product at two frequencies A ratio of trigonometric sums	Expand the numerator in powers of $x - r$ Split into $x^2 + 1$ and $x^2 - 1$; divide by x^2 $\frac{B}{2}(2x + b)$ plus a constant, then $\arctan$ Reduction formula for I_m ; power rule on the x halves Complete the square and shift the bounds Multiply by the conjugate Power reduction to $\cos 2x, \cos 4x$ $1 + \cos x = 2 \cos^2 \frac{x}{2}$, with a modulus Product-to-sum, then integrate term by term Sum-to-product; cancel the shared factor	Cover-up with j derivatives for A_{m-j} Read $x \pm x^{-1}$ as the new variable Compute Δ ; it may factor after all Numerator alignment, repeatedly Keep the $1/\sqrt{a}$ out front Keep the pair together when integrating Half angle if a radical is present Check the sign on the actual interval Write odd powers as $\cos \theta, \cos 3\theta$ Watch the sign of the half-difference

The substitution families

What this chapter is for. Substitution is not one technique. It is a small taxonomy, and almost all of the work of using it is deciding which family a given integrand belongs to before you write anything down. There are five families: differential spotting, where the derivative of an inner function is already present; the three trigonometric substitutions, selected by which of $\sqrt{a^2 - x^2}$, $\sqrt{a^2 + x^2}$, $\sqrt{x^2 - a^2}$ you are looking at, together with their hyperbolic alternatives; the rationalising substitutions, which include Euler's three choices for a general quadratic radical and the plain device of setting u equal to the radical itself; the Weierstrass substitution, which rationalises any rational function of $\sin x$ and $\cos x$ at a price; and the involutions $x \mapsto 1/x$ and $x \mapsto a/x$, which do not simplify the integrand at all but map the integral to a relative of itself. This chapter teaches the recognition, the mechanics, and one discipline that prevents more errors than any other: for a definite integral, transform the limits at the same instant you transform the variable.

Recognising the differential

The change of variables theorem says that if g is continuously differentiable on $[a, b]$ and F is continuous on the image, then

$$\int_a^b F(g(x))g'(x) dx = \int_{g(a)}^{g(b)} F(u) du.$$

Everything in this chapter is that formula, used in one direction or the other. Used left to right it is simplification: a composite disappears and the integral becomes a table entry. Used right to left it is manufacture: you introduce a g that was not there, in order that a radical or a trigonometric expression become algebraic. The two directions feel completely different in practice, and the families below are organised by which direction they use.

It helps to know in advance which direction a given problem wants. Left to right, the integrand contains a composite and a spare factor, and the substitution removes both: this is the situation in which a substitution makes the integrand shorter. Right to left, the integrand contains no composite at all, only a radical or a trigonometric expression that is not in the table, and the substitution makes the integrand longer in the hope that the longer form is one you can finish. A solver who expects every substitution to shorten things will never try $x = a \sec \theta$, and a solver who reaches for a manufactured substitution when a spare factor was sitting in plain sight will do five times the necessary work.

TRANSFORM THE LIMITS AT ONCE

For a definite integral, write the new bounds in the same stroke as the substitution, before simplifying anything. A substitution written as $u = \sin^2 x$, $x : 0 \rightarrow \pi/4$ becomes $u : 0 \rightarrow 1/2$ costs one extra symbol and removes the single most common error in the subject, which is integrating in u and then evaluating at the old bounds. It also removes the need to undo the substitution at the end, which for $x = a \sin \theta$ or $t = \tan \frac{x}{2}$ is the messiest step in the calculation. The one price is that you must check the substitution is monotone on the interval, since $g(a)$ and $g(b)$ do not determine the integral if g turns around in between.

20.1 Spotting the differential

EASY

CUE A composite $F(g(x))$ multiplied by something proportional to $g'(x)$. The recurring pairs are $\sin 2x$ beside $\sin^2 x$, x beside x^2 , $\sec^2 x$ beside $\tan x$, $1/x$ beside $\ln x$, and e^x beside $e^x + c$.

METHOD Set $u = g(x)$, compute $du = g'(x) dx$ explicitly, solve for dx rather than eyeballing it, and transform the bounds immediately. If the available factor is a constant multiple of g' , that constant comes out in front; if it is anything else, this is not the family you are in.

WORKED EXAMPLE

$$\int_0^{\pi/4} \frac{\sin 2x}{\sin^2 x + 2} dx.$$

The candidate inner function is $u = \sin^2 x$, whose derivative is $2 \sin x \cos x = \sin 2x$, exactly the numerator. The bounds go $x : 0 \rightarrow \pi/4$, so $u : 0 \rightarrow \frac{1}{2}$, and

$$\int_0^{1/2} \frac{du}{u+2} = [\ln(u+2)]_0^{1/2} = \ln \frac{5}{4} \approx 0.223144.$$

A case where the constant is not 1:

$$\int_0^1 \frac{x}{x^4+1} dx \xrightarrow{u=x^2, du=2x dx} \frac{1}{2} \int_0^1 \frac{du}{u^2+1} = \frac{1}{2} \cdot \frac{\pi}{4} = \frac{\pi}{8} \approx 0.392699.$$

The x in the numerator is half of du/dx ; had the numerator been x^2 there would be no substitution here at all, and the integral would need partial fractions over $x^4+1 = (x^2+\sqrt{2}x+1)(x^2-\sqrt{2}x+1)$.

WHERE THIS GOES WRONG

Carrying the old bounds into the new variable, which turns $\int_0^{1/2}$ into $\int_0^{\pi/4}$ and produces a number that is wrong but plausible. The second failure is a lost constant: when $du = k g'(x) dx$, the reciprocal $1/k$ must appear, and eyeballing $dx = du$ is how it disappears. Solve for dx in writing, every time.

SEE ALSO The logarithmic form; the Trojan substitution; exponential substitution.

One special case of differential spotting is common enough, and disguised often enough, to be worth separating. It is the case where the composite is a logarithm, so that the whole integrand is a quotient whose numerator is the derivative of its denominator.

EASY

20.2 The logarithmic form $\int g'/g$

CUE A quotient in which the numerator is exactly, or a constant multiple of, the derivative of the denominator. The disguise is usually one multiplication away: clearing a $\sec x$ or a reciprocal reveals the match.

METHOD Confirm by differentiating the denominator and comparing with the numerator, not by pattern matching on shapes. Then

$$\int \frac{g'(x)}{g(x)} dx = \ln |g(x)| + C.$$

The absolute value matters only when g changes sign, and if it does inside the interval the integral is improper and $\ln |g|$ is not an antiderivative across the zero.

WORKED EXAMPLE

The plain form: $\int_1^2 \frac{2x+1}{x^2+x+1} dx = [\ln(x^2+x+1)]_1^2 = \ln \frac{7}{3} \approx 0.847298$, the numerator being precisely the derivative of the denominator.

The disguised form is more instructive. Consider

$$\int_0^\pi \frac{e^x \sec x + 1}{e^x \sec x + \tan x} dx.$$

Nothing matches. Multiply numerator and denominator by $\cos x$, which is legitimate wherever $\cos x \neq 0$ and extends by continuity across $x = \pi/2$:

$$\frac{e^x \sec x + 1}{e^x \sec x + \tan x} = \frac{e^x + \cos x}{e^x + \sin x}.$$

Now the numerator is $\frac{d}{dx}(e^x + \sin x)$, and since $e^x + \sin x > 0$ on $[0, \pi]$ the modulus is unnecessary:

$$[\ln(e^x + \sin x)]_0^\pi = \ln e^\pi - \ln 1 = \pi,$$

confirmed numerically as 3.141592653590.

Two more, to fix the shape in the eye. First, $\int_0^{\pi/3} \tan x dx = -\int_0^{\pi/3} \frac{-\sin x}{\cos x} dx = -[\ln |\cos x|]_0^{\pi/3} = \ln 2$, where the minus sign is the whole difficulty and is supplied by $g = \cos x$. Second, $\int_1^e \frac{(\ln x)^2}{x} dx = [\frac{1}{3}(\ln x)^3]_1^e = \frac{1}{3}$, which is differential spotting rather than the logarithmic form: here $1/x$ is g' for $g = \ln x$ and the outer function is a square, not a reciprocal. The distinction matters because $\int \frac{dx}{x \ln x} = \ln |\ln x|$ while $\int \frac{\ln x}{x} dx = \frac{1}{2}(\ln x)^2$, and the two are confused constantly.

Omitting the absolute value and then integrating across a zero of g . For $\int_{-1}^1 \frac{2x}{x^2-4} dx$ the denominator never vanishes and $\ln|x^2-4|$ is fine; for $\int_1^3$ the same expression the denominator vanishes at $x=2$ and the integral diverges, though the naive evaluation $\ln 5 - \ln 3$ returns a finite number without complaint.

SEE ALSO Spotting the differential; partial fractions with distinct linear factors; the Trojan substitution.

Differential spotting fails in an interesting way when the factor in front of the composite is a *sum*, no term of which is the derivative of the inner function, while the sum as a whole is. Solvers trained to match single factors miss these completely, and splitting the sum makes both halves intractable.

HARD

20.3 The Trojan substitution

CUE A bracketed sum multiplying a composite, where no individual term looks like a derivative: $(\tan x + \tan^3 x)$ beside $e^{\tan^2 x}$, $(\ln x + 1)$ beside x^x , $(1 - x^{-2})$ beside a function of $x + 1/x$.

METHOD Differentiate the plausible inner function and compare against the *whole* bracket. The characteristic algebra is that $\frac{d}{dx} \tan^2 x = 2 \tan x \sec^2 x = 2 \tan x (1 + \tan^2 x) = 2(\tan x + \tan^3 x)$: the sum is the differential, and the factor of 2 is the only thing to keep track of.

WORKED EXAMPLE

$$\int_0^{\pi/4} (\tan x + \tan^3 x) e^{\tan^2 x} dx.$$

Put $u = \tan^2 x$, so $du = 2(\tan x + \tan^3 x) dx$ and $u : 0 \rightarrow 1$. The integral is

$$\frac{1}{2} \int_0^1 e^u du = \frac{e-1}{2} \approx 0.859141,$$

which quadrature confirms. The other classic in the family:

$$\int_1^e (\ln x + 1) x^x dx = [x^x]_1^e = e^e - 1 \approx 14.154262,$$

because $x^x = e^{x \ln x}$ has derivative $x^x (\ln x + 1)$. Neither $\int \ln x \cdot x^x dx$ nor $\int x^x dx$ is elementary on its own.

The hardest member of the family is the one built on $u = x + \frac{1}{x}$, whose differential is $(1 - \frac{1}{x^2}) dx$. The bracket is invisible until you divide numerator and denominator by a power of x :

$$\int_1^2 \frac{x^2 - 1}{(x^2 + 1)\sqrt{x^4 + 1}} dx = \int_1^2 \frac{1 - x^{-2}}{(x + x^{-1})\sqrt{x^2 + x^{-2}}} dx,$$

having divided the numerator by x , the first factor of the denominator by x , and the radical by x^2 . Now $x^2 + x^{-2} = u^2 - 2$, so with $u : 2 \rightarrow \frac{5}{2}$,

$$\int_2^{5/2} \frac{du}{u\sqrt{u^2 - 2}} = \frac{1}{\sqrt{2}} \left[\arccos \frac{\sqrt{2}}{u} \right]_2^{5/2} = \frac{1}{\sqrt{2}} \left(\arccos \frac{2\sqrt{2}}{5} - \frac{\pi}{4} \right) \approx 0.130202,$$

the antiderivative being the secant-substitution collapse $\frac{du}{u\sqrt{u^2 - a^2}} = \frac{d\theta}{a}$ from earlier in this chapter.

WHERE THIS GOES WRONG

Splitting the bracket and attacking the pieces separately. In both examples above each piece is individually non-elementary, so the split does not merely lengthen the work, it destroys the problem. Whenever a bracketed sum multiplies an awkward composite, differentiate the composite's argument before doing anything else.

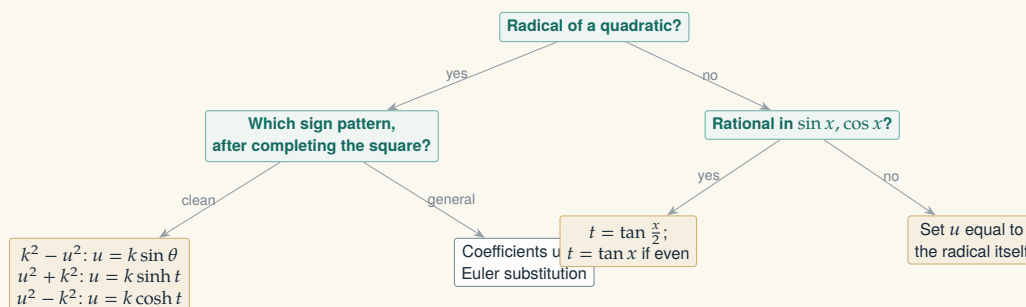
SEE ALSO Spotting the differential; the logarithmic form; rationalising a radical by substitution.

The three trigonometric substitutions, and their hyperbolic rivals

The remaining families run the change of variables formula backwards. You introduce a new variable in order to make an irrational expression rational, and the choice is dictated entirely by the shape of the radical. Three shapes, three substitutions, and the identity that each one exploits is the Pythagorean identity in one of its three forms: $1 - \sin^2 = \cos^2$,

$1 + \tan^2 = \sec^2$, $\sec^2 - 1 = \tan^2$. Once the quadratic under the radical has been completed and shifted by the previous chapter's method, every radical of a quadratic is in one of the three shapes, so the classification is exhaustive.

The tree below adds the two shapes that are not radicals of quadratics at all: a rational function of sin and cos, which needs Weierstrass, and a radical of a linear or fractional-power expression, which needs only that you name the radical.



The three trigonometric rows of that tree are worth setting out in full, because the whole of this section is contained in them and because the last column, the range restriction, is what most solvers omit and later regret.

Radical	Put	Radical becomes	dx	Range
$\sqrt{a^2 - x^2}$	$x = a \sin \theta$	$a \cos \theta$	$a \cos \theta d\theta$	$\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$
$\sqrt{a^2 + x^2}$	$x = a \tan \theta$	$a \sec \theta$	$a \sec^2 \theta d\theta$	$\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$
$\sqrt{x^2 - a^2}$	$x = a \sec \theta$	$a \tan \theta$	$a \sec \theta \tan \theta d\theta$	$\theta \in [0, \frac{\pi}{2}), x \geq a$
$\sqrt{a^2 + x^2}$	$x = a \sinh t$	$a \cosh t$	$a \cosh t dt$	$t \in \mathbb{R}$, no restriction
$\sqrt{x^2 - a^2}$	$x = a \cosh t$	$a \sinh t$	$a \sinh t dt$	$t \geq 0, x \geq a$

THE RANGE COLUMN IS NOT DECORATION

Each substitution is stated with a range because each is a change of variables, and a change of variables must be invertible on the interval in question. The ranges in the table are exactly the intervals on which sin, tan and sec are strictly monotone, so each has a genuine inverse there and the transformed bounds are unambiguous. Three consequences follow, and they are the three errors this section exists to prevent. Since $\cos \theta \geq 0$ on $[-\frac{\pi}{2}, \frac{\pi}{2}]$, the identity $\sqrt{a^2 - x^2} = a \cos \theta$ holds without a modulus, and only there. Since $\sec \theta \geq 1$ on $[0, \frac{\pi}{2})$, the secant substitution reaches only $x \geq a$, and the branch $x \leq -a$ needs $\theta \in (\frac{\pi}{2}, \pi]$ with $\sqrt{x^2 - a^2} = -a \tan \theta$. And since $\tan \theta$ sweeps all of $\mathbb{R}$ as θ crosses $(-\frac{\pi}{2}, \frac{\pi}{2})$, the tangent substitution alone imposes no restriction on x at all, which is why it is the only one of the three that never needs a case split.

Read the table as a statement about which identity is being used. The first row is $1 - \sin^2 = \cos^2$, the second $1 + \tan^2 = \sec^2$, the third $\sec^2 - 1 = \tan^2$, and the last two are $\cosh^2 - \sinh^2 = 1$ read in its two directions. There is no fourth shape, because after completing the square every quadratic is $\pm u^2 \pm k^2$ and the case with both signs negative has no real square root at all.

EASY

20.4 Sine substitution for $\sqrt{a^2 - x^2}$

CUE $\sqrt{a^2 - x^2}$, or any power of $a^2 - x^2$ with a half-integer exponent, on an interval inside $[-a, a]$.

METHOD Put $x = a \sin \theta$ with $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, so that $dx = a \cos \theta d\theta$ and $\sqrt{a^2 - x^2} = a |\cos \theta| = a \cos \theta$, the modulus being removable precisely because of the principal range. The integrand becomes a polynomial in sin and cos, finished by power reduction.

WORKED EXAMPLE

$$\int_0^1 \sqrt{1 - x^2} dx = \int_0^{\pi/2} \cos^2 \theta d\theta = \frac{\pi}{4},$$

the bounds having moved as $x : 0 \rightarrow 1$ gives $\theta : 0 \rightarrow \pi/2$. Two variants that show the pattern:

$$\int_0^1 \frac{x^2}{\sqrt{1 - x^2}} dx = \int_0^{\pi/2} \sin^2 \theta d\theta = \frac{\pi}{4}, \quad \int_0^1 x^2 \sqrt{1 - x^2} dx = \int_0^{\pi/2} \sin^2 \theta \cos^2 \theta d\theta = \frac{\pi}{16} \approx 0.196350,$$

the last by $\sin^2 \theta \cos^2 \theta = \frac{1}{4} \sin^2 2\theta = \frac{1}{8}(1 - \cos 4\theta)$. In every case the dx supplied one factor of $a \cos \theta$ and the radical supplied another, which is why the powers of cosine come out even. An odd half-integer power behaves the same way, only with more cosines:

$$\int_0^a (a^2 - x^2)^{3/2} dx = a^4 \int_0^{\pi/2} \cos^4 \theta d\theta = \frac{3\pi a^4}{16},$$

three factors of $a \cos \theta$ from the radical and one from dx . At $a = 2$ this is $3\pi \approx 9.424778$, which quadrature confirms. The shape rarely arrives already centred. When it does not, complete the square first, exactly as the previous chapter prescribes:

$$\int_0^1 \sqrt{2x - x^2} dx = \int_0^1 \sqrt{1 - (x-1)^2} dx \xrightarrow{u=x-1} \int_{-1}^0 \sqrt{1 - u^2} du = \int_{-\pi/2}^0 \cos^2 \theta d\theta = \frac{\pi}{4},$$

the geometric reading being that this is a quarter of the unit disc. Two substitutions were made and both moved their bounds at the moment they were written.

WHERE THIS GOES WRONG

Dropping the modulus outside the principal range. If a problem forces θ past $\pi/2$, then $\sqrt{a^2 \cos^2 \theta} = a|\cos \theta|$ and the sign flips. The other standard error appears when the integral is indefinite: undoing the substitution requires $\sin 2\theta = 2 \sin \theta \cos \theta = \frac{2x\sqrt{a^2 - x^2}}{a^2}$, and writing θ where $\sin 2\theta$ belongs loses the algebraic term entirely.

SEE ALSO Tangent substitution; completing the square; hyperbolic substitution.

The second shape has no bounded interval and no modulus problem, which makes it the easiest of the three, and it also covers integer powers of $x^2 + a^2$ where no factor of x is available to absorb into a differential.

EASY

20.5 Tangent substitution for $a^2 + x^2$

CUE Any power of $x^2 + a^2$, especially a half-integer power, with no x in the numerator to serve as part of du .

METHOD Put $x = a \tan \theta$, so $dx = a \sec^2 \theta d\theta$ and $a^2 + x^2 = a^2 \sec^2 \theta$. The secants cancel against each other and what remains is a plain sine or cosine integral. If $\tan^2 \theta$ survives, rewrite it as $\sec^2 \theta - 1$ rather than converting to sines and cosines.

WORKED EXAMPLE

$$\int_0^1 \frac{dx}{(1+x^2)^{3/2}} = \int_0^{\pi/4} \frac{\sec^2 \theta}{\sec^3 \theta} d\theta = \int_0^{\pi/4} \cos \theta d\theta = \frac{1}{\sqrt{2}} \approx 0.707107.$$

With an integer power the cancellation leaves an even power of cosine:

$$\int_0^{\sqrt{3}} \frac{dx}{(1+x^2)^2} = \int_0^{\pi/3} \cos^2 \theta d\theta = \left[\frac{\theta}{2} + \frac{\sin 2\theta}{4} \right]_0^{\pi/3} = \frac{\pi}{6} + \frac{\sqrt{3}}{8} \approx 0.740105.$$

Reading the same computation backwards gives the standard antiderivative $\int \frac{dx}{(1+x^2)^2} = \frac{1}{2} \left(\frac{x}{1+x^2} + \arctan x \right) + C$, which is worth memorising because it appears in every partial-fraction problem with a repeated quadratic.

On an unbounded interval the substitution is at its best, because the improperness disappears entirely:

$$\int_0^\infty \frac{dx}{(1+x^2)^2} = \int_0^{\pi/2} \cos^2 \theta d\theta = \frac{\pi}{4} \approx 0.785398.$$

The upper bound $\theta = \pi/2$ is a genuine number and the integrand $\cos^2 \theta$ is continuous there, so nothing about the transformed problem remembers that the original ran to infinity.

WHERE THIS GOES WRONG

Cancelling one power of secant too many by forgetting that dx contributed $\sec^2 \theta$. Count the powers explicitly: $(a^2 + x^2)^{-n}$ contributes $\sec^{-2n}$, the differential contributes $\sec^{+2}$, so the net is $\cos^{2n-2} \theta$. The second point is the bound: on $[0, \infty)$ the new upper limit is $\pi/2$, a finite number, so no improper integral survives the substitution.

SEE ALSO Sine substitution; numerator alignment; hyperbolic substitution.

The third shape is the awkward one. The secant substitution works, but its inverse is two-valued and the interval $|x| \geq a$ has two components, so it demands care that the other two do not.

20.6 Secant substitution for $\sqrt{x^2 - a^2}$

CUE $\sqrt{x^2 - a^2}$ with $x \geq a$, especially in the combinations $\frac{\sqrt{x^2 - a^2}}{x}$ and $\frac{1}{x^n \sqrt{x^2 - a^2}}$.

METHOD Put $x = a \sec \theta$ with $\theta \in [0, \frac{\pi}{2})$, so $dx = a \sec \theta \tan \theta d\theta$ and $\sqrt{x^2 - a^2} = a \tan \theta$. The two signature simplifications are

$$\frac{dx}{x\sqrt{x^2 - a^2}} = \frac{d\theta}{a}, \quad \frac{\sqrt{x^2 - a^2}}{x} dx = a \tan^2 \theta d\theta,$$

and the first of these is why the family is worth a separate name: a factor of x in the denominator, which ruins the other substitutions, is exactly what secant wants.

WORKED EXAMPLE

$$\int_1^\infty \frac{dx}{x^5 \sqrt{x^2 - 1}}.$$

With $x = \sec \theta$ the bounds run $\theta : 0 \rightarrow \pi/2$, and the integrand becomes

$$\frac{\sec \theta \tan \theta}{\sec^5 \theta \tan \theta} d\theta = \cos^4 \theta d\theta,$$

so the value is $\int_0^{\pi/2} \cos^4 \theta d\theta = \frac{3\pi}{16} \approx 0.589049$ by power reduction. Note that an improper integral in x became a proper one in θ .

A finite version, showing the bounds at work: for $\int_{\sqrt{2}}^2 \frac{dx}{x^3 \sqrt{x^2 - 1}}$, the bounds go $\theta : \pi/4 \rightarrow \pi/3$ and the integrand becomes $\cos^2 \theta d\theta$, giving

$$\left[\frac{\theta}{2} + \frac{\sin 2\theta}{4} \right]_{\pi/4}^{\pi/3} = \frac{\pi}{24} + \frac{\sqrt{3}}{8} - \frac{1}{4} \approx 0.097406.$$

WHERE THIS GOES WRONG

Applying one secant substitution across an interval that straddles the gap $(-a, a)$. For $x \leq -a$ the correct simplification is $\sqrt{x^2 - a^2} = -a \tan \theta$, so an interval such as $[-2, -1]$ needs its own branch and an interval such as $[-2, 2]$ is not an integral of a real function at all. Split, or use the symmetry, but never substitute through the gap.

SEE ALSO Hyperbolic substitution; trigonometric identity pre-processing; completing the square.

The second and third shapes have hyperbolic alternatives, and there are three specific reasons to prefer them rather than one general one. Knowing the reasons is what turns this from a matter of taste into a decision.

20.7 Hyperbolic substitution

CUE $\sqrt{x^2 + a^2}$ or $\sqrt{x^2 - a^2}$, especially on an unbounded interval, or when the expected answer is a logarithm rather than a trigonometric expression, or when the integrand already contains cosh, sinh or sech.

METHOD Put $x = a \sinh t$, giving $\sqrt{x^2 + a^2} = a \cosh t$ and $dx = a \cosh t dt$; or $x = a \cosh t$, giving $\sqrt{x^2 - a^2} = a \sinh t$ and $dx = a \sinh t dt$. The two quotable results are

$$\int \frac{dx}{\sqrt{x^2 + a^2}} = \operatorname{arsinh} \frac{x}{a} + C = \ln(x + \sqrt{x^2 + a^2}) + C', \quad \int \frac{dx}{\sqrt{x^2 - a^2}} = \ln(x + \sqrt{x^2 - a^2}) + C.$$

Three reasons to prefer these over tangent and secant: $\cosh t > 0$ for all real t , so there is never a modulus to track; the substitution is monotone on the whole line, so an unbounded interval maps to an unbounded interval without any endpoint reaching a place where the substitution degenerates; and the answer comes out in logarithms, which is the form a rational or algebraic problem usually wants.

WORKED EXAMPLE

$$\int_0^1 \frac{dx}{\sqrt{x^2 + 1}} = \operatorname{arsinh} 1 = \ln(1 + \sqrt{2}) \approx 0.881374.$$

With the substitution carried through a harder integrand, $x = \sinh t$ turns $\sqrt{x^2 + 1} dx$ into $\cosh^2 t dt = \frac{1}{2}(1 + \cosh 2t) dt$, so

$$\int_0^1 \sqrt{x^2 + 1} dx = \left[\frac{t}{2} + \frac{\sinh 2t}{4} \right]_0^T = \frac{T}{2} + \frac{\sinh T \cosh T}{2} = \frac{\ln(1 + \sqrt{2}) + \sqrt{2}}{2} \approx 1.147794,$$

where $T = \operatorname{arsinh} 1$, so $\sinh T = 1$ and $\cosh T = \sqrt{2}$. The tangent substitution reaches the same number through $\int \sec^3 \theta d\theta$, which requires a cyclic integration by parts; the hyperbolic route needs only $\cosh^2 = \frac{1}{2}(1 + \cosh 2t)$.

The third advantage, monotonicity on an unbounded interval, shows itself here:

$$\int_0^\infty \frac{dx}{(x^2 + 1)^{3/2}} = \int_0^\infty \frac{\cosh t dt}{\cosh^3 t} = \int_0^\infty \operatorname{sech}^2 t dt = [\tanh t]_0^\infty = 1.$$

The transformed integral is still improper, but its integrand decays like $4e^{-2t}$ and the antiderivative is a table entry with a finite limit. The tangent substitution reaches the same answer as $\int_0^{\pi/2} \cos \theta d\theta$, which is arguably tidier; the point is that neither substitution needed a modulus, a branch, or a split, and $x = \sinh t$ never will, whatever the interval.

WHERE THIS GOES WRONG

Treating arcosh as though it were odd. It is real only for $x \geq a$ and its logarithmic form $\ln(x + \sqrt{x^2 - a^2})$ is valid on that branch alone; for $x \leq -a$ the antiderivative is $-\ln(|x| + \sqrt{x^2 - a^2})$. The compensating virtue is that arsinh has no such restriction at all, which is why $\sqrt{x^2 + a^2}$ should essentially always be handled hyperbolically.

SEE ALSO Secant substitution; completing the square; Euler substitutions.

One radical, three substitutions

The choice between secant, cosine hyperbolic, and simply naming the radical is best settled by watching all three act on the same integral. Take

$$I = \int_1^2 \frac{\sqrt{x^2 - 1}}{x} dx.$$

By secant. Put $x = \sec \theta$, so $\sqrt{x^2 - 1} = \tan \theta$ and $dx = \sec \theta \tan \theta d\theta$, with $\theta : 0 \rightarrow \pi/3$. The integrand becomes $\tan^2 \theta d\theta = (\sec^2 \theta - 1) d\theta$, so

$$I = [\tan \theta - \theta]_0^{\pi/3} = \sqrt{3} - \frac{\pi}{3} \approx 0.684853.$$

By cosine hyperbolic. Put $x = \cosh t$, so $\sqrt{x^2 - 1} = \sinh t$ and $dx = \sinh t dt$. The integrand becomes $\frac{\sinh^2 t}{\cosh t} dt = (\cosh t - \operatorname{sech} t) dt$, whose antiderivative is $\sinh t - \arctan(\sinh t)$. Since $\sinh t = \sqrt{x^2 - 1}$ and $\arctan \sqrt{x^2 - 1} = \operatorname{arcsec} x$ for $x \geq 1$, this is the same antiderivative in different clothes, and $I = \sqrt{3} - \frac{\pi}{3}$ again.

By naming the radical. Put $u = \sqrt{x^2 - 1}$, so $u^2 = x^2 - 1$, $u du = x dx$, and $u : 0 \rightarrow \sqrt{3}$. Then

$$\frac{\sqrt{x^2 - 1}}{x} dx = \frac{u}{x} \cdot \frac{u du}{x} = \frac{u^2}{u^2 + 1} du = \left(1 - \frac{1}{u^2 + 1}\right) du,$$

so $I = [u - \arctan u]_0^{\sqrt{3}} = \sqrt{3} - \frac{\pi}{3}$, in four lines and with no trigonometry at all.

The third route was shortest, and that is not an accident: whenever the radical appears to an odd power alongside a compensating x , naming it is better than any trigonometric substitution. The secant route is best when the radical appears with a bare x^n in the denominator, since $\frac{dx}{x\sqrt{x^2 - a^2}}$ collapses to $d\theta/a$. And the hyperbolic route is best when the interval is unbounded or the expected answer is logarithmic. Three tools, three regimes, and knowing the regimes is the whole content of the choice.

Rationalising: Euler, Weierstrass, and naming the radical

Trigonometric and hyperbolic substitutions convert an algebraic integrand into a trigonometric one, which is progress only if the resulting trigonometric integral is easy. Sometimes it is not, and then the better move is to go the other way: choose a substitution that makes the integrand a *rational function*, which the previous chapter guarantees can always be finished. Euler's substitutions do this for a radical of a general quadratic; the Weierstrass substitution does it for any rational function of sin and cos; and for a radical of something linear or fractional, naming the radical does it in one line.

20.8 Euler substitutions for a general quadratic radical

HARD

CUE $\sqrt{ax^2 + bx + c}$ multiplied by a rational function, where completing the square is legal but produces a trigonometric integral you do not want.

METHOD Three choices, at least one of which always applies. (i) If $a > 0$, set $\sqrt{ax^2 + bx + c} = t - \sqrt{a}x$. (ii) If $c > 0$, set $\sqrt{ax^2 + bx + c} = xt + \sqrt{c}$. (iii) If the quadratic has real roots $r_1 \neq r_2$, set $\sqrt{a(x - r_1)(x - r_2)} = t(x - r_1)$. In each case, squaring cancels the leading term and leaves a linear equation for x in terms of t , so x , dx and the radical are all rational in t and the whole integrand becomes a rational function.

WHICH OF THE THREE TO USE

The choices are not interchangeable, and the rule is short. Use (iii) whenever the quadratic has real roots, because it is the only one of the three that keeps the radical of one sign across the interval. Otherwise the quadratic is definite, and if it is positive definite both (i) and (ii) apply: take (ii) when the interval has an endpoint at or near $x = 0$, since $t = \frac{\sqrt{ax^2 + bx + c} - \sqrt{c}}{x}$ is then the well behaved formula, and (i) otherwise, since $t = x\sqrt{a} + \sqrt{ax^2 + bx + c}$ is well behaved everywhere else and is the tidiest algebra of the three. If the quadratic is negative definite there is no real radical and no integral to compute. That exhausts the cases, which is why the three substitutions are stated as three and not as a list to be searched.

WORKED EXAMPLE

$$\int_0^1 \frac{dx}{(x+1)\sqrt{x^2+x+1}}.$$

Here $c = 1 > 0$, so take choice (ii): $\sqrt{x^2 + x + 1} = xt + 1$. Squaring, $x^2 + x + 1 = x^2t^2 + 2xt + 1$, so $x + 1 = xt^2 + 2t$ and

$$x = \frac{2t-1}{1-t^2}, \quad \sqrt{x^2+x+1} = xt+1 = \frac{t^2-t+1}{1-t^2}, \quad dx = \frac{2(t^2-t+1)}{(1-t^2)^2} dt.$$

Also $x+1 = \frac{2t-t^2}{1-t^2} = \frac{t(2-t)}{1-t^2}$. The bounds follow from $t = \frac{\sqrt{x^2+x+1}-1}{x}$, so $x = 0$ gives $t = \frac{1}{2}$ (by the limit) and $x = 1$ gives $t = \sqrt{3} - 1$. Assembling, everything except $\frac{2dt}{t(2-t)}$ cancels, and partial fractions give

$$\int_{1/2}^{\sqrt{3}-1} \frac{2dt}{t(2-t)} = \left[\ln \frac{t}{2-t} \right]_{1/2}^{\sqrt{3}-1} = \ln \frac{\sqrt{3}-1}{3-\sqrt{3}} + \ln 3 = \ln \frac{1}{\sqrt{3}} + \ln 3 = \frac{1}{2} \ln 3 \approx 0.549306,$$

using $3 - \sqrt{3} = \sqrt{3}(\sqrt{3} - 1)$. Quadrature on the original integrand returns 0.549306144334, and $\frac{1}{2} \ln 3 = 0.549306144334$.

The first choice is worth seeing once as well, because it is the one that applies whenever $a > 0$ and because it produces the tidiest algebra of the three. With $\sqrt{x^2 + 1} = t - x$, squaring gives $1 = t^2 - 2tx$, so

$$x = \frac{t^2-1}{2t}, \quad dx = \frac{t^2+1}{2t^2} dt, \quad x + \sqrt{x^2+1} = t,$$

and the bounds follow from $t = x + \sqrt{x^2+1}$, running $t : 1 \rightarrow 1 + \sqrt{2}$ as $x : 0 \rightarrow 1$. Hence

$$\int_0^1 \frac{dx}{x + \sqrt{x^2+1}} = \int_1^{1+\sqrt{2}} \frac{t^2+1}{2t^3} dt = \frac{1}{2} \left[\ln t - \frac{1}{2t^2} \right]_1^{1+\sqrt{2}} = \frac{\sqrt{2}-1 + \ln(1+\sqrt{2})}{2} \approx 0.647794,$$

using $(1 + \sqrt{2})^2 = 3 + 2\sqrt{2}$ and $\frac{1}{3 + 2\sqrt{2}} = 3 - 2\sqrt{2}$. The substitution $t = x + \sqrt{x^2+1}$ is of course $e^{\operatorname{arsinh} x}$, which is why the answer contains a logarithm and why the hyperbolic substitution would have reached the same place.

The third choice is the one to reach for when the quadratic has real roots, because then the other two produce a radical that changes sign inside the interval while this one does not. Take $\sqrt{x^2 - 3x + 2} = \sqrt{(x-1)(x-2)}$ and set it equal to $t(x-1)$. Squaring and cancelling one factor of $x-1$ gives $x-2 = t^2(x-1)$, so

$$x = \frac{2-t^2}{1-t^2}, \quad x-1 = \frac{1}{1-t^2}, \quad dx = \frac{2t}{(1-t^2)^2} dt, \quad \sqrt{x^2 - 3x + 2} = \frac{t}{1-t^2},$$

and therefore $\frac{dx}{\sqrt{x^2 - 3x + 2}} = \frac{2 dt}{1-t^2}$, whose antiderivative is $\ln \frac{1+t}{1-t}$. With $t = \sqrt{\frac{x-2}{x-1}}$ the bounds $x : 3 \rightarrow 5$ become $t : \frac{1}{\sqrt{2}} \rightarrow \frac{\sqrt{3}}{2}$, so

$$\int_3^5 \frac{dx}{\sqrt{x^2 - 3x + 2}} = \ln(7 + 4\sqrt{3}) - \ln(3 + 2\sqrt{2}) = 2 \ln \frac{2 + \sqrt{3}}{1 + \sqrt{2}} \approx 0.871169.$$

The same substitution earns its keep when a rational factor is present and completing the square would leave a secant integral. For instance $\sqrt{x-x^2} = \sqrt{x(1-x)}$ has roots 0 and 1; setting $\sqrt{x(1-x)} = tx$ gives $x = \frac{1}{1+t^2}$, $dx = \frac{-2t dt}{(1+t^2)^2}$, and $\frac{dx}{\sqrt{x-x^2}} = \frac{-2 dt}{1+t^2}$. The bounds run backwards, $x : 0 \rightarrow 1$ giving $t : \infty \rightarrow 0$, and the two minus signs cancel:

$$\int_0^1 \frac{dx}{(2-x)\sqrt{x-x^2}} = \int_0^\infty \frac{2 dt}{1+t^2} = \frac{\pi}{\sqrt{2}} \approx 2.221441,$$

since $2-x = \frac{1+t^2}{1+t^2}$ cancels one factor of $1+t^2$. The general statement, worth carrying, is that $\int_0^1 \frac{dx}{(b-x)\sqrt{x(1-x)}} = \frac{\pi}{\sqrt{b(b-1)}}$ for $b > 1$; at $b = 2$ that is $\pi/\sqrt{2}$ and at $b = 3$ it is $\pi/\sqrt{6} \approx 1.282550$, both of which quadrature confirms.

WHERE THIS GOES WRONG

The bounds. Each Euler substitution is a Möbius map in disguise, so it has a pole, and the branch of the square root fixed by the choice determines which side of that pole you are on. Compute the new bounds from the explicit formula for t in terms of x , never by solving the quadratic afresh at each endpoint, and check that the interval in t does not contain the pole.

SEE ALSO Rationalising a radical by substitution; completing the square; the Weierstrass substitution.

Euler rationalises an algebraic integrand. The corresponding universal tool on the trigonometric side is the Weierstrass substitution, which is guaranteed to work and, for that reason, is usually the wrong first choice: a guarantee that applies to everything gives no information, and the price it charges is a rational function of higher degree than any identity-based route would have produced.

MEDIUM

20.9 The Weierstrass substitution $t = \tan(x/2)$

CUE A rational function of $\sin x$ and $\cos x$ that has resisted every identity: $\frac{1}{a+b\cos x}, \frac{1}{\sin x + \cos x + 1}, \frac{1}{a\sin x + b\cos x + c}$.

METHOD With $t = \tan \frac{x}{2}$,

$$\sin x = \frac{2t}{1+t^2}, \quad \cos x = \frac{1-t^2}{1+t^2}, \quad \tan x = \frac{2t}{1-t^2}, \quad dx = \frac{2 dt}{1+t^2},$$

together with the derived forms

$$\sec x = \frac{1+t^2}{1-t^2}, \quad \csc x = \frac{1+t^2}{2t}, \quad \cot x = \frac{1-t^2}{2t}, \quad 1 + \cos x = \frac{2}{1+t^2}, \quad 1 - \cos x = \frac{2t^2}{1+t^2},$$

and $x = 2 \arctan t$ for the return trip. Every one of these is a rational function of t , which is the content of the substitution: the field generated by $\sin x$ and $\cos x$ over the rationals is the field of rational functions of t , so the integrand becomes rational in t unconditionally. The last two entries are worth noticing separately, because they say that the half-angle identities of the previous chapter and this substitution are the same fact written twice. When the integrand is invariant under $x \mapsto x + \pi$, which happens exactly when only even powers of $\sin$ and $\cos$ occur or when the integrand is a function of $\tan x$, prefer the cheaper $t = \tan x$ with $\sin^2 x = \frac{t^2}{1+t^2}$, $\cos^2 x = \frac{1}{1+t^2}$, $dx = \frac{dt}{1+t^2}$, which halves the degree.

$$\int_0^{\pi/2} \frac{dx}{2 + \cos x}.$$

With $t = \tan \frac{x}{2}$ the bounds go $t : 0 \rightarrow 1$, and $2 + \cos x = \frac{2(1+t^2) + (1-t^2)}{1+t^2} = \frac{3+t^2}{1+t^2}$, so the $1+t^2$ from dx cancels:

$$\int_0^1 \frac{2dt}{3+t^2} = \frac{2}{\sqrt{3}} \left[\arctan \frac{t}{\sqrt{3}} \right]_0^1 = \frac{2}{\sqrt{3}} \cdot \frac{\pi}{6} = \frac{\pi\sqrt{3}}{9} \approx 0.604600.$$

A case where the rational function is even simpler than the original:

$$\int_0^{\pi/2} \frac{dx}{1 + \sin x + \cos x} = \int_0^1 \frac{1+t^2}{2+2t} \cdot \frac{2dt}{1+t^2} = \int_0^1 \frac{dt}{1+t} = \ln 2 \approx 0.693147.$$

And the cheaper substitution where it applies: for $\int_0^{\pi/2} \frac{dx}{1 + \sin^2 x}$ only even powers occur, so $t = \tan x$ gives $\int_0^\infty \frac{dt}{1+2t^2} = \frac{\pi}{2\sqrt{2}}$, against a quartic in t had you used the half-angle form.

WHAT WEIERSTRASS COSTS

The substitution always works, and that is the reason to try it last. It converts a first-degree expression in $\sin$ and $\cos$ into a second-degree rational function, and a second-degree expression into a fourth-degree one, so the price of the guarantee is paid in partial fractions. Compare the two routes on $\int_0^{\pi/2} \frac{dx}{1 + \sin^2 x}$. The half-angle substitution gives $\int_0^1 \frac{2(1+t^2)}{t^4 + 6t^2 + 1} dt$, a quartic denominator needing two irreducible quadratic factors. The substitution $t = \tan x$, legitimate because only even powers occur, gives $\int_0^\infty \frac{dt}{1+2t^2} = \frac{\pi}{2\sqrt{2}}$, one line. Before reaching for the half angle, ask whether an identity from the previous chapter, or the cheaper $t = \tan x$, will do the job.

One more, in which the rational function factors and the answer comes out of partial fractions rather than an arctangent:

$$\int_0^{\pi/2} \frac{dx}{3 + 5 \sin x} = \int_0^1 \frac{2dt}{3t^2 + 10t + 3} = \int_0^1 \frac{2dt}{(3t+1)(t+3)} = \left[\frac{1}{4} \ln \frac{3t+1}{t+3} \right]_0^1 = \frac{1}{4} \ln 3 \approx 0.274653.$$

Notice that the denominator $a + b \sin x$ with $|b| > |a|$ produces a rational function with real roots and hence logarithms, while $|b| < |a|$ produces an irreducible quadratic and hence an arctangent. The discriminant of the transformed quadratic is exactly $4(b^2 - a^2)$ up to a positive factor, so the sign of $b^2 - a^2$ predicts the shape of the answer before the substitution is performed.

The cosine version is worth carrying as a formula, because it is the single most quoted consequence of the substitution.

For $a > |b|$ the denominator $a + b \cos x$ becomes $\frac{(a+b) + (a-b)t^2}{1+t^2}$, so the $1+t^2$ from dx cancels and

$$\int \frac{dx}{a + b \cos x} = \frac{2}{\sqrt{a^2 - b^2}} \arctan \left(\sqrt{\frac{a-b}{a+b}} \tan \frac{x}{2} \right) + C, \quad \int_0^\pi \frac{dx}{a + b \cos x} = \frac{\pi}{\sqrt{a^2 - b^2}}.$$

The definite form over a half period is clean because $\tan \frac{x}{2}$ runs from 0 to $+\infty$ and the arctangent contributes $\pi/2$. Two checks: at $a = 3, b = 2$ the half-period value is $\pi/\sqrt{5} \approx 1.404963$, and at $a = 5, b = 3$ it is $\pi/4$, both confirmed by quadrature. The condition $a > |b|$ is not decoration. If $|b| > a$ the denominator vanishes somewhere in $[0, \pi]$, the integral is divergent, and the formula returns the square root of a negative number, which is the algebra telling you so.

WHERE THIS GOES WRONG

The substitution $t = \tan \frac{x}{2}$ has a pole at $x = \pi$. Over a full period $[0, 2\pi]$ the map sends both endpoints to $t = 0$, so a mechanical application returns $\int_0^0 = 0$ for an integral that is strictly positive. The fix is either to split at π and take one-sided limits, or to use periodicity to move the window to $[-\pi, \pi]$ and treat both endpoints as improper. Check for the pole before substituting, not after getting zero.

The repair, carried out. For $\int_0^{2\pi} \frac{dx}{2 + \cos x}$ split at the pole and treat each half by the half-period formula. On $[0, \pi]$ the value is $\pi/\sqrt{4-1} = \pi/\sqrt{3}$, the substituted bound $t = \tan \frac{x}{2}$ running from 0 to $+\infty$ rather than to a finite number. On $[\pi, 2\pi]$ the reflection $x \mapsto 2\pi - x$ maps the integral to the first half, so it contributes $\pi/\sqrt{3}$ again, and

$$\int_0^{2\pi} \frac{dx}{2 + \cos x} = \frac{2\pi}{\sqrt{3}} \approx 3.627599.$$

Compare the mechanical answer of 0. The discipline that catches this is the same one that governs the whole chapter: before transforming the bounds, check that the substitution is monotone and finite across the interval. Here $\tan \frac{x}{2}$ is monotone on $(0, \pi)$ and on $(\pi, 2\pi)$ but not across π , and the two facts together are the entire diagnosis.

SEE ALSO Product to sum, and sum to product; trigonometric identity pre-processing; Euler substitutions.

Euler and Weierstrass are the heavy machinery. The light version, and the one you will use twenty times as often, applies when the radical is of something linear or of a fractional power: there is nothing to complete and nothing to rationalise by trigonometry, and the right move is simply to give the radical a name.

MEDIUM

20.10 Rationalising a radical by substitution, and peeling nested radicals

CUE An n th root of a linear expression, a fractional power of x , a radical inside a radical, or two different radicals sharing a hidden common radicand.

METHOD Set u^n equal to the radicand, so that the radical becomes u and dx becomes polynomial in u . For layered radicals, peel one layer per substitution, from the outside inwards, and stack the substitutions; each layer typically simplifies the next. The bounds transform at each layer, and because an even root forces $u \geq 0$ you must check that the map is increasing before assuming the orientation is preserved.

WORKED EXAMPLE

The one-layer case: $u = \sqrt{x}$, $x = u^2$, $dx = 2u du$, and

$$\int_0^1 \frac{dx}{1 + \sqrt{x}} = \int_0^1 \frac{2u du}{1 + u} = 2 \int_0^1 \left(1 - \frac{1}{1 + u}\right) du = 2(1 - \ln 2) \approx 0.613706.$$

Note what happened: an irrational integrand became an improper rational function, which the previous chapter's division step then finished.

Two layers, with the substitutions stacked. For $\int_0^1 \sqrt{1 - t^{1/3}} dt$, put $t = u^3$ so $dt = 3u^2 du$ and $u : 0 \rightarrow 1$:

$$3 \int_0^1 u^2 \sqrt{1 - u} du.$$

Now $v = 1 - u$ gives $3 \int_0^1 (1 - v)^2 \sqrt{v} dv = 3 \int_0^1 (v^{1/2} - 2v^{3/2} + v^{5/2}) dv = 3\left(\frac{2}{3} - \frac{4}{5} + \frac{2}{7}\right) = \frac{16}{35} \approx 0.457143$.

A shorter one in the same style: $\int_0^1 \sqrt{1 - \sqrt{x}} dx$ becomes, with $u = \sqrt{x}$, $2 \int_0^1 u \sqrt{1 - u} du$, and with $v = 1 - u$ that is $2 \int_0^1 (1 - v) \sqrt{v} dv = 2\left(\frac{2}{3} - \frac{2}{5}\right) = \frac{8}{15}$.

When several fractional powers of x appear together, take n to be their least common denominator, so that one substitution clears all of them at once. For $\int_0^1 \frac{dx}{1 + x^{1/3}}$ that is $n = 3$, giving $x = u^3$, $dx = 3u^2 du$ and

$$3 \int_0^1 \frac{u^2}{1 + u} du = 3 \int_0^1 \left(u - 1 + \frac{1}{1 + u}\right) du = 3\left(\frac{1}{2} - 1 + \ln 2\right) = 3 \ln 2 - \frac{3}{2} \approx 0.579442,$$

where the middle step is the previous chapter's polynomial division applied to the improper fraction the substitution created. With $x^{1/2}$ and $x^{1/3}$ both present the right choice is $n = 6$, and choosing $n = 2$ or $n = 3$ leaves one radical standing.

The full three-layer ladder, worth doing once in its entirety:

$$\int_0^1 \arcsin \sqrt{1 - \sqrt{x}} dx.$$

Peel from the outside in. First $u = \sqrt{x}$, $dx = 2u du$, giving $2 \int_0^1 u \arcsin \sqrt{1 - u} du$. Second $v = 1 - u$, which reverses the bounds and cancels the minus sign, giving $2 \int_0^1 (1 - v) \arcsin \sqrt{v} dv$. Third $w = \sqrt{v}$, $dv = 2w dw$, which finally clears the radical from inside the arcsine:

$$4 \int_0^1 (w - w^3) \arcsin w dw.$$

Now integrate by parts once on each piece. Since $\int_0^1 \frac{w^2}{\sqrt{1 - w^2}} dw = \frac{\pi}{4}$ and $\int_0^1 \frac{w^4}{\sqrt{1 - w^2}} dw = \frac{3\pi}{16}$, both by $w = \sin \theta$,

$$\int_0^1 w \arcsin w dw = \frac{\pi}{4} - \frac{1}{2} \cdot \frac{\pi}{4} = \frac{\pi}{8}, \quad \int_0^1 w^3 \arcsin w dw = \frac{\pi}{8} - \frac{1}{4} \cdot \frac{3\pi}{16} = \frac{5\pi}{64},$$

so the answer is $4(\frac{\pi}{8} - \frac{5\pi}{64}) = \frac{3\pi}{16} \approx 0.589049$, confirmed by quadrature. Three substitutions and two integrations by parts, and every substitution was forced: at each stage exactly one radical was outermost and exactly one choice of u removed it.

WHERE THIS GOES WRONG

Peeling from the inside out. Substituting for the inner radical first typically leaves the outer radical containing a polynomial of higher degree than before, which is strictly worse. Work outwards in, and stop as soon as the integrand is rational or a beta integral. The second hazard is orientation: with $u^2 = x$ the choice $u = -\sqrt{x}$ is also a solution of the defining equation, and taking it silently reverses the bounds.

SEE ALSO Euler substitutions; conjugate multiplication; exponential substitution.

Exponentials, and substitutions that map the integral to itself

Two families remain. The first is the exponential substitution, which is differential spotting applied to a base other than e and which turns any rational function of a^x into a rational function outright. The second is different in kind from everything else in the chapter: the involutions $x \mapsto 1/x$ and $x \mapsto a/x$ do not simplify the integrand at all. They produce a second expression for the same integral, and it is the combination of the two that gives the answer.

EASY

20.11 Exponential substitution $u = a^x$

CUE An integrand rational in e^x or in a^x , a base appearing alongside its own square (as 45^x does alongside 2025^x), or the combinations $e^x \pm e^{-x}$.

METHOD Put $u = a^x$, so $du = u \ln a \, dx$ and therefore $dx = \frac{du}{u \ln a}$. Every a^{kx} becomes u^k , and the $\ln a$ appears exactly once, as an overall constant. The extra $1/u$ contributed by dx raises the degree of the denominator by one, which is what makes the resulting rational function proper.

WORKED EXAMPLE

Since $2025 = 45^2$, the substitution $u = 45^x$ collapses the following completely:

$$\int_0^1 \frac{45^x}{2025^x + 1} dx = \frac{1}{\ln 45} \int_1^{45} \frac{u}{u^2 + 1} \cdot \frac{du}{u} = \frac{1}{\ln 45} \int_1^{45} \frac{du}{u^2 + 1} = \frac{\arctan 45 - \frac{\pi}{4}}{\ln 45} \approx 0.200485.$$

A second, showing the extra $1/u$ doing its work:

$$\int_0^{\ln 2} \frac{dx}{1 + e^x} = \int_1^2 \frac{1}{1 + u} \cdot \frac{du}{u} = \left[\ln \frac{u}{1 + u} \right]_1^2 = \ln \frac{4}{3} \approx 0.287682.$$

Here the substitution alone would have given $\frac{1}{1 + u}$, which is not what the integral is; the Jacobian supplied the second factor and turned it into a partial-fraction problem.

A third, where a bare e^x in the numerator absorbs the Jacobian and the substitution is pure differential spotting:

$$\int_0^{\ln 3} \frac{e^x}{e^{2x} + 1} dx = \int_1^3 \frac{du}{u^2 + 1} = \arctan 3 - \frac{\pi}{4} \approx 0.463648.$$

The rule of thumb is that a numerator containing exactly one power of e^x makes the substitution free, while a numerator without one forces the $1/u$ into the integrand and raises the degree.

The $e^x \pm e^{-x}$ pattern is the same substitution wearing a hyperbolic hat. Multiplying above and below by e^x turns

$\frac{1}{e^x + e^{-x}}$ into $\frac{e^x}{e^{2x} + 1}$, which is the free case again:

$$\int_0^\infty \frac{dx}{e^x + e^{-x}} = \int_1^\infty \frac{du}{u^2 + 1} = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4} \approx 0.785398,$$

and over the whole line the same computation gives $\pi/2$. Written hyperbolically this is $\frac{1}{2} \int_0^\infty \operatorname{sech} x \, dx = \frac{1}{2} [2 \arctan e^x]_0^\infty$, which is why $\arctan e^x$ is the antiderivative of $\frac{1}{2} \operatorname{sech} x$ and why the substitution $u = e^x$ is the right one for every integrand built from sech , $\tanh$ or csch .

Dropping the $1/u$ from dx . It changes the degree of the rational function and therefore the whole partial-fraction structure, and the resulting answer is typically off by exactly one logarithm. The second point concerns bounds: on $(-\infty, b]$ the new lower bound is $u = 0$, where the rational integrand may have a pole that the original integrand did not appear to have.

SEE ALSO Partial fractions with distinct linear factors; reciprocal substitution; spotting the differential.

The last family is the one that changes how you think about substitution. Nothing gets simpler; the integral is mapped to a different-looking integral with the same value, and you then solve for the common value. This is the elementary end of a set of ideas that the symmetry chapter develops in full.

HARD

20.12 Reciprocal substitution $x \mapsto 1/x$ and $x \mapsto a/x$

CUE An integrand on $(0, \infty)$ built from $\ln x$ and a palindromic denominator such as $x^2 + x + 1$ or $x^2 + a^2$; bounds that are reciprocals of each other, as $[\frac{1}{2}, 2]$ is; or $x + \frac{1}{x}$ appearing anywhere in the integrand.

METHOD Substitute $x = a/u$, so $dx = -\frac{a}{u^2} du$. The minus sign and the reversal of the bounds cancel each other. The interval $(0, \infty)$ maps to itself, and a finite interval $[c, a/c]$ maps to itself; a logarithm splits, since $\ln(a/u) = \ln a - \ln u$, and a palindromic denominator is reproduced up to a power of u . Add the transformed copy to the original and solve for the integral.

WORKED EXAMPLE

The pure antisymmetric case:

$$I = \int_0^\infty \frac{\ln x}{1+x^2} dx.$$

Under $x = 1/u$ the integrand becomes $\frac{-\ln u}{1+u^{-2}} \cdot \frac{du}{u^2} = \frac{-\ln u}{u^2+1} du$, so $I = -I$ and $I = 0$. Convergence must be checked first, and it holds because $\ln x/(1+x^2)$ is $O(\ln x)$ near 0 and $O(x^{-2} \ln x)$ at infinity. The same argument gives $\int_0^\infty \frac{\ln x}{x^2+x+1} dx = 0$, since x^2+x+1 is palindromic, and quadrature confirms both to twelve digits.

Now the scaled involution, whose job is to reduce a general problem to the one just solved:

$$J(a) = \int_0^\infty \frac{\ln x}{x^2+a^2} dx, \quad a > 0.$$

The involution that fixes this denominator is $x = a/u$, not $x = 1/u$. Then $x^2 + a^2 = \frac{a^2(1+u^2)}{u^2}$ and $dx = -\frac{a}{u^2} du$, so the two powers of u^2 cancel and

$$J(a) = \frac{1}{a} \int_0^\infty \frac{\ln a - \ln u}{1+u^2} du = \frac{\ln a}{a} \cdot \frac{\pi}{2} - \frac{1}{a} \underbrace{\int_0^\infty \frac{\ln u}{1+u^2} du}_{=0} = \frac{\pi \ln a}{2a}.$$

The scaled involution did not produce an equation for $J(a)$ in terms of itself; it produced $J(a)$ in terms of $J(1)$, and $J(1)$ was the antisymmetric case above. At $a = 2$ this gives $\frac{\pi \ln 2}{4} \approx 0.544397$, which quadrature confirms as 0.544396522576.

A version on a finite interval. Since $[\frac{1}{2}, 2]$ is fixed by $x \mapsto 1/x$, and $\arctan \frac{1}{x} = \frac{\pi}{2} - \arctan x$ for $x > 0$,

$$K = \int_{1/2}^2 \frac{\arctan x}{x} dx = \int_{1/2}^2 \frac{\arctan(1/u)}{u} du \implies 2K = \frac{\pi}{2} \int_{1/2}^2 \frac{du}{u} = \frac{\pi}{2} \ln 4,$$

so $K = \frac{\pi \ln 2}{2} \approx 1.088793$, which is exactly what quadrature returns. Not one antiderivative was computed.

The cleanest demonstration of the whole idea, and the one worth remembering, is that for every real α

$$\int_0^\infty \frac{dx}{(1+x^2)(1+x^\alpha)} = \frac{\pi}{4}.$$

Under $x = 1/u$ the factor $\frac{1}{1+x^\alpha}$ becomes $\frac{u^\alpha}{1+u^\alpha}$ while the rest is unchanged, so adding the two expressions makes the second factor sum to 1 and leaves $2I = \int_0^\infty \frac{du}{1+u^2} = \frac{\pi}{2}$. Quadrature at $\alpha = 3$ and at $\alpha = \sqrt{2}$ both return 0.785398163397, and the independence of α is the signature of every integral in this family: the parameter is decoration, because the involution removes it.

Applying the fold to a divergent integral. The argument “ $I = -I$, so $I = 0$ ” is valid only if I exists, and for $\int_0^\infty \frac{\ln x}{x} dx$ the same manipulation “proves” that a divergent integral is zero. Establish convergence first, separately on $(0, 1]$ and $[1, \infty)$. The second point is the scale: for $x^2 + a^2$ the involution that works is $x \mapsto a/x$, not $x \mapsto 1/x$, and using the wrong one produces a true but useless identity.

SEE ALSO Exponential substitution; conjugate multiplication; the logarithmic form.

Reciprocal pairing: folding the half line onto an interval

The involution has a second use, quite separate from producing an equation for the integral. Because $x \mapsto 1/x$ exchanges $(0, 1)$ with $(1, \infty)$, every integral over the half line can be folded onto a bounded interval:

$$\int_0^\infty f(x) dx = \int_0^1 \left[f(x) + \frac{1}{x^2} f\left(\frac{1}{x}\right) \right] dx.$$

Nothing has been assumed about f beyond convergence. The fold is worth performing for two reasons. It converts an improper integral into a proper one, which is what makes numerical confirmation of a claimed closed form trivial; and it exposes the symmetry of f , because the bracket collapses whenever f has one.

The collapse has two forms. If $x^{-2}f(1/x) = f(x)$, call f self-reciprocal, and the two halves of the half line contribute equally:

$$\int_0^\infty f = 2 \int_0^1 f.$$

Both $\frac{1}{1+x^2}$ and $\frac{1}{x^2+x+1}$ are self-reciprocal, the second because its denominator is palindromic. So the previous chapter's computation $\int_0^1 \frac{dx}{x^2+x+1} = \frac{\pi\sqrt{3}}{9}$ immediately gives $\int_0^\infty \frac{dx}{x^2+x+1} = \frac{2\pi\sqrt{3}}{9} \approx 1.209200$, which the direct antiderivative $\frac{2}{\sqrt{3}} \arctan \frac{2x+1}{\sqrt{3}}$ confirms.

If instead $x^{-2}f(1/x) = -f(x)$, call f antireciprocal; the bracket vanishes identically and the integral is 0. That is exactly the mechanism behind $\int_0^\infty \frac{\ln x}{1+x^2} dx = 0$, and the fold says something the bare argument does not: the two halves are not merely cancelling, they are exact negatives of each other. Numerically,

$$\int_0^1 \frac{\ln x}{1+x^2} dx = -G \approx -0.915966, \quad \int_1^\infty \frac{\ln x}{1+x^2} dx = +G,$$

where G is Catalan's constant. Neither half has an elementary closed form; their sum has the elementary value 0, and the involution is the only reason you can know that without evaluating either one. This is the pattern to expect from every symmetry argument: the symmetry gives you the combination, never the pieces.

A family-recognition drill

For each integrand, name the family and the substitution before computing anything. Every one of these is decidable from the shape alone.

- (i) xe^{x^2} . Differential spotting, $u = x^2$: the x is half of du/dx and the composite is e^u .
- (ii) $\frac{1}{x^2\sqrt{4-x^2}}$. Sine substitution with $a = 2$; the x^2 below becomes $4\sin^2\theta$ and the integrand collapses to $\frac{1}{4}\csc^2\theta$. On $[1, 2]$ the value is $\frac{\sqrt{3}}{4}$.
- (iii) $\frac{1}{x\sqrt{x^2-9}}$. Secant with $a = 3$; this is exactly the signature collapse, so the antiderivative is $\frac{1}{3}\operatorname{arcsec} \frac{x}{3}$ and no further work is needed.
- (iv) $\sqrt{x^2+2x+5}$. Complete the square to $\sqrt{(x+1)^2+4}$, then $x+1 = 2\sinh t$. Hyperbolic, not tangent, because the answer will contain a logarithm.
- (v) $\frac{1}{2+3\cos x}$. Weierstrass. Since $|b| > |a|$ the transformed denominator $5-t^2$ has real roots, so expect logarithms: on $[0, \frac{\pi}{2}]$ the value is $\frac{1}{\sqrt{5}} \ln \frac{\sqrt{5}+1}{\sqrt{5}-1} \approx 0.430409$.

- (vi) $\frac{1}{x^{1/2} + x^{1/3}}$. Name the radical with $n = \text{lcm}(2, 3) = 6$, so $x = u^6$ and everything becomes rational at once.
- (vii) $\frac{\ln x}{x^2 + 9}$ on $(0, \infty)$. Reciprocal, and the correct scale is $x \mapsto 3/x$, giving $\frac{\pi \ln 3}{6} \approx 0.575232$.
- (viii) $\frac{\sec^2 x}{1 + \tan^3 x}$. Differential spotting again, $u = \tan x$; the family is decided by the $\sec^2 x$, which is du/dx and nothing else.
- (ix) $\frac{x^3}{\sqrt{1+x^2}}$. Name the radical, $u = \sqrt{1+x^2}$, so $u du = x dx$ and $x^2 = u^2 - 1$: the integrand becomes $(u^2 - 1) du$. On $[0, 1]$ the value is $[\frac{1}{3}u^3 - u]_1^{\sqrt{2}} = \frac{2-\sqrt{2}}{3} \approx 0.195262$. A trigonometric substitution would work and would take four times as long, because an odd power of x beside the radical is always the naming case.
- (x) $\frac{1}{x \ln x \ln \ln x}$. The logarithmic form, twice over: $u = \ln \ln x$ has $du = \frac{dx}{x \ln x}$, so the integrand is du/u and the antiderivative is $\ln |\ln \ln x|$. Stacked logarithms always peel this way, one $\ln$ per layer, and the pattern continues for as many layers as the problem cares to build.

Items (ii) and (iii) are the pair worth dwelling on. Both have a power of x in the denominator beside a radical, and the family is decided entirely by whether the radical is $\sqrt{a^2 - x^2}$ or $\sqrt{x^2 - a^2}$. Nothing else about the two integrands differs.

The cost of transforming the limits last

The discipline stated at the top of the chapter is easy to agree with and easy to abandon under pressure, so it is worth pricing. Take $\int_0^1 x^2 \sqrt{1-x^2} dx$ and do it both ways.

Bounds moved at once. Put $x = \sin \theta$, $\theta : 0 \rightarrow \pi/2$. The integrand becomes $\sin^2 \theta \cos^2 \theta d\theta = \frac{1}{8}(1 - \cos 4\theta) d\theta$, and

$$\int_0^{\pi/2} \frac{1 - \cos 4\theta}{8} d\theta = \frac{1}{8} \cdot \frac{\pi}{2} = \frac{\pi}{16}.$$

Three lines, and the $\cos 4\theta$ term dies at both ends without being evaluated.

Bounds moved last. The same computation produces the antiderivative $\frac{\theta}{8} - \frac{\sin 4\theta}{32}$, and now it must be returned to x . That requires $\theta = \arcsin x$, then $\sin 4\theta = 2 \sin 2\theta \cos 2\theta$, then $\sin 2\theta = 2x\sqrt{1-x^2}$ and $\cos 2\theta = 1 - 2x^2$, giving

$$F(x) = \frac{\arcsin x}{8} - \frac{x\sqrt{1-x^2}(1-2x^2)}{8},$$

which is correct, as differentiating it confirms, and which took four separate identities to assemble. Each of them is a chance to lose a factor of two or a sign, and none of them was needed: $F(1) - F(0) = \pi/16$ is precisely the number the first route produced without ever writing F down.

The moral generalises past this example. For an indefinite integral you have no choice, and the back-substitution is part of the work. For a definite one the back-substitution is optional, and it is the single longest and least reliable step in the entire calculation. Declining it is free.

When the substitution turns around

The change of variables theorem asks that g be continuously differentiable, and for the definite form with transformed bounds it is safest to insist further that g be monotone on $[a, b]$. The reason is not pedantry. Consider $\int_{-1}^1 x^2 dx$, which is $\frac{2}{3}$, and attempt $u = x^2$. Both bounds map to $u = 1$, so a mechanical application returns $\int_1^1 = 0$. The error is that $dx = \frac{du}{2\sqrt{u}}$

holds only for $x > 0$; on $x < 0$ the correct relation is $dx = -\frac{du}{2\sqrt{u}}$, and the two halves must be handled separately.

The same failure produces the Weierstrass trap noted above, and it is the reason the secant substitution must not straddle the gap $(-a, a)$. The diagnostic is easy: if the two transformed bounds are equal, or if the transformed integral is negative when the integrand was positive, the substitution turned around somewhere inside the interval. Split at the turning point, substitute separately on each piece, and add.

Read as a whole, the taxonomy answers one question: what do you want the integrand to become? If you want it to become a table entry, spot the differential. If you want a radical of a quadratic to become trigonometric, use sine, tangent or secant according to the sign pattern, and prefer the hyperbolic version whenever the interval is unbounded or the

expected answer is a logarithm. If you want any of it to become rational, use Euler for an algebraic radical, Weierstrass for a trigonometric one, and the plain naming substitution for anything with a root of a linear expression. And if you want the integral to become a copy of itself, use the involutions. In every case, transform the bounds in the same stroke as the variable, and check monotonicity on the interval before you do.

THE RULE, WITHOUT EXCEPTIONS

Transform the limits at the moment you transform the variable. Not after simplifying, not after finding the antiderivative, not at the end. Write the substitution and its bounds on the same line, as one object: $u = \sin^2 x$, $x : 0 \rightarrow \pi/4$, $u : 0 \rightarrow \frac{1}{2}$. The rule costs one line and buys four things. It eliminates the commonest error in the subject, evaluating a u antiderivative at x bounds. It eliminates the second commonest, the back-substitution of a double angle. It forces you to notice a substitution that turns around, because the two bounds come out equal or in the wrong order. And it forces you to notice a pole, because a bound comes out infinite or the map jumps. Every trap in this chapter is a failure of one of those four checks, and all four are performed by the same line of writing.

Recognition matrix

If you see	Reach for	If that fails
$F(g(x))$ beside a multiple of $g'(x)$	$u = g(x)$; solve for dx ; move the bounds	Check the factor really is g' up to a constant
Numerator equal to the derivative of the denominator	$\ln g $, after confirming by differentiation	Multiply through by $\cos x$ or a reciprocal first
A bracketed sum beside an awkward composite	Differentiate the composite's argument; compare	Never split the bracket
$\sqrt{a^2 - x^2}$	$x = a \sin \theta$, $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$	Complete the square first, then $u = k \sin \theta$
$(x^2 + a^2)^{\pm n}$ with no spare x	$x = a \tan \theta$; count the powers of $\sec$	$x = a \sinh t$ if the answer wants logarithms
$\sqrt{x^2 - a^2}$, or $x^n \sqrt{x^2 - a^2}$ below	$x = a \sec \theta$ on one branch at a time	$x = a \cosh t$; never straddle $(-a, a)$
$\sqrt{x^2 + a^2}$ on an unbounded interval	Hyperbolic: no modulus, no wrapping	Read the answer off as a logarithm
$\sqrt{ax^2 + bx + c}$ times a rational function	Euler: $t - \sqrt{ax}$, or $xt + \sqrt{c}$, or $t(x - r_1)$	Complete the square, then a trig substitution
A rational function of $\sin x$ and $\cos x$	$t = \tan \frac{x}{2}$; watch the pole at $x = \pi$	$t = \tan x$ if only even powers appear
n th linear, or $x^{p/q}$	$u^n =$ the radicand; dx becomes polynomial	Peel layered radicals from the outside in
An integrand rational in a^x	$u = a^x$; remember $dx = du/(u \ln a)$	Look for $a^{2x} = (a^x)^2$ first
$\ln x$ over a palindromic denominator on $(0, \infty)$	$x \mapsto 1/x$; the integral maps to its negative	Confirm convergence before concluding 0
$x^2 + a^2$ below, with $\ln x$ above	$x \mapsto a/x$, not $x \mapsto 1/x$	Solve $2J = \ln a \int \dots$ for J
Bounds that are reciprocals, as $[\frac{1}{b}, b]$	$x \mapsto 1/x$ fixes the interval; add the copies	Use a co-function identity on the numerator
An improper integral over $(0, \infty)$	Fold: $\int_0^1 [f(x) + x^{-2}f(1/x)] dx$	Test for self-reciprocal or antireciprocal
$\frac{1}{a + b \cos x}$ with $a > b $	$t = \tan \frac{x}{2}$; over $[0, \pi]$ it is $\pi/\sqrt{a^2 - b^2}$	Check $a > b $, or the integral diverges
Any definite integral at all	Transform the bounds in the same stroke	Verify the substitution is monotone there

Integration by parts and its systems

What this chapter is for. Integration by parts is usually taught as a single move: pick u , pick dv , apply the formula, hope the new integral is easier. That is the smallest thing it does. Parts is a machine with three distinct running modes, and knowing which mode a problem wants is most of the skill. Run it *forwards*, repeatedly, when one factor is a polynomial that differentiates to zero; the bookkeeping is then a table, not a sequence of substitutions. Run it *in a cycle*, when two applications return the original integrand, so that the unknown integral appears on both sides of an equation you can solve algebraically. Run it *once, symbolically*, on an integrand carrying an integer parameter n , to obtain a reduction formula relating I_n to I_{n-1} or I_{n-2} , and then iterate the recursion instead of the integral. This chapter separates the three modes, gives each its own cue and its own characteristic failure, and adds the two structural tricks that make the machine much wider than it first appears: taking $dv = dx$ so that u is the entire integrand, and choosing the additive constant in v so that the remaining integral collapses. By the end you should be able to look at a product and say, before writing anything, which mode you are about to run and what the answer will look like.

One identity, three machines

Everything here comes from the product rule, integrated. From $(uv)' = u'v + uv'$ we get

$$\int_a^b u \, dv = [uv]_a^b - \int_a^b v \, du,$$

which is not a method for computing integrals so much as a method for *trading* one integral for another. The trade is favourable exactly when $v \, du$ is simpler than $u \, dv$, and the whole art is in arranging that.

Three things can happen after the trade, and they correspond to the three modes. The new integral can be strictly simpler, in which case repeating the trade eventually terminates: this is the forwards mode, and it is what happens when u is a polynomial, because differentiation drives a polynomial to zero in finitely many steps. The new integral can be the *same* as the old one up to a constant multiple, in which case nothing has been simplified but an equation has been created: this is the cyclic mode. Or the new integral can be a member of the same family with a smaller parameter, in which case you have a recursion rather than an answer: this is the reduction mode. There is no fourth outcome worth naming. If the trade produces something genuinely unrelated and no simpler, you chose u wrongly.

A word on the boundary term before anything else. In an indefinite computation $[uv]$ is just a term you carry along; in a definite computation it is a number, and it is the piece most often lost. When an endpoint is improper it must be evaluated as a limit, and it is frequently exactly the term that cancels a divergence elsewhere. Never write parts on a definite integral without writing the brackets and their limits at the same moment.

CORE

21.1 The parts identity and the choice of u

CUE A product of two factors drawn from different families: a polynomial against an exponential, a polynomial against a sine or cosine, or anything at all against a logarithm or an inverse trigonometric function. The tell is that one factor gets simpler when differentiated and the other does not get worse when integrated.

METHOD Apply

$$\int_a^b u \, dv = [uv]_a^b - \int_a^b v \, du.$$

Choose u to be the factor you want to *differentiate away*. The mnemonic LIATE orders the families Logarithmic, Inverse trigonometric, Algebraic, Trigonometric, Exponential, and says: whichever family appears first in that list becomes u , and the rest becomes dv . The ordering is not arbitrary. Logarithms and inverse trigonometric functions differentiate into algebraic functions, which is a large gain; polynomials differentiate down by one degree, a modest gain; trigonometric and exponential factors differentiate to themselves, no gain at all, so they belong in the dv slot where they must be antidifferentiated instead.

$$\int_0^{\pi/2} x^2 \sin x \, dx.$$

LIATE puts the algebraic factor first, so $u = x^2$, $dv = \sin x \, dx$, $v = -\cos x$:

$$\int_0^{\pi/2} x^2 \sin x \, dx = [-x^2 \cos x]_0^{\pi/2} + 2 \int_0^{\pi/2} x \cos x \, dx = 0 + 2 \int_0^{\pi/2} x \cos x \, dx.$$

The remaining integral is the same shape with the degree lowered, so parts again with $u = x$, $dv = \cos x \, dx$:

$$2([x \sin x]_0^{\pi/2} - \int_0^{\pi/2} \sin x \, dx) = 2\left(\frac{\pi}{2} - 1\right) = \pi - 2.$$

Two applications, because the polynomial had degree two. That count is predictable before you start, and it is what the next technique automates.

WHERE LIATE FAILS, HONESTLY

LIATE is a heuristic about *families*, and it is blind to composition. It fails in three identifiable ways. First, when a factor is a composite whose inner derivative is hiding in the other factor: in $\int x^3 e^{x^2} \, dx$, LIATE says $u = x^3$ (algebraic before exponential), but e^{x^2} has no elementary antiderivative, so that split is not merely suboptimal, it is illegal. The right split is $u = x^2$ and $dv = xe^{x^2} \, dx$, which is a substitution wearing a parts costume; the answer is $\frac{1}{2}$ on $[0, 1]$. Second, when the LIATE-preferred dv has no closed-form antiderivative at all: $\int_0^1 x^2 e^{-x^2} \, dx$ must be split as $u = x$, $dv = xe^{-x^2} \, dx$, giving $-\frac{1}{2e} + \frac{\sqrt{\pi}}{4} \operatorname{erf}(1) \approx 0.189472$. Third, when you do not actually want a simpler integral but a *recursion*, as with $\int \sec^n x \, dx$, where the productive split puts $\sec^{n-2} x$ in the u slot even though LIATE has no opinion about splitting one trigonometric factor against another. Treat LIATE as a default that costs nothing to check and that you overrule the moment you see a composition.

WHERE THIS GOES WRONG

Evaluating $[uv]$ at one endpoint only, or dropping it because it “looks like it vanishes”. In $\int_0^{\pi/2} x^2 \sin x \, dx$ the bracket really is zero at both ends, but that is a fact about $\cos(\pi/2) = 0$, not a general licence. When an endpoint is improper the bracket is a limit: in $\int_0^1 \ln x \, dx$ the term $[x \ln x]_0^1$ is $0 - \lim_{x \rightarrow 0^+} x \ln x = 0$, and asserting that without the limit is guessing.

SEE ALSO Tabular integration; parts with $dv = dx$; cyclic integration by parts; engineering the antiderivative v .

The worked example above took two applications for a quadratic, and a cubic would take three, a quartic four. Each application regenerates the same bracket-minus-integral shape with one lower degree, and the only thing that changes is the sign and which antiderivative of the trigonometric factor is in play. That is precisely the situation where a table beats a narrative.

21.2 Tabular integration

MEDIUM

CUE A polynomial (any degree, but the method earns its keep from degree three upwards) multiplied by a factor you can antidifferentiate repeatedly and cheaply: e^{ax} , $\sin ax$, $\cos ax$, $\sinh ax$, $\cosh ax$.

METHOD Make two columns. In the left column write u and differentiate downwards until you reach 0. In the right column write the other factor and antidifferentiate downwards the same number of times. Then read off the products along the diagonals, alternating in sign:

$$\int u \, dv = uv_1 - u'v_2 + u''v_3 - u'''v_4 + \cdots,$$

where v_1 is the first antiderivative, v_2 the second, and so on. It is n applications of parts with the bookkeeping done once instead of n times.

WORKED EXAMPLE

$$\int_0^1 x^3 e^{-x} \, dx.$$

Left column: $x^3, 3x^2, 6x, 6, 0$. Right column, antidifferentiating e^{-x} : $-e^{-x}, e^{-x}, -e^{-x}, e^{-x}$. Diagonals with alternating signs:

$$-x^3 e^{-x} - 3x^2 e^{-x} - 6x e^{-x} - 6e^{-x} = -e^{-x}(x^3 + 3x^2 + 6x + 6).$$

The two minus signs (one from the alternation, one from antidifferentiating e^{-x}) reinforce at every step, which is why every term here is negative. Evaluating,

$$\int_0^1 x^3 e^{-x} dx = \left[-e^{-x}(x^3 + 3x^2 + 6x + 6) \right]_0^1 = 6 - \frac{16}{e} \approx 0.113929.$$

The same table with x^4 gives $24 - 65/e \approx 0.087836$, and the pattern of coefficients $1, n, n(n-1), \dots$ is just the falling factorials appearing on the left column.

SECOND INSTANCE, TRIGONOMETRIC

$$\int_0^{\pi/2} x^3 \cos x dx.$$

Left: $x^3, 3x^2, 6x, 6, 0$. Right: $\sin x, -\cos x, -\sin x, \cos x$. Alternating diagonals give the antiderivative

$$x^3 \sin x + 3x^2 \cos x - 6x \sin x - 6 \cos x,$$

where the signs now cycle with period four rather than reinforcing. At $x = \pi/2$ this is $\frac{\pi^3}{8} - 3\pi$; at $x = 0$ it is -6 . Hence the integral is $\frac{\pi^3}{8} - 3\pi + 6 \approx 0.451007$.

WHERE THIS GOES WRONG

Two failures, both mechanical. The alternation of signs is the first: the terms are $+, -, +, -$ reading down, and this is independent of any signs generated inside the right column, so with e^{-x} you are combining two independent sign patterns and must track both. The second is termination. The table only closes when the left column reaches 0, which requires u to be a polynomial. With $u = \ln x$ or $u = \arctan x$ the left column never terminates, and tabular integration is simply the wrong machine; use single-step parts or the cyclic mode instead.

SEE ALSO The parts identity and the choice of u ; the factorial ladder; cyclic integration by parts.

When u is the whole integrand

Everything so far assumed a product. The most useful structural observation about parts is that you do not need one. Any integrand f is the product $f(x) \cdot 1$, and taking $dv = dx$ turns parts into a machine for integrating functions that have no obvious antiderivative but a very easy derivative. That describes exactly the logarithms and the inverse trigonometric functions, which is why they sit at the front of LIATE: they are the functions you always want in the u slot, and if there is nothing else in the integrand, you manufacture a dv out of the constant 1.

21.3 Parts with $dv = dx$

EASY

CUE A *single* function with an easy derivative and no memorable antiderivative: $\ln x$, $\arctan x$, $\arcsin x$, $\arccos x$, $\ln(x + \sqrt{x^2 + 1})$, $\operatorname{arcosh} x$. Also any such function multiplied by a polynomial, where the same logic applies with $dv = x^k dx$.

METHOD Write the integrand as $f(x) \cdot 1$ and take $u = f$, $dv = dx$, so $v = x$:

$$\int_a^b f(x) dx = [xf(x)]_a^b - \int_a^b xf'(x) dx.$$

The point is that f' is algebraic for every function on the cue list, so the traded integral is a rational function or a rational function of a radical, and elementary.

WORKED EXAMPLE

Three at once, each in one line.

$$\int_0^1 \ln x dx = [x \ln x]_0^1 - \int_0^1 1 dx = 0 - 1 = -1,$$

the bracket at 0 being $\lim_{x \rightarrow 0^+} x \ln x = 0$.

$$\int_0^1 \arctan x dx = [x \arctan x]_0^1 - \int_0^1 \frac{x}{1+x^2} dx = \frac{\pi}{4} - \frac{\ln 2}{2} \approx 0.438825.$$

$$\int_0^1 \arcsin x dx = [x \arcsin x]_0^1 - \int_0^1 \frac{x}{\sqrt{1-x^2}} dx = \frac{\pi}{2} - 1 \approx 0.570796.$$

In each case the traded integral was engineered by the derivative: $\frac{x}{1+x^2}$ is a logarithmic derivative, $\frac{x}{\sqrt{1-x^2}}$ is a power rule after $w = 1 - x^2$.

WITH A POLYNOMIAL ATTACHED

$$\int_0^1 x \arctan x \, dx.$$

Now $dv = x \, dx$, $v = \frac{x^2}{2}$, and

$$\int_0^1 x \arctan x \, dx = \left[\frac{x^2}{2} \arctan x \right]_0^1 - \frac{1}{2} \int_0^1 \frac{x^2}{1+x^2} \, dx = \frac{\pi}{8} - \frac{1}{2} \int_0^1 \left(1 - \frac{1}{1+x^2} \right) \, dx.$$

That last integral is $1 - \frac{\pi}{4}$, so the answer is $\frac{\pi}{8} - \frac{1}{2} + \frac{\pi}{8} = \frac{\pi}{4} - \frac{1}{2} \approx 0.285398$. Note the long division step: $\frac{x^2}{1+x^2}$ is improper as a rational function and must be reduced before anything else happens.

WHERE THIS GOES WRONG

With an improper endpoint the term $xf(x)$ can diverge even though the integral converges. That is a signal to choose a different v , not to abandon the method: $v = x$ is only one of infinitely many antiderivatives of dx , and $v = x - c$ is available for any c . Choosing c to kill the divergence is the subject of the next technique.

SEE ALSO Engineering the antiderivative v ; inverse-function integration; the logarithm ladder.

That last remark is the hinge of this section. In $\int u \, dv$ the function v is determined only up to an additive constant, and the constant is free. Almost every textbook silently takes the constant to be zero. Choosing it deliberately is one of the highest-yield habits in the whole subject, because the traded integral $\int v \, du$ depends on the choice, and a well chosen constant can make it collapse entirely.

21.4 Engineering the antiderivative v

MEDIUM

CUE Parts has left you with $\int \frac{v}{g(x)} \, dx$ where v *almost* cancels the denominator: the denominator is $x + 1$ but $v = x$; the denominator is $(x + a)^2$ but v is the wrong shift; the denominator is $x^2 - 1$ but $v = \frac{x^2}{2}$.

METHOD Replace v by $v + c$ and choose c so that $v + c$ shares a factor with the denominator produced by du . Concretely, for $\int \ln(x+1) \, dx$ take $v = x + 1$ rather than $v = x$; for $\int \frac{\ln x}{(x+a)^2} \, dx$ take $v = -\frac{1}{x+a} + \frac{1}{a} = \frac{x}{a(x+a)}$, which carries a factor x to cancel the $\frac{1}{x}$ from du . The remaining integral then has no logarithm in it at all, and usually no partial fractions either.

WORKED EXAMPLE

$$\int_0^1 \ln(1+x) \, dx.$$

With $v = x$ this becomes $[x \ln(1+x)]_0^1 - \int_0^1 \frac{x}{1+x} \, dx$, and the last integral needs the division $\frac{x}{1+x} = 1 - \frac{1}{1+x}$. With $v = x + 1$ instead,

$$\int_0^1 \ln(1+x) \, dx = [(x+1) \ln(1+x)]_0^1 - \int_0^1 \frac{x+1}{1+x} \, dx = 2 \ln 2 - 1 \approx 0.386294.$$

The traded integrand is the constant 1. No division, no partial fractions, one line.

THE CASE THAT MAKES THE TECHNIQUE INDISPENSABLE

$$\int_0^1 \frac{\ln x}{(1+x)^2} \, dx.$$

Take $u = \ln x$ and $dv = \frac{dx}{(1+x)^2}$. The obvious antiderivative is $v = -\frac{1}{1+x}$, and it leads to $\int_0^1 \frac{dx}{x(1+x)}$, which diverges at 0; so does the bracket, and the two divergences cancel only after careful limiting. Add the constant 1 instead:

$$v = -\frac{1}{1+x} + 1 = \frac{x}{1+x}.$$

Now the bracket is $\left[\frac{x \ln x}{1+x}\right]_0^1 = 0 - 0 = 0$, both ends finite because $x \ln x \rightarrow 0$, and the traded integral is

$$-\int_0^1 \frac{x}{1+x} \cdot \frac{1}{x} dx = -\int_0^1 \frac{dx}{1+x} = -\ln 2.$$

So $\int_0^1 \frac{\ln x}{(1+x)^2} dx = -\ln 2 \approx -0.693147$, obtained without ever meeting a divergent piece.

WHERE THIS GOES WRONG

Shifting v but forgetting that the bracket $[uv]_a^b$ changes too. Both halves of the identity contain v , and both must be recomputed with the new one. The error is silent: you get a finite, clean, wrong number. A cheap audit is to check that the two versions agree in an easy case, or simply to differentiate your final antiderivative.

SEE ALSO Parts with $dv = dx$; the parts identity and the choice of u ; inverse-function integration.

Running the cycle

The forwards mode terminates because differentiation destroys polynomials. When neither factor can be destroyed, parts does not terminate, and the naive reaction is that the method has failed. It has not. If two applications return you to the original integral multiplied by some constant λ , you have not computed the integral but you have written down a linear equation satisfied by it, and provided $\lambda \neq 1$ that equation has a unique solution. This is the second mode, and it is the reason $\int e^x \sin x dx$ and $\int \sec^3 x dx$ are elementary at all.

CORE

21.5 Cyclic integration by parts

CUE Two applications of parts return the original integrand up to a constant multiple. The standard members: $e^{ax} \sin bx$, $e^{ax} \cos bx$, $\sin(\ln x)$, $\cos(\ln x)$, $\sec^3 x$, and anything of the form e^{ax} times a solution of a second-order constant-coefficient equation.

METHOD Name the integral. Set $I = \int u dv$, apply parts twice *keeping the same family in the u slot both times*, and you will arrive at

$$I = (\text{boundary terms}) + \lambda I \quad \Rightarrow \quad I = \frac{\text{boundary terms}}{1 - \lambda}.$$

For $e^{ax} \sin bx$ the constant is $\lambda = -b^2/a^2$, so $1 - \lambda = (a^2 + b^2)/a^2$ and the denominator $a^2 + b^2$ that appears in every table entry for these integrals is exactly $1 - \lambda$ in disguise.

WORKED EXAMPLE

$$I = \int_0^\pi e^x \sin x dx.$$

Take $u = \sin x$, $dv = e^x dx$ both times. First pass:

$$I = [e^x \sin x]_0^\pi - \int_0^\pi e^x \cos x dx = 0 - \int_0^\pi e^x \cos x dx.$$

Second pass on the new integral, again with the trigonometric factor as u :

$$\int_0^\pi e^x \cos x dx = [e^x \cos x]_0^\pi + \int_0^\pi e^x \sin x dx = (-e^\pi - 1) + I.$$

Substituting, $I = e^\pi + 1 - I$, so $2I = e^\pi + 1$ and

$$I = \frac{e^\pi + 1}{2} \approx 12.070346,$$

consistent with the antiderivative $\frac{1}{2}e^x(\sin x - \cos x)$. Here $\lambda = -1$ and $1 - \lambda = 2$.

A CYCLE WITH $\lambda \neq -1$

$$\int_0^{\pi/2} e^x \cos x dx = [e^x \cos x]_0^{\pi/2} + \int_0^{\pi/2} e^x \sin x dx,$$

and one more pass on the right gives $[e^x \sin x]_0^{\pi/2}$ minus the original. So $I = (0 - 1) + e^{\pi/2} - I$, giving $I = \frac{e^{\pi/2} - 1}{2} \approx 1.905239$. The general antiderivative is $\frac{e^{ax}(a \cos bx + b \sin bx)}{a^2 + b^2}$, and it is worth deriving once and then quoting.

TWO WAYS TO AVOID THE CYCLE ENTIRELY

The cycle is elegant but it is not compulsory. For $e^{ax} \cos bx$ you may instead complexify: $e^{ax} \cos bx = \operatorname{Re} e^{(a+ib)x}$, and a single exponential antidifferentiates in one step to $\frac{e^{(a+ib)x}}{a+ib}$, whose real part is $\frac{e^{ax}(a \cos bx + b \sin bx)}{a^2 + b^2}$. The denominator $a^2 + b^2$ that the cycle produced as $1 - \lambda$ appears here as $|a + ib|^2$, which is what it always was. Second, a logarithmic argument can be substituted away before parts is applied at all: $\int_1^{e^\pi} \sin(\ln x) dx$ becomes, under $x = e^t$, the integral $\int_0^\pi e^t \sin t dt = \frac{e^\pi + 1}{2}$ already computed above. Reach for the substitution first; it turns an unfamiliar cycle into a familiar one.

WHERE THIS GOES WRONG

Choosing u inconsistently. If you take $u = \sin x$ on the first pass and $u = e^x$ on the second, the second pass simply undoes the first and you obtain the true but vacuous statement $I = I$. Keep the same family in the u slot throughout. Separately: $\lambda = 1$ means the method has failed on this integrand, not that I is infinite. See the next technique.

SEE ALSO Diagnosing the vacuous cycle; the secant ladder; complexifying the exponential; reduction formulae by parts.

The equation $I = B + \lambda I$ is only useful when $1 - \lambda \neq 0$, and the sign bookkeeping that produces λ is where cyclic parts is most often botched. It deserves to be isolated, because a solver who does not know the diagnosis will conclude that a perfectly ordinary integral is impossible.

21.6 Diagnosing the vacuous cycle

MEDIUM

CUE You have run parts twice, the original integral has reappeared, and either the two sides cancel identically ($\lambda = 1$) or you have simply written $I = I$.

METHOD There are exactly two diagnoses, and they have different cures. If you switched which factor plays u between the two passes, the cycle is an artefact and the cure is to redo the second pass with the same choice as the first. If you were consistent and still got $\lambda = 1$, the two passes are genuinely inverse to each other on this integrand and parts alone cannot evaluate it; the cure is a different technique. The arithmetic test is quick: track the sign contributed at each pass. Parts contributes one explicit minus sign, differentiating $\sin$ then $\cos$ contributes another, and e^x contributes none, so the total is $(-1) \cdot (-1) \cdot (-1) = -1$ for $e^x \sin x$, giving $\lambda = -1$ and $1 - \lambda = 2$. If your product of signs is $+1$, you are about to write $I = B + I$, and either $B = 0$ (vacuous) or you have made an error.

WORKED EXAMPLE

The vacuous run, shown deliberately. Take $I = \int e^x \sin x dx$ with $u = \sin x$, $dv = e^x dx$:

$$I = e^x \sin x - \int e^x \cos x dx.$$

Now switch: on the second pass take $u = e^x$, $dv = \cos x dx$, so

$$\int e^x \cos x dx = e^x \sin x - \int e^x \sin x dx = e^x \sin x - I.$$

Substituting back: $I = e^x \sin x - e^x \sin x + I = I$. True and useless. Every step is correct; the choice was wrong. Redo the second pass with $u = \cos x$ and the same computation yields $2I = e^x(\sin x - \cos x)$.

EQUATION OR IDENTITY

The distinction is worth stating flatly. Parts run twice always produces $I = B + \lambda I$. That is an *equation* determining I when $\lambda \neq 1$, and an *identity* carrying no information when $\lambda = 1$ and $B = 0$. The value of λ is fixed by the integrand and by your choice of u , not by luck, and you can predict it before doing any work by counting minus signs. For $e^{ax} \sin bx$ with the trigonometric factor as u , $\lambda = -b^2/a^2$; for the same integrand with the exponential as u , $\lambda = +1$ and you get nothing.

WHERE THIS GOES WRONG

Concluding that a cycle with $\lambda = 1$ means the integral diverges. It means your two passes composed to the identity map. $\int e^x \sin x dx$ is perfectly finite; the vacuous run above proves nothing about it.

SEE ALSO Cyclic integration by parts; the secant ladder; complexifying the exponential.

Running the ladder

The third mode is the most powerful and the least often taught as a mode. Instead of applying parts to a specific integral, apply it to a whole family indexed by an integer n , and read the result as a recursion. You do not obtain a closed form; you obtain a machine that produces one, and the machine costs a single application of parts no matter how large n is. The four families that matter are powers of sine or cosine, powers of secant, x^n against an exponential, and powers of a logarithm. Each has its own base case, and mixing base cases is the standard way to get a factor of $\pi/2$ wrong.

CORE

21.7 Reduction formulae by parts

CUE An integrand carrying an integer exponent n , where one application of parts visibly lowers n by one or two: $\sin^n x$, $\cos^n x$, $\sec^n x$, $x^n e^x$, $(\ln x)^n$, $(1+x^2)^{-n}$, $(1-x^2)^n$.

METHOD Split off *one* copy of the repeated factor as dv and leave $n-1$ copies in u , or split off two if that is what the derivative structure wants. Apply parts once, symbolically, then use a Pythagorean or algebraic identity to re-express the result in terms of I_{n-2} (or I_{n-1}). You obtain

$$I_n = (\text{boundary}) + c_n I_{n-2},$$

and you then iterate the recursion, not the integral. Establish the two base cases I_0 and I_1 before iterating, and check the parity of n to know which one you will land on.

WORKED EXAMPLE

The sine ladder, derived once. With $u = \sin^{n-1} x$ and $dv = \sin x \, dx$,

$$I_n = \int \sin^n x \, dx = -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x \cos^2 x \, dx.$$

Replace $\cos^2 x = 1 - \sin^2 x$: the integral splits into $(n-1)I_{n-2} - (n-1)I_n$, so

$$nI_n = -\sin^{n-1} x \cos x + (n-1)I_{n-2}, \quad I_n = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} I_{n-2}.$$

The single algebraic move that made this work was writing $\cos^2$ in terms of $\sin^2$ so that the new integral is in the *same* family. That move is the whole technique; everything else is parts.

THE THREE QUESTIONS TO ASK OF ANY LADDER

Before iterating, answer these. (i) Does the recursion step by one or by two? Sine and secant step by two, $x^n e^x$ and $(\ln x)^n$ step by one. (ii) Does the boundary term vanish on the given interval? On $[0, \pi/2]$ the sine boundary $-\sin^{n-1} x \cos x$ vanishes at both ends for $n \geq 2$; on $[0, \pi/4]$ it does not, and you must carry it. (iii) Which base case do you land on? A two-step recursion from even n lands on I_0 , from odd n on I_1 , and those two numbers are usually wildly different.

WHERE THIS GOES WRONG

Iterating the recursion while forgetting that the boundary term is itself n -dependent. On an interval where it does not vanish, the closed form is a *sum* of boundary terms, one per rung, each multiplied by the accumulated product of the c_k . Writing $I_n = c_n c_{n-2} \cdots I_0$ and adding a single boundary term at the end is wrong by several terms.

PARTS AT AN IMPROPER ENDPOINT

Ladders very often live on intervals with a singular endpoint, and the boundary term is then a limit that must be taken, not a value that may be substituted. The model computation:

$$\int_0^1 \frac{\ln x}{\sqrt{x}} \, dx = \left[2\sqrt{x} \ln x \right]_0^1 - \int_0^1 \frac{2\sqrt{x}}{x} \, dx = 0 - 2 \int_0^1 x^{-1/2} \, dx = -4.$$

The bracket at 0 is $\lim_{x \rightarrow 0^+} 2\sqrt{x} \ln x = 0$, which holds because $\sqrt{x}$ beats $\ln x$; the traded integral is convergent for the same reason. Both facts need checking, and the discipline is to check them before, not after. When the bracket diverges and the integral does not, shift v by a constant, exactly as in the engineered- v technique.

SEE ALSO Wallis' formulae; the secant ladder; the factorial ladder; the logarithm ladder.

The sine ladder pays its largest dividend on $[0, \pi/2]$, where the boundary term is identically zero and the recursion becomes purely numerical. The resulting closed forms are old enough to have a name, and they are worth memorising because they appear whenever a trigonometric substitution has been applied to a definite integral.

21.8 Wallis' formulae

CUE $\int_0^{\pi/2} \sin^n x \, dx$ or $\int_0^{\pi/2} \cos^n x \, dx$, or anything that becomes one of these after $x = a \sin \theta$: powers of $\sqrt{a^2 - x^2}$, and $\int_0^1 (1 - x^2)^m \, dx$.

METHOD On $[0, \pi/2]$ the boundary term in the sine ladder vanishes, so $I_n = \frac{n-1}{n} I_{n-2}$ with $I_0 = \frac{\pi}{2}$ and $I_1 = 1$. Iterating,

$$\int_0^{\pi/2} \sin^{2m} \theta \, d\theta = \frac{(2m-1)!!}{(2m)!!} \cdot \frac{\pi}{2}, \quad \int_0^{\pi/2} \sin^{2m+1} \theta \, d\theta = \frac{(2m)!!}{(2m+1)!!},$$

and the same values for $\cos^n$, since $\theta \mapsto \frac{\pi}{2} - \theta$ exchanges the two. The double factorial $(2m)!! = 2 \cdot 4 \cdots 2m$ and $(2m-1)!! = 1 \cdot 3 \cdots (2m-1)$.

WORKED EXAMPLE

$$\int_0^{\pi/2} \sin^6 \theta \, d\theta = \frac{5 \cdot 3 \cdot 1}{6 \cdot 4 \cdot 2} \cdot \frac{\pi}{2} = \frac{5\pi}{32} \approx 0.490874, \quad \int_0^{\pi/2} \sin^7 \theta \, d\theta = \frac{6 \cdot 4 \cdot 2}{7 \cdot 5 \cdot 3} = \frac{16}{35} \approx 0.457143.$$

An application that does not look trigonometric: $\int_0^1 (1 - x^2)^3 \, dx$. Put $x = \sin \theta$. Then $(1 - x^2)^3 = \cos^6 \theta$ and $dx = \cos \theta \, d\theta$, so the exponent is $6 + 1 = 7$, odd, and the integral is $\int_0^{\pi/2} \cos^7 \theta \, d\theta = \frac{16}{35}$ with no π in it. The Jacobian supplying the seventh factor is exactly the sort of detail that flips you from one base case to the other. Expanding $(1 - x^2)^3$ and integrating term by term gives $1 - 1 + \frac{3}{5} - \frac{1}{7} = \frac{16}{35}$, which is the audit.

WALLIS REACHED THROUGH A SUBSTITUTION

$$\int_0^2 x^2 \sqrt{4 - x^2} \, dx.$$

Put $x = 2 \sin \theta$, so $\sqrt{4 - x^2} = 2 \cos \theta$ and $dx = 2 \cos \theta \, d\theta$, giving

$$\int_0^{\pi/2} 4 \sin^2 \theta \cdot 2 \cos \theta \cdot 2 \cos \theta \, d\theta = 16 \int_0^{\pi/2} \sin^2 \theta \cos^2 \theta \, d\theta = 4 \int_0^{\pi/2} \sin^2 2\theta \, d\theta = 4 \cdot \frac{\pi}{4} = \pi.$$

The intermediate step used $\sin \theta \cos \theta = \frac{1}{2} \sin 2\theta$ to reduce a mixed product to a single even power, which is always available and always worth doing before reaching for the double factorials. Every integral of the form $\int_0^a x^m (a^2 - x^2)^{k/2} \, dx$ becomes a Wallis integral this way, with total exponent $m + k + 1$.

WHERE THIS GOES WRONG

The factor $\pi/2$ appears for *even* powers only, because only the even ladder bottoms out at $I_0 = \frac{\pi}{2}$; odd powers bottom out at $I_1 = 1$ and give a rational number. Landing on the wrong base case is the single commonest error in this chapter, and it is detectable by inspection: if your answer to $\int_0^{\pi/2} \sin^7$ contains π , it is wrong.

SEE ALSO Reduction formulae by parts; sine substitution; the Beta function.

Secant is the awkward relative. It obeys a two-step recursion like sine, but its base case $\int \sec x \, dx = \ln |\sec x + \tan x|$ is not a number you can guess, and the first interesting rung, $\int \sec^3$, is simultaneously a reduction and a cycle. That double character is exactly why it is worth its own block.

21.9 The secant ladder, $\int \sec^n x \, dx$

CUE Odd powers of $\sec x$, or any integrand that becomes one after $x = a \tan \theta$: $\int \sqrt{a^2 + x^2} \, dx$ produces $\sec^3$, and $\int (a^2 + x^2)^{3/2} \, dx$ produces $\sec^5$.

METHOD Split $\sec^n x = \sec^{n-2} x \cdot \sec^2 x$ with $u = \sec^{n-2} x$ and $dv = \sec^2 x \, dx$, so $v = \tan x$. Then use $\tan^2 = \sec^2 - 1$:

$$I_n = \sec^{n-2} x \tan x - (n-2) \int \sec^{n-2} x \tan^2 x \, dx = \sec^{n-2} x \tan x - (n-2) I_n + (n-2) I_{n-2},$$

so

$$I_n = \frac{\sec^{n-2} x \tan x}{n-1} + \frac{n-2}{n-1} I_{n-2}, \quad I_1 = \ln |\sec x + \tan x|, \quad I_2 = \tan x.$$

Notice that the derivation is a cycle, not a pure reduction: I_n appeared on the right and had to be solved for. That is why the denominators are $n-1$ rather than n .

$$\int_0^{\pi/4} \sec^3 x \, dx = \frac{1}{2} [\sec x \tan x]_0^{\pi/4} + \frac{1}{2} \int_0^{\pi/4} \sec x \, dx = \frac{\sqrt{2}}{2} + \frac{1}{2} \ln(\sqrt{2} + 1),$$

that is, $\frac{\sqrt{2} + \ln(1 + \sqrt{2})}{2} \approx 1.147794$. One rung higher,

$$\int_0^{\pi/4} \sec^5 x \, dx = \frac{[\sec^3 x \tan x]_0^{\pi/4}}{4} + \frac{3}{4} \int_0^{\pi/4} \sec^3 x \, dx = \frac{2\sqrt{2}}{4} + \frac{3}{4} \cdot 1.147794 \approx 1.567952.$$

The boundary term here does *not* vanish, unlike the sine case on $[0, \pi/2]$, so each rung contributes its own bracket and the answer is a sum over rungs.

WHERE THIS GOES WRONG

Treating $\sec^3$ as if it were a Wallis case and expecting the boundary to drop. It never does on a finite interval away from the origin, and on $[0, \pi/2]$ the integral diverges outright because $\sec$ blows up. Also: the even ladder terminates at $I_2 = \tan x$ and is elementary, so $\int_0^{\pi/4} \sec^4 = [\tan x + \frac{\tan^3 x}{3}]_0^{\pi/4} = \frac{4}{3}$ with no logarithm; only odd powers need I_1 .

SEE ALSO Cyclic integration by parts; reduction formulae by parts; tangent substitution.

The next two ladders are one-step rather than two-step, and they are the ones where the closed form is transparent enough to be quoted directly. They also make a good pair, because they are the same recursion read in two directions: $x^n e^x$ generates factorials by counting down, and $(\ln x)^n$ generates the same factorials with alternating signs.

EASY

21.10 The factorial ladder, $\int x^n e^x \, dx$

CUE x^n times e^{ax} , especially on $[0, \infty)$ where the whole thing collapses to a factorial, or where n is symbolic and a table is not available.

METHOD With $u = x^n$ and $dv = e^x \, dx$, one pass gives the one-step recursion

$$I_n = \int x^n e^x \, dx = x^n e^x - n I_{n-1}, \quad I_0 = e^x.$$

Unwinding it yields $I_n = e^x \sum_{k=0}^n (-1)^k \frac{n!}{(n-k)!} x^{n-k}$. On $[0, \infty)$ with e^{-x} instead, the boundary vanishes at both ends and the recursion is bare: $\int_0^\infty x^n e^{-x} \, dx = n \int_0^\infty x^{n-1} e^{-x} \, dx = n!$, which is the definition of $\Gamma(n+1)$ and the reason factorials appear in integral tables at all.

WORKED EXAMPLE

$$\int_0^\infty x^5 e^{-x} \, dx = 5! = 120,$$

each pass killing the boundary term because $x^k e^{-x} \rightarrow 0$ at infinity and $x^k \rightarrow 0$ at the origin for $k \geq 1$. On a finite interval the boundary survives at every rung: unwinding for $n = 3$ on $[0, 1]$ with e^{-x} gives $[-e^{-x}(x^3 + 3x^2 + 6x + 6)]_0^1 = 6 - \frac{16}{e} \approx 0.113929$, which is the tabular computation from earlier in this chapter, now derived as a recursion rather than read off a table. The two routes are the same computation; the ladder is the one that survives when n is a symbol.

WHERE THIS GOES WRONG

Forgetting the sign that e^{-x} contributes at every rung. With e^{+x} the recursion is $I_n = x^n e^x - n I_{n-1}$; with e^{-x} it is $I_n = -x^n e^{-x} + n I_{n-1}$, and the signs of the unwound sum are then all the same rather than alternating. Deriving the recursion afresh takes ten seconds and is safer than recalling which version you memorised.

SEE ALSO Tabular integration; the logarithm ladder; the Gamma function.

21.11 The logarithm ladder, $\int (\ln x)^n dx$

CUE A power of a logarithm, alone or against a power of x : $(\ln x)^n$, $x^m(\ln x)^n$, or $\ln^n(1/x)$ on $(0, 1)$, which is the same thing with the sign flipped.

METHOD Take $dv = dx$, the trick from earlier in this chapter, and $u = (\ln x)^n$. Since $\frac{d}{dx}(\ln x)^n = \frac{n(\ln x)^{n-1}}{x}$, the factor x from v cancels the $\frac{1}{x}$ exactly:

$$J_n = \int (\ln x)^n dx = x(\ln x)^n - nJ_{n-1}, \quad J_0 = x.$$

This is the factorial ladder again after $x = e^t$, which is why the answers are factorials. On $(0, 1)$ the boundary vanishes at both ends and

$$\int_0^1 (\ln x)^n dx = -n \int_0^1 (\ln x)^{n-1} dx = (-1)^n n!.$$

WORKED EXAMPLE

On $[1, e]$ the boundary is a clean e at every rung: $J_n(e) - J_n(1) = e - nJ_{n-1}$, so with $J_0 = e - 1$,

$$J_1 = 1, \quad J_2 = e - 2, \quad J_3 = e - 3(e - 2) = 6 - 2e \approx 0.563436.$$

On $(0, 1)$ instead, $\int_0^1 (\ln x)^4 dx = 4! = 24$, positive because the exponent is even; $\int_0^1 (\ln x)^3 dx = -6$. The check is worth doing once: substitute $x = e^{-t}$, and $\int_0^1 (\ln x)^n dx$ becomes $(-1)^n \int_0^\infty t^n e^{-t} dt = (-1)^n n!$, which is the factorial ladder in disguise.

WHERE THIS GOES WRONG

Assuming the bracket $[x(\ln x)^n]$ vanishes at $x = 0$ without argument. It does, because $x(\ln x)^n \rightarrow 0$ for every fixed n , but that is a statement about $t^n e^{-t}$ at infinity and deserves the substitution rather than a shrug. For $x^m(\ln x)^n$ with $m \leq -1$ it is false, and those integrals diverge.

SEE ALSO The factorial ladder; parts with $dv = dx$; the Gamma function.

The last ladder is the one that is not usually recognised as a ladder at all, because the repeated factor sits in a denominator. It matters out of proportion to its appearance, since $\int (1+x^2)^{-n} dx$ is what every tangent substitution eventually produces, and because its recursion is derived by the $dv = dx$ trick rather than by splitting off a copy of the repeated factor.

21.12 The rational ladder, $\int (1+x^2)^{-n} dx$

CUE A power of $1+x^2$ (or a^2+x^2) in a denominator with integer exponent $n \geq 2$. This is the standard residue of a tangent substitution, of a partial-fraction decomposition with a repeated irreducible quadratic factor, and of Cauchy-type kernels.

METHOD There is no factor to split off, so take $u = (1+x^2)^{-n}$ and $dv = dx$. Then $v = x$ and $du = -2nx(1+x^2)^{-n-1} dx$, so

$$I_n = \frac{x}{(1+x^2)^n} + 2n \int \frac{x^2}{(1+x^2)^{n+1}} dx.$$

Write $x^2 = (1+x^2) - 1$ to split the last integral into $I_n - I_{n+1}$, giving $I_{n+1} = \frac{1}{2n} \cdot \frac{x}{(1+x^2)^n} + \frac{2n-1}{2n} I_n$, with $I_1 = \arctan x$.

Note that the recursion runs *upwards*: parts on I_n produces I_{n+1} , and you read it backwards to descend.

WORKED EXAMPLE

On $[0, \infty)$ the boundary term vanishes at both ends for $n \geq 1$, so the recursion is purely numerical, $I_{n+1} = \frac{2n-1}{2n} I_n$ with $I_1 = \frac{\pi}{2}$:

$$\int_0^\infty \frac{dx}{(1+x^2)^2} = \frac{\pi}{4}, \quad \int_0^\infty \frac{dx}{(1+x^2)^3} = \frac{3\pi}{16}, \quad \int_0^\infty \frac{dx}{(1+x^2)^4} = \frac{5\pi}{32}.$$

Those are the Wallis numbers, and they should be: $x = \tan \theta$ turns $\int_0^\infty (1+x^2)^{-n} dx$ into $\int_0^{\pi/2} \cos^{2n-2} \theta d\theta$. On a finite interval the bracket survives:

$$\int_0^1 \frac{dx}{(1+x^2)^2} = \frac{1}{2} \left[\frac{x}{1+x^2} \right]_0^1 + \frac{1}{2} [\arctan x]_0^1 = \frac{1}{4} + \frac{\pi}{8} \approx 0.642699.$$

Reading the recursion in the wrong direction. Parts applied to I_n delivers a relation between I_n and I_{n+1} , so to compute I_5 you start at I_1 and climb, not the reverse. Solvers who expect a reduction formula to descend will write I_5 in terms of I_6 and never terminate.

SEE ALSO Wallis' formulae; reduction formulae by parts; tangent substitution.

The four ladders are worth seeing side by side, because what distinguishes them is not the algebra of the parts step but the base case and the fate of the boundary term, and those are the two things that get mixed up.

Family	Take $dv =$	Step	Base cases	Boundary vanishes on
$\sin^n x, \cos^n x$	$\sin x \, dx$	$n \rightarrow n-2$	$I_0 = \frac{\pi}{2}, I_1 = 1$	$[0, \frac{\pi}{2}]$
$\sec^n x$	$\sec^2 x \, dx$	$n \rightarrow n-2$	$I_1 = \ln \sec + \tan , I_2 = \tan$	nowhere useful
$x^n e^{-x}$	$e^{-x} \, dx$	$n \rightarrow n-1$	$I_0 = 1$	$[0, \infty)$
$(\ln x)^n$	dx	$n \rightarrow n-1$	$J_0 = 1$	$(0, 1)$
$(1+x^2)^{-n}$	dx	$n \rightarrow n+1$	$I_1 = \arctan x$	$[0, \infty)$

Read the fourth column as the answer to the question “which number will my answer be a multiple of”. A ladder ending at $I_0 = \frac{\pi}{2}$ produces π ; one ending at $I_1 = 1$ produces a rational number; the secant ladder ends at a logarithm and produces one. The fifth column is the answer to “how many boundary terms do I have to carry”. On the good intervals the answer is none, and the ladder is pure arithmetic; everywhere else the answer is one per rung, and the closed form is a sum.

Running it sideways

There is one further way to use the parts identity that is not a mode of iteration at all. Read $\int u \, dv + \int v \, du = [uv]$ as a statement relating the integral of a function to the integral of its inverse, and it becomes a tool for problems where the integrand is defined implicitly and cannot be written down.

HARD

21.13 Inverse-function integration

CUE The integrand is f^{-1} for a function f you can handle; or the problem defines g implicitly by $f(g(x)) = x$ and asks for $\int g$; or you are handed both f and its inverse and asked for a combination of their integrals.

METHOD Two equivalent routes. Substituting $x = f(t)$ turns $\int g(x) \, dx$ into $\int t f'(t) \, dt$, which is parts waiting to happen. Or quote Young's identity directly:

$$\int_a^b f^{-1}(x) \, dx = b f^{-1}(b) - a f^{-1}(a) - \int_{f^{-1}(a)}^{f^{-1}(b)} f(t) \, dt.$$

Geometrically this is the rectangle-minus-region picture: the area under f^{-1} over $[a, b]$ and the area under f over the corresponding t -interval together tile a difference of rectangles. Note that $\int \ln x \, dx$ and $\int \arcsin x \, dx$ are the special cases $f = e^x$ and $f = \sin$, so the technique of taking $dv = dx$ is Young's identity with the bookkeeping hidden.

WORKED EXAMPLE

Let g be the inverse of $f(t) = t2^t$ on $[0, 1]$, so $g : [0, 2] \rightarrow [0, 1]$ with $g(0) = 0$ and $g(2) = 1$ (from $t2^t = 2$, which forces $t = 1$: you must *solve*, not guess, and monotonicity of $t2^t$ guarantees uniqueness). Young's identity gives

$$\int_0^2 g(x) \, dx = 2 \cdot 1 - 0 \cdot 0 - \int_0^1 t 2^t \, dt.$$

The remaining integral is ordinary parts with $u = t$, $dv = 2^t \, dt$, $v = 2^t / \ln 2$:

$$\int_0^1 t 2^t \, dt = \left[\frac{t 2^t}{\ln 2} \right]_0^1 - \frac{1}{\ln 2} \int_0^1 2^t \, dt = \frac{2}{\ln 2} - \frac{1}{\ln^2 2}.$$

Hence

$$\int_0^2 g(x) \, dx = 2 - \frac{2}{\ln 2} + \frac{1}{\ln^2 2} \approx 1.195979,$$

and g was never written down, because it cannot be.

$\int_0^1 \arcsin x \, dx$ with $f = \sin$ on $[0, \pi/2]$: Young gives $1 \cdot \frac{\pi}{2} - 0 - \int_0^{\pi/2} \sin t \, dt = \frac{\pi}{2} - 1$, matching the computation by $dv = dx$ earlier. When both routes are available they agree, and the identity is the faster one when f is easy and f^{-1} is not.

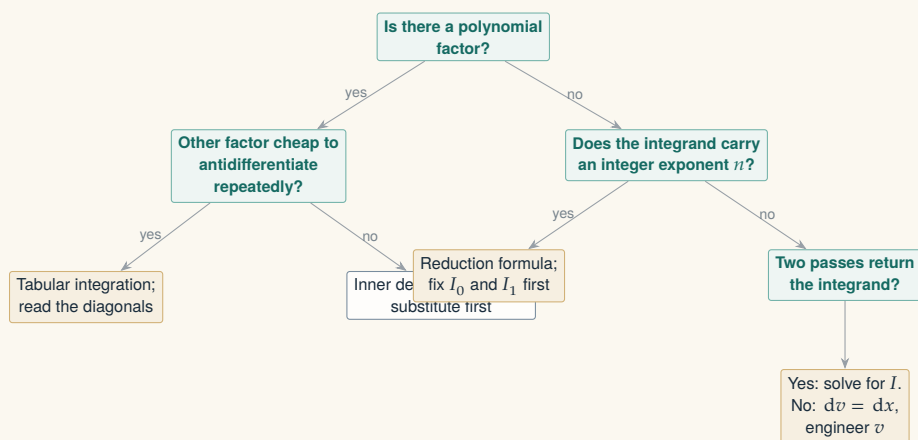
WHERE THIS GOES WRONG

Two hypotheses, both easy to skip. First, f must be monotone on the relevant range, or f^{-1} does not exist and the identity is meaningless. Second, the new limits are $f^{-1}(a)$ and $f^{-1}(b)$, and they must be obtained by solving an equation; guessing from the shape of f is how a correct method produces a wrong number.

SEE ALSO Parts with $dv = dx$; engineering the antiderivative v ; reciprocal substitution.

Choosing which machine

The three modes are distinguished by a single question asked after the first application: what does the new integral look like relative to the old one. Smaller means run forwards. Same means solve. Same family, smaller index, means recurse. The tree below routes on the shape of the integrand rather than on the outcome, so that you can make the decision before spending the first application.



Two cautions about the tree. The left branch is not the easy branch: a polynomial factor guarantees termination but says nothing about how many rungs, and a degree-six polynomial against e^{-x} is seven lines of table. The right branch is where the judgement lives, and the bottom right box hides a real decision, because the same integrand can often be attacked either as a cycle or as a ladder. $\int \sec^3$ is the canonical example: derived as a rung of the secant ladder it needs the identity $\tan^2 = \sec^2 - 1$, and derived as a cycle it needs the same identity at the same moment. They are the same computation seen from two sides, and it does not matter which name you give it.

The last thing to internalise is that parts is rarely the first move. It is the move for products that resist everything else, and a solver who reaches for it before simplifying, before checking for an inner derivative, and before looking for a reverse product rule will produce three correct and pointless pages. When the integrand is $e^x(f(x) + f'(x))$, the answer is $e^x f(x)$ and parts is a detour. When the integrand is $x^3 e^{x^2}$, the answer comes from $w = x^2$ and parts is at best a costume. Ask what the integrand was built from before asking how to take it apart.

Recognition matrix

If you see	Reach for	If that fails
$P(x)e^{ax}$, $P(x) \sin ax$, P a polynomial x^n against e^{-x} on $[0, \infty)$ $\ln x$, $\arctan x$, $\arcsin x$ alone $\ln(x+c)$ or $\ln(g(x))$ against dx $\frac{\ln x}{(x+a)^2}$	Tabular integration, $\deg P + 1$ rows The factorial ladder; answer is $n!$ $dv = dx$, so u is the whole integrand Take $v = x + c$, not $v = x$ Shift v to $\frac{x}{a(x+a)}$ so x cancels	Complexify: $e^{(a+ib)x}$ $\Gamma(n+1)$ and its shifts Young's identity for f^{-1} Substitute $w = x + c$ first Partial fractions after parts

If you see	Reach for	If that fails
$e^{ax} \sin bx, e^{ax} \cos bx$ $\sin(\ln x), \cos(\ln x)$ A cycle giving $I = I$ $\sin^n x, \cos^n x$ on $[0, \frac{\pi}{2}]$ $\sec^n x, n$ odd $(\ln x)^n$ on $(0, 1)$ $(1 + x^2)^{-n}$ $x^3 e^{x^2}, x^5 \sin(x^2)$ $e^x(f(x) + f'(x))$ f^{-1} or an implicitly defined g	Cyclic parts; divide by $a^2 + b^2$ $x = e^t$, then cyclic parts You switched u ; redo the second pass Wallis: $\frac{(n-1)!!}{n!!}$, times $\frac{\pi}{2}$ if n even Secant ladder, $dv = \sec^2 x \, dx$ Logarithm ladder; answer is $(-1)^n n!$ Reduction by parts, $dv = dx$ Substitute the inner function first Reverse product rule: answer $e^x f(x)$ Young's identity; solve for the new limits	Take real part of $e^{(a+ib)x}$ Cyclic parts directly, $dv = dx$ If consistent, parts cannot do it Beta function $\frac{1}{2} B(\frac{n+1}{2}, \frac{1}{2})$ $t = \sin x$, then partial fractions $x = e^{-t}$, then Γ $x = \tan \theta$, then Wallis on $\cos^{2n-2}$ Parts on the reduced integral Parts, which returns the same thing $x = f(t)$, then ordinary parts

Symmetry and transformation of definite integrals

What this chapter is for. A definite integral is not an antiderivative evaluated at two points. It is a number attached to a function and an interval, and the pairing carries information that the antiderivative alone does not have. Symmetry is how you extract that information, and the extraction is often complete: many of the integrals in this chapter have no elementary antiderivative and yet have a one-line exact value. The master move is always the same. Write the integral a second time under a substitution that maps the interval to itself, then add the two expressions or subtract them. If the substitution sends the integrand to its own negative, the sum is zero. If it sends the integrand to a constant minus itself, the sum is a rectangle. If it merely permutes the pieces, you have halved the work. This chapter works through the standard involutions in order of how often they pay: parity on $[-a, a]$, reflection $x \mapsto a + b - x$ on any interval, the sine and cosine exchange on $[0, \frac{\pi}{2}]$, translation by a period, splitting at a sign change, and $x \mapsto 1/x$ on $(0, \infty)$ with its powerful specialisation, Glasser's master theorem. Running through all of them is one structural fact worth stating at the outset: when a symmetry argument closes, the answer very often does not depend on a parameter that looked essential to the problem, and noticing that in advance is a reliable way to tell that a symmetry argument is the intended route.

The fold

Every technique in this chapter is an instance of one manoeuvre. Let $\varphi : [a, b] \rightarrow [a, b]$ be a substitution mapping the interval onto itself. Then

$$I = \int_a^b f(x) \, dx = \int_a^b f(\varphi(x)) |\varphi'(x)| \, dx,$$

and adding the two expressions gives $2I = \int_a^b [f(x) + f(\varphi(x)) |\varphi'(x)|] \, dx$. Call this the *fold*. It is worth nothing at all unless the bracket is simpler than f , and the entire skill of this chapter is recognising in advance when it will be. There are only a few φ worth trying, because there are only a few natural involutions of an interval: $x \mapsto -x$ and $x \mapsto a + b - x$ on a finite interval, $x \mapsto x + T$ on a periodic domain, and $x \mapsto 1/x$ or $x \mapsto a/x$ on $(0, \infty)$.

Three outcomes are worth naming. If the bracket is identically zero, $I = 0$ and you have finished before starting. If the bracket is a constant c , then $I = \frac{c(b-a)}{2}$, a rectangle, again with no antiderivative anywhere in sight. If the bracket is merely a nicer function than f , you have traded an integral for an easier one, which is the ordinary case: King's rule usually lands here, converting $x K(x)$ into $\frac{a+b}{2} K(x)$.

THE PARAMETER THAT DOES NOT MATTER

When a fold closes, the resulting answer is a functional of the *symmetry*, not of the integrand. That is why $\int_0^{\pi/2} \frac{dx}{1 + \tan^p x} = \frac{\pi}{4}$ for every p , why $\int_0^\infty \frac{dx}{(1+x^2)(1+x^a)} = \frac{\pi}{4}$ for every real a , and why $\int_0^\infty \frac{x^2 dx}{x^4 - (2c-1)x^2 + c^2} = \frac{\pi}{2}$ for every $c > 0$. Turn the observation around and use it diagnostically: if a problem carries a parameter that appears nowhere in the requested answer, or an exponent chosen to look deliberately absurd, a fold is being asked for. Conversely, if you produce an answer that still contains the decorative parameter, you have not used the symmetry.

The candidate maps are few enough to tabulate, and the table is the working checklist for the rest of the chapter. Column three is the only one that requires thought.

Domain	Map φ	$ \varphi' $	Test the fold on
$[-a, a]$	$x \mapsto -x$	1	$f(x) + f(-x)$: parity
$[a, b]$	$x \mapsto a + b - x$	1	$f(x) + f(a + b - x)$: constant?
$[0, \frac{\pi}{2}]$	$x \mapsto \frac{\pi}{2} - x$	1	$\sin \leftrightarrow \cos, \tan \leftrightarrow \cot$
Period T	$x \mapsto x + \frac{T}{2}$	1	$f(x) + f(x + \frac{T}{2})$
$(0, \infty)$	$x \mapsto \frac{1}{x}$	$\frac{1}{x^2}$	$f(x) + \frac{1}{x^2}f(\frac{1}{x})$
$(0, \infty)$	$x \mapsto \frac{a}{x}$	$\frac{a}{x^2}$	Glasser: $u = x - \frac{a}{x}$
$(0, \infty)$	$x \mapsto \lambda x$	λ	Scale invariance of $\frac{dx}{x}$

The last row is not an involution, and it is included because scaling is the one transformation of the half-line that is not self-inverse yet still produces exact answers, by generating a one-parameter family from a single known value rather than by folding.

Parity, and how to expose it

Parity is the cheapest symmetry and the first thing to test. It is also the one most often present in disguise, because a problem designed around parity will rarely hand you the interval $[-a, a]$ centred where you need it, and will rarely present the integrand already factored into pieces of definite parity.

22.1 Even and odd parity on a symmetric interval

CORE

CUE Bounds $[-a, a]$, and an integrand that factors into pieces each of which is visibly even or visibly odd. The characteristic decoration is a high odd power multiplied by an even weight such as $\sqrt{a^2 - x^2}$ or e^{-x^2} , put there to make the integral look laborious.

METHOD Classify every factor as even, odd, or neither. The product of two odds is even, odd times even is odd. Then

$$\int_{-a}^a f_{\text{odd}}(x) dx = 0, \quad \int_{-a}^a f_{\text{even}}(x) dx = 2 \int_0^a f_{\text{even}}(x) dx.$$

Discard the odd part before computing anything. When the integrand is a sum, split it first: the two halves have independent parities and only the even half survives.

WORKED EXAMPLE

$$\int_{-2}^2 \left(x^3 \cos \frac{x}{2} + \frac{1}{2} \right) \sqrt{4 - x^2} dx.$$

The weight $\sqrt{4 - x^2}$ is even. In the bracket, $x^3 \cos \frac{x}{2}$ is odd times even, hence odd, so its product with the weight is odd and contributes nothing. Only the constant $\frac{1}{2}$ survives:

$$\frac{1}{2} \int_{-2}^2 \sqrt{4 - x^2} dx = \frac{1}{2} \cdot \frac{\pi \cdot 2^2}{2} = \pi.$$

The remaining integral is the area of a semicircle of radius 2, read off geometrically rather than computed. Total: $\pi \approx 3.141593$. Every symbol in $x^3 \cos \frac{x}{2}$ was decoration.

WHERE THIS GOES WRONG

The odd part must be *separately integrable* for the cancellation to be legitimate. If it has a non-integrable singularity at an interior point, the two halves are $+\infty$ and $-\infty$ and their difference is a principal value, not zero: $\int_{-1}^1 \frac{dx}{x}$ does not equal 0, it does not exist. Separately, watch the functions that break naive parity bookkeeping: $|x|$ is even but not smooth, $x^{2/3}$ is even while $x^{1/3}$ is odd, and a real cube root implemented as $\exp(\frac{1}{3} \ln x)$ is not defined for $x < 0$ at all.

THE PARITY OF THE BUILDING BLOCKS

Worth having by heart, because parity of a product is decided factor by factor and a single misclassification flips the answer. Even: $\cos$, $\cosh$, sech , $|x|$, x^{2k} , e^{-x^2} , $\ln|x|$, and any $g(x^2)$. Odd: $\sin$, $\sinh$, $\tan$, $\arctan$, arsinh , sgn , x^{2k+1} , and $\ln(\frac{1+x}{1-x})$. Neither, and therefore the factors that destroy an otherwise clean argument: e^x , $\frac{1}{1+e^x}$, $\arccos$, and every polynomial with both even and odd terms. Note that e^x is not merely “not odd”; it is the standard way a problem is disguised, since $e^x = \cosh x + \sinh x$ splits into an even and an odd part, and only one of them survives the integration.

SEE ALSO Recentring to expose parity; disguised zero; splitting at a sign change; Cauchy principal value.

Parity as stated needs the interval centred at the origin, and problems are rarely so obliging. But the fold does not care where the centre is: any interval $[a, b]$ has a midpoint, and translating it to the origin costs one substitution. Making that translation a reflex is what turns parity from an occasional gift into a routine test.

22.2 Recentring to expose parity

EASY

CUE An interval $[a, b]$ whose midpoint $m = \frac{a+b}{2}$ appears somewhere in the integrand, or an integrand built out of $x - m$, or a trigonometric integrand on $[0, \pi]$ where the natural centre is $\pi/2$.

METHOD Substitute $t = x - m$, which sends $[a, b]$ to $[-h, h]$ with $h = \frac{b-a}{2}$, and then test parity in t . The useful trigonometric instance is $t = x - \frac{\pi}{2}$ on $[0, \pi]$, under which $\sin x = \cos t$ (even) and $\cos x = -\sin t$ (odd), so every trigonometric integrand on $[0, \pi]$ has a definite parity about the midpoint as soon as you write it in t .

WORKED EXAMPLE

$$\int_0^2 \ln\left(\frac{x}{2-x}\right) dx.$$

The midpoint is 1; put $t = x - 1$, so the integrand becomes $\ln \frac{1+t}{1-t}$ on $[-1, 1]$. That is odd in t , and it is integrable at both endpoints because $\ln$ has an integrable singularity. Hence the integral is 0. The two endpoint singularities are genuine but harmless, and the answer required no antiderivative.

A second, purely trigonometric instance: $\int_0^\pi \sin^3 x \cos x dx$. With $t = x - \frac{\pi}{2}$ the integrand is $\cos^3 t \cdot (-\sin t)$, odd, so the value is 0. Confirmed by the antiderivative $\frac{1}{4} \sin^4 x$, which returns to 0 at $x = \pi$.

WHERE THIS GOES WRONG

Recentring and then testing parity in the *old* variable. After $t = x - 1$ the question is whether $f(1+t)$ is odd in t , which is the statement $f(1+t) = -f(1-t)$, not $f(-x) = -f(x)$. Write the substituted integrand out in full before judging it; the shape of the original is not evidence.

SEE ALSO Even and odd parity on a symmetric interval; King's rule; splitting at a sign change.

Reflection

Parity is the special case of reflection where the fixed constant happens to be zero. The general reflection $x \mapsto a + b - x$ maps $[a, b]$ to itself for every a and b , has Jacobian of absolute value 1, and requires nothing of the integrand at all. It is therefore always available, which makes it the single most useful move in this chapter and also the one most often applied without checking that it helps.

22.3 King's rule, $x \mapsto a + b - x$

CORE

CUE A definite integral on $[a, b]$ carrying a stray linear factor x against a kernel that looks reflection-invariant, or a quotient whose numerator and denominator swap under the reflection. Most often the interval is $[0, \pi]$, $[0, 1]$, or $[0, \frac{\pi}{2}]$.

METHOD The identity is free:

$$\int_a^b f(x) dx = \int_a^b f(a+b-x) dx.$$

Add the two forms. The productive special case: if the kernel satisfies $K(a+b-x) = K(x)$, then

$$\int_a^b xK(x) dx = \frac{a+b}{2} \int_a^b K(x) dx,$$

because $x + (a+b-x) = a+b$ is constant. The linear weight is replaced by the midpoint. This is the whole content of King's rule in its commonest use: it removes an x that had no business being there.

WORKED EXAMPLE

$$I = \int_0^\pi \frac{x \sin x}{1 + \cos^2 x} dx.$$

The kernel is $K(x) = \frac{\sin x}{1 + \cos^2 x}$. Check invariance: $\sin(\pi - x) = \sin x$ and $\cos^2(\pi - x) = \cos^2 x$, so $K(\pi - x) = K(x)$ and the rule applies with $a + b = \pi$:

$$I = \frac{\pi}{2} \int_0^\pi \frac{\sin x}{1 + \cos^2 x} dx = \frac{\pi}{2} [-\arctan(\cos x)]_0^\pi = \frac{\pi}{2} \left(\frac{\pi}{4} + \frac{\pi}{4} \right) = \frac{\pi^2}{4} \approx 2.467401.$$

Without the rule you would be integrating $x \sin x / (1 + \cos^2 x)$ by parts and meeting a dilogarithm.

THE SAME MOVE ON A RATIONAL INTEGRAND

$$\int_0^\pi \frac{x}{1 + \sin x} dx.$$

Again $\sin(\pi - x) = \sin x$, so the kernel is invariant and $I = \frac{\pi}{2} \int_0^\pi \frac{dx}{1 + \sin x}$. The remaining integral yields to $t = \tan \frac{x}{2}$: it becomes $\int_0^\infty \frac{2 dt}{(1+t)^2} = 2$. Hence $I = \pi \approx 3.141593$.

WHERE THIS GOES WRONG

Assuming the kernel is invariant without testing it on the actual denominator. $\cos^2 \frac{x}{2}$ is *not* invariant under $x \mapsto \pi - x$: it becomes $\sin^2 \frac{x}{2}$. Reflecting anyway produces a clean-looking wrong answer, which is worse than a mess. Substitute $\pi - x$ into the denominator symbol by symbol and simplify before you commit.

SEE ALSO Constant-sum reflection; complementary sine and cosine; mirror symmetry about an interior point; recentring to expose parity.

King's rule as stated needs an invariant kernel and a linear weight. The richer family is the one where the integrand and its reflection are not equal but *sum to a constant*. Then the fold gives the answer outright, with no residual integral of any kind, and this is where the parameter-independence phenomenon is most spectacular.

CORE

22.4 Constant-sum reflection, $f(x) + f(a + b - x) = c$

CUE A quotient $\frac{g(x)}{g(x) + \tilde{g}(x)}$ where the reflection swaps g and $\tilde{g}$; a denominator of the form $1 + \varphi(x)$ where $\varphi(x)\varphi(a + b - x) = 1$; the Fermi weight $\frac{1}{1 + e^x}$ or $\frac{1}{1 + b^x}$ on an interval symmetric about the point where the exponent vanishes.

METHOD If $f(x) + f(a + b - x) = c$ identically on $[a, b]$, then folding gives $2I = c(b - a)$, so

$$\int_a^b f(x) dx = \frac{c(b - a)}{2}.$$

No antiderivative is needed and none may exist. The Fermi instance is worth memorising in its own right: since $\frac{1}{1 + e^x} + \frac{1}{1 + e^{-x}} = 1$,

$$\int_{-a}^a \frac{g(x)}{1 + e^x} dx = \int_0^a g(x) dx \quad \text{for even } g,$$

which converts a two-sided integral with an ugly weight into a one-sided integral with no weight at all.

WORKED EXAMPLE

$$\int_1^3 \frac{\sqrt{x}}{\sqrt{x} + \sqrt{4 - x}} dx.$$

Here $a + b = 4$, and reflection sends $\sqrt{x} \mapsto \sqrt{4 - x}$, so $f(x) + f(4 - x) = 1$ identically. Hence $I = \frac{1 \cdot (3 - 1)}{2} = 1$ exactly. The integrand is not elementary in any useful sense; the answer is a rectangle.

The Fermi case, with a parameter that vanishes:

$$\int_{-1}^1 \frac{x^2}{1 + e^x} dx = \int_0^1 x^2 dx = \frac{1}{3}.$$

And with a general base, $\int_3^7 \frac{dx}{1 + 2^{x-5}}$: the interval is symmetric about 5, where the exponent vanishes, and $\frac{1}{1 + 2^t} + \frac{1}{1 + 2^{-t}} = 1$, so the value is $\frac{7-3}{2} = 2$. The base 2 never appears in the answer.

$$\int_0^1 \frac{\ln(1+x)}{1+x^2} dx.$$

No reflection of $[0, 1]$ helps directly. Substitute $x = \tan \theta$ first, which turns the integral into $\int_0^{\pi/4} \ln(1 + \tan \theta) d\theta$ and puts you on an interval whose reflection is $\theta \mapsto \frac{\pi}{4} - \theta$. Under that reflection,

$$1 + \tan\left(\frac{\pi}{4} - \theta\right) = 1 + \frac{1 - \tan \theta}{1 + \tan \theta} = \frac{2}{1 + \tan \theta},$$

so the two integrands sum to $\ln(1 + \tan \theta) + \ln 2 - \ln(1 + \tan \theta) = \ln 2$, a constant. Hence

$$2I = \frac{\pi}{4} \ln 2, \quad I = \frac{\pi \ln 2}{8} \approx 0.272198.$$

The lesson is that the constant-sum structure is often one substitution away rather than absent: choose the substitution that makes the interval reflect into itself *and* makes the reflection act simply on the integrand.

WHERE THIS GOES WRONG

The identity must hold *identically* on the interval, not at a convenient sample point. Verifying $f(x) + f(a + b - x) = c$ at the midpoint is no verification at all, since the midpoint is a fixed point of the reflection and the identity there reduces to $2f(m) = c$, which merely defines c . Substitute the symbol $a + b - x$ and simplify.

SEE ALSO King's rule; complementary sine and cosine; disguised zero.

On $[0, \frac{\pi}{2}]$ the reflection $x \mapsto \frac{\pi}{2} - x$ has a name of its own because of what it does to the trigonometric functions rather than to the interval: it exchanges sine and cosine. That makes an entire class of symmetrically built integrands satisfy the constant-sum hypothesis automatically.

MEDIUM

22.5 Complementary sine and cosine on $[0, \frac{\pi}{2}]$

CUE An integrand on $[0, \frac{\pi}{2}]$ built symmetrically from $\sin x$ and $\cos x$: $\frac{\sin^p x}{\sin^p x + \cos^p x}$, $\frac{1}{1 + \tan^p x}$, $\ln \sin x$ played against $\ln \cos x$, or any $\frac{h(\sin x)}{h(\sin x) + h(\cos x)}$.

METHOD Under $x \mapsto \frac{\pi}{2} - x$ the pair $(\sin, \cos)$ is exchanged and $\tan \mapsto \cot$. Adding the original to its reflection frequently produces an integrand identically equal to 1, whereupon $I = \frac{\pi}{4}$. For the logarithmic pair, the reflection shows that $\int_0^{\pi/2} \ln \sin x dx$ and $\int_0^{\pi/2} \ln \cos x dx$ are equal, and their sum is computed from $\sin x \cos x = \frac{1}{2} \sin 2x$.

WORKED EXAMPLE

$$I = \int_0^{\pi/2} \frac{dx}{1 + \tan^{\sqrt{2}} x}.$$

Reflecting, $\tan^{\sqrt{2}} \mapsto \cot^{\sqrt{2}}$, and

$$\frac{1}{1 + \tan^p x} + \frac{1}{1 + \cot^p x} = \frac{1}{1 + \tan^p x} + \frac{\tan^p x}{\tan^p x + 1} = 1.$$

Hence $2I = \frac{\pi}{2}$ and $I = \frac{\pi}{4} \approx 0.785398$, independent of the exponent. The same computation gives

$$\int_0^{\pi/2} \frac{\sin^{3/2} x}{\sin^{3/2} x + \cos^{3/2} x} dx = \frac{\pi}{4}.$$

THE LOG-SINE INTEGRAL

Let $L = \int_0^{\pi/2} \ln \sin x dx$. Reflection gives $L = \int_0^{\pi/2} \ln \cos x dx$, so

$$2L = \int_0^{\pi/2} \ln(\sin x \cos x) dx = \int_0^{\pi/2} \ln \frac{\sin 2x}{2} dx = -\frac{\pi}{2} \ln 2 + \frac{1}{2} \int_0^{\pi} \ln \sin u du,$$

using $u = 2x$. The last integral is $2L$ by symmetry of $\sin$ about $\frac{\pi}{2}$, so $2L = -\frac{\pi}{2} \ln 2 + L$, giving

$$L = -\frac{\pi}{2} \ln 2 \approx -1.088793.$$

Two folds, one reflection and one doubling, and a self-referential equation exactly like a cyclic integration by parts.

The log-sine computation used a reflection and a doubling in succession, and that pairing is worth naming, because a single fold rarely closes on logarithmic integrands. The immediate dividend: $\int_0^\pi \ln \sin x \, dx = 2 \int_0^{\pi/2} \ln \sin x \, dx = -\pi \ln 2 \approx -2.177586$, by mirror symmetry about $\frac{\pi}{2}$ applied to the value just computed. The same two folds evaluate $\int_0^{\pi/2} \ln \tan x \, dx$ instantly: it is $\int \ln \sin - \int \ln \cos = 0$, since the reflection exchanges the two.

WHERE THIS GOES WRONG

Pattern-matching on the appearance of symmetry rather than on the identity. A mismatched pair such as $\frac{\sin^{\cos x} x}{\sin^{\cos x} x + \cos^{\sin x} x}$ still integrates to $\frac{\pi}{4}$, but only because the reflection genuinely sends the integrand f to $1 - f$; and an integrand that merely *looks* symmetric may not. Verify $f(x) + f(\frac{\pi}{2} - x) = 1$ by substitution, every time.

SEE ALSO Constant-sum reflection; King's rule; mirror symmetry about an interior point.

There is one more reflection worth separating out. When the symmetry point is interior to the interval and the integrand is invariant rather than anti-invariant, the fold does not give the answer, it halves the domain. That is a smaller gain than a rectangle but it is available far more often, and it is what makes $[0, \pi]$ and $[0, 2\pi]$ integrals of $\sin$ tractable.

EASY

22.6 Mirror symmetry about an interior point

CUE An integrand on $[0, \pi]$ depending on x only through $\sin x$, or on $[0, 2\pi]$ through $\cos x$, or any f with $f(a+b-x) = f(x)$ where the half-interval is easier.

METHOD If $f(a+b-x) = f(x)$ then the two halves of $[a, b]$ contribute equally:

$$\int_a^b f(x) \, dx = 2 \int_a^{(a+b)/2} f(x) \, dx.$$

For $[0, \pi]$ and an integrand $f(\sin x)$ this reads $\int_0^\pi f(\sin x) \, dx = 2 \int_0^{\pi/2} f(\sin x) \, dx$, which puts you on the interval where the sine and cosine exchange is available and where Wallis' formulae live.

WORKED EXAMPLE

$$\int_0^\pi \frac{dx}{1 + \sin^2 x} = 2 \int_0^{\pi/2} \frac{dx}{1 + \sin^2 x} = 2 \cdot \frac{\pi}{2\sqrt{2}} = \frac{\pi}{\sqrt{2}} \approx 2.221441.$$

The half-interval value comes from $t = \tan x$, under which $\frac{dx}{1 + \sin^2 x} = \frac{dt}{1 + 2t^2}$ and the integral over $t \in (0, \infty)$ is $\frac{\pi}{2\sqrt{2}}$. Doing this on $[0, \pi]$ directly is unpleasant, because $\tan$ has a pole at the midpoint; the mirror argument moves the pole to an endpoint, where it is harmless.

WHERE THIS GOES WRONG

Applying the halving when the symmetry is about the wrong point. On $[0, 2\pi]$ an integrand in $\sin x$ is symmetric about $\frac{\pi}{2}$ and about $\frac{3\pi}{2}$, not about π ; the correct statement is $\int_0^{2\pi} f(\sin x) \, dx = 2 \int_{-\pi/2}^{\pi/2} f(\sin x) \, dx$ after a shift, and blindly writing $2 \int_0^\pi$ is right for $f(\sin x)$ only by the accident that $\int_\pi^{2\pi} f(\sin x) \, dx = \int_0^\pi f(-\sin x) \, dx$, which differs unless f is even.

SEE ALSO King's rule; periodicity and window shifting; Wallis' formulae.

Translation

Reflection is an involution; translation is not, but on a periodic domain it is still a symmetry of the interval, and it generates two distinct manoeuvres. The first is bookkeeping: an integral over many periods is a multiple of an integral over one, and the window may be slid anywhere. The second is a genuine fold: shifting by half a period often sends the integrand to a partner that sums with it to something trivial.

22.7 Periodicity and window shifting

CUE A periodic integrand over an interval that is a whole number of periods, or over a window whose placement is inconvenient, or an absolute value of a periodic function.

METHOD For f of period T ,

$$\int_a^{a+T} f(x) dx = \int_0^T f(x) dx \quad \text{for every } a, \quad \int_0^{nT} f(x) dx = n \int_0^T f(x) dx.$$

Use the first to slide a window off a singularity of an intended substitution (the Weierstrass substitution $t = \tan \frac{x}{2}$ has a pole at $x = \pi$, and sliding the window is how you avoid it). Use the second to reduce n periods to one. And use the half-period shift $x \mapsto x + \frac{T}{2}$ as a fold when the integrand has a sign symmetry.

WORKED EXAMPLE

$$\int_0^{10\pi} |\sin x| dx = 10 \int_0^\pi \sin x dx = 20,$$

since $|\sin x|$ has period π , not 2π . A harder instance, using the half-period shift as a fold:

$$I = \int_0^{2\pi} \frac{dx}{1 + e^{\sin x}}.$$

The integrand has period 2π , so $I = \int_0^{2\pi} \frac{dx}{1 + e^{\sin(x+\pi)}} = \int_0^{2\pi} \frac{dx}{1 + e^{-\sin x}}$. Adding, the two integrands sum to 1 identically, so $2I = 2\pi$ and $I = \pi \approx 3.141593$. The translation supplied the constant-sum hypothesis that reflection could not.

SLIDING A WINDOW OFF A POLE

The freedom to place the window is not cosmetic. The Weierstrass substitution $t = \tan \frac{x}{2}$ sends $x = \pi$ to $t = \infty$, so applying it to $\int_0^{2\pi} \frac{dx}{2 + \cos x}$ as written puts a pole of the substitution strictly inside the interval, and the naive computation returns 0. Slide the window first: the integrand has period 2π , so the integral equals $\int_{-\pi}^\pi \frac{dx}{2 + \cos x}$, where the pole is now at both endpoints and the substitution is a genuine bijection onto $\mathbb{R}$. It gives $\int_{-\infty}^\infty \frac{2 dt}{3 + t^2} = \frac{2\pi}{\sqrt{3}} \approx 3.627599$. One line of translation converted a wrong answer into a right one.

WHERE THIS GOES WRONG

Two period errors dominate. The interval must be an exact whole number of periods, or the leftover fragment must be integrated separately: $\int_0^7 |\sin x| dx$ is $2[7/\pi]$ plus a genuine fragment, not $\frac{7}{\pi} \cdot 2$. And the period of a transformed function is not the period of the original: $|\sin x|$ and $\sin^2 x$ both have period π , and using 2π doubles the answer.

SEE ALSO Mirror symmetry about an interior point; splitting at a sign change; Lobachevsky's formula; Weierstrass substitution.

Absolute values arrived in that last example without comment, and they deserve their own treatment, because they are the standard way a problem forces you to break the interval before any symmetry argument can start. Splitting is the least glamorous technique in this chapter and the one whose omission causes the most wrong answers.

MEDIUM

22.8 Splitting at a symmetry or sign-change point

CUE An absolute value, a max or min of two or more functions, a nested radical such as $\sqrt{x + \sqrt{1 - 2x}}$, a floor or fractional part, or any piecewise definition.

METHOD Locate *every* point in $[a, b]$ where the inner expression changes sign or the active branch changes. Split the integral there, resolve the absolute value with the correct sign on each piece, and then treat each piece independently: each will usually admit its own substitution. For max and min of several functions, the crossing points are the solutions of all the pairwise equations, and you must then determine which function is active on each subinterval.

WORKED EXAMPLE

$$\int_0^\pi |\cos x| dx = \int_0^{\pi/2} \cos x dx - \int_{\pi/2}^\pi \cos x dx = 1 + 1 = 2.$$

A case where the split point is not the midpoint:

$$\int_0^3 |x^2 - 2x| dx = \int_0^2 (2x - x^2) dx + \int_2^3 (x^2 - 2x) dx = \frac{4}{3} + \frac{4}{3} = \frac{8}{3},$$

the two pieces having lengths 2 and 1 and, by coincidence, equal values. A case where the split is disguised by an identity:

$$\int_0^{2\pi} \sqrt{1 - \cos 2x} dx = \int_0^{2\pi} \sqrt{2 \sin^2 x} dx = \sqrt{2} \int_0^{2\pi} |\sin x| dx = \sqrt{2} \cdot 4 = 4\sqrt{2} \approx 5.656854.$$

The step that everything depends on is $\sqrt{\sin^2 x} = |\sin x|$ and not $\sin x$. Writing $\sin x$ gives 0, which is not merely wrong but wrong in a way that looks like a symmetry argument succeeding.

A MINIMUM, SPLIT THEN FOLDED

$$\int_0^{\pi/2} \min(\sin x, \cos x) dx.$$

The two curves cross where $\tan x = 1$, that is at $x = \frac{\pi}{4}$, and $\sin$ is the smaller one to the left of it. So

$$\int_0^{\pi/4} \sin x dx + \int_{\pi/4}^{\pi/2} \cos x dx = \left(1 - \frac{\sqrt{2}}{2}\right) + \left(1 - \frac{\sqrt{2}}{2}\right) = 2 - \sqrt{2} \approx 0.585786.$$

The two pieces are equal, and not by accident: the reflection $x \mapsto \frac{\pi}{2} - x$ exchanges $\sin$ and $\cos$, hence fixes the function $\min(\sin, \cos)$, so the interval folds about $\frac{\pi}{4}$ and one piece would have sufficed. Splitting and folding are complementary here: the split identifies the branches, the fold halves the work.

WHERE THIS GOES WRONG

Missing a crossing. For $\min(\sin x, \cos x, \tan x)$ on $[\frac{3\pi}{4}, \frac{5\pi}{4}]$ there are three pairwise equations to solve, not one, and the active branch can change without any two of the curves meeting at the obvious place. Never assume the split point is the midpoint: for $|x^2 - 2x|$ on $[0, 3]$ it is $x = 2$, and the two pieces have quite different lengths.

SEE ALSO Even and odd parity on a symmetric interval; periodicity and window shifting; absolute value and the modulus in the logarithm.

Inversion

The half-line $(0, \infty)$ has its own involution, $x \mapsto 1/x$, and it is exactly as fundamental there as $x \mapsto -x$ is on $\mathbb{R}$. Indeed it is the same involution: the substitution $x = e^t$ identifies $(0, \infty)$ with $\mathbb{R}$ and carries $x \mapsto 1/x$ to $t \mapsto -t$. Everything in this section is a consequence of that identification, and moving back and forth between the multiplicative and additive pictures is the main skill.

HARD

22.9 Inversion, $x \mapsto 1/x$ on $(0, \infty)$

CUE An integral over $(0, \infty)$ whose integrand is built from a self-reciprocal denominator such as $1 + x^2$ or $x + \frac{1}{x}$, weighted by a power x^s or by $\ln^k x$.

METHOD Under $x = 1/u$ the interval maps to itself with $dx = -\frac{du}{u^2}$, the sign absorbed by the reversal of limits. Compute I in the new variable and compare with the original: either the two expressions add, halving the work, or they are negatives of each other, forcing $I = 0$. The key transformation rules are $\ln(1/u) = -\ln u$ and $1 + (1/u)^2 = \frac{1 + u^2}{u^2}$, the latter being why $\frac{dx}{1 + x^2}$ is invariant.

WORKED EXAMPLE

$$I = \int_0^\infty \frac{dx}{(1 + x^2)(1 + x^a)}.$$

Put $x = 1/u$: $\frac{dx}{1+x^2}$ is invariant, and $\frac{1}{1+u^{-a}} = \frac{u^a}{1+u^a}$. So

$$I = \int_0^\infty \frac{u^a}{(1+u^2)(1+u^a)} du.$$

Adding this to the original, the two numerators are 1 and u^a over the same denominator, so they combine to 1:

$$2I = \int_0^\infty \frac{du}{1+u^2} = \frac{\pi}{2}, \quad I = \frac{\pi}{4} \approx 0.785398,$$

for every real a , positive or negative. This is the constant-sum fold transported to the multiplicative group.

THE VANISHING CASE, AND A SCALING COMPANION

$\int_0^\infty \frac{\ln x}{1+x^2} dx$: inversion sends $\ln x \mapsto -\ln u$ and leaves $\frac{dx}{1+x^2}$ alone, so $I = -I$ and $I = 0$.

Combine that with a scaling to get a family: in $\int_0^\infty \frac{\ln x}{x^2+a^2} dx$ put $x = at$, giving $\frac{1}{a} \int_0^\infty \frac{\ln a + \ln t}{1+t^2} dt$. The $\ln t$ piece vanishes by the inversion just performed, and the $\ln a$ piece is $\ln a \cdot \frac{\pi}{2}$, so

$$\int_0^\infty \frac{\ln x}{x^2+a^2} dx = \frac{\pi \ln a}{2a}.$$

At $a = 3$ this is $\frac{\pi \ln 3}{6} \approx 0.575232$. Scaling and inversion together generate the whole family from one vanishing integral.

WHERE THIS GOES WRONG

The fold is legitimate only when both halves converge separately. Applying it to $\int_0^\infty \frac{dx}{x}$ “proves” that a divergent integral equals itself, which is true and worthless; applying it to a conditionally convergent integral and then splitting the sum can produce a finite wrong number. Check convergence at 0 and at ∞ before folding.

SEE ALSO Exponential substitution and multiplicative parity; scaling on the half line; Glasser’s master theorem; disguised zero.

The scaling that appeared at the end of that example deserves separating out, because it is a different kind of tool. Inversion is an involution and produces a single number; scaling is a one-parameter group and produces a whole family of numbers from one. The two are almost always used together, and the reason is structural: on $(0, \infty)$ the natural measure is $\frac{dx}{x}$, which is invariant under scaling, and the natural symmetry of that measure is inversion.

MEDIUM

22.10 Scaling, $x \mapsto \lambda x$ on $(0, \infty)$

CUE A half-line integral carrying a positive parameter inside the integrand: $a^2 + x^2$, e^{-ax} , $1 + x^n$ with a power weight, or a logarithm of a scaled argument.

METHOD Substitute $x = \lambda t$ and pull the parameter out. Two consequences do most of the work. First, the measure $\frac{dx}{x}$ is scale-invariant, so an integrand of the form $g(x) \frac{dx}{x}$ gives $\int_0^\infty g(\lambda x) \frac{dx}{x} = \int_0^\infty g(x) \frac{dx}{x}$ for every $\lambda > 0$: the parameter cannot appear in the answer. Second, $\ln(\lambda t) = \ln \lambda + \ln t$, so scaling converts a logarithm into a constant plus a logarithm, and the constant comes out of the integral while the $\ln t$ piece is frequently killed by inversion.

WORKED EXAMPLE

$$\int_0^\infty \frac{\ln(ax)}{x^2+b^2} dx.$$

Put $x = bt$: the integral becomes $\frac{1}{b} \int_0^\infty \frac{\ln(ab) + \ln t}{1+t^2} dt$. The $\ln t$ piece vanishes by inversion, and $\int_0^\infty \frac{dt}{1+t^2} = \frac{\pi}{2}$, so

$$\int_0^\infty \frac{\ln(ax)}{x^2+b^2} dx = \frac{\pi \ln(ab)}{2b}.$$

At $a = 2$, $b = 3$ this is $\frac{\pi \ln 6}{6} \approx 0.938163$. Notice that the answer depends on a and b only through the product ab , which is exactly the invariance the scaling predicted.

$$\int_0^\infty \frac{x^{s-1}}{1+x^n} dx.$$

Write this as $\int_0^\infty \frac{x^s}{1+x^n} \cdot \frac{dx}{x}$ and substitute $x = t^{1/n}$, which turns it into $\frac{1}{n} \int_0^\infty \frac{t^{s/n}}{1+t} \cdot \frac{dt}{t}$. Scale-invariance of $\frac{dt}{t}$ says the answer can depend on s and n only through the ratio s/n and the overall factor $\frac{1}{n}$, before any evaluation is attempted. The value is $\frac{\pi}{n \sin(\pi s/n)}$; at $s = 1, n = 3$ that is $\frac{\pi}{3 \sin(\pi/3)} \approx 1.209200$, and at $s = 2, n = 5$ it is ≈ 0.660653 . The structural prediction was free; only the constant needed work.

WHERE THIS GOES WRONG

Scaling changes the measure, and a missing Jacobian is silent. Under $x = bt$, $dx = b dt$ and $x^2 + b^2 = b^2(1+t^2)$, so the net factor is $\frac{1}{b}$, not 1 and not b . Also, scaling requires $\lambda > 0$: with $\lambda < 0$ the half-line maps to the negative axis and the limits reverse.

SEE ALSO Inversion on the positive half line; splitting the half line at one; Glasser's master theorem.

Both inversion and scaling are easier to run, and far easier to see, once the multiplicative structure of the half-line has been made additive. That conversion is a single substitution, and it is worth performing as a matter of course whenever a half-line integral has a reciprocal symmetry in it: under it, scaling becomes translation and inversion becomes reflection, which are symmetries the eye is already trained on.

MEDIUM

22.11 $x = e^t$: turning inversion into parity

CUE An integral on $(0, \infty)$ containing $\ln x$, or a denominator invariant under $x \mapsto 1/x$, or powers x^s with s a parameter. Any integral where you have already noticed that $x \mapsto 1/x$ does something useful.

METHOD Substitute $x = e^t$, so $dx = e^t dt$, $\ln x = t$, and $(0, \infty)$ becomes $\mathbb{R}$. The inversion $x \mapsto 1/x$ becomes $t \mapsto -t$, so every reciprocal symmetry becomes ordinary parity and every self-reciprocal denominator becomes a hyperbolic cosine. The characteristic collapse is

$$\frac{e^t}{1+e^{2t}} = \frac{1}{2 \cosh t}, \quad \frac{e^{t/2}}{1+e^t} = \frac{1}{2 \cosh(t/2)}.$$

WORKED EXAMPLE

$$\int_0^\infty \frac{\ln^2 x}{1+x^2} dx.$$

With $x = e^t$ this becomes $\int_{-\infty}^\infty \frac{t^2 e^t}{1+e^{2t}} dt = \int_{-\infty}^\infty \frac{t^2}{2 \cosh t} dt$, manifestly even, and equal to $\int_0^\infty \frac{t^2}{\cosh t} dt = \frac{\pi^3}{8} \approx 3.875785$.

The same substitution shows instantly why the odd-power version vanishes: $\int_0^\infty \frac{\ln x}{1+x^2} dx$ becomes $\int_{-\infty}^\infty \frac{t dt}{2 \cosh t}$, odd times even, hence 0. The parity is visible in t and invisible in x .

WHERE THIS GOES WRONG

Forgetting the Jacobian e^t , which is exactly the factor that converts $1+e^{2t}$ into $2 \cosh t$ and makes the parity visible. Without it the integrand is neither even nor odd and the whole point is lost. Write $dx = e^t dt$ before touching the rest.

SEE ALSO Inversion on the positive half line; disguised zero; Glasser's master theorem.

Inversion has a second mode of use, and it is the one to reach for when the fold over the whole half-line does not close. The point $x = 1$ is the fixed point of $x \mapsto 1/x$, so it cuts $(0, \infty)$ into two pieces that the involution exchanges. Comparing the pieces is weaker than folding the whole, but it is available in cases where folding is not, and it converts a half-line problem into a statement about $(0, 1)$ alone.

22.12 Splitting the half-line at $x = 1$

CUE A half-line integral where the fold $x \mapsto 1/x$ almost closes but not quite, or where you want to compare $(0, 1)$ against $(1, \infty)$, or where the integrand's behaviour genuinely changes across $x = 1$ (a power weight x^s that is singular at one end and decaying at the other).

METHOD Write $\int_0^\infty = \int_0^1 + \int_1^\infty$ and apply $x = 1/u$ to the second piece only, which lands it on $(0, 1)$. You obtain

$$\int_0^\infty f(x) dx = \int_0^1 \left[f(x) + \frac{1}{x^2} f\left(\frac{1}{x}\right) \right] dx,$$

a single integral over a compact interval, with a bracketed integrand that is often much better behaved than either piece. If the two terms cancel identically, the original integral is zero; if they coincide, it is twice one of them.

WORKED EXAMPLE

$$\int_0^\infty \frac{\ln x}{(1+x)^2} dx.$$

On $(1, \infty)$ put $x = 1/u$: then $\ln x = -\ln u$, $(1+x)^2 = \frac{(1+u)^2}{u^2}$ and $dx = \frac{du}{u^2}$, so the piece becomes $-\int_0^1 \frac{\ln u}{(1+u)^2} du$, exactly the negative of the first piece. The integral is 0.

The same split applied to $\frac{\ln x}{1+x^2}$ shows something sharper. The two halves are

$$\int_0^1 \frac{\ln x}{1+x^2} dx = -G \approx -0.915966, \quad \int_1^\infty \frac{\ln x}{1+x^2} dx = +G,$$

where G is Catalan's constant, which has no known closed form in terms of elementary constants. The whole integral is 0 regardless, and the inaccessible constant cancels. This is worth internalising: a symmetry argument can deliver an exact value for an integral whose *pieces* are individually beyond reach.

WHERE THIS GOES WRONG

Splitting a divergent integral and cancelling the halves. If $\int_0^1$ and $\int_1^\infty$ each diverge, their formal difference is meaningless, and the bracketed form above is the only honest statement: it converges precisely when the bracket is integrable on $(0, 1)$, which is a stronger condition than the two pieces separately being finite in the principal-value sense.

SEE ALSO Inversion on the positive half line; scaling on the half line; disguised zero.

The most striking consequence of inversion is a theorem that looks, on first reading, too strong to be true: for a large class of substitutions built out of $x - \frac{a}{x}$, the substitution does not change the value of the integral at all.

HARD

22.13 Glasser's master theorem

CUE The combination $x - \frac{a}{x}$ hidden inside the integrand, usually disguised as a quartic denominator. The standard disguise is

$$x^4 - (2c-1)x^2 + c^2 = x^2 \left[\left(x - \frac{c}{x} \right)^2 + 1 \right],$$

so a quartic in x^2 with an x^2 available in the numerator is the signature.

METHOD For $a > 0$ and F even,

$$\int_0^\infty F\left(x - \frac{a}{x}\right) dx = \int_0^\infty F(u) du.$$

The proof is the fold: under $x = a/t$ the argument $x - \frac{a}{x}$ becomes $-(t - \frac{a}{t})$, so evenness of F gives $I = \int_0^\infty F(t - \frac{a}{t}) \frac{a}{t^2} dt$, and adding the two expressions produces $2I = \int_0^\infty F(x - \frac{a}{x}) (1 + \frac{a}{x^2}) dx$, which is exactly $\int_{-\infty}^\infty F(u) du$ under $u = x - \frac{a}{x}$. The odd part of the Jacobian is what integrates away. Glasser's general form allows $u = x - \sum_n \frac{a_n}{x - b_n}$ with every $a_n > 0$, over the whole line.

WORKED EXAMPLE

Two direct applications, with $F(u) = e^{-u^2}$ and $F(u) = \frac{1}{u^2 + 1}$:

$$\int_0^\infty \exp\left[-\left(x - \frac{1}{x}\right)^2\right] dx = \int_0^\infty e^{-u^2} du = \frac{\sqrt{\pi}}{2} \approx 0.886227,$$

$$\int_0^\infty \frac{dx}{(x - \frac{1}{x})^2 + 1} = \int_0^\infty \frac{du}{u^2 + 1} = \frac{\pi}{2} \approx 1.570796.$$

Now the disguised version. Using the quartic identity above,

$$\int_0^\infty \frac{x^2 dx}{x^4 - (2c-1)x^2 + c^2} = \int_0^\infty \frac{dx}{(x - \frac{c}{x})^2 + 1} = \frac{\pi}{2}$$

for every $c > 0$. At $c = 3$ the denominator is $x^4 - 5x^2 + 9$, which has no obvious symmetry at all, and the answer is still $\frac{\pi}{2}$. This is the parameter-independence signature in its purest form.

THE GENERAL FORM, AND WHY IT IS BELIEVABLE

Glasser's theorem in full says that for $a_n > 0$,

$$\int_{-\infty}^\infty F\left(x - \sum_{n=1}^N \frac{a_n}{x - b_n}\right) dx = \int_{-\infty}^\infty F(u) du,$$

with no evenness required, because the domain is now the whole line. The reason it works: the map $u(x) = x - \sum a_n(x - b_n)^{-1}$ is increasing on each of the $N + 1$ intervals between its poles, and each branch carries $(-\infty, \infty)$ onto $(-\infty, \infty)$ once. Summing the $N + 1$ branches reconstructs $\int F(u) du$ exactly once for each, and the change-of-variables factors are what the sum of branches supplies. A concrete instance with two poles:

$$\int_{-\infty}^\infty \frac{dx}{1 + (x - \frac{1}{x-1} - \frac{1}{x+1})^2} = \int_{-\infty}^\infty \frac{du}{1 + u^2} = \pi.$$

The inner expression simplifies to $\frac{x^3 - 3x}{x^2 - 1}$, so the integrand is the rational function $\frac{(x^2 - 1)^2}{(x^2 - 1)^2 + (x^3 - 3x)^2}$, degree four over degree six, vanishing to second order at $x = \pm 1$. Nothing about it suggests π .

WHERE THIS GOES WRONG

F must be even, or the odd part of the Jacobian survives and the theorem is false as stated. And every a_n must be strictly positive: a negative coefficient destroys the monotonicity of $u(x)$ on each branch, the map is no longer a bijection onto $\mathbb{R}$, and the conclusion fails. Check the sign before quoting the theorem.

SEE ALSO Cauchy and Schlomilch's transformation; inversion on the positive half line; reciprocal substitution.

Glasser's theorem has an older two-parameter relative that is used more often in practice, because it handles the scaled combination $ax - \frac{b}{x}$ and therefore covers integrands where the reciprocal symmetry is present but not at unit scale.

HARD

22.14 The Cauchy and Schlomilch transformation

CUE A squared combination $(ax - \frac{b}{x})^2$ inside a function, with $a \neq 1$: the tell is a Gaussian or a rational function whose argument is $a^2x^2 - 2ab + \frac{b^2}{x^2}$.

METHOD For $a, b > 0$ and any integrable f ,

$$\int_0^\infty f\left(\left(ax - \frac{b}{x}\right)^2\right) dx = \frac{1}{a} \int_0^\infty f(y^2) dy.$$

The scale a appears as an overall factor and b disappears entirely. Derive it from Glasser by first substituting $x \mapsto x/a$, which converts $ax - \frac{b}{x}$ into $x - \frac{ab}{x}$ at the cost of a factor $\frac{1}{a}$.

WORKED EXAMPLE

$$\int_0^\infty \exp\left[-\left(2x - \frac{3}{x}\right)^2\right] dx = \frac{1}{2} \int_0^\infty e^{-y^2} dy = \frac{\sqrt{\pi}}{4} \approx 0.443113.$$

Expanding the exponent as $-4x^2 + 12 - \frac{9}{x^2}$ shows what has been avoided: the integrand is $e^{12} \exp(-4x^2 - 9x^{-2})$, and the constant e^{12} is what makes the direct approach look hopeless. The transformation absorbs it without comment, because the cross term $-2ab$ was part of the perfect square all along.

WHERE THIS GOES WRONG

Both a and b must be positive, and the argument must be the *square* of the combination, not the combination itself: f is applied to y^2 on the right, so an odd function of $ax - \frac{b}{x}$ is outside the scope and must be handled by Glasser's version with an explicit evenness check.

SEE ALSO Glasser's master theorem; inversion on the positive half line; Gaussian integrals.

Zero, and the fold that only looks impossible

The most valuable thing a symmetry argument can tell you is that the answer is zero, because the alternative is an evaluation that may not be possible at all. Problems in this family are constructed to be intimidating: the integrand is assembled from special functions, unusual powers, and awkward domains, and none of it matters. The correct first move on any monstrous definite integral is to spend thirty seconds testing the standard involutions for anti-invariance.

22.15 Disguised zero

HARD

CUE A monstrous integrand on a symmetric or reciprocal-symmetric domain; two special-function terms of equal magnitude and opposite sign; a ratio whose numerator and denominator are the same integral with the limits reversed; an odd-looking factor such as $\ln x$ or $\sin$ multiplying something with matching symmetry.

METHOD Before computing anything, test whether some involution of the domain sends the integrand to its own negative. The standard involutions, in the order worth trying: $x \mapsto -x$ on $[-a, a]$, $x \mapsto a + b - x$ on $[a, b]$, $x \mapsto 1/x$ on $(0, \infty)$, and $x = e^{2y}$ followed by $y \mapsto -y$. If the integrand is odd under one of them and is absolutely integrable, the answer is 0 and there is nothing left to do.

WORKED EXAMPLE

$$\int_0^\infty \frac{\ln x (\pi^2 + \ln^2 x)}{(1+x)\sqrt{x}} dx.$$

Substitute $x = e^{2y}$, so $dx = 2e^{2y} dy$, $\sqrt{x} = e^y$, and $\ln x = 2y$. The integrand becomes

$$\frac{2y(\pi^2 + 4y^2)}{(1 + e^{2y})e^y} \cdot 2e^{2y} dy = \frac{4y(\pi^2 + 4y^2)}{e^{-y} + e^y} dy = \frac{2y(\pi^2 + 4y^2) dy}{\cosh y}.$$

That is an odd polynomial in y divided by an even function, hence odd on $\mathbb{R}$, and it decays exponentially, so the integral converges absolutely and equals 0.

A SECOND, ON A FINITE INTERVAL

$$\int_{-\pi/2}^{\pi/2} \ln\left(\frac{2 + \sin x}{2 - \sin x}\right) dx.$$

The interval is symmetric about 0 and $\sin$ is odd, so $x \mapsto -x$ exchanges numerator and denominator, sending the logarithm to its own negative. The integrand is bounded (the argument stays in $[\frac{1}{3}, 3]$), so integrability is not in question and the value is 0. Nothing about the constant 2 mattered; any $c > 1$ gives the same answer, which is the parameter-independence signature again.

ONE POWER DESTROYS IT

The same integrand with $(1+x)^2$ in place of $(1+x)$ is *not* zero. Under $x = e^{2y}$ the same substitution now gives, using $(1 + e^{2y})^2 = 4e^{2y} \cosh^2 y$,

$$\frac{4y(\pi^2 + 4y^2)e^y}{(1 + e^{2y})^2} dy = \frac{y(\pi^2 + 4y^2)e^{-y}}{\cosh^2 y} dy,$$

and the stray factor e^{-y} destroys the parity: an odd polynomial times a function that is neither even nor odd, over an even one. The true value is

$$\int_0^\infty \frac{\ln x (\pi^2 + \ln^2 x)}{(1+x)^2\sqrt{x}} dx = -4\pi^3 \approx -124.025107,$$

a long way from zero. The lesson is that the fold is a property of the exact exponent, not of the general shape, and the only safe procedure is to perform the substitution and look at what you get.

WHERE THIS GOES WRONG

“Odd on a symmetric interval” is not enough when the integral is only conditionally convergent: then the two halves are $+\infty$ and $-\infty$, the object is a principal value rather than an integral, and calling it zero is a claim about a regularisation rather than about the integral. State which you mean.

SEE ALSO Even and odd parity on a symmetric interval; inversion on the positive half line; exponential substitution and multiplicative parity; Cauchy principal value.

The last technique in the chapter is a fold of a different kind: instead of mapping the interval to itself, it maps an infinite interval onto a compact one by exploiting the periodicity of one factor against the decay of another. It is the natural place to end, because it shows that “symmetry” need not mean an involution.

HARD

22.16 Lobachevsky's formula

CUE $\frac{\sin x}{x}$ or $\frac{\sin^2 x}{x^2}$ multiplied by a π -periodic function, integrated over $(0, \infty)$.

METHOD Two statements, with different hypotheses. If f is π -periodic then

$$\int_0^\infty f(x) \frac{\sin^2 x}{x^2} dx = \int_0^{\pi/2} f(x) dx;$$

if in addition $f(\pi - x) = f(x)$, that is, f is symmetric about $\frac{\pi}{2}$, then also

$$\int_0^\infty f(x) \frac{\sin x}{x} dx = \int_0^{\pi/2} f(x) dx.$$

The mechanism is to cut $(0, \infty)$ into intervals of length π , translate each back to $[0, \pi]$ where f repeats, and sum the resulting series $\sum (x + n\pi)^{-2}$ into $\csc^2$; the whole half-line folds onto a quarter period.

WORKED EXAMPLE

Taking $f = 1$ in the first formula gives $\int_0^\infty \left(\frac{\sin x}{x}\right)^2 dx = \frac{\pi}{2}$, and in the second, $\int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2}$.

Taking $f(x) = \sin^2 x$, which is π -periodic and symmetric about $\frac{\pi}{2}$, the second formula gives

$$\int_0^\infty \frac{\sin^3 x}{x} dx = \int_0^{\pi/2} \sin^2 x dx = \frac{\pi}{4},$$

and the first gives $\int_0^\infty \frac{\sin^4 x}{x^2} dx = \frac{\pi}{4}$ as well.

A harder choice: $f(x) = \frac{1}{1 + \sin^2 x}$ is π -periodic, so

$$\int_0^\infty \frac{\sin^2 x}{x^2(1 + \sin^2 x)} dx = \int_0^{\pi/2} \frac{dx}{1 + \sin^2 x} = \frac{\pi}{2\sqrt{2}} \approx 1.110721.$$

No other method available at this level touches that integral.

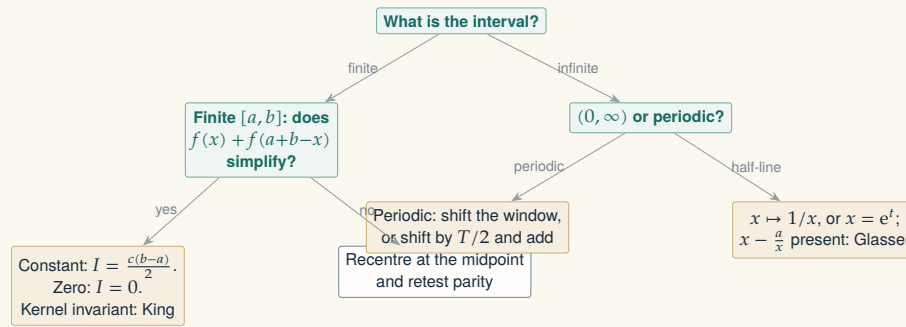
WHERE THIS GOES WRONG

The symmetry hypothesis in the second formula is not decoration. Take $f(x) = \sin 2x$, which is π -periodic but antisymmetric about $\frac{\pi}{2}$. The formula would predict $\int_0^{\pi/2} \sin 2x dx = 1$, whereas the true value of $\int_0^\infty \frac{\sin 2x \sin x}{x} dx$ is $\frac{1}{2} \ln 3 \approx 0.549306$, obtained from Frullani after the product-to-sum identity. A clean wrong answer, produced by applying the right theorem to the wrong symmetry class. Also note the companion value $\int_0^\infty (\sin x/x)^4 dx = \frac{\pi}{3}$, which is *not* a Lobachevsky consequence, since $\sin^2 x/x^2$ is not periodic; it comes from Parseval or from repeated integration by parts.

SEE ALSO Periodicity and window shifting; the Dirichlet integral; mirror symmetry about an interior point.

Choosing the involution

The decision here is faster than in most chapters, because the candidate substitutions are determined by the interval rather than by the integrand. A finite interval offers reflection about its midpoint and nothing else; a periodic domain offers translation; the half-line offers inversion. What the integrand determines is only whether the fold is worth performing, and that is a question about the $\sum f(x) + f(\varphi(x)) |\varphi'(x)|$, which takes one line to compute.



The tree omits one branch deliberately, and it is the branch to take when every box fails: split the interval. An integral that has no symmetry as a whole very often has one on each piece, and the splitting points are the same ones you would find for an absolute value, namely the zeros and the branch changes of the inner expressions. $\int_0^\infty$ split at $x = 1$ is the canonical instance, because the two halves are exchanged by $x \mapsto 1/x$; the fold that seemed unavailable on $(0, \infty)$ becomes a comparison between $(0, 1)$ and $(1, \infty)$.

One final habit, worth more than any individual technique in this chapter. When a definite integral is presented with a parameter that plays no visible role, an exponent chosen to be ugly, or a large decorative term added to the integrand, the problem is telling you that a fold is intended and that the decoration is designed to cancel. Answer the question “what would have to cancel for this to be clean” before answering “how do I integrate this”. In this family of problems those are different questions, and the first one is much easier.

AUDIT THE FOLD NUMERICALLY

Every exact value in this chapter can be checked to three decimal places in under a minute with any numerical quadrature. Do it, because the characteristic failure here is not a wrong method but a symmetry that does not quite hold: a kernel that is invariant under $x \mapsto \pi - x$ except for one factor, an exponent that is $\frac{1}{2}$ rather than 1, an interval that is 0.99 periods rather than 1. All of those produce a clean, plausible, wrong closed form, and none of them is visible in the algebra. The two useful diagnostics: evaluate the claimed identity $f(x) + f(\varphi(x)) | \varphi'(x) |$ at three or four random interior points and see whether it really is constant, and compare the final number against a coarse quadrature. If the identity fails at a random point, no amount of rearrangement will save the argument.

Recognition matrix

If you see	Reach for	If that fails
Bounds $[-a, a]$, factored integrand	Classify parities; discard the odd part	Check integrability of the odd part
Bounds $[a, b]$ with $\frac{a+b}{2}$ visible	Recentre with $t = x - \frac{a+b}{2}$	Reflect instead: $x \mapsto a + b - x$
$xK(x)$ with $K(a+b-x) = K(x)$	King: replace x by $\frac{a+b}{2}$	Test the kernel symbol by symbol
$\frac{g(x)}{g(x) + g(a+b-x)}$	Constant-sum fold: $I = \frac{b-a}{2}$	Verify the identity, do not assume it
$\frac{1}{1 + \tan^p x}$ on $[0, \frac{\pi}{2}]$	Sine and cosine exchange; $I = \frac{\pi}{4}$	Weierstrass substitution $t = \tan x$
$\frac{g(x)}{1 + e^x}$ on $[-a, a]$, g even	Fermi fold: $I = \int_0^a g$	Same with base b : $\frac{1}{1+b^x}$
$\ln \sin x$ or $\ln \cos x$ on $[0, \frac{\pi}{2}]$	Reflect, add, then $u = 2x$; get $-\frac{\pi}{2} \ln 2$	Fourier series of $\ln \sin$
$f(\sin x)$ on $[0, \pi]$	Halve: $2 \int_0^{\pi/2}$, then Wallis	Recentre at $\frac{\pi}{2}$ for parity
Periodic f over nT	$n \int_0^T f$; slide the window freely	Check the true period first
Periodic f with a sign symmetry	Shift by $T/2$ and add	Split at the sign changes
$ \cdot $, max, min, floor, radicals	Split at every crossing, resolve signs	Solve all pairwise equations
$\ln x$ or x^s against $1 + x^2$ on $(0, \infty)$	$x \mapsto 1/x$; add or cancel	$x = e^t$, then parity in t
Reciprocal-symmetric denominator	$x = e^t$, expect 2 cosh	Split at $x = 1$ and compare halves
A fold on $(0, \infty)$ that nearly closes	Split at $x = 1$; map $(1, \infty)$ back by $1/u$	Bracketed form on $(0, 1)$
A positive parameter inside the integrand	Scale it out: $x = \lambda t$, watch the Jacobian	dx/x is scale-invariant
$\ln(ax)$ against $x^2 + b^2$	Scale by b , then kill $\ln t$ by inversion	Answer depends on ab only
$x^4 - (2c-1)x^2 + c^2$ downstairs	Glasser: $(x - \frac{c}{x})^2 + 1$, answer $\frac{\pi}{2}$	Check F is even
$(ax - \frac{b}{x})^2$ inside f	Cauchy and Schlömilch: $\frac{1}{a} \int_0^\infty f(y^2)$	Complete the square first
A monstrous integrand, symmetric domain	Test the four involutions for oddness	Principal value if only conditional
$\frac{\sin x}{x}$ or $\frac{\sin^2 x}{x^2}$ times π -periodic f	Lobachevsky: $\int_0^{\pi/2} f$	Check $f(\pi - x) = f(x)$ for the first form

If you see	Reach for	If that fails
A parameter absent from the answer	A fold is intended; find the involution	Differentiate under the integral sign

Parameter methods: Feynman, Frullani, Leibniz

What this chapter is for. The most powerful move in definite integration is to make the problem harder on purpose. Given an integral with no free constant in it, you invent one, producing a whole family of integrals of which the original is a single member. You then differentiate with respect to the invented constant, which turns the integrand into something you can actually do, integrate the result back up, and pin the constant of integration by evaluating the family at a value of the parameter where the answer is obvious. The chapter treats this as a six step protocol and insists on the two steps every student drops: proving that the derivative may be taken inside the integral, and choosing an anchor value at which the family is genuinely known. Around that core sit its relatives: the Leibniz rule when the limits themselves move, Frullani's formula for a difference of two scalings, parameter differentiation of the Beta function, and the reverse move of replacing a blocking factor by an integral representation. At the end you should be able to look at an integral with no parameter and see where to put one.

Making the problem harder on purpose

An indefinite integral is a search for an antiderivative. A definite integral is not: it is a single number, and the number can often be reached without ever finding an antiderivative. Parameter methods are the largest family of such routes. They work because differentiation with respect to a parameter is a different operation from differentiation with respect to x , and it acts on the integrand in a way that can destroy exactly the feature that made the integral impossible. A $\ln x$ in a denominator is immovable under any substitution; it is produced for free by $\partial_a x^a$. A factor $1/x$ that ruins convergence at the origin is produced for free by ∂_a acting on an exponent. A $\ln \sin x$ that no trigonometric identity will absorb is produced for free by ∂_p acting on $\sin^{2p-1} x$.

The price is that you must solve a family of problems rather than one. That is almost always a bargain, because the family carries information that the single member does not. A family satisfies a differential equation in a , and a differential equation plus one boundary value determines everything. The single member satisfies nothing.

The names attached to these methods are historical accidents worth knowing so that the literature is readable. Differentiation under the integral sign is Leibniz's rule, and the version with moving limits is the one Leibniz actually proved; it acquired Feynman's name from a single paragraph in his memoirs describing how a book of tricks he read as a boy let him evaluate integrals nobody else in the room could. Frullani's integral dates from 1821 and was made rigorous by Hardy a century later, which is why its hypotheses come in two versions. Schwinger's parametrisation is the youngest of the group and comes from quantum field theory, where products of many denominators are the normal case. None of the attributions matters mathematically, but all four names appear in problem statements.

There is also a psychological obstacle worth naming. Generalising a problem feels like retreat, and students trained on antiderivatives will spend an hour on substitutions before they will spend a minute inventing a parameter. The correct instinct is the opposite: as soon as an integral has a definite value but no visible antiderivative, ask what family it belongs to. Almost every named definite integral in the subject is the special value of a family, and the family is usually easier than the value.

Here is the protocol in full. Steps 4 and 6 are the ones that get skipped, and they are the ones that produce wrong answers.

THE SIX STEPS

1. Introduce a parameter a , so that $I = I(a_*)$ for a specific a_* , and $I(a)$ is a family. 2. Differentiate: $I'(a) = \int \frac{\partial f}{\partial a} dx$. If this is not easier than I , you inserted the parameter in the wrong place; go back to step 1. 3. Evaluate $I'(a)$ in closed form. 4. Justify the interchange, on a closed parameter interval containing a_* : exhibit g with $\left| \frac{\partial f}{\partial a} \right| \leq g(x)$ for all a in that interval and $\int g < \infty$. 5. Integrate back: $I(a) = \int I'(a) da + C$. 6. Fix C from an anchor a_0 at which $I(a_0)$ is known, and check that a_0 lies in the region where steps 2 to 4 were valid.

The first technique is the protocol itself, executed on an integral that is genuinely inaccessible without it.

23.1 Differentiation under the integral sign

CUE A constant in the integrand that could obviously have been a variable: $\ln(a+x)$, e^{-ax} , x^a , $\arctan(ax)$, $(1+a^2x^2)$. Equivalently, a specific number in the problem which, if replaced by a letter, would make the integral trivial at some other value of that letter.

METHOD Set $I(a) = \int_c^d f(x, a) dx$, compute

$$I'(a) = \int_c^d \frac{\partial f}{\partial a} dx,$$

evaluate it, integrate back in a , and fix the constant at an anchor. The whole method rests on ∂_a producing an integrand that is elementary even though f was not.

WORKED EXAMPLE

$$I(a) = \int_0^1 \ln\left(\frac{a+x}{a-x}\right) \frac{dx}{x\sqrt{1-x^2}}, \quad a > 1.$$

Nothing here is an antiderivative problem. Differentiate instead. Since $\partial_a [\ln(a+x) - \ln(a-x)] = \frac{1}{a+x} - \frac{1}{a-x} = \frac{-2x}{a^2-x^2}$, the awkward $1/x$ cancels:

$$I'(a) = -2 \int_0^1 \frac{dx}{(a^2-x^2)\sqrt{1-x^2}} = -2 \int_0^{\pi/2} \frac{d\theta}{a^2 - \sin^2 \theta},$$

using $x = \sin \theta$. Divide numerator and denominator by $\cos^2 \theta$ and put $t = \tan \theta$: the standard value is $\int_0^{\pi/2} \frac{d\theta}{p+q \sin^2 \theta} = \frac{\pi}{2\sqrt{p(p+q)}}$, so with $p = a^2$, $q = -1$,

$$I'(a) = -\frac{\pi}{a\sqrt{a^2-1}}.$$

Now integrate back. On $a > 1$, $\frac{d}{da} \arccos(1/a) = \frac{1}{a\sqrt{a^2-1}}$, so $I(a) = -\pi \arccos(1/a) + C$. The anchor is $a \rightarrow \infty$: the integrand tends to 0 pointwise and is dominated by an integrable function, so $I(\infty) = 0$, and since $\arccos 0 = \frac{\pi}{2}$ we get $C = \frac{\pi^2}{2}$. Hence

$$I(a) = \frac{\pi^2}{2} - \pi \arccos \frac{1}{a}, \quad \text{and at } a = 2, \quad I(2) = \frac{\pi^2}{2} - \frac{\pi^2}{3} = \frac{\pi^2}{6} \approx 1.644934.$$

The licence for step 2 is visible: for $a \geq a_0 > 1$, $|\partial_a f| = \frac{2}{(a^2-x^2)\sqrt{1-x^2}} \leq \frac{2}{(a_0^2-1)\sqrt{1-x^2}}$, and that dominating function has finite integral $\frac{\pi}{a_0^2-1}$ on $[0, 1]$. Checking the general formula at a second point, $a = \frac{3}{2}$ gives $\frac{\pi^2}{2} - \pi \arccos \frac{2}{3} \approx 2.292507$, which quadrature confirms.

WHERE THIS GOES WRONG

Two failures, both at step 6. The first is forgetting C altogether and reporting $-\pi \arccos(1/a)$, which is off by $\pi^2/2$ and negative. The second is anchoring at a value where the family does not exist: in the example above $a = 1$ looks tempting because $\arccos 1 = 0$, but the integral diverges there and the derivative formula breaks down, so no information can be extracted from it.

SEE ALSO Choosing where to insert the parameter; the parameter ODE; Beta and Gamma parameter differentiation.

The example was chosen so that the parameter was already present. The real skill is the opposite case, where the integral has no constant in it at all and you must manufacture one. This is where the method becomes a genuine art, because a bad insertion makes $I'(a)$ harder than I and there is no warning bell.

23.2 Choosing where to insert the parameter

CUE A hard integral with no free constant: $\int_0^1 \frac{x-1}{\ln x} dx$, $\int_0^\infty \frac{\sin x}{x} dx$, $\int_0^1 \frac{\arctan x}{x} dx$. There is nothing to differentiate with respect to until you put something there.

METHOD Insert the parameter where ∂_a kills the obstruction. Three insertions cover most cases. Replace an exponent by a : $\partial_a x^a = x^a \ln x$, which cancels a $\ln x$ downstairs. Attach a damping factor e^{-ax} : ∂_a brings down $-x$, which cancels a $1/x$ and simultaneously repairs convergence at infinity. Generalise a coefficient already in the problem, as $\arctan x \mapsto \arctan(ax)$. Then check step 2 before committing: if $\partial_a f$ is not visibly easier, choose again.

$$\int_0^1 \frac{x-1}{\ln x} dx.$$

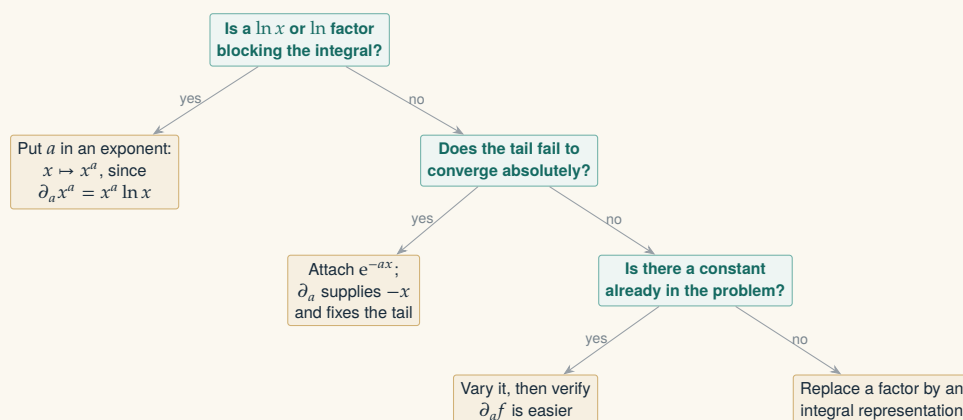
The obstruction is $\ln x$ in the denominator, so the insertion goes in the exponent: put $I(a) = \int_0^1 \frac{x^a-1}{\ln x} dx$, which recovers the target at $a = 1$. Then

$$I'(a) = \int_0^1 \frac{x^a \ln x}{\ln x} dx = \int_0^1 x^a dx = \frac{1}{a+1},$$

so $I(a) = \ln(a+1) + C$. The anchor is $a = 0$, where the numerator is identically zero and hence $I(0) = 0$, forcing $C = 0$. Therefore $I(a) = \ln(a+1)$ and the answer is $\ln 2 \approx 0.693147$. The same computation with two parameters gives $\int_0^1 \frac{x^a-x^b}{\ln x} dx = \ln \frac{a+1}{b+1}$; at $a = 3, b = 1$ this predicts $\ln 2$ again, which quadrature confirms.

Contrast a bad insertion on the same problem. Writing $J(a) = \int_0^1 \frac{x-1}{a \ln^2(ax)} dx$ also recovers the target at $a = 1$, but ∂_a produces $-\frac{x-1}{a \ln^2(ax)}$, which has a worse singularity than the original and no closed form. The parameter went into the right function and the wrong slot.

A third possibility is worse than either, because it looks like progress. For $\int_0^1 \frac{\arctan x}{x} dx$ the natural insertion is $K(a) = \int_0^1 \frac{\arctan(ax)}{x} dx$, and indeed $K'(a) = \int_0^1 \frac{dx}{1+a^2x^2} = \frac{\arctan a}{a}$, which is elementary. But integrating back gives $K(1) = \int_0^1 \frac{\arctan a}{a} da$, which is the original integral again. The parameter was inserted along the direction the integral is invariant under, so the protocol returns an identity rather than a value. The correct diagnosis is that this integral is not a parameter problem at all: its value is Catalan's constant G , reached by series expansion in the next chapter.



The tree routes on the obstruction, not on the integrand, and that is the point. Two integrals that look nothing alike take the same branch if the same feature is blocking them, and the same integral takes different branches depending on which of its features you decide is the obstacle. Read the leftmost branch first: a logarithm in a denominator is the most common blocker and the exponent insertion is the most reliable cure. The rightmost leaf is a genuine fallback rather than a technique of last resort; it appears again as the first technique of the next chapter, where the added integral is turned into an added sum.

WHERE THIS GOES WRONG

Inserting the parameter so that $I'(a)$ is harder than $I(a)$, and then persevering. There is no rule that a family must be easier than its member. The second trap is assuming the anchor: $I(0) = 0$ held above because $x^0 - 1$ vanishes identically, but in $\int_0^\infty e^{-ax} \frac{\sin x}{x} dx$ the value at $a = 0$ is precisely what you are trying to find, and using it as an anchor is circular.

SEE ALSO Differentiation under the integral sign; Frullani's integral; Schwinger parametrisation.

INSERTIONS THAT REPAY THEMSELVES

Four insertions account for most successful applications, and each is chosen for what ∂_a produces rather than for what looks natural. Exponent, $x \mapsto x^a$, produces $\ln x$ and is the answer whenever a logarithm sits in a denominator. Damping, $f \mapsto e^{-ax}f$, produces a factor $-x$ and is the answer whenever a $1/x$ blocks convergence or an oscillatory tail refuses to settle. Scale inside a bounded function, $\arctan x \mapsto \arctan(ax)$ or $\ln(1+x) \mapsto \ln(1+ax)$, produces a rational function and is the answer when the obstruction is the transcendental function itself. Quadratic weight, $e^{-x^2} \mapsto e^{-ax^2}$, produces $-x^2$ and converts a Gaussian family into a power of a . If none of the four applies, the integrand probably wants an integral representation instead of a derivative.

A parameter that enters only through a scaling can usually be removed before any differentiation, and removing it is free information. If $f(x, a) = g(ax)$ on $(0, \infty)$, then $x \mapsto x/a$ gives $I(a) = \frac{1}{a} \int_0^\infty g$, so I is determined up to one number and there is nothing to differentiate. The useful cases are the ones where scaling almost works: $\int_0^\infty e^{-ax^2} dx = \frac{1}{2} \sqrt{\pi/a}$ by scaling alone, while $\int_0^\infty e^{-ax^2} \cos(bx) dx$ needs one genuine differentiation because two parameters cannot both be scaled away. Always check for a scaling first: it costs one substitution and can retire the problem.

Licensing the interchange, and choosing the anchor

Steps 4 and 6 are not bureaucracy. They are where parameter methods actually fail, and a reader who treats them as decoration will eventually produce a confident wrong number. Take them in turn.

The interchange $\frac{d}{da} \int f = \int \partial_a f$ is a limit swap, and limit swaps require a hypothesis. The usable version is this: if $f(x, a)$ and $\partial_a f(x, a)$ are continuous on $[c, d] \times [\alpha, \beta]$, or more generally if there is a single integrable g with $|\partial_a f(x, a)| \leq g(x)$ for every $a \in [\alpha, \beta]$, then I is differentiable on $[\alpha, \beta]$ and the swap is valid there. Two features of that statement are load bearing. The interval in a must be closed, because the theorem is proved by applying dominated convergence to difference quotients and those need a uniform bound. And the dominating function must not depend on a : a bound like $|\partial_a f| \leq h(x, a)$ with $\int h(\cdot, a) < \infty$ for each fixed a proves nothing at all.

In practice you produce g by bounding the a dependent parts away from their bad values, which is exactly why the licence is stated on a subinterval $[\alpha, \beta]$ rather than on the whole range of a for which $I(a)$ happens to converge. The subinterval is then enlarged by covering: since differentiability is local, a valid licence on every $[\alpha, \beta]$ inside an open interval gives a valid formula on the open interval. What it never gives you is the endpoint, and the endpoint is usually where the anchor lives.

WHERE THE LICENCE TYPICALLY FAILS

Almost always at an endpoint of the parameter range, and almost always the endpoint you wanted to use as an anchor. In $I(a) = \int_0^\infty e^{-ax} \frac{\sin x}{x} dx$ the derivative is $I'(a) = -\int_0^\infty e^{-ax} \sin x dx = -\frac{1}{1+a^2}$, and the dominating function $e^{-a_0 x}$ exists for $a \geq a_0 > 0$ but not for $a \geq 0$: at $a = 0$ the differentiated integral $\int_0^\infty \sin x dx$ does not converge at all. So the ODE is valid only on $a > 0$, giving $I(a) = \frac{\pi}{2} - \arctan a$ after anchoring at $I(\infty) = 0$, and the value at $a = 0$ must be obtained by a *separate* argument: I is continuous at 0 from the right, because $\int_R^\infty e^{-ax} \frac{\sin x}{x} dx$ is small uniformly in $a \geq 0$ by Dirichlet's test. Only then may you write $\int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2}$.

Now the anchor. You need one value a_0 at which $I(a_0)$ is known independently, and it must lie in the closure of the region where the derivative computation was licensed. Three anchors cover nearly everything. The first is a value at which the integrand vanishes identically, so that $I(a_0) = 0$ for trivial reasons; $a = 0$ in $\int_0^1 \frac{x^a - 1}{\ln x} dx$ is this case. The second is $a \rightarrow \infty$ with an exponential or algebraic decay that lets dominated convergence give $I(\infty) = 0$. The third is a value at which the family collapses to a standard integral you already know.

A worked illustration of the second anchor, which also shows a clean partial fraction step. With $I(a) = \int_0^\infty \frac{\ln(1+a^2x^2)}{1+x^2} dx$ for $a \geq 0$,

$$I'(a) = \int_0^\infty \frac{2ax^2}{(1+a^2x^2)(1+x^2)} dx = \frac{2a}{a^2-1} \int_0^\infty \left[\frac{1}{1+x^2} - \frac{1}{1+a^2x^2} \right] dx = \frac{2a}{a^2-1} \left(\frac{\pi}{2} - \frac{\pi}{2a} \right) = \frac{\pi}{1+a},$$

where the apparently singular factor $1/(a^2-1)$ cancels against $\frac{\pi}{2} - \frac{\pi}{2a}$. Anchoring at $a = 0$, where the integrand is identically zero, gives $I(a) = \pi \ln(1+a)$; at $a = 3$ this predicts $\pi \ln 4 \approx 4.355172$, which quadrature returns. The two parameter version costs nothing extra:

$$\int_0^\infty \frac{\ln(1+a^2x^2)}{b^2+x^2} dx = \frac{\pi}{b} \ln(1+ab),$$

checked at $a = 2, b = 3$ against the numerical value 2.037752.

Not every family has a single formula, and the anchor is what tells you so. Consider $L(a) = \int_0^\pi \ln(1 - 2a \cos x + a^2) dx$. Differentiating and using the substitution $t = \tan \frac{x}{2}$ gives $L'(a) = 0$ for $|a| < 1$ and $L'(a) = \frac{2\pi}{a}$ for $|a| > 1$, so L is constant on each of the two regions separately. The anchor $a = 0$ gives $L \equiv 0$ on $|a| < 1$; the anchor $a \rightarrow \infty$, where $\ln(1 - 2a \cos x + a^2) = 2 \ln |a| + O(1/a)$, gives $L(a) = 2\pi \ln |a|$ on $|a| > 1$. Numerically $L(0.5) = 0$ and $L(3) = 2\pi \ln 3 \approx 6.902785$, both confirmed. A single anchor would have produced one of the two branches and silently misreported the other.

Two licences are in circulation and they are not the same. Uniform convergence of $\int_c^d$ as $d \rightarrow \infty$, uniformly in a on a closed interval, is what you use when the trouble is the range of integration: it is proved by a tail estimate and it is the natural tool for oscillatory integrands, where Dirichlet's test supplies the uniformity. A dominating function is what you use when the trouble is the integrand: it is proved by an inequality and it is the natural tool for singularities at a finite endpoint. Neither implies the other. When both are available, prefer domination, because it also gives continuity of I in a for free, and continuity is what lets you push the answer to the endpoint of the parameter range.

An anchor that is not an endpoint

The third kind of anchor, a value at which the family collapses to something known, is the one people forget exists, and it is often the cleanest. Consider

$$I(a) = \int_0^{\pi/2} \ln(a^2 \cos^2 x + b^2 \sin^2 x) dx, \quad a, b > 0,$$

with b held fixed. Differentiating in a and writing $D = a^2 \cos^2 x + b^2 \sin^2 x$, the identity $b^2 - D = (b^2 - a^2) \cos^2 x$ turns $\frac{\cos^2 x}{D}$ into $\frac{1}{b^2 - a^2} \left[\frac{b^2}{D} - 1 \right]$, and with the standard value $\int_0^{\pi/2} \frac{dx}{D} = \frac{\pi}{2ab}$,

$$I'(a) = 2a \int_0^{\pi/2} \frac{\cos^2 x}{D} dx = \frac{2a}{b^2 - a^2} \left(\frac{\pi b}{2a} - \frac{\pi}{2} \right) = \frac{\pi}{a + b}.$$

So $I(a) = \pi \ln(a + b) + C$. Neither $a = 0$ nor $a \rightarrow \infty$ is usable: at $a = 0$ the integrand has a logarithmic singularity at $x = 0$ and the derivative formula fails there, and as $a \rightarrow \infty$ the integral diverges. But at $a = b$ the family collapses, since $D \equiv b^2$ and $I(b) = \pi \ln b$. That forces $C = -\pi \ln 2$ and

$$\int_0^{\pi/2} \ln(a^2 \cos^2 x + b^2 \sin^2 x) dx = \pi \ln \frac{a + b}{2},$$

checked at $a = 2, b = 5$ against the numerical value 3.935671. At $a = b = 1$ it degenerates correctly to $\int_0^{\pi/2} \ln 1 dx = 0$, and setting $a = 1, b \rightarrow 0$ recovers $\int_0^{\pi/2} \ln \cos x dx = -\frac{\pi}{2} \ln 2$, a limit that is worth taking as a sanity check on the whole formula.

The protocol run end to end

It is worth watching all six steps once, in order, on a problem where none of them is trivial. Take

$$T = \int_0^{\pi/2} \frac{\arctan(3 \tan x)}{\tan x} dx.$$

Step 1. The constant 3 is the obvious letter. Put $T(a) = \int_0^{\pi/2} \frac{\arctan(a \tan x)}{\tan x} dx$, so that $T = T(3)$.

Step 2. Differentiating in a multiplies the integrand by $\frac{\tan x}{1 + a^2 \tan^2 x}$ and cancels the $\tan x$ downstairs, leaving

$$T'(a) = \int_0^{\pi/2} \frac{dx}{1 + a^2 \tan^2 x} = \int_0^{\pi/2} \frac{\cos^2 x}{\cos^2 x + a^2 \sin^2 x} dx.$$

That is elementary where the original was not, so the insertion is the right one.

Step 3. Write $D = \cos^2 x + a^2 \sin^2 x$ and note $a^2 - D = (a^2 - 1) \cos^2 x$, so $\frac{\cos^2 x}{D} = \frac{1}{a^2 - 1} \left[\frac{a^2}{D} - 1 \right]$. Since $\int_0^{\pi/2} \frac{dx}{D} = \frac{\pi}{2a}$,

$$T'(a) = \frac{1}{a^2 - 1} \left(\frac{\pi a}{2} - \frac{\pi}{2} \right) = \frac{\pi}{2(a + 1)}.$$

Step 4. On $0 \leq a \leq A$ the integrand of T' is bounded by 1, an integrable function on a set of finite measure, uniformly in a . The licence holds on every such interval, hence on $[0, \infty)$.

Step 5. Integrating, $T(a) = \frac{\pi}{2} \ln(1 + a) + C$.

Step 6. At $a = 0$ the integrand is identically zero, so $T(0) = 0$ and $C = 0$. Therefore $T(a) = \frac{\pi}{2} \ln(1 + a)$ and

$$T = \frac{\pi}{2} \ln 4 = \pi \ln 2 \approx 2.177586,$$

which quadrature confirms. Notice that step 4 was cheap here precisely because step 3 produced a bounded integrand; whenever the differentiated integrand is bounded on a finite interval, the licence is automatic and should be stated in one line rather than skipped.

When one parameter is not enough, use two. The rule is that $\partial_a \partial_b I$ must be easier than either single derivative, and the reconstruction then needs two anchors: integrate in b and fix a function of a , then integrate in a and fix a number. Equality of mixed partials gives a free consistency check that is worth performing: compute $\partial_a \partial_b$ and $\partial_b \partial_a$ separately and confirm they agree. If they do not, one of the two interchanges was illegitimate, and you have found the error before it reached the answer.

A gallery of parameter families

The following families are worth knowing by sight, because a large fraction of hard definite integrals are members of one of them. Each is proved by the six step protocol, and each has been checked numerically at a specific parameter value.

$$\begin{aligned} \int_0^\infty e^{-ax} \frac{\sin(bx)}{x} dx &= \arctan \frac{b}{a}, & \int_0^\infty \frac{e^{-ax} - e^{-bx}}{x} dx &= \ln \frac{b}{a} \quad (a, b > 0). \\ \int_0^\infty \frac{\ln(1 + a^2 x^2)}{b^2 + x^2} dx &= \frac{\pi}{b} \ln(1 + ab), & \int_0^\infty \frac{\arctan(ax)}{x(1 + x^2)} dx &= \frac{\pi}{2} \ln(1 + a). \\ \int_0^{\pi/2} \frac{\arctan(a \tan x)}{\tan x} dx &= \frac{\pi}{2} \ln(1 + a), & \int_0^{\pi/2} \ln(a^2 \cos^2 x + b^2 \sin^2 x) dx &= \pi \ln \frac{a+b}{2}. \\ \int_0^\infty e^{-ax^2} \cos(bx) dx &= \frac{1}{2} \sqrt{\frac{\pi}{a}} e^{-b^2/4a}, & \int_0^\infty \frac{\sin(ax)}{x(1 + x^2)} dx &= \frac{\pi}{2} (1 - e^{-a}) \quad (a > 0). \end{aligned}$$

Three observations are worth extracting. First, $\ln(1 + a)$ appears whenever $I'(a)$ is $\frac{c}{1+a}$, which happens whenever the differentiated integrand is a rational function whose partial fractions telescope; this is the single most common shape in the whole subject. Second, several of these families are the same family in disguise: $x = \tan \theta$ carries the second line into the third. Third, the last entry has a restriction $a > 0$ that is not cosmetic, since the integrand is odd in a , and a formula quoted without its restriction will be used at $a < 0$ and will be wrong by a sign.

When the derivative reproduces the integral

Sometimes $I'(a)$ is not an integral you can evaluate, but integration by parts in x turns it back into $I(a)$ itself. That is not a failure. It is a first order linear differential equation in the parameter, and one boundary value solves it. This is the single most elegant corner of the subject, because the integral is determined by its own structure rather than by any antiderivative, and the calculation never touches the function whose integral you want.

23.3 The parameter ODE

HARD

CUE $I'(a)$ resists direct evaluation, but the differentiated integrand is the original integrand multiplied by x , or by a factor that integration by parts can convert back. Gaussian times trigonometric is the archetype, as are integrands built from e^{-x^2} , $e^{-x}x^a$, and $\operatorname{sech} x$.

METHOD Derive a relation $I'(a) = \lambda(a)I(a) + \mu(a)$ by parts in x , discarding boundary terms only after checking them. Solve the linear ODE with an integrating factor, and fix the single constant from the undamped or undeformed integral $I(0)$.

WORKED EXAMPLE

$$F(t) = \int_0^\infty e^{-x^2} \cos(tx) dx.$$

Differentiating in t gives $F'(t) = -\int_0^\infty x e^{-x^2} \sin(tx) dx$, licensed because $|x e^{-x^2} \sin(tx)| \leq x e^{-x^2}$ uniformly in t , and $x e^{-x^2}$ is integrable on $(0, \infty)$. Integrate by parts with $u = \sin(tx)$ and $dv = x e^{-x^2} dx$, so $v = -\frac{1}{2} e^{-x^2}$:

$$F'(t) = \left[\frac{1}{2} e^{-x^2} \sin(tx) \right]_0^\infty - \frac{t}{2} \int_0^\infty e^{-x^2} \cos(tx) dx = -\frac{t}{2} F(t).$$

The boundary term vanishes at both ends, at infinity because e^{-x^2} crushes the bounded sine and at zero because $\sin 0 = 0$. Solving, $F(t) = F(0) e^{-t^2/4}$, and $F(0) = \frac{\sqrt{\pi}}{2}$ is the Gaussian integral, so

$$\int_0^\infty e^{-x^2} \cos(tx) dx = \frac{\sqrt{\pi}}{2} e^{-t^2/4}.$$

The formula is a machine. Since $\cos^3 \theta = \frac{3}{4} \cos \theta + \frac{1}{4} \cos 3\theta$ gives $\cos^3(2x) = \frac{3}{4} \cos 2x + \frac{1}{4} \cos 6x$, reading off $t = 2$ and $t = 6$ yields

$$\int_0^\infty e^{-x^2} \cos^3(2x) dx = \frac{\sqrt{\pi}}{2} \left(\frac{3}{4} e^{-1} + \frac{1}{4} e^{-9} \right) \approx 0.244546,$$

confirmed to twelve places by quadrature. Any finite trigonometric polynomial in $\cos$ can be integrated against e^{-x^2} this way, and the only work is the product to sum reduction.

A second instance, where the ODE comes from a substitution rather than from parts. Let $G(t) = \int_0^\infty e^{-x^2 - t^2/x^2} dx$ for $t \geq 0$. Differentiating,

$$G'(t) = -2t \int_0^\infty \frac{1}{x^2} e^{-x^2 - t^2/x^2} dx,$$

and the substitution $u = t/x$, which sends dx/x^2 to du/t and leaves the exponent invariant, turns the right side into $-2G(t)$. So $G' = -2G$, and $G(0) = \frac{\sqrt{\pi}}{2}$ gives

$$\int_0^\infty e^{-x^2 - t^2/x^2} dx = \frac{\sqrt{\pi}}{2} e^{-2t},$$

confirmed at $t = \frac{1}{2}$ against the numerical value 0.326025. The self reciprocal substitution $x \mapsto t/x$ is what made the ODE appear, which is a general lesson: an integral invariant under a substitution up to a factor will satisfy a differential equation in the parameter of that substitution.

WHERE THIS GOES WRONG

Discarding the boundary term without looking at it. Here $[\frac{1}{2} e^{-x^2} \sin(tx)]_0^\infty$ vanishes, but in the same computation with e^{-x} in place of e^{-x^2} and a growing test function it does not, and the resulting ODE is simply false. The second trap is solving $I' = \lambda I$ and forgetting that a nonhomogeneous term $\mu(a)$ requires an integrating factor rather than a bare exponential.

SEE ALSO Differentiation under the integral sign; Gaussian and error function integrals; cyclic integration by parts.

The ODE route is worth trying whenever the parameter sits in a place where ∂_a multiplies the integrand by x and where a factor of the integrand is the derivative of something simple. That is a narrow condition, but the family it covers is large: shifting $\cos(tx)$ to $\cosh(tx)$ turns the sign around and gives $\int_0^\infty e^{-x^2} \cosh(tx) dx = \frac{\sqrt{\pi}}{2} e^{t^2/4}$, and replacing e^{-x^2} by $e^{-x^2/2}$ rescales the result rather than changing its shape. When the ODE is second order rather than first, you are in the theory of special functions and the parameter method has handed the problem to Bessel or to Airy, which is usually the right outcome.

Frullani: two scales, one function

The next technique looks at first like a different subject, but it is the same idea with the parameter differentiation already performed once and for all. A Frullani integrand compares one function at two different scales. Each half diverges logarithmically at one end, the difference converges, and the answer depends only on the two limiting values of f and on the ratio of the scales. Nothing about the shape of f in between matters at all, which is the surprising part and the reason the formula is worth memorising.

MEDIUM

23.4 Frullani's integral

CUE $\frac{f(ax) - f(bx)}{x}$ on $(0, \infty)$: the same function at two scales, over x . The signature is that each term alone would give a divergent $\int \frac{dx}{x}$ near 0 or near ∞ , and only the difference is finite.

METHOD If f is continuous on $[0, \infty)$ and both $f(0^+)$ and $f(\infty)$ exist and are finite, then for $a, b > 0$

$$\int_0^\infty \frac{f(ax) - f(bx)}{x} dx = [f(0^+) - f(\infty)] \ln \frac{b}{a}.$$

If instead $f(\infty)$ does not exist but $\int_A^\infty \frac{f(x)}{x} dx$ converges for some $A > 0$, the same formula holds with $f(\infty)$ replaced by 0. Those are the honest hypotheses; the formula is false without one of them.

The exponential case is the one to know by heart. With $f(x) = e^{-x}$, $f(0) = 1$ and $f(\infty) = 0$, so

$$\int_0^\infty \frac{e^{-x} - e^{-3x}}{x} dx = \ln 3 \approx 1.098612.$$

It is also a parameter computation in disguise. Set $I(a) = \int_0^\infty \frac{e^{-ax} - e^{-3x}}{x} dx$; then $I'(a) = -\int_0^\infty e^{-ax} dx = -\frac{1}{a}$, so $I(a) = -\ln a + C$, and the anchor $I(3) = 0$ gives $C = \ln 3$ and $I(a) = \ln(3/a)$.

Two more, chosen to exercise the two hypothesis clauses. With $f = \arctan$, $f(0) = 0$ and $f(\infty) = \frac{\pi}{2}$, so

$$\int_0^\infty \frac{\arctan(\pi x) - \arctan x}{x} dx = \left(0 - \frac{\pi}{2}\right) \ln \frac{1}{\pi} = \frac{\pi}{2} \ln \pi \approx 1.798137.$$

With $f = \cos$ there is no limit at infinity, but $\int_A^\infty \frac{\cos x}{x} dx$ converges by Dirichlet's test, so the second clause applies and $\int_0^\infty \frac{\cos(ax) - \cos(bx)}{x} dx = \ln \frac{b}{a}$; at $a = 1$, $b = 3$ this is $\ln 3$ again, and the numerical partial integrals do converge to 1.09861 once the oscillatory tail is averaged.

WHY ONLY THE ENDPOINTS MATTER

Cut the integral at ε and R and substitute $u = ax$ in the first term and $u = bx$ in the second. Both become $\int f(u) \frac{du}{u}$ over shifted ranges, and the difference is

$$\int_{a\varepsilon}^{b\varepsilon} \frac{f(u)}{u} du - \int_{aR}^{bR} \frac{f(u)}{u} du.$$

On the first range f is close to $f(0^+)$ and $\int \frac{du}{u} = \ln \frac{b}{a}$; on the second f is close to $f(\infty)$ and the same logarithm appears. Everything between ε and R has cancelled exactly. That is the whole proof, and it explains both the form of the answer and why the hypotheses are about limits only.

Read from right to left, Frullani is a representation rather than an evaluation, and in that direction it manufactures γ . Setting $f = e^{-\cdot}$ and rearranging gives $\ln x = \int_0^\infty \frac{e^{-t} - e^{-xt}}{t} dt$, and multiplying by e^{-x} and integrating in x produces

$$\int_0^\infty e^{-x} \ln x dx = \int_0^\infty \frac{1}{t} \left(e^{-t} - \frac{1}{1+t} \right) dt = -\gamma \approx -0.577216,$$

the inner x integral having been done first, which is legitimate here because the double integrand is absolutely integrable. The same object is $\Gamma'(1) = \psi(1)$, so the Euler constant enters the subject as the derivative of the Gamma function at 1 and as a Frullani integral at the same time. That coincidence is why γ turns up in parameter differentiation of Beta and Gamma integrals and almost nowhere else.

The hypotheses, tested

The formula has two hypothesis clauses and it is worth seeing all three cases that can arise. First, both limits exist: $f = e^{-\cdot}$, $f = \frac{1}{1+\cdot}$, $f = \arctan$, and the formula applies as stated. The middle case, $f(x) = \frac{1}{1+x}$, gives $\int_0^\infty \left(\frac{1}{1+x} - \frac{1}{1+5x} \right) \frac{dx}{x} = \ln 5 \approx 1.609438$, which is a useful sanity check because the integrand is elementary and the antiderivative can be written down.

Second, $f(\infty)$ does not exist but $\int_A^\infty \frac{f}{x}$ converges. Both $\cos$ and $\sin$ are here. For $\cos$ the first clause value $f(0) = 1$ survives and the answer is $\ln \frac{b}{a}$. For $\sin$ we have $f(0) = 0$, so the answer is 0, and indeed $\int_0^\infty \frac{\sin(ax) - \sin(bx)}{x} dx = \frac{\pi}{2} - \frac{\pi}{2} = 0$ for any $a, b > 0$: each half converges separately, which is the exceptional situation in this family and is why the $\sin$ case is a poor example to learn from.

Third, neither clause holds, and the integral diverges. This is the case that punishes formula matching, because the shape of the integrand is right and the answer produced is finite. The example in the trap below is the standard one.

WHERE THIS GOES WRONG

Splitting the difference into two integrals. Both are infinite, so the split is not a manipulation of numbers at all; it is legal only inside a cut off $\int_\varepsilon^R$, where the divergences appear as $\ln$ terms that cancel. The other trap is applying the formula when neither hypothesis holds: for $f(x) = \ln(1+x)$ we have $f(0) = 0$, so the naive answer is 0, but $f(\infty) = \infty$ and $\int^\infty \frac{f}{x}$ diverges, and in fact $\int_0^\infty \frac{\ln(1+ax) - \ln(1+bx)}{x} dx$ diverges because the integrand behaves like $\frac{\ln(a/b)}{x}$ at infinity.

SEE ALSO Choosing where to insert the parameter; epsilon regularisation of a divergent pair; Schwinger parametrisation.

When the limits move

So far the parameter has lived inside the integrand. It can also live in the limits, and then differentiation picks up boundary contributions. The general rule is Leibniz's, and its practical importance is less that it evaluates integrals directly than that it lets you build a function out of an integral, show its derivative is zero, and conclude that the integral is constant. That is a genuinely different mode of argument, and it produces the Gaussian integral out of nothing.

23.5 Leibniz rule with variable limits

MEDIUM

CUE Limits that depend on the differentiation variable, or a problem that asks for $\frac{d}{dx}$ of an integral, or an integral appearing inside a larger expression whose derivative you want.

METHOD

$$\frac{d}{dx} \int_{u(x)}^{v(x)} f(x, t) dt = f(x, v(x))v'(x) - f(x, u(x))u'(x) + \int_{u(x)}^{v(x)} \frac{\partial f}{\partial x} dt.$$

All three terms are needed whenever f carries the outer variable as well. Two of them are pure boundary bookkeeping; the third is differentiation under the sign and needs its own licence.

WORKED EXAMPLE

The Gaussian integral by a constancy argument. Define

$$\Phi(x) = \left(\int_0^x e^{-t^2} dt \right)^2 + \int_0^1 \frac{e^{-x^2(1+t^2)}}{1+t^2} dt.$$

The first piece has a variable upper limit and no x inside the integrand, so the chain rule and Leibniz give $2e^{-x^2} \int_0^x e^{-t^2} dt$. The second piece has fixed limits and x inside, so only the third Leibniz term survives:

$$\frac{d}{dx} \int_0^1 \frac{e^{-x^2(1+t^2)}}{1+t^2} dt = \int_0^1 \frac{-2x(1+t^2)e^{-x^2(1+t^2)}}{1+t^2} dt = -2xe^{-x^2} \int_0^1 e^{-x^2 t^2} dt,$$

and substituting $u = xt$ turns the last integral into $\frac{1}{x} \int_0^x e^{-u^2} du$. The two derivatives cancel exactly, so $\Phi' \equiv 0$ and Φ is constant. Evaluate at $x = 0$: the first term is 0 and the second is $\int_0^1 \frac{dt}{1+t^2} = \frac{\pi}{4}$. Let $x \rightarrow \infty$: the second term is dominated by e^{-x^2} and vanishes. Hence

$$\left(\int_0^\infty e^{-t^2} dt \right)^2 = \frac{\pi}{4}, \quad \int_0^\infty e^{-t^2} dt = \frac{\sqrt{\pi}}{2} \approx 0.886227.$$

No antiderivative was found, and no polar coordinates were used.

The second standard use is structural rather than numerical: Leibniz turns a repeated integral into a single one. If f is continuous, then

$$\frac{d}{dx} \int_0^x (x-t)f(t) dt = \underbrace{(x-x)f(x)}_{\text{boundary}} \cdot 1 + \int_0^x f(t) dt = \int_0^x f(t) dt,$$

the boundary term vanishing because the integrand itself vanishes at $t = x$. Iterating gives Cauchy's formula for repeated integration,

$$\underbrace{\int_0^x \int_0^{t_1} \cdots \int_0^{t_{n-1}} f(t_n) dt_n \cdots dt_1}_{n \text{ times}} = \frac{1}{(n-1)!} \int_0^x (x-t)^{n-1} f(t) dt,$$

which is the reason an n fold antiderivative is a single convolution and the starting point for fractional integration. Both facts come out of the three term rule and nothing else.

WHERE THIS GOES WRONG

Dropping the $\partial f / \partial x$ term because the limits were the interesting part. In the Gaussian argument that term is the entire content: without it Φ' is not zero and the argument collapses. A second and subtler failure is a moving limit that drags across a singularity of f , where the rule does not apply at all and the derivative can appear finite while the integral is not differentiable.

SEE ALSO Differentiation under the integral sign; the parameter ODE; perfect derivative disguise.

The constancy argument generalises into a method. Whenever a problem gives you an integral together with a relation it satisfies, differentiate the relation: an integral equation of the form $y(x) = g(x) + \int_0^x K(x, t)y(t) dt$ becomes, after one application of the three term rule, the differential equation $y'(x) = g'(x) + K(x, x)y(x) + \int_0^x \partial_x K y dt$, and when K does not

depend on x the integral term disappears entirely. That is the standard route from a Volterra equation to an initial value problem, and it is the same computation as the one that produced $\Phi' \equiv 0$ above, run in the other direction.

Parameters that live inside named functions

The last two techniques exploit the fact that some parameters are already installed, with their derivatives already tabulated. The Beta function is a two parameter family whose derivatives are logarithms; the Gamma integral representation is a one parameter family that converts denominators into exponentials. Both are parameter differentiation, but the hard analytic work has been done once and recorded in the digamma function, so what remains is bookkeeping and a table of special values.

HARD

23.6 Beta and Gamma parameter differentiation

CUE An integrand of Beta shape, $x^{p-1}(1-x)^{q-1}$ or $\sin^{2p-1}x \cos^{2q-1}x$, multiplied by $\ln x$, $\ln(1-x)$, $\ln \sin x$ or $\ln \cos x$. Logarithms of exactly those four kinds are what ∂_p and ∂_q produce, and nothing else does.

METHOD Start from

$$B(p, q) = \int_0^1 t^{p-1} (1-t)^{q-1} dt = 2 \int_0^{\pi/2} \sin^{2p-1} x \cos^{2q-1} x dx = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}.$$

Then ∂_p inserts $\ln t$ in the algebraic form and $2 \ln \sin x$ in the trigonometric one, and

$$\partial_p \partial_q B = B \{ [\psi(p) - \psi(p+q)] [\psi(q) - \psi(p+q)] - \psi'(p+q) \}.$$

WORKED EXAMPLE

$$J = \int_0^{\pi/2} \frac{\ln \sin x \ln \cos x}{\sqrt{\tan x}} dx.$$

Write $\tan^{-1/2} x = \sin^{-1/2} x \cos^{1/2} x$, so $2p-1 = -\frac{1}{2}$ and $2q-1 = \frac{1}{2}$, giving $p = \frac{1}{4}$, $q = \frac{3}{4}$ and $p+q = 1$. Since $\partial_p \partial_q$ applied to $2 \int \sin^{2p-1} \cos^{2q-1}$ inserts $4 \ln \sin x \ln \cos x$, we have $J = \frac{1}{8} \partial_p \partial_q B$ at that point. Now $B(\frac{1}{4}, \frac{3}{4}) = \Gamma(\frac{1}{4})\Gamma(\frac{3}{4}) = \frac{\pi}{\sin(\pi/4)} = \pi\sqrt{2}$ by Euler reflection, and with $\psi(1) = -\gamma$, $\psi(\frac{1}{4}) = -\gamma - \frac{\pi}{2} - 3 \ln 2$, $\psi(\frac{3}{4}) = -\gamma + \frac{\pi}{2} - 3 \ln 2$,

$$[\psi(\frac{1}{4}) - \psi(1)][\psi(\frac{3}{4}) - \psi(1)] = (-\frac{\pi}{2} - 3 \ln 2)(\frac{\pi}{2} - 3 \ln 2) = 9 \ln^2 2 - \frac{\pi^2}{4}.$$

Subtracting $\psi'(1) = \frac{\pi^2}{6}$ and multiplying by $B/8$,

$$J = \frac{\pi\sqrt{2}}{8} \left(9 \ln^2 2 - \frac{5\pi^2}{12} \right) \approx 0.117593,$$

which quadrature confirms to fifteen places.

The algebraic form of the Beta function is used the same way and is often simpler, because the logarithms it produces are $\ln t$ and $\ln(1-t)$ rather than $\ln \sin$ and $\ln \cos$. A single derivative gives

$$\int_0^1 t^{p-1} (1-t)^{q-1} \ln t dt = B(p, q) [\psi(p) - \psi(p+q)],$$

so, for instance, $\int_0^1 \frac{\ln t}{\sqrt{t(1-t)}} dt = B(\frac{1}{2}, \frac{1}{2}) [\psi(\frac{1}{2}) - \psi(1)] = \pi(-2 \ln 2) = -2\pi \ln 2 \approx -4.355172$, which quadrature confirms.

The corresponding second derivative in the same variable inserts $\ln^2 t$ and brings in ψ' , and the mixed derivative is the formula quoted above. Three derivatives is where the method stops being pleasant and the polygamma bookkeeping starts to dominate; past that point, expand instead.

THE SPECIAL VALUES YOU NEED

Parameter differentiation of B always lands on ψ at the arguments you started from, so the table is short. $\psi(1) = -\gamma$, $\psi(\frac{1}{2}) = -\gamma - 2 \ln 2$, $\psi(\frac{1}{4}) = -\gamma - \frac{\pi}{2} - 3 \ln 2$, $\psi(\frac{3}{4}) = -\gamma + \frac{\pi}{2} - 3 \ln 2$, and $\psi(n) = -\gamma + H_{n-1}$. For the trigamma, $\psi'(1) = \frac{\pi^2}{6}$, $\psi'(\frac{1}{2}) = \frac{\pi^2}{2}$. Two structural facts generate the rest: the reflection $\psi(1-z) - \psi(z) = \pi \cot \pi z$, and the recurrence $\psi(z+1) = \psi(z) + \frac{1}{z}$. Notice that γ cancels out of every difference $\psi(p) - \psi(p+q)$, which is why final answers contain π and $\ln 2$ but not γ .

The factor of 2 in the trigonometric Beta form propagates into every derivative, so a double derivative carries 4 and the prefactor is $\frac{1}{8}$, not $\frac{1}{2}$. The other reliable slip is sign: $\psi(\frac{1}{4}) - \psi(1)$ is negative and $\psi(\frac{3}{4}) - \psi(1)$ is negative too, since $3 \ln 2 > \frac{\pi}{2}$, so their product is positive and the $-\psi'(p+q)$ term is what makes the bracket small.

SEE ALSO The Beta function; digamma and polygamma values; differentiation under the integral sign.

The final technique runs the machinery backwards. Instead of differentiating with respect to a parameter, you introduce an integral over one, chosen so that the new double integral factorises after the order is swapped. It costs you a dimension and buys you a product structure, which is a good trade whenever the obstruction is a denominator that refuses to split. It is also the bridge to the next chapter, where swapping orders becomes the main event.

HARD

23.7 Schwinger parametrisation

CUE A product of denominators, a denominator raised to a non-integer power, or a $\ln x$ or $1/x$ factor blocking a transform you would otherwise recognise.

METHOD Replace the obstruction by an integral and exchange the order. The four representations worth carrying are

$$\frac{1}{A^s} = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} e^{-At} dt, \quad \frac{1}{AB} = \int_0^1 \frac{dt}{[tA + (1-t)B]^2},$$

$$\frac{1}{x} = \int_0^\infty e^{-xt} dt, \quad \ln x = \int_0^\infty \frac{e^{-t} - e^{-xt}}{t} dt,$$

the last of which is Frullani read from right to left.

WORKED EXAMPLE

$$\int_0^\infty \frac{dx}{(1+x^2)^2}.$$

Apply the first representation with $A = 1 + x^2$ and $s = 2$, so $\frac{1}{(1+x^2)^2} = \int_0^\infty t e^{-(1+x^2)t} dt$. The double integrand is positive, so Tonelli licenses the swap unconditionally:

$$\int_0^\infty \int_0^\infty t e^{-t} e^{-tx^2} dt dx = \int_0^\infty t e^{-t} \left(\frac{1}{2} \sqrt{\frac{\pi}{t}} \right) dt = \frac{\sqrt{\pi}}{2} \int_0^\infty t^{1/2} e^{-t} dt = \frac{\sqrt{\pi}}{2} \Gamma\left(\frac{3}{2}\right) = \frac{\pi}{4}.$$

The point is structural: the representation turned a rational function of x into a Gaussian in x and a Gamma in t , and each factor was then standard.

The second representation handles products of unrelated denominators. With $A = x^2 + a^2$ and $B = x^2 + b^2$, the combined denominator is $x^2 + M^2$ with $M^2 = ta^2 + (1-t)b^2$, and $\int_{-\infty}^\infty \frac{dx}{(x^2 + M^2)^2} = \frac{\pi}{2M^3}$. Hence

$$\int_{-\infty}^\infty \frac{dx}{(x^2 + a^2)(x^2 + b^2)} = \frac{\pi}{2} \int_0^1 \frac{dt}{[ta^2 + (1-t)b^2]^{3/2}} = \frac{\pi}{2(a^2 - b^2)} \left[\frac{2}{b} - \frac{2}{a} \right] = \frac{\pi}{ab(a+b)},$$

and at $a = 2, b = 3$ this is $\pi/30 \approx 0.104720$, which quadrature confirms. Partial fractions would also work here; the parametrisation is what survives when the powers are not integers.

The same representation proves a formula that would otherwise need a substitution and a Beta table. For $0 < s < c$,

$$\int_0^\infty \frac{x^{s-1}}{(1+x)^c} dx = \frac{1}{\Gamma(c)} \int_0^\infty \int_0^\infty x^{s-1} t^{c-1} e^{-t} e^{-xt} dt dx = \frac{\Gamma(s)}{\Gamma(c)} \int_0^\infty t^{c-s-1} e^{-t} dt = B(s, c-s),$$

using $\int_0^\infty x^{s-1} e^{-xt} dx = \Gamma(s) t^{-s}$ in the middle step, and Tonelli throughout because everything is positive. At $s = \frac{1}{2}$, $c = 2$ this gives $B(\frac{1}{2}, \frac{3}{2}) = \frac{\pi}{2}$, and at $s = 0.7, c = 2.5$ it gives 0.909464, both confirmed numerically.

WHERE THIS GOES WRONG

Swapping the order because it is convenient. Tonelli applies only to a nonnegative integrand; for a signed one you owe an absolute integrability check, and if the double integral is only conditionally convergent the two orders can differ. The second trap is using $\frac{1}{A^s} = \frac{1}{\Gamma(s)} \int t^{s-1} e^{-At} dt$ when A is not positive, where the t integral diverges outright.

WHICH REPRESENTATION TO REACH FOR

Match the representation to the obstruction, not to the integrand. A denominator raised to an awkward power wants the Gamma representation, because it converts the power into a factor t^{s-1} that the outer integral can absorb. A product of two unrelated denominators wants the Feynman representation, because it fuses them into one denominator at the cost of a single extra variable.

A bare $1/x$ or $\ln x$ wants the exponential representations, because those trade a singular factor for an exponential that most integrands already like. In every case the test is the same: after the swap, is the inner integral a standard one? If not, you have added a dimension for nothing.

SEE ALSO Frullani's integral; justifying the swap; double integral and double series.

When the family closes on itself

Earlier, the family $\int_0^1 \frac{\arctan(ax)}{x} dx$ reproduced itself and the protocol returned nothing. Self reference is not always fatal; sometimes it is the solution, because a linear equation in the unknown is as good as a formula. Take

$$I(a) = \int_0^1 \frac{\ln(1+ax)}{1+x^2} dx, \quad \text{target } I(1).$$

Differentiating and decomposing $\frac{x}{(1+ax)(1+x^2)} = \frac{1}{1+a^2} \left[\frac{-a}{1+ax} + \frac{x+a}{1+x^2} \right]$,

$$I'(a) = \frac{1}{1+a^2} \left[-\ln(1+a) + \frac{\ln 2}{2} + \frac{\pi a}{4} \right].$$

Now integrate from 0 to 1, using $I(0) = 0$ and the two elementary values $\int_0^1 \frac{da}{1+a^2} = \frac{\pi}{4}$ and $\int_0^1 \frac{a da}{1+a^2} = \frac{\ln 2}{2}$:

$$I(1) = -\int_0^1 \frac{\ln(1+a)}{1+a^2} da + \frac{\ln 2}{2} \cdot \frac{\pi}{4} + \frac{\pi}{4} \cdot \frac{\ln 2}{2}.$$

The first term on the right is $-I(1)$, since the integration variable is a dummy. Hence $2I(1) = \frac{\pi \ln 2}{4}$ and

$$\int_0^1 \frac{\ln(1+x)}{1+x^2} dx = \frac{\pi \ln 2}{8} \approx 0.272198,$$

confirmed numerically. The difference between this and the Catalan failure is exactly the sign with which the family reappears: a coefficient of -1 gives a solvable equation, a coefficient of $+1$ gives the tautology $I = I$. Always finish the calculation far enough to see which one you have.

The seven techniques of this chapter are one technique seen from seven angles. In each, an integral is embedded in a family, and the family is easier because it has a derivative. Feynman differentiates the family; the parameter ODE differentiates it and recognises the result as the family again; Frullani is the special case in which the differentiation has already been done in general; Leibniz extends the derivative to moving limits; Beta differentiation uses a family whose derivatives are tabulated; and Schwinger builds the family by integration instead of finding it by inspection. What follows in the next chapter is the same idea with a sum in place of the parameter integral.

Two limits of the method are worth stating plainly. Parameter differentiation cannot help when the integral is invariant along every direction you can insert a parameter, as the Catalan example above showed; the symptom is an identity rather than a value, and the cure is a different chapter. And it cannot help when $I'(a)$ has no closed form, which is the usual situation for integrands built from more than one transcendental factor. In that case the honest move is to stop after step 3 and read $I'(a)$ as a definition of a new function, which is how the polylogarithms and the elliptic integrals were discovered in the first place.

CONFIRM BEFORE YOU COMMIT

Every closed form in this chapter is a two line numerical check: evaluate the family at two or three values of the parameter, compare against the formula, and only then believe it. The characteristic error in parameter work is not a wrong method but a lost constant of integration, and a lost constant shows up as an offset that is the same at every a . If your formula is wrong by a quantity that does not vary with a , the mistake is at step 6; if it is wrong by a quantity that does vary, the mistake is at step 3. Two evaluations distinguish the two cases in less time than rereading the derivation.

Recognition matrix

If you see	Reach for	If that fails
A constant that could be a letter $x^a - x^b$ or any $\ln x$ downstairs $\frac{\ln x}{\sin x}, \frac{1 - \cos x}{x^2}$ on $(0, \infty)$	Differentiate under the sign; anchor where it is trivial Parameter in the exponent, $\partial_a x^a = x^a \ln x$ Damping factor e^{-ax} , then let $a \rightarrow 0^+$	Insert a second parameter $\ln x = \int_0^\infty \frac{e^{-t} - e^{-xt}}{t} dt$ Justify $a \rightarrow 0^+$ separately

If you see	Reach for	If that fails
$\arctan(ax)$ or $\ln(1+ax)$ over x $I'(a)$ reproduces $I(a)$ after parts $e^{-x^2} \cos(tx)$ or $e^{-x^2} \cosh(tx)$ $\frac{f(ax)-f(bx)}{x}$ on $(0, \infty)$ A pair that each diverge logarithmically Limits depending on x An n fold repeated integral $x^{p-1}(1-x)^{q-1}$ times a logarithm $\sin^m x \cos^n x$ times $\ln \sin x$ A product of two denominators $(1+x^2)^{-s}$ with s not an integer An answer that is off by a constant	Differentiate in a ; the result is rational Solve the first order ODE in a $F' = \mp \frac{t}{2} F$, anchor at $F(0) = \frac{\sqrt{\pi}}{2}$ Frullani, $[f(0) - f(\infty)] \ln \frac{b}{a}$ Cut off at ε and R , cancel, then take limits Leibniz with all three terms Cauchy: $\frac{1}{(n-1)!} \int_0^x (x-t)^{n-1} f$ ∂_p or ∂_q of $B(p, q)$ Trigonometric Beta; remember the $\frac{1}{8}$ $\frac{1}{AB} = \int_0^1 \frac{dt}{[tA+(1-t)B]^2}$ $\frac{1}{A^s} = \frac{1}{\Gamma(s)} \int t^{s-1} e^{-At} dt$ Recheck the anchor, step 6	Frullani, if it is a difference Check the boundary term first Complete the square in the exponent Check the second hypothesis, $\int^\infty f/x$ Frullani with the Cesaro reading Build Φ and show $\Phi' \equiv 0$ Differentiate to check Digamma values at quarter integers Reflection $x \mapsto \frac{\pi}{2} - x$ Partial fractions in x^2 Substitute $x = \tan \theta$, then Beta Recompute I' at one numeric a

Series and interchange

What this chapter is for. When no antiderivative exists, expand. A geometric expansion of $1/(1 \pm x)$, a binomial expansion, or a plain Maclaurin series converts one impossible integral into infinitely many possible ones, and the sum of those is the answer. This is why the closed forms in hard definite integration are so often $\zeta(2)$, $\zeta(3)$, $\ln 2$, Catalan's constant G , or a polylogarithm: those are simply what $\sum n^{-s}$ and its relatives are called. The chapter gives the expand and integrate termwise protocol, the small dictionary of moments that lets you do the general term, the dictionary of constants that lets you recognise the sum, and, at equal length, the theorems that license the interchange. That last part is not ceremony. There is a worked example here where an unjustified swap produces a finite answer of the wrong sign, and another where two orders of a double integral give $\pi/4$ and $-\pi/4$.

Expand and integrate termwise

The strategy is a trade. You give up a closed form for the integrand and accept an infinite sum, betting that each individual term is elementary and that the sum is a constant you can name. The bet pays off surprisingly often, because the integrands that resist antidifferentiation are exactly the ones built out of $\ln$, $\arctan$ and $1/(1 \pm x^n)$, and those have clean series with rational coefficients.

The protocol has five steps, and the fifth is the one that matters.

THE FIVE STEPS

1. Identify the factor to expand and the region where the expansion is valid.
2. Expand it, keeping the index bookkeeping explicit.
3. Integrate the general term in closed form. Almost always this is the log power moment $\int_0^1 x^a \ln^b x \, dx$ or the Gamma moment $\int_0^\infty x^a e^{-nx} \, dx$.
4. Sum the resulting series and name the constant.
5. Justify the interchange $\int \sum = \sum \int$ by monotone convergence, dominated convergence, or Tonelli, and check the answer numerically.

There is a structural reason why the constants come out the way they do. Termwise integration on $[0, 1]$ against a power x^n produces $\frac{1}{n+1}$, and against $x^n \ln^b x$ produces $\frac{b!}{(n+1)^{b+1}}$ up to sign. So a single expansion delivers $\sum n^{-1}$ shapes, an expansion with one logarithm present delivers $\sum n^{-2}$, and two logarithms deliver $\sum n^{-3}$. Every logarithm in the integrand is worth one unit of weight in the answer. Counting logarithms before starting tells you which zeta value to expect, and an answer of the wrong weight is a sign of an arithmetic slip rather than a surprising identity.

The same accounting works on the half line, where the moment is $\int_0^\infty x^{s-1} e^{-nx} \, dx = \Gamma(s)n^{-s}$: the power of x in the integrand sets the weight directly. This is why $\int_0^\infty \frac{x^3}{e^x - 1} \, dx$ is a $\zeta(4)$ problem and could not have been anything else.

Geometric expansion is where to start, because it is the one expansion whose terms are all of one sign on the natural domain, which makes step 5 free.

24.1 Geometric expansion

CORE

CUE $\frac{1}{1 \pm x^n}$ on $[0, 1]$, or $\frac{1}{1 - (xy)^a}$ on the unit square, or a Bose factor $\frac{1}{e^x - 1}$, $\frac{1}{e^x + 1}$, $\frac{1}{\sinh^2 x}$ on $(0, \infty)$. Any denominator that is 1 minus something small.

METHOD Use $\frac{1}{1-u} = \sum_{k \geq 0} u^k$ for $|u| < 1$. On the half line the useful forms are

$$\frac{1}{e^{cx} - 1} = \sum_{n \geq 1} e^{-cnx}, \quad \frac{1}{e^{cx} + 1} = \sum_{n \geq 1} (-1)^{n-1} e^{-cnx}, \quad \frac{1}{\sinh^2 u} = 4 \sum_{n \geq 1} n e^{-2nu}.$$

Integrate termwise. On $[0, 1]$ each term is a power, on $(0, \infty)$ each term is a Gamma moment $\int_0^\infty x^{s-1} e^{-nx} \, dx = \Gamma(s)n^{-s}$.

Two instances, one on each domain. First,

$$\int_0^1 \frac{-\ln(1-x)}{x} dx = \int_0^1 \sum_{n \geq 1} \frac{x^{n-1}}{n} dx = \sum_{n \geq 1} \frac{1}{n} \cdot \frac{1}{n} = \zeta(2) = \frac{\pi^2}{6} \approx 1.644934.$$

Every term is nonnegative on $[0, 1]$, so monotone convergence licenses the swap outright; no epsilon argument is needed.

Second, the Bose integral. Expanding $\frac{1}{e^x - 1} = \sum_{n \geq 1} e^{-nx}$ and using $\int_0^\infty x^3 e^{-nx} dx = \frac{6}{n^4}$,

$$\int_0^\infty \frac{x^3}{e^x - 1} dx = 6 \sum_{n \geq 1} \frac{1}{n^4} = 6 \zeta(4) = \frac{\pi^4}{15} \approx 6.493939.$$

The general statement is $\int_0^\infty \frac{x^{s-1}}{e^x - 1} dx = \Gamma(s) \zeta(s)$, and with the alternating expansion $\int_0^\infty \frac{x^{s-1}}{e^x + 1} dx = \Gamma(s) \eta(s)$; at $s = 2$ the latter gives $\eta(2) = \frac{\pi^2}{12} \approx 0.822467$.

The hyperbolic expansions are the same idea after one algebraic step. From $\frac{1}{\sinh u} = \frac{2e^{-u}}{1 - e^{-2u}} = 2 \sum_{k \geq 0} e^{-(2k+1)u}$ and $\int_0^\infty u e^{-(2k+1)u} du = (2k+1)^{-2}$,

$$\int_0^\infty \frac{u}{\sinh u} du = 2 \sum_{k \geq 0} \frac{1}{(2k+1)^2} = 2 \cdot \frac{3}{4} \zeta(2) = \frac{\pi^2}{4} \approx 2.467401,$$

and differentiating the same expansion in u gives $\frac{1}{\sinh^2 u} = 4 \sum_{n \geq 1} n e^{-2nu}$, whence $\int_0^\infty \frac{u^2}{\sinh^2 u} du = 4 \sum_{n \geq 1} n \cdot \frac{1}{4n^3} = \zeta(2)$. Both are confirmed numerically.

WHERE THIS GOES WRONG

The expansion fails at the endpoint $|u| = 1$, and on $[0, 1]$ that endpoint is in the domain. When the terms are all nonnegative this costs nothing, because monotone convergence does not care about a single point. When they alternate it can cost everything, and the correct route is to work on $[0, 1 - \varepsilon]$, bound the tail, and let $\varepsilon \rightarrow 0$.

THE THREE MOMENTS

Termwise integration is only as good as your ability to do the general term, and in practice three moments cover almost everything. On $[0, 1]$, $\int_0^1 x^a \ln^b x dx = \frac{(-1)^b b!}{(a+1)^{b+1}}$. On $(0, \infty)$, $\int_0^\infty x^{s-1} e^{-nx} dx = \Gamma(s) n^{-s}$. On $[0, \frac{\pi}{2}]$, $2 \int_0^{\pi/2} \sin^{2p-1} x \cos^{2q-1} x dx = B(p, q)$. Before expanding, decide which of the three your general term will be; if it is none of them, the expansion is probably the wrong move and a substitution should come first.

SEE ALSO Termwise integration of a Maclaurin series; the log power moment; justifying the swap.

Geometric expansion handles denominators. For everything else the expandable factor is a named Maclaurin series, and the difficulty moves from convergence to bookkeeping: matching the index of the series against the index of the resulting sum.

24.2 Termwise integration of a Maclaurin series

MEDIUM

CUE An integrand with no accessible antiderivative but a standard series: $\arctan x$, $\arcsin x$, $\ln(1+x)$, $e^{x \ln x}$, or a Cauchy product such as $(\arctan x)^2$.

METHOD Expand, integrate each power against whatever else is present, and recognise the sum. The self-referential case is worth isolating: $x^x = e^{x \ln x} = \sum_{k \geq 0} \frac{(x \ln x)^k}{k!}$ turns a transcendental integrand into log power moments.

WORKED EXAMPLE

The Sophomore's dream. Expanding and applying $\int_0^1 x^k \ln^k x dx = \frac{(-1)^k k!}{(k+1)^{k+1}}$,

$$\int_0^1 x^x dx = \sum_{k \geq 0} \frac{1}{k!} \cdot \frac{(-1)^k k!}{(k+1)^{k+1}} = \sum_{k \geq 0} \frac{(-1)^k}{(k+1)^{k+1}} = \sum_{n \geq 1} \frac{(-1)^{n-1}}{n^n} \approx 0.783431,$$

and the same computation with x^{-x} flips every sign, giving $\int_0^1 x^{-x} dx = \sum_{n \geq 1} n^{-n} \approx 1.291286$. The reindexing $n = k + 1$ is the only place to go wrong, and it is where most people do.

A second instance, with Catalan's constant as the payoff:

$$\int_0^1 \frac{\arctan x}{x} dx = \sum_{k \geq 0} \frac{(-1)^k}{2k+1} \int_0^1 x^{2k} dx = \sum_{k \geq 0} \frac{(-1)^k}{(2k+1)^2} = G \approx 0.915966.$$

Here the terms alternate, so monotone convergence is unavailable; the licence comes from the alternating series bound $|\sum_{k \leq K} \frac{(-1)^k x^{2k}}{2k+1}| \leq 1$ on $[0, 1]$ uniformly in K , which is an integrable dominating function.

A binomial expansion works the same way, with $\binom{2n}{n}$ in place of a factorial. From $(1-u)^{-1/2} = \sum_{n \geq 0} \binom{2n}{n} \frac{u^n}{4^n}$ one gets $\arcsin x = \sum_{n \geq 0} \frac{\binom{2n}{n}}{4^n (2n+1)} x^{2n+1}$, so

$$\int_0^1 \frac{\arcsin x}{x} dx = \sum_{n \geq 0} \frac{\binom{2n}{n}}{4^n (2n+1)^2} = \frac{\pi}{2} \ln 2 \approx 1.088793.$$

The closed form is easier to obtain by parts, since $\int_0^1 \frac{\arcsin x}{x} dx = -\int_0^1 \frac{\ln x}{\sqrt{1-x^2}} dx$ and the right side is a Beta derivative, but the series is the check: summing forty terms reproduces 1.08879. That pairing is typical, and it is the practical reason to know both routes.

THE SAME DREAM ON THE OTHER DOMAIN

The Sophomore's dream can also be done with the Gamma moment, and comparing the two routes is a good test of whether the machinery is understood. Substituting $x = e^{-u}$ carries $[0, 1]$ to $(0, \infty)$ and turns $x^x dx$ into $e^{-ue^{-u}} e^{-u} du$. Expanding the outer exponential and using $\int_0^\infty u^k e^{-(k+1)u} du = \frac{k!}{(k+1)^{k+1}}$,

$$\int_0^1 x^x dx = \sum_{k \geq 0} \frac{(-1)^k}{k!} \cdot \frac{k!}{(k+1)^{k+1}} = \sum_{k \geq 0} \frac{(-1)^k}{(k+1)^{k+1}},$$

the same series with the same index shift. The factorials cancel in both routes, one against the log power moment and one against the Gamma moment, which is not a coincidence: the substitution $x = e^{-u}$ is precisely what carries one moment into the other. The series itself converges ferociously, since n^{-n} beats any geometric rate: six terms already give 0.7834305, correct to seven places.

WHERE THIS GOES WRONG

Mis-indexing when the k th series term produces the $(k+1)$ th sum term, as in the Sophomore's dream. Write both indices down explicitly rather than trusting the pattern. The second trap is expanding a factor whose series only converges on part of the interval of integration and then integrating over all of it.

SEE ALSO Geometric expansion; the log power moment; recognising the emergent constant.

Expansions on the half line

On $(0, \infty)$ the expandable object is almost always a function of e^{-x} , and the moment is the Gamma moment. Two immediate consequences, each one line:

$$\int_0^\infty \ln(1 - e^{-x}) dx = -\sum_{n \geq 1} \frac{1}{n} \int_0^\infty e^{-nx} dx = -\zeta(2) = -\frac{\pi^2}{6},$$

$$\int_0^\infty \ln(1 + e^{-x}) dx = \sum_{n \geq 1} \frac{(-1)^{n-1}}{n^2} = \eta(2) = \frac{\pi^2}{12} \approx 0.822467.$$

The first has terms of one sign and needs nothing; the second alternates and needs the uniform bound on partial sums, which the alternating series estimate supplies. Raising the power of x raises the weight in the expected way: $\int_0^\infty \frac{x^2}{e^x + 1} dx = \Gamma(3)\eta(3) = 2 \cdot \frac{3}{4}\zeta(3) = \frac{3}{2}\zeta(3) \approx 1.803085$.

A useful habit on this domain is to substitute $u = e^{-x}$ before doing anything else. It carries $(0, \infty)$ to $(0, 1)$, turns dx into $-\frac{du}{u}$, and converts a Bose factor into $\frac{1}{1-u}$, so a half line problem becomes a unit interval problem and the Gamma moment becomes the log power moment. Whichever domain you find easier, one substitution takes you there.

WHEN NOT TO EXPAND

Expansion is not free, and three signs say to try something else first. If the integrand has a symmetry, use it: $\int_0^{\pi/2} \frac{dx}{1 + \tan^{\sqrt{2}} x}$ falls to the reflection $x \mapsto \frac{\pi}{2} - x$ in one line and to no series at all. If a substitution makes the integrand rational, make it: partial fractions give an exact answer where a series gives an approximation you then have to recognise. And if the expansion produces a general

term you cannot integrate, stop; a series whose terms are as hard as the original problem has cost you a convergence obligation and bought you nothing.

The moment that does the work

Every termwise expansion on $[0, 1]$ eventually asks for the integral of a power times a power of a logarithm. That one formula is the engine of the whole chapter, and it is worth knowing not as a lookup but as a derivation, because the derivation tells you the sign.

24.3 The log power moment

EASY

CUE $x^a \ln^b x$ anywhere on $[0, 1]$, typically appearing after a series expansion has been performed and the general term written out.

METHOD

$$\int_0^1 x^a \ln^b x \, dx = \frac{(-1)^b b!}{(a+1)^{b+1}}, \quad a > -1, b \in \mathbb{N}_0.$$

The fastest derivation is parameter differentiation: $\int_0^1 x^a \, dx = \frac{1}{a+1}$, and ∂_a^b inserts $\ln^b x$ on the left and produces $\frac{(-1)^b b!}{(a+1)^{b+1}}$ on the right. The sign is therefore not something to memorise; it is b differentiations of a reciprocal.

WORKED EXAMPLE

Directly: $\int_0^1 x^2 \ln^3 x \, dx = \frac{(-1)^3 3!}{3^4} = -\frac{6}{81} = -\frac{2}{27} \approx -0.074074$.

In use, with $b = 2$ and $a = 2n$, the moment is $\frac{2}{(2n+1)^3}$, which is the engine behind every $\zeta(3)$ that appears in this subject:

$$\int_0^1 \frac{\ln^2 x}{1-x^2} \, dx = \sum_{n \geq 0} \int_0^1 x^{2n} \ln^2 x \, dx = 2 \sum_{n \geq 0} \frac{1}{(2n+1)^3} = 2 \cdot \frac{7}{8} \zeta(3) = \frac{7}{4} \zeta(3) \approx 2.103600.$$

The factor $\frac{7}{8}$ is the odd index part of $\zeta(3)$, namely $(1 - 2^{-3})\zeta(3)$. All terms are nonnegative, so the swap is licensed by monotone convergence.

The same moment against the three standard denominators gives a small table worth carrying:

$$\int_0^1 \frac{\ln^2 x}{1-x} \, dx = 2\zeta(3), \quad \int_0^1 \frac{\ln^2 x}{1+x} \, dx = 2\eta(3) = \frac{3}{2}\zeta(3), \quad \int_0^1 \frac{\ln^2 x}{1+x^2} \, dx = 2\beta(3) = \frac{\pi^3}{16},$$

with numerical values 2.404114, 1.803085 and 1.937892. The three differ only in which subfamily of integers the sum runs over, and reading off the right one is the entire difficulty.

WHERE THIS GOES WRONG

Applying the formula with $a \leq -1$, where the integral diverges: $\int_0^1 x^{-1} \ln x \, dx$ is not -1 , it is infinite. And on $[0, 1]$ the logarithm is negative, so an odd b gives a negative answer; a positive result for odd b is a sign error, not a surprise.

THE WEIGHT LADDER

Each extra logarithm in the integrand raises the weight of the answer by one. Weight 1 answers are $\ln 2$ and $\frac{\pi}{4}$; weight 2 answers are $\zeta(2)$, $\eta(2)$, G and $\ln^2 2$; weight 3 answers are $\zeta(3)$, $\beta(3)$, $\pi^2 \ln 2$ and $\ln^3 2$; weight 4 answers are $\zeta(4)$, $\text{Li}_4(\frac{1}{2})$ and their companions. The rule is not a theorem, but it is a reliable audit: an integral with two logarithms whose reported answer is a weight 2 constant has lost a factor somewhere, and one whose answer mixes weights is almost certainly wrong, since the pieces of a genuine evaluation are homogeneous.

SEE ALSO Termwise integration of a Maclaurin series; geometric expansion; recognising the emergent constant.

Justifying the swap

Everything above contained a step of the form $\int \sum = \sum \int$. That step is a theorem, not a notation, and it is false in general. This section states the three usable theorems and then does something more useful: it exhibits two concrete failures, one for a series and one for a double integral, in which the unjustified interchange returns a perfectly finite number that is simply wrong.

It helps to know what the theorems are actually saying. All three are statements about a double integral over a product space, one of whose factors may be a counting measure on the integers, in which case “integral” means “sum”. Tonelli says that for a nonnegative integrand the double integral and both iterated integrals are the same extended real number, with no finiteness hypothesis at all. Fubini says that if the double integral of the absolute value is finite, the same conclusion holds for the signed integrand. Dominated convergence is the version for a limit of partial sums rather than a sum, and it is what you use when the natural object is $\lim_N \int \sum_{n \leq N}$. In every case the hypothesis is checked on $|f|$, never on f , and that is the whole content.

CORE

24.4 Justifying the swap

CUE Any step that moves an integral sign past a summation sign, or that reverses the order of two integrals. If you wrote one, you owe a reason.

METHOD Three tools, in increasing cost. *Monotone convergence*: if $f_n \geq 0$, then $\int \sum f_n = \sum \int f_n$ unconditionally, both sides possibly infinite. *Tonelli*: if the double integrand is nonnegative and measurable, both iterated integrals agree with the double integral, again unconditionally. *Fubini and dominated convergence*, for signed integrands: you must first show $\sum \int |f_n| < \infty$, or exhibit an integrable g with $|\sum_{n \leq N} f_n| \leq g$ for every N . There is no fourth tool, and there is no version in which the swap is automatic.

WORKED EXAMPLE

A series where the swap fails and gives a finite wrong answer. On $[0, 1]$ put $g_n(x) = n^2 x e^{-n^2 x^2}$ and $f_n = g_n - g_{n+1}$. For each fixed $x > 0$ the partial sums telescope, $\sum_{n=1}^N f_n(x) = x e^{-x^2} - (N+1)^2 x e^{-(N+1)^2 x^2} \rightarrow x e^{-x^2}$, so

$$\int_0^1 \sum_{n \geq 1} f_n(x) dx = \int_0^1 x e^{-x^2} dx = \frac{1 - e^{-1}}{2} \approx 0.316060.$$

Now integrate first. Since $\int_0^1 g_n = \frac{1}{2}(1 - e^{-n^2})$, we get $\int_0^1 f_n = \frac{1}{2}(e^{-(n+1)^2} - e^{-n^2})$, and that telescopes to

$$\sum_{n \geq 1} \int_0^1 f_n(x) dx = -\frac{e^{-1}}{2} \approx -0.183940.$$

The two answers differ, and not by a little: they differ in sign. Numerically both values are exactly reproduced by quadrature. What went wrong is visible: g_n is a spike of fixed mass $\frac{1}{2}$ that migrates to $x = 0$, so no integrable g dominates the partial sums and dominated convergence does not apply.

The same phenomenon for a double integral. With $f(x, y) = \frac{x^2 - y^2}{(x^2 + y^2)^2} = \partial_y \frac{y}{x^2 + y^2}$ on the unit square,

$$\int_0^1 \left(\int_0^1 f dy \right) dx = \int_0^1 \frac{dx}{1+x^2} = \frac{\pi}{4}, \quad \int_0^1 \left(\int_0^1 f dx \right) dy = -\int_0^1 \frac{dy}{1+y^2} = -\frac{\pi}{4},$$

the second following from the first by the antisymmetry $f(y, x) = -f(x, y)$. Both iterated integrals exist and are finite; they disagree because $\iint |f| = \infty$, so Fubini never applied.

The discrete version is even shorter, and worth carrying because it is impossible to misremember. Let $a_{nn} = 1$, $a_{n,n+1} = -1$, and $a_{mn} = 0$ otherwise. Summing along rows, every row sums to $1 - 1 = 0$, so $\sum_m \sum_n a_{mn} = 0$. Summing along columns, column 1 contains only $a_{11} = 1$ and every later column contains $a_{nn} = 1$ and $a_{n-1,n} = -1$, so $\sum_n \sum_m a_{mn} = 1$. Two orders, two answers, and the array is as simple as an array can be.

WHAT TO ACTUALLY CHECK

In practice you run one of three checks and record which. If the terms have one sign on the domain, write “all terms nonnegative, monotone convergence” and move on. If they alternate, look for a uniform bound on the partial sums, which for an alternating series with decreasing terms is just the first term. If neither holds, compute $\sum \int |f_n|$; if it diverges, do not swap, and instead work on a subinterval where the tail can be estimated by hand.

ABEL'S THEOREM IS NOT AN INTERCHANGE THEOREM

Abel's theorem says that if $\sum a_n$ converges then $\sum a_n x^n \rightarrow \sum a_n$ as $x \rightarrow 1^-$. It is constantly invoked in this subject and it is constantly misused, because it says nothing whatever about exchanging an integral with a sum. What it does is repair the endpoint after a correct interchange on $[0, 1 - \varepsilon]$: you prove the identity on the open interval, where uniform convergence on compact subsets is available, and then let $x \rightarrow 1^-$ on both sides, using Abel on the series side and dominated or monotone convergence on the integral side. Two theorems, two jobs. Citing Abel in place of the interchange itself is the most common piece of hand waving in the literature on these integrals.

Treating a convergent answer as evidence that the swap was legal. Both failures above return finite numbers. The only evidence is a theorem, or a numerical check of the original integral against the summed answer, which is why step 5 of the protocol ends with a numerical comparison.

SEE ALSO Geometric expansion; double integral and double series; convergence tests for series.

When the terms alternate and the endpoint is in the domain, there is a technique that never needs a convergence theorem at all. Write the geometric series as a finite sum plus an exact remainder,

$$\frac{1}{1+x} = \sum_{k=0}^{N-1} (-x)^k + \frac{(-x)^N}{1+x},$$

integrate the identity, which is legal because it is a statement about finitely many terms, and bound the remainder: $|\int_0^1 \frac{(-x)^N}{1+x} dx| \leq \int_0^1 x^N dx = \frac{1}{N+1} \rightarrow 0$. That gives $\sum_{k \geq 0} \frac{(-1)^k}{k+1} = \ln 2$ with no appeal to dominated convergence, Abel's theorem, or uniform convergence. The same device works for every geometric expansion on $[0, 1]$ and for most Maclaurin series with an explicit Lagrange remainder. It is more work than quoting a theorem and it is completely airtight, which makes it the right move whenever the theorem is in doubt.

Two integrals are easier than one

The interchange theorems are usually thought of as a tax on an expansion. They are better thought of as a tool in their own right: if swapping the order of two integrals is what makes a problem easy, then a problem with only one integral in it should be given a second one on purpose. The representations of Chapter 23 supply the second integral; this section supplies the reason to want it.

There is a second reason to want the extra dimension, independent of any swap. A double integral over a symmetric region can be averaged with its own reflection, and the average is often elementary where neither half is. If R is symmetric under $(x, y) \mapsto (y, x)$ then $\iint_R f = \frac{1}{2} \iint_R [f(x, y) + f(y, x)]$, and the symmetrised integrand frequently collapses: $\frac{x}{x+y}$ becomes $\frac{1}{2}$, and $\frac{\ln x}{\ln x + \ln y}$ becomes $\frac{1}{2}$ likewise. Lifting a one dimensional problem into two dimensions in order to symmetrise it is a distinct move from lifting it in order to swap, and the two are worth keeping separate in your head even though both begin the same way.

24.5 Double integral and double series

HARD

CUE A single integral that resists everything, but whose integrand is a y integral in disguise: $\ln(1-x)$, $\frac{1}{x}$, $\frac{x^a - x^b}{\ln x}$, $\arctan x$. Also any integral over $[0, 1]^2$ of $\frac{1}{1-xy}$ shape.

METHOD Lift to two dimensions, then do one of three things. Swap the order and perform the easy integral first. Expand geometrically into a double series and sum. Or change variables to decouple the region, for which the workhorse maps are $u = x + y, v = \frac{y}{x+y}$ with Jacobian u , and Calabi's map $x = \frac{\sin u}{\cos v}, y = \frac{\sin v}{\cos u}$ sending an open triangle onto the unit square.

Start with the deliberate lift. To evaluate $\int_0^1 \frac{x-1}{\ln x} dx$, note that $\frac{x-1}{\ln x} = \int_0^1 x^t dt$, since the t integral of $x^t = e^{t \ln x}$ is $\frac{x-1}{\ln x}$. The integrand is nonnegative on $[0, 1]^2$, so Tonelli permits the swap:

$$\int_0^1 \int_0^1 x^t dt dx = \int_0^1 \int_0^1 x^t dx dt = \int_0^1 \frac{dt}{t+1} = \ln 2.$$

The same integral was done in Chapter 23 by differentiating under the sign; here nothing was differentiated at all. Now the classical square. Expanding geometrically,

$$\int_0^1 \int_0^1 \frac{dx dy}{1-xy} = \sum_{n \geq 0} \int_0^1 \int_0^1 (xy)^n dx dy = \sum_{n \geq 0} \frac{1}{(n+1)^2} = \zeta(2) = \frac{\pi^2}{6},$$

legitimate because every term is positive. Restricting to odd powers gives $\int_0^1 \int_0^1 \frac{dx dy}{1-x^2y^2} = \frac{3}{4}\zeta(2) = \frac{\pi^2}{8} \approx 1.233701$, and Calabi's map turns that double integral into the area of the triangle $u, v > 0, u + v < \frac{\pi}{2}$, which is $\frac{\pi^2}{8}$ directly. One level up, $\frac{-\ln(1-u)}{1-u} = \sum_{n \geq 1} H_n u^n$ with $u = xy$ gives

$$\int_0^1 \int_0^1 \frac{-\ln(1-xy)}{1-xy} dx dy = \sum_{n \geq 1} \frac{H_n}{(n+1)^2} = \sum_{m \geq 1} \frac{H_m}{m^2} - \zeta(3) = 2\zeta(3) - \zeta(3) = \zeta(3) \approx 1.202057,$$

using the Euler sum $\sum H_m m^{-2} = 2\zeta(3)$ and the shift $H_{m-1} = H_m - \frac{1}{m}$. Note that the answer is $\zeta(3)$ and not $2\zeta(3)$: the shift of the index by one is exactly what removes the extra $\zeta(3)$, and dropping it is the standard way to be wrong by a factor of two here.

The lift also proves Frullani. With $\frac{1}{x} = \int_0^\infty e^{-xt} dt$ and $a, b > 0$,

$$\int_0^\infty \frac{e^{-ax} - e^{-bx}}{x} dx = \int_0^\infty \int_0^\infty (e^{-(a+t)x} - e^{-(b+t)x}) dt dx = \int_0^\infty \left(\frac{1}{a+t} - \frac{1}{b+t} \right) dt = \ln \frac{b}{a},$$

the last integral being a convergent difference of two divergent logarithms, evaluated as $\lim_{T \rightarrow \infty} \ln \frac{(a+T)b}{(b+T)a}$. At $a = 1$, $b = 3$ this is $\ln 3 \approx 1.098612$. The double integrand is not of one sign, but it is absolutely integrable, so Fubini applies; that check takes one line and is the difference between a proof and an assertion.

WHERE THIS GOES WRONG

Getting the transformed *region* wrong. The Jacobian is the easy half of a change of variables; drawing the image of the boundary is where the technique actually fails, and a wrong region gives a wrong number with no symptom. The second trap is lifting an integral that is only conditionally convergent, such as $\int_0^\infty \frac{\sin x}{x} dx$ via $\frac{1}{x} = \int_0^\infty e^{-xt} dt$: the swap gives the correct $\int_0^\infty \frac{dt}{1+t^2} = \frac{\pi}{2}$, but $\iint |\sin x| e^{-xt}$ is infinite, so Fubini does not license it and a separate limiting argument is required.

SYMMETRISE, THEN INTEGRATE

The averaging move is worth stating as a formula because it is so easy to forget it exists. On the unit square,

$$\int_0^1 \int_0^1 f(x, y) dx dy = \frac{1}{2} \int_0^1 \int_0^1 [f(x, y) + f(y, x)] dx dy,$$

which is an identity, not a theorem, since the two orders integrate the same function over the same region. Its value is that the bracket can be far simpler than either term: with $f = \frac{x^a}{x^a + y^a}$ the bracket is identically 1 and the answer is $\frac{1}{2}$ for every a , a result that is invisible from either half alone.

SEE ALSO Schwinger parametrisation; justifying the swap; symmetrisation by swapping variables.

THE SECOND INTEGRAL, FROM STOCK

Manufacturing the extra integral is a lookup, not an invention. The four that recur are

$$\frac{1}{n} = \int_0^1 t^{n-1} dt, \quad \frac{1}{n^2} = -\int_0^1 t^{n-1} \ln t dt, \quad \frac{1}{n^s} = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} e^{-nt} dt, \quad H_n = \int_0^1 \frac{1-t^n}{1-t} dt.$$

Substituting one of these into a sum and swapping turns the sum geometric, which is the whole trick behind evaluating $\sum \frac{(-1)^n}{3n+1}$ and its relatives. Running the same table backwards turns an integral into a sum. Which direction you use depends only on which of the two objects you would rather be holding at the end.

A lift that needs a limit

The honest treatment of one famous case is worth writing out, because it is the standard example of a swap that is correct but not licensed by Fubini. Using $\frac{1}{x} = \int_0^\infty e^{-xt} dt$ on $\int_0^\infty \frac{\sin x}{x} dx$ produces $\int_0^\infty \frac{dt}{1+t^2} = \frac{\pi}{2}$, and the answer is right, but $\iint |\sin x| e^{-xt} dt dx = \int_0^\infty \frac{|\sin x|}{x} dx = \infty$, so the theorem does not apply.

The repair is to swap on a finite x range, where absolute integrability does hold, and take the limit afterwards. For fixed R ,

$$\int_0^R \frac{\sin x}{x} dx = \int_0^\infty \left[\int_0^R e^{-xt} \sin x dx \right] dt = \int_0^\infty \frac{1 - e^{-Rt}(\cos R + t \sin R)}{1+t^2} dt,$$

the swap now being legitimate because $\int_0^R \int_0^\infty |\sin x| e^{-xt} dt dx = \int_0^R \frac{|\sin x|}{x} dx < \infty$. The correction term is bounded by $\int_0^\infty e^{-Rt}(1+t) dt = \frac{1}{R} + \frac{1}{R^2}$, since $\frac{1+t}{1+t^2} \leq 1+t$, so it vanishes as $R \rightarrow \infty$ and

$$\int_0^\infty \frac{\sin x}{x} dx = \int_0^\infty \frac{dt}{1+t^2} = \frac{\pi}{2}.$$

Three extra lines converted an illegal manipulation into a proof. That is the usual price, and it is almost always this cheap: truncate, swap where the theorem applies, estimate the tail.

Running the lift backwards: a binomial sum

The table above is usually read from the integral towards the sum. Read the other way, it evaluates sums that no summation technique touches. Take

$$\Sigma = \sum_{n \geq 1} \frac{1}{n^2 \binom{2n}{n}}.$$

The Beta function supplies a representation of the reciprocal binomial: since $B(n, n+1) = \frac{(n-1)!n!}{(2n)!} = \frac{1}{n \binom{2n}{n}}$, we have $\frac{1}{\binom{2n}{n}} = n \int_0^1 x^{n-1} (1-x)^n dx$. Substituting and swapping, which is licensed because every term is positive on $[0, 1]$,

$$\Sigma = \sum_{n \geq 1} \frac{1}{n} \int_0^1 x^{n-1} (1-x)^n dx = \int_0^1 \frac{1}{x} \sum_{n \geq 1} \frac{[x(1-x)]^n}{n} dx = - \int_0^1 \frac{\ln(1-x+x^2)}{x} dx.$$

The remaining integral is a standard one, and both sides evaluate to $\frac{\pi^2}{18} \approx 0.548311$, which matches the numerical sum to fifteen places. The companion sum $\sum_{n \geq 1} \frac{1}{n \binom{2n}{n}} = \frac{\pi\sqrt{3}}{9} \approx 0.604600$ comes from the same representation with one fewer factor of $\frac{1}{n}$.

What made this work was that the binomial coefficient, which is an obstruction to summation, is a Beta integral, which is not an obstruction to anything. That is the same observation that drove the whole of the previous chapter, applied to a sum instead of an integral.

Changing the region instead of the integrand

The third route in the method above deserves its own treatment, because it is where a double integral stops being a bookkeeping device and starts being a geometric argument. Restricting the double geometric series to odd powers gives

$$\int_0^1 \int_0^1 \frac{dx dy}{1-x^2y^2} = \sum_{k \geq 0} \frac{1}{(2k+1)^2} = \frac{3}{4} \zeta(2),$$

so evaluating the left side by any means evaluates $\zeta(2)$. Calabi's substitution

$$x = \frac{\sin u}{\cos v}, \quad y = \frac{\sin v}{\cos u}$$

maps the open triangle $u > 0, v > 0, u+v < \frac{\pi}{2}$ bijectively onto the open unit square, and its Jacobian is exactly $1-x^2y^2$. The integrand therefore cancels the Jacobian and the double integral is the plain area of the triangle,

$$\frac{3}{4} \zeta(2) = \frac{1}{2} \left(\frac{\pi}{2} \right)^2 = \frac{\pi^2}{8} \approx 1.233701, \quad \text{hence} \quad \zeta(2) = \frac{\pi^2}{6}.$$

This is a complete and elementary proof of Euler's identity, and the only analytic step in it is the geometric expansion, which is licensed by positivity. What made it work is that the change of variables was chosen to kill the integrand rather than to simplify the region, which is the opposite of the usual instinct and worth remembering as an option.

The companion map $u = x+y, v = \frac{y}{x+y}$, with Jacobian u , does the reverse: it takes a triangle to a rectangle and separates a symmetric integrand into a product. Between them the two maps handle most integrals over the unit square whose integrand depends on xy or on $x+y$.

Full periods and orthogonality

On a bounded interval the natural expansion is in powers. On a full period the natural expansion is in cosines, and it has a property powers do not: distinct modes are orthogonal, so a double sum collapses to a single one on integration. This is what makes logarithmic trigonometric integrals tractable.

The two expansions used below are worth deriving once rather than memorising. Both come from the same source: for $|r| < 1$,

$$-\ln(1-re^{ix}) = \sum_{n \geq 1} \frac{r^n e^{inx}}{n},$$

whose real part at $r = 1$ is $-\ln(2 \sin \frac{x}{2})$ after the identity $|1 - e^{ix}| = 2 \sin \frac{x}{2}$ on $(0, 2\pi)$, and whose companion, obtained by summing the geometric series $\sum_{n \geq 1} r^n e^{inx}$ and taking twice the real part plus one, is the Poisson kernel. So one generating function produces both the logarithm and the kernel, which is why they pair so well under an integral.

24.6 Fourier kernels and orthogonality

HARD

CUE $\ln(2 \sin \frac{x}{2})$ or $\ln(2 \cos \frac{x}{2})$, or a Poisson kernel shape $\frac{1}{a - \cos x}$, integrated over a full period.

METHOD Expand both factors in cosine series and integrate over the period, where $\int_0^{2\pi} \cos(nx) \cos(mx) dx = \pi \delta_{nm}$ for $n, m \geq 1$ kills every cross term. The two expansions to know are

$$\ln\left(2 \sin \frac{x}{2}\right) = -\sum_{n \geq 1} \frac{\cos nx}{n} \quad \text{on } (0, 2\pi), \quad \frac{1 - r^2}{1 - 2r \cos x + r^2} = 1 + 2 \sum_{n \geq 1} r^n \cos nx.$$

WORKED EXAMPLE

Take $r = \frac{1}{2}$, so $1 - 2r \cos x + r^2 = \frac{5}{4} - \cos x$ and $1 - r^2 = \frac{3}{4}$, giving $\frac{1}{\frac{5}{4} - \cos x} = \frac{4}{3} \left(1 + 2 \sum_{n \geq 1} 2^{-n} \cos nx\right)$. Multiply by the logarithm series and integrate over $[0, 2\pi]$. The constant 1 pairs with every $\cos mx$ to give zero, and orthogonality keeps only $n = m$:

$$\int_0^{2\pi} \frac{\ln(2 \sin \frac{x}{2})}{\frac{5}{4} - \cos x} dx = \frac{4}{3} \cdot 2\pi \sum_{n \geq 1} 2^{-n} \left(-\frac{1}{n}\right) = -\frac{8\pi}{3} \ln 2 \approx -5.806896,$$

since $\sum_{n \geq 1} \frac{2^{-n}}{n} = \ln 2$. Quadrature agrees to thirteen places.

The same expansion against a different weight produces an odd zeta value. Using $\int_0^\pi x \cos nx dx = \frac{(-1)^n - 1}{n^2}$,

$$\int_0^\pi x \ln\left(2 \sin \frac{x}{2}\right) dx = -\sum_{n \geq 1} \frac{(-1)^n - 1}{n^3} = 2 \sum_{n \text{ odd}} \frac{1}{n^3} = \frac{7}{4} \zeta(3) \approx 2.103600.$$

Note that this integral runs over half a period, and the cross terms did not need to vanish because the weight was integrated against each mode separately.

The cheapest consequence of the same series is the log sine integral itself. On $(0, \pi)$ the doubled form reads $\ln(2 \sin x) = -\sum_{n \geq 1} \frac{\cos 2nx}{n}$, and $\int_0^{\pi/2} \cos 2nx dx = \frac{\sin n\pi}{2n} = 0$ for every $n \geq 1$, so every term integrates to zero and

$$\int_0^{\pi/2} \ln(2 \sin x) dx = 0, \quad \text{that is} \quad \int_0^{\pi/2} \ln \sin x dx = -\frac{\pi}{2} \ln 2 \approx -1.088793.$$

The same computation with $\cos$ in place of $\sin$ gives the same value, as the reflection $x \mapsto \frac{\pi}{2} - x$ requires. It is worth noticing how little was used: not orthogonality, but the accident that each mode has zero mean over the quarter period.

Squaring the logarithm turns the single sum into a double sum, and orthogonality is then doing real work rather than bookkeeping. On $(0, \pi)$, $\ln(2 \sin x) = -\sum_{n \geq 1} \frac{\cos 2nx}{n}$, so

$$\int_0^{\pi/2} \ln^2(2 \sin x) dx = \sum_{m, n \geq 1} \frac{1}{mn} \int_0^{\pi/2} \cos 2mx \cos 2nx dx = \frac{\pi}{4} \sum_{n \geq 1} \frac{1}{n^2} = \frac{\pi^3}{24} \approx 1.291928,$$

because $\int_0^{\pi/2} \cos 2mx \cos 2nx dx = \frac{\pi}{4} \delta_{mn}$ on the quarter period: the cross terms integrate to zero there for the same reason the single terms did. The companion over a full half period, $\int_0^\pi \ln^2(2 \cos \frac{x}{2}) dx = \frac{\pi^3}{12} \approx 2.583856$, is the same computation rescaled. Notice the weight accounting: two logarithms, and a weight three answer.

WHEN THE PERIOD IS PARTIAL

Over less than a full period the modes are no longer orthogonal in general, and what survives is a named function rather than a constant. The relevant one is the Clausen function $\text{Cl}_2(\theta) = \sum_{n \geq 1} \frac{\sin n\theta}{n^2}$, which satisfies $\int_0^\theta \ln(2 \sin \frac{x}{2}) dx = -\text{Cl}_2(\theta)$ and takes the values $\text{Cl}_2(\pi) = 0$ and $\text{Cl}_2(\frac{\pi}{2}) = G$. An integral over a partial period that refuses to close is usually a Clausen value in disguise, and reporting it as such is a complete answer, not a failure.

The $\ln \sin$ series converges only conditionally and only on $(0, 2\pi)$. Over a partial period the orthogonality relation fails and cross terms survive, which is where Clausen functions come from. Integrating $\ln(2 \sin \frac{x}{2})$ over $(0, \theta)$ gives $-\text{Cl}_2(\theta)$, not a rational multiple of anything.

SEE ALSO Log sine integrals and the Clausen function; product to sum identities; geometric expansion.

Reading the answer

A termwise integration ends with a series, and the last step is naming it. This is a dictionary problem, and the dictionary is short, but two entries in it are confused so regularly that the confusion deserves its own warning.

Naming is not merely cosmetic. A closed form is checkable, comparable against other derivations, and usable as an input to the next problem, whereas an unnamed series is none of those things. It is also the step at which an error becomes visible: a wrong factor of two in the integration is invisible in a series but obvious when the series is $\frac{\pi^2}{12}$ and you expected $\frac{\pi^2}{6}$. Learn the dictionary in both directions, from series to constant and from constant back to the integral that generates it, because recognition failures in this subject are almost always failures in the second direction.

EASY

24.7 Recognising the emergent constant

CUE A sum of the form $\sum n^{-s}$, $\sum (-1)^{n-1} n^{-s}$, $\sum (-1)^k (2k+1)^{-s}$, $\sum (2n+1)^{-s}$, or $\sum \frac{1}{n(n+1)}$ left standing at the end of a computation.

METHOD Know the dictionary:

$$\zeta(2) = \frac{\pi^2}{6}, \quad \zeta(4) = \frac{\pi^4}{90}, \quad \eta(s) = (1 - 2^{1-s})\zeta(s), \quad \sum_{n \text{ odd}} n^{-s} = (1 - 2^{-s})\zeta(s),$$

$$\eta(1) = \ln 2, \quad \eta(2) = \frac{\pi^2}{12}, \quad \beta(2) = G \approx 0.915966, \quad \text{Li}_4(1) - \text{Li}_4(-1) = \frac{15}{8}\zeta(4).$$

No closed form is known for $\zeta(3)$ or any odd zeta value, so landing on one is information, not an error.

WORKED EXAMPLE

$$\int_0^1 \frac{\ln x \ln(1-x)}{x} dx.$$

Expand the second logarithm: $-\ln(1-x) = \sum_{n \geq 1} \frac{x^n}{n}$, so the integrand is $-\sum_{n \geq 1} \frac{x^{n-1} \ln x}{n}$. The log power moment with $b = 1$ gives $\int_0^1 x^{n-1} \ln x dx = -\frac{1}{n^2}$, hence

$$\int_0^1 \frac{\ln x \ln(1-x)}{x} dx = \sum_{n \geq 1} \frac{1}{n} \cdot \frac{1}{n^2} = \zeta(3) \approx 1.202057.$$

Every term is nonnegative on $[0, 1]$, since $\ln x$ and $\ln(1-x)$ are both negative there.

An alternating cousin, to fix the ζ against η distinction: $\frac{1}{1+x} = \sum_{n \geq 0} (-1)^n x^n$ gives

$$\int_0^1 \frac{\ln x}{1+x} dx = \sum_{n \geq 0} (-1)^n \left(-\frac{1}{(n+1)^2} \right) = -\eta(2) = -\frac{\pi^2}{12} \approx -0.822467,$$

which is exactly half of $-\zeta(2)$ and not $-\zeta(2)$ itself. One weight higher, the same expansion with $\ln^2 x$ gives $\int_0^1 \frac{\ln^2 x}{1+x} dx = 2\eta(3) = \frac{3}{2}\zeta(3) \approx 1.803085$, and the appearance of an odd zeta value there is not a failure: no closed form exists, and $\frac{3}{2}\zeta(3)$ is the answer.

WHERE THIS GOES WRONG

Reporting $\eta(s)$ as $\zeta(s)$. The alternating sum is $\eta(2) = \frac{\pi^2}{12}$, half of $\zeta(2)$, and the factor of two is the single most common wrong answer in this chapter. The companion slip is writing $\beta(2) = \frac{\pi^2}{8}$: the alternating sum over odd squares is Catalan's constant G , which is not known to be a rational multiple of π^2 , whereas the non alternating sum over odd squares is $\frac{\pi^2}{8}$.

When the sum does not collapse to a named constant, name it anyway. $\text{Li}_s(z) = \sum_{n \geq 1} \frac{z^n}{n^s}$ specialises to $\text{Li}_s(1) = \zeta(s)$, $\text{Li}_s(-1) = -\eta(s)$, $\text{Li}_1(z) = -\ln(1-z)$, and $\text{Li}_2(\frac{1}{2}) = \frac{\pi^2}{12} - \frac{\ln^2 2}{2}$. Two identities do most of the work: the Landen reflection $\text{Li}_2(z) + \text{Li}_2(1-z) = \frac{\pi^2}{6} - \ln z \ln(1-z)$, and the duplication $\text{Li}_s(z) + \text{Li}_s(-z) = 2^{1-s} \text{Li}_s(z^2)$. An answer expressed in polylogarithms at $\frac{1}{2}$, -1 and 1 is a finished answer; the residual urge to reduce it further is what leads to invented closed forms for $\zeta(3)$.

SEE ALSO Catalan's constant; the polylogarithm ladder; the log power moment.

Where each constant comes from

It is worth knowing the standard integral source of each constant, because recognition then runs in both directions and an unexpected constant becomes a diagnostic rather than a puzzle.

$\ln 2$ comes from $\sum \frac{(-1)^{n-1}}{n}$, and therefore from any expansion of $\frac{1}{1+x}$ integrated once on $[0, 1]$; the archetype is $\int_0^1 \frac{dx}{1+x}$.

$\frac{\pi}{4}$ comes from $\sum \frac{(-1)^k}{2k+1}$, so from $\frac{1}{1+x^2}$; a π with no logarithm attached almost always entered this way.

$\zeta(2)$ comes from one logarithm against $\frac{1}{1-x}$, as in $\int_0^1 \frac{-\ln(1-x)}{x} dx$, and its alternating partner $\eta(2) = \frac{\pi^2}{12}$ from $\int_0^1 \frac{-\ln x}{1+x} dx$.

Catalan's constant $G = \beta(2)$ comes from one logarithm against $\frac{1}{1+x^2}$, that is from odd indices with alternating signs. The two cleanest sources are $\int_0^1 \frac{\arctan x}{x} dx = G$ and $\int_0^1 \frac{\ln x}{1+x^2} dx = -G$, and the trigonometric form $\int_0^{\pi/4} \ln(\tan x) dx = -G$ is the same integral after $x = \tan \theta$. All three agree numerically at 0.915966.

$\zeta(3)$ comes from two logarithms, whether as $\ln^2 x$ against $\frac{1}{1-x}$ or as $\ln x \ln(1-x)$ against $\frac{1}{x}$, and $\beta(3) = \frac{\pi^3}{32}$ is its odd alternating cousin.

$\text{Li}_2(\frac{1}{2}) = \frac{\pi^2}{12} - \frac{\ln^2 2}{2} \approx 0.582241$ is what appears when the geometric ratio is $\frac{1}{2}$ rather than 1, which happens after any substitution that halves an interval.

SUMS YOU HAVE NOT MET

If the series does not match the dictionary, try three things before giving up. Split by parity, since many sums are a ζ and an η added together. Look for a harmonic number, since $\sum \frac{H_n}{n^2} = 2\zeta(3)$ and $\sum \frac{H_n}{n^3} = \frac{\pi^4}{72}$ close most Euler sums of low weight. And check whether the general term is a Cauchy product, which it is whenever the integrand contained a square of a logarithm or a product of two expandable factors; recognising the product lets you sum the two factors separately.

The final technique is the largest generalisation of expand and integrate: it says that if you know the coefficients of an expansion as a function of the index, the Mellin transform is that same function evaluated at a negative argument. It is powerful, it is often the fastest route, and it is the easiest thing in this chapter to abuse.

HARD

24.8 Ramanujan's master theorem

CUE A Mellin shaped integral $\int_0^\infty x^{s-1} \varphi(x) dx$ where φ has a Maclaurin expansion whose coefficients are a recognisable function of the index.

METHOD If $\varphi(x) = \sum_{k \geq 0} \frac{\phi(k)}{k!} (-x)^k$ with ϕ analytic and of controlled growth in a right half plane, then

$$\int_0^\infty x^{s-1} \varphi(x) dx = \Gamma(s) \phi(-s).$$

The content is that you continue the coefficient function to a negative argument rather than summing anything.

WORKED EXAMPLE

Two applications, one safe and one delicate. Safe: $\frac{1}{1+x} = \sum_{k \geq 0} (-x)^k$, so $\phi(k) = k!$, and continuing $k!$ to $\Gamma(k+1)$ gives

$$\int_0^\infty \frac{x^{s-1}}{1+x} dx = \Gamma(s) \Gamma(1-s) = \frac{\pi}{\sin \pi s}, \quad 0 < s < 1,$$

which at $s = \frac{1}{3}$ predicts $\pi / \sin \frac{\pi}{3} \approx 3.627599$, confirmed by quadrature to eight places.

Delicate: $\cos x = \sum_{k \geq 0} \frac{(-1)^k x^{2k}}{(2k)!}$ has ϕ supported on even indices, and the correct continuation is $\phi(k) = \cos \frac{\pi k}{2}$, giving

$$\int_0^\infty x^{s-1} \cos x dx = \Gamma(s) \cos \frac{\pi s}{2}, \quad 0 < s < 1.$$

At $s = \frac{1}{2}$ this is $\Gamma(\frac{1}{2}) \cos \frac{\pi}{4} = \sqrt{\pi} \cdot \frac{\sqrt{2}}{2} = \sqrt{\frac{\pi}{2}} \approx 1.253314$, the Fresnel value. The integral converges only conditionally, so the number must be read as a limit of $\int_0^R$ along $R \rightarrow \infty$; averaging the partial integrals over successive half periods reproduces 1.2533141 numerically.

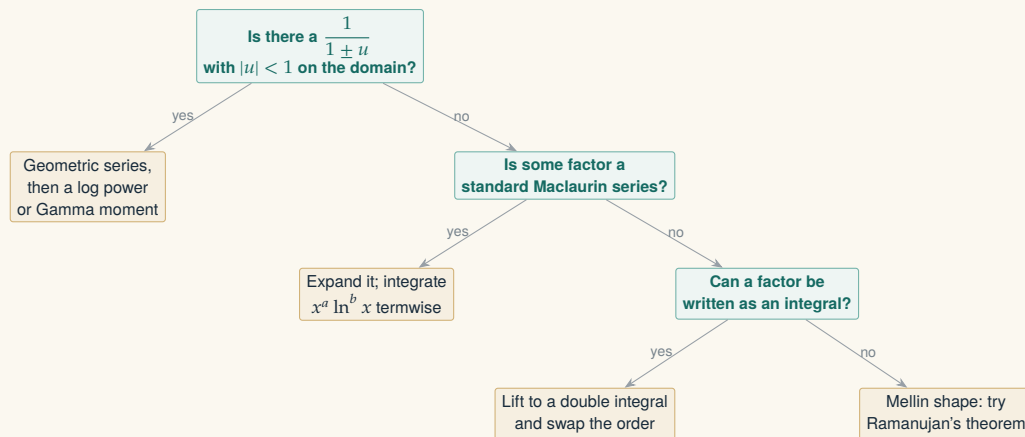
The theorem is best understood as a formal statement made rigorous by a contour argument: if you write ϕ as an inverse Mellin transform and slide the contour, the residues at the poles of Γ reproduce the Maclaurin coefficients, and the theorem is the statement that the correspondence runs both ways. That picture explains both its power and its fragility. It is exact when ϕ extends to a function with controlled growth in a vertical strip, and it is meaningless when the extension is not unique, which is the usual failure: $\phi(k) = \cos \frac{\pi k}{2}$ and $\phi(k) = \cos \frac{\pi k}{2} + \sin \pi k$ agree at every nonnegative integer and disagree everywhere else, and only the first satisfies the growth hypothesis. Carlson's theorem is what makes the choice canonical.

WHERE THIS GOES WRONG

Applying the theorem without checking growth. The hypotheses require ϕ analytic in a strip and bounded by $Ce^{-\delta|z|}$ type estimates; without them the formula returns confident nonsense, and several classical “paradoxes” are exactly this. When the integral is only conditionally convergent, the output is a regularised value and should be labelled as one.

SEE ALSO Euler reflection and the Mellin transform; the Gamma function moment; Fresnel integrals.

In practice the theorem earns its place as a fast route to a conjecture rather than as a proof. Apply it, get a candidate closed form in seconds, then verify the candidate numerically and, if it matters, re-derive it by expansion and interchange with the licences written out. That workflow uses the theorem for what it is genuinely good at, which is guessing correctly, and leaves the burden of proof with methods whose hypotheses you can actually check on the page.



The tree routes on the shape of the integrand, and its branches are ordered by cost rather than by cleverness. Note that the left branch is not always the cheapest: a geometric expansion on $[0, 1]$ is free of convergence worries only when the terms keep one sign, and an alternating expansion at the endpoint can cost more work than a direct lift to two dimensions.

Assembling the pieces

One extended example, to show the protocol running with every part in place. Evaluate

$$S = \int_0^1 \frac{\ln^2(1-x)}{x} dx.$$

Neither factor is expandable by itself in a useful way, but the square of a logarithm is a Cauchy product. From $-\ln(1-x) = \sum_{n \geq 1} \frac{x^n}{n}$, squaring and collecting the coefficient of x^n gives

$$\ln^2(1-x) = \sum_{n \geq 2} \left(\sum_{j=1}^{n-1} \frac{1}{j(n-j)} \right) x^n = \sum_{n \geq 2} \frac{2H_{n-1}}{n} x^n,$$

the inner sum collapsing because $\frac{1}{j(n-j)} = \frac{1}{n}(\frac{1}{j} + \frac{1}{n-j})$. All coefficients are positive and $x > 0$, so monotone convergence licenses the swap without further comment. Integrating termwise against $\frac{1}{x}$,

$$S = \sum_{n \geq 2} \frac{2H_{n-1}}{n} \cdot \frac{1}{n} = 2 \sum_{n \geq 1} \frac{H_n - \frac{1}{n}}{n^2} = 2[2\zeta(3) - \zeta(3)] = 2\zeta(3) \approx 2.404114,$$

using Euler's sum $\sum H_n n^{-2} = 2\zeta(3)$. Quadrature returns 2.4041138, so the answer stands.

Three checks were available and all three pass. The weight count: two logarithms plus the $\frac{1}{x}$ predicts weight 3, and $\zeta(3)$ has weight 3. The index shift: $H_{n-1} = H_n - \frac{1}{n}$ was applied explicitly rather than assumed, which is where the factor of two lives. And the numerical comparison, which costs a few seconds. A calculation that survives all three is almost certainly right; one that survives none of them is not an answer, it is a guess.

For contrast, the closely related $\int_0^1 \frac{\ln x \ln(1+x)}{x} dx$ expands to $-\sum_{n \geq 1} \frac{(-1)^{n-1}}{n} \cdot \frac{1}{n^2} = -\eta(3) = -\frac{3}{4}\zeta(3) \approx -0.901543$, and $\int_0^1 \frac{\ln(1-x) \ln^2 x}{x} dx = -2\zeta(4) = -\frac{\pi^4}{45} \approx -2.164646$. The first has an alternating expansion and needs the uniform partial sum bound; the second is of one sign and needs nothing. Same family, different licences.

THE ORDER IN WHICH TO WORK

Do the algebra before the analysis. Expand, integrate the general term, sum, and name the constant first, treating every interchange as provisional; then go back and discharge the licences. Working in that order costs nothing, because the licence you need is determined by the expansion you actually used, and it saves the common waste of proving a careful convergence result about a series you then abandon. What it must not become is an excuse to skip the second pass. Write the provisional answer, run the numerical comparison, and only then decide whether the interchange needs monotone convergence, a uniform bound, or a truncation argument.

CONFIRM BEFORE YOU COMMIT

Every closed form in this chapter can be checked in under a minute: evaluate the original integral numerically and compare with the summed answer. The characteristic error is not a wrong method but a factor of two, arising from an index shift, from confusing ζ with η , or from an odd or even restriction applied to the wrong half of a sum. All three show up immediately in the second decimal place. If the numerical value is exactly twice or exactly half your answer, do not look for a subtle analytic mistake; look for the index.

The chapter has two halves that look different and are not. Expanding an integrand into a series and lifting an integral into a double integral are the same operation performed against two different measures, counting measure in the first case and Lebesgue measure in the second, and the theorem that licenses each is the same theorem. That is why the failures look alike: a spike of fixed mass escaping to the edge of the domain in the series example, and an unsigned singularity at the corner in the double integral example. Both are failures of absolute integrability, and both are invisible in the final number.

Recognition matrix

If you see	Reach for	If that fails
$\frac{1}{1+x^n}$ on $[0, 1]$	Geometric expansion, then $\int_0^1 x^a \ln^b x$	Substitute $x^n = u$, then Beta
$\frac{1}{e^x - 1}$ or $\frac{1}{e^x + 1}$ on $(0, \infty)$	$\sum e^{-nx}$, giving $\Gamma(s)\zeta(s)$ or $\Gamma(s)\eta(s)$	Contour integration
$\ln(1-x)$, $\arctan x$, $\arcsin x$ over x	Maclaurin series, integrate termwise	Parameter differentiation
x^x , x^{-x} , $e^{x \ln x}$	Expand e^u ; each term is a log power moment	No closed form; report the series
$x^a \ln^b x$ on $[0, 1]$	$\frac{(-1)^b b!}{(a+1)^{b+1}}$, with $a > -1$	Differentiate $\int_0^1 x^a$ in a
A swap of $\int$ and $\sum$	Sign check first: nonnegative means free	Bound $\sum \int f_n $
Two orders of a double integral	Tonelli if nonnegative, else Fubini	Compute both and compare
$\frac{1}{1-xy}$ on the unit square	Double geometric series, giving $\zeta(2)$	Change variables to a triangle
$\frac{x^a - x^b}{\ln x}$	Write it as $\int_a^b x^t dt$ and swap	Differentiate under the sign
$\ln(2 \sin \frac{x}{2})$ over a full period	Cosine series and orthogonality	Clausen function on a partial period
$\sum (-1)^{n-1} n^{-s}$	$\eta(s) = (1 - 2^{1-s})\zeta(s)$	Check against ζ , factor of two
$\sum (-1)^k (2k+1)^{-2}$	Catalan's constant G	Not a multiple of π^2
$\int_0^\infty x^{s-1} \varphi(x) dx$, φ with nice coefficients	Ramanujan's master theorem, $\Gamma(s)\phi(-s)$	Verify growth, or expand directly

If you see	Reach for	If that fails
$\ln(1 \pm e^{-x})$ on $(0, \infty)$	Expand in e^{-nx} ; Gamma moment	Substitute $u = e^{-x}$
$\ln^2(1 - x)$ or a product of two series	Cauchy product; expect H_n coefficients	Euler sums of weight three
Two logarithms in the integrand	Expect weight three: $\zeta(3)$, $\beta(3)$, $\pi^2 \ln 2$	Recount; a weight mismatch is an error
$\binom{2n}{n}$ downstairs in a sum	$B(n, n+1)$ representation, then swap	Generating function of $\binom{2n}{n}$
An alternating expansion at $x = 1$	Finite sum plus explicit remainder	Work on $[0, 1 - \varepsilon]$, then Abel

The special-functions ladder

What this chapter is for. Past a certain level of difficulty a definite integral simply does not have an elementary answer, and continuing to hunt for one is the mistake. The skill changes. Instead of searching for an antiderivative you ask which named function or named constant the integral is a disguised instance of, and then you work with that object's functional equations. Those equations are the real content: the reflection formula $\Gamma(s)\Gamma(1-s) = \pi/\sin \pi s$ evaluates an entire family of half-line integrals in one line, the recurrence $\psi(z+1) = \psi(z) + 1/z$ turns a harmonic sum into a closed form, and the derivative relation $\text{Li}'_{s+1}(x) = \text{Li}_s(x)/x$ climbs a whole ladder of logarithmic integrals. This chapter builds that ladder rung by rung, starting from Gamma and Beta, passing through digamma and the polylogarithm, and ending at the elliptic integrals, where even the special functions of one variable give out. For each rung you get three things: the defining integral, the two or three identities that actually get used in practice, and the recognition cue that tells you which rung you are standing on. At the end you should be able to look at an unfamiliar definite integral and say, before computing anything, which named object its answer will be built from.

What a closed form is, past the elementary

A student trained only on antiderivatives has a single test for success: can I write the answer with $\exp$, $\ln$, powers, and the trigonometric functions? By that test $\int_0^1 e^{-x^2} dx$ has no answer, $\int_0^\infty \frac{\sin x}{x} dx$ has no answer, and $\int_0^1 \frac{dx}{\sqrt{1-x^4}}$ has no answer. All three statements are false in any useful sense. The first equals $\frac{\sqrt{\pi}}{2} \text{erf}(1)$, the second equals $\frac{\pi}{2}$, and the third equals $\Gamma(\frac{1}{4})^2/(4\sqrt{2\pi})$. What has happened is that the alphabet of acceptable answers grew.

The right way to think about the enlargement is this. A special function is a function defined by an integral or a series that is important enough to have been tabulated, named, and equipped with identities. Once it has identities, it can be computed with, and computing with it is exactly what integration at this level consists of. The functions worth carrying are few, and they come in a definite order of generality.

Rung	Object	Defining integral	The identity that does the work
1	$\Gamma(s)$	$\int_0^\infty x^{s-1} e^{-x} dx$	$\Gamma(s+1) = s\Gamma(s)$
2	Γ reflection	(same)	$\Gamma(s)\Gamma(1-s) = \pi/\sin \pi s$
3	$B(p, q)$	$\int_0^1 t^{p-1} (1-t)^{q-1} dt$	$B = \Gamma/\Gamma$
4	$\psi(z)$	Γ'/Γ	$\psi(z+1) = \psi(z) + \frac{1}{z}$
5	$\text{Li}_s(x)$	$\int_0^x \text{Li}_{s-1}(t) \frac{dt}{t}$	$\text{Li}_s(1) = \zeta(s)$; Landen
6	erf , Fresnel	$\int_0^z e^{-t^2} dt$	Complete the square
7	Si , Ci	$\int_0^\theta \frac{\sin t}{t} dt$	$\text{Si}(\infty) = \frac{\pi}{2}$
8	$\text{Cl}_2(\theta)$	$-\int_0^\theta \ln 2 \sin \frac{t}{2} dt$	Duplication in θ
9	K , E	$\int_0^{\pi/2} (1 - k^2 \sin^2 \theta)^{\mp 1/2} d\theta$	Landen and AGM

One unifying remark before the details. Almost every entry in that table is a *Mellin transform*, that is, an integral of the form $\int_0^\infty x^{s-1} f(x) dx$ with s a free parameter. Gamma is the Mellin transform of e^{-x} , Beta is the Mellin transform of $(1+x)^{-1}$ up to a change of variable, the zeta representation is the Mellin transform of $(e^x - 1)^{-1}$, and the Fresnel values are the Mellin transform of $\sin$ at $s = \frac{1}{2}$. That is why the same three moves keep working: substitute to expose a pure power against a fixed kernel, read off the parameter s , and check the strip of s for which the integral converges. If you internalise nothing else from this chapter, internalise the habit of asking “what is the free power, and what is the kernel”.

The ordering is not arbitrary. Rungs 1 through 4 are all the same function seen from different angles, since Beta is a ratio of Gammas and digamma is a logarithmic derivative of Gamma. Rung 5 is the systematic home of every integral built from $\ln$ against a rational function. Rungs 6 and 7 are what oscillation and Gaussian decay produce when the interval is cut short. Rung 8 is trigonometric logarithms. Rung 9 is the first genuinely new object, because K and E are functions of a modulus that no Gamma identity evaluates except at special points.

The practical skill here is almost entirely recognition. Nobody derives the reflection formula in the middle of a problem. What you do is look at $\int_0^\infty \frac{x^{-2/3}}{1+x} dx$, see a fractional power against a self-reciprocal denominator on the half-line, and write down $\pi/\sin(\pi/3)$. The identities go on a card; the cues go into your reflexes. When a problem resists, the productive question is not “what substitution have I missed” but “which named constant is this answer going to contain”. If you can guess the constant, you can usually find the route backwards from it.

One more orientation point, about how these objects are actually used. Almost no real problem is a single lookup. A typical hard integral is a chain: a substitution turns it into a Beta function, the Beta function becomes a Gamma quotient, one of the Gammas has argument $\frac{3}{4}$, reflection converts that into $\Gamma(\frac{1}{4})$, and the answer comes out as $\Gamma(\frac{1}{4})^2$ over something. Three named identities, none of them hard, applied in sequence. The difficulty in this material is never any single step; it is knowing which of the four or five available identities to apply next, and that is decided by looking at what is blocking you, not by looking at the original integral.

Gamma, and the three identities that evaluate everything

Gamma is where every route through this material begins, and it is worth being clear about why one function is doing so much work. The reason is that e^{-x} against a power is the unique combination that is stable under both scaling and the recurrence: substituting $x \mapsto \lambda x$ changes the integral only by a power of λ , and integrating by parts changes the parameter only by one. Those two invariances are what turn a single integral into a function with a functional equation, and a functional equation is what lets you evaluate rather than merely define. Every other rung of the ladder inherits the same two properties in some form.

The Gamma function is the interpolation of the factorial, and for our purposes it is a machine that eats any integral of the shape “power times exponential of a power on the half-line”. That is a much larger family than it sounds, because substitutions turn radicals, logarithms and Gaussians into that shape routinely. Start with the machine itself.

CORE

25.1 Gamma-function moment

CUE Any power of x multiplied by an exponential of a power of x , integrated over $(0, \infty)$: $\int_0^\infty x^m e^{-\lambda x^k} dx$. The disguises are a base other than e , as in 2^{-x^2} , and a radical, as in $e^{-\sqrt{x}}$, which is the case $k = \frac{1}{2}$.

METHOD Substitute $u = \lambda x^k$, which converts everything into the defining integral $\Gamma(s) = \int_0^\infty u^{s-1} e^{-u} du$. The result is worth memorising in one line:

$$\int_0^\infty x^m e^{-\lambda x^k} dx = \frac{1}{k \lambda^{(m+1)/k}} \Gamma\left(\frac{m+1}{k}\right), \quad \frac{m+1}{k} > 0, \lambda > 0.$$

Then reduce the argument with $\Gamma(z+1) = z\Gamma(z)$ until it lands in $(0, 1]$, and finish with $\Gamma(n) = (n-1)!$ or $\Gamma(\frac{1}{2}) = \sqrt{\pi}$. A base other than e is handled by $a^{-x^k} = e^{-x^k \ln a}$, so $\lambda = \ln a$.

WORKED EXAMPLE

Two instances, the second showing where the base goes.

$$\int_0^\infty x^4 e^{-x^3} dx = \frac{1}{3} \Gamma\left(\frac{5}{3}\right) \approx 0.300915,$$

directly from the formula with $m = 4, k = 3, \lambda = 1$. Reduce further if you like: $\Gamma(\frac{5}{3}) = \frac{2}{3} \Gamma(\frac{2}{3})$.

Now $\int_{-\infty}^\infty |x|^3 2^{-x^2} dx$. The integrand is even, so this is $2 \int_0^\infty x^3 e^{-x^2 \ln 2} dx$. Here $m = 3, k = 2, \lambda = \ln 2$, so $(m+1)/k = 2$ and the formula gives

$$2 \cdot \frac{1}{2(\ln 2)^2} \Gamma(2) = \frac{1}{\ln^2 2} \approx 2.081369.$$

The exponent on $\ln 2$ is 2, not 1, because the scale enters as $\lambda^{(m+1)/k}$.

WHERE THIS GOES WRONG

Two failures, both cheap to avoid. First, a non- e base contributes $\ln a$ to *every* power of the scale, so $(\ln 2)^2$ downstairs in the example above, not $\ln 2$; check the exponent by dimensional reasoning on x . Second, Γ demands $\frac{m+1}{k} > 0$. If it is not, the integral diverges at the origin and the formula still produces a finite-looking number, which is the worst possible failure mode. Test both endpoints before you use the answer: x^m at 0 and the exponential at ∞ .

SEE ALSO The Beta function; Gaussian and error function; Euler reflection and the Mellin transform; Zeta values and Apéry's constant.

Whatever comes out of a substitution, the argument of Γ can be brought to a standard place by three moves, applied in this order. Shift by integers with $\Gamma(z+1) = z\Gamma(z)$ until the argument lies in $(0, 1]$. If it is now $\frac{1}{2}$, you are done: $\Gamma(\frac{1}{2}) = \sqrt{\pi}$. If it is a rational $\frac{p}{q}$ and $\frac{q-p}{q}$ also appears, pair them with reflection to get $\pi/\sin(\pi p/q)$. If both z and $z + \frac{1}{2}$ appear, use duplication. Anything left over, such as a lone $\Gamma(\frac{1}{3})$ or $\Gamma(\frac{1}{4})$, does not reduce further and is part of the answer. Recognising that last case early saves more time than any other habit in this chapter.

The recurrence $\Gamma(z+1) = z\Gamma(z)$ moves the argument by integers, which is enough to reduce any rational argument to a representative in $(0, 1]$ but not enough to *evaluate* that representative. Two further identities do the evaluating, and between them they close off the whole family of half-line integrals with fractional powers. The first is the reflection formula, which is by a wide margin the most valuable single fact in this chapter.

HARD

25.2 Euler reflection and the Mellin transform

CUE A fractional power x^{s-1} against a denominator like $1+x^n$ on $(0, \infty)$; equivalently, an integrand invariant in shape under $x \mapsto 1/x$ with a non-integer exponent floating on top. Also any integral you have already reduced to $\Gamma(\alpha)\Gamma(1-\alpha)$.

METHOD The reflection formula is

$$\Gamma(s)\Gamma(1-s) = \frac{\pi}{\sin \pi s}, \quad 0 < s < 1,$$

and combining it with the Beta evaluation of the Mellin transform of $(1+x^n)^{-1}$ gives the master formula

$$\int_0^\infty \frac{x^{s-1}}{1+x^n} dx = \frac{\pi}{n \sin(\pi s/n)}, \quad 0 < s < n.$$

Everything on the half-line with a single self-reciprocal denominator is a case of this. For a compound denominator, partial-fraction it into pieces of this shape first, then apply the formula termwise.

WORKED EXAMPLE

The plain case first:

$$\int_0^\infty \frac{dx}{1+x^3} = \frac{\pi}{3 \sin(\pi/3)} = \frac{\pi}{3} \cdot \frac{2}{\sqrt{3}} = \frac{2\pi}{3\sqrt{3}} \approx 1.209200,$$

using $s = 1, n = 3$. A fractional numerator is no harder:

$$\int_0^\infty \frac{x^{-2/3}}{1+x} dx = \frac{\pi}{\sin(\pi/3)} = \frac{2\pi}{\sqrt{3}} \approx 3.627599,$$

with $s = \frac{1}{3}, n = 1$. Notice that neither integral has an elementary antiderivative worth writing down; the first one does, by partial fractions, and it takes half a page.

The same formula run backwards evaluates a Gamma product. Since $\int_0^1 (1-x^3)^{-1/3} dx = \frac{1}{3}B(\frac{1}{3}, \frac{2}{3}) = \frac{1}{3}\Gamma(\frac{1}{3})\Gamma(\frac{2}{3})$, and reflection gives $\Gamma(\frac{1}{3})\Gamma(\frac{2}{3}) = \pi/\sin(\pi/3)$, the value is again $2\pi/(3\sqrt{3})$.

A DENOMINATOR THAT IS NOT $1+x^n$

The same Beta-plus-reflection route handles any power of the denominator. Substituting $t = \frac{x}{1+x}$, or equivalently reading the Mellin transform of $(1+x)^{-\mu}$ directly,

$$\int_0^\infty \frac{x^{s-1}}{(1+x)^\mu} dx = B(s, \mu-s) = \frac{\Gamma(s)\Gamma(\mu-s)}{\Gamma(\mu)}, \quad 0 < s < \mu.$$

At $\mu = 2$ this gives $\Gamma(s)\Gamma(2-s) = (1-s)\Gamma(s)\Gamma(1-s) = \pi(1-s)/\sin \pi s$, so

$$\int_0^\infty \frac{x^{-0.4}}{(1+x)^2} dx = \frac{0.4\pi}{\sin(0.6\pi)} \approx 1.3213064,$$

confirmed by quadrature to eleven digits. The recurrence did the shifting and reflection did the evaluating, which is the standard division of labour.

WHERE THIS GOES WRONG

The convergence strip is the whole hypothesis and it is invisible in the formula. Outside $0 < s < n$ the right-hand side still evaluates to a perfectly pleasant number while the integral diverges. Check the two endpoints separately: near 0 the integrand behaves like x^{s-1} , which needs $s > 0$; near ∞ it behaves like x^{s-1-n} , which needs $s < n$. A second trap: the formula is for $1+x^n$, not $1-x^n$. The latter has a pole on the path and its principal value is $\frac{\pi}{n} \cot(\pi s/n)$, a different function entirely.

SEE ALSO The Beta function; Gamma-function moment; Legendre duplication; Quotable residue results.

Reflection relates Γ at s to Γ at $1-s$. The other structural identity relates Γ at z and $z + \frac{1}{2}$ to Γ at $2z$, and it is what you reach for whenever half-integer arguments appear alongside integer ones. It is also the cleanest illustration in the chapter of a functional equation being *produced* by a symmetry substitution rather than quoted.

MEDIUM

25.3 Legendre duplication, $\Gamma(z)\Gamma(z + \frac{1}{2})$

CUE A Gamma quotient in which both z and $z + \frac{1}{2}$ appear, or a Beta function with equal arguments $B(p, p)$, or a central binomial coefficient $\binom{2n}{n}$ that you want as a Gamma ratio. Also any Wallis-type product you want in closed form.

METHOD The identity is

$$\Gamma(z)\Gamma\left(z + \frac{1}{2}\right) = 2^{1-2z} \sqrt{\pi} \Gamma(2z).$$

Use it in whichever direction removes the half-integer. The general multiplication theorem

$$\prod_{j=0}^{n-1} \Gamma\left(z + \frac{j}{n}\right) = (2\pi)^{(n-1)/2} n^{\frac{1}{2}-nz} \Gamma(nz)$$

is the same statement for n factors, and the case $n = 2$ is duplication.

WORKED EXAMPLE

Duplication is exactly the statement that $B(p, p)$ has a symmetry. Compute

$$B(p, p) = \int_0^1 [x(1-x)]^{p-1} dx$$

by recentring at the midpoint: put $x = \frac{1+t}{2}$, so $x(1-x) = \frac{1-t^2}{4}$ and $dx = \frac{dt}{2}$, giving

$$B(p, p) = \frac{4^{1-p}}{2} \int_{-1}^1 (1-t^2)^{p-1} dt = 4^{1-p} \int_0^1 (1-t^2)^{p-1} dt = 4^{1-p} \cdot \frac{1}{2} B\left(\frac{1}{2}, p\right),$$

the last step by $u = t^2$. Writing both Betas as Gamma quotients,

$$\frac{\Gamma(p)^2}{\Gamma(2p)} = \frac{2^{1-2p} \sqrt{\pi} \Gamma(p)}{\Gamma(p + \frac{1}{2})},$$

which rearranges to the duplication formula. Checked at $p = 0.63$: both sides equal 1.3386477.

WHERE THIS GOES WRONG

The factor 2^{1-2z} , not 2^{-2z} and not 2^{2-2z} . It is pinned by the single test $z = \frac{1}{2}$, where the identity reads $\Gamma(\frac{1}{2})\Gamma(1) = 2^0 \sqrt{\pi} \Gamma(1)$, so evaluate at $z = \frac{1}{2}$ every time you write it from memory rather than trusting the exponent.

SEE ALSO The Beta function; Euler reflection and the Mellin transform; Reduction formulae and Wallis.

Reflection and duplication are identities about Γ alone. The Beta function is what you actually meet, because the natural home of a two-parameter power weight is the unit interval and not the half-line. Beta is a ratio of Gammas and therefore adds no new information, but it adds an enormous amount of recognition value: once you can spot a Beta integrand, the Gamma machinery applies to bounded intervals too.

CORE

25.4 The Beta function

CUE A weight $x^{p-1}(1-x)^{q-1}$ on $[0, 1]$, possibly after a substitution that maps your interval to $[0, 1]$; or $\sin^m x \cos^n x$ on $[0, \frac{\pi}{2}]$ with exponents that are not both non-negative integers. A radical such as $\sqrt{1-x}$ or $(1-x^3)^{-1/3}$ is a Beta weight with a fractional exponent.

METHOD Two forms, and the relation between them is the whole technique:

$$B(p, q) = \int_0^1 t^{p-1} (1-t)^{q-1} dt = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}, \quad \int_0^{\pi/2} \sin^{2p-1} x \cos^{2q-1} x dx = \frac{1}{2} B(p, q).$$

The trigonometric form follows from the algebraic one by $t = \sin^2 x$. For a general power, $\int_0^1 x^a (1-x^b)^c dx = \frac{1}{b} B\left(\frac{a+1}{b}, c+1\right)$ via $u = x^b$; this one substitution covers most of what you will meet. Half-integer Gammas then come down to $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ by the recurrence.

An algebraic case and a trigonometric one.

$$\int_0^1 x^3 \sqrt{1-x} \, dx = B\left(4, \frac{3}{2}\right) = \frac{\Gamma(4)\Gamma(\frac{3}{2})}{\Gamma(\frac{11}{2})} = \frac{6 \cdot \frac{\sqrt{\pi}}{2}}{\frac{945}{32}\sqrt{\pi}} = \frac{32}{315} \approx 0.101587,$$

using $\Gamma(\frac{11}{2}) = \frac{9}{2} \cdot \frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi} = \frac{945}{32} \sqrt{\pi}$.

Now a fractional trigonometric exponent, where no elementary method applies at all:

$$\int_0^{\pi/2} \sqrt{\sin x} \, dx.$$

Match $2p-1 = \frac{1}{2}$ and $2q-1 = 0$, so $p = \frac{3}{4}$, $q = \frac{1}{2}$, and the value is

$$\frac{1}{2} B\left(\frac{3}{4}, \frac{1}{2}\right) = \frac{\Gamma(\frac{3}{4})\Gamma(\frac{1}{2})}{2\Gamma(\frac{5}{4})} = \frac{\Gamma(\frac{3}{4})\sqrt{\pi}}{2 \cdot \frac{1}{4}\Gamma(\frac{1}{4})} = \frac{2\sqrt{\pi}\Gamma(\frac{3}{4})}{\Gamma(\frac{1}{4})}.$$

Reflection turns $\Gamma(\frac{3}{4})$ into $\pi\sqrt{2}/\Gamma(\frac{1}{4})$, so the answer is $2\sqrt{2\pi^3}/\Gamma(\frac{1}{4})^2 \approx 1.198140$. Two named identities, no antiderivative.

THE GENERAL POWER FORM, WHICH COVERS MOST SIGHTINGS

One substitution handles the whole family $x^m(1+x^n)^{-\mu}$ on the half-line. Put $u = \frac{x^n}{1+x^n}$, or simply combine the two previous results, to get

$$\int_0^\infty \frac{x^m}{(1+x^n)^\mu} \, dx = \frac{1}{n} B\left(\frac{m+1}{n}, \mu - \frac{m+1}{n}\right).$$

With $m = 2$, $n = 3$, $\mu = \frac{5}{2}$ this is $\frac{1}{3} B(1, \frac{3}{2}) = \frac{1}{3} \cdot \frac{2}{3} = \frac{2}{9} \approx 0.222222$, matching quadrature. The case $\mu = 1$ recovers the reflection master formula, so this one line contains everything in this section as a special case; memorise it in preference to the special cases.

WHERE THIS GOES WRONG

The exponent offsets, which are the single commonest error in this whole group. It is $p-1$ and $q-1$ in the algebraic form but $2p-1$ and $2q-1$ in the trigonometric form. Reading $\int_0^{\pi/2} \sin^4 x \, dx$ as $\frac{1}{2} B(4, \frac{1}{2})$ instead of $\frac{1}{2} B(\frac{5}{2}, \frac{1}{2})$ gives a clean wrong answer. Write down the two matching equations $2p-1 = m$, $2q-1 = n$ explicitly every time; do not do it in your head.

SEE ALSO Gamma-function moment; Legendre duplication; Reduction formulae and Wallis; Elliptic integrals K and E .

Digamma: what happens when you differentiate in the parameter

Gamma and Beta evaluate integrals whose integrands are pure powers. The moment a logarithm appears, differentiate the parameter: $\partial_p x^{p-1} = x^{p-1} \ln x$, so every $\ln x$ in the integrand is one p -derivative of a Beta or Gamma integral. Differentiating Γ produces Γ' , and the useful normalisation is the logarithmic derivative, because that is what appears when you differentiate a Beta *quotient*.

HARD

25.5 Digamma and polygamma

CUE A logarithm multiplying a power weight; a harmonic number H_n in a series you are trying to sum; the combination $\frac{1-x^{a-1}}{1-x}$ on $[0, 1]$; or an answer that visibly “wants” to be $-\gamma$ plus something clean.

METHOD Define $\psi(z) = \Gamma'(z)/\Gamma(z)$. Four facts do all the work:

$$\psi(1) = -\gamma, \quad \psi(z+1) = \psi(z) + \frac{1}{z}, \quad \psi(1-z) - \psi(z) = \pi \cot \pi z,$$

together with the integral representation

$$\int_0^1 \frac{1-x^{a-1}}{1-x} \, dx = \psi(a) + \gamma.$$

The recurrence gives $\psi(n+1) = H_n - \gamma$, which is the bridge between digamma and harmonic sums. Differentiating again gives the polygammas: $\psi_1(n+1) = \zeta(2) - \sum_{k=1}^n k^{-2}$, and in general $\psi_m(1) = (-1)^{m+1} m! \zeta(m+1)$.

Take $a = \frac{1}{2}$ in the representation. Then $\psi(\frac{1}{2}) = \psi(1) - 2 = -\gamma - 2 \ln 2$ by the reflection value, so

$$\int_0^1 \frac{1 - x^{-1/2}}{1 - x} dx = \psi\left(\frac{1}{2}\right) + \gamma = -2 \ln 2 \approx -1.386294.$$

A rational argument needs Gauss's theorem, which reflection plus the recurrence generates for small denominators.

At $a = \frac{1}{3}$,

$$\int_0^1 \frac{1 - x^{-2/3}}{1 - x} dx = \psi\left(\frac{1}{3}\right) + \gamma = -\frac{3}{2} \ln 3 - \frac{\pi}{2\sqrt{3}} \approx -2.554818,$$

which is confirmed to twelve digits by quadrature. Similarly $\psi(\frac{1}{4}) + \gamma = -\frac{\pi}{2} - 3 \ln 2$.

THE PARAMETER DERIVATIVE IN ACTION

Where digamma really earns its keep is as the derivative of a Beta function. Differentiate $\int_0^1 x^{p-1} dx = \frac{1}{p}$ in p to get $\int_0^1 x^{p-1} \ln x dx = -\frac{1}{p^2}$, the single most used identity in the whole of parameter differentiation. Integrating that back in p between two values gives the Frullani-type result

$$\int_0^1 \frac{x^a - x^b}{\ln x} dx = \ln \frac{a+1}{b+1},$$

so $\int_0^1 \frac{x^3 - x}{\ln x} dx = \ln 2 \approx 0.693147$, confirmed by quadrature.

One level up, differentiate the Beta function itself:

$$\frac{\partial}{\partial p} B(p, q) = B(p, q) [\psi(p) - \psi(p+q)],$$

which evaluates $\int_0^1 x^{p-1} (1-x)^{q-1} \ln x dx$ in one line. At $p = q = 1$ that reads $\int_0^1 \ln x dx = \psi(1) - \psi(2) = -1$, the correct answer, which is the check to run whenever you write the formula down.

WHERE THIS GOES WRONG

Both halves of $\frac{1 - x^{a-1}}{1 - x}$ blow up at $x = 1$ individually; only the combination converges, so you may not split the integral and evaluate the pieces. And ψ at a general rational argument is not reachable from the recurrence alone: it needs Gauss's digamma theorem, which supplies the cot and $\ln$ terms visible in the $a = \frac{1}{3}$ answer above. Quoting $\psi(\frac{1}{3}) = -\gamma - 3 \ln 3$ or any other guess without the $\pi \cot$ piece is a standard way to lose a problem.

SEE ALSO The Beta function; Euler-Mascheroni γ ; Zeta values and Apéry's constant; Feynman differentiation.

The polylogarithm ladder

Digamma handles one logarithm against a power. Two or more logarithms, or a logarithm against $\frac{1}{1-x}$, produce something Gamma cannot express, and the right object is the polylogarithm. It is worth understanding structurally rather than as a list: Li_s is defined so that dividing by x and integrating raises s by one, which makes the whole family a ladder that you climb one rung per integration.

25.6 Polylogarithm ladder

CUE $\frac{\ln(1-x)}{x}$, $\frac{\ln^k x}{1 \pm x}$, $\frac{\text{Li}_n(x)}{x}$, or a series $\sum x^n/n^s$ emerging from a termwise integration. Any integral on $[0, 1]$ whose integrand is a logarithm over a linear denominator lands here.

METHOD The definition and the ladder step:

$$\text{Li}_s(x) = \sum_{n \geq 1} \frac{x^n}{n^s}, \quad \frac{d}{dx} \text{Li}_{s+1}(x) = \frac{\text{Li}_s(x)}{x}, \quad \text{Li}_1(x) = -\ln(1-x).$$

So $\int_0^1 \frac{\text{Li}_s(x)}{x} dx = \text{Li}_{s+1}(1) = \zeta(s+1)$. At the endpoints, $\text{Li}_s(1) = \zeta(s)$ and $\text{Li}_s(-1) = -\eta(s) = -(1-2^{1-s})\zeta(s)$. The two reduction identities worth carrying are Landen,

$$\text{Li}_2(x) + \text{Li}_2(1-x) = \frac{\pi^2}{6} - \ln x \ln(1-x),$$

and inversion, $\text{Li}_2(-x) + \text{Li}_2(-\frac{1}{x}) = -\frac{\pi^2}{6} - \frac{1}{2} \ln^2 x$. Between them they move any dilogarithm argument into $[0, \frac{1}{2}]$, where the series converges fast.

WORKED EXAMPLE

The base case is the whole idea in miniature:

$$\int_0^1 \frac{\ln(1-x)}{x} dx = -\int_0^1 \frac{\text{Li}_1(x)}{x} dx = -\text{Li}_2(1) = -\frac{\pi^2}{6} \approx -1.644934.$$

Climbing two rungs at once,

$$\int_0^1 \frac{\text{Li}_3(x) - \text{Li}_3(-x)}{x} dx = \text{Li}_4(1) - \text{Li}_4(-1) = \zeta(4) + \eta(4) = \frac{15}{8}\zeta(4) = \frac{\pi^4}{48} \approx 2.029356,$$

since $\eta(4) = (1-2^{-3})\zeta(4) = \frac{7}{8}\zeta(4)$.

A harder one, which is the standard route to $\zeta(3)$. Expand $\ln(1-x) = -\sum x^n/n$ in

$$\int_0^1 \frac{\ln x \ln(1-x)}{x} dx = -\sum_{n \geq 1} \frac{1}{n} \int_0^1 x^{n-1} \ln x dx = \sum_{n \geq 1} \frac{1}{n^3} = \zeta(3) \approx 1.202057,$$

using $\int_0^1 x^{n-1} \ln x dx = -1/n^2$. Each of the two logarithms contributed one power of n downstairs; that is the arithmetic of the ladder.

Change the sign inside the logarithm and the same expansion produces the alternating counterparts:

$$\int_0^1 \frac{\ln(1+x)}{x} dx = \eta(2) = \frac{\pi^2}{12} \approx 0.822467, \quad \int_0^1 \frac{\ln x \ln(1+x)}{x} dx = -\eta(3) = -\frac{3}{4}\zeta(3) \approx -0.901543.$$

Both are verified numerically to twelve digits. The rule to take away: a $\ln(1-x)$ gives ζ , a $\ln(1+x)$ gives η , and each additional $\ln x$ raises the order by one.

WHERE THIS GOES WRONG

Two sign traps. Li_s has a branch cut along $[1, \infty)$, and the inversion identity changes form across it, so applying the real version to $x > 1$ produces a real answer for something with an imaginary part. And $\text{Li}_n(-1) = -\eta(n) = -(1-2^{1-n})\zeta(n)$: the minus sign and the η -to- ζ conversion are separate opportunities to slip. Test the conversion at $n=2$, where it must give $-\pi^2/12$.

SEE ALSO Zeta values and Apéry's constant; Log-sine integrals and the Clausen function; Catalan's constant G ; Geometric expansion.

HOW TO ACTUALLY COMPUTE A DILOGARITHM

The defining series $\sum x^n/n^2$ converges for $|x| \leq 1$ but slowly near $x=1$, where you need thousands of terms for six digits. The standard recipe is to move the argument first. If $x > \frac{1}{2}$, use Landen, $\text{Li}_2(x) = \frac{\pi^2}{6} - \ln x \ln(1-x) - \text{Li}_2(1-x)$, which sends x to $1-x$ and therefore into $[0, \frac{1}{2}]$. If $|x| > 1$, use inversion first to get inside the unit disc. Then sum the series, which on $[0, \frac{1}{2}]$ gains at least one digit every three terms. Check: $\text{Li}_2(0.9) = 1.2997147230$ by direct summation, and Landen reproduces it from $\text{Li}_2(0.1)$, which needs eight terms.

The values at the top of the ladder are zeta values, and they appear so often that they deserve their own recognition cue. The pattern to internalise is that a k -fold logarithm against a geometric denominator produces $\zeta(k+1)$, with a rational or η -type factor in front decided by the sign in the denominator.

25.7 Zeta values and Apéry's constant

CUE $\frac{x^{s-1}}{e^x - 1}$ or $\frac{x^{s-1}}{e^x + 1}$ on $(0, \infty)$; $\frac{\ln^k x}{1 \pm x}$ on $[0, 1]$; a double or triple sum with denominators $n^2 m$ or n^3 ; a Planck-type or Fermi-type integrand out of physics.

METHOD Expand the denominator as a geometric series and integrate termwise. The two master formulas are

$$\int_0^\infty \frac{x^{s-1}}{e^x - 1} dx = \Gamma(s)\zeta(s), \quad \int_0^\infty \frac{x^{s-1}}{e^x + 1} dx = \Gamma(s)\eta(s) = (1 - 2^{1-s})\Gamma(s)\zeta(s),$$

both valid for $s > 1$. The even values are elementary in π : $\zeta(2) = \frac{\pi^2}{6}$, $\zeta(4) = \frac{\pi^4}{90}$, $\zeta(6) = \frac{\pi^6}{945}$. The odd values are not. Apéry's constant $\zeta(3) = 1.2020569032 \dots$ is irrational and has no known expression in π and elementary constants; when it appears, it is the answer.

WORKED EXAMPLE

The Planck integral, and its two logarithmic cousins:

$$\int_0^\infty \frac{x^2}{e^x - 1} dx = \Gamma(3)\zeta(3) = 2\zeta(3) \approx 2.404114.$$

Substituting $x = -\ln t$ in the same integral turns it into $\int_0^1 \frac{\ln^2 t}{1-t} dt$, which therefore also equals $2\zeta(3)$. Flipping the sign in the denominator brings in η :

$$\int_0^1 \frac{\ln^2 t}{1+t} dt = \Gamma(3)\eta(3) = 2\left(1 - \frac{1}{4}\right)\zeta(3) = \frac{3}{2}\zeta(3) \approx 1.803085,$$

and splitting into even and odd n instead gives $\int_0^1 \frac{\ln^2 t}{1-t^2} dt = \frac{7}{4}\zeta(3) \approx 2.103600$. All three are the same expansion with different bookkeeping.

WHERE THIS GOES WRONG

The factor $\Gamma(s)$ is easy to drop, because at $s = 2$ it equals 1 and the formula looks like a pure zeta value. At $s = 3$ it is 2 and at $s = 4$ it is 6, and dropping it halves or sixths the answer. Separately: $\zeta(s)$ diverges at $s = 1$, so the representation fails there; $\int_0^\infty \frac{dx}{e^x - 1}$ is divergent at the origin even though the η version, $\int_0^\infty \frac{dx}{e^x + 1} = \ln 2$, is perfectly finite.

SEE ALSO Polylogarithm ladder; Gamma-function moment; Digamma and polygamma; Geometric expansion.

TRANSCENDENTAL WEIGHT

There is a grading on these constants that predicts the shape of an answer before you compute it. Assign weight 1 to π and to $\ln 2$, weight 2 to π^2 , $\ln^2 2$ and Li_2 values, weight 3 to $\zeta(3)$, $\pi^2 \ln 2$ and $\ln^3 2$, and in general weight k to $\zeta(k)$ and to any product whose weights sum to k . An integral on $[0, 1]$ whose integrand carries k logarithms almost always evaluates to a rational combination of weight- k constants and nothing else. So $\int_0^1 \frac{\ln^2 x}{1+x} dx$ must be a multiple of $\zeta(3)$, and it is; $\int_0^{\pi/2} \ln^2 \sin x dx$ must be a combination of $\pi \ln^2 2$ and π^3 , and it is. Use the grading as a filter on candidate answers: an answer of mixed weight is nearly always an algebra error.

The Gaussian family, and what happens on a finite interval

Gaussian integrals over the whole line are exact; over a finite interval they are not, and the closed form is the error function. Understanding that pair correctly is what stops you wasting time. The same phenomenon repeats one level up with oscillatory Gaussians, where the answer over the whole half-line is exact and the Fresnel integrals are the finite-interval version.

25.8 Gaussian and error function

CUE e^{-x^2} , or any e^{-ax^2+bx+c} , on the whole line, the half-line, or a finite interval. Also $e^{-x^2/2}$ with a probabilistic dressing, which is the same integral with $a = \frac{1}{2}$.

METHOD Over the whole line the value is exact:

$$\int_{-\infty}^{\infty} e^{-ax^2} dx = \sqrt{\frac{\pi}{a}}, \quad \int_0^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2},$$

the first proved by squaring and passing to polar coordinates. A linear term is removed by completing the square, which costs a constant factor and a shift of the (infinite) interval:

$$\int_{-\infty}^{\infty} e^{-ax^2+bx} dx = \sqrt{\frac{\pi}{a}} e^{b^2/(4a)}.$$

Over a finite interval there is no elementary antiderivative and the closed form is $\text{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-t^2} dt$, with $\text{erf}(\infty) = 1$ and $\text{erfc} = 1 - \text{erf}$.

WORKED EXAMPLE

Complete the square with $a = 2$, $b = 3$:

$$\int_{-\infty}^{\infty} e^{-2x^2+3x} dx = \sqrt{\frac{\pi}{2}} e^{9/8} \approx 3.860479,$$

confirmed by quadrature to fifteen digits. On a finite interval the answer stops being elementary and starts being erf:

$$\int_0^1 e^{-x^2} dx = \frac{\sqrt{\pi}}{2} \text{erf}(1) \approx 0.746824,$$

and there is nothing further to do; $\text{erf}(1) = 0.842700793$ is a tabulated number. A power weight sends you back to Γ rather than erf: $\int_0^{\infty} \sqrt{x} e^{-x^2} dx = \frac{1}{2} \Gamma(\frac{3}{4}) \approx 0.612708$.

A GAUSSIAN AGAINST AN OSCILLATION

Multiply the Gaussian by a cosine and the answer is still a Gaussian, in the frequency. Set $F(b) = \int_0^{\infty} e^{-x^2} \cos(bx) dx$ and differentiate under the integral: $F'(b) = -\int_0^{\infty} x e^{-x^2} \sin(bx) dx$, which integrates by parts (with $u = \sin bx$, $dv = x e^{-x^2} dx$) to $-\frac{b}{2} F(b)$. That is a first-order linear ODE with $F(0) = \frac{\sqrt{\pi}}{2}$, so

$$\int_0^{\infty} e^{-x^2} \cos(bx) dx = \frac{\sqrt{\pi}}{2} e^{-b^2/4},$$

which at $b = 1.3$ gives 0.5808386697, matching quadrature to fifteen digits. The Fourier transform of a Gaussian is a Gaussian, and this is the shortest honest proof of it.

WHERE THIS GOES WRONG

Hunting for an elementary antiderivative on a finite interval. There is none, by Liouville's theorem, and erf is the closed form: writing it is finishing, not giving up. The other standard slip is the factor of two: $\int_{-\infty}^{\infty} = \sqrt{\pi/a}$ while $\int_0^{\infty}$ is half that. And when completing the square, the shift $x \mapsto x - \frac{b}{2a}$ is harmless only because the interval is infinite; on a finite interval it moves the endpoints and the answer becomes a difference of two erf values.

SEE ALSO Gamma-function moment; Fresnel integrals; The parameter ODE; Rotating the contour without contours.

Replace the decaying exponential e^{-x^2} by the oscillating e^{ix^2} and the integral still converges over the half-line, but for a completely different reason: not decay but cancellation. The values are the Fresnel integrals, and they are the first place in this chapter where convergence is only conditional, which changes what manipulations are legal.

25.9 Fresnel integrals

CUE $\sin(x^2)$ or $\cos(x^2)$ on $(0, \infty)$; more generally $\sin(x^p)$ for $p > 1$; or $\frac{\sin u}{\sqrt{u}}$, which is what $u = x^2$ produces.

METHOD Substitute $u = x^p$ to reduce to a Mellin transform of the sine, then quote

$$\int_0^\infty u^{s-1} \sin u \, du = \Gamma(s) \sin \frac{\pi s}{2}, \quad \int_0^\infty u^{s-1} \cos u \, du = \Gamma(s) \cos \frac{\pi s}{2}, \quad 0 < s < 1.$$

For $p = 2$ this gives the classical pair

$$\int_0^\infty \sin(x^2) \, dx = \int_0^\infty \cos(x^2) \, dx = \frac{1}{2} \sqrt{\frac{\pi}{2}} \approx 0.626657.$$

The general statement is $\int_0^\infty \sin(x^p) \, dx = \Gamma(1 + \frac{1}{p}) \sin \frac{\pi}{2p}$.

WORKED EXAMPLE

Take $p = 2$ and do it by hand. With $u = x^2$, $dx = \frac{du}{2\sqrt{u}}$, so

$$\int_0^\infty \sin(x^2) \, dx = \frac{1}{2} \int_0^\infty u^{-1/2} \sin u \, du = \frac{1}{2} \Gamma\left(\frac{1}{2}\right) \sin \frac{\pi}{4} = \frac{\sqrt{\pi}}{2} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2} \sqrt{\frac{\pi}{2}},$$

matching 0.6266571 to seven digits under accelerated numerical summation of the half-period pieces.

For $p = 3$ the same formula gives $\int_0^\infty \sin(x^3) \, dx = \Gamma(\frac{4}{3}) \sin \frac{\pi}{6} = \frac{1}{2} \Gamma(\frac{4}{3}) \approx 0.446490$, again confirmed numerically.

WHERE THIS GOES WRONG

These converge only conditionally, so every step must respect that. No absolute-value comparison test, no naive Tonelli, no interchanging a limit with the integral without a separate argument. Practically: naive numerical quadrature on the raw $\sin(x^2)$ over a truncated range does *not* converge to the answer, because the tail contributes at the same order as your truncation error. Substitute $u = x^2$ first, or sum the alternating half-period contributions with an averaging acceleration, before you trust a numerical check.

SEE ALSO Gaussian and error function; Rotating the contour without contours; Oscillatory convergence; Named-function answers.

Oscillation divided by x : the sine-integral family

The Fresnel integrals converge because the phase accelerates. The other standard mechanism is a slowly decaying amplitude against a fixed frequency, and its type specimen is $\frac{\sin x}{x}$. This is the most famous conditionally convergent integral in analysis, and its finite-interval version is a named function that many readers refuse to accept as an answer.

25.10 Dirichlet integral, Si and Ci

CUE $\frac{\sin x}{x}$ over an infinite range, or $\frac{\sin(\text{anything})}{\text{that same thing}}$ after a substitution; also bounds that are engineered to be absurdly large, which is a signal that the intended answer is a special-function value rather than a limit.

METHOD The Dirichlet value is

$$\int_0^\infty \frac{\sin x}{x} \, dx = \frac{\pi}{2},$$

derived by inserting a convergence factor: set $I(a) = \int_0^\infty e^{-ax} \frac{\sin x}{x} \, dx$, differentiate under the integral to get $I'(a) = -\frac{1}{1+a^2}$, integrate to $I(a) = \arctan \frac{1}{a}$, and let $a \rightarrow 0^+$. For finite bounds, name the answer:

$$\text{Si}(z) = \int_0^z \frac{\sin t}{t} \, dt, \quad \text{Ci}(z) = -\int_z^\infty \frac{\cos t}{t} \, dt = \gamma + \ln z + \int_0^z \frac{\cos t - 1}{t} \, dt.$$

WORKED EXAMPLE

The substitution that manufactures a sine integral out of nothing. In $\int_{-\infty}^{20} \sin(\pi e^x) \, dx$ put $u = e^x$, so $dx = du/u$ and the range becomes $(0, e^{20})$:

$$\int_{-\infty}^{20} \sin(\pi e^x) \, dx = \int_0^{e^{20}} \frac{\sin(\pi u)}{u} \, du = \text{Si}(\pi e^{20}) \approx 1.5707963270.$$

The special function is the closed form. The enormous upper limit is a joke at the reader's expense: $\text{Si}(\pi e^{20})$ agrees with $\frac{\pi}{2} = 1.5707963268$ to nine decimal places, but it is not equal to it.

The damping trick also handles the finite- a family directly: $\int_0^\infty e^{-0.7x} \frac{\sin x}{x} dx = \arctan \frac{1}{0.7} \approx 0.960070$.

Three relatives worth having by sight, all obtained by product-to-sum followed by the Dirichlet value:

$$\int_0^\infty \frac{\sin^2 x}{x^2} dx = \frac{\pi}{2}, \quad \int_0^\infty \frac{\sin^3 x}{x} dx = \frac{\pi}{4}, \quad \int_0^\infty \frac{\cos ax - \cos bx}{x} dx = \ln \frac{b}{a}.$$

The last is Frullani, and it is the cosine integral in disguise: the individual pieces are Ci-divergent at the origin and only the difference survives, which is exactly the γ mechanism of a later section. At $a = 1, b = 2$ it gives $\ln 2 \approx 0.693147$.

WHERE THIS GOES WRONG

$\int_0^\infty \left| \frac{\sin x}{x} \right| dx = \infty$, so the integral is conditionally and not absolutely convergent, and dominated convergence is unavailable for the $a \rightarrow 0^+$ step. That limit needs Abel's theorem or an explicit estimate on the oscillation, not a wave of the hand. A separate trap: Ci has a logarithmic singularity at the origin and $\text{Ci}(z) \rightarrow -\infty$ as $z \rightarrow 0^+$, so $\int_0^z \frac{\cos t}{t} dt$ is divergent while $\int_0^z \frac{\sin t}{t} dt$ is not. The two look symmetric and are not.

SEE ALSO Lobachevsky's formula; Feynman differentiation; Fresnel integrals; Frullani's integral.

CHECKING A CONDITIONALLY CONVERGENT VALUE

The integrals of the last two techniques share a hazard: naive numerical quadrature on a truncated range does not converge to their values, because the tail contributes at the same order as the truncation. Two reliable fixes. Substitute first, so that $\int_0^\infty \sin(x^2) dx$ becomes $\frac{1}{2} \int_0^\infty u^{-1/2} \sin u du$, where the amplitude decays. Or integrate over consecutive half-periods, forming an alternating series of the pieces, and accelerate it by repeated averaging; twenty averaging passes over two hundred half-periods gives twelve digits for $\int_0^\infty \frac{\sin x}{x} dx$ and for the Fresnel values. Every conditionally convergent number quoted in this chapter was checked that way, not by truncation.

Logarithms of trigonometric functions

The next family has nothing to do with oscillation. It arises whenever a logarithm meets a sine or cosine, which happens constantly, because $\ln \sin$ is what falls out of $\ln|1 - e^{i\theta}|$ and therefore out of any Fourier computation. The base value is elementary to derive and worth being able to reproduce in three lines.

HARD

25.11 Log-sine integrals and the Clausen function

CUE $\ln \sin x, \ln \cos x, \ln \tan x, \ln(2 \sin \frac{x}{2})$, or a squared or cubed version, over $[0, \frac{\pi}{2}]$, $[0, \pi]$ or a partial range.

METHOD The base value comes from a fold. Writing $L = \int_0^{\pi/2} \ln \sin x dx$, complementarity gives $L = \int_0^{\pi/2} \ln \cos x dx$, so $2L = \int_0^{\pi/2} \ln(\frac{1}{2} \sin 2x) dx = -\frac{\pi}{2} \ln 2 + L$, hence

$$\int_0^{\pi/2} \ln \sin x dx = \int_0^{\pi/2} \ln \cos x dx = -\frac{\pi}{2} \ln 2, \quad \int_0^\pi \ln \sin x dx = -\pi \ln 2.$$

The second moment needs the Fourier series $\ln(2 \sin \frac{\theta}{2}) = -\sum_{n \geq 1} \frac{\cos n\theta}{n}$ and gives

$$\int_0^{\pi/2} \ln^2(\sin x) dx = \frac{\pi}{2} \left(\ln^2 2 + \frac{\pi^2}{12} \right) \approx 2.046622.$$

Partial ranges are not elementary: they define the Clausen function $\text{Cl}_2(\theta) = -\int_0^\theta \ln|2 \sin \frac{t}{2}| dt = \sum_{n \geq 1} \frac{\sin n\theta}{n^2}$.

WORKED EXAMPLE

The normalising 2 is exactly what makes the full period vanish:

$$\int_0^{2\pi} \ln(2 \sin \frac{x}{2}) dx = 0,$$

which is the n -integrated form of the Fourier series above, every $\int_0^{2\pi} \cos n\theta d\theta$ being zero. Numerically the quadrature returns -7×10^{-15} .

Two Clausen values worth knowing: $\text{Cl}_2(\frac{\pi}{2}) = G = 0.9159655942$, Catalan's constant, and the maximum $\text{Cl}_2(\frac{\pi}{3}) = 1.0149416064$, which is the largest value the function attains. And a mixed moment that combines this section with the last: $\int_0^{\pi/2} x \ln(\sin x) dx = \frac{7}{16}\zeta(3) - \frac{\pi^2}{8} \ln 2 \approx -0.329236$.

The cross moment is the cleanest illustration of the weight grading. Since $\ln \sin x \ln \cos x$ carries two logarithms, the answer must be a combination of $\pi \ln^2 2$ and π^3 , and it is:

$$\int_0^{\pi/2} \ln(\sin x) \ln(\cos x) dx = \frac{\pi}{2} \ln^2 2 - \frac{\pi^3}{48} \approx 0.108730.$$

Expanding $\ln^2(\sin x \cos x) = \ln^2 \sin x + 2 \ln \sin x \ln \cos x + \ln^2 \cos x$ and using $\sin x \cos x = \frac{1}{2} \sin 2x$ reduces this to the two moments already computed.

WHERE THIS GOES WRONG

The integrand is $-\infty$ at the endpoint, so every manipulation is on an improper integral and the fold above needs the convergence checked once, at the start. The "2" inside $\ln(2 \sin \frac{x}{2})$ is not decorative: it is the normalisation that makes the mean value zero, and dropping it shifts every Clausen value by a multiple of $\ln 2$. Finally, $\int_0^\pi \ln \sin x dx = -\pi \ln 2$ is twice the quarter-range value, not the same as it; the two ranges give different numbers because the interval doubled, not because the integrand changed.

SEE ALSO Catalan's constant G ; Polylogarithm ladder; Fourier series kernels; Zeta values and Apéry's constant.

The constants that keep appearing

There is a reason these three sit together rather than being scattered through the chapter. Each is a *terminal* constant: an answer, not a step. Everything else in this chapter reduces, given enough identities, to π , $\ln 2$, Gamma values at rational points, and these three. Knowing which numbers are terminal is what tells you when to stop, and stopping at the right moment is a large fraction of the skill.

Three numbers turn up again and again in this part of the subject, and each has a characteristic generating mechanism. Learning to recognise the mechanism is what lets you stop early instead of grinding for a closed form that does not exist. Catalan's constant comes from odd squares with alternating signs, Euler's γ comes from the difference of two divergences, and $\zeta(3)$ comes from a threefold logarithmic weight. None of the three simplifies further.

25.12 Catalan's constant G

MEDIUM

CUE $\frac{\arctan x}{x}$ on $[0, 1]$, $\frac{\ln x}{1+x^2}$ on $[0, 1]$, $\int_0^{\pi/4} \ln(\cot x) dx$, $\frac{x}{\sin x}$ on $[0, \frac{\pi}{2}]$, or any sum that reduces to $\sum (-1)^k (2k+1)^{-2}$.

METHOD The constant is $G = \beta(2) = \sum_{k \geq 0} \frac{(-1)^k}{(2k+1)^2} \approx 0.9159655942$. It appears whenever an alternating odd-denominator series, that is an arctan or a Li at i , is integrated once more against $\frac{dx}{x}$ or $\ln x dx$, because $\int_0^1 x^{2k} dx = \frac{1}{2k+1}$ supplies the second odd denominator. Equivalently $G = \text{Cl}_2(\frac{\pi}{2})$.

WORKED EXAMPLE

Expand and integrate termwise:

$$\int_0^1 \frac{\arctan x}{x} dx = \int_0^1 \sum_{k \geq 0} \frac{(-1)^k x^{2k}}{2k+1} dx = \sum_{k \geq 0} \frac{(-1)^k}{(2k+1)^2} = G \approx 0.915966.$$

The same constant, three more ways, all verified numerically to fifteen digits:

$$\int_0^1 \frac{-\ln x}{1+x^2} dx = G, \quad \int_0^{\pi/4} \ln(\cot x) dx = G, \quad \int_0^{\pi/2} \frac{x}{\sin x} dx = 2G.$$

The second is the first after $x = \tan \theta$; the third comes from the same arctan series against a different weight.

A case where G appears with a companion, which is the usual situation: $\int_0^1 \frac{\ln(1+x^2)}{1+x^2} dx = \frac{\pi}{2} \ln 2 - G \approx 0.172827$.

Substituting $x = \tan \theta$ turns it into $-2 \int_0^{\pi/4} \ln \cos \theta d\theta$, so it sits at the junction of this section and the log-sine section, and the two constants that appear are exactly the two that the weight grading permits.

G has no known expression in π , $\ln 2$ and zeta values, and it is not known whether it is irrational. Grinding to reduce it further wastes the problem. Recognise it, write it, and stop. The converse trap is naming it too early: $\int_0^1 \frac{\arctan x}{1+x^2} dx$ looks like the same family but is elementary, equal to $\frac{1}{2} \arctan^2(1) = \frac{\pi^2}{32}$, because the weight is the derivative of the argument.

SEE ALSO Log-sine integrals and the Clausen function; Polylogarithm ladder; Zeta values and Apéry's constant.

Catalan's constant is generated by a series. The Euler-Mascheroni constant is generated by a subtraction, and that difference in mechanism is the whole reason it behaves differently: it appears in problems where two things separately diverge.

MEDIUM

25.13 Euler-Mascheroni γ

CUE $e^{-x} \ln x$ on the half-line; $\ln \ln \frac{1}{x}$ on $[0, 1]$; a bracketed pair of divergent terms such as $\frac{1}{\ln x} + \frac{1}{1-x}$; or the difference between a harmonic-type divergence and a logarithmic one.

METHOD The definition that matters for integration is $\gamma = -\Gamma'(1) = -\psi(1)$, that is,

$$\gamma = -\int_0^\infty e^{-x} \ln x \, dx \approx 0.5772156649.$$

Two equivalent forms cover most sightings:

$$\int_0^1 \ln \ln \frac{1}{x} \, dx = -\gamma, \quad \int_0^1 \left(\frac{1}{\ln x} + \frac{1}{1-x} \right) dx = \gamma,$$

the first being the previous one after $x = e^{-t}$, the second a genuine cancellation of two logarithmic divergences at $x = 1$.

WORKED EXAMPLE

Verify the pair form directly. Near $x = 1$ write $x = 1 - \varepsilon$: then $\frac{1}{\ln x} \approx -\frac{1}{\varepsilon} - \frac{1}{2}$ and $\frac{1}{1-x} = \frac{1}{\varepsilon}$, so the $\frac{1}{\varepsilon}$ poles cancel and the integrand tends to $-\frac{1}{2}$. The integral therefore converges, and quadrature gives 0.5772156649, matching γ to ten digits.

The second moment brings in $\zeta(2)$ alongside γ , and is worth knowing because it is how γ^2 enters problems at all:

$$\int_0^\infty e^{-x} \ln^2 x \, dx = \gamma^2 + \frac{\pi^2}{6} \approx 1.978112,$$

which is $\Gamma''(1)$, and follows from $\Gamma''(1) = \Gamma(1)[\psi(1)^2 + \psi_1(1)]$ with $\psi(1) = -\gamma$ and $\psi_1(1) = \zeta(2)$.

A cue in disguise: $\int_0^1 \ln \ln \frac{1}{x} \, dx$. Put $x = e^{-t}$, so $dx = -e^{-t} dt$ and the integral becomes $\int_0^\infty e^{-t} \ln t \, dt = -\gamma \approx -0.577216$. Whenever $\ln \ln$ appears on $[0, 1]$, that substitution is the move.

WHERE THIS GOES WRONG

Splitting the pair. Each of $\int_0^1 \frac{dx}{\ln x}$ and $\int_0^1 \frac{dx}{1-x}$ diverges logarithmically at $x = 1$, so the difference is meaningful only if the two cut-offs are taken *simultaneously*: regularise as $\int_0^{1-\varepsilon}$ of the whole bracket and then let $\varepsilon \rightarrow 0$. Cutting the two terms at different points changes the answer by any constant you like.

SEE ALSO Digamma and polygamma; Frullani's integral; Epsilon-regularisation of a divergent pair.

Where the ladder runs out

Everything so far has been, in the end, Gamma and its derivatives plus the polylogarithm. That covers a startling amount of ground, but it stops at a definite frontier: a square root of a cubic or quartic. There the integral defines a genuinely new function of the coefficients, and no rearrangement brings it back.

25.14 Elliptic integrals K and E

CUE $\frac{1}{\sqrt{1-k^2 \sin^2 \theta}}$; $\frac{1}{\sqrt{(1-x^2)(1-k^2 x^2)}}$; $\frac{1}{\sqrt{1-x^4}}$ or any square root of a cubic or quartic with distinct roots; the exact period of a pendulum at finite amplitude; the arc length of an ellipse.

METHOD Name them and stop:

$$K(k) = \int_0^{\pi/2} \frac{d\theta}{\sqrt{1-k^2 \sin^2 \theta}}, \quad E(k) = \int_0^{\pi/2} \sqrt{1-k^2 \sin^2 \theta} d\theta.$$

A quartic under a square root is elliptic unless a special substitution reduces it. At the lemniscatic value $k^2 = \frac{1}{2}$ the elliptic integral does collapse to Gammas, which is why $\int_0^1 (1-x^4)^{-1/2} dx$ has a closed form while the general case does not.

WORKED EXAMPLE

The lemniscatic case, done through Beta rather than through elliptic tables. Put $u = x^4$:

$$\int_0^1 \frac{dx}{\sqrt{1-x^4}} = \frac{1}{4} \int_0^1 u^{-3/4} (1-u)^{-1/2} du = \frac{1}{4} B\left(\frac{1}{4}, \frac{1}{2}\right) = \frac{\Gamma(\frac{1}{4})\Gamma(\frac{1}{2})}{4\Gamma(\frac{3}{4})}.$$

Now use reflection, $\Gamma(\frac{3}{4}) = \pi\sqrt{2}/\Gamma(\frac{1}{4})$:

$$= \frac{\Gamma(\frac{1}{4})^2 \sqrt{\pi}}{4\pi\sqrt{2}} = \frac{\Gamma(\frac{1}{4})^2}{4\sqrt{2}\pi} \approx 1.311029,$$

matching quadrature to twelve digits. Equivalently this is $\frac{1}{\sqrt{2}}K(\frac{1}{\sqrt{2}})$ with $K(\frac{1}{\sqrt{2}}) = \Gamma(\frac{1}{4})^2/(4\sqrt{\pi}) \approx 1.854075$.

THE PENDULUM, WHICH IS WHERE THESE CAME FROM

A pendulum of length L released from angle θ_0 has exact period

$$T = 4\sqrt{\frac{L}{g}} K\left(\sin \frac{\theta_0}{2}\right),$$

against the small-angle prediction $2\pi\sqrt{L/g}$. At $\theta_0 = \frac{\pi}{2}$ the modulus is $\sin \frac{\pi}{4} = \frac{1}{\sqrt{2}}$, so the ratio of true period to small-angle period is

$$\frac{4K(1/\sqrt{2})}{2\pi} = \frac{2 \cdot 1.8540746773}{\pi} \approx 1.1803406,$$

an 18 percent correction. The lemniscatic modulus is not a coincidence here: $\theta_0 = \frac{\pi}{2}$ is exactly the amplitude at which the elliptic integral collapses to Gammas.

WHERE THIS GOES WRONG

Two conventions for the modulus, and software disagrees with tables. Some write $K(k)$ with k the modulus, others $K(m)$ with $m = k^2$ the parameter; most numerical libraries use m . State which you use, and sanity-check against K at a value you know. The other trap runs the other way: a cubic or quartic radical is not automatically elliptic. If the radical is $\sqrt{Q(x)}$ and the rest of the integrand is a multiple of $Q'(x)$, the integral is elementary, because $\int Q' \sqrt{Q} dx = \frac{2}{3} Q^{3/2}$. Look for that before you name anything.

SEE ALSO The Beta function; Euler reflection and the Mellin transform; Named-function answers; Sine substitution.

Elliptic integrals are the best-known place where naming is the answer, but they are not the only one. The general habit is worth stating explicitly, because it is the habit this whole chapter is trying to install: when a substitution lands you on a shape that has a name, use the name and then use its identities, rather than continuing to push symbols around.

25.15 Named-function answers

CUE A substitution lands you on we^w , on $\cos(\sin x)$, on $\int_0^z e^{-t} t^{a-1} dt$, or on an oscillatory kernel with a cosh or an Airy function in the denominator. In general: a shape that resists every elementary method but keeps reappearing.

METHOD Stop and name it. Lambert W is defined by $W(x)e^{W(x)} = x$; the Bessel function by $J_0(z) = \frac{1}{\pi} \int_0^\pi \cos(z \sin x) dx$; the incomplete Gammas by $\gamma(a, z) = \int_0^z e^{-t} t^{a-1} dt$ and $\Gamma(a, z) = \Gamma(a) - \gamma(a, z)$; and the Airy pair through the Wronskian identity $\frac{d}{dz} \arctan \frac{\text{Bi}(z)}{\text{Ai}(z)} = \frac{1/\pi}{\text{Ai}^2 + \text{Bi}^2}$. Then integrate the named object using its own identities, most often inverse-function integration or a recurrence.

WORKED EXAMPLE

Two instances. First, recognition of a Bessel integral:

$$\int_0^\pi \cos(\sin x) dx = \pi J_0(1) \approx 2.403939,$$

straight from the definition with $z = 1$. Second, an inverse function integrated by the graph-swapping identity: with $x = te^t$, so that $W(x) = t$ and $dx = (1+t)e^t dt$, the limits $x = 0$ and $x = e$ become $t = 0$ and $t = 1$, and

$$\int_0^e W(x) dx = \int_0^1 t(1+t)e^t dt = e - 1 \approx 1.718282.$$

The remaining integral is elementary because the substitution moved the difficulty into the limits.

INCOMPLETE GAMMAS, AND WHY THEY ARE NOT A COP-OUT

Truncating a Gamma integral gives the incomplete Gamma, and it obeys the same recurrence as Γ with a boundary term:

$$\gamma(a+1, z) = a\gamma(a, z) - z^a e^{-z}.$$

So $\int_0^1 \sqrt{x} e^{-x} dx = \gamma(\frac{3}{2}, 1) = \frac{1}{2}\gamma(\frac{1}{2}, 1) - e^{-1}$, and $\gamma(\frac{1}{2}, 1) = \sqrt{\pi} \text{erf}(1)$, giving

$$\int_0^1 \sqrt{x} e^{-x} dx = \frac{\sqrt{\pi}}{2} \text{erf}(1) - \frac{1}{e} \approx 0.3789447,$$

confirmed by quadrature. Notice that the incomplete Gamma reduced to erf, which is the general pattern: at half-integer a the incomplete Gammas are error functions, and at integer a they are finite sums times e^{-z} .

WHERE THIS GOES WRONG

The obvious failure is continuing to hunt for an elementary form. The subtler one runs the other way: naming an object when a cancellation was available. Two Bessel terms of equal magnitude and opposite sign cancel exactly, and a difference of incomplete Gammas often telescopes to something elementary. Evaluate and simplify before you name, then name and stop.

SEE ALSO Elliptic integrals K and E ; Inverse-function integration; Dirichlet integral, Si and Ci ; Disguised zero.

Three problems worked end to end

Individual techniques are easy to demonstrate and easy to overrate. What the chapter is really teaching is the chaining, so here are three complete problems in which no single step is difficult and the whole is well past what elementary methods reach. Read them for the order of the moves, not for the algebra.

The first shows parameter differentiation applied to a special-function formula rather than to an integral. Consider

$$I = \int_0^\infty \frac{\ln x}{1+x^3} dx.$$

There is no obvious substitution and no symmetry, but there is a family: the reflection master formula gives $\int_0^\infty \frac{x^{s-1}}{1+x^n} dx = \frac{\pi}{n \sin(\pi s/n)}$ for the whole strip $0 < s < n$, and $\ln x$ is what ∂_s of x^{s-1} produces. Differentiate both sides in s :

$$\int_0^\infty \frac{x^{s-1} \ln x}{1+x^n} dx = -\frac{\pi^2}{n^2} \cdot \frac{\cos(\pi s/n)}{\sin^2(\pi s/n)}.$$

Set $s = 1$, $n = 3$. Then $\cos \frac{\pi}{3} = \frac{1}{2}$ and $\sin^2 \frac{\pi}{3} = \frac{3}{4}$, so

$$I = -\frac{\pi^2}{9} \cdot \frac{1/2}{3/4} = -\frac{2\pi^2}{27} \approx -0.7310818075,$$

which quadrature confirms to twelve digits. The same line at $n = 4$ gives $\int_0^\infty \frac{\ln x}{1+x^4} dx = -\frac{\pi^2\sqrt{2}}{16} \approx -0.872358$. Notice what made this work: not cleverness about the integrand, but the fact that the answer was already known as a function of a parameter, and $\ln x$ is a parameter derivative.

The second problem chains Beta, digamma and trigamma to reach a log-sine moment that was quoted without proof earlier. Start from the trigonometric Beta integral in its one-parameter form,

$$f(p) = \int_0^{\pi/2} \sin^{2p-1} x dx = \frac{1}{2} B\left(p, \frac{1}{2}\right) = \frac{\sqrt{\pi} \Gamma(p)}{2\Gamma(p + \frac{1}{2})}.$$

Differentiating in p brings down a factor $2 \ln \sin x$ each time, so that

$$f'(p) = 2 \int_0^{\pi/2} \sin^{2p-1} x \ln \sin x dx, \quad f''(p) = 4 \int_0^{\pi/2} \sin^{2p-1} x \ln^2 \sin x dx.$$

The right-hand side is easiest to differentiate logarithmically:

$$\frac{f'}{f} = \psi(p) - \psi\left(p + \frac{1}{2}\right), \quad \frac{f''}{f} = \left[\psi(p) - \psi\left(p + \frac{1}{2}\right)\right]^2 + \psi_1(p) - \psi_1\left(p + \frac{1}{2}\right).$$

Evaluate at $p = \frac{1}{2}$, where $f(\frac{1}{2}) = \frac{\pi}{2}$, $\psi(\frac{1}{2}) - \psi(1) = -2 \ln 2$, and $\psi_1(\frac{1}{2}) - \psi_1(1) = \frac{\pi^2}{2} - \frac{\pi^2}{6} = \frac{\pi^2}{3}$. The first derivative gives

$$\int_0^{\pi/2} \ln \sin x dx = \frac{f'(\frac{1}{2})}{2} = \frac{1}{2} \cdot \frac{\pi}{2} (-2 \ln 2) = -\frac{\pi}{2} \ln 2,$$

recovering the base value, and the second gives

$$\int_0^{\pi/2} \ln^2 \sin x dx = \frac{f''(\frac{1}{2})}{4} = \frac{1}{4} \cdot \frac{\pi}{2} \left(4 \ln^2 2 + \frac{\pi^2}{3}\right) = \frac{\pi}{2} \ln^2 2 + \frac{\pi^3}{24} \approx 2.046622.$$

That is exactly the value quoted earlier, now derived. Four named objects appeared, and each was used for precisely one thing: Beta to get a closed form with a parameter, digamma to differentiate it once, trigamma to differentiate it twice, and the reflection value $\psi(\frac{1}{2}) = -\gamma - 2 \ln 2$ to evaluate at the point of interest.

The third problem is the second derivative of the first, and it is included to make a point about how much mileage one parametrised formula gives. Differentiating the master formula twice in s ,

$$\int_0^\infty \frac{x^{s-1} \ln^2 x}{1+x^n} dx = \frac{d^2}{ds^2} \left[\frac{\pi}{n} \csc \frac{\pi s}{n} \right] = \frac{\pi^3}{n^3} \left[\csc^3 \frac{\pi s}{n} + \csc \frac{\pi s}{n} \cot^2 \frac{\pi s}{n} \right].$$

At $s = 1$, $n = 2$ the cotangent vanishes and the cosecant is 1, so

$$\int_0^\infty \frac{\ln^2 x}{1+x^2} dx = \frac{\pi^3}{8} \approx 3.8757845850,$$

confirmed to eleven digits. The first derivative at the same point vanishes, which recovers the familiar $\int_0^\infty \frac{\ln x}{1+x^2} dx = 0$, a fact usually proved by the inversion $x \mapsto 1/x$; here it falls out as a stationary point of $\csc$. One formula, one parameter, and three separate results, two of which look like different problems entirely.

THE FOUR ERRORS THAT ACCOUNT FOR MOST FAILURES

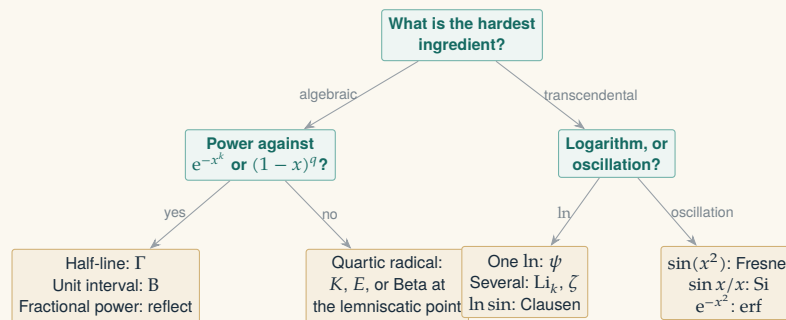
Collected here because they are worth rereading before an examination. One: applying an identity outside its strip of validity, which yields a plausible finite number for a divergent integral. Two: confusing the algebraic and trigonometric Beta conventions, $p-1$ against $2p-1$. Three: dropping the $\Gamma(s)$ factor in $\int_0^\infty \frac{x^{s-1}}{e^x-1} dx = \Gamma(s)\zeta(s)$, invisible at $s=2$ and wrong by a factor 2 at $s=3$. Four: splitting a bracketed pair of divergences, or splitting $\frac{1-x^{n-1}}{1-x}$, where only the combination converges. All four produce clean, confident, wrong answers, and all four are caught in thirty seconds by a numerical check.

A table of values, and how to choose a rung

Nearly every problem in this family is finished by one of a short list of numbers. Keeping the list where you can see it converts a hard integral into a lookup, and, more importantly, lets you recognise a target: if a numerical estimate of your integral is 0.9159655942, you know what the answer is and can work backwards to a proof.

Quantity	Exact value	Decimal
$\Gamma(\frac{1}{2})$	$\sqrt{\pi}$	1.7724538509
$\Gamma(\frac{1}{3})$	no simpler form	2.6789385347
$\Gamma(\frac{1}{4})$	no simpler form	3.6256099082
$\Gamma(\frac{3}{4})$	$\pi\sqrt{2}/\Gamma(\frac{1}{4})$	1.2254167025
$\Gamma(\frac{1}{3})\Gamma(\frac{2}{3})$	$2\pi/\sqrt{3}$	3.6275987285
$\psi(1)$	$-\gamma$	-0.5772156649
$\psi(\frac{1}{2})$	$-\gamma - 2\ln 2$	-1.9635100260
$\psi(\frac{1}{3})$	$-\gamma - \frac{3}{2}\ln 3 - \frac{\pi}{2\sqrt{3}}$	-3.1320337800
$\psi(\frac{1}{4})$	$-\gamma - \frac{\pi}{2} - 3\ln 2$	-4.2274535334
$\zeta(2)$	$\pi^2/6$	1.6449340668
$\zeta(3)$	Apery's constant	1.2020569032
$\zeta(4)$	$\pi^4/90$	1.0823232337
$\eta(2)$	$\pi^2/12$	0.8224670334
$\text{Li}_2(\frac{1}{2})$	$\frac{\pi^2}{12} - \frac{1}{2}\ln^2 2$	0.5822405265
$G = \text{Cl}_2(\frac{\pi}{2})$	no simpler form	0.9159655942
$\text{Cl}_2(\frac{\pi}{3})$	maximum of Cl_2	1.0149416064
γ	$-\Gamma'(1)$	0.5772156649
$\text{erf}(1)$	$\frac{2}{\sqrt{\pi}} \int_0^1 e^{-t^2}$	0.8427007929
$\text{Si}(\infty)$	$\pi/2$	1.5707963268
$K(\frac{1}{\sqrt{2}})$	$\Gamma(\frac{1}{4})^2/(4\sqrt{\pi})$	1.8540746773
$\int_0^\infty \sin(x^2) dx$	$\frac{1}{2}\sqrt{\pi/2}$	0.6266570687

The choice of rung is decided almost entirely by the integrand's ingredients, not by the interval, which is the reverse of the situation in the symmetry chapter. Powers and exponentials mean Gamma; powers on a bounded interval mean Beta; a logarithm means differentiate a parameter and expect digamma or a polylogarithm; oscillation means Fresnel or the sine integral; a quartic radical means elliptic.



GUESS THE CONSTANT FIRST

Before doing any work, estimate the integral numerically to four or five digits and compare against the table. This is not cheating; it is the single most efficient move available at this level. If the estimate is 2.0293561 you are looking at $\pi^4/48$, and knowing that tells you the answer is a fourth-order polylogarithm, which tells you to expand a logarithm and integrate termwise. If the estimate matches nothing, the answer is probably a product or ratio of table entries, and $\ln$ of the estimate is worth checking too. The characteristic mistake at this level is not a wrong technique but a correct technique aimed at the wrong target.

It is worth saying explicitly when this chapter is the wrong tool. A named function is the right answer only when the integral genuinely resists elementary evaluation. If the integrand is a derivative in disguise, if a symmetry fold closes it, or if a substitution makes it rational, none of the machinery here is needed and reaching for it will cost you time and introduce branch conditions you did not have to satisfy. The order of attack is always the same: simplify algebraically, test the standard folds, try the obvious substitutions, and only then ask which special function is hiding. The one exception is when a preliminary numerical estimate already matches a table entry, in which case you may as well aim straight at the target.

One closing warning about the whole chapter. The identities here are all conditional: reflection needs $0 < s < 1$, the Mellin transforms of $\sin$ and $\cos$ need $0 < s < 1$, the zeta representation needs $s > 1$, and the polylogarithm identities need the

argument off the branch cut. In elementary integration a violated hypothesis usually produces an obvious absurdity like a negative area. Here it produces a plausible number, and the only defence is to check the strip explicitly and to verify the final value numerically. Every closed form printed in this chapter was checked that way.

Recognition matrix

If you see	Reach for	If that fails
$x^m e^{-\lambda x^k}$ on $(0, \infty)$	$u = \lambda x^k; \frac{1}{k\lambda^{(m+1)/k}} \Gamma(\frac{m+1}{k})$	Check $\frac{m+1}{k} > 0$
a^{-x^2} with $a \neq e$	Write $\lambda = \ln a$, then Gamma	Watch the power of $\ln a$
$\frac{x^{s-1}}{1+x^n}$ on $(0, \infty)$	Reflection: $\frac{\pi}{n \sin(\pi s/n)}$	Verify $0 < s < n$
$\Gamma(\alpha)\Gamma(1-\alpha)$ anywhere	$\pi / \sin \pi \alpha$	Shift by the recurrence first
$\Gamma(z)$ and $\Gamma(z + \frac{1}{2})$ together	Duplication: $2^{1-2z} \sqrt{\pi} \Gamma(2z)$	Test the constant at $z = \frac{1}{2}$
$x^{p-1}(1-x)^{q-1}$ on $[0, 1]$	$B(p, q) = \Gamma(p)\Gamma(q)/\Gamma(p+q)$	Substitute to reach $[0, 1]$
$\sin^m x \cos^n x$, non-integer	$\frac{1}{2} B(\frac{m+1}{2}, \frac{n+1}{2})$	Solve $2p-1 = m$ explicitly
$x^a(1-x^b)^c$	$u = x^b; \frac{1}{b} B(\frac{a+1}{b}, c+1)$	Then reflection for fractions
A single $\ln x$ against a power	Differentiate Beta or Gamma in the parameter	Expect ψ and γ
$\frac{1-x^{a-1}}{1-x}$ on $[0, 1]$	$\psi(a) + \gamma$	Gauss's digamma theorem
Harmonic numbers in a series	$\psi(n+1) = H_n - \gamma$	Abel summation
$\frac{\ln(1-x)}{x}$ or $\frac{\text{Li}_n(x)}{x}$	Climb the ladder: Li_{n+1}	Landen or inversion
$\frac{\ln^k x}{1 \pm x}$ on $[0, 1]$	Expand; $k! \zeta(k+1)$ or $k! \eta(k+1)$	Keep the $\Gamma(s)$ factor
$\frac{x}{x^s-1}$ on $(0, \infty)$	$\Gamma(s)\zeta(s)$ or $\Gamma(s)\eta(s)$	Diverges at $s = 1$ for the minus
$e^x \mp 1$	Complete the square; $\sqrt{\pi/a} e^{b^2/4a}$	Halve it for $(0, \infty)$
e^{-ax^2+bx} on $\mathbb{R}$	$\frac{\sqrt{\pi}}{2} \text{erf}(z)$; that is the answer	No elementary form exists
e^{-x^2} on a finite interval	$u = x^p; \Gamma(1 + \frac{1}{p}) \sin \frac{\pi}{2p}$	Only conditionally convergent
$\sin(x^p)$ or $\cos(x^p)$ on $(0, \infty)$	$\frac{\pi}{2}$; damp with e^{-ax} to prove it	Finite bound: $\text{Si}(z)$
$\frac{\sin x}{x}$ to ∞	The answer is a special-function value	Substitute to expose $\frac{\sin u}{u}$
An absurdly large upper limit	Fold with $\sin 2x; -\frac{\pi}{2} \ln 2$	Fourier series of $\ln(2 \sin \frac{\theta}{2})$
$\ln \sin x$ or $\ln \cos x$ on $[0, \frac{\pi}{2}]$	Clausen $\text{Cl}_2(\theta)$	Keep the factor 2
$\ln(2 \sin \frac{x}{2})$ on a partial range	Catalan's constant G	Stop; G does not reduce
$\frac{\arctan x}{x}$ or $\frac{\ln x}{1+x^2}$ on $[0, 1]$	$x = e^{-t}$; the answer is $-\gamma$	Compare with $-\Gamma'(1)$
$\ln \ln \frac{1}{x}$ on $[0, 1]$	Regularise together; expect γ	Never split the bracket
A bracketed pair of divergences	K and E ; name and stop	Check for $\int Q' \sqrt{Q}$ first
$\sqrt{\text{cubic}}$ or $\sqrt{\text{quartic}}$	Lambert W , J_0 , Airy; use their identities	Simplify before naming
$w e^w$, $\cos(\sin x)$, Ai	Look it up in the table, then work backwards	Try ratios and logs of it
A numerical value you recognise		

Complex-flavoured technique without contours

What this chapter is for. A large class of real integrals becomes easy the moment you allow i into the working, and almost none of them require you to draw a contour. The reason is that Euler's formula $e^{i\theta} = \cos \theta + i \sin \theta$ is an algebraic identity before it is a geometric one: it lets you replace two trigonometric functions by one exponential, replace a stubborn denominator by a product over roots of unity, and replace a pair of integration-by-parts cycles by a single division. Four moves cover the ground. Complexify the exponential and take real parts at the end. Recognise a real integral as the real or imaginary part of a complex antiderivative you already know. Use the roots of unity to filter a series or to factor a denominator. Push a real parameter into the complex plane and continue analytically. This chapter teaches all four, and then draws the line honestly: it says exactly which integrals these methods cannot reach, and what a genuine residue calculation would add if you were willing to set one up.

What i buys before any geometry

There is a persistent belief that using complex numbers in an integral commits you to complex analysis: to contours, to Cauchy's theorem, to estimating the contribution of a large semicircular arc. It does not. Every technique in this chapter uses complex numbers as an *algebraic* convenience on the real line, and every one of them can be checked by differentiating a real antiderivative or by rearranging a real series. The complex numbers are scaffolding, and they come down at the end.

The one structural fact behind all of it is that the exponential function absorbs the trigonometric functions. Two real functions, $\cos bx$ and $\sin bx$, satisfying a coupled pair of differentiation rules, become one function e^{ibx} satisfying the single rule $\frac{d}{dx}e^{ibx} = ibe^{ibx}$. Differentiation and integration of the exponential are multiplication and division by a constant, and multiplication and division are much easier than solving a two-by-two linear system, which is what a cyclic integration by parts amounts to.

THREE HABITS, ONE IDENTITY

Everything downstream is a use of $e^{i\theta} = \cos \theta + i \sin \theta$ read in one of three directions. Read left to right, it turns an exponential into trigonometry: this is how you extract Re and Im at the end. Read right to left, it turns trigonometry into an exponential: this is complexification, and it collapses two integrals into one. Read as a statement about $|e^{i\theta}| = 1$, it says the unit circle is the set of $e^{i\theta}$, which is what turns a 2π -periodic trigonometric integral into an algebraic problem about a rational function on $|z| = 1$. Know which direction you are using at each step and the bookkeeping stays under control.

THE ALGEBRA OF TAKING PARTS

Re and Im are real-linear and nothing more. They commute with sums, with real scalar multiples, with differentiation in a real variable, and with integration over a real interval, which is exactly why every technique below is legitimate. They do *not* commute with multiplication, with division, with taking a power, or with any nonlinear function. The practical consequence is a single rule that prevents most errors in this chapter: carry the complex expression all the way to the end, simplify it to the form $u + iv$ with u and v real, and only then read off the part you want. Every intermediate step stays complex.

Collapsing two integrals into one

The first move is the cheapest and the most frequently useful. Any integrand of the form “exponential times a sine or cosine” is the real or imaginary part of a single exponential, and single exponentials integrate in one step.

26.1 Complexify the exponential

CUE $e^{ax} \cos bx$ or $e^{ax} \sin bx$; a product of trigonometric functions with different frequencies that you would otherwise attack with product-to-sum; any integrand where two rounds of integration by parts would return you to the original with a changed coefficient.

METHOD Write the trigonometric factor as $\operatorname{Re} e^{ibx}$ or $\operatorname{Im} e^{ibx}$, integrate the single exponential, and take the part at the very end:

$$\int e^{(a+ib)x} dx = \frac{e^{(a+ib)x}}{a+ib}, \quad \frac{1}{a+ib} = \frac{a-ib}{a^2+b^2}.$$

Rationalising the denominator is the whole computation. The real part gives $\int e^{ax} \cos bx dx$ and the imaginary part gives $\int e^{ax} \sin bx dx$, both at once, which is the point: you get two answers for the price of one.

WORKED EXAMPLE

The indefinite pair, in three lines. With $a = 2$, $b = 3$,

$$\int e^{(2+3i)x} dx = \frac{e^{2x}(\cos 3x + i \sin 3x)}{2+3i} = \frac{e^{2x}(\cos 3x + i \sin 3x)(2-3i)}{13}.$$

Taking the real part,

$$\int e^{2x} \cos 3x dx = \frac{e^{2x}(2 \cos 3x + 3 \sin 3x)}{13} + C,$$

and the imaginary part gives $\int e^{2x} \sin 3x dx = e^{2x}(2 \sin 3x - 3 \cos 3x)/13 + C$. Differentiating the first back numerically at $x = 0.7$ returns -2.04725 , which is $e^{1.4} \cos 2.1$ to five digits.

The definite version on the half-line is even shorter, because the antiderivative vanishes at ∞ when $a < 0$:

$$\int_0^\infty e^{-x} \cos x dx = \operatorname{Re} \left[\frac{-1}{-1+i} \right] = \operatorname{Re} \frac{1}{1-i} = \operatorname{Re} \frac{1+i}{2} = \frac{1}{2},$$

and simultaneously $\int_0^\infty e^{-x} \sin x dx = \frac{1}{2}$. With $a = -3$, $b = 4$ the same line gives $\frac{3}{25}$ and $\frac{4}{25}$.

POWERS OF A COSINE, WITHOUT PRODUCT-TO-SUM

The same identity linearises trigonometric powers mechanically. Since $2 \cos x = e^{ix} + e^{-ix}$,

$$16 \cos^4 x = (e^{ix} + e^{-ix})^4 = e^{4ix} + 4e^{2ix} + 6 + 4e^{-2ix} + e^{-4ix} = 2 \cos 4x + 8 \cos 2x + 6,$$

by the binomial theorem and pairing conjugate terms. Hence $\cos^4 x = \frac{1}{8} \cos 4x + \frac{1}{2} \cos 2x + \frac{3}{8}$, and

$$\int_0^{\pi/2} \cos^4 x dx = \frac{1}{8} \cdot 0 + \frac{1}{2} \cdot 0 + \frac{3}{8} \cdot \frac{\pi}{2} = \frac{3\pi}{16} \approx 0.589049.$$

The binomial coefficients *are* the Fourier coefficients. This scales to any power without a table of product-to-sum identities, and the odd-power case works identically with $2i \sin x = e^{ix} - e^{-ix}$, where the alternating signs come from the powers of i .

WHERE THIS GOES WRONG

Taking the real part before the algebra is finished. Re is additive but not multiplicative: $\operatorname{Re}(zw) \neq \operatorname{Re}(z) \operatorname{Re}(w)$, so you may not simplify $\frac{e^{ibx}}{a+ib}$ by taking real parts of numerator and denominator separately. Rationalise first, then take the part. The second trap is at the infinite endpoint: $e^{(a+ib)x} \rightarrow 0$ as $x \rightarrow \infty$ requires $a < 0$, that is $\operatorname{Re}(a+ib) < 0$. The factor e^{ibx} has modulus 1 and never helps convergence; it merely fails to hurt.

SEE ALSO Cyclic integration by parts; Complex-parameter Gamma and Laplace transforms; Real and imaginary parts of a complex antiderivative; Product-to-sum.

Complexification collapses a pair of integrals into one. The dual move takes a complex identity you already know and reads off two real identities from it. This is more powerful than it sounds, because the complex identities worth knowing are few and their real shadows are numerous.

26.2 Real and imaginary parts of a complex antiderivative

CUE A real integral or sum that looks like half of something: $\sum \frac{\cos n\theta}{n}$, $\sum \frac{\sin n\theta}{n^2}$, $\int \frac{dx}{x^2+a^2}$ with the answer wanted as a logarithm rather than an arctangent, or a real series whose coefficients are those of a familiar complex function.

METHOD Identify the complex function whose series or antiderivative you know, evaluate it at a point on the unit circle or on the real axis, and separate. The workhorse is the logarithm: from $-\ln(1-z) = \sum_{n \geq 1} \frac{z^n}{n}$ with $z = e^{i\theta}$, and $1 - e^{i\theta} = -2i \sin \frac{\theta}{2} e^{i\theta/2}$, one separation gives both

$$\sum_{n \geq 1} \frac{\cos n\theta}{n} = -\ln\left(2 \sin \frac{\theta}{2}\right), \quad \sum_{n \geq 1} \frac{\sin n\theta}{n} = \frac{\pi - \theta}{2} \quad (0 < \theta < 2\pi).$$

The same reading one rung up the polylogarithm ladder gives $\operatorname{Re} \operatorname{Li}_2(e^{i\theta}) = \frac{\pi^2}{6} - \frac{\pi\theta}{2} + \frac{\theta^2}{4}$ and $\operatorname{Im} \operatorname{Li}_2(e^{i\theta}) = \operatorname{Cl}_2(\theta)$, the Clausen function.

WORKED EXAMPLE

Check the separation at $\theta = 1.1$. Summing 400000 terms gives $\sum \frac{\cos n\theta}{n} = -0.0443739$ against $-\ln(2 \sin 0.55) = -0.0443752$, and $\sum \frac{\sin n\theta}{n} = 1.0207943$ against $(\pi - 1.1)/2 = 1.0207963$; the small gaps are the expected $O(1/N)$ truncation of a conditionally convergent series.

Now integrate the first identity from 0 to θ term by term. The left side becomes $\sum \frac{\sin n\theta}{n^2}$ and the right becomes $-\int_0^\theta \ln(2 \sin \frac{t}{2}) dt$, so

$$\operatorname{Cl}_2(\theta) = -\int_0^\theta \ln\left(2 \sin \frac{t}{2}\right) dt = \sum_{n \geq 1} \frac{\sin n\theta}{n^2},$$

which is exactly the Clausen function of the previous chapter, obtained here in two lines from a complex logarithm. At $\theta = \frac{\pi}{2}$ it gives Catalan's constant, 0.9159655942. The second-order identity checks at $\theta = 1.1$: the series gives 0.2195581074 and the quadratic gives 0.2195581074.

POISSON'S INTEGRAL, FROM THE SAME LOGARITHM

Take the real part before summing rather than after. Since $|1 - re^{i\theta}|^2 = 1 - 2r \cos \theta + r^2$,

$$\int_0^\pi \ln(1 - 2r \cos \theta + r^2) d\theta = 2 \int_0^\pi \operatorname{Re} \ln(1 - re^{i\theta}) d\theta.$$

For $|r| < 1$ expand $\ln(1 - rz) = -\sum r^n z^n/n$, which converges uniformly, and integrate term by term: every $\int_0^\pi \cos n\theta d\theta$ vanishes, so the whole integral is 0. For $|r| > 1$ factor out the larger term, $1 - re^{i\theta} = -re^{i\theta}(1 - r^{-1}e^{-i\theta})$, and the same argument leaves only the $\ln|r|$ piece:

$$\int_0^\pi \ln(1 - 2r \cos \theta + r^2) d\theta = \begin{cases} 0, & |r| < 1 \\ 2\pi \ln|r|, & |r| > 1. \end{cases}$$

Quadrature gives 5×10^{-17} at $r = 0.9$ and 6.9027845904 at $r = 3$, against $2\pi \ln 3 = 6.9027845904$. The discontinuity in the formula is not a discontinuity in the integral; it is the two different factorisations meeting at $|r| = 1$, where the integral is 0 and the integrand has a logarithmic singularity.

WHERE THIS GOES WRONG

The branch of the logarithm. Writing $1 - e^{i\theta} = -2i \sin \frac{\theta}{2} e^{i\theta/2}$ and then taking $\ln$ requires you to fix $\arg$, and the correct choice on $(0, 2\pi)$ produces $\frac{\pi - \theta}{2}$, a sawtooth that jumps at $\theta = 0$. Applying the formula at θ outside $(0, 2\pi)$ without reducing modulo 2π gives an answer off by a multiple of π . The series is 2π -periodic; the closed form as written is not.

SEE ALSO Polylogarithm ladder; Log-sine integrals and the Clausen function; Complexify the exponential; Fourier series kernels.

Roots of unity

The third move uses complex numbers not to combine two functions but to select. The n -th roots of unity sum to zero unless the exponent is a multiple of n , and that single fact both filters series and factors denominators. It is the most algebraic of the four moves and requires no analysis whatsoever.

26.3 Roots of unity as a filter and as a factorisation

CUE A sum restricted to indices in an arithmetic progression: $\sum_{n \equiv r \pmod{m}} a_n$, or $\sum \frac{1}{(3k)!}$, or $\sum_k \binom{N}{3k}$. On the integration side: a denominator $1 \pm x^n$ with $n \geq 3$ that you need in partial fractions.

METHOD Let $\omega = e^{2\pi i/m}$. The orthogonality relation

$$\frac{1}{m} \sum_{j=0}^{m-1} \omega^{j(n-r)} = \begin{cases} 1, & n \equiv r \pmod{m} \\ 0, & \text{otherwise} \end{cases}$$

converts a restricted sum into m unrestricted ones: $\sum_{n \equiv r} a_n = \frac{1}{m} \sum_{j=0}^{m-1} \omega^{-jr} \sum_n a_n \omega^{jn}$. For factorisation, the roots of $1 + x^n$ are $e^{i(2k+1)\pi/n}$, and pairing each root with its conjugate gives the real quadratic factorisation

$$1 + x^n = \prod_k \left(x^2 - 2x \cos \frac{(2k+1)\pi}{n} + 1 \right)$$

over conjugate pairs, with a linear factor $x + 1$ left over when n is odd. Partial-fraction over those real quadratics; each piece is an arctangent and a logarithm.

WORKED EXAMPLE

Filtering a series. With $m = 3$ and $a_n = 1/n!$, take $r = 0$:

$$\sum_{k \geq 0} \frac{1}{(3k)!} = \frac{1}{3} \sum_{j=0}^2 e^{\omega^j} = \frac{1}{3} \left(e + 2e^{-1/2} \cos \frac{\sqrt{3}}{2} \right) \approx 1.1680583134,$$

because $\omega = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$ and the two non-real terms are conjugates. Direct summation of the left side agrees to sixteen digits.

Factoring a denominator. For $n = 3$, the roots of $1 + x^3$ are -1 and $e^{\pm i\pi/3}$, so $1 + x^3 = (1 + x)(x^2 - x + 1)$ and partial fractions give

$$\int_0^1 \frac{dx}{1 + x^3} = \frac{1}{3} \int_0^1 \frac{dx}{1 + x} + \frac{1}{3} \int_0^1 \frac{2 - x}{x^2 - x + 1} dx = \frac{\ln 2}{3} + \frac{\pi}{3\sqrt{3}} \approx 0.835649.$$

For $n = 6$ the same recipe gives $1 + x^6 = (x^2 - \sqrt{3}x + 1)(x^2 + 1)(x^2 + \sqrt{3}x + 1)$, checked numerically at $x = 1.7$, and integrating over $(0, \infty)$ recovers $\pi/3$, in agreement with the reflection formula $\pi/(6 \sin \frac{\pi}{6})$.

The filter works on any generating function, not just the exponential. Taking $a_n = \binom{N}{n}$, whose generating function is $(1 + z)^N$,

$$\sum_{n \equiv r \pmod{m}} \binom{N}{n} = \frac{1}{m} \sum_{j=0}^{m-1} \omega^{-jr} (1 + \omega^j)^N.$$

At $N = 10$, $m = 3$ this gives 341, 341 and 342 for $r = 0, 1, 2$, summing correctly to $2^{10} = 1024$. The imaginary parts cancel exactly, as they must, and watching them cancel is a good check that the sign of the exponent is right.

WHERE THIS GOES WRONG

Two bookkeeping errors. First, ω^{-jr} and not ω^{jr} : the sign of the exponent in the filter is what selects the residue class r rather than $-r$, and at $r = 0$ the error is invisible, which is why it survives to the case where it matters. Second, the roots of $1 + x^n$ are the *odd* $2n$ -th roots of unity, $e^{i(2k+1)\pi/n}$, not the n -th roots of unity; using the latter factors $1 - x^n$ instead and gives a denominator with a pole inside your interval.

SEE ALSO Geometric expansion; Unit-circle rational-trig formula; Euler reflection and the Mellin transform; Partial fractions over the reals.

The same roots-of-unity picture, taken continuously rather than discretely, says that a 2π -periodic trigonometric integral is an integral around the unit circle. That observation converts an entire family of trigonometric integrals into algebra, and it is worth having both as a memorised formula and as a mechanism.

26.4 Unit-circle rational-trig formula

CUE $\int_0^{2\pi} \frac{d\theta}{a+b\cos\theta}$, or any rational function of $\cos\theta$ and $\sin\theta$ integrated over a full period. The full period is essential; a partial range does not close up.

METHOD Either quote the closed form

$$\int_0^{2\pi} \frac{d\theta}{a+b\cos\theta} = \frac{2\pi}{\sqrt{a^2-b^2}} \quad (a > |b|), \quad \int_0^{2\pi} \frac{d\theta}{(a+b\cos\theta)^2} = \frac{2\pi a}{(a^2-b^2)^{3/2}},$$

the second being the first differentiated in a ; or set $z = e^{i\theta}$, so that $\cos\theta = \frac{1}{2}(z+z^{-1})$, $\sin\theta = \frac{1}{2i}(z-z^{-1})$ and $d\theta = \frac{dz}{iz}$, turning the integral into a rational function on $|z|=1$ whose value is $2\pi i$ times the sum of the residues at the poles *inside* the circle. The second route is a contour integral in name only: the contour is the interval you were already given.

WORKED EXAMPLE

The plain case: $\int_0^{2\pi} \frac{d\theta}{5+4\cos\theta} = \frac{2\pi}{\sqrt{25-16}} = \frac{2\pi}{3} \approx 2.094395$.

A case where the formula alone is not enough. For $J = \int_0^{2\pi} \frac{\cos^2\theta}{5-4\cos\theta} d\theta$, write $\cos^2\theta = \frac{1}{2}(1+\cos 2\theta)$ and reduce the numerator against the denominator by polynomial division in $\cos\theta$. Setting $c = \cos\theta$,

$$\frac{c^2}{5-4c} = -\frac{c}{4} - \frac{5}{16} + \frac{25}{16} \cdot \frac{1}{5-4c},$$

so $J = -\frac{1}{4} \int_0^{2\pi} \cos\theta d\theta - \frac{5}{16} \cdot 2\pi + \frac{25}{16} \cdot \frac{2\pi}{3}$. The first integral vanishes, and

$$J = -\frac{5\pi}{8} + \frac{25\pi}{24} = \frac{-15\pi + 25\pi}{24} = \frac{5\pi}{12} \approx 1.308997,$$

matching quadrature to fifteen digits. Reduce first, quote second: that order keeps the algebra trivial. The companion $\int_0^{2\pi} \frac{\sin^2\theta}{5-4\cos\theta} d\theta = \frac{\pi}{4}$ follows from $\sin^2 = 1 - \cos^2$ and the two values just computed.

THE SUBSTITUTION ITSELF, ONCE, SO YOU CAN REBUILD THE FORMULA

Put $z = e^{i\theta}$ in $\int_0^{2\pi} \frac{d\theta}{a+b\cos\theta}$. Then $a+b\cos\theta = a + \frac{b}{2}(z+z^{-1}) = \frac{bz^2+2az+b}{2z}$, and $d\theta = \frac{dz}{iz}$, so the integral becomes

$$\oint_{|z|=1} \frac{2z}{bz^2+2az+b} \cdot \frac{dz}{iz} = \frac{2}{i} \oint \frac{dz}{b(z-z_+)(z-z_-)}, \quad z_{\pm} = \frac{-a \pm \sqrt{a^2-b^2}}{b}.$$

Since $z_+z_- = 1$, exactly one root lies inside the unit circle when $a > |b|$, namely z_+ . The residue there is $\frac{1}{b(z_+-z_-)} = \frac{1}{2\sqrt{a^2-b^2}}$, and multiplying by $2\pi i \cdot \frac{2}{i}$ gives $\frac{2\pi}{\sqrt{a^2-b^2}}$. Two facts did all the work: the product of the roots is 1, and the difference of the roots is $2\sqrt{a^2-b^2}/b$.

WHERE THIS GOES WRONG

The hypothesis $a > |b|$ is not cosmetic. If $|b| > a$ the denominator vanishes somewhere on $[0, 2\pi]$, the integral does not exist as a Lebesgue integral, and the formula returns $2\pi/\sqrt{\text{negative}}$, which should be your warning. The correct object there is a principal value. Also, when using the z -substitution, only the poles with $|z| < 1$ contribute; the two roots of $bz^2+2az+b=0$ multiply to 1, so exactly one of them is inside, and picking the wrong one flips the sign of the answer.

SEE ALSO Weierstrass substitution; Quotable residue results; Roots of unity as a filter; Fourier series kernels.

Results worth quoting rather than rebuilding

The techniques so far have all been constructive: you do something to the integrand and the answer appears. The next group is different in kind. These are results that you look up in your own memory, and the case for memorising them is worth making explicitly, because reflexive derivation from first principles is a habit that costs more than it saves at this level.

There is a small set of real integrals whose standard derivation is a contour argument and whose values are worth memorising outright. Memorising them is not laziness. Each one takes ten minutes to derive properly and two seconds to recall, and in every case the derivation teaches you nothing you did not already know after the first time.

26.5 Quotable residue results

CUE A rational function, or a rational function times $\cos ax$ or $\sin ax$, integrated over the whole real line, with the denominator decaying at least like x^2 .

METHOD Learn the following four and the pattern behind them:

$$\int_{-\infty}^{\infty} \frac{dx}{1+x^2} = \pi, \quad \int_{-\infty}^{\infty} \frac{dx}{(1+x^2)^2} = \frac{\pi}{2}, \quad \int_{-\infty}^{\infty} \frac{\cos x}{1+x^2} dx = \frac{\pi}{e}, \quad \int_{-\infty}^{\infty} \frac{\cos ax}{x^2+b^2} dx = \frac{\pi}{b} e^{-ab}$$

for $a, b > 0$. The pattern: a simple pole at $x = ib$ contributes π/b , and an oscillation $\cos ax$ contributes the damping factor e^{-ab} , because e^{iax} evaluated at $x = ib$ is e^{-ab} . Higher powers of the denominator come from differentiating in b , which is why the second value is half the first.

WORKED EXAMPLE

The first two are elementary and should be checked, not quoted: $\int \frac{dx}{1+x^2} = \arctan x$ gives π , and $x = \tan \theta$ turns the second into $\int_{-\pi/2}^{\pi/2} \cos^2 \theta d\theta = \frac{\pi}{2}$. The third is not elementary, and $\pi/e \approx 1.1557273498$ is worth knowing by sight. Differentiating in the parameter generates the rest of the family. From $\int_{-\infty}^{\infty} \frac{\cos ax}{x^2+b^2} dx = \frac{\pi}{b} e^{-ab}$ with $b = 3$, $a = 2$, the value is $\frac{\pi}{3} e^{-6} \approx 0.0025957$; and differentiating the b -form once in b gives $\int_{-\infty}^{\infty} \frac{\cos ax}{(x^2+b^2)^2} dx = \frac{\pi(1+ab)}{2b^3} e^{-ab}$.

THE THIRD VALUE, DERIVED WITHOUT A CONTOUR

It is worth seeing that even π/e is reachable by real methods, so that the memorised value is not a black box. Write the denominator as a Schwinger integral, $\frac{1}{1+x^2} = \int_0^{\infty} e^{-(1+x^2)t} dt$, and exchange the order of integration, which is legitimate because everything is absolutely convergent once the cosine is bounded by 1:

$$\int_{-\infty}^{\infty} \frac{\cos ax}{1+x^2} dx = \int_0^{\infty} e^{-t} \left[\int_{-\infty}^{\infty} e^{-tx^2} \cos(ax) dx \right] dt = \int_0^{\infty} e^{-t} \sqrt{\frac{\pi}{t}} e^{-a^2/4t} dt,$$

the inner integral being the Gaussian-against-a-cosine value of the previous chapter. The remaining integral is a Glasser-type one, $\int_0^{\infty} t^{-1/2} e^{-t-a^2/4t} dt = \sqrt{\pi} e^{-a}$ for $a > 0$, giving πe^{-a} and hence π/e at $a = 1$. Checked numerically at $a = 1$ and $a = 2.3$. Two Gaussian facts and one exchange of order; no contour anywhere.

WHERE THIS GOES WRONG

$\int_{-\infty}^{\infty} \frac{\sin x}{1+x^2} dx = 0$ by parity, not by any residue computation; quoting a residue answer for it is a sign you are not looking at the integrand. Conversely $\int_{-\infty}^{\infty} \frac{\sin x}{x} dx$ genuinely needs an indented contour, because the pole sits *on* the axis, and the naive residue recipe gives twice the right answer. Finally the damping is $e^{-|a|b}$: the formula as written assumes $a > 0$, and for $a < 0$ the cosine is unchanged while a careless application of e^{-ab} would give exponential growth.

SEE ALSO Unit-circle rational-trig formula; Cauchy principal value; Complex-parameter Gamma and Laplace transforms; Tangent substitution.

WHEN THE ANSWER MUST BE REAL

A useful audit at the end of any complexified computation: ask whether the answer was forced to be real, and check that it is. If the original integrand is real on the whole interval, the value is real, so any surviving imaginary part is an error. Conversely, if you complexified and the imaginary part did *not* vanish, you have computed a second integral for free, and it is worth writing down: the same working that gives $\int_0^{\infty} e^{-x} \cos x dx$ gives $\int_0^{\infty} e^{-x} \sin x dx$ at no cost. Complexification is a two-for-one deal, and half of it is usually thrown away by accident.

Pushing a real parameter into the complex plane

The last two techniques are the deepest of the four moves, and they are the ones that come closest to real complex analysis without crossing over. The idea is that a formula proved for real parameters often remains true for complex ones, because both sides are analytic; you then choose a complex value of the parameter that makes the real integral you wanted fall out as a real or imaginary part.

26.6 Complex-parameter Gamma and Laplace transforms

CUE A Gamma-type moment married to an oscillation: $x^m e^{-x} \sin x$, $x^m e^{-x} \cos x \ln x$, or any $x^{s-1} e^{-px}$ with p complex. Equivalently, a Laplace transform evaluated at a complex argument.

METHOD Absorb the oscillation into the exponent. Since $e^{-x} \cos x + i e^{-x} \sin x = e^{-(1-i)x}$,

$$\int_0^\infty x^{s-1} e^{-px} dx = \frac{\Gamma(s)}{p^s} \quad (\operatorname{Re} p > 0),$$

and for $p = 1 - i$ write $p^s = 2^{s/2} e^{-i\pi s/4}$ using the principal argument $\arg(1 - i) = -\frac{\pi}{4}$. Take real or imaginary parts at the end. Differentiating the identity in s inserts a factor $\ln x$ on the left and brings in ψ on the right, which is how logarithms enter.

WORKED EXAMPLE

The clean case first. With $s = 2$ and $p = 1 - i$, $\Gamma(2) = 1$ and $(1 - i)^2 = -2i$, so

$$\int_0^\infty x e^{-x} \sin x dx = \operatorname{Im} \frac{1}{-2i} = \operatorname{Im} \frac{i}{2} = \frac{1}{2},$$

and the companion real part gives $\int_0^\infty x e^{-x} \cos x dx = 0$. At $s = 3$, $(1 - i)^3 = -2 - 2i$ and $\Gamma(3) = 2$, so $2/(-2 - 2i) = -\frac{1}{2} + \frac{1}{2}i$, giving $\int_0^\infty x^2 e^{-x} \cos x dx = -\frac{1}{2}$ and $\int_0^\infty x^2 e^{-x} \sin x dx = \frac{1}{2}$.

Now insert a logarithm by differentiating in s at $s = 1$. Since $\frac{d}{ds}[\Gamma(s)p^{-s}] = \Gamma(s)p^{-s}(\psi(s) - \ln p)$ and $\psi(1) = -\gamma$, $\ln(1 - i) = \frac{1}{2} \ln 2 - \frac{i\pi}{4}$,

$$\int_0^\infty e^{-(1-i)x} \ln x dx = \frac{-\gamma - \frac{1}{2} \ln 2 + \frac{i\pi}{4}}{1 - i},$$

whose imaginary part, after multiplying by $\frac{1+i}{2}$, gives

$$\int_0^\infty e^{-x} \ln x \sin x dx = \frac{1}{2} \left(\frac{\pi}{4} - \gamma - \frac{\ln 2}{2} \right) \approx -0.0691955459,$$

confirmed by quadrature to fourteen digits.

WHERE THIS GOES WRONG

The branch of p^s . For $p = 1 - i$ the principal argument is $-\frac{\pi}{4}$; taking $+\frac{7\pi}{4}$ instead is the same complex number but a different power, and the answer changes silently by a factor $e^{2\pi i s}$, which is 1 only for integer s . Fix $\arg p \in (-\pi, \pi]$ once and use it consistently. Validity also needs $\operatorname{Re} p > 0$ strictly: at $\operatorname{Re} p = 0$ the integral is only conditionally convergent and the identity has to be justified as a limit, which is the subject of the next technique.

SEE ALSO Gamma-function moment; Complexify the exponential; Digamma and polygamma; Rotating the contour without contours.

The previous technique needed $\operatorname{Re} p > 0$. The most interesting integrals in this family sit exactly on the boundary $\operatorname{Re} p = 0$, where the integrand no longer decays and convergence is only conditional. Reaching them is the one place in this chapter where you have to say something genuinely analytic, and it is worth saying carefully, because the shortcut version of this argument is a standard way to get a right answer for a wrong reason.

26.7 Rotating the contour without contours

CUE A Gaussian or Gamma integral in which replacing a real parameter by an imaginary one would produce the answer you want. The type case is Fresnel: you know $\int_0^\infty e^{-ax^2} dx$ and you want $\int_0^\infty e^{ix^2} dx$.

METHOD Establish the identity on the open right half-plane, where both sides are analytic and the integral converges absolutely:

$$\int_0^\infty e^{-ax^2} dx = \frac{1}{2} \sqrt{\frac{\pi}{a}}, \quad \operatorname{Re} a > 0,$$

with the principal branch of $\sqrt{a}$. Both sides are analytic in a on that open set and continuous up to the boundary $\operatorname{Re} a = 0$ away from the origin, so the identity extends to the boundary by continuity. Setting $a = -i$, for which $\sqrt{-i} = e^{-i\pi/4}$, gives

$$\int_0^\infty e^{ix^2} dx = \frac{1}{2} \sqrt{\frac{\pi}{-i}} = \frac{1}{2} \sqrt{\frac{\pi}{2}} (1 + i),$$

whose real and imaginary parts are the Fresnel integrals.

Separating the last display,

$$\int_0^\infty \cos(x^2) dx = \int_0^\infty \sin(x^2) dx = \frac{1}{2} \sqrt{\frac{\pi}{2}} \approx 0.6266570687,$$

and the numerical value $\frac{1}{2} \sqrt{\pi/(-i)} = 0.6266571 + 0.6266571i$ confirms the branch choice. Note what the argument bought: the Fresnel integrals are conditionally convergent and no elementary substitution reaches them, yet the whole derivation is two lines of analytic continuation from an integral every reader already knows.

The same rotation applied to $\int_0^\infty x^{s-1} e^{-px} dx = \Gamma(s) p^{-s}$ at $p = -i$ recovers the Mellin transforms of sine and cosine used in the previous chapter: $\int_0^\infty x^{s-1} \sin x dx = \Gamma(s) \sin \frac{\pi s}{2}$ for $0 < s < 1$, checked at $s = \frac{1}{2}$ where both sides equal 1.2533141373.

WHERE THIS GOES WRONG

The continuity up to the boundary is the entire content of the argument, and it is what people skip. Simply substituting $a = -i$ into a formula proved for $a > 0$ is not a proof; it is a guess that happens to be right here and is wrong in cases where the boundary behaviour is worse. Concretely, the extension fails at $a = 0$ itself, and it fails for $\int_0^\infty e^{-ax} dx$ on the imaginary axis, where the boundary integral does not converge at all. The branch also decides the sign: $\sqrt{-i}$ is $e^{-i\pi/4}$, not $e^{3i\pi/4}$, and the wrong choice negates both Fresnel integrals.

SEE ALSO Fresnel integrals; Gaussian and error function; Complex-parameter Gamma and Laplace transforms; Oscillatory convergence.

Where this stops, and what a contour would add

It is worth being precise about the boundary, because the techniques above are so effective that they can create the impression that contour integration is never needed. That impression is false, and the cases where it fails have a common shape: the complex function you would like to integrate has a branch cut, or a pole on the path, or you need a relation between values at many points rather than at one.

Four classes lie outside what this chapter can do. First, integrals of $\ln x$ or x^{s-1} against a rational function on $(0, \infty)$ where the exponent is irrational or the rational function has many poles. The keyhole contour handles all of them uniformly by exploiting the jump of x^{s-1} across the cut; the reflection formula of the previous chapter is the special case that has been pre-digested for you. Second, principal values, where a pole sits on the axis. The indentation gives a clean half-residue rule, $\text{p.v.} \int = \pi i \sum \text{Res}_{\text{on axis}} + 2\pi i \sum \text{Res}_{\text{upper}}$, and there is no real-variable substitute of comparable generality. Third, rational integrands whose denominator has high degree: residues cost one limit per pole, while partial fractions costs a linear system. Fourth, summation of series by $\pi \cot \pi z$, which converts $\sum f(n)$ into a residue computation and is how $\zeta(2n)$ is evaluated in closed form.

It helps to see the two routes on the same integral, at a size where both are short. Take $\int_{-\infty}^\infty \frac{dx}{1+x^4}$. The contour-free route factors the denominator over conjugate root pairs, $1+x^4 = (x^2 - \sqrt{2}x + 1)(x^2 + \sqrt{2}x + 1)$, partial-fractions into two quadratics, completes the square in each, and integrates two arctangents plus two logarithms whose contributions cancel by symmetry. It takes most of a page and every step is elementary. The residue route observes that the poles in the upper half plane are $z = e^{i\pi/4}$ and $z = e^{3i\pi/4}$, that at a simple root of $z^4 + 1$ the residue of $(1+z^4)^{-1}$ is $\frac{1}{4z^3} = -\frac{z}{4}$, and therefore that the answer is

$$2\pi i \cdot \left(-\frac{e^{i\pi/4} + e^{3i\pi/4}}{4} \right) = 2\pi i \cdot \left(-\frac{i\sqrt{2}}{4} \right) = \frac{\pi}{\sqrt{2}} \approx 2.2214414691.$$

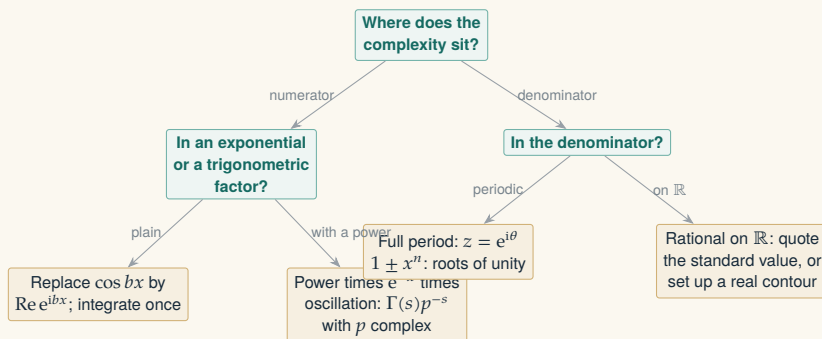
Three lines, once the arc estimate is taken as known. Both answers agree, and the arithmetic also agrees with the reflection formula $2 \cdot \frac{\pi}{4 \sin(\pi/4)}$ from the previous chapter, which is a third route again. The comparison is the honest summary of this chapter: for one integral, pick whichever route you can execute without error; for a family, the residue calculation is the one that generalises.

A FAIR ACCOUNT OF RESIDUES

What the residue theorem actually adds is *uniformity*. Every technique in this chapter is a trick keyed to a shape: complexification needs an exponential, the unit-circle formula needs a full period, the roots-of-unity filter needs a cyclic structure. The residue theorem needs only that the integrand extend meromorphically and that the arc contribution vanish, and it then computes the answer by a local calculation at each singularity. The price is real: you must choose a contour, prove the arc estimate (usually Jordan's lemma or an *ML* bound), track branch cuts, and be honest about poles on the path. For a single integral of the kind in

this chapter, the contour-free route is faster. For a family of integrals, or for an integrand with a branch point, the contour pays for itself immediately. Knowing both, and knowing which is cheaper today, is the actual skill.

It is also worth knowing precisely what makes a contour argument work, so that you can tell at a glance whether one is available. Two estimates do almost all of it. The ML bound says that if $|f| \leq M$ on an arc of length L then the integral over that arc is at most ML ; applied to a semicircle of radius R with f decaying like R^{-2} , the arc contributes $O(R^{-1})$ and vanishes. That is why rational integrands need the denominator to exceed the numerator by two degrees. Jordan's lemma does better for oscillatory integrands: if $f \rightarrow 0$ uniformly on the upper semicircle then $\int f(z)e^{iaz} dz \rightarrow 0$ for $a > 0$, because e^{iaz} decays exponentially off the real axis. That is why $\frac{\cos x}{1+x^2}$ needs only one degree of decay while $\frac{1}{1+x^2}$ needs two, and why the answer carries the factor e^{-ab} . If neither estimate applies, the contour route is not available either, and the honest move is a real-variable one: damping, parameter differentiation, or a symmetry fold.



There is a final habit worth carrying out of this chapter, and it is a discipline rather than a technique: everything here is checkable. A complexified antiderivative can be differentiated back and compared numerically at a random point. A roots-of-unity factorisation can be evaluated at $x = 1.7$ and compared with the original polynomial. An analytic continuation can be tested by approaching the boundary along a path, computing $\frac{1}{2}\sqrt{\pi/a}$ at $a = 0.01 - i$ and at $a = 0.001 - i$ and watching it converge. Complex algebra is exactly the kind of manipulation where a sign error produces a plausible answer, and exactly the kind where a thirty-second numerical check catches it.

Recognition matrix

If you see	Reach for	If that fails
$e^{ax} \cos bx$ or $e^{ax} \sin bx$	Integrate $e^{(a+ib)x}$, take the part last	Cyclic integration by parts
A parts cycle that returns to the start	Complexify instead; divide by $a + ib$	Solve the two-by-two system
A product of trigonometric functions	Write each as $e^{i\theta}$ and multiply	Product-to-sum identities
$\sum \frac{\cos n\theta}{n}$ or $\sum \frac{\sin n\theta}{n}$	Real and imaginary parts of $-\ln(1 - e^{i\theta})$	Fix the branch on $(0, 2\pi)$
$\sum \frac{\sin n\theta}{n^2}$	$\text{Im Li}_2(e^{i\theta}) = \text{Cl}_2(\theta)$	Integrate the n^{-1} identity
A sum over $n \equiv r \pmod{m}$	Roots-of-unity filter with $\omega = e^{2\pi i/m}$	Check the sign of the exponent
$1 + x^n$ downstairs, $n \geq 3$	Factor over conjugate root pairs	Reflection formula on $(0, \infty)$
$\int_0^{2\pi}$ of a rational trigonometric function	$z = e^{i\theta}$, or quote $\frac{2\pi}{\sqrt{a^2 - b^2}}$	Reduce the numerator first
$(a + b \cos \theta)^{-2}$ over a period	Differentiate the $(a + b \cos \theta)^{-1}$ formula in a	Weierstrass substitution
$a < b $ in $a + b \cos \theta$	Pole on the interval: principal value	Do not use the closed form
$\frac{\cos ax}{x^2 + b^2}$ on $\mathbb{R}$	$\frac{\pi}{b} e^{- a b}$	Differentiate in b for higher powers
$\frac{x^2 + b^2}{\sin ax}$ on $\mathbb{R}$	Zero by parity; look again	Parity before machinery
$\frac{x^2 + b^2}{\sin x}$ on $\mathbb{R}$	Pole on the axis: indented contour or damping	Feynman with e^{-ax}
$x^m e^{-x} \sin x$	$\Gamma(s)/(1-i)^s$, principal branch	Repeated parts
$\ln x$ against $e^{-x} \sin x$	Differentiate $\Gamma(s)p^{-s}$ in s ; expect ψ and γ	Frullani
e^{ix^2} , $\cos(x^2)$, $\sin(x^2)$	Continue $\frac{1}{2}\sqrt{\pi/a}$ to $a = -i$	Mellin transform of $\sin$
A formula proved for real $a > 0$	Continue analytically; check the boundary	State where analyticity holds
x^{s-1} with irrational s on $(0, \infty)$	Keyhole contour, or quote reflection	Beta and Gamma
A denominator of degree six or more	Residues at the poles, not partial fractions	Symmetry or substitution first

If you see	Reach for	If that fails
A series you want as an integral Any complex manipulation at all	$\pi \cot \pi z$ summation Differentiate back, or check at a random point	Abel or Euler-Maclaurin Numerical quadrature

Improper integrals, convergence, and engineered endpoints

What this chapter is for. Every integral in this chapter asks two questions, and they must be answered in that order. The first is whether the integral exists at all. The second is what it equals. Answering the second without the first is the commonest way to write a confident wrong answer in all of integration, because the machinery of antidifferentiation is perfectly happy to produce a finite number for a divergent object: the fundamental theorem applied across a pole gives $\int_{-1}^1 x^{-2} dx = -2$, a value that is not merely inaccurate but meaningless, and nothing in the algebra warns you. So the chapter begins with a taxonomy of impropriety (infinite interval, unbounded integrand, both at once, and the interior singularity that hides in plain sight), then builds the convergence machinery: comparison, limit comparison, the p -test at zero and at infinity, and the separate regime of oscillatory integrands that converge without converging absolutely. It then treats the two respectable ways to attach a number to a divergent object, the Cauchy principal value and the cancellation of a matched pair of logarithmic divergences, and it insists that both be labelled as what they are. The chapter closes on a different skill entirely: reading engineered endpoints. When a problem hands you a limit of integration written as $\sqrt{e^2 - 1}$ or $\log_8 e$, that number is a message about the intended substitution, and decoding it is faster than any amount of pattern-matching on the integrand.

Two questions, in order

Write $\int_a^b f(x) dx$ and the notation quietly asserts three things: that $[a, b]$ is a bounded interval, that f is bounded on it, and that the Riemann or Lebesgue construction therefore applies directly. An integral is *improper* when one of those assertions fails. It is then not an integral at all until you have defined it as a limit, and the definition is a genuine part of the answer rather than a formality to be waved at.

There are exactly three ways an assertion can fail, and a fourth case that is a combination.

Type	Symptom	Definition	Model case
I	Infinite bound	$\lim_{R \rightarrow \infty} \int_a^R f$	$\int_1^\infty x^{-p} dx$
II	Endpoint singularity	$\lim_{\epsilon \rightarrow 0^+} \int_{a+\epsilon}^b f$	$\int_0^1 x^{-p} dx$
III	Interior singularity at c	$\int_a^c + \int_c^b$, both required	$\int_{-1}^1 x^{-2} dx$
Mixed	Both ends offend	Split at any interior point	$\int_0^\infty \frac{x^{s-1}}{1+x^n} dx$

The table also settles a question of notation that causes real confusion. An improper integral of Type I and an improper integral of Type II are the same object viewed through a substitution: $x = 1/u$ carries $\int_1^\infty x^{-p} dx$ to $\int_0^1 u^{p-2} du$, and the two p -tests are one p -test seen from two sides. This is more than an observation. It is a working technique, because a singularity at 0 is often easier to analyse after being pushed out to ∞ by $x = e^{-t}$ or $x = 1/u$, and an infinite range is often easier to integrate numerically after being pulled in to a finite one by the same map.

Type III is the dangerous one, because it is the only type with no visible marker. An infinite bound announces itself. An endpoint singularity is usually obvious from the presence of $\ln x$ at 0 or $\sqrt{1-x^2}$ downstairs at 1. An interior pole is invisible unless you look for it: $\int_0^3 \frac{dx}{x-1}$ looks like an ordinary rational integral, and its integrand is finite at both endpoints. The habit that prevents this failure is mechanical. Before writing an antiderivative, solve “denominator = 0” and “argument of $\ln$ or of an even root = 0” on the closed interval, and check whether any solution lies inside.

THE TWO QUESTIONS

Does it converge? is a question about the integrand’s behaviour near the offending points only, and it is answered by comparison with a power. **What is it?** is a question about the whole interval, and it is answered by the techniques of every other chapter. The first question is cheap: it takes one line per bad point. The second is expensive. Doing them out of order means paying the

expensive one before knowing whether it has an answer, and worse, it means that a divergent integral will present itself to you as a finite number with no indication that anything went wrong.

The mixed case deserves a word because it is the standard shape of the integrals that matter. An integral over $(0, \infty)$ of a rational or algebraic integrand is almost always improper at both ends, and the two ends impose *independent* conditions.

The archetype is $\int_0^\infty \frac{x^{s-1}}{1+x^n} dx$: near 0 the integrand behaves like x^{s-1} , which needs $s > 0$; near ∞ it behaves like x^{s-1-n} , which needs $s-1-n < -1$, that is $s < n$. The verdict is the conjunction, $0 < s < n$, and a candidate closed form asserted outside that strip is not a value but a formal expression. This is why the convergence step is not pedantry: it is what tells you the domain on which your answer is an answer.

Endpoints as limits

The first technique is a discipline rather than a trick, and it is the one on which everything else in the chapter rests. Its content is that the offending endpoint must be replaced by a variable and the limit taken at the end, after the antiderivative has been evaluated symbolically.

27.1 Endpoint limit discipline

CORE

CUE An infinite bound; an integrand undefined at an endpoint ($\ln x$ at 0, $\frac{1}{\sqrt{1-x^2}}$ at 1, $x^{-1/2}$ at 0); or a zero of a denominator strictly inside the interval.

METHOD Replace every offending endpoint by a limit *before* manipulating:

$$\int_a^b f(x) dx = \lim_{\varepsilon \rightarrow 0^+} \int_{a+\varepsilon}^b f(x) dx, \quad \int_a^\infty f(x) dx = \lim_{R \rightarrow \infty} \int_a^R f(x) dx.$$

For a singularity at an interior point c , split into $\int_a^c + \int_c^b$ and require *both* pieces to converge independently, with independent cut-offs. Before splitting, decide whether the singularity is genuine: if f extends continuously across c (as $\frac{\sin x}{x}$ does at 0, or $\frac{\ln x}{1-x}$ at 1), the integral is proper and no split is needed.

WORKED EXAMPLE

$$\int_0^1 \ln x dx = \lim_{\varepsilon \rightarrow 0^+} [x \ln x - x]_\varepsilon^1 = \lim_{\varepsilon \rightarrow 0^+} (-1 - \varepsilon \ln \varepsilon + \varepsilon) = -1,$$

because $\varepsilon \ln \varepsilon \rightarrow 0$. The limit is not decoration: the antiderivative $x \ln x - x$ is undefined at $x = 0$, and the value -1 exists only as a limit of the truncated integrals.

A mixed case, both ends offending:

$$\int_0^\infty \frac{dx}{\sqrt{x}(1+x)}.$$

Near 0 the integrand is $\sim x^{-1/2}$, integrable; near ∞ it is $\sim x^{-3/2}$, integrable. So the object exists. Now evaluate: $x = t^2$ gives $\int_0^\infty \frac{2 dt}{1+t^2} = \pi \approx 3.141593$. Two lines of convergence, one line of evaluation.

A REMOVABLE SINGULARITY, CORRECTLY NOT SPLIT

$$\int_0^1 \frac{\ln x}{1-x} dx.$$

The denominator vanishes at $x = 1$, so the reflex is to split. But $\ln x = \ln(1 - (1-x)) \sim -(1-x)$ near 1, so the integrand tends to -1 there and the point is removable. At the other end $\ln x$ has an integrable singularity.

Hence the integral is improper only at 0 and converges. Expanding $\frac{1}{1-x} = \sum_{n \geq 0} x^n$ and integrating termwise gives $\sum_{n \geq 0} \frac{-1}{(n+1)^2} = -\frac{\pi^2}{6} \approx -1.644934$.

$$\int_0^2 \frac{dx}{\sqrt{|x-1|}}.$$

The integrand blows up at the interior point $x = 1$, so the object is Type III and the fundamental theorem may not be applied across it. Split and take two independent limits:

$$\lim_{\varepsilon \rightarrow 0^+} \int_0^{1-\varepsilon} \frac{dx}{\sqrt{1-x}} = \lim_{\varepsilon \rightarrow 0^+} [-2\sqrt{1-x}]_0^{1-\varepsilon} = 2, \quad \lim_{\delta \rightarrow 0^+} \int_{1+\delta}^2 \frac{dx}{\sqrt{x-1}} = 2,$$

and both exist, so the integral converges to 4. The exponent is what saves it: $p = \frac{1}{2} < 1$ on each side. Replace the square root by a first power and both halves become $+\infty$, so the integral diverges even though the picture looks almost identical.

WHERE THIS GOES WRONG

Applying the fundamental theorem across an interior pole. The antiderivative of x^{-2} is $-x^{-1}$, and $[-x^{-1}]_{-1}^1 = -1 - 1 = -2$, a negative number for a strictly positive integrand: the sign alone should convict it. The integral diverges, since $\int_0^1 x^{-2} dx = +\infty$. The second form of the same error is splitting at c but tying the two cut-offs together, writing $\lim_{\varepsilon \rightarrow 0} [\int_a^{c-\varepsilon} + \int_{c+\varepsilon}^b]$ and calling the result the integral. That limit is the principal value, which is a different and weaker object; the integral requires the two limits to exist *separately*.

SEE ALSO Comparison and limit comparison; Cauchy principal value; splitting at a symmetry or sign-change point; parity on a symmetric interval.

The discipline tells you to take a limit. It does not tell you whether the limit exists, and for anything but the simplest integrands you cannot find out by computing the antiderivative, because there may not be one. What is needed is a test that inspects only the local behaviour at the bad point, and that is exactly what comparison provides.

Deciding convergence

Convergence of an improper integral is decided entirely by the integrand's rate of blow-up or decay near each offending point, and rates are measured against powers. Two facts carry almost all of the load, and they point in opposite directions, which is why they are so often confused.

THE p -TEST, BOTH ENDS

$$\int_1^\infty \frac{dx}{x^p} \text{ converges } \iff p > 1, \quad \int_0^1 \frac{dx}{x^p} \text{ converges } \iff p < 1.$$

At infinity you need *fast* decay; at the origin you need *mild* blow-up. The exponent $p = 1$ fails at both ends, and $\int_0^\infty \frac{dx}{x}$ diverges twice over. Logarithms are weaker than any power and therefore do not move the verdict except exactly at $p = 1$, where they decide it: $\int_2^\infty \frac{dx}{x \ln^q x}$ converges iff $q > 1$, while $\int_1^\infty \frac{\ln^k x}{x^p} dx$ converges iff $p > 1$ for every fixed k . Near the origin the corresponding statement is that $x^{-p} \ln^k x$ is integrable for every k when $p < 1$, and for no k when $p \geq 1$.

The reason the p -test suffices is that it is applied through comparison, and comparison only needs an inequality near the bad point. Everything away from the singularity is a proper integral of a bounded function and contributes a finite amount that cannot change the verdict. It follows that convergence is a purely local property: you may modify f arbitrarily on any interval bounded away from the bad points without changing the answer to the first question, and that is why the analysis reduces to matching a power at each of at most two places.

A small reference library makes the matching instantaneous. Every entry below is worth knowing as a fact rather than rederiving, and every one of them is the comparison function for a large class of problems.

Integral	Verdict	Value when convergent
$\int_0^1 x^{-p} dx$	$p < 1$	$\frac{1}{1-p}$
$\int_1^\infty x^{-p} dx$	$p > 1$	$\frac{1}{p-1}$
$\int_0^1 \ln^k x dx$	always	$(-1)^k k!$
$\int_0^{1/2} \frac{dx}{x \ln^2 x}$	converges	$\frac{1}{\ln 2} \approx 1.442695$
$\int_2^\infty \frac{dx}{x \ln^q x}$	$q > 1$	$\frac{\ln^{1-q} 2}{q-1}$
$\int_0^\infty x^{s-1} e^{-x} dx$	$s > 0$	$\Gamma(s)$
$\int_0^1 \frac{dx}{\sqrt{1-x^2}}$	converges	$\frac{\pi}{2}$
$\int_0^\infty \frac{\sin x}{x} dx$	conditional	$\frac{\pi}{2}$

The fourth row repays a second look, because it is the one that surprises. Near the origin $\frac{1}{x \ln^2 x}$ blows up like x^{-1} up to a logarithmic factor, and x^{-1} diverges; yet the squared logarithm downstairs is enough to rescue it, since the substitution $u = \ln x$ turns the integral into $\int_{-\infty}^{-\ln 2} u^{-2} du$, which converges at $-\infty$. The general principle: at the exact threshold $p = 1$ the logarithms decide, and they decide by the same $q > 1$ rule one level down.

CORE

27.2 Comparison and limit comparison

CUE You are about to assert a value for an integral over an unbounded domain, or with an unbounded integrand, and you have not yet checked that the object exists.

METHOD Isolate each bad point. Near it, find the dominant behaviour of $|f|$ and compare with a power. Direct comparison: if $0 \leq f \leq g$ near the bad point and $\int g$ converges, so does $\int f$; if $f \geq g \geq 0$ and $\int g$ diverges, so does $\int f$. Limit comparison is usually faster and needs no inequality:

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = L \in (0, \infty) \implies \int f \text{ and } \int g \text{ share their verdict.}$$

Choose g to be the power that matches the leading behaviour: strip off every factor that tends to a nonzero constant. Then test each bad point separately and take the conjunction.

WORKED EXAMPLE

$$\int_0^\infty \frac{x^{s-1}}{1+x^n} dx, \quad n > 0.$$

Two bad points. At 0: $\frac{x^{s-1}}{1+x^n}/x^{s-1} \rightarrow 1$, so the piece $\int_0^1$ behaves like $\int_0^1 x^{s-1} dx$, convergent iff $1-s < 1$, that is $s > 0$. At ∞ : $\frac{x^{s-1}}{1+x^n}/x^{s-1-n} \rightarrow 1$, so $\int_1^\infty$ behaves like $\int_1^\infty x^{s-1-n} dx$, convergent iff $n-s+1 > 1$, that is $s < n$. The integral converges precisely for $0 < s < n$, and on that strip its value is $\frac{\pi}{n \sin(\pi s/n)}$. At $s = \frac{3}{2}$, $n = 4$ this reads $\frac{\pi}{4 \sin(3\pi/8)} \approx 0.850109$, matching quadrature to ten digits.

CHOOSING THE COMPARISON POWER BY INSPECTION

$\int_1^\infty \frac{dx}{\sqrt{x^3+1}}$: the radicand is dominated by x^3 , so the integrand is $\sim x^{-3/2}$ and the limit-comparison ratio tends to 1.

Since $\frac{3}{2} > 1$, it converges.

$\int_0^1 \frac{dx}{\sqrt{x-x}}$: at 0 the dominant term downstairs is $\sqrt{x}$, giving $\sim x^{-1/2}$ with $p = \frac{1}{2} < 1$, convergent. But there is a second bad point: the denominator also vanishes at $x = 1$, where $\sqrt{x-x} = \sqrt{x}(1-\sqrt{x}) \sim \frac{1}{2}(1-x)$, giving $p = 1$ and divergence. The integral diverges, and it diverges at the end nobody looked at.

$\int_1^\infty x^{100} e^{-\sqrt{x}} dx$: no power test applies directly, and none is needed. Exponential decay beats every power, so compare with x^{-2} : the ratio $x^{102} e^{-\sqrt{x}} \rightarrow 0$, which gives $f \leq x^{-2}$ eventually and convergence. The rule to internalise is the

hierarchy $\ln^k x \ll x^a \ll e^{bx}$ for all k and all $a, b > 0$; against an exponential, every polynomial and logarithmic decoration is invisible to the convergence question.

WHERE THIS GOES WRONG

Testing one endpoint and forgetting the other. A power that is comfortable at ∞ is usually the one that kills you at 0, precisely because the two conditions run in opposite directions, and the second worked example above shows the failure in its purest form: the visible singularity converges and the invisible one does not. The second habitual error is comparing $|f|$ when f changes sign and then concluding *divergence* from the divergence of $\int |f|$. Comparison proves convergence of $\int f$ from convergence of $\int |f|$, never the reverse.

SEE ALSO Endpoint limit discipline; oscillatory convergence and Dirichlet's test; Euler reflection and the Mellin transform.

Comparison is a statement about $|f|$, so it establishes *absolute* convergence, and absolute convergence is what every interchange theorem in analysis wants: dominated convergence, Tonelli, termwise integration of a series, differentiation under the integral sign. When $\int |f|$ diverges but $\int f$ converges anyway, the integral still has a value, but it has lost the right to be manipulated freely, and that distinction is the subject of the next technique.

Absolute versus conditional

An integral converges absolutely when $\int |f|$ converges. It converges conditionally when $\int f$ converges and $\int |f|$ does not. The prototype is $\int_0^\infty \frac{\sin x}{x} dx$, which equals $\frac{\pi}{2}$ while $\int_0^\infty \left| \frac{\sin x}{x} \right| dx = \infty$: on the n -th half-period the absolute integrand contributes at least $\frac{1}{\pi(n+1)}$, and the harmonic series diverges. The convergence of the signed integral comes from cancellation between successive humps, and cancellation is fragile. Rearranging it, bounding it, or passing a limit through it can destroy it.

27.3 Oscillatory convergence and Dirichlet's test

HARD

CUE $\frac{\sin x}{x}$, $\frac{\cos x}{\sqrt{x}}$, $\sin(x^2)$, $\frac{\sin x}{\ln x}$: an integrand that decays too slowly for absolute convergence but oscillates with a bounded primitive.

METHOD Dirichlet's test. If g is monotone on $[a, \infty)$ with $g(x) \rightarrow 0$, and the partial integrals $\int_a^X f$ are bounded uniformly in X , then $\int_a^\infty fg$ converges. In practice there are two working routes. Integrate by parts once to move the decay into the denominator, converting a conditionally convergent integral into an absolutely convergent one; or split at the zeros of the oscillating factor and treat the result as an alternating series with decreasing terms.

WORKED EXAMPLE

$\int_1^\infty \frac{\sin x}{x} dx$. Take $f = \sin x$, whose partial integrals $[-\cos x]_1^X$ are bounded by 2, and $g = \frac{1}{x}$, monotone to 0. Dirichlet gives convergence. Integrating by parts exhibits the mechanism:

$$\int_1^R \frac{\sin x}{x} dx = \left[\frac{-\cos x}{x} \right]_1^R - \int_1^R \frac{\cos x}{x^2} dx,$$

and the remaining integrand is $O(x^{-2})$, absolutely convergent. Together with $\int_0^1 \frac{\sin x}{x} dx$, which is proper because the singularity is removable, the total is $\frac{\pi}{2} \approx 1.570796$.

OSCILLATION MANUFACTURED BY A SUBSTITUTION

$\int_0^\infty \sin(x^2) dx$ has an integrand that does not decay at all. Substitute $u = x^2$:

$$\int_0^\infty \sin(x^2) dx = \int_0^\infty \frac{\sin u}{2\sqrt{u}} du,$$

which is exactly the Dirichlet shape with $g(u) = \frac{1}{2\sqrt{u}}$ monotone to 0. It converges conditionally, to $\frac{1}{2}\sqrt{\frac{\pi}{2}} \approx 0.626657$. The decay was hidden in the Jacobian, not in the integrand: the reason $\sin(x^2)$ converges is that its oscillations become infinitely fast, so each hump has vanishing width.

Applying dominated convergence, Tonelli, or an absolute comparison to a conditionally convergent integral. Every one of those theorems has an integrable dominating function in its hypotheses, and no such function exists here. Concretely: swapping $\int_0^\infty$ with $\sum$ in $\int_0^\infty \frac{\sin x}{x} dx$ after expanding $\frac{1}{x}$ in some series, or interchanging the order in a double integral to derive $\frac{\pi}{2}$, is only legitimate if you first insert a regulator such as e^{-sx} , do the interchange for $s > 0$, and let $s \rightarrow 0^+$ with an Abel-summation argument at the end. A naive numerical quadrature is equally unsafe: truncating at R leaves a tail of size $\sim \frac{1}{R}$, not $\sim e^{-R}$.

SEE ALSO Comparison and limit comparison; the Dirichlet integral, Si and Ci; Fresnel integrals; justifying the swap.

The practical consequence of conditional convergence is a rule about what you are allowed to do next, and it is worth stating as a checklist because the temptation to forget it is strong. With absolute convergence you may expand the integrand in a series and integrate termwise, swap the integral with a limit or a sum, swap the order of a double integral, and differentiate under the integral sign with only a local dominating bound to check. With conditional convergence you may do none of those without a separate argument. The standard repair is to insert a regulator that restores absolute convergence, perform the manipulation, and remove the regulator at the end: replace $\int_0^\infty \frac{\sin x}{x} dx$ by $\int_0^\infty e^{-sx} \frac{\sin x}{x} dx = \arctan \frac{1}{s}$, which is absolutely convergent for $s > 0$ and legal to differentiate, then let $s \rightarrow 0^+$ to recover $\frac{\pi}{2}$. The regulator is doing exactly the job that the ϵ did in the endpoint discipline: it makes the object well defined while the algebra happens.

Conditional convergence is the mildest form of misbehaviour: the integral exists, it merely needs careful handling. The next two sections deal with objects that do not exist at all, and with the two conventions under which they are nevertheless assigned numbers.

Assigning a value to a divergent object

There are exactly two respectable ways to produce a finite number from an integral that diverges, and both work by imposing a cancellation that the integral itself does not have. The first, the Cauchy principal value, cancels a singularity against its own mirror image by approaching it symmetrically. The second cancels one divergence against another divergence elsewhere in the same expression. Both are conventions, and the obligation in both cases is identical: say which one you used.

Neither convention is arbitrary, and it is worth knowing why, because the reason is what licenses their use in applications. The principal value is the unique symmetric regularisation, and it is the one that arises naturally from contour integration: indenting a contour around a simple pole on the real axis with a small semicircle contributes $\pm i\pi \operatorname{Res}$, and the remaining real integral is exactly the principal value. It is also the definition under which the Hilbert transform, the Kramers-Kronig relations, and the Sokhotski-Plemelj formula $\frac{1}{x \mp i0} = \text{p.v.} \frac{1}{x} \pm i\pi \delta(x)$ are true. The matched-pair regularisation is likewise not a choice but a bookkeeping device: the object being computed is genuinely convergent, and the divergent halves are an artefact of how it was written. That distinction matters. In the first case you are reporting a new object; in the second you are reporting the original integral, computed carefully.

HARD

27.4 Cauchy principal value

CUE A simple pole strictly inside the interval, so that the integrand behaves like $\frac{A}{x-c}$ near c ; or an integral over $\mathbb{R}$ of something like $\frac{x}{1+x^2}$ that decays like $\frac{1}{x}$ with opposite signs at the two ends.

METHOD Approach the singularity symmetrically and take one limit rather than two:

$$\text{p.v.} \int_a^b f(x) dx = \lim_{\epsilon \rightarrow 0^+} \left[\int_a^{c-\epsilon} f(x) dx + \int_{c+\epsilon}^b f(x) dx \right], \quad \text{p.v.} \int_{-\infty}^{\infty} f = \lim_{R \rightarrow \infty} \int_{-R}^R f.$$

The two cut-offs must shrink at the *same rate*, and that single requirement is what makes the odd part of the singularity cancel. A principal value exists for a simple pole (residue-type singularity) and for a $\frac{1}{x}$ tail; it does not exist for a double pole, because $\frac{1}{(x-c)^2}$ is even about c and the two sides reinforce instead of cancelling.

WORKED EXAMPLE

$$\text{p.v.} \int_0^2 \frac{dx}{x-1} = \lim_{\epsilon \rightarrow 0^+} \left[[\ln|x-1|]_0^{1-\epsilon} + [\ln|x-1|]_{1+\epsilon}^2 \right] = \lim_{\epsilon \rightarrow 0^+} [\ln \epsilon + (0 - \ln \epsilon)] = 0.$$

The ordinary integral does not exist: $\int_0^1 \frac{dx}{x-1} = -\infty$ and $\int_1^2 \frac{dx}{x-1} = +\infty$. The principal value is 0, and it is 0 because the two infinities are the same infinity approached from opposite sides.

The tail version: p. v. $\int_{-\infty}^{\infty} \frac{x}{1+x^2} dx = \lim_{R \rightarrow \infty} \left[\frac{1}{2} \ln(1+x^2) \right]_{-R}^R = 0$, while $\int_0^{\infty} \frac{x}{1+x^2} dx = +\infty$ on its own.

A TRIGONOMETRIC PRINCIPAL VALUE

$$\text{p. v.} \int_0^{\pi} \tan x \, dx = 0.$$

The singularity is at $x = \frac{\pi}{2}$, interior, and simple: near it $\tan x \sim \frac{-1}{x - \pi/2}$. With symmetric cut-offs,

$$\int_0^{\pi/2-\varepsilon} \tan x \, dx = -\ln \cos\left(\frac{\pi}{2} - \varepsilon\right) = -\ln \sin \varepsilon, \quad \int_{\pi/2+\varepsilon}^{\pi} \tan x \, dx = \ln \sin \varepsilon,$$

and the two cancel exactly for every ε , so the limit is 0. Note that they cancel *before* the limit, which is the signature of a clean principal value: no residual constant is left behind. The ordinary integral, of course, does not exist.

WHERE THIS GOES WRONG

Two errors, and the second is subtle. First, reporting a principal value as if it were a convergent integral. It is a different object, and it is not additive over subintervals in the way an integral is; a numerical routine that happens to straddle the pole symmetrically will “confirm” it, which proves nothing. Second, the symmetry requirement is not cosmetic. Cutting at $1 - \varepsilon$ on the left and $1 + 2\varepsilon$ on the right in the example above gives $\ln \varepsilon - \ln(2\varepsilon) = -\ln 2$, not 0, and every rate you choose gives a different answer. The principal value is the number attached to the *symmetric* rate, and that choice must be stated.

SEE ALSO Endpoint limit discipline; regularising a matched pair of divergences; disguised zero; quotable residue results.

The principal value cancels a singularity against itself. The richer situation, and the one that produces most of the famous constants, is an integrand that is visibly a difference of two divergent pieces, each of which you can integrate exactly. There the cancellation is between two different divergences, and the technique is to make them share a cut-off so the infinite parts can be subtracted before the limit is taken.

HARD

27.5 Regularising a matched pair of divergences

CUE An integrand that is a difference of two pieces each of which diverges on its own: $\frac{1}{\ln x} + \frac{1}{1-x}$ on $[0, 1]$, $\sec x - \tan x$ at $\frac{\pi}{2}$, $\frac{f(ax) - f(bx)}{x}$ at both ends, $\frac{1 - x^{a-1}}{1-x}$ at 1.

METHOD Introduce a common cut-off, integrate each divergent half exactly, and watch the $\ln \varepsilon$ terms cancel before letting $\varepsilon \rightarrow 0$:

$$\int_a^b (u - v) = \lim_{\varepsilon \rightarrow 0^+} \left[\int_a^{b-\varepsilon} u - \int_a^{b-\varepsilon} v \right],$$

with the *same* ε in both. The parametric alternative is often cleaner: introduce a regulator s that makes each half converge, evaluate as an analytic function of s , and take the limit of the difference. The finite remainder is where the interesting constant lives.

WORKED EXAMPLE

$$\int_0^1 \left(\frac{1}{\ln x} + \frac{1}{1-x} \right) dx = \gamma \approx 0.577216.$$

Each half diverges logarithmically at $x = 1$: near 1, $\ln x \sim -(1-x)$, so $\frac{1}{\ln x} \sim \frac{-1}{1-x}$ and the two terms cancel to leading order, leaving a bounded integrand. Cut both at $1 - \varepsilon$. The second half is $-\ln \varepsilon$ exactly. The first, after $x = e^{-t}$, is $-\int_{\varepsilon'}^{\infty} \frac{e^{-t}}{t} dt$ with $\varepsilon' \sim \varepsilon$, whose expansion is $\ln \varepsilon + \gamma + O(\varepsilon)$ up to sign. The logarithms cancel and γ is the residue. Quadrature on the combined integrand returns 0.5772156649, which is γ to ten places.

THE SAME CANCELLATION, VISIBLE IN ONE LINE

$$\int_0^{\pi/2} (\sec x - \tan x) \, dx.$$

Both $\int \sec$ and $\int \tan$ diverge at $\frac{\pi}{2}$. But $\sec x - \tan x = \frac{1 - \sin x}{\cos x}$, and the antiderivatives combine: $\ln|\sec x + \tan x| + \ln|\cos x| = \ln|1 + \sin x|$. Hence the value is $[\ln(1 + \sin x)]_0^{\pi/2} = \ln 2 \approx 0.693147$. Here the regularisation is invisible because the two logarithms were combined symbolically before the endpoint was reached, which is the cheapest form of the technique: do the subtraction inside the antiderivative.

The parametric form, for reference: $\int_0^1 \frac{1 - x^{a-1}}{1 - x} dx = \psi(a) + \gamma$. At $a = 3$ the integrand is $1 + x$ and the value is $\frac{3}{2}$, matching $\psi(3) + \gamma = \frac{3}{2}$.

WHERE THIS GOES WRONG

Cutting the two halves off at different points. If you truncate $\int \frac{dx}{\ln x}$ at $1 - \varepsilon$ and $\int \frac{dx}{1 - x}$ at $1 - 2\varepsilon$, the surviving $\ln 2$ contaminates the answer, and since $\ln 2$ is exactly the size of a plausible constant, the result looks reasonable. This is the mechanism behind almost every “off by $\ln 2$ ” error in this family. The same applies to Frullani: $\int_0^\infty \frac{f(ax) - f(bx)}{x} dx = (f(0) - f(\infty)) \ln \frac{b}{a}$ requires the two pieces to be cut at the same x , not at the same argument of f .

SEE ALSO Frullani’s integral; the Euler-Mascheroni constant; Cauchy principal value; conjugate multiplication.

Engineered endpoints

The rest of this chapter is about a different kind of reading. In every technique so far the bounds were given by the problem and the work was in the integrand. There is a large family of problems where the reverse is true: the integrand is routine, and the bounds have been chosen with such deliberate strangeness that they constitute an instruction. A limit of integration written as $\sqrt{e^2 - 1}$, or $\log_8 e$, or $\ln^2(2e)$, is not a number the problem needed. It is a number that becomes clean under one particular substitution, and identifying which one is faster than any inspection of the integrand.

READ THE BOUND AS A MESSAGE

Ask, of any strange bound β : which standard substitution sends β to something memorable? $\beta = \sqrt{e^2 - 1}$ satisfies $1 + \beta^2 = e^2$, so $u = 1 + x^2$ lands on e^2 and $\ln u$ lands on 2. $\beta = \log_8 e$ satisfies $\frac{1}{\beta} = \ln 8$, so $u = \frac{1}{x}$ lands on $\ln 8$ and $e^{1/x}$ lands on 8. $\beta = \ln(\pi/4)$ satisfies $e^\beta = \frac{\pi}{4}$, so $u = e^x$ lands on a special angle. $\beta = \ln^2(2e)$ satisfies $\sqrt{\beta} = 1 + \ln 2$, so $u = \sqrt{x}$ lands on $\ln(2e)$ and e^u lands on $2e$. In each case the substitution is then confirmed by the integrand, but it was proposed by the bound.

27.6 Endpoint engineering

EASY

CUE Bounds written strangely when a plain number would have served: $\sqrt{e^2 - 1}$, $\log_8 e$ and $\log_2 e$, $\ln^2(2e)$, $\ln(\pi/4)$ to $\ln(\pi/2)$, $\tan^{-1} 3$, $\sqrt{3}/3$. The oddness of the bound is the hint, and it is usually the only hint.

METHOD Ask what the candidate substitution does to *that number*, and transform the bounds symbolically. Never evaluate a transformed bound numerically before simplifying: 0.4809 tells you nothing and $\ln 8$ tells you everything. If the transformed bounds are clean and the transformed integrand is routine, you have found the intended route; if the bounds go clean and the integrand does not, you have found the wrong substitution and should try the next candidate on the same bound.

WORKED EXAMPLE

$$\int_{\log_8 e}^{\log_2 e} \frac{e^{1/x}}{x^2(e^{1/x} + 1)} dx.$$

The bounds are $\frac{1}{\ln 8}$ and $\frac{1}{\ln 2}$, screaming for $u = \frac{1}{x}$. Then $du = -\frac{dx}{x^2}$, the bounds become $\ln 8$ and $\ln 2$ with the orientation reversed, and

$$I = \int_{\ln 2}^{\ln 8} \frac{e^u}{e^u + 1} du = [\ln(1 + e^u)]_{\ln 2}^{\ln 8} = \ln 9 - \ln 3 = \ln 3 \approx 1.098612.$$

Every piece of the difficulty was in reading $\log_8 e$ as $\frac{1}{\ln 8}$. Quadrature on the original confirms 1.0986122887.

$$\int_0^{\sqrt{e^2-1}} \frac{x \ln(1+x^2)}{1+x^2} dx.$$

The bound says $u = 1 + x^2$, which runs from 1 to e^2 . With $du = 2x dx$,

$$I = \frac{1}{2} \int_1^{e^2} \frac{\ln u}{u} du = \frac{1}{4} [\ln^2 u]_1^{e^2} = \frac{2^2}{4} = 1.$$

Next, $\int_{\ln(\pi/4)}^{\ln(\pi/2)} e^x \cot(e^x) dx$: the bounds say $u = e^x$, running from $\frac{\pi}{4}$ to $\frac{\pi}{2}$, so $I = \int_{\pi/4}^{\pi/2} \cot u du = [\ln \sin u]_{\pi/4}^{\pi/2} = 0 - \ln \frac{1}{\sqrt{2}} = \frac{1}{2} \ln 2 \approx 0.346574$.

Last, $\int_1^{\ln^2(2e)} \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$: the squared logarithm says $u = \sqrt{x}$, running from 1 to $\ln(2e) = 1 + \ln 2$, and $dx = 2u du$ cancels the $\frac{1}{\sqrt{x}}$. So $I = 2 \int_1^{1+\ln 2} e^u du = 2(e^{1+\ln 2} - e) = 2(2e - e) = 2e \approx 5.436564$.

AN INVERSE-TRIGONOMETRIC BOUND

$$\int_0^{\arctan 3} \frac{\sec^2 x}{(1 + \tan x)^2} dx.$$

A bound of $\arctan 3$ is an instruction to set $u = \tan x$, which sends it to 3; the $\sec^2 x dx$ in the numerator is then exactly du . So

$$I = \int_0^3 \frac{du}{(1+u)^2} = \left[\frac{-1}{1+u} \right]_0^3 = 1 - \frac{1}{4} = \frac{3}{4}.$$

The same reading applies to $\arcsin$, arsinh and arcosh bounds: each names the value that its own inverse substitution lands on, and the integrand should then contain the matching derivative factor. If it does not, you have misread the bound.

WHERE THIS GOES WRONG

Evaluating the transformed bound as a decimal instead of a symbol. The whole design of these problems is that the transformed bound is a recognisable constant, and a decimal destroys the recognition; $\ln 9 - \ln 3$ is instantly $\ln 3$, while $2.1972 - 1.0986$ is a subtraction you may or may not notice collapses. The second failure is trusting the bound over the integrand: a clean bound under a substitution that leaves the integrand worse is a coincidence, not a signal, and you should back out rather than push on. And when the substitution is not injective on the interval (as $u = x^2$ is not on an interval containing 0), transformed bounds can coincide and silently annihilate the integral.

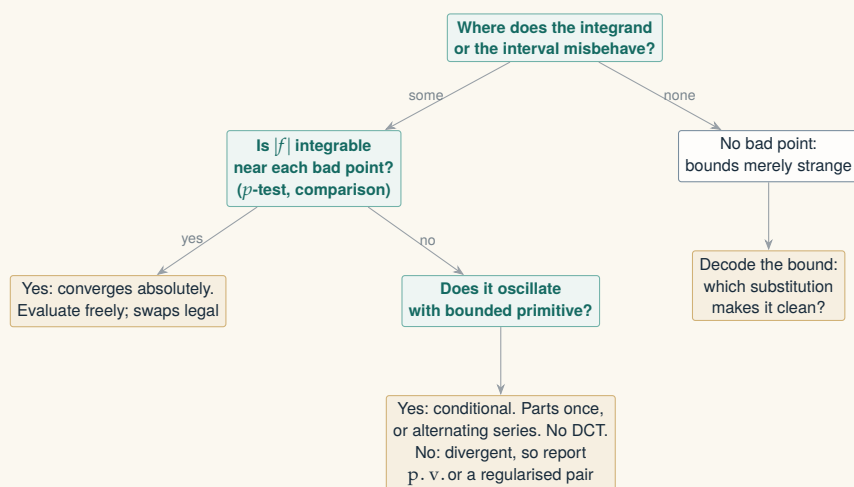
SEE ALSO Substitution by spotting the differential; exponent and logarithm law collapse; perfect-derivative disguise.

The habit generalises past the specific bounds listed above. Whenever a problem contains a number that is more complicated than it needs to be, treat the complication as a specification. A coefficient of $\frac{\sqrt{3}}{3}$ rather than 0.577 is $\tan \frac{\pi}{6}$; a bound of $\frac{1}{\sqrt{2}}$ against an $\arcsin$ is $\sin \frac{\pi}{4}$; an upper limit of e^π against a $\ln$ is a request for π ; a limit of $\ln(1 + \sqrt{2})$ is $\operatorname{arsinh} 1$. In every case the author has written the answer to “where does the substitution land” into the problem statement, because there is no other reason to write the number that way. This reading takes ten seconds and frequently determines the entire solution, which makes it the highest-yield habit in the chapter.

Endpoint engineering has a converse worth stating, because it closes the loop with the first half of this chapter. Sometimes the strange bound is strange because it is the point where the integrand becomes improper, and the design is not a substitution at all but a test of whether you notice. A bound of 1 against a $\sqrt{1-x^2}$ downstairs, a bound of 0 against $\ln x$, a bound of $\frac{\pi}{2}$ against $\tan x$: these are engineered too, and what they engineer is a limit. The reading protocol is the same in both cases. Look at the bound, ask what the integrand does there, and answer before integrating.

Choosing the route

The decision procedure for an improper integral is short, and it is short because the first two steps are forced. Locate the bad points; classify each one. Only then does the integrand’s structure matter.



MAKING AN IMPROPER INTEGRAL SAFE FOR A MACHINE

Numerical quadrature is a poor judge of improperness, so transform before you compute. An endpoint singularity of the form x^{-p} is removed by $x = u^{1/(1-p)}$, or generically by the tanh-sinh map $x = \tanh(\frac{\pi}{2} \sinh t)$, which pushes both endpoints to $t = \pm\infty$ and makes the transformed integrand decay doubly exponentially. An infinite range is pulled to a finite one by $x = \frac{u}{1-u}$ or $x = e^t$. A conditionally convergent oscillatory tail should be split at the zeros of the oscillating factor and summed as an alternating series, with the tail bounded by the first omitted hump rather than by a truncation guess. Applying these before quadrature is not a refinement; without them the returned digits are unrelated to the integral.

Two habits make the tree unnecessary in most cases. The first is to solve for the zeros of every denominator and every logarithm argument on the closed interval before doing anything else, which takes a few seconds and catches the interior pole that no amount of algebraic fluency will catch later. The second is to write down, for each bad point, the single power x^{-p} that matches the local behaviour, and to read the verdict off the p -test. Two powers and a conjunction is the entire convergence analysis of most integrals worth writing down.

WHAT TO REPORT

A convergent integral gets a number. A conditionally convergent one gets a number and a sentence saying that it is conditional, because that sentence is what licenses or forbids the reader's next manipulation. A divergent one gets the word "diverges", optionally followed by a principal value *explicitly labelled* as such. The one thing that is never acceptable is a bare number for a divergent object, and the reason it is never acceptable is that such a number is usually correct-looking: -2 for $\int_{-1}^1 x^{-2} dx$, 0 for $\int_{-1}^1 \frac{dx}{x}$, $\ln 2$ for a mismatched regularisation. Wrong answers in this chapter do not look wrong.

Recognition matrix

If you see	Reach for	If that fails
An infinite bound	$\lim_{R \rightarrow \infty} \int_a^R$; test decay against x^{-p} , $p > 1$	Substitute $x = 1/u$ to move ∞ to 0
$\ln x$ or x^{-p} at an endpoint	$\lim_{\varepsilon \rightarrow 0^+}$; test $p < 1$ at the origin	$x = e^{-t}$ turns 0 into ∞
A denominator zero inside $[a, b]$	Split at c ; both halves must converge	Simple pole: report a p. v.
$\frac{\sin x}{x}$ at 0 , $\frac{\ln x}{1-x}$ at 1	Removable: the integral is proper	Expand in series to confirm
$\frac{x}{x^{s-1}}$ on $(0, \infty)$	Two p -tests: $s > 0$ and $s < n$	Value $\frac{\pi}{n \sin(\pi s/n)}$ on the strip
$\frac{1}{1+x^n}$ near ∞	Log p -test: converges iff $q > 1$	$u = \ln x$
$\frac{x \ln^q x}{\sqrt{x^3+1}}$ or similar downstairs	Limit comparison with the leading power	Strip constants, keep the exponent
$\frac{\sin x}{x}$, $\frac{\cos x}{\sqrt{x}}$ on $(0, \infty)$	Dirichlet: conditional convergence	Parts once to gain x^{-2}
$\sin(x^2)$, $\cos(x^2)$	$u = x^2$; decay hides in the Jacobian	Fresnel values $\frac{1}{2}\sqrt{\pi/2}$
$\frac{A}{x - x^c}$ with c interior	Symmetric cut-off: p. v. exists	Double pole: no p. v.
$\frac{1}{1+x^2}$ on $\mathbb{R}$	p. v. = 0 by odd symmetry of the tail	State that it is a p. v.
$\frac{1}{\ln x} + \frac{1}{1-x}$	Common cut-off; logs cancel; answer γ	Parametric regulator s
$\sec x - \tan x$ at $\frac{\pi}{2}$	Combine antiderivatives first: $\ln(1 + \sin x)$	Weierstrass $t = \tan \frac{x}{2}$

If you see	Reach for	If that fails
$\frac{f(ax) - f(bx)}{1 - x^{\frac{x}{a-1}}}$ on $[0, 1]$	Frullani with a shared cut-off	Parameter differentiation
Bound $\sqrt{e^2 - 1}$	$\psi(a) + \gamma$	Termwise, then telescope
Bound $\log_b e$	$u = 1 + x^2$ lands on e^2	Then $\ln u$ lands on 2
Bound $\ln(\pi/4)$ or $\ln(\pi/2)$	$u = 1/x$ lands on $\ln b$	Then $e^{1/x}$ lands on b
Bound $\ln^2(2e)$	$u = e^x$ lands on a special angle	Check the integrand carries e^x
A bound that is exactly a singularity	$u = \sqrt{x}$ lands on $1 + \ln 2$	Then e^u lands on $2e$
A clean answer from a blind antiderivative	It is a limit, not a substitution	Classify, then take the limit
	Check for an interior pole	Sign test: positive integrand, negative value

Integrals that collapse

What this chapter is for. This is the mirror image of the chapter on special functions. There, an integral that looked routine turned out to require the dilogarithm or the Gamma function. Here, an integral that looks impossible turns out to require nothing at all: the integrand is exactly the derivative of something you can write down, or it telescopes, or it is a constant wearing a costume, or an odd factor on a symmetric interval annihilates it, or the only work is a case split on an absolute value. In every case the entire difficulty is noticing, and once you have noticed there is no integration to do. The species are few enough to enumerate, and the enumeration is the point of the chapter: perfect derivatives (of a product, of a quotient, of a power tower), telescoping sums and shells, disguised constants and disguised zeros, step functions that only need their level sets, and the one genuinely dangerous species, an integrand whose naive antiderivative is discontinuous inside the interval and therefore wrong. The chapter ends with a diagnostic sweep that catches all of them in under a minute, and the sweep is worth more than any individual technique in it, because these problems are designed so that the only alternative to noticing is an hour of futile work.

The sweep

The problems in this chapter are adversarial. They are written so that a competent integrator, reading left to right, will start integrating; and the integration will not terminate, because there is nothing there to integrate. The defence is a fixed sequence of cheap tests applied *before* any method is chosen. Each test costs a few seconds, and each one is a question about structure rather than about technique.

THE ONE-MINUTE SWEEP

- 1. Evaluate the integrand at two points.** Equal values on a non-constant-looking expression means a disguised constant. Cost: two arithmetic evaluations.
 - 2. Test parity about the midpoint.** On $[-a, a]$, or after $t = x - \frac{a+b}{2}$: is any factor odd against an even remainder?
 - 3. Look for a quotient with a squared denominator** and a numerator of unexpectedly low degree. That is $(u/v)'$.
 - 4. Look for a sum of unrelated pieces** where one factor is a power tower or a stray logarithm. That is F' found by logarithmic differentiation.
 - 5. Check whether the interval decomposes.** Floors, fractional parts, absolute values, max, min, sgn: the integrand is smooth on each level set.
 - 6. Check the domain and the branch.** Even roots, odd roots, arcsec, arctan of something that crosses a pole, and every denominator zero inside the interval.
- If all six fail, the integral is genuine and belongs to another chapter.

The sweep is ordered by cost, not by frequency. Step 1 is first because it is nearly free and because its payoff is total: a disguised constant reduces the problem to multiplication. Step 6 is last because it requires the most thought, and because it is the only step that catches an error rather than an opportunity.

It helps to understand how these problems are manufactured, because the construction is always the same and always leaves fingerprints. The author starts with the answer, not with the question. They pick an F , differentiate it, and then obscure the result: expand a product so that the two halves of the product rule land in separate terms; write $x^{1/x-2}$ instead of $\frac{x^{1/x}}{x^2}$ so the power tower is invisible; insert a decorative factor that is odd on a symmetric interval; wrap a constant in an identity that hides it. Each of those moves leaves a residue. A sum of terms with unrelated shapes is the fingerprint of a product rule expanded. An exponent written as a difference is the fingerprint of a quotient. A high odd power multiplied by an even weight is the fingerprint of a parity kill. A parameter that appears in the problem and could not possibly appear in the answer is the fingerprint of a cancellation. Reading fingerprints is faster than reading integrands, and it is what the sweep operationalises.

One warning about the sweep before the techniques begin. Its purpose is to *detect*, not to prove. A sample point that matches is evidence for a disguised constant, and a squared denominator is evidence for a quotient rule, but every one of the six tests must be followed by a symbolic confirmation: differentiate the guessed F and compare, or establish the identity that makes the integrand constant. Detection is cheap and unreliable; confirmation is cheap and reliable; and the whole value of the sweep lies in doing the first so that you know which of the second to attempt.

Derivatives wearing a costume

The largest species by far is the integrand that is literally F' for a writable F . What makes these hard is that the standard differentiation rules are being read backwards, and the product rule read backwards does not look like a product: it looks like a sum of two unrelated terms. The two techniques below are the two rules most often disguised, and they are worth separating because their cues are different.

28.1 Perfect-derivative disguise

CORE

CUE An integrand that is a *sum* of pieces which do not obviously belong together, especially when one of them is a power tower f^g such as $x^{1/x}$, x^x , $(\sec x)^{\sin x}$, or when a stray logarithm appears in a product where nothing else suggests one.

METHOD Guess F from the most complicated factor. Compute F' by logarithmic differentiation,

$$F' = F \cdot (\ln F)', \quad \text{so for } F = u^v: \quad F' = u^v \left(v' \ln u + \frac{vu'}{u} \right),$$

and compare with the whole integrand. If it matches, the answer is $F(b) - F(a)$ and no integration happens at all. The tell for a power tower is that its derivative always carries the factor u^v itself, so the integrand contains a copy of the very expression you are trying to integrate.

WORKED EXAMPLE

$$\int_0^2 x^{\frac{1}{x}-2} (1 - \ln x) \, dx.$$

The exponent $\frac{1}{x} - 2$ is $\frac{1}{x}$ minus 2, which suggests $F = x^{1/x}$ divided by x^2 . Confirm: $\ln F = \frac{\ln x}{x}$, so $(\ln F)' = \frac{1 - \ln x}{x^2}$ and

$$F' = x^{1/x} \cdot \frac{1 - \ln x}{x^2} = x^{\frac{1}{x}-2} (1 - \ln x),$$

which is the integrand exactly. Hence $I = [x^{1/x}]_0^2 = 2^{1/2} - 0 = \sqrt{2} \approx 1.414214$. Quadrature returns 1.4142135624.

THE PLAINER SIBLING

$$\int_1^2 x^x (1 + \ln x) \, dx.$$

Here $F = x^x$ has $\ln F = x \ln x$ and $(\ln F)' = 1 + \ln x$, so $F' = x^x (1 + \ln x)$ is the integrand. The value is $[x^x]_1^2 = 4 - 1 = 3$. The pattern is worth memorising in the abstract: whenever you see u^v multiplied by something, differentiate u^v once and see whether that something is $(v \ln u)'$.

A TRIGONOMETRIC POWER TOWER

$$\int_0^{\pi/3} (\sec x)^{\sin x} (\cos x \ln \sec x + \sin x \tan x) \, dx.$$

The bracket is unrecognisable until you notice that the leading factor is a power tower. Set $F = (\sec x)^{\sin x}$, so $\ln F = \sin x \ln \sec x$ and

$$(\ln F)' = \cos x \ln \sec x + \sin x \cdot \frac{\sec x \tan x}{\sec x} = \cos x \ln \sec x + \sin x \tan x,$$

which is exactly the bracket. Hence F' is the integrand and

$$I = [(\sec x)^{\sin x}]_0^{\pi/3} = 2^{\sqrt{3}/2} - 1 \approx 0.822635,$$

confirmed by quadrature to ten digits. The bracket was constructed by differentiating; the only way to read it is to differentiate the same thing.

WHERE THIS GOES WRONG

The lower endpoint. In the first example $x^{1/x} = \exp(\frac{\ln x}{x}) \rightarrow \exp(-\infty) = 0$ as $x \rightarrow 0^+$, because $\frac{\ln x}{x} \rightarrow -\infty$ faster than any polynomial: at $x = 10^{-8}$ the value is already below $10^{-8 \times 10^7}$. That limit must be argued, not assumed, and the integral is genuinely improper at 0. The second failure mode is matching the guessed F' against only part of the integrand: if F' reproduces one term and leaves a remainder, you have not found a perfect derivative, you have found an integration by parts, and the remainder still has to be integrated.

SEE ALSO Reverse product and quotient rule; logarithmic differentiation; engineering the antiderivative v ; constant or zero integrand in disguise.

Power towers announce themselves. The quotient rule read backwards is quieter, because a quotient with a squared denominator is exactly what partial fractions and trigonometric substitution both produce, so the shape does not by itself suggest that anything special is happening. The diagnostic is arithmetic rather than visual: compare the degree of the numerator against the degree you would expect from an honest partial-fraction problem.

28.2 Reverse product and quotient rule

CORE

CUE An integrand of the form $\frac{u'v - uv'}{v^2}$, or $f'g + fg'$, or a numerator that looks like a Wronskian $fg' - gf'$ sitting over $f^2 + g^2$.

METHOD Recognise the three shapes:

$$\frac{d}{dx} \frac{u}{v} = \frac{u'v - uv'}{v^2}, \quad \frac{d}{dx}(uv) = u'v + uv', \quad \frac{d}{dx} \arctan \frac{g}{f} = \frac{fg' - gf'}{f^2 + g^2}.$$

The third is the powerful one, because $f^2 + g^2$ downstairs is a shape that no standard method handles well and that the arctan form handles instantly. Diagnostically: a squared denominator whose numerator has *lower* degree than a partial-fraction problem would produce is the tell for the first shape; a sum of two squares downstairs with a difference of cross terms upstairs is the tell for the third.

WORKED EXAMPLE

$$\int_0^1 \frac{xe^x}{(1+x)^2} dx.$$

Partial fractions is useless because of the e^x , and parts produces another integral of the same difficulty. Instead take $u = e^x$, $v = 1 + x$:

$$\frac{d}{dx} \frac{e^x}{1+x} = \frac{e^x(1+x) - e^x}{(1+x)^2} = \frac{xe^x}{(1+x)^2},$$

the integrand exactly. So $I = \left[\frac{e^x}{1+x} \right]_0^1 = \frac{e}{2} - 1 \approx 0.359141$.

THE WRONSKIAN SHAPE

$$\int_0^{\pi/2} \frac{x \cos x - \sin x}{x^2 + \sin^2 x} dx.$$

The denominator is $f^2 + g^2$ with $f = x$, $g = \sin x$; the numerator is $fg' - gf' = x \cos x - \sin x$. Hence the integrand is $\frac{d}{dx} \arctan \frac{\sin x}{x}$ and

$$I = \left[\arctan \frac{\sin x}{x} \right]_0^{\pi/2} = \arctan \frac{2}{\pi} - \arctan 1 = \arctan \frac{2}{\pi} - \frac{\pi}{4} \approx -0.218487.$$

At the lower limit $\frac{\sin x}{x} \rightarrow 1$, so the endpoint is removable rather than singular. The same pattern with $f = Ai$, $g = Bi$ and their constant Wronskian $\frac{1}{\pi}$ gives $\int \frac{dx}{Ai^2 + Bi^2} = \pi \arctan \frac{Bi}{Ai}$, an integral with no business being elementary.

THE PRODUCT RULE, SPLIT ACROSS TWO TERMS

$$\int_0^{\pi/4} (x \sec^2 x + \tan x) dx.$$

The two terms look unrelated: the first invites parts, the second is a standard logarithm. But $\frac{d}{dx}(x \tan x) = \tan x + x \sec^2 x$ is exactly the sum, so

$$I = [x \tan x]_0^{\pi/4} = \frac{\pi}{4} \cdot 1 - 0 = \frac{\pi}{4} \approx 0.785398.$$

The diagnostic that finds this in five seconds: whenever an integrand is a sum of two terms and one of them is u times the derivative of v while the other is v times the derivative of u , the answer is $[uv]_a^b$. Doing it the long way produces $[x \tan x] - \int \tan x$ from parts, and then the second term cancels the leftover, which is the same answer arrived at with three times the writing.

Attacking with partial fractions or parts and never testing the structure. The cost of the test is one differentiation of the obvious candidate $\frac{u}{v}$, which takes ten seconds; the cost of missing it is unbounded. The second trap is a sign: $\frac{d}{dx} \frac{u}{v}$ has $u'v - uv'$ upstairs, and the reversed order $uv' - u'v$ integrates to $-\frac{u}{v}$. Since both forms look equally natural on the page, check which one you actually have before writing the answer.

SEE ALSO Perfect-derivative disguise; integration by parts, cyclic; engineering the antiderivative v ; Leibniz rule with variable limits.

Nothing there at all

Step 1 of the sweep catches the most extreme species: an integrand which is not a function of x in any meaningful sense, because every apparent dependence cancels. These problems are built by taking an identity, hiding it, and then asking for an integral, and the identity is usually one of a small stock: $\arctan x + \arctan \frac{1}{x} = \frac{\pi}{2}$ for $x > 0$, $\log_a b \cdot \log_b c = \log_a c$, a fixed-point equation for a nested radical, Euler's reflection formula for a product of Gamma values, or a determinant of matched antiderivatives.

28.3 Constant or zero integrand in disguise

MEDIUM

CUE An intimidating integrand with no free structure: mixed logarithm bases, an infinitely iterated radical, a determinant whose entries are antiderivatives of related functions, a sum of inverse trigonometric functions of reciprocal arguments, a product of shifted Gamma or double-factorial expressions inside a logarithm.

METHOD Evaluate the integrand at a sample point. Then evaluate it at a second. If the two agree, suspect a constant and prove it: chase the identity rather than the integral. Fixed-point equations collapse iterated radicals ($y = \sqrt{x+y}$ or $y = \sqrt{xy}$); change of base collapses mixed logarithms; the Wronskian-style cancellation collapses determinants; and complementary-angle identities collapse sums of inverse trigonometric functions. If the constant is 0, look first for an odd factor against an even one on a symmetric interval.

WORKED EXAMPLE

$$\int_2^4 (\log_x 8)(\log_2 x) dx.$$

Sample at $x = 3$: $\log_3 8 \cdot \log_2 3 = \frac{\ln 8}{\ln 3} \cdot \frac{\ln 3}{\ln 2} = \frac{\ln 8}{\ln 2} = 3$. Sample at $x = 4$: also 3. The $\ln x$ cancels identically, so the integrand is the constant 3 and $I = 3 \cdot 2 = 6$. Nothing was integrated.

The same move on a determinant. With

$$M(x) = \begin{pmatrix} \int_0^x \cos t \, dt & \int_0^x \sin 2t \, dt \\ \int_0^x dt & \int_0^x (t \cos t + \sin t) \, dt \end{pmatrix} = \begin{pmatrix} \sin x & \sin^2 x \\ x & x \sin x \end{pmatrix},$$

using $\int_0^x \sin 2t \, dt = \frac{1 - \cos 2x}{2} = \sin^2 x$ and $\int_0^x (t \cos t + \sin t) \, dt = [t \sin t]_0^x = x \sin x$, the determinant is $x \sin^2 x - x \sin^2 x = 0$ for every x . So $\int_a^b \det M(x) \, dx = 0$ on any interval whatever.

A DISGUISED ZERO BY PARITY, AND A DISGUISED $\frac{\pi}{2}$

$$\int_{-2}^2 \ln\left(\frac{3+x}{3-x}\right) \sqrt{4-x^2} \, dx = 0,$$

because $\ln \frac{3+x}{3-x}$ is odd, $\sqrt{4-x^2}$ is even, and the product is odd on a symmetric interval; quadrature returns 1.1×10^{-16} . The weight was chosen to look like an area computation and the logarithm to look like a partial-fraction problem, and neither has to be touched.

$$\int_0^1 \left(\arctan x + \arctan \frac{1}{x} \right) dx = \int_0^1 \frac{\pi}{2} dx = \frac{\pi}{2} \approx 1.570796.$$

Both terms separately require integration by parts; their sum is a constant on $(0, \infty)$, and the singularity of $\arctan \frac{1}{x}$ at 0 is removable in the sum.

Not checking numerically. One evaluation at $x = 1$ costs nothing and settles the question, and without it a reader will spend the whole allotted time integrating a constant. The converse trap: two sample points agreeing is evidence, not proof. $\ln \frac{3+x}{3-x} \sqrt{4-x^2}$ agrees at $x = 1$ and $x = -1$ in absolute value but is not constant, and $\sin(\pi x)$ agrees at every integer. Sample at irrational-looking points, and once you believe the claim, prove the identity.

SEE ALSO Even and odd parity on a symmetric interval; algebraic camouflage collapse; self-similar collapse; numerical cross-check as verification discipline.

THE STOCK IDENTITIES BEHIND A DISGUISED CONSTANT

There are not many, and knowing them turns step 1 of the sweep from a numerical experiment into a recognition. Inverse trigonometric: $\arctan x + \arctan \frac{1}{x} = \frac{\pi}{2}$ for $x > 0$ and $-\frac{\pi}{2}$ for $x < 0$; $\arcsin x + \arccos x = \frac{\pi}{2}$; $\arctan a + \arctan b = \arctan \frac{a+b}{1-ab}$ modulo π . Logarithmic: $\log_a b \cdot \log_b c = \log_a c$, so any chain of mixed bases collapses to the endpoints. Gamma: Euler reflection $\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin \pi z}$ turns a product of matched Gamma values into π over a sine. Hyperbolic: $\cosh^2 - \sinh^2 = 1$, and $\operatorname{arsinh} x = \ln(x + \sqrt{x^2 + 1})$, which is how a logarithm of a radical becomes an odd function. Fixed points: $y = \sqrt{xy}$ gives $y = x$, and $y = \sqrt{x+y}$ gives $y = \frac{1+\sqrt{1+4x}}{2}$, so an infinite radical is never infinite and is often trivial. Finally, any 2×2 determinant whose columns are proportional collapses to 0, which is what makes matched antiderivatives such a convenient disguise.

Cut the domain and the difficulty disappears

The next species does not collapse pointwise; it collapses after the interval is chopped. Absolute values, floors, fractional parts, signs, maxima and minima all define a partition of $[a, b]$ into pieces on which the integrand is a smooth, usually trivial, function. The work is entirely in finding the partition, and the failures are entirely off-by-one errors at its boundaries.

28.4 Absolute value, sign, and the log's modulus

EASY

CUE $|f(x)|$, $\operatorname{sgn} f(x)$, $\sqrt{x^2}$, $\sqrt{f(x)^2}$, or an odd-looking integrand on a symmetric interval where the bars have been inserted deliberately.

METHOD Bars change the parity, and that is usually the whole point: $|x|^3$ is even where x^3 is odd, so an integral you expected to vanish is instead $2 \int_0^a$. Three rules cover the species. First, $\sqrt{f^2} = |f|$, never f . Second, $\int \frac{dx}{x} = \ln|x|$, and the modulus matters whenever the argument can be negative. Third, resolve the bars by splitting at every zero of the inner expression and choosing the sign on each piece.

WORKED EXAMPLE

$$\int_{-\infty}^{\infty} |x|^3 2^{-x^2} dx.$$

Without the bars this is odd and vanishes. With them the integrand is even, so

$$I = 2 \int_0^{\infty} x^3 2^{-x^2} dx = 2 \cdot \frac{1}{2} \int_0^{\infty} u e^{-u \ln 2} du = \frac{\Gamma(2)}{\ln^2 2} = \frac{1}{\ln^2 2} \approx 2.081369,$$

substituting $u = x^2$ and then recognising a Gamma integral with rate $\ln 2$. The bars rescued the integral from being 0, and the entire content of the problem is noticing that.

A RADICAL THAT IS SECRETLY AN ABSOLUTE VALUE, AND A DOMAIN CHECK

$$\int_{-1}^1 \sqrt{x^{13} + x^8} dx.$$

First the domain: $x^{13} + x^8 = x^8(x^5 + 1)$, and on $[-1, 1]$ both factors are nonnegative, so the radicand is nonnegative and the integrand is real throughout. Now $\sqrt{x^8} = |x|^4 = x^4$, because 8 is even and the fourth power is nonnegative, so the integrand is $x^4 \sqrt{1 + x^5}$ with no bars needed. Substituting $u = 1 + x^5$, $du = 5x^4 dx$:

$$I = \frac{1}{5} \int_0^2 \sqrt{u} du = \frac{2}{15} \cdot 2^{3/2} = \frac{4\sqrt{2}}{15} \approx 0.377124.$$

Had the exponent been x^7 rather than x^8 the radicand would have been negative on part of $[-1, 0)$ and the integral would not exist as a real object.

Cancelling $\ln|g|$ across a zero of g : the antiderivative $\ln|g|$ is only valid on an interval where g does not vanish, so an interior zero of g makes the integral improper and the difference of logarithms meaningless. And $\sqrt{x^2} = x$ is the single most productive wrong step in this chapter: it converts $\int_0^{2\pi} \sqrt{1 - \cos 2x} \, dx = 4\sqrt{2}$ into 0, and the 0 looks like a symmetry argument succeeding.

SEE ALSO Splitting at a symmetry or sign-change point; even and odd parity on a symmetric interval; branch and domain traps; floor, fractional part, and step integrands.

Absolute values split the interval into two or three pieces. Floors and fractional parts split it into infinitely many, and the resulting infinite family is the point: the sum over pieces is a series, and the series is designed to telescope or to be a recognisable zeta value. The shells are usually easier to describe than the integrand.

MEDIUM

28.5 Floor, fractional part, and step integrands

CUE $\lfloor x \rfloor$, $\{x\}$, $\lfloor 1/x \rfloor$, $\lfloor \sqrt{f(x)} \rfloor$, $\lfloor x^{1/3} \rfloor$, or any piecewise-constant weight multiplying something smooth.

METHOD Partition the domain into the level sets of the step function and integrate the now-smooth integrand on each. The three shell families worth memorising:

$$\lfloor x \rfloor = n \text{ on } [n, n+1), \quad \left\lfloor \frac{1}{x} \right\rfloor = n \text{ on } \left(\frac{1}{n+1}, \frac{1}{n} \right], \text{ length } \frac{1}{n(n+1)}, \quad \lfloor x^{1/3} \rfloor = n \text{ on } [n^3, (n+1)^3), \text{ length } 3n^2 + 3n + 1.$$

Then sum the resulting series. The reciprocal family is the productive one, because $\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$ telescopes, and because it converts an integral on $(0, 1]$ into a series indexed by shells that shrink quadratically.

WORKED EXAMPLE

$$\int_0^1 \left\{ \frac{1}{x} \right\} dx = 1 - \gamma \approx 0.422784.$$

Substitute $u = \frac{1}{x}$: the integral becomes $\int_1^\infty \frac{\{u\}}{u^2} du$. On $[k, k+1)$ we have $\{u\} = u - k$, so

$$\int_1^N \frac{\{u\}}{u^2} du = \sum_{k=1}^{N-1} \int_k^{k+1} \left(\frac{1}{u} - \frac{k}{u^2} \right) du = \ln N - \sum_{k=1}^{N-1} \frac{1}{k+1} \rightarrow 1 - \gamma,$$

using $\int_k^{k+1} \frac{k}{u^2} du = \frac{1}{k+1}$ and the definition of γ as $\lim(\sum_{k=1}^N \frac{1}{k} - \ln N)$. Direct shell-summing of the partial integrals confirms 0.4227818.

A CUBE-ROOT SHELL, FINITE AND EXACT

$$\int_0^{64} \lfloor x^{1/3} \rfloor dx = \sum_{n=0}^3 n((n+1)^3 - n^3) = \sum_{n=0}^3 n(3n^2 + 3n + 1) = 0 + 7 + 38 + 111 = 156.$$

The upper limit $64 = 4^3$ was chosen so that the last shell is complete; that is endpoint engineering in the service of a step function. Quadrature with break points at 1, 8, 27 returns exactly 156.

WHERE THIS GOES WRONG

Off-by-one at the shell boundaries. $\lfloor 1/x \rfloor = n$ on the *half-open* interval $(\frac{1}{n+1}, \frac{1}{n}]$, and getting the direction wrong shifts the whole series by one index. The second error is an incomplete last shell: if the upper limit is not an integer (for $\lfloor x \rfloor$) or a perfect cube (for $\lfloor x^{1/3} \rfloor$), the final shell is truncated and contributes a fragment that must be computed separately. Third, check convergence before summing: $\int_0^1 \lfloor 1/x \rfloor dx = \sum_{n \geq 1} \frac{n}{n(n+1)} = \sum_{n \geq 1} \frac{1}{n+1}$ diverges, even though every shell is finite and the picture looks harmless.

SEE ALSO Telescoping integrands and shell decomposition; absolute value, sign, and the log's modulus; periodicity and window shifting; splitting at a symmetry or sign-change point.

Shell decomposition produced a series that telescoped. That is not a coincidence of the floor function: it is the general design principle behind a whole family, in which either the domain decomposes and the resulting series telescopes, or the integrand itself is an infinite product or sum that collapses before any integration begins.

28.6 Telescoping integrands and shell decomposition

CUE An infinite product such as $\prod_{n \geq 0} (1 + x^{2^n})$, an infinite sum of terms with $\frac{1}{n(n+1)}$ -type weights, a difference $g(x+1) - g(x)$ inside the integrand, or a floor function partitioning the domain into shells on which the integrand repeats up to sign.

METHOD Two routes, and it is worth deciding which one you are on. If the *integrand* telescopes, collapse it first: the binary identity

$$\prod_{n=0}^{\infty} (1 + x^{2^n}) = \frac{1}{1-x} \quad (|x| < 1)$$

follows from $(1-x) \prod_{n=0}^N (1+x^{2^n}) = 1-x^{2^{N+1}}$. If the *domain* decomposes, integrate on each shell and telescope the resulting series using $\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$ or a partial-fraction split of the shell weight.

WORKED EXAMPLE

$$\int_0^{1/2} (1+x)(1+x^2)(1+x^4)(1+x^8) \cdots dx = \int_0^{1/2} \frac{dx}{1-x} = [-\ln(1-x)]_0^{1/2} = \ln 2 \approx 0.693147.$$

The product is a geometric series in disguise: multiplying by $(1-x)$ collapses it pairwise to $1-x^{2^{N+1}}$, which tends to 1 on $[0, \frac{1}{2}]$. A partial product with sixty factors integrates numerically to 0.6931471806.

A SHELL DECOMPOSITION THAT TELESOPES

$$\int_1^{\infty} \frac{\pi \sin(\pi x)}{[x]([x]+1)} dx.$$

On $[n, n+1)$ the weight is the constant $\frac{1}{n(n+1)}$ and $\int_n^{n+1} \pi \sin(\pi x) dx = [-\cos(\pi x)]_n^{n+1} = 2(-1)^n$. So

$$I = \sum_{n \geq 1} \frac{2(-1)^n}{n(n+1)} = 2 \sum_{n \geq 1} (-1)^n \left(\frac{1}{n} - \frac{1}{n+1} \right) = 2[-\ln 2 - (\ln 2 - 1)] = 2 - 4 \ln 2 \approx -0.772589.$$

The two alternating sums are $\sum (-1)^n/n = -\ln 2$ and $\sum_{n \geq 1} (-1)^n/(n+1) = \ln 2 - 1$ after reindexing. Numerical shell-summing to $n = 400$ gives -0.7725950 , converging slowly as an alternating series must.

WHERE THIS GOES WRONG

Reindexing. Starting the binary product at $n = 1$ instead of $n = 0$ drops the factor $(1+x)$ and turns $\frac{1}{1-x}$ into $\frac{1}{1-x^2}$, a different answer entirely; and in the shell sum, starting at $n = 0$ divides by zero. Write the first two shells out explicitly before summing. The second trap is convergence: a telescoping series whose terms do not tend to zero telescopes to nothing useful, and an infinite product converges only where it converges, so $\prod (1+x^{2^n}) = \frac{1}{1-x}$ is false at $x = 1$ and meaningless beyond.

SEE ALSO Floor, fractional part, and step integrands; partial fractions with distinct linear factors; series truncation with a tail bound; recognising the emergent constant.

Where the naive antiderivative is wrong

Everything so far has been an opportunity. This section is a hazard. There is a species of problem in which an antiderivative exists, is correct as an antiderivative, and nevertheless gives the wrong number when evaluated at the two endpoints, because it is discontinuous somewhere inside the interval. The fundamental theorem requires F to be continuous on $[a, b]$ and to satisfy $F' = f$ throughout; an F that jumps has violated the first hypothesis, and no amount of correctness in the differentiation saves it.

Three constructions produce such an F almost automatically, and between them they account for nearly every instance. The first is an arctan whose argument passes through $\pm\infty$: $\arctan(k \tan x)$ jumps by π at every odd multiple of $\frac{\pi}{2}$, and integrands of the form $\frac{1}{a + b \sin^2 x}$ produce exactly that antiderivative. The second is $\ln |g|$ where g has an interior zero: the antiderivative goes to $-\infty$ on one side and comes back from $-\infty$ on the other, and in fact the integral is improper there rather than merely discontinuous. The third is a substitution that is not injective, of which $t = \tan \frac{x}{2}$ across $x = \pi$ is the standard example. In all three cases the failure is silent, and in all three the same diagnostic catches it: evaluate the candidate F at a dozen points across $[a, b]$ and look for a jump.

28.7 Branch and domain traps

CUE A real cube root of a negative number; $\operatorname{arcsec}; \sqrt{x^2}$; an arctan whose argument runs to $\pm\infty$ inside the interval; arcosh evaluated below 1; a Weierstrass substitution $t = \tan \frac{x}{2}$ on an interval containing π ; or any exponent manipulation that changes the domain.

METHOD Establish the domain and the branch of every factor before manipulating, and check the candidate antiderivative for continuity on the closed interval. If F jumps at an interior point c , integrate over $[a, c]$ and $[c, b]$ separately and add the one-sided limits:

$$\int_a^b f = [F]_a^{c^-} + [F]_{c^+}^b,$$

which differs from $F(b) - F(a)$ by exactly the jump. For real odd roots, note that $x^{-1/3}$ preserves sign, so $(x^{-1/3})^{-3} = x$ is genuinely odd on $[-a, a]$; the principal complex cube root does not preserve sign and would give a different answer.

WORKED EXAMPLE

$$\int_0^\pi \frac{dx}{1 + 3 \sin^2 x}.$$

Dividing through by $\cos^2 x$ and substituting $t = \tan x$ gives the antiderivative $F(x) = \frac{1}{2} \arctan(2 \tan x)$, and F' is the integrand wherever $\tan x$ is defined. But $F(\pi) - F(0) = 0$, which is absurd for a strictly positive integrand. The cause is the jump at $x = \frac{\pi}{2}$, where $\tan x$ blows up and F drops from $\frac{\pi}{4}$ to $-\frac{\pi}{4}$. Splitting there:

$$I = \left[\frac{1}{2} \arctan(2 \tan x) \right]_0^{\pi/2^-} + \left[\frac{1}{2} \arctan(2 \tan x) \right]_{\pi/2^+}^\pi = \frac{\pi}{4} + \frac{\pi}{4} = \frac{\pi}{2} \approx 1.570796,$$

which quadrature confirms. The correct value exceeds the naive one by exactly the jump $\frac{\pi}{2}$.

TWO BRANCH CHECKS THAT GO THE OTHER WAY

$$\int_{-1}^1 (x^{-1/3})^{-3} dx.$$

Read as real functions, $x^{-1/3}$ is the real cube root of $\frac{1}{x}$ and preserves sign, so $(x^{-1/3})^{-3} = x$ on $[-1, 1] \setminus \{0\}$, the integrand is odd, and the integral is 0. A computer algebra system using the principal complex branch will disagree, because for $x < 0$ it returns a root with argument $\frac{\pi}{3}$ rather than π .

Second: $\frac{d}{dx} \operatorname{arcsec} x = \frac{1}{|x| \sqrt{x^2 - 1}}$, not $\frac{1}{x \sqrt{x^2 - 1}}$, because arcsec has range $[0, \pi] \setminus \{\frac{\pi}{2}\}$ and is increasing on both

branches. Numerically at $x = -2$ the derivative is $+0.2886751$, which is $\frac{1}{2\sqrt{3}}$ and positive, confirming the modulus. Dropping it flips the sign of every integral over a negative interval.

THE SUBSTITUTION ITSELF IS DISCONTINUOUS

$$\int_0^{2\pi} \frac{dx}{2 + \cos x}.$$

The Weierstrass substitution $t = \tan \frac{x}{2}$ turns this into $\int \frac{2 dt}{3 + t^2}$, and applying it with bounds $t = 0$ to $t = \tan \pi = 0$ returns 0, again absurd for a positive integrand. The fault is not the antiderivative but the substitution: $t = \tan \frac{x}{2}$ has a pole at $x = \pi$, which is strictly inside $[0, 2\pi]$, so it is not a bijection onto $\mathbb{R}$ there. Slide the window using periodicity to $[-\pi, \pi]$, where the pole sits at both endpoints and the map is a genuine bijection:

$$\int_{-\pi}^{\pi} \frac{dx}{2 + \cos x} = \int_{-\infty}^{\infty} \frac{2 dt}{3 + t^2} = \frac{2\pi}{\sqrt{3}} \approx 3.627599.$$

The lesson generalises: a substitution must be monotone and finite on the closed interval, and checking that costs one glance at where it blows up.

WHERE THIS GOES WRONG

Assuming the algebra system has thought about it. Symbolic systems default to principal complex branches for fractional powers and to formal antiderivatives that may jump, and they will disagree with the real-analysis answer in exactly the cases this technique is about. The practical protocol: plot or sample the candidate antiderivative at ten points across the interval, and if it jumps, split there. A cheaper version of the same check is the sign test used above, since a positive integrand cannot have a nonpositive integral.

SEE ALSO Absolute value, sign, and the log's modulus; inverse and co-function collapse; Weierstrass substitution; endpoint limit discipline.

A limit that only looks like an integral

The last species is different in kind. Here the object is a limit of a family of integrals, the family concentrates on a set of measure zero, and the answer is read off the geometry of the concentration rather than obtained by integrating anything. Recognising the family is the whole technique, and the families are few.

It belongs in this chapter for the same reason as everything else here: the integral for any fixed n is genuinely hard, and the limit is trivial, so a reader who evaluates first and takes the limit second will not finish. The structural cue is a parameter appearing in a specific pattern, namely a prefactor and a matching power inside a kernel that is a function of n times something vanishing. Written abstractly, $K_n(u) = nK(nu)$ with $\int K = 1$, which is the definition of an approximate identity. Once you see that shape the rest is arithmetic on the zeros of the inner function.

28.8 Nascent delta and limiting kernels

HARD

CUE A parameter $n \rightarrow \infty$ multiplying a kernel that concentrates: $\frac{n}{1+n^2u^2}$, $\sqrt{\frac{n}{\pi}}e^{-nu^2}$, $\frac{\sin(nu)}{\pi u}$, $\frac{n}{2}e^{-n|u|}$, composed with an inner function that has isolated zeros.

METHOD Identify the kernel and its total mass. As $n \rightarrow \infty$ the integral localises at the zeros of the inner function, and near a simple zero x_0 of g the substitution $u = g(x)$ is locally linear with $du \approx |g'(x_0)| dx$, so the contribution is $\frac{\text{mass}}{|g'(x_0)|}$. Sum over the real zeros lying inside the interval. For the Lorentzian $\frac{n}{1+n^2u^2}$ the mass is π ; for the Gaussian and the Fejer kernels it is 1.

WORKED EXAMPLE

$$\lim_{n \rightarrow \infty} \int_{-\infty}^{\infty} \frac{n}{1+n^2(x^4-1)^2} dx.$$

The inner function $g(x) = x^4 - 1$ has real zeros $x = \pm 1$, both simple, with $g'(x) = 4x^3$ so $|g'(\pm 1)| = 4$. The Lorentzian has mass π , so the limit is

$$\frac{\pi}{4} + \frac{\pi}{4} = \frac{\pi}{2} \approx 1.570796.$$

At $n = 10^3$ careful quadrature (with break points placed at the spikes) gives 1.571973; at $n = 10^5$, 1.570808. The convergence is $O(n^{-1/2})$ in this instance, which is why a lazy quadrature at large n misses the spikes entirely and returns nearly zero.

A GAUSSIAN KERNEL, AND THE SAME ARITHMETIC

$$\lim_{n \rightarrow \infty} \int_{-2}^2 \sqrt{\frac{n}{\pi}} e^{-n(x^2-1)^2} dx.$$

The Gaussian family $\sqrt{n/\pi} e^{-nu^2}$ has total mass 1, not π . The inner function again has simple zeros at ± 1 with $|g'| = 2$, so the limit is $\frac{1}{2} + \frac{1}{2} = 1$. Quadrature with break points at the spikes gives 1.0018959 at $n = 100$ and 1.0000188 at $n = 10^4$, converging like n^{-1} . Notice that the only thing that changed from the Lorentzian case is the mass; the localisation arithmetic is identical, which is why it is worth memorising the masses (π for $\frac{n}{1+n^2u^2}$, 1 for the Gaussian, the Fejer and the Poisson kernels) rather than the formulas.

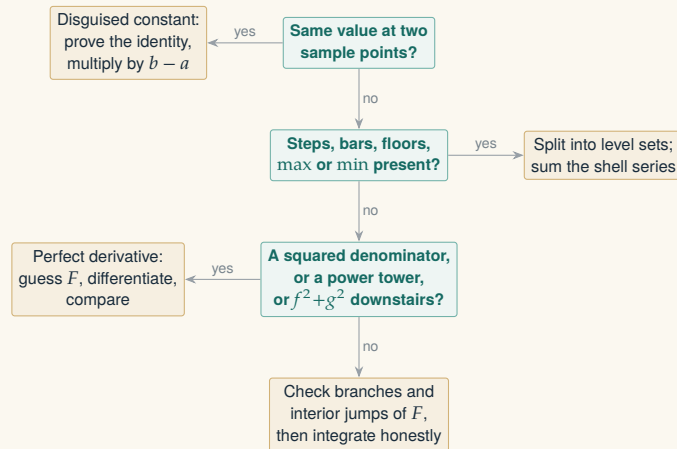
WHERE THIS GOES WRONG

Double roots. If $g'(x_0) = 0$ the local model is quadratic rather than linear, the scaling changes, and the contribution is not $\frac{\pi}{|g'|}$; it can be infinite. Check that every zero in range is simple before applying the formula. Second, only zeros *inside* the interval count, and a zero exactly at an endpoint contributes half. Third, complex zeros of g contribute nothing: $x^4 - 1$ has four roots but only two of them are real, and using all four doubles the answer.

SEE ALSO Laplace's method and Watson's lemma; bounding by envelopes; nascent delta as a distributional limit; named-function answers.

Running the sweep

The six tests of the sweep map onto the eight techniques of this chapter, and the mapping is worth having as a picture rather than a list, because in practice you run the tests in parallel: a single reading of the integrand should light up at most one of them.



The final leaf is not a technique but a warning, and it belongs at the bottom because it applies to every integral you ever compute, not only to the ones in this chapter. Before evaluating an antiderivative at two endpoints, confirm that it is continuous between them. That single check catches the $\tan \frac{x}{2}$ pole, the $\arctan(2 \tan x)$ jump, the interior zero under a $\ln |\cdot|$, and the branch cut of a fractional power, which between them account for most of the confidently wrong definite integrals in circulation.

WHY THESE ARE WORTH PRACTISING

An integral that collapses is not a curiosity. It is a test of whether you read structure before reaching for method, and reading structure first is exactly the habit that pays on integrals that do not collapse. The reader who tests parity, samples the integrand, and checks the domain out of reflex will lose ten seconds on an honest problem and gain an hour on a rigged one. The reader who begins by choosing a technique will do the reverse. Every problem in this chapter was built by someone who knew which of those two readers they were writing for.

Recognition matrix

If you see	Reach for	If that fails
$x^x, x^{1/x}, u^v$ times a sum	Logarithmic differentiation: is it F' ?	Check both endpoints for improperness
A stray $(1 \pm \ln x)$ in a product	Guess $F = x^x$ or $x^{1/x}$ and differentiate	Parts, with $\ln x$ as the part
low degree	Reverse quotient rule: try $\frac{u}{v}$	Partial fractions after all
$(\text{something})^2$	$\arctan \frac{g}{f}$; the Wronskian shape	Divide through by f^2 , then $t = g/f$
$\frac{fg' - g'f}{f^2 + g^2}$	Reverse product rule: answer $[fg]_a^b$	Parts once and compare
$f'g + fg'$ split across two terms	Change of base; expect a constant	Sample the integrand twice
Mixed logarithm bases	Fixed-point equation $y = \sqrt{x+y}$ or $y = \sqrt{xy}$	Sample, then solve for y
Nested or iterated radical	Evaluate the entries; expect 0	Row-reduce symbolically
A determinant of antiderivatives	Constant $\frac{\pi}{2}$ on $(0, \infty)$	Differentiate to confirm
$\arctan x + \arctan \frac{1}{x}$	Integral is 0; check integrability	Recentre first if not symmetric
Odd factor, even weight, $[-a, a]$	Bars flip parity: $2 \int_0^a$	Split at 0 and add
$ x ^{\text{odd}}$ on $[-a, a]$	It is $ f $; split at the zeros of f	Never write $\sqrt{f^2} = f$
$\sqrt{f(x)^2}$	Factor out the even power; check sign	Domain first, substitution second
$\sqrt{x^{13} + x^8}$ -type radicand	Shells $(\frac{1}{n+1}, \frac{1}{n}]$, length $\frac{1}{n(n+1)}$	Telescope the series
$[1/x]$ on $(0, 1]$	$\{u\} = u - [u]$; expect γ	Sum shells to N , then take $N \rightarrow \infty$
$\{x\}$ or $\{1/x\}$	Shells of length $3n^2 + 3n + 1, 2n + 1$	Watch for an incomplete last shell
$[x^{1/3}], [\sqrt{x}]$	Telescopes to $\frac{1}{1-x}$ on $ x < 1$	Multiply by $(1-x)$ to see it
$\prod_{n \geq 0} (1 + x^{2^n})$		

If you see	Reach for	If that fails
Shell weight $\frac{1}{n(n+1)}$	Partial fractions, then telescope	Alternating: expect $\ln 2$
$\arctan(2 \tan x)$ or $\tan \frac{x}{2}$ inside	Check for a jump at $\frac{\pi}{2}$ or π	Split and add one-sided limits
arcsec , arcosh , odd roots	Range and sign conventions; keep $ x $	Sample the derivative numerically
Positive integrand, nonpositive answer	An antiderivative jumped, or a pole is inside	Sign test before reporting
$\frac{n}{1 + n^2 g(x)^2}$ with $n \rightarrow \infty$	Mass π over $ g' $ at each real simple zero	Double root: rescale locally

Bounding, estimation, and numerical technique

What this chapter is for. Sometimes the right answer to “evaluate” is “bound”. A great many integrals have no closed form, a great many questions about integrals do not ask for one (compare these two numbers; show that this is positive; how does I_n behave for large n), and a great many closed forms that do exist are worth less than a tight inequality with a proof attached. This chapter collects the machinery for those situations, in increasing order of how much apparatus it requires: pointwise envelopes and the mean value theorem for integrals, then the two inequalities that handle products and convexity (Cauchy-Schwarz and Jensen), then the correspondence between sums and integrals in both directions, then quadrature proper (trapezoid and Simpson with their error terms, Gaussian nodes, Richardson extrapolation), then series truncation with an honest tail bound, then asymptotics by Laplace’s method. It ends with the technique that should be applied to everything else in this book: numerical cross-checking as a discipline. Running through the chapter is a judgement you must make each time, and the chapter states it plainly at both ends. A numeric answer is the correct deliverable when the question is quantitative and no structure exists; it is a failure when structure exists and you did not see it.

When a bound is the answer

Read the question before choosing the tool. “Evaluate” asks for a number. “Show that $\int_0^1 f > \int_0^1 g$ ” asks for an inequality, and producing both integrals in closed form to compare them is usually more work and always more fragile than comparing the integrands pointwise. “Estimate to three decimals” asks for a number *and* an error bound, and a number without an error bound is not an answer to that question. “How does I_n behave as $n \rightarrow \infty$ ” asks for an asymptotic, which is a statement about a family and cannot be obtained by evaluating any single member.

The organising fact is that integration is monotone and linear, and that is nearly all you need. Monotonicity says that pointwise domination survives integration; linearity says that bounds add. Between them they convert almost any inequality between functions into an inequality between numbers, at no cost, and without any antiderivative appearing anywhere.

There is a second reason to reach for a bound even when a closed form is available, and it is worth stating because it runs against instinct. A bound carries information that a value does not: it tells you which features of the integrand the answer depends on. Knowing that $\int_0^1 e^{-x^2} dx$ lies between $\frac{2}{3}$ and $\frac{\pi}{4}$ tells you that the integral is insensitive to the fine shape of the Gaussian and is essentially determined by its values at the two ends of the interval; knowing that it equals 0.746824 tells you nothing of the kind. In applied settings, where the integrand is a model rather than a fact, that sensitivity information is frequently the deliverable and the number is not.

CORE

29.1 Bounding by envelopes

CUE The question asks you to *show* $a < I < b$, to compare two integrals, to decide a sign, or to prove convergence with a quantitative rate. Also: an integrand that is intractable but visibly trapped between two tractable ones.

METHOD Bound the integrand pointwise by something you can integrate. The basic chain, from crudest to sharpest:

$$m \leq f \leq M \text{ on } [a, b] \implies m(b-a) \leq I \leq M(b-a); \quad f \leq g \implies \int_a^b f \leq \int_a^b g.$$

The mean value theorem for integrals sharpens the first to an equality: for continuous f there is $\xi \in (a, b)$ with $I = f(\xi)(b-a)$, and in weighted form, for $w \geq 0$, $\int_a^b fw = f(\xi) \int_a^b w$. For products use Cauchy-Schwarz, $(\int fg)^2 \leq \int f^2 \int g^2$; for a convex φ use Jensen, $\varphi(\int_a^b f) \leq \int_a^b \varphi(f)$.

WORKED EXAMPLE

On $[0, 1]$, $x^2 \leq x$, hence $1 + x^2 \leq 1 + x$, hence

$$\frac{1}{1+x^2} \geq \frac{1}{1+x} \implies \frac{\pi}{4} \geq \ln 2.$$

A one-line proof of $0.785398 \geq 0.693147$ with no evaluation of either side beyond naming its antiderivative. The same move in the other direction: $1 - x^2 \leq e^{-x^2} \leq \frac{1}{1+x^2}$ on $[0, 1]$ gives

$$\frac{2}{3} \leq \int_0^1 e^{-x^2} dx \leq \frac{\pi}{4}, \quad 0.666667 \leq 0.746824 \leq 0.785398,$$

bracketing a non-elementary integral to within 0.12 using only two elementary ones.

CAUCHY-SCHWARZ AND JENSEN, ONE LINE EACH

Cauchy-Schwarz with $g = 1$: $(\int_0^1 f)^2 \leq \int_0^1 f^2$. Applied to $f = \sqrt{1+x^4}$,

$$\int_0^1 \sqrt{1+x^4} dx \leq \sqrt{\int_0^1 (1+x^4) dx} = \sqrt{\frac{6}{5}} \approx 1.095445,$$

against the true 1.089429: a 0.6% overestimate for one line of work.

Jensen with the convex $\varphi(t) = e^t$ on the probability measure dx on $[0, 1]$:

$$\int_0^1 e^{x^2} dx \geq \exp\left(\int_0^1 x^2 dx\right) = e^{1/3} \approx 1.395612,$$

against the true 1.462652. Both inequalities go the way convexity dictates, and getting the direction right is the only thing to remember: φ convex pushes the average of φ above φ of the average.

THE MEAN VALUE THEOREM DOING REAL WORK

The theorem is usually presented as an existence statement and dismissed. Its weighted form is a computational tool. For continuous f and $w \geq 0$ there is $\xi \in (a, b)$ with $\int_a^b f w = f(\xi) \int_a^b w$; taking $w = x^n$ on $[0, 1]$ gives

$$\int_0^1 x^n f(x) dx = \frac{f(\xi_n)}{n+1}, \quad \xi_n \in (0, 1).$$

Because x^n concentrates all of its mass near 1 as n grows, $\xi_n \rightarrow 1$, and therefore

$$\lim_{n \rightarrow \infty} (n+1) \int_0^1 x^n f(x) dx = f(1).$$

With $f = e^x$ the left side at $n = 1000$ is 2.715572 against $e = 2.718282$, converging like n^{-1} . An asymptotic obtained from an existence theorem, with no integration performed. The unweighted version is the same statement with $w = 1$: on $[0, 1]$, $\int_0^1 \frac{dx}{1+\xi^2} = \frac{\pi}{4}$ forces $\frac{1}{1+\xi^2} = \frac{\pi}{4}$, so $\xi = \sqrt{\frac{4}{\pi} - 1} \approx 0.522723$, which is a fact about where the average is attained rather than about the value.

WHERE THIS GOES WRONG

Bounding the integrand where it is not bounded. Near a singularity $M = \infty$ and the rectangle bound says nothing; split the interval, handle the singular piece by comparison with a power, and use the crude bound only on the rest. The second failure is a bound that is tight at one end and useless at the other: on $[0, 10]$ the estimate $e^{-x^2} \leq 1$ is exact at 0 and catastrophic beyond 2, giving $I \leq 10$ against a true value below 0.89. Split at the point where the behaviour changes and bound each piece with its own envelope.

SEE ALSO Comparison and limit comparison; series truncation with a tail bound; Laplace's method and Watson's lemma; numerical cross-check as verification discipline.

Envelopes bound an integral by other integrals. The next correspondence bounds an integral by a *sum*, and it runs in both directions: a sum you cannot evaluate becomes an integral you can, and an integral you cannot evaluate becomes a sum you can truncate. This is the oldest estimation technique there is and still the most used.

Sums and integrals

29.2 Riemann sum recognition, in both directions

CUE A limit of a sum with an n in every denominator, $\frac{1}{n} \sum_{k=1}^n f(\frac{k}{n})$ or anything reducible to it; or conversely a definite integral you want to convert into a sum in order to estimate or to recognise it.

METHOD The correspondence is

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n f\left(\frac{k}{n}\right) = \int_0^1 f(x) dx$$

for f Riemann integrable on $[0, 1]$, with the obvious variants for $[a, b]$ and for left, right, or midpoint tags. Reading it left to right evaluates limits of sums. Reading it right to left converts an integral into a sum whose partial sums are computable, which is what every quadrature rule refines. The essential manoeuvre is to factor out the width: the $\frac{1}{n}$ must appear exactly once, as Δx , and everything else must be a function of $\frac{k}{n}$ alone.

WORKED EXAMPLE

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{n+k} = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \frac{1}{1 + \frac{k}{n}} = \int_0^1 \frac{dx}{1+x} = \ln 2 \approx 0.693147.$$

The step that does the work is dividing numerator and denominator by n so that the summand becomes a function of $\frac{k}{n}$ times $\frac{1}{n}$.

A second instance where the factoring is less obvious:

$$\lim_{n \rightarrow \infty} \frac{(n!)^{1/n}}{n} = \exp\left(\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \ln \frac{k}{n}\right) = \exp\left(\int_0^1 \ln x dx\right) = e^{-1},$$

the integral being improper at 0 but convergent, with value -1 .

WHERE THIS GOES WRONG

Mis-identifying Δx . If the $\frac{1}{n}$ is counted twice (once as the width and once inside f) the limit is off by a factor of n and typically comes out as 0 or ∞ . Write the summand explicitly in the form $\frac{1}{n} g(\frac{k}{n})$ before taking any limit. Second, the correspondence needs f integrable on the *closed* interval; when f has an endpoint singularity, as $\ln x$ does at 0, the limit still holds but requires a separate argument bounding the contribution of the first few terms, and for a non-integrable singularity it fails outright.

SEE ALSO Bounding by envelopes; Newton-Cotes rules with error terms; the Euler-Maclaurin correction; self-similar collapse.

The correspondence also runs as an inequality rather than a limit, and in that form it is the integral test. For f positive and decreasing on $[1, \infty)$,

$$\int_1^{N+1} f(x) dx \leq \sum_{k=1}^N f(k) \leq f(1) + \int_1^N f(x) dx,$$

because each term $f(k)$ is trapped between the areas of the two unit rectangles flanking it. That single picture proves the convergence of $\sum n^{-p}$ for $p > 1$, bounds the harmonic sum between $\ln(N+1)$ and $1 + \ln N$, and supplies the tail bound used later in this chapter for truncated series. It is the same monotonicity argument as the envelope technique, applied to a step function rather than a smooth one.

The Riemann sum is a crude approximation to the integral, with error $O(n^{-1})$ for the left-endpoint tag. The Euler-Maclaurin formula says exactly what the error is, and in doing so converts the correspondence from a limit statement into a computational tool that works in both directions with high accuracy.

EULER-MACLAURIN

For f smooth on $[a, b]$ with a, b integers,

$$\sum_{k=a}^b f(k) = \int_a^b f(x) dx + \frac{f(a) + f(b)}{2} + \sum_{j \geq 1} \frac{B_{2j}}{(2j)!} [f^{(2j-1)}]_a^b + R,$$

with $B_2 = \frac{1}{6}$, $B_4 = -\frac{1}{30}$, $B_6 = \frac{1}{42}$. Read left to right it evaluates sums: with $f(x) = \frac{1}{x}$ it gives $\sum_{k=1}^n \frac{1}{k} = \ln n + \gamma + \frac{1}{2n} - \frac{1}{12n^2} + \dots$, which at $n = 10$ reads 2.9289674 against the true 2.9289683, five correct digits from three terms. Read right to left it evaluates integrals: the trapezoid rule is precisely the first two terms rearranged, which is why its error is $O(h^2)$ with only *even* powers of h in the expansion. That last fact is what makes Richardson extrapolation work.

Quadrature

When no closed form exists, or when the integrand is available only as data, the deliverable is a number with an error bar. The classical rules interpolate the integrand by a polynomial on each panel and integrate the interpolant; the error term is then the integration of the interpolation error, which is why every error constant carries a high derivative of f evaluated at an unknown interior point.

29.3 Newton-Cotes rules with error terms

MEDIUM

CUE No closed form is wanted or available; the integrand is given as data; or you need three or four digits fast and honestly.

METHOD With $h = \frac{b-a}{n}$ and $f_k = f(a + kh)$:

$$\int_a^b f \approx \frac{h}{2} [f_0 + 2f_1 + \cdots + 2f_{n-1} + f_n], \quad E_T = -\frac{(b-a)h^2}{12} f''(\xi),$$

$$\int_a^b f \approx \frac{h}{3} [f_0 + 4f_1 + 2f_2 + 4f_3 + \cdots + 4f_{n-1} + f_n], \quad E_S = -\frac{(b-a)h^4}{180} f^{(4)}(\xi),$$

the second requiring n even. Simpson is exact for cubics despite being built from parabolas, which is the source of its unreasonable accuracy. Bound the error by maximising the stated derivative over $[a, b]$; that bound is rigorous, unlike the difference between two successive refinements.

WORKED EXAMPLE

$\int_0^1 e^{-x^2} dx$ with $n = 4$, $h = \frac{1}{4}$. The five ordinates are $1, e^{-1/16}, e^{-1/4}, e^{-9/16}, e^{-1}$. Trapezoid gives 0.7429841; Simpson gives

$$\frac{1/4}{3} [f_0 + 4f_1 + 2f_2 + 4f_3 + f_4] = 0.7468554,$$

against the true $\frac{\sqrt{\pi}}{2} \operatorname{erf}(1) = 0.7468241$. Simpson is wrong by 3.12×10^{-5} from five evaluations; the trapezoid, from the same five, is wrong by 3.84×10^{-3} , a factor of 123 worse. The rigorous Simpson bound: $f^{(4)} = (16x^4 - 48x^2 + 12)e^{-x^2}$ has $|f^{(4)}| \leq 12$ on $[0, 1]$, so $|E_S| \leq \frac{1 \cdot (1/4)^4}{180} \cdot 12 = 2.60 \times 10^{-4}$, comfortably containing the actual error.

RICHARDSON, AND WHERE THE EXTRA DIGITS COME FROM

The trapezoid error expansion contains only even powers of h , so $T(h) = I + c_2 h^2 + c_4 h^4 + \cdots$ and the combination

$$\frac{4T(h/2) - T(h)}{3} = I + O(h^4)$$

kills the leading term. On the same integral, $T(0.5) = 0.7313703$ and $T(0.25) = 0.7429841$ give $\frac{4(0.7429841) - 0.7313703}{3} = 0.7468554$, which is *exactly* Simpson with $h = 0.25$. That identity is not a coincidence: Simpson is the first Richardson extrapolate of the trapezoid rule, and repeating the process is Romberg integration. One more level, from $T(0.25)$ and $T(0.125) = 0.7458656$, gives 0.7468261, now wrong by 2×10^{-6} .

WHERE THIS GOES WRONG

Simpson requires an *even* number of subintervals; with n odd the alternating 4, 2, 4, 2 pattern does not close and the rule silently computes something else. More seriously, every error bound above is worthless when the stated derivative is unbounded on $[a, b]$. For $\int_0^1 \frac{dx}{\sqrt{x}}$, f'' is infinite at 0 and Simpson converges at a rate governed by the singularity, not by h^4 ; the fix is to remove the singularity by substitution ($x = u^2$ here, or a tanh-sinh transformation in general) *before* applying any Newton-Cotes rule. Finally, do not use the difference of two refinements as an error estimate when the integrand is oscillatory: successive panels can agree closely and both be wrong.

SEE ALSO Riemann sum recognition, in both directions; the Euler-Maclaurin correction; numerical cross-check as verification discipline; bounding by envelopes.

Newton-Cotes fixes the nodes at equal spacing and chooses the weights. Gaussian quadrature chooses both, and the extra freedom doubles the achievable degree of exactness. With n nodes placed at the roots of the Legendre polynomial P_n (rescaled from $[-1, 1]$ to $[a, b]$) and weights chosen accordingly, the rule is exact for every polynomial of degree $2n - 1$, against $n - 1$ or $n + 1$ for the equal-spaced rules. On the running example, two-node Gauss-Legendre gives 0.7465947 from *two* evaluations, three-node gives 0.7468146, and five-node gives 0.7468241, which is the correct value to nine decimals. That is the practical reason every serious quadrature library is Gaussian at its core: for a smooth integrand, nodes are expensive and accuracy per node is what matters.

The Gaussian idea generalises by changing the weight. Choosing nodes at the roots of the Chebyshev polynomials with weight $(1 - x^2)^{-1/2}$ gives a rule tailored to integrands with inverse-square-root endpoint singularities; Gauss-Laguerre with weight e^{-x} handles $[0, \infty)$; Gauss-Hermite with e^{-x^2} handles $\mathbb{R}$. In each case the singular or slowly decaying factor is absorbed into the weight and the rule is exact for polynomials against it, which is why a well-chosen Gaussian rule can be accurate on an integrand where every equal-spaced rule fails outright. The price is that the nodes are irrational and must be looked up or computed, and that refining the rule means recomputing every node rather than reusing the old ones. Gauss-Kronrod extensions exist precisely to recover the reuse, and they are what a general-purpose adaptive routine actually runs.

Truncating a series honestly

The third route to a number is through a series. Termwise integration frequently produces a series with no closed form, or one whose closed form is itself a special value you would have to look up. Reporting a truncation is legitimate; reporting it without a tail bound is not.

MEDIUM

29.4 Series truncation with a tail bound

CUE A termwise expansion whose sum has no closed form; an alternating series that must be reported numerically; any answer of the form $\sum_{n \geq 1} a_n$ where the question wants digits.

METHOD Truncate at N and bound the tail. For an alternating series with terms decreasing in absolute value to zero, the tail is bounded by the first omitted term, which is the cheapest useful bound in mathematics. Otherwise compare the tail with a geometric series or with an integral,

$$\sum_{n > N} a_n \leq \int_N^{\infty} a(t) dt$$

for a positive and decreasing. Then report exactly as many digits as the bound justifies, and not one more.

WORKED EXAMPLE

The sophomore's dream: $\int_0^1 x^x dx = \sum_{n \geq 1} \frac{(-1)^{n-1}}{n^n}$, obtained by expanding $x^x = e^{x \ln x}$ and integrating termwise. The series alternates with rapidly decreasing terms. Six terms give

$$1 - \frac{1}{4} + \frac{1}{27} - \frac{1}{256} + \frac{1}{3125} - \frac{1}{46656} = 0.7834294,$$

and the first omitted term is $7^{-7} = 1.21 \times 10^{-6}$, so the value lies in $[0.7834294, 0.7834306]$. The true value is 0.7834305, and the actual error 1.16×10^{-6} sits just inside the bound. Report 0.78343 and stop.

AN ALTERNATING SERIES WHERE THE BOUND IS THE WHOLE ANSWER

$\int_0^1 \frac{\sin x}{x} dx = \text{Si}(1) = \sum_{k \geq 0} \frac{(-1)^k}{(2k+1)(2k+1)!}$. Three terms give $1 - \frac{1}{18} + \frac{1}{600} = 0.9461111$; the first omitted term is $\frac{1}{7 \cdot 7!} = 2.83 \times 10^{-5}$, and the true value is 0.9460831, in error by 2.80×10^{-5} . Three terms and one division supply four correct digits with a certificate.

WHERE THIS GOES WRONG

Reporting more digits than the tail bound justifies. This is the characteristic failure of the technique, and it is invisible: the extra digits are produced by the arithmetic and look exactly like the correct ones. The second trap is applying the alternating-series bound when the terms are not monotone decreasing, which happens more often than expected in series with factorials in both numerator and denominator: check monotonicity for $n \geq N$ explicitly. Third, note the difference in cost between a super-geometric series such as $\sum n^{-n}$, where six terms suffice, and a ζ -type series such as $\sum n^{-2}$, where six terms give one digit and acceleration or Euler-Maclaurin is essential.

SEE ALSO Termwise integration of a Maclaurin series; the Euler-Maclaurin correction; numerical cross-check as verification discipline; recognising the emergent constant.

When the terms decay only polynomially, truncation is hopeless and acceleration is mandatory. The cheapest acceleration is the one already in this chapter: apply Euler-Maclaurin to the *tail* rather than summing it. For $\zeta(2) = \sum n^{-2}$, ten bare terms give 1.5497677, in error by 0.095, which is one correct digit. Ten terms plus the three-term Euler-Maclaurin tail $\frac{1}{n} - \frac{1}{2n^2} + \frac{1}{6n^3}$ at $n = 10$ give

$$1.5497677 + 0.0951667 = 1.6449344,$$

in error by 3.3×10^{-7} against $\frac{\pi^2}{6} = 1.6449341$: six correct digits from the same ten terms and three lines of algebra. For alternating series that converge slowly, the Euler transform plays the same role, replacing the terms by their repeated forward differences and typically converting an $O(N^{-1})$ error into an exponentially small one. The general principle is that a slow series should never be summed directly; it should be split into a computed head and an analysed tail.

Asymptotics

The questions that no amount of quadrature answers are the ones about a family. “How does I_n behave as $n \rightarrow \infty$ ” cannot be settled by computing I_{100} , because a single number carries no information about a rate. Computing I_n for a range of n and fitting is better but still weak: it will not distinguish $n^{-1/2}$ from $n^{-1/2} \ln n$, and it gives no constant. The asymptotic argument gives both the exponent and the constant, and it gives them from the local behaviour of the integrand at one point. The technique for the commonest such family is a localisation argument, and it is the continuous analogue of the observation that a sum of positive terms is dominated by its largest.

29.5 Laplace’s method and Watson’s lemma

HARD

CUE $\int e^{-n\phi(x)} g(x) dx$ with a large parameter n ; a factorial or a Gamma value with a large argument; or any question phrased as “how does this behave for large n ”.

METHOD The integral localises at the minimum x_0 of ϕ . Expand $\phi(x) \approx \phi(x_0) + \frac{1}{2}\phi''(x_0)(x-x_0)^2$ and integrate the resulting Gaussian. For an *interior* minimum with $\phi''(x_0) > 0$,

$$\int e^{-n\phi(x)} g(x) dx \sim g(x_0) e^{-n\phi(x_0)} \sqrt{\frac{2\pi}{n\phi''(x_0)}}.$$

If the minimum sits at an endpoint, take *half* of this, because only half of the Gaussian lies in the domain. Watson’s lemma handles the degenerate case where ϕ is linear: $\int_0^\infty e^{-nt} g(t) dt \sim \sum_{k \geq 0} \frac{g^{(k)}(0)}{n^{k+1}}$.

WORKED EXAMPLE

$$\int_0^{\pi/2} e^{-n \sin^2 x} dx.$$

Here $\phi(x) = \sin^2 x$ has its minimum at the endpoint $x_0 = 0$, with $\phi(0) = 0$ and $\phi''(0) = 2$. The interior formula would give $\sqrt{2\pi/(2n)} = \sqrt{\pi/n}$; halving for the endpoint,

$$\int_0^{\pi/2} e^{-n \sin^2 x} dx \sim \frac{1}{2} \sqrt{\frac{\pi}{n}}.$$

At $n = 400$ this predicts 0.0443113 against the true 0.0443391, an error of 0.06%, and the relative error decays like n^{-1} . Forgetting the halving would give 0.0886227, wrong by a factor of two, and no amount of care in the rest of the computation would recover it.

WATSON, AND STIRLING

$\int_0^\infty \frac{e^{-nt}}{1+t} dt$. Here $g(t) = (1+t)^{-1}$ has $g(0) = 1$, $g'(0) = -1$, $g''(0) = 2$, so Watson gives $\frac{1}{n} - \frac{1}{n^2} + \frac{2}{n^3} + \dots$. At $n = 20$ this is 0.047750 against the true 0.0477185, three correct digits from three terms.

Stirling’s formula is Laplace’s method applied to $\Gamma(n+1) = \int_0^\infty t^n e^{-t} dt$: write the integrand as $e^{-(t-n \ln t)}$, whose exponent has an interior minimum at $t = n$ with second derivative n^{-1} , giving $n! \sim \sqrt{2\pi n} (n/e)^n$. At $n = 10$ this is 3598696 against $10! = 3628800$, low by 0.83%, and the relative error is $\sim \frac{1}{12n}$ exactly as the next term of the expansion predicts.

WHERE THIS GOES WRONG

The endpoint halving, as above: a minimum *at* an endpoint contributes half of the interior formula, and forgetting it doubles every answer in the family. The second trap is a degenerate minimum: if $\phi''(x_0) = 0$ the Gaussian model is wrong, the local scaling is $n^{-1/4}$ rather than $n^{-1/2}$, and the leading constant involves $\Gamma(\frac{1}{4})$ rather than $\sqrt{2\pi}$. Third, an asymptotic series is not a convergent series: adding more terms of the Watson expansion eventually makes the answer worse, and the optimal truncation is near the smallest term.

SEE ALSO Nascent delta and limiting kernels; the Gamma-function moment; bounding by envelopes; series truncation with a tail bound.

Verification

The last technique in this chapter is the one that should be applied to every other chapter. It is not a method of evaluation. It is the habit that catches the errors the methods produce, and in a subject where a wrong answer is usually a clean-looking closed form rather than a mess, it is the only reliable defence.

29.6 Numerical cross-check as verification discipline

CORE

CUE Every closed form you are about to assert. Without exception, and especially when the derivation felt elegant.

METHOD Evaluate the *original* integral numerically and compare with the claimed value to at least six digits. Choose the method to match the pathology: substitute away endpoint singularities before quadrature, transform oscillatory infinite ranges into alternating series and accelerate, place break points at every discontinuity of the integrand or of its derivative. Then run three structural checks that cost nothing: the *sign*, the *scale* (is the answer of the order of the integrand times the interval length?), and a *limiting case* of any parameter, where the integral should reduce to something you already know.

WORKED EXAMPLE

A corpus asserts $\int_0^{\pi/2} \frac{dx}{\sec x + \tan x} = 1$. Quadrature returns 0.6931471806, which is $\ln 2$, exposing the error immediately. The structural reason: $\sec x + \tan x = \frac{1 + \sin x}{\cos x}$, so the integrand is $\frac{\cos x}{1 + \sin x}$, whose antiderivative is $\ln(1 + \sin x)$, giving $\ln 2 - \ln 1 = \ln 2$. Note that the claimed 1 is not far from 0.693, which is exactly why the check is necessary: a wrong answer of the right order of magnitude is invisible to inspection.

A second, where the error is catastrophic rather than subtle: the same corpus asserts

$$\int_0^\infty \frac{\ln x (\pi^2 + \ln^2 x)}{(1+x)^2 \sqrt{x}} dx = 0,$$

presumably by an inversion argument. Splitting at $x = 1$ and integrating each half numerically gives -124.0251067 , which is $-4\pi^3 = -124.0251067$. The claimed 0 is wrong by an infinite factor, and one minute of quadrature says so.

THE THREE STRUCTURAL CHECKS IN ACTION

Suppose you have derived $\int_0^\infty \frac{x^{s-1}}{1+x^n} dx = \frac{\pi}{n \sin(\pi s/n)}$. *Sign*: the integrand is positive on the stated strip $0 < s < n$, and $\sin(\pi s/n) > 0$ there, so the sign is right. *Scale*: as $s \rightarrow 0^+$ or $s \rightarrow n^-$ the integral must blow up, since the corresponding p -test fails, and $\sin(\pi s/n) \rightarrow 0$ delivers exactly that. *Limiting case*: at $s = 1$, $n = 2$ the formula gives $\frac{\pi}{2 \sin(\pi/2)} = \frac{\pi}{2}$, which is $\int_0^\infty \frac{dx}{1+x^2}$, known independently. Three checks, no quadrature, and a formula that survives all three is very unlikely to be wrong.

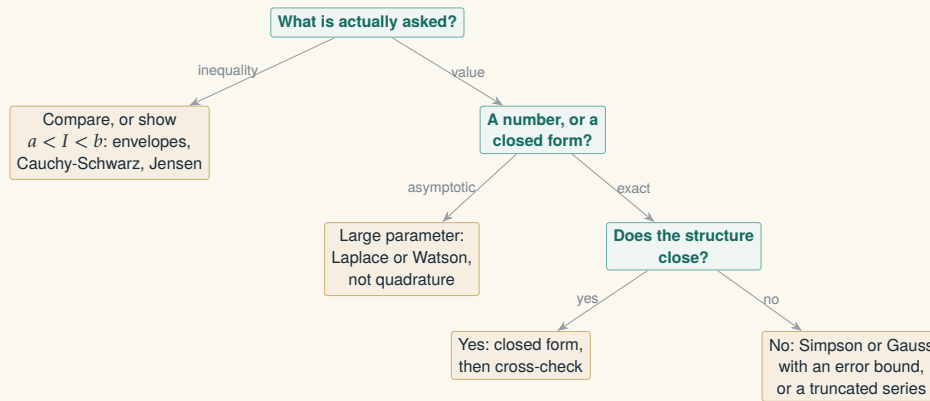
WHERE THIS GOES WRONG

Trusting a quadrature routine on a conditionally convergent or endpoint-singular integrand without transforming first. A routine that truncates $\int_0^\infty \frac{\sin x}{x} dx$ at $x = 50$ returns a number in error by $O(10^{-2})$, and a routine that samples $\int_0^1 \ln x dx$ near 0 can return anything at all; either can “confirm” a wrong closed form to two digits. Always validate the numerical method against a known value on the same interval with the same pathology before trusting it on the unknown one. And check the *original* integral, not the transformed one: verifying the last line of your own derivation cannot detect an error made in the second line.

SEE ALSO Constant or zero integrand in disguise; Newton-Cotes rules with error terms; series truncation with a tail bound; endpoint limit discipline.

Choosing the deliverable

The decision is about the question, not about the integrand, and it is usually settled by the verb.



The branch that requires judgement is the last one, and the chapter promised to say plainly when a numeric answer is right and when it is a confession. A number is the correct deliverable when the integrand is genuinely non-elementary and the question is quantitative: $\int_0^1 e^{-x^2} dx$, $\int_0^1 x^x dx$, an integral whose values arrive as measurements, or any integral over a region described by data. A number is also correct when the closed form exists but is a special value nobody can read, since 0.9460831 communicates more than an unsimplified hypergeometric expression. In both cases the number must come with an error bound, because a number without one is not an estimate but a guess.

A number is a failure when the structure was there and you did not look. The signals are the same ones this book has been cataloguing throughout: a parameter that plays no role in the answer, an interval symmetric about something, a numerator that is the derivative of a denominator, bounds engineered to land on a clean value, an integrand that is secretly a constant. Quadrature will happily return 0.785398 for an integral whose exact value is $\frac{\pi}{4}$ and never tell you that a two-line symmetry argument existed. That is why the verification technique in this chapter is stated as a check on a closed form rather than as a substitute for one: numerics confirm structure, and cannot discover it.

RIGOROUS, HEURISTIC, AND THE DIFFERENCE

An error bound is *rigorous* when it follows from a theorem plus a proved bound on the relevant quantity: $|E_S| \leq \frac{(b-a)h^4}{180} \max |f^{(4)}|$ once you have bounded $f^{(4)}$; the alternating tail bound once you have proved the terms decrease; the envelope bounds always. It is *heuristic* when it is inferred from the behaviour of the computation: the difference between two refinements, the number of digits that stopped changing, the agreement of two different routines. Heuristic bounds are usually right and occasionally catastrophically wrong, and they fail exactly where the theorem's hypotheses fail, which is to say on oscillatory integrands and near singularities. Use heuristics to decide when to stop computing and rigorous bounds to decide what to write down.

THE HONEST REPORT

Three digits with a rigorous error bound beats six digits with none, and both are beaten by an exact value that has been cross-checked numerically. When you report a number, say how it was obtained and what bounds it: "Simpson, $n = 8$, error below 2×10^{-5} " is an answer; "approximately 0.7468" is a rumour. When you report a closed form, say that you checked it and against what. The two habits together are what make a result usable by somebody else, and they cost about a minute each.

Recognition matrix

If you see	Reach for	If that fails
"Show that $a < I < b$ "	Pointwise envelopes; integrate the bounds	Split and bound each piece
"Which is larger?"	Compare integrands, not integrals	Subtract and check the sign
A weighted average shape $\int fw$	Mean value theorem: $f(\xi) \int w$	Requires $w \geq 0$
A product $\int fg$ to bound	Cauchy-Schwarz; take $g = 1$ if unsure	Holder for other exponents
φ convex applied to an average	Jensen: $\varphi(\int f) \leq \int \varphi(f)$ on $[0, 1]$	Reverse for concave φ
$\frac{1}{n} \sum f(\frac{k}{n})$ as $n \rightarrow \infty$	Riemann sum: $\int_0^1 f$	Factor out Δx exactly once
$\sum \frac{1}{n+k}, (n!)^{1/n}/n$	Divide through by n , then integrate	Take logs first if it is a product
A sum needing many digits	Euler-Maclaurin with $\frac{f(a)+f(b)}{2}$	$B_2 = \frac{1}{6}$ term next
Data, or no closed form, few digits	Simpson, n even, with $\frac{(b-a)h^4}{180} \max f^{(4)} $	Halve h and extrapolate
Two trapezoid values at $h, h/2$	Richardson: $\frac{4T(h/2) - T(h)}{3}$	Romberg: iterate the table
Smooth integrand, few evaluations	Gauss-Legendre: exact to degree $2n - 1$	Five nodes give nine digits here

If you see	Reach for	If that fails
An endpoint singularity before quadrature	Substitute it away ($x = u^2$, tanh-sinh)	Error bounds are void otherwise
An alternating series answer	Tail $\leq$ first omitted term	Verify terms decrease monotonically
A positive decreasing series	Tail $\leq \int_N^\infty a(t) dt$	Geometric bound if faster
$\sum n^{-n}$ -type terms	Six terms give six digits	ζ -type: accelerate instead
$e^{-n\phi(x)}g(x)$, n large	Laplace at the minimum of ϕ	Halve if the minimum is an endpoint
$\int_0^\infty e^{-nt}g(t) dt$, n large	Watson: $\sum g^{(k)}(0)/n^{k+1}$	Truncate near the smallest term
$n!$ or $\Gamma(n)$ for large n	Stirling $\sqrt{2\pi n}(n/e)^n$	Relative error $\sim \frac{1}{12n}$
$\phi''(x_0) = 0$ at the minimum	Rescale by $n^{-1/4}$; expect $\Gamma(\frac{1}{4})$	Not the Gaussian formula
Any closed form you just derived	Quadrature to six digits on the original	Sign, scale, limiting case
A suspiciously clean value	Check the original, not your last line	Wrong answers here look right

Differentials and differential equations

Differentials proper

What this chapter is for. A differential is the linear part of an increment, and almost every confusion about dx dissolves the moment that sentence is taken literally. If $y = f(x)$ and x is nudged by an amount Δx , the true change $\Delta y = f(x + \Delta x) - f(x)$ is some complicated thing; the differential $dy = f'(x) dx$ is the best straight-line stand-in for it, and the two agree to first order and disagree at second order. Everything in this chapter is a consequence: linear approximation is that stand-in used as an estimate, error propagation is that stand-in used as a tolerance, the total differential is that stand-in in several variables, and $du = g'(x) dx$ under a substitution is that stand-in used as a change of coordinates on the increment. By the end you should be able to produce a numerical estimate with a signed error bound, convert a measurement tolerance into a tolerance on any derived quantity, differentiate an implicit relation without ever solving it, and say precisely which manipulations of dx are legitimate and why. That last point is the one most textbooks skip, and it is the reason students distrust the notation for years.

The increment and its linear part

Fix a differentiable f and a point x . The definition of the derivative says

$$f(x + \Delta x) - f(x) = f'(x) \Delta x + \varepsilon(\Delta x) \Delta x, \quad \varepsilon(\Delta x) \rightarrow 0 \text{ as } \Delta x \rightarrow 0.$$

Read the right side as two objects rather than one. The first term is linear in Δx with a coefficient that does not depend on Δx ; the second is a remainder that vanishes faster than Δx does. We give the first term a name and a notation: for an independent variable we *define* $dx = \Delta x$, and for the dependent variable we *define*

$$dy = f'(x) dx.$$

Two things follow immediately and are worth saying out loud. First, dy is a function of two independent inputs, the base point x and the nudge dx ; it is not a number attached to x alone. Second, dy is exactly computable while Δy usually is not, which is the entire practical point: we trade an unknown quantity for a linear one and pay a second-order price.

The ratio dy/dx is then honestly a quotient of two real numbers, and it equals $f'(x)$ for every nonzero dx . This is why Leibniz notation behaves so well under the chain rule and under substitution, and it is also the precise sense in which dy/dx “is” a fraction. What is *not* legitimate is treating dx as an infinitesimal, a number smaller than every positive real. No such real number exists. Everything in this chapter can be done with dx an ordinary finite real, and everything that looks like cancelling differentials is really the chain rule or the substitution rule written compactly.

The picture is worth fixing in mind, because every technique below is a reading of it. Draw the graph of f and the tangent line at x . Step right by dx . The graph rises by Δy ; the tangent line rises by dy . The gap between them is the whole error, it is second order in dx , and its sign is the sign of f'' : for a concave function the tangent lies above the curve and $dy > \Delta y$, for a convex function the reverse. Every estimate in this chapter is the height of that tangent line, every error bound is the size of that gap, and every question about whether an estimate is high or low is answered by looking at the concavity rather than by computing anything. Halving dx halves dy and quarters the gap, which is the practical content of “second order” and the reason linear approximation is useful at all.

WHAT THE NOTATION IS, HONESTLY

There are three defensible readings of dx , and each licenses a different set of moves. (i) **Finite increment:** dx is any real number, $dy = f'(x) dx$ is a linear function of it. This is the reading used for approximation and error propagation, and it makes every equation in this chapter a true statement about real numbers. (ii) **Formal symbol under an integral or in an ODE:** dx is a bookkeeping token whose only rule is that $u = g(x)$ forces $du = g'(x) dx$, justified by the substitution rule. This is the reading used when you separate variables. (iii) **Differential form:** dx and dy are covectors and $dF = F_x dx + F_y dy$ is an exact 1-form. This is the reading that makes exactness of $M dx + N dy = 0$ a theorem rather than a coincidence. All three agree on every computation below. None of them permits “multiply both sides by dx ” as a self-standing act; that phrase is always shorthand for one of (i), (ii), or (iii).

One structural property of the notation deserves to be stated because it is what makes the whole scheme coherent, and because it is invisible if you only ever work in one variable. The expression $dz = f_x dx + f_y dy$ is true whether x and y

are independent variables or are themselves functions of some other variables s and t . In the second case dx and dy are no longer free but are given by their own differentials, and substituting them reproduces exactly the multivariable chain rule. This is called the invariance of the form of the first differential, and it is the reason you may write d of an equation without first deciding which symbols are independent. Every implicit differentiation in this chapter relies on it, and so does every substitution in the next two.

The first technique is nothing but the definition, applied on purpose. It looks too simple to need a name, and it is the technique students most reliably misuse, because they report dy where Δy was asked for and never notice the difference.

CORE

30.1 The differential dy versus the increment Δy

CUE A problem asks approximately how much y changes when x changes by a small stated amount, or supplies Δx and wants Δy without recomputing f . The word “approximately”, or a tolerance sign $\pm$, is the tell.

METHOD Set $dx = \Delta x$ and compute $dy = f'(x) dx$. Then

$$\Delta y = dy + o(dx), \quad \text{so} \quad f(x + \Delta x) \approx f(x) + f'(x) \Delta x.$$

The differential is the linear part of the increment, not the increment. Report it as an approximation, and if the question wants a bound, get the sign of the error from the sign of f'' near x .

WORKED EXAMPLE

Estimate $\sqrt{100.4}$. Take $f(x) = \sqrt{x}$, $x = 100$, $dx = 0.4$. Then $f'(100) = \frac{1}{2\sqrt{100}} = \frac{1}{20}$, so

$$dy = \frac{0.4}{20} = 0.02, \quad \sqrt{100.4} \approx 10.02.$$

The true value is $10.019980\dots$, so the differential overshoot by 2.0×10^{-5} . That sign was predictable: $f''(x) = -\frac{1}{4}x^{-3/2} < 0$, so $\sqrt{\cdot}$ is concave and its tangent line lies above the curve everywhere. The size was predictable too, from $\frac{1}{2}|f''|(dx)^2 = \frac{1}{2} \cdot \frac{1}{4000} \cdot 0.16 = 2.0 \times 10^{-5}$.

A second instance, where Δy and dy visibly part company: $y = x^{10}$ at $x = 1$ with $\Delta x = 0.02$. Here $dy = 10(0.02) = 0.2$, giving the estimate 1.2 , while $(1.02)^{10} = 1.218994\dots$. The differential captures 0.2 of the true increment 0.218994 ; the missing 0.019 is the quadratic term $\frac{1}{2} \cdot 90 \cdot (0.02)^2 = 0.018$ plus higher order.

RATES ARE DIFFERENTIALS DIVIDED BY dt

The related-rate problems of elementary calculus are this technique with both variables made functions of time. If $V = \frac{4}{3}\pi r^3$ then $dV = 4\pi r^2 dr$, and dividing by dt gives $\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$: the same identity, read as a relation between rates rather than between increments. Nothing new is needed, and setting these problems up in differential form first has a practical advantage, namely that you do not have to decide in advance which variable is independent. Take the differential of the geometric relation, then divide by dt , and the chain rule appears automatically.

WHERE THIS GOES WRONG

Reporting dy as though it were exact, and forgetting that dy depends on two variables. Students silently fix $dx = 1$, which turns dy into $f'(x)$ and destroys the units of the answer: if x is measured in centimetres and $dx = 0.4$ cm, then dy carries the units of $f' \cdot \text{cm}$, and dropping the 0.4 reports a rate where a quantity was wanted.

SEE ALSO Linear approximation and the tangent-line estimate; error propagation; Taylor polynomials, of which this is the degree-one case.

Written as a statement about f rather than about the increment, the same identity is the tangent line, and the shift of viewpoint is worth making because it tells you what to optimise. In the increment form the free choice is Δx , which the problem usually fixes. In the tangent form the free choice is the base point a , which the problem almost never fixes, and choosing a badly is how an otherwise correct estimate becomes useless.

30.2 Linear approximation and the tangent-line estimate

CUE “Estimate $f(a+h)$ ” where some nearby point has exactly computable f and f' , or the explicit instruction “use differentials to approximate”. Also any request for a first-order correction to a known value.

METHOD Use the linearisation

$$L(x) = f(a) + f'(a)(x-a), \quad f(x) - L(x) = \frac{1}{2}f''(\xi)(x-a)^2 \text{ for some } \xi \text{ between } a \text{ and } x.$$

Choose a to be the *nearest* point at which f and f' are both exact, not the most arithmetically convenient one. The error formula then does two jobs: it bounds the error, and its sign tells you whether the estimate is high or low, since $f'' < 0$ (concave) means the tangent lies above the graph.

WORKED EXAMPLE

Estimate $\ln(1.05)$. With $f = \ln$, $a = 1$: $f(1) = 0$, $f'(1) = 1$, so $L(1.05) = 0.05$. Since $f''(x) = -1/x^2 < 0$, the estimate is an overshoot, and indeed $\ln(1.05) = 0.0487902 \dots$. The bound $\frac{1}{2}|f''(\xi)|(0.05)^2 \leq \frac{1}{2}(1)(0.0025) = 0.00125$ contains the actual error 0.00121.

Estimate $\sin 31^\circ$. Convert first: $31^\circ = \frac{\pi}{6} + \frac{\pi}{180}$, so $a = \frac{\pi}{6}$ and $h = \frac{\pi}{180} = 0.0174533$. Then

$$\sin 31^\circ \approx \sin \frac{\pi}{6} + \cos \frac{\pi}{6} \cdot \frac{\pi}{180} = 0.5 + 0.8660254(0.0174533) = 0.5151150,$$

against the true 0.5150381. The estimate is high, as $\sin$ is concave on $(0, \pi)$.

Estimate $\sqrt[3]{28}$. The nearest exact point is $a = 27$, where $f(27) = 3$ and $f'(27) = \frac{1}{3}(27)^{-2/3} = \frac{1}{27}$. So $\sqrt[3]{28} \approx 3 + \frac{1}{27} = 3.037037$, against 3.036589; again high, since $x^{1/3}$ is concave for $x > 0$, and again the error 4.5×10^{-4} is of the predicted size $\frac{1}{2}|f''(27)| = \frac{1}{2} \cdot \frac{2}{9} \cdot 27^{-5/3} = 4.6 \times 10^{-4}$.

WHERE THIS GOES WRONG

Two failures, both common. First, choosing a for arithmetic comfort rather than proximity: estimating $\sqrt{104}$ from $a = 100$ is fine, but estimating $\sqrt{140}$ from $a = 100$ gives 12.0 against a true 11.832, because $(x-a)^2 = 1600$ has made the quadratic term large. Second, linearising a trigonometric function without converting to radians. The formula $d(\sin x) = \cos x dx$ is false in degrees; in degrees the derivative of $\sin$ carries a factor $\pi/180$.

SEE ALSO The differential versus the increment; error propagation; Taylor polynomials with explicit remainder.

Linear approximation stops being an approximation and starts being a lie at a rate controlled entirely by f'' , and it is worth knowing the numbers. The error is $\frac{1}{2}f''(\xi)(x-a)^2$, so halving the step quarters the error; a step ten times smaller gives an answer a hundred times better. Concretely, for $\sqrt{\cdot}$ at $a = 100$: a step of 0.4 gives an error of 2×10^{-5} , a step of 4 gives 2×10^{-3} , a step of 40 gives 0.19, and a step of 400 gives an answer, 12, that is wrong in the first digit. The linear model degrades quadratically and then catastrophically, and the only defence is to keep $|x-a|$ small, which means choosing a well. When no nearby exact point exists, the correct response is not a larger step but a second-order term: $f(a+h) \approx f(a) + f'(a)h + \frac{1}{2}f''(a)h^2$, which is the next line of the Taylor expansion and costs one more derivative.

Error, and why relative error is the natural currency

Approximation and error propagation are the same computation read in opposite directions. In approximation, dx is a step you chose and dy is the estimate you want. In error propagation, dx is an uncertainty you were handed and dy is the uncertainty you must report. Nothing changes except that absolute values appear, because a tolerance has no sign, and that the answer is usually wanted as a percentage rather than as a quantity.

30.3 Error propagation and relative error

CUE A measured quantity carries a stated tolerance ($r = 5 \pm 0.02$ cm, “accurate to within 1%”) and you are asked for the induced tolerance in something computed from it.

METHOD Absolute error is $|dy| = |f'(x)| |dx|$. Relative error is the quantity actually worth computing:

$$\left| \frac{dy}{y} \right| = \left| \frac{f'(x)}{f(x)} \right| |dx| = |(\ln |f|)'(x)| |dx|.$$

For a power law $y = Cx^n$ this collapses to $\frac{dy}{y} = n \frac{dx}{x}$: relative error multiplies by the exponent and the constant C is irrelevant. Take logs before differentiating whenever the formula is a product, quotient, or power, since $\ln$ turns all three into sums.

WORKED EXAMPLE

A sphere's radius is measured as $r = 5 \pm 0.02$ cm. Find the induced uncertainty in the volume $V = \frac{4}{3}\pi r^3$, absolutely and as a percentage.

By the power rule for relative error, $\frac{dV}{V} = 3 \frac{dr}{r} = 3 \cdot \frac{0.02}{5} = 3(0.004) = 0.012$, so 1.2%. Absolutely, $dV = 4\pi r^2 dr = 4\pi(25)(0.02) = 2\pi \approx 6.283$ cm³, against $V = \frac{500\pi}{3} \approx 523.6$ cm³; the ratio is 0.012, as promised.

The same rule read backwards answers the design question. If the volume must be known to 0.5%, then $\frac{dr}{r} \leq \frac{0.005}{3} = 0.00167$, so the radius must be measured to about 0.008 cm. Tightening a volume tolerance costs three times as much precision in the radius.

The power rule handles negative and fractional exponents without change, and that is where it earns its keep. A pendulum has period $T = 2\pi\sqrt{L/g}$, so $T = CL^{1/2}$ and $\frac{dT}{T} = \frac{1}{2} \frac{dL}{L}$: a length known to 1% gives a period known to 0.5%, and the exponent $\frac{1}{2}$ is doing all the work. Inverted to measure gravity, $g = 4\pi^2 LT^{-2}$ gives

$$\frac{dg}{g} = \frac{dL}{L} - 2 \frac{dT}{T}, \quad \left| \frac{dg}{g} \right| \leq \left| \frac{dL}{L} \right| + 2 \left| \frac{dT}{T} \right|,$$

so timing error costs twice what length error costs. That asymmetry is invisible in the formula for g and obvious in the formula for dg/g , which is the reason to compute the latter.

WHERE THIS GOES WRONG

Adding the *absolute* errors of factors in a product. For $A = \ell w$ the correct combination is $\frac{dA}{A} = \frac{d\ell}{\ell} + \frac{dw}{w}$: relative errors add, absolute errors do not. Separately, dropping the absolute value, so that a decreasing function ($f' < 0$) is reported as having a negative tolerance. A tolerance is a magnitude.

SEE ALSO The total differential of a multivariable function; the differential versus the increment; logarithmic differentiation.

One measured input is the exceptional case. Real formulas take several, each with its own tolerance, and the linear-part idea extends verbatim: the increment of $f(x, y)$ is a linear function of the pair (dx, dy) , with the two partial derivatives as coefficients. The only genuinely new question is how to combine the two contributions, and the answer depends on a modelling assumption that the problem statement, not the mathematics, decides.

MEDIUM

30.4 The total differential of a multivariable function

CUE A quantity depends on several independently measured inputs, or a problem writes dz for $z = f(x, y)$, or you need to know how a formula responds to simultaneous small changes in all of its arguments.

METHOD The linear part of the increment is

$$dz = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy, \quad \Delta z = dz + o(|dx| + |dy|).$$

For a guaranteed worst-case bound, use the triangle inequality: $|dz| \leq |f_x| |dx| + |f_y| |dy|$, which assumes the errors conspire. For independent random errors with standard deviations σ_x, σ_y , combine in quadrature instead: $\sigma_z^2 = f_x^2 \sigma_x^2 + f_y^2 \sigma_y^2$.

Two resistors in parallel satisfy $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$. Differentiate the relation as it stands, without solving for R :

$$-\frac{dR}{R^2} = -\frac{dR_1}{R_1^2} - \frac{dR_2}{R_2^2}, \quad \text{so} \quad dR = \frac{R^2}{R_1^2} dR_1 + \frac{R^2}{R_2^2} dR_2.$$

Take $R_1 = 100 \pm 1$ and $R_2 = 400 \pm 4$ ohms. Then $\frac{1}{R} = \frac{1}{100} + \frac{1}{400} = \frac{1}{80}$, so $R = 80$, and the worst-case bound is

$$|dR| \leq \frac{6400}{10000}(1) + \frac{6400}{160000}(4) = 0.64 + 0.16 = 0.80 \Omega.$$

Direct computation confirms it: replacing $(100, 400)$ by $(99, 396)$ gives $R = 79.2$ exactly, and by $(101, 404)$ gives $R = 80.8$. Note what the numbers say. Both inputs carry 1% error and the output carries $0.80/80 = 1\%$; the parallel combination is homogeneous of degree one, so it inherits the common relative error unchanged. Under the quadrature rule the answer would instead be $\sqrt{0.64^2 + 0.16^2} = 0.66 \Omega$.

A second instance, where taking logarithms first shortens the work considerably. A cylinder has $r = 2 \pm 0.01$ and $h = 5 \pm 0.02$ cm, and $V = \pi r^2 h$. Rather than computing V_r and V_h , take logs: $\ln V = \ln \pi + 2 \ln r + \ln h$, so

$$\frac{dV}{V} = 2 \frac{dr}{r} + \frac{dh}{h} = 2(0.005) + 0.004 = 0.014,$$

a worst case of 1.4%. Since $V = 20\pi = 62.832$, that is $dV = 0.880 \text{ cm}^3$, and the exact worst case, $\pi(2.01)^2(5.02) - 20\pi = 0.884$, sits just above it as the second-order terms require. The logarithmic route generalises: for any monomial $z = Cx^a y^b$, the relative errors combine as $|a|$ and $|b|$ times the input relative errors, with no differentiation at all.

WHERE THIS GOES WRONG

Using quadrature when the problem asks for a guaranteed bound ("the volume cannot be wrong by more than"), or the linear sum when it asks for a standard deviation. These give different numbers, 0.80 against 0.66 above, and only the wording of the problem decides which is wanted. A second trap: computing $R = 80$ first and then differentiating $R = \frac{R_1 R_2}{R_1 + R_2}$ by brute force. Differentiating the reciprocal relation in place is three symbols shorter and less error-prone, which is the whole reason to keep the relation implicit.

SEE ALSO Error propagation and relative error; implicit differentiation in differential form; the exactness test, which asks when a given $M dx + N dy$ is itself a total differential.

WORST CASE OR STANDARD DEVIATION

The two combination rules answer different questions and the problem statement, not the mathematics, chooses between them. "The volume cannot be wrong by more than" asks for a guaranteed bound and takes the linear sum of absolute contributions, which assumes the errors conspire to point the same way. "What is the uncertainty in the volume" in a laboratory context usually asks for a standard deviation and takes the quadrature sum, which assumes the errors are independent and random. The quadrature answer is always the smaller of the two, and for two equal contributions it is smaller by a factor $\sqrt{2}$. Reporting a quadrature figure where a bound was demanded understates the risk; reporting a bound where a standard deviation was demanded overstates the error by a factor that grows with the number of inputs.

Differentials as relations between variables

So far dy has been computed from an explicit $y = f(x)$. The notation earns its keep when there is no explicit f . An implicit relation $F(x, y) = 0$ constrains the pair (dx, dy) by a single linear equation, and that equation is all you need: solving it for the ratio gives y' , and never solving the original relation for y is the point.

30.5 Implicit differentiation in differential form

CUE A relation $F(x, y) = 0$ that cannot be solved for y in closed form, or can be solved only at the cost of radicals and case splits, together with a request for y' , a tangent line, or the location of horizontal and vertical tangents.

METHOD Apply d to both sides, treating x and y symmetrically and using the product and chain rules on every term. The result is linear in dx and dy :

$$F_x dx + F_y dy = 0 \quad \Rightarrow \quad \frac{dy}{dx} = -\frac{F_x}{F_y}.$$

Horizontal tangents are where $F_x = 0$ and $F_y \neq 0$; vertical tangents are where $F_y = 0$ and $F_x \neq 0$. Both are visible in the differential relation before any division, which is exactly why you should read it before dividing.

WORKED EXAMPLE

The folium of Descartes, $x^3 + y^3 = 6xy$. Taking d of both sides, with $d(xy) = y dx + x dy$:

$$3x^2 dx + 3y^2 dy = 6y dx + 6x dy, \quad (3x^2 - 6y) dx + (3y^2 - 6x) dy = 0,$$

so $\frac{dy}{dx} = \frac{2y - x^2}{y^2 - 2x}$. At the point $(3, 3)$, which is on the curve since $27 + 27 = 54 = 6 \cdot 9$, this gives $\frac{6 - 9}{9 - 6} = -1$: the tangent is $y = 6 - x$.

The vertical tangents come free. Set $y^2 = 2x$ and substitute into the curve: $x^3 + y^3 = 6xy$ becomes $\frac{y^6}{8} + y^3 = 3y^3$, so $y^3 = 16$ and $y = 2^{4/3}$, $x = 2^{5/3}$. Check: $x^3 + y^3 = 32 + 16 = 48$ and $6xy = 6 \cdot 2^3 = 48$. The curve has a vertical tangent at $(2^{5/3}, 2^{4/3}) \approx (3.1748, 2.5198)$, a fact that would have been painful to extract from any explicit solution for y .

A cleaner instance, showing both tangent types at once: $x^2 - xy + y^2 = 3$. Taking d gives $(2x - y) dx + (2y - x) dy = 0$, so $\frac{dy}{dx} = \frac{2x - y}{x - 2y}$. At $(1, -1)$, which lies on the curve, the slope is $\frac{3}{3} = 1$ and the tangent is $y = x - 2$. Horizontal tangents need $2x - y = 0$; substituting $y = 2x$ into the curve gives $3x^2 = 3$, so they occur at $(1, 2)$ and $(-1, -2)$. Vertical tangents need $x - 2y = 0$; substituting $x = 2y$ gives $3y^2 = 3$, so they occur at $(2, 1)$ and $(-2, -1)$. Four special points, all read off the differential relation, and the curve was never solved for y .

WHERE THIS GOES WRONG

Forgetting that y depends on x inside a product: $d(xy) = y dx + x dy$, never $y dx$ alone. And dividing by F_y without recording where it vanishes; at those points the formula for y' is not merely large, it does not exist, and the curve is doing something (a vertical tangent, a cusp, a self-intersection) that the quotient conceals. The folium passes through the origin twice, and there $F_x = F_y = 0$: the differential relation degenerates to $0 = 0$ and carries no information at all.

SEE ALSO The total differential of a multivariable function; the exactness test; differentials as substitution bookkeeping.

The implicit relation $F_x dx + F_y dy = 0$ is a constraint linking two increments. A substitution is the same kind of object read as a change of variable rather than as a constraint, and this is where the notation is most often abused. The equation $du = g'(x) dx$ is a *statement about the substitution* $u = g(x)$, derived by applying the definition of the differential to g ; it is not a fraction being cross-multiplied, and no cancellation is taking place.

CORE

30.6 Differentials as substitution bookkeeping

CUE Any integral or differential equation in which a substitution $u = g(x)$ is introduced and the dx must be converted. In a first-order ODE this is exactly how $y = vx$, $u = x + y$, or $v = y^{1-n}$ transfers the equation to a new unknown.

METHOD Write $du = g'(x) dx$, then *solve for dx before substituting anything*, so that no factor is silently absorbed:

$$dx = \frac{du}{g'(x)}, \quad g'(x) \text{ re-expressed in } u.$$

After the substitution, every occurrence of the old variable must be gone. If one survives, the substitution has not been carried out; it has been started.

WORKED EXAMPLE

Solve $\frac{dy}{dx} = (x + y)^2$ with $y(0) = 0$. The right side depends on x and y only through $u = x + y$, so substitute. The differential relation for the substitution is $du = dx + dy$, hence

$$\frac{du}{dx} = 1 + \frac{dy}{dx} = 1 + u^2.$$

This is separable: $\frac{du}{1+u^2} = dx$ gives $\arctan u = x + C$. The initial condition $y(0) = 0$ means $u(0) = 0$, so $C = 0$ and $u = \tan x$, that is,

$$y = \tan x - x, \quad y\left(\frac{\pi}{4}\right) = 1 - \frac{\pi}{4} \approx 0.214602.$$

A fourth-order Runge-Kutta integration from $x = 0$ to $x = \pi/4$ returns 0.2146018, confirming both the closed form and the branch. The solution exists only on $(-\frac{\pi}{2}, \frac{\pi}{2})$, since $u = \tan x$ escapes to infinity at the endpoints.

Note precisely what was legitimate. We did not multiply anything by dx . We wrote the differential of the function $u = x + y$ of two variables using the total differential, $du = dx + dy$, and then divided the resulting identity between real numbers by the nonzero number dx .

WHERE THIS GOES WRONG

Two errors, and both are silent. In a definite integral, converting the integrand but not the limits: $\int_0^1 xe^{x^2} dx$ with $u = x^2$ becomes $\frac{1}{2} \int_0^1 e^u du$ because u also runs from 0 to 1 here, but $\int_1^2$ would not be so forgiving. In an ODE, converting dy while leaving a stray y behind, which produces an equation in two unknowns that then gets “solved” by treating one of them as a constant.

SEE ALSO Substitution $u = ax + by + c$; the substitution $y = vx$ for homogeneous equations; recognising a differential as $d(\text{something})$.

The same bookkeeping runs an integral, and it is worth seeing the two side by side because the discipline is identical. In

$\int_1^2 \frac{x dx}{1+x^2}$, put $u = 1 + x^2$. Then $du = 2x dx$, so $x dx = \frac{1}{2} du$, and the limits move with the variable: $x = 1$ gives $u = 2$

and $x = 2$ gives $u = 5$. Hence the integral is $\frac{1}{2} \int_2^5 \frac{du}{u} = \frac{1}{2} \ln \frac{5}{2} = 0.4581454$. Nothing was cancelled and nothing was multiplied by dx ; the differential relation $du = 2x dx$ was solved for the group $x dx$ that actually appears, which is the general procedure. When the group you need is not present in the integrand, the substitution is the wrong one, and no amount of algebra on dx will fix it.

The last technique in the chapter reverses the direction of every previous one. Instead of computing the differential of a known expression, you are handed a combination of dx and dy and must recognise it as the differential of something. This is the single highest-yield pattern-matching skill in first-order differential equations, because a recognised combination turns an equation into one integration.

MEDIUM

30.7 Recognising a differential as $d(\text{something})$

CUE An equation containing the combination $x dy + y dx$, or $x dy - y dx$, or $tP' + P$, or any side that is visibly the derivative of a named expression. A hint that this is intended: the equation is not separable, not linear, and yet looks tidy.

METHOD Learn the integrable combinations by heart. Each is a product or quotient rule read backwards:

$$\begin{aligned} d(xy) &= x dy + y dx, & d\left(\frac{y}{x}\right) &= \frac{x dy - y dx}{x^2}, \\ d\left(\arctan \frac{y}{x}\right) &= \frac{x dy - y dx}{x^2 + y^2}, & d\left(\ln \frac{y}{x}\right) &= \frac{x dy - y dx}{xy}. \end{aligned}$$

The last three all begin with $x dy - y dx$; what distinguishes them is what you divide by, so identify that combination first and then look at the denominator you have been given. Two more are worth adding to the list because they turn up in exact equations: $d(\frac{1}{2} \ln(x^2 + y^2)) = \frac{x dx + y dy}{x^2 + y^2}$, and $d(x^a y^b) = x^{a-1} y^{b-1} (ay dx + bx dy)$, which is the general monomial and which contains $d(xy)$ as the case $a = b = 1$.

WORKED EXAMPLE

Solve $2y \frac{dy}{dx} = \sin x + x \cos x$ with $y(0) = 0$. Neither side needs integrating in the usual sense: the left is $\frac{d}{dx}(y^2)$ and the right is $\frac{d}{dx}(x \sin x)$, since $d(x \sin x) = \sin x dx + x \cos x dx$. Two functions with equal derivatives differ by a constant, and $y(0) = 0$ makes it zero:

$$y^2 = x \sin x.$$

A second instance, where the packaging is a quotient. Solve $x \frac{dy}{dx} - y = x^2 + y^2$. Divide by x^2 to expose the combination:

$$\frac{x dy - y dx}{x^2} = \left(1 + \frac{y^2}{x^2}\right) dx \implies \frac{d(y/x)}{1 + (y/x)^2} = dx,$$

so $\arctan \frac{y}{x} = x + C$ and $y = x \tan(x + C)$. With $y(\frac{\pi}{2}) = \frac{\pi}{2}$ we need $\tan(\frac{\pi}{2} + C) = 1$, giving $C = -\frac{\pi}{4}$ and $y = x \tan(x - \frac{\pi}{4})$; then $y(\frac{\pi}{4}) = 0$ and $y(1.2) = 0.5281360$, which matches Runge-Kutta to seven figures.

A third, the one that catches people: $t \frac{dA}{dt} - A = t^3$. Dividing by t^2 gives $\frac{d}{dt} \left(\frac{A}{t} \right) = t$, so $\frac{A}{t} = \frac{t^2}{2} + C$ and $A = \frac{t^3}{2} + Ct$.

WHERE THIS GOES WRONG

Looking only for the product rule. The three most useful combinations are quotient-rule packagings, and the giveaway is $x dy - y dx$ with a minus sign, which no product rule produces. Dividing by the wrong power is the other half of the same mistake: $x dy - y dx$ over x^2 gives $d(y/x)$, over y^2 gives $-d(x/y)$, and over xy gives $d \ln(y/x)$. Choose the one whose denominator the equation actually hands you rather than the one you remember best.

SEE ALSO Differentials as substitution bookkeeping; the exactness test; the integrating-factor machine, which manufactures the combination when it is not already present.

The combinations are few enough to tabulate, and the table is worth memorising as a unit rather than one entry at a time, because in practice you recognise the numerator first and then hunt through the possible denominators.

Combination	Divide by	Equals
$x dy + y dx$	1	$d(xy)$
$x dy + y dx$	xy	$d \ln xy $
$x dy - y dx$	x^2	$d(\frac{y}{x})$
$x dy - y dx$	y^2	$-d(\frac{x}{y})$
$x dy - y dx$	xy	$d \ln \left \frac{y}{x} \right $
$x dy - y dx$	$x^2 + y^2$	$d \arctan \frac{y}{x}$
$x dx + y dy$	$x^2 + y^2$	$d(\frac{1}{2} \ln(x^2 + y^2))$
$x dx + y dy$	$\sqrt{x^2 + y^2}$	$d\sqrt{x^2 + y^2}$
$ay dx + bx dy$	$x^{1-a} y^{1-b}$	$d(x^a y^b)$

The last row subsumes the first and is the one to reach for when the coefficients of $y dx$ and $x dy$ are different constants: an equation containing $3y dx + 2x dy$ is asking to be multiplied by $x^2 y$, after which it is $d(x^3 y^2)$. Reading the table in that direction, from a mismatched pair of coefficients to the monomial factor that repairs them, is exactly the search that the integrating-factor method automates for exact equations. Recognising the combination by eye is faster whenever it works, and it works often enough to be the first thing tried.

Which manipulations are legitimate

It is worth collecting the verdicts, because the rules are simple and the folklore around them is not. Writing $dy = f'(x) dx$ is a definition and always legal. Dividing that identity by the nonzero real number dx is legal, and gives $dy/dx = f'(x)$. Chaining is legal: if $y = f(u)$ and $u = g(x)$, then $dy = f'(u) du = f'(g(x))g'(x) dx$, and the chain rule is literally the statement that these two computations of dy agree. Writing $\int f(g(x))g'(x) dx = \int f(u) du$ is legal and is the substitution theorem, not a cancellation.

What is not legal is any move that requires dx to be nonzero and infinitely small at the same time, and any move that treats dy and dx as separately meaningful outside a ratio, an integral, or a 1-form. Separating variables, written as $h(y) dy = g(x) dx$, is a compressed instruction to integrate $h(y(x))y'(x)$ against x and apply the substitution theorem; it is legal, and the legality is a theorem about integrals rather than an algebraic identity about symbols. Finally, $d(dy)$ is not an ordinary second differential: d^2y/dx^2 is a single symbol for $(y')'$, and treating the numerator and denominator as separate powers is the one place where Leibniz notation genuinely misleads, since $\frac{d^2y}{dx^2} \neq \left(\frac{dy}{dx} \right)^2$ and the correct chain rule for second derivatives carries an extra term.

There is one more habit worth naming, because it is what converts the notation from a hazard into a tool. When a problem hands you a relation between quantities, differentiate the relation *in place* rather than solving it first. The parallel-resistor computation above is the model: differentiating $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$ takes one line, while solving for R and then differentiating the quotient takes five and introduces three opportunities for error. The same applies to implicit curves, to thermodynamic

relations such as $PV = nRT$ (whose differential $\frac{dP}{P} + \frac{dV}{V} = \frac{dT}{T}$ is the whole of the gas law's error analysis), and to any formula defined by a constraint rather than by an explicit expression. The differential of a relation is a relation between differentials, and it is almost always simpler than the thing it came from.

THE ONE-LINE TEST

Before any manipulation of dx , ask: which of the three readings am I using, and is the move licensed there? Approximation and error work is reading (i), finite increments, where every equation is an identity between real numbers. Substitution and separation are reading (ii), where the only rule is $u = g(x) \Rightarrow du = g'(x) dx$ and the justification is the substitution theorem. Exactness and path independence are reading (iii). If a manipulation is licensed in none of the three, it is not licensed.

Recognition matrix

If you see	Reach for	If that fails
"Approximately how much does y change"	$dy = f'(x) dx$ with $dx = \Delta x$	Second-order term $\frac{1}{2}f''(dx)^2$
"Estimate $f(a+h)$ ", a clean	$L(x) = f(a) + f'(a)(x-a)$	Taylor to degree two
Concavity known, error sign wanted	Sign of f'' : concave means overshoot	Lagrange remainder at ξ
A tolerance $\pm\delta$ on one input	$ dy = f' \delta$; then divide by y	Endpoint evaluation of f
A product, quotient, or power	Log first: $dy/y = (\ln f)' dx$	Term-by-term relative errors
$y = Cx^n$ with x uncertain	$\frac{dy}{y} = n \frac{dx}{x}$; C irrelevant	Direct differentiation
Several measured inputs, worst case	$ dz \leq f_x dx + f_y dy $	Evaluate at all corner points
Several inputs, random and independent	Quadrature: $\sigma_z^2 = f_x^2 \sigma_x^2 + f_y^2 \sigma_y^2$	Read the wording again
$F(x, y) = 0$, unsolvable for y	$F_x dx + F_y dy = 0$, then divide	Parametrise the curve
Vertical tangent wanted	Solve $F_y = 0$ jointly with $F = 0$	Swap roles: find $dx/dy = 0$
$d(xy)$ appearing as $y dx$	Product rule: restore $x dy$	Rewrite F before differentiating
A substitution $u = g(x)$ to be carried out	$du = g' dx$; solve for dx first	Re-express g' in terms of u
$x dy + y dx$	$d(xy)$; integrate once	Look for an integrating factor
$x dy - y dx$ over x^2	$d(y/x)$; substitute $v = y/x$	Over xy : $d \ln(y/x)$
$x dy - y dx$ over $x^2 + y^2$	$d \arctan(y/x)$	Polar coordinates
$tA' - A$ or $tP' + P$	Divide by t^2 , or read $(tP)'$	Standard integrating factor
d^2y/dx^2 treated as a square	It is not; use the chain rule with its extra term	Write $p = y'$ and reduce order

First-order equations: the classification decision

What this chapter is for. A first-order ordinary differential equation is solved by naming its type. There is no general method; there is a short list of named families, each with its own machine, and the entire difficulty of the subject at this level is deciding which family the equation in front of you belongs to. Almost every hour wasted on a first-order problem is a misclassification: an integrating factor computed for an equation that was not linear, a homogeneous substitution attempted on an equation that was not homogeneous, three pages of exactness machinery on an equation that separated in one line. This chapter gives the decision procedure three times over: as prose, as a numbered test order you can run in ninety seconds, and as a decision tree. For each family the emphasis is on the test you actually apply to the equation *as written*, which is usually not the definition. By the end you should be able to classify any first-order equation you meet, know which rearrangements to attempt before declaring a test failed, and recognise the common situation in which an equation belongs to two families at once and one of the two routes is six times shorter.

The audit

Classification is cheap and solving is expensive, so classify completely before solving at all. The audit below takes about a minute on paper and it is not optional: half of these tests can only be run on a particular arrangement of the equation, so running them in the wrong order means running them on the wrong object. In particular, separability and linearity are tests on the *derivative* form $y' = f(x, y)$, while exactness is a test on the *differential* form $M dx + N dy = 0$, and an equation can pass one while the other arrangement makes it look hopeless.

31.1 The classification audit before any solving

CORE

CUE Every first-order problem, without exception. Run this before writing a single line of solution.

METHOD Record, in this order: (i) order and degree, and whether y' appears to a power higher than one or inside a function; (ii) is it linear in y , that is $y' + P(x)y = Q(x)$; (iii) does it separate, $y' = g(x)h(y)$; (iv) written as $y' = f(x, y)$, is f homogeneous of degree zero, $f(tx, ty) = f(x, y)$; (v) written as $M dx + N dy = 0$, is $M_y = N_x$; (vi) does it match Bernoulli, Riccati, Clairaut, or Lagrange in form. Only then choose a method. Write the verdict of each test down; the discipline of writing “not linear: y^2 on the right” is what stops you from reaching for the integrating factor twenty seconds later.

WORKED EXAMPLE

Classify $\frac{dy}{dx} = \frac{2x + y}{x + 2y}$.

(ii) Not linear in y : y appears in a denominator. (iii) Not separable: the numerator and denominator are sums, and sums do not factor into a pure- x times a pure- y piece. (iv) Homogeneous of degree zero: replace (x, y) by (tx, ty) and the factor t cancels top and bottom. That verdict decides the method, and the substitution $v = y/x$ is the move. With $y = vx$, $y' = v + xv'$:

$$v + x \frac{dv}{dx} = \frac{2 + v}{1 + 2v}, \quad x \frac{dv}{dx} = \frac{2 - 2v^2}{1 + 2v}.$$

Separating and using $\frac{1 + 2v}{2(1 - v^2)} = \frac{3/4}{1 - v} - \frac{1/4}{1 + v}$ gives $-\frac{3}{4} \ln|1 - v| - \frac{1}{4} \ln|1 + v| = \ln|x| + c$, and clearing logarithms leaves the first integral

$$(x - y)^3(x + y) = C.$$

Numerical check: integrating from $(1, \frac{1}{2})$ to $x = 2$ gives $y = 1.6274956$, and $(x - y)^3(x + y) = 0.187500$ at both ends, against $(1 - \frac{1}{2})^3(1 + \frac{1}{2}) = 0.1875$.

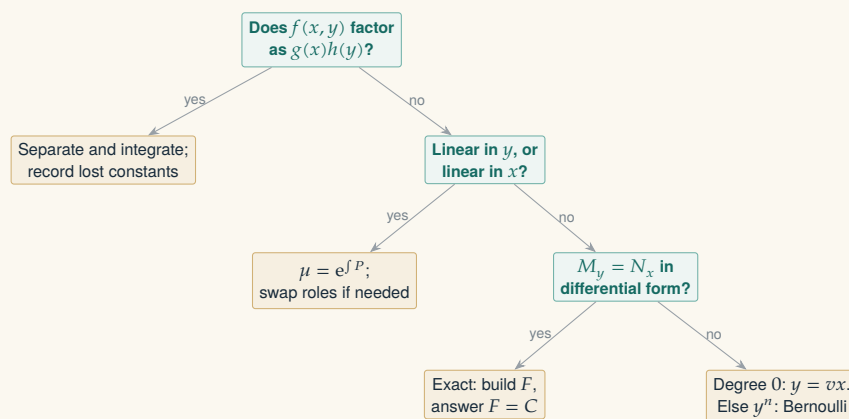
Reaching for $\mu = e^{\int P}$ on sight of a y on the left-hand side. The integrating-factor machine applies only once the equation is genuinely linear *in* y : any of y^2 , $\sqrt{y}$, e^y , $\ln y$, $1/y$, or y/y' disqualifies it, and x^2 , $\sin x$, e^x do not. The variable that must appear linearly is the unknown, not the independent one.

SEE ALSO Every technique in this chapter; the separable machine; the integrating-factor machine.

Here is the audit as an ordered checklist. The order is not arbitrary: it runs cheapest test first, and it runs the tests that can be settled by looking at the equation before the tests that require a computation.

1. **Simplify.** Factor every coefficient completely and cancel. Bloated coefficients hide two-line problems.
2. **Is x absent from the right side?** Then it is autonomous, hence separable, and the phase line answers qualitative questions without integrating.
3. **Is y absent from the right side?** Then it is a bare quadrature, $y = \int f(x) dx$.
4. **Does $f(x, y)$ factor as $g(x)h(y)$?** Separable. Cheapest real method; always test it first.
5. **Is it linear in y ?** Put it in standard form $y' + Py = Q$ and run the integrating factor.
6. **Is it linear in x instead?** Write $dx/dy + P(y)x = Q(y)$ and solve for $x(y)$.
7. **Is $f(tx, ty) = f(x, y)$?** Homogeneous of degree zero; substitute $y = vx$.
8. **Written as $M dx + N dy = 0$, is $M_y = N_x$?** Exact; find the potential F with $F = C$.
9. **Is it $y' + Py = Qy^n$?** Bernoulli; substitute $v = y^{1-n}$.
10. **Is it quadratic in y with all three coefficients present?** Riccati; you need a particular solution.
11. **Does y sit alone with a term $x \cdot y'$?** Clairaut if the rest is a function of y' only; Lagrange if the coefficient of x is a nonidentity function of y' .
12. **Does a weighting $x \sim 1$, $y \sim m$ make all terms equal?** Isobaric; substitute $y = vx^m$.
13. **Does the right side depend on x and y only through $ax + by + c$?** Substitute $u = ax + by + c$.

The same procedure, compressed to its branch points, is the tree below. It omits Riccati, isobaric, and Lagrange deliberately, since those are the rare cases you reach only after everything above has failed.



Three remarks about the tree. It is a tree of *tests*, not of methods, so a leaf tells you what the equation is rather than how much work solving it will be. It is ordered by cost, so an equation that reaches the bottom right leaf has already failed four cheaper tests and you have lost nothing by asking them. And it deliberately runs separability first even though linearity is the more powerful method, because separation is shorter whenever both apply.

Rearrangement comes before classification

Half of all misclassifications are not misjudgements about the equation but about its arrangement. Separability and linearity are tests on $y' = f(x, y)$; exactness is a test on $M dx + N dy = 0$; Clairaut and Lagrange are tests on the relation among x , y , and $p = y'$ with y isolated; and linearity in x is a test on dx/dy . An equation handed to you in one of these arrangements will fail tests belonging to the others, and the failure says nothing. Before recording any verdict, put the equation into the arrangement the test is about.

Four short recognitions make the point. First, $\frac{dy}{dx} = \frac{y}{x+y^2}$. It is not separable, not linear in y (the y^2 sits in the denominator), and not degree-zero homogeneous. Invert it: $\frac{dx}{dy} = \frac{x+y^2}{y} = \frac{x}{y} + y$, which is linear in x . With $\mu = 1/y$, $\left(\frac{x}{y}\right)' = 1$, so $x = y^2 + Cy$. Taking $C = 1$ and the point $(2, 1)$, the curve reaches $x = 6$ at $y = 2$, which numerical integration confirms to eleven figures.

Second, $2xy \frac{dy}{dx} = y^2 - x^2$. As written the derivative is not isolated and no test applies. Isolating it gives $y' = \frac{y^2 - x^2}{2xy}$, in which every term upstairs and downstairs is degree 2: degree-zero homogeneous. With $y = vx$, $xv' = -\frac{1+v^2}{2v}$, so $\ln(1+v^2) = -\ln|x| + c$ and $x^2 + y^2 = Cx$: the family of circles through the origin with centres on the x -axis. From $y(1) = 1$, $C = 2$ and $y(1.5) = \sqrt{3}/2 = 0.8660254$, confirmed numerically.

Third, $(y')^2 - xy' + y = 0$. Nothing in the audit applies to an equation quadratic in y' . Solve it for y instead: $y = xy' - (y')^2$, which is exactly the Clairaut shape with $f(p) = -p^2$.

Fourth, $\frac{dy}{dx} = -\frac{\cos y + y \cos x}{\sin x - x \sin y}$, which looks unapproachable. Cross-multiplying into differential form gives $(\cos y + y \cos x) dx + (\sin x - x \sin y) dy = 0$, and now $M_y = -\sin y + \cos x$ and $N_x = \cos x - \sin y$ agree: it is exact, with potential $F = x \cos y + y \sin x$, so the solution is

$$x \cos y + y \sin x = C.$$

Starting from $(1, 0.5)$, where $F = 1.2983181$, numerical integration to $x = 1.4$ gives $y = -0.0965647$, at which $F = 1.2983181$ again.

FOUR ARRANGEMENTS, FOUR FAMILIES OF TESTS

Derivative form $y' = f(x, y)$: separable, autonomous, degree-zero homogeneous, Bernoulli, Riccati. Standard linear form $y' + P(x)y = Q(x)$: the integrating factor. Differential form $M dx + N dy = 0$: exactness, integrating factors, the reverse product and quotient rules. Solved-for- y form $y = x\varphi(p) + \psi(p)$: Clairaut and Lagrange. Reciprocal form dx/dy : linearity in x . Converting between them is free; failing a test in the wrong arrangement costs the problem.

Tests on the shape of f

The first three tests look only at how x and y enter the right-hand side, and none of them requires a derivative to be computed. They are therefore the cheapest tests available and they should be exhausted first.

31.2 Separability test and the $g(x)h(y)$ factorisation

EASY

CUE $y' = f(x, y)$ where f factors into a pure- x piece times a pure- y piece, including the degenerate cases $f = g(x)$ alone and $f = h(y)$ alone.

METHOD Confirm the factorisation algebraically rather than by eye. The reliable test: compute $\frac{\partial}{\partial y} \ln|f|$ and check that it is free of x , or simply attempt the factorisation and see whether it closes. Then write $\frac{dy}{h(y)} = g(x) dx$ and integrate both sides, with a single constant on one side only.

WORKED EXAMPLE

$y' = x(1+y)$ with $y(0) = 0$ separates on sight: $\frac{dy}{1+y} = x dx$, so $\ln|1+y| = \frac{x^2}{2} + c$ and $y = e^{x^2/2} - 1$.

The instructive case is one that separates only after factoring. Consider $y' = xy + x + y + 1$. As a sum of four terms it looks like nothing at all; grouped, it is $x(y+1) + (y+1) = (x+1)(y+1)$, which separates:

$$\ln|1+y| = \frac{x^2}{2} + x + c, \quad y = Ae^{x^2/2+x} - 1.$$

With $y(0) = 0$, $A = 1$ and $y(1) = e^{3/2} - 1 = 3.4816891$, matching Runge-Kutta to ten figures. Note also the constant solution $y = -1$, which is $A = 0$ and was lost at the moment we divided by $1+y$.

$\frac{dy}{dx} = \frac{xe^x}{y\sqrt{1+y^2}}$ with $y(0) = 0$. The factorisation is immediate, $g(x) = xe^x$ and $h(y) = (y\sqrt{1+y^2})^{-1}$, so separability

is settled before any integration is attempted. That matters, because both integrals then need work and it would be wasteful to discover halfway through that the equation was not separable after all. Separating,

$$\int y\sqrt{1+y^2} dy = \int xe^x dx, \quad \frac{1}{3}(1+y^2)^{3/2} = (x-1)e^x + c,$$

the left by $u = 1 + y^2$ and the right by parts. From $y(0) = 0$, $\frac{1}{3} = -1 + c$, so

$$(1+y^2)^{3/2} = 3(x-1)e^x + 4,$$

and at $x = 1$ this gives $y = \sqrt{4^{2/3} - 1} = 1.2328188$. The moral is procedural: classification is a test on the *form* of the equation and is complete before any antiderivative is computed. Difficulty of integration is not evidence against separability.

WHERE THIS GOES WRONG

Declaring $y' = x + y$ separable. Sums do not separate; only products do. The trap is worth stating in its general form: separability is a property of the factored expression, and no amount of rearrangement makes $x + y$ into a product of a function of x and a function of y . Conversely, refusing to factor is how $xy + x + y + 1$ gets sent to the integrating-factor machine when one line of grouping would have finished it.

SEE ALSO The classification audit; hidden separability by factoring; autonomous recognition.

Separability fails for most equations, and the next question is whether the failure is superficial. Degree-zero homogeneity is the most common repair: an equation whose right side depends on x and y only through their ratio is separable in disguise, and the disguise is removed by one substitution. The test is a single algebraic check that costs nothing.

MEDIUM

31.3 Homogeneous-of-degree-zero test

CUE Replacing (x, y) by (tx, ty) leaves f unchanged. Equivalently, in the form $M dx + N dy = 0$, every term of M and every term of N carries the same total degree in x and y jointly.

METHOD Test $f(tx, ty) = f(x, y)$ by substituting t literally and cancelling. If it holds, f can be written as a function of $v = y/x$ alone, and $y = vx$ reduces the equation to a separable one. The degree count is the fast version: in $\frac{2x+y}{x+2y}$ every term upstairs and downstairs is degree 1, so the quotient is degree 0.

WORKED EXAMPLE

$y' = \frac{x^2 + y^2}{xy}$. Numerator and denominator are both homogeneous of degree 2, so the quotient is degree 0. Substituting $y = vx$:

$$v + x \frac{dv}{dx} = \frac{1+v^2}{v}, \quad x \frac{dv}{dx} = \frac{1+v^2-v^2}{v} = \frac{1}{v},$$

so $v dv = \frac{dx}{x}$, giving $\frac{1}{2}v^2 = \ln|x| + c$ and

$$y^2 = 2x^2 \ln|x| + Cx^2.$$

With $y(1) = 1$ we get $C = 1$ and $y(2) = \sqrt{8 \ln 2 + 4} = 3.0895271$, confirmed numerically to nine figures.

A second instance shows how little the family cares about what F looks like. In $\frac{dy}{dx} = \frac{y}{x} + \tan \frac{y}{x}$ the right side is a function of y/x by inspection, so the degree test passes without any cancelling. With $y = vx$ the v terms cancel as they always do:

$$x \frac{dv}{dx} = \tan v, \quad \cot v dv = \frac{dx}{x}, \quad \ln|\sin v| = \ln|x| + c,$$

so $\sin \frac{y}{x} = Cx$. With $y(1) = \frac{\pi}{6}$ we get $C = \frac{1}{2}$ and $y = x \arcsin \frac{x}{2}$, valid for $|x| \leq 2$; then $y(1.5) = 1.5 \arcsin 0.75 = 1.2720931$, confirmed numerically to ten figures. The lost solutions are the roots of $\tan v = 0$, that is $v = k\pi$, giving the lines $y = k\pi x$.

Confusing the two uses of the word “homogeneous”. Here it means the right-hand side has degree zero. In the linear theory it means the forcing term $Q(x)$ is zero. The two senses are unrelated, they both appear in this chapter, and $y' = y/x$ happens to satisfy both, which is exactly the sort of coincidence that entrenches the confusion. When in doubt, say “degree-zero homogeneous” or “unforced” and the ambiguity disappears.

SEE ALSO The substitution $y = vx$; isobaric equations, which are the weighted generalisation; the classification audit.

When the degree test fails, it sometimes fails only because x and y were weighted equally. Counting y with a weight $m \neq 1$ produces a larger family that behaves identically, and the weight is determined by a small linear system rather than guessed. This is the last resort among the shape tests, and it is worth knowing chiefly because equations built to defeat the standard tests are often built this way.

31.4 Isobaric and weighted-homogeneous equations

HARD

CUE The equation is not degree-zero homogeneous, but the failure looks systematic: one term is off by a fixed power of x . Assign weight 1 to x , weight m to y , weight 1 to dx , weight m to dy , and hence weight $m - 1$ to y' .

METHOD Demand that all terms carry equal weight. This is one linear equation in m ; solve it, then substitute $y = vx^m$, which makes the equation separable in (x, v) . If different pairs of terms force different values of m , the equation is not isobaric and you must stop, not average.

WORKED EXAMPLE

$\frac{dy}{dx} = \frac{y}{2x} + \frac{x^2}{2y}$. Weights: the left side is $m - 1$; the first term on the right is $m - 1$; the second is $2 - m$. Equating the first and third, $m - 1 = 2 - m$ gives $m = \frac{3}{2}$, and all three then read $\frac{1}{2}$. Substitute $y = vx^{3/2}$, so $y' = v'x^{3/2} + \frac{3}{2}vx^{1/2}$:

$$v'x^{3/2} + \frac{3}{2}vx^{1/2} = \frac{1}{2}vx^{1/2} + \frac{1}{2v}x^{1/2} \implies x \frac{dv}{dx} = \frac{1 - 2v^2}{2v},$$

which separates to $-\frac{1}{2} \ln|1 - 2v^2| = \ln|x| + c$, hence $1 - 2v^2 = Ax^{-2}$ and, since $y^2 = v^2x^3$,

$$y^2 = \frac{1}{2}x^3 + Cx.$$

With $y(1) = 1$: $C = \frac{1}{2}$, so $y(2) = \sqrt{4 + 1} = \sqrt{5} = 2.2360680$, confirmed numerically.

The shortcut worth noticing: setting $w = y^2$ turns the original equation into the linear equation $w' - \frac{w}{x} = x^2$ in two lines, because $w' = 2yy'$ absorbs both terms. Every isobaric equation of this shape is a Bernoulli equation wearing a different hat, and if you spot the Bernoulli form first you never need the weight calculation.

Finding m is quicker in practice than the description suggests, because you only need two terms to determine it and the rest are then a consistency check. Write the weight of each term as a number: $x^a y^b dx$ has weight $a + bm + 1$ and $x^a y^b dy$ has weight $a + bm + m$. Set any two equal, solve the resulting linear equation for m , and verify the remaining terms. A useful sanity check: a degree-zero homogeneous equation is the case $m = 1$, so if your weight calculation returns $m = 1$ you should have caught the equation one test earlier.

WHERE THIS GOES WRONG

Solving the weight equation from an inconsistent set of terms. If terms one and two force $m = \frac{3}{2}$ and terms one and three force $m = 2$, the equation is not isobaric; there is no correct m and no substitution $y = vx^m$ will separate it. The other trap is arithmetic: dy carries weight m , so $y' = dy/dx$ carries weight $m - 1$, not m . Getting that wrong shifts every equation by one and produces a plausible wrong m .

SEE ALSO The homogeneous-of-degree-zero test; Bernoulli recognition; the substitution $y = vx$.

Tests on how y enters algebraically

The second group of tests ignores x entirely and asks only about the algebraic degree in which the unknown appears. Linear, Bernoulli, and Riccati form a ladder: first power only, first power plus one extra power, and full quadratic. Recognising which rung you are on is usually a matter of reading the right-hand side once.

31.5 Linearity test and the standard form $y' + P(x)y = Q(x)$

CUE y and y' each appear to the first power only, with no product yy' and no y inside a transcendental function. Coefficients of x may be arbitrarily ugly; that is irrelevant to linearity.

METHOD Divide through by the coefficient of y' to reach standard form *before* computing anything. The coefficient you divide by may vanish; record those points, because they bound the interval on which the solution exists and is unique. Only then form $\mu = e^{\int P dx}$.

WORKED EXAMPLE

$\frac{dy}{dx}(x + x^2) + y(x^3 + x^4) = x^3 + x^4$ with $y(0) = 2$.

Do not reach for μ yet. Factor: $x(1 + x)y' + x^3(1 + x)y = x^3(1 + x)$. The common factor $x(1 + x)$ divides out on the whole equation, leaving

$$y' + x^2y = x^2,$$

whose integrating factor is $\mu = e^{x^3/3}$. Then $(\mu y)' = x^2 e^{x^3/3}$, so $\mu y = e^{x^3/3} + C$ and $y = 1 + Ce^{-x^3/3}$. With $y(0) = 2$, $C = 1$ and

$$y(1) = 1 + e^{-1/3} = 1.7165313,$$

which a numerical integration reproduces to ten figures. Had you not factored, you would have computed $P = \frac{x^3 + x^4}{x + x^2}$ and $\int P$ anyway, arriving at the same place after considerably more work; had you not divided at all, you would have used $P = x^3 + x^4$ and got a wrong answer with no warning.

WHERE THIS GOES WRONG

Applying $\mu = e^{\int P}$ before dividing by the leading coefficient. This silently uses the wrong P and there is no internal check that catches it. The second trap is failing to factor: coefficients like $(x + x^2)$ and $(x^3 + x^4)$ are put there precisely so that a solver who does not factor will drown. The third is ignoring the vanishing of the leading coefficient at $x = 0$ and $x = -1$; here the factor cancels and the reduced equation is fine at $x = 0$, but in general those are the endpoints of the interval of validity.

SEE ALSO The integrating-factor machine; swapping the roles of x and y ; the classification audit.

If the equation fails linearity by exactly one power of y on the right, it is not lost: the failure is repairable by substitution, and the repair is completely mechanical. This is the most valuable single recognition in first-order theory, because Bernoulli equations look nonlinear and are not.

MEDIUM

31.6 Bernoulli recognition

CUE $y' + P(x)y = Q(x)y^n$ with $n \neq 0, 1$: a linear equation spoiled by exactly one power of y on the right. The exponent n need not be an integer, and $n = \frac{1}{2}$ and $n = 2$ are the common cases.

METHOD Divide by y^n and set $v = y^{1-n}$. Since $v' = (1 - n)y^{-n}y'$, the equation becomes

$$v' + (1 - n)P(x)v = (1 - n)Q(x),$$

which is linear in v . Solve, then invert. Note that $y \equiv 0$ solves the original whenever $n > 0$ and is lost by the division; record it.

WORKED EXAMPLE

$y' + \frac{y}{x} = xy^2$ with $y(1) = \frac{1}{3}$. Here $n = 2$, so $v = y^{-1}$ and $1 - n = -1$:

$$v' - \frac{v}{x} = -x.$$

Integrating factor $\mu = 1/x$: $\left(\frac{v}{x}\right)' = -1$, so $\frac{v}{x} = -x + C$ and $v = Cx - x^2$. The condition $y(1) = \frac{1}{3}$ means $v(1) = 3$, so $C = 4$ and

$$y = \frac{1}{4x - x^2}, \quad y(3) = \frac{1}{12 - 9} = \frac{1}{3},$$

confirmed numerically. The solution exists on $(0, 4)$ and blows up at both ends, which the closed form shows and the original equation does not.

A fractional instance: $xy' = y + x^2\sqrt{y}$ with $y(1) = 1$. In standard form $y' - \frac{y}{x} = x\sqrt{y}$, so $n = \frac{1}{2}$, $v = \sqrt{y}$, and $v' - \frac{v}{2x} = \frac{x}{2}$. With $\mu = x^{-1/2}$ this gives $v = \frac{x^2}{3} + C\sqrt{x}$; from $v(1) = 1$, $C = \frac{2}{3}$, so $y = \left(\frac{x^2}{3} + \frac{2\sqrt{x}}{3}\right)^2$ and $y(4) = \left(\frac{16}{3} + \frac{4}{3}\right)^2 = \frac{400}{9} = 44.4444$, verified numerically.

WHERE THIS GOES WRONG

Sign errors in the factor $(1 - n)$, which is negative for every $n > 1$ and therefore flips the sign of both P and Q . Verify by differentiating $v = y^{1-n}$ explicitly rather than quoting the formula. And do not forget the lost solution $y \equiv 0$: for $n = 2$ it is a genuine solution of $y' + y/x = xy^2$, and no choice of C in $y = 1/(Cx - x^2)$ produces it.

SEE ALSO The integrating-factor machine; Riccati recognition; isobaric equations, which frequently reduce to Bernoulli.

Bernoulli equations are also the family most likely to be soluble by a shorter route, because the substitution $v = y^{1-n}$ is often the same thing as a reverse product rule that is already visible. Compare the two treatments of $xy' + y = x^2y^2$. As Bernoulli: divide by x to reach $y' + \frac{y}{x} = xy^2$, set $v = y^{-1}$, solve $v' - \frac{v}{x} = -x$ with $\mu = 1/x$, obtain $v = Cx - x^2$, invert. Six steps. As a product rule: the left-hand side is $(xy)'$, and the right-hand side is $(xy)^2$, so with $w = xy$ the equation is $w' = w^2$, giving $-\frac{1}{w} = x + c$ and $w = \frac{1}{C - x}$, hence $y = \frac{1}{x(C - x)}$. Two steps. With $y(1) = 1$ both give $C = 2$ and $y(1.5) = \frac{4}{3}$, which the numerics confirm. Looking for the packaged derivative before running the machine is worth the ten seconds it costs.

One more power of y and the equation becomes Riccati, at which point the mechanical guarantee disappears. Riccati equations are the first family in this chapter that is not solvable in closed form in general, and the recognition therefore has a different purpose: it tells you what extra ingredient you must find or be given.

31.7 Riccati recognition

MEDIUM

CUE $y' = q_0(x) + q_1(x)y + q_2(x)y^2$: quadratic in y , with the quadratic coefficient not identically zero. If the problem hands you one solution unprompted, that is the tell; it is not a hint, it is the missing ingredient.

METHOD With a known particular solution y_1 , substitute $y = y_1 + \frac{1}{v}$. The quadratic terms cancel identically and what remains is linear:

$$v' + (q_1 + 2q_2y_1)v = -q_2.$$

Solve for v and add back. If no particular solution is offered, hunt for one among the simple shapes: a constant, a polynomial of low degree, a/x , or ax . Substitute the ansatz and see whether the coefficients can be matched.

WORKED EXAMPLE

$y' = 1 + x^2 - 2xy + y^2$ with $y(0) = 2$. The shape is Riccati with $q_0 = 1 + x^2$, $q_1 = -2x$, $q_2 = 1$. Guess $y_1 = x$: then $y'_1 = 1$ and the right side is $1 + x^2 - 2x^2 + x^2 = 1$, so the guess works. Put $y = x + \frac{1}{v}$; since $q_1 + 2q_2y_1 = -2x + 2x = 0$, the linear equation is simply $v' = -1$, giving $v = C - x$ and

$$y = x + \frac{1}{C - x}.$$

From $y(0) = 2$ we get $C = \frac{1}{2}$, so $y(0.3) = 0.3 + \frac{1}{0.2} = 5.3$, confirmed numerically.

A case where the particular solution must be hunted: $y' = y^2 - \frac{2}{x^2}$ with $y(1) = 2$. Try $y_1 = a/x$: then $-\frac{a}{x^2} = \frac{a^2 - 2}{x^2}$, so $a^2 + a - 2 = 0$ and $a = 1$ or $a = -2$. Take $y_1 = \frac{1}{x}$. With $q_1 = 0$, $q_2 = 1$, the linear equation is $v' + \frac{2}{x}v = -1$, whose solution with $\mu = x^2$ is $v = -\frac{x}{3} + \frac{C}{x^2}$. Hence

$$y = \frac{1}{x} + \frac{3x^2}{3C - x^3},$$

and $y(1) = 2$ forces $3C = 4$, giving $y = \frac{1}{x} + \frac{3x^2}{4-x^3}$. At $x = 1.4$ this is 5.3958144, matching Runge-Kutta to eight figures.

WHERE THIS GOES WRONG

Assuming that a Riccati equation has an elementary closed form. Generically it does not: $y' = x^2 + y^2$ has no elementary solution at all, and its solutions are ratios of Bessel functions. If no particular solution is offered and none of the simple ansatzes matches, the honest answer is a series solution or a numerical one, not a longer search.

SEE ALSO Bernoulli recognition; the integrating-factor machine; series solutions about an ordinary point.

Two further facts about Riccati equations are worth carrying, because they tell you what is possible before you spend time.

First, the substitution $y = -\frac{u'}{q_2 u}$ converts any Riccati equation into a *second-order linear* equation for u ,

$$u'' - \left(q_1 + \frac{q_2'}{q_2} \right) u' + q_0 q_2 u = 0,$$

which is why Riccati equations are exactly as hard as second-order linear equations and no harder. When the second-order equation is one you recognise, the Riccati equation is solved; when it is Bessel's, as for $y' = x^2 + y^2$, the solution is a ratio of Bessel functions and there is no elementary form to find. Second, one particular solution unlocks everything, but two unlock more: if y_1 and y_2 are both known, the general solution follows by a single quadrature, and if three are known it follows by pure algebra, since the cross-ratio of any four solutions of a Riccati equation is constant in x . That last fact is the practical reason to keep hunting after finding one solution: the second one is usually cheaper than the linear equation it would otherwise cost.

A test on the differential form

Exactness is the only test in the audit that requires you to rewrite the equation before testing, and the only one that requires computing derivatives of the coefficients. It repays the effort, because when it passes the solution is immediate and needs no integration of the unknown at all.

31.8 Exactness test

MEDIUM

CUE The equation is already written, or is trivially written, as $M(x, y) dx + N(x, y) dy = 0$. Any $y' = f(x, y)$ can be put in this shape, so the real cue is that the natural arrangement makes M and N symmetric and tidy.

METHOD Compute the two mixed partials:

$$\frac{\partial M}{\partial y} \quad \text{against} \quad \frac{\partial N}{\partial x}.$$

Equality on a simply connected region means $M dx + N dy = dF$ for some potential F , and the general solution is $F(x, y) = C$. Recover F by integrating M in x , differentiating the result in y , and matching against N to pin down the function of y alone.

WORKED EXAMPLE

$2xy^3 dx + (3x^2y^2 + 4y) dy = 0$ with $y(1) = 2$. Test: $M_y = 6xy^2$ and $N_x = 6xy^2$, so the form is exact. Integrate M in x : $F = x^2y^3 + g(y)$. Differentiate in y : $F_y = 3x^2y^2 + g'(y)$, which must equal $3x^2y^2 + 4y$, so $g' = 4y$ and $g = 2y^2$. Hence

$$x^2y^3 + 2y^2 = C, \quad C = 1 \cdot 8 + 2 \cdot 4 = 16.$$

The answer is implicit and correctly left so; solving the cubic for y would be a service to nobody.

WHERE THIS GOES WRONG

Testing M_x against N_y , which is the wrong pair and which is right roughly half the time by accident. The mnemonic: each coefficient is differentiated with respect to the variable it is *not* paired with, so M , which multiplies dx , gets differentiated in y . The second trap is domain: $M_y = N_x$ guarantees a potential only on a simply connected region. On an annulus it does not, and $\frac{-y dx + x dy}{x^2 + y^2}$ is the standard counterexample, passing the test everywhere off the origin while having no single-valued potential there.

SEE ALSO Integrating factors that make a form exact; the total differential; recognising a differential as d of something.

A failed exactness test is not the end of the enquiry, and this is the one place in the audit where a failure carries useful information. If $M_y \neq N_x$, ask whether a multiplier μ can be found that makes the form exact. Two cases are cheap enough to test on sight:

$$\frac{M_y - N_x}{N} \text{ free of } y \Rightarrow \mu(x) = \exp \int \frac{M_y - N_x}{N} dx, \quad \frac{N_x - M_y}{M} \text{ free of } x \Rightarrow \mu(y) = \exp \int \frac{N_x - M_y}{M} dy.$$

Compute the first quotient; if y survives in it, compute the second; if x survives in that one too, neither single-variable factor exists and you should go back to the audit.

WORKED EXAMPLE: TWO INTEGRATING FACTORS

$(2y^2 + 3x) dx + 2xy dy = 0$. Here $M_y = 4y$ and $N_x = 2y$, so the form is not exact. The first quotient is $\frac{4y - 2y}{2xy} = \frac{1}{x}$, free of y , so $\mu = e^{\int dx/x} = x$. Multiplying through,

$$(2xy^2 + 3x^2) dx + 2x^2y dy = 0, \quad M_y = 4xy = N_x,$$

and $F = x^2y^2 + x^3 = C$. From $y(1) = 1$, $C = 2$ and $y = \sqrt{(2 - x^3)/x^2}$; at $x = 1.2$ this is 0.4346135, matching Runge-Kutta to eleven figures, and the solution ends at $x = 2^{1/3}$ where y reaches zero.

The other case: $y dx + (2x - ye^y) dy = 0$ has $M_y = 1$, $N_x = 2$, and $\frac{M_y - N_x}{N}$ is not free of y . Try the second quotient: $\frac{N_x - M_y}{M} = \frac{1}{y}$, free of x , so $\mu = y$. Then $y^2 dx + (2xy - y^2e^y) dy = 0$ is exact with

$$F = xy^2 - e^y(y^2 - 2y + 2) = C.$$

Starting from $(1, 1)$, where $F = 1 - e = -1.7182818$, integration to $x = 1.4$ gives $y = 1.3306544$, at which F is -1.7182818 again.

Degree, and equations that must be factored in y'

Item (i) of the audit asks for the degree, and it is the item most often skipped because it usually returns the answer “one”. When it does not, everything downstream changes: an equation of degree two in y' is not one differential equation but two, glued together, and no test in the audit applies until they are separated.

The procedure is to treat the equation as a polynomial in $p = y'$ and factor it. If it factors over the rationals, each factor is a first-order equation of degree one, each is classified and solved independently, and the general solution is the union of the families. Written as a single relation, the general solution is the product of the individual first integrals set equal to zero.

WORKED EXAMPLE: DEGREE TWO, FACTORABLE

$(y')^2 + 2xy' - 3x^2 = 0$. As a quadratic in p , the discriminant is $4x^2 + 12x^2 = 16x^2$, a perfect square, so

$$p = \frac{-2x \pm 4x}{2} = x \quad \text{or} \quad p = -3x,$$

equivalently $(p - x)(p + 3x) = 0$. Each branch is a bare quadrature: $y = \frac{x^2}{2} + C$ and $y = -\frac{3x^2}{2} + C$. The general solution is the pair of families, and written as one relation it is

$$\left(y - \frac{x^2}{2} - C\right)\left(y + \frac{3x^2}{2} - C\right) = 0.$$

Through every point of the plane pass exactly two solution curves, one from each family, which is what degree two means geometrically: the equation prescribes two slopes at each point rather than one, and uniqueness fails everywhere.

When the polynomial in p does not factor, three classical escapes remain, and which one applies depends on which variable can be isolated. If the equation can be solved for y , differentiate with respect to x and hope for an equation in x and p ; this is exactly the Clairaut and Lagrange manoeuvre. If it can be solved for x , differentiate with respect to y instead, obtaining an equation in y and p . If neither is possible but the equation admits a parametrisation, for instance $x = \varphi(t)$, $p = \psi(t)$, use $dy = p dx$ to recover y by a single integration. All three end with a parametric solution in p or t , which is a complete answer and should not be forced into explicit form.

For a first-order equation of degree one, the initial condition $y(x_0) = y_0$ picks out a single solution wherever the standard hypotheses hold. For degree n it picks out up to n , one for each root p of the polynomial at (x_0, y_0) , and a well-posed problem must specify the slope as well as the value. This is also where singular solutions live: the locus where two roots coincide is the discriminant curve, and it is a solution of the equation whenever it is a curve at all. The Clairaut envelope is the standard example, and it is no accident that Clairaut equations are of degree at least two in p whenever f is nonlinear.

Tests on how y' itself enters

Every family so far has had y' isolated. Two important families do not, and they are recognised not by the shape of f but by the shape of the equation as a relation among x , y , and $p = y'$. Both are solved by differentiating the equation, which is a move that appears nowhere else in the chapter and should be reserved for exactly these two shapes.

HARD

31.9 Clairaut recognition

CUE The exact shape $y = xp + f(p)$ with $p = y'$: the unknown isolated on the left, a bare product $x \cdot p$, and a function of p alone. The presence of $(y')^2$ or $\sqrt{1 + (y')^2}$ with y solved for is the usual disguise.

METHOD Differentiate the whole equation with respect to x . Since $y' = p$, the terms p cancel and what remains factors:

$$(x + f'(p)) \frac{dp}{dx} = 0.$$

The branch $p' = 0$ gives $p = C$ and hence the general solution $y = Cx + f(C)$, a family of straight lines. The branch $x + f'(p) = 0$ gives the **singular solution**, the envelope of that family, in parametric form $(x, y) = (-f'(p), -pf'(p) + f(p))$.

WORKED EXAMPLE

$y = xy' + (y')^2$. General solution: $y = Cx + C^2$. Singular: $x + 2p = 0$, so $p = -\frac{x}{2}$ and

$$y = x \left(-\frac{x}{2} \right) + \frac{x^2}{4} = -\frac{x^2}{4}.$$

Check by substitution: $y' = -\frac{x}{2}$, and $xy' + (y')^2 = -\frac{x^2}{2} + \frac{x^2}{4} = -\frac{x^2}{4} = y$. The parabola is tangent to every line $y = Cx + C^2$ and is not a member of the family.

A second, geometrically famous instance: $y = xp + \sqrt{1 + p^2}$. The lines $y = Cx + \sqrt{1 + C^2}$ are exactly the lines at distance 1 from the origin. The singular branch is $x + \frac{p}{\sqrt{1 + p^2}} = 0$, so $p = -\frac{x}{\sqrt{1 - x^2}}$ and

$$y = -\frac{x^2}{\sqrt{1 - x^2}} + \frac{1}{\sqrt{1 - x^2}} = \sqrt{1 - x^2},$$

the unit circle, which is precisely the envelope of the family of tangent lines to it. At $x = 0.6$ both sides give 0.8.

WHERE THIS GOES WRONG

Reporting only the line family and stopping. The singular solution is usually the one being asked for, it satisfies the equation, and it is not obtainable from the general solution by any choice of C . Uniqueness fails on the envelope, which is the structural reason it exists at all: through each of its points passes both the envelope and a tangent line.

SEE ALSO Lagrange equations; singular solutions and envelopes; the classification audit.

The envelope deserves a paragraph of its own, since it is the only object in this chapter that is a solution without being a member of the general family, and its existence tends to be treated as a curiosity rather than as the structural fact it is. A one-parameter family of curves $G(x, y, C) = 0$ has an envelope wherever the system $G = 0$, $\partial G / \partial C = 0$ has a solution: geometrically, the envelope is the locus of points where two infinitesimally near members of the family meet. For the Clairaut family $y = Cx + f(C)$ the second equation is $x + f'(C) = 0$, which is precisely the branch discarded when the differentiated equation was factored. So the two branches of the factorisation are not an accident of the algebra; they are the family and its envelope, and the singular solution is exactly the curve that the general solution touches everywhere and contains nowhere. Uniqueness fails at each of its points because two solution curves pass through, the envelope and the tangent line, with the same slope.

Loosening Clairaut's requirement that the coefficient of x be p itself gives the Lagrange family. The same differentiation trick applies, but the result is no longer a factorisation; it is a linear equation, and the twist is that it is linear in x as a function of p , so the roles of dependent and independent variable must be exchanged.

31.10 Lagrange and d'Alembert equations

HARD

CUE $y = x\varphi(p) + \psi(p)$ with $\varphi(p) \not\equiv p$: the Clairaut shape, but with the coefficient of x a nontrivial function of p .

METHOD Differentiate in x and rearrange, treating p as the new independent variable and x as the unknown. The result is linear:

$$\frac{dx}{dp} + \frac{\varphi'(p)}{\varphi(p) - p} x = \frac{-\psi'(p)}{\varphi(p) - p}.$$

Solve for $x(p)$ by the integrating factor, back-substitute into the original relation to get $y(p)$, and report the solution parametrically in p . Do not attempt to eliminate p unless it comes out cleanly.

WORKED EXAMPLE

$y = 2xp - p^2$, so $\varphi(p) = 2p$ and $\psi(p) = -p^2$. Differentiating with respect to x gives $p = 2p + 2xp' - 2pp'$, that is $-p = (2x - 2p)\frac{dp}{dx}$, and inverting,

$$\frac{dx}{dp} + \frac{2x}{p} = 2.$$

With $\mu = p^2$, $(xp^2)' = 2p^2$, so $xp^2 = \frac{2}{3}p^3 + C$ and

$$x = \frac{2p}{3} + \frac{C}{p^2}, \quad y = 2xp - p^2 = \frac{p^2}{3} + \frac{2C}{p}.$$

Taking $C = 0$ gives $x = \frac{2}{3}p$ and $y = \frac{1}{3}p^2 = \frac{3}{4}x^2$, a genuine solution. The parametrisation is verified by $\frac{dy/dp}{dx/dp} = p$, which is what $y' = p$ demands.

A second instance, with a genuinely two-parameter-looking answer that is not: $y = xp^2 + p^2$, so $\varphi(p) = p^2$ and $\psi(p) = p^2$. Differentiating gives $p = p^2 + 2p(x+1)\frac{dp}{dx}$, and inverting with $w = x+1$,

$$\frac{dw}{dp} = \frac{2w}{1-p}, \quad \ln|w| = -2\ln|1-p| + c, \quad w = \frac{C}{(1-p)^2}.$$

Hence $x = \frac{C}{(1-p)^2} - 1$ and $y = (x+1)p^2 = \frac{Cp^2}{(1-p)^2}$. The parametrisation is confirmed by $\frac{dy/dp}{dx/dp} = \frac{2Cp(1-p)^{-3}}{2C(1-p)^{-3}} = p$, and by direct substitution: with $C = 1$ and $p = \frac{1}{2}$ the pair is $(x, y) = (3, 1)$, and $xp^2 + p^2 = \frac{3}{4} + \frac{1}{4} = 1$. The roots of $\varphi(p) - p = p(p-1)$ give the two singular lines $y \equiv 0$ and $y = x+1$, both of which satisfy the original equation and neither of which is in the parametric family.

WHERE THIS GOES WRONG

Dividing by $\varphi(p) - p$ without checking its roots. Each root p_0 with $\varphi(p_0) = p_0$ supplies an extra straight-line solution $y = p_0x + \psi(p_0)$ that the parametric family misses. In the worked example $\varphi(p) - p = p$, whose root $p = 0$ gives the solution $y = 0$, and indeed $y \equiv 0$ satisfies $y = 2xp - p^2$.

SEE ALSO Clairaut recognition; singular solutions and envelopes; the integrating-factor machine.

Tests on what is absent

The last two recognitions are about structure rather than algebra. One notices that a variable is missing; the other notices that the equation is in the wrong orientation. Both are cheap, both are frequently decisive, and both are routinely overlooked because they ask you to look at what is not there.

31.11 Autonomous recognition

CUE $y' = f(y)$: the independent variable does not appear on the right at all. In applied problems this is the norm, since a law that does not depend explicitly on the clock is autonomous by construction.

METHOD Three facts, in order of usefulness. First, autonomous equations are always separable. Second, they are translation-invariant: if $y(x)$ solves, so does $y(x - x_0)$, so the initial time is a free parameter and never affects the shape of the solution. Third, and most important, the entire qualitative behaviour is determined by the zeros and sign pattern of f : equilibria are the roots of f , and a solution starting between two consecutive roots is monotone and converges to one of them. Do the phase line before integrating.

WORKED EXAMPLE

$y' = y(1 - y)$, the logistic equation. Equilibria at $y = 0$ and $y = 1$. On $(0, 1)$, $f > 0$, so solutions increase; on $(1, \infty)$, $f < 0$, so solutions decrease; on $(-\infty, 0)$, $f < 0$, so solutions decrease. Reading the arrows, $y = 0$ is unstable and $y = 1$ is stable, and every solution with $y(0) > 0$ tends to 1. That is the entire qualitative answer and it took no integration.

The closed form, if genuinely wanted, is $y = \frac{1}{1 + Ae^{-x}}$; with $y(0) = \frac{1}{2}$, $A = 1$ and $y(2) = \frac{1}{1 + e^{-2}} = 0.8807971$, which the numerics confirm. Note how much less this tells you than the phase line about, say, the behaviour of a solution starting at $y(0) = -0.1$, which the formula describes only after you notice that $A = -11$ makes the denominator vanish at a finite x .

WHERE THIS GOES WRONG

Grinding out the closed form when the question asks only for $\lim_{x \rightarrow \infty} y$ or for the stability of an equilibrium. The phase line answers both in one line and the closed form invites algebra errors. The complementary trap: reading stability off the sign of f' at the equilibrium without checking that $f' \neq 0$ there. If $f'(y_0) = 0$ the test is silent, and a double root such as $y' = (y - 1)^2$ gives an equilibrium that attracts from below and repels from above.

SEE ALSO The separable machine; phase-line analysis; the classification audit.

The phase line deserves one more paragraph, because it is the cheapest analytical tool in the whole subject and it is routinely ignored. To draw it: mark the roots of f on a vertical axis, determine the sign of f on each interval between them, and draw an upward arrow where $f > 0$ and a downward arrow where $f < 0$. A root with arrows pointing in from both sides is stable; out from both sides, unstable; in from one side and out from the other, semistable. That is the entire classification, and it is exact rather than approximate. For $y' = (y - 1)^2(y - 3)$ the roots are $y = 1$ (double) and $y = 3$. On $(-\infty, 1)$ and on $(1, 3)$ the sign is negative; on $(3, \infty)$ it is positive. So $y = 1$ is semistable, attracting from above and repelling from below, and $y = 3$ is unstable. The derivative test $f'(y_0)$ is silent at $y = 1$, where it vanishes, and the sign pattern is not; that is why the sign pattern is the primary tool and the derivative test is the shortcut.

MEDIUM

31.12 Swapping the roles of x and y

CUE $\frac{dy}{dx}$ is a hopeless nonlinear mess, but the reciprocal is not. The specific pattern to watch for: y appears inside transcendental functions while x appears only to the first power, which is exactly the signature of an equation that is linear in x .

METHOD Write $\frac{dx}{dy} = \frac{1}{f(x, y)}$, put it in the standard linear form $\frac{dx}{dy} + P(y)x = Q(y)$, and run the integrating-factor machine with x as the unknown function. Invert at the very end, and only if the question insists on y explicitly.

WORKED EXAMPLE

$\frac{dy}{dx} = \frac{1}{e^y - x}$. In y this is hopeless. Inverted, $\frac{dx}{dy} = e^y - x$, that is $\frac{dx}{dy} + x = e^y$, which is linear with $\mu = e^y$:

$$(xe^y)' = e^{2y}, \quad xe^y = \frac{1}{2}e^{2y} + C, \quad x = \frac{1}{2}e^y + Ce^{-y}.$$

Verify directly: $\frac{dx}{dy} = \frac{1}{2}e^y - Ce^{-y}$, while $e^y - x = e^y - \frac{1}{2}e^y - Ce^{-y}$ is the same thing. The solution is a perfectly good answer in the form $x = x(y)$; inverting it would require the Lambert function and would tell you nothing new.

A second instance, and the one whose shape you should learn to spot: $\frac{dy}{dx} = \frac{1}{2y-x}$. Any equation whose right side is the reciprocal of something linear in x is linear in x after inversion. Here $\frac{dx}{dy} + x = 2y$, so with $\mu = e^y$, $(xe^y)' = 2ye^y$ and $xe^y = 2(y-1)e^y + C$, giving

$$x = 2y - 2 + Ce^{-y}.$$

Numerically, following the solution with $C = 1$ from $y = 0.5$ to $y = 1.2$ reproduces the formula to twelve figures. Attempting this in the original orientation leads nowhere: the equation is not separable, not linear in y , not homogeneous, and not a function of any single linear combination.

WHERE THIS GOES WRONG

Solving for $x(y)$ correctly and then burning effort inverting to $y(x)$ when the quantity actually requested (a value, a constant of integration, an asymptote, an intercept) can be read off the implicit form directly. The other trap is forgetting that $dx/dy = 1/(dy/dx)$ requires $dy/dx \neq 0$; the swap loses exactly the points where the original solution has a horizontal tangent.

SEE ALSO The integrating-factor machine; the linearity test; implicit differentiation in differential form.

The audit run five times

The procedure is easier to trust once you have watched it run. Below are five equations, each with its verdict list written out as it would appear on paper. The point is not the answers, which are short, but the shape of the reasoning: a sequence of one-line verdicts terminating in a decision.

One. $\frac{dy}{dx} = \frac{y^2 + 2xy}{x^2}$, $y(1) = 1$. Separable: no, the numerator is a sum in y . Linear in y : no, y^2 . Degree-zero homogeneous: yes, everything is degree 2 over degree 2. Bernoulli: also yes, since $y' - \frac{2y}{x} = \frac{y^2}{x^2}$ with $n = 2$. Two hits, and Bernoulli is shorter: $v = y^{-1}$ gives $v' + \frac{2v}{x} = -\frac{1}{x^2}$, so $x^2v = C - x$ and

$$y = \frac{x^2}{C - x}, \quad C = 2 \text{ from } y(1) = 1, \quad y(1.5) = 4.5,$$

confirmed numerically. The solution ends at $x = 2$.

Two. $(3x^2 + 2y)dx + 2x dy = 0$, $y(1) = 1$. It is already in differential form, so test exactness first: $M_y = 2 = N_x$. Exact. $F = x^3 + 2xy = C$, so $y = \frac{C - x^3}{2x}$ with $C = 3$, and $y(2) = -1.25$. Notice that the same equation is linear $(2xy' + 2y = -3x^2)$, whose left side is $(2xy)'$ and reaches the same place; the exactness route needed no division.

Three. $\frac{dy}{dx} = e^{x+y}$, $y(0) = 0$. Separable: it looks like a sum and is not, since $e^{x+y} = e^x e^y$. Separating, $e^{-y} dy = e^x dx$ gives $e^{-y} = C - e^x$ with $C = 2$, so $y = -\ln(2 - e^x)$ and $y(0.5) = 1.0461753$, confirmed numerically. The solution exists only for $x < \ln 2$.

Four. $\frac{dy}{dx} = \frac{1}{x+y}$. Separable: no. Linear in y : no. Homogeneous: no, the constant 1 upstairs is degree 0 against degree 1 downstairs. Depends on $x+y$ only: yes, so $u = x+y$ works and gives $u - \ln|u+1| = x+C$, implicit. But run the remaining tests: linear in x ? Inverting, $\frac{dx}{dy} - x = y$, which is linear with $\mu = e^{-y}$ and yields

$$x = Ce^y - y - 1$$

in three lines, with no implicit form at all. Two routes again, and the one nobody tries is the short one.

Five. $y = xy' + \ln y'$. Nothing in the shape tests applies, because y' sits inside a logarithm. But y is isolated, there is a bare $x \cdot y'$, and the rest is a function of y' alone: Clairaut, with $f(p) = \ln p$. General solution $y = Cx + \ln C$ for $C > 0$. Singular: $x + \frac{1}{p} = 0$ gives $p = -\frac{1}{x}$ and

$$y = -1 - \ln(-x) \quad (x < 0),$$

which substitution confirms: at $x = -2$, both $xp + \ln p$ and $-1 - \ln 2$ equal -1.6931472 .

When two families both apply

The tests are not exclusive. A great many equations pass two of them, and the two routes are almost never comparable in length. The habit worth building is to finish the audit rather than stopping at the first hit, because the first hit is often not the cheapest.

The cleanest illustration is $2xy \, dx + (x^2 + 3y^2) \, dy = 0$. Exactness: $M_y = 2x$ and $N_x = 2x$, so the form is exact, $F = x^2y + y^3$, and the answer is $x^2y + y^3 = C$. Two lines. Degree-zero homogeneity: M and N are both degree 2, so $y' = -\frac{2xy}{x^2 + 3y^2}$ is

degree 0 and $y = vx$ applies. Following that route, $x \frac{dv}{dx} = -\frac{3v(1+v^2)}{1+3v^2}$, which requires the partial fraction decomposition $\frac{1+3v^2}{v(1+v^2)} = \frac{1}{v} + \frac{2v}{1+v^2}$, then $\frac{1}{3}(\ln|v| + \ln(1+v^2)) = -\ln|x| + c$, then clearing to $v(1+v^2)x^3 = K$, then unwinding $v = y/x$ to get $x^2y + y^3 = K$. Same answer, six times the work. Numerically, starting from $(1, 1)$ the value of $x^2y + y^3$ stays at 2 to eleven digits out to $x = 2$, where $y = 0.4734658$.

Several further overlaps are worth having in mind, and each carries a standing recommendation.

Autonomous and separable. Every autonomous equation is separable, so the autonomous test is really a test for whether the phase line can replace the integration entirely. Take the phase line when the question is qualitative and the closed form only when it is not.

Isobaric and Bernoulli. Every isobaric equation of the shape $y' = \frac{y}{kx} + \frac{g(x)}{y}$ is also Bernoulli with $n = -1$, since multiplying by y makes $w = y^2$ the natural unknown. The Bernoulli route needs no weight computation, so it wins. This is exactly what happened in the worked example above, where $w = y^2$ turned a weight calculation into a two-line linear equation.

Separable and linear. An equation such as $y' = 2y/x$ passes both tests. Separate it: $\ln|y| = 2\ln|x| + c$ and $y = Cx^2$. The integrating factor is correct but slower and introduces an integral $\int P$ where none was needed. In general, when separability and linearity both hold, the forcing term Q is proportional to the coefficient of y , and separation is shorter.

Homogeneous and Bernoulli. $y' = \frac{y}{x} + \frac{x}{y}$ is degree-zero homogeneous, and it is also Bernoulli with $n = -1$. The homogeneous

route via $v = y/x$ gives $xv' = 1/v$, one line; the Bernoulli route via $w = y^2$ gives $w' - \frac{2w}{x} = 2x$, also one line. Both land on $y^2 = 2x^2 \ln|x| + Cx^2$. When two routes are genuinely comparable, take the one whose integral you trust.

Exact and separable. An equation that is both, such as $(3x^2y + 2xy) \, dx + (x^3 + x^2) \, dy = 0$, should be checked for exactness first, because building F requires no division and therefore loses no solutions. Here $M_y = N_x = 3x^2 + 2x$ and $F = (x^3 + x^2)y$, so $y = \frac{C}{x^2(x+1)}$ immediately. The separable route reaches the same place after a partial fraction decomposition and a division by y that discards $y \equiv 0$.

When every test fails, the failure is itself informative and there are three honest responses. Look again for an integrating factor, including forms $\mu = x^a y^b$ and $\mu(xy)$ that the two single-variable tests miss. Look for a substitution suggested by the structure rather than by a named family: any repeated block in the equation, such as $x^2 + y^2$ or y/x^2 or e^y , is a candidate for a new variable, and the test is whether the substituted equation is shorter than the original. Failing both, the equation may simply have no elementary solution, in which case a series solution about an ordinary point or a numerical integration is the correct answer and not an admission of defeat. The equation $y' = x^2 + y^2$ is the standard reminder: it is Riccati, it is as simple as an equation can look, and its solutions are ratios of Bessel functions.

One diagnostic is worth adopting as a habit. Whenever an equation is presented with unusually bloated coefficients, or with a term that looks deliberately out of place, the problem is signalling that a factorisation or a cancellation is intended. The equation in the linearity example, with its $(x + x^2)$ and $(x^3 + x^4)$, is exactly this: the bloat exists so that a solver who does not factor will compute a genuine but useless $\int P$. Ask “what would have to cancel for this to be clean” before asking “which family is this”.

FINISH THE AUDIT

The cost of the full audit is under two minutes and the cost of a misclassification is the whole problem. Two rules follow. Run every test even after one has passed, because a second pass may be far cheaper. And when a test fails, ask whether it fails on the equation or on the arrangement: separability and linearity are tests on $y' = f(x, y)$, exactness is a test on $M \, dx + N \, dy = 0$, and Clairaut and Lagrange are tests on the relation among x , y , and p . An equation that fails a test in one arrangement may pass it in another, and rearranging costs nothing.

Recognition matrix

If you see	Reach for	If that fails
Any first-order equation at all	The full audit, in order, before solving	Rearrange and audit again
f visibly a product $g(x)h(y)$	Separate; record the roots of h	Factor the sum first
A sum of four terms, no factor visible	Group in pairs: $xy + x + y + 1 = (x + 1)(y + 1)$	Test linearity instead
x absent from the right side	Autonomous: phase line first	Separate if a formula is needed
y absent from the right side	Bare quadrature $y = \int f$	Check the integral is elementary
y and y' both to the first power	Divide by the coefficient of y' , then $\mu = e^{\int P}$	Factor the coefficients first
y transcendental, x linear	Swap: $\frac{dx}{dy} + P(y)x = Q(y)$	Leave the answer as $x(y)$
$f(tx, ty) = f(x, y)$, or equal degrees	Degree zero: $y = vx$, then separate	Try exactness on $M dx + N dy$
Terms off by a fixed power of x	Weights $x \sim 1$, $y \sim m$; solve for m ; $y = vx^m$	Bernoulli in y^2 instead
$y' + Py = Qy^n$, $n \neq 0, 1$	$v = y^{1-n}$; watch the factor $(1 - n)$	Record the lost solution $y \equiv 0$
Quadratic in y , three coefficients	Riccati: need a particular y_1	Guess a/x , ax , or a constant
$M dx + N dy = 0$ with $M_y = N_x$	Build F ; the answer is $F = C$	Integrating factor $\mu(x)$ or $\mu(y)$
$M_y \neq N_x$	$\frac{M_y - N_x}{N}$ free of y : $\mu(x)$ exists	Try $\frac{N_x - M_y}{M}$ for $\mu(y)$
$y = xp + f(p)$ exactly	Clairaut: lines $y = Cx + f(C)$, plus the envelope	Differentiate and factor
$y = x\varphi(p) + \psi(p)$, $\varphi \neq p$	Lagrange: linear in $x(p)$; report parametrically	Check roots of $\varphi(p) - p$
Right side a function of $ax + by + c$	$u = ax + by + c$; then $u' = a + bf(u)$	Do not drop the $+a$
$x dy - y dx$ present	Divide by x^2 , xy , or $x^2 + y^2$	Homogeneous substitution
Two tests both pass	Take the shorter route; usually exactness	Compare before committing
Nothing passes any test	Series solution, or numerical integration	Check for a typo in the sign

Separable equations and their disguises

What this chapter is for. Separability is the cheapest thing that can happen to a first-order equation, and it happens far more often than it appears to, because problems are built to hide it. The visible form $y' = g(x)h(y)$ is rare; the common situation is an equation that becomes separable after a grouping, an identity, or one of a very short list of substitutions. That list is the content of this chapter: $y = vx$ when the right side depends only on y/x , $u = ax + by + c$ when it depends only on that combination, and a handful of trigonometric substitutions that rationalise a stubborn side. Alongside the substitutions runs a discipline that matters more than any of them. Dividing by $h(y)$ destroys the solutions with $h(y) = 0$, and those constant solutions are frequently the equilibria that the problem is actually about. Recording them is part of the method, not an afterthought, and a solution set that omits them is wrong rather than incomplete. By the end you should be able to separate reliably, name the lost solutions before you lose them, and recognise the four or five disguises that account for nearly every hidden separable equation you will meet.

The protocol

Separation is four steps, and skipping any of them is where the marks go. Write the equation as $\frac{dy}{dx} = g(x)h(y)$. Find and record every root of h , since each root y_0 gives a constant solution $y \equiv y_0$. Divide by $h(y)$ and integrate both sides, putting a single constant on the x side. Then apply the initial condition and, separately, check whether the constant solutions you recorded are recovered by the general formula for some value of the constant, or whether they must be reported separately.

The third step deserves a comment about form. Chasing a constant through an exponential and a logarithm is where most arithmetic errors enter, and the definite form of the integration avoids all of it. Writing $\int_{y_0}^y \frac{ds}{h(s)} = \int_{x_0}^x g(t) dt$ builds the initial condition into the statement, so there is no constant to determine and no exponentiation of an unknown C to mishandle. It also makes the domain question visible: the identity holds only as long as the left-hand integral is finite and h does not vanish between y_0 and y , which is exactly the condition under which the solution exists and is unique.

The fourth step, the constant-solution check, is the one that gets skipped, and it is the one that changes answers. It is worth being explicit about why the roots of h matter. If $h(y_0) = 0$ then $y \equiv y_0$ satisfies $y' = g(x)h(y)$ identically, so it is a solution; and dividing by $h(y)$ in step three is a division by zero on that solution, which is why the general formula usually cannot see it. Sometimes the formula recovers it anyway at a limiting value of the constant, and sometimes it does not. Deciding which happened takes one line and is part of the answer.

CORE

32.1 The separable machine with an initial condition

CUE $\frac{dy}{dx} = g(x)h(y)$ together with data $y(x_0) = y_0$, in any of the equivalent arrangements $M(x) dx + N(y) dy = 0$ or $N(y)y' = M(x)$.

METHOD Prefer the definite form, which applies the initial condition automatically:

$$\int_{y_0}^y \frac{ds}{h(s)} = \int_{x_0}^x g(t) dt.$$

No constant of integration appears and none has to be chased. Solve for y explicitly only if the question needs it; an implicit relation is a complete answer. Before dividing, record every root of h : those are constant solutions, and the definite form is valid only on an interval where $h(y) \neq 0$.

WORKED EXAMPLE

$$\frac{dy}{dx} = \frac{4 + 9y^2}{e^{2x+1}} \text{ with } y(-\tfrac{1}{2}) = 0.$$

There are no roots of $h(y) = 4 + 9y^2$, so nothing is lost. In definite form,

$$\int_0^y \frac{ds}{4 + 9s^2} = \int_{-1/2}^x e^{-(2t+1)} dt.$$

The left side is $\frac{1}{6} \arctan \frac{3y}{2}$ and the right is $\frac{1}{2}(1 - e^{-(2x+1)})$, so

$$y = \frac{2}{3} \tan\left(3(1 - e^{-(2x+1)})\right).$$

Now read the formula for what it says about the domain. The tangent's argument increases from 0 toward 3 as x increases, and it crosses $\frac{\pi}{2}$ when $1 - e^{-(2x+1)} = \frac{\pi}{6}$, that is at

$$x^* = \frac{1}{2}(-1 - \ln(1 - \frac{\pi}{6})) = -0.1292526.$$

The solution blows up there and does not exist beyond it, which the differential equation itself never announced. Numerically, $y(-0.2) = 3.0205002$ from both the formula and a Runge-Kutta integration, and $y(-0.15) = 10.9963$; the agreement holds right up to the asymptote.

WHERE THIS GOES WRONG

Writing $\ln|y| = \dots + C$ and exponentiating to $y = e^C e^{\dots}$ while quietly forgetting three things: the absolute value, which means the new constant $A = \pm e^C$ ranges over all nonzero reals rather than only positive ones; the value $A = 0$, which is not of the form $\pm e^C$ and which recovers the equilibrium solution $y \equiv 0$; and the fact that a solution can cease to exist at finite x even when g and h are perfectly smooth everywhere.

SEE ALSO The separability test; hidden separability by factoring; autonomous recognition and the phase line.

The next example is the one to keep in mind whenever the constant-solution question feels pedantic, because there the constants are the whole story: they are the equilibria, they bound every other solution, and the closed form is meaningless without them.

WORKED EXAMPLE: THE LOST SOLUTIONS ARE THE ANSWER

$\frac{dy}{dx} = x(y^2 - 4)$ with $y(0) = 0$.

Roots of $h(y) = y^2 - 4$: $y = 2$ and $y = -2$, so $y \equiv 2$ and $y \equiv -2$ are constant solutions. Record them, then divide.

Partial fractions give $\frac{1}{y^2 - 4} = \frac{1}{4} \left(\frac{1}{y - 2} - \frac{1}{y + 2} \right)$, so

$$\frac{1}{4} \ln \left| \frac{y - 2}{y + 2} \right| = \frac{x^2}{2} + c.$$

At $x = 0$, $y = 0$, the left side is $\frac{1}{4} \ln 1 = 0$, so $c = 0$ and $\frac{y - 2}{y + 2} = -e^{2x^2}$, the sign fixed by the initial point. Solving,

$$y = \frac{2(1 - e^{2x^2})}{1 + e^{2x^2}} = -2 \tanh(x^2).$$

Check: $y(1.5) = -2 \tanh(2.25) = -1.9560522$, which Runge-Kutta reproduces to ten figures. Now the recorded constants pay. The solution decreases from 0 and approaches -2 but never reaches it, because $y \equiv -2$ is itself a solution and solutions of a Lipschitz equation cannot cross. That single sentence gives the long-run behaviour, the monotonicity, and the bound, and it is unavailable to anyone who divided by $y^2 - 4$ without looking. Note also that neither $y \equiv 2$ nor $y \equiv -2$ is recovered by the formula for any finite constant; both must be reported separately.

THE CONSTANT-SOLUTION CHECK IS PART OF THE METHOD

Every division by $h(y)$ is a case split, and the discarded case is a solution. Write the roots down *before* dividing, then check each against the final formula. Three outcomes are possible. The formula recovers the root at some finite constant, in which case nothing was lost. The formula recovers it only in a limit, as with $A \rightarrow 0$ in $y = Ae^{kx}$, in which case the root must be added to the solution set by hand. Or the root is a genuinely separate branch through which other solutions pass, in which case uniqueness has failed and the initial-value problem has more than one solution. The standard example of the third is $y' = y^{2/3}$ with $y(0) = 0$, which has the solutions $y \equiv 0$ and $y = \frac{1}{27}x^3$ and infinitely many hybrids that sit on zero for a while and then leave. The Lipschitz condition fails exactly at $y = 0$, and that is where the lost constant solution was.

Separability that has been hidden

The commonest disguise is not a substitution at all. It is arithmetic: an equation that separates after a grouping, a factorisation, or an elementary identity, presented in expanded form so that no factorisation is visible. This is why the audit begins with “simplify” and why the first move on any first-order equation is to factor every coefficient completely.

32.2 Hidden separability by factoring or by an identity

CUE The equation looks like a sum, or a bloated rational expression, or carries a radical whose radicand is suspiciously structured. Trigonometric coefficients that are longer than one term are almost always an identity in disguise.

METHOD Before choosing any method, factor every coefficient completely and simplify with elementary identities. Group four-term sums in pairs. Recognise the standard collapses: $\sin^2 x + \cos^2 x + \tan^2 x = \sec^2 x$; $1 + \tan^2 x = \sec^2 x$; $x + 2\sqrt{x-1} = (\sqrt{x-1} + 1)^2$; $e^{x+y} = e^x e^y$, which turns an apparent sum in the exponent into a product and separates immediately.

WORKED EXAMPLE

$\frac{dy}{dx} = \sqrt{x + 2\sqrt{x-1}}$ with $y(1) = 3$; find $y(5)$.

The radicand is a perfect square: $x + 2\sqrt{x-1} = (x-1) + 2\sqrt{x-1} + 1 = (\sqrt{x-1} + 1)^2$. On $x > 1$ the inner expression is positive, so the outer square root removes cleanly and

$$\frac{dy}{dx} = \sqrt{x-1} + 1, \quad y = \frac{2}{3}(x-1)^{3/2} + x + C.$$

From $y(1) = 3$ we get $C = 2$, so $y(5) = \frac{2}{3}(8) + 5 + 2 = \frac{37}{3} = 12.3333$, confirmed numerically.

A second instance, the four-term sum: $\frac{dy}{dx} = xy + x + y + 1$. Grouped, this is $(x+1)(y+1)$, so $\ln|1+y| = \frac{x^2}{2} + x + c$ and $y = Ae^{x^2/2+x} - 1$. The constant solution $y \equiv -1$ is the case $A = 0$. With $y(0) = 0$, $A = 1$ and $y(1) = e^{3/2} - 1 = 3.4816891$.

A third instance, where the disguise is a rational expression rather than a radical: $\frac{dy}{dx} = \frac{x + xy^2}{y + x^2y}$ with $y(0) = 1$. Both numerator and denominator factor, and the factorisation is the whole problem:

$$\frac{dy}{dx} = \frac{x(1+y^2)}{y(1+x^2)}, \quad \frac{y dy}{1+y^2} = \frac{x dx}{1+x^2},$$

so $\ln(1+y^2) = \ln(1+x^2) + c$ and $1+y^2 = C(1+x^2)$. From $y(0) = 1$, $C = 2$ and $y^2 = 1 + 2x^2$, so $y(2) = 3$, confirmed numerically. Expanded, the original right-hand side is a ratio of two-term sums and looks like nothing; factored, it is a product of a pure- x and a pure- y piece. The habit worth building is to factor numerator and denominator *separately* before judging separability, since a common factor between them would also have simplified the problem.

WHERE THIS GOES WRONG

Dropping the absolute value when un-squaring. Strictly $\sqrt{(\sqrt{x-1} + 1)^2} = |\sqrt{x-1} + 1|$, which equals $\sqrt{x-1} + 1$ only where the inner expression is nonnegative; here that is all of $x \geq 1$, but the domain must be stated rather than assumed. The general form of the trap is that $\sqrt{u^2} = |u|$, and problems are built around radicands that are perfect squares of expressions which change sign.

SEE ALSO The separability test; the classification audit; simplification before any method is chosen.

A short catalogue of the collapses worth recognising on sight, since they account for most hidden separability in practice. An exponential of a sum, $e^{x+y} = e^x e^y$, separates immediately and is the commonest disguise of all: $y' = e^{x+y}$ looks like a sum and is a product. A four-term sum with a shared binomial, $xy + x + y + 1 = (x+1)(y+1)$, is the second commonest. A radicand of the form $u \pm 2\sqrt{u-1}$ is $(\sqrt{u-1} \pm 1)^2$. A trigonometric coefficient of more than one term is nearly always a Pythagorean identity: $\sin^2 x + \cos^2 x + \tan^2 x = \sec^2 x$, and $\frac{1 - \cos 2x}{2} = \sin^2 x$. And a logarithm of a product or a power should be expanded before anything else, since $\ln(x^2 y) = 2 \ln x + \ln y$ turns a mixed expression into a sum whose separability is then visible.

The first genuine substitution handles the case where the right side is not a product but depends on x and y only through their ratio. This is the degree-zero homogeneous family, and the substitution that separates it is universal within that family: it never fails, and it always produces the same shape of separated equation.

32.3 Substitution $y = vx$ for degree-zero homogeneous equations

CUE The right side passes the degree-zero test $f(tx, ty) = f(x, y)$, or equivalently every term of M and N in $M dx + N dy = 0$ has the same total degree. Writing f in terms of y/x and finding that x disappears is the confirmation.

METHOD Put $y = vx$, so that $y' = v + x \frac{dv}{dx}$. Every degree-zero equation $y' = F(v)$ becomes

$$x \frac{dv}{dx} = F(v) - v, \quad \text{that is} \quad \frac{dv}{F(v) - v} = \frac{dx}{x},$$

which is separable in every case. The roots of $F(v) - v = 0$ are lost solutions: each root v_0 gives the straight line $y = v_0 x$ through the origin. Write $v = y/x$ rather than $y = vx$ when the equation is presented in differential form and the combination $x dy - y dx$ is already visible.

WORKED EXAMPLE

$(x^2 + y^2) dx - 2xy dy = 0$, that is $y' = \frac{x^2 + y^2}{2xy}$, with $y(1) = 0$.

Both coefficients are degree 2, so the quotient is degree 0. With $y = vx$,

$$v + x \frac{dv}{dx} = \frac{1 + v^2}{2v}, \quad x \frac{dv}{dx} = \frac{1 - v^2}{2v}.$$

Separating, $\frac{2v dv}{1 - v^2} = \frac{dx}{x}$, so $-\ln|1 - v^2| = \ln|x| + c$ and $1 - v^2 = \frac{C}{x}$. Multiplying by x^2 ,

$$x^2 - y^2 = Cx.$$

With $y(1) = 0$, $C = 1$ and $y^2 = x^2 - x$; then $y(3) = \sqrt{6} = 2.4494897$, confirmed numerically. The lost solutions are the roots of $F(v) - v = \frac{1 - v^2}{2v}$, namely $v = \pm 1$, giving the lines $y = \pm x$; and indeed $y = \pm x$ satisfies the original equation, as the case $C = 0$.

A near miss worth studying: $x \frac{dy}{dx} - y = x^2 + y^2$ is *not* degree-zero homogeneous, since $y' = \frac{y}{x} + x + \frac{y^2}{x}$ mixes degrees. Yet $y = vx$ still works, because the substitution is really an assault on the combination $x dy - y dx$ rather than on homogeneity. Substituting gives $x^2 v' = x^2(1 + v^2)$, so $v' = 1 + v^2$ exactly, $v = \tan(x + C)$, and $y = x \tan(x + C)$. With $y(\frac{\pi}{2}) = \frac{\pi}{2}$ we need $C = -\frac{\pi}{4}$, so $y = x \tan(x - \frac{\pi}{4})$, whence $y(\frac{\pi}{4}) = 0$ and $y(1.2) = 0.5281360$, matching the numerics to seven figures. Equivalently, divide the original by x^2 to expose $d(y/x) = (1 + (y/x)^2) dx$ and integrate at once. The moral: try $v = y/x$ whenever $x dy - y dx$ appears, whether or not the degree test passes.

WHERE THIS GOES WRONG

Writing $y' = v'$ instead of $y' = v + xv'$. This is the single most common error in the whole subject; it produces an equation that separates just as nicely and is simply wrong. The check costs nothing: differentiate $y = vx$ as a product. The second trap is discarding the roots of $F(v) - v$, which are the straight-line solutions through the origin and are usually the ones the geometry of the problem is about.

SEE ALSO The homogeneous-of-degree-zero test; recognising a differential as d of something; isobaric equations, the weighted generalisation.

It is worth pausing on the two ways of writing the same substitution, because they are convenient in different places. Writing $y = vx$ and differentiating gives $y' = v + xv'$, which is what you want when the equation is presented as $y' = F(y/x)$: substitute, cancel, separate. Writing $u = y/x$ instead and differentiating gives $du = \frac{x dy - y dx}{x^2}$, which is what you want when the equation is presented in differential form and the combination $x dy - y dx$ is already sitting there. The two are the same substitution and produce the same separated equation; choosing the wrong one costs an extra line of algebra, choosing the right one sometimes finishes the problem outright, as the near-miss example above shows.

The second substitution handles the complementary situation: the right side does not depend on the ratio y/x but on a fixed linear combination of x and y . This family is easy to recognise because the combination is written out in the equation, usually inside a function.

32.4 Substitution $u = ax + by + c$

CUE The right side depends on x and y only through one linear combination: $y' = f(ax + by + c)$. Look inside every function in the equation; if x and y occur only in the pattern $ax + by + c$, and nowhere else, the substitution applies.

METHOD Set $u = ax + by + c$. Then $du = a dx + b dy$, so

$$\frac{du}{dx} = a + b \frac{dy}{dx} = a + bf(u),$$

which is autonomous in u and therefore always separable. Solve for u , then recover $y = (u - ax - c)/b$. Translate the initial condition into u at the same moment as the substitution.

WORKED EXAMPLE

$\frac{dy}{dx} = \cos(x + y)$ with $y(0) = 0$. Here $u = x + y$, so $u' = 1 + \cos u = 2 \cos^2 \frac{u}{2}$ by the half-angle identity, and

$$\frac{1}{2} \sec^2 \frac{u}{2} du = dx, \quad \tan \frac{u}{2} = x + C.$$

The condition $y(0) = 0$ gives $u(0) = 0$ and $C = 0$, so $u = 2 \arctan x$ and

$$y = 2 \arctan x - x.$$

Check: $y' = \frac{2}{1+x^2} - 1 = \frac{1-x^2}{1+x^2}$, while $\cos(x+y) = \cos(2 \arctan x) = \frac{1-x^2}{1+x^2}$. They agree, and $y(2) = 2 \arctan 2 - 2 = 0.2142974$ matches the numerics. The lost solutions are the roots of $1 + \cos u = 0$, that is $u = (2k+1)\pi$, giving the straight lines $y = (2k+1)\pi - x$; these satisfy the original equation since $\cos(x+y) = -1 = y'$ on each.

A second instance with a radical: $y' = \sqrt{x+y+1}$ with $y(0) = 0$. With $u = x + y + 1$, $u' = 1 + \sqrt{u}$. Substituting $s = \sqrt{u}$ makes the integral elementary, $\int \frac{2s ds}{1+s} = 2s - 2 \ln(1+s)$, so

$$2\sqrt{u} - 2 \ln(1 + \sqrt{u}) = x + C, \quad C = 2 - 2 \ln 2 = 0.6137056$$

from $u(0) = 1$. This is implicit and should be left so; at $x = 1$ the numerics give $y = 1.4463061$, and substituting $u = x + y + 1 = 3.4463061$ into the left side returns $x + C = 1.6137056$ exactly as required.

A third instance, where the combination is a ratio rather than an argument: $\frac{dy}{dx} = \frac{x+y-1}{x+y+1}$ with $y(0) = 1$. Both numerator and denominator involve x and y only through $u = x + y$, so the substitution applies with $a = b = 1$:

$$\frac{du}{dx} = 1 + \frac{u-1}{u+1} = \frac{2u}{u+1}, \quad \frac{u+1}{2u} du = dx,$$

so $u + \ln|u| = 2x + C$. From $u(0) = 1$, $C = 1$, and the solution is the implicit relation

$$x + y + \ln(x + y) = 2x + 1, \quad \text{that is} \quad y - x + \ln(x + y) = 1.$$

At $x = 1$ numerical integration gives $y = 1.2079400$, and $u + \ln u = 3.0000000$ as required. The lost solution is the root of $2u = 0$, namely $u \equiv 0$, that is the line $y = -x$; and indeed on that line the original right-hand side is $\frac{-1}{1} = -1$, which is the slope of $y = -x$.

WHERE THIS GOES WRONG

Forgetting the $+a$ that comes from differentiating the ax part, so that $u' = bf(u)$ instead of $u' = a + bf(u)$. The resulting equation is still separable and still solvable, so nothing warns you; the answer is simply wrong. The second trap is applying the substitution when x and y also appear outside the combination, for instance in $y' = x \cos(x + y)$, where $u' = 1 + x \cos u$ is not separable and the substitution has bought nothing.

SEE ALSO Differentials as substitution bookkeeping; the substitution $y = vx$; autonomous recognition.

The two substitutions divide the territory neatly, and knowing which one applies is a matter of asking how x and y enter. If they enter only through the ratio y/x , use $y = vx$. If they enter only through a linear combination $ax + by + c$, use $u = ax + by + c$. If they enter through both, or through neither, neither substitution helps and you are back at the audit. The degenerate case $b = 0$ is worth noting: then the equation is $y' = f(ax + c)$, a bare quadrature, and no substitution is needed at all.

The three substitutions covered so far exhaust the standard repertoire for hidden separability, and it is worth having them side by side with the question each answers.

How x and y enter	Substitute	Becomes	Lost solutions
Already a product $g(x)h(y)$	nothing	$\frac{dy}{h} = g \, dx$	roots of h
Only through y/x	$y = vx$	$xv' = F(v) - v$	lines $y = v_0x$
Only through $ax + by + c$	$u = ax + by + c$	$u' = a + bf(u)$	lines $ax + by + c = u_0$
As $x \, dy - y \, dx$ over x^2	$u = y/x$	$du = (\dots) \, dx$	$x = 0$
Only through yx^{-m}	$y = vx^m$	$xv' = G(v) - mv$	curves $y = v_0x^m$

The last row is the weighted generalisation, and it is included to make the pattern visible: each substitution is chosen so that the combination through which x and y enter becomes a single new variable, after which the equation can only depend on that variable and on x , and separation follows. There is nothing to memorise beyond that principle. If you can name the combination, you can name the substitution.

Trigonometric disguises

The last family of disguises is the one where separation succeeds but the resulting integral does not look elementary. The separated equation is genuinely separable; what is hidden is the antiderivative, and the substitutions that expose it are the standard rationalising ones from integration technique, imported wholesale. The extra difficulty, and the reason this family is rated hard, is that the resulting antiderivatives are multivalued and the initial condition has to be tracked through the branches.

32.5 Trigonometric substitutions that separate

HARD

CUE A separated side containing $1 + \sin^2 y$, $\sqrt{1 - y^2}$, $1 + y^2$, $a + b \cos y$, or any expression that would call for a trigonometric or Weierstrass substitution if it were an ordinary integral.

METHOD Separate first, then treat each side as an integration problem in its own right. The standard choices: $t = \tan y$ for even powers of $\sin$ and $\cos$; $y = \sin \theta$ for $\sqrt{1 - y^2}$; $t = \tan \frac{y}{2}$ for a general rational function of $\sin y$ and $\cos y$. The one to have by heart, because it appears constantly, is

$$\int \frac{dy}{1 + \sin^2 y} = \frac{1}{\sqrt{2}} \arctan(\sqrt{2} \tan y) + \text{const},$$

obtained by dividing numerator and denominator by $\cos^2 y$ and setting $t = \tan y$. Translate the initial condition into the new variable at the same time as the integrand, so that no branch ambiguity survives.

WORKED EXAMPLE

$$\frac{dy}{dx} = (1 + \sin^2 y) \cos x \text{ with } y(0) = 0.$$

Separate: $\frac{dy}{1 + \sin^2 y} = \cos x \, dx$. The left side integrates by $t = \tan y$ as above, so

$$\frac{1}{\sqrt{2}} \arctan(\sqrt{2} \tan y) = \sin x,$$

the constant being zero because $y(0) = 0$ sends both sides to zero. Solving,

$$y = \arctan\left(\frac{\tan(\sqrt{2} \sin x)}{\sqrt{2}}\right).$$

This formula is valid for all real x , and the reason is worth seeing: the argument $\sqrt{2} \sin x$ never leaves $[-\sqrt{2}, \sqrt{2}]$ and $\sqrt{2} = 1.41421 < \frac{\pi}{2} = 1.57080$, so $\tan$ is never evaluated at a pole. Numerically, $y(3) = 0.1420605$ and $y(5) = -1.2717020$ from both the formula and a Runge-Kutta integration.

Change the forcing to $2 \cos x$ and the picture changes completely: the argument becomes $2\sqrt{2} \sin x$, which passes $\frac{\pi}{2}$, and the antiderivative $\frac{1}{\sqrt{2}} \arctan(\sqrt{2} \tan y)$ must be replaced by its continuous version

$$F(y) = \frac{1}{\sqrt{2}} \left[\arctan(\sqrt{2} \tan y) + \pi \left\lfloor \frac{y}{\pi} + \frac{1}{2} \right\rfloor \right],$$

which adds $\frac{\pi}{\sqrt{2}} = 2.2214415$ at each odd multiple of $\frac{\pi}{2}$. Checked against numerical quadrature, $F(5) - F(0) = 3.4779665$, whereas the naive single-branch expression reports -0.9649 , a value that is not even positive although the integrand is.

The other trigonometric route worth having is the one for a rational function of $\sin$ and $\cos$. If separation leaves $\int \frac{dy}{a + b \cos y}$, the Weierstrass substitution $t = \tan \frac{y}{2}$ turns it into a rational integral in t , since $\cos y = \frac{1 - t^2}{1 + t^2}$ and $dy = \frac{2 dt}{1 + t^2}$. The result splits on the sign of $a^2 - b^2$: an arctangent when $|a| > |b|$, a logarithm when $|a| < |b|$, and a rational function when $|a| = |b|$. The last case is the one to watch, because it is where the constant solutions live: if $a + b \cos y$ vanishes for some y_0 , then $y \equiv y_0$ is a constant solution of the differential equation and the integral diverges there, which is the analytic shadow of the same fact.

WHERE THIS GOES WRONG

Branch bookkeeping. The function $\arctan(\sqrt{2} \tan y)$ jumps by π at every odd multiple of $\frac{\pi}{2}$, so it is *not* an antiderivative of $\frac{1}{1 + \sin^2 y}$ across those points, even though its derivative is correct between them. Any initial condition, or any solution trajectory, that crosses $y = \frac{\pi}{2}$ requires the corrected F above. The symptom of the error is a solution that is discontinuous where the differential equation is perfectly smooth, which should never be believed.

SEE ALSO The separable machine with an initial condition; substitution $u = ax + by + c$; the Weierstrass substitution.

The simplest member of this family is worth a separate look, because it is the one where the branch question is easiest to see and hardest to remember. When both sides separate into arctangents, the solution is an addition formula in disguise, and the addition formula has a pole.

WORKED EXAMPLE: ARCTANGENT ON BOTH SIDES

$$\frac{dy}{dx} = \frac{1 + y^2}{1 + x^2} \text{ with } y(0) = 1.$$

Separating gives $\arctan y = \arctan x + C$, and $y(0) = 1$ forces $C = \frac{\pi}{4}$. Taking the tangent of both sides and using the addition formula,

$$y = \tan\left(\arctan x + \frac{\pi}{4}\right) = \frac{x + 1}{1 - x}.$$

Check: $y(0.5) = 3$ and $y(0.9) = 19$, both confirmed numerically to eleven figures. The solution exists only on $(-\infty, 1)$, blowing up at $x = 1$, and the differential equation gives no hint of this: its right-hand side is bounded by $1 + y^2$ and perfectly smooth everywhere. What produced the pole is the arctangent's range. As x increases from 0, $\arctan x + \frac{\pi}{4}$ increases toward $\frac{\pi}{2} + \frac{\pi}{4}$ and crosses $\frac{\pi}{2}$ at $x = 1$, at which point y has escaped to infinity. Every separation that produces an arctangent on the y side carries this hazard, and the check is the same each time: find where the accumulated right-hand side reaches $\frac{\pi}{2}$.

READ THE CLOSED FORM FOR ITS DOMAIN

A separable equation is smooth wherever g and h are, and its solutions can still cease to exist at a finite x . The differential equation never announces this; the closed form always does, at the poles of a tangent, the zeros of a denominator, or the branch points of a logarithm. Three instances from this chapter: $y = \frac{2}{3} \tan(3(1 - e^{-(2x+1)}))$ blows up at $x^* = -0.1292526$; $y = 1/(4x - x^2)$ from the Bernoulli family exists only on $(0, 4)$; and $y = x \tan(x - \frac{\pi}{4})$ exists only on $(-\frac{\pi}{4}, \frac{3\pi}{4})$. In every case the endpoints are invisible in the equation and explicit in the answer. Make reading the domain off the closed form the last step of every separation, and state it.

Notation disguises

A final category, cheap to handle once seen and paralysing when not. Some problems hide separability not in the algebra but in the typography, presenting the derivative in a form that has to be decoded before any classification is possible. The response is mechanical: translate to $\frac{dy}{dx}$ first, then run the audit as usual.

32.6 Notation disguises for $\frac{dy}{dx}$

CUE The derivative is buried in unusual notation: a dx -th root, a stacked exponential of differentials, a written-out limit of a difference quotient, or a phrase such as “relative growth rate” or “proportional rate of change”.

METHOD Translate mechanically before doing anything else. The standard decodings: $\sqrt[dx]{e^{dy}} = (e^{dy})^{1/dx} = e^{dy/dx}$; the relative growth rate $\frac{1}{M} \frac{dM}{d\theta} = \frac{d}{d\theta} \ln M$; “the rate of change of y is proportional to y ” is $y' = ky$; and $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ is $f'(x)$ however elaborately it is dressed. Then classify normally.

WORKED EXAMPLE

Solve $xe^{e^x} = \sqrt[dx]{e^{dy}}$ with $y(1) = e$, and find $y(e)$.

Decode the right side: it is $e^{dy/dx} = e^{y'}$. Taking logarithms of both sides,

$$y' = \ln(xe^{e^x}) = \ln x + e^x.$$

This is a bare quadrature, the most degenerate separable case. Integrating, $y = x \ln x - x + e^x + C$, and $y(1) = e$ forces $-1 + e + C = e$, so $C = 1$. Hence

$$y(e) = e \cdot 1 - e + e^e + 1 = e^e + 1 = 16.1542622,$$

confirmed by numerical integration to nine figures.

A second instance: a population satisfies $\frac{1}{M} \frac{dM}{d\theta} = \cos \theta$ with $M(0) = 2$. The left side is $\frac{d}{d\theta} \ln M$, so $\ln M = \sin \theta + \ln 2$ and $M = 2e^{\sin \theta}$: a bounded oscillation between $2e^{-1}$ and $2e$, not the runaway growth that the word “growth rate” suggests to a careless reader.

WHERE THIS GOES WRONG

Treating dy and dx as separate objects that can be manipulated outside a quotient. The disguise works precisely because it invites that: writing $\sqrt[dx]{e^{dy}}$ tempts you to take a dx -th root of something, when in fact the only meaningful object present is the ratio dy/dx . Decode to a ratio first and the temptation disappears.

SEE ALSO Differentials as substitution bookkeeping; hidden separability by factoring; the classification audit.

One further disguise belongs here even though it is verbal rather than typographic. Applied problems state the differential equation in words, and the words have standard translations that are worth running mechanically rather than interpreting. “The rate of change of y is proportional to y ” is $y' = ky$. “Proportional to the difference between y and the ambient value A ” is $y' = k(A - y)$, Newton’s law of cooling, which is autonomous with the single equilibrium $y = A$. “Proportional to the product of the number infected and the number not yet infected” is $y' = ky(N - y)$, logistic. “Inversely proportional to y ” is $y' = k/y$, which separates to $y^2 = 2kx + C$. In each case the constant k carries a sign that the wording fixes and the algebra does not, and getting that sign wrong is the commonest error in the whole applied family: cooling has $k > 0$ with the difference written $A - y$, and writing $y - A$ instead flips the stability of the equilibrium without changing anything else in the algebra.

What separation cannot do

Two limits are worth stating plainly, because both are places where a correct separation produces a wrong-looking answer and the temptation is to go back and re-derive.

The first is that separation frequently produces an implicit relation which cannot be solved for y , and that this is a finished answer rather than a stalled one. The radical example above, $2\sqrt{u} - 2\ln(1 + \sqrt{u}) = x + C$ with $u = x + y + 1$, is of this kind: no rearrangement extracts y , and none is needed, because every question one can reasonably ask (the value of y at a given x , the location of a vertical tangent, the behaviour as $x \rightarrow \infty$) is answerable from the implicit form by inspection or by one numerical root-find. Solving for y is a convenience, not a requirement, and forcing it usually costs a case split on signs that the implicit form did not need.

The second is that the antiderivatives which appear after separation need not be elementary. The equation $y' = e^{-x^2}(1 + y^2)$ separates perfectly into $\frac{dy}{1 + y^2} = e^{-x^2} dx$, and the right-hand side is $\frac{\sqrt{\pi}}{2} \operatorname{erf}(x)$, which is not elementary. The correct response is to write $\arctan y = \frac{\sqrt{\pi}}{2} \operatorname{erf}(x) + C$ and stop. Separation has done its entire job: it has reduced a differential

equation to two independent integrations. Whether those integrations have elementary answers is a separate question belonging to a different chapter, and a non-elementary antiderivative is not evidence that the separation was wrong.

Finally, a word on what the constant-solution discipline buys beyond correctness. Because solutions of a Lipschitz equation cannot cross, every constant solution is a barrier, and the barriers organise the whole solution family: they partition the strip in which the solutions live, and each non-constant solution is trapped between two consecutive barriers for its entire existence. That is why recording the roots of h first pays twice, once as a completeness check on the answer and once as a free qualitative description of every solution you did not compute.

Recognition matrix

If you see	Reach for	If that fails
$y' = g(x)h(y)$ with data $y(x_0) = y_0$	Definite form $\int_{y_0}^y ds/h = \int_{x_0}^x g$	Indefinite form, then fit C
Any division by $h(y)$	Record the roots of h first	Check them against the final formula
$\ln y $ after integrating	$A = \pm e^C$ is any nonzero real; $A = 0$ is the equilibrium	State the interval of validity
A four-term sum on the right	Group in pairs and factor	Test linearity in y
$e^{x+y}, e^{x-y}, a^{x+y}$	Split the exponential into a product	Substitute $u = x + y$
A radicand like $x + 2\sqrt{x-1}$	Complete the square inside; watch $\sqrt{u^2} = u $	State the domain explicitly
Trigonometric coefficient of two or more terms	Apply a Pythagorean identity first	Weierstrass $t = \tan(x/2)$
$f(tx, ty) = f(x, y)$	$y = vx$, then $xv' = F(v) - v$	Check $y' = v + xv'$, not v'
$x dy - y dx$ visible	$v = y/x$ directly; divide by x^2	Divide by $x^2 + y^2$ for arctan
Roots of $F(v) - v$	Straight-line solutions $y = v_0 x$; report them	They are the $C = 0$ case
$y' = f(ax + by + c)$	$u = ax + by + c$; $u' = a + bf(u)$	Do not drop the $+a$
x or y also outside the combination	The substitution fails; audit again	Try $y = vx$ or exactness
$1 + \sin^2 y$ downstairs	$t = \tan y$; get $\frac{1}{\sqrt{2}} \arctan(\sqrt{2} \tan y)$	Add $\pi \lfloor y/\pi + \frac{1}{2} \rfloor$ for continuity
$\sqrt{1 - y^2}$ on the y side	$y = \sin \theta$; watch the range of arcsin	Separate and leave implicit
Rational in $\sin y$ and $\cos y$	$t = \tan(y/2)$	Check for a hidden constant solution
A dx -th root or stacked differentials	Decode to dy/dx first, then classify	Take logarithms of both sides
"Relative growth rate"	$\frac{1}{M} \frac{dM}{d\theta} = \frac{d}{d\theta} \ln M$	Separate in $\ln M$
A closed form with $\tan$ or a denominator	Locate its poles: the solution may end there	Compare against a numerical run

Linear first-order equations and integrating factors

What this chapter is for. A first-order equation is linear when the unknown and its derivative appear only to the first power and only separately: $y' + P(x)y = Q(x)$. Every such equation is solved by the same two-line manoeuvre, and the manoeuvre is not a trick. Multiplying by $\mu = \exp(\int P dx)$ is forced: it is the unique multiplier, up to a constant factor, that converts the left side into the derivative of a single product, and once you have derived it once you never have to remember it again. This chapter derives the factor from that requirement, imposes the standard-form discipline that makes the derivation apply (divide by the coefficient of y' first, and know what that division costs you), and then generalises in two directions. Outward, to equations that are not linear at all: the same demand “make this an exact derivative” produces integrating factors depending on x alone, on y alone, or on the product xy , and turns a non-exact equation into an exact one. Inward, to equations that become linear after a change of dependent variable: Bernoulli through $u = y^{1-n}$, exponential nonlinearities through $v = e^y$, Riccati through a second-order linear equation, and integral equations through one differentiation. By the end you should be able to look at a first-order equation and say, within a few seconds, which of these it is, and you should never again write down an integrating factor you cannot derive.

The multiplier that has to exist

Start from what you want rather than from what you are told to do. You are given

$$y' + P(x)y = Q(x),$$

and the obstruction to integrating both sides is that the left-hand side is not the derivative of anything. It is y' plus a term in y , which is what the product rule produces only by accident. So ask: is there a function $\mu(x)$, not identically zero, for which $\mu(x)(y' + Py)$ is the derivative of a single product? The only candidate product is μy , since $\mu y'$ has to come from somewhere. Expand it:

$$(\mu y)' = \mu y' + \mu' y, \quad \mu(y' + Py) = \mu y' + \mu P y.$$

The two agree for every y exactly when $\mu' = \mu P$. That is a separable equation for μ with solution

$$\mu(x) = \exp\left(\int P(x) dx\right).$$

Nothing was guessed. The requirement $(\mu y)' = \mu y' + \mu P y$ has one solution family and we have written it down. Multiplying the original equation by this μ and integrating gives

$$\mu y = \int \mu Q dx + C, \quad y = \frac{1}{\mu(x)} \left(\int \mu(x) Q(x) dx + C \right).$$

Two features of that formula are worth fixing in place immediately. First, the constant of integration inside $\int P dx$ is genuinely free: replacing μ by $k\mu$ multiplies both sides of $\mu y = \int \mu Q + C$ by k and changes nothing, so take the constant to be zero every time. Second, the C sits inside the bracket, which means that after dividing by μ it is attached to $1/\mu$ and not to the whole solution. The function $C/\mu = Ce^{-\int P}$ is precisely the general solution of the homogeneous equation $y' + Py = 0$, so the structure of the answer is the structure every linear problem has: one particular solution plus the kernel.

WHY THE CONSTANT BELONGS INSIDE

Write the answer as $y = y_p + Ce^{-\int P dx}$ and you can check it in one line: the exponential kills the left side of the homogeneous equation, so it can be added to any particular solution. Write it as $y = (\int \mu Q dx)/\mu + C$ instead and the added constant is required to satisfy $C' + PC = 0$, which forces $C = 0$ unless $P \equiv 0$. That is the difference between a one-parameter family of solutions and one solution with a decorative letter attached.

For an initial value problem the tidiest statement of the answer never mentions C at all:

$$y(x) = \frac{\mu(x_0)}{\mu(x)} y(x_0) + \frac{1}{\mu(x)} \int_{x_0}^x \mu(t) Q(t) dt, \quad \mu(x) = \exp\left(\int_{x_0}^x P(s) ds\right).$$

With that choice of μ the first factor is $e^{-\int_{x_0}^x P}$ and the formula reads: the initial value decays according to the homogeneous equation, and every increment of forcing $Q(t) dt$ injected at time t decays from t onward. This is variation of parameters written out, and it is the form to use whenever Q has no elementary antiderivative, since the answer is then legitimately an integral and there is nothing further to do. It is also the form that survives into linear systems and into the Laplace transform, where the same convolution reappears.

Before any of this applies, the equation must actually be in the form $y' + Py = Q$, with the coefficient of y' equal to 1. Equations are almost never handed over that way. The discipline is: get the equation into the shape $a_1(x)y' + a_0(x)y = f(x)$, divide through by a_1 , and only then read off $P = a_0/a_1$ and $Q = f/a_1$. Skipping the division and computing $\mu = \exp(\int a_0 dx)$ is the commonest error in the whole subject, and it produces a μ that is not an integrating factor for anything. The division also has a cost, and it is the one thing about linear equations that is genuinely subtle. Wherever $a_1(x) = 0$ the division is illegal, and those points are *singular points* of the equation. The standard form is valid only on an interval free of them, and the solution you construct is a solution on that interval. At a singular point the usual existence and uniqueness guarantee is void, and the failure is visible in the answer.

The unknown must appear to the first power, undifferentiated or differentiated once, and nowhere else. So $x^3y' + e^x \sin(x)y = \ln x$ is linear, however forbidding it looks, because y and y' each appear alone and to the first power; the coefficients may be as ugly as they please. What breaks linearity is any of: a power y^n with $n \neq 1$, a product yy' or $(y')^2$, a composition $\sin y$, e^y , $\ln y$, or y inside a denominator. Each of those has its own repair in this chapter, so the diagnosis is worth making precisely: a power calls for Bernoulli, an exponential for $v = e^y$, a quadratic for Riccati, and a y downstairs usually means the equation is linear in x instead.

33.1 The $\mu = e^{\int P dx}$ machine

CUE The equation is in, or reduces by algebra to, $y' + P(x)y = Q(x)$: the unknown appears linearly, its derivative appears linearly, and nothing else involves y . Products such as yy' , powers such as y^2 , and compositions such as $\sin y$ all disqualify it.

METHOD Divide by the coefficient of y' . Compute $\mu = \exp(\int P dx)$, dropping the constant of integration. Multiply through; the left side is now $(\mu y)'$ by construction, not by inspection. Integrate once and divide:

$$y = \frac{1}{\mu(x)} \left(\int \mu(x) Q(x) dx + C \right).$$

For an initial value problem it is usually faster to keep the definite form $\mu(x)y(x) - \mu(x_0)y(x_0) = \int_{x_0}^x \mu Q$, which applies the condition without ever isolating C .

$$y' + y \tan x = \cos^2 x, \quad y(0) = 5.$$

Already in standard form, with $P = \tan x$ and $Q = \cos^2 x$. Then $\int \tan x dx = -\ln|\cos x|$, so $\mu = \sec x$ on $(-\frac{\pi}{2}, \frac{\pi}{2})$. Multiplying,

$$(y \sec x)' = \sec x \cos^2 x = \cos x, \quad y \sec x = \sin x + C, \quad y = (\sin x + C) \cos x.$$

The condition $y(0) = 5$ gives $C = 5$, so $y = (\sin x + 5) \cos x$, and $y(1) \approx 3.156160$. Note where this solution lives: $\mu = \sec x$ blows up at $x = \pm \frac{\pi}{2}$, which are exactly the singular points of the original equation, since $\tan x$ is undefined there.

$$(x^2 + 1)y' + 4xy = x, \quad y(0) = 1.$$

Divide by $x^2 + 1$, which never vanishes, so there are no singular points and the solution will be global:

$$y' + \frac{4x}{x^2 + 1}y = \frac{x}{x^2 + 1}, \quad \int \frac{4x \, dx}{x^2 + 1} = 2 \ln(x^2 + 1), \quad \mu = (x^2 + 1)^2.$$

Then $((x^2 + 1)^2 y)' = (x^2 + 1)^2 \cdot \frac{x}{x^2 + 1} = x^3 + x$, so $(x^2 + 1)^2 y = \frac{1}{4}x^4 + \frac{1}{2}x^2 + C$ and

$$y = \frac{x^4 + 2x^2 + 4C}{4(x^2 + 1)^2}.$$

With $y(0) = 1$ we get $4C = 4$, hence $y = \frac{x^4 + 2x^2 + 4}{4(x^2 + 1)^2}$, giving $y(1) = \frac{7}{16} = 0.4375$ and $y(2) = \frac{28}{100} = 0.28$. Had you used $\mu = \exp(\int 4x \, dx) = e^{2x^2}$, forgetting the division, the product $e^{2x^2}((x^2 + 1)y' + 4xy)$ would be the derivative of nothing at all.

WHERE THIS GOES WRONG

Two errors dominate, and they are unrelated. The first is placing the constant: $y = (\int \mu Q \, dx)/\mu + C$ is wrong, because the added constant must be divided by μ along with everything else. The second is Q : after dividing by a_1 , the right-hand side changes too, and using the original $f(x)$ against the new P mixes two different equations. Divide the *whole* equation, then read off both P and Q .

SEE ALSO Bernoulli via $u = y^{1-n}$; integrating factors depending on one variable; separable equations; equations linear in the other variable.

SPLIT THE FORCING, NOT THE EQUATION

Because the map $Q \mapsto y_p$ is linear, a forcing that is a sum can be handled term by term: if $y_1' + Py_1 = Q_1$ and $y_2' + Py_2 = Q_2$, then $y_1 + y_2$ solves the equation with forcing $Q_1 + Q_2$. The same μ serves all the pieces, so this costs nothing and often converts one impossible integral into two easy ones. For $y' + y = x + e^{-x}$ with $\mu = e^x$ we get $(ye^x)' = xe^x + 1$, whose two terms integrate independently to $(x - 1)e^x$ and x , giving $y = x - 1 + (x + C)e^{-x}$. Substituting back, $y' + y = 1 + e^{-x} + x - 1 = x + e^{-x}$, as required. Superposition is the only structural property linear equations have that nonlinear ones do not, and it is worth exploiting every time it appears.

One consequence of the definite form is worth having explicitly, because it is how linear equations are used in practice. If the forcing Q is piecewise continuous, with a jump at $x = c$, the solution is still perfectly well behaved: y is continuous there and only y' jumps, because $y' = Q - Py$ inherits the discontinuity directly. So you solve on $[x_0, c]$, evaluate $y(c)$, and use that value as the initial condition for the next piece. Nothing about the method changes; you simply apply it twice and glue. Solving each piece with an independent constant and forgetting to match at c is the standard error, and it produces a solution that is discontinuous where the physics is not.

The singular points deserve one worked look, because the standard form conceals them and the answer exposes them. Consider $xy' - y = x^2$. Dividing by x gives $y' - y/x = x$ with $\mu = 1/x$, so $(y/x)' = 1$ and $y = x^2 + Cx$. Every member of that family satisfies $y(0) = 0$, so the initial value problem with $y(0) = 0$ has infinitely many solutions, and the problem with $y(0) = 1$ has none: substituting $x = 0$ into $xy' - y = x^2$ forces $y(0) = 0$. Neither pathology is visible in the standard form, and both are visible in the original equation at the point where $a_1(x) = x$ vanishes. When a problem specifies an initial condition at a point where the leading coefficient dies, that is the question being asked.

Reading the equation the other way round

Linearity is a property of the equation together with a choice of which variable is the unknown, and problems are frequently set up so that the wrong choice is the obvious one. An equation that is a hopeless nonlinear mess for y as a function of x can be a two-line linear problem for x as a function of y . The reflex worth building is this: if y' appears as a quotient whose *denominator* carries the complicated y dependence while x enters only linearly, invert the derivative and start again.

The mechanism is nothing more than $\frac{dx}{dy} = 1/\frac{dy}{dx}$, valid wherever the derivative is nonzero, together with the observation that “linear” now means $\frac{dx}{dy} + P(y)x = Q(y)$. The integrating factor is $\exp(\int P(y) \, dy)$ and every other word of the previous section carries over verbatim, with the roles of the letters exchanged.

$$\frac{dy}{dx} = \frac{y}{x + y^2}, \quad y(2) = 1.$$

As an equation for y this is neither separable nor linear: the y^2 in the denominator ruins both. Invert it:

$$\frac{dx}{dy} = \frac{x + y^2}{y} = \frac{x}{y} + y, \quad \frac{dx}{dy} - \frac{1}{y}x = y.$$

That is linear in x with $P(y) = -1/y$, so $\mu = \exp(-\int dy/y) = 1/y$, and $(x/y)' = 1$. Hence $x/y = y + C$, that is $x = y^2 + Cy$. The condition $x = 2$ at $y = 1$ gives $C = 1$ and $x = y^2 + y$. Check it directly: $\frac{dx}{dy} = 2y + 1$, so $\frac{dy}{dx} = \frac{1}{2y + 1}$, while $\frac{y}{x + y^2} = \frac{y}{2y^2 + y} = \frac{1}{2y + 1}$ as required.

A second instance, where the obstruction is an exponential rather than a power:

$$\frac{dy}{dx} = \frac{1}{e^y - x} \implies \frac{dx}{dy} + x = e^y, \quad \mu = e^y, \quad (xe^y)' = e^{2y},$$

so $xe^y = \frac{1}{2}e^{2y} + C$ and $x = \frac{1}{2}e^y + Ce^{-y}$. The solution is a perfectly explicit function of y and an implicit one of x , which is the normal price of the manoeuvre and almost never a real cost.

$$\frac{dy}{dx} = \frac{y^3}{1 - 2xy^2}.$$

As an equation for y this is Bernoulli-looking but is not Bernoulli, because x appears in the denominator. Inverting,

$$\frac{dx}{dy} = \frac{1 - 2xy^2}{y^3} = \frac{1}{y^3} - \frac{2x}{y}, \quad \frac{dx}{dy} + \frac{2}{y}x = \frac{1}{y^3}.$$

This is linear with $P(y) = 2/y$, so $\mu = e^{2\ln|y|} = y^2$ and $(xy^2)' = y^{-1}$. Hence $xy^2 = \ln|y| + C$ and

$$x = \frac{\ln|y| + C}{y^2}.$$

Taking $C = 0.4$ and starting at $y = 1$, where $x = 0.4$, the trajectory reaches $x \approx 0.273287$ at $y = 2$, agreeing with the closed form to eleven places.

Notice what the diagnostic actually was. In both cases x appeared to the first power and undifferentiated, while y appeared inside a nonlinear expression; and the derivative appeared as an isolated quotient, so inverting it cost nothing. Write $M dx + N dy = 0$ if it helps: the equation is linear in y when N is free of y and M is affine in y , and linear in x when M is free of x and N is affine in x . Both tests are one glance.

Integrating factors for equations that are not linear

The derivation at the head of this chapter demanded that μ turn a specific two-term expression into a derivative. The same demand can be made of any first-order equation once it is written in differential form

$$M(x, y) dx + N(x, y) dy = 0,$$

where now “derivative” means the total differential of a potential $F(x, y)$, and the condition for one to exist is $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$. When that test fails, we look for $\mu(x, y)$ with $\mu M dx + \mu N dy = 0$ exact, which is the condition

$$\frac{\partial(\mu M)}{\partial y} = \frac{\partial(\mu N)}{\partial x}.$$

Expanded, this is a partial differential equation for μ , and in general it is harder than the equation we started with. It becomes tractable exactly when we restrict μ to depend on one thing: on x alone, on y alone, or on some combination such as xy or $x^a y^b$. Each restriction collapses the partial differential equation into an ordinary one, and each comes with a test that tells you in advance whether the restriction was legitimate. That test is the whole content of the next three techniques, and it is worth deriving once rather than memorising three times.

Suppose $\mu = \mu(x)$. Then $\frac{\partial(\mu M)}{\partial y} = \mu M_y$ while $\frac{\partial(\mu N)}{\partial x} = \mu'N + \mu N_x$, so exactness demands $\mu M_y = \mu'N + \mu N_x$, that is

$$\frac{\mu'}{\mu} = \frac{M_y - N_x}{N}.$$

The left side is a function of x only, so the right side must be too, and if it is not then no integrating factor of that restricted form exists. Run the identical computation with $\mu = \mu(y)$ and the roles of M and N swap, along with a sign: $\mu'/\mu = (N_x - M_y)/M$. There is no need to remember which numerator is which. Remember the derivation and the sign takes care of itself.

Both derivations are instances of one calculation, and doing it in general costs no more than doing it twice. Let $\mu = \mu(z)$ where $z = z(x, y)$ is any chosen combination. Then $\partial_y(\mu M) = \mu' z_y M + \mu M_y$ and $\partial_x(\mu N) = \mu' z_x N + \mu N_x$, so exactness requires

$$\frac{\mu'(z)}{\mu(z)} = \frac{M_y - N_x}{z_x N - z_y M},$$

and the whole method is: choose z , compute that quotient, and check whether it is a function of z alone. If it is, integrate; if it is not, choose another z . The standard choices and what the master formula gives for each are worth tabulating, since the third and fourth rows are exactly as easy to use as the first two and are almost never taught.

Take $\mu = \mu(z)$ with	Compute	Legitimate when
$z = x$	$\frac{M_y - N_x}{N}$	the quotient is free of y
$z = y$	$\frac{N_x - M_y}{M}$	the quotient is free of x
$z = xy$	$\frac{M_y - N_x}{yN - xM}$	the quotient depends only on xy
$z = x + y$	$\frac{M_y - N_x}{N - M}$	the quotient depends only on $x + y$
$z = x^2 + y^2$	$\frac{M_y - N_x}{2(xN - yM)}$	the quotient depends only on $x^2 + y^2$
$\mu = x^a y^b$	exponent matching	the linear system in a, b is consistent

In every case the factor is $\mu = \exp(\int (\cdot) dz)$ with the tabulated quotient as the integrand, written in terms of z . The last row is different in kind, because it fixes the shape of μ rather than its argument, and the test is algebraic rather than differential.

33.2 Integrating factor depending on x only

MEDIUM

CUE $M dx + N dy = 0$ fails the test $M_y = N_x$, and the quotient $\frac{M_y - N_x}{N}$ simplifies to an expression in which y does not appear.

METHOD Set

$$\mu(x) = \exp\left(\int \frac{M_y - N_x}{N} dx\right),$$

multiply through, confirm exactness on the new pair, and recover the potential by the two-step integration. A linear equation is the special case: for $y' + Py = Q$ written as $(Py - Q) dx + dy = 0$ we have $M_y = P$, $N_x = 0$, $N = 1$, and the recipe returns $\mu = e^{\int P dx}$ exactly as before.

WORKED EXAMPLE

$$(x + y^2) dx - 2xy dy = 0.$$

Here $M_y = 2y$ and $N_x = -2y$, so the equation is not exact and $M_y - N_x = 4y$. Divide by N :

$$\frac{M_y - N_x}{N} = \frac{4y}{-2xy} = -\frac{2}{x},$$

free of y , so the recipe applies and $\mu = \exp(-2 \ln |x|) = x^{-2}$. The rescaled equation is

$$\left(\frac{1}{x} + \frac{y^2}{x^2}\right) dx - \frac{2y}{x} dy = 0,$$

and it is exact, since both cross-derivatives equal $2y/x^2$. Integrating the first component in x gives $F = \ln|x| - \frac{y^2}{x} + g(y)$, and $F_y = -2y/x$ already matches the second component, so $g' = 0$ and the solution is

$$\ln|x| - \frac{y^2}{x} = C.$$

The sign of the y^2 term is the whole difficulty: $\int y^2 x^{-2} dx = -y^2/x$, not $+y^2/x$. Through the point $(1, 1)$ the constant is -1 , and following the trajectory numerically to $x = 2.5$ gives $y \approx 1.8708$, for which $\ln 2.5 - y^2/2.5 = -1.000000$ as it must.

A FACTOR THAT IS AN EXPONENTIAL RATHER THAN A POWER

$$(x^2 + y^2 + 2x) dx + 2y dy = 0.$$

Here $M_y = 2y$ and $N_x = 0$, so $\frac{M_y - N_x}{N} = \frac{2y}{2y} = 1$, free of y , and $\mu = e^x$. The rescaled pair is $e^x(x^2 + y^2 + 2x)$ and $2ye^x$, whose cross-derivatives are both $2ye^x$. Integrating the second in y is the shorter route: $F = y^2 e^x + h(x)$, and matching F_x against the first component gives $h' = e^x(x^2 + 2x)$, whose antiderivative is $x^2 e^x$. So

$$e^x(x^2 + y^2) = C.$$

Through $(0, 1)$ the constant is 1, and the curve reaches $y \approx 0.597102$ at $x = 0.5$, where $e^{0.5}(0.25 + y^2) = 1.000000$. The level curve turns back at $y = 0$, near $x \approx 0.7035$, which is where the solution ceases to be a function of x .

WHERE THIS GOES WRONG

Computing $\frac{M_y - N_x}{M}$, getting something that still contains y , and simplifying it into submission anyway. If the quotient is not a pure function of the single variable the recipe is about, that recipe does not apply, full stop. Try the other one before inventing anything. The second error is dividing by N in a region where N vanishes: the factor you produce is then valid only on one side of the curve $N = 0$.

SEE ALSO Integrating factor depending on y only; integrating factors of the form $x^a y^b$; recovering the potential from M and N ; the $\mu = e^{\int P dx}$ machine.

The companion recipe is not a new idea but the same one with the letters exchanged, and it is worth stating separately only because the sign reversal in the numerator catches everybody at least once. The pair should be learned together and, better, derived together: both come from $\partial_y(\mu M) = \partial_x(\mu N)$ in three lines.

MEDIUM

33.3 Integrating factor depending on y only

CUE Not exact, and $\frac{N_x - M_y}{M}$ is free of x . The numerator is the reverse of the one in the x -only recipe.

METHOD Set $\mu(y) = \exp\left(\int \frac{N_x - M_y}{M} dy\right)$ and proceed as before. The reversal is not a convention: it comes from $\mu' M = \mu(N_x - M_y)$, which is what $\partial_y(\mu M) = \partial_x(\mu N)$ becomes when μ depends on y alone.

WORKED EXAMPLE

$$y dx + (2x - ye^y) dy = 0.$$

Now $M_y = 1$ and $N_x = 2$, so $M_y - N_x = -1$ and the x -only quotient is $-1/(2x - ye^y)$, which certainly involves y . Try the other:

$$\frac{N_x - M_y}{M} = \frac{2 - 1}{y} = \frac{1}{y},$$

free of x , so $\mu = \exp(\int dy/y) = y$. Multiplying gives $y^2 dx + (2xy - y^2 e^y) dy = 0$, whose cross-derivatives are both $2y$. Integrating the first component in x : $F = xy^2 + g(y)$, so $F_y = 2xy + g'(y)$ must equal $2xy - y^2 e^y$, forcing $g' = -y^2 e^y$. Two integrations by parts give $\int y^2 e^y dy = (y^2 - 2y + 2)e^y$, hence

$$xy^2 - (y^2 - 2y + 2)e^y = C.$$

Through $(x, y) = (1, 1)$ the constant is $1 - e \approx -1.718282$; integrating $\frac{dx}{dy} = (ye^y - 2x)/y$ from $y = 1$ to $y = 2$ gives $x \approx 1.7869$, and $xy^2 - (y^2 - 2y + 2)e^y = -1.718282$ there.

A NEGATIVE POWER

$$2xy \, dx + (y^2 - 3x^2) \, dy = 0.$$

$M_y = 2x$ and $N_x = -6x$, so $M_y - N_x = 8x$ and the x -only quotient $8x/(y^2 - 3x^2)$ still carries y . The other quotient is $\frac{N_x - M_y}{M} = \frac{-8x}{2xy} = -\frac{4}{y}$, so $\mu = e^{-4 \ln|y|} = y^{-4}$. Then $F = \int 2xy^{-3} \, dx = x^2y^{-3} + g(y)$, and $F_y = -3x^2y^{-4} + g'$ must equal $y^{-2} - 3x^2y^{-4}$, so $g' = y^{-2}$ and $g = -1/y$. The solution is

$$\frac{x^2}{y^3} - \frac{1}{y} = C, \quad \text{equivalently} \quad x^2 - y^2 = Cy^3.$$

Through $(2, 1)$ the constant is 3; at $y = 1.6$ the trajectory gives $x \approx 3.853310$ and $x^2/y^3 - 1/y = 3.000000$. The special case $C = 0$ is the pair of lines $y = \pm x$, which are genuine solutions and are easy to lose.

WHERE THIS GOES WRONG

Using the x -only numerator ordering with the y -only denominator, producing $\mu = e^{-\int dy/y} = 1/y$ in the example above. That factor makes the equation *less* exact rather than more, and the error survives to the end of the calculation because the two-step integration will still produce some F ; it simply will not be a potential. Verify exactness after multiplying, every time.

SEE ALSO Integrating factor depending on x only; integrating factors of the form $x^a y^b$; recovering the potential from M and N .

When both single-variable tests fail there is still a large family of equations within reach, and it is the family in which M and N are built from monomials. The reason a monomial factor helps is structural: multiplying by $x^a y^b$ shifts every exponent in M and N by the same amounts, so the exactness condition becomes a system of linear equations in a and b rather than a differential equation in anything.

HARD

33.4 Integrating factors $x^a y^b$ and $\mu(xy)$

CUE Neither single-variable quotient is clean, but M and N are sums of monomials in x and y ; or the combination xy runs through the whole equation.

METHOD For the monomial form, posit $\mu = x^a y^b$, impose $\partial_y(\mu M) = \partial_x(\mu N)$, and match the coefficient of each distinct monomial. This gives a small linear system in a and b ; an inconsistent system is a proof that no factor of this shape exists. For the product form, the same derivation as before with $\mu = \mu(z)$, $z = xy$, gives $\mu'(xM - yN) = \mu(N_x - M_y)$, so

$$\frac{\mu'}{\mu} = \frac{M_y - N_x}{yN - xM},$$

and the factor exists in this family precisely when the right-hand side is a function of z alone.

WORKED EXAMPLE

$$(2y - 6x) \, dx + \left(3x - \frac{4x^2}{y}\right) \, dy = 0.$$

With $\mu = x^a y^b$ the two products are $\mu M = 2x^a y^{b+1} - 6x^{a+1} y^b$ and $\mu N = 3x^{a+1} y^b - 4x^{a+2} y^{b-1}$. Differentiate and compare monomial by monomial:

$$2(b+1)x^a y^b - 6bx^{a+1} y^{b-1} = 3(a+1)x^a y^b - 4(a+2)x^{a+1} y^{b-1}.$$

Matching $x^a y^b$ gives $2b + 2 = 3a + 3$; matching $x^{a+1} y^{b-1}$ gives $6b = 4a + 8$. Solving, $a = 1$ and $b = 2$, so $\mu = xy^2$. Multiplying and integrating, $F = \int (2xy^3 - 6x^2 y^2) \, dx = x^2 y^3 - 2x^3 y^2 + g(y)$, and $F_y = 3x^2 y^2 - 4x^3 y$ already equals μN , so $g' = 0$ and the solution is

$$x^2 y^3 - 2x^3 y^2 = C.$$

Through $(1, 1)$ the constant is -1 , confirmed numerically at $x = 1.6$ where $y \approx 0.68975$ and $F = -1.000000$.

$$y(1 + xy) dx - x dy = 0.$$

Here $M = y + xy^2$ and $N = -x$, so $M_y = 1 + 2xy$, $N_x = -1$, and $M_y - N_x = 2 + 2xy$. The denominator in the product recipe is $yN - xM = -xy - xy - x^2y^2 = -(2z + z^2)$ with $z = xy$, so

$$\frac{\mu'}{\mu} = -\frac{2(1+z)}{z(z+2)} = -\frac{1}{z} - \frac{1}{z+2}, \quad \mu = \frac{1}{z(z+2)} = \frac{1}{xy(xy+2)}.$$

Carrying out the two-step integration in the variable z and simplifying, the potential is $\frac{1}{2} \ln |z(z+2)| - \ln |y|$, and exponentiating twice turns the level curves into

$$x^2 + \frac{2x}{y} = C.$$

The same equation is Bernoulli in disguise, since $y' = y/x + y^2$; solving it that way with $u = y^{-1}$ gives $u' + u/x = -1$, then $xu = -\frac{1}{2}x^2 + C_1$, that is $x^2 + 2x/y = 2C_1$. Two routes, one answer, which is the cheapest possible check.

WHERE THIS GOES WRONG

Solving the exponent system and then not verifying exactness. The matching step usually gives more equations than unknowns, and if you drop one of them by accident you will get an (a, b) that satisfies part of the condition and nothing else. Substitute the factor back and compute both cross-derivatives before going on. Note also that an inconsistent system is information, not a mistake: it proves that no monomial factor works, and you should stop looking for one.

SEE ALSO Integrating factor depending on x only; integrating factor depending on y only; grouping and solving by inspection; recovering the potential from M and N .

Substitutions that linearise

The techniques so far preserve the equation and change the multiplier. The remaining family does the opposite: it leaves the multiplier problem alone and changes the unknown, choosing a new dependent variable in which the equation is linear. There are exactly three changes worth knowing, and each is matched to a specific shape of nonlinearity: a power of y , an exponential of y , and a quadratic in y with a variable coefficient.

The first is Bernoulli's, and it is not a guess either. Given $y' + Py = Qy^n$ with $n \neq 0, 1$, divide through by y^n to isolate the nonlinearity on the left:

$$y^{-n}y' + Py^{1-n} = Q.$$

Now the second term already contains y^{1-n} , so set $u = y^{1-n}$; then $u' = (1-n)y^{-n}y'$, which is exactly $(1-n)$ times the first term. Multiplying the whole equation by $(1-n)$ therefore turns it into

$$u' + (1-n)Pu = (1-n)Q,$$

linear in u . The substitution is forced by the requirement that u' reproduce $y^{-n}y'$, in the same way that μ was forced by the requirement that $(\mu y)'$ reproduce $\mu y' + \mu Py$.

MEDIUM

33.5 Bernoulli via $u = y^{1-n}$

CUE $y' + P(x)y = Q(x)y^n$ with n neither 0 nor 1. The exponent may be negative or fractional; only 0 and 1 are excluded, and for the good reason that those two cases are already linear.

METHOD Divide by y^n , substitute $u = y^{1-n}$, and solve the linear equation $u' + (1-n)Pu = (1-n)Q$ by the integrating factor $\exp((1-n) \int P dx)$. Recover $y = u^{1/(1-n)}$. For $n > 0$ the constant function $y \equiv 0$ also solves the original equation and is destroyed by the division; restore it by hand if the problem calls for all solutions.

WORKED EXAMPLE

$$y' + 2xy = xe^{-2x^2}y^4, \quad y(0) = 1.$$

Here $n = 4$, so $1 - n = -3$ and $u = y^{-3}$. Dividing by y^4 and multiplying by -3 :

$$u' - 6xu = -3xe^{-2x^2}.$$

The integrating factor is $\exp(-\int 6x \, dx) = e^{-3x^2}$, and

$$(ue^{-3x^2})' = -3xe^{-5x^2}, \quad ue^{-3x^2} = \frac{3}{10}e^{-5x^2} + C.$$

Hence $u = \frac{3}{10}e^{-2x^2} + Ce^{3x^2}$, and $u(0) = 1$ gives $C = \frac{7}{10}$. Therefore

$$y = \left(\frac{3}{10}e^{-2x^2} + \frac{7}{10}e^{3x^2}\right)^{-1/3}, \quad y(1) = \left(\frac{3}{10}e^{-2} + \frac{7}{10}e^3\right)^{-1/3} \approx 0.413925.$$

The bracket is positive for every x , so the cube root is unambiguous and the solution is global.

WHERE THIS GOES WRONG

Taking the final root without tracking sign and domain. When $1-n$ is even, $y = u^{1/(1-n)}$ has two branches and the initial condition chooses one; when $1-n$ is odd there is no ambiguity but u may pass through zero, and y blows up there. Also, n is the exponent on the right-hand side after the equation is in standard form: for $2y' + 4xy = 6xy^3$ you must divide by 2 first, and $Q = 3x$, not $6x$.

SEE ALSO The $\mu = e^{\int P \, dx}$ machine; exponential linearisation $v = e^y$; homogeneous equations via $y = vx$; Riccati and the second-order linear equation.

Bernoulli handles a power of y on the right. If the nonlinearity is an exponential of y instead, the same strategy applies with a different new variable, chosen by the same criterion: pick v so that v' absorbs the offending term. The choice $v = e^y$ gives $v' = e^y y' = vy'$, which turns a term e^{-y} into a constant, and that is precisely what is needed.

MEDIUM

33.6 Exponential linearisation $v = e^y$

CUE The nonlinearity is $e^{\pm y}$ rather than a power of y , most often as $y' = g(x) + e^{-y}$ or $y' + g(x) = h(x)e^{-y}$.

METHOD Put $v = e^y$, so that $v' = vy'$. Multiplying the equation through by v converts e^{-y} into 1 and every term $c(x)$ into $c(x)v$, giving a linear equation for v . Solve it, then $y = \ln v$, and report the interval on which $v > 0$.

WORKED EXAMPLE

$$y' = \frac{2}{t} + e^{-y}, \quad y(1) = 0.$$

Multiply by $v = e^y$: the left side becomes v' , the first term becomes $2v/t$, and the second becomes 1. So

$$v' - \frac{2}{t}v = 1, \quad \mu = t^{-2}, \quad (vt^{-2})' = t^{-2}, \quad vt^{-2} = -\frac{1}{t} + C.$$

Thus $v = -t + Ct^2$, and $v(1) = e^0 = 1$ gives $C = 2$, so $v = 2t^2 - t$ and $y = \ln(2t^2 - t)$. In particular $y(1.5) = \ln 3 \approx 1.098612$. The solution is valid on $t > \frac{1}{2}$, where $v > 0$; the initial point $t = 1$ lies in that interval and the interval of validity is $(\frac{1}{2}, \infty)$.

WHERE THIS GOES WRONG

Forgetting that $v = e^y$ is strictly positive, and reporting $y = \ln(2t^2 - t)$ as though it were defined for all t . The closed form has a logarithmic singularity where v vanishes, and the honest answer names the interval containing the initial point. A solution formula without its interval of validity is only half an answer for any equation whose coefficients are singular somewhere.

SEE ALSO Bernoulli via $u = y^{1-n}$; the $\mu = e^{\int P \, dx}$ machine; equations linear in the other variable.

The last of the three linearising substitutions is the deepest, because it does not produce a first-order linear equation at all. A Riccati equation $y' = q_0(x) + q_1(x)y + q_2(x)y^2$ is the general first-order equation quadratic in y , and it is not solvable in closed form in general. If a single particular solution is known, subtracting it reduces the problem to Bernoulli with $n = 2$ and the matter is finished in three lines. If none is known, there is still one thing to do: trade the nonlinear first-order equation for a linear second-order one.

33.7 Riccati to a second-order linear equation

CUE A Riccati equation with no particular solution supplied, especially one where the associated second-order equation is recognisable: constant coefficients, Cauchy and Euler type, or Airy.

METHOD Substitute $y = -\frac{u'}{q_2 u}$. Expanding and cancelling the $(u')^2$ terms gives

$$q_2 u'' - (q_2' + q_1 q_2) u' + q_2^2 q_0 u = 0,$$

a linear second-order equation. Solve it, then differentiate back. Since u and λu give the same y , the two constants of the second-order equation collapse to the one constant a first-order equation is entitled to.

WORKED EXAMPLE

$$y' = y^2 - \frac{2}{x^2}.$$

Here $q_2 = 1$, $q_1 = 0$, $q_0 = -2/x^2$, so the formula gives $u'' - \frac{2}{x^2} u = 0$, that is $x^2 u'' - 2u = 0$. This is a Cauchy and Euler equation: trying $u = x^r$ gives $r(r-1) = 2$, so $r = 2$ or $r = -1$ and $u = Ax^2 + Bx^{-1}$. Then

$$y = -\frac{u'}{u} = -\frac{2Ax - Bx^{-2}}{Ax^2 + Bx^{-1}} = \frac{k - 2x^3}{x(x^3 + k)}, \quad k = \frac{B}{A}.$$

Only the ratio k survives, as promised. At $k = 0$ this is $y = -2/x$, which checks instantly: $y' = 2/x^2$ and $y^2 - 2/x^2 = 4/x^2 - 2/x^2 = 2/x^2$. At $k = 1$ and $x = 1$ the solution starts at $y = -\frac{1}{2}$ and reaches $y(1.7) \approx -0.878025$, matching the closed form to six places.

WHEN THE ANSWER IS THAT THERE IS NO ANSWER

$$y' = y^2 - x.$$

Same substitution, now with $q_0 = -x$: the second-order equation is $u'' - xu = 0$, the Airy equation. Its solutions are not elementary, so the honest conclusion is that this Riccati equation has no elementary closed form, and the substitution is what proves it rather than what solves it. That is a useful outcome: it tells you to stop searching and to switch to a series solution or a numerical method.

THE SHORT ROUTE, WHEN ONE SOLUTION CAN BE GUESSED

$$y' = x^2 + 1 - y^2.$$

Guess $y_1 = x$: then $y_1' = 1$ and $x^2 + 1 - x^2 = 1$, so it solves the equation. Now put $y = x + \frac{1}{w}$. Then $y' = 1 - \frac{w'}{w^2}$, while the right-hand side becomes $1 - \frac{2x}{w} - \frac{1}{w^2}$. Cancelling the 1 and multiplying by $-w^2$,

$$w' - 2xw = 1, \quad \mu = e^{-x^2}, \quad (w\mu)^{-1} = e^{-x^2},$$

so $w\mu^{-1} = \frac{\sqrt{\pi}}{2} \operatorname{erf}(x) + C$ and

$$y = x + \frac{e^{-x^2}}{\frac{\sqrt{\pi}}{2} \operatorname{erf}(x) + C}.$$

The answer is not elementary, but it is explicit and one line long, which the second-order route would not have delivered. With $C = 1$ the solution starts at $y(0) = 1$ and reaches $y(1.2) \approx 1.331135$, matching numerical integration to eleven places. The general lesson: a Riccati equation with a known solution is a Bernoulli equation with $n = 2$ in the variable $y - y_1$, hence a linear equation in $w = 1/(y - y_1)$, and the whole reduction is three lines.

WHERE THIS GOES WRONG

Reaching for this substitution when a particular solution is available or guessable. If y_1 solves the equation, then $y = y_1 + 1/w$ turns it into a linear first-order equation for w , which is far shorter. Try the standard guesses first: a constant, a multiple of $1/x$, a linear function. Only when they all fail is the second-order route the right one.

SEE ALSO Bernoulli via $u = y^{1-n}$; the $\mu = e^{\int P dx}$ machine; second-order linear equations with variable coefficients.

Equations that become linear after one differentiation

Each of the three substitutions in the previous section was chosen by the same criterion, and the criterion is worth naming because it transfers to equations not covered here. Look at the nonlinearity, ask what new variable would have that nonlinearity as its derivative or as a factor of its derivative, and adopt it. For Qy^n the answer is $u = y^{1-n}$; for e^{-y} it is $v = e^y$; for $q_2 y^2$ it is the logarithmic derivative $y = -u'/(q_2 u)$, which is exactly the substitution that turns a quadratic term into a linear one at the cost of one order. Nothing about any of them is memorised, and if you ever meet a nonlinearity not on the list, that question is where to start.

There is one more way a linear first-order equation arrives in disguise, and it is not an algebraic disguise but an integral one. When the unknown function appears inside an integral with a variable upper limit, the Fundamental Theorem of Calculus converts the problem into a differential equation in one step, and the equation it produces is almost always linear. The only skill involved is remembering where the initial condition went.

MEDIUM

33.8 Integral equations differentiated into ODEs

CUE The unknown appears under an integral sign with a variable limit: $y(x) = A + \int_a^x K(t)y(t) dt$, or any relation in which y is defined self-referentially by such an integral.

METHOD Differentiate both sides. The Fundamental Theorem turns $\int_a^x K(t)y(t) dt$ into $K(x)y(x)$, so the relation becomes an ordinary differential equation. Then recover the initial condition by setting $x = a$ in the *original* integral relation, where the integral is zero and the value of $y(a)$ can be read straight off. Solve the resulting linear equation as usual.

WORKED EXAMPLE

$$y(x) = 3 + \int_0^x \frac{y(t)}{t^2 + 1} dt.$$

Differentiating, $y' = \frac{y}{1+x^2}$; setting $x = 0$ in the original, $y(0) = 3$. The equation is separable as well as linear, and the integrating factor $\mu = \exp(-\arctan x)$ gives $(ye^{-\arctan x})' = 0$, so $y = 3e^{\arctan x}$ and

$$y(1) = 3e^{\pi/4} \approx 6.579840.$$

A second instance where the integral carries an inhomogeneous term:

$$y(x) = 2 + \int_1^x \frac{t - y(t)}{t} dt \implies y' = 1 - \frac{y}{x}, \quad y(1) = 2.$$

Then $y' + y/x = 1$, $\mu = x$, $(xy)' = x$, $xy = \frac{1}{2}x^2 + C$, and $y(1) = 2$ gives $C = \frac{3}{2}$, so $y = \frac{x}{2} + \frac{3}{2x}$ and $y(3) = 2$ exactly.

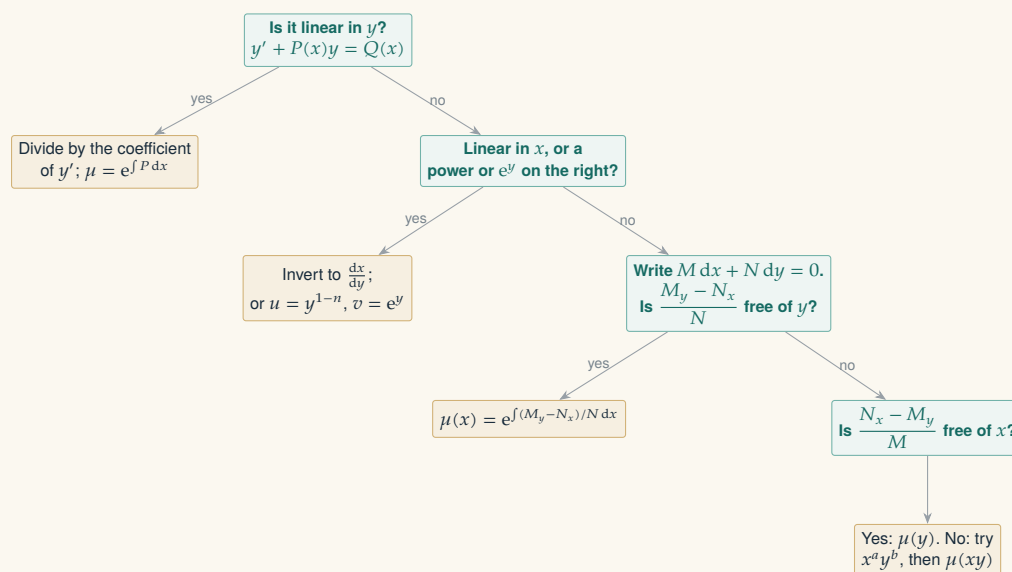
WHERE THIS GOES WRONG

Losing the initial condition. Differentiation destroys the additive constant A , so the differential equation alone has a one-parameter family of solutions and only the original integral relation knows which one is meant. Go back to $x = a$ and read $y(a) = A$ off there. A second trap: if x appears both inside the integrand and as the limit, as in $y(x) = \int_0^x (x-t)y(t) dt$, you must differentiate under the integral sign as well as at the limit, and the result is second order, not first.

SEE ALSO The $\mu = e^{\int P dx}$ machine; separable equations; Laplace transforms and convolution kernels.

Choosing the integrating factor

The decision is quick because the tests are cheap. Put the equation in differential form $M dx + N dy = 0$, or in standard form $y' + Py = Q$ if it is already linear, and then work down the list: linear in y , linear in x , exact, one-variable factor, monomial or product factor. Every step of the list costs one derivative and one division, so the whole search takes less time than a single wrong integration.



Two remarks about how to walk the tree. First, the order matters: test linearity in y before writing anything in differential form, because the linear route is shorter and because converting to $M dx + N dy = 0$ throws away the information that told you which variable was the unknown. Second, the tests are not exclusive. An equation can be linear in y , linear in x , and exact all at once, and when that happens any of the three routes is correct; take the one whose integrals you can do. The example $y(1 + xy) dx - x dy = 0$ earlier in this chapter was solvable both as a product-factor problem and as a Bernoulli equation, and agreement between the two was worth more than either derivation on its own.

The tree has a deliberate omission at the bottom. If every branch fails, the equation may still be separable, homogeneous of degree zero, or reducible by a substitution such as $u = ax + by$, and those tests belong to a different chapter; nothing in this one will help. What the tree does guarantee is that you will never multiply by a factor you cannot justify, because every leaf is the conclusion of a derivation rather than a rule to be recalled.

VERIFY THE FACTOR, NOT THE ANSWER

The characteristic failure in this chapter is not a wrong method but a factor that is nearly right: a sign reversed in a numerator, a constant of integration carried through an exponential, a division by $a_1(x)$ omitted. All of them produce a plausible μ and a plausible final formula. The check that catches every one of them costs two derivatives: after multiplying by μ , compute $\partial_y(\mu M)$ and $\partial_x(\mu N)$ and confirm they agree. For a linear equation the equivalent check is even faster: differentiate μy and see whether you get $\mu(y' + Py)$. Do this before integrating, not after, because an integration performed against the wrong factor is entirely wasted work.

AUDIT THE SOLUTION NUMERICALLY

Every closed form in this chapter can be checked to six decimal places in under a minute, and two checks between them catch essentially all errors. The algebraic one: substitute the claimed solution back into the *original* equation, not the standard form, since the division by $a_1(x)$ is itself a common place to slip. The numerical one: integrate the equation by fourth-order Runge and Kutta from the initial point to some interior point and compare with the formula. That second check is decisive for implicit answers too, where substitution is awkward: evaluate the claimed first integral $F(x, y)$ at both ends of a numerically computed trajectory and confirm that the two numbers agree. Every worked example in this chapter was verified that way, and the discrepancies quoted are the real ones, of order 10^{-11} , which is what a correct closed form against a fine step size looks like. An error in a sign or an exponent shows up in the third digit, not the eleventh.

A final habit. When an equation resists every route in the tree, go back and ask what the equation would look like if it were meant to be solved by one of them. Linear equations are built by differentiating a product, so a solvable one has a μ with a recognisable antiderivative behind it; exact equations are built by differentiating a potential, so a solvable one has cross-derivatives that agree after one cancellation. Problems in this family are constructed forwards and presented backwards, and reconstructing the forward step is usually faster than grinding through the backward one.

Recognition matrix

If you see	Reach for	If that fails
$y' + P(x)y = Q(x)$ $a_1(x)y' + a_0(x)y = f(x)$ An initial condition where a_1 vanishes y' as a quotient, y ugly downstairs $y' + Py = Qy^n$, $n \neq 0, 1$ e^{-y} on the right-hand side $y' = q_0 + q_1y + q_2y^2$, y_1 known $y' = q_0 + q_1y + q_2y^2$, nothing known $M dx + N dy = 0$, $M_y = N_x$ Not exact, $\frac{M_y - N_x}{N}$ free of y Not exact, $\frac{N_x - M_y}{M}$ free of x M, N sums of monomials xy appearing throughout $y = A + \int_a^x K(t)y(t) dt$ A factor μ you cannot derive Q a sum of unlike terms Q piecewise, with a jump at c Q with no elementary antiderivative	$\mu = e^{\int P dx}$; integrate $(\mu y)' = \mu Q$ Divide by a_1 first, then read P and Q Substitute the point into the original equation Invert: solve $\frac{dx}{dy} + P(y)x = Q(y)$ Bernoulli: $u = y^{1-n}$, then linear in u $v = e^y$; multiply through by v $y = y_1 + 1/w$: linear first order in w $y = -u'/(q_2u)$; second-order linear in u Exact: recover the potential directly $\mu(x) = \exp \int \frac{M_y - N_x}{N} dx$ $\mu(y) = \exp \int \frac{N_x - M_y}{M} dy$ Posit $\mu = x^a y^b$; match exponents $\mu(z)$ with $\frac{\mu'}{\mu} = \frac{M_y - N_x}{yN - xM}$, $z = xy$ Differentiate; read $y(a) = A$ off at $x = a$ Recompute from $\partial_y(\mu M) = \partial_x(\mu N)$ Superpose: one μ , several particular solutions Solve each piece; match y at c , not y' Leave the definite integral in the answer	Check you divided by the coefficient of y' Locate the zeros of a_1 : singular points Expect no solution or infinitely many Test whether M is free of x Restore the lost solution $y \equiv 0$ Report the interval where $v > 0$ Bernoulli with $n = 2$ on $y - y_1$ Airy or Bessel means no elementary form Recheck both partial derivatives Try the y -only recipe instead Note the reversed numerator Inconsistent system: no such factor Group the exact blocks by inspection Check whether x is also inside the integrand Verify exactness before integrating Integrate μQ term by term The definite form does this for you Variation of parameters is the same formula

Exact equations and potential functions

What this chapter is for. An exact equation is a level-set problem wearing a differential disguise. If $F(x, y) = C$ describes a family of curves, then differentiating along any one of them gives $F_x dx + F_y dy = 0$, and every exact equation is such a statement read backwards: you are handed $M dx + N dy = 0$ and asked to find the F it came from. The test $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ is not a computational rule but an existence statement, the assertion that (M, N) really is the gradient of something, and it works because mixed second partial derivatives commute. Recovering F is then a two-step integration in which the constant of the first integration is a function of the other variable, and that single observation is what separates people who can do this from people who cannot. This chapter teaches the recovery as a mechanical procedure with the check built in, shows how to choose which of M and N to integrate first, handles the case where the equation is not exact until an integrating factor makes it so, and covers the two shortcuts worth having: reading off exact groups by inspection, and spotting exactness one order up in a second-order equation. It ends where the subject continues, at conservative vector fields and path independence.

A level set in disguise

Recall the total differential. For a differentiable $F(x, y)$,

$$dF = \frac{\partial F}{\partial x} dx + \frac{\partial F}{\partial y} dy,$$

and the geometric content is that dF measures the first-order change in F produced by an arbitrary small step (dx, dy) . On a level curve $F = C$ there is no change, so a curve of the family satisfies $F_x dx + F_y dy = 0$. Turn that around. Given

$$M(x, y) dx + N(x, y) dy = 0,$$

the equation is called *exact* when there is an F with $F_x = M$ and $F_y = N$, and in that case the general solution is the implicit relation $F(x, y) = C$. No integration of the differential equation is required at all: the solution curves are the level curves of a function you construct once.

The simplest exact equations are ones nobody calls exact. A separable equation written as $f(x) dx - g(y) dy = 0$ has $M_y = 0$ and $N_x = 0$, so it passes the test trivially, and the two-step recovery degenerates to $F = \int f dx$ followed by $g(y) = -\int g dy$, which is separation of variables verbatim. Exactness is therefore the correct generalisation of separability: it is what remains of “integrate each side” when the two sides refuse to come apart. Keeping that in mind stops the procedure from feeling arbitrary, because you have been doing the degenerate case since the beginning.

The test for whether such an F exists is Clairaut’s theorem. If $F_x = M$ and $F_y = N$ with F twice continuously differentiable, then

$$\frac{\partial M}{\partial y} = \frac{\partial^2 F}{\partial y \partial x} = \frac{\partial^2 F}{\partial x \partial y} = \frac{\partial N}{\partial x},$$

so $M_y = N_x$ is necessary. It is also sufficient, provided M and N are continuously differentiable on a region with no holes in it, and the hole condition is not a technicality invented by pedants: on the punctured plane the pair $M = -y/(x^2 + y^2)$, $N = x/(x^2 + y^2)$ passes the test everywhere and has no single-valued potential, because the candidate is $\arctan(y/x)$, which gains 2π on a circuit around the origin. On any disc avoiding the origin it is exact; on the annulus it is not. For the equations in this chapter the region is a rectangle or a half-plane and the test is decisive.

THE PICTURE

Plot the level curves of F . At each point the gradient $(F_x, F_y) = (M, N)$ points across them, in the direction of steepest increase, and the solution curve is the one running perpendicular to it. The differential equation $y' = -M/N$ is just that statement written as a slope: the tangent direction (dx, dy) must have zero dot product with (M, N) , that is $M dx + N dy = 0$. Everything that looks mysterious in the algebra is obvious in this picture, including why $N = 0$ produces vertical tangents (the gradient is horizontal there) and why the solution is a family rather than a single curve (one curve per value of C).

$M_y = N_x$ is the statement that the vector field (M, N) is a gradient. Every consequence follows from that single reading. The solution is a level curve because gradients are perpendicular to level curves. The recovery works because a gradient is determined by its potential and vice versa. The failure of the test means the field has circulation, and that is why an integrating factor can rescue the situation: multiplying by μ rescales the field pointwise, which changes its circulation without changing its direction, hence without changing the solution curves.

That last sentence deserves emphasis, because it is the reason the two chapters on this material belong together. Multiplying $M dx + N dy = 0$ by any nonvanishing $\mu(x, y)$ leaves the solution set untouched: the equation $\mu M dx + \mu N dy = 0$ has exactly the same curves. So exactness is a property of the *representation*, not of the equation, and every first-order equation is exact in some representation. Finding the right one is the integrating-factor problem; recovering F once the representation is right is this chapter.

CORE

34.1 Recovering the potential F from M and N

CUE The equation is in the form $M dx + N dy = 0$ and the test $M_y = N_x$ has been confirmed by actual differentiation, not by appearance.

METHOD Two integrations with a matching step between them.

1. Integrate M with respect to x , holding y fixed: $F = \int M dx + g(y)$. The unknown is a *function of the other variable*, never a bare constant, because anything depending on y alone has zero x -derivative.
2. Differentiate that expression with respect to y and set it equal to N . Every term containing x must cancel; what remains is an equation $g'(y) = (\text{function of } y \text{ alone})$.
3. Integrate g' and write $F(x, y) = C$.

The cancellation in step 2 is the built-in check. If an x survives on the right, either the equation was not exact or a derivative was computed wrongly, and you have caught the error before doing any further work.

WORKED EXAMPLE

$$(2x + y^2) dx + (2xy + 3y^2) dy = 0.$$

Test: $M_y = 2y$ and $N_x = 2y$, so the equation is exact. Integrating M in x ,

$$F = x^2 + xy^2 + g(y).$$

Differentiating in y and matching to N : $2xy + g'(y) = 2xy + 3y^2$. The $2xy$ cancels, as it must, leaving $g' = 3y^2$ and $g = y^3$. Hence

$$x^2 + xy^2 + y^3 = C.$$

Through $(1, 1)$ the constant is 3; following the curve to $x = 1.55$ gives $y \approx 0.535286$, and $x^2 + xy^2 + y^3 = 3.000000$ there. Note that the curve has a vertical tangent where $N = y(2x + 3y)$ vanishes, at $y = 0$ and $x = \sqrt{3}$: the level curve continues past that point but y stops being a function of x , which is exactly why the implicit form is the honest answer.

WITH A TRANSCENDENTAL POTENTIAL

$$(y \cos x + 2xe^y) dx + (\sin x + x^2e^y - 1) dy = 0.$$

Both cross-derivatives equal $\cos x + 2xe^y$, so the equation is exact. Integrating M in x gives $F = y \sin x + x^2e^y + g(y)$, and then

$$F_y = \sin x + x^2e^y + g'(y) = \sin x + x^2e^y - 1 \implies g' = -1, g = -y.$$

So $y \sin x + x^2e^y - y = C$. The two x -dependent terms cancelled exactly, which is the confirmation that the first integration was right.

WHERE THIS GOES WRONG

Writing $+C$ instead of $+g(y)$ after the first integration. This is the single defining error of the technique, and it is silent: it throws away every term of the potential that depends on y alone, so in the first example above it would return $x^2 + xy^2 = C$, which satisfies nothing. The tell is that step 2 then produces an inconsistency you have no way to absorb. A related error is integrating M with respect to x but treating y as though it were a function of x ; it is not, during this step it is a constant.

SEE ALSO Integrate whichever variable is easier; grouping and solving by inspection; integrating factors depending on one variable; the total differential.

Step 1 chose M arbitrarily, and that choice is free. The procedure is symmetric in the two variables, and half the labour in a badly organised solution comes from integrating the wrong one. Which is wrong is usually obvious at a glance: one of M and N is a polynomial and the other contains a product of transcendental functions produced by the product rule, and it is the polynomial one you want.

MEDIUM

34.2 Integrate whichever variable is easier

CUE One of M , N is markedly uglier than the other, typically because the potential contains a product such as $e^{xy} \cos 2x$ whose x -derivative has two terms and whose y -derivative has one.

METHOD Start from $F = \int N \, dy + h(x)$ instead. Now the unknown is $h(x)$, and you match F_x against M . The two routes produce the same F , so choose the one that leaves you an elementary single-variable integral. As a rule of thumb, whichever of M and N has fewer terms was produced by differentiating in the variable you should integrate in.

WORKED EXAMPLE

$$2xy^3 \, dx + (3x^2y^2 + 4y) \, dy = 0.$$

Exact, since both cross-derivatives are $6xy^2$. Integrating N in y is immediate:

$$F = x^2y^3 + 2y^2 + h(x), \quad F_x = 2xy^3 + h'(x) = 2xy^3 \implies h' = 0.$$

So $x^2y^3 + 2y^2 = C$, in two lines. Going the other way costs the same here, but the next example does not have that symmetry.

WHERE THE CHOICE MATTERS

$$(ye^{xy} \cos 2x - 2e^{xy} \sin 2x + 2x) \, dx + (xe^{xy} \cos 2x - 3) \, dy = 0.$$

M has three terms and one of them is a product-rule leftover; N has two and integrates in y on sight, since $\int xe^{xy} \cos 2x \, dy = e^{xy} \cos 2x$ with x held fixed. So

$$F = e^{xy} \cos 2x - 3y + h(x).$$

Differentiating in x : $F_x = ye^{xy} \cos 2x - 2e^{xy} \sin 2x + h'(x)$, and matching to M gives $h' = 2x$, so $h = x^2$ and

$$e^{xy} \cos 2x - 3y + x^2 = C.$$

Both product-rule terms cancelled at once. Starting from M instead would have required integrating $ye^{xy} \cos 2x - 2e^{xy} \sin 2x$ with respect to x , which is only doable by recognising it as a derivative, that is, by already knowing the answer.

WHERE THIS GOES WRONG

Carrying out both routes and then adding the two expressions for F , which doubles the potential. They are two derivations of the same function, not two halves of it. If the two routes give visibly different answers, one of them contains an arithmetic error, or the two constants differ, which is harmless: $F = C$ and $F + 5 = C$ describe the same family.

SEE ALSO Recovering the potential F from M and N ; grouping and solving by inspection.

When the equation is not exact

Most equations that arrive in the form $M \, dx + N \, dy = 0$ fail the test, and failing it is not a verdict on the equation but on the way it has been written. Because multiplying by a nonvanishing μ leaves the solution curves alone, the question is whether some μ makes the representation exact, and the systematic search for one is the business of the previous chapter. Here it is worth seeing the whole cycle run once, from a failed test through the factor to the potential, because in practice the two chapters are used as a single procedure and the seam between them should be invisible.

NOT EXACT, THEN EXACT

$$(3xy + y^2) \, dx + (x^2 + xy) \, dy = 0.$$

First the test: $M_y = 3x + 2y$ and $N_x = 2x + y$, so $M_y - N_x = x + y$ and the equation is not exact. Try the x -only recipe, which asks whether $(M_y - N_x)/N$ is free of y :

$$\frac{M_y - N_x}{N} = \frac{x + y}{x^2 + xy} = \frac{x + y}{x(x + y)} = \frac{1}{x},$$

which it is, once the common factor $x + y$ is cancelled. Hence $\mu = \exp(\int dx/x) = x$, and the rescaled equation is

$$(3x^2y + xy^2) dx + (x^3 + x^2y) dy = 0.$$

Verify before proceeding: $\partial_y(3x^2y + xy^2) = 3x^2 + 2xy$ and $\partial_x(x^3 + x^2y) = 3x^2 + 2xy$. Exact. Now integrate M in x :

$$F = x^3y + \frac{1}{2}x^2y^2 + g(y), \quad F_y = x^3 + x^2y + g'(y) = x^3 + x^2y,$$

so $g' = 0$ and the solution is $x^3y + \frac{1}{2}x^2y^2 = C$, or more cleanly

$$2x^3y + x^2y^2 = C.$$

Through $(1, 1)$ the constant is 3. Following the curve to $x = 1.4$ gives $y \approx 0.468318$, and $2x^3y + x^2y^2 = 3.000000$ there; at $x = 2$ the relation $4y^2 + 16y = 3$ gives $y \approx 0.179449$. Note that the factor $\mu = x$ vanishes at $x = 0$, so the derivation is valid on $x > 0$ and on $x < 0$ separately, and the line $x = 0$ has to be examined on its own: it is itself a solution of the original equation, since M and N both vanish along $x = 0$ only where $y = 0$, and the curve $x = 0$ satisfies $dx = 0$ with $N dy = 0$.

THE SAME CYCLE WITH A y -ONLY FACTOR

$$2xy dx + \left(3x^2 + \frac{1}{y}\right) dy = 0.$$

Test: $M_y = 2x$, $N_x = 6x$, not exact. The x -only quotient is $\frac{M_y - N_x}{N} = \frac{-4x}{3x^2 + 1/y}$, which carries both variables, so try the other:

$$\frac{N_x - M_y}{M} = \frac{6x - 2x}{2xy} = \frac{2}{y},$$

free of x . Hence $\mu = \exp(\int 2 dy/y) = y^2$, and the rescaled equation $2xy^3 dx + (3x^2y^2 + y) dy = 0$ is exact, both cross-derivatives being $6xy^2$. Integrating M in x : $F = x^2y^3 + g(y)$, then $F_y = 3x^2y^2 + g' = 3x^2y^2 + y$ forces $g = \frac{1}{2}y^2$, so

$$x^2y^3 + \frac{1}{2}y^2 = C.$$

Through $(1, 1)$ the constant is $\frac{3}{2}$; at $x = 1.35$ the trajectory gives $y \approx 0.854046$ and $F = 1.500000$. The two examples in this section are the same procedure with the two recipes, and the only judgement involved was deciding which quotient to compute first.

The cancellation of $x + y$ in that computation was not luck. The recipes for one-variable integrating factors work exactly when a common factor cancels out of the quotient, so when you compute $(M_y - N_x)/N$ the first thing to do is factor the numerator and the denominator rather than expand them. An expanded quotient looks hopeless and a factored one is often a monomial.

Reading the exact structure by eye

The systematic procedure always works, and it is often slower than looking. Certain combinations of differentials are exact and recognisable on sight, and a solver who knows them can integrate an equation in one line that the procedure takes half a page to grind out. The list is short: it is essentially the derivative table of the product and quotient rules read backwards, and it is worth having in permanent memory.

Group	Is d of	Look for it when
$x dy + y dx$	xy	M and N are y and x times the same thing
$x dx + y dy$	$\frac{1}{2}(x^2 + y^2)$	M and N are x and y times the same thing
$x dy - y dx$	$\frac{y}{x}$	the equation is homogeneous of degree zero
$y dx - x dy$	$\frac{x}{y}$	the same, with the roles exchanged
$\frac{y^2}{x} dx - \frac{x^2}{y} dy$	$\ln \frac{y}{x}$	every term carries a factor xy
$\frac{x dy - y dx}{x^2 + y^2}$	$\arctan \frac{y}{x}$	$x^2 + y^2$ appears, or should
$\frac{x dx + y dy}{x^2 + y^2}$	$\frac{1}{2} \ln(x^2 + y^2)$	the same, symmetric numerator
$\frac{x dx + y dy}{\sqrt{x^2 + y^2}}$	$\sqrt{x^2 + y^2}$	a radial distance is the natural variable

Every entry is the product or quotient rule read backwards, and the four with an antisymmetric numerator are all the same identity divided by different things: $x dy - y dx$ is exact after division by x^2 , by y^2 , by xy , or by $x^2 + y^2$, and which divisor to use is decided by what else is in the equation. The skill is not remembering the list but seeing where to split the given expression so that one of these appears, and the split is usually forced by the antisymmetric combination $x dy - y dx$ or the symmetric one $x dx + y dy$.

HARD

34.3 Grouping and solving by inspection

CUE $M dx + N dy$ visibly contains one of the recognisable exact blocks, most often $x dy + y dx$ or $x dy - y dx$, with the rest of the expression a pure differential in one variable.

METHOD Split the expression into exact groups plus a single-variable remainder, integrate each group by sight, and add. When an antisymmetric block appears, look for the divisor that turns it into an exact form: x^2 , y^2 , xy , or $x^2 + y^2$, and be prepared to multiply the whole equation by that divisor's reciprocal, which is an integrating factor found by inspection rather than by recipe.

WORKED EXAMPLE

$$(x^2 + y^2 + y) dx - x dy = 0.$$

Regroup so the antisymmetric block stands alone: $(x^2 + y^2) dx + (y dx - x dy) = 0$. Divide by $x^2 + y^2$, which is the integrating factor the block is asking for:

$$dx - \frac{x dy - y dx}{x^2 + y^2} = 0 \implies d\left(x - \arctan \frac{y}{x}\right) = 0,$$

so the solution is $x - \arctan \frac{y}{x} = C$. Through $(1, 1)$ the constant is $1 - \frac{\pi}{4} \approx 0.214602$, and following the trajectory to $x = 1.7$ gives $y \approx 19.858$, at which point $x - \arctan(y/x) = 0.214602$ as required. Neither one-variable recipe would have found this factor: $(M_y - N_x)/N = -(2y + 2)/x$ involves both variables, and so does the other quotient.

THE SYMMETRIC BLOCK, WITH A CHAIN RULE ON TOP

$$(y \cos(xy) + 2x) dx + x \cos(xy) dy = 0.$$

The two terms carrying the cosine are $\cos(xy)(y dx + x dy)$, and $y dx + x dy = d(xy)$, so that pair is $\cos(u) du$ with $u = xy$, which is $d(\sin(xy))$. The remaining term is $2x dx = d(x^2)$. Hence the whole expression is $d(\sin(xy) + x^2)$ and the solution is

$$\sin(xy) + x^2 = C.$$

Confirm by differentiating: $F_x = y \cos(xy) + 2x$ and $F_y = x \cos(xy)$, both correct. The systematic method reaches the same place, but only after you notice that $\int (y \cos(xy) + 2x) dx$ requires the substitution $u = xy$ with y held fixed; seeing the group first saves that step entirely.

$$\frac{dy}{dx} = \frac{2x^3y - y^4}{x^4 - 2xy^3}, \quad y(1) = 2.$$

Cross-multiplying, $y(2x^3 - y^3) dx = x(x^3 - 2y^3) dy$. The equation is homogeneous, and the grouping route is faster: the combination $x^3 + y^3$ against xy is conserved, and one checks directly that $F = \frac{x^3 + y^3}{xy}$ has $\frac{dy}{dx} = -F_x/F_y$ equal to the right-hand side. So the first integral is

$$x^3 + y^3 = Cxy,$$

and $y(1) = 2$ gives $C = \frac{9}{2}$. If the problem then asks for the value of $\frac{x^3 + y^3}{xy}$ at some ostentatious point on the trajectory, at $x = 2026$ say, the answer is $\frac{9}{2}$ and the evaluation point is a red herring: the requested quantity is the conserved combination itself. Numerically, integrating from $(1, 2)$ to $x = 1.6$ gives $y \approx 2.333366$ and $(x^3 + y^3)/(xy) = 4.500000$.

WHERE THIS GOES WRONG

Being distracted by a large or ugly evaluation point. When what is asked for is the conserved combination, the point never matters; when it is y itself, you must actually solve the implicit relation. Read the question before computing. The second trap is dividing by a group such as $x^2 + y^2$ without noting that the division is illegal at the origin, so the resulting potential describes the solution only on a region avoiding it.

SEE ALSO Recovering the potential F from M and N ; integrating factors $x^a y^b$ and $\mu(xy)$; the total differential.

Exactness one order up

The idea generalises upward. A second-order equation is exact when one whole side of it is $\frac{d}{dx}$ of a first-order expression, and integrating once then reduces the order by one and hands the problem to the techniques of this and the previous chapter. The pattern to look for is always the product rule or the chain rule run backwards, and the search is usually a search for the multiplier that makes it appear.

34.4 Exact second-order equations and integrable combinations

HARD

CUE A second-order equation in which the terms could be a single derivative, most often after multiplying by a factor: the shapes $\mu y'' + \mu' y' = (\mu y')'$ and $e^y y'' + e^y (y')^2 = (e^y y')'$ are the two that recur.

METHOD Identify the combination and integrate once, carrying the constant. The result is a first-order equation, and the initial conditions of the original problem determine the first constant before you go any further. Then solve the first-order equation by whatever this chapter or the last one provides. The useful reflex: a term $(y')^2$ alongside y'' almost always signals that a factor of e^y or y is wanted, because those are the multipliers whose product rule generates a squared derivative.

WORKED EXAMPLE

$$y'' + (y')^2 = 2e^{-y}, \quad y(0) = y'(0) = 0.$$

The $(y')^2$ next to y'' is the signal. Multiply by e^y :

$$e^y y'' + e^y (y')^2 = 2 \implies (e^y y')' = 2 \implies e^y y' = 2x + C_1.$$

Both conditions are at $x = 0$, where $e^0 \cdot 0 = 0$, so $C_1 = 0$. Now notice that $e^y y' = (e^y)'$, so $(e^y)' = 2x$ and $e^y = x^2 + C_2$, with $C_2 = 1$ from $y(0) = 0$. Hence

$$y = \ln(x^2 + 1), \quad y(2) = \ln 5 \approx 1.609438,$$

confirmed to ten decimal places by numerical integration of the second-order system.

THE OTHER STANDARD MULTIPLIER

$$yy'' + (y')^2 = 0, \quad y(0) = 1, \quad y'(0) = 2.$$

No multiplication is needed: the left side is already $(yy')'$. So $yy' = C_1 = 2$, hence $(\frac{1}{2}y^2)' = 2$ and $y^2 = 4x + D$, with $D = 1$. Thus $y = \sqrt{4x+1}$ and $y(1) = \sqrt{5} \approx 2.236068$. The solution ceases to exist at $x = -\frac{1}{4}$, where y reaches zero and y' blows up.

WHERE THIS GOES WRONG

Integrating once and then dropping the first constant before applying the initial condition on y' . Both constants must be carried, and the first one is determined by $y'(x_0)$ together with $y(x_0)$, not by $y(x_0)$ alone. In the first example above it is $y'(0) = 0$ that kills C_1 ; had $y'(0)$ been nonzero the subsequent equation would have been $(e^y)' = 2x + C_1$ and the answer a different function entirely.

SEE ALSO Recovering the potential F from M and N ; grouping and solving by inspection; reduction of order for equations with a missing variable.

Reading the answer

An exact equation returns its solution as an implicit relation $F(x, y) = C$, and that is the answer, not a step on the way to one. Three questions then arise, and they are worth separating.

The first is whether to solve for y . Do it only when the relation is a polynomial of degree one or two in y , or otherwise obviously invertible, and only when the problem asks for a value of y rather than for the family of curves. Forcing an explicit form out of $x^2 + xy^2 + y^3 = C$ means solving a cubic and produces three branches, of which the initial condition selects one, and nothing about the geometry is clearer afterwards.

The second is which branch. When you do solve, a sign or a root choice appears, and the initial condition fixes it: substitute the initial point and keep the branch that reproduces it. The third is the interval of validity, and it is decided by where the implicit relation stops defining y as a function of x , which by the implicit function theorem is where $F_y = N$ vanishes. That is the same curve $N = 0$ on which the solution has a vertical tangent, and it is why the systematic method never needs to divide by anything.

IMPLICIT TO EXPLICIT, WITH THE BRANCH CHOSEN

The non-exact example above gave $2x^3y + x^2y^2 = C$, and through $(1, 1)$ the constant is 3. This is a quadratic in y :

$$x^2y^2 + 2x^3y - 3 = 0, \quad y = \frac{-2x^3 \pm \sqrt{4x^6 + 12x^2}}{2x^2} = \frac{-x^3 \pm \sqrt{x^6 + 3x^2}}{x^2}.$$

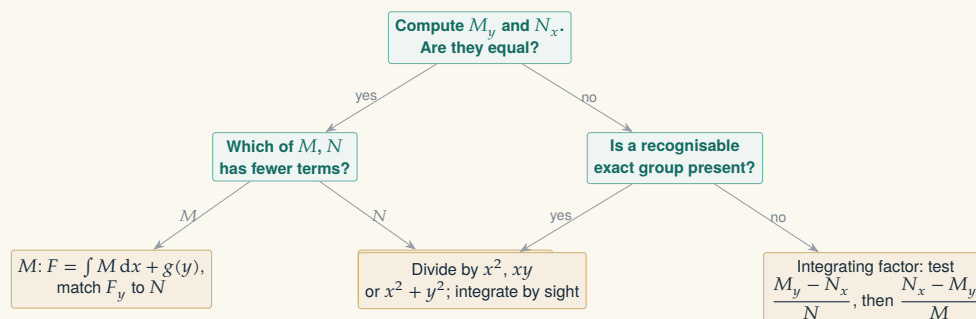
At $x = 1$ the two branches are $-1 + 2 = 1$ and $-1 - 2 = -3$, so the initial condition $y(1) = 1$ selects the upper sign. Hence

$$y = \frac{\sqrt{x^6 + 3x^2} - x^3}{x^2} \quad \text{on } x > 0,$$

and $y(2) = \frac{1}{4}(\sqrt{76} - 8) \approx 0.179449$, matching the numerical trajectory. The radicand is positive for every $x \neq 0$, so the only obstruction is $x = 0$, which is where the integrating factor $\mu = x$ died. The two facts are the same fact.

Choosing the route

Two derivatives decide everything, so compute them first and think afterwards.



The right-hand branch hands the problem back to the integrating-factor chapter and returns it exact, at which point you re-enter the tree at the top. That loop is the whole algorithm, and it terminates because the list of factor shapes to try is finite.

Where this goes next

Everything in this chapter is one-dimensional shadow of a statement about vector fields, and naming the statement makes the whole subject easier to remember. The pair (M, N) is a vector field on a region of the plane; the equation $M dx + N dy = 0$ says that the solution curve is everywhere orthogonal to it. The field is called *conservative* when it is the gradient of a scalar potential F , and three conditions are then equivalent on a region without holes: the field is a gradient, the test $M_y = N_x$ holds, and the line integral $\int_{\Gamma} M dx + N dy$ depends only on the endpoints of Γ and not on the path. Exactness of a differential equation is the first of these; the test you apply is the second; and the third is what makes the recovery procedure work, since

$$F(x, y) = \int_{(x_0, y_0)}^{(x, y)} M dx + N dy$$

is well defined precisely because the path does not matter. The two-step integration is that line integral evaluated along a staircase path: first horizontally, which contributes $\int M dx$, then vertically, which contributes the $g(y)$ you solve for. Seen that way, the “function of the other variable” stops being a rule to remember and becomes the second leg of a journey.

THE RECOVERY, RUN AS A LINE INTEGRAL

Take the first exact equation of this chapter, $(2x + y^2) dx + (2xy + 3y^2) dy = 0$, and build F from the origin along a staircase. Horizontal leg, from $(0, 0)$ to $(x, 0)$ with $dy = 0$ and $y = 0$ throughout:

$$\int_0^x (2t + 0) dt = x^2.$$

Vertical leg, from $(x, 0)$ to (x, y) with $dx = 0$ and x fixed:

$$\int_0^y (2xs + 3s^2) ds = xy^2 + y^3.$$

Adding, $F = x^2 + xy^2 + y^3$, which is exactly what the two-step procedure produced earlier, term for term. The first leg is $\int M dx$; the second leg is the $g(y)$ that the procedure solves for. Choosing the other staircase, vertical then horizontal, gives $\int_0^y 3s^2 ds = y^3$ followed by $\int_0^x (2t + y^2) dt = x^2 + xy^2$, and the same F : that agreement is path independence, demonstrated rather than asserted.

The punctured-plane example given earlier is the standard counterexample to dropping the topological hypothesis, and it is the same object in both languages: a field with zero curl and nonzero circulation, or equivalently a differential equation that is exact locally and has no global potential. It is also where the subject stops being calculus and becomes topology, which is a good place to leave it.

The physical reading is worth one sentence too, because it explains the vocabulary. If (M, N) is a force field, conservative means that the work done moving between two points does not depend on the route, so a potential energy can be defined and total energy is conserved along trajectories. The first integral of an exact second-order equation is precisely a conservation law of that kind: $yy' + \dots$ constant is a statement that something does not change, and the “constant of integration” is the value of the conserved quantity fixed by the initial data. Once you read $F = C$ as a conservation law rather than as an answer with a leftover letter, the technique in this chapter and the first-integral technique for second-order equations stop being two things.

THREE CHECKS, IN INCREASING COST

Before trusting a potential, run these in order. First, does $F_x = M$? One derivative. Second, does $F_y = N$? One more. If both hold, F is a potential and the solution is right, whatever route produced it, so a lucky guess verified this way is as good as a derivation. Third, and only if the first two are awkward to do by hand, integrate the equation numerically from a point on the curve and confirm that F has the same value at the far end. The third check catches nothing the first two miss, but it catches transcription errors, and it is the only one available when F is messy enough that differentiating it is itself error-prone.

Recognition matrix

If you see	Reach for	If that fails
$M dx + N dy = 0$, unknown type $M_y = N_x$ confirmed M ugly, N short x surviving in $g'(y)$ $M_y \neq N_x$ A common factor in $M_y - N_x$ and N $x dy + y dx$ present $x dy - y dx$ present $x dx + y dy$ present A homogeneous right-hand side A question asked at an absurd point y'' with $(y')^2$ beside it $\mu y'' + \mu' y'$ A potential you are unsure of $M_y = N_x$ but no global potential $f(x) dx = g(y) dy$ $\cos(xy)$ or e^{xy} throughout An implicit answer, y requested	Compute M_y and N_x before anything else $F = \int M dx + g(y)$, then match F_y to N Start from $F = \int N dy + h(x)$ The equation was not exact, or $M_y \neq N_x$ Integrating factor: try $(M_y - N_x)/N$ first Cancel it; the quotient is usually a monomial $d(xy)$; group and integrate by sight Divide by x^2 , xy , or $x^2 + y^2$ $\frac{1}{2} d(x^2 + y^2)$ Look for a conserved ratio such as $\frac{x^3+y^3}{xy}$ Report the conserved combination itself Multiply by e^y or by y ; look for $(\mu y')'$ $(\mu y')'$; integrate once, keep C_1 Differentiate it: check $F_x = M$ and $F_y = N$ Check for a hole: is the origin excluded? Exact for free: $M_y = N_x = 0$ Group $y dx + x dy$ and set $u = xy$ Solve, then pick the branch with the data	Rewrite $y' = f(x, y)$ as $M dx + N dy$ Integrate N in y instead Whichever has fewer terms was differentiated Recheck the first integration Factor before expanding the quotient Then $\mu = \exp \int$ of that Systematic two-step recovery $d(y/x)$, $d \ln(y/x)$, $d \arctan(y/x)$ Polar coordinates Substitute $y = vx$ Solve for y only if y is asked for Reduction of order with $p = y'$ Apply $y'(x_0)$ before continuing Numerical check along one trajectory Work on a simply connected subregion This is separation of variables Chain rule on top of a known group Validity ends where $N = 0$

Second-order linear equations with constant coefficients

What this chapter is for. Every linear equation $ay'' + by' + cy = f(x)$ with constant a, b, c is solved by one idea, applied twice. Write the equation as $L[y] = f$ where $L = aD^2 + bD + c$ and $D = \frac{d}{dx}$. Because a, b, c do not depend on x , the operator L factors exactly as the polynomial $ar^2 + br + c$ factors, and the factorisation is the entire theory: the roots of that polynomial give you the homogeneous solutions, their multiplicities tell you when to attach a factor of x , and a complex conjugate pair converts to a real oscillation by Euler's formula. That disposes of $L[y] = 0$ completely, in every case, forever. What remains is to produce one solution of $L[y] = f$, and there are exactly three ways to do it: guess a function of the same closed family as f and match coefficients, kill f with a second constant-coefficient operator and read the guess off the enlarged homogeneous problem, or integrate your way to it with the Wronskian when f is not of any nice family at all. By the end of the chapter you should be able to look at a constant-coefficient equation and know, before writing anything, which of those three you are going to use and how many undetermined constants you are committing yourself to.

The factorisation

Fix the operator $L = aD^2 + bD + c$ with $a \neq 0$. Two structural facts govern everything below, and both are consequences of linearity alone.

The first is that the solution set of $L[y] = 0$ is a vector space of dimension exactly 2. Existence and uniqueness for the initial value problem $y(x_0) = \alpha$, $y'(x_0) = \beta$ says the map sending a solution to the pair (α, β) is a bijection onto $\mathbb{R}^2$, so any two independent solutions span everything and a third is redundant. This is why a correct answer to a second-order homogeneous problem always has exactly two arbitrary constants, and why producing three is a symptom, not a style choice.

The second is the structure theorem for the forced problem. If y_p is any solution of $L[y] = f$ and y is another, then $L[y - y_p] = 0$, so $y - y_p$ lies in the two-dimensional homogeneous space. Hence

$$y = y_h + y_p, \quad y_h = C_1 y_1 + C_2 y_2,$$

and the two halves are computed by completely different machinery: y_h from the roots of a quadratic, y_p by one of the three methods of the second half of this chapter.

FIT THE DATA LAST

The single most common procedural error in this chapter is applying the initial conditions to y_h before y_p has been found. The constants C_1, C_2 are determined by the data *after* y_p is added, because y_p contributes to both $y(x_0)$ and $y'(x_0)$. Find the general solution first, complete it, and only then substitute. The same warning applies in reverse to superposition: if $f = f_1 + f_2$, solve $L[y] = f_1$ and $L[y] = f_2$ separately and add the two particular solutions, but add the homogeneous part only once.

The substitution that starts everything is $y = e^{rx}$, and the reason it works is that $De^{rx} = re^{rx}$: exponentials are eigenfunctions of differentiation, so a polynomial in D acts on them as the same polynomial in r . Formally, $L[e^{rx}] = (ar^2 + br + c)e^{rx}$, so e^{rx} solves the homogeneous equation precisely when r is a root of the *characteristic polynomial* $ar^2 + br + c$. Everything after this is bookkeeping about roots.

35.1 The characteristic equation and distinct real roots

CORE

CUE $ay'' + by' + cy = 0$ with constant a, b, c and discriminant $b^2 - 4ac > 0$. Also any problem that hands you a solution of the form $C_1 e^{r_1 x} + C_2 e^{r_2 x}$ and asks for the coefficients of the equation.

METHOD Substitute $y = e^{rx}$ to obtain $ar^2 + br + c = 0$. With distinct real roots $r_1 \neq r_2$,

$$y = C_1 e^{r_1 x} + C_2 e^{r_2 x},$$

and independence is automatic because $e^{r_1 x}$ and $e^{r_2 x}$ have different growth rates. Fit C_1, C_2 from $y(x_0)$ and $y'(x_0)$; the linear system is always solvable, since its determinant is $(r_2 - r_1)e^{(r_1 + r_2)x_0} \neq 0$.

WORKED EXAMPLE

$y'' - y' - 6y = 0$ with $y(0) = 1$, $y'(0) = 8$. The characteristic polynomial $r^2 - r - 6 = (r - 3)(r + 2)$ gives $r = 3, -2$, so $y = C_1 e^{3x} + C_2 e^{-2x}$. The data give $C_1 + C_2 = 1$ and $3C_1 - 2C_2 = 8$, hence $C_1 = 2$, $C_2 = -1$:

$$y = 2e^{3x} - e^{-2x}.$$

Check by substitution: $y'' = 18e^{3x} - 4e^{-2x}$, $y' = 6e^{3x} + 2e^{-2x}$, and $18 - 6 - 12 = 0$ on the first mode, $-4 - 2 + 6 = 0$ on the second.

THE ROOT AT THE ORIGIN

$y'' - 2y' = 0$ with $y(0) = y'(0) = \pi$. Here $c = 0$, so $r^2 - 2r = r(r - 2)$ and the roots are 0 and 2. The root $r = 0$ contributes $e^{0 \cdot x} = 1$, a genuine mode, so $y = C_1 + C_2 e^{2x}$. Then $y' = 2C_2 e^{2x}$ forces $C_2 = \pi/2$ and $C_1 = \pi/2$, so

$$y = \frac{\pi}{2}(1 + e^{2x}), \quad y(1) = \frac{\pi}{2}(1 + e^2) \approx 13.177499.$$

RUNNING THE ARGUMENT BACKWARDS

Problems often supply the solution and ask for the equation, which is the same correspondence read from right to left. If e^{-x} and e^{4x} span the solution space, the characteristic polynomial is $(r + 1)(r - 4) = r^2 - 3r - 4$ and the equation is $y'' - 3y' - 4y = 0$. If instead a single solution $3e^{-x} \cos 2x$ is given, the roots must include $-1 + 2i$ and, since the coefficients are real, its conjugate; so the polynomial is $(r + 1)^2 + 4 = r^2 + 2r + 5$ and the equation is $y'' + 2y' + 5y = 0$. The constant 3 carries no information: any nonzero multiple of a solution is a solution.

WHERE THIS GOES WRONG

Missing the root $r = 0$ when $c = 0$. The equation is still second order and still needs two constants; the constant function is one of them, and writing $y = Ce^{2x}$ alone loses a whole dimension of solution space and makes the initial value problem look overdetermined. The diagnostic is the one from the top of this section: count the arbitrary constants and compare with the order.

SEE ALSO Repeated root and the factor of x ; complex roots and the real form; higher-order characteristic polynomials.

When the discriminant vanishes the two exponentials collide, and the recipe above returns only one solution. The missing second solution is forced on us by the dimension count, and it is worth seeing where the factor of x comes from rather than memorising it, because the same factor reappears in the resonance rule for particular solutions and in the Cauchy-Euler equation of the next chapter, always for the same reason.

CORE

35.2 Repeated root and the factor of x

CUE $b^2 - 4ac = 0$, or a problem that *specifies* the solution form $(C_1 + C_2 x)e^{\lambda x}$ and asks you to recover a parameter in the equation.

METHOD With the double root $r = -b/(2a)$,

$$y = (C_1 + C_2 x)e^{rx}.$$

Two derivations, both worth knowing. Algebraically, $L = a(D - r)^2$, and $(D - r)^2[ve^{rx}] = e^{rx}v''$ by the shift rule $f(D)[e^{rx}v] = e^{rx}f(D + r)[v]$; so the homogeneous equation is just $v'' = 0$ and $v = C_1 + C_2 x$. Analytically, take the distinct-root solution $\frac{e^{r_2 x} - e^{r_1 x}}{r_2 - r_1}$, which is a legitimate combination for $r_1 \neq r_2$, and let $r_2 \rightarrow r_1$: the limit is $xe^{r_1 x}$. When a parameter is to be chosen so that the root repeats, set the discriminant to zero rather than solving the two cases separately.

WORKED EXAMPLE

$y'' - 6y' + 9y = 0$ with $y(0) = 1$, $y'(0) = 5$. The polynomial $r^2 - 6r + 9 = (r - 3)^2$ has the double root 3, so $y = (C_1 + C_2 x)e^{3x}$. Now differentiate *with the product rule*:

$$y' = C_2 e^{3x} + 3(C_1 + C_2 x)e^{3x}, \quad y'(0) = C_2 + 3C_1.$$

So $C_1 = 1$ and $C_2 + 3 = 5$, giving $C_2 = 2$ and $y = (1 + 2x)e^{3x}$.

CHOOSING A PARAMETER TO FORCE THE COLLISION

For which k does $y'' + ky' + (k + 3)y = 0$ have a repeated root? The discriminant is $k^2 - 4(k + 3) = k^2 - 4k - 12 = (k - 6)(k + 2)$, so $k = 6$ or $k = -2$. The corresponding double roots are $r = -k/2$, namely -3 and 1 . Note that the two admissible k satisfy $k_1 + k_2 = 4$ and $k_1 k_2 = -12$ by Vieta, which is faster than solving when only the sum or product is wanted.

Attaching the factor of x correctly and then differentiating as if it were not there. If you write $y = (C_1 + C_2x)e^{rx}$ and report $y'(0) = rC_1$ instead of $rC_1 + C_2$, every constant you fit afterwards is wrong. The product rule is the whole content of the repeated case; there is nothing else to get wrong, so get that right.

SEE ALSO The characteristic equation and distinct real roots; the resonance multiplier; higher-order characteristic polynomials.

Since $L = a(D - r_1)(D - r_2)$ and the two factors commute, every constant-coefficient problem can in principle be solved as two first-order problems in sequence: set $w = (D - r_2)y$, solve $a(D - r_1)w = f$ with the integrating factor e^{-r_1x} , then solve $(D - r_2)y = w$ the same way. Nobody does this in practice, because the methods of the next section are far quicker. It is worth knowing anyway for three reasons. It proves that $L[y] = f$ has a solution for every continuous f , with no assumption about the shape of f at all. It is the only route in the chapter that never needs an inspired guess. And it explains the repeated case at a glance: when $r_1 = r_2 = r$, the same integrating factor e^{-rx} is used twice, the second integration meets a constant, and $\int dx = x$ is exactly the missing factor.

A negative discriminant is not a special case of the theory, only of the arithmetic. The formula $y = C_1e^{r_1x} + C_2e^{r_2x}$ remains correct with complex r_1, r_2 and complex C_1, C_2 ; what changes is that we usually want a real answer, and Euler's formula converts one to the other in a line. The conversion is worth doing once, carefully, because it explains why the real form has a real exponential envelope multiplying a pure oscillation.

35.3 Complex roots and the real form

CUE $b^2 - 4ac < 0$. In applications: any damped oscillator, any LRC circuit below critical damping, any problem asking about an envelope, a decay rate, or a phase.

METHOD Write the roots as $r = \alpha \pm i\beta$ with

$$\alpha = -\frac{b}{2a}, \quad \beta = \frac{\sqrt{4ac - b^2}}{2a} > 0.$$

Then $e^{(\alpha \pm i\beta)x} = e^{\alpha x}(\cos \beta x \pm i \sin \beta x)$, and taking the real and imaginary parts (each of which solves the equation, because the coefficients are real) gives the real basis

$$y = e^{\alpha x}(C_1 \cos \beta x + C_2 \sin \beta x) = Ae^{\alpha x} \cos(\beta x - \varphi),$$

with $A = \sqrt{C_1^2 + C_2^2}$ and $\tan \varphi = C_2/C_1$. Use the first form to fit initial data, the second whenever the question mentions amplitude, envelope, or the location of maxima.

$y'' + 2y' + 5y = 0$ with $y(0) = 1$, $y'(0) = 1$. The roots of $r^2 + 2r + 5$ are $-1 \pm 2i$, so $\alpha = -1$, $\beta = 2$ and $y = e^{-x}(C_1 \cos 2x + C_2 \sin 2x)$. From $y(0) = 1$, $C_1 = 1$. Differentiating,

$$y' = -e^{-x}(C_1 \cos 2x + C_2 \sin 2x) + e^{-x}(-2C_1 \sin 2x + 2C_2 \cos 2x),$$

so $y'(0) = -C_1 + 2C_2 = 1$ and $C_2 = 1$. Hence $y = e^{-x}(\cos 2x + \sin 2x) = \sqrt{2}e^{-x} \cos(2x - \frac{\pi}{4})$. At $x = 0.8$ both forms return 0.436017, which is the cheapest possible check that the amplitude-phase conversion was done right.

$y'' + 4y = 0$ has roots $\pm 2i$, so $\alpha = 0$, $\beta = 2$ and $y = A \cos 2x + B \sin 2x$. With $y(0) = 3$ and $y'(0) = 5$ we get $A = 3$ and $2B = 5$, so $y = 3 \cos 2x + \frac{5}{2} \sin 2x$. The amplitude is $\sqrt{9 + \frac{25}{4}} = \frac{\sqrt{61}}{2} \approx 3.905$, and since $\alpha = 0$ that amplitude never decays. Hold on to this equation; it returns under forcing as the standard resonance example.

Computing β as $\sqrt{b^2 - 4ac}/(2a)$ and carrying a negative number under the radical, so that β comes out imaginary and the "real form" is secretly a pair of exponentials. Take $\beta = \sqrt{4ac - b^2}/(2a)$, positive by construction. Second trap: dropping the envelope. The solution of $y'' + 2y' + 5y = 0$ is not $C_1 \cos 2x + C_2 \sin 2x$; the factor e^{-x} is where all the damping lives, and without it the answer is periodic when it should decay.

SEE ALSO The characteristic equation and distinct real roots; repeated root and the factor of x ; the resonance multiplier.

For $my'' + cy' + ky = 0$ with $m, c, k > 0$ the sign of the discriminant $c^2 - 4mk$ decides the qualitative behaviour, and the standard vocabulary is worth attaching to it. Positive: two distinct negative real roots, *overdamped*, and the mass returns to equilibrium crossing it at most once. Zero: a negative double root, *critically damped*, which is the fastest possible return and the reason a door closer is tuned to $c = 2\sqrt{mk}$. Negative: a conjugate pair with $\alpha = -c/(2m) < 0$, *underdamped*, oscillating inside the envelope $e^{-cx/(2m)}$. In all three cases both roots have negative real part, so every solution decays; the general statement behind that is the Routh criterion in its smallest instance, namely that the roots of $ar^2 + br + c$ have negative real parts exactly when a, b, c are all of one sign.

$y'' + 2y' + 5y = 10 \cos x$. The homogeneous part is the underdamped $e^{-x}(C_1 \cos 2x + C_2 \sin 2x)$ of the previous example, and the forcing exponent i is not a root, so the trial $A \cos x + B \sin x$ needs no multiplier:

$$(-A + 2B + 5A) \cos x + (-B - 2A + 5B) \sin x = (4A + 2B) \cos x + (4B - 2A) \sin x = 10 \cos x.$$

So $A = 2B$ and $10B = 10$, giving $y_p = 2 \cos x + \sin x = \sqrt{5} \cos(x - \varphi)$ with $\tan \varphi = \frac{1}{2}$. The full solution is $e^{-x}(C_1 \cos 2x + C_2 \sin 2x) + 2 \cos x + \sin x$, and the vocabulary is worth attaching: y_h is the *transient*, which dies out at a rate set by the equation and not by the forcing, and y_p is the *steady state*, which persists and oscillates at the forcing frequency, not the natural one. Initial conditions affect only the transient. That is why a driven damped system eventually forgets how it was started, and why the amplitude $\sqrt{5}$ of the steady state is the only number a long-run measurement can report.

Nothing above used the fact that the order is two. Raise the order and the same three cases recur, now possibly mixed within a single equation, and the only new bookkeeping is that a root may repeat more than twice and that a complex pair may itself repeat.

35.4 Higher-order characteristic polynomials with multiplicities

CUE A constant-coefficient equation of order three or more, typically presented already factorable by inspection or by the rational root test.

METHOD Factor the characteristic polynomial completely over $\mathbb{C}$. A real root r of multiplicity m contributes the ladder

$$e^{rx}, xe^{rx}, \dots, x^{m-1}e^{rx};$$

a complex pair $\alpha \pm i\beta$ of multiplicity m contributes $x^j e^{\alpha x} \cos \beta x$ and $x^j e^{\alpha x} \sin \beta x$ for $j = 0, \dots, m-1$. Summing the contributions must give exactly n independent solutions for an equation of order n ; check that count before you go anywhere near the initial data.

$f''' + f'' - f' - f = e^x \sin x$. Group the characteristic polynomial: $r^3 + r^2 - r - 1 = r^2(r+1) - (r+1) = (r+1)(r^2-1) = (r-1)(r+1)^2$. So $r = 1$ simple and $r = -1$ double, giving three solutions as required:

$$f_h = Ae^x + (B + Cx)e^{-x}.$$

For the particular solution, put $f_p = e^x u$ and use the shift rule with $L = (D-1)(D+1)^2$, so $L[e^x u] = e^x D(D+2)^2[u] = e^x (D^3 + 4D^2 + 4D)[u]$. Taking $u = P \cos x + Q \sin x$, every second derivative flips sign: $D^2 u = -u$ and $D^3 u = -Du$, so the bracket collapses to $3Du - 4u$. Matching against $\sin x$:

$$-3P - 4Q = 1, \quad 3Q - 4P = 0 \implies P = -\frac{3}{25}, \quad Q = -\frac{4}{25},$$

$$f_p = -\frac{(3 \cos x + 4 \sin x)e^x}{25}.$$

Direct substitution confirms the residual is zero to 10^{-6} at $x = 0.2, 0.7, 1.4$.

$y'''' + 2y'' + y = 0$. The characteristic polynomial is $r^4 + 2r^2 + 1 = (r^2 + 1)^2$, so $\pm i$ are roots of multiplicity two and the ladder runs $\cos x, \sin x, x \cos x, x \sin x$:

$$y = (A + Bx) \cos x + (C + Dx) \sin x.$$

Four constants for an order-four equation, as required. Checking $y = x \cos x$: $y'' = -2 \sin x - x \cos x$ and $y'''' = 4 \sin x + x \cos x$, so $y'''' + 2y'' + y = 4 \sin x + x \cos x - 4 \sin x - 2x \cos x + x \cos x = 0$. The secular growth of $x \cos x$ is the four-dimensional version of resonance: a doubled root always produces it, whether it is real or complex.

WHERE THIS GOES WRONG

Counting a double root once, so that an order-four equation is answered with three constants. The dimension check is free and catches this every time. The second failure is arithmetic: factor by grouping or by the rational root test, but do factor. A cubic guessed at rather than factored produces an answer that satisfies nothing.

SEE ALSO Repeated root and the factor of x ; the annihilator method; the resonance multiplier.

One particular solution

With y_h known, the remaining task is to produce a single solution of $L[y] = f$. Any one will do, because the difference between two of them is homogeneous and gets absorbed into $C_1 y_1 + C_2 y_2$. That licence is what makes the methods below usable: you may discard any part of a candidate that happens to solve the homogeneous equation, and you may drop every constant of integration along the way.

The three methods differ in cost, not in validity. Undetermined coefficients is cheap but demands that f belong to a closed family under differentiation. Variation of parameters always works but pays for it with two integrals. The annihilator method is undetermined coefficients with the guessing removed. The decision tree at the end of the chapter puts them in order; here they come in the order in which they are usually learned, which is also the order of increasing generality.

35.5 Undetermined coefficients: the trial table

CORE

CUE Constant coefficients, and a forcing term assembled from polynomials, e^{ax} , $\cos bx$, $\sin bx$, and products of those. Equivalently: f is annihilated by some constant-coefficient operator, which is the same condition stated invariantly.

METHOD Guess y_p in the smallest family containing f that is closed under differentiation, substitute, and match coefficients. The families are:

Forcing $f(x)$	Trial y_p (before any resonance factor)
$P_n(x)$, degree n	$Q_n(x)$, all $n + 1$ coefficients unknown
e^{ax}	Ae^{ax}
$P_n(x)e^{ax}$	$Q_n(x)e^{ax}$
$\cos bx$ or $\sin bx$	$A \cos bx + B \sin bx$
$e^{ax} \cos bx$ or $e^{ax} \sin bx$	$e^{ax}(A \cos bx + B \sin bx)$
$P_n(x)e^{ax} \cos bx$	$e^{ax}(Q_n(x) \cos bx + R_n(x) \sin bx)$
$f_1 + f_2$	Solve separately and add

Two rules govern the table. Always carry the *full* polynomial, including terms absent from f : forcing $4x$ needs the trial $Ax + B$, not Ax . Always carry *both* trigonometric terms, because differentiation exchanges sine and cosine and a one-sided guess yields an inconsistent system.

WORKED EXAMPLE

$y'' - 3y' + 2y = 4x$. The trial is $y_p = Ax + B$, so $y'_p = A$ and $y''_p = 0$:

$$0 - 3A + 2(Ax + B) = 2Ax + (2B - 3A) = 4x.$$

Hence $A = 2$ and $B = 3$, so $y_p = 2x + 3$. The roots of $r^2 - 3r + 2$ are 1, 2, giving the general solution $y = C_1 e^x + C_2 e^{2x} + 2x + 3$.

A FULL QUADRATIC, AND A SHIFTED EXPONENTIAL

$y'' - y' - 2y = 4x^2$. Trial $y_p = Ax^2 + Bx + C$ gives

$$2A - (2Ax + B) - 2(Ax^2 + Bx + C) = -2Ax^2 + (-2A - 2B)x + (2A - B - 2C) = 4x^2,$$

so $A = -2$, then $B = -A = 2$, then $2A - B - 2C = 0$ gives $C = -3$. Thus $y_p = -2x^2 + 2x - 3$.

Now $y'' - 2y' + y = xe^{2x}$. Here $L = (D - 1)^2$ and the forcing exponent 2 is not a root, so the trial is $(Ax + B)e^{2x}$ with no extra factor. The shift rule is faster than the product rule: $(D - 1)^2[e^{2x}v] = e^{2x}(D + 1)^2[v]$, and with $v = Ax + B$,

$$(D^2 + 2D + 1)[Ax + B] = 0 + 2A + Ax + B = Ax + (2A + B).$$

Matching $Ax + (2A + B) = x$ gives $A = 1$, $B = -2$, so $y_p = (x - 2)e^{2x}$.

SUPERPOSITION ACROSS TWO FAMILIES

$y'' - 4y = 8x^2 - 2x + e^{3x}$. Split the forcing. For the polynomial part, $y_{p1} = Ax^2 + Bx + C$ gives

$$2A - 4(Ax^2 + Bx + C) = -4Ax^2 - 4Bx + (2A - 4C) = 8x^2 - 2x,$$

so $A = -2$, $B = \frac{1}{2}$, and $2A - 4C = 0$ forces $C = -1$. For the exponential part, 3 is not a root of $r^2 - 4$, so $y_{p2} = De^{3x}$ with $9D - 4D = 1$, giving $D = \frac{1}{5}$. Adding,

$$y = C_1 e^{2x} + C_2 e^{-2x} - 2x^2 + \frac{1}{2}x - 1 + \frac{1}{5}e^{3x}.$$

Note that y_h appears once, not twice. Numerical substitution of the particular part leaves a residual of 4×10^{-4} under finite differences, which is the differencing error and not an algebraic one.

WHERE THIS GOES WRONG

Guessing only $\cos$ when only $\cos$ appears in f . For $y'' + y' = \cos x$ the trial $A \cos x$ produces $-A \cos x - A \sin x = \cos x$, which has no solution; with $A \cos x + B \sin x$ it does. The second habitual error is truncating the polynomial: forcing x^2 with the trial Ax^2 fails for exactly the same reason. Include every lower-degree term even when it looks unnecessary, since the cost is one extra unknown and the alternative is an inconsistent system.

SEE ALSO The resonance multiplier; the annihilator method; variation of parameters.

COMPLEXIFY THE FORCING

For $e^{ax} \cos bx$ or $e^{ax} \sin bx$ the two-unknown trial can be replaced by a one-unknown complex trial, which is materially faster. Solve $L[z] = Fe^{(a+ib)x}$ instead; since L acts on an exponential by evaluation, $z_p = Fe^{(a+ib)x}/P(a+ib)$ whenever $P(a+ib) \neq 0$, and y_p is the real part when the forcing was the cosine, the imaginary part when it was the sine. Example: $y'' + y = 5e^{2x} \cos x$. Here $P(2+i) = (2+i)^2 + 1 = 4+4i$, so

$$z_p = \frac{5e^{(2+i)x}}{4+4i} = \frac{5(1-i)}{8}e^{2x}(\cos x + i \sin x), \quad y_p = \frac{5}{8}e^{2x}(\cos x + \sin x).$$

The condition $P(a+ib) = 0$ is exactly resonance, and there the shift rule must be used instead.

The table above fails in exactly one situation, and it fails loudly: when a term of the trial already solves the homogeneous equation, substituting it returns zero and no choice of coefficient can match a nonzero right-hand side. This is resonance, and the repair is a single, precisely statable multiplication. It is the same factor of x that appeared for the repeated root, arising for the same reason: a confluence between the forcing exponent and a characteristic root.

HARD

35.6 The resonance multiplier x^s

CUE Any term of your undetermined-coefficients trial duplicates a term of y_h . In practice: the forcing is $P(x)e^{ax}$ and a is a root of the characteristic polynomial, or the forcing is $e^{ax} \cos bx$ and $a \pm ib$ is a root. For an undamped oscillator, the forcing frequency equals the natural frequency.

METHOD Let s be the multiplicity of the forcing exponent as a root of the characteristic polynomial, where the forcing exponent means a for $P(x)e^{ax}$ and $a + ib$ for $P(x)e^{ax} \cos bx$ or $P(x)e^{ax} \sin bx$; take $s = 0$ if it is not a root at all. Then

$$y_p = x^s \times (\text{the trial from the table}),$$

with the multiplication applied to the *whole* trial, not to the offending term alone. Nothing else changes: substitute and match as before. The resulting system is always consistent, which is the practical statement of why s is the right exponent.

WORKED EXAMPLE

$y'' + 4y = 16 \sin 2x$, with $y(0) = 3$, $y'(0) = 5$. The roots are $\pm 2i$ and the forcing exponent is $0 + 2i$, a simple root, so $s = 1$ and $y_p = x(A \cos 2x + B \sin 2x)$. Differentiating twice,

$$y_p'' = -4x(A \cos 2x + B \sin 2x) + 4(-A \sin 2x + B \cos 2x),$$

so $y_p'' + 4y_p = -4A \sin 2x + 4B \cos 2x$. Matching against $16 \sin 2x$ gives $B = 0$ and $A = -4$:

$$y_p = -4x \cos 2x.$$

Adding $y_h = 3 \cos 2x + C \sin 2x$ from the earlier fit and imposing the data on the *complete* solution: $y(0) = 3$ is already satisfied, and

$$y' = -6 \sin 2x + 2C \cos 2x - 4 \cos 2x + 8x \sin 2x, \quad y'(0) = 2C - 4 = 5,$$

so $C = \frac{9}{2}$ and $y = 3 \cos 2x + \frac{9}{2} \sin 2x - 4x \cos 2x$. At $x = \pi/4$ every cosine vanishes and $y(\pi/4) = \frac{9}{2}$. The unbounded term $-4x \cos 2x$ is the physical content: an undamped oscillator driven at its natural frequency has amplitude growing linearly in time.

DOUBLE RESONANCE

$y'' - 2y' + y = e^x$. The characteristic polynomial is $(r - 1)^2$ and the forcing exponent is 1, a root of multiplicity 2, so $s = 2$ and $y_p = Ax^2e^x$. By the shift rule, $(D - 1)^2[e^x Ax^2] = e^x D^2[Ax^2] = 2Ae^x$, so $A = \frac{1}{2}$ and

$$y_p = \frac{1}{2}x^2e^x, \quad y = (C_1 + C_2x)e^x + \frac{1}{2}x^2e^x.$$

Note how the trial continues the ladder e^x, xe^x, x^2e^x exactly where y_h stopped.

RESONANCE WITH A COMPLEX PAIR

$y'' - 2y' + 5y = 8e^x \sin 2x$. The roots are $1 \pm 2i$ and the forcing exponent is $1 + 2i$, a simple root, so $s = 1$ and the trial is $xe^x(A \cos 2x + B \sin 2x)$. Write $L = (D - 1)^2 + 4$ and use the shift rule, $L[e^x v] = e^x(D^2 + 4)[v]$. For $v = xu$ with $u = A \cos 2x + B \sin 2x$, so that $u'' = -4u$, the product rule gives $(xu)'' = -4xu + 2u'$ and hence

$$(D^2 + 4)[xu] = 2u' = -4A \sin 2x + 4B \cos 2x.$$

Matching against $8 \sin 2x$: $B = 0$ and $A = -2$, so $y_p = -2xe^x \cos 2x$. Resonance against a damped or growing oscillation still produces the factor of x ; what differs from the undamped case is that here the envelope e^x dominates the linear growth entirely.

WHERE THIS GOES WRONG

Two errors, both producing an unsolvable coefficient system rather than a wrong answer, which is at least a loud failure. The first is multiplying only the term that clashes: for $y'' + 4y = 16 \sin 2x$ the trial $Ax \cos 2x + B \sin 2x$ is not a repair. The second is taking $s = 1$ when the duplicated root is itself repeated. Read s off the factorisation of the characteristic polynomial, never off the appearance of y_h .

SEE ALSO Undetermined coefficients: the trial table; the annihilator method; repeated root and the factor of x .

COUNT THE UNKNOWN BEFORE SUBSTITUTING

The number of undetermined coefficients is fixed by the forcing alone, and the resonance factor never changes it. Forcing $P_n(x)e^{ax}$ needs $n + 1$ unknowns whether s is 0, 1, or 2; forcing $P_n(x)e^{ax} \cos bx$ needs $2(n + 1)$. What x^s changes is the degree of the trial, not its dimension, and that is why the coefficient system stays square and becomes solvable rather than merely larger. Two practical consequences. If your system has more equations than unknowns, you truncated the polynomial or dropped a trigonometric partner. If it has more unknowns than equations, you multiplied by x^s and then also kept the unmultiplied terms, which are already in y_h and contribute nothing; delete them.

Undetermined coefficients, with or without the resonance factor, requires f to sit inside a finite-dimensional space closed under D . Forcings such as $\sec x$, $\tan x$, $\ln x$, and $1/x$ do not: differentiating them repeatedly generates new functions forever, so there is no finite trial to make. For those, and for every equation with variable coefficients whose homogeneous solutions you happen to know, there is a formula that produces y_p outright at the cost of two integrations.

35.7 Variation of parameters

CUE The forcing is outside the closed family: $\sec x$, $\tan x$, $\ln x$, $1/x$, e^{x^2} , or anything with a variable coefficient in front. Also any second-order linear equation, constant coefficients or not, for which y_1 and y_2 are already in hand.

METHOD Normalise the equation to $y'' + p(x)y' + q(x)y = g(x)$ by dividing through by the leading coefficient. With independent homogeneous solutions y_1, y_2 and Wronskian $W = y_1 y_2' - y_2 y_1'$,

$$y_p = -y_1 \int \frac{y_2 g}{W} dx + y_2 \int \frac{y_1 g}{W} dx.$$

The derivation is the ansatz $y_p = u_1 y_1 + u_2 y_2$ with the extra convenience constraint $u_1' y_1 + u_2' y_2 = 0$, which reduces the problem to the two-by-two system $u_1' y_1 + u_2' y_2 = 0$, $u_1' y_1' + u_2' y_2' = g$; Cramer's rule gives $u_1' = -y_2 g / W$ and $u_2' = y_1 g / W$. Drop the constants of integration: they only rebuild y_h .

WORKED EXAMPLE

$y'' + y = \sec x$ on $(-\frac{\pi}{2}, \frac{\pi}{2})$. Here $y_1 = \cos x$, $y_2 = \sin x$, and $W = \cos^2 x + \sin^2 x = 1$. The equation is already normalised, so $g = \sec x$ and

$$u_1 = -\int \sin x \sec x dx = -\int \tan x dx = \ln |\cos x|, \quad u_2 = \int \cos x \sec x dx = x.$$

Hence

$$y_p = \cos x \ln |\cos x| + x \sin x,$$

and the general solution is $C_1 \cos x + C_2 \sin x + \cos x \ln |\cos x| + x \sin x$. Substituting y_p numerically at $x = 0.2, 0.6, 1.0, 1.3$ leaves a residual below 10^{-6} . The companion result, obtained the same way, is that $y'' + y = \tan x$ has $y_p = -\cos x \ln |\sec x + \tan x|$.

REPEATED ROOT, NON-ELEMENTARY FORCING

$y'' - 2y' + y = \frac{e^x}{x}$ on $x > 0$. The homogeneous solutions are $y_1 = e^x$ and $y_2 = xe^x$, with

$$W = e^x(e^x + xe^x) - xe^x \cdot e^x = e^{2x}.$$

The equation is normalised already, so $g = e^x/x$ and

$$u_1 = -\int \frac{xe^x \cdot e^x/x}{e^{2x}} dx = -\int dx = -x, \quad u_2 = \int \frac{e^x \cdot e^x/x}{e^{2x}} dx = \ln x.$$

Then $y_p = -xe^x + xe^x \ln x$, and the first term is a homogeneous solution, so discard it:

$$y_p = xe^x \ln x.$$

No trial form could have produced this; $\ln x$ is not in any closed family.

VARIABLE COEFFICIENTS, WITH y_1, y_2 SUPPLIED

$x^2 y'' - 2xy' + 2y = x^3 e^x$ on $(0, \infty)$. The homogeneous equation is Cauchy-Euler with indicial polynomial $m(m-1) - 2m + 2 = (m-1)(m-2)$, so $y_1 = x$ and $y_2 = x^2$, and $W = x \cdot 2x - x^2 \cdot 1 = x^2$. Normalising divides by x^2 , so $g = xe^x$ and not $x^3 e^x$:

$$u_1 = -\int \frac{x^2 \cdot xe^x}{x^2} dx = -\int xe^x dx = -(x-1)e^x, \quad u_2 = \int \frac{x \cdot xe^x}{x^2} dx = e^x.$$

Hence $y_p = -x(x-1)e^x + x^2 e^x = xe^x$, and the general solution is $C_1 x + C_2 x^2 + xe^x$. The whole of the next chapter is about producing y_1 and y_2 in situations like this one.

WHERE THIS GOES WRONG

Forgetting to normalise. If the equation reads $ay'' + by' + cy = f$ with $a \neq 1$, then $g = f/a$, and using f instead scales y_p by a . Nearly every error in second-order variation of parameters traces back to this one line. Two smaller traps: the sign convention is attached to the formula as written, so do not reorder the terms from memory, and the Wronskian belongs in the denominator of both integrands, not outside them, whenever W is not constant.

SEE ALSO The Wronskian and Abel's identity; reduction of order; undetermined coefficients: the trial table.

Replacing the indefinite integrals by definite ones from x_0 gives a single formula,

$$y_p(x) = \int_{x_0}^x \frac{y_1(t)y_2(x) - y_1(x)y_2(t)}{W(t)} g(t) dt,$$

whose kernel vanishes on the diagonal $t = x$ and whose x -derivative equals 1 there. Consequently $y_p(x_0) = 0$ and $y'_p(x_0) = 0$, so the initial conditions are carried entirely by y_h and C_1, C_2 are read off the raw data with no arithmetic. This is the Green's function of the initial value problem, it is the form to use when g has no elementary antiderivative, and it makes the interval of validity explicit: the formula is good on the largest interval containing x_0 on which p, q, g are continuous and W is nonzero.

Variation of parameters needs two independent homogeneous solutions. If the coefficients are constant you always have them. If they are not, you may have only one, handed to you by the problem or guessable from the shape of the equation. One solution is enough: substituting $y = v(x)y_1$ turns the equation into a first-order equation for v' , because the terms carrying a bare v cancel by the very fact that y_1 solves the equation.

MEDIUM

35.8 Reduction of order from one known solution

CUE A second-order linear equation, coefficients arbitrary, with one solution y_1 supplied or guessable. Guessable means: try $y_1 = x, y_1 = x^m, y_1 = e^{ax}, y_1 = 1/x$ when the coefficients are polynomial, and check whether the coefficients sum to zero, which makes $y_1 = e^x$ a solution.

METHOD Set $y = v(x)y_1(x)$. Substituting into $y'' + py' + qy = 0$ gives

$$y_1 v'' + (2y_1' + py_1)v' = 0,$$

with no v term at all. Writing $w = v'$, this is first-order and separable, and integrating twice yields the closed form

$$y_2 = y_1 \int \frac{e^{-\int p dx}}{y_1^2} dx.$$

For a forced equation the same substitution leaves a first-order *linear* equation for w , so the method also produces a particular solution.

WORKED EXAMPLE

$x^2 y'' - 3xy' + 4y = 0$ with $y_1 = x^2$ given. Normalise: $p = -3/x$, so $e^{-\int p dx} = e^{3 \ln x} = x^3$ on $x > 0$. Then

$$y_2 = x^2 \int \frac{x^3}{x^4} dx = x^2 \ln x,$$

and the general solution on $(0, \infty)$ is $y = x^2(C_1 + C_2 \ln x)$. Substitution confirms it: with $y = x^2 \ln x, y' = 2x \ln x + x$ and $y'' = 2 \ln x + 3$, and $x^2(2 \ln x + 3) - 3x(2x \ln x + x) + 4x^2 \ln x = 0$ identically.

GUESSING y_1 FROM THE COEFFICIENTS

$(x+1)y'' - (x+2)y' + y = 0$. The coefficients sum to $(x+1) - (x+2) + 1 = 0$, which is precisely the condition for $y_1 = e^x$ to be a solution, since substituting e^x multiplies each coefficient by e^x and adds them. Normalising, $p = -\frac{x+2}{x+1}$, so

$$e^{-\int p dx} = \exp \int \left(1 + \frac{1}{x+1}\right) dx = (x+1)e^x,$$

and

$$y_2 = e^x \int \frac{(x+1)e^x}{e^{2x}} dx = e^x \int (x+1)e^{-x} dx = e^x(-(x+2)e^{-x}) = -(x+2).$$

Dropping the sign, $y = C_1 e^x + C_2(x+2)$ on $(-1, \infty)$. Substitution confirms both modes. The companion guesses are worth listing: coefficients summing to zero give e^x ; alternating coefficients summing to zero give e^{-x} ; a coefficient pattern with $c = 0$ gives the constant $y_1 = 1$.

REDUCTION OF ORDER ON A FORCED EQUATION

$xy'' - (x+1)y' + y = x^2$. The coefficients sum to $x - (x+1) + 1 = 0$, so $y_1 = e^x$ solves the homogeneous equation. Put $y = ve^x$; then $y' = (v' + v)e^x$ and $y'' = (v'' + 2v' + v)e^x$, and the left side collapses:

$$e^x[xv'' + (2x - x - 1)v' + (x - x - 1 + 1)v] = e^x[xv'' + (x - 1)v'],$$

with the v term gone as promised. Writing $w = v'$, the equation $xw' + (x-1)w = x^2e^{-x}$ normalises to $w' + (1 - \frac{1}{x})w = xe^{-x}$, whose integrating factor is e^x/x . Then $(e^xw/x)' = 1$, so $w = (x^2 + Ax)e^{-x}$, and integrating once more,

$$v = -(x^2 + 2x + 2)e^{-x} - A(x+1)e^{-x} + B.$$

Multiplying back by e^x and absorbing constants,

$$y = C_1e^x + C_2(x+1) - x^2.$$

One substitution produced the second homogeneous solution $x+1$ and the particular solution $-x^2$ together. Checking the last: $x(-2) - (x+1)(-2x) + (-x^2) = -2x + 2x^2 + 2x - x^2 = x^2$.

WHERE THIS GOES WRONG

Expecting a term in bare v to survive. If one does, y_1 is not a solution and the substitution was performed on a false premise; go back and check y_1 before doing any integration. The other trap is the interval: $e^{-\int p}$ and the division by y_1^2 both require $y_1 \neq 0$ and p continuous, so a zero of y_1 or a singular point of the equation cuts the domain, and $x^2 \ln x$ above is a solution on $(0, \infty)$, not on $\mathbb{R}$.

SEE ALSO Variation of parameters; the Wronskian and Abel's identity; repeated root and the factor of x .

Operator methods

Undetermined coefficients as presented above involves a guess, a resonance rule applied by hand, and a coefficient match. The guess and the rule can both be eliminated. The observation is that the class of forcings for which the trial table works is precisely the class annihilated by some constant-coefficient operator, and once such an operator is found the correct trial, resonance factor and all, falls out of a homogeneous problem.

HARD

35.9 The annihilator, or D -operator, method

CUE Constant coefficients with a forcing that is itself a solution of some constant-coefficient homogeneous equation. Useful when the resonance count is awkward, when several forcing terms interact, or when you would rather not remember the table.

METHOD Write the equation as $L[y] = f$ and find an operator A with $A[f] = 0$. The standard annihilators:

$$D^{n+1}[x^n] = 0, \quad (D-a)^{n+1}[x^n e^{ax}] = 0, \quad (D^2+b^2)[\cos bx] = 0, \\ ((D-a)^2+b^2)^{n+1}[x^n e^{ax} \cos bx] = 0.$$

Every entry of the trial table is one of these, which is the precise sense in which the table was never a list to memorise: a forcing is amenable to undetermined coefficients exactly when it is annihilated by a constant-coefficient operator, and the operator is read off the forcing by inspection. Apply A to both sides: $AL[y] = 0$, a homogeneous constant-coefficient equation whose solution space contains every solution of $L[y] = f$. Solve it, strike out the part that is already y_h , and what is left is exactly the correct trial, with the resonance factor built in. Determine its coefficients by substituting into the *original* equation.

WORKED EXAMPLE

$y'' - y = xe^x$. Here $L = D^2 - 1 = (D-1)(D+1)$ and $y_h = C_1e^x + C_2e^{-x}$. The forcing xe^x is annihilated by $A = (D-1)^2$, so

$$AL = (D-1)^3(D+1), \quad y = (C_1 + ax + bx^2)e^x + C_2e^{-x}.$$

Removing y_h leaves the trial $y_p = (ax + bx^2)e^x$, which is x times $(a + bx)e^x$: the resonance factor appeared without anybody applying a rule. Substituting via the shift rule, $(D^2 - 1)[e^xv] = e^x((D+1)^2 - 1)[v] = e^x(D^2 + 2D)[v]$ with $v = ax + bx^2$ gives $e^x(2b + 2a + 4bx)$. Matching against xe^x : $4b = 1$ and $2a + 2b = 0$, so $b = \frac{1}{4}$, $a = -\frac{1}{4}$, and

$$y_p = \frac{x(x-1)e^x}{4}.$$

$y'' + 4y = 16 \sin 2x$ again. Take $A = D^2 + 4$; then $AL = (D^2 + 4)^2$, whose roots are $\pm 2i$ each doubled, so the solution space is spanned by $\cos 2x, \sin 2x, x \cos 2x, x \sin 2x$. Strike the first two as y_h and the trial $x(A \cos 2x + B \sin 2x)$ is what remains, exactly as the resonance rule prescribed but with no rule invoked.

$y'' + y = x + \sin x$. The annihilator of x is D^2 and of $\sin x$ is $D^2 + 1$, so take their product $A = D^2(D^2 + 1)$ and

$$AL = D^2(D^2 + 1)^2,$$

whose solution space is spanned by $1, x, \cos x, \sin x, x \cos x, x \sin x$. Striking $y_h = \{\cos x, \sin x\}$ leaves the trial $a + bx + cx \cos x + dx \sin x$: the non-resonant part of the forcing produced no multiplier and the resonant part produced one, in a single stroke. Substituting, $(D^2 + 1)[a + bx] = a + bx$ matches the x , giving $a = 0, b = 1$; and $(D^2 + 1)[xu] = 2u'$ for $u = c \cos x + d \sin x$ matches $\sin x$, giving $d = 0$ and $c = -\frac{1}{2}$. Hence

$$y_p = x - \frac{1}{2}x \cos x.$$

$y'' - 2y' + y = x^2 e^x$. Here $L = (D - 1)^2$ and $A = (D - 1)^3$, so $AL = (D - 1)^5$ and the enlarged solution space is $(C_1 + C_2 x + ax^2 + bx^3 + cx^4)e^x$. Striking y_h leaves the trial $(ax^2 + bx^3 + cx^4)e^x$, and the shift rule reduces the substitution to $v'' = x^2$ with $v = ax^2 + bx^3 + cx^4$, that is $2a + 6bx + 12cx^2 = x^2$. Hence $a = b = 0, c = \frac{1}{12}$, and

$$y_p = \frac{x^4 e^x}{12}.$$

The trial table plus the resonance rule would have reached the same place, with $s = 2$ and the trial $x^2(Ax^2 + Bx + C)e^x$; the annihilator route simply never required the reader to notice that $s = 2$.

Solving $AL[y] = 0$ and reporting the whole thing as the answer. The extra constants are not free: those belonging to the trial are pinned by substitution into $L[y] = f$, and only the y_h constants are then fitted to the initial data. A second point of care: A must annihilate the entire forcing, so for $f = x + \sin x$ take $A = D^2(D^2 + 1)$, the product of the two annihilators, not either one alone.

SEE ALSO The resonance multiplier; undetermined coefficients: the trial table; operator inversion.

Treating D as an algebraic symbol is legitimate for polynomials because polynomials in D with constant coefficients commute and factor exactly as ordinary polynomials do. The temptation is to push further and invert, or even to expand a formal power series in D . Sometimes that is a genuine shortcut and sometimes it is nonsense, and the distinction is worth stating precisely.

35.10 Operator inversion and formal operator series

HARD

CUE The equation arrives as an operator identity, possibly an infinite one, such as $y + y' + y'' + \cdots = g$. Also any moment when L^{-1} applied to a simple forcing would be faster than a trial.

METHOD Three working identities. The inverse of a first-order factor is an integral,

$$\frac{1}{D - a} g = e^{ax} \int e^{-ax} g \, dx;$$

the shift rule $f(D)[e^{ax}u] = e^{ax}f(D + a)[u]$ moves an exponential past any polynomial in D ; and a geometric series in D terminates when applied to a polynomial, since D^k eventually kills it. For an infinite sum of derivatives the honest derivation avoids series algebra altogether: put $S = y + y' + y'' + \cdots$, so $S' = y' + y'' + \cdots$ and therefore $S - S' = y$. Requiring $S = g$ gives $y = g - g'$ directly, which is the statement $y = (1 - D)g$ obtained from $(1 - D)^{-1}y = g$.

$y + y' + y'' + \cdots = M_Z(t) = e^{t^2/2}$, the moment generating function of the standard normal. By the telescoping argument, $y = g - g'$ with $g = e^{t^2/2}$ and $g' = te^{t^2/2}$, so

$$y = (1 - t)e^{t^2/2}, \quad y(2) = -e^2 \approx -7.389056.$$

The operator $(1 - D)^{-1}$ was infinite; its inverse is the two-term operator $1 - D$, and it is the inverse that we actually apply.

$y'' - 3y' + 2y = e^{4x}$. Formally $y_p = \frac{1}{D^2 - 3D + 2}e^{4x}$, and since D acting on e^{4x} multiplies by 4, the operator acts as the number $16 - 12 + 2 = 6$. Hence $y_p = \frac{1}{6}e^{4x}$. The rule is: $\frac{1}{P(D)}e^{ax} = \frac{e^{ax}}{P(a)}$ whenever $P(a) \neq 0$, and $P(a) = 0$ is precisely the resonance case, where the shift rule must be used instead.

$y'' - 3y' + 2y = e^x$. Here $P(1) = 0$, so the evaluation rule fails and the factored inverse is used instead:

$$y_p = \frac{1}{D-1} \cdot \frac{1}{D-2} e^x.$$

The inner step: $\frac{1}{D-2}e^x = e^{2x} \int e^{-2x}e^x dx = e^{2x}(-e^{-x}) = -e^x$. The outer step: $\frac{1}{D-1}(-e^x) = e^x \int e^{-x}(-e^x) dx = -xe^x$. So $y_p = -xe^x$, which the resonance rule would have obtained as Axe^x with $A = -1$. Constants of integration were dropped at both steps because each would only have rebuilt part of y_h .

$y'' - y = x^3$. Write $\frac{1}{D^2 - 1} = -\frac{1}{1 - D^2} = -(1 + D^2 + D^4 + \dots)$. Applied to a cubic the series stops after two terms, since $D^4x^3 = 0$:

$$y_p = -(x^3 + D^2x^3) = -(x^3 + 6x).$$

Checking, $y_p'' - y_p = -6x + x^3 + 6x = x^3$. This is the one situation in which an infinite operator series is unambiguously safe: the operand is a polynomial, so all but finitely many terms vanish identically and no question of convergence arises. Against e^x or $1/(1+x)$ the same expansion is meaningless.

Applying formal series algebra without asking whether the manipulation has a finite meaning. The computation above is safe because the final operator $1 - D$ is a polynomial, and the telescoping derivation never sums anything; the series $\sum y^{(k)}$ for $y = (1-t)e^{t^2/2}$ does not converge pointwise, so a derivation that manipulates it as a convergent object proves nothing. Use operator series only when the result truncates, and check the truncation.

SEE ALSO The annihilator method; the resonance multiplier; variation of parameters.

Independence and the Wronskian

Everything above rested on a claim made once and never checked: that the two solutions produced are independent, so that $C_1y_1 + C_2y_2$ really is the general solution. For constant coefficients the claim is visible, since distinct exponentials, an exponential and x times it, and a sine against a cosine are all obviously independent. For variable coefficients, and for the variation-of-parameters formula, independence needs a certificate, and the certificate is a determinant.

35.11 The Wronskian and Abel's identity

CUE You must certify that two solutions are independent, or you need W for variation of parameters without computing y_1, y_2 explicitly, or a problem asks for W at one point given its value at another.

METHOD For solutions of the same equation $y'' + py' + qy = 0$ on an interval where p, q are continuous, define

$$W(x) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_2 y_1'.$$

Then $W' = -pW$, so Abel's identity holds:

$$W(x) = W(x_0) \exp\left(-\int_{x_0}^x p(t) dt\right).$$

The consequence is dichotomy: W is either identically zero on the interval or nowhere zero, and nonvanishing is equivalent to independence. Abel gives W up to a constant from the coefficient p alone. Abel is also constructive. Since $\left(\frac{y_2}{y_1}\right)' = \frac{W}{y_1^2}$, knowing W and one solution recovers the other:

$$y_2 = y_1 \int \frac{W}{y_1^2} dx,$$

which is reduction of order in one line.

WORKED EXAMPLE

$x^2 y'' - 3xy' + 4y = 0$ on $(0, \infty)$, normalised to $p = -3/x$. Abel predicts $W \propto \exp(\int 3 dx/x) = x^3$. Checking against the explicit pair $y_1 = x^2, y_2 = x^2 \ln x$ found by reduction of order:

$$W = x^2(2x \ln x + x) - x^2 \ln x \cdot 2x = x^3,$$

which is nonzero on $(0, \infty)$, so the pair is independent there and $y = x^2(C_1 + C_2 \ln x)$ is genuinely general.

RECOVERING THE SECOND SOLUTION FROM W

Take $x^2 y'' + xy' + (x^2 - \frac{1}{4})y = 0$ on $(0, \infty)$, with $y_1 = \frac{\sin x}{\sqrt{x}}$ known. Normalised, $p = 1/x$, so Abel gives $W = W_0/x$; scale y_2 so that $W_0 = 1$. Then

$$y_2 = y_1 \int \frac{W}{y_1^2} dx = \frac{\sin x}{\sqrt{x}} \int \frac{1/x}{\sin^2 x/x} dx = \frac{\sin x}{\sqrt{x}} \int \csc^2 x dx = -\frac{\sin x}{\sqrt{x}} \cot x = -\frac{\cos x}{\sqrt{x}}.$$

Dropping the sign, $y = \frac{C_1 \sin x + C_2 \cos x}{\sqrt{x}}$. The integral $\int W/y_1^2$ was elementary only because the awkward factor $1/x$ in W cancelled against the $1/x$ hidden inside y_1^2 ; that cancellation is not luck but the content of Abel's identity, which manufactures exactly the weight that makes the quotient integrable.

ABEL WITHOUT THE SOLUTIONS

For Hermite's equation $y'' - 2xy' + 2ny = 0$ we have $p = -2x$, so $W(x) = W(0)e^{x^2}$ for any pair of solutions. That single line settles independence for every n at once: if $W(0) \neq 0$, the pair is independent on all of $\mathbb{R}$, and W grows like e^{x^2} whether or not either solution is elementary.

WHERE THIS GOES WRONG

Using the converse for arbitrary functions. For a general pair, $W \equiv 0$ does *not* imply linear dependence: $f = x^2$ and $g = x|x|$ have $W \equiv 0$ on $\mathbb{R}$ yet are independent. The equivalence requires both functions to solve the same second-order linear equation with continuous coefficients on the interval in question, and that hypothesis fails for the pair above precisely because they would need to solve an equation singular at 0.

SEE ALSO Variation of parameters; reduction of order from one known solution.

BOUNDARY CONDITIONS ARE NOT INITIAL CONDITIONS

Everything asserted in this chapter about existence and uniqueness concerns data given at a single point: $y(x_0)$ and $y'(x_0)$. Two-point data behaves completely differently, and the difference is not a technicality. Consider $y'' + y = 0$. With $y(0) = 0$ and $y(\pi) = 0$ the solution is $C \sin x$ for every C : infinitely many solutions. With $y(0) = 0$ and $y(\pi) = 1$ there is none at all, since every solution vanishing at 0 is a multiple of $\sin x$ and vanishes at π too. Change the interval to $[0, \pi/2]$ and both problems become uniquely solvable. The general solution and the method of finding it are unaffected; only the last step, imposing the

data, changes character, and a boundary value problem must be checked for solvability rather than assumed to have exactly one answer.

Choosing the particular-solution method

The homogeneous half of the problem admits no choice: factor the polynomial and read off the answer. The forced half does, and the choice is decided by one question about f and one about y_h . Ask them in that order, because the first is cheap and usually terminal.

ONE COMPLETE PROBLEM, START TO FINISH

$y'' + 2y' + y = 3e^{-x} + 4x$ with $y(0) = 0$, $y'(0) = 1$.

Homogeneous part. $r^2 + 2r + 1 = (r + 1)^2$, a double root at -1 , so $y_h = (C_1 + C_2x)e^{-x}$. Two constants for a second-order equation: correct.

First forcing term. The exponent -1 is a root of multiplicity 2, so $s = 2$ and the trial is Ax^2e^{-x} . By the shift rule $(D + 1)^2[e^{-x}Ax^2] = e^{-x}D^2[Ax^2] = 2Ae^{-x}$, so $2A = 3$ and $A = \frac{3}{2}$.

Second forcing term. The exponent 0 is not a root, so the trial is $ax + b$ with no multiplier, and $0 + 2a + (ax + b) = ax + (2a + b) = 4x$ gives $a = 4$, $b = -8$.

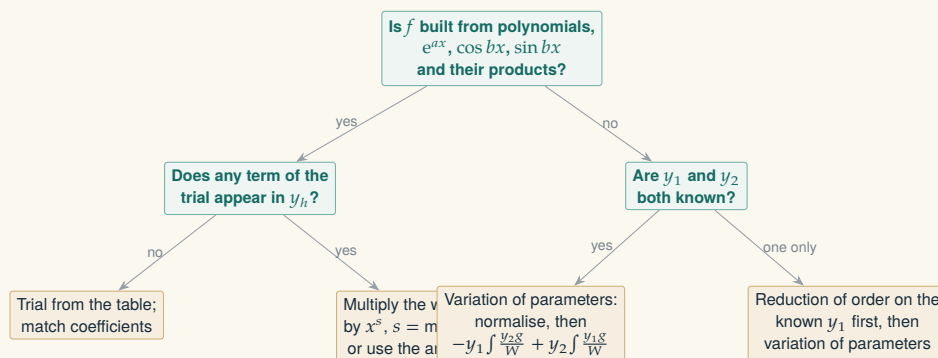
Assemble, then fit. The general solution is

$$y = (C_1 + C_2x)e^{-x} + \frac{3}{2}x^2e^{-x} + 4x - 8.$$

Only now impose the data. $y(0) = C_1 - 8 = 0$ gives $C_1 = 8$. Differentiating the whole expression, $y'(0) = C_2 - C_1 + 4 = 1$ gives $C_2 = 5$, so

$$y = \left(8 + 5x + \frac{3}{2}x^2\right)e^{-x} + 4x - 8.$$

Finite-difference substitution at $x = 0.3, 1.2, 2.5$ leaves a residual of 2×10^{-5} , and $y(0) = 0$, $y'(0) = 1$ to the accuracy of the difference quotient. The two things that would have gone wrong: fitting C_1, C_2 to y_h before adding y_p , and taking $s = 1$ instead of 2 for the first forcing term.



Three remarks the tree cannot carry. First, the left branch is almost always right when it applies: variation of parameters on $y'' - 3y' + 2y = 4x$ is correct and costs three times as much as the trial $Ax + B$. Reach for the integrals only when the trial genuinely does not exist. Second, superposition lets you take different branches for different pieces of the same forcing, so $f = x^2 + \sec x$ is handled by matching coefficients for x^2 and by the Wronskian formula for $\sec x$, with the two particular solutions simply added. Third, if the coefficients are not constant then the top question is moot: no trial table applies, and the route is reduction of order followed by variation of parameters, unless the equation is one of the reducible types of the next chapter.

WHAT EACH METHOD COSTS

For a second-order equation forced by $P_n(x)e^{ax}$, the arithmetic is easy to compare. Undetermined coefficients: one substitution, then an $(n + 1) \times (n + 1)$ triangular system solved by back-substitution, and no integration anywhere. The annihilator: one polynomial factorisation whose roots you already know, then exactly the same system, plus the benefit that the resonance exponent is never computed by hand. Variation of parameters: two integrals of products of y_1 , y_2 and g , each of which can be harder than the equation you started with, and one normalisation that is the commonest source of error in the chapter. The ordering is therefore stable: match coefficients if you can, annihilate if the resonance bookkeeping is getting awkward, integrate only when the forcing

leaves you no choice. The one exception is when y_1 and y_2 are simple and g is nasty but integrable, as with $\sec x$ against $\cos x$ and $\sin x$; there the two integrals are trivial and no trial exists at all.

THE SAME THEORY AS A FIRST-ORDER SYSTEM

Setting $v = y'$ turns $ay'' + by' + cy = 0$ into $\begin{pmatrix} y \\ v \end{pmatrix}' = \begin{pmatrix} 0 & 1 \\ -c/a & -b/a \end{pmatrix} \begin{pmatrix} y \\ v \end{pmatrix}$, whose matrix has characteristic polynomial $\lambda^2 + \frac{b}{a}\lambda + \frac{c}{a}$: the characteristic equation of this chapter, unchanged. Every case then matches an eigenvalue structure. Distinct real roots are two independent eigenvectors; a complex pair is a rotation combined with a scaling; and a repeated root is a *defective* matrix with only one eigenvector, which is why the second solution needs a generalised eigenvector and why the factor of x appears. Nothing in this chapter is lost in the translation, and one thing is gained: the same language handles equations of any order and coupled systems that are not equations of one variable at all.

TWO AUDITS THAT COST THIRTY SECONDS

Before reporting an answer, run both. The dimension audit: count arbitrary constants and compare with the order of the equation, remembering that a double root supplies two constants and not one. The substitution audit: put the claimed y_p back into the left-hand side and confirm you recover f exactly, including its coefficient. A numerical version of the second audit is even faster, and it catches the errors that algebra hides: evaluate $ay_p'' + by_p' + cy_p - f$ at three or four unrelated points with finite differences and confirm the residual is at the level of rounding. Every closed form printed in this chapter has been checked that way. The failures it detects are the characteristic ones: a dropped factor of x in a resonant case, an unnormalised g in variation of parameters, and a sign flip in W .

Recognition matrix

If you see	Reach for	If that fails
$ay'' + by' + cy = 0$, $b^2 - 4ac > 0$	$C_1 e^{r_1 x} + C_2 e^{r_2 x}$ from the roots	Recheck the factorisation
$c = 0$ in the equation	Roots 0 and $-b/a$; keep the constant mode	Count constants against the order
$b^2 - 4ac = 0$	$(C_1 + C_2 x)e^{rx}$, $r = -b/(2a)$	Shift rule: $(D - r)^2[v e^{rx}] = e^{rx} v''$
$b^2 - 4ac < 0$	$e^{ax}(C_1 \cos \beta x + C_2 \sin \beta x)$	$\beta = \sqrt{4ac - b^2}/(2a)$, positive
A question about amplitude or envelope	Amplitude-phase form $Ae^{ax} \cos(\beta x - \varphi)$	$A = \sqrt{C_1^2 + C_2^2}$
Order ≥ 3 , constant coefficients	Factor fully; ladder $x^l e^{rx}$ per multiplicity	Rational root test, then grouping
Forcing $P_n(x)$	Trial $Q_n(x)$, all $n + 1$ coefficients	Add x^s if 0 is a root
Forcing $P_n(x)e^{ax}$	Trial $Q_n(x)e^{ax}$	Shift rule beats the product rule
Forcing $e^{ax} \cos bx$ or $e^{ax} \sin bx$	Trial $e^{ax}(A \cos bx + B \sin bx)$	Both trig terms, always
Trial term already inside y_h	Multiply the whole trial by x^s	Annihilator method instead
Forcing frequency equals natural frequency	$s = 1$: expect a term linear in x	Check for damping first
Forcing exponent is a double root	$s = 2$: trial $Ax^2 e^{ax}$	Continue the ladder past y_h
Forcing $\sec x$, $\tan x$, $\ln x$, $1/x$	Variation of parameters	Normalise before reading off g
$ay'' + \dots = f$ with $a \neq 1$	Divide through: $g = f/a$	Every sign error lives here
One solution y_1 supplied	Reduction of order: $y = v y_1$	$y_2 = y_1 \int e^{-\int p} / y_1^2$
Polynomial coefficients summing to 0	Try $y_1 = e^x$, then reduce the order	Try $y_1 = x$ or $y_1 = x^m$
A bare v survives the substitution	y_1 was not a solution; recheck it	Verify y_1 before integrating
f annihilated by a known operator	Apply A , solve $AL[y] = 0$, strike y_h	Product of annihilators for sums
$\frac{1}{P(D)} e^{ax}$ with $P(a) \neq 0$	$e^{ax}/P(a)$	$P(a) = 0$: shift rule, resonance
$y + y' + y'' + \dots = g$	Telescope: $S - S' = y$, so $y = g - g'$	Do not sum the series
Independence in doubt	Wronskian $W = y_1 y_2' - y_2 y_1'$	Abel: $W \propto e^{-\int p}$
W needed but y_1, y_2 unknown	Abel from p alone, up to a constant	Fix the constant at one point
$W \equiv 0$ for two functions	Dependent only if both solve one linear ODE	x^2 and $x x $ are the counterexample
$f = f_1 + f_2$ of different types	Superpose: separate y_p , one y_h	Add the homogeneous part once

Cauchy-Euler and reducible variable-coefficient equations

What this chapter is for. There is no general method for second-order linear equations with variable coefficients, and there never will be: most of them define new transcendental functions rather than reduce to old ones. But a small, sharply defined family is not really variable-coefficient at all. Those equations are constant-coefficient equations written in the wrong coordinates, and the whole of this chapter is a catalogue of the coordinate changes that straighten them. The Cauchy-Euler equation becomes constant-coefficient under $x = e^t$; an equation missing y becomes first order under $p = y'$; an equation missing x becomes first order under $p = y'$ regarded as a function of y ; an exact equation is already the derivative of something shorter; and any linear second-order equation whatever loses its first-derivative term under one fixed substitution, which is how the shape of its solutions becomes visible. Learn to test for the five symptoms and you will convert a large fraction of the variable-coefficient problems you meet into problems you already know how to finish. Throughout, watch the leading coefficient: where it vanishes the equation is singular, the existence theorem does not apply, and the answer must be stated on an interval that excludes the bad point.

One equation in the wrong coordinates

The Cauchy-Euler equation

$$ax^2y'' + bxy' + cy = f(x), \quad a \neq 0,$$

is recognised by a matching of powers: each derivative carries exactly as many powers of x as its order. That matching is a scale invariance. Replace x by λx and every term picks up the same factor, so the homogeneous equation is unchanged in form. The natural coordinate for a scaling is a logarithm, and that is the whole idea.

Put $t = \ln x$ on $x > 0$, so $x = e^t$. Then $\frac{dt}{dx} = \frac{1}{x}$, and the chain rule gives

$$\frac{dy}{dx} = \frac{1}{x} \frac{dy}{dt}, \quad \text{so} \quad x \frac{dy}{dx} = \frac{dy}{dt}.$$

Write $\theta = x \frac{d}{dx}$ for that operator, so θ is exactly $\frac{d}{dt}$ in the new variable. For the second derivative, differentiate once more with the product rule:

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{1}{x} \frac{dy}{dt} \right) = -\frac{1}{x^2} \frac{dy}{dt} + \frac{1}{x} \cdot \frac{1}{x} \frac{d^2y}{dt^2},$$

so that

$$x^2 \frac{d^2y}{dx^2} = \frac{d^2y}{dt^2} - \frac{dy}{dt} = \theta(\theta - 1)y.$$

The equation becomes $a y_{tt} + (b - a) y_t + c y = f(e^t)$: constant coefficients, and every technique of the previous chapter applies unchanged. The general pattern continues, with $x^k \frac{d^k}{dx^k} = \theta(\theta - 1) \cdots (\theta - k + 1)$.

WHERE THE SOLUTION LIVES

The substitution $t = \ln x$ requires $x > 0$, and this is not a technicality. The leading coefficient ax^2 vanishes at $x = 0$, so the normalised equation $y'' + \frac{b}{ax}y' + \frac{c}{ax^2}y = 0$ has coefficients that blow up there. The existence and uniqueness theorem guarantees a solution only on an interval where the normalised coefficients are continuous, and $x = 0$ splits the line into two such intervals. Solve on $(0, \infty)$ or on $(-\infty, 0)$, never across. On the negative side put $t = \ln(-x)$, which replaces x''' by $|x|'''$ and $\ln x$ by $\ln|x|$. Two symptoms tell you the point is genuinely singular rather than an artefact: solutions such as x^{-2} are unbounded at 0, and solutions such as $x^2 \ln x$ are bounded but not twice differentiable there. An initial condition given at $x = 0$ for a Cauchy-Euler equation is not an initial value problem at all.

For the homogeneous equation the substitution need not be performed. Since $e^{rt} = x^r$, the exponential trial of the constant-coefficient theory becomes a power trial directly in x , and the three root cases translate one for one.

36.1 The Cauchy-Euler power trial $y = x^m$

CUE Every coefficient is a constant times x^k against the k -th derivative: $ax^2y'' + bxy' + cy = 0$. The tell is that the powers step down in lockstep with the derivative order.

METHOD Substitute $y = x^m$. Since $x^2(x^m)'' = m(m-1)x^m$ and $x(x^m)' = mx^m$, the indicial equation is

$$am(m-1) + bm + c = 0,$$

which is the characteristic polynomial of the transformed equation, $am^2 + (b-a)m + c = 0$. On $(0, \infty)$ the three cases are:

- distinct real $m_1 \neq m_2$: $y = C_1x^{m_1} + C_2x^{m_2}$;
- repeated m : $y = x^m(C_1 + C_2 \ln x)$;
- complex $m = \alpha \pm i\beta$: $y = x^\alpha(C_1 \cos(\beta \ln x) + C_2 \sin(\beta \ln x))$.

The $\ln x$ in the repeated case is the image of the xe^{rx} of the constant-coefficient theory under $x = e^t$, and the trigonometric functions of $\ln x$ are the image of $e^{\alpha t} \cos \beta t$, obtained from $x^{\alpha \pm i\beta} = x^\alpha e^{\pm i\beta \ln x}$.

WORKED EXAMPLE

$x^2y'' + 2xy' - 2y = 0$. The indicial equation is $m(m-1) + 2m - 2 = m^2 + m - 2 = (m+2)(m-1)$, with roots 1 and -2 , so

$$y = C_1x + \frac{C_2}{x^2} \quad \text{on } (0, \infty).$$

The second mode is unbounded at the origin, which is the singular point announcing itself.

Repeated root: $x^2y'' - xy' + y = 0$ gives $m(m-1) - m + 1 = (m-1)^2$, so $y = x(C_1 + C_2 \ln x)$. Substituting $y = x \ln x$ confirms it: $y' = \ln x + 1$, $y'' = 1/x$, and $x^2(1/x) - x(\ln x + 1) + x \ln x = x - x \ln x - x + x \ln x = 0$.

Complex roots: $x^2y'' + xy' + y = 0$ gives $m(m-1) + m + 1 = m^2 + 1$, so $m = \pm i$ and $y = C_1 \cos(\ln x) + C_2 \sin(\ln x)$. This solution oscillates infinitely often as $x \rightarrow 0^+$ while remaining bounded, a behaviour no constant-coefficient equation exhibits at a finite point.

THIRD ORDER, AND THE GENERAL PATTERN

$x^3y''' + 2x^2y'' - xy' + y = 0$. Use $x^k \frac{d^k}{dx^k} = \theta(\theta-1) \cdots (\theta-k+1)$ rather than substituting x^m and expanding by hand:

$$\theta(\theta-1)(\theta-2) + 2\theta(\theta-1) - \theta + 1 = \theta^3 - \theta^2 - \theta + 1 = (\theta-1)^2(\theta+1).$$

So $m = 1$ is a double root and $m = -1$ simple, giving

$$y = x(A + B \ln x) + \frac{C}{x} \quad \text{on } (0, \infty),$$

three constants for a third-order equation. Substituting each mode leaves a residual below 10^{-4} under finite differences.

WHERE THIS GOES WRONG

Writing the indicial equation as $am^2 + bm + c = 0$. The contribution of ax^2y'' is $am(m-1)$, not am^2 , so the linear coefficient is $b-a$ and not b . When $a = 1$ the error costs exactly one unit in the coefficient of m , which usually still factors and therefore still looks plausible: for $x^2y'' + 2xy' - 2y = 0$ the wrong equation $m^2 + 2m - 2 = 0$ has irrational roots and no visible alarm. Derive $m(m-1)$ each time rather than recalling the finished polynomial.

SEE ALSO The substitution $x = e^t$; disguised Cauchy-Euler in $(ax+b)$; reduction of order from one known solution.

The power trial handles the homogeneous equation and nothing else. As soon as there is forcing, and especially forcing of the form $x^k \ln^j x$, the substitution must actually be carried out, because it is in the t -variable that the resonance rule of the constant-coefficient theory can be applied. The translation dictionary is short: x^k becomes e^{kt} and $\ln x$ becomes t , so $x^k \ln^j x$ becomes $t^j e^{kt}$, which is exactly the closed family the trial table covers.

36.2 The substitution $x = e^t$ and resonance in t

CUE A Cauchy-Euler equation with forcing, above all forcing of the form $x^k \ln^j x$, and any Cauchy-Euler problem where k coincides with an indicial root.

METHOD Put $t = \ln x$ and $\theta = \frac{d}{dt}$. Using $x \frac{d}{dx} = \theta$ and $x^2 \frac{d^2}{dx^2} = \theta(\theta - 1)$ derived above, the equation becomes

$$a\theta(\theta - 1)y + b\theta y + cy = f(e^t),$$

a constant-coefficient equation in t . Solve it by the trial table with the multiplier x^s replaced by t^s , then substitute $t = \ln x$ back. Resonance in t means the forcing exponent k is an indicial root, and the resulting factor t^s appears in x as $\ln^s x$.

WORKED EXAMPLE

$x^2 y'' - 3xy' + 4y = x^2 \ln x$ with $y(1) = y'(1) = 0$. In t the operator is $\theta(\theta - 1) - 3\theta + 4 = \theta^2 - 4\theta + 4 = (\theta - 2)^2$, and the forcing $x^2 \ln x$ becomes te^{2t} . So

$$(\theta - 2)^2 y = te^{2t}.$$

The exponent 2 is a double indicial root, so write $y = e^{2t}u$ and use the shift rule: $(\theta - 2)^2[e^{2t}u] = e^{2t}\theta^2 u = e^{2t}u''$. The equation collapses to $u'' = t$, so $u = t^3/6$ up to homogeneous terms and

$$y_p = \frac{t^3 e^{2t}}{6} = \frac{x^2 \ln^3 x}{6}.$$

Both initial conditions at $x = 1$, that is $t = 0$, are satisfied by y_p alone, since y_p and y'_p each carry a factor of $\ln^2 x$; so $C_1 = C_2 = 0$ and $y = \frac{1}{6}x^2 \ln^3 x$. Then $y(e) = e^2/6 \approx 1.231509$. Direct substitution leaves a residual of order 10^{-4} under finite differences, consistent with rounding.

NO RESONANCE: THE OPERATOR ACTS AS A NUMBER

$x^2 y'' - 2y = x^3$. In t : $(\theta(\theta - 1) - 2)y = e^{3t}$, and $\theta^2 - \theta - 2 = (\theta - 2)(\theta + 1)$ has roots 2, -1. The forcing exponent 3 is not a root, so the operator acts on e^{3t} as the number $(3 - 2)(3 + 1) = 4$ and $y_p = \frac{1}{4}e^{3t} = \frac{1}{4}x^3$. The general solution on $(0, \infty)$ is

$$y = C_1 x^2 + \frac{C_2}{x} + \frac{x^3}{4}.$$

Checking: $y''_p = \frac{3x}{2}$, and $x^2 \cdot \frac{3x}{2} - 2 \cdot \frac{x^3}{4} = \frac{3x^3}{2} - \frac{x^3}{2} = x^3$.

WHERE THIS GOES WRONG

Translating the initial conditions without the chain rule. A condition $y'(x_0) = \beta$ becomes $\frac{dy}{dt}|_{t_0} = x_0 \beta$, because $\frac{dy}{dt} = x \frac{dy}{dx}$. The condition $y'(1) = 0$ above translates without a visible factor only because $x_0 = 1$; at $x_0 = e$ the factor is e and omitting it corrupts every constant. The second trap is forgetting that $\ln x$ has no meaning at $x \leq 0$: the answer $\frac{1}{6}x^2 \ln^3 x$ is a solution on $(0, \infty)$ only.

SEE ALSO The Cauchy-Euler power trial; the resonance multiplier; disguised Cauchy-Euler in $(ax + b)$.

The scale invariance that makes a Cauchy-Euler equation work is invariance under scaling about the origin. Nothing forces the centre to be the origin, and an equation that scales about $x = -b/a$ instead is the same equation shifted. Recognising it is a matter of noticing that the leading coefficient is a perfect square of a linear form and that the lower coefficients carry matching powers of that same form.

36.3 Disguised Cauchy-Euler in $(ax + b)$

CUE A quadratic leading coefficient that is the square of a linear form, with the first-derivative coefficient carrying that same linear form to the first power: $(ax + b)^2 y'' + c(ax + b)y' + dy = 0$. The leading coefficient is often written expanded, so $4x^2 + 4x + 1$ is the disguise for $(2x + 1)^2$.

METHOD Substitute $u = ax + b$. Then $\frac{d}{dx} = a \frac{d}{du}$ and $\frac{d^2}{dx^2} = a^2 \frac{d^2}{du^2}$, so the equation becomes

$$a^2 u^2 Y_{uu} + acu Y_u + dY = 0,$$

a standard Cauchy-Euler in u with the constants rescaled by powers of a . Solve by the power trial and substitute back, remembering that the singular point is now $u = 0$, that is $x = -b/a$.

$(4x^2 + 4x + 1)y'' - 2(2x + 1)y' + 4y = 0$, that is $(2x + 1)^2 y'' - 2(2x + 1)y' + 4y = 0$. With $u = 2x + 1$ and $a = 2$:

$$4u^2 Y_{uu} - 4u Y_u + 4Y = 0 \iff u^2 Y_{uu} - u Y_u + Y = 0.$$

The indicial equation is $m(m - 1) - m + 1 = (m - 1)^2 = 0$, a double root at $m = 1$, so $Y = u(A + B \ln u)$ and

$$y = (2x + 1)(A + B \ln(2x + 1)) \quad \text{on } x > -\frac{1}{2}.$$

Verify the second mode directly: with $y = (2x + 1) \ln(2x + 1)$ we get $y' = 2 \ln(2x + 1) + 2$ and $y'' = 4/(2x + 1)$, so

$$(2x + 1)^2 \frac{4}{2x + 1} - 2(2x + 1)(2 \ln(2x + 1) + 2) + 4(2x + 1) \ln(2x + 1) = 0$$

identically. At $x = \frac{e-1}{2}$ we have $2x + 1 = e$, so $y = e(A + B)$, which is why problems in this family choose that evaluation point.

WHERE THIS GOES WRONG

Losing the chain-rule factors a and a^2 . They do not cancel: in the worked example the surviving equation is $u^2 Y_{uu} - u Y_u + Y = 0$, whose double root is $m = 1$, whereas dropping the factors gives $u^2 Y_{uu} - 2u Y_u + 4Y$ and roots $\frac{3 \pm i\sqrt{7}}{2}$, a wholly different answer that still looks like a legitimate complex case. Convert the derivatives explicitly. As a check, note that the equation $(2x + 1)^2 y'' + 2(2x + 1)y' + y = 0$, which differs only in the middle sign and coefficient, yields $4m(m - 1) + 4m + 1 = 4m^2 + 1$ and hence $m = \pm \frac{i}{2}$, with solutions $\cos(\frac{1}{2} \ln(2x + 1))$ and $\sin(\frac{1}{2} \ln(2x + 1))$: small changes in the coefficients move you between root cases, so recompute the indicial equation every time.

SEE ALSO The Cauchy-Euler power trial; the substitution $x = e^t$.

Reducing the order

The Cauchy-Euler family is straightened by a change of variable. A second family is reduced instead: the order drops from two to one because one of the three ingredients x, y, y' is missing from the equation. These reductions do not require linearity, which is what makes them valuable; they are often the only route into a nonlinear second-order equation.

Which reduction applies is decided by which variable is absent, and the two are not interchangeable. If y is missing, $p = y'$ is a function of x and $y'' = p'$. If x is missing, the equation is autonomous, p is best regarded as a function of y , and $y'' = p \frac{dp}{dy}$ by the chain rule.

WHICH REDUCTION, AND WHAT THE CONSTANTS MEAN

Read the equation and ask which of x and y fails to appear. Only y absent: use $p(x)$, and the two constants enter at two successive integrations in x . Only x absent: use $p(y)$, and the first constant is the energy while the second is a translation in x , which it must be, since translating a solution of an autonomous equation gives another solution. Both absent, as in $y'' = (y')^2$: either route works, and $p(x)$ is usually the shorter. Neither absent: neither reduction applies, and you are back to the linear techniques. The commonest error is not choosing wrongly but mixing the two, writing $y'' = p p_y$ in a problem that still contains x , which produces an equation in three variables and no way forward.

MEDIUM

36.4 The missing- y reduction $p = y'$

CUE A second-order equation in which y itself never appears, only x, y' , and y'' : $F(x, y', y'') = 0$.

METHOD Set $p = y'$, so $y'' = \frac{dp}{dx}$ and the equation becomes first order in $p(x)$. Solve it by any first-order method, then recover y by a second integration. Two constants appear, one from each integration, and they must both be kept.

WORKED EXAMPLE

$xy'' + y' = 0$. With $p = y'$ this is $xp' + p = 0$, and the left side is exactly $(xp)'$, so $xp = C_1$ and $p = C_1/x$. Integrating again,

$$y = C_1 \ln|x| + C_2,$$

valid on $(0, \infty)$ or on $(-\infty, 0)$: the leading coefficient of the original equation vanishes at $x = 0$, so the two branches are separate problems and the constants need not agree across them.

$(1+x^2)y'' + 2xy' = 0$. With $p = y'$ the equation is $(1+x^2)p' + 2xp = ((1+x^2)p)' = 0$, so $(1+x^2)p = C_1$ and

$$y = C_1 \arctan x + C_2 \quad \text{on all of } \mathbb{R}.$$

Here the leading coefficient never vanishes, so a single interval carries the whole solution, in contrast with $xy'' + y' = 0$ above. Notice also that the left side was already a total derivative, which is the exactness test of the next section arriving early: with $a = 1+x^2$, $b = 2x$, $c = 0$ we have $a'' - b' + c = 2 - 2 + 0 = 0$.

$y'' = 1 + (y')^2$ with $y(0) = y'(0) = 0$. In p : $\frac{dp}{dx} = 1 + p^2$, separable, giving $\arctan p = x + A$ and $p = \tan(x + A)$. From $p(0) = 0$, $A = 0$, so $y' = \tan x$ and

$$y = -\ln |\cos x| + B, \quad B = 0, \quad y = -\ln \cos x,$$

on $(-\frac{\pi}{2}, \frac{\pi}{2})$. The interval is forced by the solution, not by the equation: y blows up as $x \rightarrow \frac{\pi}{2}^-$ even though the right-hand side is perfectly smooth everywhere. This is the standard example of finite-time blow-up in a nonlinear problem.

A flexible cable hanging under its own weight satisfies $y'' = k\sqrt{1 + (y')^2}$, with y absent. Setting $p = y'$,

$$\frac{dp}{\sqrt{1+p^2}} = k dx \implies \operatorname{arsinh} p = kx + C_1 \implies y' = \sinh(kx + C_1),$$

and one more integration gives $y = \frac{1}{k} \cosh(kx + C_1) + C_2$. The hyperbolic cosine is not assumed anywhere; it is produced by the reduction. Checking the case $C_1 = C_2 = 0$: $y' = \sinh kx$, $y'' = k \cosh kx$, and $k\sqrt{1 + \sinh^2 kx} = k \cosh kx$.

Applying $y'(x_0)$ to p correctly and then losing the second constant, so that the answer has one arbitrary constant for a second-order equation. Equally common: using the missing- x reduction here. If y is absent but x is present, p is a function of x and $y'' = p'$; writing $y'' = p \frac{dp}{dy}$ introduces a variable the equation does not contain.

SEE ALSO The missing- x reduction and the energy integral; exact second-order equations.

When x is the missing variable the equation is autonomous, and the reduction takes a different shape. Because the independent variable never appears, y itself can serve as the independent variable, and the resulting first-order equation in the phase plane is the energy integral of mechanics.

36.5 The missing- x reduction $p \frac{dp}{dy}$ and the energy integral

CUE A second-order autonomous equation: no explicit x , only y, y', y'' . The archetype is $y'' = f(y)$, a particle on a line under a force depending only on position.

METHOD Set $p = y'$ but regard p as a function of y . Then

$$y'' = \frac{dp}{dx} = \frac{dp}{dy} \frac{dy}{dx} = p \frac{dp}{dy},$$

and $F(y, p, p p_y) = 0$ is first order in $p(y)$. For $y'' = f(y)$ it integrates at once:

$$\frac{1}{2}p^2 = \int f(y) dy + C,$$

the conservation of energy, and the remaining equation $\frac{dy}{dx} = \pm \sqrt{2(\int f + C)}$ is separable.

$y'' = 2y^3$ with $y(0) = 1, y'(0) = 1$. The energy integral gives $\frac{1}{2}p^2 = \frac{1}{2}y^4 + C$, and the data $y = 1, p = 1$ force $C = 0$. Hence $p^2 = y^4$ and, taking the branch with $p > 0$ that the data select, $y' = y^2$. Separating, $-1/y = x - 1$ and

$$y = \frac{1}{1-x} \quad \text{on } (-\infty, 1).$$

RK4 integration of the original second-order system from $x = 0$ to $x = 0.5$ returns $y = 2.00000000$, matching $1/(1 - 0.5) = 2$ to eight places. The solution escapes to infinity at $x = 1$, which the energy integral predicted: p grows like y^2 , so the time to reach infinity is a convergent integral.

$y'' + \sin y = 0$ has $f(y) = -\sin y$, so $\frac{1}{2}p^2 = \cos y + C$. Every trajectory is a level curve of $\frac{1}{2}(y')^2 - \cos y$. If $C < 1$ the motion is trapped between the turning points $y = \pm \arccos(-C)$, where $p = 0$ and the sign of y' reverses; if $C > 1$ then p never vanishes and the pendulum rotates. No closed form is needed to read that off, which is the point of the energy integral.

$2yy'' = 1 + (y')^2$. The equation is autonomous but nonlinear in y as well, so the energy shortcut does not apply directly; the substitution still does. With $y'' = p \frac{dp}{dy}$,

$$2yp \frac{dp}{dy} = 1 + p^2 \implies \frac{2p dp}{1 + p^2} = \frac{dy}{y} \implies 1 + p^2 = Ay.$$

Then $y' = \sqrt{Ay - 1}$ separates to $\frac{2}{A}\sqrt{Ay - 1} = x + B$, that is

$$y = \frac{1}{A} + \frac{A}{4}(x + B)^2,$$

a family of parabolas. Checking $A = 2, B = 0$, so $y = \frac{1}{2} + \frac{1}{2}x^2$: then $y' = x, y'' = 1$, and $2yy'' = 1 + x^2 = 1 + (y')^2$.

Taking the positive square root of p^2 unconditionally. The sign of y' is fixed by the initial data and can reverse at a turning point, where $p = 0$; past that point the other branch applies. State the interval on which your branch is valid, as in the pendulum example above, and check that the initial data really do select the branch you took.

SEE ALSO The missing- y reduction; equidimensional in y ; the Cauchy-Euler power trial.

A third reduction exploits a symmetry rather than a missing variable. If replacing y by ky leaves the equation unchanged, then the equation constrains only the logarithmic derivative of y , and substituting for that derivative directly drops the order.

36.6 Equidimensional in y : the substitution $y = e^{\int v dx}$

CUE The equation is unchanged when y is replaced by ky for every constant k , which happens exactly when every term has total degree one in $\{y, y', y''\}$ jointly. The signature is a term such as yy'' balanced against $(y')^2$.

METHOD Substitute $y = e^{\int v dx}$, so that $v = y'/y$ and

$$y' = vy, \quad y'' = (v' + v^2)y.$$

Every term acquires a factor y or y^2 , which cancels by the scaling hypothesis, and the equation drops to first order in v . Recover y by one further integration, which supplies the second constant.

$yy'' - (y')^2 = 0$. Substituting, $y^2(v' + v^2) - v^2y^2 = y^2v' = 0$, so $v = C_1$ and $y = C_2 e^{C_1 x}$. The equation says the logarithmic derivative is constant, which is exactly the characterisation of the exponential.

A forced variant: $yy'' = (y')^2 + y^2$ becomes $y^2(v' + v^2) = v^2y^2 + y^2$, that is $v' = 1$, so $v = x + C_1$ and $y = C_2 \exp(\frac{1}{2}x^2 + C_1 x)$. Checking the case $C_1 = 0, C_2 = 1$: with $y = e^{x^2/2}$ we have $y' = xy$ and $y'' = (1 + x^2)y$, and $y \cdot (1 + x^2)y - x^2y^2 - y^2 = 0$.

Confusing this symmetry with the others. Scaling y alone ($y \mapsto ky$) calls for $y = e^{\int v}$; scaling both variables together ($x \mapsto \lambda x$, $y \mapsto \lambda y$) is the homogeneous case and calls for $y = xu$; scaling with different weights is the isobaric case. Test which invariance the equation actually has by substituting ky , or λx and λy , and seeing what cancels. Guessing costs a page.

SEE ALSO The missing- x reduction and the energy integral; the Cauchy-Euler power trial.

Rewriting the operator

The last family is recognised not by a missing variable but by a relation among the coefficients. Sometimes the left-hand side is already a derivative, and one integration is free. Sometimes a change of the independent or the dependent variable removes the first-derivative term and exposes the equation's real content.

36.7 Exact second-order equations

MEDIUM

CUE Variable coefficients that look as though they came from a product rule: $a(x)y'' + b(x)y' + c(x)y = f(x)$ with b close to a' . Test the identity below; it costs two derivatives.

METHOD The left-hand side is a total derivative exactly when

$$a'' - b' + c = 0,$$

and in that case

$$ay'' + by' + cy = \frac{d}{dx} [ay' + (b - a')y].$$

Verify by expanding the right side: it is $ay'' + by' + (b' - a'')y$. Integrating once gives the first-order linear equation $ay' + (b - a')y = \int f dx + C_1$, which an integrating factor finishes. If the test fails, multiply the equation by a trial function and test again; that is the integrating-factor problem for second-order equations, and it is only worth attempting when a factor such as x^k or e^{kx} is suggested by the coefficients.

WORKED EXAMPLE

$xy'' + (x + 2)y' + y = e^x$. Here $a = x$, $b = x + 2$, $c = 1$, so $a'' - b' + c = 0 - 1 + 1 = 0$ and the equation is exact. The first integral is

$$ay' + (b - a')y = xy' + (x + 1)y = e^x + C_1.$$

Divide by x and use the integrating factor $\exp \int (1 + \frac{1}{x}) dx = xe^x$:

$$(xe^x y)' = xe^x \cdot \frac{e^x + C_1}{x} = e^{2x} + C_1 e^x,$$

so $xe^x y = \frac{1}{2}e^{2x} + C_1 e^x + C_2$ and

$$y = \frac{e^x}{2x} + \frac{C_1}{x} + \frac{C_2 e^{-x}}{x} \quad \text{on } (0, \infty).$$

Two constants, as required, and both homogeneous modes are singular at $x = 0$, which is where the leading coefficient vanishes. Substitution confirms the residual vanishes for several choices of C_1, C_2 .

TWO ROUTES TO THE SAME ANSWER

$xy'' + 3y' = 0$. Exactness: $a = x$, $b = 3$, $c = 0$, so $a'' - b' + c = 0$ and the first integral is $xy' + (3 - 1)y = C_1$. Dividing by x and using the integrating factor x^2 gives $(x^2 y)' = C_1 x$, so

$$y = A + \frac{B}{x^2} \quad \text{on } (0, \infty).$$

Multiplying by x turns the same equation into $x^2 y'' + 3xy' = 0$, a Cauchy-Euler equation with indicial polynomial $m(m - 1) + 3m = m(m + 2)$ and roots 0 and -2 : the same answer by a different route. Notice, though, that the multiplied form is *not* exact, since there $a'' - b' + c = 2 - 3 + 0 = -1$. Exactness is a property of how the equation is written and not of its solution set, so multiplying through by a function can create it or destroy it. That is exactly why searching for an integrating factor is worth a try when the raw test fails.

Testing exactness with the wrong sign pattern. The condition is $a'' - b' + c = 0$, alternating in sign; $a'' + b' + c = 0$ and $a'' - b' - c = 0$ are both wrong and both easy to write. Derive it once by expanding $[ay' + (b - a')y]'$ rather than trusting memory, and note that the equation must be in its unnormalised form: dividing by a first changes a to 1 and destroys the test.

SEE ALSO The missing- y reduction; the normal form.

Exactness is a coincidence among the coefficients. The two remaining techniques manufacture a coincidence instead, by choosing a new variable so that an unwanted term cancels. Changing the independent variable can be made to kill the first-derivative term and, with luck, to turn the whole equation into one with constant coefficients; changing the dependent variable kills the first-derivative term always, at the cost of a modified potential.

HARD

36.8 Change of independent variable

CUE A variable-coefficient equation whose y' term looks as though it should cancel against the y'' term, for instance $xy'' - y' + \dots$, or one where the ratio of the coefficients suggests a natural new variable such as $t = x^2$ or $t = \ln x$.

METHOD Try $t = g(x)$. Then

$$y' = g'(x)Y_t, \quad y'' = g''(x)Y_t + (g'(x))^2 Y_{tt},$$

and the coefficient of Y_t in the transformed equation is $ag'' + bg'$ for an original $ay'' + by' + cy$. Choose g to annihilate it. For the normalised equation $y'' + py' + qy = 0$ the choice $t = \int \sqrt{q} dx$ produces constant coefficients precisely when $\frac{q' + 2pq}{q^{3/2}}$ is constant, which is the invariant test worth applying before attempting the substitution.

WORKED EXAMPLE

$xy'' - y' + 4x^3y = 0$ with $y(\sqrt{\pi}) = -1$ and $y'(\sqrt{\pi}) = 2\sqrt{\pi}$. Try $t = x^2$, so $g' = 2x$ and $g'' = 2$:

$$y' = 2xY_t, \quad y'' = 2Y_t + 4x^2Y_{tt},$$

whence $xy'' - y' = 2xY_t + 4x^3Y_{tt} - 2xY_t = 4x^3Y_{tt}$. The equation becomes $4x^3(Y_{tt} + Y) = 0$, that is

$$Y_{tt} + Y = 0, \quad Y = A \cos t + B \sin t.$$

Transfer the data at $t_0 = \pi$: $Y(\pi) = -1$ gives $-A = -1$, so $A = 1$; and $Y_t(\pi) = y'(\sqrt{\pi})/g'(\sqrt{\pi}) = 2\sqrt{\pi}/(2\sqrt{\pi}) = 1$ gives $-B = 1$, so $B = -1$. Hence $Y = \cos t - \sin t$ and $y = \cos(x^2) - \sin(x^2)$. At $x = \sqrt{\pi}/2$, $t = \pi/4$ and $y = \cos \frac{\pi}{4} - \sin \frac{\pi}{4} = 0$.

WHERE THIS GOES WRONG

Transferring the derivative data without the chain-rule factor. The relation is $y'(x_0) = g'(x_0) Y_t(t_0)$, so above $Y_t(\pi) = 1$, not $2\sqrt{\pi}$. Using the raw value gives $B = -2\sqrt{\pi}$ and an answer that satisfies the equation but not the data, which is the hardest kind of error to notice because the substitution check passes.

SEE ALSO The substitution $x = e^t$; the normal form.

The change of independent variable is opportunistic: it works when the coefficients cooperate. The change of *dependent* variable that removes the first-derivative term always works, and although it does not solve the equation it puts every second-order linear equation into a canonical shape in which two equations are equivalent exactly when one function agrees.

36.9 The normal form: removing the first-derivative term

CUE Any linear $y'' + p(x)y' + q(x)y = 0$ whose solutions you want to understand rather than merely write down: oscillation, decay envelopes, comparison with $u'' + u = 0$, or recognition that the equation is a known one in disguise.

METHOD Substitute

$$y = u(x) \exp\left(-\frac{1}{2} \int p \, dx\right).$$

The first-derivative term cancels identically and what remains is

$$u'' + I(x)u = 0, \quad I = q - \frac{1}{4}p^2 - \frac{1}{2}p',$$

where I is the *invariant* of the equation. Two equations with the same I are the same equation up to that exponential factor. If I turns out to be a constant, the equation is elementary; if $I > 0$ and bounded away from zero, solutions oscillate; if $I < 0$, they do not.

WORKED EXAMPLE

$y'' - 2xy' + (x^2 - 1)y = 0$. Here $p = -2x$ and $q = x^2 - 1$, so

$$I = (x^2 - 1) - \frac{1}{4}(4x^2) - \frac{1}{2}(-2) = x^2 - 1 - x^2 + 1 = 0.$$

The normal form is $u'' = 0$, so $u = A + Bx$, and the exponential factor is $\exp(\frac{1}{2} \int 2x \, dx) = e^{x^2/2}$. Hence

$$y = (A + Bx) e^{x^2/2}.$$

Verify the case $A = 1, B = 0$: $y' = xe^{x^2/2}$, $y'' = (1 + x^2)e^{x^2/2}$, and $(1 + x^2) - 2x^2 + (x^2 - 1) = 0$. An equation with no elementary-looking solutions turned out to have only elementary ones, and the invariant is what revealed it.

A BESSEL EQUATION IN DISGUISE

$x^2y'' + xy' + (x^2 - \frac{1}{4})y = 0$. Normalise: $p = 1/x$ and $q = 1 - \frac{1}{4x^2}$, so

$$I = 1 - \frac{1}{4x^2} - \frac{1}{4x^2} + \frac{1}{2x^2} = 1.$$

The normal form is exactly $u'' + u = 0$, and the factor is $\exp(-\frac{1}{2} \int \frac{dx}{x}) = x^{-1/2}$, so

$$y = \frac{A \cos x + B \sin x}{\sqrt{x}} \quad \text{on } (0, \infty).$$

This is Bessel's equation of order $\frac{1}{2}$, and the normal form has produced its closed solution in three lines. For Bessel of order 0, $x^2y'' + xy' + x^2y = 0$, the same computation gives $I = 1 + \frac{1}{4x^2}$, which tends to 1: the solutions are not elementary, but they oscillate with an envelope decaying like $x^{-1/2}$, and that is read straight off the substitution.

WHERE THIS GOES WRONG

Using p from the unnormalised equation. Divide by the leading coefficient first: for $x^2y'' + xy' + \dots$ the value of p is $1/x$, not x . The other slip is the sign in the exponent, which is $-\frac{1}{2} \int p$, so that the factor *decays* when $p > 0$; getting it backwards inverts the envelope and reverses every qualitative conclusion drawn from it.

SEE ALSO Change of independent variable; the substitution $x = e^t$; reduction of order from one known solution.

WHAT THE INVARIANT IS GOOD FOR

Two uses beyond solving. First, comparison: if $I(x) \geq M^2 > 0$ on an interval, then every solution of $u'' + Iu = 0$ has a zero in any subinterval of length π/M , because $u'' + M^2u = 0$ oscillates that fast and Sturm's comparison theorem transfers the zeros upward. If $I \leq 0$, no solution has two zeros at all. Since y differs from u only by a nonvanishing exponential factor, the zeros of y are exactly the zeros of u , so these conclusions apply to the original equation unchanged. Second, identification: I is unchanged by the substitution, so two equations with the same invariant are the same equation in disguise, and computing I is the quickest way to recognise a standard equation hidden behind a change of dependent variable.

Two closing observations tie the chapter together. The first is that every technique here has been a change of variable chosen to make a structural feature visible: a scaling becomes a translation under $\ln$, an absent variable becomes an absent order, a symmetry becomes a lower-order equation, a first-derivative term becomes a modified potential. Nothing was solved by force. The second is the recurring warning about the domain. A variable-coefficient equation has singular points wherever its leading coefficient vanishes, and at such points the existence theorem is silent, solutions may be unbounded or merely non-smooth, and initial data may fail to determine a solution or may determine several. Report the interval with the

answer. For the Cauchy-Euler family that interval is $(0, \infty)$ or $(-\infty, 0)$; for the disguised family it is one side of $x = -b/a$; for a nonlinear reduction it can be smaller than either, cut short by a blow-up the equation gave no warning of.

Recognition matrix

If you see	Reach for	If that fails
$ax^2y'' + bxy' + cy = 0$ Powers matching derivative orders Repeated indicial root Complex indicial root $\alpha \pm i\beta$ Cauchy-Euler with forcing $x^k \ln^j x$ k equals an indicial root $(ax + b)^2 y'' + c(ax + b)y' + dy$ y absent from the equation x absent from the equation $y'' = f(y)$ $p = 0$ somewhere on the trajectory Equation unchanged under $y \mapsto ky$ yy'' balanced against $(y')^2$ Coefficients that look like a product rule $xy'' - y' + \dots$ $(q' + 2pq)/q^{3/2}$ constant Want oscillation or envelope I constant after normalising $x^2y'' + xy' + (x^2 - \nu^2)y = 0$ Leading coefficient vanishing at x_0 Data given at $x = 0$ for Cauchy-Euler A nonlinear reduction that separates	Trial $y = x^m$; indicial $am(m-1) + bm + c$ Cauchy-Euler; expect x^m modes $y = x^m(C_1 + C_2 \ln x)$ $x^\alpha(C_1 \cos(\beta \ln x) + C_2 \sin(\beta \ln x))$ $x = e^t$; forcing becomes $t^j e^{kt}$ Resonance: expect $\ln^s x$ in y_p $u = ax + b$; keep the factors a, a^2 $p = y', y'' = p'$; first order in x $p = y', y'' = p \frac{dp}{dy}$; first order in y $\frac{1}{2}(y')^2 = \int f(y) dy + C$, then separate Turning point; the branch of $\sqrt{p^2}$ flips $y = e^{\int v dx}, y' = vy, y'' = (v' + v^2)y$ Equidimensional in y ; substitute for y'/y Test $a'' - b' + c = 0$; integrate once Change of independent variable, try $t = x^2$ $t = \int \sqrt{q} dx$ gives constant coefficients Normal form: $u'' + Iu = 0, I = q - \frac{1}{4}p^2 - \frac{1}{2}p'$ Elementary solutions; read them off Bessel; normal form gives the envelope $x^{-1/2}$ Singular point; solve on one side only Not an initial value problem; recheck Integrate, then watch for blow-up	Substitute $x = e^t$ and redo Check for a shifted centre Image of xe^{rt} under $x = e^t$ From $x^{i\beta} = e^{i\beta \ln x}$ Trial table in t , then $t = \ln x$ Shift rule on $(\theta - k)^s$ Expand $4x^2 + 4x + 1$ to spot it Two integrations, two constants Energy integral $\frac{1}{2}p^2 = \int f$ Fix the branch sign from the data State the interval of validity Compare with $y \mapsto \lambda y, x \mapsto \lambda x$ Check the degree term by term Seek an integrating factor x^k Kill Y_t : solve $ag'' + bg' = 0$ Otherwise use the normal form $y = u \exp(-\frac{1}{2} \int p)$ $I = 1$ means sin and cos Elementary only for $\nu = \frac{1}{2}$ Data at x_0 is not an IVP Ask for behaviour as $x \rightarrow 0^+$ Report the maximal interval

Systems, the phase plane, and qualitative technique

What this chapter is for. A first-order linear system $\dot{\mathbf{x}} = A\mathbf{x}$ is solved by one idea and one idea only: guess $\mathbf{x} = \mathbf{v}e^{\lambda t}$, discover that the guess works exactly when $A\mathbf{v} = \lambda\mathbf{v}$, and read every qualitative feature of every solution off the spectrum of A . Growth or decay is the sign of the real part; rotation is the imaginary part; the invariant directions are the eigenvectors. That is the whole of the linear theory, and the rest of this chapter is the discovery that it is also most of the nonlinear theory, because near an equilibrium a nonlinear field is its Jacobian plus something too small to matter. The chapter covers the eigenvalue solution in all three of its cases (real distinct, complex, defective), the matrix exponential, the trace-determinant classification of equilibria, linearisation with an honest account of the one place it goes silent, and the two low-technology tools (the phase line and the direction field) that answer qualitative questions when no closed form exists. At the end you should be able to look at a 2×2 matrix and say what the picture is before computing anything, and to look at a nonlinear field and know which of your conclusions are safe.

Why a system, and why eigenvalues

The scalar equation $\dot{y} = ay$ has solution $y = Ce^{at}$, and every question about it is answered by the sign of a . Write the same equation for a vector unknown, $\dot{\mathbf{x}} = A\mathbf{x}$ with A a constant $n \times n$ matrix, and the obvious guess is $\mathbf{x} = \mathbf{v}e^{\lambda t}$ for a constant vector $\mathbf{v}$. Substituting gives $\lambda\mathbf{v}e^{\lambda t} = A\mathbf{v}e^{\lambda t}$, and since $e^{\lambda t}$ never vanishes this is

$$A\mathbf{v} = \lambda\mathbf{v}.$$

The guess succeeds precisely on the eigenvectors. Nothing about this reasoning is two-dimensional or even finite-dimensional; it is the statement that the exponential is the eigenfunction of $\frac{d}{dt}$, transported to a space where $\frac{d}{dt}$ acts componentwise and A does the mixing.

Two consequences organise everything that follows. First, the equation is linear and homogeneous, so sums of solutions are solutions: if A has n independent eigenvectors, the n modes $\mathbf{v}_i e^{\lambda_i t}$ span the solution space and the general solution is their combination. Second, the qualitative picture is a statement about λ alone. A negative real part contracts, a positive one expands, a nonzero imaginary part rotates. Two systems with the same spectrum have the same portrait up to a linear change of coordinates, which is why a classification with only a handful of cases is possible at all.

THE SPECTRUM IS THE ANSWER

For $\dot{\mathbf{x}} = A\mathbf{x}$, every long-time question is decided by $\operatorname{Re} \lambda$ and every short-time geometric question by $\operatorname{Im} \lambda$ and the eigenvectors. All solutions tend to $\mathbf{0}$ if and only if every eigenvalue has negative real part; all solutions stay bounded if additionally the purely imaginary eigenvalues are non-defective. Compute the spectrum before you compute anything else, and in the 2×2 case compute it from the trace and the determinant, since $\lambda^2 - T\lambda + \Delta = 0$ with $T = \operatorname{tr} A$ and $\Delta = \det A$.

Nothing above used $n = 2$, and it is worth being explicit about what survives in higher dimensions, because the planar pictures that follow can leave a false impression. The mode decomposition is dimension-free: n independent eigenvectors give n modes and the general solution is their span. Stability is dimension-free: all solutions decay if and only if every eigenvalue has negative real part. What is special to the plane is the *classification*, because only in two dimensions is the list of pictures short enough to name. In three dimensions a real eigenvalue and a complex pair produce a spiral in one plane multiplied by exponential motion along a transverse line, and the useful description is exactly that product, not a new noun.

One practical consequence. Deciding stability in dimension three without computing roots is a determinant question: for $\lambda^3 + a_2\lambda^2 + a_1\lambda + a_0$ with real coefficients, every root has negative real part if and only if $a_2 > 0$, $a_0 > 0$ and $a_2a_1 > a_0$. The strict inequality is where a pair of roots crosses the imaginary axis, which is the three-dimensional analogue of $T = 0$ and the place where oscillations are born.

Before the machinery, the reduction that makes it universally applicable. Every scalar equation of order n is a first-order system in n unknowns, and the translation is mechanical. It is worth doing first because it means the eigenvalue the-

ory of this chapter and the characteristic-polynomial theory of constant-coefficient scalar equations are literally the same theorem, and because every numerical integrator on earth expects its input in first-order form.

37.1 Reducing a scalar equation to a first-order system

EASY

CUE A scalar equation of order n that you want to feed to a numerical integrator, an eigenvalue routine, or a phase-plane argument. Also any second-order equation whose *qualitative* behaviour is wanted rather than its formula.

METHOD Name the derivatives as new unknowns: $x_1 = y$, $x_2 = \dot{y}$, up to $x_n = y^{(n-1)}$. Then $\dot{x}_i = x_{i+1}$ for $i < n$ automatically, and $\dot{x}_n$ is whatever the equation says $y^{(n)}$ is. For $y^{(n)} + a_{n-1}y^{(n-1)} + \dots + a_0y = 0$ the result is $\dot{\mathbf{x}} = \mathbf{C}\mathbf{x}$ with the companion matrix

$$\mathbf{C} = \begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ -a_0 & -a_1 & -a_2 & \dots & -a_{n-1} \end{pmatrix},$$

whose characteristic polynomial is exactly $\lambda^n + a_{n-1}\lambda^{n-1} + \dots + a_0$, the original characteristic polynomial. The order matters: the coefficients occupy the *last* row when the state is ordered from y upward.

WORKED EXAMPLE

Take $\ddot{y} + 4y = 0$. Set $x_1 = y$, $x_2 = \dot{y}$; then $\dot{x}_1 = x_2$ and $\dot{x}_2 = \ddot{y} = -4y = -4x_1$, so

$$\dot{\mathbf{x}} = \begin{pmatrix} 0 & 1 \\ -4 & 0 \end{pmatrix} \mathbf{x}, \quad \det(\mathbf{C} - \lambda \mathbf{I}) = \lambda^2 + 4,$$

giving $\lambda = \pm 2i$, which is the same pair the characteristic equation $r^2 + 4 = 0$ produces. The phase portrait is therefore a family of closed orbits in the $(y, \dot{y})$ plane, which is the statement that the undamped oscillator conserves energy, obtained without ever writing $\cos 2t$.

Damping it, $\ddot{y} + \frac{1}{2}\dot{y} + 4y = 0$ gives the last row $(-4, -\frac{1}{2})$, trace $-\frac{1}{2}$, determinant 4, and $T^2 - 4\Delta = \frac{1}{4} - 16 < 0$: a stable spiral, which is underdamped decay.

WHERE THIS GOES WRONG

Ordering the state as $(\dot{y}, y)$ rather than $(y, \dot{y})$, which transposes the two off-diagonal entries and gives $\begin{pmatrix} -\frac{1}{2} & -4 \\ 1 & 0 \end{pmatrix}$ for the damped example. The eigenvalues survive the relabelling, so a check based on them detects nothing, but the phase portrait is drawn in swapped axes and every statement about direction of rotation is reversed. Fix the order once, write it down beside the matrix, and keep it.

SEE ALSO Real distinct eigenvalues; the characteristic equation and distinct real roots; Euler's method and step-by-step numerical integration.

Solving the linear system

With the reduction in hand, the linear problem is finite: find the eigenvalues, and handle each of the three things that can happen to them. Real and distinct is the generic case and the easiest. A complex conjugate pair carries the same information but must be repackaged into real functions before it is usable. A repeated eigenvalue is harmless if it still has enough eigenvectors and requires one extra construction if it does not. There is no fourth case.

37.2 Real distinct eigenvalues: the mode decomposition

CUE A first-order linear system with a constant matrix, or a pair of coupled linear equations such as $\dot{x} = x + y$, $\dot{y} = 4x + y$, which is the same thing written out.

METHOD Solve $\det(A - \lambda I) = 0$. For each real eigenvalue λ_i , solve $(A - \lambda_i I)\mathbf{v}_i = \mathbf{0}$ for an eigenvector, and form the mode $\mathbf{v}_i e^{\lambda_i t}$. The general solution is

$$\mathbf{x}(t) = \sum_i C_i \mathbf{v}_i e^{\lambda_i t}.$$

Initial data are fitted by solving the single algebraic system $\sum_i C_i \mathbf{v}_i = \mathbf{x}(0)$, that is, by expressing the initial vector in the eigenbasis. In the 2×2 case use $\lambda^2 - T\lambda + \Delta = 0$ directly; there is no reason to expand the determinant by hand.

WORKED EXAMPLE

$$A = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix}, \quad \mathbf{x}(0) = \begin{pmatrix} 3 \\ 2 \end{pmatrix}.$$

Here $T = 2$ and $\Delta = 1 - 4 = -3$, so $\lambda^2 - 2\lambda - 3 = 0$ and $\lambda = 3, -1$. For $\lambda = 3$, $(A - 3I) = \begin{pmatrix} -2 & 1 \\ 4 & -2 \end{pmatrix}$ has kernel spanned by $(1, 2)^T$; for $\lambda = -1$, $(A + I) = \begin{pmatrix} 2 & 1 \\ 4 & 2 \end{pmatrix}$ has kernel spanned by $(1, -2)^T$. So

$$\mathbf{x}(t) = C_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{3t} + C_2 \begin{pmatrix} 1 \\ -2 \end{pmatrix} e^{-t}.$$

Fitting the data: $C_1 + C_2 = 3$ and $2C_1 - 2C_2 = 2$, so $C_1 = 2$, $C_2 = 1$ and

$$\mathbf{x}(t) = 2 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{3t} + \begin{pmatrix} 1 \\ -2 \end{pmatrix} e^{-t}.$$

At $t = 0.7$ this reads (16.828925, 31.671509), confirmed to six figures by Runge-Kutta from the same data. Since $\Delta < 0$ the origin is a saddle: the line spanned by $(1, -2)^T$ is the stable direction and every other trajectory eventually runs parallel to $(1, 2)^T$.

A ZERO EIGENVALUE, AND WHAT IT MEANS

Two hundred-litre tanks exchange brine at ten litres per minute in each direction, with x and y the grams of salt in each. Then

$$\dot{x} = -\frac{1}{10}x + \frac{1}{10}y, \quad \dot{y} = \frac{1}{10}x - \frac{1}{10}y, \quad A = \frac{1}{10} \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Here $T = -\frac{1}{5}$ and $\Delta = 0$, so the eigenvalues are 0 and $-\frac{1}{5}$, with eigenvectors $(1, 1)^T$ and $(1, -1)^T$. A zero eigenvalue means a whole line of equilibria, namely the line $x = y$, and it also means a conserved quantity: the eigenvalue 0 of A pairs with the left eigenvector $(1, 1)$, so $\frac{d}{dt}(x + y) = 0$ and the total salt never changes. Starting with 20 grams in the first tank and none in the second,

$$x(t) = 10 + 10e^{-t/5}, \quad y(t) = 10 - 10e^{-t/5},$$

which at $t = 7$ is (12.465970, 7.534030), confirmed numerically. The system relaxes to equal concentrations with time constant 5 minutes, and *which* equilibrium it relaxes to is decided entirely by the conserved total.

WHERE THIS GOES WRONG

Fitting the constants componentwise, as though C_1 belonged to the first component and C_2 to the second. Both constants appear in both components. The correct step is to solve the 2×2 system $C_1 \mathbf{v}_1 + C_2 \mathbf{v}_2 = \mathbf{x}(0)$, which is a change of basis, not a matching of exponentials. A second and quieter error: normalising eigenvectors to unit length halfway through, so that the constants found earlier no longer correspond to the vectors written later.

SEE ALSO Complex eigenvalues and the real form; defective eigenvalues and generalised eigenvectors; the trace-determinant classification.

A real matrix can perfectly well have complex eigenvalues, and when it does the algebra above still works: $\mathbf{v}e^{\lambda t}$ is a genuine solution of the system, taking values in $\mathbb{C}^2$. What is wanted is a real solution, and the bridge is the observation that the real and imaginary parts of a complex solution of a *real* linear equation are each solutions in their own right. That single

remark converts one complex mode into the two real modes you need, and it explains why the conjugate eigenvalue supplies nothing new.

37.3 Complex eigenvalues and the real form

CORE

CUE The characteristic equation $\lambda^2 - T\lambda + \Delta = 0$ has negative discriminant $T^2 - 4\Delta < 0$, giving a conjugate pair $\lambda = \alpha \pm i\beta$ with $\beta > 0$.

METHOD Work with one root only, say $\lambda = \alpha + i\beta$, find a complex eigenvector $\mathbf{v} = \mathbf{p} + i\mathbf{q}$, and expand

$$\mathbf{v}e^{(\alpha+i\beta)t} = e^{\alpha t}(\mathbf{p} + i\mathbf{q})(\cos \beta t + i \sin \beta t).$$

Its real and imaginary parts,

$$\mathbf{x}_1 = e^{\alpha t}(\mathbf{p} \cos \beta t - \mathbf{q} \sin \beta t), \quad \mathbf{x}_2 = e^{\alpha t}(\mathbf{p} \sin \beta t + \mathbf{q} \cos \beta t),$$

are two independent real solutions. Geometrically the orbit is a spiral of angular rate β on an exponential envelope $e^{\alpha t}$: outward if $\alpha > 0$, inward if $\alpha < 0$, and a closed ellipse if $\alpha = 0$.

WORKED EXAMPLE

$$A = \begin{pmatrix} -1 & 2 \\ -2 & -1 \end{pmatrix}.$$

$T = -2$, $\Delta = 1 + 4 = 5$, so $\lambda^2 + 2\lambda + 5 = 0$ and $\lambda = -1 \pm 2i$. For $\lambda = -1 + 2i$,

$$A - \lambda I = \begin{pmatrix} -2i & 2 \\ -2 & -2i \end{pmatrix},$$

whose first row gives $-2iv_1 + 2v_2 = 0$, that is $\mathbf{v} = (1, i)^T$. Thus $\mathbf{p} = (1, 0)^T$ and $\mathbf{q} = (0, 1)^T$, and the two real solutions are

$$\mathbf{x}_1 = e^{-t} \begin{pmatrix} \cos 2t \\ -\sin 2t \end{pmatrix}, \quad \mathbf{x}_2 = e^{-t} \begin{pmatrix} \sin 2t \\ \cos 2t \end{pmatrix}.$$

Both were checked against Runge-Kutta at $t = 1.3$, agreeing to twelve figures. Since $\alpha = -1 < 0$ the origin is a stable spiral; since $\beta = 2$, one full turn takes time π , and the amplitude shrinks by the factor $e^{-\pi} \approx 0.0432$ per turn.

THE SAME COMPUTATION READ AS A DAMPED OSCILLATOR

The equation $\ddot{y} + \frac{1}{2}\dot{y} + 4y = 0$ has companion matrix $\begin{pmatrix} 0 & 1 \\ -4 & -\frac{1}{2} \end{pmatrix}$, with $T = -\frac{1}{2}$ and $\Delta = 4$, so $\lambda = -\frac{1}{4} \pm i\sqrt{4 - \frac{1}{16}} = -\frac{1}{4} \pm 1.984313i$. The real part fixes the decay time constant at 4 and the imaginary part fixes the damped frequency, slightly below the undamped 2. In the phase plane $(y, \dot{y})$ the orbit is a clockwise spiral, clockwise because at the point $(1, 0)^T$ the velocity is $(0, -4)^T$, pointing down. Reading these three numbers off T and Δ takes longer to describe than to do, and it is the entire qualitative content of the solution.

WHERE THIS GOES WRONG

Building four solutions by using both $\alpha + i\beta$ and $\alpha - i\beta$. The conjugate eigenvector is the conjugate of $\mathbf{v}$ and its complex solution is the conjugate of the first, so its real part is $\mathbf{x}_1$ and its imaginary part is $-\mathbf{x}_2$: nothing new, and a two-dimensional problem cannot have four independent solutions anyway. The other frequent slip is dropping the $e^{\alpha t}$ envelope when α is small, which converts a slow spiral into a false centre.

SEE ALSO Real distinct eigenvalues; the trace-determinant classification; complex roots and the real form.

Repetition is the remaining case, and it splits. A repeated eigenvalue may still have a full set of eigenvectors, in which case A is a multiple of the identity on that eigenspace and there is nothing to do. The interesting case is *defective*: the algebraic multiplicity exceeds the number of independent eigenvectors, the mode construction runs out of vectors, and a solution of a genuinely new shape is required. The shape is dictated by the scalar analogy $y = (C_1 + C_2 t)e^{rt}$, but the vector version has a term the scalar version cannot show you.

37.4 Defective eigenvalues and generalised eigenvectors

CUE A repeated root of the characteristic equation ($T^2 = 4\Delta$ in the plane) together with a matrix that is not a scalar multiple of I ; equivalently, $(A - \lambda I)$ has rank 1 rather than rank 0.

METHOD Find the eigenvector $\mathbf{v}$ from $(A - \lambda I)\mathbf{v} = \mathbf{0}$, then solve the *inhomogeneous* system

$$(A - \lambda I)\mathbf{w} = \mathbf{v}$$

for a generalised eigenvector $\mathbf{w}$. This is always solvable in the defective 2×2 case, and $\mathbf{w}$ is determined only up to adding a multiple of $\mathbf{v}$, which is harmless. The two solutions are

$$\mathbf{x}_1 = \mathbf{v}e^{\lambda t}, \quad \mathbf{x}_2 = (t\mathbf{v} + \mathbf{w})e^{\lambda t}.$$

The check is direct: $\dot{\mathbf{x}}_2 = \mathbf{v}e^{\lambda t} + \lambda(t\mathbf{v} + \mathbf{w})e^{\lambda t}$, while $A\mathbf{x}_2 = (t\lambda\mathbf{v} + A\mathbf{w})e^{\lambda t}$, and the two agree exactly because $A\mathbf{w} = \lambda\mathbf{w} + \mathbf{v}$.

WORKED EXAMPLE

$$A = \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix}.$$

$T = 4$, $\Delta = 3 + 1 = 4$, so $\lambda^2 - 4\lambda + 4 = 0$ and $\lambda = 2$ twice. Now $A - 2I = \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix}$ has rank 1, so the eigenvalue is defective; its kernel is spanned by $\mathbf{v} = (1, 1)^T$. Solve $(A - 2I)\mathbf{w} = \mathbf{v}$, that is $w_1 - w_2 = 1$, and take $\mathbf{w} = (1, 0)^T$. Hence

$$\mathbf{x}(t) = C_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{2t} + C_2 \begin{pmatrix} t+1 \\ t \end{pmatrix} e^{2t}.$$

Starting from $\mathbf{x}(0) = (1, 0)^T$ forces $C_1 = 0$, $C_2 = 1$, giving $\mathbf{x} = (t+1, t)^T e^{2t}$, which at $t = 0.9$ is $(11.494330, 5.444683)$, matched by Runge-Kutta to eleven figures. Every trajectory leaves the origin tangent to $\mathbf{v}$ and returns to that direction as $t \rightarrow \pm\infty$, because $t\mathbf{v}$ dominates $\mathbf{w}$ at large $|t|$: this is the degenerate node.

WHERE THIS GOES WRONG

Writing $t\mathbf{v}e^{\lambda t}$ alone as the second solution, by false analogy with the scalar $x e^{rx}$. Substituting it gives $\dot{\mathbf{x}} - A\mathbf{x} = \mathbf{v}e^{\lambda t} \neq \mathbf{0}$; the term $\mathbf{w}e^{\lambda t}$ exists exactly to cancel that residue, and it is *added* to $t\mathbf{v}$, never multiplied by t . Second trap: solving $(A - \lambda I)\mathbf{w} = \mathbf{v}$ with the wrong scaling of $\mathbf{v}$. If you rescale $\mathbf{v}$ after finding $\mathbf{w}$, the pair no longer satisfies the relation and $\mathbf{x}_2$ stops being a solution.

SEE ALSO Real distinct eigenvalues; the matrix exponential; repeated root and the factor of x .

The non-defective repeated case deserves one sentence so that it is not mistaken for the one above. If λ is repeated and $A - \lambda I = 0$, then $A = \lambda I$, every nonzero vector is an eigenvector, and the solution is $\mathbf{x} = e^{\lambda t}\mathbf{x}(0)$: every trajectory is a straight ray through the origin, traversed at exponential speed. This is the star node, and it is the only planar case in which the portrait has no distinguished directions at all. The test that separates it from the degenerate node is the rank of $A - \lambda I$, not the multiplicity of λ .

The three cases can be packaged into a single object. The matrix exponential is not a computational shortcut in small examples, where the mode decomposition is faster, but it is the right language for the nonhomogeneous problem, for estimating growth rates, and for any argument in which the solution operator itself, rather than one solution, is the thing being studied.

37.5 The matrix exponential

CUE You want the solution operator, a single formula $\mathbf{x}(t) = e^{At}\mathbf{x}(0)$, or the variation-of-parameters formula for a forced system.

METHOD Define $e^{At} = \sum_{k \geq 0} \frac{A^k t^k}{k!}$, which converges for every A and satisfies $\frac{d}{dt}e^{At} = Ae^{At}$ with $e^{A \cdot 0} = I$. Two ways to evaluate it. If $A = P\Lambda P^{-1}$ is diagonalisable, then $e^{At} = Pe^{\Lambda t}P^{-1}$ with $e^{\Lambda t}$ diagonal. If A has a single eigenvalue λ , split $A = \lambda I + N$ with $N = A - \lambda I$ nilpotent; since λI commutes with N ,

$$e^{At} = e^{\lambda t} \left(I + Nt + \frac{N^2 t^2}{2} + \dots \right),$$

a terminating sum. In the plane $N^2 = 0$ always, so this is just $e^{\lambda t}(I + Nt)$.

For the defective $A = \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix}$ above, $N = A - 2I = \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix}$ and $N^2 = 0$, so

$$e^{At} = e^{2t} \begin{pmatrix} 1+t & -t \\ t & 1-t \end{pmatrix}.$$

Its first column is $(1+t, t)^T e^{2t}$, exactly the solution found above from $\mathbf{x}(0) = (1, 0)^T$, as it must be.

For the diagonalisable $A = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix}$ with $P = \begin{pmatrix} 1 & 1 \\ 2 & -2 \end{pmatrix}$ and $\Lambda = \text{diag}(3, -1)$, carrying out $Pe^{\Lambda t}P^{-1}$ gives

$$e^{At} = \begin{pmatrix} \frac{e^{3t}+e^{-t}}{2} & \frac{e^{3t}-e^{-t}}{4} \\ e^{3t}-e^{-t} & \frac{e^{3t}+e^{-t}}{2} \end{pmatrix}.$$

Two audits, both cheap and both worth doing every time: at $t = 0$ the matrix is I , and differentiating at $t = 0$ returns $\begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix} = A$. Numerically at $t = 0.6$ the columns agree with Runge-Kutta from $(1, 0)^T$ and $(0, 1)^T$ to eleven figures.

WHERE THIS GOES WRONG

Writing $e^{A+B} = e^A e^B$ for matrices that do not commute. It is false in general, and the splitting trick above is legitimate only because λI commutes with everything. A related error is computing e^{At} entrywise, as though it meant the matrix of $e^{a_{ij}t}$; the exponential of a matrix is defined by the series and has nothing to do with exponentiating entries. The diagonal case is the only one where the two coincide.

SEE ALSO Defective eigenvalues and generalised eigenvectors; nonhomogeneous systems and the Duhamel formula; convolution and the transfer function.

Forcing is handled exactly as it is for scalar linear equations, by variation of parameters, and the matrix exponential makes the formula memorable. The structural point is that the forced solution is a superposition of impulse responses, one for each instant s at which the forcing acts, propagated forward for the remaining time $t-s$. That reading is worth internalising here, because it recurs verbatim as Duhamel's formula in the Laplace chapter.

37.6 Nonhomogeneous systems and the Duhamel formula

HARD

CUE $\dot{\mathbf{x}} = A\mathbf{x} + \mathbf{g}(t)$ with $\mathbf{g}$ not identically zero, or a forced higher-order equation converted to a system.

METHOD Let $\Phi(t)$ be a fundamental matrix, meaning a matrix whose columns are independent homogeneous solutions. Variation of parameters gives

$$\mathbf{x}(t) = \Phi(t) \int \Phi^{-1}(t) \mathbf{g}(t) dt.$$

With $\Phi = e^{At}$ (so $\Phi^{-1}(s) = e^{-As}$) and data at $t = 0$ this becomes the clean form

$$\mathbf{x}(t) = e^{At} \mathbf{x}_0 + \int_0^t e^{A(t-s)} \mathbf{g}(s) ds.$$

Read the integrand: at time s the forcing injects $\mathbf{g}(s) ds$, and the free flow carries it for the remaining time $t-s$.

WORKED EXAMPLE

$$\dot{\mathbf{x}} = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix} \mathbf{x} + \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \mathbf{x}(0) = \mathbf{0}.$$

The matrix is triangular, so the second component decouples: $\dot{x}_2 = 2x_2 + 1$ with $x_2(0) = 0$ gives $x_2 = \frac{e^{2t}-1}{2}$. The first component is then $\dot{x}_1 = 2x_1 + x_2$, and Duhamel gives $x_1 = \int_0^t e^{2(t-s)} x_2(s) ds$. Evaluate:

$$x_1 = \frac{1}{2} \int_0^t (e^{2t} - e^{2t-2s}) ds = \frac{te^{2t}}{2} - \frac{e^{2t}-1}{4}.$$

Audit by substitution: $\dot{x}_1 = \frac{1}{2}e^{2t} + te^{2t} - \frac{1}{2}e^{2t} = te^{2t}$, while $2x_1 + x_2 = te^{2t} - \frac{e^{2t}-1}{2} + \frac{e^{2t}-1}{2} = te^{2t}$. At $t = 1.1$ the pair is (2.957504, 4.012507), reproduced by Runge-Kutta.

Take the undamped oscillator forced at its own frequency: $\ddot{y} + y = \cos t$ with zero data, that is $\dot{x} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} x + \begin{pmatrix} 0 \\ \cos t \end{pmatrix}$. The impulse response of $\ddot{y} + y$ is $\sin t$, so Duhamel gives

$$y(t) = \int_0^t \sin(t-s) \cos s \, ds = \frac{1}{2} \int_0^t (\sin t + \sin(t-2s)) \, ds = \frac{t \sin t}{2},$$

using the product-to-sum identity and noting that the second term integrates to zero over the symmetric range. The linear growth appears by itself: no resonance rule, no multiplier t^s , no separate case. At $t = 3.4$ the formula gives -0.4344199 , reproduced by numerical integration. Compare this with undetermined coefficients, where forgetting the resonance multiplier produces a contradiction rather than an answer.

WHERE THIS GOES WRONG

Inverting Φ at the wrong argument. Inside the integral the inverse carries the *dummy* variable, $\Phi^{-1}(s)$, and the factor outside carries t ; writing $\Phi^{-1}(t)$ inside and pulling it out turns a convolution into a product and produces a function that is not a solution. Equally common: forgetting that $e^{A(t-s)} \neq e^{At}e^{As}$, the sign in the exponent is what makes the propagation forward rather than backward.

SEE ALSO The matrix exponential; variation of parameters; convolution and the transfer function.

 LIOUVILLE'S FORMULA, AND A FREE CHECK ON EVERY e^{At}

The determinant of a fundamental matrix obeys $\frac{d}{dt} \det \Phi = (\text{tr } A) \det \Phi$, so for constant A

$$\det e^{At} = e^{t \text{tr } A}.$$

In particular e^{At} is invertible for every t , which is why the Duhamel formula never divides by zero, and why solutions of a linear system exist for all time rather than blowing up in finite time as nonlinear ones can. It is also a one-line audit. For $A = \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix}$

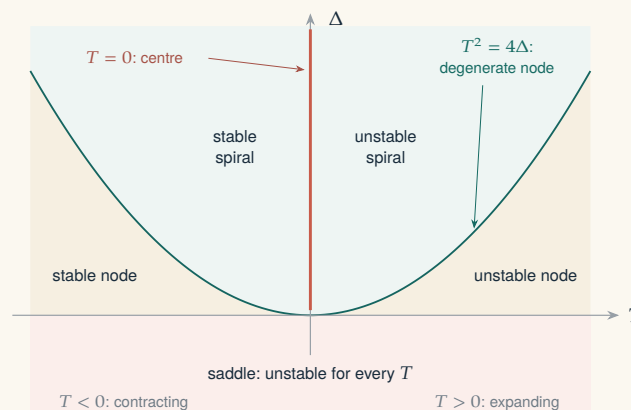
with $\text{tr } A = 4$, the computed $e^{2t} \begin{pmatrix} 1+t & -t \\ t & 1-t \end{pmatrix}$ has determinant $e^{4t}[(1+t)(1-t) + t^2] = e^{4t}$, as required. For $A = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix}$

with $\text{tr } A = 2$, the computed exponential has determinant $\frac{1}{4}[(e^{3t} + e^{-t})^2 - (e^{3t} - e^{-t})^2] = e^{2t}$. Any error in a single entry breaks this identity, and it costs two multiplications to test.

The phase plane

Stop solving and start looking. A planar autonomous system assigns a velocity vector to each point, and the solutions are the curves everywhere tangent to that field. The equilibria (points where the velocity vanishes) organise the picture, because they are the only places where a trajectory can begin or end in finite or infinite time, and the behaviour near each one is what the classification names.

For $\dot{x} = Ax$ there is exactly one equilibrium when $\det A \neq 0$, namely the origin, and its type is a function of two numbers. Write $T = \text{tr } A$ and $\Delta = \det A$, so that $\lambda^2 - T\lambda + \Delta = 0$ and $\lambda_{\pm} = \frac{1}{2}(T \pm \sqrt{T^2 - 4\Delta})$. The whole classification is the sign chart of Δ , of $T^2 - 4\Delta$, and of T , and it is best carried in the head as a picture rather than a list.



Three features of the picture deserve saying out loud. The parabola $T^2 = 4\Delta$ and the positive Δ axis are boundaries of measure zero, so the degenerate node and the centre are exactly the cases that an arbitrarily small perturbation of A destroys. The lower half plane is entirely saddle, with no dependence on T at all, because $\Delta < 0$ forces one positive and one negative eigenvalue. And the vertical line $T = 0$ separates contraction from expansion everywhere in the upper half plane, which is why $T < 0$ together with $\Delta > 0$ is the complete criterion for asymptotic stability in two dimensions.

CORE

37.7 Equilibria and the trace-determinant classification

CUE “Classify the equilibrium”, “is it stable”, “sketch the phase portrait”, or any question about long-time behaviour of a planar linear system.

METHOD Compute $T = \text{tr } A$ and $\Delta = \det A$ and read the chart. If $\Delta < 0$: saddle, unstable. If $\Delta > 0$ and $T^2 > 4\Delta$: node, stable when $T < 0$ and unstable when $T > 0$. If $\Delta > 0$ and $T^2 < 4\Delta$: spiral, stable when $T < 0$ and unstable when $T > 0$. If $\Delta > 0$ and $T = 0$: centre, with closed orbits and neutral stability. If $T^2 = 4\Delta$: degenerate or star node, stable when $T < 0$. If $\Delta = 0$ the matrix is singular and the equilibria form a line, which must be handled directly.

WORKED EXAMPLE

$A = \begin{pmatrix} -1 & 2 \\ -2 & -1 \end{pmatrix}$: $T = -2$, $\Delta = 5$, $T^2 - 4\Delta = 4 - 20 = -16 < 0$, so a spiral, and $T < 0$ makes it stable. This agrees with the explicit solution $e^{-t}(\cos 2t, -\sin 2t)^T$ found earlier.

$A = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix}$: $\Delta = -3 < 0$, saddle, and no further computation is needed to know it is unstable.

$A = \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix}$: $T = 4$, $\Delta = 4$, $T^2 = 4\Delta$ exactly, so a degenerate node, unstable since $T > 0$. Note how the chart delivers all three verdicts from six multiplications, without an eigenvector in sight.

READING A PARAMETER SWEEP OFF THE CHART

The mass-spring-damper $m\ddot{x} + c\dot{x} + kx = 0$ has companion matrix with $T = -c/m$ and $\Delta = k/m$, both of which are fixed by the physics. Since $\Delta > 0$ always and $T < 0$ whenever there is any damping, the equilibrium is stable for every $c > 0$; what the parameter changes is only which kind. The boundary $T^2 = 4\Delta$ reads $c^2/m^2 = 4k/m$, that is $c^2 = 4mk$, which is the classical critical damping. Take $m = 1$ and $k = 4$, so the critical value is $c = 4$. Then $c = 1$ gives $T^2 - 4\Delta = 1 - 16 < 0$, a stable spiral (underdamped, the system rings); $c = 5$ gives $25 - 16 > 0$, a stable node (overdamped, no ring at all); and $c = 4$ sits exactly on the parabola, the degenerate node, which is critical damping and the fastest approach to rest without overshoot. A single trajectory in the (T, Δ) plane, moving leftward along the horizontal line $\Delta = 4$ as c increases, is the whole story of damping.

WHERE THIS GOES WRONG

Applying the chart to a nonlinear system without linearising, and applying it to a *nonzero* equilibrium of a linear system without first translating that equilibrium to the origin. For $\dot{\mathbf{x}} = A\mathbf{x} + \mathbf{b}$ with $\det A \neq 0$ the equilibrium is $\mathbf{x}^* = -A^{-1}\mathbf{b}$, and the classification uses A , not the matrix of the affine map. The third habitual error is reading stability off Δ : a large positive determinant with positive trace is an unstable spiral, not a stable one.

SEE ALSO Linearising at an equilibrium; complex eigenvalues and the real form; the phase line for a scalar autonomous equation.

The chart names the equilibrium. Drawing it takes three more observations, and they are worth practising until they are automatic. For a node or a saddle, the eigenvectors are invariant lines: a trajectory starting on one stays on it forever, moving outward if the corresponding eigenvalue is positive and inward if it is negative, so sketch those two lines first and let every other trajectory be asymptotic to them. For a node, the slower mode dominates as $t \rightarrow \infty$, so all trajectories except those on the fast line arrive tangent to the *slow* eigenvector. For a spiral there are no invariant lines; get the sense of rotation by evaluating $A\mathbf{x}$ at a single convenient point, say $(1, 0)^T$, and seeing whether the resulting velocity points up (counterclockwise) or down (clockwise).

SKETCHING A LINEAR PORTRAIT IN FOUR STROKES

One: compute T and Δ and name the type. Two: if there are real eigenvalues, draw the eigenvector lines with arrows set by the sign of each λ . Three: if the eigenvalues are complex, evaluate the velocity at one point to fix the rotation sense, and choose the envelope from the sign of T . Four: fill in the remaining trajectories so that they never cross, are tangent to the slow direction at the origin for a node, and are asymptotic to the unstable direction at infinity for a saddle. The picture is then determined; there is nothing else to decide.

Now the claim that makes this chapter worth more than its linear content. Near an equilibrium $\mathbf{x}^*$ of a nonlinear field $\mathbf{F}$, Taylor's theorem gives $\mathbf{F}(\mathbf{x}^* + \mathbf{u}) = D\mathbf{F}(\mathbf{x}^*)\mathbf{u} + O(\|\mathbf{u}\|^2)$, so the deviation $\mathbf{u}$ obeys a linear system plus a remainder that is quadratically small. If the linear part is strongly hyperbolic (no eigenvalue on the imaginary axis), the remainder cannot change the verdict and the linear portrait is the local truth. If an eigenvalue sits on the imaginary axis, the remainder decides, and the linearisation tells you nothing.

CORE

37.8 Linearising at an equilibrium

CUE $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$ with $\mathbf{F}$ nonlinear, and a question about stability, classification, or a phase portrait.

METHOD Find every equilibrium by solving $\mathbf{F}(\mathbf{x}) = \mathbf{0}$, which is an algebraic system, usually by factoring each component. At each equilibrium evaluate the Jacobian

$$J = D\mathbf{F}(\mathbf{x}^*) = \begin{pmatrix} \frac{\partial F_1}{\partial x} & \frac{\partial F_1}{\partial y} \\ \frac{\partial F_2}{\partial x} & \frac{\partial F_2}{\partial y} \end{pmatrix}_{\mathbf{x}=\mathbf{x}^*}$$

and classify J by trace and determinant. The Hartman-Grobman theorem says that if no eigenvalue of J has zero real part, the nonlinear portrait near $\mathbf{x}^*$ is a continuous deformation of the linear one: saddles stay saddles, stable nodes and spirals stay stable, unstable stay unstable. Nodes and spirals may swap with each other under perturbation, so the honest conclusion in a borderline node/spiral case is “attracting” rather than a specific shape.

WORKED EXAMPLE

Two competing species,

$$\dot{x} = x(3 - x - 2y), \quad \dot{y} = y(2 - x - y).$$

Equilibria: each factor may vanish, giving $(0,0)$, $(3,0)$, $(0,2)$, and the interior point where $3 - x - 2y = 0$ and $2 - x - y = 0$, whose difference is $1 - y = 0$, so $(1,1)$. The Jacobian is

$$J(x,y) = \begin{pmatrix} 3 - 2x - 2y & -2x \\ -y & 2 - x - 2y \end{pmatrix}.$$

At $(0,0)$: $J = \text{diag}(3,2)$, $T = 5$, $\Delta = 6$, $T^2 - 4\Delta = 1 > 0$, an unstable node. At $(3,0)$: $J = \begin{pmatrix} -3 & -6 \\ 0 & -1 \end{pmatrix}$, eigenvalues -3 and -1 , a stable node. At $(0,2)$: $J = \begin{pmatrix} -1 & 0 \\ -2 & -2 \end{pmatrix}$, eigenvalues -1 and -2 , a stable node. At $(1,1)$: $J = \begin{pmatrix} -1 & -2 \\ -1 & -1 \end{pmatrix}$, $\Delta = 1 - 2 = -1 < 0$, a saddle.

The portrait is therefore bistable: the interior coexistence state is a saddle, and its stable manifold is a curve dividing the first quadrant into two basins. Runge-Kutta confirms it: starting at $(1.05, 1)$ the orbit runs to $(3,0)$, starting at $(0.95, 1)$ it runs to $(0,2)$, and the same split occurs for $(1, 0.95)$ and $(1, 1.05)$ with the destinations exchanged. A one-percent difference in initial abundance decides which species survives.

WHEN THE INCONCLUSIVE CASE CAN BE SETTLED

The undamped pendulum $\ddot{\theta} + \sin \theta = 0$ becomes $\dot{x} = y$, $\dot{y} = -\sin x$ with $x = \theta$. Equilibria are $(n\pi, 0)$, and

$$J(x,y) = \begin{pmatrix} 0 & 1 \\ -\cos x & 0 \end{pmatrix}.$$

At $(\pi, 0)$, $J = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ has $\Delta = -1 < 0$: a saddle, and being hyperbolic the verdict is safe. The inverted pendulum really is unstable, and the two eigendirections $(1, \pm 1)^T$ are the tangents to the trajectories that fall away from and creep toward the upright position.

At $(0,0)$, $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ has $T = 0$ and $\Delta = 1$: a centre, and by the previous paragraph that is inconclusive. Here it can be settled, because the system has a conserved quantity. Set $E = \frac{1}{2}y^2 + 1 - \cos x$; then

$$\dot{E} = y\dot{y} + \sin x \dot{x} = -y \sin x + y \sin x = 0.$$

E has a strict local minimum at the origin, so the level curves near it are closed, and the origin is a genuine centre. Numerically, starting from $(0.4, 0)$, $(2, 0)$ and $(3, 0)$, the value of E is preserved to fourteen figures over 9 units of

time, which is the sort of check a conserved quantity makes cheap. The moral: linearisation raises the question, and structure (energy, a Lyapunov function, a polar computation) answers it.

WHERE THIS GOES WRONG

Concluding “centre” because the Jacobian has purely imaginary eigenvalues. This is the one verdict linearisation cannot deliver. Take

$$\dot{x} = -y + ax(x^2 + y^2), \quad \dot{y} = x + ay(x^2 + y^2).$$

For every a the Jacobian at the origin is $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, a perfect centre. But in polar coordinates $\dot{r} = ar^3$ and $\dot{\theta} = 1$, so the origin is an unstable spiral for $a > 0$, a stable spiral for $a < 0$, and a centre only for $a = 0$. Starting at $r = 0.2$, the exact solution $r(t) = (r_0^{-2} - 2at)^{-1/2}$ gives $r(6) = 0.229416$ for $a = \frac{1}{2}$ and $r(6) = 0.179605$ for $a = -\frac{1}{2}$, both reproduced numerically. The linearisation is identical in all three cases. A centre in the linearisation is a licence to investigate, never a conclusion; look for a conserved quantity, a Lyapunov function, or a polar computation.

SEE ALSO The trace-determinant classification; direction fields, isoclines, and nullclines; the phase line for a scalar autonomous equation.

Linearisation is local by construction, and a phase portrait is a global object. The bridge is the nullcline picture. For a planar system, the curves $F_1 = 0$ and $F_2 = 0$ are where the velocity is purely vertical and purely horizontal respectively; they intersect exactly at the equilibria, and they cut the plane into regions in which the pair of signs $(\text{sgn } F_1, \text{sgn } F_2)$ is constant, so the velocity lies in a fixed quadrant of directions. Draw those regions, mark each with its arrow quadrant, and the global flow is visible without solving anything. In the competing-species example the nullclines are the two lines $x + 2y = 3$ and $x + y = 2$ together with the axes, and the four regions they cut from the first quadrant are exactly what forces every orbit to one of the two stable nodes.

That construction also supplies trapping regions. If every arrow on the boundary of a bounded region points inward, no trajectory can leave, and in the plane the Poincaré-Bendixson theorem then guarantees that whatever is inside is an equilibrium or a closed orbit. This is the standard route to proving that a limit cycle exists, and it is purely qualitative: no formula for the cycle is produced or needed.

One dimension, and no dimensions at all

Below the plane the geometry collapses to something even easier. A scalar autonomous equation $\dot{y} = f(y)$ has a one-dimensional state space, trajectories can only move left or right along it, and they cannot pass an equilibrium because uniqueness forbids two solutions through the same point. So the entire qualitative theory is the sign chart of f , and monotonicity is automatic: a solution of an autonomous scalar equation is monotone on its whole interval of existence. Nothing oscillates in one dimension.

EASY

37.9 The phase line for $\dot{y} = f(y)$

CUE $\dot{y} = f(y)$ with no explicit t , and a question about long-time limits, monotonicity, the number of equilibria, or which equilibrium a given initial value runs to.

METHOD Mark the zeros of f on a line. On each interval between consecutive zeros the sign of f is constant and gives the direction of motion: rightward where $f > 0$, leftward where $f < 0$. An equilibrium y^* is asymptotically stable if $f'(y^*) < 0$ and unstable if $f'(y^*) > 0$. If $f'(y^*) = 0$ the test is silent and you must read the sign of f on the two sides; equal signs mean semi-stable.

WORKED EXAMPLE

$\dot{y} = y^3 - 4y = y(y-2)(y+2)$. Equilibria $-2, 0, 2$; $f'(y) = 3y^2 - 4$, so $f'(0) = -4 < 0$ (stable) and $f'(\pm 2) = 8 > 0$ (both unstable). The sign chart confirms it: $f < 0$ on $(-\infty, -2)$ and on $(0, 2)$, $f > 0$ on $(-2, 0)$ and on $(2, \infty)$. Every initial value in $(-2, 2)$ decays to 0; anything outside runs to $\pm\infty$, and in fact in finite time, since $\dot{y} \sim y^3$ there. Numerically, $y(0) = 2.1$ blows up before $t = 0.3$ while $y(0) = 1.9$ has fallen to 3.7×10^{-5} by $t = 3$.
 $\dot{y} = y^2(1-y)$. Equilibria 0 (a double zero) and 1. At $y = 1$, $f'(1) = 2 - 3 = -1 < 0$, stable. At $y = 0$ the derivative test is silent, and the sign of f is positive on both sides, so the flow crosses 0 upward: 0 is semi-stable, attracting from below and repelling above. Numerically, $y(0) = -0.4$ creeps toward 0 from below while $y(0) = 0.3$, $y(0) = 0.9$ and $y(0) = 1.4$ all converge to 1.

Reading $f'(y^*) = 0$ as evidence of stability. It is evidence of nothing. Both $\dot{y} = -y^3$ (stable at 0) and $\dot{y} = y^3$ (unstable at 0) and $\dot{y} = y^2$ (semi-stable at 0) have $f'(0) = 0$; only the sign chart separates them. A second and worse error is concluding that a solution which increases toward an equilibrium reaches it: it never does in finite time when the equilibrium is hyperbolic, since uniqueness forbids two solutions through y^* .

SEE ALSO Direction fields, isoclines, and nullclines; autonomous recognition; the trace-determinant classification.

One thing the phase line makes visible that a formula hides: how the answer changes when a parameter moves. In $\dot{y} = ry - y^3$ the equilibria are $y = 0$ for every r , together with $y = \pm\sqrt{r}$ when $r > 0$. For $r < 0$ the only equilibrium is 0 and $f'(0) = r < 0$ makes it stable. As r passes through zero the stability of 0 flips and two new stable equilibria appear on either side. Nothing in the algebra is difficult; what matters is that the qualitative answer changed at a single value of the parameter, and that the value is exactly where the linear test went silent. Parameters worth studying are the ones that make $f'(y^*)$ vanish, which is another reason to compute that derivative even when the sign chart has already settled the question.

When the equation is not autonomous the state space is the (x, y) plane again, but now x is the independent variable and the picture is a field of slopes rather than a field of velocities. The tools are the same in spirit: find the curves along which the slope is constant, and let uniqueness do the rest.

EASY

37.10 Direction fields, isoclines, and nullclines

CUE “Sketch the solution curves”, “describe the behaviour as $x \rightarrow \infty$ ”, or any qualitative question about $y' = f(x, y)$ where no closed form is available or wanted.

METHOD Draw the isoclines $f(x, y) = k$ for a few values of k ; along each, every slope element has the same slope k . The nullcline $k = 0$ separates regions where solutions increase from regions where they decrease, and is where solution curves have horizontal tangents. Where f blows up, tangents are vertical. Then use uniqueness: wherever the hypotheses of the existence theorem hold, distinct solution curves never meet, so any special solution you can find acts as a barrier that no other solution can cross.

WORKED EXAMPLE

$y' = y - x^2$. The isoclines $y = x^2 + k$ are congruent upward parabolas, and along $y = x^2$ every tangent is horizontal. This equation is linear, so we can name the barrier exactly: $y = Ce^x + x^2 + 2x + 2$, and the choice $C = 0$ gives the parabola $y = x^2 + 2x + 2$. Solutions above it have $C > 0$ and are swept up to $+\infty$; solutions below have $C < 0$ and are swept down. Starting from $y(0) = 2$ (which is $C = 0$) the numerical solution reaches 7.25 at $x = 1.5$, exactly $1.5^2 + 3 + 2$; starting from $y(0) = 2.1$ it reaches 19.0086 at $x = 3$ against the predicted $0.1e^3 + 17$; starting from $y(0) = 1.9$ it reaches 14.9914 against $-0.1e^3 + 17$. The sketch and the formula agree.

$y' = x^2 + y^2$, $y(0) = 1$. The isoclines are the circles $x^2 + y^2 = k$, every slope is nonnegative, and the slope grows at least like y^2 , so comparison with $y' = y^2$ predicts blow-up in finite time. Numerically the solution passes 5.85 at $x = 0.8$, 14.3 at $x = 0.9$, and diverges near $x \approx 0.97$.

THE PLANAR VERSION: NULLCLINES AND ARROW QUADRANTS

Return to the competing species $\dot{x} = x(3 - x - 2y)$, $\dot{y} = y(2 - x - y)$ and sketch the first quadrant without solving anything. The x -nullclines (where $\dot{x} = 0$) are the line $x = 0$ and the line $x + 2y = 3$; the y -nullclines are $y = 0$ and $x + y = 2$. The two slanted lines cross at $(1, 1)$. Inside the triangle nearest the origin both brackets are positive, so both populations grow and the arrows point up and to the right. Cross $x + 2y = 3$ and $\dot{x}$ turns negative while $\dot{y}$ is still positive: arrows point up and to the left, driving the state toward $(0, 2)$. Cross $x + y = 2$ instead and the arrows point down and to the right, driving it toward $(3, 0)$. Beyond both lines everything decreases. Four regions, four arrow quadrants, and the bistable picture is complete before a single trajectory is drawn. On a nullcline the arrows are exactly vertical or exactly horizontal, which is what makes the boundaries easy to draw correctly.

WHERE THIS GOES WRONG

Sketching solution curves that cross. Where f and $\frac{\partial f}{\partial y}$ are continuous, two distinct solutions cannot meet, so a crossing in your picture is a proof that the picture is wrong. The related failure is drawing a solution that passes through an equilibrium of an autonomous equation, or through the nullcline transversally when the nullcline is itself a solution. Check whether the nullcline satisfies the equation before letting curves cross it: for $y' = y - x^2$ the nullcline $y = x^2$ is *not* a solution and may be crossed, while for $y' = (y - x)^2$ the line $y = x$ is not a solution either but $y = x - 1$ is.

SEE ALSO The phase line for a scalar autonomous equation; Picard-Lindelof: when a unique solution is guaranteed; Euler's method and step-by-step numerical integration.

The trapping-region idea deserves its standard example, because it is the one case where qualitative technique produces a result that no formula can. Van der Pol's equation $\ddot{x} - \mu(1 - x^2)\dot{x} + x = 0$ with $\mu > 0$ has, at the origin, the Jacobian $\begin{pmatrix} 0 & 1 \\ -1 & \mu \end{pmatrix}$ with $T = \mu > 0$ and $\Delta = 1$: an unstable spiral for $0 < \mu < 2$ and an unstable node beyond. So trajectories are pushed away from the origin. But for $|x| > 1$ the coefficient $-\mu(1 - x^2)$ is a positive damping, so large orbits lose energy and are pushed back inward. Nothing can escape and nothing can rest, and Poincare-Bendixson then forces the existence of a closed orbit. Numerically, for $\mu = 1$, trajectories started at $x = 0.05$ and at $x = 3$ both settle onto the same cycle with $\max |x| = 2.00862$; the cycle has no closed form, and the argument that it exists never needed one.

When nothing closes

Qualitative technique tells you where a solution goes; it does not tell you what it is at $x = 1$. When a number is wanted and no closed form exists, the remaining option is to march. The point of including a numerical method in a book about structure is that the arithmetic is trivial and the accounting is not: almost every error in a hand-computed Euler table is a bookkeeping error about which argument goes where.

MEDIUM

37.11 Euler's method and step-by-step numerical integration

CUE No closed form, or an explicit instruction to use a step size h , or a request for a value of the solution rather than a formula.

METHOD Euler's method is the tangent line applied repeatedly:

$$y_{k+1} = y_k + hf(x_k, y_k), \quad x_{k+1} = x_k + h.$$

Both arguments of f are the *current* point. Local truncation error is $O(h^2)$ and global error is $O(h)$, so halving h halves the error, which is poor. The classical fourth-order Runge-Kutta method, with $k_1 = f(x_k, y_k)$, $k_2 = f(x_k + \frac{h}{2}, y_k + \frac{h}{2}k_1)$, $k_3 = f(x_k + \frac{h}{2}, y_k + \frac{h}{2}k_2)$, $k_4 = f(x_k + h, y_k + hk_3)$ and $y_{k+1} = y_k + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$, has global error $O(h^4)$ at four function evaluations per step, and is the default worth remembering.

WORKED EXAMPLE

$y' = \frac{1}{3 + 3x + x^2}$, $y(0) = 1$, $h = \frac{1}{10}$, in exact rationals. First step: $f(0, 1) = \frac{1}{3}$, so

$$y_1 = 1 + \frac{1}{10} \cdot \frac{1}{3} = \frac{31}{30}.$$

Second step: $f(\frac{1}{10}, y_1) = \frac{1}{3 + \frac{3}{10} + \frac{1}{100}} = \frac{100}{331}$, so

$$y_2 = \frac{31}{30} + \frac{1}{10} \cdot \frac{100}{331} = \frac{31}{30} + \frac{10}{331} = \frac{10561}{9930},$$

already in lowest terms, since $9930 = 2 \cdot 3 \cdot 5 \cdot 331$ and 10561 is divisible by none of those.

How good is it? Here the right side depends on x only, so the exact solution is available: completing the square, $x^2 + 3x + 3 = (x + \frac{3}{2})^2 + \frac{3}{4}$, and

$$y(1) = 1 + \frac{2}{\sqrt{3}} \left(\arctan \frac{5}{\sqrt{3}} - \frac{\pi}{3} \right) = 1.219538.$$

Ten Euler steps of size 0.1 give 1.229255. Halving the step repeatedly makes the order visible:

h	Euler $y(1)$	Error	Ratio
0.1	1.2292546	9.72×10^{-3}	
0.05	1.2243482	4.81×10^{-3}	2.02
0.025	1.2219311	2.39×10^{-3}	2.01
0.0125	1.2207316	1.19×10^{-3}	2.01

The error halves when the step halves, which is global order one made visible. Runge-Kutta with $h = 0.1$ gives 1.2195381549, an error of 1.8×10^{-8} , and with $h = 0.05$ it gives 1.2195381377, an error of 1.1×10^{-9} : a factor of 16 for a halving, which is order four. Ten steps of RK4 beat ten thousand steps of Euler on this problem.

WHERE THIS GOES WRONG

Using $f(x_{k+1}, y_k)$, a hybrid that is neither Euler nor anything else, or switching to decimals partway through a problem that asks for an exact fraction. In the example above, evaluating f at x_1 but with y_0 would be invisible here (since f does not depend on y) and catastrophic in general. The structural trap is different: Euler applied to a stiff equation such as $\dot{y} = -100y$ is unstable unless $h < 0.02$, and produces oscillating garbage rather than an obviously wrong number. If successive values alternate in sign and grow, the step size is the problem, not the arithmetic.

SEE ALSO Direction fields, isoclines, and nullclines; reducing a scalar equation to a first-order system; interval of validity and finite-time blow-up.

A numerical solution answers a quantitative question and can suggest a qualitative one, but it certifies nothing on its own. It cannot distinguish a slow spiral from a centre, because both look like a closed curve over any finite time; it cannot distinguish a trajectory that converges to an equilibrium from one that passes very close and leaves later; and near a saddle it is unreliable in the worst way, since the unstable direction amplifies rounding error at exactly the rate that the true solution amplifies real perturbations. Use it to check a formula you have derived, to explore before you conjecture, and to produce numbers. Do not use it to decide stability. That is what the trace, the determinant and the conserved quantities are for.

THE ORDER OF OPERATIONS FOR A SYSTEM

Given $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$, do these in order and stop when the question is answered. One: is $\mathbf{F}$ linear? If not, solve $\mathbf{F} = \mathbf{0}$ for the equilibria and linearise at each. Two: for each matrix, compute T and Δ and read the chart, which costs six multiplications. Three: only if a formula is wanted, find eigenvectors and assemble modes. Four: if an eigenvalue of a Jacobian is purely imaginary, discard the linear verdict and look for a conserved quantity. Most exam questions are answered at step two, and most wrong answers come from starting at step three.

Two habits close the chapter. First, name the equilibrium before you solve for anything, because the name is cheap and it tells you what the formula must look like: a saddle must produce one growing and one decaying exponential, a stable spiral must produce $e^{\alpha t}$ times a sinusoid with $\alpha < 0$, and a formula that disagrees with the chart contains an arithmetic error. Second, when a problem hands you a nonlinear system and asks for a sketch, resist the urge to find trajectories. Find the equilibria, linearise, draw the nullclines, mark the arrow quadrants, and only then draw curves. The picture assembled that way is correct by construction; a picture assembled from guessed trajectories is usually wrong in a way that is invisible until someone asks what happens near the third equilibrium.

Recognition matrix

If you see	Reach for	If that fails
$\dot{\mathbf{x}} = A\mathbf{x}$, A constant	Eigenvalues from $\lambda^2 - T\lambda + \Delta = 0$	Matrix exponential e^{At}
Two coupled linear equations	Write as a matrix; same method	Eliminate one unknown, get order two
$T^2 - 4\Delta < 0$	Complex pair; take real and imaginary parts of one mode	Convert to polar; spiral of rate β
$T^2 - 4\Delta = 0$, $A \neq \lambda I$	Defective: solve $(A - \lambda I)\mathbf{w} = \mathbf{v}$	Jordan form, or $e^{\lambda t}(I + Nt)$
$\Delta < 0$	Saddle, unstable; stop	Find the two invariant lines
$\Delta > 0$, $T < 0$, $T^2 > 4\Delta$	Stable node; both modes decay	Compare the two decay rates
$T = 0 < \Delta$ in a linear system	Centre; closed orbits	Conserved quantity confirms it
$T = 0 < \Delta$ from a Jacobian	Inconclusive; do not conclude centre	Polar form, Lyapunov function
$\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$, $\mathbf{F}$ nonlinear	Solve $\mathbf{F} = \mathbf{0}$, then Jacobian at each root	Nullclines and trapping regions
Equilibria hard to solve	Factor each component separately	Intersect the nullcline curves
$\dot{\mathbf{x}} = A\mathbf{x} + \mathbf{b}$, $\mathbf{b}$ constant	Shift to $\mathbf{x}^* = -A^{-1}\mathbf{b}$; classify A	$\Delta = 0$: a whole line of equilibria
$\dot{\mathbf{x}} = A\mathbf{x} + \mathbf{g}(t)$	Duhamel: $e^{At}\mathbf{x}_0 + \int_0^t e^{A(t-s)}\mathbf{g}(s)ds$	$\Phi(t) \int \Phi^{-1}(s)\mathbf{g}(s)ds$
$\mathbf{g}$ piecewise or impulsive	Laplace transform the system	Duhamel with a step in the integrand
Scalar equation, order n	$x_i = y^{(i-1)}$; companion matrix	Characteristic polynomial is unchanged
$\dot{y} = f(y)$, no explicit t	Phase line: zeros of f , then signs	Separate variables for a formula
$f'(y^*) < 0$	Asymptotically stable	$f'(y^*) = 0$: read signs on both sides

If you see	Reach for	If that fails
$f'(y^*) = 0$	Semi-stable if f has equal signs	Higher-order term decides
$y' = f(x, y)$, sketch wanted	Isoclines $f = k$; nullcline $f = 0$	Special solutions act as barriers
Solution curves appear to cross	The sketch is wrong; uniqueness forbids it	Check $\frac{\partial f}{\partial y}$ is continuous there
$y' \geq cy^2$ for large y	Finite-time blow-up; find the interval	Comparison with a separable equation
"Use step size h "	Euler: $y_{k+1} = y_k + hf(x_k, y_k)$	RK4 if accuracy matters
Numerical values alternating and growing	Stiffness; shrink h or go implicit	Eigenvalue of large negative real part

Laplace transforms and series solutions

What this chapter is for. The methods of the preceding chapters solve a linear equation when the coefficients are constant and the forcing is a polynomial, an exponential, or a sinusoid. Two things break them. Forcing that switches on at a fixed time, or arrives as an instantaneous blow, is not in the undetermined-coefficients table and destroys the smoothness that variation of parameters quietly assumes. Coefficients that genuinely vary, with no Cauchy-Euler or reducible structure to exploit, leave no closed form to find at all. This chapter supplies one escape hatch for each. The Laplace transform converts a constant-coefficient initial-value problem with data at $t = 0$ into an algebra problem, absorbs steps and impulses without a special case, and turns the initial conditions into terms rather than constants to be fitted afterwards. Power series and the Frobenius method construct the solution coefficient by coefficient at a point, whether or not the answer has a name. The organising decision is simple and worth fixing before any of the machinery: constant coefficients with awkward forcing means Laplace, variable coefficients with no closed form means series, and if the coefficients are constant and the forcing is smooth you should not be here at all.

Two escape hatches and the rule for choosing

Both methods trade the differential equation for something algebraic, and they differ in what they algebraise. Laplace replaces the function $y(t)$ by a single new function $Y(s)$, and the whole equation collapses to one linear equation in Y ; the price is that the transform of $y^{(n)}$ needs the values $y(0), \dots, y^{(n-1)}(0)$, so the method demands data at the origin and constant coefficients (otherwise the transform of $a(t)y'$ is not $a \cdot \mathcal{L}\{y'\}$ and nothing collapses). Series replaces y by its coefficient sequence (a_n) , and the equation becomes a recurrence; the price is that you get an infinite list of numbers rather than a formula, and a radius of convergence to worry about.

The choice is therefore made by inspecting the equation and not by taste:

WHICH HATCH

Use **Laplace** when the coefficients are constant, the data are at $t = 0$, and the forcing is discontinuous, impulsive, piecewise, delayed, or left as an unspecified function $f(t)$. Use **series** when the coefficients are non-constant and not of a reducible type, or when a solution near a specific point is wanted and no closed form exists. Use **neither** when the coefficients are constant and the forcing is a polynomial times an exponential times a sinusoid: characteristic roots and undetermined coefficients are faster and give a formula directly. The two methods overlap in exactly one place, the linear t coefficient, where differentiation in s turns the transformed equation into a first-order ODE.

The transform, and the table that gets used

There is a third option that the rule above deliberately omits, and it should be named so that it is chosen rather than fallen into. If the coefficients vary, the point of interest is irregular, and no closed form is wanted, then neither hatch opens and the answer is numerical: convert to a first-order system and integrate. That is not a failure. Series methods are local and Laplace is global but structurally restricted, and there are perfectly ordinary equations that fall outside both. What matters is recognising the case in the first minute rather than the fortieth.

The transform itself is the integral

$$\mathcal{L}\{f\}(s) = F(s) = \int_0^\infty e^{-st} f(t) dt,$$

defined for s large enough that the integral converges, which for functions of exponential order is a half-line $s > a$. Nobody computes it from the definition after the first week. What gets used is a short table together with three structural rules (the two shifts and differentiation in s), and it is worth having the table genuinely memorised, because inversion is pattern-matching and you cannot match a pattern you do not know.

$f(t)$	$F(s)$	Remark
1	$\frac{1}{s}$	$s > 0$
t^n	$\frac{n!}{s^{n+1}}$	$t^p \mapsto \Gamma(p+1)/s^{p+1}$ for real $p > -1$
e^{at}	$\frac{1}{s-a}$	the first shift applied to 1
$\sin bt$	$\frac{b}{s^2 + b^2}$	numerator b , not s
$\cos bt$	$\frac{s}{s^2 + b^2}$	
$\sinh bt, \cosh bt$	$\frac{s}{s^2 - b^2}, \frac{s}{s^2 + b^2}$	sign of b^2 is the only difference
$e^{at}f(t)$	$F(s-a)$	first shift
$u_c(t)f(t-c)$	$e^{-cs}F(s)$	second shift, $c > 0$
$\delta(t-c)$	e^{-cs}	the limit of a tall thin pulse
$tf(t)$	$-F'(s)$	differentiation in s
$f'(t)$	$sF(s) - f(0)$	
$f''(t)$	$s^2F(s) - sf(0) - f'(0)$	
$(f * g)(t)$	$F(s)G(s)$	convolution, not pointwise product

Two facts about the table are worth stating rather than absorbing silently. Every entry decays as $s \rightarrow \infty$, so a proper rational function is a plausible transform and an improper one is not: if your algebra yields $Y(s) = s + \frac{1}{s}$ you have made a mistake, because no function of exponential order transforms to something unbounded. And the transform is injective on continuous functions, so “invert by recognition” is legitimate: if you find any f whose transform is Y , it is the answer.

One point of hygiene before the machine. The integral defining $F(s)$ converges when f is piecewise continuous on $[0, \infty)$ and of exponential order, meaning $|f(t)| \leq Me^{at}$ for some constants and all large t ; then $F(s)$ exists for $s > a$. Every function that arises from a constant-coefficient linear equation with forcing of exponential order satisfies this, which is why the method never has to worry about convergence in practice. What does fail the test is worth knowing: e^{t^2} grows too fast and has no transform, and t^{-1} has a non-integrable singularity at the origin, though $t^{-1/2}$ is fine and transforms to $\sqrt{\pi}/s$.

INITIAL CONDITIONS BECOME TERMS, NOT CONSTANTS

This is the structural advantage of the transform and the reason to reach for it even when the forcing is tame. In the characteristic-root method you build a general solution containing n arbitrary constants and then solve an $n \times n$ linear system to fit the data. In the transform method the data appear in the numerator the moment you transform, so the answer that comes out of the inversion already satisfies them. There is no fitting step, hence no opportunity to fit at the wrong stage, and for piecewise problems there is no matching of constants across the switching times either: continuity of y and y' is automatic.

38.1 The Laplace-transform machine for an initial-value problem

CORE

CUE A constant-coefficient linear initial-value problem with data specified at $t = 0$, especially one whose forcing is discontinuous, impulsive, delayed, or piecewise.

METHOD Transform every term. Using $\mathcal{L}\{y'\} = sY - y(0)$ and $\mathcal{L}\{y''\} = s^2Y - sy(0) - y'(0)$, the equation $ay'' + by' + cy = f(t)$ becomes

$$(as^2 + bs + c)Y(s) = F(s) + a(sy(0) + y'(0)) + by(0),$$

a linear equation in Y . Solve it, then invert. The polynomial $as^2 + bs + c$ is the characteristic polynomial with r renamed s , so the transform method and the characteristic-root method share a denominator; what differs is that the initial data enter as numerator terms, algebraically, with no constants to fit at the end.

WORKED EXAMPLE

$$y'' + 2y' + 5y = 0, \quad y(0) = 1, \quad y'(0) = -1.$$

Transforming: $s^2Y - s + 1 + 2(sY - 1) + 5Y = 0$, so $(s^2 + 2s + 5)Y = s + 1$ and

$$Y = \frac{s+1}{s^2+2s+5} = \frac{s+1}{(s+1)^2+4} \implies y = e^{-t} \cos 2t,$$

the completed square being the whole of the work. Note that no constants were fitted: the numerator $s + 1$ is the initial data.

Forced, with a polynomial right side:

$$y'' + y = t, \quad y(0) = 0, \quad y'(0) = 2.$$

Then $s^2 Y - 2 + Y = \frac{1}{s^2}$, so $Y = \frac{1}{s^2(s^2 + 1)} + \frac{2}{s^2 + 1}$. Since $\frac{1}{s^2(s^2 + 1)} = \frac{1}{s^2} - \frac{1}{s^2 + 1}$, we get $y = t - \sin t + 2 \sin t = t + \sin t$.

Check: $y'' + y = -\sin t + t + \sin t = t$, and $y'(0) = 1 + \cos 0 = 2$. Runge-Kutta at $t = 2.3$ returns 3.0457052 against $2.3 + \sin 2.3 = 3.0457052$.

A SYSTEM, WITH NO EIGENVECTORS COMPUTED

The transform does not care how many unknowns there are. Take

$$\dot{x} = x + y, \quad \dot{y} = 4x + y, \quad x(0) = 3, \quad y(0) = 2.$$

Transforming both equations gives the linear system

$$(s - 1)X - Y = 3, \quad -4X + (s - 1)Y = 2,$$

whose determinant is $(s - 1)^2 - 4 = (s - 3)(s + 1)$, the characteristic polynomial of the matrix. Cramer's rule gives

$$X = \frac{3(s - 1) + 2}{(s - 3)(s + 1)} = \frac{3s - 1}{(s - 3)(s + 1)} = \frac{2}{s - 3} + \frac{1}{s + 1},$$

$$Y = \frac{2(s - 1) + 12}{(s - 3)(s + 1)} = \frac{2s + 10}{(s - 3)(s + 1)} = \frac{4}{s - 3} - \frac{2}{s + 1},$$

so $x = 2e^{3t} + e^{-t}$ and $y = 4e^{3t} - 2e^{-t}$. This is the same solution the eigenvector method produces for this system, obtained without finding a single eigenvector: the eigenvalues appear as poles and the eigenvector combination is done for you by the partial fractions.

WHERE THIS GOES WRONG

Using Laplace when the data are given at $t = t_0 \neq 0$. The formula $\mathcal{L}\{y'\} = sY - y(0)$ refers to the value at the origin and nothing else, so data at $t = 2$ cannot be inserted. Substitute $\tau = t - t_0$ first, transform in τ , and translate back at the end. The other structural error is transforming a variable coefficient as though it were constant: $\mathcal{L}\{ty'\}$ is not $t(sY - y(0))$, and indeed is not a multiple of Y at all.

SEE ALSO Partial fractions for the inverse transform; the first shifting theorem; the second shifting theorem and Heaviside steps.

Transforming is mechanical; inverting is where the work is. Every $Y(s)$ produced by the machine above from rational forcing is a proper rational function, possibly multiplied by exponentials e^{-cs} that the second shift will strip off. So the core inversion skill is partial fractions, executed with the transform table in mind rather than with the calculus-of-integration habits that produce logarithms.

CORE

38.2 Partial fractions for the inverse transform

CUE $Y(s)$ is a proper rational function, or becomes one after an exponential factor is set aside.

METHOD Factor the denominator over the reals. A linear factor $(s - a)^m$ contributes $\frac{A_1}{s - a} + \dots + \frac{A_m}{(s - a)^m}$, inverting to $(A_1 + A_2 t + \dots + A_m \frac{t^{m-1}}{(m-1)!})e^{at}$. An irreducible quadratic $(s - a)^2 + b^2$ contributes $\frac{B(s - a) + Cb}{(s - a)^2 + b^2}$, inverting to $e^{at}(B \cos bt + C \sin bt)$. Find the constants by covering up at the roots and by matching one or two coefficients; do not expand everything.

WORKED EXAMPLE

$$y'' + 4y = e^t, \quad y(0) = y'(0) = 0.$$

The transform is $Y = \frac{1}{(s - 1)(s^2 + 4)}$. Write $Y = \frac{A}{s - 1} + \frac{Bs + C}{s^2 + 4}$, so $1 = A(s^2 + 4) + (Bs + C)(s - 1)$. Setting $s = 1$ gives $A = \frac{1}{5}$. Matching s^2 : $0 = A + B$, so $B = -\frac{1}{5}$. Matching constants: $1 = 4A - C$, so $C = -\frac{1}{5}$. Hence

$$Y = \frac{1}{5} \left(\frac{1}{s - 1} - \frac{s + 1}{s^2 + 4} \right), \quad y = \frac{1}{5} \left(e^t - \cos 2t - \frac{1}{2} \sin 2t \right).$$

Audit by substitution: $y'' + 4y = \frac{1}{5}(e^t + 4\cos 2t + 2\sin 2t) + \frac{4}{5}(e^t - \cos 2t - \frac{1}{2}\sin 2t) = e^t$, and $y(0) = y'(0) = 0$. At $t = 1.7$ the value is 1.3137032, matched by Runge-Kutta.

A repeated factor: $\frac{3s+2}{s^2(s+2)} = \frac{1}{s} + \frac{1}{s^2} - \frac{1}{s+2}$ inverts to $1 + t - e^{-2t}$.

A REPEATED IRREDUCIBLE QUADRATIC, WHERE PARTIAL FRACTIONS IS THE WRONG TOOL

Resonance produces $(s^2 + \omega^2)^2$ in the denominator, and expanding that into $\frac{As+B}{s^2 + \omega^2} + \frac{Cs+D}{(s^2 + \omega^2)^2}$ is legal but leaves you needing the inverse of the second term anyway. Go the other way, through $\mathcal{L}\{tf\} = -F'(s)$. Solve $y'' + y = \cos t$ with $y(0) = y'(0) = 0$:

$$Y = \frac{s}{(s^2 + 1)^2}.$$

Differentiate the known transform $F(s) = \frac{1}{s^2 + 1}$ of $\sin t$: $F'(s) = \frac{-2s}{(s^2 + 1)^2}$, so $\mathcal{L}\{t \sin t\} = -F'(s) = \frac{2s}{(s^2 + 1)^2}$ and therefore

$$y = \frac{t \sin t}{2}.$$

The amplitude grows linearly, which is resonance, and the growth appears as a repeated pole on the imaginary axis rather than as a special rule. The same three companions are worth deriving once and keeping: $\mathcal{L}^{-1}\{\frac{1}{(s^2 + \omega^2)^2}\} = \frac{1}{2\omega^3}(\sin \omega t - \omega t \cos \omega t)$, $\mathcal{L}^{-1}\{\frac{s}{(s^2 + \omega^2)^2}\} = \frac{t \sin \omega t}{2\omega}$, and $\mathcal{L}^{-1}\{\frac{s^2}{(s^2 + \omega^2)^2}\} = \frac{1}{2\omega}(\sin \omega t + \omega t \cos \omega t)$.

WHERE THIS GOES WRONG

Two omissions, both silent. Writing only $\frac{A}{(s-a)^m}$ for a repeated factor and dropping the lower-order terms; and putting a constant over an irreducible quadratic when it needs $Bs+C$. Both produce an answer with the right exponentials and the wrong amplitudes. A third, specific to transforms: reducing $\frac{5}{s^2+9}$ to $5 \cdot \frac{1}{s^2+9}$ and calling it $5 \sin 3t$. The table entry has numerator b , so it is $\frac{5}{3} \sin 3t$.

SEE ALSO The first shifting theorem; the Laplace-transform machine; logistic growth and the odds substitution.

COVER UP, DO NOT EXPAND

For a simple pole at $s = a$ the coefficient is $A = \lim_{s \rightarrow a} (s-a)Y(s)$, computed by deleting the factor $(s-a)$ from the denominator and evaluating what remains at $s = a$. For a pole of order m the top coefficient is found the same way and the lower ones by differentiating $(s-a)^m Y(s)$ before evaluating, $A_{m-k} = \frac{1}{k!} \frac{d^k}{ds^k} [(s-a)^m Y]_{s=a}$. For an irreducible quadratic, do not clear denominators over the whole expression; substitute the complex root and match real and imaginary parts, which gives both constants at once. Expanding everything into a polynomial identity and solving a four-by-four system works, but it is where the arithmetic errors live.

A completed square in the denominator is the signature of a damped oscillation, and it is handled by the first of the two shifting theorems. The theorem is trivial to prove (absorb e^{at} into the exponential of the defining integral) and easy to misapply, because shifting the denominator without shifting the numerator is a mistake the algebra does not object to.

38.3 The first shifting theorem, $F(s) \mapsto F(s-a)$

CORE

CUE A factor e^{at} multiplying something whose transform you know, or a denominator that completes the square to $(s-a)^2 + b^2$ or $(s-a)^m$.

METHOD The statement is

$$\mathcal{L}\{e^{at}f(t)\} = F(s-a).$$

Forward, absorb the exponential and shift the argument. Backward, complete the square in the denominator, rewrite the numerator in terms of $s-a$ as well, invert the resulting expression as a function of $u = s-a$, and multiply the result by e^{at} .

WORKED EXAMPLE

$$\mathcal{L}^{-1}\left\{\frac{s+3}{s^2+6s+13}\right\}.$$

The denominator is $(s+3)^2 + 4$ and the numerator is already $s+3$, so with $u = s+3$ the expression is $\frac{u}{u^2 + 4}$, whose inverse is $\cos 2t$. Therefore the answer is $e^{-3t} \cos 2t$.

Now the same denominator with a bare numerator, $\mathcal{L}^{-1} \left\{ \frac{s}{(s+3)^2 + 4} \right\}$. Write $s = (s+3) - 3$:

$$\frac{(s+3)}{(s+3)^2 + 4} - \frac{3}{(s+3)^2 + 4} = \frac{u}{u^2 + 4} - \frac{3}{2} \cdot \frac{2}{u^2 + 4},$$

giving $e^{-3t} (\cos 2t - \frac{3}{2} \sin 2t)$.

THE SHIFT ON A REPEATED FACTOR

Resonant forcing produces repeated factors, and the first shift handles them with no extra apparatus. Solve $y'' - 4y' + 4y = e^{2t}$ with $y(0) = y'(0) = 0$. Transforming, $(s^2 - 4s + 4)Y = \frac{1}{s-2}$, so

$$Y = \frac{1}{(s-2)^3}.$$

Since $\mathcal{L}\{t^2\} = 2/s^3$, the first shift gives $\mathcal{L}\{t^2 e^{2t}\} = 2/(s-2)^3$, hence $y = \frac{1}{2} t^2 e^{2t}$. Check: $y' = (t + t^2)e^{2t}$ and $y'' = (1 + 4t + 2t^2)e^{2t}$, so $y'' - 4y' + 4y = (1 + 4t + 2t^2 - 4t - 4t^2 + 2t^2)e^{2t} = e^{2t}$. The double resonance that would need the multiplier t^2 in undetermined coefficients appears here as the exponent 3 on a linear factor, and requires no rule of its own.

WHERE THIS GOES WRONG

Shifting the denominator and leaving the numerator alone, which in the second example above would give $e^{-3t} \cos 2t$ and lose the entire sine term. Whenever you complete a square, immediately rewrite every s in the numerator as $(s-a) + a$. A second trap: applying the shift to a numerator that already contains a shifted quantity, and shifting twice.

SEE ALSO Partial fractions for the inverse transform; the second shifting theorem and Heaviside steps; complex roots and the real form.

Forcing that is not smooth

This is the reason the transform exists. Undetermined coefficients has no entry for “zero until $t = 3$, then one”, and variation of parameters can be pushed through piecewise but demands that solutions be matched at each switching time, with two continuity conditions per switch. The transform handles all of it by a single rule, because a delay in t is a factor e^{-cs} in s , and factors are cheap.

38.4 The second shifting theorem and the step $u_c(t)$

HARD

CUE Forcing that switches on or off at a stated time (“the voltage is applied at $t = 2$ ”), a right side defined by cases, or an equation containing $u(t-c)$ explicitly.

METHOD Write the unit step $u_c(t) = u(t-c)$, which is 0 for $t < c$ and 1 for $t > c$. A function equal to g_1 before c and g_2 after is $g_1 + (g_2 - g_1)u_c$. The theorem is

$$\mathcal{L}\{u_c(t)f(t-c)\} = e^{-cs}F(s), \quad \mathcal{L}^{-1}\{e^{-cs}F(s)\} = u_c(t)f(t-c).$$

Everything hinges on the argument $t-c$. To transform $u_c(t)g(t)$, first rewrite $g(t) = g((t-c) + c)$ and expand until every occurrence of t appears as $t-c$.

WORKED EXAMPLE

$$y'' + 4y = f(t), \quad f(t) = \begin{cases} 1, & 0 \leq t < 3 \\ 0, & t \geq 3 \end{cases}, \quad y(0) = y'(0) = 0.$$

Here $f = 1 - u_3(t)$, so $F = \frac{1}{s} - \frac{e^{-3s}}{s}$ and

$$Y = \frac{1 - e^{-3s}}{s(s^2 + 4)}.$$

Since $\frac{1}{s(s^2 + 4)} = \frac{1}{4} \left(\frac{1}{s} - \frac{s}{s^2 + 4} \right)$ inverts to $\frac{1}{4}(1 - \cos 2t)$, the second shift gives

$$y = \frac{1}{4}(1 - \cos 2t) - \frac{1}{4}(1 - \cos(2t - 6))u_3(t).$$

For $t \geq 3$ the constants cancel and $y = \frac{1}{4}(\cos(2t - 6) - \cos 2t) = \frac{1}{2} \sin 3 \sin(2t - 3)$: after the forcing stops the system rings at its natural frequency with fixed amplitude $\frac{1}{2} |\sin 3| \approx 0.070560$. At $t = 2$ the formula gives 0.4134109 and at $t = 6$ it gives 0.0290791, both reproduced by numerical integration of the piecewise problem.

A RAMP THAT SATURATES

$$y' + y = f(t), \quad f(t) = \begin{cases} t, & 0 \leq t < 2 \\ 2, & t \geq 2 \end{cases}, \quad y(0) = 0.$$

The decomposition that keeps the shifts honest is $f(t) = t - (t - 2)u_2(t)$, in which the second piece is already written as a function of $t - 2$, so

$$F(s) = \frac{1}{s^2} - \frac{e^{-2s}}{s^2}, \quad Y = \frac{1}{s^2(s + 1)} - e^{-2s} \frac{1}{s^2(s + 1)}.$$

Partial fractions give $\frac{1}{s^2(s + 1)} = -\frac{1}{s} + \frac{1}{s^2} + \frac{1}{s + 1}$, inverting to $g(t) = t - 1 + e^{-t}$, so

$$y(t) = g(t) - g(t - 2)u_2(t).$$

For $t \geq 2$ the linear parts cancel and $y = 2 + e^{-t}(1 - e^2)$, a constant with an exponentially decaying correction. At $t = 1.5$ this gives 0.7231302, at $t = 4$ it gives 1.8829804, and at $t = 8$ it gives 1.9978567, all reproduced by numerical integration of the piecewise problem. The output tracks the ramp with a lag of one time constant and then relaxes to the held value 2.

WHERE THIS GOES WRONG

Transforming $u_c(t)g(t)$ as $e^{-cs}G(s)$ without re-expressing g in terms of $t - c$. For $g(t) = t$ and $c = 2$, the correct route is $t = (t - 2) + 2$, giving

$$\mathcal{L}\{u_2(t)t\} = e^{-2s} \left(\frac{1}{s^2} + \frac{2}{s} \right),$$

not e^{-2s}/s^2 . The two differ by a term of the same order as the answer. Verified numerically at $s = 1.3$: the defining integral of the ramp $f(t) = t$ on $[0, 2]$, zero after, is 0.43350007, matching $\frac{1}{s^2} - e^{-2s}(\frac{1}{s^2} + \frac{2}{s})$ and not the shorter expression.

SEE ALSO The Dirac delta and the impulse response; the first shifting theorem; RC and RL circuits.

A forcing that switches repeatedly, such as a square wave or a sawtooth, would need infinitely many steps written out. It does not, because the geometric series sums them. If f is T -periodic on $[0, \infty)$ then

$$\mathcal{L}\{f\}(s) = \frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t) dt,$$

the factor $(1 - e^{-sT})^{-1} = 1 + e^{-sT} + e^{-2sT} + \dots$ being exactly the sum of the delayed copies. Only one period has to be integrated. Inverting such a transform means expanding the geometric series again and reading the result as an infinite sum of shifted responses, which is how the ringing of a repeatedly kicked oscillator is computed.

Push the switching interval to zero while holding the total impulse fixed and the step becomes a delta. This is not a new theory but the limiting case of the last one, and it is the cheapest forcing of all to transform: the delta has transform e^{-cs} with no $F(s)$ attached, because it is the identity for convolution.

HARD

38.5 The Dirac delta and the impulse response

CUE “A hammer blow”, “an instantaneous impulse of magnitude I at time c ”, a sudden kick with no duration given, or $\delta(t - c)$ written outright.

METHOD Use $\mathcal{L}\{\delta(t - c)\} = e^{-cs}$ for $c \geq 0$, and more generally $\int_0^\infty \delta(t - c)g(t) dt = g(c)$. For $ay'' + by' + cy = I\delta(t - t_0)$ the solution is continuous at t_0 while y' jumps by I/a there. The solution of $ay'' + by' + cy = \delta(t)$ with zero data is the *impulse response* $h(t)$, and $H(s) = 1/(as^2 + bs + c)$ is the transfer function: every other forcing is handled by convolving with h .

$$y'' + 4y = \delta(t - \pi), \quad y(0) = y'(0) = 0.$$

Transforming, $(s^2 + 4)Y = e^{-\pi s}$, so $Y = \frac{e^{-\pi s}}{s^2 + 4}$ and, since $\mathcal{L}^{-1}\{(s^2 + 4)^{-1}\} = \frac{1}{2} \sin 2t$,

$$y = \frac{1}{2} \sin(2(t - \pi)) u_\pi(t) = \frac{1}{2} \sin(2t) u_\pi(t),$$

the last simplification because $\sin$ has period 2π in its argument $2t$. The system is at rest until $t = \pi$, then oscillates forever with amplitude $\frac{1}{2}$. Check the jump: $y(\pi^+) = 0$ and $y'(\pi^+) = \cos 2\pi = 1$, which is exactly the impulse magnitude divided by the leading coefficient. Integrating $y'' + 4y = 0$ forward from $(0, 1)$ at $t = \pi$ reproduces $y(5) = -0.2720106 = \frac{1}{2} \sin 10$.

WHERE THIS GOES WRONG

Expecting y itself to jump. For a second-order equation the impulse is absorbed by the highest derivative: y'' contains a delta, y' has a step, and y is continuous with a corner. If your solution has a discontinuity in y , an algebra error has occurred. Conversely, for a first-order equation $y' + ay = I\delta(t - c)$ it is y that jumps, by I ; the rule is that the jump lands on the derivative one order below the leading one.

SEE ALSO Convolution and the transfer function; the second shifting theorem and Heaviside steps; nonhomogeneous systems and the Duhamel formula.

THE TRANSFER FUNCTION SAYS EVERYTHING AT ONCE

For $ay'' + by' + cy = f$ with zero data, $H(s) = 1/(as^2 + bs + c)$ contains the whole system. Its poles are the characteristic roots, so the free behaviour is read off the denominator: poles in the left half plane mean a stable system, a pole pair on the imaginary axis means an undamped oscillator. Its value at $s = i\omega$ is the complex gain applied to a sinusoidal input of frequency ω , with $|H(i\omega)|$ the amplitude ratio and $\arg H(i\omega)$ the phase lag; the resonance peak sits near the frequency that minimises $|as^2 + bs + c|$ on that line. And its inverse transform is the impulse response. One rational function, three readings.

Once the impulse response is in hand, arbitrary forcing costs nothing extra. The transform of a product is not the product of transforms, but there is an operation on functions whose transform *is* the product, and identifying it is the single most useful theorem in the subject.

HARD

38.6 Convolution and the transfer function

CUE $Y(s)$ is a product in which one factor has no elementary inverse, or the forcing is deliberately left as an unspecified $f(t)$ and a formula valid for all f is wanted.

METHOD Define

$$(f * g)(t) = \int_0^t f(\tau)g(t - \tau) d\tau,$$

which is commutative and satisfies $\mathcal{L}\{f * g\} = F(s)G(s)$. Hence $\mathcal{L}^{-1}\{FG\} = f * g$. Applied to $ay'' + by' + cy = f$ with zero data, $Y = H(s)F(s)$ with H the transfer function, so $y = h * f$: the solution is the forcing convolved with the impulse response. For $y'' + \omega^2 y = f(t)$ this is Duhamel's formula,

$$y(t) = \frac{1}{\omega} \int_0^t \sin(\omega(t - \tau))f(\tau) d\tau.$$

WORKED EXAMPLE

For $y'' + y = f(t)$ with $y(0) = y'(0) = 0$ the impulse response is $\sin t$, so

$$y(t) = \int_0^t \sin(t - \tau)f(\tau) d\tau \quad \text{for every } f.$$

Test it on $f(\tau) = \tau$. Substituting $u = t - \tau$ turns $\int_0^t \sin(t - \tau) \tau d\tau$ into

$$\int_0^t \sin u (t - u) du = t(1 - \cos t) - [\sin u - u \cos u]_0^t = t - \sin t,$$

which is the answer the machine gave for $y'' + y = t$ with zero data. Test it on $f(\tau) = e^{-\tau}$: the convolution equals $\frac{1}{2}(e^{-t} - \cos t + \sin t)$, and at $t = 2$ that is 0.7303898, matched by Runge-Kutta on the original problem to twelve figures.

The convolution theorem is not only for differential equations. Consider the Volterra equation

$$y(t) = t + \int_0^t y(\tau) \sin(t - \tau) d\tau,$$

in which the unknown appears under an integral. The integral is exactly $y * \sin$, so transforming gives

$$Y = \frac{1}{s^2} + Y \cdot \frac{1}{s^2 + 1} \Rightarrow Y \left(1 - \frac{1}{s^2 + 1} \right) = \frac{1}{s^2} \Rightarrow Y \cdot \frac{s^2}{s^2 + 1} = \frac{1}{s^2},$$

so $Y = \frac{s^2 + 1}{s^4} = \frac{1}{s^2} + \frac{1}{s^4}$ and

$$y(t) = t + \frac{t^3}{6}.$$

Verified by quadrature: at $t = 0.8$ the right side of the original equation evaluates to 0.88533333 and $y(0.8) = 0.88533333$; at $t = 1.7$ both are 2.51883333. Nothing here was differentiated. An equation that is hard in the time domain became a linear equation in one unknown, which is the whole promise of the transform delivered in its cleanest form.

WHERE THIS GOES WRONG

Writing $(f * g)(t) = \int_0^t f(\tau)g(t - \tau) d\tau$. The reversal $t - \tau$ in the second argument is the entire content of the theorem, and without it the identity $\mathcal{L}\{f * g\} = FG$ is false. Equally: $\mathcal{L}\{fg\} \neq F \cdot G$, so you cannot transform a product of two known functions by multiplying their transforms. And the lower limit is 0, not $-\infty$; the one-sided transform only ever sees $t \geq 0$.

SEE ALSO The Dirac delta and the impulse response; nonhomogeneous systems and the Duhamel formula; variation of parameters.

The one variable coefficient the transform can absorb

The transform method dies on variable coefficients because $\mathcal{L}\{a(t)y'\}$ is not $a \cdot \mathcal{L}\{y'\}$. There is exactly one exception worth knowing, and it comes from the rule $\mathcal{L}\{tf(t)\} = -F'(s)$: a coefficient *linear* in t produces a derivative in s , so the transformed equation is a first-order ODE in Y . That is progress if and only if the original was second order.

38.7 Differentiation in s for a linear t coefficient

HARD

CUE A coefficient linear in t , as in $ty'' + (1 - t)y' + y = 0$ or $ty'' + 2y' + ty = 0$. Laguerre, Bessel, and their relatives all present this way.

METHOD Differentiate the transform:

$$\mathcal{L}\{tf(t)\} = -F'(s), \quad \mathcal{L}\{ty'\} = -\frac{d}{ds}(sY - y(0)), \quad \mathcal{L}\{ty''\} = -\frac{d}{ds}(s^2Y - sy(0) - y'(0)).$$

The result is a first-order linear equation for $Y(s)$, solved by an integrating factor. Invert at the end. Check the degree of the t coefficient first: a t^2 coefficient produces Y'' and you have gained nothing.

WORKED EXAMPLE

$$ty'' + (1 - t)y' + y = 0, \quad y(0) = 1.$$

Transform term by term. With $\mathcal{L}\{ty''\} = -(2sY + s^2Y' - y(0))$, $\mathcal{L}\{y'\} = sY - y(0)$ and $\mathcal{L}\{ty'\} = -(Y + sY')$, the equation becomes

$$-2sY - s^2Y' + y(0) + sY - y(0) + Y + sY' + Y = 0,$$

in which $y(0)$ cancels, leaving

$$(s - s^2)Y' + (2 - s)Y = 0 \Rightarrow \frac{Y'}{Y} = \frac{s - 2}{s(1 - s)} = -\frac{2}{s} - \frac{1}{1 - s}.$$

Integrating, $\ln Y = -2 \ln s + \ln(1 - s) + \text{const}$, so

$$Y = C \frac{1 - s}{s^2} = C \left(\frac{1}{s^2} - \frac{1}{s} \right), \quad y = C(t - 1).$$

Imposing $y(0) = 1$ gives $C = -1$ and $y = 1 - t$, the first Laguerre polynomial. Direct check: $t \cdot 0 + (1 - t)(-1) + (1 - t) = 0$.

$$ty'' + 2y' + ty = 0, \quad y(0) = 1.$$

This is the equation whose Frobenius treatment appears later in this chapter. Transform it. With $\mathcal{L}\{ty''\} = -(2sY + s^2Y' - y(0))$, $\mathcal{L}\{2y'\} = 2sY - 2y(0)$ and $\mathcal{L}\{ty\} = -Y'$, the equation collapses to

$$-(s^2 + 1)Y' - y(0) = 0 \implies Y' = \frac{-1}{s^2 + 1},$$

so $Y = C - \arctan s$. Now use the fact that every transform tends to 0 as $s \rightarrow \infty$: that forces $C = \frac{\pi}{2}$, so

$$Y(s) = \frac{\pi}{2} - \arctan s = \arctan \frac{1}{s}, \quad y(t) = \frac{\sin t}{t}.$$

The transform of $\sin t/t$ is indeed $\arctan(1/s)$; at $s = 1.3$ numerical quadrature of the defining integral returns 0.6556956 against $\arctan(1/1.3) = 0.6556956$. Notice what the transform did and did not find. It produced the branch that is analytic at the origin, and it was the decay condition at $s = \infty$, not an initial condition, that pinned the constant. The second solution $\cos t/t$ is unbounded at $t = 0$, has no transform of this type, and is simply invisible to the method.

WHERE THIS GOES WRONG

Two things. First, applying the method when the coefficient is quadratic or worse: $\mathcal{L}\{t^2y''\}$ involves Y'' , so a second-order equation stays second order and nothing has been gained. Second, believing you have found the general solution. In the example the point $t = 0$ is a regular singular point with a repeated indicial root, so the second solution contains $\ln t$ and has no transform of the assumed type; the transform recovers only the analytic branch. The single arbitrary constant C is a warning sign, not an accident.

SEE ALSO The Laplace-transform machine; Frobenius at a regular singular point; Frobenius with repeated roots.

Series: constructing a solution that has no name

When the coefficients vary and no substitution reduces the equation to a solved family, there may simply be no closed form. Series methods respond by lowering the ambition: instead of a formula, produce the Taylor coefficients of the solution about a chosen point, in a recurrence that generates as many as you like. This is not a defeat. The recurrence is often more informative than a formula would be, and the classical special functions are all defined this way.

Everything depends on the character of the expansion point. Write the equation in normalised form $y'' + p(x)y' + q(x)y = 0$. The point x_0 is *ordinary* if p and q are analytic there, and then a plain power series works and two free constants appear, matching the two initial conditions. It is a *regular singular* point if p or q blows up but $(x - x_0)p(x)$ and $(x - x_0)^2q(x)$ are analytic; then a power series alone fails and must be multiplied by x^r for an r determined by an auxiliary quadratic. Anything worse is irregular and outside the method.

MEDIUM

38.8 Power series about an ordinary point

CUE Polynomial or analytic coefficients, the leading coefficient nonzero at the expansion point, and no name-brand method applicable.

METHOD Substitute $y = \sum_{n \geq 0} a_n x^n$, differentiate term by term, and re-index every sum so that all of them run over the same power of x . Reading off the coefficient of x^n gives a recurrence, and a_0 and a_1 remain free, corresponding to $y(0)$ and $y'(0)$. The radius of convergence is at least the distance from x_0 to the nearest singularity of p or q in the complex plane, which you can therefore state without summing anything.

WORKED EXAMPLE

$$y'' + 3ty' - 6y = 2, \quad y(0) = y'(0) = 0.$$

With $y = \sum a_n t^n$, the three sums are $\sum (n+2)(n+1)a_{n+2}t^n$, $3 \sum na_n t^n$ and $-6 \sum a_n t^n$, already at a common power. Matching t^0 : $2a_2 - 6a_0 = 2$, and since $a_0 = 0$ this gives $a_2 = 1$. For $n \geq 1$,

$$a_{n+2} = \frac{(6 - 3n)a_n}{(n+2)(n+1)}.$$

Since $a_1 = 0$, all odd coefficients vanish. Since $a_2 = 1$ and the factor $6 - 3n$ vanishes at $n = 2$, we get $a_4 = 0$ and hence every later even coefficient is zero. The series terminates: $y = t^2$ exactly. Substituting back, $2 + 3t(2t) - 6t^2 = 2$. Runge-Kutta at $t = 1.3$ returns (1.69, 2.6) for (y, y') , that is $(1.3^2, 2 \cdot 1.3)$.

A non-terminating instance, $y'' = xy$ with $y(0) = 1, y'(0) = 0$. Matching gives $(n+2)(n+1)a_{n+2} = a_{n-1}$, so $a_2 = 0$ and

$$y = 1 + \frac{x^3}{6} + \frac{x^6}{180} + \frac{x^9}{12960} + \dots$$

Five terms give $y(1) = 1.17229997$ against 1.17229997 from Runge-Kutta. The coefficients have no closed form; the function is an Airy function, and the recurrence is the definition.

A RECURRENCE THAT IS RECOGNISABLE

$y'' + xy' + y = 0$ with $y(0) = 1, y'(0) = 0$. Substituting and matching the coefficient of x^n gives $(n+2)(n+1)a_{n+2} + na_n + a_n = 0$, that is

$$a_{n+2} = \frac{-a_n}{n+2},$$

a two-step recurrence that never mixes parities. With $a_1 = 0$ every odd coefficient vanishes, and from $a_0 = 1$ we get

$a_2 = -\frac{1}{2}, a_4 = \frac{1}{8}, a_6 = -\frac{1}{48}$, and in general $a_{2k} = \frac{(-1)^k}{2^k k!}$. That is a series you should recognise:

$$y = \sum_{k \geq 0} \frac{(-1)^k}{k!} \left(\frac{x^2}{2} \right)^k = e^{-x^2/2}.$$

Confirm directly: $y' = -xe^{-x^2/2}$ and $y'' = (x^2 - 1)e^{-x^2/2}$, so $y'' + xy' + y = (x^2 - 1 - x^2 + 1)e^{-x^2/2} = 0$. At $x = 1.4$ both the closed form and Runge-Kutta give 0.37531110. The lesson is worth the space: always look at the recurrence before summing it, because a factorial in the denominator and alternating signs almost always name an exponential, a sine, or a Bessel function.

WHERE THIS GOES WRONG

Comparing coefficients before re-indexing. The sum $\sum n(n-1)a_n x^{n-2}$ and the sum $\sum a_n x^n$ are indexed on different powers, and matching them term by term relates a_n to itself instead of to a_{n-2} . Shift one index so that both run over x^n , and note that the lower limits move too: $\sum_{n \geq 2} n(n-1)a_n x^{n-2} = \sum_{n \geq 0} (n+2)(n+1)a_{n+2} x^n$. A second trap: forgetting that a nonzero right side contributes only at the powers it actually contains, so the recurrence is inhomogeneous at finitely many n and homogeneous thereafter.

SEE ALSO Frobenius at a regular singular point; the standard expansion table; ratio and root tests.

THE RADIUS IS A GEOMETRY QUESTION, NOT A SERIES QUESTION

For $y'' + p(x)y' + q(x)y = 0$ with p and q analytic at x_0 , every power-series solution about x_0 converges at least on the disc reaching the nearest singularity of p or q in the complex plane. Two consequences. For Legendre's equation $(1-x^2)y'' - 2xy' + \lambda y = 0$ the singularities are at ± 1 , so a series about 0 has radius at least 1 and cannot be expected to say anything at the endpoints, which is exactly where the boundary conditions of the classical problem live. And for $(1+x^2)y'' + y = 0$ the coefficients are perfectly well behaved for every real x , yet the radius about 0 is 1, because the singularities sit at $\pm i$. A real-variable inspection would miss that entirely; look for complex zeros of the leading coefficient.

At a regular singular point the plain series fails for a reason that is easy to see: the equation forces the leading behaviour to be a power x^r with r not necessarily an integer, and no power series can produce $\sqrt{x}$. Frobenius fixes this by putting the power in by hand and letting the equation decide what it is. The quadratic that decides is the same one that Cauchy-Euler equations produce, which is no coincidence: near the singular point the equation is a Cauchy-Euler equation perturbed by analytic corrections.

38.9 Frobenius at a regular singular point

CUE The leading coefficient vanishes at the expansion point, but $x^2q(x)$ and $x^2q(x)$ are analytic there (taking $x_0 = 0$ after a shift).

METHOD Write $p_0 = \lim_{x \rightarrow 0} xp(x)$ and $q_0 = \lim_{x \rightarrow 0} x^2q(x)$. Substitute $y = x^r \sum_{n \geq 0} a_n x^n$ with $a_0 \neq 0$. The lowest power of x gives the **indicial equation**

$$r(r-1) + p_0 r + q_0 = 0,$$

whose roots $r_1 \geq r_2$ are the admissible exponents. For each root, the remaining powers give a recurrence for a_n in terms of earlier coefficients. The larger root always works; the smaller root works unless the roots coincide or differ by an integer.

WORKED EXAMPLE

$$2xy'' + y' + y = 0.$$

Normalising, $p = \frac{1}{2x}$ and $q = \frac{1}{2x}$, so $p_0 = \frac{1}{2}$ and $q_0 = 0$, and the indicial equation is $r(r-1) + \frac{1}{2}r = r(r - \frac{1}{2}) = 0$ with roots $r = 0, \frac{1}{2}$. Substituting $y = \sum a_n x^{n+r}$ and collecting the coefficient of x^{n+r-1} gives

$$a_n(n+r)(2n+2r-1) = -a_{n-1}, \quad \text{that is} \quad a_n = \frac{-a_{n-1}}{(n+r)(2n+2r-1)}.$$

For $r = 0$ this is $a_n = -a_{n-1}/(n(2n-1))$, giving $1, -1, \frac{1}{6}, -\frac{1}{90}, \dots$ and

$$y_1 = 1 - x + \frac{x^2}{6} - \frac{x^3}{90} + \dots = \cos \sqrt{2x}.$$

For $r = \frac{1}{2}$ it is $a_n = -a_{n-1}/(n(2n+1))$, giving

$$y_2 = \sqrt{x} \left(1 - \frac{x}{3} + \frac{x^2}{30} - \frac{x^3}{630} + \dots \right) = \frac{\sin \sqrt{2x}}{\sqrt{2}}.$$

Both closed forms follow from the substitution $t = \sqrt{2x}$, which turns the equation into $\ddot{y} + y = 0$, and both were checked against twenty-five terms of the recurrence at $x = 0.3, 1, 2.5$, agreeing to fifteen figures.

THE GENERIC CASE: ROOTS THAT ARE NOT AN INTEGER APART

$$2x^2y'' + 3xy' - (1+x)y = 0.$$

Dividing by $2x^2$ gives $p_0 = \frac{3}{2}$ and $q_0 = -\frac{1}{2}$, so the indicial equation is $r(r-1) + \frac{3}{2}r - \frac{1}{2} = 0$, that is $2r^2 + r - 1 = (2r-1)(r+1) = 0$ with roots $r = \frac{1}{2}$ and $r = -1$. Their difference is $\frac{3}{2}$, not an integer, so both branches go through cleanly. Substituting $y = \sum a_n x^{n+r}$ and writing $u = n+r$, the coefficient of x^{n+r} is

$$a_n(2u^2 + u - 1) = a_{n-1}, \quad \text{that is} \quad a_n = \frac{a_{n-1}}{(2(n+r) - 1)(n+r+1)}.$$

For $r = \frac{1}{2}$ the denominator is $n(2n+3)$, giving $1, \frac{1}{5}, \frac{1}{70}, \frac{1}{1890}, \dots$ and

$$y_1 = \sqrt{x} \left(1 + \frac{x}{5} + \frac{x^2}{70} + \frac{x^3}{1890} + \dots \right).$$

For $r = -1$ it is $n(2n-3)$, giving $1, -1, -\frac{1}{2}, -\frac{1}{18}, -\frac{1}{360}, \dots$ and

$$y_2 = \frac{1}{x} \left(1 - x - \frac{x^2}{2} - \frac{x^3}{18} - \frac{x^4}{360} - \dots \right).$$

Neither has a closed form. Substituting forty terms of each back into the equation leaves a residual below 2×10^{-7} at $x = 0.6$ and $x = 1.4$, which is the finite-difference error of the check. Note that $n(2n-3)$ never vanishes for a positive integer n , which is the arithmetic fact behind “the roots differ by $\frac{3}{2}$, so nothing breaks”.

WHERE THIS GOES WRONG

Assuming $a_0 \neq 0$ and then finding that the recurrence forces $a_0 = 0$. That means either the wrong indicial root was used or the point is irregular, not that the method failed. Also: the two roots generally have *different* recurrences, since r appears in the denominator; computing one recurrence and reusing it for the other root is the commonest way to produce a second “solution” that solves nothing. Finally, check that the point really is regular singular. For $x^3y'' + y = 0$, $x^2q = 1/x$ is not analytic, and no Frobenius series exists.

SEE ALSO Frobenius with repeated roots; the Cauchy-Euler power trial; power series about an ordinary point.

The canonical instance of all of this is Bessel's equation,

$$x^2 y'' + xy' + (x^2 - \nu^2)y = 0,$$

for which $p_0 = 1$ and $q_0 = -\nu^2$, so the indicial equation is $r^2 - \nu^2 = 0$ with roots $\pm \nu$. The larger root generates

$$J_\nu(x) = \sum_{k \geq 0} \frac{(-1)^k}{k! \Gamma(k + \nu + 1)} \left(\frac{x}{2}\right)^{2k + \nu},$$

and the whole taxonomy of the method is visible in the single parameter ν . Non-integer ν with 2ν not an integer: the roots differ by a non-integer and both branches work, giving J_ν and $J_{-\nu}$. Half-integer ν : the roots differ by an integer, yet no logarithm appears, and indeed $J_{1/2} = \sqrt{2/(\pi x)} \sin x$ and $J_{-1/2} = \sqrt{2/(\pi x)} \cos x$ are both elementary. Integer ν : the roots differ by an even integer, $J_{-n} = (-1)^n J_n$ is not independent, and a logarithm is forced, producing Y_n . Every case in the next block is a value of ν .

The two exceptional cases have to be treated because they are not rare: they are exactly what happens for Bessel's equation of integer order and for every equation whose singular behaviour is a repeated power. The mechanism in both is the same. A logarithm appears whenever the second exponent cannot support an independent series of its own, and it appears because $\frac{d}{dr} x^r = x^r \ln x$ is what a repeated root of the indicial equation produces, exactly as te^{rt} arises from a repeated characteristic root.

It is worth seeing where the logarithm comes from rather than accepting it as a rule. Write L for the differential operator and let $y(x, r) = x^r \sum a_n(r) x^n$ be the Frobenius series with the recurrence solved formally as a function of r , without imposing the indicial equation. Then $L[y(x, r)] = a_0 I(r) x^r$, where I is the indicial polynomial: everything cancels except the lowest term. If r_1 is a double root of I , then $I(r_1) = I'(r_1) = 0$, so differentiating with respect to r gives $L[\frac{\partial y}{\partial r}] = 0$ at $r = r_1$ as well. And $\frac{\partial}{\partial r} x^r = x^r \ln x$, so the second solution is $y_1 \ln x$ plus the series obtained by differentiating the coefficients. The logarithm is the derivative of the exponent, exactly as te^{rt} is the derivative of e^{rt} with respect to the repeated characteristic root.

HARD

38.10 Frobenius with repeated roots or roots differing by an integer

CUE The indicial roots coincide, or $r_1 - r_2 = N$ is a positive integer.

METHOD Build y_1 from the larger root r_1 as usual. Then:

- **Equal roots.** The second solution necessarily contains a logarithm: $y_2 = y_1 \ln x + x^{r_1} \sum_{n \geq 1} b_n x^n$, and substituting produces a recurrence for b_n whose right side involves the already known a_n .
- **Roots differing by N .** Run the recurrence for r_2 . At $n = N$ the coefficient of a_N vanishes. If the right side also vanishes, a_N is free, no logarithm is needed, and the resulting series contains y_1 inside it. If the right side does not vanish, take $y_2 = C y_1 \ln x + x^{r_2} \sum_{n \geq 0} b_n x^n$ and solve for C along with the b_n .

EQUAL ROOTS

$$xy'' + y' - y = 0.$$

Here $p_0 = 1$ and $q_0 = 0$, so the indicial equation is $r^2 = 0$: a double root at $r = 0$. The recurrence from the coefficient of x^{n-1} is $n^2 a_n = a_{n-1}$, so $a_n = 1/(n!)^2$ and

$$y_1 = \sum_{n \geq 0} \frac{x^n}{(n!)^2},$$

which happens to be $I_0(2\sqrt{x})$. For the second solution put $y_2 = y_1 \ln x + \sum_{n \geq 1} b_n x^n$. Since $x(u \ln x)'' + (u \ln x)' - (u \ln x) = \ln x \cdot (xu'' + u' - u) + 2u'$, the bracket vanishes and the substitution leaves $2y_1' + \sum (n^2 b_n - b_{n-1}) x^{n-1} = 0$, that is

$$n^2 b_n - b_{n-1} = \frac{-2n}{(n!)^2}, \quad b_1 = -2, \quad b_2 = -\frac{3}{4}, \quad b_3 = -\frac{11}{108}, \quad \dots$$

Substituting the truncated y_2 back into the equation numerically leaves a residual below 10^{-7} at $x = 0.8$ and $x = 1.3$, which is the discretisation error of the check and not a defect of the series.

ROOTS DIFFERING BY AN INTEGER, WITH NO LOGARITHM

$$xy'' + 2y' + xy = 0.$$

Now $p_0 = 2$, $q_0 = 0$, indicial $r(r+1) = 0$, roots $r_1 = 0$ and $r_2 = -1$ differing by $N = 1$. The recurrence is $a_n(n+r)(n+r+1) = -a_{n-2}$. For $r = -1$ it reads $n(n-1)a_n = -a_{n-2}$, and at $n = 1$ the left side is $0 \cdot a_1$ while the right side is $-a_{-1} = 0$. The equation $0 = 0$ is satisfiable, a_1 is free, and no logarithm is required. The two solutions are

$$y_1 = \frac{\sin x}{x}, \quad y_2 = \frac{\cos x}{x},$$

as the substitution $w = xy$ confirms: the equation becomes $w'' + w = 0$. Both were checked by numerical integration from $x = 0.4$ to $x = 1.9$.

ROOTS DIFFERING BY AN INTEGER, WITH THE LOGARITHM FORCED

$$xy'' - y = 0.$$

Here $p_0 = 0$ and $q_0 = 0$, so the indicial equation is $r(r-1) = 0$ with roots 1 and 0, again differing by $N = 1$. Substituting $y = \sum a_n x^{n+r}$ and collecting x^{n+r-1} gives

$$a_n(n+r)(n+r-1) = a_{n-1}.$$

For the larger root $r = 1$ this is $a_n = \frac{a_{n-1}}{n(n+1)}$, so

$$y_1 = x \left(1 + \frac{x}{2} + \frac{x^2}{12} + \frac{x^3}{144} + \cdots \right).$$

For $r = 0$ it is $n(n-1)a_n = a_{n-1}$, and at $n = 1$ that reads $0 \cdot a_1 = a_0$. Since $a_0 \neq 0$ by assumption, this is impossible: contrast the previous example, where the same position gave $0 = 0$. The logarithm is therefore unavoidable. Writing $y_2 = y_1 \ln x + \sum_{n \geq 0} b_n x^n$ and using $x(u \ln x)'' - u \ln x = \ln x (xu'' - u) + 2u' - \frac{u}{x}$, the bracket vanishes and what remains is

$$n(n-1)b_n - b_{n-1} = -(2n-1)c_n, \quad y_1 = \sum_{m \geq 1} c_m x^m,$$

giving $b_0 = 1$, b_1 free (it merely adds a multiple of y_1 , so set it to 0), $b_2 = -\frac{3}{4}$, $b_3 = -\frac{7}{36}$, $b_4 = -\frac{35}{1728}$. Substituting thirty terms of y_2 back into $xy'' - y$ leaves a residual of order 10^{-8} at $x = 0.5$ and $x = 1.2$, the size of the finite-difference error in the check. The single arithmetic fact that decided the whole structure was whether $0 \cdot a_N$ was being set equal to zero or to something else.

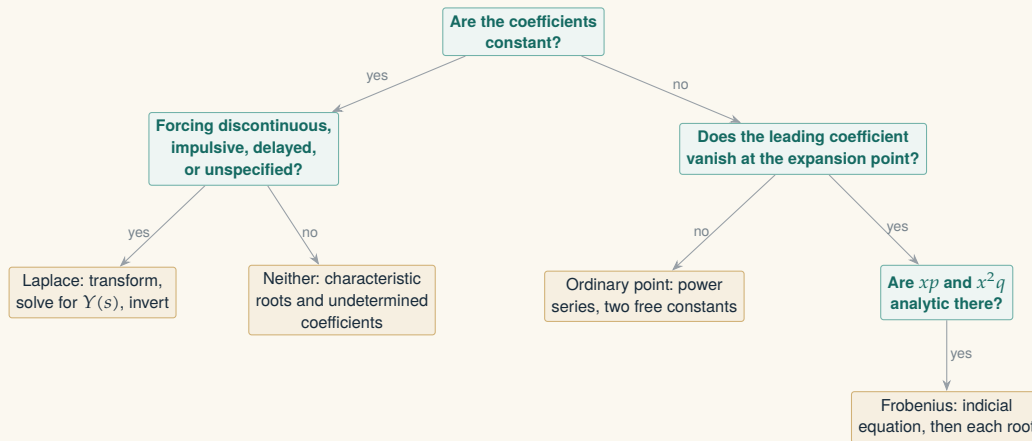
WHERE THIS GOES WRONG

Assuming a logarithm is always needed when the roots differ by an integer. As the second example shows, the obstruction is an equation of the form $0 \cdot a_N = (\text{something})$, and when that something is zero the series goes through cleanly. Test it before importing the log machinery. The reverse error is also common: for equal roots the logarithm is *never* optional, and no choice of a_0 avoids it.

SEE ALSO Frobenius at a regular singular point; the Cauchy-Euler power trial; forcing that is exactly an eigenfunction.

Laplace, series, or neither

The decision is made from the coefficients first and the forcing second, and never from how difficult the equation looks. The tree below is the whole rule.



One branch is missing on purpose, the “no” out of the last test. An irregular singular point admits neither a power series nor a Frobenius series, and the honest answers there are asymptotic expansion, a numerical solution, or a change of variable that moves the singularity. Recognising the case is the useful part: for $x^3y'' + y = 0$ the quantity $x^2q = 1/x$ is not analytic at 0, so no amount of persistence with $y = x^r \sum a_n x^n$ will succeed, and time spent on it is time lost.

Two further remarks on the tree. The left half assumes data at $t = 0$; data anywhere else must be translated there first, and if the problem gives boundary data at two points instead, neither Laplace nor the initial-value machinery applies and the problem belongs to the eigenvalue chapter. And the right half is silent about which point to expand at, which is usually dictated: expand where the data are given, or where the behaviour is asked about, and if that point is singular you are in Frobenius territory whether you like it or not.

THE TWO LIMIT THEOREMS, AND WHAT THEY AUDIT

Two identities read the endpoints of y straight off Y , without inverting. The initial-value theorem says $y(0^+) = \lim_{s \rightarrow \infty} sY(s)$, always, for any transform of a function continuous at the origin. The final-value theorem says $\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} sY(s)$, but only when every pole of $sY(s)$ lies strictly in the left half plane; applied to $Y = 1/(s^2 + 1)$ it would predict that $\sin t$ has a limit, which is the standard way the hypothesis is discovered. Used correctly they are the cheapest possible check on a long calculation. For the ramp example above, $sY = (1 - e^{-2s})/(s(s + 1))$ tends to 0 as $s \rightarrow \infty$, matching $y(0) = 0$, and tends to 2 as $s \rightarrow 0$, matching the saturated value.

AUDIT THE TRANSFORM, NOT THE ALGEBRA

Every step in a Laplace calculation is reversible, and each has a cheap check. After transforming, confirm that $Y(s) \rightarrow 0$ as $s \rightarrow \infty$; a transform that does not decay is wrong. After partial fractions, evaluate both sides at one convenient value of s that is not a pole. After inverting, substitute the answer into the original equation and check both initial conditions; this catches every sign error in the shifting theorems, which are otherwise invisible. For series, the analogous audit is to sum ten terms numerically and compare against a Runge-Kutta solution started just off the singular point. Both audits take a minute and both catch the errors that reading cannot.

One closing observation about the two halves of this chapter, since they are usually taught as unrelated. Both replace a differential equation by an algebraic object indexed by a new variable: Laplace by a function of s , series by a sequence indexed by n . Both turn differentiation into multiplication, by s in one case and by a shift of index in the other. Both encode the initial conditions in the object itself rather than in constants fitted at the end. And both are blind in the same way, to solutions that are not of the assumed shape: Laplace cannot see a solution of super-exponential growth or one singular at the origin, and a Frobenius series cannot see a solution at an irregular point. When a method returns fewer solutions than the order of the equation, that is not a mistake in the algebra. It is the method reporting the boundary of what it can represent.

Recognition matrix

If you see	Reach for	If that fails
Constant coefficients, data at $t = 0$	Laplace: $\mathcal{L}\{y''\} = s^2 Y - sy(0) - y'(0)$	Characteristic roots if forcing is smooth
Data at $t = t_0 \neq 0$	Substitute $\tau = t - t_0$ first	Then transform in τ
$Y(s)$ proper rational	Partial fractions, then the table	Complete the square, first shift
$(s - a)^2 + b^2$ downstairs	First shift; rewrite the numerator in $s - a$	$e^{at}(B \cos bt + C \sin bt)$
Repeated linear factor $(s - a)^m$	All m terms; gives $t^k e^{at}/k!$	Cover-up at $s = a$ for the top one
e^{-cs} in $Y(s)$	Second shift: $u_c(t)f(t - c)$	Rewrite f in terms of $t - c$
Forcing defined by cases	$g_1 + (g_2 - g_1)u_c(t)$	Solve piecewise and match at c
"Applied at $t = c$ ", "switched off"	Heaviside step, then $e^{-cs}F(s)$	Check the argument is $t - c$
"Hammer blow", "impulse I "	$\mathcal{L}\{\delta(t - c)\} = e^{-cs}$	y' jumps by I/a , y does not
$Y = H(s)F(s)$, f unspecified	Convolution $h * f$; Duhamel	Impulse response first
$\int_0^t f(\tau)g(t - \tau) d\tau$ appears	Recognise a convolution; transform to FG	Interchange the roles of f and g
An integral equation for y	Transform it; convolution becomes a product	Differentiate to get an ODE
Coefficient linear in t	$\mathcal{L}\{tf\} = -F'(s)$; first-order ODE in s	Frobenius instead
Coefficient quadratic in t	Not Laplace; use series	Y'' appears, no progress
Analytic coefficients, x_0 ordinary	$y = \sum a_n x^n$; re-index; recurrence	Two free constants a_0, a_1
Leading coefficient vanishes at x_0	Test $x^p, x^2 q$ analytic; Frobenius	Irregular: asymptotics or numerics
Indicial roots distinct, not integer apart	Two independent Frobenius series	Different recurrence for each root
Indicial roots equal	$y_2 = y_1 \ln x + x^r \sum b_n x^n$	The log is not optional
Roots differing by an integer N	Run the small root; check $0 \cdot a_N$	If the right side is nonzero, add $Cy_1 \ln x$
Recurrence forces $a_0 = 0$	Wrong indicial root, or irregular point	Recheck p_0 and q_0
Series terminates	The solution is a polynomial; verify directly	Check the vanishing factor in the recurrence
Boundary data at two points	Neither method; eigenvalue problem	Separation of variables

Modelling archetypes and their closed forms

What this chapter is for. A modelling problem is two problems stacked on top of each other, and only the second one is calculus. The first is translation: reading a paragraph of English about coffee, brine, capacitors or falling bodies and producing a differential equation with the right variables, the right signs and the right units. The second is solving that equation, and by this point in the book you can already solve every equation the first stage will hand you, because there are only about six of them. Almost all the marks in this subject are lost in the translation, and almost all of the translation is one instruction: name the quantity that is accumulating, then write *rate of change equals rate in minus rate out* and refuse to write anything else until both terms are expressed in the named quantity. This chapter works through the archetypes one by one. For each you get the physical statement, the equation it becomes, the closed form, and the phrase in the problem's wording that tells you which archetype you are looking at. By the end you should be able to read a modelling problem and write down its solution family before you have finished reading the second sentence, and to know which of the standard sign errors you are at risk of committing.

The translation step

Every model in this chapter is produced by the same four decisions, made in this order, and it is worth making them explicitly the first fifty times.

One: name the state variable and its units. Not “temperature” but $T(t)$ in degrees Celsius; not “salt” but $A(t)$ in kilograms of dissolved salt, as distinct from $c(t)$ in kilograms per litre. Half of all mixing errors are a confusion between an amount and a concentration, and naming the units prevents it.

Two: write the balance law. For anything that accumulates, $\frac{dx}{dt} = (\text{in}) - (\text{out})$. For anything mechanical, $F = ma$ with every force listed and signed against a stated positive direction. For a circuit, Kirchhoff's voltage law around the loop.

Three: express every term in the state variable and t alone. This is the step that produces the equation. An outflow term that still mentions concentration is not finished; a drag term that still mentions the distance fallen is not finished.

Four: read off the type. Separable, first-order linear, or a linear system. There is no fourth option anywhere in this chapter.

UNITS ARE A FREE PROOF

Every constant that appears in a model carries units, and checking them catches structural errors that no amount of rereading will. In $T' = -k(T - T_a)$ the left side is degrees per minute and $T - T_a$ is degrees, so k is per minute and $1/k$ is a time. In $LI' + RI = E$ every term is a voltage, so L/R has units of time and $\tau = L/R$ is forced; nobody who checks this ever writes $\tau = R/L$. In $\frac{dA}{dt} = rc_{\text{in}} - \frac{r_{\text{out}}}{V}A$ the second term is (litres per minute) times (kilograms per litre), which is kilograms per minute, matching the left side; the wrong version $r_{\text{out}}A$ would be kilogram-litres per minute and is dead on arrival. Do this check before you integrate, not after.

The first three archetypes all reduce to the same equation, $y' = \lambda y$ or a shifted version of it, and they are grouped here for that reason. What distinguishes them in practice is not the mathematics but which quantity turns out to be the one decaying exponentially.

The exponential family

The equation $y' = \lambda y$ is the entire content of the next three techniques. What varies is the substitution that produces it. In decay it is already there. In cooling you must subtract the ambient temperature first. In logistic growth you must form the odds. Each of those is a one-line change of variable, and recognising which one to make is the whole recognition problem.

39.1 Radioactive decay, half-life, and dating

CUE The phrases “half-life”, “decays at a rate proportional to the amount present”, “carbon dating”, or any quantity described by a fixed percentage loss per fixed time.

METHOD $N' = -\lambda N$ gives $N = N_0 e^{-\lambda t}$, with $t_{1/2} = \frac{\ln 2}{\lambda}$. In practice do not compute λ at all. Write

$$N(t) = N_0 2^{-t/t_{1/2}},$$

which is the same function with the data already substituted, and answer every question directly from it. Inverting, a sample at fraction f of its original content has age $t = t_{1/2} \log_2(1/f)$.

WORKED EXAMPLE

A sample retains 25% of its original carbon-14, whose half-life is 5730 years. Since $0.25 = 2^{-2}$, the sample is exactly two half-lives old: $t = 11460$ years, with no logarithm required. If instead the sample retains 62%, the same formula gives

$$t = 5730 \cdot \frac{\ln(1/0.62)}{\ln 2} = 5730 \times 0.689661 = 3951.8 \text{ years,}$$

so roughly 3950 years, which is as much precision as a radiocarbon measurement supports.

DECAY AGAINST A SOURCE

A short-lived isotope is produced in a reactor at a constant rate R atoms per second and decays with constant λ . Now $N' = R - \lambda N$, first-order linear, and the substitution $U = N - R/\lambda$ removes the constant: $U' = -\lambda U$, so

$$N(t) = \frac{R}{\lambda} + \left(N_0 - \frac{R}{\lambda}\right)e^{-\lambda t}.$$

The equilibrium $N_\infty = R/\lambda$ is called *secular equilibrium*, and the approach to it has exactly the half-life of the isotope: after one half-life the gap to equilibrium has halved. Starting from $N_0 = 0$, the inventory reaches 90% of equilibrium after $\log_2(10) \approx 3.32$ half-lives.

WHERE THIS GOES WRONG

Reporting the decay constant when the half-life was asked for, or the reverse. They differ by the factor $\ln 2 \approx 0.693$, so both answers are the same order of magnitude and neither looks wrong. Attach units the moment you compute either one: λ is per year, $t_{1/2}$ is years, and $\lambda t_{1/2} = \ln 2$ is the relation to quote.

SEE ALSO Newton's law of cooling and the equal-time ratio; logistic growth and the odds substitution; compartment models.

Cooling is decay with an offset, and the offset is where the difficulty lives. A cooling problem that tells you the ambient temperature is a decay problem in disguise and takes one line. A cooling problem that withholds the ambient temperature looks impossible and is not, because three readings at equal spacing determine it algebraically.

39.2 Newton's law of cooling and the equal-time ratio

CUE A body at temperature T in surroundings at T_a , with two or three temperature readings at stated times. Also: “how long until it reaches”, “what was the temperature before the measurements began”, “estimate the room temperature”.

METHOD The law is $T' = -k(T - T_a)$. Do not solve it in T . Set $U = T - T_a$, the *excess*, and note $U' = -kU$, so

$$U(t) = U_0 e^{-kt}, \quad T(t) = T_a + (T_0 - T_a)e^{-kt}.$$

The operational consequence is that over any two intervals of equal length the excess is multiplied by the same factor $\rho = e^{-k\Delta}$. So if readings are taken at equally spaced times, the excesses form a geometric progression, and k never has to be computed. If T_a is unknown, three equally spaced readings T_0, T_1, T_2 determine it: the condition $(T_1 - T_a)^2 = (T_0 - T_a)(T_2 - T_a)$ expands to a linear equation, whose solution is

$$T_a = \frac{T_0 T_2 - T_1^2}{T_0 + T_2 - 2T_1}.$$

A body is measured at 80° , then 60° ten minutes later, then 48° after a further ten minutes. The ambient temperature is not given. Apply the formula:

$$T_a = \frac{80 \cdot 48 - 60^2}{80 + 48 - 120} = \frac{3840 - 3600}{8} = 30^\circ.$$

Check it against the geometric progression: the excesses are 50, 30, 18, and $30/50 = 18/30 = \frac{3}{5}$, so $\rho = \frac{3}{5}$ per ten minutes. Everything else now follows without solving for k . Ten minutes later the excess is $18 \cdot \frac{3}{5} = 10.8$, so $T(30) = 40.8^\circ$. And the time to reach 35° , an excess of 5, satisfies $50\rho^{t/10} = 5$, so

$$\frac{t}{10} = \frac{\ln 10}{\ln(5/3)} = \frac{2.302585}{0.510826} = 4.5075, \quad t \approx 45.1 \text{ minutes.}$$

Coffee at 90° in a 20° room is 70° after ten minutes. The excess went from 70 to 50, so $\rho = \frac{5}{7}$ per ten minutes, and after twenty minutes the excess is $70 \cdot (\frac{5}{7})^2 = \frac{250}{7}$. Hence

$$T(20) = 20 + \frac{250}{7} = \frac{390}{7} \approx 55.71^\circ.$$

Note what was not done: $k = \frac{1}{10} \ln \frac{7}{5} \approx 0.033647$ was never needed, and computing it would have introduced a rounding error into an exact answer.

Applying the ratio to T rather than to $T - T_a$. In the coffee example, $70/90$ is not the decay factor and $90 \cdot (7/9)^2 \approx 54.4$ is not the answer, though it is close enough to look right. Temperatures do not decay geometrically; excess temperatures do. The tell is that the wrong method predicts $T \rightarrow 0$, which is a statement about absolute zero that no coffee cup has ever made.

SEE ALSO Radioactive decay and half-life; RC and RL circuits; compartment models.

Unbounded exponential growth is a model with a short shelf life, since every real population runs into a resource ceiling. The standard repair multiplies the growth rate by a factor that vanishes at the ceiling, and the resulting equation is nonlinear but still solvable in closed form, because a single substitution linearises it.

39.3 Logistic growth and the odds substitution

CUE $P' = kP(K - P)$ or $P' = rP(1 - \frac{P}{K})$: growth proportional to the population *and* to the room remaining. In words: “grows exponentially at first but levels off at a carrying capacity”.

METHOD The two forms are the same equation with $r = kK$. Partial fractions,

$$\frac{1}{P(K - P)} = \frac{1}{K} \left(\frac{1}{P} + \frac{1}{K - P} \right),$$

show that the correct variable is the *odds* $W = \frac{P}{K - P}$, which satisfies $W' = rW$ exactly. So the odds grow purely exponentially, $W = W_0 e^{rt}$, and unwinding gives

$$P(t) = \frac{KP_0}{P_0 + (K - P_0)e^{-rt}} = \frac{K}{1 + W_0^{-1}e^{-rt}}.$$

Work in odds throughout: every data point converts to an odds value, every question converts back at the end, and no logarithm is ever taken of a difference.

A deer population has carrying capacity $K = 500$, with $N(0) = 100$ and $N(4) = 200$. When does it reach 350?

Convert the data to odds: $W_0 = \frac{100}{400} = \frac{1}{4}$ and $W(4) = \frac{200}{300} = \frac{2}{3}$. The odds multiplied by $\frac{2/3}{1/4} = \frac{8}{3}$ over four years, so $e^{4r} = \frac{8}{3}$ and $r = \frac{1}{4} \ln \frac{8}{3} = 0.245207$ per year. The target $N = 350$ is odds $\frac{350}{150} = \frac{7}{3}$, which is $\frac{7/3}{1/4} = \frac{28}{3}$ times the initial odds. Hence

$$t = \frac{\ln(28/3)}{r} = 4 \frac{\ln(28/3)}{\ln(8/3)} = 4 \times \frac{2.233592}{0.980829} = 9.109 \text{ years.}$$

A fourth-order Runge-Kutta integration of $N' = rN(1 - N/500)$ from $N(0) = 100$ returns $N(9.108995) = 350.000$, confirming both r and t .

THE INFLECTION POINT IS THE FASTEST GROWTH

Differentiating $P' = rP(1 - P/K)$ gives $P'' = rP'(1 - 2P/K)$, which vanishes at $P = K/2$. So the population curve has its steepest point exactly at half capacity, with maximum growth rate $rK/4$. This is worth remembering as a data-fitting shortcut: if a problem states the year of fastest growth, it has told you the year at which $P = K/2$, hence that $W = 1$, hence that $e^{rt} = W_0^{-1}$ at that time. In the deer example $W = 1$ at $t = \ln 4/r = 5.653$ years, when the herd passes 250.

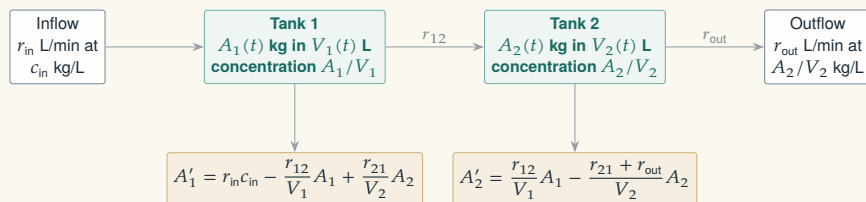
WHERE THIS GOES WRONG

Calibrating the parameters in the wrong order, or believing that two data points determine all three of P_0 , r and K . They do not. With K given, two readings fix W_0 and r and the problem is closed; with K unknown you need a third condition, and the cleanest one is the inflection time. The second trap is sign: writing $P' = rP(1 - K/P)$ instead of $1 - P/K$ inverts the whole model and predicts extinction.

SEE ALSO Radioactive decay and half-life; Bernoulli equations; compartment models.

Bookkeeping models: rate in minus rate out

The exponential family had a single reservoir with no explicit flows. The models in this section make the flows explicit, and the extra difficulty is entirely one of accounting: what enters carries a concentration you are told, and what leaves carries the concentration currently inside, which is a function of the unknown. The figure below is the whole subject.



Delete Tank 2 and you have the single-tank problem; keep it and you have a compartment model. Nothing else changes. The volumes are bookkeeping too: $V_i(t) = V_i(0) + (\text{flow in} - \text{flow out})t$, and a tank whose volume changes has a genuinely non-constant coefficient in front of A_i , which is the difference between a five-line problem and a fifteen-line one.

CORE

39.4 Mixing and tank problems

CUE Brine, pollutant, or dye entering a well-stirred tank at a stated rate and concentration, and leaving at a stated rate. The words “well-stirred” or “thoroughly mixed” are there to license the assumption that the outflow concentration equals the tank average.

METHOD Track the *amount* $A(t)$, not the concentration. Then

$$\frac{dA}{dt} = r_{\text{in}}c_{\text{in}} - \frac{r_{\text{out}}}{V(t)}A, \quad V(t) = V_0 + (r_{\text{in}} - r_{\text{out}})t.$$

This is first-order linear, always. If $r_{\text{in}} = r_{\text{out}}$ the coefficient is constant and the answer is a steady state plus a decaying transient. If they differ, the integrating factor is a power of $V(t)$: with $r_{\text{in}} - r_{\text{out}} = \delta \neq 0$ the coefficient is $\frac{r_{\text{out}}}{V_0 + \delta t}$ and the factor is $(V_0 + \delta t)^{r_{\text{out}}/\delta}$.

CONSTANT VOLUME

A 200 L tank holds pure water. Brine at 0.4 kg/L enters at 5 L/min and the mixture leaves at 5 L/min. Here $V \equiv 200$, so

$$A' = 2 - \frac{A}{40}, \quad A(0) = 0.$$

The steady state is $A_{\infty} = 80$ kg, which is the tank at the inflow concentration, and the transient decays with time constant 40 minutes:

$$A(t) = 80(1 - e^{-t/40}).$$

Note the two sanity checks: the limiting concentration $80/200 = 0.4$ kg/L must equal c_{in} , and the time constant $V/r = 200/5 = 40$ minutes is the time to replace the tank's volume once.

CHANGING VOLUME

A 100 L tank holds pure water. Brine at 1 kg/L enters at 6 L/min; the well-stirred mixture is drawn off at 4 L/min. Find the salt content when the volume reaches 200 L.

The volume grows at 2 L/min, so $V(t) = 100 + 2t$ and the tank reaches 200 L at $t = 50$ min. The balance is

$$\frac{dA}{dt} = 6 - \frac{4A}{100 + 2t} = 6 - \frac{2A}{50 + t},$$

so with $\mu = (50 + t)^2$,

$$[(50 + t)^2 A]' = 6(50 + t)^2 \Rightarrow (50 + t)^2 A = 2(50 + t)^3 + C.$$

The condition $A(0) = 0$ gives $C = -2 \cdot 50^3 = -250000$, hence

$$A(t) = 2(50 + t) - \frac{250000}{(50 + t)^2}, \quad A(50) = 200 - \frac{250000}{10000} = 175 \text{ kg}.$$

Runge-Kutta integration of the original equation returns $A(50) = 175.000$. The concentration at that moment is $175/200 = 0.875$ kg/L, still short of the inflow value 1, as it must be.

WHERE THIS GOES WRONG

Writing the outflow term as $r_{out}c_{in}$, or as $r_{out}A$. What leaves carries the tank's *current* concentration $A/V(t)$, so the term is $r_{out}A/V(t)$; the first error makes the equation trivially separable and wrong, the second is dimensionally impossible. The third and subtlest error is holding V fixed when the rates differ: that turns a variable-coefficient equation into a constant-coefficient one and produces a solution that approaches a finite limit when the true one grows without bound.

SEE ALSO Compartment models; first-order linear equations and the integrating factor; Torricelli draining.

Two tanks exchanging fluid, a drug moving between blood and tissue, and a chain of lakes are all the same object: several reservoirs, one balance equation each, and a linear system. The system machinery of the earlier chapters applies unchanged, and the only genuinely new material is a structural check that catches setup errors before any eigenvalue is computed.

HARD

39.5 Compartment models

CUE Two or more reservoirs exchanging material: connected tanks, a two-compartment pharmacokinetic model, a lake system, a tracer in blood and tissue.

METHOD Write one balance equation per compartment, each in the form (in) minus (out), with every flow expressed as a rate times the source compartment's concentration. The result is

$$\mathbf{x}' = A\mathbf{x} + \mathbf{b},$$

solved by the eigenvalue methods for linear systems. The equilibrium is $\mathbf{x}^* = -A^{-1}\mathbf{b}$ when A is invertible, and the transient decays at rates set by the eigenvalues of A . Before solving, run the conservation check: if the model is closed (nothing enters or leaves), the column sums of A must be zero, $\lambda = 0$ must be an eigenvalue, and the corresponding conserved quantity is the total.

WORKED EXAMPLE

Two 100 L tanks are connected so that 10 L/min flows from the first to the second and 10 L/min back. Tank 1 starts with 20 kg of salt, tank 2 with none. With $x = A_1$ and $y = A_2$,

$$x' = \frac{10}{100}y - \frac{10}{100}x = 0.1(y - x), \quad y' = 0.1(x - y),$$

so $A = 0.1 \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$. The column sums are zero, as required for a closed system, and the eigenvalues are 0 and -0.2 . Rather than diagonalising, use the two combinations the eigenvalues point to. The sum obeys $(x + y)' = 0$, so $x + y \equiv 20$. The difference obeys $(x - y)' = -0.2(x - y)$, so $x - y = 20e^{-0.2t}$. Adding and halving,

$$x(t) = 10 + 10e^{-0.2t}, \quad y(t) = 10 - 10e^{-0.2t}.$$

At $t = 7$ minutes this gives $x = 12.4660$, $y = 7.5340$, which Runge-Kutta reproduces to six figures. The general lesson is that a symmetric two-compartment system is always solved by sum and difference, and the relaxation rate is $2r/V$, twice what a single-tank intuition would suggest, because both tanks are moving toward each other.

A TWO-COMPARTMENT DRUG MODEL

A drug is infused into the blood at a constant rate I , exchanges with tissue at rates k_{12} and k_{21} , and is cleared from the blood at rate k_e . Writing b and q for the blood and tissue amounts,

$$b' = I - (k_{12} + k_e)b + k_{21}q, \quad q' = k_{12}b - k_{21}q.$$

This system is *not* closed, so $\lambda = 0$ is not an eigenvalue and there is a genuine steady state. Setting both derivatives to zero, the second equation gives $q^* = \frac{k_{12}}{k_{21}}b^*$, and substituting into the first collapses the exchange terms entirely: $I = k_e b^*$, so $b^* = I/k_e$ and $q^* = \frac{k_{12}I}{k_{21}k_e}$. The steady blood level depends only on the infusion rate and the clearance rate, not on the exchange rates at all, which is why the tissue compartment changes how fast a drug reaches its plateau but not where the plateau sits.

WHERE THIS GOES WRONG

Omitting the return flow, or writing it with the wrong compartment's concentration. The flow from tank 2 into tank 1 is $r_{21} \cdot A_2/V_2$, using tank 2's concentration, and putting A_1/V_1 there produces a matrix whose column sums are not zero. Run the conservation check on every closed model: it costs one addition per column and catches almost every sign and index error in the setup.

SEE ALSO Mixing and tank problems; linear systems and the matrix exponential; Newton's law of cooling.

Draining under gravity breaks the pattern of the previous two techniques in one respect: the outflow rate is not given but determined by the physics, and it depends on the state. The result is separable rather than linear, and it produces the chapter's first solution that reaches its terminal state in finite time.

MEDIUM

39.6 Torricelli draining and square-root emptying

CUE A tank draining through a hole in the bottom, with the outflow speed proportional to $\sqrt{h}$; equivalently any statement that the efflux velocity is $\sqrt{2gh}$.

METHOD Volume balance with cross-sectional area $A(h)$ at height h and hole area a :

$$A(h) \frac{dh}{dt} = -ca\sqrt{2gh}.$$

For a tank of constant cross-section this is $h' = -k\sqrt{h}$, separable, and the useful form of the answer is that $\sqrt{h}$ is *affine* in time:

$$2\sqrt{h} = 2\sqrt{H} - kt, \quad h(t) = \left(\sqrt{H} - \frac{k}{2}t\right)^2,$$

so the tank empties at the finite time $t_{\text{empty}} = \frac{2\sqrt{H}}{k}$. Plot $\sqrt{h}$ against t to fit k from data: it is a straight line, and its intercept is the emptying time.

WORKED EXAMPLE

A tank of constant cross-section, initially 4 m deep, obeys $h' = -\frac{1}{2}\sqrt{h}$. Then $\sqrt{h} = 2 - \frac{1}{4}t$, so

$$h(t) = \left(2 - \frac{t}{4}\right)^2, \quad t_{\text{empty}} = 8 \text{ minutes},$$

and at $t = 4$ the depth is $h = 1$ m, not 2 m: half the time drains three quarters of the depth. Numerical integration with the right-hand side cut off at $h = 0$ agrees, and confirms that h stays at 0 after $t = 8$.

NON-CONSTANT CROSS-SECTION

A hemispherical bowl of radius R drains through a hole at the bottom. At depth h the free surface is a circle of radius $\sqrt{R^2 - (R - h)^2} = \sqrt{2Rh - h^2}$, so $A(h) = \pi(2Rh - h^2)$ and

$$\pi(2Rh - h^2) \frac{dh}{dt} = -k\sqrt{h} \implies \frac{\pi}{k}(2Rh^{1/2} - h^{3/2}) dh = -dt.$$

Integrating from $h = R$ down to 0,

$$t_{\text{empty}} = \frac{\pi}{k} \left[\frac{4R}{3} h^{3/2} - \frac{2}{5} h^{5/2} \right]_0^R = \frac{\pi}{k} \left(\frac{4}{3} - \frac{2}{5} \right) R^{5/2} = \frac{14\pi R^{5/2}}{15k}.$$

The lesson generalises: the geometry enters only through $A(h)$, and every such problem is the single integral $\int A(h)h^{-1/2} dh$.

WHERE THIS GOES WRONG

Using the formula past the emptying time. The expression $(\sqrt{H} - \frac{k}{2}t)^2$ is a perfectly good parabola that turns around and increases after t_{empty} , describing a tank that spontaneously refills. The solution of the physical problem is that formula on $[0, t_{\text{empty}}]$ and $h \equiv 0$ afterwards, and the two pieces genuinely both solve the equation from the moment they meet: uniqueness fails at $h = 0$ because $\sqrt{h}$ is not Lipschitz there. That failure is the subject of the next chapter, and this is where a reader first meets it in the wild.

SEE ALSO Mixing and tank problems; separable equations; solutions lost by dividing.

Mechanical and electrical archetypes

The models so far accumulated a substance. The next three conserve or balance something else: charge around a loop, momentum against gravity and drag, momentum against an exhaust stream. The mathematics repeats, but the recognition cues are entirely different, and so are the standard errors.

CORE

39.7 RC and RL circuits

CUE A single-loop circuit with exactly one energy-storage element (a capacitor or an inductor) and a source. Words such as “time constant”, “steady state current”, “transient”.

METHOD Kirchhoff’s voltage law gives a first-order linear equation in each case:

$$L \frac{dI}{dt} + RI = E(t), \quad \tau = \frac{L}{R}; \quad R \frac{dQ}{dt} + \frac{Q}{C} = E(t), \quad \tau = RC.$$

For a constant source the answer is always *steady state plus decaying transient*, and you can write it without integrating: identify the final value, identify the initial value, and interpolate exponentially,

$$X(t) = X_{\infty} + (X_0 - X_{\infty})e^{-t/\tau}.$$

For a sinusoidal source use the steady-state method of the constant-coefficient chapter: the amplitude of the current in an RL loop driven by $E_0 \sin \omega t$ is $E_0 / \sqrt{R^2 + \omega^2 L^2}$, and the phase lag is $\arctan(\omega L / R)$.

WORKED EXAMPLE

$R = 12 \Omega$, $L = 4 \text{ H}$, $E = 60 \text{ V}$, $I(0) = 0$. The equation is $4I' + 12I = 60$, that is $I' + 3I = 15$, so $\tau = L/R = \frac{1}{3} \text{ s}$ and $I_{\infty} = E/R = 5 \text{ A}$. Interpolating,

$$I(t) = 5(1 - e^{-3t}).$$

The current reaches 99% of its final value after $t = \tau \ln 100 \approx 1.535 \text{ s}$, which is the useful engineering statement: five time constants is 99.3%, and that is the origin of the rule of thumb.

DISCHARGE, AND READING τ OFF A GRAPH

A capacitor charged to Q_0 is discharged through a resistor: $RQ' + Q/C = 0$, so $Q = Q_0 e^{-t/RC}$. With $R = 100 \Omega$ and $C = 200 \mu\text{F}$, $\tau = RC = 100 \times 2 \times 10^{-4} = 0.02 \text{ s}$. After 50 ms the charge is $Q_0 e^{-2.5} = 0.0821 Q_0$. Conversely, given a measured decay from Q_0 to $Q_0/2$ in 13.9 ms, $\tau = 13.9 / \ln 2 = 20.05 \text{ ms}$, recovering RC without knowing either component. The half-life language of radioactive decay transfers verbatim.

WHERE THIS GOES WRONG

Forming the time constant by the wrong combination. For RC it is the *product* RC and for RL it is the *quotient* L/R , and the two are easy to swap under pressure. The unit check settles it in three seconds: ohms times farads is seconds, henries divided by ohms is seconds, and R/L or C/R is neither. The second trap is applying $I = \frac{E}{R}(1 - e^{-t/\tau})$ when $I(0) \neq 0$; use the interpolation form, which handles any initial value.

SEE ALSO Mixing and tank problems; Newton’s law of cooling; first-order linear equations.

A falling body with resistance is the archetype that teaches sign discipline, because the drag force reverses with the motion and the two natural resistance laws produce two entirely different functions. Both are separable; only one is also linear.

MEDIUM

39.8 Falling bodies with drag and terminal velocity

CUE “Terminal velocity”, “air resistance proportional to speed” or “to the square of the speed”, a parachute, a raindrop, a body launched upward against resistance.

METHOD Take down as positive and write $mv' = mg - F_{\text{drag}}$.

Linear drag, $F = bv$. The equation $mv' = mg - bv$ is both linear and separable, with terminal velocity $v_{\infty} = mg/b$ and

$$v(t) = v_{\infty}(1 - e^{-bt/m}) = v_{\infty}(1 - e^{-gt/v_{\infty}}).$$

Quadratic drag, $F = bv^2$. Now $mv' = mg - bv^2$ separates with $v_{\infty} = \sqrt{mg/b}$ to give

$$v(t) = v_{\infty} \tanh\left(\frac{gt}{v_{\infty}}\right), \quad y(t) = \frac{v_{\infty}^2}{g} \ln \cosh\left(\frac{gt}{v_{\infty}}\right).$$

In both cases v_{∞} is read off without solving anything, by setting $v' = 0$ in the equation. Do that first: it is one line and it is frequently the entire question.

WORKED EXAMPLE

$v' = 9.8 - 0.2v$ with $v(0) = 0$. Setting $v' = 0$ gives $v_{\infty} = 9.8/0.2 = 49$ m/s immediately, and the time constant is $m/b = 1/0.2 = 5$ s, so

$$v(t) = 49(1 - e^{-t/5}), \quad v(5) = 49(1 - e^{-1}) = 30.9739 \text{ m/s}.$$

Runge-Kutta integration returns 30.973907, agreeing to six decimals.

QUADRATIC DRAG, AND WHY THE APPROACH IS FASTER

A skydiver with $b/m = 0.002 \text{ m}^{-1}$ has $v_{\infty} = \sqrt{9.8/0.002} = \sqrt{4900} = 70$ m/s. Then

$$v(t) = 70 \tanh(0.14t), \quad v(10) = 70 \tanh(1.4) = 70 \times 0.885352 = 61.97 \text{ m/s},$$

which is 88.5% of terminal after ten seconds. Compare a linear-drag model tuned to the same terminal speed: $v = 70(1 - e^{-gt/70}) = 70(1 - e^{-0.14t})$ reaches only $70(1 - e^{-1.4}) = 52.75$ m/s, or 75.3%. Quadratic drag is weak at low speed and therefore lets the body accelerate closer to terminal before biting, so $\tanh$ approaches its asymptote faster than the exponential does. If a problem gives you the fraction of terminal velocity at a stated time, that fraction alone tells you which drag law was intended.

WHERE THIS GOES WRONG

Keeping the form $-bv^2$ while the object is moving *upward*. Drag opposes motion, so the honest force is $-bv|v|$, and on the way up, with down still positive, $v < 0$ and the drag term is $+bv^2$: the equation becomes $mv' = mg + bv^2$ on the ascent and $mv' = mg - bv^2$ on the descent. These are genuinely different equations with genuinely different solutions ($\tan$ on the way up, $\tanh$ on the way down), and one consequence is that a projectile launched with speed u returns with speed strictly less than u . Solve the two phases separately and match at the apex.

SEE ALSO The Tsiolkovsky rocket equation; RC and RL circuits; separable equations.

The last mechanical archetype is the one where the naive translation is wrong, and the reason it is wrong is worth more than the formula. When mass leaves the body, $F = ma$ is no longer the correct statement of Newton's law, and the correct starting point is a momentum balance over the body plus the matter it has just ejected.

39.9 The Tsiolkovsky rocket equation

CUE A body whose mass changes as it moves: a rocket burning fuel, a chain being picked up off a table, a raindrop accreting vapour.

METHOD Work with momentum over a short interval, comparing the system before and after the ejection. In free space, with exhaust speed u relative to the rocket, conservation of momentum gives $m dv = -u dm$, which separates immediately:

$$v - v_0 = u \ln \frac{m_0}{m}.$$

Under constant gravity and a constant burn rate $|\dot{m}|$, the same balance gives $m \frac{dv}{dt} = -u\dot{m} - mg$, whose solution is the rocket equation minus a gravity loss:

$$v(t) = u \ln \frac{m_0}{m(t)} - gt.$$

WORKED EXAMPLE

A single stage with exhaust speed $u = 3000$ m/s is to deliver a velocity change of 9000 m/s. Then $\ln(m_0/m_f) = 3$, so $m_0/m_f = e^3 = 20.0855$ and the final mass is 4.98% of the initial mass. Since structural mass alone is typically eight to ten percent of a stage, this calculation is the reason single-stage-to-orbit vehicles are hard and staging exists: the logarithm makes velocity expensive in a way no engineering margin can absorb.

The special case worth memorising is $\Delta v = u$, which needs $m_f = m_0/e$, so the fuel fraction is

$$1 - \frac{1}{e} = 0.632121 = 63.21\%.$$

WHERE THIS GOES WRONG

Starting from $F = \frac{d(mv)}{dt}$ and expanding it as $m\dot{v} + \dot{m}v$. That expression is not frame-independent when mass crosses the system boundary: evaluate it in a frame moving with the rocket and you get a different answer, which is proof that it is not a physical law. The correct relation uses the exhaust velocity relative to the rocket, and the resulting thrust term is $-u\dot{m}$ with $\dot{m} < 0$, hence positive. Any derivation that never mentions the word “relative” has gone wrong.

SEE ALSO Falling bodies with drag; separable equations; logarithmic differentiation.

Models that arrive as geometry

The remaining two archetypes are not physical balances at all. Their statements are geometric conditions on a curve, and the translation step converts a condition about tangent lines into a condition about y' . That conversion is the whole difficulty, and the equations that result are usually easy.

39.10 Orthogonal trajectories

CUE “Find the family of curves cutting the given family at right angles”, or a physical pairing that is always orthogonal: isotherms against heat-flow lines, equipotentials against field lines, streamlines against velocity potentials.

METHOD Three steps, and the middle one is the one people skip.

1. Differentiate the family $F(x, y, C) = 0$ implicitly.
2. Eliminate C between the original equation and its derivative, producing a slope field $y' = f(x, y)$ in x and y alone.
3. Solve $y' = -1/f(x, y)$.

The elimination is what makes the slope field a property of the geometry rather than of the labelling, and until it is done the negative reciprocal means nothing.

WORKED EXAMPLE

The family $y = Cx^2$ of parabolas through the origin. Differentiating, $y' = 2Cx$; eliminating $C = y/x^2$ gives the slope field

$$y' = \frac{2y}{x}.$$

The orthogonal family therefore solves $y' = -\frac{x}{2y}$, which separates: $2y \, dy = -x \, dx$, so $y^2 + \frac{1}{2}x^2 = k$, a family of ellipses centred at the origin with their major axes along $y = 0$. Every one of them meets every parabola at right angles, and the two families together are the standard picture of a dipole-like flow.

WHEN LOG-DIFFERENTIATION DOES THE ELIMINATION FOR YOU

Take $e^y \tan\left(\frac{x}{2} + \frac{\pi}{4}\right) = C$. Taking logarithms first turns the product into a sum and makes C vanish under differentiation without any algebra:

$$y + \ln \tan\left(\frac{x}{2} + \frac{\pi}{4}\right) = \ln C \implies y' + \frac{\frac{1}{2} \sec^2 u}{\tan u} = 0, \quad u = \frac{x}{2} + \frac{\pi}{4}.$$

Now $\frac{\sec^2 u}{2 \tan u} = \frac{1}{2 \sin u \cos u} = \frac{1}{\sin 2u} = \frac{1}{\sin(x + \frac{\pi}{2})} = \sec x$, so the slope field is $y' = -\sec x$ with no C anywhere.

The orthogonal family solves $y' = \cos x$, hence

$$y = \sin x + C_1.$$

Numerically differentiating the original family at $x = 0.7$ with $C = 2$ returns -1.30729 , and $-\sec(0.7) = -1.30729$: the elimination is verified before the reciprocal is taken.

WHERE THIS GOES WRONG

Taking the negative reciprocal before eliminating C . The expression $y' = 2Cx$ describes one curve per value of C , and $-1/(2Cx)$ is the perpendicular direction along that one curve, not a field on the plane; solving it produces a one-parameter family for each C , which is one parameter too many and answers nothing. A second, quieter trap: points where $f = 0$ or f is infinite are where the two families have horizontal and vertical tangents, and the orthogonal equation is singular there. Note them rather than dividing through them.

SEE ALSO Pursuit curves and the tractrix; exact equations and integrating factors; implicit general solutions.

Pursuit problems are the same kind of translation with a moving constraint instead of a static one. The constraint is always that the tangent to the curve points at something, and turning that sentence into an equation in y' is the entire modelling step.

HARD

39.11 Pursuit curves and the tractrix

CUE One object always heading directly at another; an object dragged by a taut string of fixed length; a dog chasing along a bank; a missile with pure pursuit guidance.

METHOD Write the heading condition as a slope, then differentiate to eliminate the pursued object's position.

Tractrix. The tangent segment from the curve down to the x -axis has constant length a . If the curve is $y(x)$ then the horizontal run of that segment is $\sqrt{a^2 - y^2}$, so

$$\frac{dy}{dx} = -\frac{y}{\sqrt{a^2 - y^2}},$$

which separates. The solution, with the cusp at $(0, a)$, is

$$x = a \ln \frac{a + \sqrt{a^2 - y^2}}{y} - \sqrt{a^2 - y^2}.$$

True pursuit. Target moving up the y -axis from the origin at speed v , pursuer starting at $(a, 0)$ with speed w , always heading at the target. Writing $k = v/w$ and eliminating time gives

$$x \frac{d^2 y}{dx^2} = k \sqrt{1 + \left(\frac{dy}{dx}\right)^2},$$

a second-order equation in which y does not appear, so $p = y'$ reduces it to first order.

Differentiating the closed form with respect to y and writing $s = \sqrt{a^2 - y^2}$,

$$\frac{dx}{dy} = a \left(\frac{-y/s}{a+s} - \frac{1}{y} \right) + \frac{y}{s} = \frac{-ay^2 - as(a+s) + y^2(a+s)}{ys(a+s)} = \frac{-s^2(a+s)}{ys(a+s)} = -\frac{s}{y},$$

using $y^2 = a^2 - s^2$ to collapse the numerator. Inverting, $\frac{dy}{dx} = -y/s$, which is the equation. Numerically, at $y = 0.6$ with $a = 1$ the closed form gives $\frac{dy}{dx} = -0.75$ and $-y/\sqrt{1-y^2} = -0.6/0.8 = -0.75$. The curve meets the y -axis at $y = a$ with a vertical cusp and approaches the x -axis asymptotically as $y \rightarrow 0$.

PURSUIT, AND WHERE THE CAPTURE HAPPENS

Take the pursuit equation with $p = y'$ and $p(a) = 0$, since at $t = 0$ the target is at the origin and the pursuer at $(a, 0)$ is heading along the x -axis. Then

$$x \frac{dp}{dx} = k \sqrt{1+p^2} \Rightarrow \frac{dp}{\sqrt{1+p^2}} = k \frac{dx}{x} \Rightarrow \operatorname{arcsinh} p = k \ln \frac{x}{a},$$

so $p = \sinh(k \ln(x/a)) = \frac{1}{2}[(x/a)^k - (x/a)^{-k}]$. Integrating once more and imposing $y(a) = 0$,

$$y = \frac{a}{2} \left[\frac{(x/a)^{k+1}}{k+1} - \frac{(x/a)^{1-k}}{1-k} \right] + \frac{ak}{1-k^2}.$$

If $k < 1$, so the pursuer is faster, both powers vanish as $x \rightarrow 0$ and capture occurs on the y -axis at

$$y_{\text{capture}} = \frac{ak}{1-k^2}.$$

With $k = \frac{1}{2}$ and $a = 1$ that is $\frac{1/2}{3/4} = \frac{2}{3}$. If $k = 1$ the second term integrates to a logarithm instead and $y \rightarrow \infty$: the pursuer closes on the target but never catches it, converging to a fixed lag of $a/2$. If $k > 1$ the pursuer is slower and never gets there at all.

WHERE THIS GOES WRONG

Setting up the geometry with the wrong sign, so that the pursuer flees. In the tractrix the trailing object is always *behind*, so y decreases as x increases and $y' < 0$; a derivation returning $y' = +y/\sqrt{a^2 - y^2}$ has the string pushing rather than pulling. Sketch the configuration and check the sign of y' against the picture before integrating, because both signs integrate perfectly well and only one describes the problem.

SEE ALSO Orthogonal trajectories; reduction of order when the dependent variable is absent; separable equations.

THE SEVEN SHAPES, AND HOW TO TELL THEM APART IN ONE READING

Every closed form in this chapter is one of seven functions, and the wording of the problem picks it out before any algebra. *Pure exponential* $N_0 e^{-\lambda t}$: proportional decay with nothing else happening. *Shifted exponential* $X_\infty + (X_0 - X_\infty)e^{-t/\tau}$: anything approaching an equilibrium, which covers cooling, RC and RL with a constant source, constant-volume mixing, and decay against a source. *Logistic* $K/(1 + Ae^{-rt})$: growth against a ceiling. *Hyperbolic tangent* $v_\infty \tanh(gt/v_\infty)$: quadratic drag. *Rational or power* $2(50 + t) - c(50 + t)^{-2}$: mixing with changing volume, where the integrating factor is a power of V . *Quadratic reaching zero* $(\sqrt{H} - \frac{k}{2}t)^2$: gravity-driven draining, and the only member of the list that finishes in finite time. *Logarithm* $u \ln(m_0/m)$: variable mass. If your answer is not one of these seven and the problem was not a compartment system, recheck the translation rather than the integration.

STEADY STATE FIRST, ALWAYS

In every autonomous model here, setting the derivative to zero and solving the resulting algebraic equation costs one line and yields the terminal value: T_a for cooling, K for logistic, $c_{\text{in}}V$ for constant-volume mixing, E/R for the RL current, $\sqrt{mg/b}$ for quadratic drag, $-A^{-1}\mathbf{b}$ for a compartment system. Do this before solving, for three reasons. It is often the entire question, since “find the terminal velocity” and “find the limiting concentration” never require the transient. It gives you the substitution $U = X - X_\infty$ that reduces the equation to the homogeneous one. And it is a check on the closed form you eventually produce: if $\lim_{t \rightarrow \infty} X(t)$ does not match the equilibrium you computed in one line, the solution is wrong and you know it before the marker does.

Recognition matrix

If the problem says	Write	Then
"Half-life", "carbon dating"	$N = N_0 2^{-t/t_{1/2}}$	Never compute λ
"Proportional to the amount present"	$N' = -\lambda N$, pure exponential	Check whether a source term is also present
Constant production and decay	$N' = R - \lambda N$; equilibrium R/λ	Substitute $U = N - R/\lambda$
Cooling or heating toward surroundings	$U = T - T_a$ decays exponentially	Ratios of <i>excess</i> , never of T
Three readings, ambient unknown	$T_a = \frac{T_0 T_2 - T_1^2}{T_0 + T_2 - 2T_1}$	Verify the excesses are geometric
Two readings at equal spacing	Excess multiplies by a fixed ρ	Answer in powers of ρ ; skip k
"Levels off at a carrying capacity"	Logistic; odds $\frac{P}{K-P}$ grow as e^{rt}	$P = \frac{K}{1 + W_0^{-1} e^{-rt}}$
"Fastest growth occurred in year t_1 "	That is the inflection: $P(t_1) = K/2$	Use it as the third data point
Well-stirred tank, brine in and out	$A' = r_{in} c_{in} - \frac{r_{out}}{V(t)} A$	Track amount, not concentration
Inflow rate $\neq$ outflow rate	$V(t) = V_0 + \delta t$; factor $(V_0 + \delta t)^{r_{out}/\delta}$	Note the time the tank fills or empties
Two or more connected reservoirs	One balance equation each; $\mathbf{x}' = A\mathbf{x} + \mathbf{b}$	Closed system: column sums 0, $\lambda = 0$
Symmetric two-tank exchange	Solve by sum and difference	Relaxation rate $2r/V$
Drug infused, blood and tissue	$b^* = I/k_e$, independent of exchange	Exchange sets the speed, not the plateau
Draining through a hole	$A(h)h' = -ka\sqrt{h}$; $\sqrt{h}$ affine in t	Stop at t_{empty} ; then $h \equiv 0$
"Time constant", one storage element	RL: $\tau = L/R$; RC: $\tau = RC$	Unit check: both must be seconds
Step source, any first-order model	$X_\infty + (X_0 - X_\infty)e^{-t/\tau}$	No integration needed
Sinusoidal source, RL loop	Amplitude $E_0/\sqrt{R^2 + \omega^2 L^2}$	Phase lag $\arctan(\omega L/R)$
"Terminal velocity"	Set $v' = 0$ in the equation	One line; often the whole question
Resistance proportional to v	$v = v_\infty(1 - e^{-gt/v_\infty})$, $v_\infty = mg/b$	Linear <i>and</i> separable
Resistance proportional to v^2	$v = v_\infty \tanh(gt/v_\infty)$, $v_\infty = \sqrt{mg/b}$	Distance: $\frac{v_\infty^2}{g} \ln \cosh$
Body thrown <i>upward</i> against drag	Drag is $-b v v $: two different equations	$\tan$ up, $\tanh$ down; match at apex
Mass changes as the body moves	$m dv = -u dm$, so $v = u \ln(m_0/m)$	Never expand $\frac{d(mv)}{dt}$
Rocket under gravity, constant burn	$v = u \ln(m_0/m) - gt$	The $-gt$ is the gravity loss
"Cuts the family at right angles"	Differentiate, <i>eliminate</i> C , then $y' \mapsto -1/y'$	A leftover C voids the answer
Family is a product of factors	Log-differentiate: C disappears free	Then simplify before inverting
"Always heading directly at"	Tangent points at the target; then $p = y'$	Capture at $ak/(1 - k^2)$ if $k < 1$
Tangent segment of fixed length a	Tractrix: $y' = -y/\sqrt{a^2 - y^2}$	Check the sign against a sketch

Existence, uniqueness, and lost solutions

What this chapter is for. Every previous chapter in this part produced a formula. This one asks whether the formula is an answer. Three separate things can go wrong, and they go wrong at three different points in the working. A solution can fail to exist, or fail to be unique, so that the question itself was malformed. A solution can exist on a smaller interval than the equation suggests, so that a perfectly smooth problem blows up in finite time and your closed form is valid only up to that instant. And a solution can be quietly destroyed by your own algebra, most often by dividing through by an expression that is allowed to vanish. We cover Picard and Lindelof and what Lipschitz continuity actually buys, the interval of validity, singular and envelope solutions, the discipline of applying an initial condition at the right moment, and then the boundary-value flavour, where the count of solutions is zero, one, or infinite depending on a sign, and where that trichotomy generates the eigenvalue problems that the whole of separation of variables rests on.

When a solution exists, and when it is the only one

An initial-value problem is a promise that the future is determined by the present. The promise is not free. It holds under hypotheses that are easy to state and easy to check, and it fails in a way that is easy to recognise once you have seen it once.

40.1 Picard and Lindelof: continuity gives existence, Lipschitz gives uniqueness

MEDIUM

CUE Any initial-value problem $y' = f(x, y)$, $y(x_0) = y_0$, where you are asked whether the solution is unique, or where the answer you found does not look like the only possibility. The strongest cue is a right-hand side with a fractional power of y , such as $y^{1/3}$ or $\sqrt{|y|}$, at a point where $y = 0$.

METHOD Continuity of f near (x_0, y_0) gives at least one solution on some interval. Uniqueness needs more: f must be Lipschitz in y , meaning $|f(x, y_1) - f(x, y_2)| \leq L|y_1 - y_2|$ near the point, and in practice you check the sufficient condition that $\frac{\partial f}{\partial y}$ exists and is bounded there. The two hypotheses do different jobs and it is worth keeping them apart.

$$f \text{ continuous} \implies \text{a solution exists}; \quad \frac{\partial f}{\partial y} \text{ bounded near } (x_0, y_0) \implies \text{it is the only one.}$$

WORKED EXAMPLE

Consider $y' = 3y^{2/3}$ with $y(0) = 0$. Separating gives $y^{-2/3} dy = 3 dx$, so $3y^{1/3} = 3x + C$ and $y = (x + c)^3$. With the initial condition, $y = x^3$. But $y \equiv 0$ also solves the problem, and so does the hybrid

$$y(x) = \begin{cases} 0, & x \leq a, \\ (x - a)^3, & x > a, \end{cases}$$

for every $a \geq 0$, which is continuously differentiable at $x = a$ because both the value and the slope vanish there. The problem has infinitely many solutions. Nothing is wrong with the separation; what is wrong is the question. Here $f(x, y) = 3y^{2/3}$ is continuous everywhere, so existence holds, but $\frac{\partial f}{\partial y} = 2y^{-1/3}$ is unbounded as $y \rightarrow 0$, so uniqueness fails at exactly the point the initial condition names.

WHERE THIS GOES WRONG

Testing the Lipschitz condition at a generic point instead of at the initial point. The failure in the example above is invisible at $y = 1$ and total at $y = 0$. Check the hypothesis where the initial condition sits, not where the algebra is comfortable.

SEE ALSO Lost solutions from division; singular and envelope solutions; the interval of validity.

The theorem is local, and the word local carries the second failure mode. It promises a solution on some interval around x_0 , and says nothing about how long that interval is. For linear equations the interval is as long as the coefficients are

continuous, which makes linear problems well behaved and slightly misleading. For nonlinear problems the solution can end while the equation is still perfectly smooth.

MEDIUM

40.2 The interval of validity

CUE A nonlinear initial-value problem whose closed-form solution has a denominator, a logarithm, or an even root that can go bad at a finite x , even though f itself is a polynomial. Also any question phrased as “for how long” or “on what interval is the solution defined”.

METHOD Solve, then find the largest open interval containing x_0 on which the closed form is defined and satisfies the equation. The answer depends on the initial condition, not only on the equation, which is the point that surprises people.

WORKED EXAMPLE

Take $y' = y^2$, $y(0) = y_0 > 0$. Separation gives $-1/y = x + C$, so $y = y_0/(1 - y_0x)$. The right-hand side of the equation is a polynomial, defined and smooth on the whole plane, and yet the solution blows up at $x = 1/y_0$ and exists only on $(-\infty, 1/y_0)$. Doubling the initial value halves the lifetime. If instead $y_0 < 0$ the solution exists for all $x > 0$ and decays to zero, so the same equation is globally solvable or not depending entirely on where you start.

WHERE THIS GOES WRONG

Reporting the closed form without its interval, then evaluating it on the far side of the singularity. The formula $y_0/(1 - y_0x)$ is a perfectly good function at $x = 2/y_0$; it is simply not the solution there, since no solution exists there.

SEE ALSO Picard and Lindelof; implicit general solutions; separable equations and the constant-solution check.

The solutions your algebra destroyed

The commonest error in this whole part is not a mistake in calculus. It is a division.

EASY

40.3 Lost solutions from division

CUE Any separation in which you divided by a function of y , or any cancellation of a factor that is not identically nonzero. The signature is a general solution that visibly cannot produce a constant that the equation obviously admits.

METHOD Before dividing by $g(y)$, solve $g(y) = 0$. Each root $y = r$ gives a constant function; check whether it satisfies the original equation, and if it does, record it as a solution of the problem. It will usually not be recoverable from the general solution by any choice of the constant, which is precisely why it must be recorded separately.

WORKED EXAMPLE

Solve $\frac{dy}{dx} = y(1 - y)$. Dividing by $y(1 - y)$ and integrating gives the logistic family $y = (1 + Ae^{-x})^{-1}$. But the division required $y \neq 0$ and $y \neq 1$, and both constants satisfy the equation. The solution $y \equiv 1$ is the limit $A \rightarrow 0$ and might be forgiven as a boundary case; the solution $y \equiv 0$ is not obtainable from the family for any finite A , because the family is never zero. The full solution set is the logistic family together with $y \equiv 0$ and $y \equiv 1$.

WHERE THIS GOES WRONG

Writing $|y|$ inside the logarithm and then dropping the absolute value when exponentiating without noting that the sign is absorbed into the arbitrary constant. That step is legitimate, but only because $\pm e^C$ ranges over all nonzero reals, and it is the zero that gets lost.

SEE ALSO Separable equations and the constant-solution check; singular and envelope solutions; equilibria on the phase line.

HARD

40.4 Singular and envelope solutions

CUE A first-order equation that is nonlinear in y' , most visibly a Clairaut or Lagrange equation, whose general solution is a family of straight lines. Also any problem where the general solution is a one-parameter family of curves and the question asks for “all solutions”.

METHOD A singular solution is one not obtainable from the general solution by any value of the constant. For a family $F(x, y, C) = 0$ the candidate is the envelope, found by eliminating C between $F = 0$ and $\frac{\partial F}{\partial C} = 0$; then verify by substitution that the eliminant really solves the equation. Geometrically the envelope is tangent to every member of the family, which is why it satisfies the same first-order relation at each of its points.

The Clairaut equation $y = xp + p^2$, where $p = y'$, has general solution $y = Cx + C^2$, a family of lines. Differentiating with respect to C gives $0 = x + 2C$, so $C = -x/2$, and substituting back gives $y = -x^2/2 + x^2/4 = -x^2/4$. Check: $y' = -x/2$, and $xy' + (y')^2 = -x^2/2 + x^2/4 = -x^2/4 = y$. The parabola is a genuine solution and belongs to no line of the family.

Producing the eliminant and declaring victory without substituting it back. The envelope construction is a necessary condition on the candidate, not a proof that it solves anything; for Lagrange equations it frequently produces curves that fail.

SEE ALSO Clairaut and Lagrange equations; lost solutions from division; implicit general solutions.

Where in the working the condition goes

An initial condition is a constraint on a solution, and the order in which you impose it changes how much work the rest of the problem takes and, in a few cases, whether the answer is right at all.

40.5 Applying the initial condition at the right stage

CUE A separable or exact equation with an initial condition, where the general solution is implicit or involves a logarithm, and where the constant is awkward to isolate.

METHOD Impose the condition at the earliest point at which it is legitimate, which is normally immediately after the first integration, and prefer definite integrals with the initial point as the lower limit. Writing

$$\int_{y_0}^y \frac{dt}{g(t)} = \int_{x_0}^x h(s) ds$$

eliminates the constant entirely and removes both the sign ambiguity of the logarithm and the temptation to solve for y before you need to.

Imposing the condition after squaring, after exponentiating, or after taking a root. Each of those steps can introduce a branch, and the condition is what selects the branch; used afterwards, it cannot. If the general solution is $y^2 = x^2 + C$ and $y(0) = -3$, the answer is $y = -\sqrt{x^2 + 9}$, and a solver who fixed $C = 9$ and then wrote the positive root has produced a function that does not satisfy the condition it was given.

SEE ALSO Implicit general solutions; the interval of validity; lost solutions from division.

40.6 Leaving the general solution implicit

CUE An answer of the form $F(x, y) = C$ where solving for y requires a root, a Lambert function, or a case split that the problem never asked for.

METHOD Leave it implicit, state the interval, and if an initial condition is given, substitute it to fix C rather than to solve for y . An implicit relation together with an initial point is a complete answer; a wrong explicit branch is not. Solve explicitly only when the question asks for a value of y , when you need to differentiate the answer, or when the branch is forced and easy to name.

The exact equation $(2xy + \cos y) dx + (x^2 - x \sin y + 2y) dy = 0$ has potential $F = x^2y + x \cos y + y^2$, so the general solution is $x^2y + x \cos y + y^2 = C$. With $y(1) = 0$ we get $C = 1$. There is no elementary way to solve for y , and none is needed: the relation, plus the point, plus the observation that $\frac{\partial F}{\partial y} = x^2 - x \sin y + 2y$ equals 1 at $(1, 0)$ and so is nonzero nearby, tells you a unique differentiable branch through that point exists by the implicit function theorem.

SEE ALSO Exact equations and potential functions; applying the initial condition at the right stage.

Boundary conditions and the eigenvalue trichotomy

Replacing two conditions at one point by one condition at each of two points changes the character of the problem completely. Existence and uniqueness stop being generic. The same equation, with the same boundary points, can have no

solution, exactly one, or a whole family, and which of the three happens is decided by an arithmetic condition on a parameter.

40.7 Counting solutions of a two-point boundary-value problem

MEDIUM

CUE Two conditions imposed at different values of the independent variable, typically $y(0) = \alpha$ and $y(L) = \beta$.

METHOD Write the general solution with its two constants, impose both conditions, and read off the resulting 2×2 linear system. If the coefficient matrix is invertible there is exactly one solution. If it is singular, there is either no solution or a one-parameter family, decided by whether the right-hand side lies in the column space.

WORKED EXAMPLE

For $y'' + y = 0$ the general solution is $y = A \cos x + B \sin x$. Impose $y(0) = 0$ and $y(\pi) = \beta$. The first gives $A = 0$; the second gives $B \sin \pi = 0 = \beta$. So if $\beta \neq 0$ there is no solution at all, and if $\beta = 0$ every $y = B \sin x$ works, an infinite family. Move the right endpoint to $\pi/2$ and the same equation has exactly one solution $y = \beta \sin x$ for every β . The equation did not change; the interval did.

WHERE THIS GOES WRONG

Assuming that because the equation is linear and smooth, a solution must exist. That intuition is imported from initial-value problems and is simply false here.

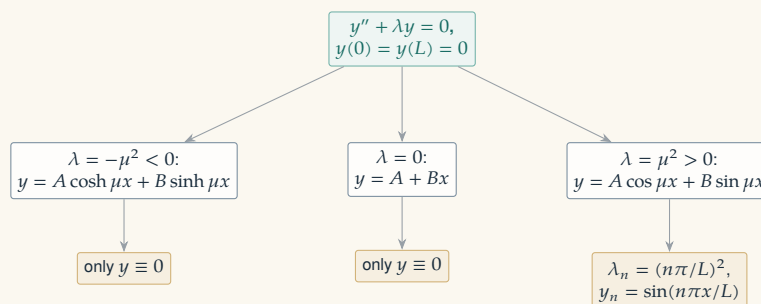
SEE ALSO The three sign cases for the separation constant; Sturm and Liouville problems.

40.8 The three sign cases for the separation constant

MEDIUM

CUE An eigenvalue problem $y'' + \lambda y = 0$ with homogeneous boundary conditions, arising from separation of variables in a partial differential equation, and the instruction to find all λ for which a nontrivial solution exists.

METHOD Do not assume a sign. Treat $\lambda < 0$, $\lambda = 0$ and $\lambda > 0$ as three separate computations, because the general solution has a different shape in each. Only in the oscillatory case do the boundary conditions have infinitely many chances to be satisfied, which is why the nontrivial eigenvalues live there.



WORKED EXAMPLE

With $y(0) = y(L) = 0$: in the negative case $y(0) = 0$ forces $A = 0$ and $y(L) = B \sinh \mu L = 0$ forces $B = 0$, since $\sinh$ vanishes only at zero. In the zero case $y = A + Bx$ with both endpoints zero forces $A = B = 0$. In the positive case $y(0) = 0$ forces $A = 0$ and $y(L) = B \sin \mu L = 0$ is satisfied without killing B whenever $\mu L = n\pi$. Hence $\lambda_n = n^2 \pi^2 / L^2$ with eigenfunctions $\sin(n\pi x / L)$, $n = 1, 2, 3, \dots$

WHERE THIS GOES WRONG

Writing the negative case as $y = A \cos \mu x + B \sin \mu x$ with imaginary μ and then treating $\sin$ as though it still had zeros. Use the real hyperbolic form and the argument is immediate.

SEE ALSO Counting solutions of a two-point boundary-value problem; Sturm and Liouville problems.

40.9 Sturm and Liouville structure

CUE A second-order eigenvalue problem with variable coefficients, written or writable as $(p(x)y')' + q(x)y + \lambda w(x)y = 0$ on $[a, b]$ with separated homogeneous boundary conditions, and $p, w > 0$ on the open interval.

METHOD Put the equation into that self-adjoint form first, multiplying by the integrating factor $\mu = \frac{1}{p_2} \exp\left(\int \frac{p_1}{p_2} dx\right)$ if it is given as $p_2 y'' + p_1 y' + p_0 y + \lambda w y = 0$. The structure then guarantees three facts you may use without proof: the eigenvalues are real and form an increasing sequence tending to infinity, the n th eigenfunction has exactly $n - 1$ interior zeros, and eigenfunctions for distinct eigenvalues are orthogonal with respect to the weight,

$$\int_a^b y_m(x) y_n(x) w(x) dx = 0 \quad (m \neq n).$$

That orthogonality is the entire reason expansions in eigenfunctions work: it is what lets you extract a coefficient by integrating against a single mode.

WORKED EXAMPLE

Bessel's equation $x^2 y'' + xy' + (\lambda x^2 - n^2)y = 0$ on $(0, b)$ becomes, after dividing by x , the self-adjoint form $(xy')' - \frac{n^2}{x}y + \lambda xy = 0$, so $p = x$, $q = -n^2/x$ and the weight is $w = x$. The orthogonality relation for the eigenfunctions $J_n(\sqrt{\lambda_k} x)$ therefore carries a factor of x in the integrand, which is exactly the factor that appears in the Fourier and Bessel coefficient formula and which is otherwise mysterious.

WHERE THIS GOES WRONG

Applying the plain unweighted orthogonality $\int y_m y_n = 0$ to an equation whose weight is not 1. It is false, and the coefficients extracted from it are wrong.

SEE ALSO The three sign cases for the separation constant; series solutions about a regular singular point.

Recognition matrix

If you see	Reach for	If that fails
f has a fractional power of y	Check Lipschitz at the initial point	Expect a family of solutions
Nonlinear, closed form has a pole	State the interval of validity	Solve for the blow-up time
You divided by $g(y)$	Solve $g(y) = 0$; test each constant	Compare against the general family
General solution is a line family	Envelope: eliminate C from $F = \partial_C F = 0$	Verify by substitution
Implicit answer, ugly to invert	Leave it implicit; fix C from the point	Use the implicit function theorem
Condition given, root or square taken	Impose the condition before the branch	Definite integrals from (x_0, y_0)
Conditions at two different points	Build the 2×2 system; test invertibility	Singular: zero or infinitely many
$y'' + \lambda y = 0$, homogeneous ends	Three sign cases, separately	Oscillatory case carries the spectrum
Variable coefficients, eigenvalue asked	Self-adjoint form; read off p and w	Orthogonality is weighted by w
Expansion coefficients look wrong	Check the weight in the inner product	Recompute the normalisation

PART VI

Logic and puzzle technique

Formal logic as a working tool

What this chapter is for. Almost every logic puzzle arrives written in English, and English is ambiguous in ways that matter: “if” sometimes means “if and only if”, “or” sometimes excludes, “not both” is routinely read as “neither”. The formal apparatus of propositional and predicate logic earns its place here for one reason only, which is that it converts those sentences into objects you can compute with. A truth table is a finite computation. A contrapositive is a rewrite that preserves meaning exactly. De Morgan’s laws and the quantifier negation rules are a mechanical procedure for finding out what it would take to refute a claim, which is very often the only usable handle a problem offers. This chapter covers exhaustion over assignments and the point at which it stops being practical, the four forms a conditional can take and which of them you actually own, negation normal form for compound and quantified statements, the material conditional together with the two traps it sets, and the discipline of separating necessary from sufficient when reading a problem statement. By the end you should be able to take a paragraph of puzzle English, write down the exact propositional content, and know before starting whether the question is one of satisfiability, of entailment, or of optimality.

Exhaustion, and where it stops

The cheapest thing you can do with a set of propositional constraints is to try every assignment. It is worth being precise about what that computation returns, because three different questions are routinely asked of the same table and they have three different answers.

Fix atomic propositions $p_1, \dots, p_n$ and constraints $C_1, \dots, C_m$, and write $\varphi = C_1 \wedge \dots \wedge C_m$. The table of 2^n rows partitions the assignments into those satisfying φ (call them the *survivors*) and those falsifying it. *Satisfiability* asks whether at least one survivor exists. *Entailment*, written $\varphi \models \psi$, asks whether every survivor also satisfies ψ ; this is not a question about any single row. *Determination* asks whether all survivors agree on the value of some particular atom, which is the question a puzzle means when it asks “who is the knight?”. A puzzle can be satisfiable and yet determine nothing, and the difference between those two outcomes is the difference between a well-posed puzzle and a badly posed one.

EASY

41.1 Truth-table exhaustion

CUE A finite number of atomic propositions, in practice $n \leq 4$ by hand and $n \leq 20$ by machine, tied together by compound constraints, and a question of the form “is this consistent”, “does this follow”, or “which assignments survive”. Also any puzzle whose entire content is the joint satisfiability of a list of English sentences.

METHOD Translate each sentence into a formula over the atoms, enumerate all 2^n assignments, and keep the rows on which every constraint evaluates to true. Read the answer off the surviving set: nonempty means consistent, a single row means fully determined, and $\varphi \models \psi$ holds exactly when no survivor falsifies ψ . The one operation that is never legitimate is inspecting a favourable row and stopping.

WORKED EXAMPLE

Three relays p, q, r are wired so that $p \rightarrow q$, $q \rightarrow r$, and $\neg(p \wedge r)$. Which configurations are possible, and does it follow that p is off?

Take the cases on p . If p is true then q is true by the first constraint and r is true by the second, so $p \wedge r$ holds and the third constraint fails. Hence p is false in every survivor, and with p false the first constraint is automatically satisfied, leaving $q \rightarrow r$ as the only live requirement. The survivors are therefore

$$(p, q, r) = (F, F, F), \quad (F, F, T), \quad (F, T, T),$$

three rows out of eight, confirmed by enumerating all eight assignments in `python3`. So $\varphi \models \neg p$: the relay p is off in every consistent configuration. Note what does *not* follow. Two survivors have q false and one has q true, so $\varphi \not\models \neg q$ and $\varphi \not\models q$: the state of q is undetermined, and any answer that names it is wrong.

Two failures, and they are opposite. The first is exhibiting one consistent row and reporting it as the answer, which establishes satisfiability and nothing more. The second is finding one row on which the conclusion happens to hold and calling that entailment; entailment is a statement about every survivor, so it is refuted by a single survivor and confirmed by none.

SEE ALSO De Morgan normalisation; the Boolean encoding of a truthful or lying speaker; exhaustive casework with a discipline.

The table is a complete decision procedure, which is exactly why it is the wrong tool as soon as the problem has any structure. Doubling n squares the work, so $n = 30$ is already out of reach for a table you write out, and even at $n = 6$ the sixty-four rows are a transcription exercise in which arithmetic slips are more likely than insight. Every remaining technique in this chapter is a way of not building the table: a rewrite that halves the case count, a normalisation that makes the branching visible, or a reading discipline that stops you from formalising the wrong sentence in the first place. The single most valuable of those rewrites is the one that turns a conditional around.

Rewriting a conditional

A conditional $A \rightarrow B$ has three relatives, and puzzles are built on the fact that only one of them is equivalent to it. The equivalence is not a curiosity. It is the standard move for turning an unusable hypothesis into a usable one, because a problem is almost always engineered so that exactly one of A and B is the concrete side.

A	B	$A \rightarrow B$ original	$B \rightarrow A$ converse	$\neg A \rightarrow \neg B$ inverse	$\neg B \rightarrow \neg A$ contrapositive
T	T	T	T	T	T
T	F	F	T	T	F
F	T	T	F	F	T
F	F	T	T	T	T

Read the table by columns, not by rows. The first and fourth columns are identical, which is the whole content of the contrapositive rule. The second and third are identical to each other and differ from the first in the two middle rows, which is why a puzzle that hands you the converse has handed you nothing about the original. And the first column is false in exactly one row, a fact so consequential that it gets its own technique later in this chapter.

EASY

41.2 Contrapositive, converse, inverse

CUE A hypothesis of the form “if A then B ”, and a problem that gives you information about B or about $\neg B$ rather than about A . The moment a puzzle offers you the consequent, stop and ask which direction you actually own.

METHOD Only $\neg B \rightarrow \neg A$ is equivalent to $A \rightarrow B$; the converse $B \rightarrow A$ and the inverse $\neg A \rightarrow \neg B$ are equivalent to each other and to neither. So a rule “ A implies B ” is tested by looking at things that are A and at things that are *not* B , and by nothing else. In symbols,

$$A \rightarrow B \equiv \neg B \rightarrow \neg A \equiv \neg A \vee B,$$

and the last form is the one to use when you want to case-split rather than chain.

WORKED EXAMPLE

A warehouse rule states: every crate bearing a red seal contains glass. Four crates sit on the dock. You can see that the first has a red seal, that the second has a blue seal, that the third contains glass, and that the fourth contains steel; in each case the other fact is hidden. Which crates must be opened or unsealed to test the rule?

Write A for “red seal” and B for “contains glass”; the rule is $A \rightarrow B$. Crate one shows A , so it tests the rule directly: if it holds steel the rule is dead. Crate four shows $\neg B$, so it tests the contrapositive $\neg B \rightarrow \neg A$: if its seal is red the rule is dead. Crate two shows $\neg A$ and crate three shows B , and by the table above neither row can falsify $A \rightarrow B$; whatever they hide, the rule survives. Answer: crates one and four, and only those.

The instructive part is why crate three is so tempting. Opening it can only confirm the rule, and confirmation is not a test. A rule of the form $A \rightarrow B$ is refutable in exactly one way, by an A that is not a B , so the crates worth touching are precisely those that could be that counterexample.

Inspecting the case that could confirm the rule instead of the case that could refute it. This is the most reliable single failure in conditional reasoning, and it survives knowing the truth table, because the pull is toward the row that makes the rule look good. The mechanical guard: before choosing what to examine, write the unique falsifying row A true and B false, and examine only what could occupy it.

SEE ALSO Necessary versus sufficient; proof by contrapositive; the material conditional and vacuous truth.

Everything about the conditional follows from the observation that $A \rightarrow B$ is false in exactly one of its four rows. To *refute* a conditional, produce that row. To *use* a conditional, note that it forbids that row and nothing else. To *prove* a conditional, rule that row out, which you may do by assuming A and deriving B or by assuming $\neg B$ and deriving $\neg A$; those are the two ways of walking into the same forbidden cell. A great many arguments that feel like different methods are this one observation approached from different sides.

Pushing a negation inward

The rewrites of the previous section handle a negation sitting outside a conditional. Negations also sit outside conjunctions, disjunctions and quantifiers, and there the rule is uniform: push the negation in one connective at a time, flipping each connective as it passes, until every negation sits on an atom. The resulting form is called negation normal form, and its virtue is that the case structure of the problem becomes visible on the page instead of remaining hidden inside a $\neg$.

41.3 De Morgan normalisation

EASY

CUE A negated compound, in any of its English disguises: “it is not the case that both”, “none of them”, “at most one of these holds”, “neither”. Also any moment when you want to case-split but the statement offers you no visible disjunction to split on.

METHOD Apply

$$\neg(A \wedge B) \equiv \neg A \vee \neg B, \quad \neg(A \vee B) \equiv \neg A \wedge \neg B,$$

repeatedly, together with $\neg(A \rightarrow B) \equiv A \wedge \neg B$ and $\neg\neg A \equiv A$, until no negation has a compound in its scope. Then read the branching directly: a top-level $\vee$ is a case split, a top-level $\wedge$ is a list of facts you may all use at once.

A label on a box reads: “it is not true both that the prize is in this box and that this label is honest.” Normalise.

Let P be “the prize is in this box” and H “this label is honest”. The label asserts $\neg(P \wedge H)$, which normalises to $\neg P \vee \neg H$: either the prize is elsewhere or the label lies, possibly both. That is two branches, and it is worth seeing that the sentence has not been weakened by the rewrite, only made usable. Now add the self-referential constraint from the next chapter’s encoding, $H \leftrightarrow \neg(P \wedge H)$. If H is true then $\neg(P \wedge H)$ is true, so $P \wedge H$ is false, so P is false. If H is false then $\neg(P \wedge H)$ is false, so $P \wedge H$ is true, so H is true, a contradiction. Answer: the label is honest and the prize is elsewhere, the unique surviving row of four.

“At most one of A_1, A_2, A_3 holds” is a negated existential over pairs: $\neg\exists i < j (A_i \wedge A_j)$, which normalises to $\bigwedge_{i < j} (\neg A_i \vee \neg A_j)$, three clauses. “At least two hold” is its negation, $\bigvee_{i < j} (A_i \wedge A_j)$, three cases. “Exactly one holds” is the conjunction of “at most one” with $A_1 \vee A_2 \vee A_3$. Written this way the three phrases become three different pieces of syntax and stop being confusable, which is the entire practical point of normalising.

Reading “not both” as “neither”. The De Morgan output is an inclusive disjunction, and that inclusive $\vee$ is exactly what makes the puzzle branch instead of collapse. The mirror error is reading “neither” as “not both”, which throws away a constraint: $\neg A \wedge \neg B$ is strictly stronger than $\neg A \vee \neg B$.

SEE ALSO Quantifier negation; truth-table exhaustion; existential claims about the speakers themselves.

The same procedure runs over quantifiers, and there it does more work, because a quantified statement usually cannot be attacked head-on while its negation often has a one-object proof. Knowing exactly what the negation says is therefore not bookkeeping; it is the step that decides whether you are looking for a construction or for an argument over an infinite domain.

41.4 Quantifier negation

CUE Any statement built from “all”, “some”, “no”, “at least one”, “exactly one”, which you need to refute, or which appears inside a negation, or whose failure is the constructive lever the problem wants.

METHOD Push the negation inward one quantifier at a time, flipping each as it passes and leaving the order untouched:

$$\neg \forall x P(x) \equiv \exists x \neg P(x), \quad \neg \exists x P(x) \equiv \forall x \neg P(x).$$

For nested quantifiers apply this outside-in, so that $\neg \forall x \exists y R(x, y) \equiv \exists x \forall y \neg R(x, y)$. The asymmetry is the payoff: refuting a universal costs exactly one witness, whereas refuting an existential costs a proof over the whole domain. Read the negated form first and let it tell you which of those two jobs you have.

WORKED EXAMPLE

Claim: in every red-blue colouring of the edges of the complete graph on five vertices there is a triangle with all three edges the same colour. Negate it. The negation is: there exists a colouring in which every triangle has two colours among its edges. That is an existential over a finite set, so one explicit colouring settles the matter.

Label the vertices 0, 1, 2, 3, 4 and colour the edge $\{i, j\}$ red when $j - i \equiv \pm 1 \pmod{5}$ and blue otherwise. The red edges form a pentagon and the blue edges form the pentagram; neither is a graph containing a triangle, since both are five-cycles. So no triangle is monochromatic and the claim is false. Exhaustive search over all $2^{10} = 1024$ colourings in `python3` confirms this and returns 12 such colourings in all.

Now run the same negation on six vertices. There the negation asserts a colouring of $\binom{6}{2} = 15$ edges with no monochromatic triangle, and exhaustive search over all $2^{15} = 32768$ colourings returns none. So on six vertices the universal claim is true, and the negation has told us that no witness exists rather than exhibiting one. Answer: false for five vertices, true for six.

WHERE THIS GOES WRONG

Two errors, both silent. The first is negating $\forall x P(x)$ to $\forall x \neg P(x)$, the strong denial: “not everything is prime” is not “everything is composite”. The second is reordering quantifiers while negating. The statements $\forall x \exists y R(x, y)$ and $\exists y \forall x R(x, y)$ are genuinely different, the second being strictly stronger, and a negation that shuffles them is answering a different question.

SEE ALSO De Morgan normalisation; the extremal principle; proof by contradiction.

ORDER OF QUANTIFIERS, IN ONE EXAMPLE

Over the positive integers, $\forall m \exists n (n > m)$ is true, since $n = m + 1$ works for each m separately. Swapping the quantifiers gives $\exists n \forall m (n > m)$, which asserts a single integer exceeding every integer and is false. Nothing changed but the order, and the truth value flipped. In puzzle English the distinction hides inside the word “a”: “for every lock there is a key” and “there is a key for every lock” are the two readings, and the second is the one a careless problem statement means by accident. When a problem says “there is”, ask immediately whether the object is allowed to depend on the earlier variables.

What “if” actually asserts

The connective $\rightarrow$ was defined above by a column of a truth table, and that definition has two consequences that regularly surprise readers. The first is that a conditional with a false antecedent is true, no matter what the consequent says. The second is that English “if” frequently means something stronger than $\rightarrow$, so that formalising a sentence as $A \rightarrow B$ can lose information the problem intended to give you. Both consequences are exploitable, and both are traps.

41.5 The material conditional and vacuous truth

CUE A conditional whose antecedent might be false: a speaker’s conditional claim, a rule quantified over a set that may turn out to be empty, or a hypothesis of the form “if the first box is empty then ...”.

METHOD Take $A \rightarrow B$ to mean $\neg A \vee B$, true in every row but A true with B false. Two working consequences. A conditional is established for free once you know its antecedent fails, which is *vacuous truth*: a rule such as “every element of S is red” is true when S is empty. And an assertion of $A \rightarrow B$ by someone whose truthfulness is itself in question can force A to be true, because the assertion is cheap when A fails and expensive when it holds.

WORKED EXAMPLE

On an island where each inhabitant is either a knight, who always speaks truly, or a knave, who always speaks falsely, one inhabitant says: “if I am a knight, then this box holds gold.” What follows?

Write k for “the speaker is a knight” and G for “the box holds gold”. The speaker’s assertion is $S = (k \rightarrow G)$, and the island rule is that the speaker is a knight exactly when the assertion is true, so $k \leftrightarrow S$. Suppose k is false. Then $S = (F \rightarrow G) = T$, so $k \leftrightarrow S$ reads $F \leftrightarrow T$, which fails. So k is true, hence S is true, hence $G = (k \rightarrow G)$ with k true gives G true. Answer: the speaker is a knight and the box holds gold, the unique consistent row among the four, verified by enumerating all four assignments in `python3`.

Change one word and everything changes. If the speaker instead says “if I am a knave, then this box holds gold”, the assertion is $\neg k \rightarrow G$, and enumeration returns three consistent rows: knight with gold, knight without gold, and knave without gold. Nothing at all is determined. The asymmetry is pure vacuous truth: “if I am a knight” has an antecedent that a knight cannot escape, whereas “if I am a knave” has an antecedent that a knight discharges for free.

WHERE THIS GOES WRONG

Importing relevance into $\rightarrow$. The sentence “if the moon is cheese then $2 + 2 = 5$ ” is true, and no amount of discomfort changes that; there is no causal content in the connective at all. The mirror trap is the second one this section promised: English “if” is very often biconditional. “You may enter if you have a ticket” is normally meant as “if and only if”. When a puzzle says “if”, decide from the context whether the converse was also intended, and if the problem turns on the difference, say which reading you took.

SEE ALSO Truth-table exhaustion; the Boolean encoding of a truthful or lying speaker; necessary versus sufficient.

VACUOUS TRUTH IS NOT A LOOPHOLE

“Every purple crate on this dock contains gold” is true when no purple crate is present, and this is a consequence of the definition rather than a lawyer’s reading of it. The reason to define $\rightarrow$ this way is that the alternative destroys induction and quantification: we want $\forall n (n > 100 \rightarrow P(n))$ to be checkable by looking only at $n > 100$, and we want a statement about the empty set to be harmless rather than meaningless. In puzzles the practical consequences are two. A universal statement over a possibly empty range gives you nothing until you have shown the range is nonempty. And a claim you cannot refute may be unrefutable because it is vacuous, in which case the useful move is to attack the emptiness rather than the claim.

Vacuous truth is what happens when a conditional’s antecedent is empty. The remaining reading discipline concerns which side of a conditional a piece of English has landed on, and it is the one place in this chapter where the cost of a mistake is an entire wrong answer rather than a lost step.

41.6 Necessary versus sufficient

MEDIUM

CUE The words “only if”, “provided that”, “it suffices that”, “one must”, “no more than”. Also, and this is the important case, any problem asking for a *minimum* number of tests, questions, weighings or moves, where a bound and a construction are two separate obligations.

METHOD Fix the dictionary and use it mechanically. “ A only if B ” means $A \rightarrow B$, so B is *necessary* for A . “ A if B ” means $B \rightarrow A$, so B is *sufficient* for A . “If and only if” asserts both directions. The consequence for optimisation problems is a discipline: a claim that the answer is N decomposes into a *sufficiency* half, an explicit procedure using N steps, and a *necessity* half, an argument that no procedure using $N - 1$ steps can work. Neither half implies the other and an answer with only one half is incomplete.

WORKED EXAMPLE

A polynomial p has unknown degree and unknown coefficients, all of them non-negative integers. You may ask for the value $p(k)$ at any integer k of your choosing, and you learn the answer before choosing the next k . How many questions do you need?

Sufficiency: two suffice. Ask $p(1)$. Since the coefficients are non-negative integers, $p(1) = S$ is their sum, so every coefficient lies in $\{0, 1, \dots, S\}$. Now ask $p(S + 1)$. Because each coefficient is strictly less than the base $S + 1$, the value $p(S + 1) = \sum_i c_i (S + 1)^i$ is precisely the base- $(S + 1)$ representation of that integer, and the coefficients are its digits. Reading the digits recovers p completely. Random testing of this procedure on four thousand polynomials of degree up to six with coefficients up to forty recovers every one of them exactly, verified in `python3`.

Necessity: one does not suffice. A single question is a single k fixed in advance. Whatever k you choose, the two distinct polynomials $p(x) = x^2$ and $q(x) = kx$ both have non-negative integer coefficients and both return k^2 , so no single value can separate them. Answer: exactly two, and note that the two halves used completely different ideas, the first a construction and the second a collision.

Producing a clever procedure and calling it optimal. A construction proves an upper bound on the cost, never a lower one, and the sentence “I cannot see how to do better” is not an argument. The opposite failure is equally common in written solutions: a slick counting bound with no strategy exhibited, so that the claimed minimum may not be achievable at all. Write both halves down separately and label them, because a solution that silently merges them usually contains only one.

SEE ALSO Contrapositive, converse, inverse; the extremal principle; the material conditional and vacuous truth.

Six English constructions cover most of what puzzle statements do with conditionals, and it is worth having them memorised as a table rather than reasoning them out each time. “ A implies B ”, “ A only if B ”, “ B whenever A ”, “ A is sufficient for B ” and “ B is necessary for A ” all mean $A \rightarrow B$. “ A if B ”, “ B implies A ” and “ A is necessary for B ” all mean $B \rightarrow A$. The two lists are exchanged by swapping A and B , and the single phrase that catches people out is “only if”, which points the arrow the opposite way from the way it reads. When a problem statement mixes several of these in one paragraph, translate every sentence into arrow form before doing anything else; the mixture is usually the difficulty.

Recognition matrix

If you see	Reach for	If that fails
Three or four atoms, several constraints	Enumerate 2^n rows; keep the survivors	Case on the most constrained atom
“Does this follow?”	Entailment: check <i>every</i> survivor	One counterexample row refutes it
“Is this possible?”	Satisfiability: one survivor suffices	Zero survivors means impossible
A unique survivor	Fully determined; report it	Several survivors: report only the shared part
Information given about B , rule is $A \rightarrow B$	Contrapositive $\neg B \rightarrow \neg A$	You do not own the converse
Asked which case to test	The unique false row: A true, B false	Ignore rows that can only confirm
“Not both”	$\neg A \vee \neg B$: two branches	Not “neither”, which is $\neg A \wedge \neg B$
“None of them”	$\neg A_1 \wedge \dots \wedge \neg A_n$	A list of facts, not a case split
“At most one”	$\bigwedge_{i < j} (\neg A_i \vee \neg A_j)$	Add a disjunction for “exactly one”
A negated compound	Push $\neg$ inward to negation normal form	Then read the top-level connective
Refute “for all”	Exhibit one witness with $\neg P$	Search a small finite case first
Refute “there exists”	Prove $\neg P$ over the whole domain	An invariant or a parity argument
Nested $\forall \exists$	Negate outside-in, keep the order	Ask whether y may depend on x
A conditional with a doubtful antecedent	$\neg A \vee B$; check the vacuous branch	The rule is free when A fails
“If I am a knight then ...”	$k \leftrightarrow (k \rightarrow S)$ forces k and S	The knave version determines nothing
A rule over a possibly empty set	Vacuously true; attack the emptiness	Show the range is nonempty first
“Only if”	$A \rightarrow B$, arrow away from the “only”	Reverses the naive reading
“If” in a permission or a promise	Suspect a hidden biconditional	State which reading you adopted
“Minimum number of ...”	Construction plus lower bound, separately	One half alone is not an answer
A clever strategy, no bound	You have sufficiency only	Find a collision of two inputs

Proof technique as puzzle technique

What this chapter is for. The standard proof methods are usually taught as ways of writing down an argument you have already found. That is not what they are. Each one is a search strategy: it tells you what to assume, what to aim at, and therefore where to look. Contradiction hands you an extra hypothesis for free. Contraposition swaps an unusable starting point for a usable one. Induction converts a statement about all n into a statement about one step. The extremal principle manufactures an object with a property nothing in the problem mentioned. Choosing among them is the same recognition skill the calculus parts of this book train, applied to a different kind of object, and the cue is almost always the logical shape of the claim rather than its subject matter.

Forward, backward, and sideways

42.1 Direct construction and forward chaining

EASY

CUE A claim of the form “there exists an object with property P ”, or a claim whose hypothesis is concrete and whose conclusion is a computation away. Also any problem that asks for a number, a configuration, or a strategy rather than for an impossibility.

METHOD Start from the hypothesis and push forward, keeping every derived fact written down. When the target is an existence claim, build the object explicitly; a construction is a complete proof of existence and is usually shorter than any argument about it. Keep a separate list of what you have used, because an unused hypothesis in a competition problem is nearly always a signal that the intended route is different.

WORKED EXAMPLE

Show that every integer $n \geq 8$ is a sum of threes and fives. Construct: $8 = 3 + 5$, $9 = 3 + 3 + 3$, $10 = 5 + 5$. Given a representation of n , add a three to get one of $n + 3$. Since the three base cases cover the residues modulo three, induction on the step of three covers everything above. The construction is the proof.

WHERE THIS GOES WRONG

Chaining forward past the target. Solvers routinely derive the conclusion, fail to notice, and continue until the chain closes on something circular. Write the target at the top of the page and check each new line against it.

SEE ALSO Induction: ordinary, strong, and structural; exhaustive casework with a discipline.

42.2 Proof by contrapositive

CUE An implication $A \rightarrow B$ where A is a negative or a non-constructive hypothesis (“ n is not a perfect square”, “the set is not connected”) and $\neg B$ is concrete. The general cue: you cannot get a grip on A , but you can get a grip on $\neg B$.

METHOD Prove $\neg B \rightarrow \neg A$ instead. The two statements are logically identical, so nothing is lost, and the direction that starts from a concrete object is the direction you can compute in. This is the single highest-yield rewrite in elementary proof, and it costs one line to try.

WORKED EXAMPLE

Show that if n^2 is even then n is even. The hypothesis gives you $n^2 = 2k$, from which extracting information about n requires a factorisation argument. The contrapositive is: if n is odd then n^2 is odd. Now $n = 2m + 1$ is concrete, and $n^2 = 4m^2 + 4m + 1 = 2(2m^2 + 2m) + 1$ finishes it in one line.

WHERE THIS GOES WRONG

Proving the converse $B \rightarrow A$ or the inverse $\neg A \rightarrow \neg B$ by accident. Only the contrapositive is equivalent. Write both statements out before you start rather than trusting the flip.

SEE ALSO Rewriting a conditional; proof by contradiction; quantifier negation.

42.3 Proof by contradiction

CUE A claim of impossibility, of irrationality, of uniqueness, or of infinitude. Also the shape “no such object exists”, which offers nothing to work with until you assume one does.

METHOD Assume the negation, which gives you a concrete object to manipulate, and derive a contradiction with a hypothesis, a known fact, or the assumption itself. The value is entirely in the extra hypothesis: assuming a counterexample exists turns a statement about all objects into a statement about one object you can name.

WORKED EXAMPLE

Show that $\log_2 3$ is irrational. Suppose $\log_2 3 = p/q$ in lowest terms with $q > 0$. Then $2^p = 3^q$. The left side is even for $p \geq 1$ and the right side is odd, and $p = 0$ gives $1 = 3^q$, impossible for $q > 0$. The assumption is untenable, so no such p, q exist.

WHERE THIS GOES WRONG

Reaching for contradiction where a direct argument is shorter. A proof that assumes $\neg B$, never uses the assumption, and derives B is a direct proof wearing a costume, and the costume hides errors. If you finish and find the assumption unused, rewrite it forward.

SEE ALSO Well-ordering and the minimal counterexample; proof by contrapositive; the extremal principle.

Splitting the problem, and climbing it

42.4 Exhaustive casework with a discipline

CUE A finite classification is available and the argument is easy inside each class: residues modulo a small number, parity, the sign of a quantity, which of several objects is largest, or which of a small set of positions an item occupies.

METHOD Choose the partition before you start, and state it. The two obligations are exhaustiveness (the cases cover everything) and, ideally, disjointness (no configuration is counted twice), and both should be visible on the page. Then look for symmetry: a good partition usually lets you dispose of several cases with the phrase “by the same argument with the roles exchanged”, and a partition that does not is often the wrong partition.

WORKED EXAMPLE

Show that $n(n+1)(2n+1)$ is divisible by 6 for every integer n . Divisibility by 2: of n and $n+1$ one is even. Divisibility by 3: take $n \equiv 0, 1, 2 \pmod{3}$. If $n \equiv 0$ the first factor is divisible by three; if $n \equiv 2$ then $n+1 \equiv 0$; if $n \equiv 1$ then $2n+1 \equiv 3 \equiv 0$. Three exhaustive, disjoint cases, each one line, and 2 and 3 are coprime.

A partition that is exhaustive on the surface but not underneath. “Either $x > y$ or $x < y$ ” silently omits equality; “either the largest element is in A or in B ” omits the possibility of a tie. Check the boundary of every case split.

SEE ALSO Truth-table exhaustion; residue classes and modular reduction; when hand enumeration is the right method.

42.5 Induction: ordinary, strong, and structural

MEDIUM

CUE A statement indexed by a natural number, or by any object built up from smaller objects of the same kind: a formula for a sum, a claim about a recursively defined sequence, a property of every tree or every expression.

METHOD Ordinary induction assumes $P(n)$ and proves $P(n+1)$. Strong induction assumes $P(k)$ for all $k < n$ and proves $P(n)$, and is the right form whenever the recursive step reaches back more than one place, as it does for the Fibonacci recurrence, for prime factorisation, and for anything defined by splitting into two smaller pieces. Structural induction is the same idea over a recursively defined set: prove the property for the base constructors and prove it is preserved by each building operation.

THE INDUCTIVE STEP IS WHERE THE WORK IS

State the hypothesis you are assuming and the target you are proving, in full, before manipulating anything. Most failed inductions are failures to notice that the step needs more than $P(n)$, and the repair is almost always to strengthen the statement being proved rather than to work harder on the step.

WORKED EXAMPLE

Prove $\sum_{k=1}^n k^3 = \left(\frac{n(n+1)}{2}\right)^2$. Base $n = 1$: both sides are 1. Step: assume the identity for n . Then

$$\sum_{k=1}^{n+1} k^3 = \frac{n^2(n+1)^2}{4} + (n+1)^3 = \frac{(n+1)^2(n^2+4n+4)}{4} = \left(\frac{(n+1)(n+2)}{2}\right)^2,$$

which is the statement for $n+1$.

STRENGTHENING THE HYPOTHESIS

Try to prove by induction that $\sum_{k=1}^n 1/k^2 < 2$. The step fails: knowing the partial sum is below 2 says nothing about the next one. Prove the stronger claim $\sum_{k=1}^n 1/k^2 \leq 2 - 1/n$ instead. Base $n = 1$ gives $1 \leq 1$. Step: $2 - \frac{1}{n} + \frac{1}{(n+1)^2} \leq 2 - \frac{1}{n+1}$ reduces to $\frac{1}{(n+1)^2} \leq \frac{1}{n} - \frac{1}{n+1} = \frac{1}{n(n+1)}$, which is true. The stronger statement is easier to prove, because it carries more into the step.

WHERE THIS GOES WRONG

Verifying a base case that the step does not actually reach. If the step is valid only for $n \geq 3$, a base at $n = 1$ proves nothing about $n = 4$. Check the smallest n for which the step's algebra is legitimate.

SEE ALSO Well-ordering and the minimal counterexample; recursion, state DP, and generating functions; monovariants and the termination argument.

42.6 Well-ordering and the minimal counterexample

MEDIUM

CUE A statement about all positive integers, or about all finite configurations, where the inductive step is awkward to phrase but where a smallest bad case would be easy to attack. Also the phrase “show that this process always terminates”.

METHOD Suppose the claim fails, and let n be the smallest index at which it fails; well-ordering guarantees such an n exists. Then derive a smaller failure, contradicting minimality. This is logically equivalent to strong induction, and it is often much easier to write, because it gives you a concrete object with an extra property (nothing smaller is bad) rather than an abstract hypothesis.

WORKED EXAMPLE

Show that every integer $n > 1$ has a prime factor. Suppose not, and let n be the smallest counterexample. Then n is not prime, so $n = ab$ with $1 < a < n$. By minimality a has a prime factor p , and $p \mid a \mid n$. Contradiction.

Applying well-ordering to a set that is not a set of non-negative integers. The principle fails for the positive rationals and for the integers, and a descent argument in those settings must be recast in terms of an integer quantity such as a numerator, a denominator, or an exponent.

SEE ALSO Vieta jumping and infinite descent; induction; monovariants and the termination argument.

Manufacturing an object

42.7 The extremal principle

HARD

CUE A configuration problem with no obvious starting point, especially one that asserts something must exist, must be equal, or cannot happen, in a finite arrangement of points, people, numbers, or edges.

METHOD Consider the extreme: the largest, the smallest, the leftmost, the pair at minimum distance, the vertex of highest degree, the configuration with fewest crossings. Extremality is a free hypothesis, because whatever you do to the extremal object must not produce something more extreme, and that impossibility is the contradiction. The whole art is choosing the right quantity to extremise.

WORKED EXAMPLE

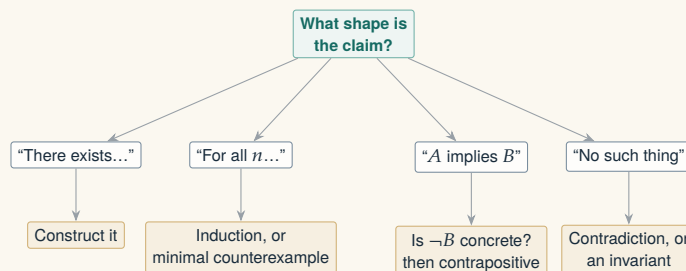
Finitely many points in the plane, not all collinear, and every line through two of them passes through a third. Show this is impossible. Among all pairs consisting of a point P and a line ℓ through two of the points with $P \notin \ell$, choose the pair minimising the distance from P to ℓ ; finiteness makes this possible. By hypothesis ℓ carries at least three of the points, so two of them lie on the same side of the foot of the perpendicular from P ; call the nearer one B and the further one C . The distance from B to the line PC is strictly less than the distance from P to ℓ , contradicting minimality.

WHERE THIS GOES WRONG

Extremising a quantity that the operation does not decrease. The argument only closes if your move provably produces a strictly more extreme configuration; check the strictness, since ties are where these proofs fail.

SEE ALSO Monovariants and the termination argument; extremal configurations in geometry; the minimal counterexample.

Choosing the method



The tree is a first move, not a decision. Two refinements matter in practice. If the claim is an impossibility and the objects are produced by a repeated operation, skip contradiction and look for an invariant, since that is what a proof by contradiction in this setting will end up constructing anyway. And if the claim is about all n but the step reaches back further than one, take the strong form immediately rather than discovering the need halfway through.

Recognition matrix

If you see	Reach for	If that fails
"There exists a configuration"	Construct one explicitly	Counting or probabilistic existence
Hypothesis is a negation	Contrapositive	Contradiction
"Irrational", "infinitely many"	Contradiction	Descent on an integer
"Impossible", with a process	An invariant	Colouring or parity
A statement indexed by n	Induction	Strengthen the statement
The step reaches back more than one	Strong induction	Minimal counterexample
Recursively built objects	Structural induction	Induction on size
Finite arrangement, no entry point	Extremal principle	Pigeonhole
Small finite classification available	Casework, stated and exhaustive	Symmetry to halve the cases
Induction step fails, claim seems true	Prove a stronger claim	Check the base the step reaches
Unused hypothesis at the end	Wrong route; find where it enters	Re-read for a hidden constraint

Knights, knaves, and self-reference

What this chapter is for. A puzzle in which people make claims about who is lying looks like a test of verbal cleverness and is nothing of the kind. Once each speaker is given a Boolean variable and each utterance is turned into a biconditional, the puzzle becomes a small system of equations, and solving it is mechanical. This chapter gives the encoding, the consistency check that solves the system, the family of existential claims that behave differently from the rest, and the metapuzzles, where the information is carried not by what was said but by the fact that somebody was unable to answer. The technique generalises well past its recreational origin: any problem in which the reliability of a source is itself unknown has this shape.

The encoding

43.1 The standard Boolean encoding

EASY

CUE Speakers who are known to be either always truthful or always lying, making statements about themselves or about each other. Also any variant with a third type who alternates or answers arbitrarily, which is handled by adding a variable rather than by changing the method.

METHOD Give speaker i a Boolean k_i , true when the speaker is truthful. If speaker i utters the statement S , the content of the utterance is exactly

$$k_i \leftrightarrow S.$$

This one line is the whole technique. A truthful speaker's statement is true, and a lying speaker's statement is false, and the biconditional says both at once. Conjoin the biconditionals from every utterance and solve the resulting propositional system.

WHY A BICONDITIONAL AND NOT AN IMPLICATION

Writing $k_i \rightarrow S$ records only half the information: it captures that knights tell the truth and forgets that knaves lie. The forgotten half is usually the half that determines the answer, because it is what makes a knave's statement informative.

WORKED EXAMPLE

Two islanders. A says "I am a knave". Encode: $k_A \leftrightarrow \neg k_A$, which is unsatisfiable. So no islander can say this, and a puzzle containing it is malformed. Change it to A says " B is a knave", B says " A and I are of the same type". The system is $k_A \leftrightarrow \neg k_B$ and $k_B \leftrightarrow (k_A \leftrightarrow k_B)$. From the first, $k_B = \neg k_A$. Substitute into the second: $\neg k_A \leftrightarrow (k_A \leftrightarrow \neg k_A)$, and since $k_A \leftrightarrow \neg k_A$ is always false, this reads $\neg k_A \leftrightarrow F$, giving k_A true and k_B false. Verified by enumerating all four assignments.

WHERE THIS GOES WRONG

Encoding what a speaker means rather than what the speaker says. "I am not a knight" is $\neg k_i$, not k_i ; the negation belongs inside S , and the truthfulness belongs in the biconditional. Keeping the two layers separate is the single discipline that makes these puzzles routine.

SEE ALSO Consistency checking as the solution method; truth-table exhaustion; existential claims.

43.2 Consistency checking as the solution method

EASY

CUE A system of biconditionals with three or more speakers, where substitution is getting messy.

METHOD Case on one variable, propagate, and reject the branches that produce a contradiction. With n speakers there are 2^n branches, but the first constraint usually kills half of them immediately, so choose to case on the speaker who appears in the most utterances. The surviving branches are the answer; report the whole surviving set, not the first branch you find.

Three islanders. A : “exactly one of us is a knight”. B : “ A is a knave”. C : “ B and I are not both knaves”.

Case k_B true: then A is a knave, so A ’s statement is false and the number of knights is not one. C ’s statement $\neg(\neg k_B \wedge \neg k_C)$ is true because k_B holds, so k_C must be true. Knights are B and C , two of them, and “not one” is satisfied. Consistent.

Case k_B false: then A is a knight, so exactly one is a knight, so C is a knave. Then C ’s statement is false, meaning B and C are both knaves, which is true. But a knave’s statement must be false. Contradiction.

Unique solution: A knave, B and C knights. Confirmed by exhaustive enumeration of all eight assignments.

WHERE THIS GOES WRONG

Stopping at the first consistent branch. A puzzle is only well posed if exactly one branch survives, and checking the others is how you find out that it does. A second surviving branch means the intended answer was underdetermined, and that is worth reporting rather than hiding.

SEE ALSO The standard Boolean encoding; truth-table exhaustion; branch-and-contradict.

Statements that behave differently

43.3 Existential claims and the asymmetry they create

MEDIUM

CUE A statement of the form “at least one of us is a knave”, “someone here is lying”, or “not all of us are knights”.

METHOD These are the workhorses of the genre because their negations are universal and therefore very strong. Suppose speaker i says “at least one of us is a knave”, that is $S = \neg(k_1 \wedge \dots \wedge k_n)$. If i were a knave then S is false, so everyone including i is a knight, contradicting that i is a knave. So i is a knight and S is true, which then forces at least one knave among the others. One utterance determines the speaker outright and constrains the rest.

WORKED EXAMPLE

Four islanders, and D says “at least one of us is a knave”. By the argument above D is a knight, and at least one of A, B, C is a knave. If additionally A says “all four of us are knights”, then A ’s statement contradicts D ’s, so A is a knave, and the two utterances together are consistent with any assignment of B and C : four surviving branches. The puzzle needs a third utterance to be well posed.

WHERE THIS GOES WRONG

Reading “at least one of us” as excluding the speaker. It does not, and the self-inclusion is exactly what makes the argument work. When a puzzle intends exclusion it says “one of the others”.

SEE ALSO Quantifier negation; the standard Boolean encoding; consistency checking.

43.4 Reasoning from an agent’s inability to answer

HARD

CUE A puzzle in which a participant says “I do not know”, or fails to act, or in which the same question is asked repeatedly and produces the same non-answer until suddenly it does not. The information is carried by the failure.

METHOD Model each participant’s knowledge as the set of worlds consistent with what that participant has observed. A declaration of ignorance eliminates every world in which that participant would have known, which is a constraint on the world visible to everybody else. Iterate: each round of announced ignorance shrinks the common set, and the puzzle resolves at the round where exactly one world survives.

WORKED EXAMPLE

Two players each wear a hat with a positive integer, each sees only the other’s, and they are told the two numbers are consecutive. They are asked in turn whether they know their own number, and both say no, repeatedly, until one says yes.

A player seeing 1 knows immediately: their own number must be 2. So the first “no” eliminates the worlds in which the opponent sees 1, that is, the worlds in which the first speaker wears 1. A player seeing 2 can then reason: my number is 1 or 3, and if it were 1 the opponent would already have answered, so it is 3. Each round eliminates one more level, and a player wearing n answers on round $n - 1$ or so, depending on the ordering. The chain of ignorance is a ladder, and its rungs are the numbers.

Forgetting that ignorance announcements are public and therefore compound. The second “no” is not the same statement as the first: it is made with the first already known to both, and it eliminates a different set of worlds. Track the common knowledge, not just the private observations.

SEE ALSO Consistency checking; states, transitions, and absorbing reasoning; symmetry arguments.

Recognition matrix

If you see	Reach for	If that fails
Speaker i utters S	Record $k_i \leftrightarrow S$	Never $k_i \rightarrow S$ alone
“I am a knave”	Unsatisfiable; the puzzle is malformed	Re-read for a third type
“At least one of us is a knave”	Speaker is a knight; a knave exists elsewhere	Check whether “us” includes the speaker
“All of us are knights”	Speaker is a knave	Then at least one other is too
Three or more speakers	Case on the most-mentioned variable	Enumerate all 2^n rows
Two branches survive	Underdetermined; say so	Look for an unencoded sentence
“I do not know”	Eliminate worlds where they would know	Iterate over rounds
Repeated identical questions	Each round is new common knowledge	Count the rungs of the ladder
A third type who alternates	Add a variable, keep the method	Case on the position in the cycle

Invariants and monovariants

What this chapter is for. This is the highest-yield idea in combinatorial problem solving, and it deserves the claim. Whenever a problem describes a process, a move, an operation repeated any number of times, there is a question worth asking before anything else: what does the move fail to change? If some quantity is preserved, then every reachable state agrees with the start on that quantity, and any target that disagrees is unreachable. A search that could run forever collapses to one line of arithmetic. The companion idea, the monovariant, is a quantity that moves only one way; it cannot prove unreachability but it proves termination, and it bounds how long a process can run. The difficult part is never using an invariant. It is finding one, and this chapter is organised around that search.

Finding the invariant

There is a reliable order in which to look. Parity first, because it is free and it works surprisingly often. Then a colouring, because a colouring is a parity argument with more colours and it is the natural tool when the moves are geometric. Then a weighting, when the objects have positions or sizes that the move relates. Then an algebraic combination, typically a sum or a product taken modulo a small number. If all four fail, the problem is probably not an invariant problem, and the chapter on counting or the chapter on games is the right place to look next.

44.1 Parity invariant

EASY

CUE A move that changes several quantities at once by amounts summing to an even number, or that flips signs, or that swaps two objects. Any question of the form “can this configuration become that one”, where the two configurations look similar.

METHOD Find a quantity whose parity the move preserves: a count of objects of one kind, a sum, the number of inversions in an arrangement, the number of objects in a distinguished region. Then compare the parity of the start with the parity of the target. Different parity means unreachable, and the proof is complete.

WORKED EXAMPLE

A board carries the numbers 1 through $2n$. A move deletes two numbers a, b and writes $|a - b|$. After $2n - 1$ moves one number remains. Is it even or odd? The move replaces $a + b$ by $|a - b|$, and $a + b$ and $|a - b|$ always have the same parity, so the parity of the total sum is invariant. The initial sum is $1 + 2 + \dots + 2n = n(2n + 1)$, whose parity is that of n . So the final number is even exactly when n is even. For $n = 3$ the initial sum is 21, odd, so the survivor is odd whatever moves are made, which is easy to confirm by random simulation.

WHERE THIS GOES WRONG

Verifying the invariant on one move and assuming it holds for all. Check every move type the problem allows, including the degenerate ones, and check what happens at the boundary of the board or the end of the list.

SEE ALSO Colouring arguments; multiplicative and other algebraic invariants; tiling impossibility by colouring.

44.2 Colouring arguments: chessboard and modulo k

MEDIUM

CUE A geometric or grid problem in which a piece covers a fixed shape, or moves by a fixed step, and you are asked whether a covering or a tour is possible. Also any tiling question.

METHOD Colour the cells so that every legal placement or every legal move covers a fixed multiset of colours. Then count colours in the region and compare. Two colours in a chequered pattern handle dominoes and rook or bishop moves; colouring by column index modulo k handles $1 \times k$ tiles; diagonal colourings handle knight-like steps. The colouring is chosen to make the move's colour signature constant, and inventing it is the entire creative act.

WORKED EXAMPLE

Remove two opposite corners from an 8×8 board and try to tile the remaining 62 squares with 31 dominoes. Colour the board normally. Every domino covers one black and one white square, so a tiling needs equal counts. Opposite corners share a colour, so the mutilated board has 32 of one colour and 30 of the other. No tiling exists.

A COLOURING WITH MORE THAN TWO COLOURS

Can a 10×10 board be tiled by 1×4 pieces? Colour cell (i, j) with $(i + 2j) \bmod 4$. Every horizontal piece covers the four residues once each; every vertical piece covers $(i + 2j)$, $(i + 1 + 2j)$, $(i + 2 + 2j)$, $(i + 3 + 2j)$, again all four residues once each. So a tiling requires the four colour classes to be equal in size, at 25 each. Counting the classes on the 10×10 board gives 25, 25, 25, 25, so this colouring does not obstruct. The standard four-colouring by $(i + j) \bmod 4$ does not obstruct either, and in fact a tiling exists, which is the honest outcome: a colouring that balances proves nothing, and you must then construct.

WHERE THIS GOES WRONG

Concluding possibility from a colouring that balances. A colouring argument is a one-way test: imbalance proves impossibility, balance proves nothing at all. When the count comes out even, you still owe a construction.

SEE ALSO Parity invariant; weighting arguments; tiling impossibility by colouring.

HARD

44.3 Weighting arguments

CUE A move that relates positions with different values, most characteristically a jump in which a piece at position p moves to position $p + 2$ while a piece at $p + 1$ is removed. Also any problem asking how far something can reach.

METHOD Assign a weight $w(c)$ to each cell so that the move never increases the total weight, then compare the target's weight with the finite total available. The classic choice is a geometric weighting $w = \varphi^d$ where d is the distance to the target and φ is the golden ratio, chosen exactly because $\varphi^2 = \varphi + 1$ makes a jump weight-preserving.

WORKED EXAMPLE

Conway's soldiers. Pegs fill the half-plane $y \leq 0$ of a grid, and a move jumps a peg over an adjacent peg into an empty cell two away, removing the jumped peg. How far above the line can a peg reach? Weight cell c by $\varphi^{d(c)}$ where d is the taxicab distance to the target cell and $\varphi = (\sqrt{5} - 1)/2$ satisfies $\varphi^2 + \varphi = 1$. A jump toward the target preserves weight, and any other jump strictly decreases it. Summing the weights of the entire filled half-plane converges, and the total is exactly 1, the weight of a peg sitting on the target itself. Reaching level 5 would require a weight strictly greater than the whole half-plane supplies, so level 5 is unreachable no matter how many pegs are used.

WHERE THIS GOES WRONG

Choosing a weighting that is preserved by the move but whose total over the initial configuration diverges. The argument needs a finite bound to compare against, so the geometric decay is not decoration.

SEE ALSO Potential functions; monovariants; colouring arguments.

44.4 Multiplicative and other algebraic invariants

CUE Moves that multiply, that change signs, or that act on a tuple by a linear substitution. Also processes on numbers where the sum is visibly not invariant.

METHOD Try the product of the entries, the product of the signs, the sum of squares, the greatest common divisor, or a linear combination $\sum a_i x_i$ with coefficients chosen so that the move's effect cancels. The last of these is a small linear algebra problem: write the move as a vector of changes and find a functional that annihilates it.

WORKED EXAMPLE

Numbers a, b, c on a board; a move replaces them by $a + b - c, b + c - a, c + a - b$ in some order. The sum $a + b + c$ is invariant, since the three new numbers sum to $a + b + c$. So starting from $(1, 2, 3)$ you can never reach $(2, 3, 5)$, whose sum is 10 rather than 6, whatever sequence of moves is used.

WHERE THIS GOES WRONG

Finding an invariant and forgetting to check that the target genuinely violates it. If start and target agree on every invariant you found, you have learned nothing and must either find another invariant or construct a path.

SEE ALSO Parity invariant; weighting arguments; residue classes and modular reduction.

Quantities that move one way

44.5 Monovariants and the termination argument

CUE The question is not “can this be reached” but “must this stop”, “how many moves at most”, or “what does it look like when no move is available”.

METHOD Find a quantity that strictly decreases (or strictly increases) with every legal move and that is bounded, and preferably integer-valued. A strictly decreasing non-negative integer cannot decrease forever, so the process terminates, and the bound on the quantity bounds the number of moves. Then characterise the terminal states: they are exactly the configurations in which no move is legal, and identifying them is often the actual answer.

WORKED EXAMPLE

People sit around a table; if any two adjacent people are ordered wrongly by height, they may swap. Does this stop? The number of inversions, meaning pairs out of order, strictly decreases by one at every legal swap, and is a non-negative integer bounded by $\binom{n}{2}$. So the process halts after at most $\binom{n}{2}$ swaps, and it halts only when no adjacent pair is inverted, which for a linear order means fully sorted.

WHERE THIS GOES WRONG

A monovariant that decreases weakly rather than strictly. “Does not increase” permits an infinite loop among states of equal value, so either sharpen the quantity or add a tie-breaking second component and argue lexicographically.

SEE ALSO Potential functions; well-ordering and the minimal counterexample; winning and losing positions.

44.6 Potential functions

CUE A process where no single obvious quantity is monotone, but a weighted combination might be. Also amortised bounds, where individual moves can be expensive but the total is not.

METHOD Design Φ on states so that every move decreases Φ by at least a fixed positive amount, or so that the actual cost of a move plus the change in Φ is bounded. Then the total number of moves is at most $(\Phi_{\text{start}} - \Phi_{\text{min}}) / \delta$. Designing Φ is the same craft as designing a weighting, and the usual candidates are sums of squares, sums of a convex function of the parts, and entropy-like expressions.

WORKED EXAMPLE

Chip-firing on a line of n cells: a cell holding at least two chips may send one chip to each neighbour. Take $\Phi = \sum_i c_i i^2$ where c_i is the chip count at cell i . Firing at i changes Φ by $(i-1)^2 + (i+1)^2 - 2i^2 = 2$, so Φ strictly increases by exactly 2 per firing while remaining bounded above by n^2 times the total chip count. The process therefore terminates, and the number of firings is determined by the change in Φ alone, independent of the order in which cells are fired.

SEE ALSO Monovariants; weighting arguments; the extremal principle.

Recognition matrix

If you see	Reach for	If that fails
"Can state X become state Y "	An invariant separating them	Construct a path
A move changing two things at once	Parity of the sum	Sum modulo a small k
Tiles on a grid	Colouring matched to the tile	Weighted colouring
$1 \times k$ tiles	Colour by index modulo k	Diagonal colouring
Jumps of length two	Geometric weighting with $\varphi^2 = \varphi + 1$	Sum of squares
Signs flipping	Product of the signs	Count of negatives modulo 2
Linear substitution on a tuple	A functional annihilating the change	Eigenvector of the move matrix
"Show the process stops"	A strictly decreasing integer	Lexicographic pair
"How many moves at most"	Bound the monovariant's range	Potential function
Order matters, apparently	Check whether Φ change is order-free	Abelian property
Colouring balances	Nothing proved; construct	Try a finer colouring

Pigeonhole and its strong forms

What this chapter is for. The pigeonhole principle is the most trivial theorem in this book and one of the most useful, and the gap between those two facts is where the technique lives. Stating it takes one line: if more than k objects are placed into k classes, some class holds at least two. Using it takes judgment, because the problem never tells you what the objects are or what the classes are, and choosing them is the entire creative act. This chapter drills that choice. The recognition cue is sharp and worth memorising: when a problem asks you to show that some object with a property must exist, without giving you any way to identify which one, pigeonhole is the first thing to try.

The principle and its strengthenings

45.1 Basic pigeonhole

EASY

CUE An existence claim with no constructive route: “show that two of them must share”, “prove some pair differs by”, “there are always two with the same”. The number of objects usually exceeds some natural classification by exactly one, which is a deliberate signal.

METHOD Name the pigeons and the holes explicitly, in that order, and count both. If n objects go into m classes with $n > m$, some class holds at least two. The proof is by contradiction and is never worth writing out; what must be written out is the map from objects to classes.

WORKED EXAMPLE

Among any five points chosen inside a unit square, two are at distance at most $\sqrt{2}/2$. Divide the square into four closed quadrant squares of side $1/2$. Five points, four squares, so two points lie in the same small square, and the diameter of a square of side $1/2$ is $\sqrt{2}/2$.

WHERE THIS GOES WRONG

Choosing holes that do not cover everything, or that overlap in a way that breaks the count. If a point lands on a boundary line, decide once and for all which square owns it; the argument survives, but only if the assignment is a genuine function.

SEE ALSO Generalised pigeonhole; pigeonhole on residues and differences; the extremal principle.

45.2 Generalised pigeonhole

EASY

CUE The conclusion asks for three or more objects sharing a class, or asks for a class with at least a stated number of members, or the problem gives a ratio rather than a plus-one.

METHOD With n objects in m classes, some class holds at least $\lceil n/m \rceil$ objects, and some class holds at most $\lfloor n/m \rfloor$. Both halves are useful; the second is the one people forget, and it is what proves that something must be small.

WORKED EXAMPLE

A group of 100 people, each born in one of the twelve months. Some month contains at least $\lceil 100/12 \rceil = 9$ of them. And some month contains at most $\lfloor 100/12 \rfloor = 8$, so the distribution cannot be uniform, which is occasionally the point.

WHERE THIS GOES WRONG

Rounding the wrong way. The guaranteed maximum uses the ceiling and the guaranteed minimum uses the floor, and swapping them produces a claim that is false whenever $m \nmid n$.

SEE ALSO Basic pigeonhole; averaging arguments; double counting.

45.3 Infinite pigeonhole

CUE Infinitely many objects distributed into finitely many classes, or a sequence from which you need an infinite subsequence with a shared property. Common in analysis proofs and in Ramsey-flavoured combinatorics.

METHOD If infinitely many objects fall into finitely many classes, at least one class is infinite. Iterating this is how one extracts nested subsequences: apply it, restrict to the infinite class, apply it again with a new classification, and repeat. Finitely many iterations are free; infinitely many need a diagonal argument.

WORKED EXAMPLE

Every sequence of real numbers has a monotone subsequence. Call an index n a peak if $a_n \geq a_m$ for all $m > n$. Either there are infinitely many peaks, and the peaks form a non-increasing subsequence, or there are finitely many, and past the last peak every index is followed by a strictly larger term, giving an increasing subsequence. The two cases are exhaustive, which is pigeonhole with two holes over an infinite set.

SEE ALSO Erdos and Szekeres; the Bolzano and Weierstrass argument; casework with a discipline.

Choosing the holes

The examples above chose obvious holes. The interesting problems do not, and there are three constructions that recur often enough to be worth naming as techniques in their own right: residues, intervals, and sums.

45.4 Pigeonhole on residues and differences

CUE A conclusion about divisibility, about a difference being a multiple of something, or about a subset summing to a multiple of n .

METHOD Take the holes to be residue classes modulo n and the pigeons to be a cleverly chosen set of $n + 1$ numbers, very often the partial sums of the given sequence. Two partial sums in the same class have a difference divisible by n , and a difference of partial sums is a contiguous block, which is usually exactly what the problem wanted.

WORKED EXAMPLE

Given any n integers $a_1, \dots, a_n$, some nonempty consecutive block sums to a multiple of n . Form the $n + 1$ partial sums $S_0 = 0, S_k = a_1 + \dots + a_k$. These are $n + 1$ numbers in n residue classes modulo n , so $S_i \equiv S_j$ for some $i < j$, and then $a_{i+1} + \dots + a_j = S_j - S_i \equiv 0 \pmod{n}$. Note that S_0 is doing essential work: without it the count is n and the argument fails.

WHERE THIS GOES WRONG

Forgetting the empty partial sum. It is the extra pigeon, and it is what lets the chosen block start at the first element.

SEE ALSO Residue classes and modular reduction; basic pigeonhole; double counting.

45.5 Pigeonhole on intervals and on structure

CUE A conclusion about two objects being close, about an approximation by a rational with bounded denominator, or about a monochromatic structure appearing in any colouring.

METHOD Partition the ambient space into finitely many small pieces and let the objects be points of that space. For approximation problems the classic construction takes the fractional parts $\{k\alpha\}$ for $k = 0, \dots, N$ as the pigeons and the N intervals $[j/N, (j+1)/N)$ as the holes.

WORKED EXAMPLE

Dirichlet's approximation theorem. For irrational α and any N , there are integers p, q with $1 \leq q \leq N$ and $|\alpha - p/q| < 1/(qN)$. The $N+1$ fractional parts $\{0 \cdot \alpha\}, \dots, \{N\alpha\}$ lie in N intervals of length $1/N$, so two of them, say at indices $k < l$, lie in the same interval. Then $|(l-k)\alpha - m| < 1/N$ for the appropriate integer m , and dividing by $q = l-k \leq N$ gives the claim. This single argument is the source of the continued-fraction convergents being as good as they are.

WHERE THIS GOES WRONG

Using half-open intervals inconsistently, so that a point at an endpoint is counted twice or not at all. Fix the convention before counting.

SEE ALSO Pigeonhole on residues; Erdos and Szekeres; extremal configurations in geometry.

45.6 Erdos and Szekeres, and pigeonhole with structure

CUE A claim that any sufficiently long sequence, or any sufficiently large colouring, contains a structured piece: a monotone run, a monochromatic triangle, a repeated pattern.

METHOD Attach to each object a label that encodes the structure you want, then pigeonhole on labels. For monotone subsequences, label position i with the pair (length of the longest increasing subsequence ending at i , length of the longest decreasing subsequence ending at i). Distinct positions get distinct labels, so a sequence of length exceeding the number of available labels forces a long run.

WORKED EXAMPLE

Any sequence of $mn+1$ distinct reals contains an increasing subsequence of length $m+1$ or a decreasing subsequence of length $n+1$. Suppose not. Then every label (x_i, y_i) has $1 \leq x_i \leq m$ and $1 \leq y_i \leq n$, giving at most mn labels for $mn+1$ positions, so two positions $i < j$ share a label. But if $a_i < a_j$ then $x_j \geq x_i + 1$, and if $a_i > a_j$ then $y_j \geq y_i + 1$; either way the labels differ. Contradiction.

RAMSEY, THE SMALLEST CASE

Among any six people, three are mutual acquaintances or three are mutual strangers. Fix a person P . Of the five others, by generalised pigeonhole at least $\lceil 5/2 \rceil = 3$ stand in the same relation to P , say acquainted. If any two of those three know each other, they and P form an acquainted triple; if none do, those three are mutual strangers.

SEE ALSO Pigeonhole on intervals; the extremal principle; double counting.

Recognition matrix

If you see	Reach for	If that fails
"Two of them must share"	Basic pigeonhole; name the holes	Averaging
"At least three share"	Generalised form, $\lceil n/m \rceil$	Count the classes again
"Some class is small"	The floor half, $\lfloor n/m \rfloor$	Complementary counting
$n+1$ objects, n natural classes	The plus-one is the signal	Find the intended classification
Divisibility of a block sum	Partial sums modulo n , including S_0	Residues of the terms
"Two are within distance d "	Partition into cells of diameter d	Finer grid, then generalised form
Rational approximation	Fractional parts into N intervals	Continued fractions
Infinitely many, finitely many types	One type occurs infinitely often	Iterate, then diagonalise
Monotone subsequence	Label by longest run ending here	Peaks and non-peaks
Monochromatic structure	Fix a vertex; pigeonhole its edges	Induct on the Ramsey bound
No constructive handle at all	Suspect pigeonhole	Extremal principle

Counting technique

What this chapter is for. Counting is a translation problem. There is a small stock of sets whose size everyone knows, and every counting technique is a way of exhibiting the set in front of you as one of them in disguise. A bijection says the two sets are the same set relabelled. Double counting says one set has two descriptions and therefore one size with two formulas. Inclusion and exclusion says an overcount can be repaired systematically. Complementary counting says the easier set to count is the one you were not asked about. Recursion says the set decomposes into smaller copies of itself. This chapter treats each as a recognition problem, and every count in it has been checked against brute-force enumeration for small parameters.

Making the set into a set you know

46.1 Bijective counting

MEDIUM

CUE Two sets that ought to have the same size, or a set whose elements are naturally encoded as strings, paths, subsets, or arrangements. Also any answer that comes out as a familiar binomial coefficient, which is a hint that a bijection was available.

METHOD Exhibit an explicit map, verify it is injective by describing the inverse, and count the target instead. A bijection is the strongest kind of counting argument because it survives generalisation: the same map usually answers the next four questions in the family.

WORKED EXAMPLE

Count the lattice paths from $(0, 0)$ to (m, n) using unit steps right and up. Encode a path as the string of its steps, a word of length $m + n$ in which m letters are R. The map from paths to such words is a bijection, and choosing which positions hold the R's gives $\binom{m+n}{m}$. For $m = 3, n = 4$ this is 35, confirmed by enumerating all 2^7 words and keeping those with three R's.

WHERE THIS GOES WRONG

Describing a map and never checking injectivity. The usual failure is a map that forgets an ordering, so that several objects share an image; the repair is either to divide by the size of the fibre, which requires the fibres to be equal in size, or to enrich the target.

SEE ALSO Stars and bars; reflection and the ballot problem; double counting.

46.2 Double counting

CUE An identity to prove, especially one with binomial coefficients on both sides; or a quantity that can be totalled by rows and by columns; or a problem about incidences between two kinds of object.

METHOD Find a single set and count it two ways. Equating the two counts proves the identity, and the proof carries an explanation that an algebraic manipulation does not. The standard frame is a table: sum the entries by row and then by column.

WORKED EXAMPLE

Prove $\sum_{k=0}^n k \binom{n}{k} = n2^{n-1}$. Count the pairs (S, x) where $S \subseteq \{1, \dots, n\}$ and $x \in S$. By choosing S first, the count is $\sum_k k \binom{n}{k}$. By choosing x first, there are n choices, and then any of the 2^{n-1} subsets of the remaining elements may be adjoined, giving $n2^{n-1}$. Checked by direct summation for n up to 12.

WHERE THIS GOES WRONG

Counting two different sets. Write down the set once, in words, and then count that same sentence twice; if the two descriptions do not both describe it, the identity you prove is not the one you wanted.

SEE ALSO Bijective counting; the handshake identity; Burnside and symmetry counting.

Repairing an overcount

46.3 Inclusion and exclusion

CUE A count of objects avoiding several forbidden properties, where each property alone is easy to count but they overlap. The words “at least one”, “none of”, and “exactly k of” are all cues.

METHOD Add the sizes of the sets, subtract the pairwise intersections, add the triples, and so on:

$$|A_1 \cup \dots \cup A_n| = \sum_{\emptyset \neq S \subseteq [n]} (-1)^{|S|+1} \left| \bigcap_{i \in S} A_i \right|.$$

The sign discipline is the whole difficulty, and the practical form is: count the complement, so that the alternating sum runs over “how many of the bad properties hold”.

WORKED EXAMPLE

Derangements. Let A_i be the permutations of $[n]$ fixing i . Then $|\bigcap_{i \in S} A_i| = (n - |S|)!$, so

$$D_n = n! \sum_{k=0}^n \frac{(-1)^k}{k!},$$

giving $D_4 = 9$ and $D_5 = 44$, both confirmed by exhaustive enumeration. The ratio $D_n/n!$ tends to $1/e$, which is why the probability that a random permutation has no fixed point is close to 0.3679 for every n beyond about six.

WHERE THIS GOES WRONG

Applying the formula when the intersections do not have a size depending only on $|S|$. In the derangement case they do, which is what collapses 2^n terms into $n + 1$; in general you must sum over the subsets themselves.

SEE ALSO Complementary counting; recursion and generating functions; Burnside.

46.4 Complementary counting

CUE The words “at least one” in the question. Also any count where the forbidden configurations are few and the permitted ones are many.

METHOD Count everything and subtract what fails. The test for whether this is the right move is simple: compare the number of cases in the direct count with the number in the complement, and take the smaller.

WORKED EXAMPLE

How many five-card hands from a standard deck contain at least one ace? Directly this is a four-case sum. By complement it is $\binom{52}{5} - \binom{48}{5} = 2598960 - 1712304 = 886656$.

SEE ALSO Inclusion and exclusion; bijective counting.

Distributions and recursions

46.5 Stars and bars, compositions, and the hockey stick

CUE Distributing identical objects into distinct boxes, or counting non-negative integer solutions of $x_1 + \dots + x_k = n$, or a sum of binomial coefficients along a diagonal.

METHOD Non-negative solutions number $\binom{n+k-1}{k-1}$; positive solutions number $\binom{n-1}{k-1}$. The proof is the bijection with arrangements of n stars and $k-1$ bars. The hockey-stick identity $\sum_{i=r}^n \binom{i}{r} = \binom{n+1}{r+1}$ is the same idea read along a diagonal of Pascal’s triangle, and it is the standard way to collapse a sum of binomials.

WORKED EXAMPLE

Non-negative integer solutions of $x_1 + x_2 + x_3 = 7$: $\binom{9}{2} = 36$, confirmed by enumeration. With the constraint $x_1 \geq 2$, substitute $x'_1 = x_1 - 2$ to get $x'_1 + x_2 + x_3 = 5$ and $\binom{7}{2} = 21$. Upper bounds are handled by inclusion and exclusion on the violated constraints rather than by substitution.

WHERE THIS GOES WRONG

Using the identical-objects formula for distinguishable objects. Seven distinct balls into three distinct boxes is 3^7 , not $\binom{9}{2}$, and the two are confused whenever the problem statement is read quickly.

SEE ALSO Bijective counting; inclusion and exclusion.

46.6 Recursion, state, and generating functions

CUE A count parameterised by n where the objects of size n decompose into objects of smaller size, especially with a forbidden local pattern (“no two adjacent”, “no three in a row”).

METHOD Define the count with enough state to make the recursion close. The commonest error is too little state, and the fix is to split by what the last element was. Then either solve the recurrence by the characteristic equation, or encode it as a generating function $F(x) = \sum a_n x^n$, translate the recursion into a functional equation, and extract coefficients.

WORKED EXAMPLE

Binary strings of length n with no two consecutive ones. Let a_n be the count. A valid string ends in 0, preceded by any valid string of length $n-1$, or ends in 01, preceded by any valid string of length $n-2$. So $a_n = a_{n-1} + a_{n-2}$ with $a_1 = 2$, $a_2 = 3$, giving the Fibonacci numbers shifted: $a_n = F_{n+2}$. For $n = 6$ this predicts 21, confirmed by enumerating all 64 strings.

THE SAME COUNT BY GENERATING FUNCTION

Every valid string is a sequence of blocks, each block being 0 or 10, with an optional leading 1. The generating function is therefore $\frac{1+x}{1-x-x^2}$, whose denominator reproduces the Fibonacci recurrence directly. Reading the structure off the object and reading the recurrence off the structure are the same step.

WHERE THIS GOES WRONG

A recursion whose base cases do not match the convention used in the step. Compute a_1 and a_2 by hand, then check a_3 against the recursion before trusting anything.

SEE ALSO Induction; reflection and the ballot problem; series evaluation by generating function.

46.7 Reflection, the ballot problem, and Catalan structure

CUE Paths constrained to stay on one side of a line, balanced bracket sequences, non-crossing configurations, binary trees, triangulations of a polygon. The number $\binom{2n}{n}/(n+1)$ appearing anywhere is the signature.

METHOD The reflection principle counts bad paths by reflecting the portion before the first violation, which biject bad paths to unconstrained paths with shifted endpoints. Subtracting gives

$$C_n = \frac{1}{n+1} \binom{2n}{n},$$

the Catalan numbers 1, 1, 2, 5, 14, 42, 132, ... Recognising a Catalan family is worth more than rederiving the formula, because the same numbers answer at least a dozen superficially unrelated questions.

WORKED EXAMPLE

Balanced strings of n pairs of brackets. Encode an open bracket as an up step and a close as a down step. Balanced means the path ends at zero and never goes below it. Total paths $\binom{2n}{n}$; bad paths reflect to paths ending at -2 , of which there are $\binom{2n}{n+1}$. The difference is $\binom{2n}{n} - \binom{2n}{n+1} = C_n$. For $n = 4$ this gives 14, confirmed by enumerating all 70 balanced-length strings and testing each prefix.

SEE ALSO Bijective counting; recursion and generating functions.

46.8 Burnside and symmetry counting

CUE Counting configurations up to a symmetry: colourings of a necklace up to rotation, of a cube's faces up to rotation, of a board up to reflection. The cue is the phrase "considered the same if".

METHOD The number of orbits is the average number of fixed points of the group elements:

$$|X/G| = \frac{1}{|G|} \sum_{g \in G} |\text{Fix}(g)|.$$

The work is entirely in computing $|\text{Fix}(g)|$ for each symmetry, which for a rotation by d positions out of n with c colours is $c^{\gcd(d,n)}$.

WORKED EXAMPLE

Necklaces of six beads in three colours, up to rotation. The six rotations have $\gcd(d, 6) = 6, 1, 2, 3, 2, 1$ for $d = 0, \dots, 5$, so the fixed-point counts are $3^6, 3, 3^2, 3^3, 3^2, 3$, totalling $729 + 3 + 9 + 27 + 9 + 3 = 780$, and $780/6 = 130$. Confirmed by generating all 729 colourings and counting rotation classes.

WHERE THIS GOES WRONG

Dividing the total by the group size without averaging. That shortcut is correct only when no configuration has a nontrivial symmetry, which is exactly the case the problem is usually designed to break.

SEE ALSO Double counting; bijective counting; symmetry reduction.

Recognition matrix

If you see	Reach for	If that fails
Two sets that should match	An explicit bijection	Count both directly
A binomial identity to prove	Double counting one set	Induction on n
"At least one"	Complementary counting	Inclusion and exclusion
Several overlapping forbidden traits	Inclusion and exclusion	Group by how many hold
No fixed points, no repeats	Derangements, $D_n = n! \sum (-1)^k / k!$	Recursion $D_n = (n-1)(D_{n-1} + D_{n-2})$
Identical items, distinct boxes	Stars and bars	Substitute away lower bounds
Distinct items, distinct boxes	k^n , or surjections by IE	Exponential generating function
A diagonal sum of binomials	Hockey stick	Absorb and telescope
"No two adjacent"	Recursion split by the last element	Transfer matrix
Paths staying above a line	Reflection; expect Catalan	Ballot formula
$\binom{2n}{n}/(n+1)$ appears	You are in a Catalan family	Find the bracket encoding
"Considered the same if rotated"	Burnside averaging	Fix a canonical representative

Deduction grids and constraint propagation

What this chapter is for. The attribute puzzle, in which five houses have five owners with five drinks and five pets and a list of clues relates them, is a constraint satisfaction problem, and solving one by hand efficiently means running, by hand, the algorithm a solver would run. That algorithm has four parts: represent the state so that no deduction is lost, propagate every consequence of every new fact immediately, recognise the two kinds of forced move, and, when propagation stalls, branch at the point where branching is cheapest. This chapter is that algorithm. The same machinery solves scheduling puzzles, Sudoku, seating arrangements, and any problem in which a bijection between finite sets must be pinned down by relational clues.

Representation

47.1 The elimination grid

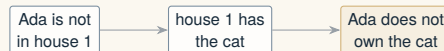
EASY

CUE Several categories of the same size, a bijection between them, and clues that are mostly negative or relational.

METHOD Draw a grid with one row per value of one category and one column per value of another, one such block for every pair of categories. Mark a cell with a cross when the pairing is excluded and a tick when it is forced. Never hold a deduction only in your head: the entire efficiency of the method comes from the fact that a written cross participates in later reasoning and a remembered one does not.

TWO RULES THAT FIRE AUTOMATICALLY

A tick in a cell crosses out the rest of its row and the rest of its column, because the relation is a bijection. And a row or column with exactly one surviving cell converts that cell to a tick. Applying these two rules to exhaustion after every single new mark is what propagation means.



The figure is the move that makes grids worth drawing. Two facts in different blocks combine through the shared category to give a fact in a third block, and the transitive step is invisible unless both parents are written down. Most stalled grids are stalled because a cross in one block was never carried across.

WHERE THIS GOES WRONG

Recording only the positive information. The crosses carry most of the content, and a grid with ticks alone loses the ability to make transitive deductions at all.

SEE ALSO Constraint propagation; forced-move reasoning.

47.2 Constraint propagation

EASY

CUE Any point at which you have just learned something.

METHOD Maintain, for each variable, the set of values it can still take. Every new fact removes values; every removal may reduce another variable's set to a single value, which is a new fact. Loop until nothing changes. Positional clues of the form "X is immediately left of Y" should be converted into explicit domain restrictions at the start: in a row of four, "immediately left" already forbids X from position 4 and Y from position 1, and those two crosses are free.

WHERE THIS GOES WRONG

Propagating one consequence and moving on. The loop must run to a fixed point, because the second and third rounds are where the useful deductions usually appear.

SEE ALSO The elimination grid; minimal-remaining-values branching.

47.3 Forced moves: the naked single and the hidden single

CUE Propagation has stalled and you are looking for the next deduction.

METHOD Two distinct patterns produce a forced assignment and they must both be checked, because scanning for one and not the other is the commonest reason a solvable puzzle looks stuck. A *naked single* is a variable whose domain has shrunk to one value: read it off. A *hidden single* is a value that can be placed in only one variable, even though that variable still has several candidates: it must go there. The naked single is found by scanning variables; the hidden single is found by scanning values, and that second scan is the one people skip.

SEE ALSO Constraint propagation; branch and contradict.

Working a puzzle

Four houses stand in a row, numbered 1 to 4 from left to right. Each is occupied by one of Ada, Bo, Cy and Di, who each drink one of tea, coffee, cocoa and juice, and keep one of a cat, a dog, a finch and a gecko. The clues are these.

- (1) Ada lives immediately to the left of the coffee drinker.
- (2) The dog is in house 1 or house 4.
- (3) Cy drinks cocoa.
- (4) Bo lives somewhere to the right of the cat.
- (5) The juice drinker keeps the finch.
- (6) Di lives in house 2.
- (7) The gecko keeper drinks tea.
- (8) The cat keeper does not drink coffee.
- (9) The juice drinker lives to the left of the tea drinker.

WORKED EXAMPLE

Start with the free domain restrictions. Clue (1) puts Ada in 1, 2 or 3 and coffee in 2, 3 or 4; clue (6) puts Di in 2, so Ada is in 1 or 3, and coffee is in 2 or 4. Clue (4) puts the cat in 1, 2 or 3 and Bo in 2, 3 or 4, and since Di holds 2, Bo is in 3 or 4.

Now combine (3) with (1). If Ada were in 3 then coffee is in 4, so Bo is in 4 and Cy in 1, giving cocoa to house 1. If Ada were in 1 then coffee is in 2, so Di drinks coffee, and Cy is in 3 or 4 with cocoa there. Keep both branches open and use the pet clues.

Clue (7) ties gecko to tea and clue (5) ties finch to juice, so cocoa and coffee are held by the cat and the dog in some order. Clue (8) forbids cat with coffee, so the cat keeper drinks cocoa and the dog keeper drinks coffee. By (3), Cy keeps the cat.

Take the second branch first: Ada in 1, coffee in 2, so Di keeps the dog in house 2, contradicting clue (2). That branch dies, and the whole puzzle collapses to the first.

So Ada is in 3, coffee is in 4, Bo is in 4 with the dog, and Cy is in 1 with the cat and cocoa. Houses 2 and 3 hold juice and tea in some order, and clue (9) puts juice left of tea, so Di in house 2 drinks juice and keeps the finch, and Ada in house 3 drinks tea and keeps the gecko. Every clue checks. Exhaustive search over all $24^3 = 13824$ assignments confirms this is the unique solution.

Notice where the work actually happened. The two pet clues (5) and (7), combined with (8), determined the entire drink-to-pet block before a single house was fixed, and that block then did all the remaining work. Looking for a pair of categories that the clues over-determine, and solving that block first, is the practical version of the branching heuristic in the next section.

When propagation stalls

47.4 Minimal-remaining-values branching

CUE The fixed point has been reached and no cell is forced, but the puzzle is not solved.

METHOD Branch, and branch where the branching factor is smallest: choose the variable with the fewest remaining candidates. Two candidates means two subproblems, and one of them will usually die within a few propagation steps. Branching on a variable with five candidates multiplies your work by five for the same information.

WHERE THIS GOES WRONG

Branching before propagation has genuinely reached a fixed point. Every deduction you could have made for free becomes a deduction you now make twice, once in each branch.

SEE ALSO Branch and contradict; exhaustive casework with a discipline.

47.5 Branch and contradict

CUE A variable with exactly two candidates, and a suspicion that one of them is impossible.

METHOD Assume one candidate, propagate aggressively, and look for a contradiction. If one appears, the other candidate is forced and you have gained a fact without the bookkeeping of a genuine case split. If no contradiction appears in a dozen steps, abandon the trial and keep both branches properly. The technique in the worked puzzle above, where assuming Ada in house 1 died against clue (2), is exactly this.

WHERE THIS GOES WRONG

Carrying marks made during a failed trial back into the main grid. Use a separate sheet, or mark trial deductions in a way you can erase; a contaminated grid is worse than no grid.

SEE ALSO Minimal-remaining-values branching; proof by contradiction.

Recognition matrix

If you see	Reach for	If that fails
Equal-sized categories, relational clues	An elimination grid, all pairs	Domain lists per variable
"Immediately left of"	Free crosses at both ends	Enumerate the possible positions
"Somewhere to the right of"	Cross the extreme cells of both	Combine with a second ordering clue
A new tick placed	Cross its row and its column	Then carry across to other blocks
A row with one survivor	Naked single: assign it	Scan values for a hidden single
A value with one home	Hidden single: assign it	Scan variables for a naked single
Two blocks over-determined	Solve that pair first	Look for a clue chain linking them
Propagation stalled	Branch on the smallest domain	Trial one candidate for contradiction
Two candidates left	Branch and contradict	Keep both branches cleanly separated
Two full solutions found	The clue set is incomplete	Re-read for an unencoded clue

Combinatorial games and strategy

What this chapter is for. A finite two-player game with no chance and no hidden information is, in principle, already solved: work backwards from the terminal positions and every position is labelled a win or a loss for the player to move. In practice that computation is too large to run by hand, and the craft is finding the pattern in the losing positions so that the label can be read off a position directly. This chapter covers backward induction and the two-rule characterisation it produces, the Nim theory and why exclusive-or of all the heap sizes is the right invariant, Grundy values for games that split, and the two non-constructive arguments, pairing and strategy stealing, that prove somebody wins without ever exhibiting a move.

Backward induction and the pattern in the losses

48.1 Winning and losing positions by backward induction

EASY

CUE A game with a finite set of positions, alternating moves, and a player who loses when unable to move. The question is “who wins” or “what is the winning first move”.

METHOD Label positions from the end. A position is a loss for the player to move, an $\mathcal{P}$ -position, exactly when every legal move leads to a $\mathcal{W}$ -position; it is a win exactly when some legal move leads to a $\mathcal{P}$ -position. Compute the labels for small positions, look for the pattern, and then prove the pattern by verifying those two closure conditions. That verification, not the table, is the proof.

THE TWO CONDITIONS THAT CHARACTERISE A LOSING SET

A set L of positions is exactly the set of losing positions if and only if every terminal position is in L , no move from a position in L leads to a position in L , and from every position outside L some move leads into L . Guessing L from a table and then checking these two lines is the standard proof, and it needs no induction hypothesis to be stated.

WORKED EXAMPLE

A pile of n counters; a move removes 1, 3 or 4; the player who cannot move loses. Computing the labels for $n = 0, 1, 2, \dots$ gives losing positions $0, 2, 7, 9, 14, 16, \dots$, which is the set of n congruent to 0 or 2 modulo 7. Verify: from $n \equiv 0$ the moves reach $n - 1 \equiv 6$, $n - 3 \equiv 4$, $n - 4 \equiv 3$, none of which is 0 or 2; from $n \equiv 2$ they reach 1, 6, 5, again none; and from each of 1, 3, 4, 5, 6 modulo 7 some legal subtraction lands on 0 or 2. So the losing positions are exactly $n \equiv 0, 2 \pmod{7}$, confirmed by direct computation up to $n = 20$.

WHERE THIS GOES WRONG

Reading a pattern off four values. Compute at least two full periods before conjecturing, and always finish with the closure check; subtraction games in particular are eventually periodic but frequently have an irregular prefix.

SEE ALSO Grundy values and the mex rule; Nim and the exclusive-or rule; induction.

Nim, and why exclusive-or

48.2 Nim and the exclusive-or rule

MEDIUM

CUE Several independent heaps, a move affecting exactly one of them, and the player unable to move losing.

METHOD Compute the *Nim-sum*, the bitwise exclusive-or of the heap sizes. The position is losing for the player to move exactly when the Nim-sum is zero. To win from a nonzero Nim-sum s , find a heap h with $h \oplus s < h$, which exists because the leading bit of s must appear in some heap, and reduce that heap to $h \oplus s$.

The argument is the closure check again, in two lines. From a position with Nim-sum zero, any move changes exactly one heap, and changing one term of an exclusive-or necessarily changes the total, so every move leaves a nonzero Nim-sum. From a position with Nim-sum $s \neq 0$, the move described above produces heaps whose exclusive-or is $s \oplus s = 0$. So zero Nim-sum is closed under moving out of, and reachable from everything else, which is exactly the characterisation of the losing set.

WORKED EXAMPLE

Heaps of 3, 5 and 7. The Nim-sum is $3 \oplus 5 \oplus 7 = 1$, nonzero, so the first player wins. The heap to change is one with $h \oplus 1 < h$, and $7 \oplus 1 = 6 < 7$, so removing one counter from the heap of seven leaves 3, 5, 6 with Nim-sum 0. Heaps of 1, 2, 3 have Nim-sum 0 and are a loss for the player to move.

WHERE THIS GOES WRONG

Adding the heap sizes instead of exclusive-oring them. The ordinary sum is not invariant under the losing set and gives the wrong answer as soon as three heaps are unequal.

SEE ALSO Grundy values and the mex rule; winning and losing positions.

HARD

48.3 Grundy values and the mex rule

CUE A game that decomposes into independent components, where a move acts on one component, and the components are not themselves Nim heaps. Also any game where a move splits a position into two.

METHOD Assign each position a Grundy value G , defined recursively as the least non-negative integer not among the Grundy values of its options:

$$G(x) = \text{mex}\{G(y) : x \rightarrow y\}.$$

A position is losing exactly when $G = 0$. The Sprague and Grundy theorem says a disjoint sum of games has Grundy value equal to the exclusive-or of the components' values, so every such game is equivalent to a single Nim heap of that size, and Nim theory then applies unchanged.

WORKED EXAMPLE

For the subtraction game with moves $\{1, 3, 4\}$ the Grundy values for $n = 0, 1, \dots, 12$ are

$$0, 1, 0, 1, 2, 3, 2, 0, 1, 0, 1, 2, 3,$$

periodic with period 7, and the zeros sit at $n \equiv 0, 2 \pmod{7}$ as computed before. Now play three such piles at once, of sizes 5, 6 and 11. Their Grundy values are 3, 2 and 2, whose exclusive-or is $3 \oplus 2 \oplus 2 = 3$, nonzero, so the first player wins by moving in the pile of five to reach a pile of Grundy value $2 \oplus 2 = 0$, that is, to a size congruent to 0 or 2 modulo 7: reduce five to two.

WHERE THIS GOES WRONG

Taking the exclusive-or of the heap *sizes* rather than of their Grundy values. The two coincide only for ordinary Nim, where $G(n) = n$.

SEE ALSO Nim and the exclusive-or rule; backward induction.

Winning without a formula

MEDIUM

48.4 Symmetry and pairing strategies

CUE A game on a symmetric board, or a game whose moves come in natural pairs, especially one where the second player is asked to win, or where the total number of moves is fixed and its parity decides the outcome.

METHOD Have one player answer every move with its mirror image, or with the partner in a fixed pairing of the resources. The strategy is legal as long as the mirror move is always available when needed, and that availability is the only thing to verify. A pairing argument proves a win without ever computing a Grundy value.

WORKED EXAMPLE

Two players alternately place non-overlapping identical coins anywhere on a circular table; the player who cannot place loses. The first player wins by placing the first coin at the exact centre and thereafter mirroring the opponent through the centre. If the opponent's move was legal, its reflection is legal too, because the table is symmetric and

the centre is already occupied, so no reflected coin can overlap the first one or itself. The first player therefore always has a reply and cannot be the one to run out.

WHERE THIS GOES WRONG
A mirroring strategy on a board with a fixed point that is not occupied. If the centre cell is free, the opponent can play it and destroy the pairing. Occupying or eliminating the fixed point is what the first move is for.

SEE ALSO Strategy stealing; parity invariant; the extremal principle.

48.5 Strategy stealing

HARD

CUE A symmetric game in which having an extra move can never hurt, and the question asks only whether the second player can win, not how anyone wins.

METHOD Suppose the second player had a winning strategy. The first player makes an arbitrary move, then adopts that strategy, treating their own extra piece as irrelevant. Since the extra move can never be a disadvantage, the first player now wins, contradicting the assumption that both players cannot win. Therefore the second player has no winning strategy, so with a draw excluded the first player wins. Note carefully what this argument delivers: existence, and nothing else. It never tells you a single move.

WORKED EXAMPLE
In the game of Chomp, players alternately take a cell from a rectangular grid together with everything above and to the right of it, and the player forced to take the bottom-left poisoned cell loses. The first player wins. Suppose the second player had a winning reply to the first player's move at the top-right corner. That reply is itself a position the first player could have created directly on move one, and could then follow the same strategy. So the second player cannot have a winning strategy, and since the game is finite with no draws, the first player does. No winning first move for a general rectangle is known.

WHERE THIS GOES WRONG
Using strategy stealing in a game where an extra move can hurt. Many games are not monotone in that sense, and the argument silently fails; the monotonicity is a hypothesis to check, not a convention.

SEE ALSO Symmetry and pairing strategies; proof by contradiction.

Recognition matrix

If you see	Reach for	If that fails
One pile, a fixed move set	Tabulate labels; find the period	Grundy values, same table
Several independent heaps	Nim-sum by exclusive-or	Grundy values, then exclusive-or
Components that are not heaps	Grundy value by the mex rule	Sprague and Grundy sum
A conjectured losing set	Verify the two closure conditions	Extend the table two periods
A symmetric board	Mirror strategy; handle the fixed point	Pair the resources explicitly
"Show the second player wins"	Pairing, not strategy stealing	Parity of the total move count
"Show the first player wins", no move asked	Strategy stealing	Check that an extra move cannot hurt
A fixed total number of moves	Parity decides it	Count the moves exactly
Move splits a position in two	Grundy of the parts, exclusive-ored	Recompute the mex table

Probability and expectation

What this chapter is for. One tool dominates this subject, and it is worth stating before anything else: expectation is linear, always, with no hypothesis whatever about independence. That single fact solves problems whose direct computation is hopeless, because it lets you break a complicated random quantity into a sum of indicators, compute each indicator's probability in isolation, and add. Everything else in the chapter is either a way of computing one of those probabilities (conditioning, Bayes, symmetry) or a way of handling quantities that are not sums (states and transitions, tail sums). The chapter also treats the classical paradoxes seriously, because each of them fails in the same way: the intuition is answering a correctly posed question about a different sample space, and naming that sample space is the repair.

The master tool

49.1 Linearity of expectation

EASY

CUE A random quantity that counts something: the number of fixed points, of matches, of records, of empty boxes, of adjacent pairs. Any question beginning “what is the expected number of”.

METHOD Write $X = X_1 + \dots + X_n$ and use

$$\mathbb{E}[X] = \mathbb{E}[X_1] + \dots + \mathbb{E}[X_n],$$

which holds whether or not the X_i are independent, and whether or not they are even defined on comprehensible events. The distribution of X is irrelevant and usually much harder than the answer.

WORKED EXAMPLE

A random permutation of n objects: what is the expected number of fixed points? Let X_i be 1 if i is fixed. Then $\mathbb{E}[X_i] = 1/n$ by symmetry, so $\mathbb{E}[X] = n \cdot (1/n) = 1$, independent of n . The indicators are dependent (knowing $n - 1$ points are fixed forces the last), and it does not matter. Verified by averaging over all 120 permutations of five objects, which gives exactly 1.

WHERE THIS GOES WRONG

Trying to compute the distribution first. The number of fixed points has a messy distribution built out of derangement numbers; the mean does not need any of it.

SEE ALSO Indicator random variables; symmetry arguments in probability; inclusion and exclusion.

49.2 Indicator random variables as a protocol

EASY

CUE Any application of linearity. This is the mechanical half of the previous technique and deserves to be run as a checklist.

METHOD Four steps, in order. Name the thing being counted. Introduce one indicator per potential contribution, saying exactly what event it indicates. Compute each indicator's probability, using symmetry wherever the situation is exchangeable. Sum. The step people skip is the second, and skipping it is how an index gets counted twice.

WORKED EXAMPLE

Coupon collector. With n distinct coupons, the expected number of purchases to collect all of them is not a sum of indicators over coupons but a sum over *stages*: once $k - 1$ distinct coupons are held, the number of further purchases to get a new one is geometric with success probability $(n - k + 1)/n$, so its expectation is $n/(n - k + 1)$. Summing over k ,

$$\mathbb{E}[T] = n \sum_{j=1}^n \frac{1}{j} = nH_n,$$

which for $n = 6$ is 14.7 exactly. Note that the decomposition is into stages, not into coupons; choosing the right decomposition is the whole of step two.

SEE ALSO Linearity of expectation; tail-sum expectation; states and transitions.

Conditioning

49.3 Conditional probability and Bayes as a mechanical update

MEDIUM

CUE New information arrives and a probability must be revised. Also any problem in which a rare condition is tested by an imperfect test.

METHOD Write the update as odds, not as fractions, because the odds form removes the denominator that causes most errors:

$$\frac{\mathbb{P}(H | E)}{\mathbb{P}(\neg H | E)} = \frac{\mathbb{P}(H)}{\mathbb{P}(\neg H)} \times \frac{\mathbb{P}(E | H)}{\mathbb{P}(E | \neg H)}.$$

posterior odds prior odds likelihood ratio

Then convert back to a probability at the end if the question asks for one.

WORKED EXAMPLE

A condition affects one person in a thousand. A test has a 99% detection rate and a 2% false positive rate. A random person tests positive. Prior odds 1 : 999. Likelihood ratio $0.99/0.02 = 49.5$. Posterior odds 49.5 : 999, that is about 1 : 20.2, so the probability is roughly 0.0472. The test is informative, multiplying the odds by fifty, and the person is still very probably unaffected, because fifty is small compared with a thousand.

WHERE THIS GOES WRONG

Confusing $\mathbb{P}(E | H)$ with $\mathbb{P}(H | E)$. The detection rate is a statement about the test given the condition, and reporting it as the answer inverts the conditioning entirely.

SEE ALSO Repairing the sample space; symmetry arguments.

49.4 Repairing the sample space: the anatomy of a paradox

MEDIUM

CUE An answer that everybody finds obvious and that is wrong, or a problem where two defensible answers exist. The universal cue is a phrase like “you are told that at least one” or “the host then opens”, where the mechanism generating the information is left implicit.

METHOD Every classical paradox in this subject has the same anatomy. The intuitive answer is the correct answer to a well-posed question about a different sample space, and the fix is to write down explicitly what random experiment generated the information you were given. Do that before computing anything.

THREE DOORS

A prize sits behind one of three doors. You choose one; the host, who knows where the prize is and always opens a different door with no prize behind it, opens one; you may switch. Switching wins with probability $2/3$. The intuition that says $1/2$ is answering the question “a door has been eliminated at random, what now”, which would indeed give $1/2$. The host’s knowledge is what breaks the symmetry: the door he opens is not a random door, and his choice carries information about the two doors you did not pick. If instead the host opens a door at random and it happens to be empty, the answer really is $1/2$, and the two versions differ only in a sentence about the host.

TWO CHILDREN

A family has two children. Told only that at least one is a boy, the probability both are boys is $1/3$, since the sample space $\{BB, BG, GB, GG\}$ loses only GG . Told instead that the elder is a boy, the answer is $1/2$. The two pieces of information are different, and the difference is which subset of the four equally likely families survives. Verified by enumeration.

With 23 people the probability that some pair shares a birthday is $1 - \prod_{k=0}^{22} (365 - k)/365 \approx 0.5073$. The intuition that expects a much larger group is comparing against the wrong count: there are $\binom{23}{2} = 253$ pairs, not 23 people, and the relevant quantity is the number of opportunities for a collision.

Arguing about the answer instead of about the sample space. Two people who disagree about one of these problems are almost always modelling different experiments, and until that is settled no amount of arithmetic helps.

SEE ALSO Conditional probability and Bayes; symmetry arguments; complementary counting.

MEDIUM

49.5 Symmetry arguments in probability

CUE Exchangeable objects, a uniformly random arrangement, or a question about a particular position among many identical ones.

METHOD If a relabelling of the objects leaves the probability model unchanged, then quantities related by that relabelling have equal probability. This converts a question about a specific position into an averaging argument and often gives the answer with no computation at all.

A deck is shuffled uniformly. What is the probability that the ace of spades lies immediately above the ace of hearts? By symmetry every ordered pair of distinct positions is equally likely for the two cards, so the answer is the number of adjacent ordered pairs over the total: $51/(52 \cdot 51) = 1/52$. Equivalently, consider the 52 possible positions of the hearts ace relative to the spades ace.

SEE ALSO Linearity of expectation; repairing the sample space.

Quantities that are not sums

HARD

49.6 States, transitions, and absorbing reasoning

CUE An expected time until something happens, a probability of reaching one outcome before another, or a process with memory. The words “on average how long” are the strongest cue.

METHOD Define E_i as the expected time to absorption from state i , write one equation per state by conditioning on the first step, and solve the linear system. The equation is always of the form

$$E_i = 1 + \sum_j p_{ij} E_j,$$

with $E = 0$ at the absorbing states. Choosing the right state space is the whole difficulty, and the right state space is the smallest one in which the process is Markov.

Expected number of fair coin tosses until the pattern HH appears. States: no progress, one H, done. From no progress, $E_0 = 1 + \frac{1}{2}E_1 + \frac{1}{2}E_0$. From one H, $E_1 = 1 + \frac{1}{2} \cdot 0 + \frac{1}{2}E_0$. Solving gives $E_0 = 6$. For the pattern HT the same method gives 4. The asymmetry surprises people and is entirely explained by what a failure costs: a failed HH attempt (seeing T) throws away all progress, while a failed HT attempt (seeing H) leaves you exactly where you were.

A walker on $0, 1, \dots, N$ moves left or right with probability $\frac{1}{2}$, absorbed at both ends. The probability of absorption at N starting from k is k/N by the martingale (or by solving the linear recurrence $p_k = \frac{1}{2}p_{k-1} + \frac{1}{2}p_{k+1}$), and the expected time to absorption is $k(N - k)$. With $N = 5$ and $k = 2$ this gives an expected 6 steps.

A state space that is not Markov. If the answer depends on how you arrived at a state, the state is under-specified, and the system of equations you write down will be inconsistent or simply wrong.

SEE ALSO Tail-sum expectation; linearity of expectation.

49.7 Tail-sum expectation and hitting times

CUE A non-negative integer random variable whose tail probabilities $\mathbb{P}(X \geq k)$ are easier to compute than its point probabilities. Common for maxima, for waiting times, and for records.

METHOD Use

$$\mathbb{E}[X] = \sum_{k \geq 1} \mathbb{P}(X \geq k),$$

which for maxima is far easier than the distribution, since $\mathbb{P}(\max \geq k) = 1 - \mathbb{P}(\text{all} < k)$ is a product.

WORKED EXAMPLE

Roll a fair die repeatedly until a six appears. Then $\mathbb{P}(X \geq k) = (5/6)^{k-1}$, so $\mathbb{E}[X] = \sum_{k \geq 1} (5/6)^{k-1} = 6$, recovering the geometric mean without differentiating a series.

SEE ALSO States and transitions; indicator random variables.

HARD

49.8 The probabilistic method

CUE An existence claim about a large combinatorial structure: a colouring with no monochromatic piece, a set with no three in arithmetic progression, a tournament with a property.

METHOD Put a probability measure on the candidate objects and show that a random one has the desired property with positive probability, or that the expected number of defects is less than one. Either conclusion proves that a good object exists, without constructing one.

WORKED EXAMPLE

Any 2-colouring question of the following shape: colour the edges of the complete graph on n vertices at random. The expected number of monochromatic copies of K_k is $\binom{n}{k} 2^{1-\binom{k}{2}}$. If that quantity is less than 1, some colouring has none, which gives the classical lower bound on the Ramsey number. For $k = 3$ and $n = 5$ the expectation is $10 \cdot 2^{-2} = 2.5$, too large to conclude; for $k = 4$ and $n = 17$ it is $2380 \cdot 2^{-5} \approx 74$, also too large, which is why the useful bounds appear only for larger k .

WHERE THIS GOES WRONG

Concluding that the random object almost always works. Positive probability is all you get, and it is all you need; claiming more is claiming something the argument does not support.

SEE ALSO Linearity of expectation; pigeonhole with structure; the extremal principle.

Recognition matrix

If you see	Reach for	If that fails
"Expected number of"	Indicators, then linearity	Decompose into stages
Dependence looks fatal	Linearity does not care	Check the indicator definitions
Collect all n types	Stages; $\mathbb{E} = nH_n$	Tail-sum formula
A test result revises a belief	Odds form of Bayes	Tree of the four outcomes
"At least one is a boy"	Enumerate the sample space	Ask what generated the information
A host who knows and chooses	The choice carries information	Compare the random-host variant
Two defensible answers	The experiment is under-specified	Write the mechanism down
Exchangeable objects	Symmetry: all positions alike	Count ordered pairs
"On average how long until"	States, one equation per state	Check the state is Markov
Waiting for a pattern	Progress states, first-step conditioning	Correlation of the pattern with itself
Maximum of several variables	Tail sum with a product	Order statistics
"Show such an object exists"	Random object, expected defects < 1	Explicit construction

Number-theoretic puzzle technique

What this chapter is for. Most number-theoretic puzzles are asking you to choose a modulus, and almost all of the difficulty is in that choice. A good modulus makes the problem collapse to a one-line check; a bad one leaves it exactly as opaque as before. So this chapter is organised around the choice itself. Look at the exponents and you are being pointed at Fermat, Euler, and multiplicative order. Look at the digits and you are being pointed at 9 and 11. Look at a claim about squares and you are being pointed at 3, 4 and 8, since squares occupy only $\{0, 1\}$ modulo 3 and 4 and only $\{0, 1, 4\}$ modulo 8. Look at an equation with a product plus linear terms and you are being pointed at factoring, not at congruences at all. The rest of the chapter covers the constructions that go beyond a single modulus: the Chinese remainder theorem, base representation, valuations, and the two descent arguments.

Choosing the modulus

50.1 Residue classes and modular reduction

EASY

CUE An assertion that some expression is always divisible by m , or never equal to something, or that a Diophantine equation has no solutions.

METHOD Reduce the whole statement modulo a small m and check the finitely many residue classes. Choose m so that the awkward terms become constant: powers become periodic, squares occupy a small set, and a coefficient divisible by m disappears entirely. For non-existence proofs the standard first attempts are $m = 3, 4, 8, 9$ and 16.

WORKED EXAMPLE

Show that $n^5 - n$ is divisible by 30 for every integer n . Factor $30 = 2 \cdot 3 \cdot 5$ and treat each prime separately. Modulo 2 and modulo 3, Fermat gives $n^2 \equiv n$ and $n^3 \equiv n$, so $n^5 \equiv n$ in both. Modulo 5, Fermat gives $n^5 \equiv n$ directly. Three coprime moduli, so $30 \mid n^5 - n$, verified for all n from -50 to 50.

THE QUADRATIC RESIDUES WORTH MEMORISING

Modulo 3 a square is 0 or 1. Modulo 4 a square is 0 or 1. Modulo 8 a square is 0, 1 or 4. Modulo 9 a cube is 0, 1 or 8. These four lines settle a surprising fraction of all impossibility questions about sums of squares and cubes.

WHERE THIS GOES WRONG

Choosing a modulus that does not separate the cases. If every residue survives, you have learned nothing and must try another m rather than concluding the statement is true.

SEE ALSO Digit sums and casting out nines; last digits by multiplicative order; base representation.

50.2 Digit sums, casting out nines, and digital roots

EASY

CUE The problem mentions the digits of a number in base ten, or asks about a number formed by concatenation, or asserts a divisibility by 9 or 11.

METHOD Since $10 \equiv 1 \pmod{9}$, a number is congruent to its digit sum modulo 9; since $10 \equiv -1 \pmod{11}$, it is congruent to its alternating digit sum modulo 11. Both facts turn a statement about digits into a congruence, which is the only way digits ever become tractable.

WORKED EXAMPLE

Let N be the number formed by writing 1, 2, ..., 2026 in order. Is N divisible by 9? The digit sum of N is the sum of the digit sums of the parts, which modulo 9 is $\sum_{k=1}^{2026} k = 2026 \cdot 2027/2 = 2053351$. That number's digit sum is 19, whose digit sum is 10, so $N \equiv 1 \pmod{9}$ and N is not divisible by 9.

SEE ALSO Residue classes; base representation.

50.3 Last k digits: multiplicative order, Fermat, and Euler

CUE A large exponent and a question about the last digit, the last two digits, or a remainder modulo a composite.

METHOD Compute the multiplicative order of the base modulo m , which divides $\varphi(m)$, and reduce the exponent modulo that order. Euler's theorem gives $a^{\varphi(m)} \equiv 1 \pmod{m}$ when $\gcd(a, m) = 1$, and Fermat is the prime case. When the base and modulus share a factor, split m into prime powers and recombine by the Chinese remainder theorem, since Euler does not apply to the shared part.

WORKED EXAMPLE

Find the last two digits of 7^{222} . Here $m = 100$ and $\varphi(100) = 40$, so the order divides 40; in fact $7^4 = 2401 \equiv 1 \pmod{100}$, so the order is 4. Then $222 \equiv 2 \pmod{4}$ and $7^{222} \equiv 7^2 = 49 \pmod{100}$. Confirmed directly.

WHERE THIS GOES WRONG

Reducing the exponent modulo m instead of modulo the order. The exponent lives in a different ring from the base, and confusing the two is the single commonest error in this whole area.

SEE ALSO Chinese Remainder Theorem; residue classes; lifting the exponent.

Constructions beyond one modulus

50.4 The Chinese Remainder Theorem

CUE Simultaneous congruences with pairwise coprime moduli, or a divisibility by a composite that factors, or the phrase “find the smallest positive integer such that”.

METHOD For pairwise coprime $m_1, \dots, m_r$, the system $x \equiv a_i \pmod{m_i}$ has a unique solution modulo $M = \prod m_i$. In practice solve two congruences at a time by substitution, which is faster by hand than the general formula and less error-prone. The theorem is also a licence to work one prime at a time and recombine, which is how most divisibility problems should be organised.

WORKED EXAMPLE

Find x with $x \equiv 2 \pmod{3}$, $x \equiv 3 \pmod{5}$, $x \equiv 2 \pmod{7}$. From the first, $x = 3t + 2$; substituting into the second gives $3t \equiv 1 \pmod{5}$, so $t \equiv 2 \pmod{5}$ and $x = 15s + 8$. The third gives $15s + 8 \equiv 2 \pmod{7}$, that is $s \equiv 1 \pmod{7}$, so $x = 105u + 23$. The smallest positive solution is 23, and the solution set is 23 modulo 105, confirmed by search.

WHERE THIS GOES WRONG

Applying the theorem with moduli that are not coprime. The system may still be solvable, but only if the congruences agree on every common factor, and the modulus of the answer is the least common multiple rather than the product.

SEE ALSO Last digits by multiplicative order; residue classes.

50.5 Base representation

CUE Digits in a base other than ten, a weighing or measuring problem, a problem about binary operations, or a sum in which each term is used at most once.

METHOD Write the quantity in the base the problem is secretly using. Sums of distinct powers of two are binary strings; sums of powers of three with coefficients $-1, 0, 1$ are balanced ternary, which is what makes the classical weighing problem work; and questions about carrying digits are questions about the base representation of a sum.

WORKED EXAMPLE

With weights placed on both pans of a balance, which sets of weights can measure every integer mass from 1 to 40? Balanced ternary gives every integer in that range a representation $\sum \epsilon_i 3^i$ with $\epsilon_i \in \{-1, 0, 1\}$, so the weights 1, 3, 9, 27 suffice, with $\epsilon = -1$ meaning the weight joins the object's pan. Four weights, and $(3^4 - 1)/2 = 40$ is exactly the largest reachable mass, so the set is optimal.

SEE ALSO The information-theoretic bound for weighings; digit sums.

HARD

50.6 Lifting the exponent and p -adic valuations

CUE An exact power of a prime dividing an expression, the phrase “the largest k such that p^k divides”, or factorials and binomial coefficients with divisibility conditions.

METHOD Write $v_p(n)$ for the exponent of p in n . Legendre’s formula gives $v_p(n!) = \sum_{i \geq 1} \lfloor n/p^i \rfloor$. The lifting-the-exponent lemma gives, for odd p with $p \mid a - b$ and $p \nmid a, b$,

$$v_p(a^n - b^n) = v_p(a - b) + v_p(n),$$

with a separate statement at $p = 2$ that carries an extra term. These two formulas answer nearly every exact-power question that appears in practice.

WORKED EXAMPLE

The exponent of 2 in $100!$ is $50 + 25 + 12 + 6 + 3 + 1 = 97$, so 2^{97} divides $100!$ and 2^{98} does not. And $v_3(10^9 - 1) = v_3(9) + v_3(9) = 2 + 2 = 4$, so 81 divides $10^9 - 1$ and 243 does not.

SEE ALSO Residue classes; Chinese Remainder Theorem.

Factoring, and going downwards

EASY

50.7 Factoring the constraint

CUE A Diophantine equation containing a product of the unknowns together with linear terms, or a divisor-counting question. The shape $xy + ax + by = c$ is the signature.

METHOD Add the constant that completes the product:

$$xy + ax + by = c \iff (x + b)(y + a) = c + ab.$$

Then enumerate the divisors of the right-hand side. This converts an equation with infinitely many candidate solutions into a finite list, and it is almost always faster than any congruence argument on the same equation.

WORKED EXAMPLE

Solve $xy + 3x + 2y = 12$ over the integers. Add 6: $(x + 2)(y + 3) = 18$. The divisor pairs of 18, positive and negative, give the full solution set, which includes $(x, y) = (1, 3), (0, 6), (-1, 15), (-5, -9), (-8, -6), (-11, -5)$ and the rest, twelve solutions in all, found by enumerating divisors and confirmed by direct search.

WHERE THIS GOES WRONG

Enumerating only the positive divisors when the problem allows negative integers. Half the solutions live there.

SEE ALSO Residue classes; Vieta jumping.

HARD

50.8 Vieta jumping and infinite descent

CUE A symmetric Diophantine condition, most characteristically a divisibility $ab + c \mid a^2 + b^2 + d$, or a claim that a certain quotient is always a perfect square. Also any equation where a solution can be transformed into a smaller one.

METHOD Fix the value k of the quotient and regard the equation as a quadratic in one variable. The two roots satisfy Vieta’s relations, so from one solution (a, b) you obtain another, $(b, \text{second root})$, and the second root is an integer. Show it is smaller, and descend to a minimal solution, which must have a special form you can analyse directly.

WORKED EXAMPLE

Suppose $ab + 1 \mid a^2 + b^2$ and write $a^2 + b^2 = k(ab + 1)$. Fix k and take the solution with $a + b$ minimal, with $a \geq b$. Then a is a root of $x^2 - kbx + (b^2 - k) = 0$, whose other root is $a' = kb - a = (b^2 - k)/a$. It is an integer, it is non-negative, and it is smaller than a , so by minimality $b = 0$, whence $k = a^2$ is a perfect square. Brute force over $a, b \leq 60$ finds only $k \in \{1, 4, 9\}$, all squares, as the argument predicts.

WHERE THIS GOES WRONG

Descending without checking that the new solution is still admissible. The jump must preserve non-negativity as well as integrality, and the boundary case where the new root is zero is the base of the descent rather than a failure of it.

SEE ALSO Well-ordering and the minimal counterexample; factoring the constraint.

Recognition matrix

If you see	Reach for	If that fails
"Always divisible by m "	Factor m ; one prime at a time	Induction on n
"No integer solutions"	Try mod 3, 4, 8, 9, 16	Descent or factoring
A claim about squares	Squares mod 4 are 0, 1; mod 8 are 0, 1, 4	Mod 3 for sums of two squares
Digits in base ten	Digit sum mod 9; alternating mod 11	Write the base expansion
A large exponent	Multiplicative order, then reduce it	Euler, then CRT on prime powers
Base and modulus share a factor	Split into prime powers	Euler fails on the shared part
Simultaneous congruences	CRT two at a time by substitution	Check coprimality first
Weights on both pans	Balanced ternary	Information bound
"Largest k with $p^k \mid$ "	Legendre for factorials; LTE otherwise	Direct valuation count
$xy + ax + by = c$	Add ab and factor	Enumerate divisors, both signs
Symmetric divisibility, quotient asked	Vieta jumping	Bound the minimal solution
A solution generates a smaller one	Infinite descent	Well-ordering

Weighing, measuring, crossing, and information

What this chapter is for. A large family of puzzles is decided before you design a strategy, by an information count. A balance with three outcomes, used three times, distinguishes at most 27 situations, so a puzzle that asks you to identify one of 12 coins *and* its direction of error, which is 24 situations, is telling you that the bound is nearly tight and that no weighing may be wasted. Reading that bound first turns an open-ended search into a design problem with a specification. The chapter covers the information bound and how to use it, pouring problems and their arithmetic, state-space search done honestly, comparison lower bounds, the distributed-strategy family in which a parity code carries information nobody individually has, and then the geometric side: area invariants, tilings, symmetry, and extremal configurations.

Counting the information first

51.1 The information-theoretic bound

MEDIUM

CUE A puzzle that limits the number of tests, queries, weighings or questions, and asks whether a given number suffices.

METHOD Count the outcomes. If each test has b possible results and you are allowed t of them, the entire procedure can distinguish at most b^t situations, so a problem with N possible truths requires $t \geq \log_b N$. Use the bound in both directions: it proves impossibility when $N > b^t$, and when N is just under b^t it tells you the strategy must split the possibilities as evenly as the arithmetic allows, which is a strong design constraint.

WORKED EXAMPLE

Twelve coins, one counterfeit of unknown weight direction, a balance, three weighings. There are 24 possible truths and $3^3 = 27$ outcomes, so three weighings might suffice and no weighing may be badly unbalanced. The first weighing must therefore split 24 into three groups as near to eight each as possible, which forces four coins against four, leaving four aside: eight truths in each pan outcome and eight in the balanced outcome. Every subsequent weighing is chosen by the same rule, and the strategy essentially writes itself once the balance requirement is recognised.

WHERE THIS GOES WRONG

Counting the objects rather than the truths. Twelve coins is not twelve possibilities, because the direction of the error is part of what must be determined; getting this count wrong is what makes the puzzle look impossible or trivially easy.

SEE ALSO Comparison and sorting lower bounds; base representation; the pigeonhole principle.

51.2 Comparison and sorting lower bounds

MEDIUM

CUE Any question of the form “how many comparisons are needed to”.

METHOD A comparison has two outcomes, so t comparisons distinguish at most 2^t orderings. Sorting n items must distinguish $n!$ of them, giving $t \geq \log_2 n! \approx n \log_2 n - n \log_2 e$, which is the classical bound. Finding just the maximum needs only $n - 1$ comparisons, since every non-maximum must lose at least once and each comparison produces one loser.

WORKED EXAMPLE

Find both the largest and the second largest of n items with fewest comparisons. A knockout tournament finds the largest in $n - 1$ comparisons; the second largest must have lost directly to the largest, and only $\lceil \log_2 n \rceil$ items did, so a second tournament among those costs $\lceil \log_2 n \rceil - 1$. Total $n + \lceil \log_2 n \rceil - 2$, which matches the information-style lower bound for this problem.

SEE ALSO The information-theoretic bound; double counting.

Arithmetic and state spaces

51.3 Pouring problems via greatest common divisor

CUE Two or more unmarked containers of known capacities and a target volume.

METHOD Every reachable volume is an integer combination of the capacities, so with capacities a and b the reachable amounts are exactly the multiples of $\gcd(a, b)$ that do not exceed $\max(a, b)$. Bezout's identity gives the coefficients, and the sign of each coefficient tells you which jar is being filled and which emptied.

WORKED EXAMPLE

Jugs of 5 and 3 litres, target 4. Since $\gcd(5, 3) = 1$ divides 4, it is reachable, and $4 = 3 \cdot 3 - 5$ suggests filling the three-litre jug three times and emptying the five-litre jug once. Concretely: fill the three, pour into the five; fill the three again, top up the five (which takes two), leaving one in the three-litre jug; empty the five, pour the one across, fill the three and add it, giving four.

WHERE THIS GOES WRONG

Assuming a target below the smaller capacity is easier. Reachability depends only on divisibility by the greatest common divisor, not on size, and 4 from $\{5, 3\}$ is no harder than 2.

SEE ALSO Base representation; explicit state-space search.

51.4 Explicit state-space search

CUE Crossings, jumps, sliding pieces, or any puzzle with a small set of configurations and a legality rule on transitions.

METHOD Define the state as the smallest description that determines the future, list the legal transitions, and search breadth-first, which finds a shortest solution and proves it is shortest. Prune with an invariant if one exists: an invariant that separates the start from the goal turns the search into a one-line impossibility proof.

WORKED EXAMPLE

Three couples must cross a river in a boat holding two, with the constraint that no person may be with someone else's partner unless their own is present. The state is the triple of counts on the near bank plus the boat's position. Breadth-first search over the legal states finds a shortest solution in eleven crossings, and the same search with four couples and a two-seat boat finds none, which is the honest way to establish that the four-couple version is unsolvable without a larger boat.

SEE ALSO Pouring problems; parity invariant; branch and contradict.

51.5 Distributed strategies and correlated errors

CUE Several participants who cannot communicate after the puzzle begins, each seeing part of the situation, who must guess simultaneously. The question asks for a strategy guaranteeing a good outcome, and the naive analysis says the guesses are independent so nothing better than chance is possible.

METHOD The naive analysis is wrong because a strategy may correlate the errors. Nobody's individual guess beats chance, and that is not the objective: the objective is to concentrate all the wrong guesses onto a few situations so that the remaining situations are entirely right. A parity code is the standard device, since it lets each participant compute what the others' parity implies about their own unseen value.

WORKED EXAMPLE

Each of n people receives a hat, black or white, chosen by independent fair coins, and sees everyone else's but not their own. All guess simultaneously; the team wins if everyone is correct. Without coordination the chance is 2^{-n} . Agree in advance that everybody assumes the total number of black hats is even; then each person deduces their own hat from the parity of what they see. This wins exactly when the true parity is even, that is with probability $1/2$, for every n . All the losses have been concentrated on the odd-parity configurations, and each individual is still right exactly half the time.

WHERE THIS GOES WRONG

Concluding from "each guess is individually a coin flip" that the team cannot do better. Individual marginals say nothing about the joint distribution, and the whole technique lives in that gap.

SEE ALSO Linearity of expectation; parity invariant; the information bound.

Geometry and dissection

MEDIUM

51.6 Area and lattice invariants

CUE A dissection, a rearrangement, or a claim about a polygon with vertices at lattice points.

METHOD Area is invariant under cutting and reassembling, which settles most rearrangement puzzles immediately, including the ones designed to look as though area has changed. For lattice polygons, Pick's theorem

$$A = I + \frac{B}{2} - 1,$$

with I interior and B boundary lattice points, converts an area statement into a counting statement and back. A standard consequence: no equilateral triangle has all three vertices on the integer lattice, since Pick forces a rational area while the equilateral formula forces an irrational multiple of a square length.

WHERE THIS GOES WRONG

Reading a picture instead of computing. The classical missing-square dissections work entirely because two slopes that look equal are not, and only arithmetic detects it.

SEE ALSO Tiling impossibility by colouring; symmetry reduction.

MEDIUM

51.7 Tiling impossibility by colouring

CUE A region and a tile shape, and the question whether the region can be covered exactly.

METHOD Colour so that each placement of the tile covers a fixed multiset of colours, then count. This is the same technique as the colouring invariants of the invariants chapter, and it is listed again here because in geometric dress the colouring is usually driven by the tile's dimensions: a $1 \times k$ tile wants colouring by index modulo k , an L-tromino wants a colouring that breaks the 2×2 symmetry, and a T-tetromino wants the chessboard.

WORKED EXAMPLE

Can 25 T-tetrominoes tile a 10×10 board? Chequer the board, which then has 50 squares of each colour. Every T covers three squares of one colour and one of the other. Suppose k of the tiles cover three black squares and the remaining $25 - k$ cover three white. Counting black squares gives $3k + (25 - k) = 2k + 25 = 50$, so $2k = 25$, which has no integer solution. No tiling exists.

SEE ALSO Colouring arguments; area and lattice invariants.

HARD

51.8 Symmetry reduction and extremal configurations

CUE A geometric optimisation, a claim that some configuration is best possible, or a counting problem over configurations that are equivalent under a symmetry group.

METHOD For optimisation, argue that an optimal configuration may be assumed symmetric, usually by a smoothing argument: replace an asymmetric pair by its symmetrised version and show the objective does not worsen. For counting, quotient by the symmetry group using the averaging formula. For existence and impossibility, take the extremal object and derive a contradiction from a local improvement.

WORKED EXAMPLE

Among all triangles inscribed in a given circle, the equilateral has the largest area. Fix two vertices; the third should maximise its distance from that chord, so it sits at the midpoint of the arc, making the triangle isosceles on that base. Applying the same argument to each of the three sides in turn forces all three to be equal. The smoothing step never decreases the area, so the equilateral triangle is optimal.

WHERE THIS GOES WRONG

Assuming the optimum is symmetric without a smoothing argument. It is true remarkably often and false often enough to matter, and the argument that establishes it is usually two lines.

SEE ALSO The extremal principle; Burnside and symmetry counting.

Recognition matrix

If you see	Reach for	If that fails
A limited number of tests	Count truths against b^t	Design for an even split
A balance, three outcomes	Powers of three; balanced ternary	Count directions of error too
"How many comparisons"	$\log_2$ of the number of orderings	Tournament construction
Unmarked jugs, a target volume	Divisibility by the gcd; Bezout	Explicit state search
Crossings, jumps, sliding pieces	Breadth-first over states	An invariant to prove impossibility
Simultaneous guesses, no talking	A parity code; correlate the errors	Concentrate losses on few cases
"Each guess is a coin flip"	Marginals do not bound the joint	Design the correlation
A dissection that gains area	Compute; the slopes differ	Pick's theorem
Lattice polygon, area asked	Pick: $A = I + B/2 - 1$	Shoelace formula
A region and one tile shape	Colour to match the tile	Weighted colouring
Optimal configuration sought	Smoothing toward symmetry	Extremal principle
Configurations up to symmetry	Burnside averaging	Canonical form

Choosing the right weapon

What this chapter is for. Eleven chapters of technique are useless if you cannot tell, looking at an unfamiliar puzzle, which of them applies. This chapter is the triage layer. It gives the five diagnostic questions that route a problem to a chapter, the specific signs that a puzzle is an invariant problem wearing a disguise, the discipline of small cases as both a discovery tool and a check, and an honest account of when brute-force enumeration by hand is the correct method rather than an admission of defeat. It closes the content of the book by returning to the claim Chapter 1 opened with.

Five questions, asked in order

52.1 Technique triage: reading the cue before reading the problem

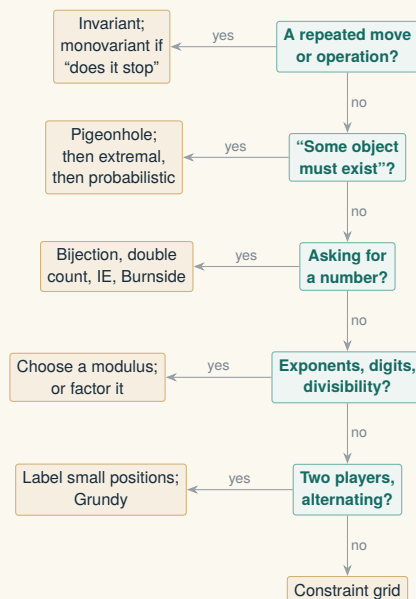
CORE

CUE Any unfamiliar puzzle.

METHOD Ask five questions before attempting anything. Each one, answered yes, names a chapter.

1. **Is there a process?** Something repeated, a move, an operation applied any number of times. If yes, ask what the move fails to change, and go to invariants. If the question is whether it stops, go to monovariants.
2. **Does it assert that something must exist, without saying which?** “Some two of them share”, “there is always a pair”. If yes, go to pigeonhole, and if pigeonhole fails, to the extremal principle or the probabilistic method.
3. **Is it a count?** If yes, ask what standard set it bijects to, and go to counting. A count with overlapping forbidden traits goes to inclusion and exclusion; a count up to a symmetry goes to Burnside.
4. **Is there a hidden modulus?** Exponents, digits, divisibility, or an impossibility claim about integers. If yes, go to number theory and choose the modulus from the shape of the expression.
5. **Is it finite and adversarial?** Two players, alternating, no chance. If yes, go to games, and start by labelling small positions.

If none of the five fires, the puzzle is probably a constraint satisfaction problem in disguise, and the grid chapter applies. That fallback is more often right than it sounds, because scheduling, seating, ordering and identification puzzles all reduce to it.



Answering the questions out of order. Question one is first because an invariant, when it exists, is both the shortest argument and the one that makes every other approach unnecessary, and a solver who reaches for counting first will often produce a correct but enormous calculation of something that a parity check settles in a line.

SEE ALSO Detecting a disguised invariant; sanity-checking by small cases.

The disguise that matters most

52.2 Detecting a disguised invariant

HARD

CUE A problem that mentions no process at all, but whose objects are naturally produced by one. Three specific disguises account for most cases.

METHOD The first disguise is a *reachability question phrased as an equality*: “can the numbers a, b, c be transformed into x, y, z ” is obviously a process, but “is there a sequence of operations after which all entries are equal” hides the target. Write the target explicitly and compare invariants.

The second is a *tiling or covering question*, which is a process in which pieces are laid down one at a time; the invariant is a colouring, and the chapter on invariants applies verbatim.

The third, and the one people miss, is a *claim about a final state*: “show that the last number on the board is odd”, “prove that the process ends with a single pile”. Any statement about the terminal configuration of a repeated operation is an invariant question, whether or not the word “invariant” or even the word “process” appears.

WORKED EXAMPLE

A set of numbers is written down; repeatedly, two numbers are erased and their sum written. Prove that the final number does not depend on the choices made. Nothing in the statement is phrased as an invariant, and yet the sum is one: it is unchanged by every move, so the terminal value equals the initial sum. Contrast with the version in which $|a - b|$ is written instead, where the sum is not invariant but its parity is, and the terminal value is therefore determined only up to parity, which is exactly what that version’s question asks about.

SEE ALSO Parity invariant; monovariants; technique triage.

Small cases, and honest enumeration

52.3 Sanity-checking by small cases

CORE

CUE Every problem, at two moments: before solving, to find the pattern, and after solving, to check the formula.

METHOD Compute the answer for $n = 1, 2, 3$ and, if it is cheap, 4. Use the values twice. Before solving, they suggest the shape of the answer and often the mechanism; a sequence that starts 1, 2, 5, 14 is telling you Catalan and therefore telling you to look for a bracket encoding. After solving, they test the formula, and a formula that fails at $n = 1$ is wrong or is missing a hypothesis. Small cases are also where degenerate behaviour lives, so a formula that works from $n = 3$ onward and fails at $n = 2$ usually indicates a case the argument silently assumed away.

WHERE THIS GOES WRONG

Checking only a case that the derivation used. If the formula was fitted to $n = 1, 2, 3$, those three values test nothing. Verify at a value the derivation never touched.

SEE ALSO Induction; recursion and generating functions; verification.

52.4 When hand enumeration is the right method

CUE A finite search space small enough to exhaust, and a problem whose structure has resisted two or three genuine attempts at a technique.

METHOD Enumerate, but enumerate with a discipline: fix an ordering of the cases, use symmetry to cut the space before starting, and record what has been eliminated rather than only what has survived. Enumeration is the right method when the space is small (a few dozen cases), when the answer is a specific configuration rather than a general formula, or when you need one concrete example to see the pattern. It is the wrong method when the space grows with a parameter that the problem leaves free, because then you are proving a special case of a general claim.

ENUMERATION AS A STEP, NOT AN ANSWER

The most productive use of enumeration is not to answer the question but to generate data for the previous technique. Exhaust the small cases, look at the sequence, conjecture, then prove. That loop is how most of the results in this part were originally found.

SEE ALSO Sanity-checking by small cases; exhaustive casework with a discipline; branch and contradict.

What the book has been arguing

The first chapter claimed that a hard problem is hard because of what you fail to notice in the first thirty seconds, and that difficulty is layers of camouflage rather than depth of technique. Fifty chapters later that claim can be stated more precisely, because we now have the catalogue of what the camouflage hides.

In every part of this book the same structure appeared. A small number of genuine ideas, perhaps a dozen per domain, are dressed in a large number of surfaces. The integral chapters found that reflection, parameter differentiation and series expansion account for most of what looks impossible, and that the exponent which appears to make a problem hard is very often the exponent that cancels. The derivative chapters found that the difficulty in a hard derivative is always algebraic, never differential. The limit chapters found that dominance and expansion do the work that L'Hopital appears to do. The differential-equation chapters found that classification is the whole game. And this part found that invariants, pigeonhole, bijection, modulus and backward induction cover nearly the entire ground of combinatorial puzzles.

Two consequences follow, and they are the reason the book is organised by recognition cue rather than by subject.

The first is that expertise in this material is mostly a matter of catalogue size. A solver who has seen forty surfaces of the reflection trick recognises the forty-first instantly, not because they compute faster but because the surface is no longer camouflage. That is a practical claim, and it makes practice a matter of collecting surfaces deliberately rather than of doing many problems of the kind you already solve.

The second is that the recognition cue is a better organising principle than the technique name. You will never be told which chapter a problem belongs to. What you will be given is a surface: an exponent that looks arbitrary, an interval that happens to be symmetric, a process with a move, a count that resists direct attack. This book is a map from those surfaces to the methods that dissolve them, and the recognition matrices at the end of each chapter are that map in its most compressed form. They are the part worth returning to.

The techniques will not change. The surfaces will keep changing, which is the whole pleasure of the subject.

Recognition matrix

If you see	Reach for	If that fails
A repeated move	What does it not change?	What does it change one way?
A claim about a final state	An invariant, disguised	A monovariant for termination
A tiling or covering	A colouring matched to the piece	Area or weighting
"Some pair must"	Pigeonhole; name the holes	Extremal principle
"There exists a configuration"	Construct it	Probabilistic method
A count that resists	Biject to a standard set	Double count, or recurse
Overlapping forbidden traits	Inclusion and exclusion	Complementary count
Exponents or digits	Choose a modulus	Factor the constraint
Two players, no chance	Label small positions	Grundy values
Categories and relational clues	Elimination grid	Branch on the smallest domain
Nothing fires	Constraint satisfaction	Enumerate small cases and look
A formula you just derived	Test it at an untouched n	Check $n = 1$ and $n = 2$
A sequence 1, 2, 5, 14, 42	Catalan; find the bracket encoding	Search the recursion
A search space of a few dozen	Enumerate with a discipline	Symmetry-reduce first

Rapid reference

The consolidated recognition matrices

What this chapter is for. Every chapter of this book ends with a recognition matrix scoped to its own material. This chapter collects the highest-value rows into five cross-domain tables, one per part, so that a reader with a problem in front of them and no idea which chapter it belongs to has a single place to look. The tables are deliberately terse. Each row is a surface feature, the move it should trigger, and the fallback if that move fails; the chapter that owns the move is where the reasoning lives. Read the leftmost column only, until something matches.

Limits, sequences and series

Surface	First move	Fallback
A quotient at infinity	Divide by the dominant term	Compare growth classes
0/0 with smooth pieces	Expand both to matching order	L'Hopital, hypotheses checked
1^∞	$\exp(\lim g \ln f)$	Substitute $u = f - 1$
$\infty - \infty$	Common denominator, or conjugate	Factor the dominant term
$0 \cdot \infty$	Move one factor into a denominator	Expand the small factor
A radical difference	Multiply by the conjugate	Binomial expansion
A sum with $1/n$ and k/n	Riemann sum; read off the integral	Bound above and below
A product with n factors	Take logs; look for a Riemann sum	Stirling
Difference quotient in disguise	It is $f'(a)$; identify f	Mean value theorem
A recursion $a_{n+1} = f(a_n)$	Prove convergence, then solve $L = f(L)$	Monotone plus bounded
$ f' < 1$ near the fixed point	Contraction; convergence is automatic	Cobweb picture
Factorials or n th powers	Ratio or root test	Stirling, then compare
An algebraic general term	Limit comparison with a p -series	Integral test
Alternating and decreasing	Alternating test; error is the next term	Check absolute convergence
A telescoping candidate	Partial fractions, then cancel	Difference of a known sequence
$\sum nx^n$ or similar	Differentiate the geometric series	Generating function
Two variables, limit at the origin	Two paths first, then polar squeeze	Bound by r times something
L'Hopital cycles	Stop; expand instead	Divide by the dominant term
A floor or fractional part	Split into cases on the interval	Squeeze between consecutive integers

Derivatives

Surface	First move	Fallback
A logarithm of a product	Log laws first, then differentiate	Logarithmic differentiation
A product of many factors	Logarithmic differentiation	Generalised product rule
$f(x)^{g(x)}$	Write as $e^{g \ln f}$	Logarithmic differentiation
A quotient with a sum on top	Split into separate terms	Quotient rule as a last resort
Trigonometric mess	Simplify with identities first	Check for a constant function
A relation, not a function	Implicit differentiation	Solve for a branch locally
Inverse function value asked	$(f^{-1})'(b) = 1/f'(a)$	Implicit differentiation
$x(t)$ and $y(t)$	$\frac{dy}{dx} = \dot{y}/\dot{x}$; second derivative needs care	Eliminate the parameter
An n th derivative	Find the orbit, or read a Maclaurin coefficient	Leibniz, most terms vanish
A variable limit of integration	Fundamental theorem plus chain rule	Full Leibniz rule
A piecewise definition	Match value, then match slope	One-sided derivatives
Rates with time	Relate first, differentiate second	Substitute values last

Surface	First move	Fallback
Optimisation with a constraint	Reduce to one variable, state the domain	Test endpoints too
Several variables	Draw the dependency graph	Sum over paths
Derivative asked only at a point	Evaluate the pieces there first	Count what vanishes
An enormous expression	Suspect it is constant	Differentiate one factor only

Integrals

Surface	First move	Fallback
Anything at all	Simplify as raw algebra first	Then classify
A rational function	Divide if improper, then partial fractions	Complete the square
$\sqrt{a^2 - x^2}$	$x = a \sin \theta$	Hyperbolic or Euler substitution
$\sqrt{x^2 \pm a^2}$	$x = a \sinh u$ or $a \cosh u$	Trigonometric substitution
A rational function of sin and cos	$t = \tan(x/2)$	Symmetry first; it is often cheaper
$g'(x)/g(x)$	$\ln g $	Substitute $u = g$
Polynomial times exponential	Tabular parts	Reduction formula
The integral reappears after parts	Solve algebraically for it	Check the sign carefully
A symmetric interval	Split into even and odd parts	Pair x with $-x$
$1/(f(x) + f(a + b - x))$	Reflection; add the two copies	Parameter differentiation
$[0, \pi/2]$ with sin and cos	Complementarity $x \mapsto \pi/2 - x$	Beta function
$(0, \infty)$ and a reciprocal structure	$x \mapsto 1/x$	Split at $x = 1$
An arbitrary-looking exponent	Suspect it cancels; test the reflection	Introduce it as a parameter
A parameter already present	Differentiate under the sign	Frullani
No parameter, no antiderivative	Introduce one deliberately	Expand in a series
In or arctan over a polynomial	Series expansion, then term by term	Feynman
Γ , ζ or a polylogarithm emerges	Use its functional equations	Look up the special value
$\cos bx$ or $\sin bx$ with e^{ax}	Complexify to one exponential	Cyclic parts
An infinite limit	Test convergence before evaluating	Principal value if symmetric
An integrand that looks impossible	Check for a perfect derivative	Check for an odd factor
Absolute value, floor, or sign	Split at the breakpoints	Geometric reading
No closed form wanted	Bound, or quadrature with an error term	Euler and Maclaurin

Differentials and differential equations

Surface	First move	Fallback
"Approximately how much does"	$dy = f'(x) dx$	Second-order term if needed
First order, any form	Run the classification order	Try a substitution
Variables split by inspection	Separate; record lost constants	Substitute $v = y/x$
Linear in y	Integrating factor $e^{\int P}$	Linear in x instead
$y' + Py = Qy^n$	$u = y^{1-n}$	Recognise as separable if $P = 0$
$M dx + N dy = 0$	Test $M_y = N_x$	Find an integrating factor
Second order, constant coefficients	Characteristic polynomial	Undetermined coefficients
Forcing matches a root	Multiply the trial by x^s	Variation of parameters
Forcing is a quotient or sec	Variation of parameters	Reduction of order
$x^2 y'' + axy' + by = 0$	Trial $y = x^r$	Substitute $x = e^t$
Missing y , or missing x	Reduce order with $p = y'$	Exact second order
A system	Eigenvalues and eigenvectors	Matrix exponential
Discontinuous or impulsive forcing	Laplace transform	Solve piecewise and match
Variable coefficients, no closed form	Power series or Frobenius	Numerical solution
A word problem	Name the rate, write the balance	Match against a standard model
A fractional power of y	Check uniqueness at the initial point	Expect a family
The closed form has a pole	State the interval of validity	Solve for the blow-up time

Surface	First move	Fallback
Conditions at two points $y'' + \lambda y = 0$, homogeneous ends	Build the linear system; test invertibility Three sign cases, separately	Zero or infinitely many Read off the eigenvalues

Logic and puzzles

Surface	First move	Fallback
A repeated move or operation "Can X become Y"	What does it not change? Find an invariant separating them	A monovariant, for termination Construct a path
A tile and a region "Some two of them must"	Colour to match the tile Pigeonhole; name pigeons and holes	Weighting or area Extremal principle
"At least k share"	Generalised pigeonhole, $\lceil n/m \rceil$	Averaging
A count to prove equal "At least one" in a count	Bijection, or double counting Complementary counting	Induction Inclusion and exclusion
Identical items into boxes	Stars and bars	Generating function
Configurations up to symmetry	Burnside averaging	Canonical representative
Divisibility or digits	Choose the modulus from the shape	Factor the constraint
A large exponent $xy + ax + by = c$	Multiplicative order Add ab and factor	Euler, then CRT Enumerate divisors, both signs
A solution generating a smaller one Speakers who lie or tell the truth	Descent, or Vieta jumping $k_i \leftrightarrow S$ for each utterance	Well-ordering Case on the busiest variable
"I do not know"	Eliminate worlds where they would know	Iterate the rounds
Categories and relational clues	Elimination grid, all pairs	Branch on the smallest domain
Two players, alternating, finite	Label small positions	Grundy values and mex
Independent heaps	Nim-sum by exclusive-or	Grundy first if not heaps
A symmetric board "Expected number of"	Mirror strategy; handle the centre Indicators, then linearity	Pairing Stages, not objects
New evidence revises a belief	Odds form of Bayes	Enumerate the four outcomes
Two defensible probabilities "On average how long until"	The experiment is under-specified States, one equation each	Write the mechanism down Check the state is Markov
A limited number of tests	Count truths against b^t	Design for an even split
Simultaneous guesses, no talking	Correlate the errors with a parity code	Concentrate the losses

The final checking protocol

What this chapter is for. Chapter 2 argued that verification is a technique rather than a virtue, and gave the ladder of checks. This chapter is the operational version: a protocol short enough to run on every problem, organised by what kind of answer you are holding. It is deliberately mechanical. The point of a protocol is that it catches errors you did not suspect, which means it must be applied without first deciding whether it is needed.

The universal five

Run these on any answer, in any domain, before writing it down. They cost under a minute between them.

- 1. Sign and magnitude.** Is the answer the right sign? Is it plausibly the right size? A definite integral of a positive integrand that comes out negative is wrong, and a probability outside $[0, 1]$ is wrong, and both errors are common enough to be worth one second of attention.
- 2. Units and dimensions.** If the problem has physical content, do the units match? If it has a parameter, does the answer scale with it correctly? An answer that should double when a length doubles, and does not, is wrong.
- 3. Degenerate cases.** Set a parameter to 0, to 1, or to infinity. The answer should degenerate to something you can check independently. This is the single most productive check in the list.
- 4. One numerical evaluation.** Put in a concrete value and compare against a direct computation. Two or three matching digits is enough to catch essentially every algebraic slip.
- 5. Symmetry.** If the problem is symmetric in two quantities, the answer must be. If it is invariant under a substitution, the answer must be too.

Checks by the shape of the answer

You are holding	The check that catches an error in it	The error it catches
An antiderivative	Differentiate it	A missed chain-rule factor
A definite integral	Numerical quadrature at the given bounds	Bounds not transformed
An integral with a $\ln$	Is the argument's sign controlled on the interval?	Missing absolute value
An integral after substitution	Do the new bounds match the new variable?	Bounds transformed at the end
A trigonometric substitution result	Is θ in the range the substitution allows?	Wrong branch of the inverse
A parameter-differentiation answer	Set the parameter to its trivial value	Wrong constant of integration
A derivative	Complex-step or central difference at a point	Sign slip in a product
An n th derivative formula	Test it at $n = 1$ and $n = 2$	Off-by-one in the pattern
A limit	Evaluate at three points approaching	Expansion truncated too early
A limit from L'Hopital	Were both hypotheses actually true?	Applied to a non-indeterminate form
A series sum	Partial sums against the closed form	Index shifted by one
A convergence verdict	Does the general term tend to zero?	Test misapplied
A rearranged series	Is it absolutely convergent?	Conditional rearrangement
An ODE solution	Substitute it back into the equation	Sign in the integrating factor
A general solution	Does it have the right number of constants?	A lost solution from division
An ODE with an initial condition	Does it satisfy the condition, as written?	Condition applied after a root
A closed-form model answer	Does it degenerate correctly as $t \rightarrow \infty$?	Wrong sign in an exponent
A count	Brute force it for $n = 1, 2, 3, 4$	Over- or undercounting by symmetry
A probability	Is it in $[0, 1]$? Does it sum to one over all cases?	A missing or duplicated case
An expectation	Simulate, or enumerate exactly for small n	Wrong indicator decomposition
A game verdict	Verify the two closure conditions	A pattern read from too few values
An invariant argument	Check the invariant on every move type	A move you did not consider

You are holding	The check that catches an error in it	The error it catches
A pigeonhole argument	Do the holes cover everything, exactly once?	An object with no hole
A number-theoretic claim	Enumerate the residues	A modulus that separates nothing
A grid-puzzle solution	Does it satisfy every clue, re-read?	A clue used in the wrong direction
A proof by induction	Does the step reach the base you checked?	Base case below the step's validity
Any answer at all	Test it at a value the derivation never used	A formula fitted to its own data

The three questions to ask when the check fails

A failed check is information, and there is a productive order in which to use it.

First, *where does the discrepancy start?* Evaluate both the answer and an independent computation at an intermediate stage. Binary-search the working: check the halfway line, then the quarter or three-quarter line. Three or four evaluations locate the bad step in a page of algebra, which is far faster than re-reading.

Second, *is the error in the answer or in the check?* A numerical check can be wrong: quadrature on an oscillatory integrand, finite differences on a badly conditioned derivative, and Monte Carlo with too few samples all produce confident nonsense. When the analytic answer is clean and the numeric one is not, suspect the numerics first, and re-run the check by a different method before rewriting the solution.

Third, *does the failure reveal a missing hypothesis rather than a mistake?* An answer that is correct on part of the domain and wrong elsewhere is usually not an algebra error at all. It is a branch, an interval of validity, or a case split that the working assumed away. That failure mode is the most valuable of the three, because the repair improves the answer rather than merely correcting it.

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